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Graphs #1 Graphs By Pattern · Python Only · 5 questions p.16	Graphs #128 Longest Consecutive SequenceMED LeetCode · SimplyLeet · WalkCC · LeetCode Array · Hash Table · Union Find Lists: Grind 169 CodeSignal Time Complexity: O(n) Space Complexity: O(n) Problem Given an unsorted array of integers nums, return the length of the longest consecutive elements sequence. You must write an algorithm that runs in O(n) time. Example 1: Input: nums = [100,4,200,1,3,2] Output: 4 Explanation: The longest consecutive elements sequence is [1, 2, 3, 4]. Therefore its length is 4. Example 2: Input: nums = [0,3,7,2,5,8,4,6,0,1] Output: 9 Example 3: Input: nums = [1,0,1,2] Output: 3 Constraints: Graphs Sheet 2 / 317 · LeetCode Mastery By Pattern · Python Only · 5-up Python p.17	Graphs • 0 <= nums.length <= 105 • -109 <= nums[i] <= 109 Brute Force (Python) # Brute Force Approach class Solution: def longestConsecutive(self, nums): # Brute Force O(n^2) – for each number extend its sequence num_set = set(nums) best = 0 for num in nums: length = 0 while num + length in num_set: length += 1 best = max(best, length) return best Community Solutions (Python) • WalkCC class Solution: def longestConsecutive(self, nums: List[int]) -> int: ans = 0 seen = set(nums) for num in seen: # `num` is the start of a sequence. if num - 1 in seen: continue length = 0 Graphs p.18

```

while num in seen:
    num += 1
    length += 1
ans = max(ans, length)

return ans

```

• LeetCode

```

class Solution:
    def longestConsecutive(self, nums: List[int]) -> int:
        s = set(nums)
        ans = 0
        d = defaultdict(int)
        for x in nums:
            y = x
            while y in s:
                s.remove(y)
                y += 1

            d[x] = d[y] + y - x
            ans = max(ans, d[x])
        return ans

```

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```

# Python solution for Longest Consecutive Sequence

def longestConsecutive(nums):
    # Create a set from the list for O(1) look-up times
    num_set = set(nums)
    longest_streak = 0

    # Iterate through each number in the set
    for num in num_set:
        # Only start counting if 'num-1' is not present in the set

        if num - 1 not in num_set:
            current_num = num
            current_streak = 1

            # Count consecutive numbers starting from current_num
            while current_num + 1 in num_set:
                current_num += 1
                current_streak += 1

            # Update the longest streak if current one is longer

            longest_streak = max(longest_streak, current_streak)

        return longest_streak

# Example usage:
print(longestConsecutive([100, 4, 200, 1, 3, 2])) #
Output: 4

```

• LC.ca

```

class Solution:
    def longestConsecutive(self, nums: List[int]) -> int:
        s = set(nums)
        ans = 0
        for x in nums:
            if x - 1 not in s:
                y = x + 1
                while y in s:
                    y += 1
                ans = max(ans, y - x)

        return ans

```

Graphs

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Graphs

p.20

Graphs

p.21

Graphs

#277 Find the Celebrity

MED

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Two Pointers · Graph · Interactive

Lists: Premium 98

Time Complexity: $O(n)$ - One pass for candidate selection and one for validation. Space Complexity: $O(1)$ - Only constant extra space is used.

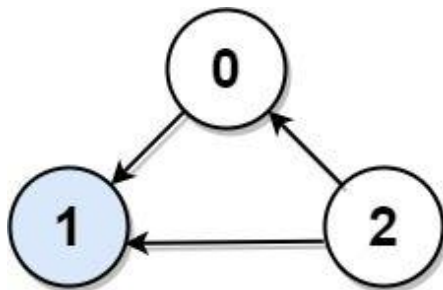
Problem

Suppose you are at a party with n people labeled from 0 to $n - 1$ and among them, there may exist one celebrity. The definition of a celebrity is that all the other $n - 1$ people know the celebrity, but the celebrity does not know any of them.

Now you want to find out who the celebrity is or verify that there is not one. You are only allowed to ask questions like: "Hi, A. Do you know B?" to get information about whether A knows B. You need to find out the celebrity (or verify there is not one) by asking as few questions as possible (in the asymptotic sense).

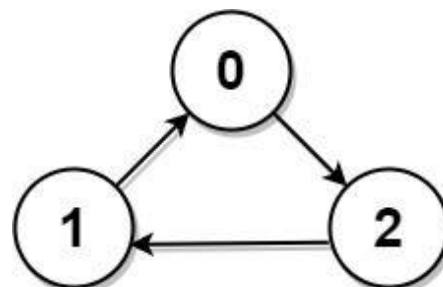
You are given an integer n and a helper function `bool knows(a, b)` that tells you whether a knows b. Implement a function `int findCelebrity(n)`. There will be exactly one celebrity if they are at the party.

Return the celebrity's label if there is a celebrity at the party. If there is no celebrity, return `-1`.
Note that the $n \times n$ 2D array `graph` given as input is not directly available to you, and instead only accessible through the helper function `knows`. `graph[i][j] == 1` represents person i knows person j , whereas `graph[i][j] == 0` represents person j does not know person i .
Example 1:



Input: `graph = [[1,1,0],[0,1,0],[1,1,1]]`
Output: 1
Explanation: There are three persons labeled with 0, 1 and 2. `graph[i][j] = 1` means person i knows person j , otherwise `graph[i][j] = 0` means person i does not know person j . The celebrity is the person labeled as 1 because both 0 and 2 know him but 1 does not know anybody.

Example 2:



Input: `graph = [[1,0,1],[1,1,0],[0,1,1]]`
Output: -1
Explanation: There is no celebrity.

Constraints:

- $n == \text{graph.length} == \text{graph}[i].\text{length}$

Graphs

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Graphs

p.23

Graphs

p.24

- $2 \leq n \leq 100$
- `graph[i][j]` is 0 or 1.
- `graph[i][i] == 1`

Follow up: If the maximum number of allowed calls to the API `knows` is $3 * n$, could you find a solution without exceeding the maximum number of calls?

Brute Force (Python)

```

# Brute Force Approach
class Solution:
    def findCelebrity(self, n):
        # Brute Force - try all paths via exhaustive DFS
        visited = set()
        def dfs(node):
            if node == dst: return True
            visited.add(node)
            for nxt in graph.get(node, []):
                if nxt not in visited and dfs(nxt):
                    return True
            return False
        return dfs(src)

        visited.remove(node)
        return False
        return dfs(src)

```

Community Solutions (Python)

• WalkCC

```

# The knows API is already defined for you.
# Returns a bool, whether a knows b
# def knows(a: int, b: int) -> bool:

```

```

class Solution:
    def findCelebrity(self, n: int) -> int:
        candidate = 0

        # Everyone knows the celebrity.

        for i in range(1, n):
            if knows(candidate, i):
                candidate = i

        # The candidate knows nobody and everyone knows the celebrity.
        for i in range(n):
            if i < candidate and knows(candidate, i) or not knows(i, candidate):
                return -1
            if i > candidate and not knows(i, candidate):
                return -1

        return candidate

```

• LeetCode

```

# The knows API is already defined for you.
# return a bool, whether a knows b
# def knows(a: int, b: int) -> bool:

```

```

class Solution:
    def findCelebrity(self, n: int) -> int:
        ans = 0
        for i in range(1, n):
            if knows(ans, i):

                ans = i
        for i in range(n):
            if ans != i:
                if knows(ans, i) or not knows(i, ans):
                    return -1
        return ans

```

• SimplyLeet

```

# Function definition for findCelebrity using the helper knows(a, b) API.
def findCelebrity(n):
    # Step 1: Candidate Selection.
    candidate = 0
    for i in range(1, n):
        # If candidate knows i, then candidate cannot be the celebrity.
        if knows(candidate, i):
            candidate = i # Select new candidate.

    # Step 2: Verification.

```

Graphs

p.25

Graphs

p.26

Graphs

p.27

```
# Check if candidate does not know any other person.
for i in range(n):
    if i != candidate:
        # Either candidate knows someone or someone
        # doesn't know candidate.
        if knows(candidate, i) or not knows(i,
candidate):
            return -1
return candidate

# Example helper function (for testing, this is normally
provided in the problem context).
def knows(a, b):

    # This function should return True if person a knows
    person b, otherwise False.
    pass
```

LC.ca

```
# The knows API is already defined for you.
# return a bool, whether a knows b
# def knows(a: int, b: int) -> bool:

class Solution:
    def findCelebrity(self, n: int) -> int:
        ans = 0
        for i in range(1, n):
            if knows(ans, i):
```

Graphs

p.28

```
ans = i
for i in range(n):
    if ans != i:
        if knows(ans, i) or not knows(i, ans):
            return -1
return ans
```

Graphs

p.29

Graphs

#1136 Parallel Courses

MED

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Graph · Topological Sort

Lists: Premium 98

Time Complexity: $O(n + m)$ where n is the number of courses and m is the length of relations (prerequisites).
Space Complexity: $O(n + m)$ for storing the graph, in-degree counts, and the processing queue

Problem

You are given an integer n , which indicates that there are n courses labeled from 1 to n . You are also given an array relations where $relations[i] = [prevCoursei, nextCoursei]$, representing a prerequisite relationship between course $prevCoursei$ and course $nextCoursei$: course $prevCoursei$ has to be taken before course $nextCoursei$.

In one semester, you can take any number of courses as long as you have taken all the prerequisites in the previous semester for the courses you are taking. Return the minimum number of semesters needed to take all courses. If there is no way to take all the courses, return -1 .

Example 1:

- All the pairs $[prevCoursei, nextCoursei]$ are unique.
- Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def minimumSemesters(self, numCourses, relations):
        # Brute Force - try all paths via exhaustive DFS
        visited = set()
        def dfs(node):
            if node == dst: return True
            visited.add(node)
            for nxt in graph.get(node, []):
                if nxt not in visited and dfs(nxt):
                    return True
            visited.remove(node)
            return False
        return dfs(src)
```

Community Solutions (Python)

• WalkCC

```
from enum import Enum
```

```
class State(Enum):
    INIT = 0
    VISITING = 1
    VISITED = 2
```

```
class Solution:
```

Graphs

p.33

Input: $n = 3$, $relations = [[1,3],[2,3]]$
Output: 2
Explanation: The figure above represents the given graph.
In the first semester, you can take courses 1 and 2.
In the second semester, you can take course 3.

Example 2:

Input: $n = 3$, $relations = [[1,2],[2,3],[3,1]]$
Output: -1
Explanation: No course can be studied because they are prerequisites of each other.

Constraints:

- $1 \leq n \leq 5000$
- $1 \leq relations.length \leq 5000$
- $relations[i].length == 2$
- $1 \leq prevCoursei, nextCoursei \leq n$
- $prevCoursei \neq nextCoursei$

Graphs

p.32

```
def minimumSemesters(self, n: int, relations:
List[List[int]]) -> int:
    graph = [[] for _ in range(n)]
    states = [State.INIT] * n
    depth = [1] * n
```

```
for u, v in relations:
    graph[u - 1].append(v - 1)
```

```
def hasCycle(u: int) -> bool:
    if states[u] == State.VISITING:
```

```
        return True
    if states[u] == State.VISITED:
        return False
    states[u] = State.VISITING
    for v in graph[u]:
        if hasCycle(v):
            return True
        depth[u] = max(depth[u], 1 + depth[v])
    states[u] = State.VISITED
    return False
```

```
if any(hasCycle(i) for i in range(n)):
    return -1
return max(depth)
```

Graphs

p.34

```
class Solution:
    def minimumSemesters(self, n: int, relations:
List[List[int]]) -> int:
        graph = [[] for _ in range(n)]
        inDegrees = [0] * n

        # Build the graph.
        for u, v in relations:
            graph[u - 1].append(v - 1)
            inDegrees[v - 1] += 1
```

```
# Perform topological sorting.
q = collections.deque([i for i, d in
enumerate(inDegrees) if d == 0])
```

```
step = 0
while q:
    for _ in range(len(q)):
        u = q.popleft()
        n -= 1
        for v in graph[u]:
            inDegrees[v] -= 1
```

```
        if inDegrees[v] == 0:
            q.append(v)
        step += 1
    return step if n == 0 else -1
```

• LeetDoocs

Graphs

p.35

```
class Solution:
    def minimumSemesters(self, n: int, relations:
List[List[int]]) -> int:
        g = defaultdict(list)
        indeg = [0] * n
        for prev, nxt in relations:
            prev, nxt = prev - 1, nxt - 1
            g[prev].append(nxt)
            indeg[nxt] += 1
        q = deque(i for i, v in enumerate(indeg) if v ==
0)
        ans = 0

        while q:
            ans += 1
            for _ in range(len(q)):
                i = q.popleft()
                n -= 1
                for j in g[i]:
                    indeg[j] -= 1
                    if indeg[j] == 0:
                        q.append(j)
            return -1 if n else ans
```

• SimplyLeet

Graphs

p.36

Python Implementation

```
from collections import deque

def minimumSemesters(numCourses, relations):
    # Build graph and in-degree dictionary.
    graph = {i: [] for i in range(1, numCourses + 1)}
    in_degree = {i: 0 for i in range(1, numCourses + 1)}

    # Construct graph from relations.

    for pre, course in relations:
        graph[pre].append(course)
        in_degree[course] += 1

    # Initialize queue with all courses having no prerequisites.
    queue = deque()
    for course in range(1, numCourses + 1):
        if in_degree[course] == 0:
            queue.append(course)

    semesters = 0
    takenCourses = 0

    # Process each level (semester) using BFS.
    while queue:
        semesters += 1
        current_level_size = len(queue)
        for _ in range(current_level_size):
            course = queue.popleft()
            takenCourses += 1

            for neighbor in graph[course]:
                in_degree[neighbor] -= 1
                if in_degree[neighbor] == 0:
                    queue.append(neighbor)

    # If not all courses have been taken, a cycle exists.
    return semesters if takenCourses == numCourses else -1
```

Graphs

p.37

```
# Decrement in-degree for neighboring
courses.
for neighbor in graph[course]:
    in_degree[neighbor] -= 1
    if in_degree[neighbor] == 0:
        queue.append(neighbor)

# If not all courses have been taken, a cycle exists.
return semesters if takenCourses == numCourses else -1

# Example usage:

print(minimumSemesters(3, [[1, 3], [2, 3]])) # Expected
output: 2
```

• LC.ca

```
class Solution:
    def minimumSemesters(self, n: int, relations:
List[List[int]]) -> int:
        g = defaultdict(list)
        indeg = [0] * n
        for prev, nxt in relations:
            prev, nxt = prev - 1, nxt - 1
            g[prev].append(nxt)
            indeg[nxt] += 1
        q = deque(i for i, v in enumerate(indeg) if v ==
0)
        ans = 0
```

Graphs

p.38

```
while q:
    ans += 1
    for _ in range(len(q)):
        i = q.popleft()
        n -= 1
        for j in g[i]:
            indeg[j] -= 1
            if indeg[j] == 0:
                q.append(j)
    return -1 if n else ans
```

Graphs

p.39

Graphs

#305 Number of Islands II

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Union Find · Matrix

Lists: Premium 98

Time Complexity: $O(K * \alpha(m * n))$ where K is the number of operations and α is the inverse Ackermann function (almost constant time per union/find). Space Complexity: $O(m * n)$ for storing the union-find st

Problem

You are given an empty 2D binary grid grid of size m x n. The grid represents a map where 0's represent water and 1's represent land. Initially, all the cells of grid are water cells (i.e., all the cells are 0's).

We may perform an add land operation which turns the water at position into a land. You are given an array positions where positions[i] = [ri, ci] is the position (ri, ci) at which we should operate the ith operation.

Return an array of integers answer where answer[i] is the number of islands after turning the cell (ri, ci) into a land.

An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are all surrounded

Graphs

p.40

by water.

Example 1:

Graphs

p.41

Input: m = 3, n = 3, positions = [[0,0],[0,1],[1,2],[2,1]]

Output: [1,1,2,3]

Explanation:

Initially, the 2d grid is filled with water.

- Operation #1: addLand(0, 0) turns the water at grid[0][0] into a land. We have 1 island.

- Operation #2: addLand(0, 1) turns the water at grid[0][1] into a land. We still have 1 island.

- Operation #3: addLand(1, 2) turns the water at grid[1][2] into a land. We have 2 islands.

- Operation #4: addLand(2, 1) turns the water at grid[2][1] into a land. We have 3 islands.

Example 2:

Input: m = 1, n = 1, positions = [[0,0]]

Output: [1]

Constraints:

- 1 <= m, n, positions.length <= 104
- 1 <= m * n <= 104
- positions[i].length == 2
- 0 <= ri < m
- 0 <= ci < n

Follow up: Could you solve it in time complexity $O(k \log(mn))$, where k == positions.length?

Brute Force (Python)

Graphs

p.42

```
# Brute Force Approach
class Solution:
    def numIslands2(self, m, n, positions):
        # Brute Force - try all paths via exhaustive DFS
        visited = set()
        def dfs(node):
            if node == dst: return True
            visited.add(node)
            for nxt in graph.get(node, []):
                if nxt not in visited and dfs(nxt):
                    return True
        return True

        visited.remove(node)
        return False
    return dfs(src)
```

Community Solutions (Python)

• WalkCC

```
class UnionFind:
    def __init__(self, n: int):
        self.id = [-1] * n
        self.rank = [0] * n

    def unionByRank(self, u: int, v: int) -> None:
        i = self.find(u)
        j = self.find(v)
        if i == j:
            return
```

Graphs

p.43

```
if self.rank[i] < self.rank[j]:
    self.id[i] = j
elif self.rank[i] > self.rank[j]:
    self.id[j] = i
else:
    self.id[i] = j
    self.rank[j] += 1

def find(self, u: int) -> int:
    if self.id[u] != u:
        self.id[u] = self.find(self.id[u])
    return self.id[u]
```

class Solution:

```
def numIslands2(
self,
m: int,
n: int,
positions: list[list[int]],

) -> list[int]:
    DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0))
    ans = []
    seen = [[False] * n for _ in range(m)]
    uf = UnionFind(m * n)
    count = 0

    def getId(i: int, j: int, n: int) -> int:
        return i * n + j
```

Graphs

p.44

```
for i, j in positions:
    if seen[i][j]:
        ans.append(count)
        continue
    seen[i][j] = True
    id = getId(i, j, n)
    uf.id[id] = id
    count += 1
    for dx, dy in DIRS:
        x = i + dx
        y = j + dy
        if x < 0 or x == m or y < 0 or y == n:
            continue
        neighborId = getId(x, y, n)
        if uf.id[neighborId] == -1: # water
            continue
        currentParent = uf.find(id)
        neighborParent = uf.find(neighborId)
        if currentParent != neighborParent:
            uf.unionByRank(currentParent, neighborParent)
```

count -= 1

ans.append(count)

return ans

• SimplyLeet

Graphs

p.45

```
# Python implementation using Union Find with path
compression and union by rank.
class UnionFind:
    def __init__(self, size):
        # parent: stores parent of each node. -1
        indicates water.
        self.parent = [-1] * size
        self.rank = [0] * size
        self.count = 0 # number of islands

    def find(self, x):
        # Find with path compression.

        if self.parent[x] != x:
            self.parent[x] = self.find(self.parent[x])
        return self.parent[x]

    def union(self, x, y):
        # Union the sets; if union is successful, reduce
        island count.
        rootX = self.find(x)
        rootY = self.find(y)
        if rootX == rootY:
            return False
```

Graphs

p.46

```
# union by rank.
if self.rank[rootX] < self.rank[rootY]:
    self.parent[rootX] = rootY
elif self.rank[rootX] > self.rank[rootY]:
    self.parent[rootY] = rootX
else:
    self.parent[rootY] = rootX
    self.rank[rootX] += 1
self.count -= 1
return True

class Solution:
    def numIslands2(self, m, n, positions):
        uf = UnionFind(m * n)
        result = []
        directions = [(0, 1), (1, 0), (0, -1), (-1, 0)]

        for r, c in positions:
            idx = r * n + c
            # If the position is already land, skip.
```

Graphs

p.47

```
if uf.parent[idx] != -1:
    result.append(uf.count)
    continue

# Mark this cell as land forming a new
island.
uf.parent[idx] = idx
uf.count += 1

# Union with all 4 neighbors if they are
already land.
for dr, dc in directions:

    nr, nc = r + dr, c + dc
    nidx = nr * n + nc
    if 0 <= nr < m and 0 <= nc < n and
uf.parent[nidx] != -1:
        uf.union(idx, nidx)
        result.append(uf.count)
    return result

# Example run:
# s = Solution()
# print(s.numIslands2(3, 3, [[0,0],[0,1],[1,2],[2,1]]))
```

• LC.ca

Graphs

p.48

```
class Solution:
    def numIslands2(self, m: int, n: int, positions:
List[List[int]]) -> List[int]:
        def check(i, j):
            return 0 <= i < m and 0 <= j < n and
            grid[i][j] == 1

        def find(x):
            if p[x] != x:
                p[x] = find(p[x])
            return p[x]

        p = list(range(m * n))
        grid = [[0] * n for _ in range(m)]
        res = []
        cur = 0 # current island count
        for i, j in positions:
            if grid[i][j] == 1: # already counted, same
            as previous island count
                res.append(cur)
                continue
            grid[i][j] = 1
            cur += 1
```

Graphs

p.49

```
for x, y in [(-1, 0), (1, 0), (0, -1), (0,
1)]:
    if check(i + x, j + y) and find(i * n + j)
!= find((i + x) * n + j + y):
        p[find(i * n + j)] = find((i + x) * n
+ j + y)
        cur -= 1
        # 1 0 1
        # 0 1 0
        # for above, if setting 1st-row
        # cur will -1 for 3 times
        # but cur += 1 after grid[i][j] = 1
        so cur final result is 1
        res.append(cur)

    return res

#####

class UnionFind(object):
    def __init__(self, m, n):
        self.dad = [i for i in range(m * n)]
        self.rank = [0 for _ in range(m * n)]

    def find(self, x):
```

Graphs

p.50

```
if self.dad[x] != x:
    self.dad[x] = self.find(self.dad[x])
    return self.dad[x]

def union(self, xy): # xy is a tuple (x,y), with x and y
value inside
    x, y = map(self.find, xy)
    if x == y:
        return False
    if self.rank[x] > self.rank[y]:
        self.dad[y] = x

    elif self.rank[x] < self.rank[y]:
        self.dad[x] = y
    else:
        self.dad[y] = x # search to the left, to find
parent
        self.rank[x] += 1 # now x a parent, so +1 its rank
    return True

class Solution(object):
    def numIslands2(self, m, n, positions):
```

Graphs

p.51

```
"""
:type m: int
:type n: int
:type positions: List[List[int]]
:rtype: List[int]
"""
uf = UnionFind(m, n)
ans = 0
ret = []
dirs = [(0, -1), (0, 1), (1, 0), (-1, 0)]

grid = set()
for i, j in positions:
    ans += 1
    grid |= {(i, j)}
    for di, dj in dirs:
        ni, nj = i + di, j + dj
        if 0 <= ni < m and 0 <= nj < n and (ni, nj) in
grid:
            if uf.union((ni * n + nj, i * n + j)):
                ans -= 1
            ret.append(ans)

    return ret

#####

class UnionFind:
    def __init__(self, n: int):
        self.p = list(range(n))
        self.size = [1] * n
```

Graphs

p.52

```
def find(self, x: int):
    if self.p[x] != x:
        self.p[x] = self.find(self.p[x])
    return self.p[x]

def union(self, a: int, b: int) -> bool:
    pa, pb = self.find(a - 1), self.find(b - 1)
    if pa == pb:
        return False

    if self.size[pa] > self.size[pb]:
        self.p[pb] = pa
        self.size[pa] += self.size[pb]
    else:
        self.p[pa] = pb
        self.size[pb] += self.size[pa]
    return True

class Solution:

    def numIslands2(self, m: int, n: int, positions:
List[List[int]]) -> List[int]:
        uf = UnionFind(m * n)
        grid = [[0] * n for _ in range(m)]
        ans = []
        dirs = (-1, 0, 1, 0, -1)
        cnt = 0
        for i, j in positions:
            if grid[i][j]:
                ans.append(cnt)
                continue
```

Graphs

p.53

```
grid[i][j] = 1
cnt += 1
for a, b in pairwise(dirs):
    x, y = i + a, j + b
    if (
        0 <= x < m
        and 0 <= y < n
        and grid[x][y]
        and uf.union(i * n + j, x * n + y)
    ):

        cnt -= 1
        ans.append(cnt)
    return ans
```

Graphs

p.54

Input: str1 = "aabcc", str2 = "ccdee"
Output: true
Explanation: Convert 'c' to 'e' then 'b' to 'd' then 'a' to 'c'. Note that the order of conversions matter.

p.55

```
class Solution:
    def canConvert(self, str1: str, str2: str) -> bool:
        if str1 == str2:
            return True

        mappings = {}

        for a, b in zip(str1, str2):
            if mappings.get(a, b) != b:
                return False

        mappings[a] = b

# No letter in the str1 maps to > 1 letter in the str2
# and there is at
# least one temporary letter can break any loops.
return len(set(str2)) < 26
```

- p.56

```
# Python solution with line-by-line comments
def canConvert(str1: str, str2: str) -> bool:
    # If both strings are equal, no conversion is needed.
    if str1 == str2:
        return True

    # Dictionary to hold the mapping from characters in
    str1 to characters in str2
    mapping = {}

    # Iterate through the characters of both strings
    simultaneously.
```

p.57

```
# Example Usage:
#print(canConvert("aabcc", "ccdee")) # Expected True
#print(canConvert("leetcode", "codeleet")) # Expected
False
```

- p.58

```
elif d[a] != b:
    return False
return True
```

p.59

We first insert all the numbers into a hash set to enable $O(1)$ membership checks. For each number in the set, we check if it is the start of a sequence by ensuring that the number minus one is not in the set. If it is the beginning, we then count all consecutive numbers following it, updating the maximum sequence length when necessary. This method guarantees that we only count each number once, satisfying the $O(n)$ time requirement.

p.61

The approach consists of two main steps:

- Candidate Selection: Iterate through the people and maintain a candidate. For every person i , if the current candidate knows i , then set candidate = i . This works because if candidate knows i , candidate cannot be the celebrity.
- Verification: Check that the candidate does not know any other person and that every other person knows the candidate. If both conditions hold, candidate is the celebrity; otherwise, return -1.

Graphs Sheet 7 / 317 · LeetMastery By Pattern · Python Only · 9-Up Portrait p.62

We solve the problem using a topological sort with Kahn's algorithm. First, construct a graph representation using an adjacency list and compute the in-degree for

p.63

each course. Initialize a queue with courses that have no prerequisites (in-degree == 0). Then, process courses level by level. Each level processed represents one semester. For every course taken in that semester, decrease the in-degree of all its neighbors. Add any neighbor with an in-degree that becomes zero to the queue for the next semester. If, after processing, the total courses taken is less than n, then a cycle exists and return -1. Otherwise, the number of semesters processed is the answer.

#305 Number of Islands II [Hard]

Key Insights

- Use Union Find (Disjoint Set Union) to efficiently manage connected components (islands).
- Map 2D grid coordinates to a 1D index.
- When adding a new land, check its 4 neighbors (up, down, left, right) and perform unions if they are also lands.
- Avoid duplicate processing if the same cell is added multiple times.
- Maintain a counter for islands that is updated during each union operation.

Space & Time

Time Complexity: O(K * α(m))

Graphs p.64

exists at least one character not used in str2 (i.e., distinct characters in str2 is less than 26), the transformation is possible. The important trick is identifying the need for a free character when the target mapping is “full” (i.e., covers all 26 letters). Without a free character, you cannot avoid cycles where a temporary intermediate is necessary.

Greedy

#2 Greedy

By Pattern · Python Only · 6 questions

Graphs p.67

```
class Solution:
    def longestPalindrome(self, s: str) -> int:
        odd = defaultdict(int)
        cnt = 0
        for c in s:
            odd[c] ^= 1
            cnt += 1 if odd[c] else -1
        return len(s) - cnt + 1 if cnt else len(s)
```

• SimplyLeet

```
# Python solution for the Longest Palindrome problem

def longestPalindrome(s: str) -> int:
    # Create a dictionary to store the frequency of each character
    char_counts = {}
    for char in s:
        # Increment the count for the character
        char_counts[char] = char_counts.get(char, 0) + 1

    # Variable to store the length of the palindrome
```

Greedy p.70

n)) where K is the number of operations and α is the inverse Ackermann function (almost constant time per union/find).
Space Complexity: O(mn) for storing the union-find structure.

Solution

The solution relies on the Union Find data structure to dynamically merge islands. When a new land cell is added:
– If the cell is already land, record the current island count. – Otherwise, mark the cell as land and increment the island count. – Check the four neighboring cells; if any neighbor is land, attempt to union the current cell with that neighbor. – If a union actually happens (i.e., the two cells were in separate islands), decrement the island counter. – Output the island count after each add-land operation. The key trick is converting 2D indices to a 1D representation (r * n + c) and using path compression along with union by rank in the Union Find implementation for optimal performance.

#1153 String Transforms Into Another String [Hard]

Key Insights

- Each character mapping from str1 to str2 must be consistent; one character in str1 always maps to the same character in str2.

Graphs p.65

Greedy

#409 Longest PalindromeEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Greedy

Lists: Grind 169

Time Complexity: O(n), where n is the length of the string. Space Complexity: O(1) since the frequency counter will have at most 52 keys (considering both uppercase and lowercase letters).

Problem

Given a string s which consists of lowercase or uppercase letters, return the length of the longest palindrome that can be built with those letters. Letters are case sensitive, for example, "Aa" is not considered a palindrome.

Example 1:

```
Input: s = "abcccd"
Output: 7
Explanation: One longest palindrome that can be built is "dccaccd", whose length is 7.
```

Example 2:

Greedy p.68

```
palindrome_length = 0
# Flag to indicate whether we can add one more character in the middle for an odd count
center_added = False

# Iterate through the character counts
for count in char_counts.values():
    # Add the largest even number from count to palindrome_length
    palindrome_length += (count // 2) * 2
    # If count is odd and we haven't added the center character yet
    if count % 2 == 1 and not center_added:

        palindrome_length += 1
        center_added = True

    return palindrome_length

# Example usage:
print(longestPalindrome("abcccd"))
```

• LC.ca

```
class Solution:
    def longestPalindrome(self, s: str) -> int:
        cnt = Counter(s)
        ans = 0
        for v in cnt.values():
            ans += v - (v & 1)
            ans += (ans & 1 ^ 1) and (v & 1)
        return ans
```

Greedy Sheet 8 / 317 · LeetMastery By Pattern · Python Only · 9-up Portrait

- A bijective mapping is not necessary; multiple characters in str1 can map to the same character in str2.
- The order of conversions matters. When there is a cycle (e.g., mapping 'a'-'>'b' and 'b'-'>'a'), an intermediate unused character might be required to break the cycle.
- If str2 uses all 26 letters, no temporary character is available to break mapping cycles. In this case, transformation is possible only if str1 is already equal to str2.

Space & Time

Time Complexity: O(n), where n is the length of the strings, because we traverse the strings to build and check the mapping.
Space Complexity: O(1) or O(26), which is constant space, to store the mapping for the 26 lowercase English letters.

Solution

The approach is as follows:

– If str1 equals str2, return true immediately. – Build a mapping from each character in str1 to the corresponding character in str2. While creating the mapping, check that each character maps consistently. – Count the number of distinct characters in str2. If this number is 26 and str1 is not already equal to str2, it is impossible to perform the transformation because there is no spare character available as a temporary placeholder. – If the mapping is consistent and there

Graphs p.66

```
Input: s = "a"
Output: 1
Explanation: The longest palindrome that can be built is "a", whose length is 1.
```

Constraints:

- 1 <= s.length <= 2000
- s consists of lowercase and/or uppercase English letters only.

Community Solutions (Python)

• WalkCC

```
class Solution:
    def longestPalindrome(self, s: str) -> int:
        ans = 0
        count = collections.Counter(s)

        for c in count.values():
            ans += c if c % 2 == 0 else c - 1

        hasOddCount = any(c % 2 == 1 for c in count.values())
        return ans + hasOddCount
```

• LeetDoocs

```
class Solution:
    def longestPalindrome(self, s: str) -> int:
        cnt = Counter(s)
        ans = sum(v // 2 * 2 for v in cnt.values())
        ans += int(ans < len(s))
        return ans
```

Greedy p.69

Greedy

#134 Gas StationMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Greedy

Lists: Grind 169

Time Complexity: O(n), where n is the number of gas stations, as we traverse the list only once. Space Complexity: O(1), as no extra space is used other than a few variables.

Problem

There are n gas stations along a circular route, where the amount of gas at the ith station is gas[i]. You have a car with an unlimited gas tank and it costs cost[i] of gas to travel from the ith station to its next (i + 1)th station. You begin the journey with an empty tank at one of the gas stations.

Given two integer arrays gas and cost, return the starting gas station's index if you can travel around the circuit once in the clockwise direction, otherwise return -1. If there exists a solution, it is guaranteed to be unique.

Example 1:

Greedy p.72

Greedy

#280 Wiggle SortMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Greedy · Sorting

Lists: Premium 98

Time Complexity: O(n) — Only one pass through the array is needed. Space Complexity: O(1) — The modifications are done in-place without extra data structures.

Problem

Given an integer array nums, reorder it such that nums[0] <= nums[1] >= nums[2] <= nums[3].... You may assume the input array always has a valid answer.

Example 1:

Input: nums = [3,5,2,1,6,4]

Output: [3,5,1,6,2,4]

Explanation: [1,6,2,5,3,4] is also accepted.

Example 2:

Input: nums = [6,6,5,6,3,8]

Output: [6,6,5,6,3,8]

Constraints:

- 1 <= nums.length <= 5 * 104

Greedyp.82

- 0 <= nums[i] <= 104
- It is guaranteed that there will be an answer for the given input nums.

Follow up: Could you solve the problem in O(n) time complexity?

Brute Force (Python)

Brute Force Approach

class Solution:

def wiggleSort(self, nums):

Brute Force O(n^2) - check all subsets/orderings exhaustively

from itertools import permutations

best = float('inf')

for perm in permutations(nums):

cost = sum(perm)

best = min(best, cost)

return best

Community Solutions (Python)

- WalkCC

class Solution:

def wiggleSort(self, nums: List[int]) -> None:

1. If i is even, then nums[i] <= nums[i - 1].

2. If i is odd, then nums[i] >= nums[i - 1].

for i in range(1, len(nums)):

if (i % 2 == 0 and nums[i] > nums[i - 1] or

i % 2 == 1 and nums[i] < nums[i - 1]):

nums[i], nums[i - 1] = nums[i - 1], nums[i]

- LeetCode

Greedyp.83

class Solution:

def wiggleSort(self, nums: List[int]) -> None:

"""

Do not return anything, modify nums in-place instead.

"""

for i in range(1, len(nums)):

if (i % 2 == 1 and nums[i] < nums[i - 1]) or (

i % 2 == 0 and nums[i] > nums[i - 1])

:

nums[i], nums[i - 1] = nums[i - 1],

nums[i]

- SimplyLeet

Python solution for Wiggle Sort

def wiggleSort(nums):

Iterate over the list starting from index 1

for i in range(1, len(nums)):

Check the condition based on the parity of the index

if i % 2 == 1:

For odd index, if previous element is greater, swap them

if nums[i] < nums[i-1]:

nums[i], nums[i-1] = nums[i-1], nums[i]

else:

Greedyp.84

For even index, if previous element is smaller, swap them

if nums[i] > nums[i-1]:

nums[i], nums[i-1] = nums[i-1], nums[i]

return nums

Example usage:

nums = [3,5,2,1,6,4]

print(wiggleSort(nums))

- LC.ca

'''

math prove:

example [3, 5, 2, 1]

when reaching 2, all good

then next, is 1,

so here comes the question, will the number after 2 violating the wiggle rule?

answer is no.

in example, when reaching 2:

Greedyp.85

* if [3, 5, 2, 10], so 10 is bigger than 2, then no swap needed according to the if () conditions

* if [3, 5, 2, 1], so previous round, guaranteed that 2 is smaller than 5.

And, now a swap needed meaning the number is smaller than 2, i.e. must be smaller than 5, so wiggle rule is still good

'''

class Solution:

def wiggleSort(self, nums: List[int]) -> None:

"""

Do not return anything, modify nums in-place instead.

"""

for i in range(1, len(nums)):

if (i % 2 == 1 and nums[i] < nums[i - 1]) or (

i % 2 == 0 and nums[i] > nums[i - 1])

:

nums[i], nums[i - 1] = nums[i - 1],

nums[i]

#####

class Solution: # sort list first

Greedyp.86

def wiggleSort(self, nums: List[int]) -> None:

nums.sort() # Sort the list in ascending order

n = len(nums)

for i in range(1, n - 1, 2):

nums[i], nums[i + 1] = nums[i + 1], nums[i] # Swap adjacent elements

"""

If the list has an odd length, the last element will be compared with the second-to-last element and swapped if necessary to form a wiggle sequence.

"""

if n % 2 == 0 or nums[-1] < nums[-2]:

nums[-1], nums[-2] = nums[-2], nums[-1]

#####

import random

class Solution:

def wiggleSort(self, nums: List[int]) -> None:

"""

Greedyp.87

Do not return anything, modify nums in-place instead.

"""

if len(nums) <= 1:

return

Find the median using quickselect

n = len(nums)

mid = self.quickselect(nums, 0, n-1, (n-1)//2)

Create an empty array with the same length as nums

ans = [None] * n

Assign the median to every other element

i = 0

j = n-1 if n % 2 == 1 else n-2

for k in range(n):

if nums[k] < mid:

ans[j] = nums[k]

j -= 2

elif nums[k] > mid:

Greedyp.88

ans[i] = nums[k]

i += 2

else:

ans[j+1] = mid

Copy the result back to nums

nums[:] = ans

def quickselect(self, nums, left, right, k):

"""

Returns the k-th smallest element in nums[left:right+1].

"""

if left == right:

return nums[left]

Partition the subarray around a pivot element

pivot_idx = self.partition(nums, left, right)

Recursively select on the left or right subarray

if k == pivot_idx:

Greedyp.89

return nums[k]

elif k < pivot_idx:

return self.quickselect(nums, left,

pivot_idx-1, k)

else:

return self.quickselect(nums, pivot_idx+1,

right, k)

def partition(self, nums, left, right):

"""

Partitions the subarray nums[left:right+1] using the first element as the pivot.

Returns the final index of the pivot element.

"""

pivot = nums[left]

i, j = left+1, right

while i <= j:

if nums[i] <= pivot:

i += 1

elif nums[j] > pivot:

j -= 1

else:

nums[i], nums[j] = nums[j], nums[i]

nums[left], nums[j] = nums[j], nums[left]

return j

Greedyp.90

Sheet 10 / 317 · LeetCode Mastery By Pattern · Python Only · 9-Up Portrait

Greedy

#954 Array of Doubled PairsMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Greedy · Sorting

Lists: CodeSignal

Time Complexity: O(n log n) due to sorting. Space Complexity: O(n) for storing the frequency map.

Problem

Given an integer array of even length arr, return true if it is possible to reorder arr such that arr[2 * i + 1] = 2 * arr[2 * i] for every 0 <= i < len(arr) / 2, or false otherwise.

Example 1:

Input: arr = [3,1,3,6]
Output: false

Example 2:

Input: arr = [2,1,2,6]
Output: false

Example 3:

Input: arr = [4,-2,2,-4]
Output: true
Explanation: We can take two groups, [-2,-4] and [2,4] to form [-2,-4,2,4] or [2,4,-2,-4].

Greedyp.91

Constraints:

- 2 <= arr.length <= 3 * 104
- arr.length is even.
- 105 <= arr[i] <= 105

Brute Force (Python)

Brute Force Approach
class Solution:
 def canReorderDoubled(self, arr):
 # Brute Force O(n^2) - check all
 subsets/orderings exhaustively
 from itertools import permutations
 best = float('inf')
 for perm in permutations(arr):
 cost = sum(perm)
 best = min(best, cost)
 return best

Community Solutions (Python)

- WalkCC

class Solution:
 def canReorderDoubled(self, arr: list[int]) -> bool:
 count = collections.Counter(arr)

 for key in sorted(count, key=abs):
 if count[key] > count[2 * key]:
 return False
 count[2 * key] -= count[key]

 return True

Greedyp.92

- LeetCode

class Solution:
 def canReorderDoubled(self, arr: List[int]) -> bool:
 freq = Counter(arr)
 if freq[0] & 1:
 return False
 for x in sorted(freq, key=abs):
 if freq[x < 1] < freq[x]:
 return False
 freq[x < 1] -= freq[x]
 return True

- SimplyLeet

from collections import Counter

def canReorderDoubled(arr):
 # Count frequency for each element in arr
 count = Counter(arr)
 # Sort the array based on the absolute value of elements
 for x in sorted(arr, key=abs):
 # If no x is left to be paired, skip to the next element
 if count[x] == 0:
 continue

Greedyp.93

Check if double the current number exists in sufficient frequency
if count[2 * x] < count[x]:
 return False
Decrement the frequency for the double of x
count[2 * x] -= count[x]
return True

Example usage:
print(canReorderDoubled([4,-2,2,-4])) # Expected Output: True

- LC.ca

class Solution:
 def canReorderDoubled(self, arr: List[int]) -> bool:
 freq = Counter(arr)
 if freq[0] & 1:
 return False
 for x in sorted(freq, key=abs):
 if freq[x < 1] < freq[x]:
 return False
 freq[x < 1] -= freq[x]
 return True

Greedyp.94

Greedy

#2131 Longest Palindrome by Concatenating Two Letter WordsMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · String · Greedy · Counting

Lists: CodeSignal

Time Complexity: O(n), where n is the number of words, since we iterate through the list and perform constant time operations per word. Space Complexity: O(n), to store counts of words in the hash map

Problem

You are given an array of strings words. Each element of words consists of two lowercase English letters.

Create the longest possible palindrome by selecting some elements from words and concatenating them in any order. Each element can be selected at most once.

Return the length of the longest palindrome that you can create. If it is impossible to create any palindrome, return 0.

A palindrome is a string that reads the same forward and backward.

Example 1:

Greedyp.95

Input: words = ["lc","cl","gg"]
Output: 6
Explanation: One longest palindrome is "lc" + "gg" + "cl" = "lccgcl", of length 6.
Note that "clggcl" is another longest palindrome that can be created.

Example 2:

Input: words = ["ab","ty","yt","lc","cl","ab"]
Output: 8
Explanation: One longest palindrome is "ty" + "lc" + "cl" + "yt" = "tylcclyt", of length 8.
Note that "lcytycl" is another longest palindrome that can be created.

Example 3:

Input: words = ["cc","ll","xx"]
Output: 2
Explanation: One longest palindrome is "cc", of length 2.
Note that "ll" is another longest palindrome that can be created, and so is "xx".

Constraints:

- 1 <= words.length <= 105
- words[i].length == 2
- words[i] consists of lowercase English letters.

Brute Force (Python)

Greedyp.96

Brute Force Approach
class Solution:
 def longestPalindrome(self, words):
 # Brute Force O(n^2) - check all
 subsets/orderings exhaustively
 from itertools import permutations
 best = float('inf')
 for perm in permutations(words):
 cost = sum(perm)
 best = min(best, cost)
 return best

Community Solutions (Python)

- WalkCC

class Solution:
 def longestPalindrome(self, words: List[str]) -> int:
 ans = 0
 count = [0] * 26 for _ in range(26)]

 for a, b in words:
 i = ord(a) - ord('a')
 j = ord(b) - ord('a')
 if count[j][i]:
 ans += 4

Greedyp.97

count[j][i] -= 1
else:
 count[i][j] += 1

for i in range(26):
 if count[i][i]:
 return ans + 2

return ans

- LeetCode

class Solution:
 def longestPalindrome(self, words: List[str]) -> int:
 cnt = Counter(words)
 ans = x = 0
 for k, v in cnt.items():
 if k[0] == k[1]:
 x += v & 1
 ans += v // 2 * 2 * 2
 else:
 ans += min(v, cnt[k[::-1]]) * 2

 ans += 2 if x else 0
 return ans

- SimplyLeet

Greedyp.98

Python solution
from collections import Counter

class Solution:
 def longestPalindrome(self, words):
 # Count the frequency of each word
 word_count = Counter(words)
 length = 0 # Total words used count (each adds 2 letters)
 center_used = False

 for word, count in list(word_count.items()):
 reverse = word[::-1] # Get the reverse of the current word

 if word == reverse:
 # For words that are palindromic themselves (e.g. "gg")
 pairs = count // 2
 length += pairs * 2 # Each pair adds 2 words (4 letters)
 # If an extra word exists and center is not used, add one word in the center
 if count % 2 == 1 and not center_used:
 length += 1

Greedyp.99

<pre> center_used = True else: # For words that are not self-palindromic. if reverse in word_count: # We only count each pair once, so use minimum and then set the reverse count to 0. pair_count = min(count, word_count[reverse]) length += pair_count * 2 # each pair contributes two words # Set reverse count to 0 to avoid double counting later. word_count[reverse] = 0 return length * 2 # Each word is of length 2 # Example usage: # sol = Solution() # print(sol.longestPalindrome(["lc", "cl", "gg"])) # Output: 6</pre> • LC.ca <pre>class Solution: def longestPalindrome(self, words: List[str]) -> int: cnt = Counter(words) ans = x = 0 for k, v in cnt.items(): if k[0] == k[1]: x += v & 1 ans += v // 2 * 2 * 2 else: ans += min(v, cnt[k[::-1]]) * 2</pre> Greedy p.100	<pre>ans += 2 if x else 0 return ans</pre> Greedy p.101	Greedy p.102
>> Quick Review — Greedy 6 questions · insights · complexity · solution #409 Longest Palindrome [Easy] Key Insights - Count the frequency of each character. - For each character, you can use pairs (even count or odd count minus one) to form the palindrome. - If any character has an odd frequency, you can add one extra character as the center of the palindrome. Space & Time Time Complexity: O(n), where n is the length of the string. Space Complexity: O(1) since the frequency counter will have at most 52 keys (considering both uppercase and lowercase letters). Solution The solution uses a hash table to count the frequency of each character. For every character, the maximum number of characters that can be used in a palindrome is determined by taking the largest even number that is less than or equal to its frequency (which is frequency – (frequency % 2)). If there exists any character with an odd frequency, one of these can be used as the center of the palindrome, adding an extra character to the total Greedy p.103	length. This greedy approach efficiently computes the length of the largest possible palindrome from the available characters. #134 Gas Station [Medium] Key Insights - If the total amount of gas available is less than the total travel cost, then completing the circuit is impossible. - A greedy approach can be used by keeping track of the current surplus of gas. If the surplus becomes negative at any station, the journey cannot start from any station before that point. - Restart the journey from the next station whenever the surplus goes negative. - The problem guarantees that if a solution exists, it is unique. Space & Time Time Complexity: O(n), where n is the number of gas stations, as we traverse the list only once. Space Complexity: O(1), as no extra space is used other than a few variables. Solution The solution involves iterating over the list of stations while maintaining two variables: one for the current gas surplus and one for the total gas surplus. If at any Greedy p.104	station the current surplus drops below zero, the current starting station is not viable and we set the starting station to the next index and reset the current surplus. After one pass, if the total gas surplus is non-negative, then the identified station is the valid starting point. #179 Largest Number [Medium] Key Insights - The order in which numbers appear drastically affects the concatenated result. - A custom sorting strategy is required where two numbers are compared by checking which concatenation (first-second vs second-first) yields a larger number. - Edge case: When the numbers are all zeros, the result should be "0" instead of several leading zeros. Space & Time Time Complexity: O(n log n * k), where n is the number of elements in the array and k is the maximum number of digits in a number (due to string comparisons in the custom sort). Space Complexity: O(n * k) for storing and processing the strings. Solution We solve the problem by converting all integers to strings so that we can use string concatenation to Greedy p.105
compare the results. A custom comparator is defined that, given two number strings, compares the concatenated results in both orders. By sorting the array using this comparator in descending order, we ensure that placing the highest contributing number first yields the largest possible number. After sorting, a check is performed to return "0" if the highest number is "0". Finally, the sorted strings are concatenated to form the answer. #280 Wiggle Sort [Medium] Key Insights - The alternating "wiggle" requirement can be satisfied by ensuring two conditions: - For every odd index i, nums[i] should be greater than or equal to nums[i-1]. - For every even index i (where i > 0), nums[i] should be less than or equal to nums[i-1]. - A single pass through the array can ensure these conditions by swapping adjacent elements when the condition is violated. - This greedy approach works because the local adjustments guarantee the overall wiggle pattern without needing to sort the entire array. Space & Time Time Complexity: O(n) — Only one pass through the array is needed. Greedy p.106	Space Complexity: O(1) — The modifications are done in-place without extra data structures. Solution We can achieve a wiggle sort in one pass by iterating through the array and, at each index i (starting from 1), checking if the pair (nums[i-1], nums[i]) satisfies the wiggle condition. If i is odd, nums[i] should be at least nums[i-1]; if not, we swap them. Conversely, if i is even and nums[i] is greater than nums[i-1], then we swap them. These local swaps adjust the order incrementally while ensuring that the overall wiggle pattern is maintained. This greedy approach works because only two adjacent numbers are involved in any violation and fixing them locally does not disturb the rest of the order. #954 Array of Doubled Pairs [Medium] Key Insights - Sort the array based on the absolute value of the elements to handle negative numbers correctly. - Use a hash table (or frequency counter) to track the occurrence of each element. - For each number in the sorted list, if the number is still available in the frequency map, try to pair it with its double. - Decrement the appropriate counts and validate that it is possible to find sufficient doubles for all numbers. Greedy p.107	- Special case: when dealing with zeros, pairing works within the same value. Space & Time Time Complexity: O(n log n) due to sorting. Space Complexity: O(n) for storing the frequency map. Solution The solution starts by counting the frequency of each element in the array. Sorting the array by absolute value ensures that when we process an element, any potential pair (i.e., its double) comes later in the sequence. For each number, if there are available occurrences, we try to match it with its double. If there are not enough doubles available, the answer is false. If all elements can be paired appropriately by decrementing their counts in the frequency map, the function returns true. #2131 Longest Palindrome by Concatenating Two Letter Words [Medium] Key Insights - Pair words with their reverse (e.g., "ab" with "ba") to form a palindrome. - Handle words that are palindromic by themselves (like "gg" or "aa") differently; they can be used in pairs and one extra can be placed in the center. - Use a hash map (or dictionary) to count frequencies of each word. Greedy p.108

- For non-palindromic word pairs, consider each pair only once to avoid double counting.
- For self-palindromic words, count how many pairs can be used and whether an extra word can be placed in the middle to maximize length.

Space & Time

Time Complexity: O(n), where n is the number of words, since we iterate through the list and perform constant time operations per word.
Space Complexity: O(n), to store counts of words in the hash map.

Solution

The approach is to count the occurrences of each word using a hash map. For each word:
– If the word is not self-palindromic, check if its reverse exists and form pairs by considering the minimum count between the word and its reverse. – If the word is self-palindromic, form as many pairs as possible (using count // 2) and note if there is any word left; you can use one extra self-palindromic word as the center of the palindrome. Finally, compute the total length by multiplying the number of words used by 2, since each word contributes 2 characters.

JavaScript

Greedy p.109

#3 JavaScript

By Pattern · Python Only · 7 questions

JavaScript p.110

JavaScript

#2627 Debounce

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

JavaScript

Lists: Premium 98

Time Complexity: O(1) per call, as each invocation only clears and sets a timer. Space Complexity: O(1), since only a single timer reference is maintained.

Problem

Given a function fn and a time in milliseconds t, return a debounced version of that function.
A debounced function is a function whose execution is delayed by t milliseconds and whose execution is cancelled if it is called again within that window of time. The debounced function should also receive the passed parameters.
For example, let's say t = 50ms, and the function was called at 30ms, 60ms, and 100ms.
The first 2 function calls would be cancelled, and the 3rd function call would be executed at 150ms.
If instead t = 35ms, The 1st call would be cancelled, the 2nd would be executed at 95ms, and the 3rd would be executed at 135ms.

```
type F = (...p: any[]) => any;

function debounce(fn: F, t: number): F {
  let timeout: ReturnType<typeof setTimeout> |
    undefined;

  return function (...args) {
    if (timeout !== undefined) {
      clearTimeout(timeout);
    }
    timeout = setTimeout(() => {
      fn.apply(this, args);
    }, t);
  };
}

/**
 * const log = debounce(console.log, 100);
 * log('Hello'); // cancelled
 * log('Hello'); // cancelled
 * log('Hello'); // Logged at t=100ms
 */
```

• LC.ca

JavaScript p.114

The above diagram shows how debounce will transform events. Each rectangle represents 100ms and the debounce time is 400ms. Each color represents a different set of inputs.
Please solve it without using lodash's `_debounce()` function.

Example 1:

Input:
t = 50
calls = [
 {"t": 50, inputs: [1]},
 {"t": 75, inputs: [2]}
]
Output: [{"t": 125, inputs: [2]}]
Explanation:

Example 2:

Input:
t = 20
calls = [
 {"t": 50, inputs: [1]},
 {"t": 100, inputs: [2]}
]
Output: [{"t": 70, inputs: [1]}, {"t": 120, inputs: [2]}]
Explanation:

Example 3:

JavaScript p.112

- Constraints:
- 0 <= t <= 1000
 - 1 <= calls.length <= 10
 - 0 <= calls[i].t <= 1000
 - 0 <= calls[i].inputs.length <= 10

Community Solutions (JavaScript / TypeScript)

- WalkCC

```
type F = (...args: number[]) => void;

function debounce(fn: F, t: number): F {
  let timeout: ReturnType<typeof setTimeout> | undefined;
  return function (...args) {
    clearTimeout(timeout);
    timeout = setTimeout(() => fn(...args), t);
  };
}
```
- LeetCode

JavaScript p.113

```
type F = (...p: any[]) => any;

function debounce(fn: F, t: number): F {
  let timeout: ReturnType<typeof setTimeout> |
    undefined;

  return function (...args) {
    if (timeout !== undefined) {
      clearTimeout(timeout);
    }
    timeout = setTimeout(() => {
      fn.apply(this, args);
    }, t);
  };
}

/**
 * const log = debounce(console.log, 100);
 * log('Hello'); // cancelled
 * log('Hello'); // cancelled
 * log('Hello'); // Logged at t=100ms
 */
```

JavaScript p.115

JavaScript

#2628 JSON Deep Equal

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

JavaScript

Lists: Premium 98

Time Complexity: O(N) in the worst-case, where N is the total number of elements/values in the JSON structure. Space Complexity: O(N) due to the recursion call stack in the worst-case scenario of deep

Problem

Given two values o1 and o2, return a boolean value indicating whether two values, o1 and o2, are deeply equal.

For two values to be deeply equal, the following conditions must be met:

- If both values are primitive types, they are deeply equal if they pass the === equality check.
- If both values are arrays, they are deeply equal if they have the same elements in the same order, and each element is also deeply equal according to these conditions.
- If both values are objects, they are deeply equal if they have the same keys, and the associated values for each key are also deeply equal according to these conditions.

JavaScript p.116

You may assume both values are the output of JSON.parse. In other words, they are valid JSON.
Please solve it without using lodash's `_isEqual()` function

Example 1:

Input: o1 = {"x":1,"y":2}, o2 = {"x":1,"y":2}
Output: true
Explanation: The keys and values match exactly.

Example 2:

Input: o1 = {"y":2,"x":1}, o2 = {"x":1,"y":2}
Output: true
Explanation: Although the keys are in a different order, they still match exactly.

Example 3:

Input: o1 = {"x":null,"L":[1,2,3]}, o2 = {"x":null,"L":["1","2","3"]}
Output: false
Explanation: The array of numbers is different from the array of strings.

Example 4:

Input: o1 = true, o2 = false
Output: false
Explanation: true !== false

Constraints:

- 1 <= JSON.stringify(o1).length <= 105
- 1 <= JSON.stringify(o2).length <= 105

JavaScript p.117

• maxNestingDepth <= 1000

Community Solutions (JavaScript / TypeScript)

• WalkCC

```
type JSONValue =
| null
| boolean
| number
| string
| JSONValue[]
| { [key: string]: JSONValue };

function areDeeplyEqual(o1: JSONValue, o2: JSONValue):
boolean {
  if (o1 === o2) {

    return true;
  }
  if (o1 === null || o1 === undefined || o2 === null || o2
=== undefined) {
    return false;
  }
  if (typeof o1 !== 'object' || typeof o2 !== 'object') {
    return false;
  }
  if (Array.isArray(o1) !== Array.isArray(o2)) {
    return false;
  }
}
```

```
}
if (Object.keys(o1).length !== Object.keys(o2).length) {
  return false;
}
for (const key in o1) {
  if (!areDeeplyEqual(o1[key], o2[key])) {
    return false;
  }
}
return true;
}
```

JavaScript

#2630 Memoize

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

JavaScript

Lists: Premium 98

Time Complexity: O(1) average for each call assuming efficient hashing of arguments. Space Complexity: O(n) for caching n distinct input combinations.

Problem

Given a function fn, return a memoized version of that function.

A memoized function is a function that will never be called twice with the same inputs. Instead it will return a cached value.

fn can be any function and there are no constraints on what type of values it accepts. Inputs are considered identical if they are === to each other.

Example 1:

```
Input:
getInputs = () => [[2,2],[2,2],[1,2]]
fn = function (a, b) { return a + b; }
Output: [{"val":4,"calls":1},{val":4,"calls":1},{val":3,"calls":2}]
Explanation:
const inputs = getInputs();
const memoized = memoize(fn);
for (const arr of inputs) {
```

Example 2:

```
Input:
getInputs = () => [{},{},{}],[{},{},{}],[{},{},{}]]
fn = function (a, b) { return ({...a, ...b}); }
Output: [{"val":{},"calls":1},{val":{},"calls":2},{val":{},"calls":3}]
Explanation:
Merging two empty objects will always result in an empty object. It may seem like there should only be 1 call to fn() because of cache-hits, however none of those objects are === to each other.
```

Example 3:

```
Input:
getInputs = () => { const o = {}; return
[[o,o],[o,o],[o,o]]; }
fn = function (a, b) { return ({...a, ...b}); }
Output: [{"val":{},"calls":1},{val":{},"calls":1},{val":{},"calls":1}]
Explanation:
Merging two empty objects will always result in an empty object. The 2nd and 3rd third function calls result in a cache-hit. This is because every object passed in is identical.
```

Constraints:

- 1 <= inputs.length <= 105
- 0 <= inputs.flat().length <= 105
- inputs[i][j] !== NaN

Community Solutions (JavaScript / TypeScript)

• LeetDoocs

```
type Fn = (...params: any) => any;

function memoize(fn: Fn): Fn {
  const idxMap: Map<string, number> = new Map();
  const cache: Map<string, any> = new Map();

  const getIdx = (obj: any): number => {
    if (!idxMap.has(obj)) {
      idxMap.set(obj, idxMap.size);
    }
  }
}
```

```
return idxMap.get(obj)!;
});

return function (...params: any) {
  const key = params.map(getIdx).join(',');
  if (!cache.has(key)) {
    cache.set(key, fn(...params));
  }
  return cache.get(key)!;
};

/**
 * let callCount = 0;
 * const memoizedFn = memoize(function (a, b) {
 *   callCount += 1;
 *   return a + b;
 * })
 * memoizedFn(2, 3) // 5
 * memoizedFn(2, 3) // 5
 *
 * console.log(callCount) // 1
 */
```

• LC.ca

```
type Fn = (...params: any) => any;

function memoize(fn: Fn): Fn {
  const idxMap: Map<string, number> = new Map();
  const cache: Map<string, any> = new Map();

  const getIdx = (obj: any): number => {
    if (!idxMap.has(obj)) {
      idxMap.set(obj, idxMap.size);
    }
  }

  return idxMap.get(obj)!;
};

return function (...params: any) {
  const key = params.map(getIdx).join(',');
  if (!cache.has(key)) {
    cache.set(key, fn(...params));
  }
  return cache.get(key)!;
};
}
```

```
/**
 * let callCount = 0;
 * const memoizedFn = memoize(function (a, b) {
 *   callCount += 1;
 *   return a + b;
 * })
 * memoizedFn(2, 3) // 5
 * memoizedFn(2, 3) // 5
```

```
* console.log(callCount) // 1
*/
```

JavaScript

#2633 Convert Object to JSON String

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

JavaScript

Lists: Premium 98

Time Complexity: O(n), where n is the total number of elements/characters in the JSON structure. Space Complexity: O(n) due to the recursion stack and the constructed output string.

Problem

Given a value, return a valid JSON string of that value. The value can be a string, number, array, object, boolean, or null. The returned string should not include extra spaces. The order of keys should be the same as the order returned by Object.keys().

Please solve it without using the built-in JSON.stringify method.

Example 1:

```
Input: object = {"y":1,"x":2}
Output: {"y":1,"x":2}
Explanation:
Return the JSON representation.
Note that the order of keys should be the same as the order returned by Object.keys().
```


Example 2:

Input: object = {"a":"str","b":-12,"c":true,"d":null}
Output: {"a":"str","b":-12,"c":true,"d":null}
Explanation:
The primitives of JSON are strings, numbers, booleans, and null.

Example 3:

Input: object = {"key":{"a":1,"b":[{"},null,"Hello"]}}
Output: {"key":{"a":1,"b":[{"},null,"Hello"]}}
Explanation:
Objects and arrays can include other objects and arrays.

Example 4:

Input: object = true
Output: true
Explanation:
Primitive types are valid inputs.

Constraints:

- value is a valid JSON value
- 1 <= JSON.stringify(object).length <= 105
- maxNestingLevel <= 1000
- all strings contain only alphanumeric characters

Community Solutions (JavaScript / TypeScript)

- WalkCC

```
type JSONValue =
  | null
  | boolean
  | number
  | string
  | JSONValue[]
  | { [key: string]: JSONValue };
```

```
function jsonStringify(object: JSONValue): string {
  if (object === null) {
```

```
    return 'null';
  }
  if (typeof object === 'boolean' || typeof object ===
'number') {
    return String(object);
  }
  if (typeof object === 'string') {
    return `${object}`;
  }
  if (Array.isArray(object)) {
    const elems = object.map((elem) =>
jsonStringify(elem));
```

```
    return `[${elems.join(',')}]`;
  }
  const pairs = Object.keys(object).map(
(key) => `${key}:${jsonStringify(object[key])}`
);
  return `[${pairs.join(',')}]`;
}
```

JavaScript

#2636 Promise Pool

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

JavaScript · Concurrency

Lists: Premium 98

Time Complexity: O(m), where m is the number of functions to execute. Space Complexity: O(n) for the active pool of concurrent promises.

Problem

Given an array of asynchronous functions and a pool limit n, return an asynchronous function promisePool. It should return a promise that resolves when all the input functions resolve. Pool limit is defined as the maximum number promises that can be pending at once. promisePool should begin execution of as many functions as possible and continue executing new functions when old promises resolve. promisePool should execute functions[i] then functions[i + 1] then functions[i + 2], etc. When the last promise resolves, promisePool should also resolve.

For example, if n = 1, promisePool will execute one function at a time in series. However, if n = 2, it first executes two functions. When either of the two functions

resolve, a 3rd function should be executed (if available), and so on until there are no functions left to execute. You can assume all functions never reject. It is acceptable for promisePool to return a promise that resolves any value.

Example 1:

Input:
functions = [
 () => new Promise(res => setTimeout(res, 300)),
 () => new Promise(res => setTimeout(res, 400)),
 () => new Promise(res => setTimeout(res, 200))
,
n = 2
Output: [[300,400,500],500]

Example 2:

Input:
functions = [
 () => new Promise(res => setTimeout(res, 300)),
 () => new Promise(res => setTimeout(res, 400)),
 () => new Promise(res => setTimeout(res, 200))
,
n = 5
Output: [[300,400,200],400]

Example 3:

Input:
functions = [
 () => new Promise(res => setTimeout(res, 300)),
 () => new Promise(res => setTimeout(res, 400)),
 () => new Promise(res => setTimeout(res, 200))
,
n = 1
Output: [[300,700,900],900]

Constraints:

- 0 <= functions.length <= 10
- 1 <= n <= 10

Community Solutions (JavaScript / TypeScript)

- WalkCC

```
type F = () => Promise<any>;

function promisePool(functions: F[], n: number):
Promise<any> {
  const next = () => functions[n++]?.().then(next);
  return Promise.all(functions.slice(0, n).map((f) =>
f().then(next)));
}
```

- LeetDoocs

```
type F = () => Promise<any>;

function promisePool(functions: F[], n: number):
Promise<any> {
  const wrappers = functions.map(fn => async () => {
    await fn();
    const nxt = waiting.shift();
    nxt && (await nxt());
  });

  const running = wrappers.slice(0, n);
```

```
  const waiting = wrappers.slice(n);
  return Promise.all(running.map(fn => fn()));
}
```

- LC.ca

```
type F = () => Promise<any>;

function promisePool(functions: F[], n: number):
Promise<any> {
  const wrappers = functions.map(fn => async () => {
    await fn();
    const nxt = waiting.shift();
    const nxt && (await nxt());
  });

  const running = wrappers.slice(0, n);
```

```
  const waiting = wrappers.slice(n);
  return Promise.all(running.map(fn => fn()));
}
```

JavaScript

#2675 Array of Objects to Matrix

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

JavaScript · Array

Lists: Premium 98

Problem

Write a function that converts an array of objects arr into a matrix m.

arr is an array of objects or arrays. Each item in the array can be deeply nested with child arrays and child objects. It can also contain numbers, strings, booleans, and null values.

The first row m should be the column names. If there is no nesting, the column names are the unique keys within the objects. If there is nesting, the column names are the respective paths in the object separated by ".".

Each of the remaining rows corresponds to an object in arr. Each value in the matrix corresponds to a value in an object. If a given object doesn't contain a value for a given column, the cell should contain an empty string "".

The columns in the matrix should be in lexicographically ascending order.

Example 1:

Input:
arr = [
 {"b": 1, "a": 2},
 {"b": 3, "a": 4}
,
Output:
[
 ["a", "b"],

Example 2:

Input:
arr = [
 {"a": 1, "b": 2},
 {"c": 3, "d": 4},
 {}
,
Output:
[
 ["a", "b"],

Example 3:

Input:
arr = [
 {"a": {"b": 1, "c": 2}},
 {"a": {"b": 3, "d": 4}}
,
Output:
[
 ["a.b", "a.c", "a.d"],

Example 4:

Input:
arr = [
 [{"a": null}],
 [{"b": true}],
 [{"c": "x"}]
,
Output:
[

Example 5:

Input:
arr = [
 {},
 {},
 {}
,
Output:
[

Constraints:

- arr is a valid JSON array
- 1 <= arr.length <= 1000
- unique keys <= 1000

Community Solutions (JavaScript / TypeScript)

- WalkCC

```
function jsonToMatrix(arr: any[]): (string | number | boolean | null)[][] {
  const isObject = (o: any) => o !== null && typeof o === 'object';

  // Returns the keys of a JSON-like object by recursively unwrapping the nests.
  const getKeys = (json: any): string[] => {
    if (!isObject(json)) {
      return [''];
    }
    return Object.keys(json).reduce((acc: string[], currKey: string) => {
      return (
        acc.push(
          ...getKeys(json[currKey]).map((nextKey: string) =>
            nextKey === '' ? currKey :
            `${currKey}.${nextKey}`
          )
        ),
        acc
      );
    }, []);
  };
};
```

JavaScript

p.136

```
const sortedKeys: string[] = [
  ...arr.reduce((acc: Set<string>, curr: any) => {
    getKeys(curr).forEach((key: string) =>
      acc.add(key));
    return acc;
  }, new Set<string>()),
].sort();

// Returns the value of `obj` keyed by `nestedKey`.
const getValue = (
  obj: any,

  nestedKey: string
): string | number | boolean | null => {
  let value: any = obj;
  for (const key of nestedKey.split('.')) {
    if (!isObject(value) || !(key in value)) {
      return '';
    }
    value = value[key];
  }
  return isObject(value) ? '' : value;
};
```

JavaScript

p.137

```
};

const matrix: (string | number | boolean | null)[][] =
[sortedKeys];
arr.forEach((obj: any) => {
  matrix.push(
    sortedKeys.map((nestedKey: string) => getValue(obj,
      nestedKey))
  );
});

return matrix;
}

• LC.ca

function jsonToMatrix(arr: any[]): (string | number | boolean | null)[][] {
  const dfs = (key: string, obj: any) => {
    if (
      typeof obj === 'number' ||
      typeof obj === 'string' ||
      typeof obj === 'boolean' ||
      obj === null
    ) {
      return [ [key]: obj ];
    }
  };
};
```

JavaScript

p.138

```
const res: any[] = [];
for (const [k, v] of Object.entries(obj)) {
  const newKey = key ? `${key}.${k}` : `${k}`;
  res.push(dfs(newKey, v));
}
return res.flat();
};

const kv = arr.map(obj => dfs('', obj));
const keys = [

  ...new Set(
    kv
      .flat()
      .map(obj => Object.keys(obj))
      .flat(),
  ),
].sort();
const ans: any[] = [keys];
for (const row of kv) {
  const newRow: any[] = [];

  for (const key of keys) {
    const v = row.find(r =>
      r.hasOwnProperty(key))?.[key];
    newRow.push(v === undefined ? '' : v);
  }
  ans.push(newRow);
}
return ans;
};
```

JavaScript

p.139

JavaScript

#2700 Differences Between Two Objects

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

JavaScript

Lists: Premium 98

Time Complexity: O(n) in average, where n is the number of keys/elements in the common parts of the structures. Space Complexity: O(n) due to recursion and storage needed for the output differences.

Problem

Write a function that accepts two deeply nested objects or arrays obj1 and obj2 and returns a new object representing their differences.

The function should compare the properties of the two objects and identify any changes. The returned object should only contains keys where the value is different from obj1 to obj2.

For each changed key, the value should be represented as an array [obj1 value, obj2 value]. Keys that exist in one object but not in the other should not be included in the returned object. The end result should be a deeply nested object where each leaf value is a difference array.

When comparing two arrays, the indices of the arrays are considered to be their keys.

JavaScript

p.140

You may assume that both objects are the output of JSON.parse.

Example 1:

```
Input:
obj1 = {}
obj2 = {
  "a": 1,
  "b": 2
}
Output: {}
Explanation: There were no modifications made to obj1.
New keys "a" and "b" appear in obj2, but keys that are
added or removed should be ignored.
```

Example 2:

```
Input:
obj1 = {
  "a": 1,
  "v": 3,
  "x": [],
  "z": {
    "a": null
  }
}
```

Example 3:

JavaScript

p.141

```
Input:
obj1 = {
  "a": 5,
  "v": 6,
  "z": [1, 2, 4, [2, 5, 7]]
}
obj2 = {
  "a": 5,
```

Example 4:

```
Input:
obj1 = {
  "a": {"b": 1},
}
obj2 = {
  "a": [5],
}
Output:
```

Example 5:

```
Input:
obj1 = {
  "a": [1, 2, {}],
  "b": false
}
obj2 = {
  "b": false,
  "a": [1, 2, {}]
```

Constraints:

- obj1 and obj2 are valid JSON objects or arrays
- 2 <= JSON.stringify(obj1).length <= 104

JavaScript

p.142

- 2 <= JSON.stringify(obj2).length <= 104

Community Solutions (JavaScript / TypeScript)

• WalkCC

```
type JSONValue =
  | null
  | boolean
  | number
  | string
  | string[]
  | JSONValue[]
  | { [key: string]: JSONValue };
type Obj = Record<string, JSONValue> | Array<JSONValue>;

function objDiff(obj1: Obj, obj2: Obj): Obj {

  if (obj1 === obj2) {
    return {};
  }
  if (obj1 === null || obj2 === null) {
    return [obj1, obj2];
  }
  if (typeof obj1 !== 'object' || typeof obj2 !==
    'object') {
    return [obj1, obj2];
  }
  if (Array.isArray(obj1) !== Array.isArray(obj2)) {
```

JavaScript

p.143

JavaScript

p.144

	>> Quick Review — JavaScript 7 questions · insights · complexity · solution #2627 Debounce [Medium] Key Insights - Each call to the debounced function resets the timer. - Only the most recent function call (that isn't subsequently cancelled) will eventually execute. - Passing of parameters is handled by storing the arguments of the latest call. - The solution leverages timers (or equivalent scheduling mechanisms) and cancellation techniques. Space & Time Time Complexity: O(1) per call, as each invocation only clears and sets a timer. Space Complexity: O(1), since only a single timer reference is maintained. Solution The solution creates a wrapper function that maintains an internal timer reference. On each call to the debounced function, if a timer exists, it is cancelled. A new timer is then set to execute the original function after t milliseconds with the provided arguments. This ensures that only the last invocation runs if multiple JavaScriptp.145	calls occur within the delay period. #2628 JSON Deep Equal [Medium] Key Insights - Use recursion to compare nested structures. - For primitive types, a simple equality check (=== in JavaScript, == in Python, etc.) suffices. - For arrays, verify both have the same length and compare each element recursively. - For objects, ensure both have the same keys (order does not matter) then recursively compare the corresponding values. - Handle cases where one operand is null or differs by type early to avoid unnecessary recursion. Space & Time Time Complexity: O(N) in the worst-case, where N is the total number of elements/values in the JSON structure. Space Complexity: O(N) due to the recursion call stack in the worst-case scenario of deeply nested objects or arrays. Solution We solve this problem using a recursive approach. The algorithm first checks if both values are strictly equal, which handles primitives immediately. For arrays and objects, it verifies that the sizes (length for arrays and number of keys for objects) match. Then it recurses into JavaScriptp.147
each element (for arrays) or checks each key-value pair (for objects) to ensure that they are also deeply equal. For objects, the order of keys is irrelevant, so we compare key sets. This method naturally handles nested structures by delegating equality checks to the same function. #2630 Memoize [Medium] Key Insights - Cache previously computed results using a data structure keyed by the function arguments. - Ensure the key uniquely represents the argument order; simple tuple (or equivalent) suffices. - Use closures (or object state) to maintain both the cache and a call count tracking actual function invocations. - Expose a method (getCallCount) on the memoized function to report the number of times the original function was actually called. - The memoized version works for functions that might take one or more parameters. Space & Time Time Complexity: O(1) average for each call assuming efficient hashing of arguments. Space Complexity: O(n) for caching n distinct input combinations. JavaScriptp.148	Solution We use a hash-based cache (a dictionary or map) to store results for every unique set of arguments. Upon each call, we check if the key (constructed from the input arguments) exists in the cache. If not, we increment a counter, call the original function, store the result in the cache, and return it. Otherwise, we return the cached result. The memoized function is augmented with a getCallCount method to report the number of actual calls made to the original function. #2633 Convert Object to JSON String [Medium] Key Insights - The solution must support all JSON primitives: strings, numbers, booleans, and null. - Composite types such as arrays and objects require recursive traversal for proper serializing. - For strings, wrap them in double quotes. - For arrays, recursively process every element and join them with commas between square brackets. - For objects, iterate keys in insertion order and combine key-value pairs (key must be a string in double quotes) with commas between curly braces. - Use recursion carefully to handle nested structures without adding unnecessary spaces. Space & Time JavaScriptp.149	Time Complexity: O(n), where n is the total number of elements/characters in the JSON structure. Space Complexity: O(n) due to the recursion stack and the constructed output string. Solution We use a recursive approach that inspects the type of the value and processes it accordingly: - If the value is null, output "null". - If the value is a boolean or number, convert it directly to its string representation. - For a string, ensure it is enclosed in double quotes. - For an array, iterate over each element, serialize each recursively, join them with commas, and enclose in square brackets. - For an object, iterate over its keys in insertion order (or using Object.keys in JS / LinkedHashMap in Java), serialize both key (wrapped in double quotes) and its corresponding value, join with commas, and enclose in curly braces. This approach leverages recursion to handle arbitrarily nested objects and arrays while ensuring there are no extra spaces in the final string. #2636 Promise Pool [Medium] Key Insights - Use a concurrency control mechanism to limit the number of actively running promises. - Maintain an index or pointer to track which function from the input array should be executed next. JavaScriptp.150
- Upon the resolution of any promise, start a new one if available until all functions have been processed. - The overall task is complete when there are no functions left and all running promises have resolved. Space & Time Time Complexity: O(m), where m is the number of functions to execute. Space Complexity: O(n) for the active pool of concurrent promises. Solution We can solve the problem by maintaining an index that tracks which asynchronous function to execute next. We define a helper routine (or executor) that: - Checks if there is a function left to run. - If yes, increments the index and awaits the result of running that function. - Once the function completes, the routine recursively calls itself to pick up the next function. We start n such routines (to respect the pool limit) and use Promise.all (or the equivalent in other languages) to wait until all concurrent routines have finished executing. This structure ensures that we never run more than n promises concurrently and that a new promise starts as soon as one finishes. #2675 Array of Objects to Matrix [Medium] Key Insights JavaScriptp.151	- Flatten each object or array into dot-separated paths such as `a.b` or `a.0.c`. - Use DFS so nested objects and nested arrays are handled uniformly. - For each row, store a map from flattened path to value. - Collect every path seen across all rows into a set of column names. - Sort the column names lexicographically to build the header row. - For each flattened row map, output values in header order and use an empty string when a column is missing. Space & Time Time Complexity: O(T + R * C log C), where T is the total number of nested nodes visited, R is the number of rows, and C is the number of unique flattened columns. Space Complexity: O(R * C + C) in the worst case for the flattened row maps and the set of all column names. Solution We solve the problem by flattening every input row first. A DFS walks through objects and arrays, extending the current path with either the property name or the array index. When the DFS reaches a primitive value or `null`, it stores that value under the built path. While flattening each row, we also collect every path into a global set of column names. After all rows are processed, we sort the JavaScriptp.152	columns lexicographically, place them in the first row of the matrix, and then build each remaining row by reading values from that row's flattened map, using `` for missing paths. #2700 Differences Between Two Objects [Medium] Key Insights - Use recursion to traverse nested objects and arrays. - Compare only keys that exist in both objects. - When encountering two values of different types or unequal primitives, record the difference as [value from obj1, value from obj2]. - When both values are objects or arrays, recursively compute their differences and include them only if the resulting nested diff is non-empty. - Arrays are treated like objects with indices as keys. - The solution requires careful type checking and handling of recursive structures. Space & Time Time Complexity: O(n) in average, where n is the number of keys/elements in the common parts of the structures. Space Complexity: O(n) due to recursion and storage needed for the output differences. Solution We solve the problem using a recursive function that compares keys present in both input objects. For each JavaScriptp.153

common key:
– If both values are objects (or both are arrays), recursively compute their differences. – If the recursive call returns a non-empty object, include that key in the result. – If the two values are of different types or are primitives that differ, add the key with a difference array [value from obj1, value from obj2]. – Ignore keys that exist only in one object. This recursive approach naturally handles the deeply nested structure while ensuring only common keys with different values are captured.

String

#4 String

By Pattern · Python Only · 8 questions

JavaScript

p.154

```
# Function to check if ransom note can be constructed from magazine
def canConstruct(ransomNote: str, magazine: str) -> bool:
    # Create a dictionary to hold the frequency of each character in magazine
    char_frequency = {}
    for char in magazine:
        char_frequency[char] = char_frequency.get(char, 0) + 1

    # Check each character in ransomNote against the frequency map
    for char in ransomNote:
        if char_frequency.get(char, 0) == 0:
            return False # Character not available or used up
        char_frequency[char] -= 1 # Use the character

    return True

# Example usage:
if __name__ == "__main__":
    print(canConstruct("aa", "aab")) # Expected output: True
```

• LC.ca

String

p.157

Notice that a word is always similar to itself, also notice that the similarity relation is not transitive. For example, if the words a and b are similar, and the words b and c are similar, a and c are not necessarily similar.

Example 1:

```
Input: sentence1 = ["great","acting","skills"],
sentence2 = ["fine","drama","talent"], similarPairs = [
["great","fine"],["drama","acting"],["skills","talent"]]
Output: true
Explanation: The two sentences have the same length and each word i of sentence1 is also similar to the corresponding word in sentence2.
```

Example 2:

```
Input: sentence1 = ["great"], sentence2 = ["great"],
similarPairs = []
Output: true
Explanation: A word is similar to itself.
```

Example 3:

```
Input: sentence1 = ["great"], sentence2 = ["doubleplus","good"], similarPairs = [
["great","doubleplus"]]
Output: false
Explanation: As they don't have the same length, we return false.
```

Constraints:

- 1 <= sentence1.length, sentence2.length <= 1000
- 1 <= sentence1[i].length, sentence2[i].length <= 20

String

p.160

String

#383 Ransom Note

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Counting

Lists: Grind 169

Time Complexity: $O(n + m)$, where n is the length of magazine and m is the length of ransomNote. Space Complexity: $O(1)$, since the extra space required for the hash table is limited to at most 26 entries

Problem

Given two strings ransomNote and magazine, return true if ransomNote can be constructed by using the letters from magazine and false otherwise.

Each letter in magazine can only be used once in ransomNote.

Example 1:

```
Input: ransomNote = "a", magazine = "b"
Output: false
```

Example 2:

```
Input: ransomNote = "aa", magazine = "ab"
Output: false
```

Example 3:

String

p.155

```
class Solution:
    def canConstruct(self, ransomNote: str, magazine: str) -> bool:
        cnt = Counter(magazine)
        for c in ransomNote:
            cnt[c] -= 1
            if cnt[c] < 0:
                return False
        return True
```

String

p.158

- sentence1[i] and sentence2[i] consist of English letters.
- 0 <= similarPairs.length <= 1000
- similarPairs[i].length == 2
- 1 <= xi.length, yi.length <= 20
- xi and yi consist of lower-case and upper-case English letters.
- All the pairs (xi, yi) are distinct.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def areSentencesSimilar(self, sentence1, sentence2, similarPairs):
        # Brute Force  $O(n^2)$  - check all substrings
        n = len(sentence1)
        best = ''
        for i in range(n):
            for j in range(i + 1, n + 1):
                sub = sentence1[i:j]
                if len(sub) > len(best): best = sub

        return best
```

Community Solutions (Python)

• WalkCC

String

p.161

Sheet 18 / 317 · LeetCode By Pattern · Python Only · 9-up Portrait

```
Input: ransomNote = "aa", magazine = "aab"
Output: true
```

Constraints:

- 1 <= ransomNote.length, magazine.length <= 105
- ransomNote and magazine consist of lowercase English letters.

Community Solutions (Python)

• WalkCC

```
class Solution:
    def canConstruct(self, ransomNote: str, magazine: str) -> bool:
        count1 = collections.Counter(ransomNote)
        count2 = collections.Counter(magazine)
        return all(count1[c] <= count2[c] for c in string.ascii_lowercase)
```

• LeetCode

```
class Solution:
    def canConstruct(self, ransomNote: str, magazine: str) -> bool:
        cnt = Counter(magazine)
        for c in ransomNote:
            cnt[c] -= 1
            if cnt[c] < 0:
                return False
        return True
```

• SimplyLeet

String

p.156

String

#734 Sentence Similarity

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · String

Lists: Premium 98

Time Complexity: $O(n + m)$, where n is the number of words in the sentences and m is the number of similar pairs. Space Complexity: $O(m)$, for storing the similar pairs in a hash table.

Problem

We can represent a sentence as an array of words, for example, the sentence "I am happy with leetcode" can be represented as arr = ["I","am","happy","with","leetcode"].

Given two sentences sentence1 and sentence2 each represented as a string array and given an array of string pairs similarPairs where similarPairs[i] = [xi, yi] indicates that the two words xi and yi are similar.

Return true if sentence1 and sentence2 are similar, or false if they are not similar.

Two sentences are similar if:

- They have the same length (i.e., the same number of words)
- sentence1[i] and sentence2[i] are similar.

String

p.159

```
class Solution:
    def areSentencesSimilar(self, sentence1: list[str], sentence2: list[str], similarPairs: list[list[str]]):
        ) -> bool:
            if len(sentence1) != len(sentence2):
                return False
```

```
# map[key] := all the similar words of key
map = collections.defaultdict(set)
```

```
for a, b in similarPairs:
    map[a].add(b)
    map[b].add(a)
```

```
for word1, word2 in zip(sentence1, sentence2):
    if word1 == word2:
        continue
```

```
if word1 not in map:
    return False
if word2 not in map[word1]:
    return False
```

return True

• LeetCode

String

p.162

Example 3:
Input: s = "1337c0d3"
Output: 1337
Explanation:

Step 1: "1337c0d3" (no characters read because there is no leading whitespace)
^
Step 2: "1337c0d3" (no characters read because there is neither a '-' nor '+')
^
Step 3: "1337c0d3" ("1337" is read in; reading stops because the next character is a non-digit)
^

Example 4:
Input: s = "0-1"
Output: 0
Explanation:

Step 1: "0-1" (no characters read because there is no leading whitespace)
^
Step 2: "0-1" (no characters read because there is neither a '-' nor '+')
^
Step 3: "0-1" ("0" is read in; reading stops because the next character is a non-digit)
^

Example 5:
Input: s = "words and 987"

Stringp.172

Output: 0
Explanation:
Reading stops at the first non-digit character 'w'.
Constraints:

- 0 <= s.length <= 200
- s consists of English letters (lower-case and upper-case), digits (0-9), '-', '+', '-', and '!'.

Community Solutions (Python)

- WalkCC

```
class Solution:
    def myAtoi(self, s: str) -> int:
        s = s.strip()
        if not s:
            return 0

        sign = -1 if s[0] == '-' else 1
        if s[0] in ['-','+']:
            s = s[1:]

        num = 0

        for c in s:
            if not c.isdigit():
                break
            num = num * 10 + int(c)
            if sign * num <= -2**31:
                return -2**31
            if sign * num >= 2**31 - 1:
                return 2**31 - 1
```

Stringp.173

```
        return sign * num

• LeetCode

class Solution:
    def myAtoi(self, s: str) -> int:
        if not s:
            return 0
        n = len(s)
        if n == 0:
            return 0
        i = 0
        while s[i] == ' ':
            i += 1

        if i == n:
            return 0
        sign = -1 if s[i] == '-' else 1
        if s[i] in ['-','+']:
            i += 1
        res, flag = 0, (2**31 - 1) // 10
        while i < n:
            if not s[i].isdigit():
                break
            c = int(s[i])

            if res > flag or (res == flag and c > 7):
                return -2**31
            res = res * 10 + c
            i += 1
        return sign * res
```

Stringp.174

- SimplyLeet

```
# Python solution for the myAtoi problem.
def myAtoi(s: str) -> int:
    INT_MAX = 2**31 - 1
    INT_MIN = -2**31

    i, n = 0, len(s)
    # Skip leading whitespace.
    while i < n and s[i] == ' ':
        i += 1

    # Handle sign.
    sign = 1
    if i < n and (s[i] == '+' or s[i] == '-'):
        if s[i] == '-':
            sign = -1
        i += 1

    result = 0
    # Convert digits and build the integer.
    while i < n and s[i].isdigit():

        digit = int(s[i])
        # Check for overflow or underflow.
        if result > (INT_MAX - digit) // 10:
            return INT_MAX if sign == 1 else INT_MIN
        result = result * 10 + digit
        i += 1

    return sign * result

# Example usage:
```

Stringp.175

```
print(myAtoi("42"))
print(myAtoi("-042"))
print(myAtoi("1337c0d3"))
print(myAtoi("0-1"))
print(myAtoi("words and 987"))

• LC.ca

class Solution:
    def myAtoi(self, s: str) -> int:
        if not s:
            return 0
        n = len(s)
        if n == 0:
            return 0
        i = 0
        while s[i] == ' ':
            i += 1

        # only contains blank space
        if i == n:
            return 0
        sign = -1 if s[i] == '-' else 1
        if s[i] in ['-','+']:
            i += 1
        res, flag = 0, (2**31 - 1) // 10
        while i < n:
            # not a number, exit the loop
            if not s[i].isdigit():
                break
            c = int(s[i])

            if res > flag or (res == flag and c > 7):
                return -2**31
            res = res * 10 + c
            i += 1
        return sign * res
```

Stringp.176

```
        break
        c = int(s[i])
        # if overflows
        if res > flag or (res == flag and c > 7):
            return 2**31 - 1 if sign > 0 else
            -2**31
        res = res * 10 + c
        i += 1
    return sign * res
```

Stringp.177

String

#249 Group Shifted Strings

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · String

Lists: Premium 98

Time Complexity: $O(n * m)$ where n is the number of strings and m is the average length of the strings (to compute each string's key). Space Complexity: $O(n * m)$ to store the keys and grouping in the h

Problem

Perform the following shift operations on a string:

- Right shift: Replace every letter with the successive letter of the English alphabet, where 'z' is replaced by 'a'. For example, "abc" can be right-shifted to "bcd" or "xyz" can be right-shifted to "yza".
- Left shift: Replace every letter with the preceding letter of the English alphabet, where 'a' is replaced by 'z'. For example, "bcd" can be left-shifted to "abc" or "yza" can be left-shifted to "xyz".

We can keep shifting the string in both directions to form an endless shifting sequence.

- For example, shift "abc" to form the sequence: ... <-> "abc" <-> "bcd" <-> ... <-> "xyz" <-> "yza" <-> ... <-> "zab" <-> "abc" <-> ...

Stringp.178

You are given an array of strings strings, group together all strings[i] that belong to the same shifting sequence. You may return the answer in any order.

Example 1:

Input: strings = ["abc","bcd","acef","xyz","az","ba","a","z"]

Output: [["acef"],["a","z"],["abc","bcd","xyz"],["az","ba"]]

Example 2:

Input: strings = ["a"]

Output: [["a"]]

Constraints:

- 1 <= strings.length <= 200
- 1 <= strings[i].length <= 50
- strings[i] consists of lowercase English letters.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def group_shifted_strings(self, strings):
        # Brute Force O(n^2) - check all substrings
        n = len(strings)
        best = ''
        for i in range(n):
            for j in range(i + 1, n + 1):
                sub = strings[i:j]
                if len(sub) > len(best): best = sub

        return best
```

Community Solutions (Python)

Stringp.179

- WalkCC

```
class Solution:
    def groupStrings(self, strings: List[str]) -> List[List[str]]:
        keyToStrings = collections.defaultdict(List)

        def getKey(s: str) -> str:
            """
            Returns the key of 's' by pairwise calculation of differences.
            e.g. getKey("abc") -> "1,1" because diff(a, b) = 1 and diff(b, c) = 1.
            """
            diffs = []

            for i in range(1, len(s)):
                diff = (ord(s[i]) - ord(s[i - 1]) + 26) % 26
                diffs.append(str(diff))

            return ','.join(diffs)

        for s in strings:
            keyToStrings[getKey(s)].append(s)

        return keyToStrings.values()
```

- LeetCode

Stringp.180


```
class Solution:
    def groupStrings(self, strings: List[str]) ->
List[List[str]]:
    g = defaultdict(list)
    for s in strings:
        diff = ord(s[0]) - ord("a")
        t = []
        for c in s:
            c = ord(c) - diff
            if c < ord("a"):
```

```
                c += 26
            t.append(chr(c))
        g["".join(t)].append(s)
    return list(g.values())
```

• SimplyLeet

Python solution for grouping shifted strings

```
def group_shifted_strings(strings):
    # Helper function to generate a unique key for each
    string based on differences
    def get_key(s):
        # Single character strings return a fixed key
        if len(s) == 1:
            return "0"
        # Compute differences modulo 26 for each adjacent
        characters
        key = []
```

String

p.181

```
        for i in range(1, len(s)):
            diff = (ord(s[i]) - ord(s[i-1]) + 26) % 26
            key.append(str(diff))
        # Use tuple interleaving or join to form a string
        key
        return ','.join(key)

    groups = {}
    # Group strings by their computed key
    for s in strings:
        key = get_key(s)
```

```
        if key not in groups:
            groups[key] = []
        groups[key].append(s)
    # Return grouped strings as a list of lists
    return list(groups.values())

# Example usage:
print(group_shifted_strings(["abc","bcd","acef","xyz","az
", "ba","a","z"]))
```

• LC.ca

```
'''
>>> a = defaultdict(list)
>>> b = defaultdict(list)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: first argument must be callable or None

>>> ord('c')
99
```

String

p.182

```
>>> ord('c') - ord('a')
2
>>> ord('d') - 2
98
>>> chr(98)
'b'

>>> a = {"s":1, "t":2, "x":3}
>>> a

{'s': 1, 't': 2, 'x': 3}
>>> a.items()
dict_items([('s', 1), ('t', 2), ('x', 3)])
>>> a.values()
dict_values([1, 2, 3])
>>> list(a.values())
[1, 2, 3]
...
```

from collections import defaultdict

```
class Solution:
    def groupStrings(self, strings: List[str]) ->
List[List[str]]:
    mp = defaultdict(list)
    for s in strings:
        t = []
        diff = ord(s[0]) - ord('a')
        for c in s:
            d = ord(c) - diff
            if d < ord('a'):
```

String

p.183

```
        d += 26
        t.append(chr(d))
        k = ''.join(t)
        mp[k].append(s)
    return list(mp.values())
```

String

p.184

String

#271 Encode and Decode Strings

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · String · Design

Lists: Grind 169 | Premium 98

Time Complexity: O(n), where n is the total number of characters across all strings. Space Complexity: O(n), required for the encoded string and the resulting decoded list.

Problem

Design an algorithm to encode a list of strings to a string. The encoded string is then sent over the network and is decoded back to the original list of strings.

Machine 1 (sender) has the function:

```
string encode(vector<string> strs) {
// ... your code
return encoded_string;
}
```

Machine 2 (receiver) has the function:

```
vector<string> decode(string s) {
//... your code
return strs;
}
```

So Machine 1 does:

String

p.185

```
string encoded_string = encode(strs);
```

and Machine 2 does:

```
vector<string> strs2 = decode(encoded_string);
```

strs2 in Machine 2 should be the same as strs in Machine 1.

Implement the encode and decode methods.

You are not allowed to solve the problem using any serialize methods (such as eval).

Example 1:

```
Input: dummy_input = ["Hello","World"]
Output: ["Hello","World"]
Explanation:
Machine 1:
Codec encoder = new Codec();
String msg = encoder.encode(strs);
Machine 1 ----msg----> Machine 2
```

Example 2:

```
Input: dummy_input = [""]
Output: [""]
```

Constraints:

- 1 <= strs.length <= 200
- 0 <= strs[i].length <= 200
- strs[i] contains any possible characters out of 256 valid ASCII characters.

Follow up: Could you write a generalized algorithm to work on any possible set of characters?

String

p.186

Community Solutions (Python)

• WalkCC

```
class Codec:
    def encode(self, strs: list[str]) -> str:
        """Encodes a list of strings to a single string."""
        return ''.join(str(len(s)) + '/' + s for s in strs)
```

```
    def decode(self, s: str) -> list[str]:
        """Decodes a single string to a list of strings."""
        decoded = []
```

```
        i = 0
```

```
        while i < len(s):
            slash = s.find('/', i)
            length = int(s[i:slash])
            i = slash + length + 1
            decoded.append(s[slash + 1:i])
```

```
        return decoded
```

• LeetDoocs

```
class Codec:
    def encode(self, strs: List[str]) -> str:
        """Encodes a list of strings to a single
        string."""
        ans = []
        for s in strs:
            ans.append('{:4}'.format(len(s)) + s)
        return ''.join(ans)

    def decode(self, s: str) -> List[str]:
        """Decodes a single string to a list of
        strings."""
```

```
        ans = []
        i, n = 0, len(s)
        while i < n:
            size = int(s[i : i + 4])
            i += 4
            ans.append(s[i : i + size])
            i += size
        return ans
```

```
# Your Codec object will be instantiated and called as
such:
# codec = Codec()
# codec.decode(codec.encode(strs))
```

• SimplyLeet

String

p.188

String

p.189

<pre># Convert the substring from i to j into an integer length. length = int(s[i:j]) # Extract the actual string using the obtained length. string = s[j+1:j+1+length] decoded.append(string) # Move the pointer to the start of the next encoded segment. i = j + 1 + length return decoded # Example usage: # codec = Codec() # encoded_str = codec.encode(["Hello", "World"]) # decoded_list = codec.decode(encoded_str)</pre> • LC.ca <pre>''' >>> s = 'abcde' >>> '{:4}'.format(len(s)) + s ' 5abcde' >>> >>> '{:4}'.format(333) ' 333' >>> >>> '{:4}'.format(1234) '1234'</pre>	p.190	<pre>>>> '{:4}'.format(12345) '12345' >>> '{0:4}'.format(12345) '12345' >>> '{1:4}'.format(12345) Traceback (most recent call last): File "<stdin>", line 1, in <module> IndexError: Replacement index 1 out of range for positional args tuple >>> '{1:4}'.format(12345, 678910) '678910' >>> '{:4}'.format(12345, 678910) '12345' >>> '{0:4}'.format(12345, 678910) '12345' ''' class Codec: # no splitter def encode(self, strs: List[str]) -> str: ""Encodes a list of strings to a single string."" ans = [] def decode(self, s: str) -> List[str]: ""Decodes a single string to a list of strings."" ans = [] i, n = 0, len(s) while i < n: size = int(s[i : i + 4]) # so here potential overflow, if larger than int-max, need to clarify assumption with interviewer ans.append(s[i : i + size]) # note: exclusive for right index i += size return ans # Your Codec object will be instantiated and called as such: # codec = Codec() # codec.decode(codec.encode(strs))</pre>	p.191	<pre>for s in strs: ans.append('{:4}'.format(len(s)) + s) return ''.join(ans) def decode(self, s: str) -> List[str]: ""Decodes a single string to a list of strings."" ans = [] i, n = 0, len(s) while i < n: size = int(s[i : i + 4]) # so here potential overflow, if larger than int-max, need to clarify assumption with interviewer i += 4 ans.append(s[i : i + size]) # note: exclusive for right index i += size return ans # Your Codec object will be instantiated and called as such: # codec = Codec() # codec.decode(codec.encode(strs))</pre>	p.192
---	-------	---	-------	---	-------

<pre>##### class Codec: def encode(self, strs: List[str]) -> str: encoded = "" for s in strs: # cover for input: ["abcd", "####"] encoded += str(len(s)) + "#" + s.replace("#", "\$\$") + "\$\$" return encoded def decode(self, s: str) -> List[str]: decoded = [] i = 0 while i < len(s): j = s.find("#", i) length = int(s[i:j]) decoded.append(s[j+1:j+1+length].replace("\$\$", "#")) i = j + 1 + length + 1</pre>		<pre> return decoded ##### class Codec: # wrong compared with above, for input: ["abcd", "####"] def encode(self, strs): encoded = "" for s in strs: encoded += str(len(s)) + "#" + s return encoded def decode(self, s): decoded = [] i = 0 while i < len(s): delimiter = s.index("#", i) length = int(s[i:delimiter]) decoded.append(s[delimiter+1:delimiter+1+length]) i = delimiter + 1 + length return decoded</pre>		String #635 Design Log Storage SystemMED LeetDoocs · SimplyLeet · WalkCC · LeetCode Hash Table · String · Design · Ordered Set Lists: Denny Zhang Time Complexity: O(N) per retrieval query, where N is the number of logs stored (since each log is checked). In the worst case, all log entries are examined. Space Complexity: O(N) where N is the numb Problem You are given several logs, where each log contains a unique ID and timestamp. Timestamp is a string that has the following format: Year:Month:Day:Hour:Minute:Second, for example, 2017:01:01:23:59:59. All domains are zero-padded decimal numbers. Implement the LogSystem class: • LogSystem() Initializes the LogSystem object. • void put(int id, string timestamp) Stores the given log (id, timestamp) in your storage system. • int[] retrieve(string start, string end, string granularity) Returns the IDs of the logs whose timestamps are within the range from start to end inclusive. start and end all have the same format as timestamp, and granularity means how precise the	
String	p.193	String	p.194	String	p.195

range should be (i.e. to the exact Day, Minute, etc.). For example, start = "2017:01:01:23:59:59", end = "2017:01:02:23:59:59", and granularity = "Day" means that we need to find the logs within the inclusive range from Jan. 1st 2017 to Jan. 2nd 2017, and the Hour, Minute, and Second for each log entry can be ignored. Example 1: <pre>Input ["LogSystem", "put", "put", "put", "retrieve", "retrieve"] [[], [1, "2017:01:01:23:59:59"], [2, "2017:01:01:22:59:59"], [3, "2016:01:01:00:00:00"], ["2016:01:01:01:01:01", "2017:01:01:23:00:00", "Year"], ["2016:01:01:01:01:01", "2017:01:01:23:00:00", "Hour"]] Output [null, null, null, null, [3, 2, 1], [2, 1]] Explanation LogSystem logSystem = new LogSystem();</pre> Constraints: • 1 <= id <= 500 • 2000 <= Year <= 2017 • 1 <= Month <= 12 • 1 <= Day <= 31 • 0 <= Hour <= 23 • 0 <= Minute, Second <= 59 • granularity is one of the values ["Year", "Month", "Day", "Hour", "Minute", "Second"]. • At most 500 calls will be made to put and retrieve.		Community Solutions (Python) • WalkCC<pre>class LogSystem: def __init__(self): self.granularityToIndices = {'Year': 4, 'Month': 7, 'Day': 10, 'Hour': 13, 'Minute': 16, 'Second': 19} self.idAndTimestamps = [] def put(self, id: int, timestamp: str) -> None: self.idAndTimestamps.append((id, timestamp)) def retrieve(self, start: str, end: str, granularity: str) -> List[int]: index = self.granularityToIndices[granularity] s = start[:index] e = end[:index] return [id for id, timestamp in self.idAndTimestamps if s <= timestamp[:index] <= e]</pre> • LeetDoocs		class LogSystem: def __init__(self): self.logs = [] self.d = { "Year": 4, "Month": 7, "Day": 10, "Hour": 13, "Minute": 16, "Second": 19, } def put(self, id: int, timestamp: str) -> None: self.logs.append((id, timestamp)) def retrieve(self, start: str, end: str, granularity: str) -> List[int]: i = self.d[granularity] return [id for id, ts in self.logs if start[:i] <= ts[:i] <= end[:i]] # Your LogSystem object will be instantiated and called as such: # obj = LogSystem() # obj.put(id,timestamp) # param_2 = obj.retrieve(start,end,granularity) • SimplyLeet	
String	p.196	String	p.197	String	p.198

```
class LogSystem:
    def __init__(self):
        # List to store logs as (timestamp, id) tuples
        self.logs = []
        # Mapping granularity to index for the substring
        # (fixed format: Year:Month:Day:Hour:Minute:Second)
        self.granularity_to_index = {
            "Year": 4,
            "Month": 7,
            "Day": 10,
            "Hour": 13,

            "Minute": 16,
            "Second": 19
        }

    def put(self, id, timestamp):
        # Append the log as a tuple
        self.logs.append((timestamp, id))

    def retrieve(self, start, end, granularity):
        # Determine the slicing index based on
        granularity
```

String

p.199

```
        idx = self.granularity_to_index[granularity]
        # Slice the given start and end timestamps
        start_slice = start[:idx]
        end_slice = end[:idx]
        result = []
        # Iterate over all logs and check if the
        timestamp slice is within the range
        for ts, log_id in self.logs:
            ts_slice = ts[:idx]
            if start_slice <= ts_slice <= end_slice:
                result.append(log_id)

        return result
```

```
# Example usage:
# logSystem = LogSystem()
# logSystem.put(1, "2017:01:01:23:59:59")
# logSystem.put(2, "2017:01:01:22:59:59")
# logSystem.put(3, "2016:01:01:00:00:00")
# print(logSystem.retrieve("2016:01:01:01:01:01",
# "2017:01:01:23:00:00", "Year")) # Output: [3, 2, 1]
# print(logSystem.retrieve("2016:01:01:01:01:01",
# "2017:01:01:23:00:00", "Hour")) # Output: [2, 1]
```

String

p.200

String

#1138 Alphabet Board Path

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String

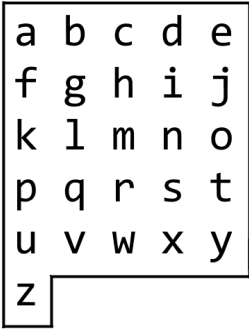
Lists: Denny Zhang

Time Complexity: O(N) where N is the length of target (each move is processed in constant time) Space Complexity: O(N) for the resulting string of moves

Problem

On an alphabet board, we start at position (0, 0), corresponding to character board[0][0].

Here, board = ["abcde", "fghij", "klmno", "pqrst", "uvwxy", "z"], as shown in the diagram below.



We may make the following moves:

- 'U' moves our position up one row, if the position exists on the board;
- 'D' moves our position down one row, if the position exists on the board;
- 'L' moves our position left one column, if the position exists on the board;
- 'R' moves our position right one column, if the position exists on the board;
- '!' adds the character board[r][c] at our current position (r, c) to the answer.

String

p.202

(Here, the only positions that exist on the board are positions with letters on them.)

Return a sequence of moves that makes our answer equal to target in the minimum number of moves. You may return any path that does so.

Example 1:

```
Input: target = "leet"

Output: "DDR!UURRR!!DDD!"
```

Example 2:

```
Input: target = "code"

Output: "RR!DDRR!UUL!R!"
```

Constraints:

- 1 <= target.length <= 100
- target consists only of English lowercase letters.

Community Solutions (Python)

- LeetDoocs

String

p.203

```
class Solution:
    def alphabetBoardPath(self, target: str) -> str:
        i = j = 0
        ans = []
        for c in target:
            v = ord(c) - ord("a")
            x, y = v // 5, v % 5
            while j > y:
                j -= 1
                ans.append("L")

            while i > x:
                i -= 1
                ans.append("U")
            while j < y:
                j += 1
                ans.append("R")
            while i < x:
                i += 1
                ans.append("D")
            ans.append("!")

        return "".join(ans)
```

- SimplyLeet

String

p.204

```
# Python solution with clear variable names and inline
comments.
class Solution:
    def alphabetBoardPath(self, target: str) -> str:
        # Starting position at the top-left corner (0,0)
        current_row, current_col = 0, 0
        result = []

        # Helper function to convert letter to its (row,
        col) coordinates
        def getPosition(letter):
            index = ord(letter) - ord('a')

            return index // 5, index % 5

        for char in target:
            target_row, target_col = getPosition(char)
            # If target character is 'z', move
            horizontally first then vertically
            if char == 'z':
                # Move left if needed
                while current_col > target_col:
                    result.append("L")
                    current_col -= 1
```

String

p.205

```
        # Move right if needed
        while current_col < target_col:
            result.append("R")
            current_col += 1
        # Then move down if needed
        while current_row < target_row:
            result.append("D")
            current_row += 1
        # Then move up if needed
        while current_row > target_row:
            result.append("U")
            current_row -= 1

        else:
            # For all other characters, move
            vertically first
            while current_row > target_row:
                result.append("U")
                current_row -= 1
            while current_row < target_row:
                result.append("D")
                current_row += 1

        # Then move horizontally
        while current_col > target_col:
            result.append("L")
            current_col -= 1
        while current_col < target_col:
            result.append("R")
            current_col += 1

        # Append pick-up action for the letter
        result.append("!")
        return "".join(result)
```

String

p.206

- LC.ca

```
class Solution:
    def alphabetBoardPath(self, target: str) -> str:
        i = j = 0
        ans = []
        for c in target:
            v = ord(c) - ord("a")
            x, y = v // 5, v % 5
            while j > y:
                j -= 1
                ans.append("L")

            while i > x:
                i -= 1
                ans.append("U")
            while j < y:
                j += 1
                ans.append("R")
            while i < x:
                i += 1
                ans.append("D")
            ans.append("!")

        return "".join(ans)
```

String

p.207

	>> Quick Review — String 8 questions · insights · complexity · solution #383 Ransom Note [Easy] Key Insights - Use a hash table (or dictionary) to store the frequency of each character present in magazine. - Iterate over every character in ransomNote and check if the corresponding count in the hash table is available and sufficient. - Decrease the character count for every match; if any character count falls below zero, return false. - If all characters in ransomNote can be accounted for, return true. Space & Time Time Complexity: $O(n + m)$, where n is the length of magazine and m is the length of ransomNote. Space Complexity: $O(1)$, since the extra space required for the hash table is limited to at most 26 entries for lowercase English letters. Solution We construct a frequency map for the magazine string using a hash table. This frequency map records how many times each letter appears in magazine. Then we Stringp.208	iterate through the ransomNote string, and for every letter, we check the corresponding frequency in the map. If a letter is not found or its frequency is zero, we immediately return false as it means the letter cannot be used to build the note. Otherwise, we decrement the frequency count by one. If we successfully process all characters in ransomNote, then it can be constructed from magazine, and we return true. #734 Sentence Similarity [Easy] Key Insights - First, check if both sentences have the same length. - A word is always similar to itself. - Similarity is defined only by the given pairs; it is not transitive. - Use a hash table (dictionary) to store bi-directional similar pairs for quick look-ups. Space & Time Time Complexity: $O(n + m)$, where n is the number of words in the sentences and m is the number of similar pairs. Space Complexity: $O(m)$, for storing the similar pairs in a hash table. Solution – Check if the two sentences have the same length; if not, return false. – Build a hash table where each word Stringp.210
maps to a set of words it is similar to. For each pair $[x, y]$ in similarPairs, add y to the set for x and x to the set for y. – Iterate over the words of both sentences simultaneously. For each corresponding pair, if the words are identical, continue; otherwise, check if one word appears in the similar set of the other. – If all pairs meet the similarity condition, return true; otherwise, return false. #1165 Single Row Keyboard [Easy] Key Insights - Create a mapping (hash table/dictionary) from each character to its index on the keyboard for $O(1)$ lookups. - The cost to type each character is the absolute difference between the current position and the target character's position on the keyboard. - Start from index 0 and sum up the distances moved for each successive character in the word. Space & Time Time Complexity: $O(n)$, where n is the length of the word. Space Complexity: $O(1)$ (constant space for a mapping of 26 characters). Solution We solve the problem by first building a dictionary that maps each character in the keyboard layout to its Stringp.211	corresponding index. We then initialize a variable to track the current finger position (starting at 0). For each character in the word, we look up its index, calculate the absolute difference from the current position, add that distance to the total time, and update the current position. This approach uses a single pass over the word with constant time lookups, ensuring an efficient solution. #8 String to Integer (atoi) [Medium] Key Insights - Skip leading spaces in the string. - Detect and handle the optional sign indicator ('+' or '-'). - Read and convert digit characters until a non-digit is encountered. - Handle cases with no digits or leading zeros. - Clamp the result to fit within the 32-bit signed integer range. Space & Time Time Complexity: $O(n)$ where n is the length of the string. Space Complexity: $O(1)$ as only a constant amount of extra space is used. Solution Stringp.212	We iterate through the string to first eliminate any leading whitespace. Next, we check for a sign character to determine if the resulting integer should be negative. We then convert each consecutive digit into an integer, all the while updating the result and checking for potential overflow. If an overflow is detected, we immediately return the appropriate maximum or minimum 32-bit signed integer boundary value. This approach ensures that we strictly follow the conversion rules and efficiently handle edge cases. #249 Group Shifted Strings [Medium] Key Insights - All strings in the same group have the same relative differences between adjacent characters. - Normalize each string's pattern by converting it to a key representation based on the differences between consecutive letters. - A single character string forms its own group as there are no differences to compute. Space & Time Time Complexity: $O(n * m)$ where n is the number of strings and m is the average length of the strings (to compute each string's key). Space Complexity: $O(n * m)$ to store the keys and grouping in the hash table. Stringp.213
Solution The solution involves iterating over each string and generating a unique key that represents the relative differences between adjacent characters. This key is calculated by computing the difference (mod 26) between every consecutive pair in the string. Strings that share the same key can be grouped together since they can be shifted into one another. A hash table (or dictionary) is used to store and group the strings by their computed key. This approach handles edge cases like single-character strings separately. #271 Encode and Decode Strings [Medium] Key Insights - Use length-prefix encoding for each string to avoid ambiguity. - Append a delimiter (e.g., "#") after the length to separate it from the string content. - This method ensures that even if the original strings contain special characters, including the delimiter, the decoding remains unambiguous. - The approach handles edge cases, such as empty strings. Space & Time Time Complexity: $O(n)$, where n is the total number of characters across all strings. Stringp.214	Space Complexity: $O(n)$, required for the encoded string and the resulting decoded list. Solution We encode by iterating over each string and building a new string that contains the length of the string, a delimiter (e.g., "#"), and then the string itself. This way, during decoding, we can safely determine how many characters to extract by first reading until the delimiter to get the length, and then reading that number of characters. This length-prefix approach avoids conflicts with potential characters in the strings. #635 Design Log Storage System [Medium] Key Insights - Store logs in a simple data structure such as a list or array. - Convert the timestamps into strings; due to fixed-length and zero padding, lexicographical string comparisons are valid. - Use a mapping to determine the slice length for each granularity, e.g. "Year" corresponds to the first 4 characters, "Month" corresponds to the first 7 characters, etc. - During retrieval, compare the relevant substring of each stored timestamp against the start and end query timestamp slices. Stringp.215	The inclusive nature of the query simplifies checking conditions. Space & Time Time Complexity: $O(N)$ per retrieval query, where N is the number of logs stored (since each log is checked). In the worst case, all log entries are examined. Space Complexity: $O(N)$ where N is the number of logs stored. Solution We implement a LogSystem class that supports putting log entries and retrieving them using a specified granularity. – Data Structures: Use a list/array to store log entries as tuples (or objects) containing the timestamp and log ID. A dictionary/map is used to store the mapping from granularity string to the substring index (e.g., "Year" => 4, "Month" => 7, etc.). – For the put() method, simply append the new log entry into the list. – For the retrieve() method, compute the substring length based on the provided granularity. Then iterate over all stored logs and compare the sliced timestamp against the sliced start and end query values. If it falls within the bounds, include the log ID in the result. – Since the timestamps have fixed width with leading zeros, slicing and lexicographical comparisons are both valid and simple. Stringp.216

#1138 Alphabet Board Path [Medium]

Key Insights

- Determine the row and column for each letter (using integer division and modulus).
- Because "z" is in a special row that only contains one column, the move order is critical to avoid invalid positions.
- For target letter "z", perform horizontal moves before vertical moves; for all other letters, do vertical moves before horizontal moves.
- Build the answer step-by-step by moving from your current position to each target letter.

Space & Time

Time Complexity: O(N) where N is the length of target (each move is processed in constant time)
Space Complexity: O(N) for the resulting string of moves

Solution

The approach calculates the destination coordinates for each character. For each letter in the target:
– Convert the letter to its board coordinates (row and col). For example, for letter c, row = (ord(c) - ord('a')) // 5 and col = (ord(c) - ord('a')) % 5. Note that "z" (the 26th letter) always maps to (5, 0). – To move from the current position (curRow, curCol) to the target's (targetRow, targetCol), decide on the order of moves: If the target letter is "z", first move horizontally (left or right) then

String p.217

vertically (up or down) to avoid stepping into invalid regions from the extra row. For any other letter, move vertically (up or down) first, then horizontally (left or right). – Append "!" to the moves after reaching each letter. – Update the current position to the letter's coordinates.
This method ensures you only traverse valid cells on the board and minimizes moves by taking the optimal coordinate difference each step.

Sliding Window

#5 Sliding Window

By Pattern · Python Only · 9 questions

String p.218

Sliding Window

#3 Longest Substring Without Repeating Characters

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Sliding Window

Lists: Grind 169

Time Complexity: O(n), where n is the length of the input string. Space Complexity: O(min(n, m)), where m is the size of the character set (constant for fixed character sets).

Problem

Given a string s, find the length of the longest substring without duplicate characters.

Example 1:

Input: s = "abcabcbb"
Output: 3
Explanation: The answer is "abc", with the length of 3. Note that "bca" and "cab" are also correct answers.

Example 2:

Input: s = "bbbbb"
Output: 1
Explanation: The answer is "b", with the length of 1.

Example 3:

Sliding Window p.219

Input: s = "pwwkew"
Output: 3
Explanation: The answer is "wke", with the length of 3. Notice that the answer must be a substring, "pwke" is a subsequence and not a substring.

Constraints:

- 0 <= s.length <= 5 * 104
- s consists of English letters, digits, symbols and spaces.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def length_of_longest_substring(self, s):
        # Brute Force O(n^2) - enumerate every starting index; expand until repeat
        res = 0
        for i in range(len(s)):
            seen = set()
            for j in range(i, len(s)):
                if s[j] in seen:
                    break

            seen.add(s[j])
            res = max(res, j - i + 1)
        return res
```

Community Solutions (Python)

- WalkCC

Sliding Window p.220

```
class Solution:
    def lengthOfLongestSubstring(self, s: str) -> int:
        ans = 0
        count = collections.Counter()

        l = 0
        for r, c in enumerate(s):
            count[c] += 1
            while count[c] > 1:
                count[s[l]] -= 1
                l += 1
            ans = max(ans, r - l + 1)

        return ans
```

- LeetDoocs

```
class Solution:
    def lengthOfLongestSubstring(self, s: str) -> int:
        cnt = Counter()
        ans = l = 0
        for r, c in enumerate(s):
            cnt[c] += 1
            while cnt[c] > 1:
                cnt[s[l]] -= 1
                l += 1
            ans = max(ans, r - l + 1)

        return ans
```

- SimplyLeet

Sliding Window p.221

```
# Function to find the length of the longest substring
without repeating characters
def length_of_longest_substring(s):
    # Dictionary to store the most recent index of each character
    char_index = {}
    # Initialize starting index of current window and max length
    left = 0
    max_length = 0

    # Iterate over the string with the right pointer
    for right, char in enumerate(s):
```

```
        # If char is in dictionary and its index is
        within the current window
        if char in char_index and char_index[char] >=
        left:
            # Move the left pointer to one position after
            the last occurrence of char
            left = char_index[char] + 1
            # Update the latest index of the char
            char_index[char] = right
            # Update max_length with the current window size
        if larger
            max_length = max(max_length, right - left + 1)

    return max_length
```

```
# Example usage:
# print(length_of_longest_substring("abcabcbb")) #
Expected output: 3
```

Sliding Window p.222

- LC.ca

```
class Solution:
    def lengthOfLongestSubstring(self, s): # map for index
        d = {} # value => its index
        i = 0
        ans = 0
        for j, c in enumerate(s):
            if c in d:
                # mast max check i, example "abba"
                i = max(i, d[c] + 1)
            d[c] = j

        ans = max(ans, j - i + 1)
        return ans
```

```
class Solution:
    def lengthOfLongestSubstring(self, s: str) -> int:

        if not s:
            return 0
```

```
        l = 0 # left
        r = 0 # right

        result = 0
        isFoundInWindow = [False] * 256
```

```
        while r < len(s):

            while isFoundInWindow[ord(s[r])]:
                isFoundInWindow[ord(s[l])] = False
```

Sliding Window p.223

```
        l += 1

        isFoundInWindow[ord(s[r])] = True

        # check before shrink
        result = max(result, r - l + 1)

        r += 1

        return result
```

```
class Solution: # extra space for set()
    def lengthOfLongestSubstring(self, s: str) -> int:

        d = collections.defaultdict(int)
        start = 0
        ans = 0
        for i,c in enumerate(s):
            # shrink
            while d[c] > 0:
```

Sliding Window p.224

```
        d[s[start]] -= 1 # not s[start], not
        start

        start += 1
        ans = max(ans, i - start + 1)
        d[c] = 1
        return ans
```

#####

```
class Solution(object):
    def lengthOfLongestSubstring(self, s): # no extra data
    structure
```

```
        """"
        :type s: str
        :rtype: int
        """"
        d = collections.defaultdict(int)
        l = ans = 0
        for i, c in enumerate(s):
            while l > 0 and d[c] > 0:
                d[s[i - l]] -= 1
                l -= 1
```

Sliding Window p.225

```
d[c] += 1
l += 1
ans = max(ans, l)
return ans

#####

class Solution:
    def lengthOfLongestSubstring(self, s: str) -> int:
        ss = set()

        i = ans = 0
        for j, c in enumerate(s):
            while c in ss:
                ss.remove(s[i])
                i += 1
            ss.add(c)
            ans = max(ans, j - i + 1)
        return ans
```

Sliding Window

#159 Longest Substring with At Most Two Distinct Characters

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Sliding Window

Lists: Premium 98

Time Complexity: $O(n)$ where n is the length of the string, since each character is processed at most twice.

Space Complexity: $O(1)$ because the hash map holds at most 3 entries (usually up to 2 distinct)

Problem

Given a string s , return the length of the longest substring that contains at most two distinct characters.

Example 1:

Input: $s = \text{"eceba"}$
Output: 3
Explanation: The substring is "ece" which its length is 3.

Example 2:

Input: $s = \text{"ccaabbb"}$
Output: 5
Explanation: The substring is "aabbb" which its length is 5.

Constraints:

- $1 \leq s.length \leq 105$
- s consists of English letters.

Community Solutions (Python)

```
• WalkCC

class Solution:
    def lengthOfLongestSubstringTwoDistinct(self, s: str) -> int:
        ans = 0
        distinct = 0
        count = [0] * 128

        l = 0
        for r, c in enumerate(s):
            count[ord(c)] += 1
            if count[ord(c)] == 1:
                distinct += 1
                while distinct == 3:
                    count[ord(s[l])] -= 1
                    if count[ord(s[l])] == 0:
                        distinct -= 1
                    l += 1
            ans = max(ans, r - l + 1)

        return ans

• LeetDoocs
```

```
class Solution:
    def lengthOfLongestSubstringTwoDistinct(self, s: str) -> int:
        cnt = Counter()
        ans = j = 0
        for i, c in enumerate(s):
            cnt[c] += 1
            while len(cnt) > 2:
                cnt[s[j]] -= 1
                if cnt[s[j]] == 0:
                    cnt.pop(s[j])
                    j += 1
            ans = max(ans, i - j + 1)
        return ans
```

```
• SimplyLeet

# Python solution for Longest Substring with At Most Two Distinct Characters
class Solution:
    def lengthOfLongestSubstringTwoDistinct(self, s: str) -> int:
        left = 0 # Initialize left pointer of the window
        max_length = 0 # Variable to store the maximum window size found
        char_count = {} # Dictionary to count characters in the current window

        for right, char in enumerate(s):
            # Add the current character to the window (increase count)
            char_count[char] = char_count.get(char, 0) + 1

            while len(char_count) > 2:
                # Remove the leftmost character until we have at most 2 distinct characters
                char_count[s[left]] -= 1
                if char_count[s[left]] == 0:
                    del char_count[s[left]]
                    left += 1

            max_length = max(max_length, right - left + 1)

        return max_length
```

```
# If there are more than two distinct characters, shrink the window from the left
while len(char_count) > 2:
    left_char = s[left]
    char_count[left_char] -= 1
    if char_count[left_char] == 0:
        del char_count[left_char]
    left += 1

# Update max_length if current window is larger
max_length = max(max_length, right - left + 1)

return max_length
```

```
• LC.ca

'''
>>> from collections import Counter
>>> c = Counter()
>>> c
Counter()
>>> c['a'] += 1
>>> c
Counter({'a': 1})
'''
```

```
>>> c
Counter({'a': 1, 'c': 1, 'b': 1})
>>> c.pop(2)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
KeyError: 2
>>>
>>> c.pop('c')
1
>>> c
Counter({'a': 1, 'b': 1})
'''

from collections import Counter

class Solution:
    def lengthOfLongestSubstringTwoDistinct(self, s: str) -> int:
        cnt = Counter() # or, cnt = defaultdict(int)
        ans = i = 0
        for j, c in enumerate(s):
            cnt[c] += 1
            while len(cnt) > 2: # shrink
                cnt[s[i]] -= 1
                if cnt[s[i]] == 0:
                    cnt.pop(s[i])
                    i += 1
            ans = max(ans, j - i + 1)
        return ans
```

Sliding Window

#340 Longest Substring with At Most K Distinct Characters

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Sliding Window

Lists: Premium 98

Time Complexity: $O(n)$, where n is the length of the string, as each character is processed at most twice (once when expanding the window and once when contracting it). Space Complexity: $O(\min(m, k))$,

Problem

Given a string s and an integer k , return the length of the longest substring of s that contains at most k distinct characters.

Example 1:

Input: $s = \text{"eceba"}$, $k = 2$
Output: 3
Explanation: The substring is "ece" with length 3.

Example 2:

Input: $s = \text{"aa"}$, $k = 1$
Output: 2
Explanation: The substring is "aa" with length 2.

Constraints:

- $1 \leq s.length \leq 5 * 10^4$

Sliding Window

#340 Longest Substring with At Most K Distinct Characters

MED

```
• WalkCC

class Solution:
    def lengthOfLongestSubstringKDistinct(self, s: str, k: int) -> int:
        ans = 0
        distinct = 0
        count = collections.Counter()

        l = 0
        for r, c in enumerate(s):
            count[c] += 1
            if count[c] == 1:
                distinct += 1
                while distinct == k + 1:
                    count[s[l]] -= 1
                    if count[s[l]] == 0:
                        distinct -= 1
                    l += 1
            ans = max(ans, r - l + 1)

        return ans

• LeetDoocs
```

```
class Solution:
    def lengthOfLongestSubstringKDistinct(self, s: str, k: int) -> int:
        l = 0
        cnt = Counter()
        for c in s:
            cnt[c] += 1
            if len(cnt) > k:
                cnt[s[l]] -= 1
                if cnt[s[l]] == 0:
                    del cnt[s[l]]
                l += 1
        return len(s) - l

• SimplyLeet

# Python solution for Longest Substring with At Most K Distinct Characters

def longest_substring_with_k_distinct(s: str, k: int) -> int:
    # Handle edge case: if k is 0, no valid substring exists
    if k == 0:
        return 0

    # Dictionary to store frequency of characters in the current window
    char_count = {}
    left = 0 # Left pointer for the window
```



```
max_length = 0 # Variable to track the maximum length found

# Iterate through the string using the right pointer
for right in range(len(s)):
    # Add/update the count of the current character
    # in the dictionary
    current_char = s[right]
    char_count[current_char] = char_count.get(current_char, 0) + 1

    # If window contains more than k distinct
    # characters, shrink from the left
    while len(char_count) > k:
        left_char = s[left]
        char_count[left_char] -= 1
        # Remove the character count if it becomes 0
        if char_count[left_char] == 0:
            del char_count[left_char]
        left += 1

    # Update the maximum length of the valid window
    # found so far
    max_length = max(max_length, right - left + 1)

return max_length

# Example usage:
# s = "eceba", k = 2
# print(longest_substring_with_k_distinct(s, 2)) #
# Output: 3
```

• LC.ca

```
class Solution:
    def lengthOfLongestSubstringTwoDistinct(self, s: str)
    -> int:
        cnt = Counter() # or, cnt = defaultdict(int)
        ans = i = 0
        for j, c in enumerate(s):
            cnt[c] += 1
            while len(cnt) > k:
                cnt[s[i]] -= 1
                if cnt[s[i]] == 0:
                    cnt.pop(s[i])
                i += 1
            ans = max(ans, j - i + 1)
        return ans
```

Sliding Window

#424 Longest Repeating Character Replacement MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Sliding Window

Lists: Grind 169

Time Complexity: O(n) where n is the length of the string, since the window traverses the string in linear time. Space Complexity: O(1) because the frequency count for letters (A–Z) has a constant siz

Problem

You are given a string s and an integer k. You can choose any character of the string and change it to any other uppercase English character. You can perform this operation at most k times.

Return the length of the longest substring containing the same letter you can get after performing the above operations.

Example 1:

Input: s = "ABAB", k = 2
Output: 4
Explanation: Replace the two 'A's with two 'B's or vice versa.

Example 2:

```
Input: s = "AABABBA", k = 1
Output: 4
Explanation: Replace the one 'A' in the middle with 'B' and form "AABBBBB".
The substring "BBBB" has the longest repeating letters, which is 4.
There may exists other ways to achieve this answer too.
```

- Constraints:
- 1 <= s.length <= 105
 - s consists of only uppercase English letters.
 - 0 <= k <= s.length

Community Solutions (Python)

```
class Solution:
    def characterReplacement(self, s: str, k: int) -> int:
        ans = 0
        maxCount = 0
        count = collections.Counter()

        l = 0
        for r, c in enumerate(s):
            count[c] += 1
            maxCount = max(maxCount, count[c])

            while maxCount + k < r - l + 1:
                count[s[l]] -= 1
                l += 1

            return r - l + 1
```

• WalkCC

```
while maxCount + k < r - l + 1:
    count[s[l]] -= 1
    l += 1
ans = max(ans, r - l + 1)

return ans

class Solution:
    def characterReplacement(self, s: str, k: int) -> int:
        maxCount = 0
        count = collections.Counter()

        # l and r track the maximum window instead of the valid window.
        l = 0
        for r, c in enumerate(s):
            count[c] += 1
            maxCount = max(maxCount, count[c])

            while maxCount + k < r - l + 1:
                count[s[l]] -= 1
                l += 1

            return r - l + 1
```

• LeetDoocs

```
class Solution:
    def characterReplacement(self, s: str, k: int) -> int:
        cnt = Counter()
        l = mx = 0
        for r, c in enumerate(s):
            cnt[c] += 1
            mx = max(mx, cnt[c])
            if r - l + 1 - mx > k:
                cnt[s[l]] -= 1
                l += 1

        return len(s) - l

# Python solution using the sliding window technique
def characterReplacement(s: str, k: int) -> int:
    # Dictionary to store frequency of characters in the current window
    char_counts = {}
    max_freq = 0 # Track the maximum frequency of a single character in the window
    max_length = 0
    left = 0 # Left pointer of the sliding window

    # Expand the sliding window with the right pointer
    for right in range(len(s)):
```

• SimplyLeet

```
# Count the current character
char_counts[s[right]] = char_counts.get(s[right], 0) + 1

# Update the max frequency in the current window
max_freq = max(max_freq, char_counts[s[right]])

# Check if the number of replacements needed
# (current window size) - (count of most frequent character) > k
if (right - left + 1) - max_freq > k:
    # If condition fails, shrink the window by moving left pointer
    char_counts[s[left]] -= 1
    left += 1

# Update the maximum length found so far
max_length = max(max_length, right - left + 1)

return max_length

# Example usage:
# print(characterReplacement("AABABBA", 1)) # Expected output: 4
```

• LC.ca

```
class Solution:
    def characterReplacement(self, s: str, k: int) -> int:
        counter = [0] * 26
        i = j = maxCnt = 0
        while i < len(s):
            counter[ord(s[i]) - ord('A')] += 1
            maxCnt = max(maxCnt, counter[ord(s[i]) - ord('A')])
            if i - j + 1 > maxCnt + k:
                counter[ord(s[j]) - ord('A')] -= 1
                j += 1
            i += 1
        return i - j
```

Sliding Window

#438 Find All Anagrams in a String MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Sliding Window

Lists: Grind 169

Time Complexity: O(n) where n is the length of s. Space Complexity: O(1) since we use fixed-size frequency arrays or hash maps.

Problem

Given two strings s and p, return an array of all the start indices of p's anagrams in s. You may return the answer in any order.

Example 1:

Input: s = "cbaebabacd", p = "abc"
Output: [0,6]
Explanation:
The substring with start index = 0 is "cba", which is an anagram of "abc".
The substring with start index = 6 is "bac", which is an anagram of "abc".

Example 2:

Input: s = "abab", p = "ab"
Output: [0,1,2]
Explanation:
The substring with start index = 0 is "ab", which is an anagram of "ab".
The substring with start index = 1 is "ba", which is an anagram of "ab".
The substring with start index = 2 is "ab", which is an anagram of "ab".

Constraints:

- 1 <= s.length, p.length <= 3 * 104
- s and p consist of lowercase English letters.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def findAnagrams(self, s: str, p: str):
        # Brute Force O(n^2) - enumerate all windows
        n = len(s)
        best = 0
        for l in range(n):
            window_state = 0
            for r in range(l, n):
                window_state += s[r]

            best = max(best, r - l + 1)
        return best
```

Community Solutions (Python)

- WalkCC

```
class Solution:
    def findAnagrams(self, s: str, p: str) -> list[int]:
        ans = []
        count = collections.Counter(p)
        required = len(p)

        for r, c in enumerate(s):
            count[c] -= 1
            if count[c] >= 0:
                required -= 1

            if r >= len(p):
                count[s[r - len(p)]] += 1
                if count[s[r - len(p)]] > 0:
                    required += 1
                if required == 0:
                    ans.append(r - len(p) + 1)

        return ans
```

- LeetCode

```
class Solution:
    def findAnagrams(self, s: str, p: str) -> List[int]:
        m, n = len(s), len(p)
        ans = []
        if m < n:
            return ans
        cnt1 = Counter(p)
        cnt2 = Counter(s[: n - 1])
        for i in range(n - 1, m):
            cnt2[s[i]] += 1
```

```
if cnt1 == cnt2:
    ans.append(i - n + 1)
cnt2[s[i - n + 1]] -= 1
return ans
```

- SimplyLeet

```
# Define the function to find all anagram indices in the string
def findAnagrams(s: str, p: str):
    # Import Counter for frequency counting
    from collections import Counter
    # List to store the starting indices of all anagrams
    result = []
    # Length of string p
    p_length = len(p)
    # Frequency count for characters in p
    p_count = Counter(p)
```

```
# Frequency count for the current window in s
s_count = Counter()

# Iterate over the string s
for i in range(len(s)):
    # Add current character to the window count
    s_count[s[i]] += 1

    # Remove the leftmost character from window if
    window size exceeds p_length
    if i >= p_length:
```

Sliding Window

p.244

Sliding Window

p.245

Sliding Window

p.246

```
if s_count[s[i - p_length]] == 1:
    del s_count[s[i - p_length]] # Remove key
    completely if count becomes 0
else:
    s_count[s[i - p_length]] -= 1

# Compare window count with p_count when window
size is equal to p_length
if s_count == p_count:
    result.append(i - p_length + 1)

return result
```

```
# Example usage:
print(findAnagrams("cbaebabacd", "abc")) # Expected
output: [0, 6]
```

- LC.ca

```
class Solution:
    def findAnagrams(self, s: str, p: str) -> List[int]:
        m, n = len(s), len(p)
        ans = []
        if m < n:
            return ans
        cnt1 = Counter(p)
        cnt2 = Counter(s[: n - 1])
        for i in range(n - 1, m):
            cnt2[s[i]] += 1
```

Sliding Window

p.247

```
if cnt1 == cnt2:
    ans.append(i - n + 1)
cnt2[s[i - n + 1]] -= 1
return ans
```

Sliding Window

p.248

Sliding Window

#487 Max Consecutive Ones II

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming · Sliding Window

Lists: Premium 98

Time Complexity: O(n) where n is the number of elements in the array. Space Complexity: O(1).

Problem

Given a binary array nums, return the maximum number of consecutive 1's in the array if you can flip at most one 0.

Example 1:

```
Input: nums = [1,0,1,1,0]
Output: 4
Explanation:
- If we flip the first zero, nums becomes [1,1,1,1,0]
  and we have 4 consecutive ones.
- If we flip the second zero, nums becomes [1,0,1,1,1]
  and we have 3 consecutive ones.
The max number of consecutive ones is 4.
```

Example 2:

Sliding Window

p.249

```
Input: nums = [1,0,1,1,0,1]
Output: 4
Explanation:
- If we flip the first zero, nums becomes [1,1,1,1,0,1]
  and we have 4 consecutive ones.
- If we flip the second zero, nums becomes
  [1,0,1,1,1,1] and we have 4 consecutive ones.
The max number of consecutive ones is 4.
```

Constraints:

- 1 <= nums.length <= 105
- nums[i] is either 0 or 1.

Follow up: What if the input numbers come in one by one as an infinite stream? In other words, you can't store all numbers coming from the stream as it's too large to hold in memory. Could you solve it efficiently?

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def findMaxConsecutiveOnes(self, nums):
        # Brute Force O(n^2) - enumerate all windows
        n = len(nums)
        best = 0
        for l in range(n):
            window_state = 0
            for r in range(l, n):
                window_state += nums[r]

            best = max(best, r - l + 1)
        return best
```

Sliding Window

p.250

Community Solutions (Python)

- WalkCC

```
class Solution:
    def findMaxConsecutiveOnes(self, nums: list[int]) -> int:
        ans = 0
        zeros = 0

        l = 0
        for r, num in enumerate(nums):
            if num == 0:
                zeros += 1
                while zeros == 2:

                    if nums[l] == 0:
                        zeros -= 1
                        l += 1
                    ans = max(ans, r - l + 1)

        return ans
```

- LeetCode

```
class Solution:
    def findMaxConsecutiveOnes(self, nums: List[int]) -> int:
        l = cnt = 0
        for x in nums:
            cnt += x ^ 1
            if cnt > 1:
                cnt -= nums[l] ^ 1
                l += 1
        return len(nums) - l
```

- SimplyLeet

```
# Function to find the maximum consecutive ones with at
most one zero flip
def findMaxConsecutiveOnes(nums):
    max_length = 0 # Tracks the maximum length of the
window
    left = 0 # Left pointer for the sliding window
    zero_count = 0 # Counter for zeros in the current
window

    # Iterate with right pointer over the entire array
    for right in range(len(nums)):
        # If current element is zero, increase the
zero_count
        if nums[right] == 0:
```

Sliding Window

p.251

Sliding Window

p.252

```
        zero_count += 1

        # If window invalid (more than one zero), shrink
        from left
        while zero_count > 1:
            if nums[left] == 0:
                zero_count -= 1
                left += 1

        # Update max_length if current window is larger
        max_length = max(max_length, right - left + 1)

    return max_length

# Example usage:
# nums = [1,0,1,1,0]
# print(findMaxConsecutiveOnes(nums)) # Expected output:
# 4
```

```
• LC.ca

from collections import deque

class Solution:
    def findMaxConsecutiveOnes(self, nums: List[int]) ->
    int:

        res, zero, left, k = 0, 0, 0, 1

        for right in range(len(nums)):
            if nums[right] == 0:
                zero += 1
```

Sliding Window p.253

```
        while zero > k:
            if nums[left] == 0:
                zero -= 1
                left += 1
            res = max(res, right - left + 1) # max-check
        every for iteration

        return res

class Solution: # followup, extra space
    def findMaxConsecutiveOnes(self, nums: List[int]) ->
    int:

        res, left, k = 0, 0, 1
        q = deque()

        for right in range(len(nums)):
            if nums[right] == 0:
                q.append(right)
                if len(q) > k:
                    left = q.popleft() + 1
            res = max(res, right - left + 1) # max-check
        every for iteration
```

Sliding Window p.254

```
        return res

#####

# maintain a sliding window of size 2, that keeps track
of the counts.
class Solution: # follow up, optimized
    def findMaxConsecutiveOnes(self, nums: List[int]) ->
    int:

        count = 0 # Count of consecutive ones
        prev_count = 0 # Count of consecutive ones before
        a zero

        max_count = 0 # Maximum count of consecutive ones

        for num in nums:
            if num == 1:
                count += 1
                prev_count += 1
            else:
                count = prev_count + 1
                prev_count = 0
```

Sliding Window p.255

```
        max_count = max(max_count, count)

        # Reset the count and prev_count if we
        encounter two zeros in a row
        if num == 0 and prev_count == 0:
            count = 0
            prev_count = 0

        return max_count

#####

class Solution:
    def findMaxConsecutiveOnes(self, nums: List[int]) ->
    int:

        l = r = 0
        k = 1
        while r < len(nums):
            if nums[r] == 0:
                k -= 1
            if k < 0:
                if nums[l] == 0:
```

Sliding Window p.256

```
                k += 1
                l += 1
                r += 1
            return r - l

#####

class Solution(object):
    def __init__(self):
        self.ans = 0

        self.count = 0
        self.lastCount = 0

        def findMaxConsecutiveOnes(self, nums):
            """"
            :type nums: List[int]
            :rtype: int
            """
            for num in nums:
                self.readNum(num) # stream the input

        return self.ans

    def readNum(self, num):
        """"
        :type nums: int
        """
        if num == 1:
            self.count += 1
        else:
            self.count = self.count - self.lastCount + 1
```

Sliding Window p.257

```
        self.lastCount = self.count
        self.ans = max(self.ans, self.count)
```

Sliding Window p.258

Sliding Window

#1100 Find K–Length Substrings with No Repeated Characters MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Sliding Window

Lists: Premium 98

Time Complexity: O(n) where n is the length of s, as each character is visited at most twice. Space Complexity: O(1) since the frequency tracking is over a fixed set (26 lowercase letters).

Problem

Given a string s and an integer k, return the number of substrings in s of length k with no repeated characters.

Example 1:

Input: s = "havefunonleetcode", k = 5
Output: 6
Explanation: There are 6 substrings they are: 'havef', 'avefu', 'vefun', 'efuno', 'etcod', 'tcode'.

Example 2:

Input: s = "home", k = 5
Output: 0
Explanation: Notice k can be larger than the length of s. In this case, it is not possible to find any substring.

Sliding Window p.259

Constraints:

• 1 <= s.length <= 104

• s consists of lowercase English letters.

• 1 <= k <= 104

Community Solutions (Python)

• WalkCC

class Solution:
 def numKLenSubstrNoRepeats(self, s: str, k: int) ->
 int:

 ans = 0
 unique = 0
 count = collections.Counter()

 for i, c in enumerate(s):
 count[c] += 1
 if count[c] == 1:
 unique += 1

 if i >= k:
 count[s[i - k]] -= 1
 if count[s[i - k]] == 0:
 unique -= 1
 if unique == k:
 ans += 1

 return ans

• LeetDoocs

Sliding Window p.260

class Solution:
 def numKLenSubstrNoRepeats(self, s: str, k: int) ->
 int:

 cnt = Counter(s[:k])
 ans = int(len(cnt) == k)
 for i in range(k, len(s)):
 cnt[s[i]] += 1
 cnt[s[i - k]] -= 1
 if cnt[s[i - k]] == 0:
 cnt.pop(s[i - k])
 ans += int(len(cnt) == k)

 return ans

• SimplyLeet

def numKLenSubstrNoRepeats(s: str, k: int) -> int:
 # If k is larger than the string length, no valid
 substring exists.
 if k > len(s):
 return 0

 count = 0
 left = 0
 # Dictionary to store the frequency of characters in
 the current window.
 char_count = {}

Sliding Window p.261

```
# Iterate over the string using right pointer.
for right in range(len(s)):
    # Update frequency count for the current
    character.
    char_count[s[right]] = char_count.get(s[right],
0) + 1

    # If current character frequency > 1, slide
    window from the left to remove duplicates.
    while char_count[s[right]] > 1:
        char_count[s[left]] -= 1
        left += 1

    # When window size equals k, we have a valid
    substring.
    if right - left + 1 == k:
        count += 1
        # Slide the window by removing the leftmost
        character.
        char_count[s[left]] -= 1
        left += 1
    return count

# Example usage:
# print(numKLenSubstrNoRepeats("havefunonleetcode", 5))
#-> 6
```

• LC.ca

Sliding Window

p.262

```
class Solution:
    def numKLenSubstrNoRepeats(self, s: str, k: int) ->
    int:
        n = len(s)
        if k > n or k > 26:
            return 0
        ans = j = 0
        cnt = Counter()
        for i, c in enumerate(s):
            cnt[c] += 1
            while cnt[c] > 1 or i - j + 1 > k:

                cnt[s[j]] -= 1
                j += 1
                ans += i - j + 1 == k
            return ans
```

Sliding Window

p.263

Sliding Window

#1248 Count Number of Nice Subarrays

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Math · Sliding Window · Prefix Sum
Lists: CodeSignal

Time Complexity: $O(n)$, where n is the number of elements in `nums`, because we traverse the array once.
Space Complexity: $O(n)$, in the worst case where each prefix sum is unique and stored in the hash `t`

Problem

Given an array of integers `nums` and an integer `k`. A continuous subarray is called nice if there are `k` odd numbers on it.

Return the number of nice sub-arrays.

Example 1:

```
Input: nums = [1,1,2,1,1], k = 3
Output: 2
Explanation: The only sub-arrays with 3 odd numbers are
[1,1,2,1] and [1,2,1,1].
```

Example 2:

```
Input: nums = [2,4,6], k = 1
Output: 0
Explanation: There are no odd numbers in the array.
```

Sliding Window

p.264

Example 3:

```
Input: nums = [2,2,2,1,2,2,1,2,2,2], k = 2
Output: 16
```

Constraints:

- $1 \leq \text{nums.length} \leq 50000$
- $1 \leq \text{nums}[i] \leq 10^5$
- $1 \leq k \leq \text{nums.length}$

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def numberOfSubarrays(self, nums, k):
        # Brute Force  $O(n^2)$  - enumerate all windows
        n = len(nums)
        best = 0
        for l in range(n):
            window_state = 0
            for r in range(l, n):
                window_state += nums[r]

                best = max(best, r - l + 1)
        return best
```

Community Solutions (Python)

• WalkCC

Sliding Window

p.265

```
class Solution:
    def numberOfSubarrays(self, nums: List[int], k: int) ->
    int:
        def numberOfSubarraysAtMost(k: int) -> int:
            ans = 0
            l = 0
            r = 0

            while r <= len(nums):
                if k >= 0:
                    ans += r - l

                    if r == len(nums):
                        break
                    if nums[r] & 1:
                        k -= 1
                        r += 1
                    else:
                        if nums[l] & 1:
                            k += 1
                            l += 1
                        l += 1
                    return ans

            return numberOfSubarraysAtMost(k) -
            numberOfSubarraysAtMost(k - 1)
```

• LeetDoocs

Sliding Window

p.266

```
class Solution:
    def numberOfSubarrays(self, nums: List[int], k: int)
    -> int:
        cnt = Counter({0: 1})
        ans = t = 0
        for v in nums:
            t += v & 1
            ans += cnt[t - k]
            cnt[t] += 1
        return ans
```

• SimplyLeet

Python solution with line-by-line comments

```
def numberOfSubarrays(nums, k):
    # Initialize dictionary to store frequency of prefix
    sums
    prefix_count = {0: 1} # There is one way to have zero
    odd numbers initially.
    count = 0 # This will store the final count of nice
    subarrays
    odd_sum = 0 # Prefix sum representing the count of
    odd numbers so far

    # Iterate through each number in the array
    for num in nums:
```

Sliding Window

p.267

```
# If the number is odd, increment the odd_count
prefix sum
if num % 2 == 1:
    odd_sum += 1

# If there is a prefix such that (current odd_sum
- prefix) equals k,
# then we found subarrays ending here with k odd
numbers
count += prefix_count.get(odd_sum - k, 0)

# Update the frequency for the current odd_sum in
the hash table
prefix_count[odd_sum] = prefix_count.get(odd_sum,
0) + 1

# Return the total number of nice subarrays
return count

# Example usage:
# nums = [1,1,2,1,1]
# k = 3
# print(numberOfSubarrays(nums, k)) # Expected output is
2
```

• LC.ca

Sliding Window

p.268

```
class Solution:
    def numberOfSubarrays(self, nums: List[int], k: int)
    -> int:
        cnt = Counter({0: 1})
        ans = t = 0
        for v in nums:
            t += v & 1
            ans += cnt[t - k]
            cnt[t] += 1
        return ans
```

Sliding Window

p.269

Sliding Window

#76 Minimum Window Substring

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Sliding Window
Lists: Grind 169

Time Complexity: $O(m + n)$, where m is the length of `s` and n is the length of `t`. Space Complexity: $O(m + n)$ in the worst case due to the storage for the hash maps.

Problem

Given two strings `s` and `t` of lengths `m` and `n` respectively, return the minimum window substring of `s` such that every character in `t` (including duplicates) is included in the window. If there is no such substring, return the empty string `""`.

The testcases will be generated such that the answer is unique.

Example 1:

```
Input: s = "ADOBECODEBANC", t = "BANC"
Output: "BANC"
Explanation: The minimum window substring "BANC"
includes 'A', 'B', and 'C' from string t.
```

Example 2:

Sliding Window

p.270

Input: s = "a", t = "a"
Output: "a"
Explanation: The entire string s is the minimum window.

Example 3:

Input: s = "a", t = "aa"
Output: ""
Explanation: Both 'a's from t must be included in the window.
Since the largest window of s only has one 'a', return empty string.

Constraints:

- m == s.length
- n == t.length
- 1 <= m, n <= 105
- s and t consist of uppercase and lowercase English letters.

Follow up: Could you find an algorithm that runs in O(m + n) time?

Community Solutions (Python)

- WalkCC

```
class Solution:
    def minWindow(self, s: str, t: str) -> str:
        count = collections.Counter(t)
        required = len(t)
        bestLeft = -1
        minLength = len(s) + 1

        l = 0
        for r, c in enumerate(s):
            count[c] -= 1

            if count[c] >= 0:
                required -= 1
                while required == 0:
                    if r - l + 1 < minLength:
                        bestLeft = l
                        minLength = r - l + 1
                    count[s[l]] += 1
                    if count[s[l]] > 0:
                        required += 1
                    l += 1

        return '' if bestLeft == -1 else s[bestLeft: bestLeft + minLength]
```

- LeetDoocs

```
class Solution:
    def minWindow(self, s: str, t: str) -> str:
        need = Counter(t)
        window = Counter()
        cnt = l = 0
        k, mi = -1, inf
        for r, c in enumerate(s):
            window[c] += 1
            if need[c] >= window[c]:
                cnt += 1

            while cnt == len(t):
                if r - l + 1 < mi:
                    mi = r - l + 1
                    k = l
                if need[s[l]] >= window[s[l]]:
                    cnt -= 1
                window[s[l]] -= 1
                l += 1
            return "" if k < 0 else s[k : k + mi]
```

- SimplyLeet

```
# Python implementation of Minimum Window Substring solution

def minWindow(s: str, t: str) -> str:
    # If t is longer than s, no solution exists.
    if len(t) > len(s):
        return ""

    # Dictionary to store the frequency of each character in t.
    t_freq = {}
    for char in t:
        t_freq[char] = t_freq.get(char, 0) + 1

    # Dictionary to keep track of the characters in the current window.
    window_freq = {}

    required = len(t_freq) # Unique characters in t to be matched
    formed = 0 # Number of unique characters in the current window that meet required frequency

    # (window_length, left, right)
    answer = (float('inf'), None, None)
```

```
l = 0 # Left pointer
# Start sliding the window with the right pointer.
for r, char in enumerate(s):
    # Add the current character to the window counter.
    window_freq[char] = window_freq.get(char, 0) + 1

    # Check if the current character's frequency in the window matches that in t.
    if char in t_freq and window_freq[char] == t_freq[char]:
        formed += 1

    # Try and contract the window if all characters are matched.
    while l <= r and formed == required:
        character = s[l]

        # Update answer if the current window is smaller than the previous best.
        if r - l + 1 < answer[0]:
            answer = (r - l + 1, l, r)

        # The character at the left is going to be removed from the window.
```

```
window_freq[character] -= 1
if character in t_freq and window_freq[character] < t_freq[character]:
    formed -= 1

# Move the left pointer ahead.
l += 1

# Return the smallest window, if found.
return "" if answer[0] == float('inf') else s[answer[1]:answer[2]+1]
```

```
# Example usage:
if __name__ == "__main__":
    s = "ADOBECODEBANC"
    t = "ABC"
    print(minWindow(s, t)) # Expected output: "BANC"
```

- LC.ca

```
'''
>>> deq = collections.deque([])
>>> deq.append(11)
>>> deq.append(22)
>>> deq.append(33)
>>>
>>> deq[0]
11
'''
```

```
import collections

class Solution:
    def minWindow(self, s: str, t: str) -> str:
        cnt = 0
        need = collections.Counter(t)
        start, end = len(s), 3 * len(s) # Arbitrary 3, just make sure end-start is larger than input s length, for later min-check
        d = {}
        # Using queue to store indexes, good for a large amount of API calls
        deq = collections.deque([])

        for i, c in enumerate(s):
            if c in need:
                deq.append(i)
                d[c] = d.get(c, 0) + 1
                # '=' also +1, because it's increased already one line above :)
                if d[c] <= need[c]:
                    cnt += 1
                while deq and d[s[deq[0]]] > need[s[deq[0]]]:
                    d[s[deq.popleft()]] -= 1
                    if cnt == len(t) and deq[-1] - deq[0] < end - start:
```

```
start, end = deq[0], deq[-1]
return s[start:end + 1]

#####

from collections import Counter

class Solution:
    def minWindow(self, s: str, t: str) -> str:
        ans = ''

        m, n = len(s), len(t)
        if m < n:
            return ans
        need = Counter(t)
        window = Counter()
        i, cnt, mi = 0, 0, inf
        for j, c in enumerate(s):
            window[c] += 1
            if need[c] >= window[c]: # >= , because 1
                line above already +1
                cnt += 1
```

```
while cnt == n:
    if j - i + 1 < mi: # in while
        mi = j - i + 1
        ans = s[i : j + 1]
    c = s[i]
    if need[c] >= window[c]: # char in window but not in need, need[c]=0, window[c]=1..2..
        cnt -= 1
        window[c] -= 1
        i += 1
    return ans
```

	>> Quick Review — Sliding Window 9 questions · insights · complexity · solution #3 Longest Substring Without Repeating Characters [Medium] Key Insights - Use the sliding window technique to maintain a window of unique characters. - Use a hash table (or dictionary) to quickly check for duplicate characters and update window boundaries. - The window is expanded by moving a pointer (right) and contracted by moving another pointer (left) when duplicates are encountered. - Time complexity can be maintained at O(n) by ensuring each character is processed a limited number of times. Space & Time Time Complexity: O(n), where n is the length of the input string. Space Complexity: O(min(n, m)), where m is the size of the character set (constant for fixed character sets). Solution The solution uses a sliding window technique with two pointers, left and right, to keep track of the current Sliding Windowp.280	substring without repeating characters. A hash table (dictionary or map) is used to store the most recent index of each character encountered. As the right pointer moves forward, if a duplicate character is found within the current window, the left pointer is advanced to the position right after where the duplicate was last seen. This ensures that the window always contains unique characters. The maximum length is updated throughout the process. #159 Longest Substring with At Most Two Distinct Characters [Medium] Key Insights - Use a sliding window to maintain a contiguous substring. - Keep track of character frequencies within the current window using a hash map or dictionary. - When the window contains more than two distinct characters, shrink the window from the left until only two remain. - Update the maximum length each time the window satisfies the condition. Space & Time Time Complexity: O(n) where n is the length of the string, since each character is processed at most twice. Sliding Windowp.282
Space Complexity: O(1) because the hash map holds at most 3 entries (usually up to 2 distinct characters). Solution We use the sliding window technique with two pointers (left and right) to represent the current window. A hash table/dictionary records the count of each character inside the window. As we extend the right pointer, we update the count. If the hash table contains more than two distinct characters, we start moving the left pointer to reduce the number of distinct characters until we have at most two distinct keys. The length of the current valid window is then used to update our maximum length answer. #340 Longest Substring with At Most K Distinct Characters [Medium] Key Insights - Use the sliding window technique to efficiently scan through the string. - Maintain a hash table (or dictionary) that keeps track of the frequency of characters in the current window. - When the number of distinct characters exceeds k, increment the left pointer to shrink the window until the condition is met. - Update the maximum length of valid substrings throughout the process. Sliding Windowp.283	Space & Time Time Complexity: O(n), where n is the length of the string, as each character is processed at most twice (once when expanding the window and once when contracting it). Space Complexity: O(min(m, k)), where m is the size of the charset (or number of unique characters in s), since at most k+1 keys are stored. Solution This problem is solved using the sliding window technique. The algorithm keeps track of a window defined by two indices (left and right pointers). A hash map is used to count the frequency of each character in the current window. As the right pointer extends the window, the count of the character is incremented. When the count of distinct keys in the map exceeds k, the left pointer is moved forward while decrementing the count of the leftmost character, until the condition is restored. The maximum window length encountered is recorded and returned as the result. #424 Longest Repeating Character Replacement [Medium] Key Insights Use a Sliding Window algorithm to keep track of a contiguous substring. Sliding Windowp.284	- Maintain a frequency count of letters in the current window. - Keep track of the count of the most frequent character in the window. - If the current window size minus the maximum frequency is greater than k, shrink the window. - The maximum window size seen during the process is the answer. Space & Time Time Complexity: O(n) where n is the length of the string, since the window traverses the string in linear time. Space Complexity: O(1) because the frequency count for letters (A-Z) has a constant size. Solution The problem is solved using the sliding window technique. We initialize two pointers representing the left and right bounds of the current window. As we expand the window by moving the right pointer, we update the frequency count of the letter at the right pointer and track the letter that appears most frequently. The key observation is that the number of characters that need to be replaced in the current window is the window's size minus the count of the most frequently occurring letter. If this number exceeds k, we increment the left pointer to shrink the window until the condition is satisfied again. The maximum size Sliding Windowp.285
of the window during the process gives the answer. #438 Find All Anagrams in a String [Medium] Key Insights - Use a sliding window of length equal to p over s. - Maintain frequency counts of characters in p and within the current window in s. - Compare frequency counts to determine if the current window is an anagram. - Update the window efficiently by adding one character and removing the character that goes out of the window. Space & Time Time Complexity: O(n) where n is the length of s. Space Complexity: O(1) since we use fixed-size frequency arrays or hash maps. Solution We use the sliding window technique to iterate over s with a window size equal to the length of p. First, count the frequencies of characters in p. Then, iterate over s and update a frequency count for the current window. When the window size equals the length of p, compare the two frequency counts. If they match, record the start index. To slide the window efficiently, increment the count for the incoming character and decrement the count for the outgoing character. This way, we achieve an O(n) solution that is optimal for the problem Sliding Windowp.286	constraints. #487 Max Consecutive Ones II [Medium] Key Insights - Utilize a sliding window to maintain the current segment. - The window can contain at most one 0. - Expand the window with the right pointer and contract it with the left pointer when the constraint is violated. - The maximum window size observed during this process is the desired result. Space & Time Time Complexity: O(n) where n is the number of elements in the array. Space Complexity: O(1). Solution We use a sliding window approach to efficiently find the longest contiguous subarray that contains at most one 0. Two pointers (left and right) define the window. As we iterate with the right pointer, counting zeros in the window, we adjust the left pointer when the count of zeros exceeds one. The length of the valid window gives us the number of consecutive 1's (considering one flipped 0), and we update the maximum length accordingly. Sliding Windowp.287	#1100 Find K-Length Substrings with No Repeated Characters [Medium] Key Insights - Use a sliding window of fixed size k. - Track character counts within the current window using a hash table (or frequency array). - When a duplicate is found in the window, slide the window from the left to remove duplicates. - Count a valid window when its size equals k and all characters are unique. - Optimize by adjusting the window immediately after counting a valid substring. Space & Time Time Complexity: O(n) where n is the length of s, as each character is visited at most twice. Space Complexity: O(1) since the frequency tracking is over a fixed set (26 lowercase letters). Solution We use a sliding window approach with two pointers: left and right. A hash table (or frequency array) tracks the occurrences of characters in the window. As we move the right pointer, we update the frequency and check for duplicates. If a duplicate is encountered, we increment the left pointer until the duplicate is removed. Once the window size exactly equals k, we have a valid substring, so we increment our count and then slide the Sliding Windowp.288

window forward. This ensures that every possible k-length substring is examined once.

#1248 Count Number of Nice Subarrays [Medium]

Key Insights

- Convert the problem to counting subarrays with a specific "prefix sum" value where the sum represents the count of odd numbers.
- Use a hash table (or dictionary) to store the frequency of prefix sums encountered.
- The number of nice subarrays ending at the current index equals the frequency of (current prefix sum – k) seen so far.
- This approach allows you to find the answer in one pass over the array.

Space & Time

Time Complexity: O(n), where n is the number of elements in nums, because we traverse the array once.
Space Complexity: O(n), in the worst case where each prefix sum is unique and stored in the hash table.

Solution

We solve the problem using a prefix sum approach. First, we iterate through the array and convert it into a prefix sum representing the cumulative count of odd numbers encountered. As we compute the prefix sums, we use a hash table to record how many times each prefix sum

Sliding Window p.289

appears. For every current prefix sum value, we check how many times we have seen a prefix sum equal to (current prefix sum – k). The idea is that if prefix[j] – prefix[i] equals k, then between indices i+1 and j, there are exactly k odd numbers. This gives us the count of valid subarrays ending at the current index.
This method works efficiently as it requires only a single pass through the array and maintains a running frequency count using the hash table.

#76 Minimum Window Substring [Hard]

Key Insights

- Use a sliding window approach to dynamically adjust the window boundaries.
- Utilize a hash table (or dictionary) to keep track of the frequency of characters required (from t) and the frequency within the current window.
- Expand the window until it satisfies the requirements, then contract to minimize the window.
- Efficiently update counts and check valid window with two pointers (left and right).

Space & Time

Time Complexity: O(m + n), where m is the length of s and n is the length of t.
Space Complexity: O(m + n) in the worst case due to the storage for the hash maps.

Sliding Window p.290

Solution

We use a sliding window technique with two pointers representing the window's boundaries. First, create a frequency map of characters from t. Then, expand the right pointer to include characters in the window until the window contains all required characters (meeting or exceeding the counts in t). Once the requirement is met, try contracting the window by moving the left pointer to reduce its size while still maintaining a valid window. Throughout the process, if a valid window is found that is smaller than the previously recorded minimum window, update the minimum. The solution leans heavily on hash tables to track counts and compares them against the required counts from t.

Sorting

#6 Sorting

By Pattern · Python Only · 9 questions

Sliding Window p.291

Sorting

#169 Majority Element

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Divide and Conquer · Sorting · Counting

Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(1)

Problem

Given an array nums of size n, return the majority element.

The majority element is the element that appears more than n / 2 times. You may assume that the majority element always exists in the array.

Example 1:

Input: nums = [3,2,3]
Output: 3

Example 2:

Input: nums = [2,2,1,1,1,2,2]
Output: 2

Constraints:

- n == nums.length
- 1 <= n <= 5 * 10⁴
- -109 <= nums[i] <= 109

Sorting p.292

- The input is generated such that a majority element will exist in the array.

Follow-up: Could you solve the problem in linear time and in O(1) space?

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def majorityElement(self, nums):
        # Brute Force O(n^2) - for each element count its frequency
        n = len(nums)
        for num in nums:
            if nums.count(num) > n // 2:
                return num
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def majorityElement(self, nums: list[int]) -> int:
        ans = None
        count = 0

        for num in nums:
            if count == 0:
                ans = num
            count += (1 if num == ans else -1)

        return ans
```

• LeetDoocs

Sorting p.293

```
class Solution:
    def majorityElement(self, nums: List[int]) -> int:
        cnt = m = 0
        for x in nums:
            if cnt == 0:
                m, cnt = x, 1
            else:
                cnt += 1 if m == x else -1
        return m
```

• SimplyLeet

Python implementation of the Boyer-Moore Voting Algorithm

```
def majorityElement(nums):
    # Initialize candidate and counter
    candidate = None
    count = 0

    # Iterate over each number in the input list
    for num in nums:
        # If count is 0, choose the current number as the candidate
```

Sorting p.294

```
if count == 0:
    candidate = num
# Increment count if the current number equals the candidate, otherwise decrement
count += 1 if num == candidate else -1
# Return the candidate, which is the majority element
return candidate
```

Example usage
print(majorityElement([2,2,1,1,1,2,2])) # Output: 2

• LC.ca

```
class Solution:
    def majorityElement(self, nums: List[int]) -> int:
        cnt = m = 0
        for x in nums:
            if cnt == 0:
                m, cnt = x, 1
            else:
                cnt += 1 if m == x else -1
        return m
```

Sorting p.295

Sorting

#217 Contains Duplicate

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Sorting

Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(n)

Problem

Given an integer array nums, return true if any value appears at least twice in the array, and return false if every element is distinct.

Example 1:

Input: nums = [1,2,3,1]

Output: true

Explanation:

The element 1 occurs at the indices 0 and 3.

Example 2:

Input: nums = [1,2,3,4]

Output: false

Explanation:

All elements are distinct.

Example 3:

Input: nums = [1,1,1,3,3,4,3,2,4,2]

Output: true

Sorting p.296

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Constraints:

- 1 <= nums.length <= 10⁵
- -109 <= nums[i] <= 109

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def contains_duplicate(self, nums):
        # Brute Force O(n^2) - compare every pair
        for i in range(len(nums)):
            for j in range(i + 1, len(nums)):
                if nums[i] == nums[j]:
                    return True
        return False
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def containsDuplicate(self, nums: list[int]) -> bool:
        return len(nums) != len(set(nums))
```

• LeetDoocs

```
class Solution:
    def containsDuplicate(self, nums: List[int]) -> bool:
        return any(a == b for a, b in pairwise(sorted(nums)))
```

• SimplyLeet

Sorting p.297

# Function to determine if there are any duplicates in the list. def contains_duplicate(nums): # Create a set to keep track of numbers seen so far. seen = set() # Iterate over each number in the list. for num in nums: # If num is already in seen, duplicate found. if num in seen: return True # Add num to the set. seen.add(num) # No duplicates found. return False # Example usage: if __name__ == "__main__": nums = [1, 2, 3, 1] print(contains_duplicate(nums)) # Expected output: True • LC.ca class Solution: def containsDuplicate(self, nums: List[int]) -> bool: return len(set(nums)) < len(nums) ##### class Solution(object): def containsDuplicate(self, nums): """" :type nums: List[int] Sortingp.298	:rtype: bool """" nums.sort() for i in range(0, len(nums) - 1): if nums[i] == nums[i + 1]: return True return False Sortingp.299	Sorting #242 Valid AnagramEAS LeetDoocs · SimplyLeet · WalkCC · LeetCode Hash Table · String · Sorting Lists: Grind 169 Time Complexity: O(n), where n is the length of the strings (using a hash table for counting). Space Complexity: O(1) if we use a fixed-size array (for 26 letters), otherwise O(n) for a hash table wit Problem Given two strings s and t, return true if t is an anagram of s, and false otherwise. Example 1: Input: s = "anagram", t = "nagaram" Output: true Example 2: Input: s = "rat", t = "car" Output: false Constraints: • 1 <= s.length, t.length <= 5 * 10⁴ • s and t consist of lowercase English letters. Follow up: What if the inputs contain Unicode characters? How would you adapt your solution to such a case? Sortingp.300
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Community Solutions (Python) • WalkCC class Solution: def isAnagram(self, s: str, t: str) -> bool: if len(s) != len(t): return False count = collections.Counter(s) count.subtract(collections.Counter(t)) return all(freq == 0 for freq in count.values()) • LeetDoocs class Solution: def isAnagram(self, s: str, t: str) -> bool: if len(s) != len(t): return False cnt = Counter(s) for c in t: cnt[c] -= 1 if cnt[c] < 0: return False return True class Solution: def isAnagram(self, s: str, t: str) -> bool: return Counter(s) == Counter(t) • SimplyLeet Sortingp.301	# Python solution for Valid Anagram def isAnagram(s: str, t: str) -> bool: # Check if lengths differ; if so, they can't be anagrams if len(s) != len(t): return False # Create a dictionary to count frequency of each character in s char_count = {} for char in s: # Increase the count of the character char_count[char] = char_count.get(char, 0) + 1 # Decrease the count for each character in t for char in t: if char not in char_count or char_count[char] == 0: # Character not found or count mismatch, not an anagram return False char_count[char] -= 1 # If all counts are zero, it's a valid anagram return True # Example usage: print(isAnagram("anagram", "nagaram")) # Output: True print(isAnagram("rat", "car")) # Output: False • LC.ca Sortingp.302	class Solution: def isAnagram(self, s: str, t: str) -> bool: if len(s) != len(t): return False cnt = Counter(s) for c in t: cnt[c] -= 1 if cnt[c] < 0: return False return True Sortingp.303
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Sorting #252 Meeting RoomsEAS LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Sorting Lists: Grind 169 Premium 98 Time Complexity: O(n log n) due to sorting the intervals. Space Complexity: O(1) if sorting is done in-place; otherwise O(n). Problem Given an array of meeting time intervals where intervals[i] = [starti, endi], determine if a person could attend all meetings. Example 1: Input: intervals = [[0,30],[5,10],[15,20]] Output: false Example 2: Input: intervals = [[7,10],[2,4]] Output: true Constraints: • 0 <= intervals.length <= 104 • intervals[i].length == 2 • 0 <= starti < endi <= 106 Sortingp.304	Brute Force (Python) # Brute Force Approach class Solution: def canAttendMeetings(self, intervals): # Brute Force O(n^2) - check every pair of intervals for overlap for i in range(len(intervals)): for j in range(i + 1, len(intervals)): a, b = intervals[i], intervals[j] if a[0] < b[1] and b[0] < a[1]: # overlap return False return True Community Solutions (Python) • WalkCC class Solution: def canAttendMeetings(self, intervals: list[list[int]]) -> bool: intervals.sort() for i in range(1, len(intervals)): if intervals[i - 1][1] > intervals[i][0]: return False return True • LeetDoocs Sortingp.305	class Solution: def canAttendMeetings(self, intervals: List[List[int]]) -> bool: intervals.sort() return all(a[1] <= b[0] for a, b in pairwise(intervals)) • SimplyLeet # Python solution for checking if one can attend all meetings def canAttendMeetings(intervals): # Sort intervals based on start time intervals.sort(key=lambda x: x[0]) # Iterate through intervals to find any overlap for i in range(1, len(intervals)): # Check if the previous meeting ends after the current meeting starts if intervals[i-1][1] > intervals[i][0]: return False return True # Example usage: print(canAttendMeetings([[0, 30], [5, 10], [15, 20]])) # Expected output: False print(canAttendMeetings([[7, 10], [2, 4]])) # Expected output: True • LC.ca Sortingp.306
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class Solution: def canAttendMeetings(self, intervals: List[List[int]]) -> bool: intervals.sort() return all(a[1] <= b[0] for a, b in pairwise(intervals)) ##### from itertools import pairwise class Solution: def canAttendMeetings(self, intervals: List[List[int]]) -> bool: intervals.sort(key=lambda x: x[0]) return not any(a[1] > b[0] for a,b in pairwise(intervals)) # any is True, then conflict found ##### class Solution: def canAttendMeetings(self, intervals: List[List[int]]) -> bool: intervals.sort(key=lambda x: x[0]) for i in range(len(intervals) - 1):	if intervals[i][1] > intervals[i + 1][0]: return False return True ##### # Definition for an interval. # class Interval(object): # def __init__(self, s=0, e=0): # self.start = s # self.end = e class Solution(object): def canAttendMeetings(self, intervals): """ :type intervals: List[Interval] :rtype: bool """ intervals = sorted(intervals, key=lambda x: x.start) for i in range(1, len(intervals)): if intervals[i].start < intervals[i - 1].end: return False return True	Sorting #1086 Largest Unique NumberEAS LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Hash Table · Sorting Lists: Premium 98 Time Complexity: O(n), where n is the number of elements in nums. Space Complexity: O(n), for the hash table storing frequency counts. Problem Given a list of the scores of different students, items, where items[i] = [IDi, scorei] represents one score from a student with IDi, calculate each student's top five average. Return the answer as an array of pairs result, where result[j] = [IDj, topFiveAveragej] represents the student with IDj and their top five average. Sort result by IDj in increasing order. A student's top five average is calculated by taking the sum of their top five scores and dividing it by 5 using integer division. Example 1:
Sortingp.307	Sortingp.308	Sortingp.309

Input: items = [[1,91],[1,92],[2,93],[2,97],[1,60],[2,77],[1,65],[1,87],[1,100],[2,100],[2,76]] Output: [[1,87],[2,88]] Explanation: The student with ID = 1 got scores 91, 92, 60, 65, 87, and 100. Their top five average is (100 + 92 + 91 + 87 + 65) / 5 = 87. The student with ID = 2 got scores 93, 97, 77, 100, and 76. Their top five average is (100 + 97 + 93 + 77 + 76) / 5 = 88.6, but with integer division their average converts to 88. Example 2: Input: items = [[1,100],[7,100],[1,100],[7,100],[1,100],[7,100],[1,100],[7,100],[1,100],[7,100]] Output: [[1,100],[7,100]] Constraints: • 1 <= items.length <= 1000 • items[i].length == 2 • 1 <= IDi <= 1000 • 0 <= scorei <= 100 • For each IDi, there will be at least five scores. Brute Force (Python)	# Brute Force Approach class Solution: def largestUniqueNumber(self, nums): # Brute Force O(n^2) - bubble sort arr = list(nums) n = len(arr) for i in range(n): for j in range(n - i - 1): if arr[j] > arr[j + 1]: arr[j], arr[j + 1] = arr[j + 1], arr[j] return arr Community Solutions (Python) • WalkCC class Solution: def highFive(self, items: list[list[int]]) -> list[list[int]]: idToScores = collections.defaultdict(list) for id, score in items: heapq.heappush(idToScores[id], score) if len(idToScores[id]) > 5: heapq.heappop(idToScores[id]) return [[id, sum(scores) // 5] for id, scores in sorted(idToScores.items())] • LeetDoocs	class Solution: def highFive(self, items: List[List[int]]) -> List[List[int]]: d = defaultdict(list) m = 0 for i, x in items: d[i].append(x) m = max(m, i) ans = [] for i in range(1, m + 1): if xs := d[i]: avg = sum(nlargest(5, xs)) // 5 ans.append([i, avg]) return ans • LC.ca class Solution: def highFive(self, items: List[List[int]]) -> List[List[int]]: d = defaultdict(list) m = 0 for i, x in items: d[i].append(x) m = max(m, i) ans = [] for i in range(1, m + 1): if xs := d[i]: avg = sum(nlargest(5, xs)) // 5 ans.append([i, avg]) return ans
Sortingp.310	Sortingp.311	Sortingp.312

Sorting #1122 Relative Sort ArrayEAS LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Hash Table · Sorting · Counting Sort Lists: Denny Zhang Time Complexity: O(n + m log m) where n is the length of arr1 and m is the number of leftover elements not in arr2. Given that element values are bounded (0 to 1000), a counting sort can make this nea Problem Given two arrays arr1 and arr2, the elements of arr2 are distinct, and all elements in arr2 are also in arr1. Sort the elements of arr1 such that the relative ordering of items in arr1 are the same as in arr2. Elements that do not appear in arr2 should be placed at the end of arr1 in ascending order. Example 1: Input: arr1 = [2,3,1,3,2,4,6,7,9,2,19], arr2 = [2,1,4,3,9,6] Output: [2,2,2,1,4,3,3,9,6,7,19] Example 2: Input: arr1 = [28,6,22,8,44,17], arr2 = [22,28,8,6] Output: [22,28,8,6,17,44]	Constraints: • 1 <= arr1.length, arr2.length <= 1000 • 0 <= arr1[i], arr2[i] <= 1000 • All the elements of arr2 are distinct. • Each arr2[i] is in arr1. Brute Force (Python) # Brute Force Approach class Solution: def relativeSortArray(self, arr1, arr2): # Brute Force O(n^2) - bubble sort arr = list(arr1) n = len(arr) for i in range(n): for j in range(n - i - 1): if arr[j] > arr[j + 1]: arr[j], arr[j + 1] = arr[j + 1], arr[j] return arr Community Solutions (Python) • WalkCC	class Solution: def relativeSortArray(self, arr1: list[int], arr2: list[int]) -> list[int]: ans = [] count = [0] * 1001 for a in arr1: count[a] += 1 for a in arr2: while count[a] > 0: ans.append(a) count[a] -= 1 for num in range(1001): for _ in range(count[num]): ans.append(num) return ans • LeetDoocs class Solution: def relativeSortArray(self, arr1: List[int], arr2: List[int]) -> List[int]: pos = {x: i for i, x in enumerate(arr2)} return sorted(arr1, key=lambda x: pos.get(x, 1000 + x)) • SimplyLeet
Sortingp.313	Sortingp.314 Sheet 35 / 317 · LeetMastery By Pattern · Python Only · 5-Up Portrait	Sortingp.315

<pre># Python solution using a frequency dictionary def relativeSortArray(arr1, arr2): # Count frequency of each element in arr1 count = {} for num in arr1: count[num] = count.get(num, 0) + 1 result = [] # Process elements defined in arr2: append each # element count times for num in arr2: if num in count: result.extend([num] * count[num]) # Remove the element from the count del count[num] # Process remaining numbers: sort them in ascending # order and append remaining = [] for num, freq in count.items(): remaining.extend([num] * freq) result.extend(sorted(remaining)) return result # Example usage: print(relativeSortArray([2,3,1,3,2,4,6,7,9,2,19], [2,1,4,3,9,6]))</pre> • LC.ca	<pre>class Solution: def relativeSortArray(self, arr1: List[int], arr2: List[int]) -> List[int]: pos = {x: i for i, x in enumerate(arr2)} return sorted(arr1, key=lambda x: pos.get(x, 1000 + x))</pre>	Sorting #49 Group AnagramsMED LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Hash Table · String · Sorting Lists: Grind 169 Time Complexity: $O(n * k \log k)$, where n is the number of strings and k is the maximum length of a string (due to sorting each string). Space Complexity: $O(n * k)$ for storing the grouped strings in th Problem Given an array of strings $strs$, group the anagrams together. You can return the answer in any order. Example 1: Input: $strs = ["eat","tea","tan","ate","nat","bat"]$ Output: $[["bat"],["nat","tan"],["ate","eat","tea"]]$ Explanation: - There is no string in $strs$ that can be rearranged to form "bat". - The strings "nat" and "tan" are anagrams as they can be rearranged to form each other. - The strings "ate", "eat", and "tea" are anagrams as they can be rearranged to form each other. Example 2: Input: $strs = [""]$
Sortingp.316	Sortingp.317	Sortingp.318

Output: $[[""]]$ Example 3: Input: $strs = ["a"]$ Output: $[["a"]]$ Constraints: $1 \leq str.length \leq 104$ $0 \leq str[i].length \leq 100$ $strs[i]$ consists of lowercase English letters. Brute Force (Python) <pre># Brute Force Approach class Solution: def groupAnagrams(self, strs): # Brute Force $O(n^2 \cdot k \log k)$ - compare each # pair by sorting n = len(strs) used = [False] * n result = [] for i in range(n): if used[i]: continue group = [strs[i]] used[i] = True for j in range(i + 1, n): if not used[j] and sorted(strs[i]) == sorted(strs[j]): group.append(strs[j]) used[j] = True result.append(group) return result</pre>	Community Solutions (Python) • WalkCC <pre>class Solution: def groupAnagrams(self, strs: List[str]) -> List[List[str]]: dict = collections.defaultdict(List) for str in strs: key = ''.join(sorted(str)) dict[key].append(str) return dict.values()</pre> • LeetDoocs <pre>class Solution: def groupAnagrams(self, strs: List[str]) -> List[List[str]]: d = defaultdict(List) for s in strs: k = ''.join(sorted(s)) d[k].append(s) return list(d.values())</pre> • SimplyLeet	<pre># Python solution for the grouping anagrams from collections import defaultdict def groupAnagrams(strs): # Create a dictionary to hold lists of anagrams; # default to an empty list for new keys anagrams = defaultdict(list) # Iterate over each string in the input list for s in strs: # Sort the string to use as a key; sorted returns # a list so join back to string key = ''.join(sorted(s)) # Append the original string under the sorted key anagrams[key].append(s) # Return the grouped anagrams as a list of lists return list(anagrams.values()) # Example usage: if __name__ == "__main__": strs = ["eat","tea","tan","ate","nat","bat"] result = groupAnagrams(strs) print(result)</pre> • LC.ca
Sortingp.319	Sortingp.320	Sortingp.321

<pre>... >>> from collections import defaultdict >>> d = defaultdict(list) >>> d["a"]=1 >>> d["b"]=2 >>> d defaultdict(<class 'list'>, {'a': 1, 'b': 2}) >>> d.values() dict_values([1, 2]) >>> list(d.values()) [1, 2] ### sorted(str) will return a list of chars >>> a = "sdfddxyz" >>> sorted(a) ['d', 'd', 'd', 'f', 's', 'x', 'y', 'z'] >>> ''.join(sorted(a)) 'dddfsxyz' ...</pre> <pre>class Solution: def groupAnagrams(self, strs: List[str]) -> List[List[str]]: d = defaultdict(List) for s in strs: k = ''.join(sorted(s)) d[k].append(s) return list(d.values())</pre>	Sorting #56 Merge IntervalsMED LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Sorting Lists: Grind 169 Time Complexity: $O(n \log n)$ due to sorting the intervals. Space Complexity: $O(n)$ to store the merged intervals. Problem Given an array of intervals where $intervals[i] = [starti, endi]$, merge all overlapping intervals, and return an array of the non-overlapping intervals that cover all the intervals in the input. Example 1: Input: $intervals = [[1,3],[2,6],[8,10],[15,18]]$ Output: $[[1,6],[8,10],[15,18]]$ Explanation: Since intervals $[1,3]$ and $[2,6]$ overlap, merge them into $[1,6]$. Example 2: Input: $intervals = [[1,4],[4,5]]$ Output: $[[1,5]]$ Explanation: Intervals $[1,4]$ and $[4,5]$ are considered overlapping. Example 3:	Sorting Input: $intervals = [[4,7],[1,4]]$ Output: $[[1,7]]$ Explanation: Intervals $[1,4]$ and $[4,7]$ are considered overlapping. Constraints: $1 \leq intervals.length \leq 104$ $intervals[i].length == 2$ $0 \leq starti \leq endi \leq 104$ Brute Force (Python) <pre># Brute Force Approach class Solution: def merge(self, intervals): # Brute Force $O(n^2 \log n)$ - repeatedly merge any # two overlapping intervals intervals.sort(key=lambda x: x[0]) merged = True while merged: merged = False result = [intervals[0]] for iv in intervals[1:]: if iv[0] <= result[-1][1]: result[-1][1] = max(result[-1][1], iv[1]) merged = True else: result.append(iv) intervals = result return intervals</pre>
Sortingp.322	Sortingp.323	Sortingp.324

Community Solutions (Python)

```
• WalkCC

class Solution:
    def merge(self, intervals: List[List[int]]) -> List[List[int]]:
        ans = []

        for interval in sorted(intervals):
            if not ans or ans[-1][1] < interval[0]:
                ans.append(interval)
            else:
                ans[-1][1] = max(ans[-1][1], interval[1])

        return ans
```

```
• LeetCode

class Solution:
    def merge(self, intervals: List[List[int]]) -> List[List[int]]:
        intervals.sort()
        ans = []
        st, ed = intervals[0]
        for s, e in intervals[1:]:
            if ed < s:
                ans.append([st, ed])
                st, ed = s, e
            else:
                ed = max(ed, e)
        ans.append([st, ed])
        return ans
```

```
• SimplyLeet

def merge(intervals):
    # Edge case: if the list is empty, return an empty list
    if not intervals:
        return []
    # Sort intervals based on the starting time
    intervals.sort(key=lambda x: x[0])
    merged = [] # List to store merged intervals

    # Iterate over each interval in the sorted list
    for interval in intervals:

        # If merged list is not empty and the current interval
        # overlaps with the last one in merged, merge them
        if merged and interval[0] <= merged[-1][1]:
            # Update the end of the last interval in merged if needed
            merged[-1][1] = max(merged[-1][1], interval[1])
        else:
            # No overlap, so add the current interval to merged
            merged.append(interval)

    return merged

# Example usage:
print(merge([[1,3],[2,6],[8,10],[15,18]]))
```

```
# Definition for an interval.
# class Interval(object):
#     def __init__(self, s=0, e=0):
#         self.start = s
#         self.end = e

class Solution(object):
    def merge(self, intervals):
        """
        :type intervals: List[Interval]
        :rtype: List[Interval]
        """
        ans = []
        for intv in sorted(intervals, key=lambda x: x.start):
            if ans and ans[-1].end >= intv.start:
                ans[-1].end = max(ans[-1].end, intv.end)
            else:
                ans.append(intv)
        return ans

#####

'''
>>> intervals = [[111,222],[1,3],[2,6],[8,10],[15,18]]
>>> intervals.sort()
>>> intervals
[[1, 3], [2, 6], [8, 10], [15, 18], [111, 222]]
'''

class Solution:
    def merge(self, intervals: List[List[int]]) -> List[List[int]]:
```

```
intervals.sort() # default sort also ok
ans = [intervals[0]]
for s, e in intervals[1:]:
    if ans[-1][1] < s:
        ans.append([s, e])
    else:
        ans[-1][1] = max(ans[-1][1], e)
return ans
```

Sorting #1152 Analyze User Website Visit Pattern MED

LeetCode · SimplyLeet · WalkCC · LeetCode
Array · Hash Table · Sorting
Lists: Premium 98
Time Complexity: $O(U * L^3)$ where U is the number of users and L is the maximum length of visits per user (since we generate combinations of 3 visits per user).
Sorting all events costs $O(N \log N)$ for
Problem
You are given two string arrays username and website and an integer array timestamp. All the given arrays are of the same length and the tuple [username[i], website[i], timestamp[i]] indicates that the user username[i] visited the website website[i] at time timestamp[i].
A pattern is a list of three websites (not necessarily distinct).
• For example, ["home", "away", "love"], ["leetcode", "love", "leetcode"], and ["luffy", "luffy", "luffy"] are all patterns.
The score of a pattern is the number of users that visited all the websites in the pattern in the same order they appeared in the pattern.

• For example, if the pattern is ["home", "away", "love"], the score is the number of users x such that x visited "home" then visited "away" and visited "love" after that.
• Similarly, if the pattern is ["leetcode", "love", "leetcode"], the score is the number of users x such that x visited "leetcode" then visited "love" and visited "leetcode" one more time after that.
• Also, if the pattern is ["luffy", "luffy", "luffy"], the score is the number of users x such that x visited "luffy" three different times at different timestamps.
Return the pattern with the largest score. If there is more than one pattern with the same largest score, return the lexicographically smallest such pattern.
Note that the websites in a pattern do not need to be visited contiguously, they only need to be visited in the order they appeared in the pattern.
Example 1:

```
Input: username = ["joe","joe","joe","james","james","james","james","mary","mary","mary"], timestamp = [1,2,3,4,5,6,7,8,9,10], website = ["home","about","career","home","cart","maps","home","home","about","career"]
Output: ["home","about","career"]
Explanation: The tuples in this example are:
["joe","home",1],["joe","about",2],["joe","career",3],["james","home",4],["james","cart",5],["james","maps",6],["james","home",7],["mary","home",8],["mary","about",9], and ["mary","career",10].
The pattern ("home", "about", "career") has score 2 (joe and mary).
The pattern ("home", "cart", "maps") has score 1 (james).
The pattern ("home", "cart", "home") has score 1 (james).
The pattern ("home", "maps", "home") has score 1 (james).
```

Example 2:
Input: username = ["ua","ua","ua","ub","ub","ub"], timestamp = [1,2,3,4,5,6], website = ["a","b","a","a","b","c"]
Output: ["a","b","a"]
Constraints:
• 3 <= username.length <= 50
• 1 <= username[i].length <= 10
• timestamp.length == username.length
• 1 <= timestamp[i] <= 109

```
• website.length == username.length
• 1 <= website[i].length <= 10
• username[i] and website[i] consist of lowercase English letters.
• It is guaranteed that there is at least one user who visited at least three websites.
• All the tuples [username[i], timestamp[i], website[i]] are unique.

Brute Force (Python)

# Brute Force Approach
class Solution:
    def mostVisitedPattern(self, username, timestamp, website):
        # Brute Force  $O(n^2)$  - bubble sort
        arr = list(zip(username, timestamp, website))
        n = len(arr)
        for i in range(n):
            for j in range(n - i - 1):
                if arr[j] > arr[j + 1]:
                    arr[j], arr[j + 1] = arr[j + 1], arr[j]

        return arr
```

Community Solutions (Python)
• WalkCC

```
class Solution:
    def mostVisitedPattern(self, username, timestamp, website):
        """
        :type username: List[str]
        :type timestamp: List[int]
        :type website: List[str]
        :rtype: List[str]
        """
        userToSites = collections.defaultdict(list)

        # Sort websites of each user by timestamp.

        for user, _, site in sorted(zip(username, timestamp, website), key=lambda x: x[1]):
            userToSites[user].append(site)

        # For each of three websites, count its frequency.
        patternCount = collections.Counter()

        for user, sites in userToSites.items():
            patternCount.update(Counter(set(itertools.combinations(sites, 3))))

        return max(sorted(patternCount), key=patternCount.get)
```

• LeetCode

<pre>class Solution: def mostVisitedPattern(self, username: List[str], timestamp: List[int], website: List[str]) -> List[str]: d = defaultdict(list) for user, _, site in sorted(zip(username, timestamp, website), key=lambda x: x[1]): d[user].append(site) cnt = Counter() for sites in d.values(): m = len(sites) s = set() if m > 2: for i in range(m - 2): for j in range(i + 1, m - 1): for k in range(j + 1, m): s.add((sites[i], sites[j], sites[k])) for t in s: cnt[t] += 1 return sorted(cnt.items(), key=lambda x: (-x[1], x[0]))[0][0]</pre> • SimplyLeet	<pre># Python code solution for Analyze User Website Visit Pattern from collections import defaultdict from itertools import combinations def mostVisitedPattern(username, timestamp, website): # Combine the logs into a list of tuples and sort by timestamp. logs = sorted(zip(timestamp, username, website)) # Dictionary to hold each user's list of website visits user_visits = defaultdict(list) for time, user, site in logs: user_visits[user].append(site) # Dictionary to map a pattern (tuple of 3 websites) to set of users who visited that pattern pattern_users = defaultdict(set) # For each user, generate all unique 3-sequences from their visit list and update pattern_users for user, sites in user_visits.items(): # Use set to avoid counting duplicate patterns from the same user multiple times</pre>	<pre>seen_patterns = set() if len(sites) < 3: continue # Generate all 3-sequence combinations (order matters since visits are in time order) for pattern in combinations(sites, 3): if pattern not in seen_patterns: seen_patterns.add(pattern) pattern_users[pattern].add(user) # Find the pattern with the highest score (i.e., most users) # In case of a tie, lexicographically smallest pattern wins. max_count = 0 result = None for pattern, users in pattern_users.items(): count = len(users) if count > max_count or (count == max_count and (result is None or pattern < result)): max_count = count result = pattern return list(result)</pre>
Sortingp.334	Sortingp.335	Sortingp.336

<pre># Example usage: print(mostVisitedPattern(["joe","joe","joe","james","james","james","james","m ary","mary","mary"], [1,2,3,4,5,6,7,8,9,10], ["home","about","career","home","cart","maps","home", "home","about","career"])) • LC.ca class Solution: def mostVisitedPattern(self, username: List[str], timestamp: List[int], website: List[str]) -> List[str]: d = defaultdict(list) for user, _, site in sorted(zip(username, timestamp, website), key=lambda x: x[1]): d[user].append(site)</pre>	<pre>cnt = Counter() for sites in d.values(): m = len(sites) s = set() if m > 2: for i in range(m - 2): for j in range(i + 1, m - 1): for k in range(j + 1, m): s.add((sites[i], sites[j], sites[k])) for t in s: cnt[t] += 1 return sorted(cnt.items(), key=lambda x: (-x[1], x[0]))[0][0]</pre>	
Sortingp.337	Sortingp.338	Sortingp.339

>> Quick Review — Sorting 9 questions · insights · complexity · solution #169 Majority Element [Easy] Key Insights • The majority element appears more than half the length of the array. • A simple hash table approach can count occurrences, but a more efficient solution exists. • The Boyer-Moore Voting Algorithm finds the majority in O(n) time and O(1) space. • The algorithm maintains a candidate and a counter, adjusting the candidate when the count drops to zero. Space & Time Time Complexity: O(n) Space Complexity: O(1) Solution We use the Boyer-Moore Voting Algorithm to identify the majority element. The algorithm iterates through the array using two variables: one for the candidate and one for the count. Initially, we set the candidate to an arbitrary value and count to 0. For each element, if the count is 0, we update the candidate to the current element. Then, if the current element is the same as the	candidate, we increment the count; otherwise, we decrement it. Since the majority element appears more than n/2 times, it will remain as the candidate by the end of the iteration. #217 Contains Duplicate [Easy] Key Insights • A hash set can be used to keep track of seen numbers. • Iterating once through the array provides an efficient solution. • If a number is already in the set, a duplicate is found. • Alternative approaches like sorting have a higher time complexity (O(n log n)). Space & Time Time Complexity: O(n) Space Complexity: O(n) Solution The solution uses a hash set to track the numbers seen while iterating through the array. For each number in the array: - Check if it exists in the hash set. - If it does, return true immediately as a duplicate is found. - Otherwise, add the number to the hash set. If the iteration completes without finding any duplicates, return false. This approach efficiently determines if any duplicates exist in a single pass.	#242 Valid Anagram [Easy] Key Insights • If s and t have different lengths, they cannot be anagrams. • An anagram requires exactly the same character frequency for both strings. • Using a hash table (or frequency array when limited to 26 lowercase letters) provides an efficient solution. • An alternative approach is to sort both strings and compare them. • For Unicode inputs, a hash table is more adaptable than a fixed-size frequency array. Space & Time Time Complexity: O(n), where n is the length of the strings (using a hash table for counting). Space Complexity: O(1) if we use a fixed-size array (for 26 letters), otherwise O(n) for a hash table with a larger character set. Solution We use a hash table (dictionary) to count the frequency of each character in the first string s. Then, we iterate over string t and decrement the count for each character. If at any point a character count goes negative or a character is missing in the hash table, t cannot be an anagram of s. Finally, if all counts return to zero, the strings are anagrams. This approach is efficient and
Sortingp.340	Sortingp.341	Sortingp.342

handles the problem in linear time. #252 Meeting Rooms [Easy] Key Insights - Sort the intervals based on their starting times. - Compare each interval's end time with the next interval's start time. - Detect overlap if the end time of one meeting exceeds the start time of the following meeting. - If no overlaps are detected, all meetings can be attended. Space & Time Time Complexity: O(n log n) due to sorting the intervals. Space Complexity: O(1) if sorting is done in-place; otherwise O(n). Solution The solution starts by sorting the intervals in ascending order by their start times. After sorting, iterate through the intervals and for each consecutive pair, check if the previous meeting's end time exceeds the next meeting's start time. An overlap indicates that the person cannot attend all meetings. The primary data structure used is an array (or list) for holding the intervals. Sorting makes it easier to compare adjacent intervals for potential overlaps. Sortingp.343	#1086 Largest Unique Number [Easy] Key Insights - Use a hash table (or dictionary) to count the frequency of each number. - Iterate over the hash table to find the numbers that appear exactly once. - Identify the maximum number among the unique numbers. - Return -1 if no unique number exists. Space & Time Time Complexity: O(n), where n is the number of elements in nums. Space Complexity: O(n), for the hash table storing frequency counts. Solution The approach starts by initializing a hash table to keep track of the frequency of each element in the array. Once the frequency count is complete, we traverse the hash table to collect numbers that appear exactly once. Finally, we determine the maximum number among these unique elements. If there is no unique element, we return -1. This method is efficient because the frequency count and the search for the maximum unique element both run in linear time relative to the input size. #1122 Relative Sort Array [Easy] Sortingp.344	Key Insights - Use a hash table (or array for counting given the value constraints) to count the occurrences of each element in arr1. - Iterate through arr2 and for each element, append it to the result based on its count from the hash table. - Remove the used elements from the hash table. - Sort the remaining elements (those not in arr2) in ascending order and append them to the result. Space & Time Time Complexity: O(n + m log m) where n is the length of arr1 and m is the number of leftover elements not in arr2. Given that element values are bounded (0 to 1000), a counting sort can make this nearly O(n). Space Complexity: O(n) for storing the frequency counts and the result array. Solution The solution leverages a frequency counter (hash table) to count how many times each element appears in arr1. We then process arr2 to add each number in its given order according to its frequency. Finally, we sort the remaining numbers that are not in arr2 and append them to the result array. This approach ensures that the relative order specified by arr2 is maintained and that the other elements are sorted in ascending order. Sortingp.345
#49 Group Anagrams [Medium] Key Insights - Anagrams have the same characters when sorted alphabetically. - Sorting each string provides a unique key for grouping. - Use a hash table (dictionary) to map each sorted key to a list of its original strings. - Finally, gather all the lists from the dictionary to form the result. Space & Time Time Complexity: O(n * k log k), where n is the number of strings and k is the maximum length of a string (due to sorting each string). Space Complexity: O(n * k) for storing the grouped strings in the hash table. Solution The primary idea is to iterate through each string, sort it to obtain a key that represents its character composition, and then group the original string under this key in a hash table. After processing all strings, the values of the hash table will be the groups of anagrams. This approach efficiently groups anagrams by leveraging sorting and a dictionary for constant-time lookups. Sortingp.346	#56 Merge Intervals [Medium] Key Insights - Sort the intervals based on their start values. - Use a pointer or iteration to compare the current interval with the last merged interval. - If the current interval overlaps with the previous one, merge them by updating the end time. - Otherwise, add the current interval to the results as a new merged interval. Space & Time Time Complexity: O(n log n) due to sorting the intervals. Space Complexity: O(n) to store the merged intervals. Solution First, sort the array of intervals based on the starting time. Then, iterate through the sorted list and check if the current interval overlaps with the last interval in the merged list. If an overlap is detected (i.e., the start of the current interval is less than or equal to the end of the last merged interval), update the last merged interval's end to be the maximum of both ends. If there is no overlap, simply add the current interval to the merged list. This approach efficiently merges overlapping intervals and ensures that all intervals are processed in a single pass after sorting. #1152 Analyze User Website Visit Pattern [Medium] Sortingp.347	Key Insights - Combine the inputs into a single list of events and sort them by timestamp. - Group the visits by each user, preserving the sorted order. - Generate all unique combinations of 3-sequences for each user (order matters and non-contiguous visits are allowed). - Use a dictionary (or hash map) to map each 3-sequence pattern to the set of users who visited that pattern. - The problem constraints are small, so generating combinations per user (even using triple nested loops) is acceptable. - When multiple patterns have the same frequency, choose the lexicographically smallest one. Space & Time Time Complexity: O(U * L^3) where U is the number of users and L is the maximum length of visits per user (since we generate combinations of 3 visits per user). Sorting all events costs O(N log N) for N total events. Space Complexity: O(P) where P is the number of unique 3-sequence patterns stored in the dictionary, plus O(N) for grouping the events. Solution Sortingp.348
#7 Bit Manipulation By Pattern · Python Only · 10 questions Sortingp.349	Bit Manipulation #67 Add Binary EAS LeetCode · SimplyLeet · WalkCC · LeetCode Math · String · Bit Manipulation · Simulation Lists: Grind 169 Time Complexity: O(max(n, m)), where n and m are the lengths of the two binary strings. Space Complexity: O(max(n, m)) for storing the resulting binary string. Problem Given two binary strings a and b, return their sum as a binary string. Example 1: Input: a = "11", b = "1" Output: "100" Example 2: Input: a = "1010", b = "1011" Output: "10101" Constraints: 1 <= a.length, b.length <= 104 - a and b consist only of '0' or '1' characters. - Each string does not contain leading zeros except for the zero itself. Bit Manipulationp.350	Community Solutions (Python) • WalkCC class Solution: def addBinary(self, a: str, b: str) -> str: ans = [] carry = 0 i = len(a) - 1 j = len(b) - 1 while i >= 0 or j >= 0 or carry: if i >= 0: carry += int(a[i]) i -= 1 if j >= 0: carry += int(b[j]) j -= 1 ans.append(str(carry % 2)) carry //= 2 return ''.join(reversed(ans)) • LeetCode Bit Manipulationp.351

class Solution: def addBinary(self, a: str, b: str) -> str: ans = [] i, j, carry = len(a) - 1, len(b) - 1, 0 while i >= 0 or j >= 0 or carry: carry += (0 if i < 0 else int(a[i])) + (0 if j < 0 else int(b[j])) carry, v = divmod(carry, 2) ans.append(str(v)) i, j = i - 1, j - 1 return "".join(ans[::-1]) • SimplyLeet # Function to add two binary strings def addBinary(a: str, b: str) -> str: i = len(a) - 1 # Pointer for string a j = len(b) - 1 # Pointer for string b carry = 0 # Initialize carry to 0 result = [] # List to store binary digits # Loop until both strings are processed and no carry remains while i >= 0 or j >= 0 or carry: total_sum = carry # Start with the carry from previous step Bit Manipulationp.352	if i >= 0: # If there is a digit left in a total_sum += int(a[i]) # Convert character to integer and add i -= 1 # Move to the next digit if j >= 0: # If there is a digit left in b total_sum += int(b[j]) # Convert character to integer and add j -= 1 # Move to the next digit # Append the remainder of total_sum divided by 2 (0 or 1) result.append(str(total_sum % 2)) carry = total_sum // 2 # Update the carry # Reverse the result since we built it backwards and join to form a string return ''.join(result[::-1]) # Sample test cases print(addBinary("11", "1")) # Expected output: "100" print(addBinary("1010", "1011")) # Expected output: "10101" • LC.ca Bit Manipulationp.353	class Solution: def addBinary(self, a: str, b: str) -> str: ans = [] i, j, carry = len(a) - 1, len(b) - 1, 0 while i >= 0 or j >= 0 or carry: carry += (0 if i < 0 else int(a[i])) + (0 if j < 0 else int(b[j])) carry, v = divmod(carry, 2) ans.append(str(v)) i, j = i - 1, j - 1 return "".join(ans[::-1]) Bit Manipulationp.354
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Bit Manipulation #136 Single NumberEAS LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Bit Manipulation Lists: Grind 169 Time Complexity: O(n) - We only traverse the array once. Space Complexity: O(1) - Only a single extra variable is used regardless of input size. Problem Given a non-empty array of integers nums, every element appears twice except for one. Find that single one. You must implement a solution with a linear runtime complexity and use only constant extra space. Example 1: Input: nums = [2,2,1] Output: 1 Example 2: Input: nums = [4,1,2,1,2] Output: 4 Example 3: Input: nums = [1] Output: 1 Bit Manipulationp.355	Constraints: • 1 <= nums.length <= 3 * 10⁴ • -3 * 10⁴ <= nums[i] <= 3 * 10⁴ • Each element in the array appears twice except for one element which appears only once. Brute Force (Python) # Brute Force Approach class Solution: def singleNumber(self, nums): # Brute Force O(n^2) - for each element, count its occurrences for num in nums: if nums.count(num) == 1: return num Community Solutions (Python) • WalkCC class Solution: def singleNumber(self, nums: List[int]) -> int: return functools.reduce(operator.xor, nums, 0) • LeetDoocs class Solution: def singleNumber(self, nums: List[int]) -> int: return reduce(xor, nums) • SimplyLeet Bit Manipulationp.356	# Function to find the single number in the list def singleNumber(nums): # Initialize result to 0; any number XOR 0 is the number itself result = 0 # Iterate through every number in the list for num in nums: # XOR the current result with the current number result ^= num # Using XOR to cancel duplicates # Return the unique number that did not cancel out return result # Example usage nums = [4, 1, 2, 1, 2] print(singleNumber(nums)) # Output should be 4 • LC.ca ''' >>> from functools import reduce >>> reduce(lambda x, y: x ^ y, [3, 5, 3]) 5 ''' from functools import reduce class Solution: def singleNumber(self, nums: List[int]) -> int: Bit Manipulationp.357
--	--	---

return reduce(lambda x, y: x ^ y, nums) ##### class Solution(object): def singleNumber(self, nums): '''' :type nums: List[int] :rtype: int '''' for i in range(1, len(nums)): nums[0] ^= nums[i] return nums[0] Bit Manipulationp.358	Bit Manipulation #190 Reverse BitsEAS LeetDoocs · SimplyLeet · WalkCC · LeetCode Divide and Conquer · Bit Manipulation Lists: Grind 169 Time Complexity: O(1) - We always process 32 bits regardless of the input. Space Complexity: O(1) - Only constant extra space is used. Problem Reverse bits of a given 32 bits signed integer. Example 1: Input: n = 43261596 Output: 964176192 Explanation: Example 2: Input: n = 2147483644 Output: 1073741822 Explanation: Constraints: • 0 <= n <= 2³¹ - 2 • n is even. Follow up: If this function is called many times, how would you optimize it? Bit Manipulationp.359	Community Solutions (Python) • WalkCC class Solution: def reverseBits(self, n: int) -> int: ans = 0 for i in range(32): if n >> i & 1: ans = 1 << 31 - i return ans • LeetDoocs class Solution: def reverseBits(self, n: int) -> int: ans = 0 for i in range(32): ans = (n & 1) << (31 - i) n >>= 1 return ans • SimplyLeet Bit Manipulationp.360
--	---	---

Function to reverse bits of a 32-bit binary string.
def reverse_bits(n: str) -> int:
 # Reverse the binary string using slicing.
 reversed_binary = n[::-1]
 # Convert the reversed binary string to an integer.
 result = int(reversed_binary, 2)
 return result

Example usage:
binary_input = "00000010100101000001111010011100"

print(reverse_bits(binary_input)) # Expected output:
964176192

LC.ca

class Solution:
 def reverseBits(self, n: int) -> int:
 res = 0
 for i in range(32):
 res |= (n & 1) << (31 - i)
 n >>= 1
 return res

Bit Manipulationp.361

Bit Manipulation

#191 Number of 1 BitsEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Divide and Conquer · Bit Manipulation

Lists: Grind 169

Time Complexity: O(k), where k is the number of set bits, worst-case O(log n) (or O(1) considering 32-bit integers). Space Complexity: O(1).

Problem

Given a positive integer n, write a function that returns the number of set bits in its binary representation (also known as the Hamming weight).

Example 1:
Input: n = 11
Output: 3
Explanation:
The input binary string 1011 has a total of three set bits.

Example 2:
Input: n = 128
Output: 1
Explanation:
The input binary string 10000000 has a total of one set bit.

Bit Manipulationp.362

Example 3:
Input: n = 2147483645
Output: 30
Explanation:
The input binary string 111111111111111111111111111101 has a total of thirty set bits.
Constraints:
 • 1 <= n <= 231 - 1
Follow up: If this function is called many times, how would you optimize it?

Community Solutions (Python)

WalkCC

class Solution:
 def hammingWeight(self, n: int) -> int:
 ans = 0

 for i in range(32):
 if (n >> i) & 1:
 ans += 1

 return ans

class Solution:
 def hammingWeight(self, n: int) -> int:
 return n.bit_count()

LeetDoocs

Bit Manipulationp.363

class Solution:
 def hammingWeight(self, n: int) -> int:
 ans = 0
 while n:
 n &= n - 1
 ans += 1
 return ans

SimplyLeet

Function to count the number of set bits in a given integer n
def hammingWeight(n: int) -> int:
 count = 0 # Initialize count of set bits
 while n:
 # Remove the lowest set bit from n
 n &= n - 1
 count += 1 # Increment count for each set bit removed
 return count

Example usage:

print(hammingWeight(11)) # Output: 3

LC.ca

class Solution:
 def hammingWeight(self, n: int) -> int:
 ans = 0
 while n:
 n &= n - 1
 ans += 1
 return ans

Bit Manipulationp.364

Bit Manipulation

#268 Missing NumberEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Math · Binary Search · Bit Manipulation

Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(1)

Problem

Given an array nums containing n distinct numbers in the range [0, n], return the only number in the range that is missing from the array.

Example 1:
Input: nums = [3,0,1]
Output: 2
Explanation:
n = 3 since there are 3 numbers, so all numbers are in the range [0,3]. 2 is the missing number in the range since it does not appear in nums.

Example 2:
Input: nums = [0,1]
Output: 2
Explanation:
n = 2 since there are 2 numbers, so all numbers are in the range [0,2]. 2 is the missing number in the range

Bit Manipulationp.365

since it does not appear in nums.
Example 3:
Input: nums = [9,6,4,2,3,5,7,0,1]
Output: 8
Explanation:
n = 9 since there are 9 numbers, so all numbers are in the range [0,9]. 8 is the missing number in the range since it does not appear in nums.
Constraints:
 • n == nums.length
 • 1 <= n <= 104
 • 0 <= nums[i] <= n
 • All the numbers of nums are unique.
Follow up: Could you implement a solution using only O(1) extra space complexity and O(n) runtime complexity?

Brute Force (Python)

Brute Force Approach
class Solution:
 def missingNumber(self, nums):
 # Brute Force O(n^2) - for each 0..n check if it's in nums
 for i in range(len(nums) + 1):
 if i not in nums:
 return i

Community Solutions (Python)

Bit Manipulationp.366

WalkCC

class Solution:
 def missingNumber(self, nums: List[int]) -> int:
 ans = len(nums)

 for i, num in enumerate(nums):
 ans ^= i ^ num

 return ans

LeetDoocs

class Solution:
 def missingNumber(self, nums: List[int]) -> int:
 return reduce(xor, (i ^ v for i, v in enumerate(nums, 1)))

class Solution:
 def missingNumber(self, nums: List[int]) -> int:
 n = len(nums)
 return (1 + n) * n // 2 - sum(nums)

SimplyLeet

Bit Manipulationp.367

Function to find the missing number in the array
def missingNumber(nums):
 # Calculate n, where n is the length of the nums array
 n = len(nums)
 # Compute the expected total sum using the arithmetic sum formula for numbers 0 to n
 expected_sum = n * (n + 1) // 2
 # Compute the actual sum of the numbers in the array
 actual_sum = sum(nums)
 # The missing number is the difference between expected and actual sums
 return expected_sum - actual_sum

Example Usage
print(missingNumber([3, 0, 1])) # Output: 2

LC.ca

class Solution:
 def missingNumber(self, nums: List[int]) -> int:
 return reduce(xor, (i ^ v for i, v in enumerate(nums, 1)))

Bit Manipulationp.368

Bit Manipulation

#338 Counting BitsEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Dynamic Programming · Bit Manipulation

Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(n)

Problem

Given an integer n, return an array ans of length n + 1 such that for each i (0 <= i <= n), ans[i] is the number of 1's in the binary representation of i.

Example 1:
Input: n = 2
Output: [0,1,1]
Explanation:
0 -> 0
1 -> 1
2 -> 10

Example 2:

Bit Manipulationp.369

Sheet 41 / 317 · LeetMastery By Pattern · Python Only · 9-Up Portrait

```
Input: n = 5
Output: [0,1,1,2,1,2]
Explanation:
0 --> 0
1 --> 1
2 --> 10
3 --> 11
4 --> 100
```

Constraints:

- $0 \leq n \leq 105$

Follow up:

- It is very easy to come up with a solution with a runtime of $O(n \log n)$. Can you do it in linear time $O(n)$ and possibly in a single pass?
- Can you do it without using any built-in function (i.e., like `__builtin_popcount` in C++)?

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def countBits(self, n):
        # Brute Force  $O(n \cdot 32)$  - count 1-bits for each
        # number 0..n individually
        def count_ones(x):
            c = 0
            while x:
                c += x & 1
                x >>= 1
            return c
```

Bit Manipulation

p.370

```
return [count_ones(i) for i in range(n + 1)]
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def countBits(self, n: int) -> list[int]:
        # f(i) := i's number of 1s in bitmask
        # f(i) = f(i / 2) + i % 2
        ans = [0] * (n + 1)

        for i in range(1, n + 1):
            ans[i] = ans[i // 2] + (i & 1)

        return ans
```

• LeetCode

```
class Solution:
    def countBits(self, n: int) -> List[int]:
        return [i.bit_count() for i in range(n + 1)]
```

```
class Solution:
    def countBits(self, n: int) -> List[int]:
        ans = [0] * (n + 1)
        for i in range(1, n + 1):
            ans[i] = ans[i & (i - 1)] + 1
        return ans
```

• SimplyLeet

Bit Manipulation

p.371

```
def countBits(n):
    # Create a list to store the number of 1's for each
    # number from 0 to n
    dp = [0] * (n + 1)
    # Loop over each number starting from 1, since dp[0]
    # is already 0
    for i in range(1, n + 1):
        # dp[i] is computed as the count for i//2 plus 1
        # if i is odd (i & 1)
        dp[i] = dp[i >> 1] + (i & 1)
    return dp
```

Example usage:

```
# print(countBits(5)) -> [0, 1, 1, 2, 1, 2]
```

• LC.ca

```
class Solution:
    def countBits(self, n: int) -> List[int]:
        ans = [0] * (n + 1)
        for i in range(1, n + 1):
            ans[i] = ans[i & (i - 1)] + 1
        return ans
```

Bit Manipulation

p.372

Bit Manipulation

#78 Subsets

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Backtracking · Bit Manipulation

Lists: Grind 169

Time Complexity: $O(2^n \cdot n)$ - There are 2^n subsets and creating each subset may take up to $O(n)$ time. Space Complexity: $O(2^n \cdot n)$ - Space for storing all subsets. The recursion stack has a maximum d

Problem

Given an integer array `nums` of unique elements, return all possible subsets (the power set).

The solution set must not contain duplicate subsets.

Return the solution in any order.

Example 1:

```
Input: nums = [1,2,3]
Output: [[],[1],[2],[1,2],[3],[1,3],[2,3],[1,2,3]]
```

Example 2:

```
Input: nums = [0]
Output: [[],[0]]
```

Constraints:

- $1 \leq \text{nums.length} \leq 10$
- $-10 \leq \text{nums}[i] \leq 10$

Bit Manipulation

p.373

- All the numbers of `nums` are unique.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def subsets(self, nums):
        # Brute Force  $O(2^n \cdot n)$  - bitmask over all  $2^n$ 
        # subsets
        result = []
        for mask in range(1 << len(nums)):
            sub = [nums[i] for i in range(len(nums)) if
                mask & (1 << i)]
            result.append(sub)
        return result
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def subsets(self, nums: list[int]) -> list[list[int]]:
        ans = []

        def dfs(s: int, path: list[int]) -> None:
            ans.append(path)

            for i in range(s, len(nums)):
                dfs(i + 1, path + [nums[i]])

        dfs(0, [])
        return ans
```

• LeetCode

Bit Manipulation

p.374

```
class Solution:
    def subsets(self, nums: List[int]) ->
    List[List[int]]:
        def dfs(i: int):
            if i == len(nums):
                ans.append(t[:])
                return
            dfs(i + 1)
            t.append(nums[i])
            dfs(i + 1)
            t.pop()
```

```
ans = []
t = []
dfs(0)
return ans
```

• SimplyLeet

```
# Define the function to generate all subsets using
# backtracking.
def subsets(nums):
    # This list will store all the subsets
    result = []

    # Recursive helper function to generate subsets.
    def backtrack(start, current):
        # Append a copy of the current subset to the
        # final result.
        result.append(current.copy())
        # Iterate over the possible next elements
        # starting from index 'start'
```

Bit Manipulation

p.375

```
for i in range(start, len(nums)):
    # Include nums[i] in the current subset.
    current.append(nums[i])
    # Recurse with the next starting index.
    backtrack(i + 1, current)
    # Backtrack: remove the last element to
    # explore other possibilities.
    current.pop()
```

```
# Start the recursion with an empty path.
backtrack(0, [])
```

```
return result
```

```
# Example usage:
print(subsets([1, 2, 3]))
```

• LC.ca

from typing import List

```
class Solution: # bfs
    def subsets(self, nums: List[int]) ->
    List[List[int]]:
        if not nums:
            return [[]]
```

```
        nums.sort() # sort() not necessary if no
        # duplicates
        result = [[]]
```

Bit Manipulation

p.376

```
for num in nums:
    result += [subset + [num] for subset in
    result]

    return result
```

```
class Solution: # dfs
    def subsets(self, nums: List[int]) ->
    List[List[int]]:
        def dfs(u, t):
            ans.append(t[:]) # or, t.copy()

            for i in range(u, len(nums)):
                t.append(nums[i])
                dfs(i + 1, t)
                t.pop()
```

```
ans = []
nums.sort() # sort() not necessary if no
# duplicates
dfs(0, [])
return ans
```

Bit Manipulation

p.377

```
class Solution: # dfs, just pass down the final path
    def subsets(self, nums: List[int]) ->
    List[List[int]]:
        def dfs(nums, index, path, ans):
            ans.append(path)
            [dfs(nums, i + 1, path + [nums[i]], ans) for
            i in range(index, len(nums))]
```

```
ans = []
dfs(nums, 0, [], ans)
return ans
```

```
class Solution: # dfs, but running slower, since need to
    # reference from parent method for 'nums' and 'ans'
    def subsets(self, nums: List[int]) ->
    List[List[int]]:
        def dfs(index, path):
            ans.append(path)
            [dfs(i + 1, path + [nums[i]]) for i in
            range(index, len(nums))]
```

```
ans = []
dfs(0, [])
```

```
return ans
```

Bit Manipulation

p.378

Bit Manipulation

#90 Subsets II

MED

LeetCode · [SimplyLeet](#) · [WalkCC](#) · [LeetCode](#)
Array · [Backtracking](#) · [Bit Manipulation](#)
Lists: [Denny Zhang](#)
Time Complexity: $O(2^n \cdot n)$ since each subset creation may take $O(n)$ in the worst case. Space Complexity: $O(2^n \cdot n)$ for storing the subsets, plus $O(n)$ for recursion stack.
Problem
Given an integer array `nums` that may contain duplicates, return all possible subsets (the power set). The solution set must not contain duplicate subsets. Return the solution in any order.
Example 1:

Input: `nums = [1,2,2]`
Output: `[], [1], [1,2], [1,2,2], [2], [2,2]`

Example 2:

Input: `nums = [0]`
Output: `[], [0]`

Constraints:

- `1 <= nums.length <= 10`
- `-10 <= nums[i] <= 10`

Bit Manipulation

p.379

Brute Force (Python)

Brute Force Approach
class Solution:
 def subsetsWithDup(self, nums):
 # Brute Force $O(2^n \cdot n \log n)$ – bitmask with deduplication via set
 result = set()
 nums.sort()
 for mask in range(1 << len(nums)):
 sub = tuple(nums[i] for i in range(len(nums)) if mask & (1 << i))
 result.add(sub)
 return [list(s) for s in result]

Community Solutions (Python)

• [WalkCC](#)
class Solution:
 def subsetsWithDup(self, nums: List[int]) -> List[List[int]]:
 ans = []

 def dfs(s: int, path: List[int]) -> None:
 ans.append(path)
 if s == len(nums):
 return

 for i in range(s, len(nums)):

Bit Manipulation

p.380

```
if i > s and nums[i] == nums[i - 1]:  
    continue  
dfs(i + 1, path + [nums[i]])  
  
nums.sort()  
dfs(0, [])  
return ans
```

• [LeetCode](#)
class Solution:
 def subsetsWithDup(self, nums: List[int]) -> List[List[int]]:
 def dfs(i: int):
 if i == len(nums):
 ans.append(t[:])
 return
 t.append(nums[i])
 dfs(i + 1)
 x = t.pop()
 while i + 1 < len(nums) and nums[i + 1] == x:

 i += 1
 dfs(i + 1)

 nums.sort()
 ans = []
 t = []
 dfs(0)
 return ans

• [SimplyLeet](#)

Bit Manipulation

p.381

def subsetsWithDup(nums):
 # Sort the numbers to handle duplicates
 nums.sort()
 result = [] # To store all subsets
 subset = [] # Current subset being built

 def backtrack(start):
 # Append a copy of the current subset to the result
 result.append(list(subset))
 # Iterate over possible elements to include

 for i in range(start, len(nums)):
 # If the current element is a duplicate and not at the start of this loop, skip it
 if i > start and nums[i] == nums[i - 1]:
 continue
 # Include current element into the subset
 subset.append(nums[i])
 # Recurse with next start index
 backtrack(i + 1)
 # Backtrack and remove the last element added
 subset.pop()

 backtrack(0)
 return result

Example usage:
print(subsetsWithDup([1, 2, 2]))

• [LC.ca](#)

Bit Manipulation

p.382

class Solution:
 def subsetsWithDup(self, nums: List[int]) -> List[List[int]]:
 def dfs(u, t):
 ans.append(t[:]) # or, t.copy()
 for i in range(u, len(nums)):
 # also good for [1,5,5], second '5' will be skipped here if
 # but second '5' is always covered, because i==u when ans=[1,5] and i=2
 if i != u and nums[i] == nums[i - 1]:
 continue
 t.append(nums[i])

 dfs(i + 1, t)
 t.pop()

 ans = []
 nums.sort()
 dfs(0, [])
 return ans

 # iteration

Bit Manipulation

p.383

class Solution:
 def subsetsWithDup(self, nums: List[int]) -> List[List[int]]:
 if not nums:
 return [[]]

 nums.sort() # Sorting is necessary to handle duplicates.
 result = [[]]
 start_idx = 0 # Start index for the new subsets
 new_subsets_size = 0

 for i in range(len(nums)):
 size = len(result)
 # If the current number is the same as the previous one,
 # we only want to extend the subsets added in the previous iteration.
 if i > 0 and nums[i] == nums[i - 1]:
 start_idx = size - new_subsets_size
 else:
 start_idx = 0

 new_subsets_size = 0 # Reset the size of newly created subsets

Bit Manipulation

p.384

Bit Manipulation

#287 Find the Duplicate Number

MED

LeetCode · [SimplyLeet](#) · [WalkCC](#) · [LeetCode](#)
Array · [Two Pointers](#) · [Binary Search](#) · [Bit Manipulation](#)
Lists: [Grind 169](#)
Time Complexity: $O(n)$ Space Complexity: $O(1)$
Problem
Given an array of integers `nums` containing $n + 1$ integers where each integer is in the range $[1, n]$ inclusive. There is only one repeated number in `nums`, return this repeated number. You must solve the problem without modifying the array `nums` and using only constant extra space.
Example 1:

Input: `nums = [1,3,4,2,2]`
Output: `2`

Example 2:

Input: `nums = [3,1,3,4,2]`
Output: `3`

Example 3:

Bit Manipulation

p.385

Bit Manipulation

Input: `nums = [3,3,3,3,3]`
Output: `3`
Constraints:

- `1 <= n <= 105`
- `nums.length == n + 1`
- `1 <= nums[i] <= n`
- All the integers in `nums` appear only once except for precisely one integer which appears two or more times.

Follow up:

- How can we prove that at least one duplicate number must exist in `nums`?
- Can you solve the problem in linear runtime complexity?

Brute Force (Python)
Brute Force Approach
class Solution:
 def findDuplicate(self, nums):
 # Brute Force $O(n^2)$ – for each number count its occurrences
 for i in range(len(nums)):
 count = 0
 for j in range(len(nums)):
 if nums[j] == nums[i]:
 count += 1
 if count > 1:

 return nums[i]

Bit Manipulation

p.386

Bit Manipulation

Input: `nums = [3,3,3,3,3]`
Output: `3`
Constraints:

- `1 <= n <= 105`
- `nums.length == n + 1`
- `1 <= nums[i] <= n`
- All the integers in `nums` appear only once except for precisely one integer which appears two or more times.

Follow up:

- How can we prove that at least one duplicate number must exist in `nums`?
- Can you solve the problem in linear runtime complexity?

Brute Force (Python)
Brute Force Approach
class Solution:
 def findDuplicate(self, nums):
 # Brute Force $O(n^2)$ – for each number count its occurrences
 for i in range(len(nums)):
 count = 0
 for j in range(len(nums)):
 if nums[j] == nums[i]:
 count += 1
 if count > 1:

 return nums[i]

Bit Manipulation

p.387

Community Solutions (Python)

• WalkCC

```
class Solution:
    def findDuplicate(self, nums: List[int]) -> int:
        slow = nums[nums[0]]
        fast = nums[nums[nums[0]]]

        while slow != fast:
            slow = nums[slow]
            fast = nums[nums[fast]]

        slow = nums[0]

        while slow != fast:
            slow = nums[slow]
            fast = nums[fast]

        return slow
```

• LeetCode

```
class Solution:
    def findDuplicate(self, nums: List[int]) -> int:
        def f(x: int) -> bool:
            return sum(v <= x for v in nums) > x

        return bisect_left(range(len(nums)), True, key=f)
```

• SimplyLeet

```
# Function to find the duplicate number using Floyd's
Tortoise and Hare algorithm.
def findDuplicate(nums):
    # Phase 1: Find the intersection point inside the
    cycle.
    tortoise = nums[0] # Initialize tortoise pointer.
    hare = nums[0] # Initialize hare pointer.
    while True:
        tortoise = nums[tortoise] # Move tortoise one
        step.
        hare = nums[nums[hare]] # Move hare two steps.
        if tortoise == hare: # If pointers meet, cycle is
        detected.
            break

    # Phase 2: Find the entrance to the cycle.
    pointer = nums[0] # Initialize a new pointer from the
    start.
    while pointer != tortoise:
        pointer = nums[pointer] # Move pointer one step.
        tortoise = nums[tortoise] # Move tortoise one
        step.
    return pointer # The duplicate number.

# Example usage:
if __name__ == "__main__":

    sample = [1, 3, 4, 2, 2]
    print(findDuplicate(sample)) # Expected output: 2
```

• LC.ca

```
class Solution:
    def findDuplicate(self, nums: List[int]) -> int:
        def f(x: int) -> bool:
            return sum(v <= x for v in nums) > x

        return bisect_left(range(len(nums)), True, key=f)
```

Bit Manipulation

#1506 Find Root of N-Ary Tree

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Hash Table · Bit Manipulation · Tree · DFS

Lists: Premium 98

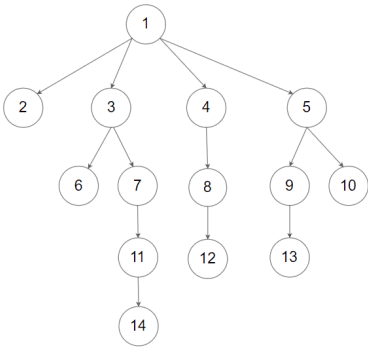
Time Complexity: O(n) where n is the number of nodes, since we iterate through all nodes and their children. Space Complexity: O(1) extra space using arithmetic accumulators.

Problem

You are given all the nodes of an N-ary tree as an array of Node objects, where each node has a unique value. Return the root of the N-ary tree.

Custom testing:

An N-ary tree can be serialized as represented in its level order traversal where each group of children is separated by the null value (see examples).



For example, the above tree is serialized as [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14].

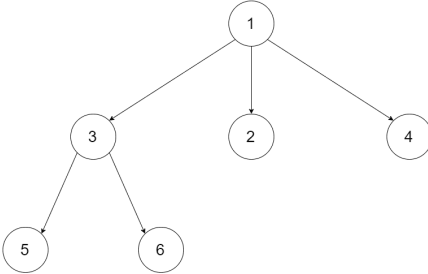
The testing will be done in the following way:

1. The input data should be provided as a serialization of the tree.
2. The driver code will construct the tree from the serialized input data and put each Node object into an array in an arbitrary order.
3. The driver code will pass the array to findRoot, and your function should find and return the root Node

object in the array.

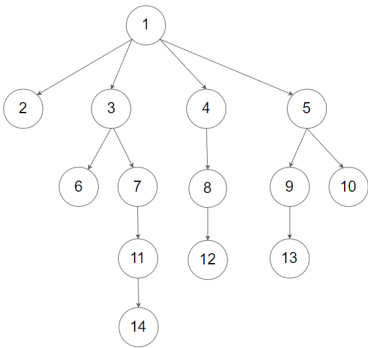
4. The driver code will take the returned Node object and serialize it. If the serialized value and the input data are the same, the test passes.

Example 1:



Input: tree = [1,null,3,2,4,null,5,6]
Output: [1,null,3,2,4,null,5,6]
Explanation: The tree from the input data is shown above.
The driver code creates the tree and gives findRoot the Node objects in an arbitrary order.
For example, the passed array could be [Node(5),Node(4),Node(3),Node(6),Node(2),Node(1)] or [Node(2),Node(6),Node(1),Node(3),Node(5),Node(4)].
The findRoot function should return the root Node(1), and the driver code will serialize it and compare with the input data.
The input data and serialized Node(1) are the same, so the test passes.

Example 2:



Input: tree = [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,11,null,12,null,13,null,14]
Output: [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,11,null,12,null,13,null,14]

Constraints:

- The total number of nodes is between [1, 5 * 10⁴].
- Each node has a unique value.

Follow up:

- Could you solve this problem in constant space complexity with a linear time algorithm?

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def findRoot(self, tree):
        # Brute Force - check each bit with a loop
        result = 0
        for i in range(32):
            bit = (n >> i) & 1
            result |= (bit << i)
        return result
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def findRoot(self, tree: List['Node']) -> 'Node':
        sum = 0

        for node in tree:
            sum ^= node.val
            for child in node.children:
                sum ^= child.val

        for node in tree:
            if node.val == sum:
                return node
```

• LeetCode

<pre> """ # Definition for a Node. class Node: def __init__(self, val=None, children=None): self.val = val self.children = children if children is not None else [] """ class Solution: def findRoot(self, tree: List['Node']) -> 'Node': x = 0 for node in tree: x ^= node.val for child in node.children: x ^= child.val return next(node for node in tree if node.val == x) </pre> • SimplyLeet	<pre> # Python implementation with detailed comments def findRoot(tree): # Initialize variables to store the sum of all node values and children values total_sum = 0 children_sum = 0 # Iterate over each node in the tree array for node in tree: total_sum += node.val # Add current node's value to total_sum # Iterate over each child of the current node and add their values to children_sum for child in node.children: children_sum += child.val # The root's value is the difference between total_sum and children_sum root_val = total_sum - children_sum # Find and return the node with the value equal to root_val for node in tree: if node.val == root_val: return node return None # This should never happen if the input tree is valid </pre> • LC.ca	<pre> """ # Definition for a Node. class Node: def __init__(self, val=None, children=None): self.val = val self.children = children if children is not None else [] """ class Solution: def findRoot(self, tree: List['Node']) -> 'Node': x = 0 for node in tree: x ^= node.val for child in node.children: x ^= child.val return next(node for node in tree if node.val == x) </pre>
Bit Manipulation p.397	>> Quick Review — Bit Manipulation 10 questions · insights · complexity · solution #67 Add Binary [Easy] Key Insights • Simulate binary addition just like you add numbers digit by digit. • Process the strings from right to left, handling the carry at each step. • Use modulo and division operations to extract the current digit and update the carry. • Append any remaining carry at the end. Space & Time Time Complexity: O(max(n, m)), where n and m are the lengths of the two binary strings. Space Complexity: O(max(n, m)) for storing the resulting binary string. Solution The strategy is to iterate over the input strings from the end to the beginning. At each step, calculate the sum of corresponding bits and the existing carry. Use the modulo operation (% 2) to determine the current digit and integer division (/ / 2) to update the carry. Once all digits have been processed, reverse the result to obtain	the final binary string. #136 Single Number [Easy] Key Insights • Using bitwise XOR: XOR-ing a number with itself cancels out to 0. • XOR is both commutative and associative, so the order of operations does not matter. • A single traversal (linear time) and constant extra space is sufficient for solving the problem. Space & Time Time Complexity: O(n) – We only traverse the array once. Space Complexity: O(1) – Only a single extra variable is used regardless of input size. Solution We can solve this problem by initializing a variable to 0 and then traversing the array while XOR-ing each element with the variable. Due to the properties of XOR, pairs of identical numbers will cancel each other out (x XOR x = 0) and the unique number will be left as the final result. This approach uses constant extra space and performs in linear time. #190 Reverse Bits [Easy] Key Insights
• The input is a binary string of fixed length (32 bits). • Reversing the bits is equivalent to reversing the string. • The reversed binary string can be converted back into its integer representation. • Since the number of bits is constant (32), the solution can be achieved in constant time and constant space. • For performance optimization when this function is called many times, caching may be applied for common intermediate results. Space & Time Time Complexity: O(1) – We always process 32 bits regardless of the input. Space Complexity: O(1) – Only constant extra space is used. Solution We solve the problem by reversing the 32-character binary string representing the unsigned integer. Once reversed, the new binary string is converted into its integer value. An alternative solution involves using bitwise operations where you iterate over each bit, shift the result to the left, and add the current bit. This approach does the reversal one bit at a time. The chosen approach leverages string reversal which is straightforward to implement and understand.	#191 Number of 1 Bits [Easy] Key Insights • The problem is essentially counting the number of 1s in the binary representation of a number. • A common efficient approach uses bit manipulation: repeatedly clear the lowest set bit until n becomes zero. • The expression n & (n – 1) clears the lowest set bit of n. • This method only iterates as many times as there are set bits, making it efficient. Space & Time Time Complexity: O(k), where k is the number of set bits, worst-case O(log n) (or O(1) considering 32-bit integers). Space Complexity: O(1). Solution To solve the problem, we use the bit manipulation trick n & (n – 1) which removes the lowest set bit from n. The algorithm is as follows: – Initialize a counter to 0. – While n is not zero, update n by n = n & (n – 1), and increment the counter. – The counter will hold the number of set bits by the end of the loop. This approach leverages efficient bit-level operations and works well even when the function is called frequently.	#268 Missing Number [Easy] Key Insights • The range of numbers is continuous from 0 to n, but one number is missing. • The sum of numbers from 0 to n can be computed using the arithmetic sum formula. • Subtracting the sum of the given array from the expected sum gives the missing number. • There is a follow-up challenge to solve the problem in O(n) time and O(1) extra space. Space & Time Time Complexity: O(n) Space Complexity: O(1) Solution We compute the expected sum of all numbers from 0 to n using the formula sum = n*(n+1)/2. Then, we compute the actual sum of the numbers in the array. The missing number is obtained by subtracting the actual sum from the expected sum. This approach uses arithmetic operations (O(1) extra space) and loops over the array once (O(n) time). #338 Counting Bits [Easy] Key Insights
Bit Manipulation p.403	Bit Manipulation p.404	Bit Manipulation p.405

- We can use dynamic programming to build the solution by utilizing the count of bits for smaller numbers.
- The relationship $dp[i] = dp[i \gg 1] + (i \& 1)$ holds, where $i \gg 1$ gives the count for i divided by 2 (dropping the least significant bit) and $(i \& 1)$ checks if the least significant bit is 1.
- This approach computes the result for each number in constant time, resulting in a linear time solution.

Space & Time
Time Complexity: $O(n)$
Space Complexity: $O(n)$

Solution
We use a dynamic programming approach where an array (or list) dp is initialized with size $n+1$. The key idea is to iterate from 1 to n and for each number, use the formula $dp[i] = dp[i \gg 1] + (i \& 1)$.
– The expression $(i \gg 1)$ effectively divides i by 2, giving the count of 1's for that half number. – The $(i \& 1)$ operation checks if i is odd, adding 1 if true. This method ensures each $dp[i]$ is computed in constant time, leading to an overall $O(n)$ time complexity.

#78 Subsets [Medium]

Key Insights

Bit Manipulation p.406

- Since there are $n+1$ elements with values only from 1 to n , by the pigeonhole principle at least one number must be repeated.
- The problem can be approached by mapping the array values to indices, forming a cycle structure.
- Floyd's Tortoise and Hare algorithm (cycle detection) can be applied to find the duplicate number in linear time and constant space.
- There is no need for extra data structures like hash sets or alterations to the original array, satisfying the space and array-modification constraints.

Space & Time
Time Complexity: $O(n)$
Space Complexity: $O(1)$

Solution
The solution leverages Floyd's Tortoise and Hare cycle detection algorithm. Think of each array element as a pointer to another index, forming a linked list with a cycle (the duplicate number creates the cycle). The algorithm works in two phases:
– Detect a meeting point within the cycle by having two pointers (tortoise and hare) move at different speeds. – Find the cycle entrance by resetting one pointer to the start and moving both pointers at the same speed. The meeting point reveals the duplicate number.

Bit Manipulation p.409

Backtracking

#17 Letter Combinations of a Phone Number [MED]

LeetCode · SimplyLeet · WalkCC · LeetCode
Hash Table · String · Backtracking
Lists: Grind 169

Time Complexity: $O(4^n \cdot n)$ where n is the length of the input string (each digit may map to up to 4 letters and adding a combination takes $O(n)$ time). Space Complexity: $O(n)$ for the recursion call stack

Problem
Given a string containing digits from 2–9 inclusive, return all possible letter combinations that the number could represent. Return the answer in any order.
A mapping of digits to letters (just like on the telephone buttons) is given below. Note that 1 does not map to any letters.

Backtracking p.412

- The problem is equivalent to generating the power set of the given set.
- A recursive backtracking approach can be used to explore both including and excluding each element.
- Iterative (bit manipulation) solutions are also possible since there are 2^n subsets in total.
- The maximum array length is small (up to 10), making exponential time solutions feasible.

Space & Time
Time Complexity: $O(2^n \cdot n)$ – There are 2^n subsets and creating each subset may take up to $O(n)$ time.
Space Complexity: $O(2^n \cdot n)$ – Space for storing all subsets. The recursion stack has a maximum depth of $O(n)$.

Solution
We solve the problem using backtracking, where we decide for each element whether to include it in the current subset or not. We recursively build all subsets starting at an empty list and at each step appending the current subset to our result list. We can also use iterative bit manipulation but here we focus on the backtracking approach. This algorithm leverages recursion and a list to maintain the current state, leading to a clear and concise implementation.

#90 Subsets II [Medium]

Bit Manipulation p.407

This approach provides an elegant solution that satisfies both linear runtime and constant space requirements.

#1506 Find Root of N-Ary Tree [Medium]

Key Insights
• Every node except the root appears as a child of another node.
• By summing all node values and then subtracting the sum of all children node values, the remaining value corresponds to the root.
• This subtraction trick allows us to identify the root in a single pass of the data structure without extra memory for tracking child nodes.

Space & Time
Time Complexity: $O(n)$ where n is the number of nodes, since we iterate through all nodes and their children.
Space Complexity: $O(1)$ extra space using arithmetic accumulators.

Solution
The solution calculates two sums: one for all nodes' values and one for all children nodes' values. The difference between these gives the value of the root node since every node's value (except the root's) is added once as a child. Finally, we iterate over the nodes to find the node with the computed root value and

Bit Manipulation p.410



Example 1:
Input: digits = "23"
Output: ["ad","ae","af","bd","be","bf","cd","ce","cf"]

Example 2:
Input: digits = "2"
Output: ["a","b","c"]

Constraints:
• $1 \leq \text{digits.length} \leq 4$
• $\text{digits}[i]$ is a digit in the range ['2', '9'].

Backtracking Sheet 46 / 317 · LeetCode Mastery By Pattern · Python Only · 5-Up Portrait p.413

Key Insights
• Sort the array first so that duplicate elements are adjacent.
• Use backtracking to generate all possible subsets.
• Skip duplicate elements in the recursion to avoid forming duplicate subsets.
• The use of sorted order and conditionally skipping elements is the trick to handle duplicates.

Space & Time
Time Complexity: $O(2^n \cdot n)$ since each subset creation may take $O(n)$ in the worst case.
Space Complexity: $O(2^n \cdot n)$ for storing the subsets, plus $O(n)$ for recursion stack.

Solution
The solution uses a backtracking technique that builds subsets incrementally. The input array is first sorted to bring duplicates together. During recursion, if the current element is the same as the previous one and it has not been used in the current recursion path, it is skipped to prevent duplicate subsets. This ensures that each subset is unique. The algorithm stores each valid subset by copying the current subset state into the result list.

#287 Find the Duplicate Number [Medium]

Key Insights

Bit Manipulation p.408

return it. This approach meets the requirement of constant space complexity and linear time.
Backtracking

#8 Backtracking

By Pattern · Python Only · 10 questions

Bit Manipulation p.411

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def letterCombinations(self, digits):
        # Brute Force O(4^n * n) - itertools.product over mapped letters
        if not digits: return []
        from itertools import product
        mapping = {
            '2': 'abc', '3': 'def', '4': 'ghi', '5': 'jkl',
            '6': 'mno', '7': 'pqrs', '8': 'tuv', '9': 'wxyz'
        }
        return [''.join(c) for c in product(*(mapping[d] for d in digits))]
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def letterCombinations(self, digits: str) -> list[str]:
        if not digits:
            return []

        digitToLetters = {
            '2': 'abc', '3': 'def', '4': 'ghi', '5': 'jkl',
            '6': 'mno', '7': 'pqrs', '8': 'tuv', '9': 'wxyz'
        }
        ans = []

        def dfs(i: int, path: list[str]) -> None:
```

Backtracking p.414

```
if i == len(digits):
    ans.append(''.join(path))
    return

for letter in digitToLetters[int(digits[i])]:
    path.append(letter)
    dfs(i + 1, path)
    path.pop()

dfs(0, [])

return ans
```

• LeetCode

```
class Solution:
    def letterCombinations(self, digits: str) -> List[str]:
        if not digits:
            return []
        d = ["abc", "def", "ghi", "jkl", "mno", "pqrs",
            "tuv", "wxyz"]
        ans = [""]
        for i in digits:
            s = d[int(i) - 2]
            ans = [a + b for a in ans for b in s]
        return ans
```

• SimplyLeet

```
# Python solution for Letter Combinations of a Phone Number

def letterCombinations(digits):
    # Mapping of digits to corresponding letters
    digit_to_letters = {
        "2": "abc",
        "3": "def",
        "4": "ghi",
        "5": "jkl",
        "6": "mno",

        "7": "pqrs",
        "8": "tuv",
        "9": "wxyz"
    }

    # If input is empty, return an empty list
    if not digits:
        return []

    result = []
```

```
# Backtracking function to generate combinations
def backtrack(index, current_combination):
    # If the current combination length equals the
    digits length,
    # add the combination to the result list.
    if index == len(digits):
        result.append(current_combination)
        return

    # Get the corresponding letters for the current
    digit

    letters = digit_to_letters[digits[index]]
    # Explore all possible letter routes for the
    current
    digit.
    for letter in letters:
        backtrack(index + 1, current_combination +
        letter)

    # Initiate backtracking starting from index 0 and
    empty combination.
    backtrack(0, "")
    return result

# Example usage:

#print(letterCombinations("23"))
```

• LC.ca

```
class Solution:
    def letterCombinations(self, digits: str) -> List[str]:
        if not digits:
            return []
        d = ["abc", "def", "ghi", "jkl", "mno", "pqrs",
            "tuv", "wxyz"]
        ans = [""]
        for i in digits:
            s = d[int(i) - 2]
            ans = [a + b for a in ans for b in s]
        return ans
```

Backtracking

#22 Generate ParenthesesMED

LeetCode · SimplyLeet · WalkCC · LeetCode

String · Dynamic Programming · Backtracking

Lists: Grind 169

Time Complexity: $O(4^n / (n^{(3/2)}))$ Space Complexity: $O(n)$ for the recursion stack, plus space for storing all valid combinations.

Problem

Given n pairs of parentheses, write a function to generate all combinations of well-formed parentheses.

Example 1:

Input: n = 3
Output: ["((()))", "(()())", "(())()", "()(())", "()()()"]

Example 2:

Input: n = 1
Output: ["()"]

Constraints:

• $1 \leq n \leq 8$

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def generateParenthesis(self, n):
        # Brute Force  $O(2^{(2n)} \cdot n)$  – generate all
        2n-length binary strings, filter valid
        def is_valid(comb):
            bal = 0
            for c in comb:
                bal += 1 if c == '(' else -1
                if bal < 0: return False
            return bal == 0

        from itertools import product
        return [''.join(c) for c in product('()',
        repeat=2*n) if is_valid(c)]
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def generateParenthesis(self, n):
        ans = []

        def dfs(l: int, r: int, s: list[str]) -> None:
            if l == 0 and r == 0:
                ans.append(''.join(s))
            if l > 0:
                s.append('(')
                dfs(l - 1, r, s)
```

```
s.pop()
if l < r:
    s.append('(')
    dfs(l, r - 1, s)
    s.pop()

dfs(n, n, [])
return ans
```

• LeetCode

```
class Solution:
    def generateParenthesis(self, n: int) -> List[str]:
        def dfs(l: int, r: int, t: str):
            if l > n or r > n or l < r:
                return
            if l == n and r == n:
                ans.append(t)
                return
            dfs(l + 1, r, t + "(")
            dfs(l, r + 1, t + ")")

        ans = []
        dfs(0, 0, "")
        return ans
```

• SimplyLeet

```
def generateParenthesis(n):
    # List to store valid combinations
    result = []

    # Helper function for backtracking
    def backtrack(current, open_count, close_count):
        # If current string reaches the maximum length,
        it's a valid combination
        if len(current) == 2 * n:
            result.append(current)
            return

        # If we can add an opening parenthesis, do it
        if open_count < n:
            backtrack(current + '(', open_count + 1,
            close_count)
        # If adding a closing parenthesis maintains the
        validity, do it
        if close_count < open_count:
            backtrack(current + ')', open_count,
            close_count + 1)

        # Start the recursion with an empty string and zero
        counts for both
        backtrack("", 0, 0)
        return result

# Example usage:
print(generateParenthesis(3))
```

• LC.ca

```
class Solution:
    def generateParenthesis(self, n: int) -> List[str]:
        def dfs(l, r, t):
            if l > n or r > n or l < r:
                return
            if l == n and r == n:
                ans.append(t)
                return
            dfs(l + 1, r, t + '(')
            dfs(l, r + 1, t + ')')

        ans = []
        dfs(0, 0, '')
        return ans

#####

class Solution(object):
    def generateParenthesis(self, n):
        """"

        :type n: int
        :rtype: List[str]
        """"

        def dfs(left, path, res, n):
            if len(path) == 2 * n:
                if left == 0:
                    res.append(''.join(path))
                return
```

```

    if left < n:
        path.append("")
        dfs(left + 1, path, res, n)
        path.pop()
    if left > 0:
        path.append("")
        dfs(left - 1, path, res, n)
        path.pop()

res = []

dfs(0, [], res, n)
return res
```

Backtracking

#39 Combination SumMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Backtracking

Lists: Grind 169

Time Complexity: $O(2^{\min(\text{target}/\min(\text{candidates}))})$ in the worst case where many combinations are possible. Space Complexity: $O(\text{target}/\min(\text{candidates}))$ for the recursion stack and temporary space used to s

Problem

Given an array of distinct integers candidates and a target integer target, return a list of all unique combinations of candidates where the chosen numbers sum to target. You may return the combinations in any order.

The same number may be chosen from candidates an unlimited number of times. Two combinations are unique if the frequency of at least one of the chosen numbers is different.

The test cases are generated such that the number of unique combinations that sum up to target is less than 150 combinations for the given input.

Example 1:

```

Input: candidates = [2,3,6,7], target = 7
Output: [[2,2,3],[7]]
Explanation:
2 and 3 are candidates, and 2 + 2 + 3 = 7. Note that 2
can be used multiple times.
7 is a candidate, and 7 = 7.
These are the only two combinations.
```

Example 2:

```

Input: candidates = [2,3,5], target = 8
Output: [[2,2,2,2],[2,3,3],[3,5]]
```

Example 3:

```

Input: candidates = [2], target = 1
Output: []
```

Constraints:

- 1 <= candidates.length <= 30
- 2 <= candidates[i] <= 40
- All elements of candidates are distinct.
- 1 <= target <= 40

Brute Force (Python)

```

# Brute Force Approach
class Solution:
    def combinationSum(self, candidates, target):
        # Brute Force  $O(n^{\min(\text{target}/\min)})$  - recursion
        without pruning
        result = []
        def backtrack(start, path, remain):
            if remain == 0:
                result.append(list(path)); return
            if remain < 0: return
            for i in range(start, len(candidates)):
                path.append(candidates[i])
                backtrack(i, path, remain - candidates[i])
                path.pop()
        backtrack(0, [], target)
        return result
```

Community Solutions (Python)

• WalkCC

```

class Solution:
    def combinationSum(self, candidates: List[int], target: int) -> List[List[int]]:
        ans = []

        def dfs(s: int, target: int, path: List[int]) -> None:
            if target < 0:
                return
            if target == 0:
                ans.append(path.clone())

        dfs(0, target, [])
```

```

        return

        for i in range(s, len(candidates)):
            path.append(candidates[i])
            dfs(i, target - candidates[i], path)
            path.pop()

        candidates.sort()
        dfs(0, target, [])
        return ans

• LeetCode
class Solution:
    def combinationSum(self, candidates: List[int], target: int) -> List[List[int]]:
        def dfs(i: int, s: int):
            if s == 0:
                ans.append(t[:])
                return
            if s < candidates[i]:
                return
            for j in range(i, len(candidates)):
                t.append(candidates[j])
                dfs(j, s - candidates[j])
                t.pop()

        candidates.sort()
        t = []
        ans = []
        dfs(0, target)
        return ans
```

• SimplyLeet

```

# Function to find all unique combinations that sum to the target
def combinationSum(candidates, target):
    # List to store all valid combinations
    result = []
    # Sort candidates to facilitate pruning of the search
    candidates.sort()

    # Recursive helper function for backtracking
    def backtrack(remaining, start, path):
        # If remaining is 0, path is a valid combination
        # so add a copy of it to result

        if remaining == 0:
            result.append(list(path))
            return
        # Loop through candidates starting from the current index
        for i in range(start, len(candidates)):
            # If candidate is greater than remaining, break out for efficiency
            if candidates[i] > remaining:
                break
            # Include candidate in the current combination
            path.append(candidates[i])
            backtrack(remaining - candidates[i], i, path)
```

```

# Recursively call backtrack with updated remaining and same index (allowing reuse)
backtrack(remaining - candidates[i], i, path)
# Backtrack by removing the candidate added in this iteration
path.pop()

# Start backtracking from index 0 with an empty path
backtrack(target, 0, [])
return result

# Example usage

print(combinationSum([2,3,6,7], 7))

• LC.ca
class Solution:
    def combinationSum(self, candidates: List[int], target: int) -> List[List[int]]:
        # i: start index for this recursion
        # s: sum
        def dfs(i, s):
            if s == target:
                ans.append(t.copy())
                return
            if s > target:
                return
```

```

        for j in range(i, len(candidates)):
            c = candidates[j]
            t.append(c)
            dfs(j, s + c)
            t.pop()

        ans = []
        t = []
        # candidates.sort() # diff from combinationSum-II, no need sorting it
        dfs(0, 0)

        return ans

#####

# dp version
class Solution_dp:
    def combinationSum(self, candidates: List[int], target: int) -> List[List[int]]:
        # for each-target (from 1 to target), its dp[i][j]
        # ==> so 3-D array dp[i][j][t]
        dp = []
```

```

        candidates.sort()

        for i in range(1, target+1):
            cur = []
            for j in range(len(candidates)):
                if candidates[j] > i:
                    break
                if candidates[j] == i:
                    one = [candidates[j]]
                    cur.append(one)

            break
            for a in dp[i - candidates[j] - 1]:
                if candidates[j] > a[0]:
                    continue
                deepCopied = a.copy()
                deepCopied.insert(0, candidates[j])
                cur.append(deepCopied)
            dp.append(cur)

        return dp[-1]

#####

class Solution(object):
    def combinationSum(self, candidates, target):
        """
        :type candidates: List[int]
        :type target: int
        :rtype: List[List[int]]
```

000000 def dfs(candidates, start, target, path, res): if target == 0: return res.append(path + []) for i in range(start, len(candidates)): if target - candidates[i] >= 0: path.append(candidates[i]) dfs(candidates, i, target - candidates[i], path, res) path.pop() res = [] dfs(candidates, 0, target, [], res) return res ##### class Solution: def combinationSum(self, candidates: List[int], target: int) -> List[List[int]]: Backtrackingp.433	def dfs(i: int, s: int): if s == 0: ans.append(t[:]) return if i >= len(candidates) or s < candidates[i]: return dfs(i + 1, s) t.append(candidates[i]) dfs(i, s - candidates[i]) t.pop() candidates.sort() t = [] ans = [] dfs(0, target) return ans Backtrackingp.434	Backtracking #46 PermutationsMED LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Backtracking Lists: Grind 169 Time Complexity: $O(n * n!)$ – There are $n!$ permutations, and constructing each permutation takes $O(n)$ time. Space Complexity: $O(n)$ – Recursion depth is limited to n, plus additional space for the current permutation. Problem Given an array nums of distinct integers, return all the possible permutations. You can return the answer in any order. Example 1: Input: nums = [1,2,3] Output: [[1,2,3],[1,3,2],[2,1,3],[2,3,1],[3,1,2],[3,2,1]] Example 2: Input: nums = [0,1] Output: [[0,1],[1,0]] Example 3: Input: nums = [1] Output: [[1]] Backtrackingp.435
Constraints: • $1 \leq \text{nums.length} \leq 6$ • $-10 \leq \text{nums}[i] \leq 10$ • All the integers of nums are unique. Brute Force (Python) # Brute Force Approach class Solution: def permute(self, nums): # Brute Force $O(n! \cdot n)$ – itertools.permutations (or full recursion) from itertools import permutations as perms return [list(p) for p in perms(nums)] Community Solutions (Python) • WalkCC class Solution: def permute(self, nums: list[int]) -> list[list[int]]: ans = [] used = [False] * len(nums) def dfs(path: list[int]) -> None: if len(path) == len(nums): ans.append(path.copy()) return Backtrackingp.436	for i, num in enumerate(nums): if used[i]: continue used[i] = True path.append(num) dfs(path) path.pop() used[i] = False dfs([]) return ans • LeetDoocs class Solution: def permute(self, nums: List[int]) -> List[List[int]]: def dfs(i: int): if i >= n: ans.append(t[:]) return for j, x in enumerate(nums): if not vis[j]: vis[j] = True t[i] = x Backtrackingp.437	dfs(i + 1) vis[j] = False n = len(nums) vis = [False] * n t = [0] * n ans = [] dfs(0) return ans • SimplyLeet def permute(nums): result = [] def backtrack(current, remaining): # If no remaining elements, current permutation is complete. if not remaining: result.append(current.copy()) return # Explore each candidate number. for i in range(len(remaining)): Backtrackingp.438
# Add the chosen number to the current permutation. current.append(remaining[i]) # Recursively generate permutations with the remaining numbers. backtrack(current, remaining[:i] + remaining[i+1:]) # Backtrack by removing the chosen number. current.pop() backtrack([], nums) return result # Example usage: print(permute([1,2,3])) • LC.ca ... remove an element by its value from a set >>> my_set = {1, 2, 3, 4, 5} >>> my_set.remove(3) >>> print(my_set) {1, 2, 4, 5} ----- Backtrackingp.439	cannot remove an element by index from a set in Python3 need to convert the set to a list >>> my_set = {1, 2, 3, 4, 5} >>> my_list = list(my_set) >>> del my_list[2] # remove the element at index 2 >>> my_set = set(my_list) >>> print(my_set) {1, 2, 4, 5} ... class Solution: # iterative def permute(self, nums: List[int]) -> List[List[int]]: res = [[]] if nums is None or len(nums) == 0: return ans for num in nums: new_res = [] Backtrackingp.440	for perm in res: for i in range(len(perm) + 1): new_perm = perm[:i] + [num] + perm[i:] new_res.append(new_perm) res = new_res return res ##### class Solution: # iterative, single_perm.insert(index, num) def permute(self, nums: List[int]) -> List[List[int]]: ans = [[]] if nums is None or len(nums) == 0: return ans for num in nums: tmp_list = [] Backtrackingp.441

```
for single_perm in ans:
    for index in range(len(single_perm) + 1):
        single_perm.insert(index, num)
        tmp_list.append(single_perm.copy())
        single_perm.pop(index)

ans = tmp_list

return ans

#####

...

In Python 3, both the discard() and remove() methods of a
set object are used to remove an element from the set,
but there is one key difference:

* discard() removes the specified element from the set if
it is present, but does nothing if the element is not
present.
* remove() removes the specified element from the set if
it is present, but raises a KeyError exception if the
element is not present.
```

```
>>> a
{33, 66, 11, 44, 22, 55}
>>> a.discard(22)
>>> a
{33, 66, 11, 44, 55}

>>> a.discard(200)
>>>
>>> a.remove(200)
Traceback (most recent call last):

  File "<stdin>", line 1, in <module>
KeyError: 200
'''

...

>>> v1 = set({})
>>> v2 = set({})
>>>
>>> v1
set({})

>>> v2
set({})
'''

class Solution:
    def permute(self, nums: List[int]) ->
List[List[int]]:
    res = []
    visited = set({})

    def dfs(nums, path, res, visited):
        if len(path) == len(nums):
```

```
res.append(path + [])
return

for i in range(0, len(nums)):
    if i not in visited:
        visited.add(i)
        path.append(nums[i])
        dfs(nums, path, res, visited)
        path.pop()
        visited.discard(i) # remove(i) will
throw exception if i not existing

dfs(nums, [], res, visited)
return res

#####

class Solution:
    def permute(self, nums: List[int]) ->
List[List[int]]:
    def dfs(i):
        if i == n:
```

```
ans.append(t[:])
return
for j in range(n):
    if not vis[j]:
        vis[j] = True
        t[i] = nums[j]
        dfs(i + 1)
        vis[j] = False

n = len(nums)

vis = [False] * n
t = [0] * n
ans = []
dfs(0)
return ans
```

Backtracking

#79 Word SearchMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · String · Backtracking · Matrix

Lists: Grind 169

Time Complexity: $O(m * n * 3^L)$, where L is the length of the word (each DFS may branch to at most 3 further cells, excluding the coming cell). Space Complexity: $O(L)$ due to recursion call stack (wher

Problem

Given an m x n grid of characters board and a string word, return true if word exists in the grid. The word can be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once.

Example 1:

A	B	C	E
S	F	C	S
A	D	E	E

Input: board = [["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]], word = "ABCCED"

Output: true

Example 2:

A	B	C	E
S	F	C	S
A	D	E	E

Input: board = [["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]], word = "SEE"

Output: true

Example 3:

A	B	C	E
S	F	C	S
A	D	E	E

Input: board = [["A","B","C","E"],["S","F","C","S"],["A","D","E","E"]], word = "ABCB"

Output: false

Constraints:

- m == board.length
- n == board[i].length
- 1 <= m, n <= 6
- 1 <= word.length <= 15
- board and word consists of only lowercase and uppercase English letters.

Follow up: Could you use search pruning to make your solution faster with a larger board?

Brute Force (Python)

Brute Force Approach

class Solution:

def exist(self, board, word):

Brute Force $O(m \cdot n \cdot 4^L)$ – DFS from every cell,

backtrack on visited

m, n = len(board), len(board[0])

def dfs(r, c, i, visited):

if i == len(word): return True

if r < 0 or r >= m or c < 0 or c >= n: return False

if (r, c) in visited or board[r][c] !=

word[i]: return False

visited.add((r, c))

found = (dfs(r+1, c, i+1, visited) or

dfs(r-1, c, i+1, visited) or

dfs(r, c+1, i+1, visited) or

dfs(r, c-1, i+1, visited))

visited.remove((r, c))

return found

for r in range(m):

for c in range(n):

if dfs(r, c, 0, set()): return True

return False

Community Solutions (Python)

- WalkCC


```

class Solution:
    def exist(self, board: List[List[str]], word: str) -> bool:
        m = len(board)
        n = len(board[0])

        def dfs(i: int, j: int, s: int) -> bool:
            if i < 0 or i == m or j < 0 or j == n:
                return False
            if board[i][j] != word[s] or board[i][j] == '*':
                return False

            if s == len(word) - 1:
                return True

            cache = board[i][j]
            board[i][j] = '*'
            isExist = (dfs(i + 1, j, s + 1) or
                       dfs(i - 1, j, s + 1) or
                       dfs(i, j + 1, s + 1) or
                       dfs(i, j - 1, s + 1))
            board[i][j] = cache

            return isExist

        return any(dfs(i, j, 0)
                   for i in range(m)
                   for j in range(n))

```

• LeetCode

Backtrackingp.451

```

class Solution:
    def exist(self, board: List[List[str]], word: str) -> bool:
        def dfs(i: int, j: int, k: int) -> bool:
            if k == len(word) - 1:
                return board[i][j] == word[k]
            if board[i][j] != word[k]:
                return False
            c = board[i][j]
            board[i][j] = "0"
            for a, b in pairwise((-1, 0, 1, 0, -1)):
                x, y = i + a, j + b
                ok = 0 <= x < m and 0 <= y < n and
                board[x][y] != "0"
                if ok and dfs(x, y, k + 1):
                    return True
            board[i][j] = c
            return False

        m, n = len(board), len(board[0])
        return any(dfs(i, j, 0) for i in range(m) for j in range(n))

```

• SimplyLeet

Backtrackingp.452

```

# Define the function to solve the Word Search problem.
def exist(board, word):
    # Get dimensions of the board.
    rows, cols = len(board), len(board[0])

    # Helper function for DFS search.
    def dfs(r, c, index):
        # If the entire word is found, return True.
        if index == len(word):
            return True

        # Check boundaries and if current cell already
        # used or not matching.
        if r < 0 or c < 0 or r >= rows or c >= cols or
        board[r][c] != word[index]:
            return False

        # Temporarily mark the cell as visited.
        temp = board[r][c]
        board[r][c] = "#"

        # Explore all four directions.

```

Backtrackingp.453

```

found = (dfs(r + 1, c, index + 1) or
         dfs(r - 1, c, index + 1) or
         dfs(r, c + 1, index + 1) or
         dfs(r, c - 1, index + 1))

# Backtrack: restore the original cell value.
board[r][c] = temp
return found

# Initiate DFS from each cell in the grid.

```

```

for i in range(rows):
    for j in range(cols):
        if dfs(i, j, 0):
            return True
return False

```

```

# Example test case
if __name__ == "__main__":
    board = [
        ["A", "B", "C", "E"],
        ["S", "F", "C", "S"],
        ["A", "D", "E", "E"]
    ]
    word = "ABCCED"
    print(exist(board, word)) # Expected output: True

```

• LC.ca

Backtrackingp.454

```

class Solution:
    def exist(self, board: List[List[str]], word: str) -> bool:
        def dfs(i, j, cur): # cur: current char count
            if cur == len(word):
                return True
            if (
                i < 0
                or i >= m
                or j < 0
                or j >= n
                or board[i][j] == '0'
                or word[cur] != board[i][j]
            ):
                return False
            t = board[i][j]
            board[i][j] = '0' # mark as visited
            for a, b in [[0, 1], [0, -1], [-1, 0], [1, 0]]:
                x, y = i + a, j + b
                if dfs(x, y, cur + 1):
                    return True

            board[i][j] = t
            return False

        m, n = len(board), len(board[0])
        return any(dfs(i, j, 0) for i in range(m) for j in range(n))

```

Backtrackingp.455

Backtracking

#113 Path Sum II

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Backtracking · Tree · DFS

Lists: Grind 169

Time Complexity: $O(N^2)$ in the worst-case scenario (where N is the number of nodes) when most nodes are leaf nodes, since each potential path copy operation can be $O(N)$. Space Complexity: $O(N)$ for the

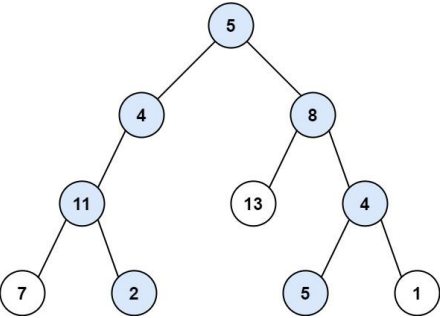
Problem

Given the root of a binary tree and an integer targetSum, return all root-to-leaf paths where the sum of the node values in the path equals targetSum. Each path should be returned as a list of the node values, not node references.

A root-to-leaf path is a path starting from the root and ending at any leaf node. A leaf is a node with no children.

Example 1:

Backtrackingp.456



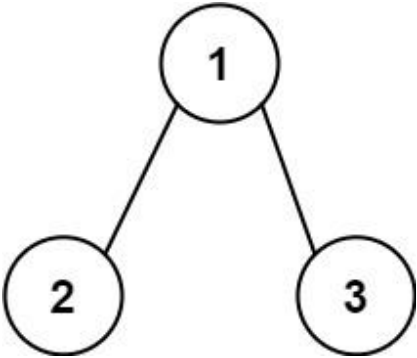
```

Input: root = [5,4,8,11,null,13,4,7,2,null,null,5,1],
targetSum = 22
Output: [[5,4,11,2],[5,8,4,5]]
Explanation: There are two paths whose sum equals
targetSum:
5 + 4 + 11 + 2 = 22
5 + 8 + 4 + 5 = 22

```

Example 2:

Backtrackingp.457



```

Input: root = [1,2,3], targetSum = 5
Output: []

```

Example 3:

```

Input: root = [1,2], targetSum = 0
Output: []

```

Constraints:

- The number of nodes in the tree is in the range [0, 5000].
- $-1000 \leq \text{Node.val} \leq 1000$
- $-1000 \leq \text{targetSum} \leq 1000$

Backtrackingp.458

Brute Force (Python)

```

# Brute Force Approach
class Solution:
    def pathSum(self, root, targetSum):
        # Brute Force - exhaustive backtracking without
        # pruning
        results = []
        def backtrack(start, path):
            results.append(List(path))
            for i in range(start, len(root)):
                path.append(root[i])
                backtrack(i + 1, path)

        path.pop()
        backtrack(0, [])
        return results

```

Community Solutions (Python)

• WalkCC

Backtrackingp.459

```
class Solution:
    def pathSum(self, root: TreeNode, summ: int) -> List[List[int]]:
        list[list[int]]:
            ans = []

            def dfs(root: TreeNode, summ: int, path: list[int]) -> None:
                if not root:
                    return
                if root.val == summ and not root.left and not root.right:
                    ans.append(path + [root.val])
                    return

                dfs(root.left, summ - root.val, path + [root.val])
                dfs(root.right, summ - root.val, path + [root.val])

            dfs(root, summ, [])
            return ans
```

- LeetCode

```
# Definition for a binary tree node.
# class TreeNode:
#     def __init__(self, val=0, left=None, right=None):
#         self.val = val
#         self.left = left
#         self.right = right
class Solution:
    def pathSum(self, root: Optional[TreeNode], targetSum: int) -> List[List[int]]:
        def dfs(root, s):
            if root is None:
```

Backtrackingp.460

```
        return
        s += root.val
        t.append(root.val)
        if root.left is None and root.right is None
        and s == targetSum:
            ans.append(t[:])
            dfs(root.left, s)
            dfs(root.right, s)
            t.pop()

        ans = []

        t = []
        dfs(root, 0)
        return ans
```

- SimplyLeet

```
# Definition for a binary tree node.
# class TreeNode:
#     def __init__(self, val=0, left=None, right=None):
#         self.val = val
#         self.left = left
#         self.right = right

def pathSum(root, targetSum):
    result = [] # This will store all valid paths
```

Backtrackingp.461

```
def dfs(node, current_path, remaining_sum):
    if not node:
        return # Base case: reached the end of a path

    # Add the current node's value to the path
    current_path.append(node.val)

    # Check if it's a leaf and if the path sum equals targetSum
    if not node.left and not node.right and node.val == remaining_sum:
        result.append(list(current_path)) # Add a copy of the current path

    else:
        # Continue the search in the left and right subtrees
        dfs(node.left, current_path, remaining_sum - node.val)
        dfs(node.right, current_path, remaining_sum - node.val)

    # Backtrack: remove the last element before returning to the caller
    current_path.pop()

    dfs(root, [], targetSum)
    return result
```

- LC.ca

Backtrackingp.462

```
# Definition for a binary tree node.
# class TreeNode:
#     def __init__(self, val=0, left=None, right=None):
#         self.val = val
#         self.left = left
#         self.right = right
class Solution:
    def pathSum(self, root: Optional[TreeNode], targetSum: int) -> List[List[int]]:
        def dfs(root, s):
            if root is None:
```

```
        return
        s += root.val
        t.append(root.val)
        if root.left is None and root.right is None
        and s == targetSum:
            ans.append(t[:])
            dfs(root.left, s)
            dfs(root.right, s)
            t.pop()

        ans = []

        t = []
        dfs(root, 0)
        return ans
```

Backtrackingp.463

```
Backtracking
```

#37 Sudoku Solver

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Backtracking · Matrix

Lists: Grind 169

Time Complexity: In the worst-case scenario, the algorithm can explore up to $O(9^n)$ possibilities where n is the number of empty cells. However, since n is at most 81 and many branches are pruned ea

Problem

Write a program to solve a Sudoku puzzle by filling the empty cells.

A sudoku solution must satisfy all of the following rules:

- Each of the digits 1–9 must occur exactly once in each row.
- Each of the digits 1–9 must occur exactly once in each column.
- Each of the digits 1–9 must occur exactly once in each of the 9 3x3 sub-boxes of the grid.

The '.' character indicates empty cells.

Example 1:

Backtrackingp.464

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

Backtrackingp.465

```
Input: board = [["5","3",".",".","7",".",".","."],[
"6",".",".","1","9","5",".","."],[
".","9","8",".",".","6",".","."],[
".","8",".",".","6",".","."],[
"4",".",".","8",".","3",".","1","7",".","."],
["2",".",".","6",".","6",".","."],
["8",".","4","1","9","5","."],
["8",".","7","4","1","9","6","3","5"],
["3","4","5","2","8","6","1","7","9"]]]
Output: [["5","3","4","6","7","8","9","1","2"],
["6","7","2","1","9","5","3","4","8"],
["1","9","8","3","4","2","5","6","7"],
["8","5","3","7","9","1","7","1","3","9","2"],
["4","8","5","6","1","9","6","1","5","3","7","2","8","4"],
["2","8","7","4","1","9","6","3","5"],
["3","4","5","2","8","6","1","7","9"]]]
Explanation: The input board is shown above and the only valid solution is shown below:
```

Constraints:

- board.length == 9
- board[i].length == 9
- board[i][j] is a digit or '.'.
- It is guaranteed that the input board has only one solution.

Brute Force (Python)

Backtrackingp.466

```
# Brute Force Approach
class Solution:
    def solveSudoku(self, board):
        # Brute Force - exhaustive backtracking without pruning

        results = []
        def backtrack(start, path):
            results.append(list(path))
            for i in range(start, len(board)):
                path.append(board[i])
                backtrack(i + 1, path)

            path.pop()
            backtrack(0, [])
            return results
```

Community Solutions (Python)

- WalkCC

```
class Solution:
    def solveSudoku(self, board: list[list[str]]) -> None:
        def isValid(row: int, col: int, c: str) -> bool:
            for i in range(9):
                if (board[i][col] == c or
                    board[row][i] == c or
                    board[3 * (row // 3) + i // 3][3 * (col // 3) + i % 3] == c):
                    return False
            return True
```

Backtrackingp.467

```
def solve(s: int) -> bool:
    if s == 81:
        return True

    i = s // 9
    j = s % 9

    if board[i][j] != '.':
        return solve(s + 1)

    for c in string.digits[1:]:
        if isValid(i, j, c):
            board[i][j] = c
            if solve(s + 1):
                board[i][j] = '.'
            return False

    return False

solve(0)
```

- SimplyLeet

```
def solveSudoku(board):
    # Check if placing the digit 'char' at board[row][col] is valid
    def is_valid(row, col, char):
        for i in range(9):
            # Check the row
            if board[row][i] == char:
                return False
            # Check the column
            if board[i][col] == char:
                return False
```

Backtrackingp.468

```
# Check the 3x3 sub-box
sub_row = 3 * (row // 3) + i // 3
sub_col = 3 * (col // 3) + i % 3
if board[sub_row][sub_col] == char:
    return False
return True

# Backtracking function to fill the board
def backtrack():
    for row in range(9):
        for col in range(9):
            if board[row][col] == '.':
                for char in map(str, range(1, 10)):
                    if is_valid(row, col, char):
                        board[row][col] = char #
                        # Backtracking function to fill the board
                        if backtrack():
                            return True # Recurse; if
                        # Backtracking function to fill the board
                        board[row][col] = '.' #
                        # Backtracking function to fill the board
                        if not valid further down
                        return False # No valid digit found;
                        # Backtracking function to fill the board
                        trigger backtracking
                        return True # Board completely filled
```

Backtracking

p.469

```
def backtrack():
    # Example usage:
    board = [
        ["5","3",".", ".", ".", "7",".", ".", ".", "."],
        ["6",".", ".", ".", "1","9","5",".", ".", "."],
        [".","9","8",".", ".", ".", ".", "6","."],
        ["8",".", ".", ".", "6",".", ".", ".", ".", "3"],
        ["4",".", ".", ".", "8",".", "3",".", ".", "1"],
        ["7",".", ".", ".", ".", "2",".", ".", ".", "6"],
        [".","6",".", ".", ".", ".", "2","8",".", "."],
        [".",".", ".", ".", "4","1","9",".", ".", "5"],
        [".",".", ".", ".", ".", "8",".", ".", ".", "9"]
    ]
    solveSudoku(board)
    print(board)

• LC.ca
class Solution:
    def solveSudoku(self, board: List[List[str]]) ->
    None:
        def dfs(k):
            nonlocal ok
            if k == len(t):
                ok = True
                return
            i, j = t[k]
            for v in range(9):
                if row[i][v] == col[j][v] == block[i //
3][j // 3][v] == False:
```

Backtracking

p.470

```
row[i][v] = col[j][v] = block[i //
3][j // 3][v] = True
board[i][j] = str(v + 1)
dfs(k + 1)
row[i][v] = col[j][v] = block[i //
3][j // 3][v] = False
if ok:
    return

row = [[False] * 9 for _ in range(9)]
col = [[False] * 9 for _ in range(9)]
block = [[[False] * 9 for _ in range(3)] for _ in
range(3)]

t = []
ok = False
for i in range(9):
    for j in range(9):
        if board[i][j] == '.':
            t.append((i, j))
        else:
            v = int(board[i][j]) - 1
            row[i][v] = col[j][v] = block[i //
3][j // 3][v] = True
            dfs(0)
```

Backtracking

p.471

Backtracking

#51 N-Queens

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Backtracking

Lists: Grind 169

Time Complexity: O(n!) in the worst-case scenario, as the solution explores permutations for queen placements. Space Complexity: O(n^2) for storing board configurations and O(n) for recursion stack an

Problem

The n-queens puzzle is the problem of placing n queens on an n x n chessboard such that no two queens attack each other.

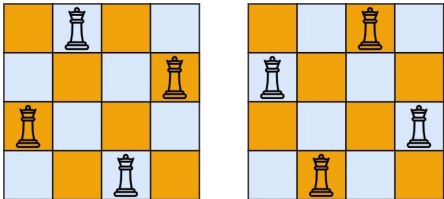
Given an integer n, return all distinct solutions to the n-queens puzzle. You may return the answer in any order.

Each solution contains a distinct board configuration of the n-queens' placement, where 'Q' and '.' both indicate a queen and an empty space, respectively.

Example 1:

Backtracking

p.472



Input: n = 4
Output: [
[".", "Q.", "...Q", "Q...", "...Q."],
["..Q.", "Q...", "...Q", "Q..."]
]
Explanation: There exist two distinct solutions to the 4-queens puzzle as shown above

Example 2:

Input: n = 1
Output: [
["Q"]
]

Constraints:

• 1 <= n <= 9

Brute Force (Python)

Backtracking

p.473

```
# Brute Force Approach
class Solution:
    def solveNQueens(self, n):
        # Brute Force O(n^n) - place a queen in every
        column each row, filter valid
        from itertools import product
        def is_valid(placement):
            for i in range(len(placement)):
                for j in range(i+1, len(placement)):
                    if placement[i]==placement[j]: return
                    if
abs(placement[i]-placement[j])==abs(i-j): return False #
diagonal

            return True
        result = []
        for cols in product(range(n), repeat=n):
            if is_valid(cols):
                board = []
                for c in cols:
                    row = '.'*c + 'Q' + '.'*(n-c-1)
                    board.append(row)
                result.append(board)
        return result
```

Community Solutions (Python)

• WalkCC

Backtracking

p.474

```
class Solution:
    def solveNQueens(self, n: int) -> list[list[str]]:
        ans = []
        cols = [False] * n
        diag1 = [False] * (2 * n - 1)
        diag2 = [False] * (2 * n - 1)

        def dfs(i: int, board: list[int]) -> None:
            if i == n:
                ans.append(board)

            return

            for j in range(n):
                if cols[j] or diag1[i + j] or diag2[j - i + n -
1]:
                    continue
                cols[j] = diag1[i + j] = diag2[j - i + n - 1] =
True
                dfs(i + 1, board + ['. ' * j + 'Q' + ' ' * (n - j -
1)])
                cols[j] = diag1[i + j] = diag2[j - i + n - 1] =
False

            dfs(0, [])

        return ans
```

• LeetDoocs

Backtracking

p.475

```
class Solution:
    def solveNQueens(self, n: int) -> List[List[str]]:
        def dfs(i: int):
            if i == n:
                ans.append(["".join(row) for row in g])
                return
            for j in range(n):
                if col[j] + dg[i + j] + udg[n - i + j] ==
0:
                    g[i][j] = "Q"
                    col[j] = dg[i + j] = udg[n - i + j] =
1

                    dfs(i + 1)
                    col[j] = dg[i + j] = udg[n - i + j] =
0
                    g[i][j] = "."

            ans = []
            g = [
                "." * n for _ in range(n)
            ]
            col = [0] * n
            dg = [0] * (n << 1)
            udg = [0] * (n << 1)
            dfs(0)

        return ans
```

• SimplyLeet

Backtracking

p.476

```
def solveNQueens(n):
    result = []

    # Helper function to backtrack
    def backtrack(row, columns, diag1, diag2, board):
        # Base case: if all rows are filled, add a copy
        of board to result
        if row == n:
            result.append(list(board))
            return

        # Try to place a queen in each column in the
        current row
        for col in range(n):
            # Check if the column or diagonals are not
            under attack
            if col in columns or (row - col) in diag1 or
(row + col) in diag2:
                continue

            # Place the queen
            board[row] = '.' * col + 'Q' + '.' * (n - col
- 1)

            # Mark the column and diagonals as occupied
            columns.add(col)
```

Backtracking

p.477

<pre> diag1.add(row - col) diag2.add(row + col) # Move to the next row backtrack(row + 1, columns, diag1, diag2, board) # Backtrack: remove the queen and unmark the sets columns.remove(col) diag1.remove(row - col) diag2.remove(row + col) # Initialize board with empty rows board = ["." * n for _ in range(n)] backtrack(0, set(), set(), set(), board) return result # Example usage: print(solveNQueens(4)) </pre> • LC.ca	<pre> class Solution: def solveNQueens(self, n: int) -> List[List[str]]: def dfs(i: int): if i == n: ans.append(''.join(row) for row in g) return for j in range(n): if col[j] + dg[i + j] + udg[n - i + j] == 0: g[i][j] = "Q" col[j] = dg[i + j] = udg[n - i + j] = 1 dfs(i + 1) col[j] = dg[i + j] = udg[n - i + j] = 0 g[i][j] = "." ans = [] g = ["." * n for _ in range(n)] col = [0] * n dg = [0] * (n << 1) udg = [0] * (n << 1) dfs(0) return ans </pre>	Backtracking #126 Word Ladder II HAR LeetDoocs · SimplyLeet · WalkCC · LeetCode Hash Table · String · Backtracking · BFS Lists: Denny Zhang Time Complexity: $O(N * L * 26)$ where N is the number of words in the wordList and L is the length of each word. This accounts for checking all possible one-letter transformations. Space Complexity: O Problem A transformation sequence from word beginWord to word endWord using a dictionary wordList is a sequence of words beginWord -> s1 -> s2 -> ... -> sk such that: • Every adjacent pair of words differs by a single letter. • Every si for $1 \leq i \leq k$ is in wordList. Note that beginWord does not need to be in wordList. • sk == endWord Given two words, beginWord and endWord, and a dictionary wordList, return all the shortest transformation sequences from beginWord to endWord, or an empty list if no such sequence exists. Each sequence should be returned as a list of the words [beginWord, s1, s2, ..., sk]. Example 1:
Input: beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log","cog"] Output: [["hit","hot","dot","dog","cog"],["hit","hot","lot","log","cog"]] Explanation: There are 2 shortest transformation sequences: "hit" -> "hot" -> "dot" -> "dog" -> "cog" "hit" -> "hot" -> "lot" -> "log" -> "cog" Example 2: Input: beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log"] Output: [] Explanation: The endWord "cog" is not in wordList, therefore there is no valid transformation sequence. Constraints: • $1 \leq \text{beginWord.length} \leq 5$ • $\text{endWord.length} == \text{beginWord.length}$ • $1 \leq \text{wordList.length} \leq 500$ • $\text{wordList}[i].\text{length} == \text{beginWord.length}$ • beginWord, endWord, and wordList[i] consist of lowercase English letters. • $\text{beginWord} \neq \text{endWord}$ • All the words in wordList are unique. • The sum of all shortest transformation sequences does not exceed 105. Brute Force (Python)	<pre> # Brute Force Approach class Solution: def findLadders(self, beginWord, endWord, wordList): # Brute Force - exhaustive backtracking without pruning results = [] def backtrack(start, path): results.append(list(path)) for i in range(start, len(beginWord)): path.append(beginWord[i]) backtrack(i + 1, path) path.pop() backtrack(0, []) return results </pre> Community Solutions (Python) • WalkCC <pre> class Solution: def findLadders(self, beginWord: str, endWord: str, wordList: List[str]) -> List[List[str]]: wordSet = set(wordList) if endWord not in wordList: return [] # {"hit": ["hot"], "hot": ["dot", "lot"], ...} graph: dict[str, list[str]] = collections.defaultdict(list) # Build the graph from the beginWord to the endWord. </pre>	<pre> if not self._bfs(beginWord, endWord, wordSet, graph): return [] ans = [] self._dfs(graph, beginWord, endWord, [beginWord], ans) return ans def _bfs(self, beginWord: str, endWord: str, wordSet: set[str], graph: dict[str, list[str]],) -> bool: currentLevelWords = {beginWord} while currentLevelWords: for word in currentLevelWords: wordSet.discard(word) nextLevelWords = set() reachEndWord = False for parent in currentLevelWords: for child in self._getChildren(parent, wordSet): if child in wordSet: nextLevelWords.add(child) graph[parent].append(child) if child == endWord: reachEndWord = True if reachEndWord: return True </pre>
<pre> currentLevelWords = nextLevelWords return False def _getChildren(self, parent: str, wordSet: set[str]) -> list[str]: children = [] s = list(parent) for i, cache in enumerate(s): for c in string.ascii_lowercase: if c == cache: continue s[i] = c child = ''.join(s) if child in wordSet: children.append(child) s[i] = cache return children def _dfs(self, graph: dict[str, list[str]], word: str, endWord: str, path: list[str], ans: list[list[str]],) -> None: if word == endWord: ans.append(path.copy()) </pre>	<pre> return for child in graph.get(word, []): path.append(child) self._dfs(graph, child, endWord, path, ans) path.pop() • SimplyLeet # Python solution for Word Ladder II from collections import defaultdict, deque def findLadders(beginWord, endWord, wordList): # Convert list to set for fast lookup wordSet = set(wordList) if endWord not in wordSet: return [] # Dictionary to hold parent pointers for backtracking: child -> list of parents parents = defaultdict(list) # Set for words visited at current BFS level level = {beginWord} # Flag to stop BFS when endWord is reached found = False while level and not found: next_level = set() # Remove words of current level from wordSet to prevent cycles </pre>	<pre> wordSet -= level for word in level: # Try all possible one-letter transformations for i in range(len(word)): for char in 'abcdefghijklmnopqrstuvwxyz': candidate = word[:i] + char + word[i+1:] if candidate in wordSet: # If candidate is the endWord, note that we have found a path if candidate == endWord: found = True # Record the current word as a parent of candidate word parents[candidate].append(word) next_level.add(candidate) # Move to the next level of the BFS level = next_level # The result list to hold all valid transformation sequences res = [] # Helper function to backtrack from endWord to beginWord using the parents dictionary </pre>

```
def backtrack(word, path):
    if word == beginWord:
        # Append reversed path to have the sequence
        from beginWord to endWord
        res.append(path[::-1])
        return
    # Recurse over all parents of the current word
    for parent in parents[word]:
        backtrack(parent, path + [parent])

if found:

    backtrack(endWord, [endWord])
    return res

# Example usage:
# print(findLadders("hit", "cog",
# ["hot","dot","dog","lot","log","cog"]))
```

LC.ca

```
...
remove() same as discard()

>>> a = set([11,22,33])
>>> a.remove(22)
>>> a
{33, 11}

>>> a = set([11,22,33])
>>> a.discard(22)
```

```
>>> a
{33, 11}

>>> a=set([1,2,3])
>>> a.discard(555)
>>> a.remove(555)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
KeyError: 555

...

class Solution:
    def findLadders(
        self, beginWord: str, endWord: str, wordList:
        List[str]
    ) -> List[List[str]]:
        # endWord to beginWord
        # better than begin to end, too many paths not in
        final result
        def dfs(path, cur):
            if cur == beginWord:
```

```
ans.append(path[::-1])
return
for precursor in prev[cur]:
    path.append(precursor)
    dfs(path, precursor)
    path.pop()

ans = []
words = set(wordList)
if endWord not in words:

    return ans
# no exception if beginWord not in set
words.discard(beginWord)
dist = {beginWord: 0}
prev = defaultdict(set)
q = deque([beginWord])
found = False
step = 0
while q and not found:
    step += 1

for _ in range(len(q), 0, -1):
    p = q.popleft()
    s = list(p)
    for i in range(len(s)):
        ch = s[i]
        for j in range(26):
            s[i] = chr(ord('a') + j)
            t = ''.join(s)
            if dist.get(t, 0) == step:
                prev[t].add(p) # repeated 3

lines below
```

```
if t not in words:
    # if above '== step' met, then t
    must be removed from words[] from previous iterations
    continue
    prev[t].add(p) # repeated 3 lines
above

words.discard(t)
q.append(t)
dist[t] = step
if endWord == t:
    found = True
s[i] = ch

if found:
    path = [endWord]
    dfs(path, endWord)
    return ans

#####

from collections import deque

class Solution(object):
```

```
def findLadders(self, beginWord, endWord, wordlist):
    """
    :type beginWord: str
    :type endWord: str
    :type wordlist: Set[str]
    :rtype: List[List[int]]
    """

    def getNbrs(src, dest, wordList):
        res = []

        for c in string.ascii_lowercase:
            for i in range(0, len(src)):
                newWord = src[:i] + c + src[i + 1:]
                if newWord == src:
                    continue
                if newWord in wordList or newWord == dest:
                    yield newWord

    def bfs(beginWord, endWord, wordList):
        distance = {beginWord: 0}

        queue = deque([beginWord])
        length = 0
        while queue:
            length += 1
            for k in range(0, len(queue)):
                top = queue.popleft()
                for nbr in getNbrs(top, endWord, wordList):
                    if nbr not in distance:
                        distance[nbr] = distance[top] + 1
                        queue.append(nbr)
```

```
return distance

def dfs(beginWord, endWord, wordList, path, res,
distance):
    if beginWord == endWord:
        res.append(path + [])
        return

    for nbr in getNbrs(beginWord, endWord, wordList):
        if distance.get(nbr, -2) + 1 ==
        distance[beginWord]:
            path.append(nbr)

    dfs(nbr, endWord, wordList, path, res,
        distance)
    path.pop()

res = []
distance = bfs(endWord, beginWord, wordList)
dfs(beginWord, endWord, wordList, [beginWord], res,
    distance)
return res
```

Backtracking

#996 Number of Squareful Arrays

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Math · Dynamic Programming · Backtracking

Lists: Denny Zhang

Time Complexity: $O(n^2 \cdot 2^n)$ in the worst-case scenario due to exploring all bitmask states with n elements and checking pairwise sums. Space Complexity: $O(n)$ for recursion and $O(n)$ for the visited f

Problem

An array is squareful if the sum of every pair of adjacent elements is a perfect square.

Given an integer array `nums`, return the number of permutations of `nums` that are squareful.

Two permutations `perm1` and `perm2` are different if there is some index i such that `perm1[i] != perm2[i]`.

Example 1:

Input: `nums = [1,17,8]`
Output: 2
Explanation: `[1,8,17]` and `[17,8,1]` are the valid permutations.

Example 2:

Input: `nums = [2,2,2]`
Output: 1

Constraints:

- $1 \leq \text{nums.length} \leq 12$
- $0 \leq \text{nums}[i] \leq 109$

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def is_square(self, num):
        # Brute Force - exhaustive backtracking without
        pruning
        results = []
        def backtrack(start, path):
            results.append(list(path))
            for i in range(start, len(num)):
                path.append(num[i])
                backtrack(i + 1, path)

        path.pop()
        backtrack(0, [])
        return results
```

Community Solutions (Python)

- WalkCC

class Solution:

def numSquarefulPerms(self, nums: list[int]) -> int:

ans = 0

used = [False] * len(nums)

def isSquare(num: int) -> bool:

root = math.isqrt(num)

return root * root == num

def dfs(path: list[int]) -> None:

nonlocal ans

if len(path) > 1 and not isSquare(path[-1] + path[-2]):

return

if len(path) == len(nums):

ans += 1

return

for i, a in enumerate(nums):

if used[i]:

continue

if i > 0 and nums[i] == nums[i - 1] and not used[i - 1]:

continue

used[i] = True

dfs(path + [a])

used[i] = False

nums.sort()

dfs([])

return ans

```
class Solution:
    def numSquarefulPerms(self, nums: List[int]) -> int:
        n = len(nums)
        f = [[0] * n for _ in range(1 << n)]
        for j in range(n):
            f[1 < j][j] = 1
        for i in range(1 << n):
            for j in range(n):
                if i >> j & 1:
                    for k in range(n):
                        if (i >> k & 1) and k != j:
                            s = nums[j] + nums[k]
                            t = int(sqrt(s))
                            if t * t == s:
                                f[i][j] += f[i ^ (1 <<
j)][k]

ans = sum(f[(1 < n) - 1][j] for j in range(n))
for v in Counter(nums).values():
    ans //= factorial(v)
return ans
```

acktracking p.496

Backtracking p.497

Backtracking p.498

Backtracking p.498

Backtracking p.499

Backtracking p.500

Backtracking p.501

Backtracking n.502

Backtracking Sheet 56 / 317 · LeetMastery By Pattern · Python Only · 9-u

Backtracking p.504

Space Complexity: $O(n)$ – Recursion depth is limited to n, plus additional space for the current permutation. Solution The solution employs a backtracking algorithm. A helper function is used to build permutations recursively. At each step, the algorithm iterates through the available numbers, adds one to the current permutation if it has not been used yet, and recurses to handle the remaining numbers. Once the current permutation matches the length of the input array, it is added to the result list. Then, by "backtracking" (removing the last number added), the algorithm explores alternative possibilities. This systematic exploration ensures that all possible permutations are generated.	- Search pruning can be applied by stopping a DFS branch as soon as the current path does not match the prefix of the target word. Space & Time Time Complexity: $O(m * n * 3^L)$, where L is the length of the word (each DFS may branch to at most 3 further cells, excluding the coming cell). Space Complexity: $O(L)$ due to recursion call stack (where L is the word length). Solution The solution uses DFS with backtracking to traverse the grid. For each cell, we: - Check if the cell matches the first character of the word. - Recursively explore the adjacent cells (up, down, left, right) while marking the current cell as visited. - If the path does not lead to a complete match, backtrack by resetting the visited cell. - The DFS stops if all characters in the word have been found in sequence. The algorithm leverages in-place modifications (or an auxiliary visited array) to avoid revisiting cells while exploring a candidate path.	- Backtracking is essential: add the current node value to the path before the recursive calls and remove it (backtrack) after exploring both subtrees. - At each leaf, check if the accumulated sum equals targetSum. - Maintain a running sum or subtract the current node's value from the target sum as you traverse. Space & Time Time Complexity: $O(N^2)$ in the worst-case scenario (where N is the number of nodes) when most nodes are leaf nodes, since each potential path copy operation can be $O(N)$. Space Complexity: $O(N)$ for the recursion stack and the path storage, where N is the height of the tree. Solution We solve the problem using a recursive DFS approach combined with backtracking. The idea is to traverse the tree while maintaining the current path and the remaining sum needed to reach targetSum. When a leaf node is reached (node with no children), we check if the remaining sum equals the leaf's value – if yes, we record a copy of the current path. Data structures and algorithmic approaches used: - Recursion for DFS: to traverse to each leaf. - List (or Array) for storing the current path. - Backtracking: to remove the last added element when stepping back up the recursion stack. - Copying the path when a valid leaf
#79 Word Search [Medium] Key Insights - Use Depth-First Search (DFS) to explore each cell in the board as a potential starting point. - Employ backtracking to mark cells as visited when exploring a path and revert them back when backtracking. - Check boundary conditions and ensure the current cell's character matches the corresponding character in the word. - Given the small grid and word sizes, a recursive solution is both feasible and straightforward.	#113 Path Sum II [Medium] Key Insights Use Depth-First Search (DFS) to explore all paths from the root to the leaves.	Time Complexity: $O(n!)$ in the worst-case scenario, as the solution explores permutations for queen placements. Space Complexity: $O(n^2)$ for storing board configurations and $O(n)$ for recursion stack and conflict-tracking sets. Solution We use a backtracking algorithm that places queens row by row. For each row, we try placing a queen in each column. Before placing, we check if the column, main diagonal (row – col), and anti-diagonal (row + col) are free using dedicated sets. If a queen can be safely placed, we mark the column and diagonals as occupied and recurse to the next row. Once all rows are processed, the board configuration is a valid solution and is added to the result. After that, we backtrack by unmarking the sets and removing the queen from the board. This systematic approach ensures all possible configurations are considered.
is reached to store it in the final result. A common pitfall is to forget to backtrack, which would result in incorrect paths being recorded. #37 Sudoku Solver [Hard] Key Insights - Use backtracking to explore digit placements for each empty cell. - Validate placements by ensuring that the chosen digit is not already present in the corresponding row, column, or 3x3 sub-box. - Employ recursion to fill the board incrementally and backtrack upon encountering invalid configurations. - Although the worst-case time complexity is exponential, the fixed board size (9x9) keeps the problem computationally manageable. Space & Time Time Complexity: In the worst-case scenario, the algorithm can explore up to $O(9^{(n)})$ possibilities where n is the number of empty cells. However, since n is at most 81 and many branches are pruned early, the practical runtime is far less. Space Complexity: $O(n)$ due to the recursion stack, where n is the number of empty cells; given the board's fixed size, this space is constant in practice. Solution	The solution uses a backtracking approach. For each empty cell on the board, we attempt to place a digit from '1' to '9'. Before placing a digit, we check if the placement is valid by ensuring that: - The digit is not already present in the same row. - The digit is not already present in the same column. - The digit is not already present in the corresponding 3x3 sub-box. If a valid digit is found, we place it on the board and recursively attempt to solve the rest of the board. If no valid digit can be placed, the algorithm backtracks by reverting the last move and trying a different digit. This recursive search continues until the board is completely filled with a valid Sudoku configuration. #51 N-Queens [Hard] Key Insights - Use backtracking to explore potential positions row by row. - Use sets to track which columns and diagonals are under attack. - Diagonals can be tracked by (row – col) and (row + col), ensuring that queens do not conflict along any diagonal. - Recursively build each valid board configuration and backtrack when a conflict arises. Space & Time	#126 Word Ladder II [Hard] Key Insights - Use Breadth-First Search (BFS) to traverse level-by-level and capture only the shortest paths. - Build a parent mapping during BFS to record which words lead to each newly discovered word. Space Complexity: $O(n)$ for recursion and $O(n)$ for the visited flag, plus potentially $O(2^n * n)$ for memoization if implemented. Solution The solution uses a backtracking algorithm to explore all permutations. The array is first sorted to handle duplicates. A helper function checks if a number is a perfect square and is used to validate that the sum of every adjacent pair is a perfect square. During DFS, a visited array is maintained to track elements included so far. Skipping duplicate elements (unless the earlier duplicate was used) is essential to avoid overcounting. This approach ensures that every valid permutation is considered exactly once. BFS
- Once the shortest path is found via BFS, use backtracking (or DFS) from endWord to reconstruct all transformation sequences. - Remove words as they are visited within a level to avoid unnecessary revisits and cycles. - The solution must handle multiple shortest paths, so maintaining a list of predecessors for each word is crucial. Space & Time Time Complexity: $O(N * L * 26)$ where N is the number of words in the wordList and L is the length of each word. This accounts for checking all possible one-letter transformations. Space Complexity: $O(N * L)$ for storing the parent mapping and maintaining the BFS queue and other auxiliary data structures. Solution We solve the problem in two major phases: - Use BFS to explore from beginWord to endWord. During BFS, record all parents for each word encountered. This ensures we capture all possible predecessors that yield a shortest transformation. - Once the BFS completes (once endWord is reached), use backtracking (DFS) beginning from the endWord and moving backwards via the parent mapping to reconstruct all transformation sequences in reverse. Finally, reverse each sequence to present it from	beginWord to endWord. This two-phase approach guarantees that only the shortest paths are reconstructed, as BFS ensures level-by-level exploration. #996 Number of Squareful Arrays [Hard] Key Insights - The problem requires ensuring adjacent pairs sum up to a perfect square. - Sorting the array helps in efficiently handling duplicates; when iterating in the backtracking, skip duplicate elements to avoid overcounting. - For each valid candidate, check that the sum with the previous number (if any) is a perfect square. - Use backtracking (DFS) to generate all valid permutations while maintaining a visited flag for each index. - Since the array may contain duplicates, extra care is taken to only use each occurrence in a controlled order (i.e., if two identical numbers exist, only the first unvisited instance can be chosen at a given recursive call). Space & Time Time Complexity: $O(N^2 * 2^n)$ in the worst-case scenario due to exploring all bitmask states with n elements and checking pairwise sums.	#9 BFS By Pattern · Python Only · 10 questions

BFS #286 Walls and Gates LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · BFS · Matrix Lists: Premium 98 Time Complexity: $O(m * n)$ where m and n are the dimensions of the grid, as each cell is processed at most once. Space Complexity: $O(m * n)$ in the worst-case scenario used by the BFS queue. Problem You are given an $m \times n$ grid rooms initialized with these three possible values. • -1 A wall or an obstacle. • 0 A gate. • INF Infinity means an empty room. We use the value $231 - 1 = 2147483647$ to represent INF as you may assume that the distance to a gate is less than 2147483647. Fill each empty room with the distance to its nearest gate. If it is impossible to reach a gate, it should be filled with INF. Example 1: BFS p.514	Input: rooms = [[2147483647,-1,0,2147483647],[2147483647,2147483647,2147483647,-1],[2147483647,-1,2147483647,-1],[0,-1,2147483647,2147483647]] Output: [[3,-1,0,1],[2,2,1,-1],[1,-1,2,-1],[0,-1,3,4]] Example 2: Input: rooms = [[-1]] Output: [[-1]] Constraints: • $m == \text{rooms.length}$ • $n == \text{rooms}[i].\text{length}$ • $1 \leq m, n \leq 250$ • $\text{rooms}[i][j]$ is -1, 0, or $231 - 1$. Brute Force (Python) BFS p.515	<pre># Brute Force Approach class Solution: def wallsAndGates(self, rooms): # Brute Force $O((m \cdot n)^2)$ - BFS from every empty room independently from collections import deque m, n = len(rooms), len(rooms[0]) INF = 2147483647 def bfs_from(sr, sc): queue = deque([(sr, sc, 0)]) visited = set() while queue: r, c, d = queue.popleft() if (r, c) in visited: continue visited.add((r, c)) if rooms[r][c] == 0: return d # reached a gate for dr, dc in [(1, 0), (-1, 0), (0, 1), (0, -1)]: nr, nc = r + dr, c + dc if 0 <= nr < m and 0 <= nc < n and rooms[nr][nc] != -1: queue.append((nr, nc, d + 1)) return INF for r in range(m): for c in range(n): if rooms[r][c] == INF: rooms[r][c] = bfs_from(r, c)</pre> Community Solutions (Python) • WalkCC BFS p.516
<pre>class Solution: def wallsAndGates(self, rooms: List[List[int]]) -> None: DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0)) INF = 2**31 - 1 m = len(rooms) n = len(rooms[0]) q = collections.deque((i, j) for i in range(m) for j in range(n) if rooms[i][j] == 0) while q: i, j = q.popleft() for dx, dy in DIRS: x = i + dx y = j + dy if x < 0 or x == m or y < 0 or y == n: continue if rooms[x][y] != INF: continue rooms[x][y] = rooms[i][j] + 1 q.append((x, y))</pre> • LeetDoocs BFS p.517	<pre>class Solution: def wallsAndGates(self, rooms: List[List[int]]) -> None: """ Do not return anything, modify rooms in-place instead. """ m, n = len(rooms), len(rooms[0]) inf = 2**31 - 1 q = deque([(i, j) for i in range(m) for j in range(n) if rooms[i][j] == 0]) d = 0 while q: d += 1 for _ in range(len(q)): i, j = q.popleft() for a, b in [(0, 1), (0, -1), (1, 0), (-1, 0)]: x, y = i + a, j + b if 0 <= x < m and 0 <= y < n and rooms[x][y] == inf: rooms[x][y] = d q.append((x, y))</pre> • SimplyLeet BFS p.518	<pre>from collections import deque def wallsAndGates(rooms): if not rooms or not rooms[0]: return rows, cols = len(rooms), len(rooms[0]) # Directions: up, down, left, right. directions = [(-1, 0), (1, 0), (0, -1), (0, 1)] queue = deque() # Enqueue all gates (cells with 0). for r in range(rows): for c in range(cols): if rooms[r][c] == 0: queue.append((r, c)) # Perform multi-source BFS. while queue: r, c = queue.popleft() for dr, dc in directions: nr, nc = r + dr, c + dc # Process neighbor if it is within bounds and is an empty room. if 0 <= nr < rows and 0 <= nc < cols and rooms[nr][nc] == 2147483647: rooms[nr][nc] = rooms[r][c] + 1 # Update distance. queue.append((nr, nc))</pre> • LC.ca BFS p.519
<pre>class Solution: def wallsAndGates(self, rooms: List[List[int]]) -> None: """ Do not return anything, modify rooms in-place instead. """ m, n = len(rooms), len(rooms[0]) inf = 2**31 - 1 q = deque([(i, j) for i in range(m) for j in range(n) if rooms[i][j] == 0]) d = 0 while q: d += 1 for _ in range(len(q)): i, j = q.popleft() for a, b in [(0, 1), (0, -1), (1, 0), (-1, 0)]: x, y = i + a, j + b # bfs so first reaching inf is always the shortest # set its value from inf to d, same as marking as visited if 0 <= x < m and 0 <= y < n and rooms[x][y] == inf: rooms[x][y] = d q.append((x, y))</pre> BFS p.520	<pre>##### class Solution: # dfs def wallsAndGates(self, rooms: List[List[int]]) -> None: if not rooms or not rooms[0]: return m, n = len(rooms), len(rooms[0]) isVisited = [[False] * n for _ in range(m)] # isVisited map shared among all gates def fill(i: int, j: int, distance: int) -> None: nonlocal m, n # rooms[i][j] <= 0 meaning == -1 or == 0, both stop if i < 0 or i >= m or j < 0 or j >= n or rooms[i][j] <= 0 or isVisited[i][j]: return rooms[i][j] = min(rooms[i][j], distance) isVisited[i][j] = True</pre> BFS p.521	<pre>fill(i - 1, j, distance + 1) fill(i, j + 1, distance + 1) fill(i + 1, j, distance + 1) fill(i, j - 1, distance + 1) isVisited[i][j] = False for i in range(m): for j in range(n): if rooms[i][j] == 0: # start from every gate fill(i, j, 0) ##### from collections import deque INF = 2147483647 class Solution(object): def wallsAndGates(self, rooms): """ :type rooms: List[List[int]] :rtype: void Do not return anything, modify rooms in-place instead. """ queue = deque([]) directions = [(1, 0), (-1, 0), (0, 1), (0, -1)] for i in range(0, len(rooms)): for j in range(0, len(rooms[0])):</pre> BFS p.522

```

        if rooms[i][j] == 0:
            queue.append((i, j))

    while queue:
        i, j = queue.popleft()
        for di, dj in directions:
            p, q = i + di, j + dj
            if 0 <= p < len(rooms) and 0 <= q < len(rooms[0])
            and rooms[p][q] == INF:
                rooms[p][q] = rooms[i][j] + 1
                queue.append((p, q))

```

BFS p.523

#322 Coin Change

LeetCode · SimplyLeet · WalkCC · LeetCode
 Array · Dynamic Programming · BFS
 Lists: Grind 169
 Time Complexity: $O(\text{amount} * n)$ where n is the number of coins. Space Complexity: $O(\text{amount})$ for the dp array.
Problem
 You are given an integer array coins representing coins of different denominations and an integer amount representing a total amount of money.
 Return the fewest number of coins that you need to make up that amount. If that amount of money cannot be made up by any combination of the coins, return -1.
 You may assume that you have an infinite number of each kind of coin.

Example 1:
 Input: coins = [1,2,5], amount = 11
 Output: 3
 Explanation: 11 = 5 + 5 + 1

Example 2:
 Input: coins = [2], amount = 3
 Output: -1

BFS p.524

Example 3:
 Input: coins = [1], amount = 0
 Output: 0

Constraints:

- 1 <= coins.length <= 12
- 1 <= coins[i] <= 231 - 1
- 0 <= amount <= 104

Brute Force (Python)

```

# Brute Force Approach
class Solution:
    def coinChange(self, coins, amount):
        # Brute Force O(amount * 5^(amount/min)) - recursion without memo
        def dp(rem):
            if rem == 0: return 0
            if rem < 0: return float('inf')
            return 1 + min(dp(rem - c) for c in coins)
        res = dp(amount)
        return res if res != float('inf') else -1

```

Community Solutions (Python)

- WalkCC

BFS p.525

```

class Solution:
    def coinChange(self, coins: List[int], amount: int) -> int:
        # dp[i] := the minimum number of coins to make up i
        dp = [0] + [amount + 1] * amount

        for coin in coins:
            for i in range(coin, amount + 1):
                dp[i] = min(dp[i], dp[i - coin] + 1)

        return -1 if dp[amount] == amount + 1 else dp[amount]

```

• LeetCode

```

class Solution:
    def coinChange(self, coins: List[int], amount: int) -> int:
        m, n = len(coins), amount
        f = [[inf] * (n + 1) for _ in range(m + 1)]
        f[0][0] = 0
        for i, x in enumerate(coins, 1):
            for j in range(n + 1):
                f[i][j] = f[i - 1][j]
                if j >= x:
                    f[i][j] = min(f[i][j], f[i][j - x] + 1)

        return -1 if f[m][n] >= inf else f[m][n]

```

• SimplyLeet

BFS p.526

```

def coinChange(coins, amount):
    # Create a dp array with amount+1 entries set to a value greater than any possible minimum number of coins.
    dp = [amount + 1] * (amount + 1)
    # Base case: 0 coins are needed to form amount 0.
    dp[0] = 0

    # Build up the dp table.
    for current_amount in range(1, amount + 1):
        for coin in coins:
            if coin <= current_amount:
                dp[current_amount] = min(dp[current_amount], dp[current_amount - coin] + 1)

    # If dp[amount] remains greater than amount, no solution was found.
    return dp[amount] if dp[amount] != amount + 1 else -1

```

• LC.ca

```

...
>>> float("inf")
inf
>>> float("inf") + 1
inf
...

class Solution(object):
    def coinChange(self, coins, amount):
        ....

```

BFS p.527

```

:type coins: List[int]
:type amount: int
:rtype: int
"""

dp = [float("inf")] * (amount + 1)
dp[0] = 0
for i in range(1, amount + 1):
    for coin in coins:
        if i - coin >= 0:
            dp[i] = min(dp[i], dp[i - coin] + 1)
    return dp[-1] if dp[-1] != float("inf") else -1

#####

class Solution:
    def coinChange(self, coins: List[int], amount: int) -> int:
        dp = [amount + 1] * (amount + 1)
        dp[0] = 0
        for coin in coins:
            for j in range(coin, amount + 1):
                dp[j] = min(dp[j], dp[j - coin] + 1)
        return -1 if dp[-1] > amount else dp[-1]

```

BFS p.528

#542 01 Matrix

LeetCode · SimplyLeet · WalkCC · LeetCode
 Array · BFS · Matrix
 Lists: Grind 169 | Denny Zhang
 Time Complexity: $O(m * n)$ since each cell is processed once. Space Complexity: $O(m * n)$ for the queue and the answer matrix.
Problem
 Given an $m * n$ binary matrix mat, return the distance of the nearest 0 for each cell.
 The distance between two cells sharing a common edge is 1.
Example 1:

BFS p.529

0	0	0
0	1	0
0	0	0

```

Input: mat = [[0,0,0],[0,1,0],[0,0,0]]
Output: [[0,0,0],[0,1,0],[0,0,0]]

```

Example 2:

BFS p.530 Sheet 59 / 317 · LeetCode Mastery By Pattern · Python Only · 3-Up Portrait

0	0	0
0	1	0
1	1	1

```

Input: mat = [[0,0,0],[0,1,0],[1,1,1]]
Output: [[0,0,0],[0,1,0],[1,2,1]]

```

Constraints:

- m == mat.length
- n == mat[i].length
- 1 <= m, n <= 104
- 1 <= m * n <= 104
- mat[i][j] is either 0 or 1.
- There is at least one 0 in mat.

BFS p.531

Note: This question is the same as 1765:
https://leetcode.com/problems/map-of-highest-peak/
Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def updateMatrix(self, mat):
        # Brute Force - BFS layer by layer
        from collections import deque
        queue = deque([0])
        visited = {0}
        steps = 0
        while queue:
            for _ in range(len(queue)):

                node = queue.popleft()
                for nb in graph[node]:
                    if nb not in visited:
                        visited.add(nb); queue.append(nb)

            steps += 1
        return steps
```

Community Solutions (Python)

- WalkCC

```
class Solution:
    def updateMatrix(self, mat: List[List[int]]) -> List[List[int]]:
        DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0))
        m = len(mat)
        n = len(mat[0])
        q = collections.deque()
        seen = [[False] * n for _ in range(m)]

        for i in range(m):
            for j in range(n):

                if mat[i][j] == 0:
                    q.append((i, j))
                    seen[i][j] = True

        while q:
            i, j = q.popleft()
            for dx, dy in DIRS:
                x = i + dx
                y = j + dy
                if x < 0 or x == m or y < 0 or y == n:

                    continue
                if seen[x][y]:
                    continue
                mat[x][y] = mat[i][j] + 1
                q.append((x, y))
                seen[x][y] = True

        return mat
```

- LeetDoocs

```
class Solution:
    def updateMatrix(self, mat: List[List[int]]) -> List[List[int]]:
        m, n = len(mat), len(mat[0])
        ans = [[-1] * n for _ in range(m)]
        q = deque()
        for i, row in enumerate(mat):
            for j, x in enumerate(row):
                if x == 0:
                    ans[i][j] = 0
                    q.append((i, j))

        dirs = (-1, 0, 1, 0, -1)
        while q:
            i, j = q.popleft()
            for a, b in pairwise(dirs):
                x, y = i + a, j + b
                if 0 <= x < m and 0 <= y < n and ans[x][y]

                    ans[x][y] = ans[i][j] + 1
                    q.append((x, y))

        return ans
```

- SimplyLeet

```
from collections import deque

def updateMatrix(mat):
    # Get the dimensions of the matrix
    rows, cols = len(mat), len(mat[0])
    # Initialize a queue for multi-source BFS
    queue = deque()

    # Initialize the matrix: use -1 to indicate unvisited cells
    for r in range(rows):

        for c in range(cols):
            if mat[r][c] == 0:
                # Enqueue all 0 cells as starting points
                queue.append((r, c))
            else:
                # Mark 1 cells as unvisited (or infinite distance)
                mat[r][c] = -1

    # Directions for neighbors (up, down, left, right)
    directions = [(1, 0), (-1, 0), (0, 1), (0, -1)]
```

```
# BFS traversal to update distances
while queue:
    r, c = queue.popleft()
    for dr, dc in directions:
        nr, nc = r + dr, c + dc
        # Check boundaries and if neighbor has not been visited
        if 0 <= nr < rows and 0 <= nc < cols and mat[nr][nc] == -1:
            # Update neighbor's distance
            mat[nr][nc] = mat[r][c] + 1

            # Enqueue the neighbor for further BFS
            queue.append((nr, nc))

    return mat

# Example usage:
# print(updateMatrix([[0,0,0],[0,1,0],[1,1,1]]))
```

- LC.ca

```
class Solution:
    def updateMatrix(self, mat: List[List[int]]) -> List[List[int]]:
        m, n = len(mat), len(mat[0])
        ans = [[-1] * n for _ in range(m)]
        q = deque()
        for i, row in enumerate(mat):
            for j, x in enumerate(row):
                if x == 0:
                    ans[i][j] = 0
                    q.append((i, j))

        dirs = (-1, 0, 1, 0, -1)
        while q:
            i, j = q.popleft()
            for a, b in pairwise(dirs):
                x, y = i + a, j + b
                if 0 <= x < m and 0 <= y < n and ans[x][y]

                    ans[x][y] = ans[i][j] + 1
                    q.append((x, y))

        return ans
```

BFS #994 Rotting Oranges MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode
Array · BFS · Matrix
Lists: Grind 169
Time Complexity: $O(m * n)$ as every cell is processed at most once. Space Complexity: $O(m * n)$ in the worst case for the queue and auxiliary storage.
Problem
You are given an $m \times n$ grid where each cell can have one of three values:
• 0 representing an empty cell,
• 1 representing a fresh orange, or
• 2 representing a rotten orange.
Every minute, any fresh orange that is 4-directionally adjacent to a rotten orange becomes rotten.
Return the minimum number of minutes that must elapse until no cell has a fresh orange. If this is impossible, return -1.
Example 1:

Minute 0

Minute 1

Minute 2

Minute 3

Minute 4

Input: grid = [[2,1,1],[1,1,0],[0,1,1]]
Output: 4

Example 2:
Input: grid = [[2,1,1],[0,1,1],[1,0,1]]
Output: -1
Explanation: The orange in the bottom left corner (row 2, column 0) is never rotten, because rotting only happens 4-directionally.

Example 3:
Input: grid = [[0,2]]
Output: 0
Explanation: Since there are already no fresh oranges at minute 0, the answer is just 0.

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 10$
- $\text{grid}[i][j]$ is 0, 1, or 2.

Brute Force (Python)

BFS

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BFS p.539

```
# Brute Force Approach
class Solution:
    def orangesRotting(self, grid):
        # Brute Force  $O((m \cdot n)^2)$  - repeatedly scan entire grid, spread rot each pass
        from copy import deepcopy
        m, n = len(grid), len(grid[0])
        minutes = 0
        while True:
            new_grid = deepcopy(grid)
            changed = False

            for r in range(m):
                for c in range(n):
                    if grid[r][c] == 2:
                        for dr, dc in [(1,0), (-1,0), (0,1), (0,-1)]:
                            nr, nc = r+dr, c+dc
                            if 0 <= nr < m and 0 <= nc < n and

                                new_grid[nr][nc]=2;

            grid[nr][nc]=1:

            changed=True
            grid = new_grid
            if not changed: break
            minutes += 1

        return minutes if not any(1 in row for row in grid) else -1
```

Community Solutions (Python)

- LeetDoocs

BFS p.540

```
class Solution:
    def orangesRotting(self, grid: List[List[int]]) ->
int:
    m, n = len(grid), len(grid[0])
    cnt = 0
    q = deque()
    for i, row in enumerate(grid):
        for j, x in enumerate(row):
            if x == 2:
                q.append((i, j))
            elif x == 1:

                cnt += 1

    ans = 0
    dirs = (-1, 0, 1, 0, -1)
    while q and cnt:
        ans += 1
        for _ in range(len(q)):
            i, j = q.popleft()
            for a, b in pairwise(dirs):
                x, y = i + a, j + b
                if 0 <= x < m and 0 <= y < n and
grid[x][y] == 1:

                    grid[x][y] = 2
                    q.append((x, y))
                    cnt -= 1
                    if cnt == 0:
                        return ans
    return -1 if cnt else 0
```

• SimplyLeet

BFSp.541

```
from collections import deque

def orangesRotting(grid):
    # Dimensions of the grid
    rows, cols = len(grid), len(grid[0])
    # Queue to perform BFS, containing tuples of (row,
col)
    rotten_queue = deque()
    fresh_count = 0

    # Step 1: Initialize the queue with all initially
rotten oranges and count fresh oranges.

    for r in range(rows):
        for c in range(cols):
            if grid[r][c] == 2:
                rotten_queue.append((r, c))
            elif grid[r][c] == 1:
                fresh_count += 1

    # If there are no fresh oranges, return 0
    immediately.
    if fresh_count == 0:
        return 0
```

BFSp.542

```
minutes_passed = 0
# Directions for adjacent cells: up, down, left,
right.
directions = [(-1,0), (1,0), (0,-1), (0,1)]

# Step 2: BFS to rot adjacent fresh oranges.
while rotten_queue:
    # Process oranges that are rotten in the current
minute.
    for _ in range(len(rotten_queue)):
        x, y = rotten_queue.popleft()

        for dx, dy in directions:
            nx, ny = x + dx, y + dy
            # Check if the neighbor is inside the
grid and is a fresh orange.
            if 0 <= nx < rows and 0 <= ny < cols and
grid[nx][ny] == 1:
                # Mark the fresh orange as rotten.
                grid[nx][ny] = 2
                fresh_count -= 1
                # Add the new rotten orange into the
queue.
                rotten_queue.append((nx, ny))
            # Increment minutes after processing one level.
```

BFSp.543

```
minutes_passed += 1

# If there are still fresh oranges left, return -1.
return minutes_passed - 1 if fresh_count == 0 else -1

# Example usage:
# grid = [[2,1,1],[1,1,0],[0,1,1]]
# print(orangesRotting(grid))
```

• LC.ca

```
class Solution:
    def orangesRotting(self, grid: List[List[int]]) ->
int:
    m, n = len(grid), len(grid[0])
    q = deque()
    cnt = 0
    for i in range(m):
        for j in range(n):
            if grid[i][j] == 2:
                q.append((i, j))
            elif grid[i][j] == 1:
```

BFSp.544

```
                cnt += 1

                ans = 0
                while q and cnt:
                    ans += 1
                    for _ in range(len(q)):
                        i, j = q.popleft()
                        for a, b in [(0, 1), (0, -1), (1, 0),
[-1, 0]]:
                            x, y = i + a, j + b
                            if 0 <= x < m and 0 <= y < n and
grid[x][y] == 1:
                                cnt -= 1

                                grid[x][y] = 2
                                q.append((x, y))
                                return ans if cnt == 0 else -1
```

BFSp.545

BFS

#1197 Minimum Knight Moves

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

BFS

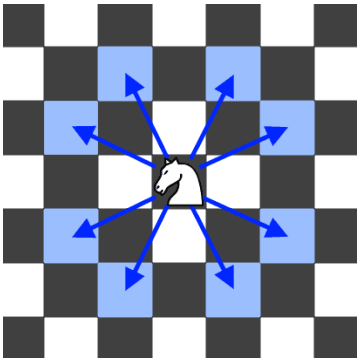
Lists: Grind 169 | Premium 98

Time Complexity: $O(n)$ where n is the number of positions processed during the BFS Space Complexity: $O(n)$ due to the storage requirements for the BFS queue and the visited set

Problem

In an infinite chess board with coordinates from $-\infty$ to $+\infty$, you have a knight at square $[0, 0]$. A knight has 8 possible moves it can make, as illustrated below. Each move is two squares in a cardinal direction, then one square in an orthogonal direction.

BFSp.546



Return the minimum number of steps needed to move the knight to the square $[x, y]$. It is guaranteed the answer exists.

Example 1:

```
Input: x = 2, y = 1
Output: 1
Explanation: [0, 0] -> [2, 1]
```

Example 2:

BFSp.547

```
Input: x = 5, y = 5
Output: 4
Explanation: [0, 0] -> [2, 1] -> [4, 2] -> [3, 4] -> [5, 5]
```

Constraints:

- $-300 \leq x, y \leq 300$
- $0 \leq |x| + |y| \leq 300$

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def minKnightMoves(self, x, y):
        # Brute Force - BFS layer by layer
        from collections import deque
        queue = deque([0])
        visited = {0}
        steps = 0
        while queue:
            for _ in range(len(queue)):
                node = queue.popleft()
                for nb in graph[node]:
                    if nb not in visited:
                        visited.add(nb); queue.append(nb)
                        steps += 1
            return steps
```

Community Solutions (Python)

• LeetDoocs

BFSp.548Sheet 61 / 317 · LeetMastery By Pattern · Python Only · 9-up Portrait

```
class Solution:
    def minKnightMoves(self, x: int, y: int) -> int:
        q = deque([(0, 0)])
        ans = 0
        vis = {(0, 0)}
        dirs = ((-2, 1), (-1, 2), (1, 2), (2, 1), (2,
-1), (1, -2), (-1, -2), (-2, -1))
        while q:
            for _ in range(len(q)):
                i, j = q.popleft()
                if (i, j) == (x, y):
```

```
                return ans
                for a, b in dirs:
                    c, d = i + a, j + b
                    if (c, d) not in vis:
                        vis.add((c, d))
                        q.append((c, d))

            ans += 1
            return -1
```

```
class Solution:
    def minKnightMoves(self, x: int, y: int) -> int:
        def extend(m1, m2, q):
            for _ in range(len(q)):
                i, j = q.popleft()
                step = m1[(i, j)]
                for a, b in (
                    (-2, 1),
                    (-1, 2),
                    (1, 2),
```

BFSp.549

```
(2, 1),
(2, -1),
(1, -2),
(-1, -2),
(-2, -1),
):
    x, y = i + a, j + b
    if (x, y) in m1:
        continue
    if (x, y) in m2:
        return step + 1 + m2[(x, y)]
    q.append((x, y))
    m1[(x, y)] = step + 1
    return -1

if (x, y) == (0, 0):
    return 0
q1, q2 = deque([(0, 0)]), deque([(x, y)])
m1, m2 = {(0, 0): 0}, {(x, y): 0}
while q1 and q2:
    t = extend(m1, m2, q1) if len(q1) <= len(q2)
    else extend(m2, m1, q2)
    if t != -1:
        return t
    return -1
```

• SimplyLeet

```
# Python solution using BFS:
from collections import deque

def minKnightMoves(x, y):
    # Normalize target coordinates using symmetry
    x, y = abs(x), abs(y)

    # Possible knight moves (dx, dy)
    moves = [(2, 1), (1, 2), (-1, 2), (-2, 1), (-2, -1),
              (-1, -2), (1, -2), (2, -1)]

    # BFS queue stores tuples of the form (current_x,
    current_y, move_count)
    queue = deque([(0, 0, 0)])

    # Set to track visited positions to prevent cycles
    visited = {(0, 0)}

    # Process BFS
    while queue:
        curr_x, curr_y, move_count = queue.popleft()
```

```
# Return the current move count if target is
reached
if curr_x == x and curr_y == y:
    return move_count

# Explore all knight moves
for dx, dy in moves:
    next_x, next_y = curr_x + dx, curr_y + dy

# Prune search space: only process positions
with valid bounds
if (next_x, next_y) not in visited and next_x
>= -1 and next_y >= -1:

    visited.add((next_x, next_y))
    queue.append((next_x, next_y, move_count
+ 1))

    return -1 # Unreachable because a solution always
exists

# Example test
#print(minKnightMoves(5, 5))
```

• LC.ca

```
class Solution:
    def minKnightMoves(self, x: int, y: int) -> int:
        q = deque([(0, 0)])
        ans = 0
        vis = {(0, 0)}
        dirs = ((-2, 1), (-1, 2), (1, 2), (2, 1), (2,
-1), (1, -2), (-1, -2), (-2, -1))
        while q:
            for _ in range(len(q)):
                i, j = q.popleft()
                if (i, j) == (x, y):
                    return ans
                for a, b in dirs:
                    c, d = i + a, j + b
                    if (c, d) not in vis:
                        vis.add((c, d))
                        q.append((c, d))
            ans += 1
        return -1
```

BFS

#1730 Shortest Path to Get Food MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · BFS · Matrix

Lists: Grind 169

Time Complexity: $O(m * n)$, where m is the number of rows and n is the number of columns since each cell is processed at most once. Space Complexity: $O(m * n)$ in the worst case due to the queue and vis

Problem

You are starving and you want to eat food as quickly as possible. You want to find the shortest path to arrive at any food cell.

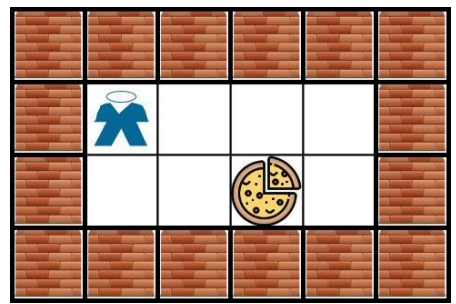
You are given an $m \times n$ character matrix, grid, of these different types of cells:

- ' $'s'$ ' is your location. There is exactly one ' $'s'$ ' cell.
- ' $'\#'$ ' is a food cell. There may be multiple food cells.
- ' $'O'$ ' is free space, and you can travel through these cells.
- ' $'X'$ ' is an obstacle, and you cannot travel through these cells.

You can travel to any adjacent cell north, east, south, or west of your current location if there is not an obstacle.

Return the length of the shortest path for you to reach any food cell. If there is no path for you to reach food, return -1.

Example 1:



Input: grid = [[["X","X","X","X","X"],["X","*", "0","0","X"],["X","0","0","#", "0","X"],["X","X","X","X","X","X"],["X","X","X","X","X"]]

Output: 3

Explanation: It takes 3 steps to reach the food.

Example 2:

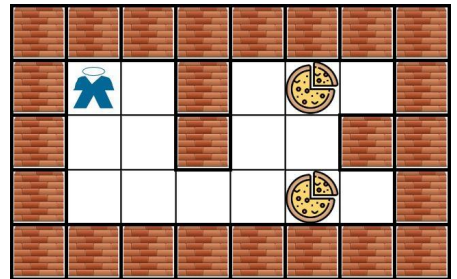


Input: grid = [[["X","X","X","X","X"],["X","*", "0","0","X"],["X","0","0","#", "0","X"],["X","X","X","X","X","X"],["X","X","X","X","X"]]

Output: -1

Explanation: It is not possible to reach the food.

Example 3:



Input: grid = [[["X","X","X","X","X"],["X","*", "0","0","X"],["X","0","0","#", "0","X"],["X","X","X","X","X","X"],["X","X","X","X","X"]]

Output: 6

Explanation: There can be multiple food cells. It only takes 6 steps to reach the bottom food.

Example 4:

Input: grid = [[["X","X","X","X","X"],["X","*", "0","0","X"],["X","0","0","#", "0","X"],["X","X","X","X","X","X"],["X","X","X","X","X"]]

Output: 5

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$
- $1 \leq m, n \leq 200$
- $\text{grid}[\text{row}][\text{col}]$ is ' $'s'$ ', ' $'X'$ ', ' $'O'$ ', or ' $'\#'$ '.
- The grid contains exactly one ' $'s'$ '.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def getFood(self, grid):
        # Brute Force - BFS layer by layer
        from collections import deque
        queue = deque([0])
        visited = {0}
        steps = 0
        while queue:
            for _ in range(len(queue)):
                node = queue.popleft()
                for nb in graph[node]:
                    if nb not in visited:
                        visited.add(nb); queue.append(nb)
            steps += 1
        return steps
```

Community Solutions (Python)

- WalkCC

<pre>class Solution: def getFood(self, grid: list[list[str]]) -> int: DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0)) m = len(grid) n = len(grid[0]) ans = 0 q = collections.deque([self._getStartLocation(grid)]) while q: for _ in range(len(q)): i, j = q.popleft() for dx, dy in DIRS: x = i + dx y = j + dy if x < 0 or x == m or y < 0 or y == n: continue if grid[x][y] == 'X': continue if grid[x][y] == '#': return ans + 1 q.append((x, y)) grid[x][y] = 'X' # Mark as visited. ans += 1 return -1 def _getStartLocation(self, grid: list[list[str]]) -> tuple[int, int]: for i, row in enumerate(grid): for j, cell in enumerate(row): if cell == '*':</pre>		<pre> return (i, j) • LeetDoocs class Solution: def getFood(self, grid: List[List[str]]) -> int: m, n = len(grid), len(grid[0]) i, j = next((i, j) for i in range(m) for j in range(n) if grid[i][j] == '*') q = deque([(i, j)]) dirs = (-1, 0, 1, 0, -1) ans = 0 while q: ans += 1 for _ in range(len(q)): i, j = q.popleft() for a, b in pairwise(dirs): x, y = i + a, j + b if 0 <= x < m and 0 <= y < n: if grid[x][y] == '#': return ans if grid[x][y] == 'O': grid[x][y] = 'X' q.append((x, y)) return -1 • SimplyLeet</pre>		<pre>from collections import deque def getFood(grid): # number of rows and columns rows = len(grid) cols = len(grid[0]) # finding the start position start = None for r in range(rows): for c in range(cols): if grid[r][c] == '*': start = (r, c) break if start is not None: break # initialize queue for BFS, with (row, col, steps) queue = deque([(start[0], start[1], 0)]) # mark visited cells to avoid reprocessing visited = [[False] * cols for _ in range(rows)] visited[start[0]][start[1]] = True # possible moves: up, right, down, left directions = [(-1, 0), (0, 1), (1, 0), (0, -1)] # begin BFS while queue: row, col, steps = queue.popleft() # if food is found, return current number of steps if grid[row][col] == '#':</pre>	
BFS	p.559	BFS	p.560	BFS	p.561

<pre> return steps # check all adjacent cells for dr, dc in directions: newRow, newCol = row + dr, col + dc # check boundaries, obstacles, and visited status if 0 <= newRow < rows and 0 <= newCol < cols and not visited[newRow][newCol] and grid[newRow][newCol] != 'X': visited[newRow][newCol] = True queue.append((newRow, newCol, steps + 1)) # if no food is reachable return -1 return -1 # Example usage: grid = [["X","X","X","X","X","X","X"], ["X","*", "0","0","0","0","X"], ["X","0","0","0","#", "0","X"], ["X","X","X","X","X","X","X"]] print(getFood(grid)) # Expected output: 3 • LC.ca</pre>		<pre>class Solution: def getFood(self, grid: List[List[str]]) -> int: m, n = len(grid), len(grid[0]) i, j = next((i, j) for i in range(m) for j in range(n) if grid[i][j] == '*') q = deque([(i, j)]) dirs = (-1, 0, 1, 0, -1) ans = 0 while q: ans += 1 for _ in range(len(q)): i, j = q.popleft() for a, b in pairwise(dirs): x, y = i + a, j + b if 0 <= x < m and 0 <= y < n: if grid[x][y] == '#': return ans if grid[x][y] == 'O': grid[x][y] = 'X' q.append((x, y)) return -1</pre>		BFS #127 Word Ladder HAR LeetDoocs · SimplyLeet · WalkCC · LeetCode Hash Table · String · BFS Lists: Grind 169 Time Complexity: $O(N * M^2)$ where N is the number of words and M is the length of each word. Space Complexity: $O(N * M)$ for the storage of intermediate mappings and the BFS queue. Problem A transformation sequence from word beginWord to word endWord using a dictionary wordList is a sequence of words beginWord -> s1 -> s2 -> ... -> sk such that: - Every adjacent pair of words differs by a single letter. - Every si for $1 \leq i \leq k$ is in wordList. Note that beginWord does not need to be in wordList. - sk == endWord Given two words, beginWord and endWord, and a dictionary wordList, return the number of words in the shortest transformation sequence from beginWord to endWord, or 0 if no such sequence exists. Example 1:	
BFS	p.562	BFS	p.563	BFS	p.564

Input: beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log","cog"] Output: 5 Explanation: One shortest transformation sequence is "hit" -> "hot" -> "dot" -> "dog" -> "cog", which is 5 words long. Example 2: Input: beginWord = "hit", endWord = "cog", wordList = ["hot","dot","dog","lot","log"] Output: 0 Explanation: The endWord "cog" is not in wordList, therefore there is no valid transformation sequence. Constraints: $1 \leq \text{beginWord.length} \leq 10$ $\text{endWord.length} == \text{beginWord.length}$ $1 \leq \text{wordList.length} \leq 5000$ $\text{wordList}[i].\text{length} == \text{beginWord.length}$ - beginWord, endWord, and wordList[i] consist of lowercase English letters. $\text{beginWord} \neq \text{endWord}$ - All the words in wordList are unique. Brute Force (Python)		# Brute Force Approach class Solution: def ladderLength(self, beginWord, endWord, wordList): # Brute Force $O(n^2 \cdot L)$ - BFS; try all word pairs each level (no pattern precomp) from collections import deque word_set = set(wordList) if endWord not in word_set: return 0 queue = deque([(beginWord, 1)]) visited = {beginWord} while queue: word, steps = queue.popleft() for candidate in list(word_set): # Check one-letter difference by comparing character by character if sum(a!=b for a,b in zip(word,candidate))==1 and len(word)==len(candidate): if candidate == endWord: return steps+1 if candidate not in visited: visited.add(candidate) queue.append((candidate, steps+1)) return 0 Community Solutions (Python) WalkCC		<pre>class Solution: def ladderLength(self, beginWord: str, endWord: str, wordList: list[str],) -> int: wordSet = set(wordList) if endWord not in wordSet: return 0 q = collections.deque([beginWord]) step = 1 while q: for _ in range(len(q)): wordList = list(q.popleft()) for i, cache in enumerate(wordList): for c in string.ascii_lowercase: wordList[i] = c word = ''.join(wordList) if word == endWord: return step + 1 if word in wordSet: q.append(word) wordSet.remove(word) wordList[i] = cache step += 1 return 0 • LeetDoocs</pre>	
BFS	p.565	BFS	p.566	BFS	p.567

• **SimplyLeet**

```
# Initialize BFS
```

- LC.ca

BFS p.568

BFS p.569

BFS p.570

p.571

p.572

BFS p.573

p.574

BFS Sheet 64 / 317 · LeetMastery By Pattern · Python Only · p.575 BFS 9-up Portrait

BFS p.576

BFS

#317 Shortest Distance from All Buildings

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · BFS · Matrix

Lists: Premium 98

Time Complexity: $O(k * m * n)$ where k is the number of buildings and m,n are the grid dimensions. Space Complexity: $O(m * n)$ due to the auxiliary matrices and visited state tracking used in BFS.

Problem

You are given an m x n grid grid of values 0, 1, or 2, where:

● each 0 marks an empty land that you can pass by freely,

● each 1 marks a building that you cannot pass through, and

● each 2 marks an obstacle that you cannot pass through.

You want to build a house on an empty land that reaches all buildings in the shortest total travel distance. You can only move up, down, left, and right.

Return the shortest travel distance for such a house. If it is not possible to build such a house according to the above rules, return -1.

BFS

p.577

The total travel distance is the sum of the distances between the houses of the friends and the meeting point.

Example 1:

1	0	2	0	1
0	0	0	0	0
0	0	1	0	0

Input: grid = [[1,0,2,0,1],[0,0,0,0,0],[0,0,1,0,0]]

Output: 7

Explanation: Given three buildings at (0,0), (0,4), (2,2), and an obstacle at (0,2). The point (1,2) is an ideal empty land to build a house, as the total travel distance of 3+3+1=7 is minimal.

So return 7.

Example 2:

BFS

p.578

Input: grid = [[1,0]]

Output: 1

Example 3:

Input: grid = [[1]]

Output: -1

Constraints:

● m == grid.length

● n == grid[i].length

● 1 <= m, n <= 50

● grid[i][j] is either 0, 1, or 2.

● There will be at least one building in the grid.

Brute Force (Python)

Brute Force Approach

class Solution:

def shortestDistance(self, grid):

Brute Force - BFS layer by layer

from collections import deque

queue = deque([0])

visited = {}

steps = 0

while queue:

for _ in range(len(queue)):

BFS

p.579

node = queue.popleft()

for nb in graph[node]:

if nb not in visited:

visited.add(nb); queue.append(nb)

steps += 1

return steps

Community Solutions (Python)

• WalkCC

class Solution:

def shortestDistance(self, grid: List[List[int]]) -> int:

DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0))

m = len(grid)

n = len(grid[0])

nBuildings = sum(a == 1 for row in grid for a in row)

ans = math.inf

dist[i][j] := the total distance of grid[i][j] (0)

to reach all the

buildings (1)

dist = [[0] * n for _ in range(m)]

BFS

p.580

reachCount[i][j] := the number of buildings (1)

grid[i][j] (0) can reach

reachCount = [[0] * n for _ in range(m)]

def bfs(row: int, col: int) -> bool:

q = collections.deque([(row, col)])

seen = {(row, col)}

seenBuildings = 1

step = 1

while q:

for _ in range(len(q)):

i, j = q.popleft()

for dx, dy in DIRS:

x = i + dx

y = j + dy

if x < 0 or x == m or y < 0 or y == n:

continue

if (x, y) in seen:

continue

seen.add((x, y))

if not grid[x][y]:

dist[x][y] += step

reachCount[x][y] += 1

q.append((x, y))

elif grid[x][y] == 1:

seenBuildings += 1

step += 1

True if all the buildings (1) are connected

return seenBuildings == nBuildings

BFS

p.581

for i in range(m):

for j in range(n):

if grid[i][j] == 1: # BFS from this building.

if not bfs(i, j):

return -1

for i in range(m):

for j in range(n):

if reachCount[i][j] == nBuildings:

ans = min(ans, dist[i][j])

return -1 if ans == math.inf else ans

• SimplyLeet

from collections import deque

def shortestDistance(grid):

if not grid or not grid[0]:

return -1

m, n = len(grid), len(grid[0])

total_buildings = 0

total_distance = [[0] * n for _ in range(m)]

reachable_count = [[0] * n for _ in range(m)]

BFS

p.582

def bfs(start_i, start_j):

visited = [[False] * n for _ in range(m)]

q = deque([(start_i, start_j, 0)])

visited[start_i][start_j] = True

while q:

i, j, dist = q.popleft()

for x, y in [(i-1, j), (i+1, j), (i, j-1), (i, j+1)]:

if 0 <= x < m and 0 <= y < n and not visited[x][y] and grid[x][y] == 0:

visited[x][y] = True

total_distance[x][y] += dist + 1

reachable_count[x][y] += 1

q.append((x, y, dist+1))

for i in range(m):

for j in range(n):

if grid[i][j] == 1:

total_buildings += 1

bfs(i, j)

BFS

p.583

min_distance = float('inf')

for i in range(m):

for j in range(n):

if grid[i][j] == 0 and reachable_count[i][j] == total_buildings:

min_distance = min(min_distance, total_distance[i][j])

return min_distance if min_distance != float('inf')

else -1

Example test:

print(shortestDistance([[1,0,2,0,1],[0,0,0,0,0],[0,0,1,0,0]]))

• LC.ca

class Solution:

def shortestDistance(self, grid: List[List[int]]) -> int:

m, n = len(grid), len(grid[0])

q = deque()

total = 0 # total building count

cnt: how many buildings can reach (x,y)

if there is a column all blockers, then (x,y) cannot reach every building

cnt = [[0] * n for _ in range(m)]

dist = [[0] * n for _ in range(m)]

for i in range(m):

BFS

p.584

for j in range(n):

if grid[i][j] == 1:

total += 1

q.append((i, j))

d = 0

vis = set() # reset for every

while q:

d += 1

for _ in range(len(q)):

r, c = q.popleft()

for a, b in [(0, 1), (0, -1), [1, 0], [-1, 0]]:

x, y = r + a, c + b

if (0 <= x < m and 0 <= y < n and grid[x][y] == 0 and (x, y) not in vis):

cnt[x][y] += 1

dist[x][y] += d

q.append((x, y))

vis.add((x, y))

ans = inf

for i in range(m):

for j in range(n):

if grid[i][j] == 0 and cnt[i][j] == total:

ans = min(ans, dist[i][j])

return -1 if ans == inf else ans

BFS

p.585

Sheet 65 / 317 · LeetMastery By Pattern · Python Only · 9-up Portrait

BFS

#773 Sliding Puzzle

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode
Array · BFS · Matrix
Lists: Denny Zhang
Time Complexity: O(1) in practice, since there are at most 720 states (6! permutations). Space Complexity: O(1) for the same reason, as the state space is fixed and limited.
Problem
On an 2 x 3 board, there are five tiles labeled from 1 to 5, and an empty square represented by 0. A move consists of choosing 0 and a 4-directionally adjacent number and swapping it.
The state of the board is solved if and only if the board is `[[1,2,3],[4,5,0]]`.
Given the puzzle board board, return the least number of moves required so that the state of the board is solved. If it is impossible for the state of the board to be solved, return -1.
Example 1:

BFS

p.586

1	2	3
4		5

Input: board = `[[1,2,3],[4,0,5]]`
Output: 1
Explanation: Swap the 0 and the 5 in one move.

Example 2:

BFS

p.587

1	2	3
5	4	

Input: board = `[[1,2,3],[5,4,0]]`
Output: -1
Explanation: No number of moves will make the board solved.

Example 3:

BFS

p.588

4	1	2
5		3

Input: board = `[[4,1,2],[5,0,3]]`
Output: 5
Explanation: 5 is the smallest number of moves that solves the board.
An example path:
After move 0: `[[4,1,2],[5,0,3]]`
After move 1: `[[4,1,2],[0,5,3]]`
After move 2: `[[0,1,2],[4,5,3]]`
After move 3: `[[1,0,2],[4,5,3]]`

Constraints:

- board.length == 2
- board[i].length == 3
- 0 <= board[i][j] <= 5
- Each value board[i][j] is unique.

BFS

p.589

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def slidingPuzzle(self, board):
        # Brute Force - BFS layer by layer
        from collections import deque
        queue = deque([0])
        visited = {0}
        steps = 0
        while queue:
            for _ in range(len(queue)):
                node = queue.popleft()
                for nb in graph[node]:
                    if nb not in visited:
                        visited.add(nb); queue.append(nb)
                        steps += 1
            return steps
```

Community Solutions (Python)

• SimplyLeet

BFS

p.590

```
import collections

def slidingPuzzle(board):
    # Flatten board into a string representation.
    start = ''.join(str(num) for row in board for num in row)
    goal = "123450"
    # Mapping of indices to their neighbors in a 2x3 board.
    neighbors = {0: [1, 3], 1: [0,2,4], 2: [1,5],
                 3: [0,4], 4: [1,3,5], 5: [2,4]}

    # Initialize BFS with the starting state and move count 0.
    q = collections.deque([(start, 0)])
    visited = set([start])

    # Process states in the queue.
    while q:
        state, moves = q.popleft()
        if state == goal:
            return moves
        zero_index = state.index('0')
```

BFS

p.591

```
# Try all valid moves by swapping 0 with its neighbours.
for neighbor in neighbors[zero_index]:
    new_state = list(state)
    new_state[zero_index], new_state[neighbor] = new_state[neighbor], new_state[zero_index]
    new_state_str = ''.join(new_state)
    if new_state_str not in visited:
        visited.add(new_state_str)
        q.append((new_state_str, moves + 1))

return -1

# Example usage:
print(slidingPuzzle([[1,2,3],[4,0,5]]))
```

• LC.ca

```
class Solution:
    def slidingPuzzle(self, board: List[List[int]]) -> int:
        m, n = 2, 3
        seq = []
        start, end = '', '123450'
        for i in range(m):
            for j in range(n):
                if board[i][j] != 0:
                    seq.append(board[i][j])
        start += str(board[i][j])
```

BFS

p.592

```
def check(seq):
    n = len(seq)
    cnt = sum(seq[i] > seq[j] for i in range(n)
              for j in range(i, n))
    return cnt % 2 == 0

def f(s):
    ans = 0
    for i in range(m * n):
        if s[i] != '0':
            num = ord(s[i]) - ord('1')
            ans += abs(i // n - num // n) + abs(i % n - num % n)
    return ans

if not check(seq):
    return -1
q = [(f(start), start)]
dist = {start: 0}
while q:
    _, state = heappop(q)
```

BFS

p.593

```
if state == end:
    return dist[state]
p1 = state.index('0')
i, j = p1 // n, p1 % n
s = list(state)
for a, b in [[0, -1], [0, 1], [1, 0], [-1, 0]]:
    x, y = i + a, j + b
    if 0 <= x < m and 0 <= y < n:
        p2 = x * n + y
        s[p1], s[p2] = s[p2], s[p1]

    next = ''.join(s)
    s[p1], s[p2] = s[p2], s[p1]
    if next not in dist or dist[next] > dist[state] + 1:
        dist[next] = dist[state] + 1
        heappush(q, (dist[next] + f(next), next))
    return -1
```

BFS

p.594

BFS #815 Bus Routes HAR LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Hash Table · BFS Lists: Grind 169 Time Complexity: $O(N * L)$ in the worst-case scenario, where N is the number of bus routes and L is the average number of stops per route. Space Complexity: $O(N * L)$ due to storage of the stop-to-buses Problem You are given an array routes representing bus routes where routes[i] is a bus route that the ith bus repeats forever. For example, if routes[0] = [1, 5, 7], this means that the 0th bus travels in the sequence 1 -> 5 -> 7 -> 1 -> 5 -> 7 -> 1 -> ... forever. You will start at the bus stop source (You are not on any bus initially), and you want to go to the bus stop target. You can travel between bus stops by buses only. Return the least number of buses you must take to travel from source to target. Return -1 if it is not possible. Example 1: BFS p.595	Input: routes = [[1,2,7],[3,6,7]], source = 1, target = 6 Output: 2 Explanation: The best strategy is take the first bus to the bus stop 7, then take the second bus to the bus stop 6. Example 2: Input: routes = [[7,12],[4,5,15],[6],[15,19],[9,12,13]], source = 15, target = 12 Output: -1 Constraints: 1 <= routes.length <= 500. 1 <= routes[i].length <= 105 - All the values of routes[i] are unique. - sum(routes[i].length) <= 105 0 <= routes[i][j] < 106 0 <= source, target < 106 Brute Force (Python) BFS p.596	# Brute Force Approach class Solution: def numBusesToDestination(self, routes, source, target): # Brute Force - BFS layer by layer from collections import deque queue = deque([0]) visited = {0} steps = 0 while queue: for _ in range(len(queue)): node = queue.popleft() for nb in graph[node]: if nb not in visited: visited.add(nb); queue.append(nb) steps += 1 return steps Community Solutions (Python) • WalkCC class Solution: def numBusesToDestination(self, routes: List[List[int]], source: int, target: int,) -> int: if source == target: <divreturn 0<="" div=""></divreturn> BFS p.597
graph = collections.defaultdict(list) usedBuses = set() for i in range(len(routes)): for route in routes[i]: graph[route].append(i) q = collections.deque([source]) step = 1 while q: for _ in range(len(q)): for bus in graph[q.popleft()]: if bus in usedBuses: continue usedBuses.add(bus) for nextRoute in routes[bus]: if nextRoute == target: return step q.append(nextRoute) step += 1 return -1 • LeetDoocs BFS p.598	class Solution: def numBusesToDestination(self, routes: List[List[int]], source: int, target: int) -> int: if source == target: return 0 g = defaultdict(list) for i, route in enumerate(routes): for stop in route: g[stop].append(i) if source not in g or target not in g: return -1 q = [(source, 0)] vis_bus = set() vis_stop = {source} for stop, bus_count in q: if stop == target: return bus_count for bus in g[stop]: if bus not in vis_bus: vis_bus.add(bus) for next_stop in routes[bus]: if next_stop not in vis_stop: vis_stop.add(next_stop) q.append((next_stop, bus_count + 1)) return -1 • SimplyLeet BFS p.599	from collections import defaultdict, deque def numBusesToDestination(routes, source, target): # If source equals target, no buses are needed. if source == target: return 0 # Build mapping: bus stop -> list of bus indices stop_to_buses = defaultdict(list) for bus_index, stops in enumerate(routes): for stop in stops: stop_to_buses[stop].append(bus_index) visited_buses = set() visited_stops = set([source]) queue = deque() # Initialize queue with all buses that include the source stop. for bus in stop_to_buses[source]: visited_buses.add(bus) queue.append((bus, 1)) # Perform BFS over bus routes. while queue: current_bus, buses_taken = queue.popleft() for stop in routes[current_bus]: if stop == target: return buses_taken if stop not in visited_stops: visited_stops.add(stop) BFS p.600
for neighbor_bus in stop_to_buses[stop]: if neighbor_bus not in visited_buses: visited_buses.add(neighbor_bus) queue.append((neighbor_bus, buses_taken + 1)) return -1 # Example usage: routes = [[1,2,7],[3,6,7]] # print(numBusesToDestination(routes, 1, 6)) • LC.ca class Solution: def numBusesToDestination(self, routes: List[List[int]], source: int, target: int) -> int: if source == target: return 0 s = [set(r) for r in routes] d = defaultdict(list) for i, r in enumerate(routes): BFS p.601	for v in r: d[v].append(i) g = defaultdict(list) for ids in d.values(): m = len(ids) for i in range(m): for j in range(i + 1, m): a, b = ids[i], ids[j] g[a].append(b) g[b].append(a) q = deque(d[source]) ans = 1 vis = set(d[source]) while q: for _ in range(len(q)): i = q.popleft() if target in s[i]: return ans for j in g[i]: if j not in vis: vis.add(j) q.append(j) ans += 1 return -1 BFS p.602	Sheet 67 / 317 · LeetMastery By Pattern · Python Only · 9-Up Portrait BFS p.603

>> Quick Review — BFS 10 questions · insights · complexity · solution #286 Walls and Gates Medium Key Insights - Use a multi-source Breadth-First Search (BFS) starting from all the gates, as this allows spreading the shortest distance to each empty room. - Instead of starting from every empty room separately, starting from all gates simultaneously helps update all distances in one pass. - Only update a room if the new calculated distance is smaller than its current value to prevent unnecessary processing. - The problem is essentially a multi-source shortest path problem on a grid. Space & Time Time Complexity: $O(m * n)$ where m and n are the dimensions of the grid, as each cell is processed at most once. Space Complexity: $O(m * n)$ in the worst-case scenario used by the BFS queue. Solution BFS p.604	The solution uses a multi-source BFS algorithm by initially enqueueing all the gate positions (cells with value 0). Then, for each cell obtained from the queue, check its four possible neighbors (up, down, left, right). If a neighbor is an empty room (INF), update its distance to be the current cell's distance plus one, and then enqueue the neighbor to process further. This ensures that each empty room gets the minimum distance from any gate. #322 Coin Change Medium Key Insights - Use a dynamic programming approach to build up a solution from smaller subproblems. - Define $dp[x]$ as the minimum number of coins required for amount x. - The recurrence relation is: $dp[x] = \min(dp[x], dp[x - coin] + 1)$ for each coin. - Initialize $dp[0] = 0$ and set other values to a number larger than the maximum possible coins (amount+1 works as an infinity substitute). - An alternative approach is through BFS, treating each amount as a node and each coin as an edge, but the DP method is more intuitive here. Space & Time Time Complexity: $O(\text{amount} * n)$ where n is the number of coins. BFS p.605	Space Complexity: $O(\text{amount})$ for the dp array. Solution We use a bottom-up dynamic programming approach. The idea is to use a one-dimensional dp array where each element at index x represents the minimum number of coins required to form the amount x. Starting from $dp[0]$ which is 0, we iterate through all amounts up to the target amount and update $dp[x]$ based on each coin denomination. If after filling the dp array the value $dp[\text{amount}]$ remains unchanged (i.e., greater than amount), it means the amount cannot be formed and we return -1. #542 01 Matrix Medium Key Insights - Treat all 0s as starting points for a multi-source BFS. - Update each neighboring cell's distance if a shorter distance is found. - Utilize a queue to process cells in order of increasing distance. - Ensure that each cell is processed only once by marking visited cells. Space & Time Time Complexity: $O(mn)$ since each cell is processed once. Space Complexity: $O(m)$ BFS p.606
n) for the queue and the answer matrix. Solution We use a multi-source Breadth-First Search (BFS) where every 0 in the matrix is initially added to the BFS queue. Each neighboring cell that is a 1 will have its distance updated to be one plus the distance of the current cell if it hasn't been visited yet (or if a smaller distance is found). This ensures that the first time a 1 is reached, it is reached by the shortest path. Data structures used include a queue (or deque) for BFS and the matrix itself for storing distances. #994 Rotting Oranges Medium Key Insights - Use Breadth-First Search (BFS) starting from all initially rotten oranges. - Process the grid level by level, where each level represents one minute. - Keep a count of fresh oranges and decrement it as they become rotten. - If after the BFS there are still fresh oranges, then it is impossible to rot every orange. Space & Time Time Complexity: $O(m * n)$ as every cell is processed at most once. BFS p.607	Space Complexity: $O(m * n)$ in the worst case for the queue and auxiliary storage. Solution We begin by scanning the grid to locate all rotten oranges and count the fresh ones. These rotten oranges are added to a queue for a BFS. For each minute (BFS level), we process all the current rotten oranges and try to infect their adjacent fresh oranges. If an adjacent cell contains a fresh orange, we mark it rotten, add it to the queue, and decrement our fresh count. We continue this process level by level (minute by minute) until there are no fresh oranges left. If, after processing, there are still fresh oranges remaining, we return -1, indicating that not all oranges can become rotten. #1197 Minimum Knight Moves Medium Key Insights - Leverage the symmetry of the chessboard by converting target coordinates to their absolute values. - Use a Breadth-First Search (BFS) because all moves have equal weight, ensuring the first encounter with the target is the minimal path. - Prune the search space by only considering positions that are likely beneficial (e.g., positions having coordinates not far less than -1). - Track visited states to avoid redundant processing. BFS p.608	- Optimize by bounding the search region relative to the target distance. Space & Time Time Complexity: $O(n)$ where n is the number of positions processed during the BFS Space Complexity: $O(n)$ due to the storage requirements for the BFS queue and the visited set Solution We solve the problem using a BFS starting from the origin. First, normalize the target by taking the absolute value of x and y. BFS is then applied by exploring all eight possible knight moves at each step. Each move generates a new position that is enqueued if it hasn't been visited and falls within a reasonable bound (e.g., coordinates ≥ -1). The search terminates as soon as the target position is reached. #1730 Shortest Path to Get Food Medium Key Insights - Use Breadth-First Search (BFS) to explore the grid level by level. - The BFS guarantees that when a food cell ('#') is reached, the current distance is the shortest. - Maintain a visited structure to avoid re-processing the same cell. BFS p.609
- Consider boundary and obstacle ('X') conditions while traversing. Space & Time Time Complexity: $O(m * n)$, where m is the number of rows and n is the number of columns since each cell is processed at most once. Space Complexity: $O(m * n)$ in the worst case due to the queue and visited matrix. Solution We use Breadth-First Search (BFS) starting from the cell marked with ". The BFS uses a queue to explore neighbors in the four allowed directions (up, down, left, right). Each level of BFS represents one step in the path. As soon as a cell containing food ('#') is reached, the current step count is returned. We also maintain a visited matrix to ensure that cells are not revisited, which could otherwise result in cycles or redundant processing. This approach ensures that the first time we encounter food, we have found the shortest possible path. #127 Word Ladder Hard Key Insights - Use Breadth-First Search (BFS) to explore transformations level by level to guarantee the shortest path. BFS p.610	- Preprocess words by generating intermediate representations (e.g., replacing each letter with a wildcard) to quickly find neighbors. - Use a visited set to avoid cycles. - Each transformation counts as a level in the BFS, so the answer is the number of levels traversed. Space & Time Time Complexity: $O(N * M^2)$ where N is the number of words and M is the length of each word. Space Complexity: $O(N * M)$ for the storage of intermediate mappings and the BFS queue. Solution We solve the problem using a BFS approach. First, we preprocess the wordList by creating a mapping of all possible intermediate states. For example, for the word "dog" and wildcard position, we generate "og", "d g", and "do* ". This helps us find all adjacent words (neighbors) that are only one letter different in constant time. We then initialize a queue with the beginWord and initiate the BFS. For each word dequeued, we explore all valid transformations using the precomputed intermediate mapping. If the endWord is reached during this process, we return the current level (transformation sequence length). Otherwise, we continue the BFS until all possibilities are exhausted. If no valid transformation sequence is found, we return 0. BFS p.611	#317 Shortest Distance from All Buildings Hard Key Insights - We need to compute the Manhattan distance from each building (cell value 1) to all empty lands (cell value 0) by performing a BFS. - Maintain two matrices: one for cumulative total distances and another for the count of buildings that reached a particular empty cell. - Ensure that only those empty cells reachable by every building are considered. - The ideal cell will have a reach count equal to the total number of buildings. - If multiple buildings cannot all reach any specific empty cell, then return -1. Space & Time Time Complexity: $O(k * m * n)$ where k is the number of buildings and m,n are the grid dimensions. Space Complexity: $O(m * n)$ due to the auxiliary matrices and visited state tracking used in BFS. Solution We first iterate through the grid to locate all buildings while counting them. For each building, we perform a BFS to compute the shortest distance to each reachable empty cell. During the BFS, we update the cumulative distance for that cell and record that it was reached by one additional building. After processing all the BFS p.612

buildings, we scan the grid to find the empty cell that is reachable from all buildings and has the minimum cumulative distance. If no such cell exists, we return -1.

#773 Sliding Puzzle [Hard]

Key Insights

- Represent the board as a string (e.g. "123450") to simplify state comparison and manipulation.
- Use Breadth First Search (BFS) to explore all possible moves since the puzzle has a small fixed state space (6! = 720 possible states).
- Precompute the valid neighbor indices for each board position to efficiently generate new states.
- Track visited states to avoid loops and redundant work.

Space & Time

Time Complexity: O(1) in practice, since there are at most 720 states (6! permutations).
Space Complexity: O(1) for the same reason, as the state space is fixed and limited.

Solution

We solve the problem using a Breadth First Search (BFS) approach. The board configuration is flattened into a string to represent states. A mapping from each index to its possible neighbors (i.e., positions to which 0 can move) is precomputed. BFS is then used to explore every

BFSp.613

valid move from the initial state while marking states as visited to avoid cycles. When the goal state ("123450") is reached, we return the number of moves taken. If BFS completes without reaching the goal, the puzzle is unsolvable and we return -1.

#815 Bus Routes [Hard]

Key Insights

- Model the problem as a graph where nodes are bus routes.
- Two bus routes are connected if they share at least one common bus stop.
- Use a hash map to quickly look up bus routes available at any bus stop.
- Employ Breadth-First Search (BFS) on the bus routes graph to find the minimum number of bus transfers.
- Mark visited bus routes and bus stops to avoid redundant work.

Space & Time

Time Complexity: O(N * L) in the worst-case scenario, where N is the number of bus routes and L is the average number of stops per route.
Space Complexity: O(N * L) due to storage of the stop-to-buses mapping and BFS auxiliary data structures.

Solution

BFSp.614

We begin by constructing a mapping from each bus stop to the list of buses passing through it. This allows quick retrieval of buses available at a given stop. Starting at the source stop, initialize the BFS queue with all buses that can be boarded at that stop. Then, for each bus route in the queue, inspect each stop along that route. If a stop is the target, return the number of buses taken so far. Otherwise, for each unvisited stop, add all adjacent bus routes (buses that pass through this stop) to the BFS queue with an incremented bus count. Continue until the queue is empty, which means the target cannot be reached.

Trie

#10 Trie

By Pattern · Python Only · 12 questions

BFSp.615

Trie

#14 Longest Common Prefix

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

String · Trie

Lists: Grind 169

Time Complexity: O(N*M) where N is the number of strings and M is the length of the shortest string. Space Complexity: O(1) additional space (ignoring output space).

Problem

Write a function to find the longest common prefix string amongst an array of strings.
If there is no common prefix, return an empty string "".

Example 1:

Input: strs = ["flower","flow","flight"]

Output: "fl"

Example 2:

Input: strs = ["dog","racecar","car"]

Output: ""

Explanation: There is no common prefix among the input strings.

Constraints:

- 1 <= strs.length <= 200
- 0 <= strs[i].length <= 200

Trip.616

- strs[i] consists of only lowercase English letters if it is non-empty.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def longestCommonPrefix(self, strs):
        # Brute Force O(n*L) - start with strs[0], trim
        until prefix of all
        if not strs: return ''
        prefix = strs[0]
        for word in strs[1:]:
            while not word.startswith(prefix):
                prefix = prefix[:-1]
            if not prefix: return ''
```

return prefix

Community Solutions (Python)

• WalkCC

```
class Solution:
    def longestCommonPrefix(self, strs: List[str]) -> str:
        if not strs:
            return ''

        for i in range(len(strs[0])):
            for j in range(1, len(strs)):
                if i == len(strs[j]) or strs[j][i] != strs[0][i]:
                    return strs[0][:i]

        return strs[0]
```

Trip.617

• LeetCode

```
class Solution:
    def longestCommonPrefix(self, strs: List[str]) -> str:
        for i in range(len(strs[0])):
            for s in strs[1:]:
                if len(s) <= i or s[i] != strs[0][i]:
                    return s[:i]
        return strs[0]
```

• SimplyLeet

```
def longestCommonPrefix(strs):
    # If no strings exist, return empty
    if not strs:
        return ""
    # Iterate over each index of the first string
    for i in range(len(strs[0])):
        current_char = strs[0][i]
        # Compare the current character with the
        corresponding one in the rest of the strings
        for s in strs[1:]:
            # If index exceeds length or characters
            differ, return the prefix found so far

            if i >= len(s) or s[i] != current_char:
                return strs[0][:i]
        # If no mismatch is found, return the complete first
        string
        return strs[0]

# Example usage:
print(longestCommonPrefix(["flower", "flow", "flight"]))
```

Trip.618

• LC.ca

```
class Solution:
    def longestCommonPrefix(self, strs: List[str]) -> str:
        for i in range(len(strs[0])):
            for s in strs[1:]:
                if len(s) <= i or s[i] != strs[0][i]:
                    return s[:i]
        return strs[0]
```

Trip.619

Trie

#139 Word Break

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · String · Dynamic Programming · Trie · Memoization

Lists: Grind 169

Time Complexity: O(n^2) in the worst-case scenario, where n is the length of s (each substring check is O(1) if using a hash set, and there could be O(n^2) checks overall). Space Complexity: O(n) for

Problem

Given a string s and a dictionary of strings wordDict, return true if s can be segmented into a space-separated sequence of one or more dictionary words.

Note that the same word in the dictionary may be reused multiple times in the segmentation.

Example 1:

Input: s = "leetcode", wordDict = ["leet","code"]
Output: true
Explanation: Return true because "leetcode" can be segmented as "leet code".

Example 2:

Trip.620

Trie

Input: s = "applepenapple", wordDict = ["apple","pen"]
Output: true
Explanation: Return true because "applepenapple" can be segmented as "apple pen apple".
Note that you are allowed to reuse a dictionary word.

Example 3:

Input: s = "catsandog", wordDict = ["cats","dog","sand","and","cat"]
Output: false

Constraints:

- 1 <= s.length <= 300
- 1 <= wordDict.length <= 1000
- 1 <= wordDict[i].length <= 20
- s and wordDict[i] consist of only lowercase English letters.
- All the strings of wordDict are unique.

Brute Force (Python)

Trip.621

```
# Brute Force Approach
class Solution:
    def wordBreak(self, s, wordDict):
        # Brute Force O(2^n) - try all partition points
        recursively
        word_set = set(wordDict)
        def can_break(start):
            if start == len(s): return True
            for end in range(start + 1, len(s) + 1):
                if s[start:end] in word_set and
can_break(end):
                return True

        return False
        return can_break(0)
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def wordBreak(self, s: str, wordDict: list[str]) ->
bool:
    wordSet = set(wordDict)

    @functools.lru_cache(None)
    def wordBreak(s: str) -> bool:
        """Returns True if s can be segmented."""
        if s in wordSet:
            return True
        return any(s[:i] in wordSet and wordBreak(s[i:]))
        for i in range(len(s)))
```

Trie

p.622

```
return wordBreak(s)
```

• LeetDoocs

```
class Solution:
    def wordBreak(self, s: str, wordDict: List[str]) ->
bool:
    words = set(wordDict)
    n = len(s)
    f = [True] + [False] * n
    for i in range(1, n + 1):
        f[i] = any(f[j] and s[j:i] in words for j in
range(i))
    return f[n]

class Trie:
    def __init__(self):
        self.children: List[Trie | None] = [None] * 26
        self.isEnd = False

    def insert(self, w: str):
        node = self
        for c in w:
            idx = ord(c) - ord('a')
            if not node.children[idx]:
```

Trie

p.623

```
node.children[idx] = Trie()
node = node.children[idx]
node.isEnd = True

class Solution:
    def wordBreak(self, s: str, wordDict: List[str]) ->
bool:
    trie = Trie()
    for w in wordDict:
        trie.insert(w)

    n = len(s)
    f = [False] * (n + 1)
    f[n] = True
    for i in range(n - 1, -1, -1):
        node = trie
        for j in range(i, n):
            idx = ord(s[j]) - ord('a')
            if not node.children[idx]:
                break
            node = node.children[idx]

            if node.isEnd and f[j + 1]:
                f[i] = True
                break

    return f[0]
```

• SimplyLeet

Trie

p.624

```
# Python solution for the Word Break problem

def wordBreak(s, wordDict):
    # Convert the list to a set for O(1) look-ups
    wordSet = set(wordDict)

    # dp[i] is True if s[0:i] can be segmented into words
    from the wordDict
    n = len(s)
    dp = [False] * (n + 1)

    # Base case: empty string is always segmentable
    dp[0] = True

    # Iterate through the string positions
    for i in range(1, n + 1):
        # Check every possible partition s[0:j] and
s[j:i]
        for j in range(i):
            # If s[0:j] can be segmented and s[j:i] is in
the dictionary
            if dp[j] and s[j:i] in wordSet:
                dp[i] = True # s[0:i] can be segmented

        break # No need to check further

    partitions
    return dp[n]

# Example usage:
s = "leetcode"
wordDict = ["leet", "code"]
print(wordBreak(s, wordDict)) # Output: True
```

Trie

p.625

• LC.ca

```
class Solution:
    def wordBreak(self, s: str, wordDict: List[str]) ->
bool:
    words = set(wordDict)
    n = len(s)

    # dp[j] meaning s from 0 to index=j-1 is
breakable
    dp = [True] + [False] * n # so actually size is
n+1
    for i in range(1, n + 1):
        dp[i] = any( (dp[j] and s[j:i] in words) for
j in range(i) )
    return dp[n]

#####

class Solution:
    def wordBreak(self, s: str, wordDict: Set[str]) ->
bool:
    if not s or not wordDict:
        return False

    length = len(s)
```

Trie

p.626

```
# construct dp, dp[i] meaning position i-1 can
from dict or not
dp[-1] meaning at last-index plus 1, checking up
to last-index
dp = [False] * (length + 1)
dp[0] = True # initiate

for i in range(length + 1):
    if not dp[i]:
        continue
    for word in wordDict:
        if i + len(word) > length:

            continue
            if s[i:i + len(word)] == word:
                dp[i + len(word)] = True

    return dp[length]

#####

class Solution:
    def wordBreak(self, s: str, wordDict: List[str]) ->
bool:
```

Trie

p.627

```
words = set(wordDict)
n = len(s)
dp = [False] * (n + 1)
dp[0] = True # i'th char in string, is good
for j in range(1, n + 1):
    for i in range(j): # starting at 0, including
dp[0]
        if dp[i] and s[i:j] in words:
            dp[j] = True # j is exclusive,
meaning True until index i-1
            break
    return dp[-1]

#####

class Trie:
    def __init__(self):
        self.children = [None] * 26
        self.is_end = False

    def insert(self, w):
```

Trie

p.628

```
node = self
for c in w:
    idx = ord(c) - ord('a')
    if node.children[idx] is None:
        node.children[idx] = Trie()
    node = node.children[idx]
node.is_end = True

def search(self, w):
    node = self

    for c in w:
        idx = ord(c) - ord('a')
        if node.children[idx] is None:
            return False
        node = node.children[idx]
    return node.is_end

class Solution:
    def wordBreak(self, s: str, wordDict: List[str]) ->
bool:
    # https://docs.python.org/3/library/functools.htm
l#functools.cache
```

Trie

p.629

```
# creating a thin wrapper around a dictionary
lookup for the function arguments
@cache
def dfs(s):
    return not s or any(trie.search(s[:i]) and
dfs(s[i:])) for i in range(1, len(s) + 1))

trie = Trie()
for w in wordDict:
    trie.insert(w)
    return dfs(s)

#####

class Solution(object):
    def wordBreak(self, s, wordDict):
        """
        :type s: str
        :type wordDict: Set[str]
        :rtype: bool
        """
        queue = [0]

        ls = len(s)
        lenList = [l for l in set(map(len, wordDict))]
        visited = [0 for _ in range(0, ls + 1)]
        while queue:
            start = queue.pop(0)
            for l in lenList:
                if s[start:start + l] in wordDict:
                    if start + l == ls:
                        return True
                    if visited[start + l] == 0:
```

Trie

p.630

```
queue.append(start + l)
visited[start + l] = 1
return False
```

Trie p.631

Trie

#208 Implement Trie (Prefix Tree) MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode
Hash Table · String · Design · Trie
Lists: Grind 169
Time Complexity:

Problem
A trie (pronounced as "try") or prefix tree is a tree data structure used to efficiently store and retrieve keys in a dataset of strings. There are various applications of this data structure, such as autocomplete and spellchecker.
Implement the Trie class:

- `Trie()` Initializes the trie object.
- `void insert(String word)` Inserts the string word into the trie.
- `boolean search(String word)` Returns true if the string word is in the trie (i.e., was inserted before), and false otherwise.
- `boolean startsWith(String prefix)` Returns true if there is a previously inserted string word that has the prefix prefix, and false otherwise.

Example 1:

Trie p.632

Input

["Trie", "insert", "search", "search", "startsWith", "insert", "search"]
[[], ["apple"], ["apple"], ["app"], ["app"], ["app"], ["app"]]
Output
[null, null, true, false, true, null, true]

Explanation

Trie trie = new Trie();

Constraints:

- $1 \leq \text{word.length}, \text{prefix.length} \leq 2000$
- word and prefix consist only of lowercase English letters.
- At most $3 * 10^4$ calls in total will be made to insert, search, and startsWith.

Community Solutions (Python)

• WalkCC

class TrieNode:
 def __init__(self):
 self.children: dict[str, TrieNode] = {}
 self.isWord = False

class Trie:
 def __init__(self):
 self.root = TrieNode()

Trie p.633

```
def insert(self, word: str) -> None:  
    node: TrieNode = self.root  
    for c in word:  
        node = node.children.setdefault(c, TrieNode())  
    node.isWord = True  
  
def search(self, word: str) -> bool:  
    node = self._find(word)  
    return node is not None and node.isWord  
  
def startsWith(self, prefix: str) -> bool:  
    return self._find(prefix) is not None  
  
def _find(self, prefix: str) -> TrieNode | None:  
    node: TrieNode = self.root  
    for c in prefix:  
        if c not in node.children:  
            return None  
        node = node.children[c]  
    return node
```

• LeetDoocs

class Trie:
 def __init__(self):
 self.children = [None] * 26
 self.is_end = False

 def insert(self, word: str) -> None:
 node = self
 for c in word:
 idx = ord(c) - ord('a')
 if node.children[idx] is None:

Trie p.634

```
        node.children[idx] = Trie()  
        node = node.children[idx]  
        node.is_end = True  
  
    def search(self, word: str) -> bool:  
        node = self._search_prefix(word)  
        return node is not None and node.is_end  
  
    def startsWith(self, prefix: str) -> bool:  
        node = self._search_prefix(prefix)  
  
        return node is not None  
  
    def _search_prefix(self, prefix: str):  
        node = self  
        for c in prefix:  
            idx = ord(c) - ord('a')  
            if node.children[idx] is None:  
                return None  
            node = node.children[idx]  
        return node  
  
# Your Trie object will be instantiated and called as  
# such:  
# obj = Trie()  
# obj.insert(word)  
# param_2 = obj.search(word)  
# param_3 = obj.startsWith(prefix)
```

• SimplyLeet

Trie p.635

```
class TrieNode:  
    def __init__(self):  
        # Dictionary to store child nodes and a flag for  
        # end of word.  
        self.children = {}  
        self.is_end_of_word = False  
  
class Trie:  
    def __init__(self):  
        # Initialize the Trie with the root node.  
        self.root = TrieNode()  
  
    def insert(self, word: str) -> None:  
        # Start from the root.  
        node = self.root  
        for char in word:  
            # If character doesn't exist, add a new  
            # TrieNode.  
            if char not in node.children:  
                node.children[char] = TrieNode()  
            node = node.children[char]  
        # Mark the end of word.
```

Trie p.636

```
        node.is_end_of_word = True  
  
    def search(self, word: str) -> bool:  
        # Traverse the Trie for each character.  
        node = self.root  
        for char in word:  
            if char not in node.children:  
                return False  
            node = node.children[char]  
        # Returns True if word ends exactly here.  
  
        return node.is_end_of_word  
  
    def startsWith(self, prefix: str) -> bool:  
        # Traverse checking for the prefix.  
        node = self.root  
        for char in prefix:  
            if char not in node.children:  
                return False  
            node = node.children[char]  
        return True
```

• LC.ca

Trie p.637

```
import collections  
  
class TrieNode:  
    def __init__(self):  
        ...  
        Usually, a Python dictionary throws a KeyError if you  
        try to get an item with a key that is not currently in the  
        dictionary.  
        The defaultdict in contrast will simply create any  
        items that you try to access  
  
        https://stackoverflow.com/questions/5900578/how-does-  
        collections-defaultdict-work  
  
        ...  
        self.child = collections.defaultdict(TrieNode)  
        self.is_word = False  
  
class Trie:  
    def __init__(self):  
        self.root = TrieNode()  
  
    def insert(self, word: str) -> None:
```

Trie p.638

```
        cur = self.root  
        for letter in word:  
            cur = cur.child[letter]  
        cur.is_word = True  
  
    def search(self, word: str) -> bool:  
        cur = self.root  
        for letter in word:  
            # cur = cur.child[letter] ==> will not work, it will  
            # creat a default node for this letter  
            cur = cur.child.get(letter)  
        if not cur:
```

Trie p.639

<pre>return False return True # Your Trie object will be instantiated and called as such: # obj = Trie() # obj.insert(word) # param_2 = obj.search(word) # param_3 = obj.startsWith(prefix) ##### class Trie: def __init__(self): self.children = [None] * 26 self.is_end = False def insert(self, word: str) -> None: node = self for c in word: idx = ord(c) - ord('a') if node.children[idx] is None: node.children[idx] = Trie() node = node.children[idx] node.is_end = True def search(self, word: str) -> bool: node = self._search_prefix(word) return node is not None and node.is_end</pre>		<pre>def startswith(self, prefix: str) -> bool: node = self._search_prefix(prefix) return node is not None def _search_prefix(self, prefix: str): node = self for c in prefix: idx = ord(c) - ord('a') if node.children[idx] is None: return None node = node.children[idx] return node # Your Trie object will be instantiated and called as such: # obj = Trie() # obj.insert(word) # param_2 = obj.search(word) # param_3 = obj.startsWith(prefix) # below is using reduce() class Trie(object): def __init__(self): T = lambda: collections.defaultdict(T) self.root = T() def insert(self, word):</pre>		<pre> reduce(dict.__getitem__, word, self.root)['#'] = True def search(self, word): return '#' in reduce(lambda cur, c: cur.get(c, {}), word, self.root) def startswith(self, prefix): return bool(reduce(lambda cur, c: cur.get(c, {}), prefix, self.root))</pre>	
Trie	p.640	Trie	p.641	Trie	p.642

Trie #211 Design Add and Search Words MED LeetDoocs · SimplyLeet · WalkCC · LeetCode String · DFS · Design · Trie Lists: Grind 169 Time Complexity: Problem Design a data structure that supports adding new words and finding if a string matches any previously added string. Implement the WordDictionary class: • WordDictionary() Initializes the object. • void addWord(word) Adds word to the data structure, it can be matched later. • bool search(word) Returns true if there is any string in the data structure that matches word or false otherwise. word may contain dots '.' where dots can be matched with any letter. Example:		Input ["WordDictionary", "addWord", "addWord", "addWord", "search", "search", "search"] [[], ["bad"], ["dad"], ["mad"], ["pad"], ["bad"], [".ad"], [".b.."]] Output [null, null, null, null, false, true, true, true] Explanation WordDictionary wordDictionary = new WordDictionary(); Constraints: • 1 <= word.length <= 25 • word in addWord consists of lowercase English letters. • word in search consist of '.' or lowercase English letters. • There will be at most 2 dots in word for search queries. • At most 104 calls will be made to addWord and search. Community Solutions (Python) • WalkCC		class TrieNode: def __init__(self): self.children: dict[str, TrieNode] = {} self.isWord = False class WordDictionary: def __init__(self): self.root = TrieNode() def addWord(self, word: str) -> None: node: TrieNode = self.root for c in word: node = node.children.setdefault(c, TrieNode()) node.isWord = True def search(self, word: str) -> bool: return self._dfs(word, 0, self.root) def _dfs(self, word: str, s: int, node: TrieNode) -> bool: if s == len(word): return node.isWord if word[s] != '.': child: TrieNode = node.children.get(word[s], None) return self._dfs(word, s + 1, child) if child else False return any(self._dfs(word, s + 1, child) for child in node.children.values()) • LeetDoocs	
Trie	p.643	Trie	p.644	Trie	p.645

<pre>class Trie: def __init__(self): self.children = [None] * 26 self.is_end = False class WordDictionary: def __init__(self): self.trie = Trie() def addWord(self, word: str) -> None: node = self.trie for c in word: idx = ord(c) - ord('a') if node.children[idx] is None: node.children[idx] = Trie() node = node.children[idx] node.is_end = True def search(self, word: str) -> bool: def search(word, node): for i in range(len(word)): c = word[i] idx = ord(c) - ord('a') if c != '.' and node.children[idx] is None: return False if c == '.': for child in node.children: if child is not None and search(word[i + 1 :], child): return True</pre>		<pre> return False node = node.children[idx] return node.is_end return search(word, self.trie) # Your WordDictionary object will be instantiated and called as such: # obj = WordDictionary() # obj.addWord(word) # param_2 = obj.search(word) • SimplyLeet class TrieNode: def __init__(self): # Dictionary to store child nodes self.children = {} # Flag to indicate if a word ends here self.is_end = False class WordDictionary: def __init__(self): # Initialize the Trie with an empty root node</pre>		<pre> self.root = TrieNode() def addWord(self, word: str) -> None: # Start at the root of the Trie node = self.root for char in word: # Create a new node if the character is not present if char not in node.children: node.children[char] = TrieNode() node = node.children[char] # Mark the end of the word node.is_end = True def search(self, word: str) -> bool: # DFS helper function def dfs(index, node): if index == len(word): return node.is_end char = word[index] if char == '.':</pre>	
Trie	p.646	Trie	p.647	Trie	p.648

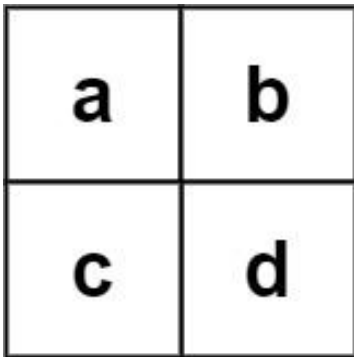
<pre> # If wildcard, search all children nodes for child in node.children.values(): if dfs(index + 1, child): return True return False else: # If character not found, word does not exist if char not in node.children: return False return dfs(index + 1, node.children[char]) return dfs(0, self.root) # Example usage: wd = WordDictionary() wd.addWord("bad") wd.addWord("dad") wd.addWord("mad") print(wd.search("pad")) # Output: False print(wd.search("bad")) # Output: True print(wd.search(".ad")) # Output: True print(wd.search("b..")) # Output: True</pre>	<pre>class Trie: def __init__(self): self.children = [None] * 26 self.is_end = False class WordDictionary: def __init__(self): self.trie = Trie() def addWord(self, word: str) -> None: node = self.trie for c in word: idx = ord(c) - ord('a') if node.children[idx] is None: node.children[idx] = Trie() node = node.children[idx] node.is_end = True def search(self, word: str) -> bool: def search(word, node): for i in range(len(word)): c = word[i] idx = ord(c) - ord('a') if c != '.' and node.children[idx] is None: return False if c == '.': for child in node.children: if child is not None and search(word[i + 1 :], child): return True</pre>	<pre> return False node = node.children[idx] return node.is_end return search(word, self.trie) # Your WordDictionary object will be instantiated and called as such: # obj = WordDictionary() # obj.addWord(word) # param_2 = obj.search(word)</pre>			
• LC.ca					
Trie	p.649	Trie	p.650	Trie	p.651

Trie #616 Add Bold Tag in String LeetCode · SimplyLeet · WalkCC · LeetCode Array · Hash Table · String · Trie Lists: Premium 98 Time Complexity: $O(n * m * k)$ where n is the length of s, m is the number of words, and k is the average length of the words (due to substring matching). Space Complexity: $O(n)$ for the auxiliary boole O(n) for the auxiliary boole Problem You are given a string s and an array of strings $words$. You should add a closed pair of bold tag $$ and $$ to wrap the substrings in s that exist in words. • If two such substrings overlap, you should wrap them together with only one pair of closed bold-tag. • If two substrings wrapped by bold tags are consecutive, you should combine them. Return s after adding the bold tags. Example 1:	Input: $s = \text{"abcxyz123"}$, $words = [\text{"abc"}, \text{"123"}]$ Output: "abcxyz123" Explanation: The two strings of words are substrings of s as following: "abcxyz123". We add $$ before each substring and $$ after each substring. Example 2: Input: $s = \text{"aaabbbb"}$, $words = [\text{"aa"}, \text{"b"}]$ Output: "aaabbbb" Explanation: "aa" appears as a substring two times: "aaabbbb" and "aaabbb". "b" appears as a substring three times: "aaabbbb", "aaabbb", and "aaabbbb". We add $$ before each substring and $$ after each substring: "aaabbbbbb". Since the first two $$'s overlap, we merge them: "aaabbbbbbb". Since now the four $$'s are consecutive, we merge them: "aaabbbb". Constraints: • $1 \leq s.length \leq 1000$ • $0 \leq words.length \leq 100$ • $1 \leq words[i].length \leq 1000$ • s and $words[i]$ consist of English letters and digits. • All the values of words are unique. Note: This question is the same as 758. Bold Words in String.	Brute Force (Python) # Brute Force Approach class Solution: def addBoldTag(self, s, words): # Brute Force $O(n * m)$ - linear scan through all words matches = [] for word in words: if word.startswith(prefix): matches.append(word) return matches Community Solutions (Python) • WalkCC class Solution: def addBoldTag(self, s: str, words: list[str]) -> str: n = len(s) ans = [] # bold[i] := True if s[i] should be bolded bold = [0] * n boldEnd = -1 # s[i:boldEnd] should be bolded for i in range(n): for word in words:			
Trie	p.652	Trie	p.653	Trie	p.654

<pre>if s[i:].startswith(word): boldEnd = max(boldEnd, i + len(word)) bold[i] = boldEnd > i # Construct the with bold tags i = 0 while i < n: if bold[i]: j = i while j < n and bold[j]: j += 1 # `s[i..j)` should be bolded. ans.append('' + s[i:j] + '') i = j else: ans.append(s[i]) i += 1 return ''.join(ans)</pre> • SimplyLeet <pre># Python Code Solution def addBoldTag(s, words): n = len(s) # Mark positions to be bolded bold = [False] * n # Iterate each word and mark its occurrences for word in words: start = s.find(word) while start != -1: for i in range(start, start + len(word)):</pre>	<pre> bold[i] = True start = s.find(word, start + 1) # Build the final result with bold tags result = [] i = 0 while i < n: if not bold[i]: result.append(s[i]) i += 1 else: result.append("") while i < n and bold[i]: result.append(s[i]) i += 1 result.append("") return "".join(result) # Example usage: # s = "abcxyz123"</pre>	<pre># words = ["abc", "123"] # print(addBoldTag(s, words)) # "abcxyz123" # Line-by-line: # 1. Initialize an array "bold" of length n to track which characters need bold formatting. # 2. For each word, use string.find to locate every occurrence and mark the corresponding indices. # 3. Traverse the string s. If the current character is marked bold and it is the start of a bold segment, # insert the opening tag before appending characters until the end of that bold segment. # 4. Append the closing tag after the bold segment and continue the process.</pre> • LC.ca <pre>class Trie: def __init__(self): self.children = [None] * 128 self.is_end = False def insert(self, word): node = self for c in word: idx = ord(c) if node.children[idx] is None:</pre>			
Trie	p.655	Trie	p.656	Trie	p.657

Trie p.658	Trie p.659	Trie p.660
Trie p.661	Trie p.662	Trie p.663
Trie p.664	Trie p.665	Trie p.666

Trie p.667	Trie p.668	Trie p.669
Trie p.670	Trie p.671	Trie p.672
Trie p.673	Trie p.674	Trie p.675



Input: board = [["a","b"],["c","d"]], words = ["abcb"]
Output: []

Constraints:

- $m == \text{board.length}$
- $n == \text{board}[i].\text{length}$
- $1 \leq m, n \leq 12$
- $\text{board}[i][j]$ is a lowercase English letter.
- $1 \leq \text{words.length} \leq 3 \times 10^4$
- $1 \leq \text{words}[i].\text{length} \leq 10$
- $\text{words}[i]$ consists of lowercase English letters.

Trie

p.685

- All the strings of words are unique.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def findWords(self, board, words):
        # Brute Force  $O(k \cdot m \cdot n \cdot 4^L)$  - run word-search
        # for each word independently
        def exist(board, word):
            def dfs(r, c, i, seen):
                if i == len(word): return True
                if r < 0 or r >= m or c < 0 or c >= n: return
                if (r,c) in seen or board[r][c] != word[i]:
                    return False
                seen.add((r,c))
                ok = any(dfs(r+dr,c+dc,i+1,seen)
                        for dr,dc in [(1,0),(-1,0),(0,1),(0,-1)])
                seen.remove((r,c))
                return ok
            return any(dfs(r,c,0,set()) for r in range(m)
                      for c in range(n))
        return [w for w in words if exist(board, w)]
```

Community Solutions (Python)

- WalkCC

Trie

p.686

```
class TrieNode:
    def __init__(self):
        self.children: dict[str, TrieNode] = {}
        self.word: str | None = None

class Solution:
    def findWords(self, board: List[List[str]], words:
    List[str]) -> List[str]:
        m = len(board)
        n = len(board[0])

        ans = []
        root = TrieNode()

        def insert(word: str) -> None:
            node = root
            for c in word:
                node = node.children.setdefault(c, TrieNode())
            node.word = word

        for word in words:
            insert(word)

        def dfs(i: int, j: int, node: TrieNode) -> None:
            if i < 0 or i == m or j < 0 or j == n:
                return
            if board[i][j] == '*':
                return

            c = board[i][j]
            if c not in node.children:
```

Trie

p.687

```
return

child = node.children[c]
if child.word:
    ans.append(child.word)
    child.word = None

board[i][j] = '*'
dfs(i + 1, j, child)
dfs(i - 1, j, child)
```

```
dfs(i, j + 1, child)
dfs(i, j - 1, child)
board[i][j] = c
```

```
for i in range(m):
    for j in range(n):
        dfs(i, j, root)
```

return ans

• SimplyLeet

```
# Define a TrieNode for building the Trie.
class TrieNode:
    def __init__(self):
        self.children = {}
        self.word = None # Stores complete word at the
    end node

class Solution:
    def findWords(self, board, words):
        # Build the Trie from the list of words.
        root = TrieNode()
```

Trie

p.688

```
for word in words:
    node = root
    for char in word:
        if char not in node.children:
            node.children[char] = TrieNode()
        node = node.children[char]
    node.word = word # Mark the end of a word

result = []
m, n = len(board), len(board[0])
```

```
# Define the DFS helper function.
def dfs(i, j, node):
    char = board[i][j]
    if char not in node.children:
        return
    next_node = node.children[char]
    if next_node.word:
        result.append(next_node.word)
        next_node.word = None # Avoid duplicate

entries
```

Trie

p.689

```
board[i][j] = '#' # Mark the current cell as
visited

# Explore all four adjacent directions.
for dx, dy in ((0, 1), (1, 0), (0, -1), (-1,
0)):
    ni, nj = i + dx, j + dy
    if 0 <= ni < m and 0 <= nj < n and
board[ni][nj] != '#':
        dfs(ni, nj, next_node)
    board[i][j] = char # Restore the original
character

# Start DFS from each cell.
```

```
for i in range(m):
    for j in range(n):
        dfs(i, j, root)
return result
```

• LC.ca

```
class Trie:
    def __init__(self):
        self.children = [None] * 26
        self.w = ''
        # minor adjust, not a boolean is_end, but the
    whole word
        # so it's easier for dfs to save the word

    def insert(self, w):
        node = self
        for c in w:
```

Trie

p.690

```
idx = ord(c) - ord('a')
if node.children[idx] is None:
    node.children[idx] = Trie()
node = node.children[idx]
node.w = w
```

```
class Solution:
    def findWords(self, board: List[List[str]], words:
    List[str]) -> List[str]:
        def dfs(node, i, j):
```

```
            idx = ord(board[i][j]) - ord('a')
            if node.children[idx] is None:
                return
            node = node.children[idx]
            if node.w:
                ans.add(node.w)
            c = board[i][j]
            board[i][j] = '0' # 0 for visited already
            for a, b in [(0, -1), (0, 1), (1, 0), (-1,
0)]:
                x, y = i + a, j + b
```

Trie

p.691

```
            if 0 <= x < m and 0 <= y < n and
board[x][y] != '0':
                dfs(node, x, y)
            board[i][y] = c
```

```
trie = Trie()
for w in words:
    trie.insert(w)
ans = set()
m, n = len(board), len(board[0])
for i in range(m):
```

```
    for j in range(n):
        dfs(trie, i, j)
    return list(ans)
```

#####

```
class Trie:
    def __init__(self):

        self.children: List[Trie | None] = [None] * 26
        self.ref: int = -1

    def insert(self, w: str, ref: int):
        node = self
        for c in w:
            idx = ord(c) - ord('a')
            if node.children[idx] is None:
                node.children[idx] = Trie()
            node = node.children[idx]
```

Trie

p.692

```
node.ref = ref

class Solution:
    def findWords(self, board: List[List[str]], words:
    List[str]) -> List[str]:
        def dfs(node: Trie, i: int, j: int):
            idx = ord(board[i][j]) - ord('a')
            if node.children[idx] is None:
                return
            node = node.children[idx]

            if node.ref >= 0:
                ans.append(words[node.ref])
                node.ref = -1
            c = board[i][j]
            board[i][j] = '#'
            for a, b in pairwise((-1, 0, 1, 0, -1)):
                x, y = i + a, j + b
                if 0 <= x < m and 0 <= y < n and
board[x][y] != '#':
                    dfs(node, x, y)
            board[i][j] = c
```

Trie

p.693

```
tree = Trie()
for i, w in enumerate(words):
    tree.insert(w, i)
m, n = len(board), len(board[0])
ans = []
for i in range(m):
    for j in range(n):
        dfs(tree, i, j)
return ans
```

Trie p.694

Trie

#336 Palindrome Pairs

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · String · Trie

Lists: Grind 169

Time Complexity: $O(n * k^2)$ in the worst case where n is the number of words and k is the maximum word length, but with the palindrome checking optimized this can be considered $O(\text{sum of word lengths})$:

Problem

You are given a 0-indexed array of unique strings words.

A palindrome pair is a pair of integers (i, j) such that:

- $0 \leq i, j < \text{words.length}$,
- $i \neq j$, and
- $\text{words}[i] + \text{words}[j]$ (the concatenation of the two strings) is a palindrome.

Return an array of all the palindrome pairs of words.

You must write an algorithm with $O(\text{sum of words}[i].\text{length})$ runtime complexity.

Example 1:

Trie p.695

```
Input: words = ["abcd","dcba","lls","s","sssll"]
Output: [[0,1],[1,0],[3,2],[2,4]]
Explanation: The palindromes are
["abccddcba","dcbaabcd","slls","llssssll"]
```

Example 2:

```
Input: words = ["bat","tab","cat"]
Output: [[0,1],[1,0]]
Explanation: The palindromes are ["battab","tabbat"]
```

Example 3:

```
Input: words = ["a",""]
Output: [[0,1],[1,0]]
Explanation: The palindromes are ["a","a"]
```

Constraints:

- $1 \leq \text{words.length} \leq 5000$
- $0 \leq \text{words}[i].\text{length} \leq 300$
- $\text{words}[i]$ consists of lowercase English letters.

Brute Force (Python)

Brute Force Approach
class Solution:
 def is_palindrome(self, s):
 # Brute Force O(n*m) - linear scan through all
 matches = []
 for word in words:
 if word.startswith(prefix):
 matches.append(word)
 return matches

Trie p.696

Community Solutions (Python)

• WalkCC

class Solution:
 def palindromePairs(self, words: list[str]) -> list[list[int]]:
 ans = []
 dict = {word[::-1]: i for i, word in enumerate(words)}

 for i, word in enumerate(words):
 if "" in dict and dict[""] != i and word == word[::-1]:
 ans.append([i, dict[""]])

 for j in range(1, len(word) + 1):
 l = word[:j]
 r = word[j:]
 if l in dict and dict[l] != i and r == r[::-1]:
 ans.append([i, dict[l]])
 if r in dict and dict[r] != i and l == l[::-1]:
 ans.append([dict[r], i])

 return ans

• LeetDoocs

Trie p.697

class Solution:
 def palindromePairs(self, words: List[str]) -> List[List[int]]:
 d = {w: i for i, w in enumerate(words)}
 ans = []
 for i, w in enumerate(words):
 for j in range(len(w) + 1):
 a, b = w[:j], w[j:]
 ra, rb = a[::-1], b[::-1]
 if ra in d and d[ra] != i and b == rb:
 ans.append([i, d[ra]])

 if j and rb in d and d[rb] != i and a == ra:
 ans.append([d[rb], i])

 return ans

• SimplyLeet

Python implementation for Palindrome Pairs

def is_palindrome(s):
 # Helper function to check if a string is a palindrome
 return s == s[::-1]

def palindromePairs(words):
 result = []
 # Create a dictionary mapping each word to its index
 word_dict = {word: i for i, word in enumerate(words)}

Trie p.698

Iterate over each word with its index
for i, word in enumerate(words):
 # For each possible split position (including edge empty string splits)
 for j in range(len(word) + 1):
 prefix = word[:j]
 suffix = word[j:]

 # If prefix is palindrome, check for reversed suffix in dictionary
 if is_palindrome(prefix):
 reversed_suffix = suffix[::-1]
 # Ensure that the reversed_suffix exists and is not the same word
 if reversed_suffix in word_dict and word_dict[reversed_suffix] != i:
 result.append([word_dict[reversed_suffix], i])

 # Check the suffix; avoid duplicate pairs when j equals len(word) because
 # the case when suffix is empty was handled in the first condition
 if j != len(word) and is_palindrome(suffix):
 reversed_prefix = prefix[::-1]
 if reversed_prefix in word_dict and word_dict[reversed_prefix] != i:

Trie p.699

result.append([i, word_dict[reversed_prefix]])

return result

Example Usage:
words = ["abcd", "dcba", "lls", "s", "sssll"]
print(palindromePairs(words)) # Expected Output: [[0,1],[1,0],[3,2],[2,4]]

• LC.ca

class Solution:
 def palindromePairs(self, words: List[str]) -> List[List[int]]:
 d = {w: i for i, w in enumerate(words)}
 ans = []
 for i, w in enumerate(words):
 for j in range(len(w) + 1):
 a, b = w[:j], w[j:]
 ra, rb = a[::-1], b[::-1]
 if ra in d and d[ra] != i and b == rb:
 ans.append([i, d[ra]])

 if j and rb in d and d[rb] != i and a == ra:
 ans.append([d[rb], i])

 return ans

Trie p.700

Trie

#588 Design In-Memory File System

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Design · Trie

Lists: Grind 169 | Premium 98

Time Complexity:

Problem

Design a data structure that simulates an in-memory file system.

Implement the FileSystem class:

- `FileSystem()` Initializes the object of the system.
- `List<String> ls(String path)` If path is a file path, returns a list that only contains this file's name.
- If path is a directory path, returns the list of file and directory names in this directory.
- If `filePath` does not exist, creates that file containing given content.
- If `filePath` already exists, appends the given content to original content.

Example 1:

Trie p.701

Operation	Output	Explanation
FileSystem fs = new FileSystem()	null	The constructor returns nothing.
fs.ls("/")	[]	Initially, directory "/" has nothing. So return empty list.
fs.mkdir("/a/b/c")	null	Create directory "a" in directory "/". Then create directory "b" in directory "a". Finally, create directory "c" in directory "b".
fs.addContentToFile("/a/b/c/d","hello")	null	Create a file named "d" with content "hello" in directory "/a/b/c".
fs.ls("/")	["a"]	Only directory "a" is in directory "/".
fs.readContentFromFile("/a/b/c/d")	"hello"	Output the file content.

Input

```
["FileSystem", "ls", "mkdir", "addContentToFile", "ls", "readContentFromFile"]
[[], ["/"], ["/a/b/c"], ["/a/b/c/d", "hello"], ["/"], ["/a/b/c/d"]]
Output
[null, [], null, null, ["a"], "hello"]
```

Explanation

```
FileSystem fileSystem = new FileSystem();
```

Constraints:

- $1 \leq \text{path.length}, \text{filePath.length} \leq 100$
- path and filePath are absolute paths which begin with '/' and do not end with '/' except that the path is just '/'.
- You can assume that all directory names and file names only contain lowercase letters, and the same names will not exist in the same directory.

Trie p.702

- You can assume that all operations will be passed valid parameters, and users will not attempt to retrieve file content or list a directory or file that does not exist.
- You can assume that the parent directory for the file in addContentToFile will exist.
- 1 <= content.length <= 50
- At most 300 calls will be made to ls, mkdir, addContentToFile, and readContentFromFile.

Community Solutions (Python)

```
• SimplyLeet
class Node:
    def __init__(self):
        # Indicates whether this node represents a file
        # or a directory
        self.is_file = False
        # For directories, mapping from name to Node
        self.children = {}
        # For files, store content
        self.content = ""

class FileSystem:
```

Trie p.703

```
def __init__(self):
    # Root directory node initialization
    self.root = Node()

def ls(self, path: str) -> List:
    # Traverse the path to obtain the target node
    node, name = self.traverse(path)
    # If the node is a file, return the single
    element list containing its name
    if node.is_file:

        return [name]
    # Otherwise, list all keys (names) in the
    directory, sorted lexicographically
    return sorted(node.children.keys())

def mkdir(self, path: str) -> None:
    # Create directory by traversing the path and
    creating nodes if necessary
    self.traverse(path)

def addContentToFile(self, filePath: str, content:
str) -> None:
    # Traverse to the file's node; if it does not
    exist, create necessary nodes along the path
```

Trie p.704

```
node, name = self.traverse(filePath)
# If not already a file, mark the node as a file
if not node.is_file:
    node.is_file = True
    node.content = ""
# Append the new content to the file's content
node.content += content

def readContentFromFile(self, filePath: str) -> str:
    # Traverse to the file's node

    node, _ = self.traverse(filePath)
    # Return the content of the file
    return node.content

def traverse(self, path: str):
    # Start from the root node
    node = self.root
    if path == "/":
        # Special case for root path
        return node, ""

    # Split the path by "/" and traverse the nodes
    parts = path.split("/")[:-1]
    for part in parts:
        # Create the node if it does not exist
        if part not in node.children:
            node.children[part] = Node()
        node = node.children[part]
    # Return the target node and its name (last part
    of the path)
    return node, parts[-1]
```

Trie p.705

```
# Example usage:
# fs = FileSystem()
# print(fs.ls("/")) # prints []
# fs.mkdir("/a/b/c")
# fs.addContentToFile("/a/b/c/d", "hello")
# print(fs.ls("/")) # prints ['a']
# print(fs.readContentFromFile("/a/b/c/d")) # prints
"hello"
```

```
• LC.ca
...
>>> "/a/b/c".split('/')
[['', 'a', 'b', 'c']]
...

class Trie:
    def __init__(self):
        self.name = None
        self.isFile = False
        self.content = []

        self.children = {}

    def insert(self, path, isFile):
        node = self
        ps = path.split('/')
        for p in ps[1:]:
            if p not in node.children:
                node.children[p] = Trie()
            node = node.children[p]
        node.isFile = isFile
```

Trie p.706

```
if isFile:
    node.name = ps[-1]
    return node

def search(self, path):
    node = self
    if path == '/':
        return node
    ps = path.split('/')
    for p in ps[1:]:

        if p not in node.children:
            return None
        node = node.children[p]
    return node

class FileSystem:
    def __init__(self):
        self.root = Trie()

    def ls(self, path: str) -> List[str]:
        node = self.root.search(path)
        if node is None:
            return []
        if node.isFile:
            return [node.name]
        return sorted(node.children.keys())

    def mkdir(self, path: str) -> None:
        self.root.insert(path, False)
```

Trie p.707

```
def addContentToFile(self, filePath: str, content:
str) -> None:
    node = self.root.insert(filePath, True)
    node.content.append(content)

def readContentFromFile(self, filePath: str) -> str:
    node = self.root.search(filePath)
    return ''.join(node.content)

# Your FileSystem object will be instantiated and called
as such:
# obj = FileSystem()
# param_1 = obj.ls(path)
# obj.mkdir(path)
# obj.addContentToFile(filePath,content)
# param_4 = obj.readContentFromFile(filePath)
```

Trie p.708

Trie

#642 Design Search Autocomplete System

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

String · Design · Trie · Data Stream · Heap

Lists: Premium 98

Time Complexity: O(L * log k) per character input query, where L is the length of the current prefix and k is 3 (constant factor), thus ensuring efficient autocompletion. Space Complexity: O(N * L), w

Problem

Design a search autocomplete system for a search engine. Users may input a sentence (at least one word and end with a special character '#').

You are given a string array sentences and an integer array times both of length n where sentences[i] is a previously typed sentence and times[i] is the corresponding number of times the sentence was typed. For each input character except '#', return the top 3 historical hot sentences that have the same prefix as the part of the sentence already typed.

Here are the specific rules:

- The hot degree for a sentence is defined as the number of times a user typed the exactly same sentence before.

Trie p.709

- The returned top 3 hot sentences should be sorted by hot degree (The first is the hottest one). If several sentences have the same hot degree, use ASCII-code order (smaller one appears first).
- If less than 3 hot sentences exist, return as many as you can.
- When the input is a special character, it means the sentence ends, and in this case, you need to return an empty list.

Implement the AutocompleteSystem class:

- AutocompleteSystem(String[] sentences, int[] times) Initializes the object with the sentences and times arrays.
- List<String> input(char c) This indicates that the user typed the character c. Returns an empty array [] if c == '#' and stores the inputted sentence in the system.
- Returns the top 3 historical hot sentences that have the same prefix as the part of the sentence already typed. If there are fewer than 3 matches, return them all.

Example 1:

Trie Sheet 79 / 317 · LeetCode Mastery By Pattern · Python Only

```
Input
["AutocompleteSystem", "input", "input", "input",
"input"]
[[["i love you", "island", "iroman", "i love
leetcode"], [5, 3, 2, 2]], [{"i"}, [" "], ["a"], ["#"]]
Output
[null, ["i love you", "island", "i love leetcode"], ["i
love you", "i love leetcode"], [], []]

Explanation
AutocompleteSystem obj = new AutocompleteSystem(["i
love you", "island", "iroman", "i love leetcode"], [5,
3, 2, 2]);
```

Constraints:

- n == sentences.length
- n == times.length
- 1 <= n <= 100
- 1 <= sentences[i].length <= 100
- 1 <= times[i] <= 50
- c is a lowercase English letter, a hash '#', or space ' '.
- Each tested sentence will be a sequence of characters c that end with the character '#'.
- Each tested sentence will have a length in the range [1, 200].
- The words in each input sentence are separated by single spaces.
- At most 5000 calls will be made to input.

Community Solutions (Python)

Trie p.711

• WalkCC

```
class TrieNode:
    def __init__(self):
        self.children: dict[str, TrieNode] = {}
        self.s: str | None = None
        self.time = 0
        self.top3: list[TrieNode] = []

    def __lt__(self, other):
        if self.time == other.time:
            return self.s < other.s

        return self.time > other.time

    def update(self, node) -> None:
        if node not in self.top3:
            self.top3.append(node)
            self.top3.sort()
            if len(self.top3) > 3:
                self.top3.pop()

class AutocompleteSystem:
    def __init__(self, sentences: list[str], times: list[int]):
        self.root = TrieNode()
        self.curr = self.root
        self.s: list[str] = []

        for sentence, time in zip(sentences, times):
            self._insert(sentence, time)

    def input(self, c: str) -> list[str]:
```

Trie p.712

```
if c == '#':
    self._insert(''.join(self.s), 1)
    self.curr = self.root
    self.s = []
    return []

self.s.append(c)

if self.curr:
    self.curr = self.curr.children.get(c, None)

if not self.curr:
    return []
return [node.s for node in self.curr.top3]

def _insert(self, sentence: str, time: int) -> None:
    node = self.root
    for c in sentence:
        node = node.children.setdefault(c, TrieNode())
        node.s = sentence
        node.time += time

    leaf = node
    node: TrieNode = self.root
    for c in sentence:
        node = node.children[c]
        node.update(leaf)
```

• SimplyLeet

Trie p.713

```
class TrieNode:
    def __init__(self):
        # Mapping from character to TrieNode
        self.children = {}
        # Dictionary mapping sentence to its frequency
        count

        self.counts = {}

class AutocompleteSystem:
    def __init__(self, sentences, times):
        self.root = TrieNode()

        self.current_node = self.root
        self.current_input = ""
        # Initialize Trie with the given sentences and their times
        for sentence, time in zip(sentences, times):
            self.addSentence(sentence, time)

    def addSentence(self, sentence, freq):
        node = self.root
        for char in sentence:
            if char not in node.children:
```

Trie p.714

```
node.children[char] = TrieNode()
node = node.children[char]
# Update sentence frequency at each node
node.counts[sentence] =
node.counts.get(sentence, 0) + freq

def input(self, c):
    results = []
    if c == '#':
        # End of input: add current sentence to Trie
        self.addSentence(self.current_input, 1)

# Reset the current input and traversal
pointer
self.current_input = ""
self.current_node = self.root
return results

self.current_input += c
# Move to the next Trie node if it exists
if self.current_node:
    self.current_node =
self.current_node.children.get(c, None)
if not self.current_node:
```

Trie p.715

```
return results

# Retrieve sentences and sort them by frequency
(descending) and lexicographical order (ascending)
data = list(self.current_node.counts.items())
data.sort(key=lambda x: (-x[1], x[0]))
for i in range(min(3, len(data))):
    results.append(data[i][0])
return results

# Example usage:

# auto = AutocompleteSystem(["i love you", "island",
                             "iroman", "i love leetcode"], [5, 3, 2, 2])
# print(auto.input("i"))
# print(auto.input(" "))
# print(auto.input("a"))
# print(auto.input("#"))
```

• LC.ca

```
class Trie:
    def __init__(self):
        # 26+1, extra 1 for space ' ' node
        self.children = [None] * 27
        self.v = 0
        self.w = ''

    def insert(self, w, t):
        node = self
        for c in w:
```

Trie p.716

```
idx = 26 if c == ' ' else ord(c) - ord('a')
if node.children[idx] is None:
    node.children[idx] = Trie()
node = node.children[idx]
node.v += t
node.w = w

def search(self, pref):
    node = self
    for c in pref:

        idx = 26 if c == ' ' else ord(c) - ord('a')
        if node.children[idx] is None:
            return None
        node = node.children[idx]
    return node

class AutocompleteSystem:
    def __init__(self, sentences: List[str], times: List[int]):
        self.trie = Trie()

        for a, b in zip(sentences, times):
            self.trie.insert(a, b)
            self.t = [] # target

    def input(self, c: str) -> List[str]:
        def dfs(node):
            if node is None:
                return
            if node.v:
                res.append((node.v, node.w))
```

Trie p.717

```
for nxt in node.children:
    dfs(nxt)
```

```
if c == '#':
    s = ''.join(self.t)
    self.trie.insert(s, 1)
    self.t = []
    return []

res = []
```

```
self.t.append(c)
node = self.trie.search(''.join(self.t))
if node is None:
    return res
dfs(node)
res.sort(key=lambda x: (-x[0], x[1]))
return [v[1] for v in res[:3]]
```

Your AutocompleteSystem object will be instantiated and called as such:

```
# obj = AutocompleteSystem(sentences, times)
# param_1 = obj.input(c)
```

Trie p.718

>> Quick Review — Trie

12 questions · insights · complexity · solution

#14 Longest Common Prefix [Easy]

Key Insights

- Compare characters of the strings one column at a time (vertical scanning).
- The longest possible prefix is limited by the length of the shortest string in the array.
- As soon as a mismatch is found, the prefix up to that point is the answer.
- Handle edge cases such as an empty input array or strings with no common prefix.

Space & Time

Time Complexity: $O(N \times M)$ where N is the number of strings and M is the length of the shortest string.

Space Complexity: $O(1)$ additional space (ignoring output space).

Solution

We use the vertical scanning approach:

- If the input array is empty, return an empty string.
- Iterate character by character over the first string.
- For each character, compare it with the corresponding character in each of the other strings.
- If a mismatch or

Trie p.719

Trie p.720

end of any string is encountered, return the substring from the start up to the current index. – If no mismatch is found, the entire first string is the common prefix. This solution ensures that we only traverse the necessary part of the strings and stops early if a discrepancy is discovered. #139 Word Break [Medium] Key Insights • Use Dynamic Programming (DP) to break the problem into subproblems. • Create a DP array where dp[i] is true if the substring s[0:i] can be segmented using words from the dictionary. • Leverage a hash set for fast look-up of dictionary words. • For every index i, check all possible previous break points j such that dp[j] is true and s[j:i] is in the wordDict. • The problem can also be approached with recursion and memoization, but DP is more straightforward for iterative implementation. Space & Time Time Complexity: O(n^2) in the worst-case scenario, where n is the length of s (each substring check is O(1) if using a hash set, and there could be O(n^2) checks	overall). Space Complexity: O(n) for the dp array and O(k) for the hash set where k is the number of words in the dictionary. Solution The solution uses a Dynamic Programming (DP) approach: – Convert wordDict to a hash set for constant time lookup. – Create a boolean dp array of size n+1 (n is the length of s), where dp[i] indicates if s[0:i] can be segmented. – Set dp[0] to true because an empty string is always segmentable. – For each index i from 1 to n, iterate through all indices j from 0 to i: if dp[j] is true and the substring s[j:i] is in the dictionary, then set dp[i] to true. Break early from the inner loop when a valid break is found. – Return dp[n] as the result. This method efficiently builds up the solution by utilizing previous results to determine if the entire string can be segmented. #208 Implement Trie (Prefix Tree) [Medium] Key Insights • Use a tree-like structure where each node represents a character. • Each node stores its children in a dictionary (or hash map) for O(1) average time access.	• Mark the end of a word with a boolean flag. • Insertion is done character by character, creating new nodes as needed. • Searching and checking prefixes follow traversals down the tree. Space & Time Time Complexity: – Insert: O(m) where m is the length of the word. – Search: O(m) – startsWith: O(m) Space Complexity: – O(N * M) in the worst-case scenario, where N is the number of words and M is the average length of the words, due to storing each character in the Trie. Solution We implement the Trie using a node-based approach. Each TrieNode contains a hash map (or dictionary) of its children and a boolean indicating if the node marks the end of a word. The Trie class uses these nodes to build out the words character by character. The insert method traverses and builds the necessary nodes; the search method verifies that all characters exist in order and that the final node marks the end of a word; the startsWith method checks if the prefix exists in the Trie. No tricky edge cases exist beyond handling null or empty inputs, as constraints guarantee valid lowercase
strings. #211 Design Add and Search Words Data Structure [Medium] Key Insights • Utilize a Trie (prefix tree) to efficiently store words. • Each node in the Trie contains a dictionary for its children and a flag indicating the end of a word. • The search operation is implemented with a depth-first search (DFS) approach. • When the search query contains a dot, explore all possible child nodes. Space & Time Time Complexity: – addWord: O(w), where w is the length of the word. – search: Worst-case O(26^w) for words with multiple wildcards, but practical performance is better given constraints. Space Complexity: O(n * w), where n is the number of words and w is the average word length. Solution The WordDictionary is implemented using a Trie. For adding, we insert each character of the word into the Trie. For searching, we use DFS. When encountering a wildcard '.', we recursively search all children nodes. This structure combined with DFS efficiently addresses	the wildcard matching and ensures good performance for multiple operations. #616 Add Bold Tag in String [Medium] Key Insights • We need to identify every occurrence of each word in s. • Overlapping or consecutive intervals should be merged to form one continuous bold region. • Use an auxiliary data structure (e.g., a boolean array) to mark positions that should be bolded. • Once the positions are marked, build the final string by inserting bold tags at the appropriate boundaries. Space & Time Time Complexity: O(n * m * k) where n is the length of s, m is the number of words, and k is the average length of the words (due to substring matching). Space Complexity: O(n) for the auxiliary boolean array used to mark bold positions. Solution The solution involves two main steps: – Identify and mark the indices in the string s that are part of any matching word. We do this by iterating over each word and marking every occurrence's start and end in a boolean array. – Construct the final string by traversing the boolean array. When encountering a	segment of consecutive true values, wrap that segment with and tags. The key idea is to merge intervals by marking every index that should be bold and then using a single pass to insert tags only when transitioning from non-bold to bold and vice versa. #692 Top K Frequent Words [Medium] Key Insights • Count the frequency of each unique word using a hash table or dictionary. • Use a min-heap (priority queue) to keep track of the top k frequent words. In the heap, the ordering is based on frequency; if frequencies are equal, order by lexicographical order (inverted condition for min-heap). • Alternatively, you can sort the unique words based on frequency and lexicographical order and then pick the top k. • For an efficient solution, use a min-heap that maintains size k achieving O(n log k) time complexity. Space & Time Time Complexity: O(n log k) when using a heap, where n is the number of unique words. Space Complexity: O(n) for storing the frequency counts and the heap. Solution
We first calculate the frequency of each word using a hash table (or dictionary). Then, we use a min-heap that stores pairs of (frequency, word). The comparator for the heap is defined such that the root of the heap is the word with the smallest frequency; if two words have the same frequency, the word with the lexicographically larger value is considered “smaller” in our ordering so that it gets popped when the heap size exceeds k. After processing all words, the heap contains the top k frequent words. Finally, the heap elements are extracted and reversed (if needed) to produce the final list sorted from highest to lowest frequency with lexicographic tie-breaking. #720 Longest Word in Dictionary [Medium] Key Insights • The word must be built character by character; hence every prefix (removing one character from the end at a time) must exist in the dictionary. • Sorting the input list can help in ensuring that smaller words (prefixes) are processed first. • Using a hash set to store valid words enables rapid prefix checks. • Lexicographical order plays a role when words have the same length. Space & Time	Time Complexity: O(n * L + n log n) where n is the number of words and L is the maximum length of a word (n * L for prefix checking, and n log n for sorting). Space Complexity: O(n * L) for storing words in the set. Solution The approach begins by sorting the list of words. Sorting is done primarily by word length and secondarily by lexicographical order so that when we iterate through the words, all possible prefixes for a word have already been considered. Initialize a set to keep track of valid words that can be built one character at a time. For each word in the sorted list, check if all its prefixes (word without the last character progressively) exist in the set. If a word qualifies, add it to the set and check if it should be the new result based on length and lexicographical order. The set lookup ensures that checking each prefix is efficient. #1166 Design File System [Medium] Key Insights • Use a data structure (like a hash map) to store paths and associated values. • To create a new path, check that the path does not already exist. • Ensure the immediate parent path exists before creation.	• When retrieving a value, simply look it up in the storage structure. • Splitting the path by "/" allows easy identification of the parent. Space & Time Time Complexity: – createPath: O(n) per call where n is the number of segments in the path (to process and validate parent path). – get: O(1) per call if using a hash map. Space Complexity: O(m * L) where m is the number of paths stored and L is the average length of a path. Solution The solution uses a hash map (dictionary) to store file paths along with their associated values. For the createPath method, the algorithm: – Checks if the path already exists. – Extracts the parent path by removing the last segment. – Validates the existence of the parent path. – Inserts the new path with its corresponding value if validation passes. For the get method, the algorithm simply returns the stored value if the path exists; otherwise, it returns -1. This approach ensures efficient lookup and validation using string manipulation and hash map operations. #212 Word Search II [Hard]
Trie p.727	Trie Sheet 81 / 317 · LeetMastery By Pattern · Python Only · 9-Up Portrait p.728	Trie p.729

Key Insights

- Use Depth-First Search (DFS) to explore possible words starting from each board cell.
- Build a Trie (prefix tree) to store the words for O(1) prefix checks and efficient pruning.
- Mark visited cells and restore them after DFS to avoid revisiting in the current search path.
- Prune search paths early if the current prefix is not part of any word in the Trie.

Space & Time

Time Complexity: $O(m * n * 4^L)$ in the worst case, where L is the maximum word length; however, using a Trie prunes unnecessary paths.

Space Complexity: $O(K)$ for the Trie storage, where K is the total number of letters in all words, plus $O(m * n)$ auxiliary space for recursion and board marking.

Solution

We first insert all words into a Trie data structure to allow quick lookups of prefixes. Then, for each cell in the board, we perform DFS. During DFS, if the current path matches a Trie node that represents a complete word, we add that word to the result list and mark it as found to avoid duplicates. We continue exploring adjacent cells (up, down, left, right) while marking the cell as visited. After exploring, we restore the original letter for further DFS calls. This combination of Trie and DFS with backtracking efficiently finds valid words on

Trie p.730

– Is: $O(k \log k)$ where k is the number of entries in the directory (due to sorting), or $O(n)$ if traversing a long path.

– mkdir, addContentToFile, readContentFromFile: $O(d)$ where d is the depth of the file path.

Space Complexity:

– $O(T)$ where T is the total number of characters stored in directory names and file contents.

Solution

The solution uses a tree-based approach where each node represents either a directory or a file. Each directory node stores its children in a hash table (or dictionary) mapping names to node objects. File nodes store their content as a string and carry a flag indicating they are files. For the Is operation, check if the node is a file—if so, return its name; otherwise, list and sort the names of all children in that directory. The mkdir operation traverses (or creates) nodes for each segment of the provided path. The addContentToFile method accesses or creates the file node and appends the content. Finally, readContentFromFile retrieves the stored content from the file node.

#642 Design Search Autocomplete System [Hard]

Key Insights

- Use a Trie to efficiently store and retrieve sentences based on their prefixes.

Trie p.733

Math

#9 Palindrome Number

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Math

Lists: Grind 169

Time Complexity: $O(\log_{10}(n))$ – Each iteration reduces the number by a factor of 10. Space Complexity: $O(1)$ – Only constant extra space is used.

Problem

Given an integer x, return true if x is a palindrome, and false otherwise.

Example 1:

Input: x = 121
Output: true
Explanation: 121 reads as 121 from left to right and from right to left.

Example 2:

Input: x = -121
Output: false
Explanation: From left to right, it reads -121. From right to left, it becomes 121-. Therefore it is not a palindrome.

Example 3:

Math p.736

the board.

#336 Palindrome Pairs [Hard]

Key Insights

- A palindrome is a string that reads the same forwards and backwards.
- For any given word, if you split it into two parts, one part (either prefix or suffix) might be a palindrome.
- If the other part's reverse exists elsewhere in the array, then that reverse can be concatenated to yield a palindrome.
- Carefully handle edge cases like empty strings, since an empty string is a palindrome by default.

Space & Time

Time Complexity: $O(n * k^2)$ in the worst case where n is the number of words and k is the maximum word length, but with the palindrome checking optimized this can be considered $O(\text{sum of word lengths})$: Each word is processed once and each split is checked in $O(k)$ time.

Space Complexity: $O(n * k)$ due to the hashmap storing all words and their indices.

Solution

We use a hashmap (dictionary) to store each word and its corresponding index. For every word, we iterate through every possible split position to divide the word into a prefix and a suffix. For each split:

Trie p.731

- Maintain a mapping at each Trie node to record sentences (or their frequencies) that pass through that node.
- Use sorting or a min-heap to select the top three sentences based on their frequency (hot degree) and lexicographical order if frequencies tie.
- On receiving '#', update the Trie with the current sentence and reset the current search state.

Space & Time

Time Complexity: $O(L * \log k)$ per character input query, where L is the length of the current prefix and k is 3 (constant factor), thus ensuring efficient autocompletion.

Space Complexity: $O(N * L)$, where N is the number of unique sentences and L is the average sentence length, for maintaining the Trie.

Solution

The solution revolves around building a Trie where each node contains:

– A mapping to its child nodes. – A hash map that records all sentences passing through the node along with their occurrence counts.

For each character entered (except '#'), the system:

– Traverses the Trie to find the current prefix node. – Retrieves stored sentences at that node and sorts them based on the following criteria: descending by frequency

Trie p.734

Input: x = 10
Output: false
Explanation: Reads 01 from right to left. Therefore it is not a palindrome.

Constraints:

- $-231 \leq x \leq 231 - 1$

Follow up: Could you solve it without converting the integer to a string?

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def isPalindrome(self, x):
        # Brute Force - iterate all candidates
        for i in range(2, n + 1):
            is_valid = all(i % j != 0 for j in range(2,
i))
            if is_valid: pass
            return 0
```

Community Solutions (Python)

• WalkCC

Math Sheet 82 / 317 · LeetMastery By Pattern · Python Only · 9-Up Portrait

– If the prefix is a palindrome, we look for the reverse of the suffix in the hashmap. If found (and not the same word), then combining the reverse with the current word forms a palindrome. – Similarly, if the suffix is a palindrome, we look for the reverse of the prefix.

We must be cautious with edge cases, especially when either part is empty, as the empty string is considered a palindrome. This method avoids checking every possible pair explicitly and leverages string reversal and palindrome checks to achieve efficiency.

#588 Design In-Memory File System [Hard]

Key Insights

- Use a tree (trie) to represent directories and files.
- Each node can be either a directory or a file; directories maintain a mapping of names to child nodes.
- For ls, if the given path is a file, return only its name. If it's a directory, return the sorted list of names of its children.
- mkdir must create all intermediate directories if they do not already exist.
- File nodes store content and support content appending.

Space & Time

Time Complexity:

Trie p.732

and ascending lexicographically if frequencies are equal.

– Returns the top three sentences from the sorted result.

When the user inputs '#', the current sentence (tracked as user input so far) is added or its frequency updated in the Trie, and the input state resets.

Math

#11 Math

By Pattern · Python Only · 12 questions

Trie p.735

```
class Solution:
    def isPalindrome(self, x: int) -> bool:
        if x < 0:
            return False

        rev = 0
        y = x

        while y:
            rev = rev * 10 + y % 10

            y //= 10

        return rev == x
```

• LeetDoocs

```
class Solution:
    def isPalindrome(self, x: int) -> bool:
        if x < 0 or (x and x % 10 == 0):
            return False
        y = 0
        while y < x:
            y = y * 10 + x % 10
            x //= 10
        return x in (y, y // 10)
```

• SimplyLeet

Math p.738

Python solution for converting an integer to its base 7 string representation.
def convertToBase7(num):
 # Special case: If num is 0, return "0"
 if num == 0:
 return "0"

 # Flag to check if the number is negative
 negative = num < 0
 # Work with the absolute value of the number
 num = abs(num)

 # List to store base 7 digits
 digits = []

 # Process the number until it becomes 0
 while num > 0:
 # Get the remainder (digit in base 7)
 remainder = num % 7
 # Append the digit as string to the list
 digits.append(str(remainder))

 # Convert the number to a string to process each digit
 original_str = str(n)
 rotated_str = ""

 # Process each digit in reverse order to simulate 180 degree rotation
 for digit in reversed(original_str):
 # If digit is not valid for rotation, return false immediately
 if digit not in rotation_map:
 return False
 # Append the rotated digit to the rotated string
 rotated_str += rotation_map[digit]

 # Convert rotated string to an integer (leading zeros are ignored)
 rotated_num = int(rotated_str)

 # The number is confusing if the rotated number is different from the original number

Mathp.748

Update num by integer division by 7
num //= 7

Reverse the list to get digits in correct order
base7 = ''.join(reversed(digits))
If original number was negative, attach the minus sign
return "-" + base7 if negative else base7

Example usage:
print(convertToBase7(100)) # Expected output: "202"

print(convertToBase7(-7)) # Expected output: "-10"

• LC.ca
class Solution:
 def convertToBase7(self, num: int) -> str:
 if num == 0:
 return '0'
 if num < 0:
 return '-' + self.convertToBase7(-num)
 ans = []
 while num:
 ans.append(str(num % 7))
 num //= 7

 return ''.join(ans[::-1])

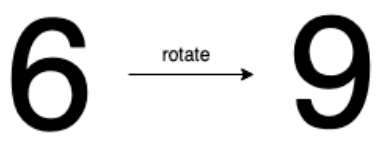
Mathp.749

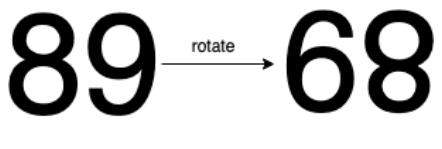
Math

#1056 Confusing NumberEAS

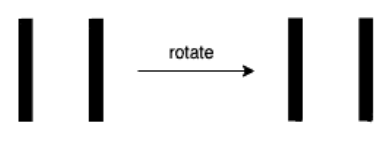
LeetDoocs · SimplyLeet · WalkCC · LeetCode
Math
Lists: Premium 98
Time Complexity: O(d), where d is the number of digits in n (approximately O(log n)). Space Complexity: O(d), needed to store the rotated number string.
Problem
A confusing number is a number that when rotated 180 degrees becomes a different number with each digit valid.
We can rotate digits of a number by 180 degrees to form new digits.
• When 0, 1, 6, 8, and 9 are rotated 180 degrees, they become 0, 1, 9, 8, and 6 respectively.
• When 2, 3, 4, 5, and 7 are rotated 180 degrees, they become invalid.
Note that after rotating a number, we can ignore leading zeros.
• For example, after rotating 8000, we have 0008 which is considered as just 8.
Given an integer n, return true if it is a confusing number, or false otherwise.

Mathp.750

Example 1:

Input: n = 6
Output: true
Explanation: We get 9 after rotating 6, 9 is a valid number, and 9 != 6.

Example 2:


Mathp.751

Input: n = 89
Output: true
Explanation: We get 68 after rotating 89, 68 is a valid number and 68 != 89.
Example 3:

Input: n = 11
Output: false
Explanation: We get 11 after rotating 11, 11 is a valid number but the value remains the same, thus 11 is not a confusing number
Constraints:
• 0 <= n <= 109
Community Solutions (Python)
• WalkCC

Mathp.752

class Solution:
 def confusingNumber(self, n: int) -> bool:
 s = str(n)
 rotated = {'0': '0', '1': '1', '6': '9', '8': '8', '9': '6'}
 rotatedNum = []

 for c in s[::-1]:
 if c not in rotated:
 return False
 rotatedNum.append(rotated[c])

 return ''.join(rotatedNum) != s

• LeetDoocs
class Solution:
 def confusingNumber(self, n: int) -> bool:
 x, y = n, 0
 d = [0, 1, -1, -1, -1, -1, 9, -1, 8, 6]
 while x:
 x, v = divmod(x, 10)
 if d[v] < 0:
 return False
 y = y * 10 + d[v]
 return y != n

• SimplyLeet

Mathp.753

Define the mapping for valid rotations
def is_confusing_number(n: int) -> bool:
 rotation_map = {'0': '0', '1': '1', '6': '9', '8': '8', '9': '6'}

 # Convert the number to a string to process each digit
 original_str = str(n)
 rotated_str = ""

 # Process each digit in reverse order to simulate 180 degree rotation
 for digit in reversed(original_str):
 # If digit is not valid for rotation, return false immediately
 if digit not in rotation_map:
 return False
 # Append the rotated digit to the rotated string
 rotated_str += rotation_map[digit]

 # Convert rotated string to an integer (leading zeros are ignored)
 rotated_num = int(rotated_str)

 # The number is confusing if the rotated number is different from the original number

Mathp.754

return rotated_num != n

Example usage:
print(is_confusing_number(6)) # Expected output: True
print(is_confusing_number(89)) # Expected output: True
print(is_confusing_number(11)) # Expected output: False

• LC.ca
class Solution:
 def confusingNumber(self, n: int) -> bool:
 x, y = n, 0
 d = [0, 1, -1, -1, -1, -1, 9, -1, 8, 6]
 while x:
 x, v = divmod(x, 10)
 if d[v] < 0:
 return False
 y = y * 10 + d[v]
 return y != n

Mathp.755

Math

#1228 Missing Number in Arithmetic ProgressionEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode
Array · Math
Lists: Premium 98
Time Complexity: O(n) Space Complexity: O(1)
Problem
In some array arr, the values were in arithmetic progression: the values arr[i + 1] - arr[i] are all equal for every 0 <= i < arr.length - 1.
A value from arr was removed that was not the first or last value in the array.
Given arr, return the removed value.
Example 1:
Input: arr = [5,7,11,13]
Output: 9
Explanation: The previous array was [5,7,9,11,13].
Example 2:
Input: arr = [15,13,12]
Output: 14
Explanation: The previous array was [15,14,13,12].
Constraints:

Mathp.756

- 3 <= arr.length <= 1000
- 0 <= arr[i] <= 105
- The given array is guaranteed to be a valid array.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def missingNumber(self, arr):
        # Brute Force - iterate all candidates
        for i in range(2, n + 1):
            is_valid = all(i % j != 0 for j in range(2,
i))
            if is_valid: pass
        return 0
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def missingNumber(self, arr: list[int]) -> int:
        n = len(arr)
        delta = (arr[-1] - arr[0]) // n
        l = 0
        r = n - 1

        while l < r:
            m = (l + r) // 2
            if arr[m] == arr[0] + m * delta:
```

```
l = m + 1
else:
    r = m

return arr[0] + l * delta
```

• LeetDoocs

```
class Solution:
    def missingNumber(self, arr: List[int]) -> int:
        return (arr[0] + arr[-1]) * (len(arr) + 1) // 2 -
sum(arr)

class Solution:
    def missingNumber(self, arr: List[int]) -> int:
        n = len(arr)
        d = (arr[-1] - arr[0]) // n
        for i in range(1, n):
            if arr[i] != arr[i - 1] + d:
                return arr[i - 1] + d
        return arr[0]
```

• SimplyLeet

```
# Function to find the missing number in the arithmetic
progression
def missingNumber(arr):
    # Calculate the expected difference using the first
and last elements
    n = len(arr)
    diff = (arr[-1] - arr[0]) // n # integer division
because values are integers
    # Iterate over the array to find the place where the
difference deviates from diff
    for i in range(len(arr) - 1):
        # If the difference between consecutive numbers
is not equal to the expected diff
        if arr[i+1] - arr[i] != diff:
            # Return the missing number which should be
current number + diff
```

```
        return arr[i] + diff
    # Return -1 if no missing number is found (should not
happen given the problem constraints)
    return -1
```

```
# Example usage:
# print(missingNumber([5,7,11,13])) # Output should be 9
```

• LC.ca

```
class Solution:
    def missingNumber(self, arr: List[int]) -> int:
        return (arr[0] + arr[-1]) * (len(arr) + 1) // 2 -
sum(arr)
```

Math

#1427 Perform String Shifts

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Math · String

Lists: Premium 98

Time Complexity: O(n + m), where n is the length of the string and m is the number of shift operations. Space Complexity: O(n) for the output string.

Problem

You are given a string s containing lowercase English letters, and a matrix shift, where shift[i] = [directioni, amounti]:

- directioni can be 0 (for left shift) or 1 (for right shift).
- amounti is the amount by which string s is to be shifted.
- A left shift by 1 means remove the first character of s and append it to the end.
- Similarly, a right shift by 1 means remove the last character of s and add it to the beginning.

Return the final string after all operations.

Example 1:

```
Input: s = "abc", shift = [[0,1],[1,2]]
Output: "cab"
Explanation:
[0,1] means shift to left by 1. "abc" -> "bca"
[1,2] means shift to right by 2. "bca" -> "cab"
```

Example 2:

```
Input: s = "abcdefg", shift = [[1,1],[1,1],[0,2],[1,3]]
Output: "efgabcd"
Explanation:
[1,1] means shift to right by 1. "abcdefg" -> "gabcdef"
[1,1] means shift to right by 1. "gabcdef" -> "fgabcde"
[0,2] means shift to left by 2. "fgabcde" -> "abcdefg"
[1,3] means shift to right by 3. "abcdefg" -> "efgabcd"
```

Constraints:

- 1 <= s.length <= 100
- s only contains lower case English letters.
- 1 <= shift.length <= 100
- shift[i].length == 2
- directioni is either 0 or 1.
- 0 <= amounti <= 100

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def stringShift(self, s, shift):
        # Brute Force - iterate all candidates
        for i in range(2, n + 1):
            is_valid = all(i % j != 0 for j in range(2,
i))
            if is_valid: pass
        return 0
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def stringShift(self, s: str, shift: list[list[int]])
-> str:
        move = 0

        for direction, amount in shift:
            if direction == 0:
                move -= amount
            else:
                move += amount
```

```
        move %= len(s)
        return s[-move:] + s[:-move]
```

• LeetDoocs

```
class Solution:
    def stringShift(self, s: str, shift: List[List[int]])
-> str:
        x = sum((b if a else -b) for a, b in shift)
        x %= len(s)
        return s[-x:] + s[:-x]
```

• SimplyLeet

```
# Python solution

def stringShift(s, shift):
    # Initialize the net shift counter
    net_shift = 0
    for direction, amount in shift:
        if direction == 0:
            net_shift -= amount # Left shift reduces
index
        else:
            net_shift += amount # Right shift increases
index
```

```
# Normalize shift to be within bounds of the string
length
net_shift %= len(s)

# A right shift can be simulated by taking the last
net_shift characters
# and concatenating them with the beginning of the
string.
return s[-net_shift:] + s[:-net_shift]

# Example usage
if __name__ == "__main__":

    s = "abcdefg"
    shift_operations = [[1,1],[1,1],[0,2],[1,3]]
    print(stringShift(s, shift_operations)) # Output:
"efgabcd"
```

• LC.ca

```
class Solution:
    def stringShift(self, s: str, shift: List[List[int]])
-> str:
        x = sum((b if a else -b) for a, b in shift)
        x %= len(s)
        return s[-x:] + s[:-x]
```

Math

#7 Reverse Integer

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Math

Lists: Grind 169

Time Complexity: O(n), where n is the number of digits in the integer. Space Complexity: O(1), only constant extra space is used.

Problem

Given a signed 32-bit integer x, return x with its digits reversed. If reversing x causes the value to go outside the signed 32-bit integer range [-231, 231 - 1], then return 0.

Assume the environment does not allow you to store 64-bit integers (signed or unsigned).

Example 1:

```
Input: x = 123
Output: 321
```

Example 2:

```
Input: x = -123
Output: -321
```

Example 3:

Input: x = 120
Output: 21

- Constraints:
- $-231 \leq x \leq 231 - 1$

Community Solutions (Python)

• WalkCC

```
class Solution:
    def reverse(self, x: int) -> int:
        ans = 0
        sign = -1 if x < 0 else 1
        x *= sign

        while x:
            ans = ans * 10 + x % 10
            x //= 10

        return 0 if ans < -2**31 or ans > 2**31 - 1 else sign * ans
```

• LeetDoocs

Math p.766

```
class Solution:
    def reverse(self, x: int) -> int:
        ans = 0
        mi, mx = -(2**31), 2**31 - 1
        while x:
            if ans < mi // 10 + 1 or ans > mx // 10:
                return 0
            y = x % 10
            if x < 0 and y > 0:
                y -= 10

            ans = ans * 10 + y
            x = (x - y) // 10
        return ans
```

• SimplyLeet

```
# Function to reverse an integer
def reverse(x: int) -> int:
    # Define the maximum and minimum limits for a 32-bit signed integer
    INT_MAX = 2**31 - 1
    INT_MIN = -2**31

    reversed_num = 0 # This will store the reversed number

    # Process each digit until x becomes 0
    while x != 0:
```

Math p.767

```
# Pop the last digit from x
# Use modulo 10 to get the last digit
# For negative numbers, modulo may yield negative
remainder = x % 10 if x > 0 else x % -10
# Remove the last digit from x by integer division
x = x // 10

# Check if appending the digit would cause overflow
if reversed_num > INT_MAX // 10 or (reversed_num == INT_MAX // 10 and pop > 7):
    return 0

if reversed_num < INT_MIN // 10 or (reversed_num == INT_MIN // 10 and pop < -8):
    return 0

# Append the digit to the reversed number
reversed_num = reversed_num * 10 + pop

return reversed_num

# Example usage:
print(reverse(123)) # Output: 321

print(reverse(-123)) # Output: -321
print(reverse(120)) # Output: 21
```

• LC.ca

Math p.768

```
class Solution:
    def reverse(self, x: int) -> int:
        ans = 0
        mi, mx = -(2**31), 2**31 - 1
        while x:
            if ans < mi // 10 + 1 or ans > mx // 10:
                return 0
            y = x % 10
            if x < 0 and y > 0:
                y -= 10

            ans = ans * 10 + y
            x = (x - y) // 10
        return ans
```

Math p.769

Math

#50 Pow(x, n)

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Math · Recursion

Lists: Grind 169

Time Complexity: $O(\log |n|)$ Space Complexity: $O(1)$ for the iterative solution, or $O(\log |n|)$ if implemented recursively

Problem

Implement `pow(x, n)`, which calculates x raised to the power n (i.e., x^n).

Example 1:

Input: $x = 2.00000$, $n = 10$
Output: 1024.00000

Example 2:

Input: $x = 2.10000$, $n = 3$
Output: 9.26100

Example 3:

Input: $x = 2.00000$, $n = -2$
Output: 0.25000
Explanation: $2^{-2} = 1/2^2 = 1/4 = 0.25$

Constraints:

- $-100.0 < x < 100.0$

Math p.770

```
class Solution:
    def myPow(self, x: float, n: int) -> float:
        def qpow(a: float, n: int) -> float:
            ans = 1
            while n:
                if n & 1:
                    ans *= a
                a *= a
                n >>= 1
            return ans
```

return qpow(x, n) if n >= 0 else 1 / qpow(x, -n)

• SimplyLeet

```
# Python implementation of pow(x, n) using exponentiation by squaring
def myPow(x, n):
    # if exponent is negative, invert x and make n positive
    if n < 0:
        x = 1 / x
        n = -n

    result = 1.0 # initialize result
    while n:
        # if the current exponent is odd, multiply result by the base
```

Math p.772

```
if n % 2:
    result *= x
    x *= x # square the base for the next iteration
    n //= 2 # divide the exponent by 2
return result
```

```
# Example usage:
print(myPow(2.0, 10)) # Output: 1024.0
print(myPow(2.1, 3)) # Output: 9.261000000000001
print(myPow(2.0, -2)) # Output: 0.25
```

• LC.ca

```
class Solution:
    def myPow(self, x: float, n: int) -> float:
        def qpow(a: float, n: int) -> float:
            ans = 1
            while n:
                if n & 1:
                    ans *= a
                a *= a
                n >>= 1
            return ans

        return qpow(x, n) if n >= 0 else 1 / qpow(x, -n)
```

Math p.773

Math

#380 Insert Delete GetRandom O(1)

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Math · Design

Lists: Grind 169

Time Complexity: $O(1)$ on average for insert, remove, and getRandom operations. Space Complexity: $O(n)$ where n is the number of elements stored in the data structure.

Problem

Implement the `RandomizedSet` class:

- `RandomizedSet()` Initializes the `RandomizedSet` object.
- `bool insert(int val)` Inserts an item `val` into the set if not present. Returns `true` if the item was not present, `false` otherwise.
- `bool remove(int val)` Removes an item `val` from the set if present. Returns `true` if the item was present, `false` otherwise.
- `int getRandom()` Returns a random element from the current set of elements (it's guaranteed that at least one element exists when this method is called). Each element must have the same probability of being returned.

You must implement the functions of the class such that each function works in average $O(1)$ time complexity.

Example 1:

Math p.774


```
Input
["RandomizedSet", "insert", "remove", "insert",
 "getRandom", "remove", "insert", "getRandom"]
[[], [1], [2], [2], [], [1], [2], []]
Output
[null, true, false, true, 2, true, false, 2]
```

Explanation

```
RandomizedSet randomizedSet = new RandomizedSet();
```

Constraints:

- `-231 <= val <= 231 - 1`
- At most $2 * 10^5$ calls will be made to `insert`, `remove`, and `getRandom`.
- There will be at least one element in the data structure when `getRandom` is called.

Community Solutions (Python)

- WalkCC

```
class RandomizedSet:
    def __init__(self):
        self.vals = []
        self.valToIndex = collections.defaultdict(int) # {val: index in vals}

    def insert(self, val: int) -> bool:
        if val in self.valToIndex:
            return False
        self.valToIndex[val] = len(self.vals)
        self.vals.append(val)
```

Math

p.775

```

    return True

def remove(self, val: int) -> bool:
    if val not in self.valToIndex:
        return False
    index = self.valToIndex[val]
    # The order of the following two lines is important
    when vals.size() == 1.
    self.valToIndex[self.vals[-1]] = index
    del self.valToIndex[val]
    self.vals[index] = self.vals[-1]

    self.vals.pop()
    return True

def getRandom(self) -> int:
    index = random.randint(0, len(self.vals) - 1)
    return self.vals[index]

```

- LeetDoocs

```
class RandomizedSet:
    def __init__(self):
        self.d = {}
        self.q = []

    def insert(self, val: int) -> bool:
        if val in self.d:
            return False
        self.d[val] = len(self.q)
        self.q.append(val)
```

Math

p.776

```

    return True

def remove(self, val: int) -> bool:
    if val not in self.d:
        return False
    i = self.d[val]
    self.d[self.q[-1]] = i
    self.q[i] = self.q[-1]
    self.q.pop()
    self.d.pop(val)

    return True

def getRandom(self) -> int:
    return choice(self.q)

```

Your RandomizedSet object will be instantiated and lled as such:

```

obj = RandomizedSet()
param_1 = obj.insert(val)
param_2 = obj.remove(val)

param_3 = obj.getRandom()

```

- **SimplyLeet**

Math

p.777

```
import random

class RandomizedSet:
    def __init__(self):
        # List to store the values
        self.values = []
        # Dictionary to map a value to its index in the list
        self.val_to_index = {}

    def insert(self, val: int) -> bool:
        # If the value is already present, do not insert.
        if val in self.val_to_index:
            return False
        # Append the value and store its index.
        self.val_to_index[val] = len(self.values)
        self.values.append(val)
        return True

    def remove(self, val: int) -> bool:
        # If the value is not present, return False.
```

Math

p.778

```

    if val not in self.val_to_index:
        return False
    # Index of element to remove.
    index = self.val_to_index[val]
    # Get the last element.
    last_element = self.values[-1]
    # Place the last element at the index of the
    element to remove.
    self.values[index] = last_element
    self.val_to_index[last_element] = index
    # Remove the last element from the list.

    self.values.pop()
    # Delete the mapping for the removed element.
    del self.val_to_index[val]
    return True

def getRandom(self) -> int:
    # Return a random element from the list.
    return random.choice(self.values)

```

- LC.ca

```
...
my_dict.pop('b') # vs del d[k]

my_dict = {'a': 1, 'b': 2, 'c': 3}
val = my_dict.pop('b')
print(my_dict) # {'a': 1, 'c': 3}
print(val) # 2

>>> item = my_dict.pop()
Traceback (most recent call last):
```

Math

p.779

```
File "<stdin>", line 1, in <module>
TypeError: pop expected at least 1 argument, got 0

>>> my_dict = {'a': 1, 'b': 2, 'c': 3}
>>> del my_dict['b']
>>> my_dict
{'a': 1, 'c': 3}
...

from collections import defaultdict

from random import choice

class RandomizedSet(object):
    def __init__(self):
        """
        Initialize your data structure here.
        """

    # value (map) -> its index -> value (list)
    self.d = {}

    self.a = [] # introduce it purely for random

    def insert(self, val):
        """
        Inserts a value to the set. Returns true if the set
        did not already contain the specified element.
        :type val: int
        :rtype: bool
        """
        if val in self.d:
            return False
```

Math

p.780

```

self.a.append(val)
self.d[val] = len(self.a) - 1
return True

def remove(self, val):
    """
    Removes a value from the set. Returns true if the set
    contained the specified element.
    :type val: int
    :rtype: bool
    """
    if val not in self.d:
        return False
    index = self.d[val]

    # process last index/val
    self.a[index] = self.a[-1]
    self.d[self.a[-1]] = index

    # process to be deleted index/val
    self.a.pop() # delete in list

    del self.d[val] # delete in dict
    # or, self.d.pop(val)
    return True

def getRandom(self):
    """
    Get a random element from the set.
    :rtype: int
    """
    return self.a[random.randrange(0, len(self.a))]

```

Math

p.781

```
# return random.choice(self.a)

#####

# Your RandomizedSet object will be instantiated and
called as such:
# obj = RandomizedSet()
# param_1 = obj.insert(val)
# param_2 = obj.remove(val)
# param_3 = obj.getRandom()

'''
>>> import random
>>> random.choice([1,2,3,4,5])
3
>>> random.choice([1,2,3,4,5])
4
>>> random.choice([1,2,3,4,5])
4
'''

class RandomizedSet:

    def __init__(self):
        self.m = {}
        self.l = []

    def insert(self, val: int) -> bool:
        if val in self.m:
            return False
        self.m[val] = len(self.l)
        self.l.append(val)
        return True
```

Math

Sheet 87 / 317 · LeetMastery By Pattern · Python Only · 9-up Portrait

Math

```
def remove(self, val: int) -> bool:
    if val not in self.m:
        return False
    idx = self.m[val]
    self.l[idx] = self.l[-1]
    self.m[self.l[-1]] = idx
    self.l.pop()
    self.m.pop(val)
    return True

def getRandom(self) -> int:
    return random.choice(self.l)
```

Math

p.783

Math

#1017 Convert to Base -2MED

LeetCode · LeetCode · LeetCode · LeetCode

Math

Lists: Denny Zhang

Time Complexity: $O(\log n)$ - the loop runs for about the number of digits in the base -2 representation. Space Complexity: $O(\log n)$ - extra space for storing the computed digits.

Problem

Given an integer n, return a binary string representing its representation in base -2.

Note that the returned string should not have leading zeros unless the string is "0".

Example 1:

Input: n = 2

Output: "110"

Explanation: $(-2)2 + (-2)1 = 2$

Example 2:

Input: n = 3

Output: "111"

Explanation: $(-2)2 + (-2)1 + (-2)0 = 3$

Example 3:

Math

p.784

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

LeetCode · LeetCode · LeetCode · LeetCode

Math

Lists: CodeSignal

Track node→color mapping and color→count mapping. For each query: if node was previously colored, decrement old color count (remove if 0). Set new color, increment its count. Answer is len(color_count)

Problem

You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.785

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

LeetCode · LeetCode · LeetCode · LeetCode

Math

Lists: CodeSignal

Track node→color mapping and color→count mapping. For each query: if node was previously colored, decrement old color count (remove if 0). Set new color, increment its count. Answer is len(color_count)

Problem

You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.786

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

LeetCode · LeetCode · LeetCode · LeetCode

Math

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You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.787

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

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Math

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Problem

You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.788

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

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Math

Lists: CodeSignal

Track node→color mapping and color→count mapping. For each query: if node was previously colored, decrement old color count (remove if 0). Set new color, increment its count. Answer is len(color_count)

Problem

You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.789

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

LeetCode · LeetCode · LeetCode · LeetCode

Math

Lists: CodeSignal

Track node→color mapping and color→count mapping. For each query: if node was previously colored, decrement old color count (remove if 0). Set new color, increment its count. Answer is len(color_count)

Problem

You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.790

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

LeetCode · LeetCode · LeetCode · LeetCode

Math

Lists: CodeSignal

Track node→color mapping and color→count mapping. For each query: if node was previously colored, decrement old color count (remove if 0). Set new color, increment its count. Answer is len(color_count)

Problem

You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.791

Math

#3160 Find the Number of Distinct Colors Among the BallsMED

LeetCode · LeetCode · LeetCode · LeetCode

Math

Lists: CodeSignal

Track node→color mapping and color→count mapping. For each query: if node was previously colored, decrement old color count (remove if 0). Set new color, increment its count. Answer is len(color_count)

Problem

You are given an integer limit and a 2D array queries of size $n \times 2$.

There are limit + 1 balls with distinct labels in the range [0, limit]. Initially, all balls are uncolored. For every query in queries that is of the form [x, y], you mark ball x with the color y. After each query, you need to find the number of colors among the balls.

Return an array result of length n, where result[i] denotes the number of colors after ith query.

Note that when answering a query, lack of a color will not be considered as a color.

Example 1:

Input: limit = 4, queries = [[1,4],[2,5],[1,3],[3,4]]

Math

p.792

```
# Update ball's color to the new color
ball_color[ball] = color
# Increase the frequency count of the new color
color_count[color] = color_count.get(color, 0) + 1

# The number of distinct colors is the number of
keys in color_count
results.append(len(color_count))

return results

# Example usage:
# limit = 4
# queries = [[1,4],[2,5],[1,3],[3,4]]
# print(distinctColors(limit, queries)) # Output:
[1,2,3]
```

```
• LC.ca

class Solution:
    def queryResults(self, limit: int, queries:
List[List[int]]) -> List[int]:
    g = {}
    cnt = Counter()
    ans = []
    for x, y in queries:
        cnt[y] += 1
        if x in g:
            cnt[g[x]] -= 1
            if cnt[g[x]] == 0:
```

```
cnt.pop(g[x])
g[x] = y
ans.append(len(cnt))
return ans
```

Math

#68 Text Justification

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · String · Simulation

Lists: CodeSignal

Time Complexity: $O(n)$ where n is the number of words, since each word is processed once. Space Complexity: $O(n)$ for storing the result and temporary buffers.

Problem

Given an array of strings words and a width maxWidth, format the text such that each line has exactly maxWidth characters and is fully (left and right) justified.

You should pack your words in a greedy approach; that is, pack as many words as you can in each line. Pad extra spaces ' ' when necessary so that each line has exactly maxWidth characters.

Extra spaces between words should be distributed as evenly as possible. If the number of spaces on a line does not divide evenly between words, the empty slots on the left will be assigned more spaces than the slots on the right.

For the last line of text, it should be left-justified, and no extra space is inserted between words.

Note:

- A word is defined as a character sequence consisting of non-space characters only.
- Each word's length is guaranteed to be greater than 0 and not exceed maxWidth.
- The input array words contains at least one word.

```
Example 1:

Input: words = ["This", "is", "an", "example", "of",
"text", "justification."], maxWidth = 16
Output:
[
    "This   is   an",
    "example of text",
    "justification.  "
]
```

```
Example 2:

Input: words =
["What","must","be","acknowledgment","shall","be"],
maxWidth = 16
Output:
[
    "What must be",
    "acknowledgment ",
    "shall be        "
]
Explanation: Note that the last line is "shall be "
instead of "shall be", because the last line must be
left-justified instead of fully-justified.
```

Example 3:

```
Input: words = ["Science","is","what","we","understand",
"well","enough","to","explain","to","a","computer.","A
rt","is","everything","else","we","do"], maxWidth = 20
Output:
[
    "Science is what we",
    "understand well",
    "enough to explain to",
    "a computer. Art is",
    "everything else we",
```

- Constraints:**
- $1 \leq \text{words.length} \leq 300$
 - $1 \leq \text{words}[i].\text{length} \leq 20$
 - words[i] consists of only English letters and symbols.
 - $1 \leq \text{maxWidth} \leq 100$
 - $\text{words}[i].\text{length} \leq \text{maxWidth}$

```
Brute Force (Python)

# Brute Force Approach
class Solution:
    def fullJustify(self, words, maxWidth):
        # Brute Force - iterate all candidates
        for i in range(2, n + 1):
            is_valid = all(i % j != 0 for j in range(2,
i))
            if is_valid: pass
            return 0
```

Community Solutions (Python)

```
• WalkCC

class Solution:
    def fullJustify(self, words: list[str], maxWidth: int)
-> list[str]:
    ans = []
    row = []
    rowLetters = 0

    for word in words:
        # If we place the word in this row, it will exceed
the maximum width.
        # Therefore, we cannot put the word in this row and
have to pad spaces
        # for each word in this row.

        if rowLetters + len(word) + len(row) > maxWidth:
            for i in range(maxWidth - rowLetters):
                row[i % (len(row) - 1 or 1)] += ' '
            ans.append(''.join(row))
            row = []
            rowLetters = 0
            row.append(word)
            rowLetters += len(word)

    return ans + [' '.join(row).ljust(maxWidth)]
```

• LeetDoocs

```
class Solution:
    def fullJustify(self, words: List[str], maxWidth:
int) -> List[str]:
    ans = []
    i, n = 0, len(words)
    while i < n:
        t = []
        cnt = len(words[i])
        t.append(words[i])
        i += 1
        while i < n and cnt + 1 + len(words[i]) <=
maxWidth:
            cnt += 1 + len(words[i])
            t.append(words[i])
            i += 1
        if i == n or len(t) == 1:
            left = ' '.join(t)
            right = ' ' * (maxWidth - len(left))
            ans.append(left + right)
            continue
        space_width = maxWidth - (cnt - len(t) + 1)
        w, m = divmod(space_width, len(t) - 1)

        row = []
        for j, s in enumerate(t[:-1]):
            row.append(s)
            row.append(' ' * (w + (1 if j < m else
0)))
            row.append(t[-1])
            ans.append(''.join(row))
            return ans
```

• SimplyLeet

```
def fullJustify(words, maxWidth):
    result = [] # Final list of justified text lines.
    current_words = [] # Current line's words.
    current_len = 0 # Total length of current line's
words (excluding spaces).

    # Process all words.
    for word in words:
        # Check if adding the next word would exceed the
maxWidth when including minimum one space between words.
        if current_words and (current_len + len(word) +
len(current_words)) > maxWidth:
            total_spaces = maxWidth - current_len # Total
spaces to distribute.

            gaps = len(current_words) - 1 # Number of
gaps between words.

            # If there is only one word, left justify.
            if gaps == 0:
                line = current_words[0] + ' ' *

            total_spaces
            else:
                # Calculate even spaces and extras to add
from the left.
                space_per_gap, extra =
divmod(total_spaces, gaps)
                line = ''
                for i in range(gaps):
```

```
# Distribute an extra space to the
leftmost gaps.
spaces = space_per_gap + (1 if i <
extra else 0)
line += current_words[i] + ' ' *
spaces
line += current_words[-1] # Append the
last word.
result.append(line)
# Reset the current line to start with the
current word.
current_words = []
current_len = 0

# Add the word to the current line.

current_words.append(word)
current_len += len(word)

# Process the last line, which is left-justified.
line = ' '.join(current_words)
line += ' ' * (maxWidth - len(line))
result.append(line)
return result

# Example usage:

words = ["This", "is", "an", "example", "of", "text",
"justification."]
maxWidth = 16
print(fullJustify(words, maxWidth))
```

• LC.ca

```

...
Explanation:

- The function `fullJustify` takes a list of words and a maximum width `maxWidth` as inputs.
- It iterates over each word, deciding whether to add the current word to the current line (`cur`) or to start a new line.
- If adding the current word to the line would exceed `maxWidth`, it justifies the current line by adding extra spaces between words as needed, then starts a new line.
- Once all words are processed, it left-justifies the last line by joining the remaining words in `cur` with a single space and then using `.ljust(maxWidth)` to ensure the line is of maximum width.
- The result is a list of strings, where each string represents a justified line of text.

```

This solution carefully handles edge cases, such as when there's only one word in the line (avoiding division by zero) and ensuring the last line is left-justified instead of fully justified.

```

...

...
>>> cur = [1]
>>> len(cur) - 1
0
>>> ( len(cur) - 1 or 1 )
1
...

```

Math p.802

```

...
# api: string.ljust(width[, fillchar])

>>> "Hello".ljust(10)
'Hello '
>>> "Hello".ljust(10, '-')
'Hello-----'
>>> "Hello".ljust(2)
'Hello'

>>> "Hello World".ljust(20)
'Hello World '
>>> "Hello World".ljust(20, '-')
'Hello World-----'

# api: string.rjust(width[, fillchar])
>>> "Hello".rjust(10)
' Hello'
>>> "Hello".rjust(10, '-')
'-----Hello'

>>> "Hello".rjust(2)
'Hello'

>>> "Hello World".rjust(20)
' Hello World'
>>> "Hello World".rjust(20, '-')
'-----Hello World'
...

```

Math p.803

```

class Solution:
    def fullJustify(self, words: List[str], maxWidth: int) -> List[str]:
        res, cur, num_of_letters = [], [], 0

        for w in words:
            # len(cur) is the space count, from last round, right one less after adding 'w'
            if num_of_letters + len(cur) + len(w) > maxWidth:
                for i in range(maxWidth - num_of_letters):
                    # The "or 1" part is for dealing with the edge case `len(cur) == 1`

                    cur[i % ( len(cur) - 1 or 1 )] += ' '
                res.append(''.join(cur))
                cur, num_of_letters = [], 0
            cur += [w]
            num_of_letters += len(w)

        return res + [ ' '.join(cur).ljust(maxWidth)]

#####

```

Math p.804

```

...
>>> 17 % 3
2
>>> divmod(17,3)
(5, 2)
...

...
>>> ['This', 'is', 'an']

```

```

['This', 'is', 'an']
>>> t = ['This', 'is', 'an']
>>> spaces = [' ', ' ', ' ']

```

```

# note: here j starts from 0
>>> for j, word in enumerate(t[1:]):
...     print(j)
...     print(word)
...
0

```

Math p.805

```

is
1
an
...

class Solution:
    def fullJustify(self, words: List[str], maxWidth: int) -> List[str]:
        def partition(n, cnt):
            base, mod = divmod(n, cnt)
            res = ['{' + ' ' * (base + (i < mod))}] for i in range(cnt)]

        return res

        ans = []
        i, n = 0, len(words)
        while i < n: # one line per iteration
            t = [words[i]]
            cnt = len(words[i])
            i += 1 # move to next word
            while i < n and cnt + 1 + len(words[i]) <= maxWidth: # greedy search for one line
                cnt += 1 + len(words[i])

```

Math p.806

```

t.append(words[i])
i += 1
if i == n or len(t) == 1:
    # this is the last line or only one (super-long) word in a line
    left = ' '.join(t)
    right = ' ' * (maxWidth - len(left))
    ans.append(f'{left}{right}')
    continue # so i != n , so only one word, no need to add spaces.
    # e.g. a word with the same length as maxWidth,
    # or, e.g. the 2nd word is super long and cannot fit in current line

    # if i==n, then will not enter next while
    words_width = cnt - (len(t) - 1) # pure words total length, with no spaces
    # spaces
    in-between these words in t[: len(t) - 1
    space_width = maxWidth - words_width
    spaces = partition(space_width, len(t) - 1)
    sb = [t[0]] # 1st word of this line, no space before it
    for j, word in enumerate(t[1:]): # last word has no following space
        sb.append(spaces[j]) # j starts from 0
        sb.append(word)
    ans.append(''.join(sb))

```

Math p.807

```

        return ans

#####

class Solution:
    def fullJustify(self, words: List[str], maxWidth: int) -> List[str]:
        left = 0
        result = []

        while left < len(words):

            right = self.findRight(left, words, maxWidth)
            result.append(self.justify(left, right, words, maxWidth))
            left = right + 1

        return result

    def findRight(self, left, words, maxWidth):
        right = left
        word_length = len(words[right])

```

Math p.808

```

        while (right + 1) < len(words) and (word_length + 1 + len(words[right+1])) <= maxWidth:
            right += 1
            word_length += 1 + len(words[right])

        return right

    def justify(self, left, right, words, maxWidth):
        if right - left == 0:
            return self.padResult(words[left], maxWidth)

        is_last_line = (right == len(words) - 1)
        num_spaces = right - left
        total_space = maxWidth - self.wordLength(left, right, words)

        space = " " if is_last_line else " " * (total_space // num_spaces)
        remainder = 0 if is_last_line else total_space % num_spaces

        result = ""
        for i in range(left, right):
            result += words[i]

```

Math p.809

```

        result += space
        if remainder > 0:
            result += " "
            remainder -= 1

        result += words[right]
        result += " " * (maxWidth - len(result))

        return result

    def wordLength(self, left, right, words):
        words_length = 0
        for i in range(left, right+1):
            words_length += len(words[i])
        return words_length

    def padResult(self, result, maxWidth):
        return result + " " * (maxWidth - len(result))

```

Math p.810

	>> Quick Review — Math 12 questions · insights · complexity · solution #9 Palindrome Number [Easy] Key Insights - Negative numbers are not palindromic. - A number ending in 0 (except 0 itself) cannot be a palindrome. - Instead of reversing the entire number (risking overflow), reverse only half of the number and compare with the other half. - For numbers with an odd number of digits, the middle digit can be disregarded. Space & Time Time Complexity: $O(\log_{10}(n))$ – Each iteration reduces the number by a factor of 10. Space Complexity: $O(1)$ – Only constant extra space is used. Solution The solution checks for obvious non-palindrome cases first (negative numbers and numbers ending with 0 that are not 0). Then, it reverses half of the digits of the number while the original number is larger than the reversed half. Finally, it compares the remaining part of	the original number with the reversed half. In the case of an odd number of digits, the middle digit is removed by dividing the reversed half by 10 before the final comparison. #13 Roman to Integer [Easy] Key Insights - Map each Roman numeral character to its integer value using a hash table or dictionary. - Loop through the string and check if the current numeral is less than the next numeral; if so, subtract its value. - Otherwise, add its value to the total. - This approach takes advantage of the subtractive notation used in Roman numerals. Space & Time Time Complexity: $O(n)$, where n is the length of the input string. Space Complexity: $O(1)$, since the mapping and additional variables use constant space. Solution We use a dictionary (or hash map) to store the values of each Roman numeral. As we iterate over the string, we compare the current numeral's value with the next numeral's value. When a numeral of smaller value appears before a numeral of larger value, we subtract
Math p.811	Math p.812	Math p.813
the current numeral's value; otherwise, we add it. This technique efficiently handles both the standard and subtractive cases in one pass through the string. #504 Base 7 [Easy] Key Insights - Handle the edge case where num is 0. - Use division and modulo operations to extract base 7 digits. - Manage negative numbers by converting to positive and reattaching the minus sign. - Build the base 7 digits in reverse order during iteration. Space & Time Time Complexity: $O(\log_7(n))$ where n is the absolute value of the number. Space Complexity: $O(\log_7(n))$ to store the digits of the number. Solution The algorithm follows these steps: - Check if num equals 0 and return "0". - Determine if the number is negative. If so, work with its absolute value but remember the negative flag. - Iteratively compute the remainder when divided by 7 (using modulo) and update the number by integer division by 7. - Append each remainder (converted to character) to a	list representing the digits. – The digits are computed in reverse order; reverse the list for the correct order. – If the original number was negative, append the "-" sign to the front. – Join the digits to form the resulting base 7 string. This approach uses basic arithmetic (division and modulo) and a list (or equivalent structure) to accumulate the computed digits. #1056 Confusing Number [Easy] Key Insights - Use a mapping dictionary for valid rotations: {0:0, 1:1, 6:9, 8:8, 9:6}. - Process the digits of n from right to left to form the rotated number. - If any digit is not present in the mapping, the number is instantly not confusing. - Compare the rotated number with the original number to check if they are different. Space & Time Time Complexity: $O(d)$, where d is the number of digits in n (approximately $O(\log n)$). Space Complexity: $O(d)$, needed to store the rotated number string. Solution	The approach is to convert the number to a string and iterate through it in reverse. For each digit, check if it is valid using the predefined mapping. If it is, append its rotated counterpart to a new string. After processing, convert the rotated string to an integer (ignoring any leading zeros). Finally, compare the rotated number to the original number. If they are different, the number is confusing. #1228 Missing Number in Arithmetic Progression [Easy] Key Insights - The complete arithmetic progression has equal differences between consecutive numbers. - The missing element can be found by calculating the expected common difference using the first and last elements. - Since exactly one number is missing, iterate through the provided array to detect where the difference deviates from the expected common difference. Space & Time Time Complexity: $O(n)$ Space Complexity: $O(1)$ Solution We first compute the expected common difference (diff) for the original arithmetic progression. Since the actual
Math p.814	Math p.815	Math p.816
progression had one more element than the current array, the formula used is: $\text{diff} = (\text{last element} - \text{first element}) / (n)$ where n is the length of the current array. Next, iterate over the array and check the difference between each pair of consecutive elements. When the actual difference deviates from diff, it indicates that the missing number is the previous element plus diff. This approach uses a simple loop and constant extra space. #1427 Perform String Shifts [Easy] Key Insights - Consolidate all shifts by calculating a net shift: subtract for left shifts and add for right shifts. - Use modulo arithmetic with the string length to avoid unnecessary full rotations. - Slice the string based on the effective shift to construct the final string. Space & Time Time Complexity: $O(n + m)$, where n is the length of the string and m is the number of shift operations. Space Complexity: $O(n)$ for the output string. Solution The solution involves computing the net shift value by iterating through the shift operations. Left shifts subtract from the net value, and right shifts add to the net value. After obtaining the cumulative shift, use	modulo with the string length to normalize the shift, ensuring it lies within bounds. The final string is constructed by slicing the string into two parts and concatenating them appropriately. This approach leverages string slicing as the primary operation for shifting. #7 Reverse Integer [Medium] Key Insights - The problem requires reversing the digits while preserving the sign. - Watch out for integer overflow due to the reversed number exceeding the 32-bit signed integer limits. - The approach involves repeatedly extracting the last digit from the number and appending it to a new reversed integer. - Overflow checks are done during each iteration before actually appending the digit. Space & Time Time Complexity: $O(n)$, where n is the number of digits in the integer. Space Complexity: $O(1)$, only constant extra space is used. Solution The solution uses a while loop to extract the last digit from the original number using modulo operation ($x \%$	10) and then removes this digit using integer division ($x // = 10$). A reversed number is constructed by multiplying the current reversed number by 10 and adding the extracted digit. Before appending each digit, the algorithm checks if the new reversed number will overflow a 32-bit signed integer by comparing it with the thresholds derived from INT_MAX and INT_MIN. The algorithm handles negative values by working with x directly since the modulo and division operations work properly with negatives in most languages. #50 Pow(x, n) [Medium] Key Insights - Use exponentiation by squaring to reduce the number of multiplications. - For a negative exponent, compute the power for n and then take the reciprocal. - Handle the edge case when n is 0 (return 1). - The algorithm works in $O(\log n)$ time complexity. Space & Time Time Complexity: $O(\log n)$ Space Complexity: $O(1)$ for the iterative solution, or $O(\log n)$ if implemented recursively Solution We use exponentiation by squaring, an algorithm that reduces the number of multiplications by repeatedly
Math p.817	Math Sheet 91 / 317 · LeetMastery By Pattern · Python Only p.818	Math 9-Up Portrait p.819

squaring the base and halving the exponent. Here's how it works:

- If n is negative, switch x to 1/x and n to -n.
- Initialize the result as 1.
- Loop while n is greater than 0: If the current n is odd, multiply the result by x. Square the base x. Divide n by 2 (using integer division). This iterative approach efficiently computes the power in logarithmic steps. The same approach can be implemented recursively.

#380 Insert Delete GetRandom O(1) [Medium]

Key Insights

- Use a dynamic array (or list) to store the elements to allow O(1) access by index.
- Use a hash table (or dictionary/map) to store the mapping from value to its index in the array.
- For removal, swap the element to be removed with the last element in the array, then delete the last element. This avoids expensive shifting of elements.
- getRandom can be implemented by generating a random index into the array.

Space & Time

Time Complexity: O(1) on average for insert, remove, and getRandom operations.
Space Complexity: O(n) where n is the number of elements stored in the data structure.

Mathp.820

- If an old color count becomes zero, remove it from the color-frequency map.
- Add the new color to the ball and increment its frequency.
- After each query, the number of distinct colors is the number of keys in the color-frequency map.

Space & Time

Time Complexity: O(q) average for q queries, since each query does only O(1) average-time hash map work.
Space Complexity: O(min(q, n)) for colored balls, plus O(q) in the worst case for distinct active colors.

Solution

We solve the problem using two hash maps. The first map stores the current color of each ball, and the second map stores the frequency of each active color. For every query, if the ball already has a color, we decrement the old color's count and remove that color from the frequency map if its count reaches zero. Then we update the ball with the new color and increment the new color's frequency. After processing the query, the number of distinct colors is simply the number of keys remaining in the frequency map.

#68 Text Justification [Hard]

Key Insights

Mathp.823

- Use a greedy approach to fill each line with as many words as possible without exceeding maxWidth.
- Calculate total space to distribute by subtracting the combined words length from maxWidth.
- If the line has more than one word, distribute extra spaces evenly; if not, pad the line with spaces at the end.
- Handle the last line separately by left-justifying the text (i.e., adding a single space between words and padding the end).

Space & Time

Time Complexity: O(n) where n is the number of words, since each word is processed once.
Space Complexity: O(n) for storing the result and temporary buffers.

Solution

The solution iterates over the list of words and groups them into lines such that the combined length of the words and minimum required spaces does not exceed maxWidth. For each completed group (line):

- If it's not the last line, calculate extra spaces required.
- If the line has multiple words, distribute the spaces evenly, adding any extra spaces from left to right.
- If the line contains a single word, append spaces to the right.
- For the last line, join words with a single space and pad the remaining width with spaces.

Mathp.824

Space Complexity: O(log|n|) – extra space for storing the computed digits.

Solution

To solve the problem, we simulate the conversion process similar to how you would convert an integer to a positive base. However, since the base here is -2, special handling of negative remainders is needed. In each iteration, we:

- Divide the current number by -2.
- Calculate the remainder. If the remainder is negative, adjust it by adding 2, and increment the quotient to compensate.
- Append the computed remainder to a list (which represents the binary digit).
- Continue until the number reduces to zero. Finally, reverse the list to form the correct string representation from the most significant to the least significant digit.

#3160 Find the Number of Distinct Colors Among the Balls [Medium]

Key Insights

- Use a hash map to store the current color of each ball that has been updated.
- Use another hash map to store how many balls currently use each color.
- When a ball is recolored, decrement the count of its old color first.

Mathp.822

Data structures used include lists (or arrays) for current words accumulation and the final result. The logic is implemented using a simulation approach to mimic text formatting.
Arrays & Hashing

#12 Arrays & Hashing

By Pattern · Python Only · 18 questions

Mathp.825

Arrays & Hashing

#1 Two Sum

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table

Lists: Grind 169

Time Complexity: O(n) – We traverse the list only once.
Space Complexity: O(n) – In the worst-case, we store all elements in the hash table.

Problem

Given an array of integers nums and an integer target, return indices of the two numbers such that they add up to target.

You may assume that each input would have exactly one solution, and you may not use the same element twice. You can return the answer in any order.

Example 1:

Input: nums = [2,7,11,15], target = 9
Output: [0,1]
Explanation: Because nums[0] + nums[1] == 9, we return [0, 1].

Example 2:

Input: nums = [3,2,4], target = 6
Output: [1,2]

Arrays & Hashingp.826

Example 3:

Input: nums = [3,3], target = 6
Output: [0,1]

Constraints:

- 2 <= nums.length <= 104
- -109 <= nums[i] <= 109
- -109 <= target <= 109
- Only one valid answer exists.

Follow-up: Can you come up with an algorithm that is less than O(n2) time complexity?

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def two_sum(self, nums, target):
        # Brute Force O(n^2) – check every pair; O(1)
        extra space
        for i in range(len(nums)):
            for j in range(i + 1, len(nums)):
                if nums[i] + nums[j] == target:
                    return [i, j]
```

Community Solutions (Python)

- WalkCC

Arrays & Hashingp.827

```
class Solution:
    def twoSum(self, nums: List[int], target: int) -> List[int]:
        numToIndex = {}

        for i, num in enumerate(nums):
            if target - num in numToIndex:
                return numToIndex[target - num], i
            numToIndex[num] = i
```

• LeetDoocs

```
class Solution:
    def twoSum(self, nums: List[int], target: int) -> List[int]:
        d = {}
        for i, x in enumerate(nums):
            if (y := target - x) in d:
                return [d[y], i]
            d[x] = i
```

• SimplyLeet

Arrays & Hashingp.828

<pre>def two_sum(nums, target): # Create a dictionary to store the number and its index index num_to_index = {} # Iterate over the list with index for index, num in enumerate(nums): # Calculate the complement needed to reach the target complement = target - num # Check if the complement is in the dictionary if complement in num_to_index: # Return the index of the complement and the current index return [num_to_index[complement], index] # Store the number with its index num_to_index[num] = index # Since the problem guarantees one solution, we don't need a return statement here. # Example usage: # print(two_sum([2, 7, 11, 15], 9)) # Output: [0, 1]</pre> Arrays & Hashingp.829	Arrays & Hashing #163 Missing RangesEAS LeetCode · SimplyLeet · WalkCC · LeetCode Array Lists: Premium 98 Time Complexity: O(n), where n is the length of the nums array, since we iterate through the array once. Space Complexity: O(1) additional space (ignoring the output list), as we only use a few extra Problem You are given an inclusive range [lower, upper] and a sorted unique integer array nums, where all elements are within the inclusive range. A number x is considered missing if x is in the range [lower, upper] and x is not in nums. Return the shortest sorted list of ranges that exactly covers all the missing numbers. That is, no element of nums is included in any of the ranges, and each missing number is covered by one of the ranges. Example 1: Arrays & Hashingp.830	<pre>Input: nums = [0,1,3,50,75], lower = 0, upper = 99 Output: [[2,2],[4,49],[51,74],[76,99]] Explanation: The ranges are: [2,2] [4,49] [51,74] [76,99]</pre> Example 2: <pre>Input: nums = [-1], lower = -1, upper = -1 Output: [] Explanation: There are no missing ranges since there are no missing numbers.</pre> Constraints: -109 <= lower <= upper <= 109 0 <= nums.length <= 100 - lower <= nums[i] <= upper - All the values of nums are unique. Brute Force (Python) Arrays & Hashingp.831
<pre># Brute Force Approach class Solution: def findMissingRanges(self, nums, lower, upper): # Brute Force O(n^2) - check all pairs without a hash map n = len(nums) for i in range(n): for j in range(i + 1, n): # process pair nums[i], nums[j] pass return [] # adjust return</pre> Community Solutions (Python) WalkCC <pre>class Solution: def findMissingRanges(self, nums: List[int], lower: int, upper: int,) -> List[List[int]]: def getRange(lo: int, hi: int) -> List[int]: if lo == hi: return [lo, lo]</pre> Arrays & Hashingp.832	<pre> return [lo, hi] if not nums: return [getRange(lower, upper)] ans = [] if nums[0] > lower: ans.append(getRange(lower, nums[0] - 1)) for prev, curr in zip(nums, nums[1:]): if curr > prev + 1: ans.append(getRange(prev + 1, curr - 1)) if nums[-1] < upper: ans.append(getRange(nums[-1] + 1, upper)) return ans</pre> LeetCode <pre>class Solution: def findMissingRanges(self, nums: List[int], lower: int, upper: int) -> List[List[int]]: n = len(nums) if n == 0: return [[lower, upper]] ans = [] if nums[0] > lower: ans.append([lower, nums[0] - 1])</pre> Arrays & Hashingp.833	<pre> for a, b in pairwise(nums): if b - a > 1: ans.append([a + 1, b - 1]) if nums[-1] < upper: ans.append([nums[-1] + 1, upper]) return ans</pre> SimplyLeet <pre># Python solution for Missing Ranges problem def findMissingRanges(nums, lower, upper): missing_ranges = [] # List to store the missing range intervals prev = lower - 1 # Initialize previous value to one less than lower # Append a sentinel at the end for easier gap handling nums.append(upper + 1) # Loop through each number, including the sentinel</pre> Arrays & Hashingp.834
<pre> for num in nums: # If there is a gap between previous and current number if num - prev >= 2: # If gap is of single number, add that number as a range if num - prev == 2: missing_ranges.append([prev + 1, prev + 1]) else: missing_ranges.append([prev + 1, num - 1]) prev = num # Update previous to current return missing_ranges # Example usage: print(findMissingRanges([0,1,3,50,75], 0, 99))</pre> LC.ca <pre>class Solution: def findMissingRanges(self, nums: List[int], lower: int, upper: int) -> List[List[int]]: n = len(nums) if n == 0: return [[lower, upper]] ans = [] if nums[0] > lower: ans.append([lower, nums[0] - 1])</pre> Arrays & Hashingp.835	<pre> for a, b in pairwise(nums): if b - a > 1: ans.append([a + 1, b - 1]) if nums[-1] < upper: ans.append([nums[-1] + 1, upper]) return ans</pre> Arrays & Hashingp.836	Arrays & Hashing #346 Moving Average from Data StreamEAS LeetCode · SimplyLeet · WalkCC · LeetCode Array · Design · Queue · Data Stream Lists: Premium 98 Time Complexity: O(1) per call to next() Space Complexity: O(window size) Problem Given a stream of integers and a window size, calculate the moving average of all integers in the sliding window. Implement the MovingAverage class: - MovingAverage(int size) Initializes the object with the size of the window size. - double next(int val) Returns the moving average of the last size values of the stream. Example 1: <pre>Input ["MovingAverage", "next", "next", "next", "next"] [[3], [1], [10], [3], [5]] Output [null, 1.0, 5.5, 4.66667, 6.0] Explanation MovingAverage movingAverage = new MovingAverage(3);</pre> Arrays & Hashingp.837

- Constraints:
- 1 <= size <= 1000
 - -105 <= val <= 105
 - At most 104 calls will be made to next.

Community Solutions (Python)

• WalkCC

```
class MovingAverage:
    def __init__(self, size: int):
        self.size = size
        self.sum = 0
        self.q = collections.deque()

    def next(self, val: int) -> float:
        if len(self.q) == self.size:
            self.sum -= self.q.popleft()
            self.sum += val

        self.q.append(val)
        return self.sum / len(self.q)
```

- LeetCode

```
class MovingAverage:

    def __init__(self, size: int):
        self.s = 0
        self.data = [0] * size
        self.cnt = 0

    def next(self, val: int) -> float:
        i = self.cnt % len(self.data)
        self.s += val - self.data[i]

        self.data[i] = val
        self.cnt += 1
        return self.s / min(self.cnt, len(self.data))

# Your MovingAverage object will be instantiated and
called as such:
# obj = MovingAverage(size)
# param_1 = obj.next(val)

class MovingAverage:
    def __init__(self, size: int):
        self.n = size
        self.s = 0
        self.q = deque()

    def next(self, val: int) -> float:
        if len(self.q) == self.n:
            self.s -= self.q.popleft()
            self.q.append(val)
```

```
self.s += val
return self.s / len(self.q)

# Your MovingAverage object will be instantiated and
called as such:
# obj = MovingAverage(size)
# param_1 = obj.next(val)

• SimplyLeet

# Python solution for Moving Average from Data Stream

from collections import deque # Import deque for
efficient queue operations

class MovingAverage:
    def __init__(self, size):
        # Initialize with the maximum window size
        self.size = size # Maximum number of elements in
the window
        self.queue = deque() # Queue to store current
window values
        self.current_sum = 0 # Running sum of the current
window
```

```
def next(self, val):
    # Add new value to the window
    if len(self.queue) == self.size:
        # If the window is full, remove the oldest
element and update the sum
        removed_val = self.queue.popleft() # Remove
the front element
        self.current_sum -= removed_val # Subtract
its value from the running sum
        # Append the new value to the queue and update
the sum
        self.queue.append(val)
        self.current_sum += val

    # Calculate and return the moving average by
dividing the running sum by the current window size
    return self.current_sum / len(self.queue)
```

• LC.ca

```
class MovingAverage:
    def __init__(self, size: int):
        self.arr = [0] * size
        self.s = 0
        self.cnt = 0

    def next(self, val: int) -> float:
        idx = self.cnt % len(self.arr) # circular array
        self.s += val - self.arr[idx]
        self.arr[idx] = val
```

```
self.cnt += 1
return self.s / min(self.cnt, len(self.arr))

# Your MovingAverage object will be instantiated and
called as such:
# obj = MovingAverage(size)
# param_1 = obj.next(val)

#####

from collections import deque

# with no sum variable, calculate on the fly
class MovingAverage:
    def __init__(self, size: int):
        self.queue = deque()
        self.size = size
        self.avg = 0

    def next(self, val: int) -> float:

        if len(self.queue) < self.size:
            self.queue.append(val)
            self.avg = sum(self.queue) / len(self.queue)
            return self.avg
        else:
            head = self.queue.popleft()
            self.queue.append(val)
            minus = head / self.size
            add = val / self.size
            self.avg = self.avg + add - minus
```

```
return self.avg

#####

from collections import deque

class MovingAverage(object):

    def __init__(self, size):
        """
        Initialize your data structure here.
        :type size: int
        """
        self.windowSize = size
        self.windowSum = 0.0
        self.data = deque([])

    def next(self, val):
        """
        :type val: int

        :rtype: float
        """
        self.windowSum += val
        data = self.data

        leftTop = 0
        if len(data) >= self.windowSize:
            leftTop = data.popleft()
            data.append(val)
```

```
self.windowSum -= leftTop
if len(data) < self.windowSize:
    return self.windowSum / len(data)
return self.windowSum / self.windowSize

# Your MovingAverage object will be instantiated and
called as such:
# obj = MovingAverage(size)
# param_1 = obj.next(val)
```

Arrays & Hashing

#760 Find Anagram Mappings

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table

Lists: Premium 98

Time Complexity: O(n) where n is the number of elements in the arrays (single pass to build the dictionary and another pass to form the mapping). Space Complexity: O(n) for storing the dictionary that

Problem

You are given two integer arrays nums1 and nums2 where nums2 is an anagram of nums1. Both arrays may contain duplicates.

Return an index mapping array mapping from nums1 to nums2 where mapping[i] = j means the ith element in nums1 appears in nums2 at index j. If there are multiple answers, return any of them.

An array a is an anagram of an array b means b is made by randomizing the order of the elements in a.

Example 1:

Input: nums1 = [12,28,46,32,50], nums2 = [50,12,32,46,28]

Output: [1,4,3,2,0]

Explanation: As mapping[0] = 1 because the 0th element of nums1 appears at nums2[1], and mapping[1] = 4 because the 1st element of nums1 appears at nums2[4], and so on.

Example 2:

Input: nums1 = [84,46], nums2 = [84,46]

Output: [0,1]

Constraints:

• 1 <= nums1.length <= 100

• nums2.length == nums1.length

• 0 <= nums1[i], nums2[i] <= 105

• nums2 is an anagram of nums1.

Brute Force (Python)

Brute Force Approach

class Solution:

def anagramMappings(self, nums1, nums2):

Brute Force O(n^2) - check all pairs without a hash map

n = len(nums1)

for i in range(n):

for j in range(i + 1, n):

process pair nums1[i], nums1[j]

pass

return [] # adjust return

Community Solutions (Python)

• WalkCC

```
class Solution:
    def anagramMappings(self, nums1: List[int], nums2: List[int]) -> List[int]:
        numToIndices = collections.defaultdict(list)

        for i, num in enumerate(nums2):
            numToIndices[num].append(i)

        return [numToIndices[num].pop() for num in nums1]
```

• LeetCode

```
class Solution:
    def anagramMappings(self, nums1: List[int], nums2: List[int]) -> List[int]:
        d = {x: i for i, x in enumerate(nums2)}
        return [d[x] for x in nums1]
```

• SimplyLeet

```
# Build a dictionary mapping each number in nums2 to a list of its indices.
def anagramMappings(nums1, nums2):
    # Dictionary to store number : list of indices
    index_dict = {}
    for index, num in enumerate(nums2):
        if num in index_dict:
            index_dict[num].append(index)
        else:
            index_dict[num] = [index]
```

```
mapping = []
# Generate the mapping for each element in nums1.
for num in nums1:
    # Pop the first available index for num and add it to mapping.
    mapping.append(index_dict[num].pop(0))
    return mapping

# Example usage:
nums1 = [12,28,46,32,50]
nums2 = [50,12,32,46,28]

print(anagramMappings(nums1, nums2)) # Output: [1,4,3,2,0] or any valid mapping
```

• LC.ca

```
class Solution:
    def anagramMappings(self, nums1: List[int], nums2: List[int]) -> List[int]:
        mapper = defaultdict(set)
        for i, num in enumerate(nums2):
            mapper[num].add(i)
        return [mapper[num].pop() for num in nums1]
```

Arrays & Hashing

#1426 Counting Elements

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table

Lists: Premium 98

Time Complexity: $O(n)$, where n is the number of elements in `arr` (set lookups are $O(1)$ on average). Space Complexity: $O(n)$, due to the use of an auxiliary set to store unique elements.

Problem

Given an integer array `arr`, count how many elements x there are, such that $x + 1$ is also in `arr`. If there are duplicates in `arr`, count them separately.

Example 1:

```
Input: arr = [1,2,3]
Output: 2
Explanation: 1 and 2 are counted cause 2 and 3 are in arr.
```

Example 2:

```
Input: arr = [1,1,3,3,5,5,7,7]
Output: 0
Explanation: No numbers are counted, cause there is no 2, 4, 6, or 8 in arr.
```

```
• 1 <= arr.length <= 1000
• 0 <= arr[i] <= 1000

Brute Force (Python)

# Brute Force Approach
class Solution:
    def countElements(self, arr):
        # Brute Force  $O(n^2)$  - check all pairs without a hash map
        n = len(arr)
        for i in range(n):
            for j in range(i + 1, n):
                # process pair arr[i], arr[j]
                pass
        return [] # adjust return
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def countElements(self, arr: List[int]) -> int:
        count = collections.Counter(arr)
        return sum(freq
                    for a, freq in count.items()
                    if count[a + 1] > 0)
```

```
class Solution:
    def countElements(self, arr: List[int]) -> int:
        cnt = Counter(arr)
        return sum(v for x, v in cnt.items() if cnt[x + 1])

• SimplyLeet

# Function to count elements where (x + 1) is in the array
def count_elements(arr):
    # Create a set of unique numbers for  $O(1)$  lookups
    unique_numbers = set(arr)
    count = 0
    # Iterate through each number in the array
    for number in arr:
        # If (number + 1) exists in the unique set, increment count
        if number + 1 in unique_numbers:
            count += 1

    return count

# Example usage:
# arr = [1, 2, 3]
# print(count_elements(arr)) # Expected output: 2

• LC.ca

class Solution:
    def countElements(self, arr: List[int]) -> int:
        cnt = Counter(arr)
        return sum(v for x, v in cnt.items() if cnt[x + 1])
```

Arrays & Hashing

#1534 Count Good Triplets

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Enumeration

Lists: CodeSignal

Time Complexity: $O(n^3)$ where n is the length of the array. Space Complexity: $O(1)$ extra space (ignoring input space).

Problem

Given an array of integers `arr`, and three integers a , b and c . You need to find the number of good triplets.

A triplet `(arr[i], arr[j], arr[k])` is good if the following conditions are true:

- $0 \leq i < j < k < \text{arr.length}$
- $|\text{arr}[i] - \text{arr}[j]| \leq a$
- $|\text{arr}[j] - \text{arr}[k]| \leq b$
- $|\text{arr}[i] - \text{arr}[k]| \leq c$

Where $|x|$ denotes the absolute value of x .

Return the number of good triplets.

Example 1:

```
Input: arr = [3,0,1,1,9,7], a = 7, b = 2, c = 3

Output: 4

Explanation: There are 4 good triplets: [(3,0,1), (3,0,1), (3,1,1), (0,1,1)].

Example 2:

Input: arr = [1,1,2,2,3], a = 0, b = 0, c = 1

Output: 0

Explanation: No triplet satisfies all conditions.
```

Constraints:

- $3 \leq \text{arr.length} \leq 100$
- $0 \leq \text{arr}[i] \leq 1000$
- $0 \leq a, b, c \leq 1000$

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def countGoodTriplets(self, arr, a, b, c):
        # Brute Force  $O(n^3)$  - check all pairs without a hash map
        n = len(arr)
        for i in range(n):
            for j in range(i + 1, n):
                for k in range(j + 1, n):
                    # process pair arr[i], arr[j]
                    pass
        return [] # adjust return
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def countGoodTriplets(self, arr: List[int], a: int, b: int, c: int) -> int:
        return sum(abs(arr[i] - arr[j]) <= a and
                    abs(arr[j] - arr[k]) <= b and
                    abs(arr[i] - arr[k]) <= c
                    for i, j, k in
                    itertools.combinations(range(len(arr)), 3))

• LeetCode

class Solution:
    def countGoodTriplets(self, arr: List[int], a: int, b: int, c: int) -> int:
        ans, n = 0, len(arr)
        for i in range(n):
            for j in range(i + 1, n):
                for k in range(j + 1, n):
                    ans += (
                        abs(arr[i] - arr[j]) <= a
                        and abs(arr[j] - arr[k]) <= b
                        and abs(arr[i] - arr[k]) <= c
                    )
        return ans

• SimplyLeet
```

```
# Python solution for Count Good Triplets
def countGoodTriplets(arr, a, b, c):
    # Initialize counter for good triplets
    good_triplets_count = 0
    n = len(arr)
    # Iterate over all triplet combinations with indices i, j, k where i < j < k
    for i in range(n):
        for j in range(i + 1, n):
            for k in range(j + 1, n):
                # Check if absolute differences satisfy the conditions

                if abs(arr[i] - arr[j]) <= a and
                    abs(arr[j] - arr[k]) <= b and abs(arr[i] - arr[k]) <= c:
                        good_triplets_count += 1
    # Return the count of good triplets
    return good_triplets_count

# Example usage:
arr = [3, 0, 1, 1, 9, 7]
a, b, c = 7, 2, 3
print(countGoodTriplets(arr, a, b, c)) # Expected output: 4

• LC.ca
```

```
class Solution:
    def countGoodTriplets(self, arr: List[int], a: int,
b: int, c: int) -> int:
    ans, n = 0, len(arr)
    for i in range(n):
        for j in range(i + 1, n):
            for k in range(j + 1, n):
                ans += (
                    abs(arr[i] - arr[j]) <= a
                    and abs(arr[j] - arr[k]) <= b
                    and abs(arr[i] - arr[k]) <= c
                )
    return ans
```

Arrays & Hashing

p.856

Arrays & Hashing

#57 Insert Interval

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array

Lists: Grind 169

Time Complexity: O(n) since we scan the list once. Space Complexity: O(n) to store the resulting list of intervals.

Problem

You are given an array of non-overlapping intervals intervals where intervals[i] = [starti, endi] represent the start and the end of the ith interval and intervals is sorted in ascending order by starti. You are also given an interval newInterval = [start, end] that represents the start and end of another interval.

Insert newInterval into intervals such that intervals is still sorted in ascending order by starti and intervals still does not have any overlapping intervals (merge overlapping intervals if necessary).

Return intervals after the insertion.

Note that you don't need to modify intervals in-place. You can make a new array and return it.

Example 1:

Arrays & Hashing

p.857

Input: intervals = [[1,3],[6,9]], newInterval = [2,5]
Output: [[1,5],[6,9]]

Example 2:

Input: intervals = [[1,2],[3,5],[6,7],[8,10],[12,16]], newInterval = [4,8]
Output: [[1,2],[3,10],[12,16]]
Explanation: Because the new interval [4,8] overlaps with [3,5],[6,7],[8,10].

Constraints:

- 0 <= intervals.length <= 104
- intervals[i].length == 2
- 0 <= starti <= endi <= 105
- intervals is sorted by starti in ascending order.
- newInterval.length == 2
- 0 <= start <= end <= 105

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def insert(self, intervals, newInterval):
        # Brute Force O(n log n) - insert, sort, then
        merge like merge-intervals
        intervals.append(newInterval)
        intervals.sort(key=lambda x: x[0])
        result = [intervals[0]]
        for iv in intervals[1:]:
            if iv[0] <= result[-1][1]:
                result[-1][1] = max(result[-1][1], iv[1])
```

Arrays & Hashing

p.858

```
else:
    result.append(iv)
return result
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def insert(self, intervals: list[list[int]],
newInterval: list[int]) -> list[list[int]]:
    n = len(intervals)
    ans = []
    i = 0

    while i < n and intervals[i][1] < newInterval[0]:
        ans.append(intervals[i])
        i += 1

    # Merge overlapping intervals.
    while i < n and intervals[i][0] <= newInterval[1]:
        newInterval[0] = min(newInterval[0],
intervals[i][0])
        newInterval[1] = max(newInterval[1],
intervals[i][1])
        i += 1

    ans.append(newInterval)

    while i < n:
```

Arrays & Hashing

p.859

```
ans.append(intervals[i])
i += 1

return ans
```

• LeetDoocs

```
class Solution:
    def insert(
        self, intervals: List[List[int]], newInterval:
List[int]
    ) -> List[List[int]]:
        def merge(intervals: List[List[int]]) ->
List[List[int]]:
            intervals.sort()
            ans = [intervals[0]]
            for s, e in intervals[1:]:
                if ans[-1][1] < s:
                    ans.append([s, e])

        else:
            ans[-1][1] = max(ans[-1][1], e)
            return ans

        intervals.append(newInterval)
        return merge(intervals)
```

• SimplyLeet

Arrays & Hashing

p.860

Python solution for Insert Interval

```
class Solution:
    def insert(self, intervals, newInterval):
        # List to store the resulting intervals
        merged_intervals = []
        i = 0
        n = len(intervals)

        # Add all intervals that end before newInterval
        starts

        while i < n and intervals[i][1] < newInterval[0]:
            merged_intervals.append(intervals[i])
            i += 1

        # Merge overlapping intervals with newInterval
        while i < n and intervals[i][0] <=
newInterval[1]:
            # Update the start and end of newInterval by
merging with overlapping intervals
            newInterval[0] = min(newInterval[0],
intervals[i][0])
            newInterval[1] = max(newInterval[1],
intervals[i][1])
            i += 1
```

Arrays & Hashing

p.861

```
# Add the merged newInterval
merged_intervals.append(newInterval)

# Add any remaining intervals that come after
newInterval
while i < n:
    merged_intervals.append(intervals[i])
    i += 1

return merged_intervals
```

• LC.ca

```
...
>>> a = [1,2,3,4,5]
>>> a[1]
2
>>> a[~1]
4
...
class Solution:
    def insert(self, intervals: List[List[int]],
newInterval: List[int]) -> List[List[int]]:
        start = newInterval[0]
```

Arrays & Hashing

p.862

```
end = newInterval[1]

left = list(filter(lambda x: x[1] < start,
intervals)) # cast via list()
right = list(filter(lambda x: x[0] > end,
intervals))

if left + right != intervals:
    start = min(start, intervals[len(left)][0]) #
note, left not -1, because index starts at 0
    end = max(end, intervals[~len(right)][1]) #
same, starting at right with index=0

return left + [[start, end]] + right
```

#####

```
class Solution: # re-use Leetcode-56's merge solution
    def insert(self, intervals: List[List[int]],
newInterval: List[int]) -> List[List[int]]:
        intervals.append(newInterval)
        return self.merge(intervals)

    def merge(self, intervals: List[List[int]]) ->
List[List[int]]:
        ans = []
```

Arrays & Hashing

p.863

```
for intv in sorted(intervals, key=lambda x:
x[0]):
    if ans and ans[-1][1] >= intv[0]:
        ans[-1][1] = max(ans[-1][1], intv[1])
    else:
        ans.append(intv)
return ans
```

#####

class Solution:

```
def insert(
    self, intervals: List[List[int]], newInterval:
List[int]
) -> List[List[int]]:
    def merge(intervals: List[List[int]]) ->
List[List[int]]:
        intervals.sort()
        ans = [intervals[0]]
        for s, e in intervals[1:]:
            if ans[-1][1] < s:
                ans.append([s, e])
            else:
                ans[-1][1] = max(ans[-1][1], e)
                return ans

        intervals.append(newInterval)
        return merge(intervals)
```

Arrays & Hashing

p.864

Arrays & Hashing

#238 Product of Array Except SelfMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Prefix Sum

Lists: Grind 169

Time Complexity: O(n) because we iterate through the array twice. Space Complexity: O(1) extra space (excluding the output array).

Problem

Given an integer array nums, return an array answer such that answer[i] is equal to the product of all the elements of nums except nums[i].

The product of any prefix or suffix of nums is guaranteed to fit in a 32-bit integer.

You must write an algorithm that runs in O(n) time and without using the division operation.

Example 1:

Input: nums = [1,2,3,4]
Output: [24,12,8,6]

Example 2:

Input: nums = [-1,1,0,-3,3]
Output: [0,0,9,0,0]

Constraints:

Arrays & Hashing

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• 2 <= nums.length <= 105

• -30 <= nums[i] <= 30

• The input is generated such that answer[i] is guaranteed to fit in a 32-bit integer.

Follow up: Can you solve the problem in O(1) extra space complexity? (The output array does not count as extra space for space complexity analysis.)

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def productExceptSelf(self, nums):
        # Brute Force O(n^2) - for each index multiply all other elements
        n = len(nums)
        result = []
        for i in range(n):
            prod = 1
            for j in range(n):
                if i != j:
                    prod *= nums[j]
            result.append(prod)
        return result
```

Community Solutions (Python)

• WalkCC

Arrays & Hashing

p.866

```
class Solution:
    def productExceptSelf(self, nums: List[int]) -> List[int]:
        n = len(nums)
        prefix = [1] * n # prefix product
        suffix = [1] * n # suffix product

        for i in range(1, n):
            prefix[i] = prefix[i - 1] * nums[i - 1]

        for i in reversed(range(n - 1)):
            suffix[i] = suffix[i + 1] * nums[i + 1]

        return [prefix[i] * suffix[i] for i in range(n)]
```

• LeetCode

class Solution:
 def productExceptSelf(self, nums: List[int]) -> List[int]:
 n = len(nums)
 ans = [0] * n
 left = right = 1
 for i, x in enumerate(nums):
 ans[i] = left
 left *= x
 for i in range(n - 1, -1, -1):
 ans[i] *= right
 right *= nums[i]
 return ans

• SimplyLeet

Arrays & Hashing

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```
# Function to compute product of array except self
def productExceptSelf(nums):
    n = len(nums)
    # Initialize the output array with 1s for prefix multiplication
    answer = [1] * n

    # Compute prefix products: for each element answer[i] holds the product of all numbers left of i
    prefix_product = 1
    for i in range(n):
        answer[i] = prefix_product # store prefix product so far
        prefix_product *= nums[i] # update the prefix product

    # Compute suffix products and multiply with prefix products in answer array
    suffix_product = 1
    for i in range(n - 1, -1, -1):
        answer[i] *= suffix_product # multiply current answer with suffix product
        suffix_product *= nums[i] # update the suffix product

    return answer

# Example usage
nums = [1, 2, 3, 4]
print(productExceptSelf(nums)) # Output: [24, 12, 8, 6]
```

• LC.ca

Arrays & Hashing

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```
class Solution: # 2 extra arrays
    def productExceptSelf(self, nums):
        if nums is None or len(nums) == 0:
            return nums

        # Product from index=0 to index=i-1
        from_left = [1] * len(nums)
        for i in range(1, len(nums)):
            from_left[i] = nums[i - 1] * from_left[i - 1]

        # Product from index=n-1 to current position
        from_right = [1] * len(nums)
        for i in range(len(nums) - 2, -1, -1):
            from_right[i] = nums[i + 1] * from_right[i + 1]

        # Calculate result
        result = [0] * len(nums)
        for i in range(len(nums)):
            result[i] = from_left[i] * from_right[i]

        return result

#####

class Solution: # follow up, no extra space
    def productExceptSelf(self, nums: List[int]) -> List[int]:
        n = len(nums)
        ans = [0] * n
        left = right = 1
        for i, v in enumerate(nums):
```

Arrays & Hashing

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```
ans[i] = left
left *= v
for i in range(n - 1, -1, -1):
    ans[i] *= right
    right *= nums[i]
return ans

#####

class Solution(object):
    def productExceptSelf(self, nums):
        """:type nums: List[int]
        :rtype: List[int]
        """
        dp = [1] * len(nums)
        for i in range(1, len(nums)):
            dp[i] = dp[i - 1] * nums[i - 1]

        prod = 1
        for i in reversed(range(0, len(nums))):
            dp[i] = dp[i] * prod
            prod *= nums[i]
        return dp
```

Arrays & Hashing

p.870

Arrays & Hashing

#281 Zigzag IteratorMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Design · Queue · Iterator

Lists: Denny Zhang

Time Complexity: O(1) per next() call on average, with an overall O(n) for n total elements. Space Complexity: O(k) for the queue structure (where k is the number of vectors), plus the storage for the

Problem

Given two vectors of integers v1 and v2, implement an iterator to return their elements alternately.

Implement the ZigzagIterator class:

• ZigzagIterator(List<int> v1, List<int> v2) initializes the object with the two vectors v1 and v2.

• boolean hasNext() returns true if the iterator still has elements, and false otherwise.

• int next() returns the current element of the iterator and moves the iterator to the next element.

Example 1:

Arrays & Hashing

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```
Input: v1 = [1,2], v2 = [3,4,5,6]
Output: [1,3,2,4,5,6]
Explanation: By calling next repeatedly until hasNext returns false, the order of elements returned by next should be: [1,3,2,4,5,6].
```

Example 2:

Input: v1 = [1], v2 = [1]
Output: [1]

Example 3:

Input: v1 = [], v2 = [1]
Output: [1]

Constraints:

• 0 <= v1.length, v2.length <= 1000

• 1 <= v1.length + v2.length <= 2000

• -231 <= v1[i], v2[i] <= 231 - 1

Follow up: What if you are given k vectors? How well can your code be extended to such cases?

Clarification for the follow-up question: The "Zigzag" order is not clearly defined and is ambiguous for k > 2 cases. If "Zigzag" does not look right to you, replace "Zigzag" with "Cyclic".

Follow-up Example:

Input: v1 = [1,2,3], v2 = [4,5,6,7], v3 = [8,9]
Output: [1,4,8,2,5,9,3,6,7]

Community Solutions (Python)

Arrays & Hashing

p.872

• WalkCC

```
class ZigzagIterator:
    def __init__(self, v1: List[int], v2: List[int]):
        def vals():
            for i in itertools.count():
                for v in v1, v2:
                    if i < len(v):
                        yield v[i]
                self.vals = vals()
                self.n = len(v1) + len(v2)

        def next(self):
            self.n -= 1
            return next(self.vals)

        def hasNext(self):
            return self.n > 0
```

• LeetCode

```
class ZigzagIterator:
    def __init__(self, v1: List[int], v2: List[int]):
        self.cur = 0
        self.size = 2
        self.indexes = [0] * self.size
        self.vectors = [v1, v2]

        def next(self) -> int:
            vector = self.vectors[self.cur]
            index = self.indexes[self.cur]
```

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<pre> res = vector[index] self.indexes[self.cur] = index + 1 self.cur = (self.cur + 1) % self.size return res def hasNext(self) -> bool: start = self.cur while self.indexes[self.cur] == len(self.vectors[self.cur]): self.cur = (self.cur + 1) % self.size if self.cur == start: return False return True # Your ZigzagIterator object will be instantiated and called as such: # i, v = ZigzagIterator(v1, v2), [] # while i.hasNext(): v.append(i.next()) </pre> • SimplyLeet <pre> class ZigzagIterator: def __init__(self, v1, v2): # Initialize the queue with non-empty vectors paired with starting index 0 from collections import deque self.queue = deque() if v1: self.queue.append((v1, 0)) if v2: self.queue.append((v2, 0)) </pre>	<pre> def next(self): # Pop from the left vec, idx = self.queue.popleft() result = vec[idx] # If there are more elements, put it back with the next index if idx + 1 < len(vec): self.queue.append((vec, idx + 1)) return result def hasNext(self): return len(self.queue) > 0 # Example usage: # v1 = [1,2]; v2 = [3,4,5,6] # iterator = ZigzagIterator(v1, v2) # while iterator.hasNext(): # print(iterator.next(), end=" ") </pre> • LC.ca <pre> class ZigzagIteratorK: # int[0] is index of vector of vectors list, int[1] is start of a vector, int[2] is end q = [] # no map needed, if move 'vectors' a class variable vectors = [] def __init__(self, vectors): self.vectors = vectors </pre>	<pre> for i in range(len(vectors)): self.q.append([i, 0, len(vectors[i])]) def next(self): current = self.q.pop(0) i = current[0] start = current[1] end = current[2] if start + 1 < end: self.q.append([i, start + 1, end]) return self.vectors[i][start] def hasNext(self): return len(self.q) != 0 ##### </pre> <pre> class ZigzagIterator: def __init__(self, v1: List[int], v2: List[int]): self.cur = 0 self.size = 2 self.indexes = [0] * self.size self.vectors = [v1, v2] def next(self) -> int: vector = self.vectors[self.cur] index = self.indexes[self.cur] </pre>
<pre> res = vector[index] self.indexes[self.cur] = index + 1 self.cur = (self.cur + 1) % self.size return res def hasNext(self) -> bool: start = self.cur while self.indexes[self.cur] == len(self.vectors[self.cur]): self.cur = (self.cur + 1) % self.size if self.cur == start: return False return True # Your ZigzagIterator object will be instantiated and called as such: # i, v = ZigzagIterator(v1, v2), [] # while i.hasNext(): v.append(i.next()) ##### </pre> <pre> ''' iter() - builtin function, return an iterator object https://docs.python.org/3/library/functions.html#iter >>> a = map(iter, [[1,2],[3,4]]) >>> >>> b = next(a) >>> b <list_iterator object at 0x101630280> </pre>	<pre> >>> next(b) 1 >>> next(b) 2 >>> next(b) Traceback (most recent call last): File "<stdin>", line 1, in <module> StopIteration </pre> <pre> >>> b = deque(map(iter, [[1,2],[3,4]])) >>> >>> c = b.popleft() >>> c <list_iterator object at 0x1016304f0> >>> c.next() Traceback (most recent call last): File "<stdin>", line 1, in <module> AttributeError: 'list_iterator' object has no attribute 'next' >>> next(c) </pre> <pre> 1 >>> next(c) 2 >>> next(c) Traceback (most recent call last): File "<stdin>", line 1, in <module> StopIteration len(s) - Return the length (the number of items) of an object https://docs.python.org/3/library/functions.html#len </pre>	<pre> ''' from collections import deque class ZigzagIterator(object): def __init__(self, v1, v2): ''' Initialize your data structure here. :type v1: List[int] :type v2: List[int] ''' self.iters = deque(map(iter, [v1, v2])) self.total = sum(map(len, [v1, v2])) def next(self): ''' :rtype: int ''' top = self.iters.popleft() try: value = top.next() except StopIteration: return self.next() self.total -= 1 if self.total != 0: self.iters.append(top) return value def hasNext(self): </pre>
<pre> ''' ''' :rtype: bool ''' return self.total > 0 # Your ZigzagIterator object will be instantiated and called as such: # i, v = ZigzagIterator(v1, v2), [] # while i.hasNext(): v.append(i.next()) </pre>	Arrays & Hashing #307 Range Sum Query – Mutable MED LeetDoocs · SimplyLeet · WalkCC · LeetCode Array · Design · Binary Indexed Tree · Segment Tree Lists: Denny Zhang Time Complexity: O(log n) for both update and sumRange operations Space Complexity: O(n) to store the additional tree structure Problem Given an integer array nums, handle multiple queries of the following types: 1. Update the value of an element in nums. 2. Calculate the sum of the elements of nums between indices left and right inclusive where left <= right. Implement the NumArray class: • NumArray(int[] nums) Initializes the object with the integer array nums. • void update(int index, int val) Updates the value of nums[index] to be val. • int sumRange(int left, int right) Returns the sum of the elements of nums between indices left and right inclusive (i.e. nums[left] + nums[left + 1] + ... + nums[right]). Example 1:	Input ["NumArray", "sumRange", "update", "sumRange"] [[[1, 3, 5]], [0, 2], [1, 2], [0, 2]] Output [null, 9, null, 8] Explanation NumArray numArray = new NumArray([1, 3, 5]); Constraints: • 1 <= nums.length <= 3 * 10⁴ • -100 <= nums[i] <= 100 • 0 <= index < nums.length • -100 <= val <= 100 • 0 <= left <= right < nums.length • At most 3 * 10⁴ calls will be made to update and sumRange. Community Solutions (Python) • WalkCC <pre> class FenwickTree: def __init__(self, n: int): self.sums = [0] * (n + 1) def add(self, i: int, delta: int) -> None: while i < len(self.sums): self.sums[i] += delta i += FenwickTree.lowbit(i) def get(self, i: int) -> int: </pre>
Arrays & Hashing p.880	Arrays & Hashing p.881	Arrays & Hashing p.882


```
summ = 0
while i > 0:
    summ += self.sums[i]
    i -= FenwickTree.lowbit(i)
return summ

@staticmethod
def lowbit(i: int) -> int:
    return i & -i

class NumArray:
    def __init__(self, nums: List[int]):
        self.nums = nums
        self.tree = FenwickTree(len(nums))
        for i, num in enumerate(nums):
            self.tree.add(i + 1, num)

    def update(self, index: int, val: int) -> None:
        self.tree.add(index + 1, val - self.nums[index])

    self.nums[index] = val

    def sumRange(self, left: int, right: int) -> int:
        return self.tree.get(right + 1) - self.tree.get(left)
```

• LeetCode

```
class BinaryIndexedTree:
    __slots__ = ["n", "c"]

    def __init__(self, n):
        self.n = n
        self.c = [0] * (n + 1)

    def update(self, x: int, delta: int):
        while x <= self.n:
            self.c[x] += delta
            x += x & -x

    def query(self, x: int) -> int:
        s = 0
        while x > 0:
            s += self.c[x]
            x -= x & -x
        return s

class NumArray:
    __slots__ = ["tree"]

    def __init__(self, nums: List[int]):
        self.tree = BinaryIndexedTree(len(nums))
        for i, v in enumerate(nums, 1):
            self.tree.update(i, v)

    def update(self, index: int, val: int) -> None:
        prev = self.sumRange(index, index)
```

```
self.tree.update(index + 1, val - prev)

def sumRange(self, left: int, right: int) -> int:
    return self.tree.query(right + 1) - self.tree.query(left)

# Your NumArray object will be instantiated and called as such:
# obj = NumArray(nums)
# obj.update(index, val)
# param_2 = obj.sumRange(left, right)

• SimplyLeet

# Python implementation using a Binary Indexed Tree (Fenwick Tree)
class NumArray:
    def __init__(self, nums):
        # Store the original array
        self.nums = nums[:]
        self.n = len(nums)
        # Initialize the BIT with 0s, size is (n+1)
        # BIT is 1-indexed
        self.tree = [0] * (self.n + 1)
        # Build the BIT: update tree for each value in the nums array
        for i, num in enumerate(nums):
```

```
self._update_tree(i, num)

def _update_tree(self, index, delta):
    # Update the BIT for index with delta change
    i = index + 1 # Convert to 1-indexed for BIT operations
    while i <= self.n:
        self.tree[i] += delta # Add delta to current tree node
        i += i & -i # Move to the next responsible node

    def update(self, index, val):
        # Calculate the difference between new value and current value
        delta = val - self.nums[index]
        self.nums[index] = val # Update the original array
        self._update_tree(index, delta) # Update the BIT to reflect this change

    def _prefix_sum(self, index):
        # Compute the prefix sum from start up to and including index
        result = 0
        i = index + 1 # Convert to 1-indexed
        while i:
```

```
result += self.tree[i]
i -= i & -i # Move to parent node
return result

def sumRange(self, left, right):
    # Return the sum of the range [left, right]
    return self._prefix_sum(right) - self._prefix_sum(left - 1) if left > 0 else self._prefix_sum(right)

# Example usage:
# numArray = NumArray([1, 3, 5])

# print(numArray.sumRange(0, 2))
# numArray.update(1, 2)
# print(numArray.sumRange(0, 2))

• LC.ca

class BinaryIndexedTree:
    def __init__(self, n):
        self.n = n
        self.c = [0] * (n + 1)

    @staticmethod
    def lowbit(x):
        return x & -x

    def update(self, x, delta):
```

```
while x <= self.n:
    self.c[x] += delta
    x += BinaryIndexedTree.lowbit(x)

def query(self, x):
    s = 0
    while x > 0:
        s += self.c[x]
        x -= BinaryIndexedTree.lowbit(x)
    return s

class NumArray:
    def __init__(self, nums: List[int]):
        self.tree = BinaryIndexedTree(len(nums))
        for i, v in enumerate(nums, 1):
            self.tree.update(i, v)

    def update(self, index: int, val: int) -> None:
        prev = self.sumRange(index, index)
```

```
self.tree.update(index + 1, val - prev)

def sumRange(self, left: int, right: int) -> int:
    return self.tree.query(right + 1) - self.tree.query(left)

# Your NumArray object will be instantiated and called as such:
# obj = NumArray(nums)
# obj.update(index, val)
# param_2 = obj.sumRange(left, right)

#####

# Segment tree node
class STNode(object):
    def __init__(self, start, end):
        self.start = start
        self.end = end
        self.total = 0
        self.left = None
```

```
self.right = None

class SegmentedTree(object):
    def __init__(self, nums, start, end):
        self.root = self.buildTree(nums, start, end)

    def buildTree(self, nums, start, end):
        if start > end:
            return None

        if start == end:
            node = STNode(start, end)
            node.total = nums[start]
            return node

        mid = start + (end - start) // 2

        root = STNode(start, end)
        root.left = self.buildTree(nums, start, mid)

        root.right = self.buildTree(nums, mid + 1, end)
        root.total = root.left.total + root.right.total
        return root

    def updateVal(self, i, val):
        def updateVal(root, i, val):
            if root.start == root.end:
                root.total = val
                return val
            mid = root.start + (root.end - root.start) // 2
```

```
if i <= mid:
    updateVal(root.left, i, val)
else:
    updateVal(root.right, i, val)

root.total = root.left.total + root.right.total
return root.total

return updateVal(self.root, i, val)

def sumRange(self, i, j):
    def rangeSum(root, start, end):
        if root.start == start and root.end == end:
            return root.total

        mid = root.start + (root.end - root.start) // 2
        if j <= mid:
            return rangeSum(root.left, start, end)
        elif i >= mid + 1:
            return rangeSum(root.right, start, end)

        else:
            return rangeSum(root.left, start, mid) + rangeSum(root.right, mid + 1, end)

    return rangeSum(self.root, i, j)

class NumArray(object):
    def __init__(self, nums):
        """
        initialize your data structure here.
```

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Arrays & Hashing p.895	Arrays & Hashing p.896	Arrays & Hashing p.897
Arrays & Hashing p.898	Arrays & Hashing p.899	Arrays & Hashing p.900

Input: nums = [0,1,1,1,1,0,0,0]
Output: 6
Explanation: [1,1,1,0,0,0] is the longest contiguous subarray with equal number of 0 and 1.

Constraints:

- 1 <= nums.length <= 105
- nums[i] is either 0 or 1.

Brute Force (Python)

```
# Brute Force Approach
class Solution:
    def findMaxLength(self, nums):
        # Brute Force O(n^2) - check all pairs without a hash map
        n = len(nums)
        for i in range(n):
            for j in range(i + 1, n):
                # process pair nums[i], nums[j]
                pass
        return [] # adjust return
```

Community Solutions (Python)

- WalkCC

```
class Solution:
    def findMaxLength(self, nums: List[int]) -> int:
        ans = 0
        prefix = 0
        prefixToIndex = {0: -1}

        for i, num in enumerate(nums):
            prefix += 1 if num else -1
            ans = max(ans, i - prefixToIndex.setdefault(prefix, i))

        return ans
```

• LeetCode

```
class Solution:
    def findMaxLength(self, nums: List[int]) -> int:
        d = {0: -1}
        ans = s = 0
        for i, x in enumerate(nums):
            s += 1 if x else -1
            if s in d:
                ans = max(ans, i - d[s])
            else:
                d[s] = i

        return ans
```

- SimplyLeet

```
# Python solution for Contiguous Array

def findMaxLength(nums):
    # Dictionary to store the earliest occurrence of each prefix sum
    prefix_indices = {0: -1}
    prefix_sum = 0
    max_length = 0

    # Iterate over the array
    for index, num in enumerate(nums):
```

```
    # Convert 0 to -1, keep 1 as is
    if num == 0:
        prefix_sum -= 1
    else:
        prefix_sum += 1

    # If the prefix sum has been seen before, a balanced subarray exists
    if prefix_sum in prefix_indices:
        # Calculate the length of the subarray
        current_length = index - prefix_indices[prefix_sum]
```

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p.901

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p.903

```
        max_length = max(max_length, current_length)
    else:
        # Store the index for this prefix sum if not already seen
        prefix_indices[prefix_sum] = index

    # Return the maximum length of balanced subarray
    return max_length
```

Example usage:
print(findMaxLength([0,1,0])) # Expected output: 2

• LC.ca

```
class Solution:
    def findMaxLength(self, nums: List[int]) -> int:
        s = ans = 0
        mp = {0: -1}
        for i, v in enumerate(nums):
            s += 1 if v == 1 else -1
            if s in mp:
                ans = max(ans, i - mp[s])
            else:
                mp[s] = i

        return ans
```

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p.904

Arrays & Hashing

#560 Subarray Sum Equals K

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Prefix Sum

Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(n)

Problem

Given an array of integers nums and an integer k, return the total number of subarrays whose sum equals to k.

A subarray is a contiguous non-empty sequence of elements within an array.

Example 1:

Input: nums = [1,1,1], k = 2
Output: 2

Example 2:

Input: nums = [1,2,3], k = 3
Output: 2

Constraints:

- 1 <= nums.length <= 2 * 10⁴
- -1000 <= nums[i] <= 1000
- -107 <= k <= 107

Brute Force (Python)

```
count += prefix_counts.get(current_sum - k, 0)
# Update the frequency count for the current sum
prefix_counts[current_sum] =
prefix_counts.get(current_sum, 0) + 1
return count
```

Example usage:
print(subarraySum([1,1,1], 2)) # Expected output: 2

• LC.ca

```
class Solution:
    def subarraySum(self, nums: List[int], k: int) -> int:
        dp = Counter({0: 1})
        ans = s = 0
        for num in nums:
            s += num
            ans += dp[s - k] # not matter if the same num like [1,1]
            dp[s] += 1
            return ans
```

#####

```
class Solution: # over time limit, not fast enough
    def subarraySum(self, nums: List[int], k: int) -> int:
        count = 0

        # sum value from 1 to i-th element
        sum_arr = [0] * (len(nums) + 1)

        # cached sum
```

Arrays & Hashing

p.908

```
# Brute Force Approach
class Solution:
    def subarraySum(self, nums, k):
        # Brute Force O(n^2) - compute sum of every contiguous subarray
        count = 0
        for i in range(len(nums)):
            total = 0
            for j in range(i, len(nums)):
                total += nums[j]
                if total == k:
```

```
                count += 1
        return count
```

Community Solutions (Python)

• WalkCC

```
class Solution:
    def subarraySum(self, nums: List[int], k: int) -> int:
        ans = 0
        prefix = 0
        count = collections.Counter({0: 1})
```

```
        for num in nums:
            prefix += num
            ans += count[prefix - k]
            count[prefix] += 1
```

```
        return ans
```

• LeetCode

Arrays & Hashing

p.906

```
class Solution:
    def subarraySum(self, nums: List[int], k: int) -> int:
        cnt = Counter({0: 1})
        ans = s = 0
        for x in nums:
            s += x
            ans += cnt[s - k]
            cnt[s] += 1
        return ans
```

• SimplyLeet

```
# Function to calculate the number of subarrays that sum to k
def subarraySum(nums, k):
    # Dictionary to store frequency of prefix sums
    prefix_counts = {0: 1} # Initialize with prefix sum 0
    current_sum = 0 # Running sum initialization
    count = 0 # To count the valid subarrays
    # Iterate over the array
    for num in nums:
        current_sum += num # Update the running (prefix) sum
        # Check if there's a prefix sum that differs from current_sum by k
```

Arrays & Hashing

p.907

```
count += prefix_counts.get(current_sum - k, 0)
# Update the frequency count for the current sum
prefix_counts[current_sum] =
prefix_counts.get(current_sum, 0) + 1
return count

# Example usage:
print(subarraySum([1,1,1], 2)) # Expected output: 2
```

• LC.ca

```
class Solution:
    def subarraySum(self, nums: List[int], k: int) -> int:
        dp = Counter({0: 1})
        ans = s = 0
        for num in nums:
            s += num
            ans += dp[s - k] # not matter if the same num like [1,1]
            dp[s] += 1
            return ans
```

#####

```
class Solution: # over time limit, not fast enough
    def subarraySum(self, nums: List[int], k: int) -> int:
        count = 0

        # sum value from 1 to i-th element
        sum_arr = [0] * (len(nums) + 1)

        # cached sum
```

Arrays & Hashing

p.908

```
for i in range(1, len(nums) + 1):
    sum_arr[i] = sum_arr[i - 1] + nums[i - 1]

for start in range(len(nums)):
    for end in range(start + 1, len(nums) + 1):
        if sum_arr[end] - sum_arr[start] == k:
            count += 1

return count
```

Arrays & Hashing

p.909

Arrays & Hashing

#1010 Pairs of Songs With Total Durations Divisible by 60

MED

LeetCode · SimplyLeetCode · WalkCC · LeetCode

Array · Hash Table · Counting

Lists: CodeSignal

Time Complexity: O(n), where n is the number of songs.
Space Complexity: O(60) → O(1), since we only need to keep track of 60 possible remainders.

Problem

You are given a list of songs where the ith song has a duration of time[i] seconds.

Return the number of pairs of songs for which their total duration in seconds is divisible by 60. Formally, we want the number of indices i, j such that i < j with (time[i] + time[j]) % 60 == 0.

Example 1:

Input: time = [30,20,150,100,40]

Output: 3

Explanation: Three pairs have a total duration divisible by 60:
(time[0] = 30, time[2] = 150): total duration 180
(time[1] = 20, time[3] = 100): total duration 120
(time[1] = 20, time[4] = 40): total duration 60

Example 2:

Arrays & Hashing

p.910

Arrays & Hashing

p.911

Arrays & Hashing

p.912

Arrays & Hashing

#1272 Remove Interval

MED

LeetCode · SimplyLeetCode · WalkCC · LeetCode

Array

Lists: Premium 98

Time Complexity: O(n), where n is the number of intervals. Space Complexity: O(n), for storing the resulting intervals.

Problem

A set of real numbers can be represented as the union of several disjoint intervals, where each interval is in the form [a, b). A real number x is in the set if one of its intervals [a, b) contains x (i.e. a ≤ x < b).

You are given a sorted list of disjoint intervals intervals representing a set of real numbers as described above, where intervals[i] = [ai, bi) represents the interval [ai, bi). You are also given another interval toBeRemoved.

Return the set of real numbers with the interval toBeRemoved removed from intervals. In other words, return the set of real numbers such that every x in the set is in intervals but not in toBeRemoved. Your answer should be a sorted list of disjoint intervals as described above.

Example 1:

Arrays & Hashing

p.913

Arrays & Hashing

p.914

Arrays & Hashing

p.915

Arrays & Hashing

#1272 Remove Interval

MED

LeetCode · SimplyLeetCode · WalkCC · LeetCode

Array

Lists: Premium 98

Time Complexity: O(n), where n is the number of intervals. Space Complexity: O(n), for storing the resulting intervals.

Problem

A set of real numbers can be represented as the union of several disjoint intervals, where each interval is in the form [a, b). A real number x is in the set if one of its intervals [a, b) contains x (i.e. a ≤ x < b).

You are given a sorted list of disjoint intervals intervals representing a set of real numbers as described above, where intervals[i] = [ai, bi) represents the interval [ai, bi). You are also given another interval toBeRemoved.

Return the set of real numbers with the interval toBeRemoved removed from intervals. In other words, return the set of real numbers such that every x in the set is in intervals but not in toBeRemoved. Your answer should be a sorted list of disjoint intervals as described above.

Example 1:

Arrays & Hashing

p.916

Arrays & Hashing

p.917

Arrays & Hashing

p.918

```
• LeetCode

class Solution:
    def removeInterval(
        self, intervals: List[List[int]], toBeRemoved:
        List[int]
    ) -> List[List[int]]:
        x, y = toBeRemoved
        ans = []
        for a, b in intervals:
            if a >= y or b <= x:
                ans.append([a, b])
            else:
                if a < x:
                    ans.append([a, x])
                if b > y:
                    ans.append([y, b])
        return ans
```

```
• SimplyLeet

def removeInterval(intervals, toBeRemoved):
    # Unpack the removal interval boundaries.
    remove_start, remove_end = toBeRemoved
    result = []

    # Iterate over each interval.
    for interval in intervals:
        start, end = interval

        # If the interval is completely outside
        toBeRemoved, add as is.
```

```
        if end <= remove_start or start >= remove_end:
            result.append([start, end])
        else:
            # If there is a non-overlapping left part,
            add it.
            if start < remove_start:
                result.append([start, remove_start])
            # If there is a non-overlapping right part,
            add it.
            if end > remove_end:
                result.append([remove_end, end])

        return result

# Example usage:
intervals = [[0,2],[3,4],[5,7]]
toBeRemoved = [1,6]
print(removeInterval(intervals, toBeRemoved)) # Expected
output: [[0,1],[6,7]]
```

```
• LC.ca

'''
>>> a
[1, 2, 3, 4, 5]
>>>
>>> x, y = a
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
ValueError: too many values to unpack
>>> x,y,z,p,q = a
>>> x
```

```
1
>>> y
2
>>> z
3
...

class Solution:
    def removeInterval(
        self, intervals: List[List[int]], toBeRemoved:
        List[int]
    ) -> List[List[int]]:

        x, y = toBeRemoved
        ans = []
        for a, b in intervals:
            if a >= y or b <= x:
                ans.append([a, b])
            else:
                if a < x:
                    ans.append([a, x]) # in this else
block, meaning b>x, so no need to append [a,min(b,x)]
                if b > y:
                    ans.append([y, b])
```

```
        # if x < a < b < y, then delete (a,b), do
nothing and skip here
        return ans

#####

class Solution:
    def removeInterval(
        self, intervals: List[List[int]], toBeRemoved:
        List[int]
    ) -> List[List[int]]:
        result = []

        for ind, intv in enumerate(intervals):
            if intv[1] < toBeRemoved[0] or toBeRemoved[1]
< intv[0]:
                result.append(intv)
            else:
                if intv[0] < toBeRemoved[0]:
                    result.append([intv[0], min(intv[1],
toBeRemoved[0])]) # min() is not needed
                if intv[1] > toBeRemoved[1]:
                    result.append([max(intv[0],
toBeRemoved[1]), intv[1]])

        return result
```

Arrays & Hashing

#1429 First Unique NumberMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Design · Queue · Data Stream

Lists: Premium 98

Time Complexity: O(1) amortized for each add and showFirstUnique call. Space Complexity: O(n), where n is the number of distinct integers encountered.

Problem

You have a queue of integers, you need to retrieve the first unique integer in the queue.

Implement the FirstUnique class:

- FirstUnique(int[] nums) Initializes the object with the numbers in the queue.
- int showFirstUnique() returns the value of the first unique integer of the queue, and returns -1 if there is no such integer.
- void add(int value) insert value to the queue.

Example 1:

```
Input:
["FirstUnique", "showFirstUnique", "add", "showFirstUnique",
"add", "showFirstUnique", "add", "showFirstUnique"]
[[[2,3,5]], [], [5], [], [2], [1], [3], []]
Output:
[null,2,null,2,null,3,null,-1]
Explanation:
FirstUnique firstUnique = new FirstUnique([2,3,5]);
firstUnique.showFirstUnique(); // return 2

Example 2:

Input:
["FirstUnique", "showFirstUnique", "add", "add", "add", "add",
"add", "showFirstUnique"]
[[[7,7,7,7,7]], [], [7], [3], [3], [7], [17], []]
Output:
[null,-1,null,null,null,null,17]
Explanation:
FirstUnique firstUnique = new
FirstUnique([7,7,7,7,7]);
firstUnique.showFirstUnique(); // return -1

Example 3:
```

```
Input:
["FirstUnique", "showFirstUnique", "add", "showFirstUnique",
"add", "showFirstUnique", "add", "showFirstUnique"]
[[[809]], [], [809], []]
Output:
[null,809,null,-1]
Explanation:
FirstUnique firstUnique = new FirstUnique([809]);
firstUnique.showFirstUnique(); // return 809
```

Constraints:

- 1 <= nums.length <= 10⁴
- 1 <= nums[i] <= 10⁸
- 1 <= value <= 10⁸
- At most 50000 calls will be made to showFirstUnique and add.

Community Solutions (Python)

```
• WalkCC

class FirstUnique:
    def __init__(self, nums: list[int]):
        self.seen = set()
        self.unique = {}
        for num in nums:
            self.add(num)

    def showFirstUnique(self) -> int:
        return next(iter(self.unique), -1)
```

```
def add(self, value: int) -> None:
    if value not in self.seen:
        self.seen.add(value)
        self.unique[value] = 1
    elif value in self.unique:
        # We have added this value before, and this is the
second time we're
        # adding it. So, erase the value from `unique`.
        self.unique.pop(value)

• LeetCode

class FirstUnique:
    def __init__(self, nums: List[int]):
        self.cnt = Counter(nums)
        self.unique = OrderedDict({v: 1 for v in nums if
self.cnt[v] == 1})

    def showFirstUnique(self) -> int:
        return -1 if not self.unique else next(v for v in
self.unique.keys())

    def add(self, value: int) -> None:
        self.cnt[value] += 1
```

```
if self.cnt[value] == 1:
    self.unique[value] = 1
elif value in self.unique:
    self.unique.pop(value)

# Your FirstUnique object will be instantiated and called
as such:
# obj = FirstUnique(nums)
# param_1 = obj.showFirstUnique()
# obj.add(value)

• SimplyLeet

import collections

class FirstUnique:
    # Initialize the object with the numbers in the
queue.
    def __init__(self, nums):
        # Using deque for efficient popleft operations.
        self.queue = collections.deque()
        # Dictionary to store frequency count.
        self.count = collections.defaultdict(int)
        # Process the initial list of numbers.
```

```
for num in nums:
    self.add(num)

# Returns the value of the first unique integer of
the queue.
def showFirstUnique(self):
    # Remove numbers from the front if they are no
longer unique.
    while self.queue and self.count[self.queue[0]] >
1:
        self.queue.popleft()
    # Return the first unique number if available.
    return self.queue[0] if self.queue else -1

# Inserts value to the queue.
def add(self, value):
    self.count[value] += 1
    # If it is the first occurrence, append it to the
queue.
    if self.count[value] == 1:
        self.queue.append(value)
```

- LC.ca

```
class FirstUnique:
    def __init__(self, nums: List[int]):
        self.cnt = Counter(nums)
        self.unique = OrderedDict({v: 1 for v in nums if
self.cnt[v] == 1})

    def showFirstUnique(self) -> int:
        return -1 if not self.unique else next(v for v in
self.unique.keys())

    def add(self, value: int) -> None:
        self.cnt[value] += 1

        if self.cnt[value] == 1:
            self.unique[value] = 1
        elif value in self.unique:
            self.unique.pop(value)

# Your FirstUnique object will be instantiated and called
as such:
# obj = FirstUnique(nums)
# param_1 = obj.showFirstUnique()
# obj.add(value)
```

Arrays & Hashing

#3159 Find Occurrences of an Element in an Array

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table

Lists: CodeSignal

Collect all indices where `nums[i] == x` into a list. For each query `q`, return `indices[q-1]` if `q <= len(indices)`, else `-1`. $O(n + m)$ time, $O(k)$ space where $k = \text{occurrences of } x$.

Problem

You are given an integer array `nums`, an integer array `queries`, and an integer `x`.

For each `queries[i]`, you need to find the index of the `queries[i]`th occurrence of `x` in the `nums` array. If there are fewer than `queries[i]` occurrences of `x`, the answer should be `-1` for that query.

Return an integer array `answer` containing the answers to all queries.

Example 1:

Input: `nums = [1,3,1,7]`, `queries = [1,3,2,4]`, `x = 1`

Output: `[0,-1,2,-1]`

Explanation:

- For the 1st query, the first occurrence of 1 is at index 0.

- For the 2nd query, there are only two occurrences of 1 in `nums`, so the answer is `-1`.
- For the 3rd query, the second occurrence of 1 is at index 2.
- For the 4th query, there are only two occurrences of 1 in `nums`, so the answer is `-1`.

Example 2:

Input: `nums = [1,2,3]`, `queries = [10]`, `x = 5`

Output: `[-1]`

Explanation:

- For the 1st query, 5 doesn't exist in `nums`, so the answer is `-1`.

- Constraints:**
- $1 \leq \text{nums.length}, \text{queries.length} \leq 105$
 - $1 \leq \text{queries}[i] \leq 105$
 - $1 \leq \text{nums}[i], x \leq 104$

Community Solutions (Python)

- WalkCC

```
class Solution:
    def occurrencesOfElement(
        self,
        nums: List[int],
        queries: List[int],
        x: int,
    ) -> List[int]:
        indices = [i for i, num in enumerate(nums) if num ==
x]
        return [indices[query - 1] if query <= len(indices)
else -1
                for query in queries]
```

- LeetDoocs
- ```
class Solution:
 def occurrencesOfElement(
 self, nums: List[int], queries: List[int], x: int
) -> List[int]:
 ids = [i for i, v in enumerate(nums) if v == x]
 return [ids[i - 1] if i - 1 < len(ids) else -1 for
i in queries]- SimplyLeet

```

```
Function to find occurrences of x at specified query
indices
def find_occurrences(nums, queries, x):
 # Precompute the indices where x occurs in nums
 positions = []
 for index, num in enumerate(nums):
 if num == x:
 positions.append(index)

 # Process each query
 results = []

 for occurrence in queries:
 # Check if the requested occurrence exists
 if occurrence <= len(positions):
 results.append(positions[occurrence - 1]) #
Convert to zero-based index
 else:
 results.append(-1)
 return results

Example usage:
nums = [1, 3, 1, 7]

queries = [1, 3, 2, 4]
x = 1
print(find_occurrences(nums, queries, x)) # Output: [0,
-1, 2, -1]
```

- LC.ca

```
class Solution:
 def occurrencesOfElement(
 self, nums: List[int], queries: List[int], x: int
) -> List[int]:
 ids = [i for i, v in enumerate(nums) if v == x]
 return [ids[i - 1] if i - 1 < len(ids) else -1 for
i in queries]
```

Arrays & Hashing

#41 First Missing Positive

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table

Lists: Grind 169

Time Complexity:  $O(n)$  as each number is swapped at most once. Space Complexity:  $O(1)$  since the rearrangement occurs in-place.

**Problem**

Given an unsorted integer array `nums`. Return the smallest positive integer that is not present in `nums`.

You must implement an algorithm that runs in  $O(n)$  time and uses  $O(1)$  auxiliary space.

**Example 1:**

Input: `nums = [1,2,0]`  
Output: 3  
Explanation: The numbers in the range `[1,2]` are all in the array.

**Example 2:**

Input: `nums = [3,4,-1,1]`  
Output: 2  
Explanation: 1 is in the array but 2 is missing.

**Example 3:**

Input: `nums = [7,8,9,11,12]`  
Output: 1  
Explanation: The smallest positive integer 1 is missing.

- Constraints:**
- $1 \leq \text{nums.length} \leq 105$
  - $-231 \leq \text{nums}[i] \leq 231 - 1$
- Brute Force (Python)**

```
Brute Force Approach
class Solution:
 def firstMissingPositive(self, nums):
 # Brute Force $O(n^2)$ - check 1,2,3,... until one
is missing
 num_set = set(nums)
 i = 1
 while i in num_set:
 i += 1
 return i
```

Community Solutions (Python)

- WalkCC



```
class Solution:
 def firstMissingPositive(self, nums: list[int]) -> int:
 n = len(nums)

 # Correct s1ot:
 # nums[i] = i + 1
 # nums[i] - 1 = i
 # nums[nums[i] - 1] = nums[i]
 for i in range(n):
 while nums[i] > 0 and nums[i] <= n and nums[nums[i]
- 1] != nums[i]:

 nums[nums[i] - 1], nums[i] = nums[i],
nums[nums[i] - 1]

 for i, num in enumerate(nums):
 if num != i + 1:
 return i + 1

 return n + 1
```

• LeetCode

```
class Solution:
 def firstMissingPositive(self, nums: List[int]) ->
int:
 n = len(nums)
 for i in range(n):
 while 1 <= nums[i] <= n and nums[i] !=
nums[nums[i] - 1]:
 j = nums[i] - 1
 nums[i], nums[j] = nums[j], nums[i]
 for i in range(n):
 if nums[i] != i + 1:
 return i + 1

 return n + 1
```

• SimplyLeet

```
Python implementation of First Missing Positive

def firstMissingPositive(nums):
 n = len(nums)
 # Rearrange numbers to their correct positions
 for i in range(n):
 while 1 <= nums[i] <= n and nums[nums[i] - 1] !=
nums[i]:
 correct_index = nums[i] - 1
 nums[i], nums[correct_index] =
nums[correct_index], nums[i]
```

```
Identify the first missing positive integer
for i in range(n):
 if nums[i] != i + 1:
 return i + 1
return n + 1

Example usage:
print(firstMissingPositive([3, 4, -1, 1])) # Output
should be 2
```

• LC.ca

```
class Solution:
 def firstMissingPositive(self, nums: List[int]) ->
int:
 def swap(i, j):
 nums[i], nums[j] = nums[j], nums[i]

 n = len(nums)
 for i in range(n):
 while 1 <= nums[i] <= n and nums[i] !=
nums[nums[i] - 1]:
 swap(i, nums[i] - 1)
 for i in range(n):

 if i + 1 != nums[i]:
 return i + 1
 return n + 1
```

>> Quick Review — Arrays & Hashing

18 questions · insights · complexity · solution

#1 Two Sum [Easy]

Key Insights

- Use a hash table to store the elements and their indices for fast lookup.
- For each element, calculate its complement: target minus the current element.
- Check if the complement exists in the hash table.
- If found, return the indices of the current element and its complement.
- This approach only requires a single pass through the array.

Space & Time

Time Complexity: O(n) – We traverse the list only once.  
Space Complexity: O(n) – In the worst-case, we store all elements in the hash table.

Solution

We use a hash table (dictionary) to map each array element to its index as we iterate. For each number, we compute the complement (target – number) and check if it already exists in our hash table. If the complement is found, we return the indices. This one-pass solution

ensures linear time complexity and avoids the need for nested loops.

#163 Missing Ranges [Easy]

Key Insights

- Use a pointer (or iterate index) to traverse the nums array while keeping track of the previous number considered.
- Check for missing numbers before the first element, between consecutive elements, and after the last element.
- When a gap is found, convert it to a range string that is either a single number (if the gap is exactly one number) or an interval.
- Handle edge cases when the nums array is empty.

Space & Time

Time Complexity: O(n), where n is the length of the nums array, since we iterate through the array once.  
Space Complexity: O(1) additional space (ignoring the output list), as we only use a few extra variables.

Solution

The solution leverages a single pass through the input array and calculates the missing ranges by comparing the expected number with the current number in nums. First, initialize a variable "prev" to one less than the lower bound so that missing numbers from the very

start of the range can be handled uniformly. For each number in the array, if there is a gap between prev and the current number (i.e., current number is at least 2 more than prev), then add the missing range (prev+1 to current number-1) to the output list. Update "prev" to the current number and finally handle any gap between the last number and the upper bound. The primary data structures used are the list for the output; the approach is iterative.

#346 Moving Average from Data Stream [Easy]

Key Insights

- Use a queue (or circular buffer) to keep track of the current window elements.
- Maintain a running sum to quickly update the window's total when a new element comes in.
- When the window is full, subtract the value of the element that falls out as a new one is added.
- Each call to next() is executed in constant time, O(1).

Space & Time

Time Complexity: O(1) per call to next()  
Space Complexity: O(window size)

Solution

The solution leverages a queue data structure to store the elements of the sliding window. A running sum variable is maintained for efficient calculation of the

moving average. When a new integer is added:  
– If the size of the queue is equal to the maximum window size, remove the oldest element and subtract its value from the running sum.  
– Add the new element to both the queue and the running sum.  
– Compute and return the moving average as the running sum divided by the number of elements in the queue. This approach ensures that each update is done in constant time without having to sum all elements in the window repeatedly.

#760 Find Anagram Mappings [Easy]

Key Insights

- Since nums2 is an anagram of nums1, every element in nums1 has a corresponding position in nums2.
- A dictionary (or hash table) can be used to map each number in nums2 to its indices.
- When iterating through nums1, we can assign the next available index from the dictionary for that number.

Space & Time

Time Complexity: O(n) where n is the number of elements in the arrays (single pass to build the dictionary and another pass to form the mapping).  
Space Complexity: O(n) for storing the dictionary that holds indices of elements in nums2.

Solution

The approach consists of two primary steps:  
– Build a dictionary to record all indices where each number appears in nums2. This dictionary maps a number to a list of indices.  
– Iterate over nums1 and for each element, remove (or pop) one index from the corresponding list in the dictionary, and add that index to the final mapping array. This guarantees that each number from nums1 is correctly mapped to one of its positions in nums2 while efficiently handling duplicates.

#1426 Counting Elements [Easy]

Key Insights

- Use a hash table or set to store unique elements from arr for constant time lookups.
- Iterate through each element in arr and check if x + 1 exists in the set.
- Count each valid occurrence, including duplicates.
- This approach efficiently handles the problem without needing nested loops.

Space & Time

Time Complexity: O(n), where n is the number of elements in arr (set lookups are O(1) on average).  
Space Complexity: O(n), due to the use of an auxiliary set to store unique elements.

Solution

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <p>The solution leverages a set to achieve constant time membership checks. First, we convert arr to a set of unique numbers. Then, iterate through arr, and for each element, check if (element + 1) exists in the set. If it does, increment the count. This approach correctly handles duplicates by counting each occurrence during iteration.</p> <p><b>#1534 Count Good Triplets [Easy]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Brute force approach using three nested loops is acceptable since the maximum array length is 100.</li> <li>Check all possible triplets and count only those meeting the specified conditions.</li> <li>Absolute difference comparisons are the key condition validations.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n^3)</math> where n is the length of the array.</p> <p>Space Complexity: <math>O(1)</math> extra space (ignoring input space).</p> <p><b>Solution</b></p> <p>We use a brute force algorithm with three nested loops, iterating over all combinations of triplets (i, j, k) ensuring <math>i &lt; j &lt; k</math>. For each triplet, we calculate the absolute differences:</p>                                                                                                          | <p><math>-  arr[i] - arr[j]  -  arr[j] - arr[k]  -  arr[i] - arr[k] </math> The triplet is counted as good if all absolute differences satisfy the conditions (<math>\leq a</math>, <math>\leq b</math>, and <math>\leq c</math> respectively). This simple approach leverages the small input size, eliminating the need for more complex data structures or optimizations.</p> <p><b>#57 Insert Interval [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The intervals array is sorted by start times.</li> <li>Three main steps are used:</li> <li>Add intervals that come completely before the new interval.</li> <li>Merge overlapping intervals with the new interval.</li> <li>Append intervals that start after the new (merged) interval.</li> <li>Iterating through the list once allows an <math>O(n)</math> solution.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n)</math> since we scan the list once.</p> <p>Space Complexity: <math>O(n)</math> to store the resulting list of intervals.</p> <p><b>Solution</b></p> <p>Traverse the intervals list and perform the following:</p> <ul style="list-style-type: none"> <li>Append all intervals that end before the new interval starts.</li> <li>For intervals that overlap with newInterval, update newInterval to merge them (using min for start</li> </ul> | <p>and max for end). <li>Append the new merged interval.</li> <li>Append all remaining intervals. This ensures that all intervals remain non-overlapping and sorted with a single pass.</li></p> <p><b>#238 Product of Array Except Self [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The product for each index can be computed by multiplying the product of all numbers to the left and the product of all numbers to the right of that index.</li> <li>Division is not allowed, so we must compute the cumulative products explicitly.</li> <li>Create two passes: one forward pass for prefix products and one backward pass for suffix products.</li> <li>Use the output array to store the cumulative product values, achieving <math>O(1)</math> extra space by not using additional arrays.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n)</math> because we iterate through the array twice.</p> <p>Space Complexity: <math>O(1)</math> extra space (excluding the output array).</p> <p><b>Solution</b></p> <p>The approach uses two passes over the input array:</p> <ul style="list-style-type: none"> <li>Initialize the answer array and fill it with prefix products. For each index i, answer[i] contains the</li> </ul> |
| <p>product of all elements before i. <li>Traverse the array from the end using a running suffix product. Update the answer array by multiplying the current suffix product with the prefix product already stored. This method leverages an output array to store intermediate results and uses a temporary variable for the suffix product, ensuring constant extra space.</li></p> <p><b>#281 Zigzag Iterator [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a FIFO (first-in-first-out) structure to maintain the order of vectors which still have remaining elements.</li> <li>For each call to next(), select the next element from the current vector and then move that vector to the end of the queue if it still contains elements.</li> <li>The approach ensures proper zigzag (or cyclic) ordering among the vectors.</li> <li>When extending to k vectors, the same approach works: initialize the queue with each non-empty vector.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(1)</math> per next() call on average, with an overall <math>O(n)</math> for n total elements.</p> <p>Space Complexity: <math>O(k)</math> for the queue structure (where k is the number of vectors), plus the storage for the input vectors.</p> | <p><b>Solution</b></p> <p>We can solve the Zigzag Iterator problem by using a queue (or deque) to hold pairs of the vector's current index and the vector itself. For each call to next(), we remove the front element from the queue, return the current item from its associated vector, and if the vector has more elements, reinsert the pair (with an updated index) into the back of the queue. This cyclic process naturally interleaves the elements from the provided vectors.</p> <p><b>#307 Range Sum Query – Mutable [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>To efficiently support both point updates and range sum queries, a specialized data structure like a Binary Indexed Tree (Fenwick Tree) or a Segment Tree is ideal.</li> <li>Both data structures provide update and sum query operations in <math>O(\log n)</math> time.</li> <li>The challenge is to maintain a mutable array while enabling fast sum computation over any specified range.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(\log n)</math> for both update and sumRange operations</p> <p>Space Complexity: <math>O(n)</math> to store the additional tree structure</p> <p><b>Solution</b></p>                                                                                                                                  | <p>We can use a Binary Indexed Tree (Fenwick Tree) to solve this problem. The Fenwick Tree allows for efficient prefix sum calculations and point updates, both in <math>O(\log n)</math> time. The approach involves:</p> <ul style="list-style-type: none"> <li>Building the Fenwick Tree using the initial array.</li> <li>On an update, computing the difference between the new value and the current value, then propagating the change appropriately in the tree.</li> <li>For the sum range query, computing the prefix sum up to right and subtracting the prefix sum up to left-1.</li> </ul> <p><b>#454 4Sum II [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a hash table to store the frequency of all possible sums from nums1 and nums2.</li> <li>For each pair from nums3 and nums4, compute the complement (negative sum) and look it up in the hash table.</li> <li>This reduces the time complexity from <math>O(n^4)</math> to <math>O(n^2)</math>.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n^2)</math> where n is the length of each array.</p> <p>Space Complexity: <math>O(n^2)</math> due to the storage of pair sums in the hash table.</p> <p><b>Solution</b></p>                                        |
| <p>The solution employs a hash table to precompute and store the frequency of sums of all pairs from the first two arrays. Then, by iterating over all pairs from the third and fourth arrays, we calculate the required complement that would result in a total sum of zero. This complement is checked in the hash table, and the frequency (number of valid pairs) is added to the result. This method efficiently finds the number of valid quadruplets without resorting to a brute-force <math>O(n^4)</math> approach.</p> <p><b>#525 Contiguous Array [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Convert 0s to -1 so that a balanced subarray (equal number of 0s and 1s) sums to zero.</li> <li>Use a prefix sum to track the cumulative sum as you iterate through the array.</li> <li>Use a hash table (or dictionary) to store the first occurrence of each prefix sum.</li> <li>When the same prefix sum is encountered again, it indicates that the subarray between these indices is balanced.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n)</math> – We iterate through the array once.</p> <p>Space Complexity: <math>O(n)</math> – In the worst case, we store an entry for each prefix sum.</p>                             | <p><b>Solution</b></p> <p>We use a technique where we convert each 0 in the array to -1. Then, we compute a running prefix sum as we iterate through the array. A prefix sum of 0 at any point indicates that the subarray from the beginning to the current index is balanced. For each prefix sum, if we have seen it before, the subarray between the previous index (where the prefix sum was first seen) and the current index sums to 0. We update the maximum length accordingly. We use a hash table to store the first index where each prefix sum occurs.</p> <p><b>#560 Subarray Sum Equals K [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Utilize the prefix sum technique to simplify sum calculation for subarrays.</li> <li>Use a hash table (or dictionary/map) to store the frequency of prefix sums encountered.</li> <li>For each element, check if (current prefix sum - k) exists in the hash table; if it does, it indicates a valid subarray.</li> <li>Initialize the hash table with {0: 1} to manage cases where a subarray's sum is exactly k.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n)</math></p> <p>Space Complexity: <math>O(n)</math></p>                                                                                                                                                | <p><b>Solution</b></p> <p>We solve the problem by maintaining a running sum (prefix sum) and using a hash table to record the number of times each sum has occurred. As we iterate through the array, we update the running sum. For each new sum, we check if (current sum - k) exists in our hash table. If it does, the value associated with (current sum - k) gives the number of subarrays ending at the current index that sum to k. This method efficiently computes the result in a single pass through the array.</p> <p><b>#1010 Pairs of Songs With Total Durations Divisible by 60 [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Each song duration's effect is determined by its remainder when divided by 60.</li> <li>For any given module value r (where <math>0 \leq r &lt; 60</math>), the complementary module needed to complete 60 is <math>(60 - r) \% 60</math>.</li> <li>Special handling is needed when r is 0 (or 30, since <math>30 + 30 = 60</math>) to avoid double-counting.</li> <li>A frequency counter for remainders can be efficiently used to count valid pairs in one pass through the list.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n)</math>, where n is the number of songs.</p>              |

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| <div> <div>Space Complexity: <math>O(60) \rightarrow O(1)</math>, since we only need to keep track of 60 possible remainders.</div> <div>Solution</div> <div> <p>We iterate over the list of song durations and calculate the remainder of each duration modulo 60. A frequency array (or hash table) is used to record how many times each remainder has been seen so far. For each song, we look up the frequency of its complementary remainder (i.e., <math>(60 - \text{current\_remainder}) \% 60</math>) already encountered. This frequency indicates how many valid pairs can be formed with the current song. After counting pairs with the current song, we update the frequency of its remainder for future pairings.</p> </div> </div> <div> <div>#1272 Remove Interval <span>Medium</span></div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Iterate over each interval and check if it overlaps with the toBeRemoved interval.</li> <li>If the current interval is completely outside of toBeRemoved, add it to the result as is.</li> <li>For overlapping intervals, compute the non-overlapping portions:</li> <li>If the left side of the interval lies before toBeRemoved, include <math>[\text{start}, \min(\text{end}, \text{toBeRemoved.start})]</math>.</li> <li>If the right side of the interval lies after toBeRemoved, include <math>[\max(\text{start}, \text{toBeRemoved.end}), \text{end}]</math>.</li> </ul> </div> </div> <div> <div>Arrays &amp; Hashing</div> <div>p.955</div> </div>                            | <div> <div>Space &amp; Time</div> <div>Time Complexity: <math>O(n)</math>, where n is the number of intervals.</div> <div>Space Complexity: <math>O(n)</math>, for storing the resulting intervals.</div> <div>Solution</div> <div> <p>The approach involves iterating through the list of sorted disjoint intervals and checking for overlap with the toBeRemoved interval. For each interval:</p> <ul style="list-style-type: none"> <li>If it is completely before or after the removal interval, add it directly to the result.</li> <li>If it overlaps with toBeRemoved, determine the remaining portion(s): The left side of the interval that does not intersect (if any). The right side of the interval that does not intersect (if any). Use simple conditional checks and list manipulation to form the final set of intervals.</li> </ul> </div> </div> <div> <div>#1429 First Unique Number <span>Medium</span></div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Use a queue to maintain the order of potential unique numbers.</li> <li>Use a hash table (or map) to count the occurrences of each number.</li> <li>When showing the first unique number, remove numbers from the front of the queue if their frequency is more than one.</li> </ul> </div> </div> <div> <div>Arrays &amp; Hashing</div> <div>p.956</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <div> <ul style="list-style-type: none"> <li>All operations (add and showFirstUnique) can be done in <math>O(1)</math> amortized time.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: <math>O(1)</math> amortized for each add and showFirstUnique call.</div> <div>Space Complexity: <math>O(n)</math>, where n is the number of distinct integers encountered.</div> </div> <div> <div>Solution</div> <div> <p>We design the FirstUnique class using two main data structures: a queue and a hash map (or dictionary). The queue preserves the order of insertion for unique candidates, while the hash map holds the frequency count for each integer.</p> <p>When a new number is added:</p> <ul style="list-style-type: none"> <li>Increment its count in the hash map.</li> <li>If the count is one, append it to the queue.</li> </ul> <p>To retrieve the first unique number:</p> <ul style="list-style-type: none"> <li>While the queue is not empty and the count for the element at the front is greater than one, remove it from the queue.</li> <li>Return the front element of the queue if it exists, otherwise return <math>-1</math>.</li> </ul> <p>This approach ensures that we are always up-to-date with the current first unique number.</p> </div> </div> <div> <div>#3159 Find Occurrences of an Element in an Array <span>Medium</span></div> <div>Arrays &amp; Hashing</div> <div>p.957</div> </div> |
| <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Scan the array once and record every index where <math>\text{nums}[i] == x</math>.</li> <li>Store those indices in order inside a positions array.</li> <li>For each query <math>k</math>, check whether the <math>k</math>-th occurrence exists.</li> <li>If it exists, return <math>\text{positions}[k - 1]</math>.</li> <li>Otherwise, return <math>-1</math>.</li> <li>This lets every query be answered in constant time after the preprocessing pass.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: <math>O(n + q)</math>, where n is the length of <code>nums</code> and q is the number of queries.</div> <div>Space Complexity: <math>O(m)</math>, where m is the number of occurrences of <code>x</code>.</div> </div> <div> <div>Solution</div> <div> <p>We solve the problem by first precomputing all occurrences of <code>x</code>. As we iterate through <code>nums</code>, every time we see <code>x</code>, we append its index to an array called <code>positions</code>. Then, for each query value <code>k</code>, we check whether <code>k</code> is within the bounds of <code>positions</code>. If it is, the answer is <code>positions[k - 1]</code>; otherwise, the required occurrence does not exist, so the answer is <code>-1</code>. This avoids rescanning the array for every query.</p> </div> </div> <div> <div>#41 First Missing Positive <span>Hard</span></div> <div>Arrays &amp; Hashing</div> <div>p.958</div> </div> | <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>The missing positive number must be in the range <math>[1, n+1]</math> where n is the length of the array.</li> <li>Use the array indices to place numbers in their correct positions (i.e., number i should be at index i-1) with in-place swapping.</li> <li>After rearrangement, scan the array: the first index where the number is not equal to index+1 indicates the missing positive integer.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: <math>O(n)</math> as each number is swapped at most once.</div> <div>Space Complexity: <math>O(1)</math> since the rearrangement occurs in-place.</div> </div> <div> <div>Solution</div> <div> <p>The approach involves reordering the array so that each positive integer in the range <math>[1, n]</math> is placed at the index corresponding to its value minus one. This means for any valid number i, <code>nums[i-1]</code> should be i. We iterate through the array and perform in-place swaps to achieve this order. After this process, another pass through the array reveals the first position where the number does not match the expected value (i+1), which is the smallest missing positive. If every position is correct, then the missing value is <code>n+1</code>.</p> </div> </div> <div> <div>Two Pointers</div> <div>Arrays &amp; Hashing</div> <div>p.959</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                  | <div> <div>#13 Two Pointers</div> <div>By Pattern · Python Only · 19 questions</div> <div>Two Pointers</div> <div>p.960</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <div> <div>Two Pointers</div> <div>#88 Merge Sorted Array <span>EAS</span></div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Array · Two Pointers · Sorting</div> <div>Lists: Denny Zhang</div> <div>Time Complexity: <math>O(m + n)</math> Space Complexity: <math>O(1)</math></div> <div>Problem</div> <div> <p>You are given two integer arrays <code>nums1</code> and <code>nums2</code>, sorted in non-decreasing order, and two integers <code>m</code> and <code>n</code>, representing the number of elements in <code>nums1</code> and <code>nums2</code> respectively.</p> <p>Merge <code>nums1</code> and <code>nums2</code> into a single array sorted in non-decreasing order.</p> <p>The final sorted array should not be returned by the function, but instead be stored inside the array <code>nums1</code>. To accommodate this, <code>nums1</code> has a length of <code>m + n</code>, where the first <code>m</code> elements denote the elements that should be merged, and the last <code>n</code> elements are set to 0 and should be ignored. <code>nums2</code> has a length of <code>n</code>.</p> <p>Example 1:</p> </div> <div>Two Pointers</div> <div>p.961</div> </div>                                                                                                                                                                                                                                                                                                                                                        | <div> <div>Input: <code>nums1 = [1,2,3,0,0]</code>, <code>m = 3</code>, <code>nums2 = [2,5,6]</code>, <code>n = 3</code></div> <div>Output: <code>[1,2,2,3,5,6]</code></div> <div>Explanation: The arrays we are merging are <code>[1,2,3]</code> and <code>[2,5,6]</code>.</div> <div>The result of the merge is <code>[1,2,2,3,5,6]</code> with the underlined elements coming from <code>nums1</code>.</div> </div> <div> <div>Example 2:</div> <div>Input: <code>nums1 = [1]</code>, <code>m = 1</code>, <code>nums2 = []</code>, <code>n = 0</code></div> <div>Output: <code>[1]</code></div> <div>Explanation: The arrays we are merging are <code>[1]</code> and <code>[]</code>.</div> <div>The result of the merge is <code>[1]</code>.</div> </div> <div> <div>Example 3:</div> <div>Input: <code>nums1 = [0]</code>, <code>m = 0</code>, <code>nums2 = [1]</code>, <code>n = 1</code></div> <div>Output: <code>[1]</code></div> <div>Explanation: The arrays we are merging are <code>[]</code> and <code>[1]</code>.</div> <div>The result of the merge is <code>[1]</code>.</div> <div>Note that because <code>m = 0</code>, there are no elements in <code>nums1</code>. The 0 is only there to ensure the merge result can fit in <code>nums1</code>.</div> </div> <div> <div>Constraints:</div> <div> <ul style="list-style-type: none"> <li><code>nums1.length == m + n</code></li> <li><code>nums2.length == n</code></li> <li><code>0 &lt;= m, n &lt;= 200</code></li> <li><code>1 &lt;= m + n &lt;= 200</code></li> <li><code>-109 &lt;= nums1[i], nums2[j] &lt;= 109</code></li> </ul> </div> <div>Follow up: Can you come up with an algorithm that runs in <math>O(m + n)</math> time?</div> </div> <div> <div>Two Pointers</div> <div>Sheet 107 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait</div> <div>p.962</div> </div> | <div> <div>Brute Force (Python)</div> <div># Brute Force Approach</div> <div>class Solution:</div> <div>def merge(self, nums1: List[int], m: int, nums2: List[int], n: int) -&gt; None:</div> <div># Brute Force <math>O(n^2)</math> - nested loops instead of two pointers</div> <div>n = len(nums1)</div> <div>for i in range(n):</div> <div>for j in range(i + 1, n):</div> <div># check <code>nums1[i]</code> and <code>nums1[j]</code></div> <div>pass</div> </div> <div> <div>Community Solutions (Python)</div> <div>• WalkCC</div> <div>class Solution:</div> <div>def merge(self, nums1: List[int], m: int, nums2: List[int], n: int) -&gt; None:</div> <div>i = m - 1 # <code>nums1</code>'s index (the actual nums)</div> <div>j = n - 1 # <code>nums2</code>'s index</div> <div>k = m + n - 1 # <code>nums1</code>'s index (the next filled position)</div> <div>while j &gt;= 0:</div> <div>if i &gt;= 0 and <code>nums1[i] &gt; nums2[j]</code>:</div> <div>nums1[k] = <code>nums1[i]</code></div> <div>k -= 1</div> </div> <div> <div>Two Pointers</div> <div>p.963</div> </div>                                                                                                                                                                                                                                                                                                                                                           |

```

 i -= 1
else:
 nums1[k] = nums2[j]
 k -= 1
 j -= 1

• LeetCode

class Solution:
 def merge(self, nums1: List[int], m: int, nums2:
List[int], n: int) -> None:
 k = m + n - 1
 i, j = m - 1, n - 1
 while j >= 0:
 if i >= 0 and nums1[i] > nums2[j]:
 nums1[k] = nums1[i]
 i -= 1
 else:
 nums1[k] = nums2[j]

 j -= 1
 k -= 1
```

• SimplyLeet

```

def merge(nums1, m, nums2, n):
 # Initialize pointers for nums1, nums2 and the tail
 index
 p1 = m - 1
 p2 = n - 1
 tail = m + n - 1

 # Merge arrays from the end by choosing the larger
 element
 while p1 >= 0 and p2 >= 0:
 if nums1[p1] > nums2[p2]:
 nums1[tail] = nums1[p1]

 p1 -= 1 # Move pointer in nums1
 else:
 nums1[tail] = nums2[p2]
 p2 -= 1 # Move pointer in nums2
 tail -= 1 # Move tail pointer

 # Copy any remaining elements from nums2 into nums1
 while p2 >= 0:
 nums1[tail] = nums2[p2]
 p2 -= 1

 tail -= 1
```

• LC.ca

```

class Solution:
 def merge(self, nums1: List[int], m: int, nums2:
List[int], n: int) -> None:

 Do not return anything, modify nums1 in-place
 instead.

 i, j, k = m - 1, n - 1, m + n - 1
 # what if j=-1 already but k is not 0 yet? e.g.
 m=[1,2,0], n=[3]
 # ==> then no more ops needed, rest of m[] already
 sorted, so this while is good enough
 while j >= 0:
 # i could be -1, eg. [7,8,9,] and [1]

 if i >= 0 and nums1[i] > nums2[j]:
 nums1[k] = nums1[i]
 i -= 1
 else:
 nums1[k] = nums2[j]
 j -= 1
 k -= 1
```

Two Pointers

#125 Valid PalindromeEAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Two Pointers · String

Lists: Grind 169

Time Complexity: O(n), where n is the length of the string, as each character is processed at most once. Space Complexity: O(1), using only a few extra variables for pointers and temporary comparisons

Problem

A phrase is a palindrome if, after converting all uppercase letters into lowercase letters and removing all non-alphanumeric characters, it reads the same forward and backward. Alphanumeric characters include letters and numbers.

Given a string s, return true if it is a palindrome, or false otherwise.

Example 1:

Input: s = "A man, a plan, a canal: Panama"  
Output: true  
Explanation: "amanaplanacanalpanama" is a palindrome.

Example 2:

Input: s = "race a car"  
Output: false  
Explanation: "raceacar" is not a palindrome.

Example 3:

Input: s = " "  
Output: true  
Explanation: s is an empty string "" after removing non-alphanumeric characters. Since an empty string reads the same forward and backward, it is a palindrome.

Constraints:

- 1 <= s.length <= 2 \* 10<sup>5</sup>
- s consists only of printable ASCII characters.

Community Solutions (Python)

• WalkCC

class Solution:
 def isPalindrome(self, s: str) -> bool:
 l = 0
 r = len(s) - 1

 while l < r:
 while l < r and not s[l].isalnum():
 l += 1
 while l < r and not s[r].isalnum():
 r -= 1

if s[l].lower() != s[r].lower():
 return False
l += 1
r -= 1

return True

• LeetCode

class Solution:
 def isPalindrome(self, s: str) -> bool:
 i, j = 0, len(s) - 1
 while i < j:
 if not s[i].isalnum():
 i += 1
 elif not s[j].isalnum():
 j -= 1
 elif s[i].lower() != s[j].lower():
 return False

 else:
 i, j = i + 1, j - 1
 return True

• SimplyLeet

```

class Solution:
 def isPalindrome(self, s: str) -> bool:
 # Initialize two pointers at the beginning and
 end of the string
 left, right = 0, len(s) - 1
 while left < right:
 # Skip non-alphanumeric characters from the
 left

 while left < right and not s[left].isalnum():
 left += 1
 # Skip non-alphanumeric characters from the
 right

 while left < right and not
s[right].isalnum():

 right -= 1
 # Compare the characters in lowercase
 if s[left].lower() != s[right].lower():
 return False
 left += 1
 right -= 1
 return True

• LC.ca
```

```

class Solution:
 def isPalindrome(self, s: str) -> bool:
 i, j = 0, len(s) - 1
 while i < j:
 if not s[i].isalnum():
 i += 1
 elif not s[j].isalnum():
 j -= 1
 elif s[i].lower() != s[j].lower():
 return False

 else:
 i, j = i + 1, j - 1
 return True
```

Two Pointers

#283 Move ZeroesEAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers

Lists: Grind 169

Time Complexity: O(n), where n is the number of elements in the array, since only one pass (or at most two simple passes) is required. Space Complexity: O(1) extra space, as the reordering is accompli

Problem

Given an integer array nums, move all 0's to the end of it while maintaining the relative order of the non-zero elements.

Note that you must do this in-place without making a copy of the array.

Example 1:

Input: nums = [0,1,0,3,12]  
Output: [1,3,12,0,0]

Example 2:

Input: nums = [0]  
Output: [0]

Constraints:

- 1 <= nums.length <= 104

• -231 <= nums[i] <= 231 - 1

Follow up: Could you minimize the total number of operations done?

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def moveZeroes(self, nums):
 # Brute Force O(n^2) - nested loops instead of
 two pointers
 n = len(nums)
 for i in range(n):
 for j in range(i + 1, n):
 # check nums[i] and nums[j]
 pass
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def moveZeroes(self, nums: List[int]) -> None:
 j = 0
 for num in nums:
 if num != 0:
 nums[j] = num
 j += 1

 for i in range(j, len(nums)):
 nums[i] = 0
```

• LeetCode

Two Pointers p.973

```
class Solution:
 def moveZeroes(self, nums: List[int]) -> None:
 k = 0
 for i, x in enumerate(nums):
 if x:
 nums[k], nums[i] = nums[i], nums[k]
 k += 1
```

• SimplyLeet

```
Moves all zeros in the list 'nums' to the end in-place.
def moveZeroes(nums):
 # 'j' is the pointer where the next non-zero element
 should be placed.
 j = 0
 # Loop through all the elements with index 'i'
 for i in range(len(nums)):
 # Check if the current element is non-zero.
 if nums[i] != 0:
 # Swap only if i is not equal to j to avoid
 unnecessary operations.
 nums[j], nums[i] = nums[i], nums[j]

 # Increment j to move to the next insertion
 index.
 j += 1

Example usage:
nums = [0, 1, 0, 3, 12]
moveZeroes(nums)
print(nums) # Expected output: [1, 3, 12, 0, 0]
```

• LC.ca

Two Pointers p.974

```
class Solution:
 def moveZeroes(self, nums: List[int]) -> None:
 i = -1
 for j, x in enumerate(nums):
 if x:
 i += 1
 nums[i], nums[j] = nums[j], nums[i]
```

Two Pointers p.975

Two Pointers

#408 Valid Word AbbreviationEAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Two Pointers · String

Lists: Premium 98

Time Complexity: O(n + m) where n is the length of the word and m is the length of the abbreviation. Space Complexity: O(1) (constant extra space)

Problem

A string can be abbreviated by replacing any number of non-adjacent, non-empty substrings with their lengths. The lengths should not have leading zeros. For example, a string such as "substitution" could be abbreviated as (but not limited to):

- "s10n" ("s ubstitutio n")
- "sub4u4" ("sub stit u tion")
- "12" ("substitution")
- "su3i1u2on" ("su bst i t u ti on")
- "substitution" (no substrings replaced)

The following are not valid abbreviations:

- "s55n" ("s ubsti tutio n", the replaced substrings are adjacent)
- "s010n" (has leading zeros)
- "s0ubstitution" (replaces an empty substring)

Two Pointers p.976

Given a string word and an abbreviation abbr, return whether the string matches the given abbreviation. A substring is a contiguous non-empty sequence of characters within a string.

Example 1:

Input: word = "internationalization", abbr = "i12iz4n"  
Output: true  
Explanation: The word "internationalization" can be abbreviated as "i12iz4n" ("i nternational iz atio n").

Example 2:

Input: word = "apple", abbr = "a2e"  
Output: false  
Explanation: The word "apple" cannot be abbreviated as "a2e".

Constraints:

- 1 <= word.length <= 20
- word consists of only lowercase English letters.
- 1 <= abbr.length <= 10
- abbr consists of lowercase English letters and digits.
- All the integers in abbr will fit in a 32-bit integer.

Community Solutions (Python)

• WalkCC

Two Pointers p.977

```
class Solution:
 def validWordAbbreviation(self, word: str, abbr: str) -> bool:
 i = 0 # word's index
 j = 0 # abbr's index

 while i < len(word) and j < len(abbr):
 if word[i] == abbr[j]:
 i += 1
 j += 1
 continue

 if not abbr[j].isdigit() or abbr[j] == '0':
 return False
 num = 0
 while j < len(abbr) and abbr[j].isdigit():
 num = num * 10 + int(abbr[j])
 j += 1
 i += num

 return i == len(word) and j == len(abbr)
```

• LeetCode

Two Pointers p.978

```
class Solution:
 def validWordAbbreviation(self, word: str, abbr: str) -> bool:
 m, n = len(word), len(abbr)
 i = j = x = 0
 while i < m and j < n:
 if abbr[j].isdigit():
 if abbr[j] == "0" and x == 0:
 return False
 x = x * 10 + int(abbr[j])
 else:
 i += x
 x = 0
 if i >= m or word[i] != abbr[j]:
 return False
 i += 1
 j += 1
 return i + x == m and j == n
```

• SimplyLeet

```
Python solution for valid word abbreviation

def validWordAbbreviation(word: str, abbr: str) -> bool:
 word_index = 0 # pointer for the word
 abbr_index = 0 # pointer for the abbreviation

 while abbr_index < len(abbr):
 # if current abbrev char is a digit
 if abbr[abbr_index].isdigit():
 # leading zero check: if the digit is '0', it
 is invalid
```

Two Pointers p.979

```
 if abbr[abbr_index] == '0':
 return False
 # build the entire number
 num = 0
 while abbr_index < len(abbr) and
abbr[abbr_index].isdigit():
 num = num * 10 + int(abbr[abbr_index])
 abbr_index += 1
 # skip num characters in word
 word_index += num
 else:

 # if letter doesn't match or word index is
 out of bounds, invalid
 if word_index >= len(word) or
word[word_index] != abbr[abbr_index]:
 return False
 word_index += 1
 abbr_index += 1

 # Valid if and only if the entire word was covered
 return word_index == len(word)

Example usage:

#print(validWordAbbreviation("internationalization",
"i12iz4n")) # Expected output: True
#print(validWordAbbreviation("apple", "a2e")) # Expected
output: False
```

• LC.ca

Two Pointers p.980

```
class Solution:
 def validWordAbbreviation(self, word: str, abbr: str) -> bool:
 m, n = len(word), len(abbr)
 i = j = x = 0
 while i < m and j < n:
 if abbr[j].isdigit():
 if abbr[j] == "0" and x == 0:
 return False
 x = x * 10 + int(abbr[j])
 else:
 i += x
 x = 0
 if i >= m or word[i] != abbr[j]:
 return False
 i += 1
 j += 1
 return i + x == m and j == n
```

Two Pointers p.981

Two Pointers

#977 Squares of a Sorted Array

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers · Sorting

Lists: Grind 169

Time Complexity: O(n) – We traverse the array only once.

Space Complexity: O(n) – We use an extra array to store the output.

Problem

Given an integer array nums sorted in non-decreasing order, return an array of the squares of each number sorted in non-decreasing order.

Example 1:

Input: nums = [-4,-1,0,3,10]  
Output: [0,1,9,16,100]  
Explanation: After squaring, the array becomes [16,1,0,9,100].  
After sorting, it becomes [0,1,9,16,100].

Example 2:

Input: nums = [-7,-3,2,3,11]  
Output: [4,9,9,49,121]

Constraints:

- 1 <= nums.length <= 104
- -104 <= nums[i] <= 104

Two Pointers

p.982

- nums is sorted in non-decreasing order.

Follow up: Squaring each element and sorting the new array is very trivial, could you find an O(n) solution using a different approach?

Brute Force (Python)

# Brute Force Approach  
class Solution:  
 def sortedSquares(self, nums):  
 # Brute Force O(n^2) – nested loops instead of two pointers  
 n = len(nums)  
 for i in range(n):  
 for j in range(i + 1, n):  
 # check nums[i] and nums[j]  
 pass

Community Solutions (Python)

• WalkCC  
class Solution:  
 def sortedSquares(self, nums: list[int]) -> list[int]:  
 n = len(nums)  
 l = 0  
 r = n - 1  
 ans = [0] \* n  
  
 while n:  
 n -= 1  
 if abs(nums[l]) > abs(nums[r]):

Two Pointers

p.983

ans[n] = nums[l] \* nums[l]  
 l += 1  
else:  
 ans[n] = nums[r] \* nums[r]  
 r -= 1  
  
 return ans

• LeetDoocs  
class Solution:  
 def sortedSquares(self, nums: List[int]) -> List[int]:  
 ans = []  
 i, j = 0, len(nums) - 1  
 while i <= j:  
 a = nums[i] \* nums[i]  
 b = nums[j] \* nums[j]  
 if a > b:  
 ans.append(a)  
 i += 1  
  
 else:  
 ans.append(b)  
 j -= 1  
 return ans[::-1]

• SimplyLeet

Two Pointers

p.984

# Function to return squares of a sorted array in non-decreasing order.  
def sortedSquares(nums):  
 # Initialize two pointers at the start and end of the array.  
 left, right = 0, len(nums) - 1  
 # Create a result array of the same length as nums.  
 result = [0] \* len(nums)  
 # Index to insert the largest square at the current position.  
 position = len(nums) - 1  
  
 # Loop until the left and right pointers cross.  
  
 while left <= right:  
 # Compare absolute values at the left and right pointers.  
 if abs(nums[left]) > abs(nums[right]):  
 result[position] = nums[left] \* nums[left]  
 left += 1 # Move left pointer rightward.  
 else:  
 result[position] = nums[right] \* nums[right]  
 right -= 1 # Move right pointer leftward.  
 position -= 1 # Move to the next insertion position.  
 return result  
  
# Example usage:  
print(sortedSquares([-4, -1, 0, 3, 10])) # Output: [0, 1, 9, 16, 100]

• LC.ca

Two Pointers

p.985

class Solution:  
 def sortedSquares(self, nums: List[int]) -> List[int]:  
 n = len(nums)  
 res = [0] \* n  
 i, j, k = 0, n - 1, n - 1  
 while i <= j:  
 if nums[i] \* nums[i] > nums[j] \* nums[j]:  
 res[k] = nums[i] \* nums[i]  
 i += 1  
 else:  
 res[k] = nums[j] \* nums[j]  
 j -= 1  
 k -= 1  
 return res

Two Pointers

p.986

Two Pointers

#11 Container With Most Water

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers · Greedy

Lists: Grind 169

Time Complexity: O(n) – The algorithm makes a single pass through the array with two pointers.

Space Complexity: O(1) – Only a few extra variables are used, regardless of the input size.

Problem

You are given an integer array height of length n. There are n vertical lines drawn such that the two endpoints of the ith line are (i, 0) and (i, height[i]).

Find two lines that together with the x-axis form a container, such that the container contains the most water.

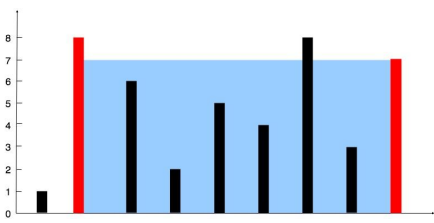
Return the maximum amount of water a container can store.

Notice that you may not slant the container.

Example 1:

Two Pointers

p.987



Input: height = [1,8,6,2,5,4,8,3,7]  
Output: 49  
Explanation: The above vertical lines are represented by array [1,8,6,2,5,4,8,3,7]. In this case, the max area of water (blue section) the container can contain is 49.

Example 2:

Input: height = [1,1]  
Output: 1

Constraints:

- n == height.length
- 2 <= n <= 105
- 0 <= height[i] <= 104

Brute Force (Python)

Two Pointers

p.988

# Brute Force Approach  
class Solution:  
 def maxArea(self, height):  
 # Brute Force O(n^2) – try every pair of lines  
 res = 0  
 for i in range(len(height)):  
 for j in range(i + 1, len(height)):  
 water = min(height[i], height[j]) \* (j - i)  
 res = max(res, water)  
 return res

Community Solutions (Python)

• WalkCC  
class Solution:  
 def maxArea(self, height: list[int]) -> int:  
 ans = 0  
 l = 0  
 r = len(height) - 1  
  
 while l < r:  
 minHeight = min(height[l], height[r])  
 ans = max(ans, minHeight \* (r - l))  
 if height[l] < height[r]:  
 l += 1  
 else:  
 r -= 1  
 return ans

• LeetDoocs

Two Pointers

p.989

class Solution:  
 def maxArea(self, height: List[int]) -> int:  
 l, r = 0, len(height) - 1  
 ans = 0  
 while l < r:  
 t = min(height[l], height[r]) \* (r - l)  
 ans = max(ans, t)  
 if height[l] < height[r]:  
 l += 1  
 else:  
 r -= 1  
 return ans

• SimplyLeet  
# Function to calculate the maximum water container area  
def maxArea(height):  
 left, right = 0, len(height) - 1 # Initialize two pointers  
 max\_water = 0 # Variable to store the maximum area  
  
 # Loop until pointers meet  
 while left < right:  
 # Calculate current area based on smaller height and distance between pointers  
 current\_area = min(height[left], height[right]) \* (right - left)  
 max\_water = max(max\_water, current\_area) # Update maximum area if current is larger

Two Pointers

p.990



# Move the pointer corresponding to the smaller height  
if height[left] < height[right]:  
    left += 1  
else:  
    right -= 1  
return max\_water

# Test the function with an example  
if \_\_name\_\_ == "\_\_main\_\_":  
  
    input\_heights = [1,8,6,2,5,4,8,3,7]  
    print(maxArea(input\_heights)) # Expected output: 49

• LC.ca

class Solution:  
    def maxArea(self, height: List[int]) -> int:  
        i, j = 0, len(height) - 1  
        ans = 0  
        while i < j:  
            t = (j - i) \* min(height[i], height[j])  
            ans = max(ans, t)  
            if height[i] < height[j]:  
                i += 1  
            else:  
                j -= 1  
        return ans

Two Pointersp.991

Two Pointers

#15 3SumMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Array · Two Pointers · Sorting  
Lists: Grind 169  
Time Complexity:  $O(n^2)$  where n is the number of elements in the array. Space Complexity:  $O(1)$  extra space (ignoring the space required for the output).

**Problem**  
Given an integer array nums, return all the triplets [nums[i], nums[j], nums[k]] such that  $i \neq j$ ,  $i \neq k$ , and  $j \neq k$ , and  $nums[i] + nums[j] + nums[k] == 0$ .  
Notice that the solution set must not contain duplicate triplets.

**Example 1:**  
Input: nums = [-1,0,1,2,-1,-4]  
Output: [[-1,-1,2],[-1,0,1]]  
Explanation:  
 $nums[0] + nums[1] + nums[2] = (-1) + 0 + 1 = 0$ .  
 $nums[1] + nums[2] + nums[4] = 0 + 1 + (-1) = 0$ .  
 $nums[0] + nums[3] + nums[4] = (-1) + 2 + (-1) = 0$ .  
The distinct triplets are [-1,0,1] and [-1,-1,2].  
Notice that the order of the output and the order of the triplets does not matter.

**Example 2:**

Two Pointersp.992

Input: nums = [0,1,1]  
Output: []  
Explanation: The only possible triplet does not sum up to 0.

Example 3:

Input: nums = [0,0,0]  
Output: [[0,0,0]]  
Explanation: The only possible triplet sums up to 0.

**Constraints:**  
•  $3 \leq \text{nums.length} \leq 3000$   
•  $-105 \leq \text{nums}[i] \leq 105$

Brute Force (Python)

# Brute Force Approach  
class Solution:  
    def threeSum(self, nums):  
        # Brute Force  $O(n^3)$  - check all triples,  
        deduplicate via set  
        nums.sort()  
        res = set()  
        for i in range(len(nums)):  
            for j in range(i + 1, len(nums)):  
                for k in range(j + 1, len(nums)):  
                    if nums[i] + nums[j] + nums[k] == 0:  
  
                        res.add((nums[i], nums[j],  
nums[k]))  
        return [list(t) for t in res]

Two Pointersp.993

## Community Solutions (Python)

• WalkCC

class Solution:  
    def threeSum(self, nums: List[int]) -> List[List[int]]:  
        if len(nums) < 3:  
            return []  
  
        ans = []  
  
        nums.sort()  
  
        for i in range(len(nums) - 2):  
  
            if i > 0 and nums[i] == nums[i - 1]:  
                continue  
            # Choose nums[i] as the first number in the triplet, then search the  
            # remaining numbers in [i + 1, n - 1].  
            l = i + 1  
            r = len(nums) - 1  
            while l < r:  
                summ = nums[i] + nums[l] + nums[r]  
                if summ == 0:  
                    ans.append((nums[i], nums[l], nums[r]))

Two Pointersp.994

l += 1  
r -= 1  
while nums[l] == nums[l - 1] and l < r:  
    l += 1  
while nums[r] == nums[r + 1] and l < r:  
    r -= 1  
elif summ < 0:  
    l += 1  
else:  
    r -= 1

return ans

• LeetDoocs

class Solution:  
    def threeSum(self, nums: List[int]) -> List[List[int]]:  
        nums.sort()  
        n = len(nums)  
        ans = []  
        for i in range(n - 2):  
            if nums[i] > 0:  
                break  
            if i and nums[i] == nums[i - 1]:  
                continue

Two Pointersp.995

j, k = i + 1, n - 1  
while j < k:  
    x = nums[i] + nums[j] + nums[k]  
    if x < 0:  
        j += 1  
    elif x > 0:  
        k -= 1  
    else:  
        ans.append([nums[i], nums[j],  
nums[k]])  
        j, k = j + 1, k - 1

while j < k and nums[j] == nums[j - 1]:  
    j += 1  
while j < k and nums[k] == nums[k + 1]:  
    k -= 1

return ans

• SimplyLeet

# Function to find all unique triplets that sum to zero.  
def threeSum(nums):  
    # Sort the array to facilitate finding duplicates and using two pointers.  
    nums.sort()  
    result = []  
    n = len(nums)  
  
    # Iterate through the array, fixing one element at a time.  
    for i in range(n - 2):  
        # Skip duplicate elements for the first element.

Two Pointersp.996

if i > 0 and nums[i] == nums[i - 1]:  
    continue

left, right = i + 1, n - 1  
while left < right:  
    current\_sum = nums[i] + nums[left] +  
nums[right]  
    # Check if the triplet adds up to zero.  
    if current\_sum == 0:  
        result.append([nums[i], nums[left],  
nums[right]])  
        # Skip duplicates for the second element.

while left < right and nums[left] ==  
nums[left + 1]:  
    left += 1  
    # Skip duplicates for the third element.  
while left < right and nums[right] ==  
nums[right - 1]:  
    right -= 1  
    left += 1  
    right -= 1  
    elif current\_sum < 0:  
        left += 1  
    else:

right -= 1

return result

# Example usage:  
# print(threeSum([-1,0,1,2,-1,-4]))

• LC.ca

Two Pointersp.997

class Solution:  
    def threeSum(self, nums: List[int]) -> List[List[int]]:  
        nums.sort()  
        n = len(nums)  
        ans = []  
        for i in range(n - 2):  
            if nums[i] > 0:  
                break  
            if i and nums[i] == nums[i - 1]:  
                continue

j, k = i + 1, n - 1  
while j < k:  
    x = nums[i] + nums[j] + nums[k]  
    if x < 0:  
        j += 1  
    elif x > 0:  
        k -= 1  
    else:  
        ans.append([nums[i], nums[j],  
nums[k]])  
        j, k = j + 1, k - 1

while j < k and nums[j] == nums[j - 1]:  
    j += 1  
while j < k and nums[k] == nums[k + 1]:  
    k -= 1

return ans

Two Pointersp.998

Two Pointers

#16 3Sum ClosestMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Array · Two Pointers · Sorting  
Lists: Grind 169  
Time Complexity:  $O(n^2)$  Space Complexity:  $O(\log n)$  to  $O(n)$  depending on the sorting algorithm used (typically in-place sort, so extra space might be minimal)

**Problem**  
Given an integer array nums of length n and an integer target, find three integers at distinct indices in nums such that the sum is closest to target.  
Return the sum of the three integers.

You may assume that each input would have exactly one solution.

**Example 1:**  
Input: nums = [-1,2,1,-4], target = 1  
Output: 2  
Explanation: The sum that is closest to the target is 2.  $(-1 + 2 + 1 = 2)$ .

**Example 2:**

Two Pointersp.999

Input: nums = [0,0,0], target = 1  
Output: 0  
Explanation: The sum that is closest to the target is 0. (0 + 0 + 0 = 0).

Constraints:

- 3 <= nums.length <= 500
- -1000 <= nums[i] <= 1000
- -104 <= target <= 104

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def threeSumClosest(self, nums, target):
 # Brute Force O(n^2) - nested loops instead of
 # two pointers
 n = len(nums)
 for i in range(n):
 for j in range(i + 1, n):
 # check nums[i] and nums[j]
 pass
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def threeSumClosest(self, nums: List[int], target: int)
 -> int:
 ans = nums[0] + nums[1] + nums[2]

 nums.sort()

 for i in range(len(nums) - 2):
 if i > 0 and nums[i] == nums[i - 1]:
 continue
 # Choose nums[i] as the first number in the
 # triplet, then search the

 # remaining numbers in [i + 1, n - 1].
 l = i + 1
 r = len(nums) - 1
 while l < r:
 summ = nums[i] + nums[l] + nums[r]
 if summ == target:
 return summ
 if abs(summ - target) < abs(ans - target):
 ans = summ
 if summ < target:

 l += 1
 else:
 r -= 1

 return ans
```

- LeetCode

```
class Solution:
 def threeSumClosest(self, nums: List[int], target:
 int) -> int:
 nums.sort()
 n = len(nums)
 ans = inf
 for i, v in enumerate(nums):
 j, k = i + 1, n - 1
 while j < k:
 t = v + nums[j] + nums[k]
 if t == target:

 return t
 if abs(t - target) < abs(ans - target):
 ans = t
 if t > target:
 k -= 1
 else:
 j += 1
 return ans
```

- SimplyLeet

```
Python solution for 3Sum Closest
def threeSumClosest(nums, target):
 # Sort the array to allow two pointer technique
 nums.sort()
 # Initialize best_sum with the sum of the first three
 elements
 best_sum = sum(nums[:3])

 # Iterate through each number as the fixed element
 for i in range(len(nums) - 2):
 left, right = i + 1, len(nums) - 1
```

Two Pointers p.1000

Two Pointers p.1001

Two Pointers p.1002

```
Use two pointers to check pairs
while left < right:
 current_sum = nums[i] + nums[left] +
 nums[right]
 # Update best_sum if current_sum is closer to
 target
 if abs(target - current_sum) < abs(target -
 best_sum):
 best_sum = current_sum
 # Move left pointer if current_sum is less
 # than target
 # to increase sum
 if current_sum < target:
 left += 1
 # Move right pointer if current_sum is
 # greater than target to decrease sum

 elif current_sum > target:
 right -= 1
 else:
 # If the current_sum exactly equals
 # target, return the target sum
 return current_sum
 # Return the closest sum found
 return best_sum

Example usage:
print(threeSumClosest([-1,2,1,-4], 1))
```

- LC.ca

```
class Solution:
 def threeSumClosest(self, nums: List[int], target:
 int) -> int:
 nums.sort()
 n = len(nums)
 ans = inf
 for i, v in enumerate(nums):
 j, k = i + 1, n - 1
 while j < k:
 t = v + nums[j] + nums[k]
 if t == target:

 return t
 if abs(t - target) < abs(ans - target):
 ans = t
 if t > target:
 k -= 1
 else:
 j += 1
 return ans
```

Two Pointers p.1004

Two Pointers

#31 Next Permutation

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers

Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(1)

Problem

A permutation of an array of integers is an arrangement of its members into a sequence or linear order.

- For example, for arr = [1,2,3], the following are all the permutations of arr: [1,2,3], [1,3,2], [2, 1, 3], [2, 3, 1], [3,1,2], [3,2,1].

The next permutation of an array of integers is the next lexicographically greater permutation of its integer. More formally, if all the permutations of the array are sorted in one container according to their lexicographical order, then the next permutation of that array is the permutation that follows it in the sorted container. If such arrangement is not possible, the array must be rearranged as the lowest possible order (i.e., sorted in ascending order).

- For example, the next permutation of arr = [1,2,3] is [1,3,2].

Two Pointers p.1005

- Similarly, the next permutation of arr = [2,3,1] is [3,1,2].
- While the next permutation of arr = [3,2,1] is [1,2,3] because [3,2,1] does not have a lexicographical larger rearrangement.

Given an array of integers nums, find the next permutation of nums.

The replacement must be in place and use only constant extra memory.

Example 1:

Input: nums = [1,2,3]  
Output: [1,3,2]

Example 2:

Input: nums = [3,2,1]  
Output: [1,2,3]

Example 3:

Input: nums = [1,1,5]  
Output: [1,5,1]

Constraints:

- 1 <= nums.length <= 100
- 0 <= nums[i] <= 100

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def nextPermutation(self, nums):
 # Brute Force O(n^2) - nested loops instead of
 # two pointers
 n = len(nums)
 for i in range(n):
 for j in range(i + 1, n):
 # check nums[i] and nums[j]
 pass
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def nextPermutation(self, nums: List[int]) -> None:
 n = len(nums)

 # From back to front, find the first number < nums[i
 + 1].
 i = n - 2
 while i >= 0:
 if nums[i] < nums[i + 1]:
 break
 i -= 1
```

Two Pointers p.1007

```
From back to front, find the first number >
nums[i], swap it with nums[i].
if i >= 0:
 for j in range(n - 1, i, -1):
 if nums[j] > nums[i]:
 nums[i], nums[j] = nums[j], nums[i]
 break

def reverse(nums: List[int], l: int, r: int) -> None:
 while l < r:

 nums[l], nums[r] = nums[r], nums[l]
 l += 1
 r -= 1

 # Reverse nums[i + 1..n - 1].
 reverse(nums, i + 1, len(nums) - 1)
```

- LeetCode

```
class Solution:
 def nextPermutation(self, nums: List[int]) -> None:
 n = len(nums)
 i = next((i for i in range(n - 2, -1, -1) if
 nums[i] < nums[i + 1]), -1)
 if ~i:
 j = next((j for j in range(n - 1, i, -1) if
 nums[j] > nums[i]))
 nums[i], nums[j] = nums[j], nums[i]
 nums[i + 1 :] = nums[i + 1 :][::-1]
```

- SimplyLeet

Two Pointers p.1008

Two Pointers p.1006

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>def nextPermutation(nums):     # Length of the array     n = len(nums)     # Find the first index from the right where the     current number is less than the next number     i = n - 2     while i &gt;= 0 and nums[i] &gt;= nums[i + 1]:         i -= 1     # If such an index is found, find the number just     larger than nums[i] from the right side to swap     if i &gt;= 0:         j = n - 1          while nums[j] &lt;= nums[i]:             j -= 1         nums[i], nums[j] = nums[j], nums[i] # Swap the         two numbers     # Reverse the subarray from i+1 to the end to get the     smallest sequence     left, right = i + 1, n - 1     while left &lt; right:         nums[left], nums[right] = nums[right], nums[left]         left += 1         right -= 1  # Example usage: nums = [1, 2, 3] nextPermutation(nums) print(nums) # Output: [1, 3, 2]</pre> <ul style="list-style-type: none"><li>LC.ca</li></ul> | <pre>... &gt;&gt;&gt; i = 3 &gt;&gt;&gt; ~i -4 &gt;&gt;&gt; bool(~i) True #####  &gt;&gt;&gt; j = -1 &gt;&gt;&gt; ~j  0 &gt;&gt;&gt; bool(~j) False #####  &gt;&gt;&gt; a = (i for i in range (10, -1, -1) if i &lt; 6) &gt;&gt;&gt; a &lt;generator object &lt;genexpr&gt; at 0x10a17eeb0&gt; &gt;&gt;&gt; next(a) 5  &gt;&gt;&gt; &gt;&gt;&gt; &gt;&gt;&gt; b = (i for i in range (10, -1, -1) if i &lt; 0) &gt;&gt;&gt; next(b) Traceback (most recent call last):   File "&lt;stdin&gt;", line 1, in &lt;module&gt; StopIteration &gt;&gt;&gt; next(b, -1) -1 &gt;&gt;&gt;</pre> | <pre>... class Solution:     def nextPermutation(self, nums: List[int]) -&gt; None:         ....         Do not return anything, modify nums in-place         instead.         ....         n = len(nums)         # next(func, -1) =&gt; default value -1         i = next((i for i in range(n - 2, -1, -1) if         nums[i] &lt; nums[i + 1]), -1)         if i != -1:              j = next((j for j in range(n - 1, i, -1) if             nums[j] &gt; nums[i]))             nums[i], nums[j] = nums[j], nums[i]             nums[i + 1:] = nums[i + 1:][:~-1]             # wrong reverse: nums[i + 1:] = nums[i + 1::-1]  #####  ... &gt;&gt;&gt; a=[1,2,3] &gt;&gt;&gt; reversed(a)</pre> |
| Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| p.1009                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | p.1010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&lt;list_reverseiterator object at 0x108458be0&gt; &gt;&gt;&gt; list(reversed(a)) [3, 2, 1]  # but, use reversed(a) to directly assign values is ok &gt;&gt;&gt; b=[4,5,6,7,8] &gt;&gt;&gt; b[:3] = reversed(a) &gt;&gt;&gt; b [3, 2, 1, 7, 8] ...</pre> <pre>class Solution:     def nextPermutation(self, nums: List[int]) -&gt; None:         if not nums:             return          # #####next         for i in range(len(nums)-2, -1, -1): # 3, 4, 5,             2, 1             if nums[i] &lt; nums[i+1]: # ##### =&gt; 4                 j = next(j for j in range(len(nums)-1, i,                 -1) if nums[j] &gt; nums[i]) # #####nums-i#### =&gt; 5                 nums[i], nums[j] = nums[j], nums[i] #                 3,5,4,2,1</pre> | <pre>            # reverse #####             # #####nums-i#####insert  position         nums[i+1:] = reversed(nums[i+1:]) # [4,2,1] reverse to [1,2,4] =&gt; 3, 5, 1, 2, 4         return          nums.reverse() # for[]return#####  #####  class Solution:     def nextPermutation(self, nums: List[int]) -&gt; None:         n = len(nums)         to_left = [i for i in range(n - 2, -1, -1) if         nums[i] &lt; nums[i + 1]]          if not to_left:             nums = nums[::-1]             return          i = max(to_left)         if i &gt;= 0:             to_right = [ j for j in range(n - 1, i, -1) if             nums[j] &gt; nums[i] ]             j = to_right[0]             nums[i], nums[j] = nums[j], nums[i]             nums[i + 1:] = nums[i + 1::-1]</pre> | <div>Two Pointers</div> <div>#75 Sort ColorsMED</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Array · Two Pointers · Sorting</div> <div>Lists: Grind 169</div> <div>Time Complexity: O(n) Space Complexity: O(1)</div> <div>Problem</div> <div>Given an array nums with n objects colored red, white, or blue, sort them in-place so that objects of the same color are adjacent, with the colors in the order red, white, and blue.</div> <div>We will use the integers 0, 1, and 2 to represent the color red, white, and blue, respectively.</div> <div>You must solve this problem without using the library's sort function.</div> <div>Example 1:</div> <div>Input: nums = [2,0,2,1,1,0]<br/>Output: [0,0,1,1,2,2]</div> <div>Example 2:</div> <div>Input: nums = [2,0,1]<br/>Output: [0,1,2]</div> <div>Constraints:</div> <ul style="list-style-type: none"><li>n == nums.length</li></ul> |
| Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| p.1012                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | p.1013                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | p.1014                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

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| <ul style="list-style-type: none"><li>1 &lt;= n &lt;= 300</li><li>nums[i] is either 0, 1, or 2.</li></ul> <p>Follow up: Could you come up with a one-pass algorithm using only constant extra space?</p> <p>Brute Force (Python)</p> <pre># Brute Force Approach class Solution:     def sortColors(self, nums):         # Brute Force O(n log n) - just sort; Dutch-flag         is the optimal O(n) approach         nums.sort()</pre> <p>Community Solutions (Python)</p> <ul style="list-style-type: none"><li>WalkCC</li></ul> <pre>class Solution:     def sortColors(self, nums: list[int]) -&gt; None:         zero = -1         one = -1         two = -1          for num in nums:             if num == 0:                 two += 1             one += 1</pre> | <pre>        zero += 1         nums[two] = 2         nums[one] = 1         nums[zero] = 0         elif num == 1:             two += 1             one += 1             nums[two] = 2             nums[one] = 1         else:              two += 1             nums[two] = 2  class Solution:     def sortColors(self, nums: list[int]) -&gt; None:         l = 0 # The next 0 should be placed in l.         r = len(nums) - 1 # The next 2 should be placed in r.          i = 0         while i &lt;= r:             if nums[i] == 0:                 nums[i], nums[l] = nums[l], nums[i]                 i += 1</pre> | <pre>        l += 1         elif nums[i] == 1:             i += 1         else:             # We may swap a 0 to index i, but we're still not             sure whether this 0             # is placed in the correct index, so we can't             move pointer i.             nums[i], nums[r] = nums[r], nums[i]             r -= 1</pre> <ul style="list-style-type: none"><li>LeetDoocs</li></ul> <pre>class Solution:     def sortColors(self, nums: List[int]) -&gt; None:         i, j, k = -1, len(nums), 0         while k &lt; j:             if nums[k] == 0:                 i += 1                 nums[i], nums[k] = nums[k], nums[i]                 k += 1             elif nums[k] == 2:                 j -= 1              nums[j], nums[k] = nums[k], nums[j]         else:             k += 1</pre> <ul style="list-style-type: none"><li>SimplyLeet</li></ul> |
| Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Two Pointers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| p.1015                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | p.1016                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | p.1017                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

```
Function to sort the colors using Dutch National Flag algorithm
def sortColors(nums):
 # Initialize pointers for the start, current, and end positions
 low, mid, high = 0, 0, len(nums) - 1

 # Process elements until mid pointer exceeds high pointer
 while mid <= high:
 # If the current element is 0 (red)
 if nums[mid] == 0:
 # Swap current element with the element at low pointer

 nums[low], nums[mid] = nums[mid], nums[low]
 # Increment both low and mid pointers
 low += 1
 mid += 1
 # If the current element is 1 (white)
 elif nums[mid] == 1:
 # Move mid pointer forward since it's already in the correct place
 mid += 1
 # If the current element is 2 (blue)
 else:
 # Swap current element with the element at high pointer
 nums[mid], nums[high] = nums[high], nums[mid]
 # Decrement high pointer
 high -= 1

 # Example usage:
 nums = [2,0,2,1,1,0]
 sortColors(nums)
 print(nums) # Output: [0,0,1,1,2,2]
```

Two Pointers

p.1018

```
Swap current element with the element at high pointer
nums[mid], nums[high] = nums[high], nums[mid]
Decrement high pointer
high -= 1

Example usage:
nums = [2,0,2,1,1,0]
sortColors(nums)
print(nums) # Output: [0,0,1,1,2,2]
```

Two Pointers

p.1019

Two Pointers

#161 One Edit Distance

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Two Pointers · String

Lists: Premium 98

Time Complexity:  $O(n)$ , where  $n$  is the length of the shorter string, as we traverse both strings at most once. Space Complexity:  $O(1)$ , using only constant extra space.

Problem

Given two strings  $s$  and  $t$ , return true if they are both one edit distance apart, otherwise return false.

A string  $s$  is said to be one distance apart from a string  $t$  if you can:

- Insert exactly one character into  $s$  to get  $t$ .
- Delete exactly one character from  $s$  to get  $t$ .
- Replace exactly one character of  $s$  with a different character to get  $t$ .

Example 1:

```
Input: s = "ab", t = "acb"
Output: true
Explanation: We can insert 'c' into s to get t.
```

Example 2:

Two Pointers

p.1020

```
Input: s = "", t = ""
Output: false
Explanation: We cannot get t from s by only one step.
```

Constraints:

- $0 \leq s.length, t.length \leq 104$
- $s$  and  $t$  consist of lowercase letters, uppercase letters, and digits.

Community Solutions (Python)

- WalkCC

```
class Solution:
 def isOneEditDistance(self, s: str, t: str) -> bool:
 m = len(s)
 n = len(t)
 if m > n: # Make sure that |s| <= |t|.
 return self.isOneEditDistance(t, s)

 for i in range(m):
 if s[i] != t[i]:
 if m == n:

 return s[i + 1:] == t[i + 1:] # Replace s[i] with t[i].
 return s[i:] == t[i + 1:] # Delete t[i].

 return m + 1 == n # Delete t[-1].
```

- LeetCode

Two Pointers

p.1021

```
class Solution:
 def isOneEditDistance(self, s: str, t: str) -> bool:
 if len(s) < len(t):
 return self.isOneEditDistance(t, s)
 m, n = len(s), len(t)
 if m - n > 1:
 return False
 for i, c in enumerate(t):
 if c != s[i]:
 return s[i + 1:] == t[i + 1:] if m == n
 else s[i + 1:] == t[i:]

 return m == n + 1
```

Two Pointers

p.1022

```
if m > n:
 return isOneEditDistance(t, s)

Initialize variables; found difference flag and indices for both strings
found_difference = False
i = j = 0

while i < m and j < n:
 if s[i] != t[j]:
 # If a difference has already been found,
 return False

 if found_difference:
 return False
 found_difference = True
 # If lengths are equal, move both pointers
 if m == n:
 i += 1
 else:
 # If characters are the same, move the pointer in the shorter string
 i += 1
 # Always move pointer for longer string
 j += 1

If no difference was found during iteration, then one extra character at the end is the edit
return found_difference or (j - i == 1)

Example usage:
#print(isOneEditDistance("ab", "acb")) # Output: True
```

Two Pointers

p.1023

- LC.ca

```
class Solution:
 def isOneEditDistance(self, s: str, t: str) -> bool:
 if len(s) < len(t): # goal is to ensure t is always the shorter one
 return self.isOneEditDistance(t, s)
 m, n = len(s), len(t)
 if m - n > 1:
 # case: abc, abcdefg, return false
 return False
 for i, c in enumerate(t):
 if c != s[i]:

 return s[i + 1:] == t[i + 1:] if m == n
 else s[i + 1:] == t[i:]
 # case: abc, abcd. return true
 # case: abc, abc. the same 2 strings should
 return false
 # case: abc, abcdefg, return false
 return m == n + 1

#####

class Solution(object):
 def isOneEditDistance(self, s, t):
```

Two Pointers

p.1024

```
"""
:type s: str
:type t: str
:rtype: bool
"""
if len(s) >= len(t):
 i = 0
 while i < len(t) and s[i] == t[i]:
 i += 1
 return s != t and (s[i + 1:] == t[i:] if len(s) != len(t) else s[i + 1:] == t[i + 1:])

else:
 return self.isOneEditDistance(t, s)
```

Two Pointers

p.1025

Two Pointers

#167 Two Sum II – Input Array Is Sorted

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers · Binary Search

Lists: Denny Zhang

Time Complexity:  $O(n)$ , where  $n$  is the number of elements in the array, because in the worst-case scenario each element is visited at most once. Space Complexity:  $O(1)$ , as no additional data structure

Problem

Given a 1-indexed array of integers numbers that is already sorted in non-decreasing order, find two numbers such that they add up to a specific target number. Let these two numbers be numbers[index1] and numbers[index2] where  $1 \leq \text{index1} < \text{index2} \leq \text{numbers.length}$ .

Return the indices of the two numbers index1 and index2, each incremented by one, as an integer array [index1, index2] of length 2.

The tests are generated such that there is exactly one solution. You may not use the same element twice. Your solution must use only constant extra space.

Example 1:

Two Pointers

p.1026

Input: numbers = [2,7,11,15], target = 9  
Output: [1,2]  
Explanation: The sum of 2 and 7 is 9. Therefore, index1 = 1, index2 = 2. We return [1, 2].

Example 2:

Input: numbers = [2,3,4], target = 6  
Output: [1,3]  
Explanation: The sum of 2 and 4 is 6. Therefore index1 = 1, index2 = 3. We return [1, 3].

Example 3:

Input: numbers = [-1,0], target = -1  
Output: [1,2]  
Explanation: The sum of -1 and 0 is -1. Therefore index1 = 1, index2 = 2. We return [1, 2].

Constraints:

- 2 <= numbers.length <= 3 \* 104
- -1000 <= numbers[i] <= 1000
- numbers is sorted in non-decreasing order.
- -1000 <= target <= 1000
- The tests are generated such that there is exactly one solution.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def twoSum(self, numbers, target):
 # Brute Force O(n^2) - check every pair (ignore
 sorted property)
 for i in range(len(numbers)):
 for j in range(i + 1, len(numbers)):
 if numbers[i] + numbers[j] == target:
 return [i + 1, j + 1]
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def twoSum(self, numbers: list[int], target: int) ->
list[int]:
 l = 0
 r = len(numbers) - 1

 while l < r:
 summ = numbers[l] + numbers[r]
 if summ == target:
 return [l + 1, r + 1]
 if summ < target:

 l += 1
 else:
 r -= 1
```

• LeetDoocs

```
class Solution:
 def twoSum(self, numbers: List[int], target: int) ->
List[int]:
 n = len(numbers)
 for i in range(n - 1):
 x = target - numbers[i]
 j = bisect_left(numbers, x, lo=i + 1)
 if j < n and numbers[j] == x:
 return [i + 1, j + 1]
```

• SimplyLeet

```
Python solution for Two Sum II - Input Array Is Sorted
def twoSum(numbers, target):
 # Initialize two pointers at the beginning and end of
 the array
 left = 0
 right = len(numbers) - 1

 # Continue until the two pointers meet
 while left < right:
 # Calculate the sum of the two pointed values
 current_sum = numbers[left] + numbers[right]
```

Two Pointers p.1027

Two Pointers p.1028

Two Pointers p.1029

```
If the sum matches the target, return the
1-indexed results
if current_sum == target:
 return [left + 1, right + 1]
If the current sum is less than the target,
move the left pointer to the right
elif current_sum < target:
 left += 1
Otherwise, move the right pointer to the left
since the current sum is too high
else:
 right -= 1
```

```
Example call
print(twoSum([2, 7, 11, 15], 9)) # Output: [1, 2]
```

• LC.ca

```
class Solution:
 def twoSum(self, numbers: List[int], target: int) ->
List[int]:
 i, j = 0, len(numbers) - 1
 while i < j:
 x = numbers[i] + numbers[j]
 if x == target:
 return [i + 1, j + 1]
 if x < target:
 i += 1
 else:
 j -= 1
```

Two Pointers p.1030

Two Pointers

#186 Reverse Words in a String II

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Two Pointers · String

Lists: Premium 98

Time Complexity: O(n) Space Complexity: O(1)

Problem

Given a character array s, reverse the order of the words. A word is defined as a sequence of non-space characters. The words in s will be separated by a single space.

Your code must solve the problem in-place, i.e. without allocating extra space.

Example 1:

```
Input: s = ["t","h","e"," ", "s","k","y"," ", "i","s"," ",
",","b","l","u","e"]
Output: ["b","l","u","e"," ", "i","s"," ", "s","k","y"," ",
",","t","h","e"]
```

Example 2:

```
Input: s = ["a"]
Output: ["a"]
```

Constraints:

- 1 <= s.length <= 105

Two Pointers p.1031

- s[i] is an English letter (uppercase or lowercase), digit, or space ' '.
- There is at least one word in s.
- s does not contain leading or trailing spaces.
- All the words in s are guaranteed to be separated by a single space.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def reverse_range(self, s, left, right):
 # Brute Force O(n^2) - nested loops instead of
 two pointers
 n = len(s)
 for i in range(n):
 for j in range(i + 1, n):
 # check s[i] and s[j]
 pass
```

Community Solutions (Python)

• WalkCC

Two Pointers p.1032

```
class Solution:
 def reverseWords(self, s: list[str]) -> None:
 def reverse(l: int, r: int) -> None:
 while l < r:
 s[l], s[r] = s[r], s[l]
 l += 1
 r -= 1

 def reverseWords(n: int) -> None:
 i = 0

 j = 0

 while i < n:
 while i < j or (i < n and s[i] == ' '): # Skip the
spaces.
 i += 1
 while j < i or (j < n and s[j] != ' '): # Skip the
spaces.
 j += 1
 reverse(i, j - 1) # Reverse the word.

 reverse(0, len(s) - 1) # Reverse the whole string.

 reverseWords(len(s)) # Reverse each word.
```

• LeetDoocs

Two Pointers p.1033

```
class Solution:
 def reverseWords(self, s: List[str]) -> None:
 def reverse(i: int, j: int):
 while i < j:
 s[i], s[j] = s[j], s[i]
 i, j = i + 1, j - 1

 i, n = 0, len(s)
 for j, c in enumerate(s):
 if c == " ":
 reverse(i, j - 1)
 i = j + 1
 elif j == n - 1:
 reverse(i, j)
 reverse(0, n - 1)
```

• SimplyLeet

```
def reverse_range(s, left, right):
 # Reverse the elements in s from index left to right.
 while left < right:
 s[left], s[right] = s[right], s[left]
 left += 1
 right -= 1

def reverseWords(s):
 # Step 1: Reverse the entire array.
 reverse_range(s, 0, len(s) - 1)
```

Two Pointers p.1034

```
n = len(s)
start = 0
Step 2: Reverse each word to correct their order.
for i in range(n + 1):
 if i == n or s[i] == ' ':
 reverse_range(s, start, i - 1)
 start = i + 1
```

• LC.ca

```
class Solution:
 def reverseWords(self, s: List[str]) -> None:
 """"
 Do not return anything, modify s in-place
 instead.
 """"

 def reverse(s, i, j):
 while i < j:
 s[i], s[j] = s[j], s[i]
 i += 1
 j -= 1

 i, j, n = 0, 0, len(s)
 while j < n:
 if s[j] == ' ':
 reverse(s, i, j - 1)
 i = j + 1
 elif j == n - 1:
 reverse(s, i, j)
 j += 1
```

Two Pointers p.1035

```
reverse(s, 0, n - 1)
```

Two Pointers

#189 Rotate ArrayMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Array · Math · Two Pointers  
Lists: Grind 169  
Time Complexity: O(n) where n is the number of elements in the array. Space Complexity: O(1) if done in-place.

**Problem**  
Given an integer array nums, rotate the array to the right by k steps, where k is non-negative.

**Example 1:**  
Input: nums = [1,2,3,4,5,6,7], k = 3  
Output: [5,6,7,1,2,3,4]  
Explanation:  
rotate 1 steps to the right: [7,1,2,3,4,5,6]  
rotate 2 steps to the right: [6,7,1,2,3,4,5]  
rotate 3 steps to the right: [5,6,7,1,2,3,4]

**Example 2:**  
Input: nums = [-1,-100,3,99], k = 2  
Output: [3,99,-1,-100]  
Explanation:  
rotate 1 steps to the right: [99,-1,-100,3]  
rotate 2 steps to the right: [3,99,-1,-100]

**Constraints:**

- 1 <= nums.length <= 105
- -231 <= nums[i] <= 231 - 1
- 0 <= k <= 105

**Follow up:**

- Try to come up with as many solutions as you can. There are at least three different ways to solve this problem.
- Could you do it in-place with O(1) extra space?

**Brute Force (Python)**

```
Brute Force Approach
class Solution:
 def rotate(self, nums, k):
 # Brute Force O(n^2) - nested loops instead of two pointers
 n = len(nums)
 for i in range(n):
 for j in range(i + 1, n):
 # check nums[i] and nums[j]
 pass
```

**Community Solutions (Python)**

- WalkCC

```
class Solution:
 def rotate(self, nums: List[int], k: int) -> None:
 k %= len(nums)
 self.reverse(nums, 0, len(nums) - 1)
 self.reverse(nums, 0, k - 1)
 self.reverse(nums, k, len(nums) - 1)

 def reverse(self, nums: List[int], l: int, r: int) -> None:
 while l < r:
 nums[l], nums[r] = nums[r], nums[l]

 l += 1
 r -= 1
```

• LeetDoocs

```
class Solution:
 def rotate(self, nums: List[int], k: int) -> None:
 def reverse(i: int, j: int):
 while i < j:
 nums[i], nums[j] = nums[j], nums[i]
 i, j = i + 1, j - 1

 n = len(nums)
 k %= n
 reverse(0, n - 1)

 reverse(0, k - 1)
 reverse(k, n - 1)
```

• SimplyLeet

```
Python solution using the reverse method

def rotate(nums, k):
 # Calculate the effective rotations needed
 n = len(nums)
 k = k % n # in case k is greater than the length of the array

 # Helper function to reverse a portion of the list in place
 def reverse(left, right):
 while left < right:

 # Swap the elements at indices left and right
 nums[left], nums[right] = nums[right], nums[left]

 left += 1
 right -= 1

 # Reverse the entire array
 reverse(0, n - 1)
 # Reverse the first k elements to restore their order
 reverse(0, k - 1)
 # Reverse the remaining n - k elements

 reverse(k, n - 1)

Example usage:
nums = [1,2,3,4,5,6,7]
rotate(nums, 3)
print(nums) # Output: [5,6,7,1,2,3,4]
```

• LC.ca

```
class Solution:
 def rotate(self, nums: List[int], k: int) -> None:
 k %= len(nums)
 nums[:] = nums[-k:] + nums[:-k]

#####

class Solution:
 def rotate(self, nums: List[int], k: int) -> None:
 """
 Do not return anything, modify nums in-place instead.
 """
 n = len(nums)
 k %= n
 if n < 2 or k == 0:
 return
 nums[:] = nums[::-1]
 nums[:k] = nums[:k][::-1]
 nums[k:] = nums[k:][::-1]

#####

class Solution(object):
 def rotate(self, nums, k):
 """
 :type nums: List[int]
 :type k: int
 :rtype: void Do not return anything, modify nums in-place instead.
 """
 if len(nums) == 0 or k == 0:
```

```
return

def reverse(start, end, s):
 while start < end:
 s[start], s[end] = s[end], s[start]
 start += 1
 end -= 1

n = len(nums) - 1
k = k % len(nums)

reverse(0, n - k, nums)
reverse(n - k + 1, n, nums)
reverse(0, n, nums)
```

Two Pointers

#532 K-diff Pairs in an ArrayMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Array · Hash Table · Two Pointers · Binary Search · Sorting  
Lists: CodeSignal  
Time Complexity: O(n), where n is the length of the array since we iterate over the array and then over the unique numbers. Space Complexity: O(n), for storing the frequency count or the unique element

**Problem**  
Given an array of integers nums and an integer k, return the number of unique k-diff pairs in the array. A k-diff pair is an integer pair (nums[i], nums[j]), where the following are true:

- 0 <= i, j < nums.length
- i != j
- |nums[i] - nums[j]| == k

Notice that |val| denotes the absolute value of val.  
**Example 1:**

Input: nums = [3,1,4,1,5], k = 2  
Output: 2  
Explanation: There are two 2-diff pairs in the array, (1, 3) and (3, 5).  
Although we have two 1s in the input, we should only return the number of unique pairs.

**Example 2:**  
Input: nums = [1,2,3,4,5], k = 1  
Output: 4  
Explanation: There are four 1-diff pairs in the array, (1, 2), (2, 3), (3, 4) and (4, 5).

**Example 3:**  
Input: nums = [1,3,1,5,4], k = 0  
Output: 1  
Explanation: There is one 0-diff pair in the array, (1, 1).

**Constraints:**

- 1 <= nums.length <= 104
- -107 <= nums[i] <= 107
- 0 <= k <= 107

**Brute Force (Python)**



```
Brute Force Approach
class Solution:
 def find_pairs(self, nums, k):
 # Brute Force O(n^2) - nested loops instead of
two pointers
 n = len(nums)
 for i in range(n):
 for j in range(i + 1, n):
 # check nums[i] and nums[j]
 pass
```

## Community Solutions (Python)

### • WalkCC

```
class Solution:
 def findPairs(self, nums: list[int], k: int) -> int:
 ans = 0
 numToIndex = {num: i for i, num in enumerate(nums)}

 for i, num in enumerate(nums):
 target = num + k
 if target in numToIndex and numToIndex[target] !=
i:
 ans += 1
 del numToIndex[target]

 return ans
```

### • LeetDoocs

Two Pointers

p.1045

```
class Solution:
 def findPairs(self, nums: List[int], k: int) -> int:
 ans = set()
 vis = set()
 for x in nums:
 if x - k in vis:
 ans.add(x - k)
 if x + k in vis:
 ans.add(x)
 vis.add(x)

 return len(ans)
```

### • SimplyLeet

```
Define a function to count k-diff pairs in the array.
def find_pairs(nums, k):
 # If k is negative, return 0 as absolute difference
 can't be negative.
 if k < 0:
 return 0

 count = 0

 # k == 0: Need to find numbers that appear more than
 once.
 if k == 0:
```

Two Pointers

p.1046

```
Create a frequency map for nums.
freq = {}
for num in nums:
 freq[num] = freq.get(num, 0) + 1
Count numbers with frequency greater than 1.
for val in freq.values():
 if val > 1:
 count += 1

else:
 # k > 0: Use a set to check for the existence of
 each number's complement.
```

```
unique_nums = set(nums)
for num in unique_nums:
 if num + k in unique_nums:
 count += 1

return count
```

```
Example usage:
print(find_pairs([3,1,4,1,5], 2)) # Expected output: 2
```

### • LC.ca

```
class Solution:
 def findPairs(self, nums: List[int], k: int) -> int:
 vis, ans = set(), set()
 for v in nums:
 if v - k in vis:
 ans.add(v - k)
 if v + k in vis:
 ans.add(v)
 vis.add(v)
 return len(ans)
```

Two Pointers

p.1047

Two Pointers

## #923 3Sum With Multiplicity

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Two Pointers · Sorting · Counting  
Lists: CodeSignal

Time Complexity:  $O(M^2)$ , where  $M = 101$  (the range of possible numbers), which is efficient given the constraints. Space Complexity:  $O(M)$  for the frequency array.

### Problem

Given an integer array `arr`, and an integer `target`, return the number of tuples  $i, j, k$  such that  $i < j < k$  and `arr[i] + arr[j] + arr[k] == target`.

As the answer can be very large, return it modulo  $10^9 + 7$ .

### Example 1:

```
Input: arr = [1,1,2,2,3,3,4,4,5,5], target = 8
Output: 20
Explanation:
Enumerating by the values (arr[i], arr[j], arr[k]):
(1, 2, 5) occurs 8 times;
(1, 3, 4) occurs 8 times;
(2, 2, 4) occurs 2 times;
(2, 3, 3) occurs 2 times.
```

### Example 2:

Two Pointers

p.1048

```
Input: arr = [1,1,2,2,2,2], target = 5
Output: 12
Explanation:
arr[i] = 1, arr[j] = arr[k] = 2 occurs 12 times:
We choose one 1 from [1,1] in 2 ways,
and two 2s from [2,2,2,2] in 6 ways.
```

### Example 3:

```
Input: arr = [2,1,3], target = 6
Output: 1
Explanation: (1, 2, 3) occurred one time in the array so
we return 1.
```

### Constraints:

- $3 \leq \text{arr.length} \leq 3000$
- $0 \leq \text{arr}[i] \leq 100$
- $0 \leq \text{target} \leq 300$

### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def threeSumMulti(self, arr, target):
 # Brute Force O(n^2) - nested loops instead of
two pointers
 n = len(arr)
 for i in range(n):
 for j in range(i + 1, n):
 # check arr[i] and arr[j]
 pass
```

### Community Solutions (Python)

Two Pointers

p.1049

```
class Solution:
 def threeSumMulti(self, arr: List[int], target: int)
-> int:
 mod = 10**9 + 7
 cnt = Counter(arr)
 ans = 0
 for j, b in enumerate(arr):
 cnt[b] -= 1
 for a in arr[:j]:
 c = target - a - b
 ans = (ans + cnt[c]) % mod

 return ans
```

### • SimplyLeet

```
MOD = 10**9 + 7

def threeSumMulti(arr, target):
 # Create a frequency array for numbers 0 to 100
 freq = [0] * 101
 for num in arr:
 freq[num] += 1

 count = 0
 # Iterate through all possible values for x, y, and
 implicitly z
```

Two Pointers

p.1051

```
for x in range(101):
 if freq[x] == 0:
 continue
 for y in range(x, 101):
 if freq[y] == 0:
 continue
 z = target - x - y
 if z < 0 or z > 100 or freq[z] == 0:
 continue
 if x == y and y == z:
```

```
All numbers are the same: nC3 =
n*(n-1)*(n-2)/6
count += freq[x] * (freq[x] - 1) *
(freq[x] - 2) // 6
elif x == y and y != z:
 # x and y are the same, z is different:
nC2 for x multiplied by count[z]
count += (freq[x] * (freq[x] - 1) // 2) *
freq[z]
elif x < y and y < z:
 # All distinct: product of counts
count += freq[x] * freq[y] * freq[z]
elif x < y and y == z:
 # x different, y and z are the same:
count[x] multiplied by nC2 for y
```

Two Pointers

p.1052

```
count += freq[x] * (freq[y] * (freq[y] -
1) // 2)
count %= MOD
return count
```

```
Example usage:
print(threeSumMulti([1,1,2,2,3,3,4,4,5,5], 8))
```

### • LC.ca

```
class Solution:
 def threeSumMulti(self, arr: List[int], target: int)
-> int:
 cnt = Counter(arr)
 ans = 0
 mod = 10**9 + 7
 for j, b in enumerate(arr):
 cnt[b] -= 1
 for i in range(j):
 a = arr[i]
 c = target - a - b

 ans = (ans + cnt[c]) % mod
 return ans
```

Two Pointers

p.1053

Two Pointers

#986 Interval List IntersectionsMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers

Lists: Denny Zhang

Time Complexity:  $O(m + n)$  where  $m$  and  $n$  are the lengths of firstList and secondList respectively. Space Complexity:  $O(1)$  extra space (excluding the space used for the output).

Problem

You are given two lists of closed intervals, firstList and secondList, where firstList[i] = [starti, endi] and secondList[j] = [startj, endj]. Each list of intervals is pairwise disjoint and in sorted order.

Return the intersection of these two interval lists.

A closed interval [a, b] (with  $a \leq b$ ) denotes the set of real numbers  $x$  with  $a \leq x \leq b$ .

The intersection of two closed intervals is a set of real numbers that are either empty or represented as a closed interval. For example, the intersection of [1, 3] and [2, 4] is [2, 3].

Example 1:

Two Pointersp.1054

A

B

ans

0

4

8

12

16

20

24

Input: firstList = [[0,2],[5,10],[13,23],[24,25]], secondList = [[1,5],[8,12],[15,24],[25,26]]

Output: [[1,2],[5,5],[8,10],[15,23],[24,24],[25,25]]

Example 2:

Input: firstList = [[1,3],[5,9]], secondList = []

Output: []

Constraints:

- $0 \leq \text{firstList.length}, \text{secondList.length} \leq 1000$
- $\text{firstList.length} + \text{secondList.length} \geq 1$
- $0 \leq \text{starti} < \text{endi} \leq 109$
- $\text{endi} < \text{starti} + 1$
- $0 \leq \text{startj} < \text{endj} \leq 109$
- $\text{endj} < \text{startj} + 1$

Brute Force (Python)

Two Pointersp.1055

# Brute Force Approach

class Solution:

def intervalIntersection(self, firstList, secondList):

# Brute Force  $O(n^2)$  - nested loops instead of two pointers

n = len(firstList)

for i in range(n):

for j in range(i + 1, n):

# check firstList[i] and firstList[j]

pass

Community Solutions (Python)

• WalkCC

class Solution:

def intervalIntersection(self, firstList: list[list[int]], secondList: list[list[int]])

-> list[list[int]]:

ans = []

i = 0

j = 0

while i < len(firstList) and j < len(secondList):

# lo := the start of the intersection

# hi := the end of the intersection

Two Pointersp.1056

lo = max(firstList[i][0], secondList[j][0])

hi = min(firstList[i][1], secondList[j][1])

if lo <= hi:

ans.append([lo, hi])

if firstList[i][1] < secondList[j][1]:

i += 1

else:

j += 1

return ans

• LeetCode

class Solution:

def intervalIntersection(self, firstList: List[List[int]], secondList: List[List[int]])

-> List[List[int]]:

i = j = 0

ans = []

while i < len(firstList) and j < len(secondList):

s1, e1, s2, e2 = \*firstList[i], \*secondList[j]

l, r = max(s1, s2), min(e1, e2)

if l <= r:

ans.append([l, r])

if e1 < e2:

i += 1

else:

j += 1

return ans

• SimplyLeet

Two Pointersp.1057

# Function to compute the interval intersections using two pointers

def intervalIntersection(firstList, secondList):

intersections = [] # List to store the resulting intersections

i, j = 0, 0 # Pointers for firstList and secondList

# Loop until either list is fully traversed

while i < len(firstList) and j < len(secondList):

start1, end1 = firstList[i]

start2, end2 = secondList[j]

# Find the intersection boundaries

start\_overlap = max(start1, start2)

end\_overlap = min(end1, end2)

# Check if there is an intersection

if start\_overlap <= end\_overlap:

intersections.append([start\_overlap, end\_overlap])

# Move the pointer that has the smaller interval ending

if end1 < end2:

Two Pointersp.1058

i += 1

else:

j += 1

return intersections

# Example usage:

firstList = [[0,2],[5,10],[13,23],[24,25]]

secondList = [[1,5],[8,12],[15,24],[25,26]]

print(intervalIntersection(firstList, secondList)) # Output: [[1,2],[5,5],[8,10],[15,23],[24,24],[25,25]]

• LC.ca

class Solution:

def intervalIntersection(self, firstList: List[List[int]], secondList: List[List[int]])

-> List[List[int]]:

i = j = 0

ans = []

while i < len(firstList) and j < len(secondList):

s1, e1, s2, e2 = \*firstList[i], \*secondList[j]

l, r = max(s1, s2), min(e1, e2)

if l <= r:

ans.append([l, r])

if e1 < e2:

i += 1

else:

j += 1

return ans

Two Pointersp.1059

Two Pointers

#1055 Shortest Way to Form StringMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Two Pointers · String · Binary Search

Lists: Premium 98

Time Complexity:  $O(n * m)$  in the worst-case, where  $n$  = length of source and  $m$  = length of target. Space Complexity:  $O(1)$  for the greedy simulation  $O(n)$  if additional data structures for binary search

Problem

A subsequence of a string is a new string that is formed from the original string by deleting some (can be none) of the characters without disturbing the relative positions of the remaining characters. (i.e., "ace" is a subsequence of "abcde" while "aec" is not).

Given two strings source and target, return the minimum number of subsequences of source such that their concatenation equals target. If the task is impossible, return -1.

Example 1:

Input: source = "abc", target = "abcbc"

Output: 2

Explanation: The target "abcbc" can be formed by "abc" and "bc", which are subsequences of source "abc".

Two Pointersp.1060

Example 2:

Input: source = "abc", target = "acdbc"

Output: -1

Explanation: The target string cannot be constructed from the subsequences of source string due to the character "d" in target string.

Example 3:

Input: source = "xyz", target = "xzyxz"

Output: 3

Explanation: The target string can be constructed as follows "xz" + "y" + "xz".

Constraints:

- $1 \leq \text{source.length}, \text{target.length} \leq 1000$
- source and target consist of lowercase English letters.

Brute Force (Python)

# Brute Force Approach

class Solution:

def shortestWay(self, source, target):

# Brute Force  $O(n^2)$  - nested loops instead of two pointers

n = len(source)

for i in range(n):

for j in range(i + 1, n):

# check source[i] and source[j]

pass

Community Solutions (Python)

Two Pointersp.1061

• LeetCode

class Solution:

def shortestWay(self, source: str, target: str) -> int:

def f(i, j):

while i < m and j < n:

if source[i] == target[j]:

j += 1

i += 1

return j

m, n = len(source), len(target)

ans = j = 0

while j < n:

k = f(0, j)

if k == j:

return -1

j = k

ans += 1

return ans

• SimplyLeet

Two Pointersp.1062

# Python solution using greedy two-pointer approach

```
def shortestWay(source, target):
 count = 0 # Count of subsequences used
 t_index = 0
 t_len = len(target)

 # Pre-check: if any character in target is not in
 source, return -1
 if set(target) - set(source):
 return -1
```

```
While there are still characters in target to match
while t_index < t_len:
 s_index = 0 # Pointer for source
 progress = False # Flag to check if we made
 progress this pass
```

```
Iterate through source and try to match target
characters
while s_index < len(source) and t_index < t_len:
 if source[s_index] == target[t_index]:
 t_index += 1 # Move to next character in
 target
```

Two Pointers

p.1063

```
 progress = True
 s_index += 1

 count += 1 # One subsequence used for this pass

 # If no progress was made, then it is impossible
 to form the target
 if not progress:
 return -1

 return count
```

```
Example usage:
print(shortestWay("abc", "abcbc")) # Output: 2
```

• LC.ca

```
class Solution:
 def shortestWay(self, source: str, target: str) ->
 int:

 def f(i, j):
 while i < m and j < n:
 if source[i] == target[j]:
 j += 1
 i += 1
 return j

 m, n = len(source), len(target)
```

Two Pointers

p.1064

```
ans = j = 0
while j < n:
 k = f(0, j)
 if k == j:
 return -1
 j = k
 ans += 1
return ans
```

Two Pointers

p.1065

Two Pointers

## #2563 Count the Number of Fair Pairs

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers · Binary Search · Sorting

Lists: CodeSignal

Time Complexity:  $O(n \log n)$ , primarily due to sorting and (for each element) binary search operations. Space Complexity:  $O(n)$  if sorting creates a copy;  $O(1)$  additional space if sorting in-place.

### Problem

Given a 0-indexed integer array `nums` of size `n` and two integers `lower` and `upper`, return the number of fair pairs.

A pair  $(i, j)$  is fair if:

- $0 \leq i < j < n$ , and
- $\text{lower} \leq \text{nums}[i] + \text{nums}[j] \leq \text{upper}$

Example 1:

```
Input: nums = [0,1,7,4,4,5], lower = 3, upper = 6
Output: 6
Explanation: There are 6 fair pairs: (0,3), (0,4),
(0,5), (1,3), (1,4), and (1,5).
```

Example 2:

Two Pointers

p.1066

```
Input: nums = [1,7,9,2,5], lower = 11, upper = 11
Output: 1
Explanation: There is a single fair pair: (2,3).
```

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $\text{nums.length} == n$
- $-109 \leq \text{nums}[i] \leq 109$
- $-109 \leq \text{lower} \leq \text{upper} \leq 109$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def countFairPairs(self, nums, lower, upper):
 # Brute Force $O(n^2)$ - nested loops instead of
 two pointers
 n = len(nums)
 for i in range(n):
 for j in range(i + 1, n):
 # check nums[i] and nums[j]
 pass
```

Community Solutions (Python)

• WalkCC

Two Pointers

p.1067

```
class Solution:
 def countFairPairs(self, nums: List[int], lower: int,
 upper: int) -> int:
 nums.sort()
 ans = 0
 for i, x in enumerate(nums):
 j = bisect_left(nums, lower - x, lo=i + 1)
 k = bisect_left(nums, upper - x + 1, lo=i +
1)
 ans += k - j
 return ans
```

• SimplyLeet

```
import bisect

def countFairPairs(nums, lower, upper):
 # Sort the array to allow binary search
 nums.sort()
 fair_pairs = 0
 n = len(nums)

 for i in range(n):
 # Find lower bound index for the required
 complement so that $\text{nums}[i] + \text{nums}[j] \geq \text{lower}$
```

Two Pointers

p.1069

```
 left = bisect.bisect_left(nums, lower - nums[i],
 i + 1, n)
 # Find upper bound index for the required
 complement so that $\text{nums}[i] + \text{nums}[j] \leq \text{upper}$
 right = bisect.bisect_right(nums, upper -
 nums[i], i + 1, n)
 fair_pairs += (right - left)

 return fair_pairs

Example usage:
print(countFairPairs([0,1,7,4,4,5], 3, 6)) # Expected
output: 6
```

• LC.ca

```
class Solution:
 def countFairPairs(self, nums: List[int], lower: int,
 upper: int) -> int:
 nums.sort()
 ans = 0
 for i, x in enumerate(nums):
 j = bisect_left(nums, lower - x, lo=i + 1)
 k = bisect_left(nums, upper - x + 1, lo=i +
1)
 ans += k - j
 return ans
```

Two Pointers

p.1070

```
 while i < j and nums[i] + nums[j] > summ:
 j -= 1
 res += j - i
 i += 1
 return res

 return countLess(upper) - countLess(lower - 1)
```

• LeetCode

Two Pointers

p.1068

Two Pointers

p.1071

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
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| <div>&gt;&gt; Quick Review — Two Pointers</div> <div>19 questions · insights · complexity · solution</div> <div>#88 Merge Sorted Array <span>[Easy]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Both arrays are already sorted in non-decreasing order.</li> <li>The merge should be done in-place into nums1.</li> <li>Start merging from the end to avoid overwriting elements in nums1.</li> <li>Use two pointers, one for nums1 and one for nums2, and a tail pointer for placement.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(m + n)</div> <div>Space Complexity: O(1)</div> </div> <div>Solution <div>We use a two-pointer approach starting from the end of both arrays. Initialize three pointers: <ul style="list-style-type: none"> <li>p1 pointing to the last valid element in nums1.</li> <li>p2 pointing to the last element in nums2.</li> <li>tail pointing to the last index of nums1.</li> </ul> While both pointers are valid, compare the elements pointed by p1 and p2, placing the larger element at the tail position. Decrement the corresponding pointer and</div> </div> <div>Two Pointersp.1072</div>                                                                                                                                                                                                                                                                | <div>the tail pointer. Finally, if any elements remain in nums2 (i.e., p2 &gt;= 0), copy them over into nums1. This solution leverages the extra space at the end of nums1 and ensures the merge is done in-place in linear time.</div> <div>#125 Valid Palindrome <span>[Easy]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Use two pointers (one starting at the beginning and one at the end) to compare characters.</li> <li>Skip characters that are not alphanumeric.</li> <li>Compare characters in a case-insensitive manner.</li> <li>Continue until the pointers meet or cross.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n), where n is the length of the string, as each character is processed at most once.</div> <div>Space Complexity: O(1), using only a few extra variables for pointers and temporary comparisons.</div> </div> <div>Solution <div>The solution employs a two-pointer approach: <ul style="list-style-type: none"> <li>Initialize two pointers: one at the start (left) and one at the end (right) of the string.</li> <li>Skip any non-alphanumeric characters using these pointers.</li> <li>Convert the remaining characters at each pointer to lowercase and compare them.</li> <li>If any mismatch is found, return false; otherwise, continue until the pointers cross.</li> <li>Return true if all corresponding</li> </ul> </div> </div> <div>Two Pointersp.1073</div> | <div>characters match. This method efficiently checks for a palindrome without needing additional space for a filtered string.</div> <div>#283 Move Zeroes <span>[Easy]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Use a two-pointer technique: one pointer to iterate through the array and another pointer to track the position for the next non-zero element.</li> <li>For each non-zero element encountered, swap it with the element at the tracking pointer.</li> <li>This strategy minimizes operations by ensuring each non-zero element is moved at most once.</li> <li>After repositioning all non-zero elements, zeros will naturally occupy the remaining positions.</li> <li>The in-place requirement ensures space complexity remains O(1).</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n), where n is the number of elements in the array, since only one pass (or at most two simple passes) is required.</div> <div>Space Complexity: O(1) extra space, as the reordering is accomplished in-place.</div> </div> <div>Solution <div>The solution leverages a two-pointer approach. The left pointer (j) keeps track of the index where the next</div> </div> <div>Two Pointersp.1074</div>                                                                                                                          |
| <div>non-zero element should be placed, while the right pointer (i) iterates through the array. Whenever a non-zero element is encountered, it is swapped with the element at index j, and j is then incremented. This effectively gathers all non-zero elements at the beginning while moving zeros to the end. A key optimization is to perform the swap only when necessary (i.e., when i and j are not the same) to reduce the number of write operations.</div> <div>#408 Valid Word Abbreviation <span>[Easy]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Use two pointers: one for looping through the word and one for the abbreviation.</li> <li>When encountering a digit in abbr, accumulate the number and skip that many characters in the word.</li> <li>Check for invalid cases such as numbers with leading zeros or consecutive digit segments that might imply adjacent skipped substrings.</li> <li>Compare letters directly when they are not digits.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n + m) where n is the length of the word and m is the length of the abbreviation.</div> <div>Space Complexity: O(1) (constant extra space)</div> </div> <div>Solution</div> <div>Two Pointersp.1075</div>                                                                                                                                                                   | <div>The approach uses two pointers to iterate over the word and abbreviation. For each character in the abbreviation: <ul style="list-style-type: none"> <li>If it is a letter, compare it with the current character in the word.</li> <li>If it is a digit, first ensure that it does not start with '0'. Then, convert the sequence of digits into a number which represents how many characters in the word to skip. After processing, both pointers should have reached the end of their respective strings for the abbreviation to be valid. This approach is efficient due to its linear pass with constant extra space.</li> </ul> </div> <div>#977 Squares of a Sorted Array <span>[Easy]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Squaring numbers can disrupt the original order since negative numbers become positive.</li> <li>A two-pointer approach can efficiently retrieve the largest square from either end of the array.</li> <li>By comparing the absolute values at the two pointers, the largest square is inserted from the back of the result array.</li> <li>This method achieves an O(n) time complexity without needing to sort the squared values afterward.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n) – We traverse the array only once.</div> <div>Space Complexity: O(n) – We use an extra array to store the output.</div> </div> <div>Two Pointersp.1076</div>         | <div>Solution <div>We use a two-pointer technique to traverse the array from both ends. Initialize pointers (left and right) at the beginning and end of the array, respectively, and an index pointer starting at the end of a new results array. At each step, compare the absolute values of the elements at the left and right pointers. The larger absolute value, when squared, is placed in the result array at the current index, and the corresponding pointer is adjusted. This continues until the left pointer exceeds the right pointer. Filling the result array from the back ensures that the array remains sorted in non-decreasing order.</div> </div> <div>#11 Container With Most Water <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>The container's area is determined by the distance between two lines multiplied by the shorter line's height.</li> <li>A brute-force approach checking all possible pairs would result in O(n^2) time complexity, which is inefficient for large inputs.</li> <li>A two-pointer approach allows us to solve the problem in O(n) time by initializing pointers at both ends and gradually moving them inward.</li> <li>By always moving the pointer corresponding to the shorter line, we are more likely to find a taller line that</li> </ul> </div> <div>Two Pointersp.1077</div> |
| <div>could potentially form a container with a larger capacity.</div> <div>Space &amp; Time <div>Time Complexity: O(n) – The algorithm makes a single pass through the array with two pointers.</div> <div>Space Complexity: O(1) – Only a few extra variables are used, regardless of the input size.</div> </div> <div>Solution <div>The optimal approach uses a two-pointer strategy: <ul style="list-style-type: none"> <li>Start with two pointers at each end of the array.</li> <li>Calculate the area formed between the two lines at these pointers.</li> <li>Update the maximum area if the calculated value is larger.</li> <li>Move the pointer pointing to the shorter line inward, as this is the only possible way to potentially increase the height of the shorter line and thus the area.</li> <li>Continue until the two pointers meet.</li> </ul> This method effectively reduces the number of comparisons and ensures that every possible pair that can yield a larger area is considered.</div> </div> <div>#15 3Sum <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Sort the array to efficiently skip duplicates and facilitate the two pointers approach.</li> <li>Use a fixed pointer for the first element, and then use two additional pointers (left and right) to find pairs that sum to the negation of the fixed element.</li> </ul> </div> <div>Two Pointersp.1078</div> | <div> <ul style="list-style-type: none"> <li>Skip duplicate elements to ensure unique triplets.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n^2) where n is the number of elements in the array.</div> <div>Space Complexity: O(1) extra space (ignoring the space required for the output).</div> </div> <div>Solution <div>We first sort the array to bring duplicates together and allow the two-pointer technique to work efficiently. Then, iterate over the array and for each element, use two pointers (one at the beginning right after the current element and one at the end of the array) to find a pair that sums with the current element to zero. Skip duplicates for the current element and within the inner loop to avoid duplicate triplets. This ensures that each valid triplet is unique and the overall runtime remains efficient.</div> </div> <div>#16 3Sum Closest <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Sort the array to make it easier to use the two pointers technique.</li> <li>Fix one element and then use two pointers (left and right) to find the best pair that, along with the fixed element, produces a sum closest to the target.</li> <li>Update the best sum whenever a closer sum is found.</li> </ul> </div> <div>Two Pointersp.1079</div>                                                                                                       | <div> <ul style="list-style-type: none"> <li>The sorted order helps eliminate unnecessary iterations and allows efficient movement of pointers.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n^2)</div> <div>Space Complexity: O(log n) to O(n) depending on the sorting algorithm used (typically in-place sort, so extra space might be minimal)</div> </div> <div>Solution <div>The solution starts by sorting the array. For each index in the array (considered as the first element of the triple), we use two pointers: one starting just after the fixed element and the other at the end of the array. We calculate the sum of the fixed element and the two pointers. If this sum is closer to the target than the current best, we update our best sum. Depending on whether the current sum is less than or greater than the target, we adjust the left or right pointer to try and get closer to the target. This two-pointer method leverages the sorted order of the array for efficient summing.</div> </div> <div>#31 Next Permutation <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>The problem is solved by identifying the first pair from the right where the left element is less than the right element.</li> </ul> </div> <div>Two Pointersp.1080</div>                                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <ul style="list-style-type: none"> <li>Once this pivot is found, find the rightmost element that is greater than this pivot.</li> <li>Swap these two elements.</li> <li>Reverse the subarray that comes after the pivot to ensure the next permutation is the next lexicographically larger sequence.</li> <li>If no such pivot exists, the array is in descending order and should be reversed to get the smallest permutation.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n)</div> <div>Space Complexity: O(1)</div> </div> <div> <div>Solution</div> <div>We start by scanning the array from the right to find the first index (i) such that nums[i] is less than nums[i+1]. This index 'i' represents the pivot point where the next permutation can be formed. If we find such an index, we then scan from the right again to locate the first number that is greater than nums[i] (let this index be j), and swap these two numbers. Finally, we reverse the subarray starting from index i+1 to the end of the array, which gives us the next permutation in lexicographical order. This approach leverages in-place modifications using basic operations like swapping and reversing, thus meeting the constant space requirement.</div> </div> <div>#75 Sort Colors [Medium]</div> <div> <div>Two Pointers</div> <div>p.1081</div> </div>                                   | <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use the Dutch National Flag algorithm to partition the array into three sections.</li> <li>Maintain three pointers: one for the next position for 0 (low), one for the next position for 2 (high), and one to traverse the array (mid).</li> <li>Swap elements appropriately to ensure that all 0's are at the beginning, all 2's are at the end, and 1's remain in the middle.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n)</div> <div>Space Complexity: O(1)</div> </div> <div> <div>Solution</div> <div>The solution uses a three-pointer approach (Dutch National Flag algorithm). Initialize three pointers: low, mid, and high. As you iterate through the array: <ul style="list-style-type: none"> <li>If the element is 0, swap it with the element at the low pointer and increment both low and mid.</li> <li>If the element is 1, simply move mid forward.</li> <li>If the element is 2, swap it with the element at the high pointer and decrement high.</li> </ul> This single pass method ensures that the array is sorted in-place with constant extra space. Key details include correctly handling swaps without losing any data and ensuring that the pointers are updated in the proper order.</div> </div> <div> <div>Two Pointers</div> <div>p.1082</div> </div> | <div>#161 One Edit Distance [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>The difference in length between s and t must be at most 1.</li> <li>When lengths are equal, the only possibility is that exactly one character is different (replacement).</li> <li>When one string is longer by one character, the difference can be fixed by one insertion (or deletion in the other direction).</li> <li>Use two pointers to compare corresponding characters and carefully handle the first difference.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n), where n is the length of the shorter string, as we traverse both strings at most once.</div> <div>Space Complexity: O(1), using only constant extra space.</div> </div> <div> <div>Solution</div> <div>The solution uses a two-pointer approach. First, we check if the length difference is greater than 1, in which case they cannot be one edit apart. For the two primary cases: <ul style="list-style-type: none"> <li>When strings are of equal length, iterate through both strings and count the number of differing characters. There should be exactly one difference.</li> <li>When lengths differ by one, use pointers for both strings. Upon encountering a difference, increment the pointer for the longer string only and ensure that no previous</li> </ul> </div> <div> <div>Two Pointers</div> <div>p.1083</div> </div> </div> |
| <div> <div>difference has been encountered. If a second discrepancy is found, return false. The key is to handle the edge case where the difference is at the end of the string.</div> </div> <div>#167 Two Sum II – Input Array Is Sorted [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>The array is sorted in non-decreasing order, which makes the two pointers technique very effective.</li> <li>Using two pointers (one at the start and one at the end) allows checking pairs and adjusting the pointers based on the sum.</li> <li>Since the problem guarantees exactly one solution and requires constant space, using two pointers satisfies both constraints.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n), where n is the number of elements in the array, because in the worst-case scenario each element is visited at most once.</div> <div>Space Complexity: O(1), as no additional data structure is used that scales with the input size.</div> </div> <div> <div>Solution</div> <div>We use the two pointers approach: <ul style="list-style-type: none"> <li>Initialize two pointers, left at the beginning (index 0) and right at the end (last index) of the array.</li> <li>While left is less than right, compute the sum of the elements at</li> </ul> </div> <div> <div>Two Pointers</div> <div>p.1084</div> </div> </div> | <div> <div>the two pointers.</div> <ul style="list-style-type: none"> <li>If the sum equals the target, return the indices as [left+1, right+1] (to adjust for the 1-indexed array).</li> <li>If the sum is less than the target, move the left pointer to the right to increase the sum.</li> <li>If the sum is greater than the target, move the right pointer to the left to decrease the sum.</li> </ul> This approach leverages the sorted property and uses constant extra space. </div> <div>#186 Reverse Words in a String II [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Reverse the entire array first to obtain the reversed order of characters.</li> <li>Then, reverse each individual word to restore the correct letter order.</li> <li>Doing so leverages the two-pointer technique for in-place reversals.</li> <li>This method accomplishes the goal in O(n) time while using O(1) extra space.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n)</div> <div>Space Complexity: O(1)</div> </div> <div> <div>Solution</div> <div>The solution employs a two-step reversal process using the two-pointer technique. First, reverse the entire character array to reverse the overall order of</div> </div> <div> <div>Two Pointers</div> <div>p.1085</div> </div>                                        | <div> <div>characters. Although this reverses both the words and the letters in each word, it sets up the scenario where each word's letters are in reverse order. The next step is to iterate through the array and locate each word (using the space character as a delimiter) and then reverse each word individually to correct its letter order. This process is done in-place without any additional data structures, conserving space.</div> </div> <div>#189 Rotate Array [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>The rotation amount k might be greater than the length of the array, so use k % n (where n is the array length) to compute the effective rotation.</li> <li>A smart in-place solution involves reversing sections of the array: <ul style="list-style-type: none"> <li>Reverse the entire array.</li> <li>Reverse the first k elements.</li> <li>Reverse the remaining n-k elements.</li> </ul> </li> <li>Alternative approaches include using extra space or performing k single-step rotations (which is less optimal).</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n) where n is the number of elements in the array.</div> <div>Space Complexity: O(1) if done in-place.</div> </div> <div> <div>Two Pointers</div> <div>p.1086</div> </div>                                                                                                                         |
| <div> <div>Solution</div> <div>We solve the problem using the "reverse" technique to perform the rotation in-place. First, we adjust k by taking k modulo the length of the array to handle cases where k exceeds the array length. Then, we reverse the entire array. Next, we reverse the first k elements to restore their correct order. Finally, we reverse the remaining n-k elements. This sequence yields the array rotated to the right by k positions while strictly using O(1) extra space.</div> </div> <div>#532 K-diff Pairs in an Array [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>For k &lt; 0, the answer is always 0 because the absolute difference cannot be negative.</li> <li>When k is 0, you are looking for numbers that appear at least twice.</li> <li>For k &gt; 0, it is efficient to use a hash set or a hash map to check if each number's complement (number + k) exists.</li> <li>Uniqueness is ensured by processing each number only once from the set of unique numbers.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n), where n is the length of the array since we iterate over the array and then over the unique numbers.</div> </div> <div> <div>Two Pointers</div> <div>p.1087</div> </div>                                                                                                               | <div> <div>Space Complexity: O(n), for storing the frequency count or the unique elements in a hash map or hash set.</div> </div> <div> <div>Solution</div> <div>The solution first checks if k is negative, returning 0 immediately if true. For k equals 0, we count numbers that appear more than once using a frequency map. For k greater than 0, a set of unique numbers is created and for each number, we check if (number + k) is also present in the set. Each valid case contributes to the result count, ensuring that only unique pairs are considered.</div> </div> <div>#923 3Sum With Multiplicity [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use a frequency counter for elements since arr[i] is in the range [0, 100].</li> <li>Iterate through all possible combinations of numbers x, y, and z with x &lt;= y &lt;= z.</li> <li>For each valid triplet (x, y, z) that sums to target, count the number of ways using combinatorial formulas: <ul style="list-style-type: none"> <li>All distinct: count[x] * count[y] * count[z].</li> <li>Two numbers the same: use nC2 for the repeated element.</li> <li>All three the same: use nC3.</li> </ul> </li> <li>Use modulo arithmetic (MOD = 10^9 + 7) at every step to avoid overflow.</li> </ul> </div> <div> <div>Two Pointers</div> <div>p.1088</div> </div>                  | <div> <div>Space &amp; Time</div> <div>Time Complexity: O(M^2), where M = 101 (the range of possible numbers), which is efficient given the constraints.</div> <div>Space Complexity: O(M) for the frequency array.</div> </div> <div> <div>Solution</div> <div>The solution first counts the frequency of each number in the array. Then it iterates through all valid triplets (x, y, z) (with x &lt;= y &lt;= z) and checks if they sum up to the target. Depending on whether x, y, and z are distinct or have duplicates, the combination count is computed accordingly: <ul style="list-style-type: none"> <li>If x, y, and z are all the same, compute the combination as nC3.</li> <li>If only two are the same, compute as nC2 multiplied by the count of the third number.</li> <li>If all are distinct, use the direct product of the counts.</li> </ul> This covers all cases while applying the modulo operation to the result.</div> </div> <div>#986 Interval List Intersections [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use two pointers to traverse both lists simultaneously.</li> <li>Determine the overlapping interval by taking the maximum of the start points and the minimum of the end points.</li> </ul> </div> <div> <div>Two Pointers</div> <div>p.1089</div> </div>                                                                                                                                  |



- If the overlapping condition (start <= end) is met, add the intersection to the result.
- Move the pointer that has the smaller end value to explore further potential intersections efficiently.
- Achieve an overall time complexity of O(m+n), where m and n are the lengths of the two interval lists.

Space & Time

Time Complexity: O(m + n) where m and n are the lengths of firstList and secondList respectively.  
Space Complexity: O(1) extra space (excluding the space used for the output).

Solution

The solution uses a two pointers approach. Initialize two indices, one for each list, and iterate through both lists. For each pair of intervals, the potential intersection is computed by:  
– Calculating the maximum of the two start values. – Calculating the minimum of the two end values. If the maximum start is less than or equal to the minimum end, there is an intersection, and it is added to the result list. The pointer corresponding to the interval with the smaller end value is then incremented since that interval cannot intersect with any further intervals from the other list. This process repeats until one of the lists is fully traversed.

#1055 Shortest Way to Form String [Medium]

Two Pointers p.1090

Key Insights

- Check if every character in target exists in source; if not, the answer is -1.
- Use a greedy two-pointer strategy: iterate over target while scanning source for matching characters.
- During each pass over source, match as many characters as possible from target.
- Count the number of passes (subsequences) required until the target is completely matched.
- An optimized approach may precompute indices for binary search; however, the basic greedy simulation is efficient for the given constraints.

Space & Time

Time Complexity: O(n \* m) in the worst-case, where n = length of source and m = length of target.  
Space Complexity: O(1) for the greedy simulation (O(n) if additional data structures for binary search are used).

Solution

The solution uses a greedy two-pointer approach. We simulate the process of reading the target by repeatedly scanning through the source string. Each scan through source is treated as one subsequence concatenated to form the target. A pointer in the target moves forward whenever a matching character is found in source. If a complete pass through source does not advance the target pointer, then it is impossible to form the target

Two Pointers p.1091

and we return -1. The algorithm counts the number of passes required until the target is fully matched.

#2563 Count the Number of Fair Pairs [Medium]

Key Insights

- Sorting the array simplifies searching for pairs with required sum limits.
- For each element, binary search is used to locate:
  - The first index where the complement makes the sum >= lower.
  - The last index where the complement makes the sum <= upper.
- This avoids the need for a nested loop through the entire array, yielding a more efficient solution.

Space & Time

Time Complexity: O(n log n), primarily due to sorting and (for each element) binary search operations.  
Space Complexity: O(n) if sorting creates a copy; O(1) additional space if sorting in-place.

Solution

The solution starts by sorting the input array. For each element at index i in the sorted array, we search for indices j (where j > i) such that the sum of nums[i] and nums[j] falls between lower and upper. Two binary search operations are performed:

Two Pointers p.1092

– To find the smallest index j where nums[i] + nums[j] >= lower. – To find the largest index j where nums[i] + nums[j] <= upper. Subtracting the two positions (and summing the counts for each i) gives the total number of fair pairs. Edge cases (like no valid pair) are automatically handled by the binary search approach.

Heap

#14 Heap

By Pattern · Python Only · 20 questions

Two Pointers p.1093

Heap

#215 Kth Largest Element in an Array

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Divide and Conquer · Sorting · Heap · Quickselect

Lists: Grind 169

Time Complexity:

Problem

Given an integer array nums and an integer k, return the kth largest element in the array.  
Note that it is the kth largest element in the sorted order, not the kth distinct element.

Can you solve it without sorting?

Example 1:

Input: nums = [3,2,1,5,6,4], k = 2  
Output: 5

Example 2:

Input: nums = [3,2,3,1,2,4,5,5,6], k = 4  
Output: 4

Constraints:

- 1 <= k <= nums.length <= 105
- -104 <= nums[i] <= 104

Heap p.1094

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findKthLargest(self, nums, k):
 # Brute Force O(n log n) - sort descending, index at k-1
 nums.sort(reverse=True)
 return nums[k - 1]
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def findKthLargest(self, nums: List[int], k: int) -> int:
 minHeap = []

 for num in nums:
 heapq.heappush(minHeap, num)
 if len(minHeap) > k:
 heapq.heappop(minHeap)

 return minHeap[0]
```

• LeetCode

Heap p.1095

```
class Solution:
 def findKthLargest(self, nums: List[int], k: int) -> int:
 def quick_sort(l: int, r: int) -> int:
 if l == r:
 return nums[l]
 i, j = l - 1, r + 1
 x = nums[(l + r) >> 1]
 while i < j:
 while i < j:
 i += 1
 if nums[i] >= x:
 break
 while i < j:
 j -= 1
 if nums[j] <= x:
 break
 if i < j:
 nums[i], nums[j] = nums[j], nums[i]
 if j < k:
 return quick_sort(j + 1, r)

 return quick_sort(l, j)
```

```
n = len(nums)
k = n - k
return quick_sort(0, n - 1)
```

• SimplyLeet

Heap p.1096

```
import heapq

def findKthLargest(nums, k):
 # Create a min-heap with the first k elements
 min_heap = nums[:k]
 heapq.heapify(min_heap) # O(k) time to build the initial heap

 # Process the remaining elements
 for num in nums[k:]:
 # If the current element is larger than the smallest in the heap, replace it
 if num > min_heap[0]:
 heapq.heapreplace(min_heap, num) # O(log k)
 # The smallest element in the heap is the kth largest overall
 return min_heap[0]
```

```
Example usage:
nums = [3, 2, 1, 5, 6, 4]
k = 2
print(findKthLargest(nums, k)) # Expected output: 5
```

• LC.ca

Heap p.1097

```
class Solution:
 def findKthLargest(self, nums: List[int], k: int) -> int:
 lo, hi = 0, len(nums) - 1

 while lo <= hi:
 pos = self.partition(nums, lo, hi)
 if pos == k - 1: # pos starting from 0, so -1
 return nums[pos] # partially sorted, desc
 elif pos > k - 1:
 hi = pos - 1
 else: # pos < k - 1
 lo = pos + 1

 return -1 # or raise exception

 def partition(self, nums: List[int], lo: int, hi: int) -> int:
 pivot, l, r = nums[lo], lo + 1, hi
 while l <= r:
 if nums[l] < pivot and nums[r] > pivot:
 # larger num at left of pivot, easier to count for k-th largest
 nums[l], nums[r] = nums[r], nums[l]
```

Heap p.1098



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| <pre>        nums[l], nums[r] = nums[r], nums[l]         l += 1         r -= 1     if nums[l] &gt;= pivot:         l += 1     if nums[r] &lt;= pivot:         r -= 1     # use nums[l] will lead to infinite looping     nums[l], nums[r] = nums[r], nums[l]     # possible there is duplicated num, but will be     covered here      return r  #####  class Solution:     def findKthLargest(self, nums: List[int], k: int) -&gt;     int:         def quick_sort(left, right, k):             if left == right:                 return nums[left]             i, j = left - 1, right + 1</pre> |        | <pre>        x = nums[(left + right) &gt;&gt; 1]         while i &lt; j:             while i:                 i += 1                 if nums[i] &gt;= x:                     break             while j:                 j -= 1                 if nums[j] &lt;= x:                     break              if i &lt; j:                 nums[i], nums[j] = nums[j], nums[i]             if j &lt; k:                 return quick_sort(j + 1, right, k)             return quick_sort(left, j, k)          n = len(nums)         return quick_sort(0, n - 1, n - k)  #####  class Solution:     def findKthLargest(self, nums: List[int], k: int) -&gt;     int:         def quick_sort(left, right, k):             if left == right:                 return nums[left]             i, j = left - 1, right + 1             x = nums[(left + right) &gt;&gt; 1]             while i &lt; j:</pre> |        | <pre>        while 1:             i += 1             if nums[i] &gt;= x:                 break         while 1:             j -= 1             if nums[j] &lt;= x:                 break         if i &lt; j:             nums[i], nums[j] = nums[j], nums[i]          if j &lt; k:             return quick_sort(j + 1, right, k)             return quick_sort(left, j, k)          n = len(nums)         return quick_sort(0, n - 1, n - k)</pre> |        |
| Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | p.1099 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | p.1100 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                 | p.1101 |

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| <div>Heap</div> <div>#253 Meeting Rooms II</div> <div>LeetCode · SimplyLeet · WalkCC · LeetCode</div> <div>Array · Two Pointers · Greedy · Sorting · Heap</div> <div>Lists: Grind 169   Premium 98</div> <div>Time Complexity: <math>O(n \log n)</math> due to sorting and each heap operation (push/pop) taking <math>O(\log n)</math>. Space Complexity: <math>O(n)</math> in the worst-case scenario when all meetings overlap, requiring all end times in t</div> <div>Problem</div> <div>Given an array of meeting time intervals intervals where intervals[i] = [starti, endi], return the minimum number of conference rooms required.</div> <div>Example 1:</div> <div>Input: intervals = [[0,30],[5,10],[15,20]]<br/>Output: 2</div> <div>Example 2:</div> <div>Input: intervals = [[7,10],[2,4]]<br/>Output: 1</div> <div>Constraints:</div> <div><ul style="list-style-type: none"><li>1 &lt;= intervals.length &lt;= 104</li><li>0 &lt;= starti &lt; endi &lt;= 106</li></ul></div> |        | <div>Brute Force (Python)</div> <div># Brute Force Approach</div> <div>class Solution:</div> <div>def minMeetingRooms(self, intervals):</div> <div># Brute Force <math>O(n^2)</math> - simulate: assign each meeting to first free room</div> <div>intervals.sort()</div> <div>rooms = [] # end times of each room</div> <div>for start, end in intervals:</div> <div># Find earliest-ending free room</div> <div>placed = False</div> <div>for i in range(len(rooms)):</div> <div>if rooms[i] &lt;= start:</div> <div>rooms[i] = end</div> <div>placed = True</div> <div>break</div> <div>if not placed:</div> <div>rooms.append(end)</div> <div>return len(rooms)</div> <div>Community Solutions (Python)</div> <div>• WalkCC</div> |        | <div>class Solution:</div> <div>def minMeetingRooms(self, intervals: list[list[int]]) -&gt; int:</div> <div>minHeap = [] # Store the end times of each room.</div> <div>for start, end in sorted(intervals):</div> <div># There's no overlap, so we can reuse the same room.</div> <div>if minHeap and start &gt;= minHeap[0]:</div> <div>heapq.heappop(minHeap)</div> <div>heapq.heappush(minHeap, end)</div> <div>return len(minHeap)</div> <div>• LeetCode</div> <div>class Solution:</div> <div>def minMeetingRooms(self, intervals: List[List[int]]) -&gt; int:</div> <div>m = max(e[1] for e in intervals)</div> <div>d = [0] * (m + 1)</div> <div>for l, r in intervals:</div> <div>d[l] += 1</div> <div>d[r] -= 1</div> <div>ans = s = 0</div> <div>for v in d:</div> <div>s += v</div> <div>ans = max(ans, s)</div> <div>return ans</div> <div>• SimplyLeet</div> |        |
| Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | p.1102 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | p.1103 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | p.1104 |

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| <pre>import heapq  def minMeetingRooms(intervals):     # Sort intervals based on start time     intervals.sort(key=lambda x: x[0])      # Initialize a min-heap to store end times of     meetings that are currently using rooms     min_heap = []      # Process each meeting interval      for interval in intervals:         start, end = interval          # If the heap is not empty and the current         meeting starts after the earliest ended meeting         if min_heap and start &gt;= min_heap[0]:             heapq.heappop(min_heap) # Reuse the room          # Allocate the current meeting end time into the         heap (new or reused room)         heapq.heappush(min_heap, end)      # The number of rooms is the number of simultaneous     meetings     return len(min_heap)  # Example usage: intervals = [[0,30],[5,10],[15,20]] print(minMeetingRooms(intervals)) # Output: 2  • LC.ca</pre> |        | <pre>... data = [3, 4, 6, 2, 1, 9, 0, 7, 5, 8]  list(accumulate(data, operator.mul)) # running product [3, 12, 72, 144, 144, 1296, 0, 0, 0, 0]  list(accumulate(data, max)) # running maximum [3, 4, 6, 6, 6, 9, 9, 9, 9, 9]  &gt;&gt;&gt; from itertools import accumulate &gt;&gt;&gt; data = [3, 4, 6, 2, 1, 9, 0, 7, 5, 8] &gt;&gt;&gt; accumulate(data) &lt;itertools.accumulate object at 0x10fd78440&gt; &gt;&gt;&gt; list(accumulate(data)) [3, 7, 13, 15, 16, 25, 25, 32, 37, 45] &gt;&gt;&gt; max(accumulate(data)) 45  https://docs.python.org/3/library/itertools.html#itertool s.accumulate</pre> |        | <pre># Amortize a 5% loan of 1000 with 4 annual payments of 90 cashflows = [1000, -90, -90, -90, -90] list(accumulate(cashflows, lambda bal, pmt: bal*1.05 + pmt)) [1000, 960.0, 918.0, 873.9000000000001, 827.5950000000001] ... class Solution:     def minMeetingRooms(self, intervals: List[List[int]]) -&gt; int:         delta = [0] * 1000010          for start, end in intervals:             delta[start] += 1             delta[end] -= 1         return max(accumulate(delta))         # why not delta.sort()?         # because accumulate() will go by order from         index 0 to index final         # just like, from sortedcontainers import         SortedDict          #####</pre> |        |
| Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | p.1105 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | p.1106 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | p.1107 |

```
class Solution(object):
 def minMeetingRooms(self, intervals):
 """
 :type intervals: List[Interval]
 :rtype: int
 """
 meetings = []
 for i in intervals:
 meetings.append((i.start, 1))
 meetings.append((i.end, 0))
```

```
 meetings.sort()
 ans = 0
 count = 0
 for meeting in meetings:
 if meeting[1] == 1:
 count += 1
 else:
 count -= 1
 ans = max(ans, count)
 return ans
```

Heap

p.1108

Heap

## #347 Top K Frequent Elements

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Divide and Conquer · Sorting · Heap · Bucket Sort · Counting · Quickselect

Lists: Denny Zhang

Time Complexity:  $O(n)$  where  $n$  is the length of the array (bucket sort and frequency counting are linear). Space Complexity:  $O(n)$  for storing the counts and buckets.

### Problem

Given an integer array `nums` and an integer `k`, return the `k` most frequent elements. You may return the answer in any order.

Example 1:

Input: `nums = [1,1,1,2,2,3]`, `k = 2`

Output: `[1,2]`

Example 2:

Input: `nums = [1]`, `k = 1`

Output: `[1]`

Example 3:

Input: `nums = [1,2,1,2,1,2,3,1,3,2]`, `k = 2`

Output: `[1,2]`

Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $-104 \leq \text{nums}[i] \leq 104$
- $k$  is in the range  $[1, \text{the number of unique elements in the array}]$ .
- It is guaranteed that the answer is unique.

Follow up: Your algorithm's time complexity must be better than  $O(n \log n)$ , where  $n$  is the array's size.

### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def topKFrequent(self, nums, k):
 # Brute Force $O(n \log n)$ - count then sort by frequency
 count = {}
 for num in nums:
 count[num] = count.get(num, 0) + 1
 return sorted(count, key=lambda x: -count[x])[:k]
```

### Community Solutions (Python)

- WalkCC

Heap

p.1110

```
class Solution:
 def topKFrequent(self, nums: List[int], k: int) -> List[int]:
 ans = []
 bucket = [[] for _ in range(len(nums) + 1)]

 for num, freq in collections.Counter(nums).items():
 bucket[freq].append(num)

 for b in reversed(bucket):
 ans += b
```

```
 if len(ans) == k:
 return ans
```

- LeetDoocs

```
class Solution:
 def topKFrequent(self, nums: List[int], k: int) -> List[int]:
 cnt = Counter(nums)
 return [x for x, _ in cnt.most_common(k)]
```

- SimplyLeet

Heap

p.1111

```
Python solution using bucket sort

def topKFrequent(nums, k):
 # Count the frequency of each element in nums
 frequency = {}
 for num in nums:
 frequency[num] = frequency.get(num, 0) + 1

 # Create buckets where index represents frequency
 n = len(nums)
```

```
 buckets = [[] for _ in range(n + 1)]
 for num, count in frequency.items():
 buckets[count].append(num)
```

```
 # Gather the k most frequent elements
 result = []
 # Iterate over buckets in reverse order (highest frequency first)
 for freq in range(n, 0, -1):
 for num in buckets[freq]:
 result.append(num)
```

```
 if len(result) == k:
 return result
```

```
Example usage:
print(topKFrequent([1,1,1,2,2,3], 2)) -> Output: [1, 2]
```

- LC.ca

Heap

p.1112

```
diff from below solution: sorted(cnt), not sorted(cnt.items())
more in
https://leetcode.ca/2017-10-22-692-Top-K-Frequent-Words/
sorted_freqs = sorted(cnt, key=lambda x: (-cnt[x], x))
```

```
 return sorted_freqs[:k]
```

```
#####
```

```
class Solution:
 def topKFrequent(self, nums: List[int], k: int) -> List[int]:
```

```
 # Count the frequency of each element in the list
 freqs = Counter(nums)
```

```
 # Sort the elements by frequency in decreasing order
```

```
 sorted_freqs = sorted(freqs.items(), key=lambda x: x[1], reverse=True)
 # also passing OJ: sorted_freqs = sorted(freqs.items(), key=lambda x: -x[1])
```

```
 # Take the top k elements
 top_k = [num for num, freq in sorted_freqs[:k]]
```

Heap

p.1114

```
 return top_k
```

```
#####
```

```
'''
>>> from heapq import heappush
>>> h = []
>>> heappush(h, (3,1))
>>> heappush(h, (1,1))
>>> heappush(h, (2,1))
```

```
>>> h
[(1, 1), (3, 1), (2, 1)]
```

```
>>> from heapq import heappop
>>> heappop(h)
(1, 1)
>>> heappop(h)
(2, 1)
>>> heappop(h)
(3, 1)
```

Heap

p.1115

```
'''

from collections import Counter

class Solution: # heap
 def topKFrequent(self, nums: List[int], k: int) -> List[int]:
 cnt = Counter(nums)
 hp = []
 for num, freq in cnt.items():
 heappush(hp, (freq, num)) # freq first, default sort by 1st element
```

```
 if len(hp) > k:
 heappop(hp)
 return [v[1] for v in hp]
```

```
#####
```

```
from typing import List
import random
$O(n \log(n))$ in the average case
$O(n^2)$ worst case
```

Heap

p.1116

```

class Solution:
 def topKFrequent(self, nums: List[int], k: int) -> List[int]:
 freq_map = {}
 for num in nums:
 freq_map[num] = freq_map.get(num, 0) + 1

 freq_list = list(freq_map.items())
 self.quick_select(freq_list, 0, len(freq_list) - 1, k)

 return [num for num, freq in freq_list[:k]]

 def quick_select(self, freq_list, start, end, k):
 if start == end:
 return

 pivot_idx = random.randint(start, end)
 pivot_freq = freq_list[pivot_idx][1]

 left = start
 right = end

```

Heap p.1117

```

while left <= right:
 while freq_list[left][1] > pivot_freq:
 left += 1
 while freq_list[right][1] < pivot_freq:
 right -= 1

 if left <= right:
 freq_list[left], freq_list[right] = freq_list[right], freq_list[left]
 left += 1

 right -= 1

 if k <= right:
 self.quick_select(freq_list, start, right, k)
 elif k >= left:
 self.quick_select(freq_list, left, end, k)

#####

...
>>> x = [1, 2, 3]
>>> x.append([4, 5])
>>> print(x)
[1, 2, 3, [4, 5]]

>>> x = [1, 2, 3]
>>> x.extend([4, 5])

```

Heap p.1118

```

>>> print(x)
[1, 2, 3, 4, 5]
...
class Solution:
 def topKFrequent(self, nums: List[int], k: int) -> List[int]:
 freq = Counter(nums)

 # not always filled, max possible frequency is
 len(nums) [1,1,1,1,...]
 buckets = [[] for _ in range(len(nums) + 1)]
 for num, freq in freq.items():

 buckets[freq].append(num)

 res = []
 for bucket in reversed(buckets):
 if bucket:
 res.extend(bucket)
 if len(res) >= k:
 break

 return res[:k]

```

Heap p.1119

```

#####

class Solution(object):
 def topKFrequent(self, nums, k):
 """
 :type nums: List[int]
 :type k: int
 :rtype: List[int]
 """

 d = {}
 res = []
 ans = []
 buckets = [[] for _ in range(len(nums) + 1)]

 for num in nums:
 d[num] = d.get(num, 0) + 1

 for key in d:
 res.append((d[key], key))

 for t in res:
 freq, key = t
 buckets[freq].append(key)

 buckets.reverse()

 for item in buckets:
 if item and k > 0:
 while item and k > 0:

```

Heap p.1120

```

ans.append(item.pop())
k -= 1
if k == 0:
 return ans

return ans

```

Heap p.1121

Heap

## #505 The Maze II

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · DFS · BFS · Graph · Matrix · Heap · Shortest Path

Lists: Premium 98

Time Complexity:  $O(m * n * \log(m * n))$ , where  $m$  and  $n$  are the maze dimensions (each cell may get processed and queued with a logarithmic cost). Space Complexity:  $O(m * n)$  for maintaining the distance

### Problem

There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction.

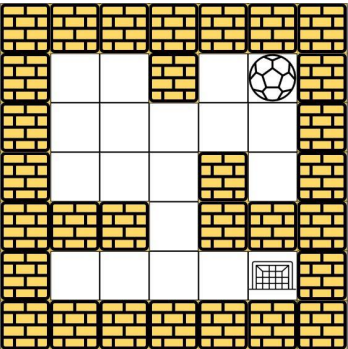
Given the  $m \times n$  maze, the ball's start position and the destination, where start = [startrow, startcol] and destination = [destinationrow, destinationcol], return the shortest distance for the ball to stop at the destination. If the ball cannot stop at destination, return -1.

The distance is the number of empty spaces traveled by the ball from the start position (excluded) to the destination (included).

Heap p.1122

You may assume that the borders of the maze are all walls (see examples).

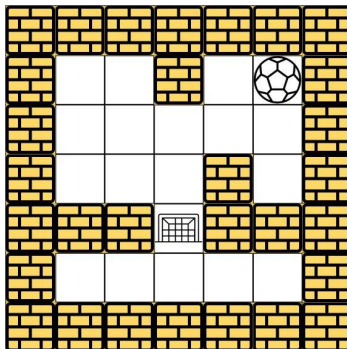
Example 1:



Input: maze = [[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]], start = [0,4], destination = [4,4]  
Output: 12  
Explanation: One possible way is : left -> down -> left -> down -> right -> down -> right.  
The length of the path is 1 + 1 + 3 + 1 + 2 + 2 + 2 = 12.

Heap p.1123

Example 2:



Input: maze = [[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]], start = [0,4], destination = [3,2]  
Output: -1  
Explanation: There is no way for the ball to stop at the destination. Notice that you can pass through the destination but you cannot stop there.

Example 3:

Heap p.1124

Input: maze = [[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]], start = [4,3], destination = [0,1]  
Output: -1

Constraints:

- $m == \text{maze.length}$
- $n == \text{maze}[i].\text{length}$
- $1 \leq m, n \leq 100$
- $\text{maze}[i][j]$  is 0 or 1.
- $\text{start.length} == 2$
- $\text{destination.length} == 2$
- $0 \leq \text{startrow}, \text{destinationrow} < m$
- $0 \leq \text{startcol}, \text{destinationcol} < n$
- Both the ball and the destination exist in an empty space, and they will not be in the same position initially.
- The maze contains at least 2 empty spaces.

Brute Force (Python)

```

Brute Force Approach
class Solution:
 def shortestDistance(self, maze, start, destination):
 # Brute Force $O(n \log n)$ - sort repeatedly
 instead of heap
 data = list(maze)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result

```

Heap p.1125

Community Solutions (Python)

• LeetCode

```
class Solution:
 def shortestDistance(
 self, maze: List[List[int]], start: List[int],
 destination: List[int]
) -> int:
 m, n = len(maze), len(maze[0])
 dirs = (-1, 0, 1, 0, -1)
 si, sj = start
 di, dj = destination
 q = deque([(si, sj)])
 dist = [[inf] * n for _ in range(m)]

 dist[si][sj] = 0
 while q:
 i, j = q.popleft()
 for a, b in pairwise(dirs):
 x, y, k = i, j, dist[i][j]
 while 0 <= x + a < m and 0 <= y + b < n
 and maze[x + a][y + b] == 0:
 x, y, k = x + a, y + b, k + 1
 if k < dist[x][y]:
 dist[x][y] = k
 q.append((x, y))

 return -1 if dist[di][dj] == inf else
 dist[di][dj]
```

• SimplyLeet

Heapp.1126

```
import heapq

def shortestDistance(maze, start, destination):
 m, n = len(maze), len(maze[0])
 # Initialize distance matrix with infinity
 dist = [[float('inf')] * n for _ in range(m)]
 dist[start[0]][start[1]] = 0
 # Priority queue (min-heap) storing tuples (distance,
 row, col)
 heap = [(0, start[0], start[1])]
 # Directions: up, down, left, right

 directions = [(-1, 0), (1, 0), (0, -1), (0, 1)]

 while heap:
 d, row, col = heapq.heappop(heap)
 # If we have reached the destination, return the
 corresponding distance
 if [row, col] == destination:
 return d
 # Skip if we already found a better path to (row,
 col)
 if d > dist[row][col]:
 continue
```

Heapp.1127

```
for dr, dc in directions:
 next_row, next_col, count = row, col, 0
 # Roll the ball until it hits a wall
 while 0 <= next_row + dr < m and 0 <=
next_col + dc < n and maze[next_row + dr][next_col + dc]
== 0:
 next_row += dr
 next_col += dc
 count += 1
 # Check if the new distance is shorter
 if d + count < dist[next_row][next_col]:
 dist[next_row][next_col] = d + count

heapq.heappush(heap,
(dist[next_row][next_col], next_row, next_col))
return -1
```

• LC.ca

```
class Solution:
 def shortestDistance(
 self, maze: List[List[int]], start: List[int],
 destination: List[int]
) -> int:
 m, n = len(maze), len(maze[0])
 dirs = (-1, 0, 1, 0, -1)
 si, sj = start
 di, dj = destination
 q = deque([(si, sj)])
 dist = [[inf] * n for _ in range(m)]
```

Heapp.1128

```
dist[si][sj] = 0
while q:
 i, j = q.popleft()
 for a, b in pairwise(dirs):
 x, y, k = i, j, dist[i][j]
 while 0 <= x + a < m and 0 <= y + b < n
 and maze[x + a][y + b] == 0:
 x, y, k = x + a, y + b, k + 1
 if k < dist[x][y]:
 dist[x][y] = k
 q.append((x, y))

return -1 if dist[di][dj] == inf else
 dist[di][dj]
```

Heapp.1129

Heap

#621 Task SchedulerMED

LeetCode · SimplyLeet · WalkCC · LeetCode  
Array · Hash Table · Greedy · Heap · Counting  
Lists: Grind 169  
Time Complexity: O(N) where N is the number of tasks (counting frequency, then iterating over constant 26 letters). Space Complexity: O(1) since the frequency count uses an array or hash map with fixe  
Problem  
You are given an array of CPU tasks, each labeled with a letter from A to Z, and a number n. Each CPU interval can be idle or allow the completion of one task. Tasks can be completed in any order, but there's a constraint: there has to be a gap of at least n intervals between two tasks with the same label.  
Return the minimum number of CPU intervals required to complete all tasks.  
Example 1:  
Input: tasks = ["A","A","A","B","B","B"], n = 2  
Output: 8  
Explanation: A possible sequence is: A -> B -> idle -> A -> B -> idle -> A -> B.

Heapp.1130

After completing task A, you must wait two intervals before doing A again. The same applies to task B. In the 3rd interval, neither A nor B can be done, so you idle. By the 4th interval, you can do A again as 2 intervals have passed.  
Example 2:  
Input: tasks = ["A","C","A","B","D","B"], n = 1  
Output: 6  
Explanation: A possible sequence is: A -> B -> C -> D -> A -> B.  
With a cooling interval of 1, you can repeat a task after just one other task.  
Example 3:  
Input: tasks = ["A","A","A","B","B","B"], n = 3  
Output: 10  
Explanation: A possible sequence is: A -> B -> idle -> idle -> A -> B -> idle -> idle -> A -> B.  
There are only two types of tasks, A and B, which need to be separated by 3 intervals. This leads to idling twice between repetitions of these tasks.  
Constraints:  
• 1 <= tasks.length <= 104  
• tasks[i] is an uppercase English letter.  
• 0 <= n <= 100  
Brute Force (Python)

Heapp.1131

```
Brute Force Approach
class Solution:
 def leastInterval(self, tasks, n):
 # Brute Force O(n * intervals) - simulate the CPU
 schedule tick by tick
 from collections import Counter
 counts = Counter(tasks)
 time = 0
 while counts:
 # Each cycle: pick up to (n+1) most-frequent
 tasks
 cycle = sorted(counts.keys(), key=lambda t:
 -counts[t])[:n+1]

 for t in cycle:
 counts[t] -= 1
 if counts[t] == 0: del counts[t]
 time += 1
 if counts: # idle the rest of the cycle
 time += max(0, n + 1 - len(cycle))
 return time
```

Community Solutions (Python)

• WalkCC

Heapp.1132

```
class Solution:
 def leastInterval(self, tasks: list[str], n: int) ->
int:
 count = collections.Counter(tasks)
 maxFreq = max(count.values())
 # Put the most frequent task in the slot first.
 maxFreqTaskOccupy = (maxFreq - 1) * (n + 1)
 # Get the number of tasks with same frequency as
 maxFreq, we'll append them after the
 # `maxFreqTaskOccupy`.
 nMaxFreq = sum(value == maxFreq for value in
count.values())
 # max(

 # the most frequent task is frequent enough to force
 some idle slots,
 # the most frequent task is not frequent enough to
 force idle slots
 #)
 return max(maxFreqTaskOccupy + nMaxFreq, len(tasks))

• LeetCode

class Solution:
 def leastInterval(self, tasks: List[str], n: int) ->
int:
 cnt = Counter(tasks)
 x = max(cnt.values())
 s = sum(v == x for v in cnt.values())
 return max(len(tasks), (x - 1) * (n + 1) + s)

• SimplyLeet
```

Heapp.1133

```
Python solution for Task Scheduler
from collections import Counter

def leastInterval(tasks, n):
 # Count the frequency of each task
 task_counts = Counter(tasks)
 # Find the maximum frequency among tasks
 max_freq = max(task_counts.values())
 # Count how many tasks have the maximum frequency
 max_count = sum(1 for count in task_counts.values()
if count == max_freq)

 # Calculate the minimum intervals required based on
 the most frequent tasks
 part_count = max_freq - 1 # Number of parts between
 the most frequent task
 part_length = n - (max_count - 1) # Effective empty
 slots per part after placing max frequency tasks
 empty_slots = part_count * part_length # Total empty
 slots to fill with idle or other tasks
 available_tasks = len(tasks) - max_freq * max_count #
 Remaining tasks to fill the empty slots
 idles = max(0, empty_slots - available_tasks) #
 Remaining idle slots if available tasks are not enough

 # The total intervals is the sum of all tasks and
 idle slots inserted
 return len(tasks) + idles
```

Heapp.1134

```
Example usage:
if __name__ == "__main__":
 tasks1 = ["A","A","A","B","B","B"]
 n1 = 2
 print(leastInterval(tasks1, n1)) # Output: 8
```

• LC.ca

```
class Solution:
 def leastInterval(self, tasks: List[str], n: int) -> int:
 cnt = Counter(tasks)
 x = max(cnt.values())
 s = sum(v == x for v in cnt.values())
 return max(len(tasks), (x - 1) * (n + 1) + s)
```

Heap

p.1135

Heap

## #658 Find K Closest Elements

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Binary Search · Sorting · Heap · Two Pointers

Lists: Grind 169

Time Complexity:  $O(\log n + k)$ , where  $O(\log n)$  for binary search and  $O(k)$  for collecting results. Space Complexity:  $O(k)$ , for storing the output.

### Problem

Given a sorted integer array `arr`, two integers `k` and `x`, return the `k` closest integers to `x` in the array. The result should also be sorted in ascending order.

An integer `a` is closer to `x` than an integer `b` if:

- $|a - x| < |b - x|$ , or
- $|a - x| == |b - x|$  and  $a < b$

Example 1:

Input: `arr = [1,2,3,4,5]`, `k = 4`, `x = 3`

Output: `[1,2,3,4]`

Example 2:

Input: `arr = [1,1,2,3,4,5]`, `k = 4`, `x = -1`

Output: `[1,1,2,3]`

Constraints:

- $1 \leq k \leq \text{arr.length}$

Heap

p.1136

- $1 \leq \text{arr.length} \leq 104$
- `arr` is sorted in ascending order.
- $-104 \leq \text{arr}[i], x \leq 104$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findClosestElements(self, arr, k, x):
 # Brute Force $O(n \log n)$ - sort repeatedly
 instead of heap
 data = list(arr)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def findClosestElements(self, arr: List[int], k: int, x: int) -> List[int]:
 l = 0
 r = len(arr) - k

 while l < r:
 m = (l + r) // 2
 if x - arr[m] <= arr[m + k] - x:
 r = m
 else:
 l = m + 1
```

Heap

p.1137

```
l = m + 1
return arr[l:l + k]
```

• LeetDoocs

```
class Solution:
 def findClosestElements(self, arr: List[int], k: int, x: int) -> List[int]:
 arr.sort(key=lambda v: abs(v - x))
 return sorted(arr[:k])
```

• SimplyLeet

```
Function to find k closest elements to x in a sorted array
def findClosestElements(arr, k, x):
 n = len(arr)
 # Initialize the binary search bounds for the starting index of the window
 left, right = 0, n - k
 # Binary search to find the best starting index
 while left < right:
 mid = left + (right - left) // 2
 # Compare the distance from x to arr[mid] and arr[mid+k]
 # If arr[mid] is farther from x than arr[mid+k], move window to the right
```

Heap

p.1138

```
if x - arr[mid] > arr[mid + k] - x:
 left = mid + 1
else:
 # Otherwise, continue searching towards the left
 right = mid
Return the k elements starting from index `left`
return arr[left:left + k]
```

```
Example usage:
arr = [1,2,3,4,5], k = 4, x = 3
```

```
print(findClosestElements(arr, 4, 3)) # Output: [1,2,3,4]
```

• LC.ca

```
class Solution:
 def findClosestElements(self, arr: List[int], k: int, x: int) -> List[int]:
 left, right = 0, len(arr) - k
 while left < right:
 mid = (left + right) // 2
 if x - arr[mid] <= arr[mid + k] - x:
 right = mid
 else:
 left = mid + 1
 return arr[left : left + k]
```

Heap

p.1139

Heap

## #787 Cheapest Flights Within K Stops

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Dynamic Programming · DFS · BFS · Graph · Heap

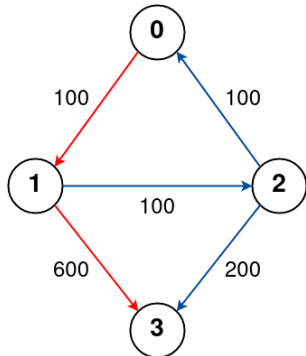
Lists: Grind 169

### Problem

There are `n` cities connected by some number of flights. You are given an array `flights` where `flights[i] = [fromi, toi, pricei]` indicates that there is a flight from city `fromi` to city `toi` with cost `pricei`.

You are also given three integers `src`, `dst`, and `k`, return the cheapest price from `src` to `dst` with at most `k` stops. If there is no such route, return `-1`.

Example 1:

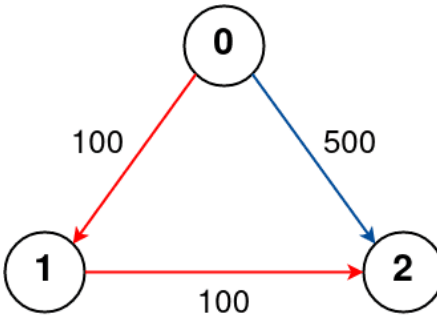


```
Input: n = 4, flights = [[0,1,100],[1,2,100],[2,0,100],[1,3,600],[2,3,200]], src = 0, dst = 3, k = 1
Output: 700
Explanation:
The graph is shown above.
The optimal path with at most 1 stop from city 0 to 3 is marked in red and has cost 100 + 600 = 700.
Note that the path through cities [0,1,2,3] is cheaper but is invalid because it uses 2 stops.
```

Example 2:

Heap

p.1141



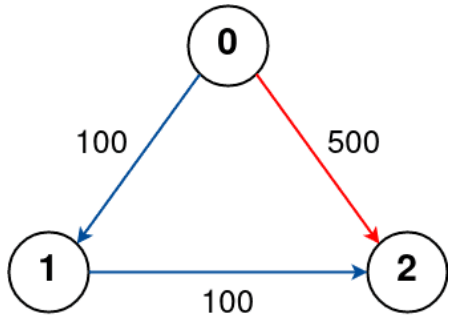
```
Input: n = 3, flights = [[0,1,100],[1,2,100],[0,2,500]], src = 0, dst = 2, k = 1
Output: 200
Explanation:
The graph is shown above.
The optimal path with at most 1 stop from city 0 to 2 is marked in red and has cost 100 + 100 = 200.
```

Example 3:

Heap

p.1142

Sheet 127 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait



```
Input: n = 3, flights = [[0,1,100],[1,2,100],[0,2,500]], src = 0, dst = 2, k = 0
Output: 500
Explanation:
The graph is shown above.
The optimal path with no stops from city 0 to 2 is marked in red and has cost 500.
```

Constraints:

- $2 \leq n \leq 100$
- $0 \leq \text{flights.length} \leq (n * (n - 1) / 2)$
- `flights[i].length == 3`
- $0 \leq \text{fromi}, \text{toi} < n$

Heap

p.1143

- fromi != toi
- 1 <= pricei <= 104
- There will not be any multiple flights between two cities.
- 0 <= src, dst, k < n
- src != dst

#### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findCheapestPrice(self, n, flights, src, dst, k):
 # Brute Force O(V^(K+2)) - DFS trying all paths
 with at most k stops
 from collections import defaultdict
 graph = defaultdict(list)
 for u,v,w in flights:
 graph[u].append((v,w))
 self.res = float('inf')
 def dfs(node, stops, cost):

 if node == dst:
 self.res = min(self.res, cost); return
 if stops > k: return
 for nei, price in graph[node]:
 if cost + price < self.res: # prune only
 dfs(nei, stops+1, cost+price)
 dfs(src, 0, 0)
 return self.res if self.res < float('inf') else -1
```

Heap p.1144

#### Community Solutions (Python)

##### • WalkCC

```
class Solution:
 def findCheapestPrice(
 self,
 n: int,
 flights: List[List[int]],
 src: int,
 dst: int,
 k: int,
) -> int:
 graph = [[] for _ in range(n)]

 for u, v, w in flights:
 graph[u].append((v, w))

 return self._dijkstra(graph, src, dst, k)

 def _dijkstra(
 self,
 graph: List[List[tuple[int, int]]],
 src: int,
```

Heap p.1145

```
 dst: int,
 k: int,
) -> int:
 dist = [[math.inf] * (k + 2) for _ in
 range(len(graph))]

 dist[src][k + 1] = 0
 minHeap = [(dist[src][k + 1], src, k + 1)] # (d, u,
 stops)

 while minHeap:
 d, u, stops = heapq.heappop(minHeap)

 if u == dst:
 return d
 if stops == 0 or d > dist[u][stops]:
 continue
 for v, w in graph[u]:
 if d + w < dist[v][stops - 1]:
 dist[v][stops - 1] = d + w
 heapq.heappush(minHeap, (dist[v][stops - 1], v,
 stops - 1))

 return -1
```

##### • LeetCode

Heap p.1146

```
class Solution:
 def findCheapestPrice(
 self, n: int, flights: List[List[int]], src: int,
 dst: int, k: int
) -> int:
 INF = 0x3F3F3F3F
 dist = [INF] * n
 dist[src] = 0
 for _ in range(k + 1):
 backup = dist.copy()
 for f, t, p in flights:
```

```
 dist[t] = min(dist[t], backup[f] + p)
 return -1 if dist[dst] == INF else dist[dst]
```

##### • SimplyLeet

```
import heapq

def findCheapestPrice(n, flights, src, dst, k):
 # Build graph: each city maps to a list of
 (neighboring city, flight price)
 graph = {i: [] for i in range(n)}
 for from_city, to_city, price in flights:
 graph[from_city].append((to_city, price))

 # Priority queue: (total_cost, current_city, stops so
 far)
 heap = [(0, src, 0)]
```

Heap p.1147

```
 # Dictionary to store the best cost to reach a city
 with a given stop count
 best = {}

 while heap:
 cost, city, stops = heapq.heappop(heap)
 # Return cost if destination is reached
 if city == dst:
 return cost
```

```
 # If the current number of stops exceeds k, skip
 further processing
 if stops > k:
 continue

 # Expand the neighboring flights
 for neighbor, price in graph[city]:
 new_cost = cost + price
 # Only proceed if this route is cheaper or
 not yet explored for this many stops
 if (neighbor, stops + 1) not in best or
 new_cost < best[(neighbor, stops + 1)]:
 best[(neighbor, stops + 1)] = new_cost
```

Heap p.1148

```
 heapq.heappush(heap, (new_cost, neighbor,
 stops + 1))
 return -1

 # Example usage:
 print(findCheapestPrice(4,
 [[0,1,100],[1,2,100],[2,0,100],[1,3,600],[2,3,200]], 0,
 3, 1))
```

##### • LC.ca

```
class Solution:
 def findCheapestPrice(
 self, n: int, flights: List[List[int]], src: int,
 dst: int, k: int
) -> int:
 INF = 0x3F3F3F3F
 dist = [INF] * n
 dist[src] = 0
 for _ in range(k + 1):
 backup = dist.copy()
 for f, t, p in flights:

 dist[t] = min(dist[t], backup[f] + p)
 return -1 if dist[dst] == INF else dist[dst]
```

Heap p.1149

#### Heap

### #973 K Closest Points to Origin

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Math · Divide and Conquer · Sorting · Heap

Lists: Grind 169

Time Complexity:  $O(n \log n)$  with a sorting approach,  $O(n \log k)$  with a heap approach, or average  $O(n)$  with a Quickselect approach. Space Complexity:  $O(n)$  if extra space is used for sorting, or  $O(k)$  fo

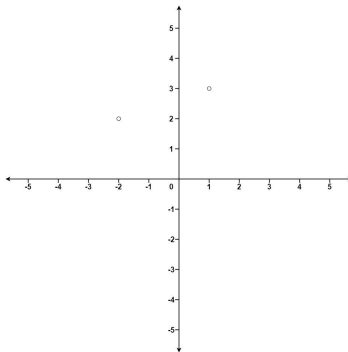
#### Problem

Given an array of points where  $\text{points}[i] = [x_i, y_i]$  represents a point on the X-Y plane and an integer k, return the k closest points to the origin (0, 0).

The distance between two points on the X-Y plane is the Euclidean distance (i.e.,  $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ ).

You may return the answer in any order. The answer is guaranteed to be unique (except for the order that it is in).

Example 1:



Input: points = [[1,3],[-2,2]], k = 1  
Output: [[-2,2]]  
Explanation:  
The distance between (1, 3) and the origin is  $\sqrt{10}$ .  
The distance between (-2, 2) and the origin is  $\sqrt{8}$ .  
Since  $\sqrt{8} < \sqrt{10}$ , (-2, 2) is closer to the origin.  
We only want the closest k = 1 points from the origin, so the answer is just [[-2,2]].

Example 2:

Heap p.1150

Heap p.1151

Heap p.1152



```

class Solution:
 def kClosest(self, points: List[List[int]], k: int) -> List[List[int]]:
 maxHeap = []

 for x, y in points:
 heapq.heappush(maxHeap, (- x * x - y * y, [x, y]))
 if len(maxHeap) > k:
 heapq.heappop(maxHeap)

 return [pair[1] for pair in maxHeap]

```

#### LeetCode

```

class Solution:
 def kClosest(self, points: List[List[int]], k: int) -> List[List[int]]:
 points.sort(key=lambda p: hypot(p[0], p[1]))
 return points[:k]

```

#### SimplyLeet

```

Python Solution
def kClosest(points, k):
 # Sort the points based on their squared Euclidean
 # distance from the origin.
 points.sort(key=lambda point: point[0]**2 +
 point[1]**2)
 # Return the first k points from the sorted list.
 return points[:k]

Example usage:
points = [[1,3], [-2,2]]
k = 1

```

Heap p.1153

```

print(kClosest(points, k))

```

#### LC.ca

```

class Solution:
 def kClosest(self, points: List[List[int]], k: int) -> List[List[int]]:
 points.sort(key=lambda p: p[0] * p[0] + p[1] *
p[1])
 return points[:k]

```

Heap p.1154

Heap

## #1057 Campus Bikes

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Greedy · Sorting · Heap

Lists: Premium 98

Time Complexity:  $O(n * m * \log(n * m))$  where  $n$  is the number of workers and  $m$  is the number of bikes. Space Complexity:  $O(n * m)$  for storing all worker-bike pairs.

### Problem

On a campus represented on the X-Y plane, there are  $n$  workers and  $m$  bikes, with  $n \leq m$ .

You are given an array `workers` of length  $n$  where `workers[i] = [xi, yi]` is the position of the  $i$ th worker. You are also given an array `bikes` of length  $m$  where `bikes[j] = [xj, yj]` is the position of the  $j$ th bike. All the given positions are unique.

Assign a bike to each worker. Among the available bikes and workers, we choose the (`workeri`, `bikej`) pair with the shortest Manhattan distance between each other and assign the bike to that worker.

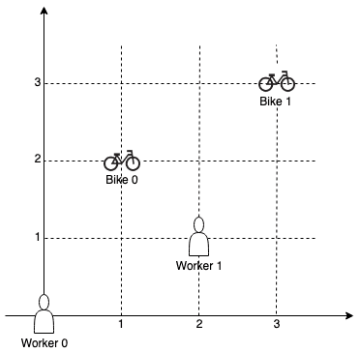
If there are multiple (`workeri`, `bikej`) pairs with the same shortest Manhattan distance, we choose the pair with the smallest worker index. If there are multiple ways to do that, we choose the pair with the smallest bike index.

Heap p.1155

Repeat this process until there are no available workers. Return an array `answer` of length  $n$ , where `answer[i]` is the index (0-indexed) of the bike that the  $i$ th worker is assigned to.

The Manhattan distance between two points  $p1$  and  $p2$  is  $\text{Manhattan}(p1, p2) = |p1.x - p2.x| + |p1.y - p2.y|$ .

Example 1:



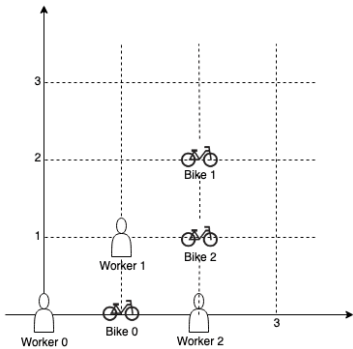
Heap p.1156

```

Input: workers = [[0,0],[2,1]], bikes = [[1,2],[3,3]]
Output: [1,0]
Explanation: Worker 1 grabs Bike 0 as they are closest
(without ties), and Worker 0 is assigned Bike 1. So the
output is [1, 0].

```

Example 2:



Heap p.1157

```

Input: workers = [[0,0],[1,1],[2,0]], bikes =
[[1,0],[2,2],[2,1]]
Output: [0,2,1]
Explanation: Worker 0 grabs Bike 0 at first. Worker 1
and Worker 2 share the same distance to Bike 2, thus
Worker 1 is assigned to Bike 2, and Worker 2 will take
Bike 1. So the output is [0,2,1].

```

Constraints:

- $n == \text{workers.length}$
- $m == \text{bikes.length}$
- $1 \leq n \leq m \leq 1000$
- $\text{workers}[i].\text{length} == \text{bikes}[j].\text{length} == 2$
- $0 \leq x_i, y_i < 1000$
- $0 \leq x_j, y_j < 1000$
- All worker and bike locations are unique.

Brute Force (Python)

```

Brute Force Approach
class Solution:
 def assign_bikes(self, workers, bikes):
 # Brute Force $O(n \log n)$ - sort repeatedly
 instead of heap
 data = list(workers)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result

```

Heap p.1158

### Community Solutions (Python)

#### WalkCC

```

class Solution:
 def assignBikes(
 self,
 workers: List[List[int]],
 bikes: List[List[int]],
) -> List[int]:
 ans = [-1] * len(workers)
 usedBikes = [False] * len(bikes)
 # buckets[k] := (i, j), where k = dist(workers[i],
bikes[j])
 buckets = [[] for _ in range(2001)]

```

```

def dist(p1: List[int], p2: List[int]) -> int:
 return abs(p1[0] - p2[0]) + abs(p1[1] - p2[1])

```

```

for i, worker in enumerate(workers):
 for j, bike in enumerate(bikes):
 buckets[dist(worker, bike)].append((i, j))

```

```

for k in range(2001):
 for i, j in buckets[k]:

```

```

 if ans[i] == -1 and not usedBikes[j]:
 ans[i] = j
 usedBikes[j] = True

```

```

 return ans

```

#### LeetCode

Heap p.1159

```

class Solution:
 def assignBikes(
 self, workers: List[List[int]], bikes:
List[List[int]]
) -> List[int]:
 n, m = len(workers), len(bikes)
 arr = []
 for i, j in product(range(n), range(m)):
 dist = abs(workers[i][0] - bikes[j][0]) +
abs(workers[i][1] - bikes[j][1])
 arr.append((dist, i, j))
 arr.sort()

```

```

 vis1 = [False] * n
 vis2 = [False] * m
 ans = [0] * n
 for _, i, j in arr:
 if not vis1[i] and not vis2[j]:
 vis1[i] = vis2[j] = True
 ans[i] = j
 return ans

```

#### SimplyLeet

Heap p.1160

#### Python solution

```

def assign_bikes(workers, bikes):
 # List to hold all possible (distance, worker, bike)
 tuples
 pairs = []
 for worker_index, (wx, wy) in enumerate(workers):
 for bike_index, (bx, by) in enumerate(bikes):
 # Calculate Manhattan distance between worker
 and bike
 distance = abs(wx - bx) + abs(wy - by)
 pairs.append((distance, worker_index,
bike_index))

```

```

 # Sort pairs based on distance, then worker index,
 then bike index
 pairs.sort()

```

```

 n = len(workers) # Number of workers
 assigned_workers = [False] * n # Track if worker is
already assigned
 assigned_bikes = [False] * len(bikes) # Track if bike
is already assigned
 result = [-1] * n # Result list to store bike
assignment for each worker

```

```

 # Iterate over the sorted pairs and assign bikes

```

Heap p.1161

```
for distance, worker_index, bike_index in pairs:
 if not assigned_workers[worker_index] and not
assigned_bikes[bike_index]:
 result[worker_index] = bike_index # Assign
bike to worker
 assigned_workers[worker_index] = True # Mark
worker as assigned
 assigned_bikes[bike_index] = True # Mark bike
as assigned
 return result

Example usage:
workers = [[0,0],[2,1]]
bikes = [[1,2],[3,3]]

print(assign_bikes(workers, bikes)) # Output: [1, 0]
```

• **LC.ca**

```
class Solution:
 def assignBikes(
 self, workers: List[List[int]], bikes:
List[List[int]]
) -> List[int]:
 n, m = len(workers), len(bikes)
 arr = []
 for i, j in product(range(n), range(m)):
 dist = abs(workers[i][0] - bikes[j][0]) +
abs(workers[i][1] - bikes[j][1])
 arr.append((dist, i, j))
 arr.sort()
```

Heapp.1162

```
vis1 = [False] * n
vis2 = [False] * m
ans = [0] * n
for _, i, j in arr:
 if not vis1[i] and not vis2[j]:
 vis1[i] = vis2[j] = True
 ans[i] = j
return ans
```

Heapp.1163

Heap

## #1167 Minimum Cost to Connect Sticks

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Greedy · Heap

Lists: Premium 98

Time Complexity:  $O(n \log n)$ , where  $n$  is the number of sticks (due to heap insertion and removal operations).  
Space Complexity:  $O(n)$ , for storing the sticks in the heap.

### Problem

You have some number of sticks with positive integer lengths. These lengths are given as an array sticks, where sticks[i] is the length of the ith stick.

You can connect any two sticks of lengths  $x$  and  $y$  into one stick by paying a cost of  $x + y$ . You must connect all the sticks until there is only one stick remaining.

Return the minimum cost of connecting all the given sticks into one stick in this way.

Example 1:

Heapp.1164

Input: sticks = [2,4,3]  
Output: 14  
Explanation: You start with sticks = [2,4,3].  
1. Combine sticks 2 and 3 for a cost of 2 + 3 = 5. Now you have sticks = [5,4].  
2. Combine sticks 5 and 4 for a cost of 5 + 4 = 9. Now you have sticks = [9].  
There is only one stick left, so you are done. The total cost is 5 + 9 = 14.

### Example 2:

Input: sticks = [1,8,3,5]  
Output: 30  
Explanation: You start with sticks = [1,8,3,5].  
1. Combine sticks 1 and 3 for a cost of 1 + 3 = 4. Now you have sticks = [4,8,5].  
2. Combine sticks 4 and 5 for a cost of 4 + 5 = 9. Now you have sticks = [9,8].  
3. Combine sticks 9 and 8 for a cost of 9 + 8 = 17. Now you have sticks = [17].  
There is only one stick left, so you are done. The total cost is 4 + 9 + 17 = 30.

### Example 3:

Input: sticks = [5]  
Output: 0  
Explanation: There is only one stick, so you don't need to do anything. The total cost is 0.

### Constraints:

- 1 <= sticks.length <= 104
- 1 <= sticks[i] <= 104

Heapp.1165

### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def connectSticks(self, sticks):
 # Brute Force $O(n \log n)$ - sort repeatedly
 instead of heap
 data = list(sticks)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result
```

### Community Solutions (Python)

• **WalkCC**

```
class Solution:
 def connectSticks(self, sticks: list[int]) -> int:
 ans = 0
 heapq.heapify(sticks)

 while len(sticks) > 1:
 x = heapq.heappop(sticks)
 y = heapq.heappop(sticks)
 ans += x + y
 heapq.heappush(sticks, x + y)

 return ans
```

• **LeetDoocs**

Heapp.1166

```
class Solution:
 def connectSticks(self, sticks: List[int]) -> int:
 heapify(sticks)
 ans = 0
 while len(sticks) > 1:
 z = heappop(sticks) + heappop(sticks)
 ans += z
 heappush(sticks, z)
 return ans
```

• **SimplyLeet**

```
import heapq

def connectSticks(sticks):
 # Initialize the heap with the provided stick lengths
 heapq.heapify(sticks)

 # Initialize total cost to 0
 total_cost = 0

 # Continue connecting sticks until there is only one left

 while len(sticks) > 1:
 # Extract the two smallest sticks
 stick1 = heapq.heappop(sticks)
 stick2 = heapq.heappop(sticks)
 # Calculate the cost to connect these two sticks
 cost = stick1 + stick2
 # Add the cost to the total cost
 total_cost += cost
 # Push the combined stick back into the heap
 heapq.heappush(sticks, cost)
```

Heapp.1167

```
return total_cost
```

# Example usage:  
# print(connectSticks([2, 4, 3])) # Output: 14

• **LC.ca**

```
class Solution:
 def connectSticks(self, sticks: List[int]) -> int:
 heapify(sticks)
 ans = 0
 while len(sticks) > 1:
 z = heappop(sticks) + heappop(sticks)
 ans += z
 heappush(sticks, z)
 return ans
```

Heapp.1168

Heap

## #1424 Diagonal Traverse II

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Sorting · Heap

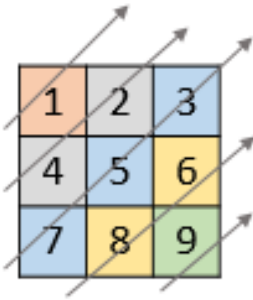
Lists: CodeSignal

Time Complexity:  $O(N)$ , where  $N$  is the total number of elements (each element is processed once).  
Space Complexity:  $O(N)$ , for storing the elements in the map and the output array.

### Problem

Given a 2D integer array nums, return all elements of nums in diagonal order as shown in the below images.

Example 1:



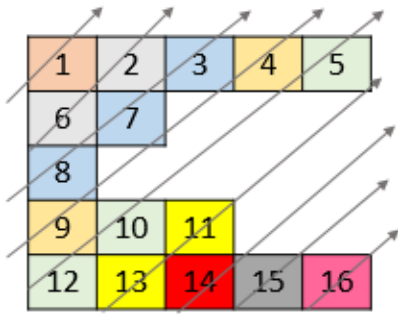
Input: nums = [[1,2,3],[4,5,6],[7,8,9]]  
Output: [1,4,2,7,5,3,8,6,9]

Example 2:

Heapp.1169

Heap

p.1170



Input: nums =  
[[1,2,3,4,5],[6,7],[8],[9,10,11],[12,13,14,15,16]]  
Output: [1,6,2,8,7,3,9,4,12,10,5,13,11,14,15,16]

#### Constraints:

- $1 \leq \text{nums.length} \leq 105$
- $1 \leq \text{nums}[i].\text{length} \leq 105$
- $1 \leq \text{sum}(\text{nums}[i].\text{length}) \leq 105$
- $1 \leq \text{nums}[i][j] \leq 105$

#### Brute Force (Python)

Heap

p.1171

```
Brute Force Approach
class Solution:
 def findDiagonalOrder(self, nums):
 # Brute Force $O(n \log n)$ - sort repeatedly
 instead of heap
 data = list(nums)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result
```

#### Community Solutions (Python)

##### • LeetCode

```
class Solution:
 def findDiagonalOrder(self, nums: List[List[int]]) -> List[int]:
 arr = []
 for i, row in enumerate(nums):
 for j, v in enumerate(row):
 arr.append((i + j, j, v))
 arr.sort()
 return [v[2] for v in arr]
```

##### • SimplyLeet

Heap

p.1172

```
def findDiagonalOrder(nums):
 # Use a dictionary to collect elements by their
 diagonal key (i + j)
 from collections import defaultdict
 diagonals = defaultdict(list)

 # Iterate over each row and its elements
 for i, row in enumerate(nums):
 for j, value in enumerate(row):
 # Insert at the beginning to achieve the
 desired order for the diagonal
 diagonals[i + j].insert(0, value)
```

```
result = []
Process the diagonals in order of increasing key
for key in sorted(diagonals.keys()):
 result.extend(diagonals[key])
return result

Example usage:
nums = [[1,2,3],[4,5,6],[7,8,9]]
print(findDiagonalOrder(nums)) # Output:
[1,4,2,7,5,3,8,6,9]
```

##### • LC.ca

Heap

p.1173

```
class Solution:
 def findDiagonalOrder(self, nums: List[List[int]]) -> List[int]:
 arr = []
 for i, row in enumerate(nums):
 for j, v in enumerate(row):
 arr.append((i + j, j, v))
 arr.sort()
 return [v[2] for v in arr]
```

Heap

p.1174

Heap

## #23 Merge k Sorted Lists

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List · Divide and Conquer · Heap · Merge Sort

Lists: Grind 169

Time Complexity:  $O(N \log k)$  where  $N$  is the total number of nodes across all lists and  $k$  is the number of lists. Space Complexity:  $O(k)$  due to the heap containing at most one node from each list.

#### Problem

You are given an array of  $k$  linked-lists lists, each linked-list is sorted in ascending order.

Merge all the linked-lists into one sorted linked-list and return it.

Example 1:

```
Input: lists = [[1,4,5],[1,3,4],[2,6]]
Output: [1,1,2,3,4,4,5,6]
Explanation: The linked-lists are:
[
 1->4->5,
 1->3->4,
 2->6
]
```

Example 2:

Heap

p.1175

Input: lists = []  
Output: []

Example 3:

Input: lists = [[]]  
Output: []

#### Constraints:

- $k == \text{lists.length}$
- $0 \leq k \leq 104$
- $0 \leq \text{lists}[i].\text{length} \leq 500$
- $-104 \leq \text{lists}[i][j] \leq 104$
- $\text{lists}[i]$  is sorted in ascending order.
- The sum of  $\text{lists}[i].\text{length}$  will not exceed 104.

#### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def mergeKLists(self, lists):
 # Brute Force $O(N \log N)$ - collect all values,
 sort, rebuild
 vals = []
 for node in lists:
 while node:
 vals.append(node.val)
 node = node.next
 vals.sort()
```

Heap

p.1176

```
dummy = ListNode(0)
curr = dummy
for v in vals:
 curr.next = ListNode(v)
 curr = curr.next
return dummy.next
```

#### Community Solutions (Python)

##### • LeetCode

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def mergeKLists(self, lists: List[Optional[ListNode]]) -> Optional[ListNode]:
 setattr(ListNode, "__lt__", lambda a, b: a.val < b.val)
 pq = [head for head in lists if head]
 heapify(pq)
```

```
dummy = cur = ListNode(0)
while pq:
 node = heappop(pq)
 if node.next:
 heappush(pq, node.next)
 cur.next = node
 cur = cur.next
return dummy.next
```

##### • SimplyLeet

Heap

p.1177

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

import heapq

class Solution:
 def mergeKLists(self, lists):

 # Create a min-heap
 min_heap = []
 # Heap entry: (node value, unique id, node) to
 handle duplicate values
 for i, node in enumerate(lists):
 if node:
 heapq.heappush(min_heap, (node.val, i, node))

 # Dummy node to form the start of merged list
 dummy = ListNode(0)
 current = dummy
```

Heap

p.1178

```
While there are elements in the heap
while min_heap:
 # Pop the smallest node from the heap
 val, idx, node = heapq.heappop(min_heap)
 # Add the smallest node to the merged list
 current.next = node
 current = current.next
 # If there is a next node in the list, push
 it into the heap
 if node.next:
 heapq.heappush(min_heap, (node.next.val, idx, node.next))

return dummy.next
```

##### • LC.ca

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def mergeKLists(self, lists: List[Optional[ListNode]]) -> Optional[ListNode]:
 setattr(ListNode, "__lt__", lambda a, b: a.val < b.val)
 pq = [head for head in lists if head]
 heapify(pq)
```

Heap

p.1179

```
dummy = cur = ListNode()
while pq:
 node = heappop(pq)
 if node.next:
 heappush(pq, node.next)
 cur.next = node
 cur = cur.next
return dummy.next

Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next

#####

import heapq

'''
__lt__(self, other) for <
__le__(self, other) for <=
__gt__(self, other) for >
__ge__(self, other) for >=
'''

'''
```

Heap p.1180

```
or create a wrapper class, without modifying existing
node class
class NodeWrapper:
 def __init__(self, node):
 self.node = node

 def __lt__(self, other):
 return self.node.val < other.node.val

'''

overwrite the comparison function, so the node can be
comparable
or else, error, TypeError: '<' not supported between
instances of 'ListNode' and 'ListNode'

define 'gt' will also work, so either lt or gt will do
the compare job for ListNode
ListNode.__gt__ = lambda x, y: (x.val > y.val)

ListNode.__lt__ = lambda x, y: (x.val < y.val)

class Solution:
```

Heap p.1181

```
def mergeKLists(self, lists:
List[Optional[ListNode]]) -> Optional[ListNode]:
 dummy = current = ListNode()
 heap = []
 for i, node in enumerate(lists):
 if node:
 # need to override
 # or else error: TypeError: '<' not
supported between instances of 'ListNode' and 'ListNode'
 heappq.heappush(heap, node)

 while heap:

 anode = heappq.heappop(heap)

 current.next = anode
 current = current.next

 if anode.next:
 heappq.heappush(heap, anode.next)

 return dummy.next
```

Heap p.1182

```
'''
tried to use node.val to order heap, but still got error
TypeError: '<' not supported between instances of
'ListNode' and 'ListNode':

heapq.heappush(heap, (node.val, node))

'''

#####

based on merge 2 lists

note for empty input
class Solution:
 def mergeKLists(self, lists:
List[Optional[ListNode]]) -> Optional[ListNode]:
 if lists is None or len(lists) == 0:
 return None
 return self.mergeKListsHelper(lists, 0,
len(lists) - 1)

 def mergeKListsHelper(self, lists:
List[Optional[ListNode]], start: int, end: int) ->
Optional[ListNode]:

 if start == end:
```

Heap p.1183

```
 return lists[start]

 mid = int((start + end) / 2)

 left = self.mergeKListsHelper(lists, start, mid)
 # print(left)

 right = self.mergeKListsHelper(lists, mid + 1,
end)
 # print(right)

 return self.mergeTwoLists(left, right)

 def mergeTwoLists(self, list1: Optional[ListNode],
list2: Optional[ListNode]) -> Optional[ListNode]:
 dummy = ListNode()
 current = dummy
 # print("merging: ", list1)
 # print("merging: ", list2)
 while list1 or list2:
 v1 = list1.val if list1 else float('inf')
 v2 = list2.val if list2 else float('inf')

 if v1 < v2:
 current.next = list1
 list1 = list1.next
 else:
 current.next = list2
 list2 = list2.next
 current = current.next

 # print("merging result: ", dummy.next)
```

Heap p.1184

```
 return dummy.next

#####

Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next

class Solution:
 def mergeKLists(self, lists: List[ListNode]) ->
ListNode:
 n = len(lists)
 if n == 0:
 return None
 for i in range(n - 1): # this is o(N) times of
merge, while if use divide-mid-merge then it's o(logN)
times of merge
 lists[i + 1] = self.mergeTwoLists(lists[i],
lists[i + 1])
 return lists[-1]

 def mergeTwoLists(self, l1: ListNode, l2: ListNode)
-> ListNode:
```

Heap p.1185

```
dummy = ListNode()
cur = dummy
while l1 and l2:
 if l1.val <= l2.val:
 cur.next = l1
 l1 = l1.next
 else:
 cur.next = l2
 l2 = l2.next
 cur = cur.next

cur.next = l1 or l2
return dummy.next

if __name__ == '__main__':

 l1 = ListNode(1)
 l1.next = ListNode(4)
 l1.next.next = ListNode(5)

 l2 = ListNode(1)
 l2.next = ListNode(3)
 l2.next.next = ListNode(4)

 l3 = ListNode(2)
 l3.next = ListNode(6)

 print(Solution().mergeKLists([l1, l2, l3]))
```

Heap p.1186

```
print(Solution().mergeKLists([]))
print(Solution().mergeKLists([[]]))
```

Heap p.1187

Heap

#218 The Skyline Problem

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Divide and Conquer · Binary Indexed Tree · Segment Tree · Line Sweep · Heap · Ordered Set

Lists: Denny Zhang

Time Complexity:  $O(n \log n)$  due to sorting events and heap operations for each building event. Space Complexity:  $O(n)$  for the event list and the heap used for processing active buildings.

Problem

A city's skyline is the outer contour of the silhouette formed by all the buildings in that city when viewed from a distance. Given the locations and heights of all the buildings, return the skyline formed by these buildings collectively.

The geometric information of each building is given in the array buildings where  $\text{buildings}[i] = [\text{lefti}, \text{righti}, \text{heighti}]$ :

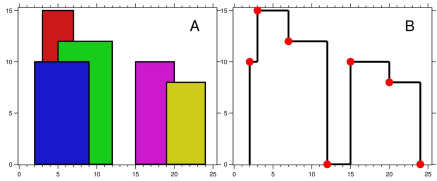
- lefti is the x coordinate of the left edge of the ith building.
- righti is the x coordinate of the right edge of the ith building.
- heighti is the height of the ith building.

Heap p.1188

You may assume all buildings are perfect rectangles grounded on an absolutely flat surface at height 0. The skyline should be represented as a list of "key points" sorted by their x-coordinate in the form `[[x1,y1],[x2,y2],...]`. Each key point is the left endpoint of some horizontal segment in the skyline except the last point in the list, which always has a y-coordinate 0 and is used to mark the skyline's termination where the rightmost building ends. Any ground between the leftmost and rightmost buildings should be part of the skyline's contour.

Note: There must be no consecutive horizontal lines of equal height in the output skyline. For instance, `[[...,[2 3],[4 5],[7 5],[11 5],[12 7],...]]` is not acceptable; the three lines of height 5 should be merged into one in the final output as such: `[[...,[2 3],[4 5],[12 7],...]]`

Example 1:



Input: buildings = `[[2,9,10],[3,7,15],[5,12,12],[15,20,10],[19,24,8]]`  
Output: `[[2,10],[3,15],[7,12],[12,0],[15,10],[20,8],[24,0]]`  
Explanation:  
Figure A shows the buildings of the input.  
Figure B shows the skyline formed by those buildings.  
The red points in figure B represent the key points in the output list.

Example 2:

Input: buildings = `[[0,2,3],[2,5,3]]`  
Output: `[[0,3],[5,0]]`

Constraints:

- `1 <= buildings.length <= 104`
- `0 <= lefti < righti <= 231 - 1`
- `1 <= heighti <= 231 - 1`
- buildings is sorted by lefti in non-decreasing order.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def getSkyline(self, buildings):
 # Brute Force O(n log n) - sort repeatedly
 # instead of heap
 data = list(buildings)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result
```

Community Solutions (Python)

```
• WalkCC
class Solution:
 def getSkyline(self, buildings: list[list[int]]) -> list[list[int]]:
 n = len(buildings)
 if n == 0:
 return []
 if n == 1:
 left, right, height = buildings[0]
 return [[left, height], [right, 0]]

 left = self.getSkyline(buildings[:n // 2])
```

```
right = self.getSkyline(buildings[n // 2:])
return self._merge(left, right)

def _merge(self, left: list[list[int]],
 right: list[list[int]]) -> list[list[int]]:
 ans = []
 i = 0 # left's index
 j = 0 # right's index
 leftY = 0
 rightY = 0

 while i < len(left) and j < len(right):
 # Choose the powith smaller x
 if left[i][0] < right[j][0]:
 leftY = left[i][1] # Update the ongoing `leftY`.
 self._addPoint(ans, left[i][0], max(left[i][1],
 rightY))
 i += 1
 else:
 rightY = right[j][1] # Update the ongoing
 `rightY`.
 self._addPoint(ans, right[j][0], max(right[j][1],
 leftY))
```

```
j += 1

while i < len(left):
 self._addPoint(ans, left[i][0], left[i][1])
 i += 1

while j < len(right):
 self._addPoint(ans, right[j][0], right[j][1])
 j += 1

return ans

def _addPoint(self, ans: list[list[int]], x: int, y:
int) -> None:
 if ans and ans[-1][0] == x:
 ans[-1][1] = y
 return
 if ans and ans[-1][1] == y:
 return
 ans.append([x, y])
```

• SimplyLeet

```
import heapq

def getSkyline(buildings):
 # List to store all events: (x, height, right)
 events = []
 # Create events for all buildings: left edge (start)
 # and right edge (end)
 for left, right, height in buildings:
 # For the start point, height is negative for
 # proper sorting (start events come before end events)
 events.append((left, -height, right))
 # For the end point, height is 0 to indicate
 # removal of building from active heap

 events.append((right, 0, 0))

 # Sort events by x coordinate.
 events.sort()

 # Max heap to store active buildings (using negative
 heights)
 result = []
 # Initialize the heap with a dummy building of height
 0 that extends infinitely
 heap = [(0, float('inf'))]
```

```
for x, neg_height, right in events:
 # Remove buildings from the heap that have ended
 # by current x.
 while heap and heap[0][1] <= x:
 heapq.heappop(heap)

 # If event is a start event (neg_height != 0),
 # push it into the heap.
 if neg_height != 0:
 heapq.heappush(heap, (neg_height, right))

 # Current highest building height is at the top
 # of the heap

 current_height = -heap[0][0]
 # If the result is empty or current height
 # differs from last recorded height, add a new key point.
 if not result or current_height != result[-1][1]:
 result.append([x, current_height])

return result

Example usage:
buildings =
[[2,9,10],[3,7,15],[5,12,12],[15,20,10],[19,24,8]]
print(getSkyline(buildings)) # Expected output:
[[2,10],[3,15],[7,12],[12,0],[15,10],[20,8],[24,0]]
```

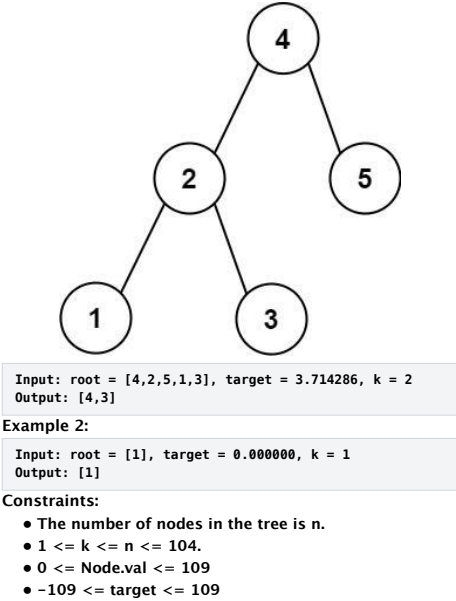
Heap

### #272 Closest Binary Search Tree Value II

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Two Pointers · Stack · Tree · DFS · BST · Heap  
Lists: Premium 98  
Time Complexity:  $O(k + \log n)$  -  $O(\log n)$  to initialize the two stacks and  $O(k)$  to collect k elements. Space Complexity:  $O(\log n)$  - due to the stack space in a balanced BST.

**Problem**  
Given the root of a binary search tree, a target value, and an integer k, return the k values in the BST that are closest to the target. You may return the answer in any order.  
You are guaranteed to have only one unique set of k values in the BST that are closest to the target.

Example 1:



Follow up: Assume that the BST is balanced. Could you solve it in less than O(n) runtime (where n = total nodes)?

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def closestKValues(self, root: TreeNode, target: float, k: int):
 # Brute Force O(n log n) - sort repeatedly
 # instead of heap
 data = list(root)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result
```

Community Solutions (Python)

```
• WalkCC
class Solution:
 def closestKValues(
 self,
 root: TreeNode | None,
 target: float,
 k: int,
) -> List[int]:
 dq = collections.deque()

 def inorder(root: TreeNode | None) -> None:
```

```
 if not root:
 return

 inorder(root.left)
 dq.append(root.val)
 inorder(root.right)

 inorder(root)

 while len(dq) > k:

 if abs(dq[0] - target) > abs(dq[-1] - target):
 dq.popleft()
 else:
 dq.pop()

 return list(dq)
```

```
• LeetDooocs
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def closestKValues(self, root: TreeNode, target: float, k: int) -> List[int]:
 def dfs(root):
 if root is None:
```

```
 return
 dfs(root.left)
 if len(q) < k:
 q.append(root.val)
 else:
 if abs(root.val - target) >= abs(q[0] - target):
 return
 q.popleft()
 q.append(root.val)
 dfs(root.right)

 q = deque()
 dfs(root)
 return list(q)
```

```
• SimplyLeet
Python solution for Closest Binary Search Tree Value II
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
```

```
 def closestKValues(self, root: TreeNode, target: float, k: int):
 # Helper function to initialize predecessors
 stack

 def initPredecessor(node):
 while node:
 # If node value is less or equal, push to
 # preds and go right
 if node.val <= target:
 preds.append(node)
 node = node.right
 else:
 node = node.left

 # Helper function to initialize successors stack
 def initSuccessor(node):
 while node:
 # If node value is greater than target,
 # push to succs and go left
 if node.val > target:
 succs.append(node)
 node = node.left
 else:
 node = node.right
```

```
 # Helper function to get next predecessor value
 def getPredecessor():
 if not preds: return None
 node = preds.pop()
 val = node.val
 # Traverse the left subtree: push nodes from
 # left child then rightmost branch
 node = node.left
 while node:
 preds.append(node)

 node = node.right
 return val

 # Helper function to get next successor value
 def getSuccessor():
 if not succs: return None
 node = succs.pop()
 val = node.val
 # Traverse the right subtree: push nodes from
 # right child then leftmost branch
 node = node.right
```

```
 while node:
 succs.append(node)
 node = node.left
 return val

 preds, succs = [], []
 # Initialize the stacks with correct boundary
 values
 initPredecessor(root)
 initSuccessor(root)

 result = []
 # If the closest value from predecessor and
 # successor are same, skip one to avoid duplicates.
 if preds and succs and preds[-1].val == succs[-1].val:
 getPredecessor()

 # Merge k closest values
 for _ in range(k):
 if not preds:
 # Only successors left
 result.append(getSuccessor())
```

```
 elif not succs:
 # Only predecessors left
 result.append(getPredecessor())
 else:
 # Compare which candidate is closer to
 # the target
 if abs(preds[-1].val - target) <= abs(succs[-1].val - target):
 result.append(getPredecessor())
 else:
 result.append(getSuccessor())
 return result

• LC.ca
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def closestKValues(self, root: TreeNode, target: float, k: int) -> List[int]:
 def dfs(root):
 if root is None:
```

```
 return
 dfs(root.left)
 if len(q) < k:
 q.append(root.val)
 else:
 if abs(root.val - target) >= abs(q[0] - target):
 return
 q.popleft()
 q.append(root.val)
 dfs(root.right)

 q = deque()
 dfs(root)
 return list(q)

 #####

 from collections import deque

 # iteration
```

```
class Solution:
 def closestKValues(self, root: TreeNode, target: float, k: int) -> List[int]:
 stack = []
 q = deque()

 # Iterative in-order traversal
 while stack or root:
 while root:
 stack.append(root)
 root = root.left

 root = stack.pop()

 # Process current node
 if len(q) < k:
 q.append(root.val)
 else:
 if abs(root.val - target) < abs(q[0] - target):
 q.popleft()
 q.append(root.val)

 else:
 break
 # Early stop if the current value is
 # farther than the first value in queue

 root = root.right

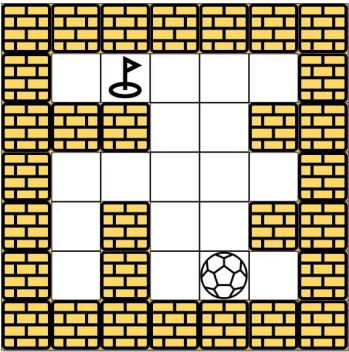
 return list(q)
```



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| <div> <div> <div>Heap</div> <div>#295 Find Median from Data Stream</div> <div>HAR</div> </div> <div> <div> <div>LeetCode · SimplyLeet · WalkCC · LeetCode</div> <div>Two Pointers · Design · Sorting · Heap</div> <div>Lists: Grind 169</div> <div>Time Complexity:</div> </div> <div> <div>Problem</div> <div> <p>The median is the middle value in an ordered integer list. If the size of the list is even, there is no middle value, and the median is the mean of the two middle values.</p> <ul style="list-style-type: none"> <li>For example, for arr = [2,3,4], the median is 3.</li> <li>For example, for arr = [2,3], the median is (2 + 3) / 2 = 2.5.</li> </ul> </div> <div> <p>Implement the MedianFinder class:</p> <ul style="list-style-type: none"> <li>MedianFinder() initializes the MedianFinder object.</li> <li>void addNum(int num) adds the integer num from the data stream to the data structure.</li> <li>double findMedian() returns the median of all elements so far. Answers within 10<sup>-5</sup> of the actual answer will be accepted.</li> </ul> </div> <div> <p>Example 1:</p> </div> </div> </div> </div> | <div> <div> <div>Input</div> <div> ["MedianFinder", "addNum", "addNum", "findMedian", "addNum", "findMedian"]<br/> [[], [1], [2], [1], [3], []]<br/> Output<br/> [null, null, null, 1.5, null, 2.0] </div> </div> <div> <div>Explanation</div> <div>MedianFinder medianFinder = new MedianFinder();</div> </div> <div> <div>Constraints:</div> <ul style="list-style-type: none"> <li>-105 &lt;= num &lt;= 105</li> <li>There will be at least one element in the data structure before calling findMedian.</li> <li>At most 5 * 10<sup>4</sup> calls will be made to addNum and findMedian.</li> </ul> </div> <div> <div>Follow up:</div> <ul style="list-style-type: none"> <li>If all integer numbers from the stream are in the range [0, 100], how would you optimize your solution?</li> <li>If 99% of all integer numbers from the stream are in the range [0, 100], how would you optimize your solution?</li> </ul> </div> <div> <div>Community Solutions (Python)</div> <ul style="list-style-type: none"> <li>WalkCC</li> </ul> </div> </div> | <div> <div> <div> <pre>class MedianFinder:     def __init__(self):         self.maxHeap = []         self.minHeap = []      def addNum(self, num: int) -&gt; None:         if not self.maxHeap or num &lt;= -self.maxHeap[0]:             heapq.heappush(self.maxHeap, -num)         else:             heapq.heappush(self.minHeap, num)      # Balance the two heaps s.t.     #  maxHeap  &gt;=  minHeap  and  maxHeap  -  minHeap  &lt;= 1.     if len(self.maxHeap) &lt; len(self.minHeap):         heapq.heappush(self.maxHeap, -heapq.heappop(self.minHeap))     elif len(self.maxHeap) - len(self.minHeap) &gt; 1:         heapq.heappush(self.minHeap, -heapq.heappop(self.maxHeap))      def findMedian(self) -&gt; float:         if len(self.maxHeap) == len(self.minHeap):              return (-self.maxHeap[0] + self.minHeap[0]) / 2.0         return -self.maxHeap[0]</pre> </div> <div> <ul style="list-style-type: none"> <li>LeetCode</li> </ul> </div> </div> </div>                              |
| <div> <div>Heap</div> <div>p.1207</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <div> <div> <div>Heap</div> <div>p.1208</div> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <div> <div> <div>Heap</div> <div>p.1209</div> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <div> <div> <div> <pre>class MedianFinder:      def __init__(self):         self.minq = []         self.maxq = []      def addNum(self, num: int) -&gt; None:         heapq.heappush(self.minq, -heapq.heappushpop(self.maxq, -num))          if len(self.minq) - len(self.maxq) &gt; 1:             heappush(self.maxq, -heappop(self.minq))      def findMedian(self) -&gt; float:         if len(self.minq) == len(self.maxq):             return (self.minq[0] - self.maxq[0]) / 2         return self.minq[0]  # Your MedianFinder object will be instantiated and called as such: # obj = MedianFinder() # obj.addNum(num)  # param_2 = obj.findMedian()</pre> </div> <div> <ul style="list-style-type: none"> <li>SimplyLeet</li> </ul> </div> </div> </div>                                                                                                                                                                                                                                                                                                                                                                              | <div> <div> <div> <pre>import heapq  class MedianFinder:     def __init__(self):         # max-heap for the lower half (store negatives to simulate max-heap)         self.lowerHalf = []         # min-heap for the upper half         self.upperHalf = []      def addNum(self, num: int) -&gt; None:          # Push to max-heap (simulate by pushing negative)         heapq.heappush(self.lowerHalf, -num)         # Ensure every element in lowerHalf is &lt;= every element in upperHalf         # Move maximum of lowerHalf to upperHalf         heapq.heappush(self.upperHalf, -heapq.heappop(self.lowerHalf))          # Balance the heaps: if upperHalf exceeds lowerHalf in size         if len(self.upperHalf) &gt; len(self.lowerHalf):             heapq.heappush(self.lowerHalf, -heapq.heappop(self.upperHalf))</pre> </div> <div> <p>Heap</p> <p>p.1211</p> </div> </div> </div>                                                                                                                                                       | <div> <div> <div> <pre>def findMedian(self) -&gt; float:     # If both heaps are balanced, median is average of both heaps' top elements     if len(self.lowerHalf) == len(self.upperHalf):         return (-self.lowerHalf[0] + self.upperHalf[0]) / 2.0     # Otherwise, median is the top element of the larger heap     return -self.lowerHalf[0]  # Example usage: # medianFinder = MedianFinder() # medianFinder.addNum(1)  # medianFinder.addNum(2) # print(medianFinder.findMedian()) # Outputs 1.5 # medianFinder.addNum(3) # print(medianFinder.findMedian()) # Outputs 2.0</pre> </div> <div> <ul style="list-style-type: none"> <li>LC.ca</li> </ul> </div> <div> <pre>from heapq import heappush, heappop  class MedianFinder:      def __init__(self):         # the smaller half of the list, max heap (invert min-heap)         self.smallerHalf = []         # the larger half of the list, min heap         self.largerHalf = []</pre> </div> <div> <p>Heap</p> <p>p.1212</p> </div> </div> </div> |
| <div> <div> <div> <pre>def addNum(self, num: int) -&gt; None:  # trick for smaller half, use -1*val # heapq in python does NOT have comparator like in Java heappush(self.smallerHalf, -num) heappush(self.largerHalf, -heappop(self.smallerHalf)) # note: not self.smallerHalf.pop()  if len(self.smallerHalf) &lt; len(self.largerHalf):     heappush(self.smallerHalf, -heappop(self.largerHalf))  def findMedian(self) -&gt; float: if len(self.smallerHalf) == len(self.largerHalf):     return (-self.smallerHalf[0] + self.largerHalf[0]) / 2.0 else:     return float(-self.smallerHalf[0])  # Your MedianFinder object will be instantiated and called as such: # obj = MedianFinder() # obj.addNum(num)  # param_2 = obj.findMedian()</pre> </div> <div> <p>Heap</p> <p>p.1213</p> </div> </div> </div>                                                                                                                                                                                                                                                                                                                                | <div> <div> <div> <pre>self.counts[num] += 1 self.total += 1  def findMedian(self) -&gt; float: if self.total % 2 == 0:     # even number of elements     middle1 = self.total // 2     middle2 = middle1 + 1     count = 0     i1 = i2 = 0      for i in range(101):         count += self.counts[i]         if count &gt;= middle1 and i1 == 0:             i1 = i         if count &gt;= middle2:             i2 = i         break     return (i1 + i2) / 2 else:     # odd number of elements      middle = self.total // 2 + 1     count = 0     for i in range(101):         count += self.counts[i]         if count &gt;= middle:             return i  #####</pre> </div> <div> <p>Heap</p> <p>p.1214</p> </div> </div> </div>                                                                                                                                                                                                                                                                                                                  | <div> <div> <div> <pre>class MedianFinder:     def __init__(self):         """         initialize your data structure here.         """         self.h1 = []         self.h2 = []      def addNum(self, num: int) -&gt; None:         heappush(self.h1, num)          heappush(self.h2, -heappop(self.h1))         if len(self.h2) - len(self.h1) &gt; 1:             heappush(self.h1, -heappop(self.h2))      def findMedian(self) -&gt; float:         if len(self.h2) &gt; len(self.h1):             return -self.h2[0]         return (self.h1[0] - self.h2[0]) / 2  # Your MedianFinder object will be instantiated and called as such:  # obj = MedianFinder() # obj.addNum(num) # param_2 = obj.findMedian()</pre> </div> <div> <p>Heap</p> <p>p.1215</p> </div> </div> </div>                                                                                                                                                                                                                               |

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| <div>Heap</div> <div>#358 Rearrange String k Distance Apart<div>HAR</div></div> <div>LeetCode · SimplyLeetCode · WalkCC · LeetCode</div> <div>Hash Table · String · Greedy · Sorting · Heap · Counting</div> <div>Lists: Premium 98</div> <div>Time Complexity: <math>O(n \log C)</math> where <math>n</math> is the length of <math>s</math> and <math>C</math> is the number of unique characters. Space Complexity: <math>O(n)</math> due to the storage used for the heap and wait queue.</div> <div>Problem</div> <div>Given a string <math>s</math> and an integer <math>k</math>, rearrange <math>s</math> such that the same characters are at least distance <math>k</math> from each other. If it is not possible to rearrange the string, return an empty string <math>""</math>.</div> <div>Example 1:</div> <div><div>Input: <math>s = "aabbcc"</math>, <math>k = 3</math></div><div>Output: <math>"abcabc"</math></div><div>Explanation: The same letters are at least a distance of 3 from each other.</div></div> <div>Example 2:</div> <div></div> <div>Heapp.1216</div> | <div></div> <div>Input: <math>s = "aaabc"</math>, <math>k = 3</math></div> <div>Output: <math>""</math></div> <div>Explanation: It is not possible to rearrange the string.</div> <div>Example 3:</div> <div><div>Input: <math>s = "aaadbcc"</math>, <math>k = 2</math></div><div>Output: <math>"abacabcd"</math></div><div>Explanation: The same letters are at least a distance of 2 from each other.</div></div> <div>Constraints:</div> <div><ul style="list-style-type: none"><li><math>1 \leq s.length \leq 3 * 10^5</math></li><li><math>s</math> consists of only lowercase English letters.</li><li><math>0 \leq k \leq s.length</math></li></ul></div> <div>Community Solutions (Python)</div> <div><ul style="list-style-type: none"><li>WalkCC</li></ul><div><pre>class Solution:     def rearrangeString(self, s: str, k: int) -&gt; str:         n = len(s)         ans = []         count = collections.Counter(s)         # valid[i] := the leftmost index i can appear         valid = collections.Counter()          def getBestLetter(index: int) -&gt; str:             """Returns the valid letter that has the most count."""</pre></div></div> <div>Heapp.1217</div> | <div></div> <div><pre>maxCount = -1 bestLetter = '*'  for c in string.ascii_lowercase:     if count[c] &gt; 0 and count[c] &gt; maxCount and index     &gt;= valid[c]:         bestLetter = c         maxCount = count[c]  return bestLetter  for i in range(n):     c = getBestLetter(i)     if c == '*':         return ''     ans.append(c)     count[c] -= 1     valid[c] = i + k  return ''.join(ans)</pre></div> <div><ul style="list-style-type: none"><li>LeetCode</li></ul><div><pre>class Solution:     def rearrangeString(self, s: str, k: int) -&gt; str:         cnt = Counter(s)         pq = [(-v, c) for c, v in cnt.items()]         heapify(pq)         q = deque()         ans = []         while pq:             v, c = heappop(pq)             ans.append(c)</pre></div></div> <div>Heapp.1218</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <div></div> <div><pre>q.append((v + 1, c)) if len(q) &gt;= k:     e = q.popleft()     if e[0]:         heappush(pq, e) return "" if len(ans) &lt; len(s) else "".join(ans)</pre></div> <div><ul style="list-style-type: none"><li>SimplyLeetCode</li></ul><div><pre>from collections import Counter, deque import heapq  def rearrangeString(s: str, k: int) -&gt; str:     if k == 0:         return s # No rearrangement needed if k is 0      # Count the frequency of each character     char_count = Counter(s)      # Build a max-heap using negative frequencies     max_heap = [(-freq, ch) for ch, freq in     char_count.items()]     heapq.heapify(max_heap)      # Queue to keep track of characters waiting for k     distance     wait_queue = deque()     result = []      # Process the heap until it's empty     while max_heap:</pre></div></div> <div>Heapp.1219</div>                                                                                                                                                                                                 | <div></div> <div><pre>freq, ch = heapq.heappop(max_heap) result.append(ch)  # Decrease frequency (remember freq is negative) freq += 1  # Add current char to the wait queue wait_queue.append((freq, ch))  # If we've placed k characters, we can re-insert the oldest one if needed  if len(wait_queue) &lt; k:     continue  freq_front, ch_front = wait_queue.popleft() if freq_front &lt; 0:     heapq.heappush(max_heap, (freq_front,     ch_front))  rearranged = ''.join(result) return rearranged if len(rearranged) == len(s) else ""</pre></div> <div><ul style="list-style-type: none"><li>LC.ca</li></ul></div> <div>Heapp.1220</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div></div> <div><pre>from heapq import heapify  class Solution:     def rearrangeString(self, s: str, k: int) -&gt; str:         h = [(-v, c) for c, v in Counter(s).items()]         heapify(h)         q = deque()         ans = []         while h:             v, c = heappop(h)              v *= -1             ans.append(c)             q.append((v - 1, c)) # enqueue even if             'v-1==0'              # not '&gt;='             # use example k=1, then q size is 2, then q             can pop one out             if len(q) &gt; k: # @note: this is avoid you             pick up same char in the same k-segment.                 w, c = q.popleft()                 if w: # w!=0                     heappush(h, (-w, c))             return "" if len(ans) != len(s) else "".join(ans)</pre></div> <div>Heapp.1221</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <div></div> <div><pre>#####  import collections import heapq  class Solution:     def rearrangeString(self, s: str, k: int) -&gt; str:         if k == 0:             return s          counts = collections.Counter(s)          pq = []         for ch, count in counts.items():             heapq.heappush(pq, (-count, ch))          sb = []         while pq:             tmp = []              d = min(k, len(s))             for i in range(d):                 if not pq:                     return ""                 count, ch = heapq.heappop(pq)                 sb.append(ch)                 if count &lt; -1: # pushed in as negated, so                 it's: if actual-positive-count &gt; 1                     tmp.append((count+1, ch))             s = s[1:]</pre></div> <div>Heapp.1222</div>                                                                                                                                                                                                                                                                       | <div></div> <div><pre>for count, ch in tmp:     heapq.heappush(pq, (count, ch))  return "".join(sb)</pre></div> <div>Heapp.1223</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div>Heap</div> <div>#499 The Maze III<div>HAR</div></div> <div>LeetCode · SimplyLeetCode · WalkCC · LeetCode</div> <div>Array · DFS · BFS · Graph · Matrix · Heap · Shortest Path</div> <div>Lists: Premium 98</div> <div>Time Complexity: <math>O(mn \log(mn))</math> where <math>m</math> and <math>n</math> are the maze dimensions Space Complexity: <math>O(mn)</math></div> <div>Problem</div> <div>There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction (must be different from last chosen direction). There is also a hole in this maze. The ball will drop into the hole if it rolls onto the hole.</div> <div>Given the <math>m \times n</math> maze, the ball's position ball and the hole's position hole, where <math>ball = [ballrow, ballcol]</math> and <math>hole = [holerow, holecol]</math>, return a string instructions of all the instructions that the ball should follow to drop in the hole with the shortest distance possible. If there are multiple valid instructions, return the lexicographically minimum one. If the ball can't drop in the hole, return "impossible".</div> <div>Heapp.1224</div> |

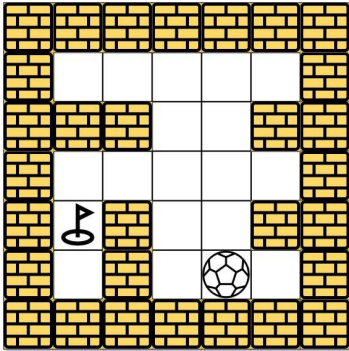
If there is a way for the ball to drop in the hole, the answer instructions should contain the characters 'u' (i.e., up), 'd' (i.e., down), 'l' (i.e., left), and 'r' (i.e., right). The distance is the number of empty spaces traveled by the ball from the start position (excluded) to the destination (included). You may assume that the borders of the maze are all walls (see examples). Example 1:



Heap p.1225

Input: maze = [[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]], ball = [4,3], hole = [0,1]  
Output: "lul"  
Explanation: There are two shortest ways for the ball to drop into the hole.  
The first way is left -> up -> left, represented by "lul".  
The second way is up -> left, represented by 'ul'.  
Both ways have shortest distance 6, but the first way is lexicographically smaller because 'l' < 'u'. So the output is "lul".

Example 2:



Input: maze = [[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]], ball = [4,3], hole = [3,0]  
Output: "impossible"  
Explanation: The ball cannot reach the hole.

Example 3:

Input: maze = [[0,0,0,0,0,0],[0,0,1,0,0,1],[0,0,0,0,1,0,0],[0,0,0,0,0,0,1]], ball = [0,4], hole = [3,5]  
Output: "dldr"

Constraints:

- m == maze.length

Heap p.1227

- n == maze[i].length
- 1 <= m, n <= 100
- maze[i][j] is 0 or 1.
- ball.length == 2
- hole.length == 2
- 0 <= ballrow, holerow <= m
- 0 <= ballcol, holecol <= n
- Both the ball and the hole exist in an empty space, and they will not be in the same position initially.
- The maze contains at least 2 empty spaces.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findShortestWay(self, maze, ball, hole):
 # Brute Force O(n log n) - sort repeatedly
 instead of heap
 data = list(maze)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result
```

Community Solutions (Python)

- WalkCC

Heap p.1228

```
class Solution:
 def findShortestWay(
 self,
 maze: List[List[int]],
 ball: List[int],
 hole: List[int],
) -> str:
 ans = 'impossible'
 minSteps = math.inf

 def dfs(i: int, j: int, dx: int, dy: int, steps: int,
 path: str):
 nonlocal ans
 nonlocal minSteps
 if steps >= minSteps:
 return

 if dx != 0 or dy != 0: # Both are zeros for the
 initial ball position.
 while (0 <= i + dx < len(maze) and 0 <= j + dy <
 len(maze[0]) and
 maze[i + dx][j + dy] != 1):
 i += dx
```

Heap p.1229

```
j += dy
steps += 1
if i == hole[0] and j == hole[1] and steps <
minSteps:
 minSteps = steps
 ans = path

if maze[i][j] == 0 or steps + 2 < maze[i][j]:
 maze[i][j] = steps + 2 # +2 because maze[i][j] ==
0 || 1.

if dx == 0:
 dfs(i, j, 1, 0, steps, path + 'd')

if dy == 0:
 dfs(i, j, 0, -1, steps, path + 'l')
if dy == 0:
 dfs(i, j, 0, 1, steps, path + 'r')
if dx == 0:
 dfs(i, j, -1, 0, steps, path + 'u')

dfs(ball[0], ball[1], 0, 0, 0, '')
return ans
```

- SimplyLeet

Heap p.1230

```
import heapq

def findShortestWay(maze, ball, hole):
 m, n = len(maze), len(maze[0])
 # Directions: sorted lexicographically as 'd' < 'l' <
 'r' < 'u'
 # since ASCII: d(100), l(108), r(114), u(117)
 directions = [('d', 1, 0), ('l', 0, -1), ('r', 0, 1),
 ('u', -1, 0)]

 # Initialize distance and path record, fill with
 (inf, '')
 dist = [[float('inf')] * n for _ in range(m)]

 pathRecord = [[''] * n for _ in range(m)]

 # Priority queue stores tuples: (distance, path, row,
 col)
 heap = [(0, '', ball[0], ball[1])]
 dist[ball[0]][ball[1]] = 0

 while heap:
 curDist, curPath, x, y = heapq.heappop(heap)
 # If we pop a state for the hole then return
 result.
 if [x, y] == hole:
```

Heap p.1231

```
return curPath
If we have a larger distance than already
recorded, continue.
if curDist > dist[x][y]:
 continue
Explore all four directions
for d, dx, dy in directions:
 nx, ny = x, y
 steps = 0
 # Roll the ball until hitting a wall or
 passing through the hole.
 # Check if ball goes through the hole during
 rolling.

 while 0 <= nx + dx < m and 0 <= ny + dy < n
 and maze[nx + dx][ny + dy] == 0:
 nx += dx
 ny += dy
 steps += 1
 if [nx, ny] == hole:
 break
 newDist = curDist + steps
 newPath = curPath + d
 # If we found a shorter route or equal
 distance but lexicographically smaller route.
 if newDist < dist[nx][ny] or (newDist ==
 dist[nx][ny] and newPath < pathRecord[nx][ny]):
```

Heap p.1232

```
dist[nx][ny] = newDist
pathRecord[nx][ny] = newPath
heapq.heappush(heap, (newDist, newPath,
nx, ny))
return "impossible"

Example usage:
maze = [[0,0,0,0,0],
 [1,1,0,0,1],
 [0,0,0,0,0],
 [0,1,0,0,1],
 [0,1,0,0,0]]

ball = [4, 3]
hole = [0, 1]
print(findShortestWay(maze, ball, hole)) # Expected
output: "lul"

LC.ca

class Solution:
 def findShortestWay(
 self, maze: List[List[int]], ball: List[int],
 hole: List[int]
) -> str:
 m, n = len(maze), len(maze[0])
 r, c = ball
 rh, ch = hole
 q = deque([(r, c)])
 dist = [[inf] * n for _ in range(m)]
 dist[r][c] = 0
```

Heap p.1233

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>path = [[None] * n for _ in range(m)] path[r][c] = '' while q:     i, j = q.popleft()     for a, b, d in [(-1, 0, 'u'), (1, 0, 'd'), (0, -1, 'l'), (0, 1, 'r')]:         x, y, step = i, j, dist[i][j]         while (             0 &lt;= x + a &lt; m             and 0 &lt;= y + b &lt; n             and maze[x + a][y + b] == 0              and (x != rh or y != ch)         ):             x, y = x + a, y + b             step += 1             if dist[x][y] &gt; step or (                 dist[x][y] == step and path[i][j] + d &lt; path[x][y]             ):                 dist[x][y] = step                 path[x][y] = path[i][j] + d                 if x != rh or y != ch:                      q.append((x, y)) return path[rh][ch] or 'impossible'</pre>                                |        | <div>Heap</div> <div>#632 Smallest Range Covering Elements from K Lists<div>HAR</div></div> <div>LeetCode · SimplyLeetCode · WalkCC · LeetCode</div> <div>Array · Hash Table · Greedy · Heap · Sliding Window · Sorting</div> <div>Lists: Grind 169</div> <div>Time Complexity: <math>O(N \log k)</math>, where <math>N</math> is the total number of elements across all lists. Space Complexity: <math>O(k)</math> for the heap, plus additional space for storing the range.</div> <div>Problem</div> <div>You have <math>k</math> lists of sorted integers in non-decreasing order. Find the smallest range that includes at least one number from each of the <math>k</math> lists. We define the range <math>[a, b]</math> is smaller than range <math>[c, d]</math> if <math>b - a &lt; d - c</math> or <math>a &lt; c</math> if <math>b - a == d - c</math>.</div> <div>Example 1:</div> <div>Input: nums = [[4,10,15,24,26],[0,9,12,20],[5,18,22,30]] Output: [20,24] Explanation: List 1: [4, 10, 15, 24,26], 24 is in range [20,24]. List 2: [0, 9, 12, 20], 20 is in range [20,24]. List 3: [5, 18, 22, 30], 22 is in range [20,24].</div>                             |        |
| Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1234 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1235 |
| <pre>class Solution:     def smallestRange(self, nums: List[List[int]]) -&gt; List[int]:         minHeap = [(row[0], i, 0) for i, row in enumerate(nums)]         heapq.heapify(minHeap)          maxRange = max(row[0] for row in nums)         minRange = heapq.nsmallest(1, minHeap)[0][0]         ans = [minRange, maxRange]          while len(minHeap) == len(nums):              num, r, c = heapq.heappop(minHeap)             if c + 1 &lt; len(nums[r]):                 heapq.heappush(minHeap, (nums[r][c + 1], r, c + 1))              maxRange = max(maxRange, nums[r][c + 1])             minRange = heapq.nsmallest(1, minHeap)[0][0]             if maxRange - minRange &lt; ans[1] - ans[0]:                 ans[0], ans[1] = minRange, maxRange          return ans</pre> <div>• LeetCode</div> |        | <pre>class Solution:     def smallestRange(self, nums: List[List[int]]) -&gt; List[int]:         t = [(x, i) for i, v in enumerate(nums) for x in v]          t.sort()         cnt = Counter()         ans = [-inf, inf]         j = 0         for i, (b, v) in enumerate(t):             cnt[v] += 1             while len(cnt) == len(nums):                  a = t[j][0]                 x = b - a - (ans[1] - ans[0])                 if x &lt; 0 or (x == 0 and a &lt; ans[0]):                     ans = [a, b]                 w = t[j][1]                 cnt[w] -= 1                 if cnt[w] == 0:                     cnt.pop(w)                     j += 1          return ans</pre> <div>• SimplyLeetCode</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1237 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1238 |
| <pre>import heapq  def smallestRange(nums):     # Initialize the min-heap and current maximum value     heap = []     current_max = float('-inf')      # Add the first element of each list to the heap     for i, lst in enumerate(nums):         value = lst[0]          heapq.heappush(heap, (value, i, 0))         current_max = max(current_max, value)      best_range = [float('-inf'), float('inf')]      # Process until one list is exhausted     while heap:         current_min, list_idx, elem_idx = heapq.heappop(heap)          # Update the best range if the current range is smaller</pre>                                                                                                                                                                                                       |        | <div>Heap</div> <div>#759 Employee Free Time<div>HAR</div></div> <div>LeetCode · SimplyLeetCode · WalkCC · LeetCode</div> <div>Array · Sorting · Heap</div> <div>Lists: Grind 169</div> <div>Time Complexity: <math>O(N \log N)</math>, where <math>N</math> is the total number of intervals (due to sorting). Space Complexity: <math>O(N)</math>, used for the flattened list and merged intervals.</div> <div>Problem</div> <div>We are given a list schedule of employees, which represents the working time for each employee. Each employee has a list of non-overlapping Intervals, and these intervals are in sorted order. Return the list of finite intervals representing common, positive-length free time for all employees, also in sorted order. (Even though we are representing Intervals in the form <math>[x, y]</math>, the objects inside are Intervals, not lists or arrays. For example, <code>schedule[0][0].start = 1</code>, <code>schedule[0][0].end = 2</code>, and <code>schedule[0][0][0]</code> is not defined). Also, we wouldn't include intervals like <math>[5, 5]</math> in our answer, as they have zero length.</div> <div>Example 1:</div> |        |
| Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1240 | Heap                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1241 |

Input: schedule = [[[1,2],[5,6]],[[1,3]],[[4,10]]]  
Output: [[3,4]]  
Explanation: There are a total of three employees, and all common free time intervals would be [-inf, 1], [3, 4], [10, inf].  
We discard any intervals that contain inf as they aren't finite.

Example 2:

Input: schedule =  
[[[1,3],[6,7]],[[2,4]],[[2,5],[9,12]]]  
Output: [[5,6],[7,9]]

Constraints:

- 1 <= schedule.length , schedule[i].length <= 50
- 0 <= schedule[i].start < schedule[i].end <= 10^8

Community Solutions (Python)

• WalkCC

```
class Solution:
 def employeeFreeTime(self, schedule: '[[Interval]]') -> 'Interval':
 ans = []
 intervals = []

 for s in schedule:
 intervals.extend(s)

 intervals.sort(key=lambda x: x.start)
```

```
prevEnd = intervals[0].end

for interval in intervals:
 if interval.start > prevEnd:
 ans.append(Interval(prevEnd, interval.start))
 prevEnd = max(prevEnd, interval.end)

return ans
```

• LeetCode

```
"""
Definition for an Interval.
class Interval:
 def __init__(self, start: int = None, end: int = None):
 self.start = start
 self.end = end
"""

class Solution:

 def employeeFreeTime(self, schedule: "[[Interval]]")
-> "[Interval]":
 intervals = []
 for e in schedule:
 intervals.extend(e)
 intervals.sort(key=lambda x: (x.start, x.end))
 merged = [intervals[0]]
 for x in intervals[1:]:
 if merged[-1].end < x.start:
 merged.append(x)
 else:
```

```
merged[-1].end = max(merged[-1].end,
x.end)
ans = []
for a, b in pairwise(merged):
 ans.append(Interval(a.end, b.start))
return ans
```

• SimplyLeet

```
Class definition for Interval
class Interval:
 def __init__(self, start, end):
 self.start = start # Start time of the interval
 self.end = end # End time of the interval

def employeeFreeTime(schedule):
 # Flatten the schedule: extract all intervals from
 each employee
 allIntervals = []
 for employee in schedule:

 for interval in employee:
 allIntervals.append(interval)

 # Sort intervals by start time
 allIntervals.sort(key=lambda interval:
interval.start)

 # Merge overlapping intervals
 merged = []
 prev = allIntervals[0]
 for interval in allIntervals[1:]:
```

```
if interval.start <= prev.end:
 # Expand the previous interval if there is an
 overlap
 prev.end = max(prev.end, interval.end)
else:
 merged.append(prev)
 prev = interval
merged.append(prev)
```

```
Collect free times from the gaps between merged
intervals
freeTimes = []
```

```
for i in range(1, len(merged)):
 # Only intervals with positive length are added
 if merged[i].start > merged[i-1].end:
 freeTimes.append(Interval(merged[i-1].end,
merged[i].start))
```

return freeTimes

```
Example usage:
schedule = [
[Interval(1,2), Interval(5,6)],
```

```
[Interval(1,3)],
[Interval(4,10)]
]
freeTimes = employeeFreeTime(schedule)
for interval in freeTimes:
print(f"[{interval.start}, {interval.end}]")
```

• LC.ca

```
"""
Definition for an Interval.
class Interval:
 def __init__(self, start: int = None, end: int = None):
 self.start = start
 self.end = end
"""

class Solution:
```

```
def employeeFreeTime(self, schedule: "[[Interval]]")
-> "[Interval]":
 intervals = []
 for e in schedule:
 intervals.extend(e)
 intervals.sort(key=lambda x: (x.start, x.end))
 merged = [intervals[0]]
 for x in intervals[1:]:
 if merged[-1].end < x.start:
 merged.append(x)
 else:
```

```
merged[-1].end = max(merged[-1].end,
x.end)
ans = []
for a, b in pairwise(merged):
 ans.append(Interval(a.end, b.start))
return ans
```

```
class Solution:
 def constrainedSubsetSum(self, nums: List[int], k: int)
-> int:
 # dp[i] := the maximum the sum of non-empty
 subsequences in nums[0..i]
 dp = [0] * len(nums)
 # dq stores dp[i - k], dp[i - k + 1], ..., dp[i - 1]
 whose values are > 0
 # in decreasing order.
 dq = collections.deque()
```

```
for i, num in enumerate(nums):
 if dq:
```

```
 dp[i] = max(dq[0], 0) + num
 else:
 dp[i] = num
 while dq and dq[-1] < dp[i]:
 dq.pop()
 dq.append(dp[i])
 if i >= k and dp[i - k] == dq[0]:
 dq.popleft()

 return max(dp)
```

• LeetCode

Input: nums = [-1,-2,-3], k = 1  
Output: -1  
Explanation: The subsequence must be non-empty, so we choose the largest number.

Example 3:

Input: nums = [10,-2,-10,-5,20], k = 2  
Output: 23  
Explanation: The subsequence is [10, -2, -5, 20].

Constraints:

- 1 <= k <= nums.length <= 105
- -104 <= nums[i] <= 104

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def constrained_subsequence_sum(self, nums, k):
 # Brute Force O(n log n) - sort repeatedly
 instead of heap
 data = list(nums)
 result = []
 for _ in range(k):
 data.sort()
 result.append(data.pop(0))
 return result
```

Community Solutions (Python)

• WalkCC

Heap

#1425 Constrained Subsequence Sum

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming · Queue · Sliding Window · Heap · Monotonic Queue

Lists: Denny Zhang

Time Complexity: O(n) Space Complexity: O(n) (O(k) extra space for the deque, with dp array using O(n))

Problem

Given an integer array nums and an integer k, return the maximum sum of a non-empty subsequence of that array such that for every two consecutive integers in the subsequence, nums[i] and nums[j], where i < j, the condition j - i <= k is satisfied.

A subsequence of an array is obtained by deleting some number of elements (can be zero) from the array, leaving the remaining elements in their original order.

Example 1:

Input: nums = [10,2,-10,5,20], k = 2  
Output: 37  
Explanation: The subsequence is [10, 2, 5, 20].

Example 2:

```
class Solution:
 def constrainedSubsetSum(self, nums: List[int], k:
int) -> int:
 q = deque([0])
 n = len(nums)
 f = [0] * n
 ans = -inf
 for i, x in enumerate(nums):
 while i - q[0] > k:
 q.popleft()
 f[i] = max(0, f[q[0]]) + x
```

```
ans = max(ans, f[i])
while q and f[q[-1]] <= f[i]:
 q.pop()
q.append(i)
return ans
```

• SimplyLeet

# Python solution using a monotonic deque for efficient window maximum computation.  
from collections import deque

```
def constrained_subsequence_sum(nums, k):
 n = len(nums)
 dp = [0] * n # dp[i] stores the maximum subsequence
 sum ending at index i
 dp[0] = nums[0]
 result = dp[0]

 # Deque to store indices in a decreasing order of dp
 values
```

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>deq = deque([0])  for i in range(1, n):     # Remove indices from the front which are out of     the window     while deq and i - deq[0] &gt; k:         deq.popleft()      # The maximum dp in the window is at the front of     the deque     max_in_window = dp[deq[0]] if dp[deq[0]] &gt; 0 else     0      # Calculate dp[i] by either starting a new     subsequence or extending the best one from the window     dp[i] = nums[i] + max_in_window      # Update overall answer     result = max(result, dp[i])      # Maintain the deque monotonically so that dp     indices are in decreasing order     while deq and dp[i] &gt;= dp[deq[-1]]:         deq.pop()     deq.append(i)  return result  # Example usage: # print(constrained_subsequence_sum([10,2,-10,5,20], 2))</pre> <p>• LC.ca</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <pre>class Solution:     def constrainedSubsetSum(self, nums: List[int], k:     int) -&gt; int:         n = len(nums)         dp = [0] * n         ans = -inf         q = deque()         for i, v in enumerate(nums):             if q and i - q[0] &gt; k:                 q.popleft()             dp[i] = max(0, 0 if not q else dp[q[0]]) + v              while q and dp[q[-1]] &lt;= dp[i]:                 q.pop()             q.append(i)         ans = max(ans, dp[i])         return ans</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Heapp.1252                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Heapp.1253                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Heapp.1254                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div>&gt;&gt; Quick Review — Heap</div> <div>20 questions · insights · complexity · solution</div> <div>#215 Kth Largest Element in an Array [Medium]</div> <div>Key Insights</div> <ul style="list-style-type: none"><li>• kth largest element can be converted to finding the element at index n-k if the array were sorted.</li><li>• A min-heap of size k is a practical method to maintain the k largest elements seen so far.</li><li>• The Quickselect algorithm can find the kth largest element in average O(n) time with in-place partitioning.</li><li>• Choose an appropriate method based on performance needs and worst-case scenarios.</li></ul> <div>Space &amp; Time</div> <p>Time Complexity:</p> <ul style="list-style-type: none"><li>– Min-Heap: O(n log k) by maintaining a heap of size k.</li><li>– Quickselect: Average O(n) but worst-case O(n^2) with poor pivot selection.</li></ul> <p>Space Complexity:</p> <ul style="list-style-type: none"><li>– Min-Heap: O(k)</li><li>– Quickselect: O(1) additional space with in-place partitioning.</li></ul> <div>Solution</div>                                                                                                                                                                 | <p>We can solve the problem using either a min-heap or the Quickselect algorithm.</p> <p>– Min-Heap Approach: Initialize a min-heap with the first k elements. Iterate through the remaining elements; if an element is greater than the heap's root (the smallest in the heap), replace the root. After processing all elements, the root of the min-heap is the kth largest element. – Quickselect Approach: Choose a pivot and partition the array so that elements larger than the pivot come before it. Determine the position of the pivot; if it matches the index corresponding to the kth largest element, return it. Recurse into the appropriate part of the array.</p> <p>Key points to consider include maintaining heap size for the first approach and proper pivot selection for the Quickselect approach.</p> <div>#253 Meeting Rooms II [Medium]</div> <div>Key Insights</div> <ul style="list-style-type: none"><li>• Sort the intervals based on start times to process meetings in order.</li><li>• Use a min-heap (priority queue) to keep track of meeting end times.</li><li>• When processing a new meeting, check if its start time is greater than or equal to the smallest end time from the heap.</li></ul>                                                  | <ul style="list-style-type: none"><li>• If so, it means a meeting room has been freed (the meeting with the smallest end time has ended) and can be reused.</li><li>• If not, allocate a new room by pushing the current meeting's end time into the heap.</li><li>• The size of the heap at any moment represents the number of rooms currently in use.</li></ul> <div>Space &amp; Time</div> <p>Time Complexity: O(n log n) due to sorting and each heap operation (push/pop) taking O(log n).</p> <p>Space Complexity: O(n) in the worst-case scenario when all meetings overlap, requiring all end times in the heap.</p> <div>Solution</div> <p>The solution involves first sorting the meeting intervals by their start times. Then, we utilize a min-heap (priority queue) to keep track of the end times of ongoing meetings. For each meeting, we compare its start time with the smallest end time in the heap:</p> <ul style="list-style-type: none"><li>– If the meeting can start after the meeting at the top of the heap has ended (i.e., its start time is greater than or equal to the smallest end), we remove that end time from the heap (freeing up a room).</li><li>– Then, we add the current meeting's end time to the heap.</li></ul> <p>By the end of processing all meetings, the size of the heap represents the minimum number of conference rooms required.</p> <div>#347 Top K Frequent Elements [Medium]</div> |
| Heapp.1255                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Heapp.1256                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Heapp.1257                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div>Key Insights</div> <ul style="list-style-type: none"><li>• Count the frequency of each element using a hash table.</li><li>• Use bucket sort to achieve linear time complexity by grouping numbers by their frequency.</li><li>• Traverse the frequency buckets from highest to lowest to collect the k most frequent elements.</li><li>• Alternative approaches include using heaps or Quickselect, but bucket sort meets the follow-up requirement of better than O(n log n) time complexity.</li></ul> <div>Space &amp; Time</div> <p>Time Complexity: O(n) where n is the length of the array (bucket sort and frequency counting are linear).</p> <p>Space Complexity: O(n) for storing the counts and buckets.</p> <div>Solution</div> <p>The solution starts by counting the frequency of each element with a hash table (or dictionary). Then, we create an array of buckets where the index represents the frequency. Each bucket holds the numbers that appear with that frequency. Finally, we iterate over the buckets in reverse order (from high frequency to low) and collect elements until we have k of them. This bucket sort approach avoids the higher complexity of sorting a list of frequencies and guarantees linear time performance.</p> | <div>#505 The Maze II [Medium]</div> <div>Key Insights</div> <ul style="list-style-type: none"><li>• The ball rolls until it hits a wall, so each move is not one step but a continuous travel in a direction.</li><li>• The problem can be modeled as finding the shortest path in a weighted graph where nodes are positions in the maze and edge weights are the distance traveled in that roll.</li><li>• A priority queue (min-heap) based algorithm, such as Dijkstra's algorithm, is ideal since each roll covers variable distances.</li><li>• Always check if the new computed distance for a cell is less than a previously recorded one before pushing it to the heap.</li></ul> <div>Space &amp; Time</div> <p>Time Complexity: O(m * n * log(m * n)), where m and n are the maze dimensions (each cell may get processed and queued with a logarithmic cost).</p> <p>Space Complexity: O(m * n) for maintaining the distance matrix and the priority queue.</p> <div>Solution</div> <p>We solve the problem using a modified version of Dijkstra's algorithm. The maze grid is treated as a graph where each cell is a node and an edge exists between the current cell and the cell where the ball stops after rolling in one direction. For every cell popped from the</p> | <p>priority queue, we roll the ball in all four directions until it hits a wall, counting the number of steps taken. If the distance calculated from this roll is shorter than a previously stored distance for the stopping cell, we update the distance and add it to the priority queue. This process continues until we either reach the destination (and return the distance) or exhaust all possibilities (returning -1).</p> <div>#621 Task Scheduler [Medium]</div> <div>Key Insights</div> <ul style="list-style-type: none"><li>• Count the frequency of each task.</li><li>• The task with the highest frequency determines the base structure.</li><li>• Identify how many tasks share the maximum frequency.</li><li>• Calculate the total time based on the formula: max(total tasks, (max frequency - 1) * (n + 1) + count of tasks with maximum frequency).</li><li>• This approach avoids simulating the entire scheduling by using a greedy strategy.</li></ul> <div>Space &amp; Time</div> <p>Time Complexity: O(N) where N is the number of tasks (counting frequency, then iterating over constant 26 letters).</p>                                                                                                                                                                                                                                                                                                       |
| Heapp.1258                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Heapp.1259                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Heapp.1260                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |



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| <p>Space Complexity: O(1) since the frequency count uses an array or hash map with fixed size (26 letters).</p> <p><b>Solution</b></p> <p>We first count the frequency of each task. The task(s) with the maximum frequency establish the minimum structure required. Compute the ideal idle slots based on the count of these maximum tasks and the cooling period n. Finally, compare this computed length with the total number of tasks to get the final answer. This greedy strategy avoids explicit simulation and leverages mathematical formulation.</p> <p><b>#658 Find K Closest Elements [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The input array is already sorted.</li> <li>• A binary search can be used to efficiently narrow down the potential starting position.</li> <li>• A two-pointers or sliding window approach is effective for selecting k consecutive elements.</li> <li>• The comparison of absolute differences (with a tie-breaker on values) is crucial.</li> <li>• Maintaining ascending order simplifies the final output.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(log n + k), where O(log n) for binary search and O(k) for collecting results.</p> <p>Heap<span style="float:right">p.1261</span></p>                                                                                                                                                                                                                     | <p>Space Complexity: O(k), for storing the output.</p> <p><b>Solution</b></p> <p>We can solve this problem by first using binary search to identify the left–most window where the k closest elements might start. The binary search operates over the possible starting indices for windows of size k (from 0 to n–k). For each middle index mid, we compare the distances between x and arr[mid] and arr[mid + k] to decide whether to move the window to the right or left. Once the correct starting position is found, return the k elements starting from that index. This approach leverages the sorted property of the array to efficiently converge on the optimal window.</p> <p><b>#787 Cheapest Flights Within K Stops [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Build the graph as an adjacency list of flights and prices.</li> <li>• Use a min-heap so the cheapest partial route is always processed first.</li> <li>• Track states as `(city, flightsUsed)` instead of only `city`, because the stop count matters.</li> <li>• For each popped state, try all outgoing flights and push the updated cost and flight count.</li> <li>• Do not expand a state once it has already used more than `k + 1` flights.</li> </ul> <p>Heap<span style="float:right">p.1262</span></p>                                                                                                               | <ul style="list-style-type: none"> <li>• Keep the best known cost for each `(city, flightsUsed)` state so worse duplicate states can be skipped.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O((V + E) * log(V * (k + 1))) for the heap–based search over `(city, stopsUsed)` states.</p> <p>Space Complexity: O(E + V * (k + 1)) for the graph plus the best–cost tracking table.</p> <p><b>Solution</b></p> <p>We solve the problem using a Dijkstra–style min-heap search, but the state must include both the city and how many flights have been used so far. We start from `(0, src, 0)` and always pop the cheapest state from the heap. For each state, if we reach `dst`, that cost is the answer. Otherwise, if we still have flights available within the `k + 1` limit, we explore all outgoing edges and push the next states into the heap. To avoid useless work, we store the best cost seen for each `(city, flightsUsed)` pair and only continue when the new state improves that record. This guarantees the cheapest valid route under the stop constraint.</p> <p><b>#973 K Closest Points to Origin [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Calculate the squared Euclidean distance (x<sup>2</sup> + y<sup>2</sup>) to avoid the overhead of square root computation.</li> </ul> <p>Heap<span style="float:right">p.1263</span></p>                                             |
| <ul style="list-style-type: none"> <li>• Sorting the points based on their distance gives an O(n log n) solution.</li> <li>• Alternatively, using a max-heap to keep track of the k closest points results in O(n log k) time, which is effective if k is much smaller than n.</li> <li>• Using Quickselect can achieve an average time complexity of O(n), though the worst–case is O(n<sup>2</sup>).</li> <li>• The problem does not require the output to be sorted in any specific order.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n log n) with a sorting approach, O(n log k) with a heap approach, or average O(n) with a Quickselect approach.</p> <p>Space Complexity: O(n) if extra space is used for sorting, or O(k) for the heap approach.</p> <p><b>Solution</b></p> <p>We can solve this problem by first computing the squared Euclidean distance for each point to avoid unnecessary square root operations. Then, we can sort the points based on their computed distance and select the first k points. Alternatively, a max-heap or Quickselect algorithm can be used:</p> <ul style="list-style-type: none"> <li>– Max-Heap: Maintain a heap of size k while iterating through the points, ensuring that only the k points with the smallest distances are kept.</li> <li>– Quickselect: Partition the array like in QuickSort to position the kth closest point in its correct position. Once partitioned, all points</li> </ul> <p>Heap<span style="float:right">p.1264</span></p> | <p>to the left of that position will be the k closest. The sorting solution is straightforward and efficient enough given the constraints.</p> <p><b>#1057 Campus Bikes [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Compute the Manhattan distance between every worker and bike pair.</li> <li>• To achieve the tie–breaking rules, sort all pairs by distance, then by worker index and bike index.</li> <li>• Once a worker is assigned a bike, skip any other pairs involving that worker or the already assigned bike.</li> <li>• The problem can also be approached using bucket sort because the Manhattan distance range is limited, but sorting is intuitive given the constraints.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n * m * log(n * m)) where n is the number of workers and m is the number of bikes.</p> <p>Space Complexity: O(n * m) for storing all worker–bike pairs.</p> <p><b>Solution</b></p> <p>Create all possible (worker, bike) pairs along with their Manhattan distances. Sort this list of pairs by:</p> <ul style="list-style-type: none"> <li>– Manhattan distance,</li> <li>– Worker index (if distances are equal),</li> <li>– Bike index (if both distance and worker index are equal).</li> </ul> <p>Heap<span style="float:right">p.1265</span></p>                                                                                            | <p>Then, iterate over the sorted list and assign bikes to workers if neither the worker nor the bike has already been assigned. This guarantees that at each step, the optimal available pairing (according to the problem's requirements) is made. This method naturally enforces the tie–breaking rules specified in the problem.</p> <p><b>#1167 Minimum Cost to Connect Sticks [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Always connect the two smallest sticks available to minimize the incremental cost.</li> <li>• A min-heap (or priority queue) is an ideal data structure to efficiently retrieve the smallest sticks.</li> <li>• The process continues until only one stick remains.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n log n), where n is the number of sticks (due to heap insertion and removal operations).</p> <p>Space Complexity: O(n), for storing the sticks in the heap.</p> <p><b>Solution</b></p> <p>The solution uses a greedy algorithm with a min-heap:</p> <ul style="list-style-type: none"> <li>– Insert all stick lengths into a min-heap.</li> <li>– While there are at least two sticks in the heap: Extract the two smallest sticks. Compute the cost of connecting them (sum the two lengths). Add this cost to the total result. Push the combined stick (with length equal to their sum)</li> </ul> <p>Heap<span style="float:right">p.1266</span></p> |
| <p>back into the heap.</p> <ul style="list-style-type: none"> <li>– Once the heap is reduced to one element, return the total cost.</li> </ul> <p>This approach ensures that at every connection, the smallest possible sticks are merged, which leads to the minimum overall cost.</p> <p><b>#1424 Diagonal Traverse II [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The diagonal key for each element can be defined as (row_index + column_index).</li> <li>• Elements in the same diagonal (with equal diagonal key) must be output in an order such that the element from the lower row comes before the one from a higher row.</li> <li>• Because the input grid may be jagged (rows with varying lengths), you must carefully iterate only over valid indices.</li> <li>• A hash table (or map) can be used to collect elements grouped by their diagonal key.</li> <li>• Inserting each element at the beginning of the list for that diagonal builds the list in the desired order without needing an extra reverse step later.</li> <li>• Finally, traverse the keys in increasing order (from 0 to maximum key) to produce the final result.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Heap<span style="float:right">p.1267</span></p>                                                                                                                                                                                                                                     | <p>Time Complexity: O(N), where N is the total number of elements (each element is processed once).</p> <p>Space Complexity: O(N), for storing the elements in the map and the output array.</p> <p><b>Solution</b></p> <p>The solution iterates over every element in nums while computing its diagonal key as (i + j). For each element, the value is inserted at the beginning of the list corresponding to that diagonal key. This ensures that when the diagonal groups are later concatenated in order of increasing key, the internal ordering within each diagonal is correct (i.e., elements appear from the bottom of the diagonal to the top). The final result is built by iterating over the keys in sorted order and appending each list of diagonal elements.</p> <p><b>#23 Merge k Sorted Lists [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a min-heap (priority queue) to efficiently extract the smallest value among the current heads of the lists.</li> <li>• By pushing the head of each non–empty list into the heap, we can pull the smallest node, then push that node's next element into the heap.</li> <li>• Alternatively, a divide and conquer approach can be used by recursively merging pairs of lists.</li> <li>• Careful management of pointers (or references) is key to constructing the new linked list.</li> </ul> <p>Heap<span style="float:right">p.1268</span></p> | <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(N log k) where N is the total number of nodes across all lists and k is the number of lists.</p> <p>Space Complexity: O(k) due to the heap containing at most one node from each list.</p> <p><b>Solution</b></p> <p>We use a min-heap to always retrieve the smallest node among the current nodes in the linked lists. First, initialize the heap with the head of each list. Then, continuously extract the smallest node from the heap, attach it to the merged list, and if the extracted node has a next node, insert that into the heap. The process repeats until the heap is empty.</p> <p><b>#218 The Skyline Problem [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a line sweep approach to process building start and end events in sorted order.</li> <li>• Differentiate between start events (which add to the skyline) and end events (which remove from the skyline).</li> <li>• Employ a max heap (or a priority queue) to efficiently track the current highest building at each sweep line position.</li> <li>• When the current maximum height changes, record the corresponding x–coordinate and new height as a key</li> </ul> <p>Heap<span style="float:right">p.1269</span></p>                                                                                                                                                                |

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| <p>point.</p> <ul style="list-style-type: none"> <li>Take care to remove expired building heights from the heap as the sweep line moves past the building's end.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n \log n)</math> due to sorting events and heap operations for each building event.</p> <p>Space Complexity: <math>O(n)</math> for the event list and the heap used for processing active buildings.</p> <p><b>Solution</b></p> <p>The solution uses the line sweep algorithm with a max heap (or priority queue) to maintain the heights of active buildings. Each building contributes two events: one when the building starts (adding its height) and one when it ends (removing its height). These events are sorted primarily by their <math>x</math>-coordinate. For start events, we use negative height values to ensure they are processed before end events at the same <math>x</math>-coordinate. During the sweep, the heap is updated by adding new building heights and removing buildings that have ended. Whenever there is a change in the top of the heap (i.e., the current maximum height), a key point is added to the result. This key point signifies a vertical change in the skyline.</p> <p><b>#272 Closest Binary Search Tree Value II [Hard]</b></p> <p><b>Key Insights</b></p>                                                          | <ul style="list-style-type: none"> <li>The BST property allows efficient narrowing down of candidate nodes.</li> <li>Using two stacks to track predecessors (values <math>\leq</math> target) and successors (values <math>&gt;</math> target) can efficiently yield the closest values.</li> <li>A modified in-order traversal helps in retrieving the next closest candidate from either side.</li> <li>Merging two sorted lists of candidate values based on absolute difference with the target is the key step.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(k + \log n) \rightarrow O(\log n)</math> to initialize the two stacks and <math>O(k)</math> to collect <math>k</math> elements.</p> <p>Space Complexity: <math>O(\log n)</math> – due to the stack space in a balanced BST.</p> <p><b>Solution</b></p> <p>We can solve the problem by splitting it into two parts:</p> <ul style="list-style-type: none"> <li>Build two stacks: The predecessor stack for nodes with values less than or equal to target. The successor stack for nodes with values greater than target. To build these stacks, traverse the BST starting from the root and push nodes along the path: if the node's value is less than or equal to target, push it to the predecessor stack and go right; if greater, push it to the successor stack and go left.</li> <li>Merge the two stacks: At each step, compare the top element of the predecessor stack (if available) and the successor stack (if available) based on their distance from the target. Pop from the stack with</li> </ul> | <p>the closer candidate and add that value to the result list. For the popped node, update the given stack by moving to the next appropriate node in the in-order traversal (for predecessor, go to left child then rightmost descendant; for successor, go to right child then leftmost descendant). Repeat until <math>k</math> values have been collected.</p> <p><b>Key Data Structures and Techniques:</b></p> <ul style="list-style-type: none"> <li>Two stacks to simulate reverse and forward in-order traversals.</li> <li>Merge technique to choose the closest candidate at each iteration.</li> </ul> <p><b>#295 Find Median from Data Stream [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use two heaps: a max-heap for the lower half and a min-heap for the upper half of numbers.</li> <li>Balancing the heaps ensures the median can be computed in <math>O(1)</math> time once numbers are added.</li> <li>Rebalance the heaps after each insertion to ensure size properties hold (difference in sizes is at most 1).</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity:</p> <ul style="list-style-type: none"> <li>addNum: <math>O(\log n)</math> per insertion (heap operations).</li> <li>findMedian: <math>O(1)</math> (accessing the top elements).</li> </ul> <p>Space Complexity:</p>                                                  |
| <p>– <math>O(n)</math>, where <math>n</math> is the number of elements added in the data stream, as they are stored in the two heaps.</p> <p><b>Solution</b></p> <p>We maintain two heaps:</p> <ul style="list-style-type: none"> <li>A max-heap (lowerHalf) for the smaller half of the numbers.</li> <li>A min-heap (upperHalf) for the larger half of the numbers.</li> </ul> <p>On inserting a new number, decide the heap membership based on the current median. After each insertion, ensure that the heaps have balanced sizes (the size difference is at most one). When computing the median: – If both heaps have the same number of elements, the median is the average of the max of lowerHalf and the min of upperHalf. – If one heap has more elements, the median is the top element of that heap.</p> <p>This method makes it efficient to dynamically update and retrieve the median as new numbers are inserted.</p> <p><b>#358 Rearrange String k Distance Apart [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The frequency of each character dictates the feasibility and placement within the rearrangement.</li> <li>A greedy approach picks the most frequent available character to maximize the chance of successful placement.</li> <li>A max-heap (priority queue) efficiently selects the highest frequency character each time.</li> </ul> | <ul style="list-style-type: none"> <li>A waiting queue ensures that once a character is used, it can't be reused until <math>k</math> positions later.</li> <li>Special case: When <math>k</math> is 0, no rearrangement is required because all orders satisfy the condition.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n \log C)</math> where <math>n</math> is the length of <math>s</math> and <math>C</math> is the number of unique characters.</p> <p>Space Complexity: <math>O(n)</math> due to the storage used for the heap and wait queue.</p> <p><b>Solution</b></p> <p>We use a greedy algorithm with a max-heap to always select the character with the highest remaining frequency that is currently available. The algorithm first counts the frequency of each character and pushes these into a max-heap (using negative frequencies for languages like Python where heapq is a min-heap). Each time a character is selected, it is appended to the result string and its frequency is decreased. This character is then placed in a waiting queue to ensure it isn't reused until <math>k</math> positions later. Once the waiting period completes, if the character still has remaining occurrences, it is reinserted into the heap. If at any point the heap is empty and the waiting queue still contains characters that haven't been reinserted while the result string is incomplete, we conclude that a valid rearrangement is impossible.</p>                                                                                                         | <p><b>#499 The Maze III [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The ball rolls continuously until hitting a wall or falling into the hole.</li> <li>A shortest path search is needed based on rolling distance (each step counts when passing an empty cell).</li> <li>Dijkstra's algorithm (or a modified BFS with a priority queue) is appropriate since we need to consider distance and lexicographic order.</li> <li>When expanding moves, simulate the roll until the ball stops — if a hole is passed en route, use that stopping point.</li> <li>Track the best (distance, path) for each cell to avoid redundant work.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(mn \log(mn))</math> where <math>m</math> and <math>n</math> are the maze dimensions</p> <p>Space Complexity: <math>O(mn)</math></p> <p><b>Solution</b></p> <p>We use a Dijkstra-like approach with a priority queue containing tuples of (distance, path, row, col). The priority queue is ordered by distance first and then by the instruction string lexicographically. For each cell, simulate rolling in each possible direction (up, down, left, right; iterated in alphabetical order so that the lexicographical property is maintained) until the ball</p>                                                                                              |
| <p>stops. If the ball encounters the hole during rolling, treat that as a valid stop immediately. For every new stopping cell, if a shorter distance or lexicographically smaller path is found, update the record and add the new state to the priority queue. If you reach the hole, return the corresponding path string. If no solution is found when the queue is empty, return "impossible".</p> <p><b>#632 Smallest Range Covering Elements from K Lists [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a min heap to maintain the smallest current element among the selected elements from each list.</li> <li>Track the current maximum among the chosen elements to easily compute the range.</li> <li>Iteratively replace the minimum element with the next element from the same list and update the current maximum.</li> <li>Stop when one of the lists is exhausted, as the range must include at least one element from each list.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(N \log k)</math>, where <math>N</math> is the total number of elements across all lists.</p> <p>Space Complexity: <math>O(k)</math> for the heap, plus additional space for storing the range.</p> <p><b>Solution</b></p>                                                                                                                    | <p>The solution uses a min heap (priority queue) that stores a tuple containing the current element, the index of the list it comes from, and its index in that list. Initially, the first element from each list is inserted into the heap and the current maximum among them is tracked. In each iteration, the smallest element is removed (heap root) which provides the current minimum, and the range <math>[current\_min, current\_max]</math> is assessed. If the list from which the minimum element was extracted has a next element, that element is inserted into the heap and the current maximum is updated if necessary. This process continues until one of the lists is exhausted.</p> <p><b>#759 Employee Free Time [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Flatten the schedule to combine all employees' intervals.</li> <li>Sort the flattened intervals by start time.</li> <li>Merge overlapping intervals to form a consolidated view of busy times.</li> <li>Identify gaps between merged intervals as the common free time.</li> <li>Ensure free intervals have strictly positive length.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(N \log N)</math>, where <math>N</math> is the total number of intervals (due to sorting).</p>                                                                                                                                                                                                                                                                              | <p>Space Complexity: <math>O(N)</math>, used for the flattened list and merged intervals.</p> <p><b>Solution</b></p> <p>The solution leverages interval merging. First, all intervals from every employee are collected and sorted by start time. By iterating through the sorted intervals, overlapping intervals are merged into one consolidated busy period. Finally, the gaps between consecutive merged intervals are identified as free periods available to all employees. This approach efficiently handles the intervals with a simple sort and a linear scan.</p> <p><b>#1425 Constrained Subsequence Sum [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use dynamic programming where <math>dp[i]</math> represents the maximum sum of a valid subsequence ending at index <math>i</math>.</li> <li>For each index <math>i</math>, consider taking <math>nums[i]</math> alone or appending it to a valid subsequence ending at an index <math>j</math> (where <math>i - j \leq k</math>) that maximizes the sum.</li> <li>To efficiently obtain the maximum <math>dp[j]</math> in a sliding window of the last <math>k</math> indices, use a monotonic (decreasing) deque.</li> <li>The idea is to maintain the window maximum and update it as we proceed through the array.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n)</math></p> |
| <p>Heap</p> <p>p.1276</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <p>Heap</p> <p>p.1277</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <p>Heap</p> <p>p.1278</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

Space Complexity: O(n) (O(k) extra space for the deque, with dp array using O(n))

Solution

We solve the problem by defining a dp array where  $dp[i] = \text{nums}[i] + \max(0, \text{maximum dp value in the window } [i-k, i-1])$ . The trick is to include  $\max(0, \dots)$  to allow starting a new subsequence when previous sums are negative. To access the maximum dp in the window in constant time, a monotonic deque is maintained that stores indices with their dp values in decreasing order. At each index, we:

- Remove elements from the front of the deque if they are out of the current window ( $i - \text{index} > k$ ).
- Use the front of the deque (if non-empty) as the maximum dp value from the previous k indices.
- Compute  $dp[i]$  and update the global maximum answer.
- Remove from the back of the deque all indices with dp values less than the current  $dp[i]$  since they are not useful.
- Insert the current index into the deque.

Matrix

Heap p.1279

#15 Matrix

By Pattern · Python Only · 20 questions

Matrix p.1280

Matrix

#422 Valid Word Square

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Matrix

Lists: Premium 98

Time Complexity:  $O(N * L)$  where N is the number of words and L is the average length of the words. Space Complexity:  $O(1)$  additional space is used for the checking procedure.

Problem

Given an array of strings words, return true if it forms a valid word square.

A sequence of strings forms a valid word square if the kth row and column read the same string, where  $0 \leq k < \max(\text{numRows}, \text{numColumns})$ .

Example 1:

Matrix p.1281

|   |   |   |   |
|---|---|---|---|
| a | b | c | d |
| b | n | r | t |
| c | r | m | y |
| d | t | y | e |

Input: words = ["abcd","bnrt","crmy","dtye"]  
Output: true  
Explanation:  
The 1st row and 1st column both read "abcd".  
The 2nd row and 2nd column both read "bnrt".  
The 3rd row and 3rd column both read "crmy".  
The 4th row and 4th column both read "dtye".  
Therefore, it is a valid word square.

Example 2:

Matrix p.1282

|   |   |   |   |
|---|---|---|---|
| a | b | c | d |
| b | n | r | t |
| c | r | m |   |
| d | t |   |   |

Input: words = ["abcd","bnrt","crm","dt"]  
Output: true  
Explanation:  
The 1st row and 1st column both read "abcd".  
The 2nd row and 2nd column both read "bnrt".  
The 3rd row and 3rd column both read "crm".  
The 4th row and 4th column both read "dt".  
Therefore, it is a valid word square.

Example 3:

Matrix p.1283

|   |   |   |   |
|---|---|---|---|
| b | a | l | l |
| a | r | e | a |
| r | e | a | d |
| l | a | d | y |

Input: words = ["ball","area","read","lady"]  
Output: false  
Explanation:  
The 3rd row reads "read" while the 3rd column reads "lead".  
Therefore, it is NOT a valid word square.

Constraints:

- $1 \leq \text{words.length} \leq 500$
- $1 \leq \text{words}[i].\text{length} \leq 500$
- $\text{words}[i]$  consists of only lowercase English letters.

Matrix p.1284

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def validWordSquare(self, words):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every cell
 m, n = len(words), len(words[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r,c) in visited or not (0<=r<m and 0<=c<n): return
 visited.add((r,c))
 for dr,dc in [(0,1),(0,-1),(1,0),(-1,0)]: flood(r+dr,c+dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

Community Solutions (Python)

- WalkCC

Matrix p.1285

```
class Solution:
 def validWordSquare(self, words: List[str]) -> bool:
 for i, word in enumerate(words):
 for j, c in enumerate(word):
 if len(words) <= j or len(words[j]) <= i: # out-of-bounds
 return False
 if c != words[j][i]:
 return False
 return True
```

• LeetDoocs

```
class Solution:
 def validWordSquare(self, words: List[str]) -> bool:
 m = len(words)
 for i, w in enumerate(words):
 for j, c in enumerate(w):
 if j >= m or i >= len(words[j]) or c != words[j][i]:
 return False
 return True
```

• SimplyLeet

Matrix p.1286

```
Function to check if the provided words form a valid word square
def validWordSquare(words):
 # Iterate over each row with its index
 for i in range(len(words)):
 # Iterate over each character in the current row
 for j in range(len(words[i])):
 # Check if the corresponding column exists
 if j >= len(words) or i >= len(words[j]) or words[i][j] != words[j][i]:
 return False
 return True
```

# Example test case  
words = ["abcd","bnrt","crmy","dtye"]  
print(validWordSquare(words)) # Expected output: True

• LC.ca

```
class Solution:
 def validWordSquare(self, words: List[str]) -> bool:
 m = len(words)
 for i, w in enumerate(words):
 for j, c in enumerate(w):
 if j >= m or i >= len(words[j]) or c != words[j][i]:
 return False
 return True
```

Matrix p.1287

Matrix

#566 Reshape the MatrixEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Matrix · Simulation

Lists: CodeSignal

Time Complexity:  $O(m \cdot n)$  as we iterate through all elements of the matrix once. Space Complexity:  $O(m \cdot n)$  for the additional list used to store flattened elements (or the new matrix).

Problem

In MATLAB, there is a handy function called reshape which can reshape an  $m \times n$  matrix into a new one with a different size  $r \times c$  keeping its original data. You are given an  $m \times n$  matrix `mat` and two integers `r` and `c` representing the number of rows and the number of columns of the wanted reshaped matrix. The reshaped matrix should be filled with all the elements of the original matrix in the same row-traversing order as they were. If the reshape operation with given parameters is possible and legal, output the new reshaped matrix; Otherwise, output the original matrix.

Example 1:

Matrix

p.1288

|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |



|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
|---|---|---|---|

Input: `mat = [[1,2],[3,4]]`, `r = 1`, `c = 4`  
Output: `[[1,2,3,4]]`

Example 2:

|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |



|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |

Input: `mat = [[1,2],[3,4]]`, `r = 2`, `c = 4`  
Output: `[[1,2],[3,4]]`

Constraints:

- `m == mat.length`
- `n == mat[i].length`
- `1 <= m, n <= 100`
- `-1000 <= mat[i][j] <= 1000`
- `1 <= r, c <= 300`

Matrix

p.1289

Brute Force (Python)

# Brute Force Approach

class Solution:

def matrixReshape(self, mat, r, c):

# Brute Force  $O(m^2 \cdot n^2)$  - flood fill from every cell

m, n = len(mat), len(mat[0])

result = 0

for r in range(m):

for c in range(n):

visited = set()

def flood(r, c):

if (r,c) in visited or not (0<=r<m and 0<=c<n): return

visited.add((r,c))

for dr,dc in [(0,1),(0,-1),(1,0),(-1,0)]: flood(r+dr,c+dc)

flood(r, c)

result = max(result, len(visited))

return result

Community Solutions (Python)

• WalkCC

Matrix

p.1290

class Solution:

def matrixReshape(self, nums: List[List[int]], r: int, c: int) -> List[List[int]]:

if nums == [] or r \* c != len(nums) \* len(nums[0]):

return nums

ans = [[0 for j in range(c)] for i in range(r)]

k = 0

for row in nums:

for num in row:

ans[k // c][k % c] = num

k += 1

return ans

• LeetDoocs

class Solution:

def matrixReshape(self, mat: List[List[int]], r: int, c: int) -> List[List[int]]:

m, n = len(mat), len(mat[0])

if m \* n != r \* c:

return mat

ans = [[0] \* c for \_ in range(r)]

for i in range(m \* n):

ans[i // c][i % c] = mat[i // n][i % n]

return ans

• SimplyLeet

Matrix

p.1291

# Function to reshape the matrix

def matrixReshape(mat, r, c):

# Calculate the dimensions of the original matrix

m = len(mat)

n = len(mat[0])

# Check if reshape is possible

if m \* n != r \* c:

return mat

# Flatten the original matrix into a list

flat\_list = []

for row in mat:

for num in row:

flat\_list.append(num)

# Create the new matrix by slicing the flat list

new\_matrix = []

for i in range(0, len(flat\_list), c):

new\_matrix.append(flat\_list[i:i+c])

return new\_matrix

# Example call:

# print(matrixReshape([[1,2],[3,4]], 1, 4))

• LC.ca

Matrix

p.1292

class Solution:

def matrixReshape(self, mat: List[List[int]], r: int, c: int) -> List[List[int]]:

m, n = len(mat), len(mat[0])

if m \* n != r \* c:

return mat

ans = [[0] \* c for \_ in range(r)]

for i in range(m \* n):

ans[i // c][i % c] = mat[i // n][i % n]

return ans

Matrix

p.1293

Matrix

#766 Toeplitz MatrixEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Matrix

Lists: CodeSignal

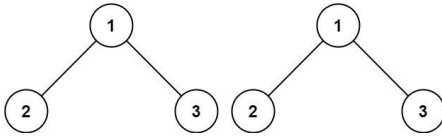
Time Complexity:  $O(m \cdot n)$  Space Complexity:  $O(1)$

Problem

Given an  $m \times n$  matrix, return true if the matrix is Toeplitz. Otherwise, return false.

A matrix is Toeplitz if every diagonal from top-left to bottom-right has the same elements.

Example 1:

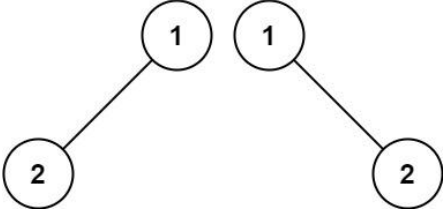


Matrix

p.1294

Input: `matrix = [[1,2,3,4],[5,1,2,3],[9,5,1,2]]`  
Output: `true`  
Explanation:  
In the above grid, the diagonals are:  
"[9]", "[5, 5]", "[1, 1, 1]", "[2, 2, 2]", "[3, 3]", "[4]".  
In each diagonal all elements are the same, so the answer is True.

Example 2:



Input: `matrix = [[1,2],[2,2]]`  
Output: `false`  
Explanation:  
The diagonal "[1, 2]" has different elements.

Constraints:

- `m == matrix.length`
- `n == matrix[i].length`
- `1 <= m, n <= 20`

Matrix

p.1295

• `0 <= matrix[i][j] <= 99`

Follow up:

- What if the matrix is stored on disk, and the memory is limited such that you can only load at most one row of the matrix into the memory at once?
- What if the matrix is so large that you can only load up a partial row into the memory at once?

Brute Force (Python)

# Brute Force Approach

class Solution:

def isToeplitzMatrix(self, matrix):

# Brute Force  $O(m^2 \cdot n^2)$  - flood fill from every cell

m, n = len(matrix), len(matrix[0])

result = 0

for r in range(m):

for c in range(n):

visited = set()

def flood(r, c):

if (r,c) in visited or not (0<=r<m and 0<=c<n): return

visited.add((r,c))

for dr,dc in [(0,1),(0,-1),(1,0),(-1,0)]: flood(r+dr,c+dc)

flood(r, c)

result = max(result, len(visited))

return result

Community Solutions (Python)

Matrix

p.1296

Sheet 144 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait

• WalkCC

```
class Solution:
 def isToeplitzMatrix(self, matrix: List[List[int]]) -> bool:
 for i in range(len(matrix) - 1):
 for j in range(len(matrix[0]) - 1):
 if matrix[i][j] != matrix[i + 1][j + 1]:
 return False
 return True
```

• LeetCode

```
class Solution:
 def isToeplitzMatrix(self, matrix: List[List[int]]) -> bool:
 m, n = len(matrix), len(matrix[0])
 for i in range(1, m):
 for j in range(1, n):
 if matrix[i][j] != matrix[i - 1][j - 1]:
 return False
 return True
```

• SimplyLeet

```
Function to determine if the matrix is Toeplitz
def isToeplitzMatrix(matrix):
 # Get the number of rows and columns in the matrix
 rows = len(matrix)
 cols = len(matrix[0])

 # Iterate starting from row 1 and column 1 because
 # first row and column are trivially Toeplitz
 for i in range(1, rows):
 for j in range(1, cols):
 # Compare current element with its top-left
 # neighbor
 if matrix[i][j] != matrix[i - 1][j - 1]:
 return False
 return True

Example usage:
matrix1 = [[1,2,3,4],[5,1,2,3],[9,5,1,2]]
print(isToeplitzMatrix(matrix1)) # Expected output: True
```

• LC.ca

```
class Solution:
 def isToeplitzMatrix(self, matrix: List[List[int]]) -> bool:
 m, n = len(matrix), len(matrix[0])
 return all(
 matrix[i][j] == matrix[i - 1][j - 1]
 for i in range(1, m)
 for j in range(1, n)
)
```

Matrix

#867 Transpose Matrix

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Matrix · Simulation

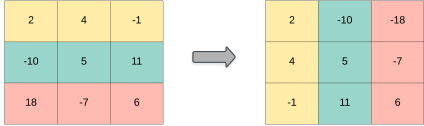
Lists: CodeSignal

Time Complexity: O(m n) Space Complexity: O(m n) for the new matrix holding the transposed values

**Problem**

Given a 2D integer array matrix, return the transpose of matrix.

The transpose of a matrix is the matrix flipped over its main diagonal, switching the matrix's row and column indices.



**Example 1:**

Input: matrix = [[1,2,3],[4,5,6],[7,8,9]]

Output: [[1,4,7],[2,5,8],[3,6,9]]

**Example 2:**

Input: matrix = [[1,2,3],[4,5,6]]

Output: [[1,4],[2,5],[3,6]]

- Constraints:
- m == matrix.length
  - n == matrix[i].length
  - 1 <= m, n <= 1000
  - 1 <= m \* n <= 105
  - 109 <= matrix[i][j] <= 109

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def transpose(self, matrix):
 # Brute Force O(m^2*n^2) - flood fill from every cell
 m, n = len(matrix), len(matrix[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
```

```
if (r,c) in visited or not (0<=r<m
and 0<=c<n): return
visited.add((r,c))
for dr,dc in
[(0,1),(0,-1),(1,0),(-1,0)]: flood(r+dr,c+dc)
flood(r, c)
result = max(result, len(visited))
return result
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def transpose(self, A: List[List[int]]) -> List[List[int]]:
 ans = [[0] * len(A) for _ in range(len(A[0]))]
 for i in range(len(A)):
 for j in range(len(A[0])):
 ans[j][i] = A[i][j]
 return ans
```

• LeetCode

```
class Solution:
 def transpose(self, matrix: List[List[int]]) -> List[List[int]]:
 return list(zip(*matrix))
```

• SimplyLeet

```
Python solution with clear comments

def transpose(matrix):
 # Number of rows in the original matrix
 num_rows = len(matrix)
 # Number of columns in the original matrix
 num_cols = len(matrix[0])

 # Create a new matrix with dimensions swapped:
 num_cols x num_rows
 transposed = [[0] * num_rows for _ in range(num_cols)]

 # Iterate through each element in the original matrix
 for i in range(num_rows):
 for j in range(num_cols):
 # Place the element at the swapped index in
 # the transposed matrix
 transposed[j][i] = matrix[i][j]

 return transposed

Example usage:

matrix = [[1, 2, 3], [4, 5, 6]]
print(transpose(matrix))

• LC.ca

class Solution:
 def transpose(self, matrix: List[List[int]]) -> List[List[int]]:
 return list(zip(*matrix))
```

Matrix

#2373 Largest Local Values in a Matrix

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Matrix

Lists: CodeSignal

Time Complexity: O(n^2) - We iterate over (n-2) x (n-2) positions, and for each position, we look at 9 elements.

Space Complexity: O(n^2) - The additional space is required for the output matrix.

**Problem**

You are given an n x n integer matrix grid.

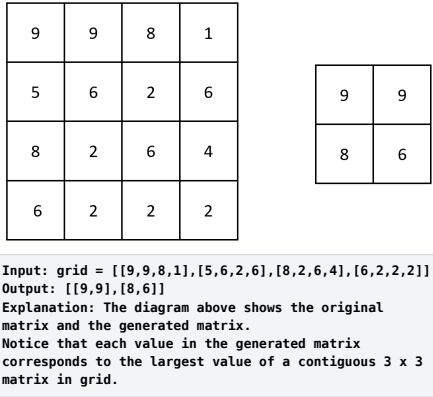
Generate an integer matrix maxLocal of size (n - 2) x (n - 2) such that:

- maxLocal[i][j] is equal to the largest value of the 3 x 3 matrix in grid centered around row i + 1 and column j + 1.

In other words, we want to find the largest value in every contiguous 3 x 3 matrix in grid.

Return the generated matrix.

**Example 1:**

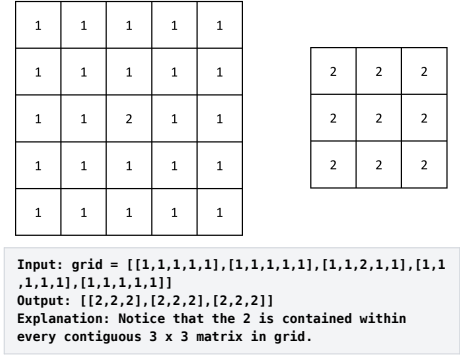


Input: grid = [[9,9,8,1],[5,6,2,6],[8,2,6,4],[6,2,2,2]]

Output: [[9,9],[8,6]]

Explanation: The diagram above shows the original matrix and the generated matrix. Notice that each value in the generated matrix corresponds to the largest value of a contiguous 3 x 3 matrix in grid.

**Example 2:**



Input: grid = [[1,1,1,1,1],[1,1,1,1,1],[1,1,2,1,1],[1,1,1,1,1],[1,1,1,1,1]]

Output: [[2,2,2],[2,2,2],[2,2,2]]

Explanation: Notice that the 2 is contained within every contiguous 3 x 3 matrix in grid.

- Constraints:
- n == grid.length == grid[i].length
  - 3 <= n <= 100
  - 1 <= grid[i][j] <= 100

Brute Force (Python)



```
Brute Force Approach
class Solution:
 def largestLocal(self, grid):
 # Brute Force O(m^2*n^2) - flood fill from every
 cell
 m, n = len(grid), len(grid[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m
 and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in
 [(0, 1), (0, -1), (1, 0), (-1, 0)]: flood(r+dr, c+dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

#### Community Solutions (Python)

- WalkCC

Matrix

p.1306

```
class Solution:
 def largestLocal(self, grid: List[List[int]]) ->
 List[List[int]]:
 n = len(grid)
 ans = [[0] * (n - 2) for _ in range(n - 2)]

 for i in range(n - 2):
 for j in range(n - 2):
 for x in range(i, i + 3):
 for y in range(j, j + 3):
 ans[i][j] = max(ans[i][j], grid[x][y])

 return ans
```

##### • LeetDoocs

```
class Solution:
 def largestLocal(self, grid: List[List[int]]) ->
 List[List[int]]:
 n = len(grid)
 ans = [[0] * (n - 2) for _ in range(n - 2)]
 for i in range(n - 2):
 for j in range(n - 2):
 ans[i][j] = max(
 grid[x][y] for x in range(i, i + 3)
 for y in range(j, j + 3)
)
 return ans
```

##### • SimplyLeet

Matrix

p.1307

```
Python solution for finding the largest local values in
a matrix

def largestLocal(grid):
 n = len(grid) # dimension of the grid
 # Initialize the result matrix with zeros, of size
 (n-2) x (n-2)
 result = [[0] * (n - 2) for _ in range(n - 2)]

 # Loop through each possible top-left index of a 3x3
 submatrix
 for i in range(n - 2):
 for j in range(n - 2):

 # Compute the maximum in the current 3x3
 submatrix
 local_max = 0 # since grid elements are >= 1,
 starting with 0 is safe
 for k in range(3):
 for l in range(3):
 local_max = max(local_max, grid[i +
 k][j + l])
 # Store the local maximum in the result
 matrix
 result[i][j] = local_max
 return result

Example usage:

grid = [[9,9,8,1],[5,6,2,6],[8,2,6,4],[6,2,2,2]]
print(largestLocal(grid))
```

##### • LC.ca

Matrix

p.1308

```
class Solution:
 def largestLocal(self, grid: List[List[int]]) ->
 List[List[int]]:
 n = len(grid)
 ans = [[0] * (n - 2) for _ in range(n - 2)]
 for i in range(n - 2):
 for j in range(n - 2):
 ans[i][j] = max(
 grid[x][y] for x in range(i, i + 3)
 for y in range(j, j + 3)
)
 return ans
```

Matrix

p.1309

Matrix

### #36 Valid Sudoku

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Matrix

Lists: Grind 169

Time Complexity: O(1) — Since the board size is fixed at 9x9. Space Complexity: O(1) — The extra space used by the sets is bounded by the fixed board size.

#### Problem

Determine if a 9 x 9 Sudoku board is valid. Only the filled cells need to be validated according to the following rules:

- Each row must contain the digits 1–9 without repetition.
- Each column must contain the digits 1–9 without repetition.
- Each of the nine 3 x 3 sub-boxes of the grid must contain the digits 1–9 without repetition.

#### Note:

- A Sudoku board (partially filled) could be valid but is not necessarily solvable.
- Only the filled cells need to be validated according to the mentioned rules.

Example 1:

Matrix

p.1310

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 5 | 3 |   |   | 7 |   |   |   |   |
| 6 |   |   | 1 | 9 | 5 |   |   |   |
|   | 9 | 8 |   |   |   |   | 6 |   |
| 8 |   |   |   | 6 |   |   |   | 3 |
| 4 |   |   | 8 |   | 3 |   |   | 1 |
| 7 |   |   |   | 2 |   |   |   | 6 |
|   | 6 |   |   |   |   | 2 | 8 |   |
|   |   |   | 4 | 1 | 9 |   |   | 5 |
|   |   |   |   | 8 |   |   | 7 | 9 |

```
Input: board =
[[["5","3",".",".","7",".",".",".","."],
["6",".",".","1","9","5",".",".","."],
[".","9","8",".",".",".","6",".","."],
["8",".",".",".","6",".",".",".","3"],
["4",".",".","8",".","3",".",".","1"],
["7",".",".","2",".","3",".",".","6"],
[".","6",".",".","2",".","8",".","."],
[".","",".","",".","",".","","."],
[".","",".","",".","",".","","."]]
```

Example 2:

Matrix

p.1311

```
Input: board =
[["8","3",".",".","7",".",".",".","."],
["6",".",".","1","9","5",".",".","."],
[".","9","8",".",".",".","6",".","."],
["8",".",".",".","6",".",".",".","3"],
["4",".",".","8",".","3",".",".","1"],
["7",".",".","2",".","3",".",".","6"],
[".","6",".",".","2",".","8",".","."],
[".","",".","",".","",".","","."],
[".","",".","",".","",".","","."]]
```

#### Constraints:

- board.length == 9
- board[i].length == 9
- board[i][j] is a digit 1–9 or '.'.

#### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def isValidSudoku(self, board):
 # Brute Force O(81) - check each row, col, box
 for duplicates
 rows = [set() for _ in range(9)]
 cols = [set() for _ in range(9)]
 boxes = [set() for _ in range(9)]
 for r in range(9):
 for c in range(9):
 val = board[r][c]
```

Matrix

p.1312

```
if val == '.': continue
box_idx = (r // 3) * 3 + c // 3
if val in rows[r] or val in cols[c] or
val in boxes[box_idx]:
 return False
rows[r].add(val); cols[c].add(val);
boxes[box_idx].add(val)
return True
```

#### Community Solutions (Python)

##### • WalkCC

```
class Solution:
 def isValidSudoku(self, board: List[List[str]]) ->
 bool:
 seen = set()

 for i in range(9):
 for j in range(9):
 c = board[i][j]
 if c == '.':
 continue
 if (c + '@row ' + str(i) in seen or
```

```
 c + '@col ' + str(j) in seen or
 c + '@box ' + str(i // 3) + str(j // 3) in
 seen):
 return False
 seen.add(c + '@row ' + str(i))
 seen.add(c + '@col ' + str(j))
 seen.add(c + '@box ' + str(i // 3) + str(j // 3))

 return True
```

Matrix

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Matrix

p.1314



```
Python solution with inline comments
def isValidSudoku(board):
 # Initialize sets for rows, columns, and boxes
 rows = [set() for _ in range(9)]
 cols = [set() for _ in range(9)]
 boxes = [set() for _ in range(9)]

 # Iterate over each cell in the board
 for r in range(9):
 for c in range(9):

 num = board[r][c]
 # Skip empty cells
 if num == '.':
 continue
 # Compute box index using integer division
 box_index = (r // 3) * 3 + (c // 3)
 # Check if the number is already present in
 # the row, column, or box
 if num in rows[r] or num in cols[c] or num in
 boxes[box_index]:
 return False
 # Add number to row, column, and box sets
```

Matrix p.1315

```
rows[r].add(num)
cols[c].add(num)
boxes[box_index].add(num)

return True

Example usage:
if __name__ == "__main__":
 board = [
 ["5","3",".", ".", ".", "7",".", ".", ".", "."],

 ["6",".", ".", "1","9","5",".", ".", ".", "."],
 [".","9","8",".", ".", ".", ".", "6",".", "."],
 ["8",".", ".", ".", "6",".", ".", ".", "3",".", "."],
 ["4",".", ".", "8",".", ".", "3",".", ".", "1","."],
 ["7",".", ".", ".", "2",".", ".", ".", "6",".", "."],
 [".","6",".", ".", ".", ".", "2","8",".", "."],
 [".",".", ".", "4",".", "1","9",".", ".", "5","."],
 [".",".", ".", ".", "8",".", ".", "7","9","."],

]
 print(isValidSudoku(board)) # Expected output: True
```

• LC.ca

Matrix p.1316

```
class Solution:
 def isValidSudoku(self, board: List[List[str]]) ->
 bool:

 row = [[False] * 9 for _ in range(9)]
 col = [[False] * 9 for _ in range(9)]
 sub = [[False] * 9 for _ in range(9)]
 for i in range(9):
 for j in range(9):
 c = board[i][j]
 if c == '.':
 continue

 num = int(c) - 1
 k = i // 3 * 3 + j // 3
 if row[i][num] or col[j][num] or
 sub[k][num]:
 return False
 row[i][num] = True
 col[j][num] = True
 sub[k][num] = True
 return True
```

Matrix p.1317

Matrix

#48 Rotate ImageMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Math · Matrix

Lists: Grind 169 | CodeSignal

Time Complexity:  $O(n^2)$  since every element is processed during the transpose and row reversal. Space Complexity:  $O(1)$  as the operations are performed in-place without extra data structures.

Problem

You are given an  $n \times n$  2D matrix representing an image, rotate the image by 90 degrees (clockwise). You have to rotate the image in-place, which means you have to modify the input 2D matrix directly. DO NOT allocate another 2D matrix and do the rotation.

Example 1:

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |



|   |   |   |
|---|---|---|
| 7 | 4 | 1 |
| 8 | 5 | 2 |
| 9 | 6 | 3 |

Matrix p.1318

Input: matrix = [[1,2,3],[4,5,6],[7,8,9]]  
Output: [[7,4,1],[8,5,2],[9,6,3]]

Example 2:

|    |    |    |    |
|----|----|----|----|
| 5  | 1  | 9  | 11 |
| 2  | 4  | 8  | 10 |
| 13 | 3  | 6  | 7  |
| 15 | 14 | 12 | 16 |



|    |    |    |    |
|----|----|----|----|
| 15 | 13 | 2  | 5  |
| 14 | 3  | 4  | 1  |
| 12 | 6  | 8  | 9  |
| 16 | 7  | 10 | 11 |

Input: matrix = [[5,1,9,11],[2,4,8,10],[13,3,6,7],[15,14,12,16]]  
Output: [[15,13,2,5],[14,3,4,1],[12,6,8,9],[16,7,10,11]]

Constraints:

- $n == \text{matrix.length} == \text{matrix}[i].\text{length}$
- $1 \leq n \leq 20$
- $-1000 \leq \text{matrix}[i][j] \leq 1000$

Brute Force (Python)

Matrix p.1319

# Brute Force Approach

class Solution:

def rotate(self, matrix):

# Brute Force  $O(n^2)$  space - copy to new matrix with rotated indices

n = len(matrix)

copy = [row[:] for row in matrix]

for r in range(n):

for c in range(n):

matrix[c][n-1-r] = copy[r][c]

Community Solutions (Python)

• WalkCC

class Solution:

def rotate(self, matrix: List[List[int]]) -> None:

matrix.reverse()

for i, j in

itertools.combinations(range(len(matrix)), 2):

matrix[i][j], matrix[j][i] = matrix[j][i], matrix[i][j]

class Solution:

def rotate(self, matrix: List[List[int]]) -> None:

for mn in range(len(matrix) // 2):

mx = len(matrix) - mn - 1

for i in range(mn, mx):

offset = i - mn

top = matrix[mn][i]

matrix[mn][i] = matrix[mx - offset][mn]

matrix[mx - offset][mn] = matrix[mx][mx - offset]

matrix[mx][mx - offset] = matrix[i][mx]

Matrix p.1320

matrix[i][mx] = top

• LeetDoocs

class Solution:

def rotate(self, matrix: List[List[int]]) -> None:

n = len(matrix)

for i in range(n >> 1):

for j in range(n):

matrix[i][j], matrix[n - i - 1][j] =

matrix[n - i - 1][j], matrix[i][j]

for i in range(n):

for j in range(i):

matrix[i][j], matrix[j][i] =

matrix[j][i], matrix[i][j]

• SimplyLeet

def rotate(matrix):

n = len(matrix)

# Transpose the matrix by swapping elements across the diagonal

for i in range(n):

for j in range(i, n):

matrix[i][j], matrix[j][i] = matrix[j][i], matrix[i][j]

# Reverse each row to complete the clockwise rotation

for i in range(n):

matrix[i].reverse()

# Example usage:

matrix = [[1,2,3],[4,5,6],[7,8,9]]

rotate(matrix)

print(matrix) # Outputs: [[7,4,1],[8,5,2],[9,6,3]]

Matrix p.1321

• LC.ca

class Solution(object):

def rotate(self, matrix):

"""

:type matrix: List[List[int]]

:rtype: void Do not return anything, modify matrix in-place instead.

"""

if len(matrix) == 0:

return

h = len(matrix)

w = len(matrix[0])

for i in range(0, h): # mirror

for j in range(0, w // 2):

matrix[i][j], matrix[i][w - j - 1] = matrix[i][w - j - 1], matrix[i][j]

for i in range(0, h): # transpos

for j in range(0, w - 1 - i):

matrix[i][j], matrix[w - 1 - j][h - 1 - i] =

matrix[w - 1 - j][h - 1 - i], matrix[i][j]

#####

Matrix p.1322

```
class Solution:
 def rotate(self, matrix: List[List[int]]) -> None:
 n = len(matrix)
 for i in range(n >> 1):
 for j in range(n):
 matrix[i][j], matrix[n - i - 1][j] =
 matrix[n - i - 1][j], matrix[i][j]
 for i in range(n):
 for j in range(i):
 matrix[i][j], matrix[j][i] =
 matrix[j][i], matrix[i][j]
```

Matrix

Matrix

Matrix

p.1323

Matrix

#54 Spiral Matrix

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Matrix · Simulation

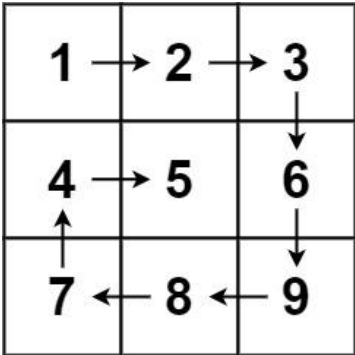
Lists: Grind 169 | CodeSignal

Time Complexity:  $O(m * n)$  since each element is visited exactly once. Space Complexity:  $O(1)$  extra space (excluding the space required for the output list).

**Problem**

Given an  $m \times n$  matrix, return all elements of the matrix in spiral order.

**Example 1:**

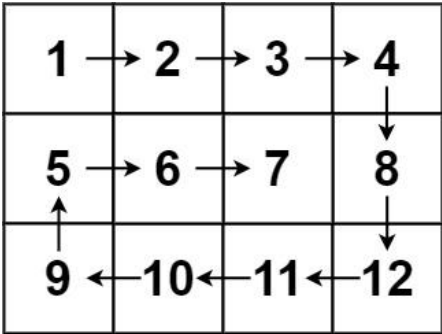


Input: matrix = [[1,2,3],[4,5,6],[7,8,9]]  
Output: [1,2,3,6,9,8,7,4,5]

Example 2:

Matrix p.1324

Matrix p.1325



Input: matrix = [[1,2,3,4],[5,6,7,8],[9,10,11,12]]  
Output: [1,2,3,4,8,12,11,10,9,5,6,7]

- Constraints:
- $m == \text{matrix.length}$
  - $n == \text{matrix}[i].\text{length}$
  - $1 \leq m, n \leq 10$
  - $-100 \leq \text{matrix}[i][j] \leq 100$

Brute Force (Python)

Matrix p.1326

```
Brute Force Approach
class Solution:
 def spiralOrder(self, matrix):
 # Brute Force O(m*n) - track visited cells,
 # simulate turning
 if not matrix: return []
 m, n = len(matrix), len(matrix[0])
 directions = [(0,1),(1,0),(0,-1),(-1,0)] # right
 down left up
 visited = [[False]*n for _ in range(m)]
 r=c=d=0; result=[]
 for _ in range(m*n):
 result.append(matrix[r][c])
 visited[r][c]=True
 dr,dc=directions[d]
 nr,nc=r+dr,c+dc
 if 0<=nr<m and 0<=nc<n and not
 visited[nr][nc]:
 r,nc=nr,nc
 else:
 d=(d+1)%4; dr,dc=directions[d]; r+=dr;
 c+=dc
 return result
```

Community Solutions (Python)

- WalkCC

Matrix p.1327

```
class Solution:
 def spiralOrder(self, matrix: List[List[int]]) -> List[int]:
 if not matrix:
 return []

 m = len(matrix)
 n = len(matrix[0])
 ans = []
 r1 = 0
 c1 = 0

 r2 = m - 1
 c2 = n - 1

 # Repeatedly add matrix[r1..r2][c1..c2] to `ans`.
 while len(ans) < m * n:
 j = c1
 while j <= c2 and len(ans) < m * n:
 ans.append(matrix[r1][j])
 j += 1
 i = r1 + 1

 while i <= r2 - 1 and len(ans) < m * n:
 ans.append(matrix[i][c2])
 i += 1

 j = c2
 while j >= c1 and len(ans) < m * n:
 ans.append(matrix[r2][j])
 j -= 1
 i = r2 - 1
 while i >= r1 + 1 and len(ans) < m * n:
 ans.append(matrix[i][c1])
```

Matrix p.1328

```
i -= 1
r1 += 1
c1 += 1
r2 -= 1
c2 -= 1

return ans
```

• LeetCode

```
class Solution:
 def spiralOrder(self, matrix: List[List[int]]) -> List[int]:
 m, n = len(matrix), len(matrix[0])
 dirs = (0, 1, 0, -1, 0)
 vis = [[False] * n for _ in range(m)]
 i = j = k = 0
 ans = []
 for _ in range(m * n):
 ans.append(matrix[i][j])
 vis[i][j] = True

 x, y = i + dirs[k], j + dirs[k + 1]
 if x < 0 or x >= m or y < 0 or y >= n or
 vis[x][y]:
 k = (k + 1) % 4
 i += dirs[k]
 j += dirs[k + 1]
 return ans
```

- SimplyLeet

Matrix p.1329

```
def spiralOrder(matrix):
 # Initialize the boundaries of the matrix
 top, bottom = 0, len(matrix) - 1
 left, right = 0, len(matrix[0]) - 1
 result = [] # This will store the spiral order

 # Loop until the pointers cross each other
 while top <= bottom and left <= right:
 # Traverse from left to right on the current top
 row = top
 for col in range(left, right + 1):
 result.append(matrix[top][col])
 top += 1 # Move the top boundary down

 # Traverse downwards on the current right column
 for row in range(top, bottom + 1):
 result.append(matrix[row][right])
 right -= 1 # Move the right boundary to the left

 # Traverse from right to left on the current
 bottom row, if valid
 if top <= bottom:
```

Matrix p.1330

```
for col in range(right, left - 1, -1):
 result.append(matrix[bottom][col])
 bottom -= 1 # Move the bottom boundary up

Traverse upwards on the current left column, if
valid
if left <= right:
 for row in range(bottom, top - 1, -1):
 result.append(matrix[row][left])
 left += 1 # Move the left boundary right

return result

Example usage:
print(spiralOrder([[1,2,3],[4,5,6],[7,8,9]]))

• LC.ca

class Solution:
 def spiralOrder(self, matrix: List[List[int]]) -> List[int]:
 m, n = len(matrix), len(matrix[0])
 dirs = (0, 1, 0, -1, 0)
 i = j = k = 0
 ans = []
 vis = set()
 for _ in range(m * n):
 ans.append(matrix[i][j])
 vis.add((i, j))
```

Matrix p.1331

```
x, y = i + dirs[k], j + dirs[k + 1]
if not 0 <= x < m or not 0 <= y < n or (x, y)
in vis:
 k = (k + 1) % 4
 i = i + dirs[k]
 j = j + dirs[k + 1]
return ans
```

Matrix p.1332

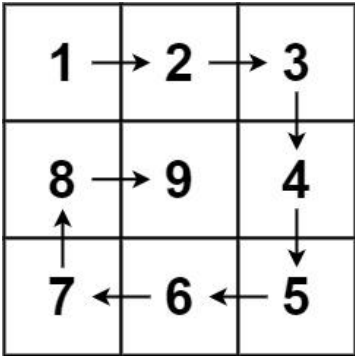
Matrix

#59 Spiral Matrix II

MED

LeetCode · SimplyLeet · WalkCC · LeetCode  
Array · Matrix · Simulation  
Lists: CodeSignal  
Time Complexity:  $O(n^2)$  Space Complexity:  $O(n^2)$   
(required for the output matrix)  
**Problem**  
Given a positive integer  $n$ , generate an  $n \times n$  matrix filled with elements from 1 to  $n^2$  in spiral order.  
Example 1:

Matrixp.1333



Input:  $n = 3$   
Output:  $[[1,2,3],[8,9,4],[7,6,5]]$

Example 2:  
Input:  $n = 1$   
Output:  $[[1]]$

Constraints:

- $1 \leq n \leq 20$

Brute Force (Python)

Matrixp.1334

# Brute Force Approach  
class Solution:  
 def generateMatrix(self, n):  
 # Brute Force  $O(m^2 \cdot n^2)$  - flood fill from every cell  
 m, n = len(n), len(n[0])  
 result = 0  
 for r in range(m):  
 for c in range(n):  
 visited = set()  
 def flood(r, c):  
 if (r,c) in visited or not (0<=r<m and 0<=c<n): return  
 visited.add((r,c))  
 for dr,dc in [(0,1),(0,-1),(1,0),(-1,0)]: flood(r+dr,c+dc)  
 flood(r, c)  
 result = max(result, len(visited))  
 return result

Community Solutions (Python)

- WalkCC

Matrixp.1335

```
class Solution:
 def generateMatrix(self, n: int) -> List[List[int]]:
 ans = [[0] * n for _ in range(n)]
 count = 1

 for mn in range(n // 2):
 mx = n - mn - 1
 for i in range(mn, mx):
 ans[mn][i] = count
 count += 1
 for i in range(mx, mn, -1):
 ans[mx][i] = count
 count += 1
 for i in range(mx, mn, -1):
 ans[i][mn] = count
 count += 1
 for i in range(mn, mx):
 ans[i][mx] = count
 count += 1

 if n % 2 == 1:
 ans[n // 2][n // 2] = count

 return ans
```

• LeetCode

Matrixp.1336

```
class Solution:
 def generateMatrix(self, n: int) -> List[List[int]]:
 ans = [[0] * n for _ in range(n)]
 dirs = (0, 1, 0, -1, 0)
 i = j = k = 0
 for v in range(1, n * n + 1):
 ans[i][j] = v
 x, y = i + dirs[k], j + dirs[k + 1]
 if x < 0 or x >= n or y < 0 or y >= n or
 k = (k + 1) % 4
 i, j = i + dirs[k], j + dirs[k + 1]
 return ans
```

• SimplyLeet  
# Function to generate spiral matrix of size  $n \times n$   
def generateMatrix(n):  
 # Initialize an  $n \times n$  matrix with zeros  
 matrix = [[0] \* n for \_ in range(n)]  
 # Initialize the starting number to fill  
 num = 1  
 # Set the initial boundaries for rows and columns  
 top, bottom = 0, n - 1  
 left, right = 0, n - 1

Matrixp.1337

```
Continue the loop until the boundaries are valid
while top <= bottom and left <= right:
 # Traverse from left to right along the top row
 for j in range(left, right + 1):
 matrix[top][j] = num
 num += 1
 top += 1 # Move the top boundary down

 # Traverse from top to bottom along the right
 column
 for i in range(top, bottom + 1):
 matrix[i][right] = num
 num += 1
 right -= 1 # Move the right boundary left

 # Traverse from right to left along the bottom
 row if still in valid bounds
 if top <= bottom:
 for j in range(right, left - 1, -1):
 matrix[bottom][j] = num
 num += 1
 bottom -= 1 # Move the bottom boundary up
```

Matrixp.1338

```
Traverse from bottom to top along the left
column if still in valid bounds
if left <= right:
 for i in range(bottom, top - 1, -1):
 matrix[i][left] = num
 num += 1
 left += 1 # Move the left boundary right

return matrix
```

# Example usage:  
print(generateMatrix(3))

• LC.ca  
class Solution:  
 def generateMatrix(self, n: int) -> List[List[int]]:  
 ans = [[0] \* n for \_ in range(n)]  
 dirs = ((0, 1), (1, 0), (0, -1), (-1, 0))  
 i = j = k = 0  
 for v in range(1, n \* n + 1):  
 ans[i][j] = v  
 x, y = i + dirs[k][0], j + dirs[k][1]  
 if x < 0 or y < 0 or x >= n or y >= n or  
 k = (k + 1) % 4  
 x, y = i + dirs[k][0], j + dirs[k][1]  
 i, j = x, y  
 return ans

Matrixp.1339

Matrix

#64 Minimum Path Sum

MED

LeetCode · SimplyLeet · WalkCC · LeetCode  
Array · Dynamic Programming · Matrix  
Lists: Denny Zhang  
Time Complexity:  $O(m * n)$ , where  $m$  and  $n$  are the number of rows and columns. Space Complexity:  $O(m * n)$  for the DP table (this can be optimized to  $O(n)$  with in-place computations).  
**Problem**  
Given a  $m \times n$  grid filled with non-negative numbers, find a path from top left to bottom right, which minimizes the sum of all numbers along its path.  
Note: You can only move either down or right at any point in time.  
Example 1:

Matrixp.1340



Input: grid =  $[[1,3,1],[1,5,1],[4,2,1]]$   
Output: 7  
Explanation: Because the path  $1 \rightarrow 3 \rightarrow 1 \rightarrow 1 \rightarrow 1$  minimizes the sum.

Example 2:  
Input: grid =  $[[1,2,3],[4,5,6]]$   
Output: 12

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid}[i].\text{length}$

Matrixp.1341

- 1 <= m, n <= 200
- 0 <= grid[i][j] <= 200

#### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def minPathSum(self, grid):
 # Brute Force O(m^2*n^2) - flood fill from every
 cell
 m, n = len(grid), len(grid[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0<=r<m
 and 0<=c<n): return
 visited.add((r,c))
 for dr,dc in
 [(0,1),(0,-1),(1,0),(-1,0)]: flood(r+dr,c+dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

#### Community Solutions (Python)

- WalkCC

Matrix

p.1342

```
class Solution:
 def minPathSum(self, grid: list[list[int]]) -> int:
 m = len(grid)
 n = len(grid[0])

 for i in range(m):
 for j in range(n):
 if i > 0 and j > 0:
 grid[i][j] += min(grid[i - 1][j], grid[i][j - 1])

 elif i > 0:
 grid[i][0] += grid[i - 1][0]
 elif j > 0:
 grid[0][j] += grid[0][j - 1]

 return grid[m - 1][n - 1]
```

- LeetCode

```
class Solution:
 def minPathSum(self, grid: List[List[int]]) -> int:
 m, n = len(grid), len(grid[0])
 f = [[0] * n for _ in range(m)]
 f[0][0] = grid[0][0]
 for i in range(1, m):
 f[i][0] = f[i - 1][0] + grid[i][0]
 for j in range(1, n):
 f[0][j] = f[0][j - 1] + grid[0][j]
 for i in range(1, m):
```

Matrix

p.1343

```
for j in range(1, n):
 f[i][j] = min(f[i - 1][j], f[i][j - 1]) +
 grid[i][j]
return f[-1][-1]
```

- SimplyLeet

```
Python solution using dynamic programming
def minPathSum(grid):
 # Number of rows and columns in the grid
 rows = len(grid)
 cols = len(grid[0])

 # Initialize the DP table directly in grid to save
 space
 # Fill the first row by accumulating the sum from the
 left
 for j in range(1, cols):
 grid[0][j] += grid[0][j-1]

 # Fill the first column by accumulating the sum above
 for i in range(1, rows):
 grid[i][0] += grid[i-1][0]

 # Compute the minimum path sum for the rest of the
 cells
 for i in range(1, rows):
 for j in range(1, cols):
 # The cell value is added to the minimum of
 the top or left cell
 grid[i][j] += min(grid[i-1][j], grid[i][j-1])
```

Matrix

p.1344

```
The bottom-right cell contains the answer
return grid[rows-1][cols-1]

Example usage
example_grid = [[1,3,1],[1,5,1],[4,2,1]]
print(minPathSum(example_grid)) # Output should be 7
```

- LC.ca

```
class Solution:
 def minPathSum(self, grid: List[List[int]]) -> int:
 m, n = len(grid), len(grid[0])
 f = [[0] * n for _ in range(m)]
 f[0][0] = grid[0][0]
 for i in range(1, m):
 f[i][0] = f[i - 1][0] + grid[i][0]
 for j in range(1, n):
 f[0][j] = f[0][j - 1] + grid[0][j]
 for i in range(1, m):

 for j in range(1, n):
 f[i][j] = min(f[i - 1][j], f[i][j - 1]) +
 grid[i][j]
 return f[-1][-1]
```

Matrix

p.1345

Matrix

### #73 Set Matrix Zeroes

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Matrix

Lists: Grind 169

Time Complexity: O(m \* n) Space Complexity: O(1)  
(in-place, using only constant extra space)

#### Problem

Given an m x n integer matrix matrix, if an element is 0, set its entire row and column to 0's.

You must do it in place.

Example 1:

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 2 | 3 | 7 | 4 | 1 |
| 4 | 5 | 6 | 8 | 5 | 2 |
| 7 | 8 | 9 | 9 | 6 | 3 |

Input: matrix = [[1,1,1],[1,0,1],[1,1,1]]  
Output: [[1,0,1],[0,0,0],[1,0,1]]

Example 2:

Matrix

p.1346

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 5  | 1  | 9  | 11 | 15 | 13 | 2  | 5  |
| 2  | 4  | 8  | 10 | 14 | 3  | 4  | 1  |
| 13 | 3  | 6  | 7  | 12 | 6  | 8  | 9  |
| 15 | 14 | 12 | 16 | 16 | 7  | 10 | 11 |

Input: matrix = [[0,1,2,0],[3,4,5,2],[1,3,1,5]]  
Output: [[0,0,0,0],[0,4,5,0],[0,3,1,0]]

#### Constraints:

- m == matrix.length
- n == matrix[0].length
- 1 <= m, n <= 200
- -231 <= matrix[i][j] <= 231 - 1

#### Follow up:

- A straightforward solution using O(mn) space is probably a bad idea.
- A simple improvement uses O(m + n) space, but still not the best solution.
- Could you devise a constant space solution?

#### Brute Force (Python)

Matrix

p.1347

```
Brute Force Approach
class Solution:
 def setZeroes(self, matrix):
 # Brute Force O(m*n*(m+n)) - record zero
 positions, then zero out rows/cols
 zeros = [(r,c) for r in range(len(matrix)) for c
 in range(len(matrix[0]))
 if matrix[r][c]==0]
 for r,c in zeros:
 for col in range(len(matrix[0])):
 matrix[r][col]=0
 for row in range(len(matrix)):
 matrix[row][c]=0
```

#### Community Solutions (Python)

- WalkCC

```
class Solution:
 def setZeroes(self, matrix: list[list[int]]) -> None:
 m = len(matrix)
 n = len(matrix[0])
 shouldFillFirstRow = 0 in matrix[0]
 shouldFillFirstCol = 0 in list(zip(*matrix))[0]

 # Store the information in the first row and the
 first column.
 for i in range(1, m):
 for j in range(1, n):
```

Matrix

p.1348

```
if matrix[i][j] == 0:
 matrix[i][0] = 0
 matrix[0][j] = 0

Fill 0s for the matrix except the first row and the
first column.
for i in range(1, m):
 for j in range(1, n):
 if matrix[i][0] == 0 or matrix[0][j] == 0:
 matrix[i][j] = 0

Fill 0s for the first row if needed.
if shouldFillFirstRow:
 matrix[0] = [0] * n

Fill 0s for the first column if needed.
if shouldFillFirstCol:
 for row in matrix:
 row[0] = 0
```

- LeetCode

```
class Solution:
 def setZeroes(self, matrix: List[List[int]]) -> None:
 m, n = len(matrix), len(matrix[0])
 row = [False] * m
 col = [False] * n
 for i in range(m):
 for j in range(n):
 if matrix[i][j] == 0:
 row[i] = col[j] = True
 for i in range(m):
```

Matrix

p.1349

```
for j in range(n):
 if row[i] or col[j]:
 matrix[i][j] = 0
```

- SimplyLeet

# Python solution for the Set Matrix Zeroes problem

```
def setZeroes(matrix):
 # Step 1: Determine the dimensions of matrix
 m = len(matrix)
 n = len(matrix[0])

 # Determine if first row or first column should be
 zeroed
 first_row_zero = any(matrix[0][j] == 0 for j in
 range(n))
 first_col_zero = any(matrix[i][0] == 0 for i in
 range(m))

 # Step 2: Use first row and column as markers
 for i in range(1, m):
 for j in range(1, n):
 if matrix[i][j] == 0:
 matrix[i][0] = 0 # mark the first element
 matrix[0][j] = 0 # mark the first element
 of the row
 of the column

 # Step 3: Zero out cells based on the markers in
 first row and column
 for i in range(1, m):
```

Matrix

p.1350

```
for j in range(1, n):
 if matrix[i][0] == 0 or matrix[0][j] == 0:
 matrix[i][j] = 0

Step 4: Zero out the first row if needed
if first_row_zero:
 for j in range(n):
 matrix[0][j] = 0

Step 5: Zero out the first column if needed

if first_col_zero:
 for i in range(m):
 matrix[i][0] = 0

Example usage:
matrix = [[1,1,1],[1,0,1],[1,1,1]]
setZeroes(matrix)
print(matrix)
```

```
• LC.ca

class Solution:
 def setZeroes(self, matrix: List[List[int]]) -> None:
 m, n = len(matrix), len(matrix[0])
 i0 = any(v == 0 for v in matrix[0])
 j0 = any(matrix[i][0] == 0 for i in range(m))
 for i in range(1, m):
 for j in range(1, n):
 if matrix[i][j] == 0:
 matrix[i][0] = matrix[0][j] = 0
 for i in range(1, m):
 for j in range(1, n):
 if matrix[i][0] == 0 or matrix[0][j] == 0:
 matrix[i][j] = 0
```

```
for j in range(1, n):
 if matrix[i][0] == 0 or matrix[0][j] == 0:
 matrix[i][j] = 0

if i0:
 for j in range(n):
 matrix[0][j] = 0
if j0:
 for i in range(m):
 matrix[i][0] = 0
```

Matrix

#74 Search a 2D Matrix MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Array · Binary Search · Matrix  
Lists: Grind 169  
Time Complexity:  $O(\log(m * n))$  Space Complexity:  $O(1)$

**Problem**  
You are given an  $m \times n$  integer matrix `matrix` with the following two properties:

- Each row is sorted in non-decreasing order.
- The first integer of each row is greater than the last integer of the previous row.

Given an integer `target`, return `true` if `target` is in `matrix` or `false` otherwise.

You must write a solution in  $O(\log(m * n))$  time complexity.

Example 1:

|    |    |    |    |
|----|----|----|----|
| 1  | 3  | 5  | 7  |
| 10 | 11 | 16 | 20 |
| 23 | 30 | 34 | 60 |

Input: `matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]]`, `target = 3`  
Output: `true`

Example 2:

|    |    |    |    |
|----|----|----|----|
| 5  | 1  | 9  | 11 |
| 2  | 4  | 8  | 10 |
| 13 | 3  | 6  | 7  |
| 15 | 14 | 12 | 16 |



|    |    |    |    |
|----|----|----|----|
| 15 | 13 | 2  | 5  |
| 14 | 3  | 4  | 1  |
| 12 | 6  | 8  | 9  |
| 16 | 7  | 10 | 11 |

Input: `matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]]`, `target = 13`  
Output: `false`

Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix}[i].\text{length}$
- $1 \leq m, n \leq 100$
- $-104 \leq \text{matrix}[i][j], \text{target} \leq 104$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def searchMatrix(self, matrix, target):
 # Brute Force $O(m \cdot n)$ - scan every cell
 for row in matrix:
 for val in row:
 if val == target:
 return True
 return False
```

### Community Solutions (Python)

```
• WalkCC

class Solution:
 def searchMatrix(self, matrix: List[List[int]], target: int) -> bool:
 if not matrix:
 return False

 m = len(matrix)
 n = len(matrix[0])
 l = 0
 r = m * n

 while l < r:
 mid = (l + r) // 2
 i = mid // n
 j = mid % n
 if matrix[i][j] == target:
 return True
 if matrix[i][j] < target:
 l = mid + 1
 else:
 r = mid

 return False

• LeetDoocs
```

```
class Solution:
 def searchMatrix(self, matrix: List[List[int]], target: int) -> bool:
 m, n = len(matrix), len(matrix[0])
 left, right = 0, m * n - 1
 while left < right:
 mid = (left + right) >> 1
 x, y = divmod(mid, n)
 if matrix[x][y] >= target:
 right = mid
 else:
 left = mid + 1
 return matrix[left // n][left % n] == target
```

```
• SimplyLeet

def searchMatrix(matrix, target):
 # Get number of rows and columns
 if not matrix or not matrix[0]:
 return False

 rows, cols = len(matrix), len(matrix[0])
 left, right = 0, rows * cols - 1

 # Perform binary search over the virtual flattened array
 while left <= right:
```

```
Find the middle index
mid = (left + right) // 2
Map the middle index to row and column in the matrix
row = mid // cols
col = mid % cols
mid_value = matrix[row][col]

Check if the mid_value equals the target
if mid_value == target:
 return True

elif mid_value < target:
 left = mid + 1 # Move to the right subarray
else:
 right = mid - 1 # Move to the left subarray

Target not found
return False

Example usage:
matrix = [[1,3,5,7],[10,11,16,20],[23,30,34,60]]

target = 3
print(searchMatrix(matrix, target))
```

• LC.ca

```
class Solution:
 def searchMatrix(self, matrix: List[List[int]], target: int) -> bool:
 m, n = len(matrix), len(matrix[0])
 left, right = 0, m * n - 1
 while left <= right: # must be <=, not <, for
 matrx=matrix[left:right+1],target=1
 mid = (left + right) >> 1
 x, y = divmod(mid, n) # note: divide column
 count

 if matrix[x][y] == target:
 return True
 elif matrix[x][y] > target:

 right = mid - 1
 else:
 left = mid + 1
 return False

#####

class Solution:
 def searchMatrix(self, matrix: List[List[int]], target: int) -> bool:
 m, n = len(matrix), len(matrix[0])
```

```

left, right = 0, m * n - 1
while left < right:
 mid = (left + right) >> 1
 x, y = divmod(mid, n)
 if matrix[x][y] >= target:
 right = mid
 else:
 left = mid + 1
return matrix[left // n][left % n] == target

```

Matrix

p.1360

Matrix

## #221 Maximal Square

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming · Matrix

Lists: Grind 169

Time Complexity:  $O(m * n)$ , where  $m$  is the number of rows and  $n$  is the number of columns. Space Complexity:  $O(m * n)$  for the DP table (this can be optimized to  $O(n)$  with space reduction techniques).

### Problem

Given an  $m \times n$  binary matrix filled with 0's and 1's, find the largest square containing only 1's and return its area.

Example 1:

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 | 0 |

Input: matrix = `[["1","0","1","0","0"],["1","0","1","1","1"],["1","1","1","1","1"],["1","0","0","1","0"]]`  
Output: 4

Example 2:

Matrix

p.1362

|   |   |
|---|---|
| 0 | 1 |
| 1 | 0 |

Input: matrix = `[["0","1"],["1","0"]]`  
Output: 1

Example 3:

Input: matrix = `[["0"]]`  
Output: 0

Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix[i].length}$
- $1 \leq m, n \leq 300$
- $\text{matrix[i][j]}$  is '0' or '1'.

Matrix

p.1363

### Brute Force (Python)

```

Brute Force Approach
class Solution:
 def maximalSquare(self, matrix):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every cell
 m, n = len(matrix), len(matrix[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in [(0, 1), (0, -1), (1, 0), (-1, 0)]:
 flood(r + dr, c + dc)
 result = max(result, len(visited))
 return result

```

### Community Solutions (Python)

- WalkCC

Matrix

p.1364

```

class Solution:
 def maximalSquare(self, matrix: List[List[str]]) -> int:
 m = len(matrix)
 n = len(matrix[0])
 dp = [[0] * n for _ in range(m)]
 maxLength = 0

 for i in range(m):
 for j in range(n):
 if i == 0 or j == 0 or matrix[i][j] == '0':
 dp[i][j] = 1 if matrix[i][j] == '1' else 0
 else:
 dp[i][j] = min(dp[i - 1][j - 1], dp[i - 1][j], dp[i][j - 1]) + 1
 maxLength = max(maxLength, dp[i][j])

 return maxLength * maxLength

```

- LeetCode

```

class Solution:
 def maximalSquare(self, matrix: List[List[str]]) -> int:
 m, n = len(matrix), len(matrix[0])
 dp = [[0] * (n + 1) for _ in range(m + 1)]
 mx = 0
 for i in range(m):
 for j in range(n):
 if matrix[i][j] == '1':
 dp[i + 1][j + 1] = min(dp[i][j + 1], dp[i + 1][j], dp[i][j]) + 1
 mx = max(mx, dp[i + 1][j + 1])
 return mx * mx

```

Matrix

p.1365

```
return mx * mx
```

- SimplyLeet

# Python implementation of the Maximal Square problem

```

def maximalSquare(matrix):
 # Check if matrix is empty
 if not matrix or not matrix[0]:
 return 0
 rows = len(matrix)
 cols = len(matrix[0])
 # DP table with extra row and col for ease of boundary conditions
 dp = [[0] * (cols + 1) for _ in range(rows + 1)]

```

```

 max_side = 0
 # Traverse through each cell in the matrix
 for i in range(1, rows + 1):
 for j in range(1, cols + 1):
 # If the current cell is '1'
 if matrix[i - 1][j - 1] == '1':
 # Compute the minimum value among the top, left, and top-left neighbors and add 1
 dp[i][j] = min(dp[i - 1][j], dp[i][j - 1], dp[i - 1][j - 1]) + 1
 max_side = max(max_side, dp[i][j])
 # The area of the square is the square of the side length

```

Matrix

p.1366

```
return max_side * max_side
```

# Example usage:

```

matrix = [
 ["1", "0", "1", "0", "0"],
 ["1", "0", "1", "1", "1"],
 ["1", "1", "1", "1", "1"],
 ["1", "0", "0", "1", "0"]
]
print(maximalSquare(matrix))

```

- LC.ca

```

'''
eg. a 10*10 square with full of 1s
this square move 1 line down
this square move 1 line right

so there needs an extra single 1 at bottom right, to make it a larger full square
'''
class Solution:
 def maximalSquare(self, matrix: List[List[str]]) -> int:
 m, n = len(matrix), len(matrix[0])

```

Matrix

p.1367

Matrix

p.1368



Matrix

#311 Sparse Matrix MultiplicationMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Matrix

Lists: Premium 98

Time Complexity:  $O(m * k * n)$  in the worst-case, but significantly lower when many elements are zero. Space Complexity:  $O(m * n)$  for the resulting matrix.

Problem

Given two sparse matrices mat1 of size  $m \times k$  and mat2 of size  $k \times n$ , return the result of mat1 x mat2. You may assume that multiplication is always possible.

Example 1:

|    |   |   |
|----|---|---|
| 1  | 0 | 0 |
| -1 | 0 | 3 |

|   |   |   |
|---|---|---|
| 7 | 0 | 0 |
| 0 | 0 | 0 |
| 0 | 0 | 1 |

|    |   |   |
|----|---|---|
| 7  | 0 | 0 |
| -7 | 0 | 3 |

Input: mat1 = [[1,0,0],[-1,0,3]], mat2 = [[7,0,0],[0,0,0],[0,0,1]]

Output: [[7,0,0],[-7,0,3]]

Example 2:

Matrix

p.1369

Input: mat1 = [[0]], mat2 = [[0]]

Output: [[0]]

Constraints:

- $m == \text{mat1.length}$
- $k == \text{mat1}[i].\text{length} == \text{mat2.length}$
- $n == \text{mat2}[i].\text{length}$
- $1 \leq m, n, k \leq 100$
- $-100 \leq \text{mat1}[i][j], \text{mat2}[i][j] \leq 100$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def sparseMatrixMultiplication(self, mat1, mat2):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every
 cell
 m, n = len(mat1), len(mat1[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m
 and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in [(0, 1), (0, -1), (1, 0), (-1, 0)]:
 flood(r + dr, c + dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

Matrix

p.1370

Community Solutions (Python)

- WalkCC

```
class Solution:
 def multiply(self, mat1: List[List[int]], mat2: List[List[int]]) -> List[List[int]]:
 m = len(mat1)
 n = len(mat2)
 l = len(mat2[0])
 ans = [[0] * l for _ in range(m)]

 for i in range(m):
 for j in range(l):

 for k in range(n):
 ans[i][j] += mat1[i][k] * mat2[k][j]

 return ans
```

- LeetCode

```
class Solution:
 def multiply(self, mat1: List[List[int]], mat2: List[List[int]]) -> List[List[int]]:
 m, n = len(mat1), len(mat2[0])
 ans = [[0] * n for _ in range(m)]
 for i in range(m):
 for j in range(n):
 for k in range(len(mat2)):
 ans[i][j] += mat1[i][k] * mat2[k][j]
 return ans
```

- SimplyLeet

Matrix

p.1371

```
def sparseMatrixMultiplication(mat1, mat2):
 # Get dimensions of mat1 and mat2
 m, k = len(mat1), len(mat1[0])
 n = len(mat2[0])
 # Initialize result matrix with zeros
 result = [[0] * n for _ in range(m)]

 # Iterate through each row in mat1
 for i in range(m):
 # For each element in row i of mat1
 for p in range(k):
 if mat1[i][p] != 0: # Only consider non-zero
 element in mat1
 # Multiply with corresponding elements in
 the row of mat2
 for j in range(n):
 if mat2[p][j] != 0: # Only consider
 non-zero element in mat2
 result[i][j] += mat1[i][p] *
 mat2[p][j]
 return result

Example usage:
print(sparseMatrixMultiplication([[1,0,0],[-1,0,3]],
[[7,0,0],[0,0,0],[0,0,1]]))
```

- LC.ca

Matrix

p.1372

```
class Solution:
 def multiply(self, mat1: List[List[int]], mat2: List[List[int]]) -> List[List[int]]:
 def f(mat: List[List[int]]) -> List[List[int]]:
 g = [[0] for _ in range(len(mat))]
 for i, row in enumerate(mat):
 for j, x in enumerate(row):
 if x:
 g[i].append((j, x))
 return g

 g1 = f(mat1)
 g2 = f(mat2)
 m, n = len(mat1), len(mat2[0])
 ans = [[0] * n for _ in range(m)]
 for i in range(m):
 for k, x in g1[i]:
 for j, y in g2[k]:
 ans[i][j] += x * y
 return ans
```

Matrix

p.1373

Matrix

#498 Diagonal TraverseMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Matrix · Simulation

Lists: CodeSignal

Time Complexity:  $O(m * n)$  where m is the number of rows and n is the number of columns, as every element is processed once. Space Complexity:  $O(m * n)$  for the output array (excluding the input matrix)

Problem

Given an  $m \times n$  matrix mat, return an array of all the elements of the array in a diagonal order.

Example 1:

Matrix

p.1374

Input: mat = [[1,2,3],[4,5,6],[7,8,9]]

Output: [1,2,4,7,5,3,6,8,9]

Example 2:

Input: mat = [[1,2],[3,4]]

Output: [1,2,3,4]

Constraints:

- $m == \text{mat.length}$
- $n == \text{mat}[i].\text{length}$
- $1 \leq m, n \leq 104$
- $1 \leq m * n \leq 104$

Matrix

p.1375

- $-105 \leq \text{mat}[i][j] \leq 105$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findDiagonalOrder(self, mat):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every
 cell
 m, n = len(mat), len(mat[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m
 and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in [(0, 1), (0, -1), (1, 0), (-1, 0)]:
 flood(r + dr, c + dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

Community Solutions (Python)

- LeetCode

Matrix

p.1376

```
class Solution:
 def findDiagonalOrder(self, mat: List[List[int]]) -> List[int]:
 m, n = len(mat), len(mat[0])
 ans = []
 for k in range(m + n - 1):
 t = []
 i = 0 if k < n else k - n + 1
 j = k if k < n else n - 1
 while i < m and j >= 0:
 t.append(mat[i][j])

 i += 1
 j -= 1
 if k % 2 == 0:
 t = t[::-1]
 ans.extend(t)
 return ans
```

- SimplyLeet

```
def findDiagonalOrder(mat):
 if not mat or not mat[0]:
 return []

 m, n = len(mat), len(mat[0])
 result = []
 row, col = 0, 0
 direction = 1 # 1 means upward, -1 means downward

 while len(result) < m * n:
```

Matrix

p.1377

```
result.append(mat[row][col])
if direction == 1: # moving upward
 if col == n - 1: # Hit the right boundary
 row += 1
 direction = -1
 elif row == 0: # Hit the top boundary
 col += 1
 direction = -1
 else:
 row -= 1
 col += 1
else: # moving downward
 if row == m - 1: # Hit the bottom boundary
 col += 1
 direction = 1
 elif col == 0: # Hit the left boundary
 row += 1
 direction = 1
 else:
 row += 1
 col -= 1

return result

Example usage:
print(findDiagonalOrder([[1,2,3],[4,5,6],[7,8,9]]))
```

• LC.ca

Matrix

p.1378

```
class Solution:
 def findDiagonalOrder(self, mat: List[List[int]]) -> List[int]:
 m, n = len(mat), len(mat[0])
 ans = []
 for k in range(m + n - 1):
 t = []
 i = 0 if k < n else k - n + 1
 j = k if k < n else n - 1
 while i < m and j >= 0:
 t.append(mat[i][j])
 i += 1
 j -= 1
 if k % 2 == 0:
 t = t[::-1]
 ans.extend(t)
 return ans
```

Matrix

p.1379

Matrix

## #531 Lonely Pixel I

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Matrix

Lists: Premium 98

Time Complexity:  $O(m * n)$ , where  $m$  and  $n$  are the dimensions of the picture. Space Complexity:  $O(m + n)$ , used to store the row and column counts.

### Problem

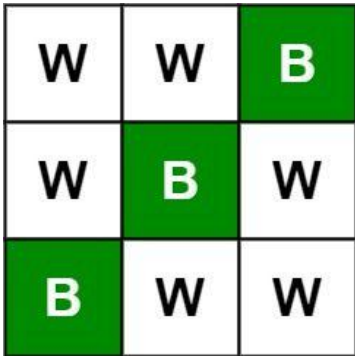
Given an  $m \times n$  picture consisting of black 'B' and white 'W' pixels, return the number of black lonely pixels.

A black lonely pixel is a character 'B' that located at a specific position where the same row and same column don't have any other black pixels.

Example 1:

Matrix

p.1380

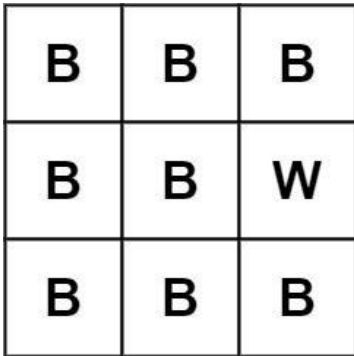


Input: picture =  
[["W","W","B"],["W","B","W"],["B","W","W"]]  
Output: 3  
Explanation: All the three 'B's are black lonely pixels.

Example 2:

Matrix

p.1381



Input: picture =  
[["B","B","B"],["B","B","W"],["B","B","B"]]  
Output: 0

### Constraints:

- $m == \text{picture.length}$
- $n == \text{picture[i].length}$
- $1 \leq m, n \leq 500$
- $\text{picture[i][j]}$  is 'W' or 'B'.

Brute Force (Python)

Matrix

p.1382

```
Brute Force Approach
class Solution:
 def findLonelyPixel(self, picture):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every cell
 m, n = len(picture), len(picture[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in [(0,1), (0,-1), (1,0), (-1,0)]: flood(r+dr, c+dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

### Community Solutions (Python)

• WalkCC

Matrix

p.1383

```
class Solution:
 def findLonelyPixel(self, picture: List[List[str]]) -> int:
 m = len(picture)
 n = len(picture[0])
 ans = 0
 rows = [0] * m # rows[i] := the number of B's in rows i
 cols = [0] * n # cols[j] := the number of B's in cols j

 for i in range(m):
 for j in range(n):
 if picture[i][j] == 'B':
 rows[i] += 1
 cols[j] += 1

 for i in range(m):
 if rows[i] == 1: # Only have to examine the rows if rows[i] == 1.
 for j in range(n):
 # After meeting a 'B' in this rows, break and search the next row.
 if picture[i][j] == 'B':
 if cols[j] == 1:
 ans += 1
 break
 return ans
```

• LeetDoocs

Matrix

p.1384

```
class Solution:
 def findLonelyPixel(self, picture: List[List[str]]) -> int:
 rows = [0] * len(picture)
 cols = [0] * len(picture[0])
 for i, row in enumerate(picture):
 for j, x in enumerate(row):
 if x == "B":
 rows[i] += 1
 cols[j] += 1

 ans = 0

 for i, row in enumerate(picture):
 for j, x in enumerate(row):
 if x == "B" and rows[i] == 1 and cols[j] == 1:
 ans += 1
 return ans
```

### • SimplyLeet

```
Function to find the number of lonely black pixels in a picture.
def findLonelyPixel(picture):
 # Get dimensions of the picture
 rows = len(picture)
 cols = len(picture[0]) if rows > 0 else 0

 # Initialize row and column counts with zeros
 rowCount = [0] * rows
 colCount = [0] * cols
```

Matrix

Sheet 154 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait

p.1385

Matrix

p.1386

```
class Solution:
 def findLonelyPixel(self, picture: List[List[str]])
-> int:
 m, n = len(picture), len(picture[0])
 rows, cols = [0] * m, [0] * n
 for i in range(m):
 for j in range(n):
 if picture[i][j] == 'B':
 rows[i] += 1
 cols[j] += 1

 res = 0

 for i in range(m):
 if rows[i] == 1:
 for j in range(n):
 if picture[i][j] == 'B' and cols[j]
== 1:
 res += 1
 break

 return res
```

Matrix p.1387

Matrix

#723 Candy Crush

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Two Pointers · Matrix · Simulation

Lists: Premium 98

Time Complexity:  $O((m \times n)^2)$  in the worst case since each iteration includes scanning the  $m \times n$  grid and then applying gravity. Space Complexity:  $O(m \times n)$  for maintaining additional markers or flags (or

Problem

This question is about implementing a basic elimination algorithm for Candy Crush.

Given an  $m \times n$  integer array board representing the grid of candy where  $board[i][j]$  represents the type of candy. A value of  $board[i][j] == 0$  represents that the cell is empty.

The given board represents the state of the game following the player's move. Now, you need to restore the board to a stable state by crushing candies according to the following rules:

- If three or more candies of the same type are adjacent vertically or horizontally, crush them all at the same time – these positions become empty.
- After crushing all candies simultaneously, if an empty space on the board has candies on top of itself, then

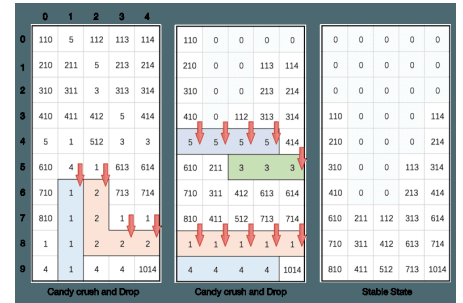
Matrix p.1388

these candies will drop until they hit a candy or bottom at the same time. No new candies will drop outside the top boundary.

- After the above steps, there may exist more candies that can be crushed. If so, you need to repeat the above steps.
- If there does not exist more candies that can be crushed (i.e., the board is stable), then return the current board.

You need to perform the above rules until the board becomes stable, then return the stable board.

Example 1:



Matrix p.1389

```
Input: board = [[110,5,112,113,114],[210,211,5,213,214],
,[310,311,3,313,314],[410,411,412,5,414],[5,1,512,3,3],
,[610,4,1,613,614],[710,1,2,713,714],[810,1,2,1,1],[1,1,
,2,2,2],[4,1,4,4,1014]]
Output: [[0,0,0,0,0],[0,0,0,0,0],[0,0,0,0,0],[110,0,0,0,
,114],[210,0,0,0,214],[310,0,0,113,314],[410,0,0,213,41
4],[610,211,112,313,614],[710,311,412,613,714],[810,411
,512,713,1014]]
Example 2:
```

```
Input: board = [[1,3,5,5,2],[3,4,3,3,1],[3,2,4,5,2],[2,
4,4,5,5],[1,4,4,1,1]]
Output: [[1,3,0,0,0],[3,4,0,5,2],[3,2,0,3,1],[2,4,0,5,2
],[1,4,3,1,1]]
```

Constraints:

- $m == \text{board.length}$
- $n == \text{board}[i].\text{length}$
- $3 \leq m, n \leq 50$
- $1 \leq \text{board}[i][j] \leq 2000$

Brute Force (Python)

Matrix p.1390

```
Brute Force Approach
class Solution:
 def candyCrush(self, board):
 # Brute Force $O(m^2 \times n^2)$ – flood fill from every
cell
 m, n = len(board), len(board[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r,c) in visited or not (0<=r<m
and 0<=c<n): return
 visited.add((r,c))
 for dr,dc in
[(0,1),(0,-1),(1,0),(-1,0)]: flood(r+dr,c+dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

Community Solutions (Python)

- SimplyLeet

Matrix p.1391

```
If no candies need to be crushed, the board is
stable; break out of the loop.
if not crush_set:
 break

Set all marked positions to 0 (simulate
crushing)
for i, j in crush_set:
 board[i][j] = 0

Apply gravity: For each column, move uncrushed
candies downwards

for j in range(n):
 # Pointer for placing the next candy from
bottom-up
 write_index = m - 1
 # Traverse each column from bottom to top
 for i in range(m - 1, -1, -1):
 if board[i][j] != 0:
 board[write_index][j] = board[i][j]
 write_index -= 1
 # Fill remaining positions with 0
 for i in range(write_index, -1, -1):
```

Matrix p.1393

```
board[i][j] = 0

return board

Example usage:
board = [[110,5,112,113,114],[210,211,5,213,214],[310,311
,3,313,314],
,[410,411,412,5,414],[5,1,512,3,3],[610,4,1,613,6
14],
,[710,1,2,713,714],[810,1,2,1,1],[1,1,2,2,2],[4,1
,4,4,1014]]
print(candyCrush(board))

• LC.ca
class Solution:
 def candyCrush(self, board: List[List[int]]) ->
List[List[int]]:
 m, n = len(board), len(board[0])
 run = True
 while run:
 run = False
 for i in range(m):
 for j in range(n - 2):
 if (
 board[i][j] != 0
 and abs(board[i][j]) ==
abs(board[i][j + 1])
 and abs(board[i][j]) ==
abs(board[i][j + 2])
):
 run = True
 board[i][j] = board[i][j + 1] =
board[i][j + 2] = -abs(
board[i][j]
)
 for j in range(n):
 for i in range(m - 2):
 if (
 board[i][j] != 0
 and abs(board[i][j]) ==
abs(board[i + 1][j])
 and abs(board[i][j]) ==
abs(board[i + 2][j])
):
 run = True
 board[i][j] = board[i + 1][j] =
board[i + 2][j] = -abs(
board[i][j]
)
 if run:
 for j in range(n):
```

Matrix p.1394 Sheet 155 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait

```
abs(board[i][j + 1])
 and abs(board[i][j]) ==
abs(board[i][j + 2])
):
 run = True
 board[i][j] = board[i][j + 1] =
board[i][j + 2] = -abs(
board[i][j]
)
 for j in range(n):
 for i in range(m - 2):
 if (
 board[i][j] != 0
 and abs(board[i][j]) ==
abs(board[i + 1][j])
 and abs(board[i][j]) ==
abs(board[i + 2][j])
):
 run = True
 board[i][j] = board[i + 1][j] =
board[i + 2][j] = -abs(
board[i][j]
)
 if run:
 for j in range(n):
```

Matrix p.1395

```
curr = m - 1
for i in range(m - 1, -1, -1):
 if board[i][j] > 0:
 board[curr][j] = board[i][j]
 curr -= 1
while curr > -1:
 board[curr][j] = 0
 curr -= 1

return board
```

Matrix

#835 Image OverlapMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Matrix

Lists: CodeSignal

Time Complexity:  $O(n^4)$  in the worst-case when using a nested iteration over positions for both matrices, though the average situation is typically better since not all cells are 1. Space Complexity:

Problem

You are given two images, `img1` and `img2`, represented as binary, square matrices of size  $n \times n$ . A binary matrix has only 0s and 1s as values.

We translate one image however we choose by sliding all the 1 bits left, right, up, and/or down any number of units. We then place it on top of the other image. We can then calculate the overlap by counting the number of positions that have a 1 in both images.

Note also that a translation does not include any kind of rotation. Any 1 bits that are translated outside of the matrix borders are erased.

Return the largest possible overlap.

Example 1:

|   |   |   |
|---|---|---|
| 1 | 1 | 0 |
| 0 | 1 | 0 |
| 0 | 1 | 0 |

|   |   |   |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 0 | 0 | 1 |

Input: `img1 = [[1,1,0],[0,1,0],[0,1,0]]`, `img2 = [[0,0,0],[0,1,1],[0,0,1]]`  
Output: 3  
Explanation: We translate `img1` to right by 1 unit and down by 1 unit.

The number of positions that have a 1 in both images is 3 (shown in red).

Example 2:

Input: `img1 = [[1]]`, `img2 = [[1]]`  
Output: 1

Example 3:

Input: `img1 = [[0]]`, `img2 = [[0]]`  
Output: 0

Constraints:

- $n == \text{img1.length} == \text{img1[i].length}$

Matrix p.1398

- $n == \text{img2.length} == \text{img2[i].length}$
- $1 \leq n \leq 30$
- `img1[i][j]` is either 0 or 1.
- `img2[i][j]` is either 0 or 1.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def largestOverlap(self, img1, img2):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every cell
 m, n = len(img1), len(img1[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in [(0, 1), (0, -1), (1, 0), (-1, 0)]:
 flood(r + dr, c + dc)
 result = max(result, len(visited))
 return result
```

Community Solutions (Python)

- WalkCC

Matrix p.1399

```
class Solution:
 def largestOverlap(self, img1: List[List[int]], img2: List[List[int]]) -> int:
 MAGIC = 100
 ones1 = [(i, j)
 for i, row in enumerate(img1)
 for j, num in enumerate(row)
 if num == 1]
 ones2 = [(i, j)
 for i, row in enumerate(img2)
 for j, num in enumerate(row)
 if num == 1]
 offsetCount = collections.Counter()
 for ax, ay in ones1:
 for bx, by in ones2:
 offsetCount[(ax - bx) * MAGIC + (ay - by)] += 1
 return max(offsetCount.values()) if offsetCount else 0
```

- LeetDoocs

Matrix p.1400

```
class Solution:
 def largestOverlap(self, img1: List[List[int]], img2: List[List[int]]) -> int:
 n = len(img1)
 cnt = Counter()
 for i in range(n):
 for j in range(n):
 if img1[i][j]:
 for h in range(n):
 for k in range(n):
 if img2[h][k]:
 cnt[(i - h, j - k)] += 1
 return max(cnt.values()) if cnt else 0
```

- SimplyLeet

```
def largestOverlap(img1, img2):
 # Get the dimension of the square images
 n = len(img1)

 # Gather positions (coordinates) where there is a 1 in img1 and img2
 ones_img1 = []
 ones_img2 = []
 for i in range(n):
 for j in range(n):
 if img1[i][j] == 1:
```

Matrix p.1401

```
ones_img1.append((i, j))
if img2[i][j] == 1:
 ones_img2.append((i, j))

Dictionary to count translation shifts that align 1's of img1 with img2
translation_count = {}
max_overlap = 0

Iterate over every pair of 1's from both images
for (x1, y1) in ones_img1:
 for (x2, y2) in ones_img2:
 # Calculate the translation vector needed to align img1's 1 with img2's 1
 delta = (x2 - x1, y2 - y1)
 # Update the count for this translation
 vector = translation_count[delta] = translation_count.get(delta, 0) + 1
 # Update max_overlap if we find a higher count
 count = max_overlap = max(max_overlap, translation_count[delta])

 return max_overlap

Example usage:
img1 = [[1,1,0],[0,1,0],[0,1,0]]
img2 = [[0,0,0],[0,1,1],[0,0,1]]
print(largestOverlap(img1, img2)) # Expected output: 3
```

- LC.ca

Matrix p.1402

```
class Solution:
 def largestOverlap(self, img1: List[List[int]], img2: List[List[int]]) -> int:
 n = len(img1)
 cnt = Counter()
 for i in range(n):
 for j in range(n):
 if img1[i][j]:
 for h in range(n):
 for k in range(n):
 if img2[h][k]:
 cnt[(i - h, j - k)] += 1
 return max(cnt.values()) if cnt else 0
```

Matrix p.1403

Matrix

#1198 Find Smallest Common Element in All RowsMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Binary Search · Matrix · Counting

Lists: Premium 98

Time Complexity:  $O(m * n * \log(n))$  where  $m$  is the number of rows and  $n$  is the number of columns (binary search in each row for each candidate from the first row). Space Complexity:  $O(1)$  (only a few au

Problem

Given an  $m \times n$  matrix `mat` where every row is sorted in strictly increasing order, return the smallest common element in all rows.

If there is no common element, return `-1`.

Example 1:

Input: `mat = [[1,2,3,4,5],[2,4,5,8,10],[3,5,7,9,11],[1,3,5,7,9]]`  
Output: 5

Example 2:

Input: `mat = [[1,2,3],[2,3,4],[2,3,5]]`  
Output: 2

Constraints:

- $m == \text{mat.length}$

Matrix p.1404

- $n == \text{mat}[i].\text{length}$
- $1 \leq m, n \leq 500$
- $1 \leq \text{mat}[i][j] \leq 104$
- $\text{mat}[i]$  is sorted in strictly increasing order.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def smallestCommonElement(self, mat):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every cell
 m, n = len(mat), len(mat[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in [(0, 1), (0, -1), (1, 0), (-1, 0)]: flood(r+dr, c+dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def smallestCommonElement(self, mat: List[List[int]])
 -> int:
 MAX = 10000
 count = [0] * (MAX + 1)

 for row in mat:
 for a in row:
 count[a] += 1
 if count[a] == len(mat):
 return a

 return -1
```

- LeetDoocs

```
class Solution:
 def smallestCommonElement(self, mat: List[List[int]])
 -> int:
 cnt = Counter()
 for row in mat:
 for x in row:
 cnt[x] += 1
 if cnt[x] == len(mat):
 return x

 return -1
```

- SimplyLeet

```
def smallestCommonElement(mat):
 # Get number of rows and columns
 m = len(mat)
 n = len(mat[0])

 # Helper function: binary search for target in a row
 def binary_search(row, target):
 left, right = 0, n - 1
 while left <= right:
 mid = (left + right) // 2

 if row[mid] == target:
 return True
 elif row[mid] < target:
 left = mid + 1
 else:
 right = mid - 1
 return False

 # Iterate over each candidate in the first row
 for candidate in mat[0]:

 is_common = True
 # Check each subsequent row for the candidate
 using binary search
 for i in range(1, m):
 if not binary_search(mat[i], candidate):
 is_common = False
 break
 if is_common:
 return candidate
 return -1
```

```
Example usage:
print(smallestCommonElement([[1,2,3,4,5],[2,4,5,8,10],[3,5,7,9,11],[1,3,5,7,9]]))
```

- LC.ca

```
class Solution:
 def smallestCommonElement(self, mat: List[List[int]])
 -> int:
 cnt = Counter()
 for row in mat:
 for x in row:
 cnt[x] += 1
 if cnt[x] == len(mat):
 return x

 return -1
```

Matrix

#1074 Number of Submatrices That Sum to Target

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Matrix · Prefix Sum

Lists: Denny Zhang

Time Complexity:  $O(n^2 \cdot m)$ , where  $n$  is the number of rows and  $m$  is the number of columns. Space Complexity:  $O(m)$ , used by the hash table for storing cumulative sums along the columns.

Problem

Given a matrix and a target, return the number of non-empty submatrices that sum to target.

A submatrix  $x1, y1, x2, y2$  is the set of all cells  $\text{matrix}[x][y]$  with  $x1 \leq x \leq x2$  and  $y1 \leq y \leq y2$ .

Two submatrices  $(x1, y1, x2, y2)$  and  $(x1', y1', x2', y2')$  are different if they have some coordinate that is different: for example, if  $x1 \neq x1'$ .

Example 1:

|   |   |   |
|---|---|---|
| 0 | 1 | 0 |
| 1 | 1 | 1 |
| 0 | 1 | 0 |

Input:  $\text{matrix} = [[0,1,0],[1,1,1],[0,1,0]]$ ,  $\text{target} = 0$   
Output: 4  
Explanation: The four  $1 \times 1$  submatrices that only contain 0.

Example 2:

Input:  $\text{matrix} = [[1,-1],[-1,1]]$ ,  $\text{target} = 0$   
Output: 5  
Explanation: The two  $1 \times 2$  submatrices, plus the two  $2 \times 1$  submatrices, plus the  $2 \times 2$  submatrix.

Example 3:

```
Input: matrix = [[904]], target = 0
Output: 0
```

Constraints:

- $1 \leq \text{matrix.length} \leq 100$
- $1 \leq \text{matrix}[0].\text{length} \leq 100$
- $-1000 \leq \text{matrix}[i][j] \leq 1000$
- $-10^8 \leq \text{target} \leq 10^8$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def numSubmatrixSumTarget(self, matrix, target):
 # Brute Force $O(m^2 \cdot n^2)$ - flood fill from every cell
 m, n = len(matrix), len(matrix[0])
 result = 0
 for r in range(m):
 for c in range(n):
 visited = set()
 def flood(r, c):
 if (r, c) in visited or not (0 <= r < m and 0 <= c < n): return
 visited.add((r, c))
 for dr, dc in [(0, 1), (0, -1), (1, 0), (-1, 0)]: flood(r+dr, c+dc)
 flood(r, c)
 result = max(result, len(visited))
 return result
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def numSubmatrixSumTarget(self, matrix: List[List[int]], target: int) -> int:
 m = len(matrix)
 n = len(matrix[0])
 ans = 0

 # Transfer each row in the matrix to the prefix sum.
 for row in matrix:
 for i in range(1, n):
 row[i] += row[i - 1]

 for baseCol in range(n):
 for j in range(baseCol, n):
 prefixCount = collections.Counter({0: 1})
 summ = 0
 for i in range(m):
 if baseCol > 0:
 summ -= matrix[i][baseCol - 1]
 summ += matrix[i][j]
 ans += prefixCount[summ - target]

 prefixCount[summ] += 1

 return ans
```

- LeetDoocs

```
class Solution:
 def numSubmatrixSumTarget(self, matrix: List[List[int]], target: int) -> int:
 def f(nums: List[int]) -> int:
 d = defaultdict(int)
 d[0] = 1
 cnt = s = 0
 for x in nums:
 s += x
 cnt += d[s - target]
 d[s] += 1

 return cnt

 m, n = len(matrix), len(matrix[0])
 ans = 0
 for i in range(m):
 col = [0] * n
 for j in range(i, m):
 for k in range(n):
 col[k] += matrix[j][k]
 ans += f(col)

 return ans
```

- SimplyLeet

|                                                                  |                                                                    |                                                                     |
|------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| <div data-bbox="43 44 496 506"></div> <div>Matrixp.1414</div>    | <div data-bbox="583 44 1037 485"></div> <div>Matrixp.1415</div>    | <div data-bbox="1123 44 1578 474"></div> <div>Matrixp.1416</div>    |
| <div data-bbox="43 743 496 1356"></div> <div>Matrixp.1417</div>  | <div data-bbox="583 743 1037 1320"></div> <div>Matrixp.1418</div>  | <div data-bbox="1123 743 1578 1377"></div> <div>Matrixp.1419</div>  |
| <div data-bbox="43 1442 496 2076"></div> <div>Matrixp.1420</div> | <div data-bbox="583 1442 1037 2076"></div> <div>Matrixp.1421</div> | <div data-bbox="1123 1442 1578 2076"></div> <div>Matrixp.1422</div> |



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| <div> <div>nested loops (one for rows and one for columns) to traverse each possible starting position of a 3 x 3 window. The simplicity of the solution comes from the fixed 3x3 block size, which enables constant-time maximum computation for each block.</div> <div>#36 Valid Sudoku [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Validate each row by checking for duplicate numbers while ignoring empty cells.</li> <li>Validate each column in the same way.</li> <li>Divide the board into nine 3x3 sub-boxes and check each for duplicate numbers.</li> <li>Use hash sets (or equivalent data structures) to keep track of seen digits for rows, columns, and boxes.</li> <li>The board size is fixed (9x9), so the overall time complexity is constant with respect to input size.</li> </ul> <div>Space &amp; Time</div> <div>Time Complexity: O(1) — Since the board size is fixed at 9x9.</div> <div>Space Complexity: O(1) — The extra space used by the sets is bounded by the fixed board size.</div> <div>Solution</div> <div>The solution uses three main data structures: an array of sets for rows, an array of sets for columns, and an array of sets for 3x3 sub-boxes. For each cell, if it is not</div> </div> <div> <div>Matrix</div> <div>p.1423</div> </div> </div>                                                                                                                                    | <div> <div>empty, check if the number is already present in the corresponding row, column, or box. If found, return false immediately. Otherwise, add the number to the respective sets. Compute the index for the 3x3 sub-box using the formula: (row_index // 3) * 3 + (col_index // 3). If no duplicates are found after traversing the entire board, the board is considered valid.</div> <div>#48 Rotate Image [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>The rotation can be achieved in two steps: first transpose the matrix and then reverse each row.</li> <li>Transposing the matrix swaps matrix[i][j] with matrix[j][i], converting rows into columns.</li> <li>Reversing each row post-transposition yields the desired clockwise rotation.</li> <li>This approach works in-place with O(1) extra space.</li> </ul> <div>Space &amp; Time</div> <div>Time Complexity: O(n^2) since every element is processed during the transpose and row reversal.</div> <div>Space Complexity: O(1) as the operations are performed in-place without extra data structures.</div> <div>Solution</div> <div>The solution leverages a two-step in-place transformation:</div> </div> <div> <div>Matrix</div> <div>p.1424</div> </div> </div>                                                                                          | <div> <div>– Transpose the matrix: Iterate through the matrix and swap each element (i, j) with element (j, i) for j &gt;= i. – Reverse each row: After the transpose, each row is reversed to rotate the matrix 90 degrees clockwise. The key trick is to realize that a transpose followed by a reverse of each row achieves the rotation without needing extra space.</div> <div>#54 Spiral Matrix [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use four boundaries: top, bottom, left, and right to track the current layer.</li> <li>Traverse right across the top row, then down the right column, then left on the bottom row, and finally up the left column.</li> <li>After each direction, adjust the corresponding boundary to move inward.</li> <li>Continue the traversal until the boundaries cross.</li> </ul> <div>Space &amp; Time</div> <div>Time Complexity: O(m * n) since each element is visited exactly once.</div> <div>Space Complexity: O(1) extra space (excluding the space required for the output list).</div> <div>Solution</div> <div>The solution uses a simulation approach with boundary pointers (top, bottom, left, right). At each step, the</div> </div> <div> <div>Matrix</div> <div>p.1425</div> </div> </div>                                                                                                                                             |
| <div> <div>boundaries indicate the current submatrix to traverse. You iterate as follows: – Traverse from left to right along the top boundary. – Traverse downward along the right boundary. – If the top boundary has not passed the bottom, traverse from right to left along the bottom boundary. – If the left boundary has not passed the right, traverse upward along the left boundary. After each traversal, the boundaries are adjusted inward. This continues until all elements have been visited.</div> <div>#59 Spiral Matrix II [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use four pointers (top, bottom, left, right) to keep track of the matrix boundaries.</li> <li>Traverse the matrix in a spiral: move right along the top row, then down the right column, left along the bottom row, and finally up the left column.</li> <li>Update the boundary pointers after each complete traversal of one side.</li> <li>Continue until all numbers from 1 to n^2 are filled into the matrix.</li> </ul> <div>Space &amp; Time</div> <div>Time Complexity: O(n^2)</div> <div>Space Complexity: O(n^2) (required for the output matrix)</div> <div>Solution</div> </div> <div> <div>Matrix</div> <div>p.1426</div> </div> </div>                                                                                                                                                                                    | <div> <div>We simulate the spiral movement by initializing four pointers: top, bottom, left, and right, which denote the current boundaries of the matrix that have not yet been filled. For each iteration, we: – Traverse from left to right along the top boundary, then increment the top pointer. – Traverse from top to bottom along the right boundary, then decrement the right pointer. – If there are rows remaining, traverse from right to left along the bottom boundary, then decrement the bottom pointer. – If there are columns remaining, traverse from bottom to top along the left boundary, then increment the left pointer. This algorithm ensures that the matrix is filled in a spiral order until all numbers are placed.</div> <div>#64 Minimum Path Sum [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>The problem can be solved using Dynamic Programming.</li> <li>The optimal solution for each cell depends on the minimum path sum of the cell above it and the cell to its left.</li> <li>Since you can only move right or down, only two adjacent cells affect the current cell's result.</li> <li>Edge cases include the first row and first column, which have only one direction of movement.</li> </ul> <div>Space &amp; Time</div> </div> <div> <div>Matrix</div> <div>p.1427</div> </div> </div> | <div> <div>Time Complexity: O(m * n), where m and n are the number of rows and columns.</div> <div>Space Complexity: O(m * n) for the DP table (this can be optimized to O(n) with in-place computations).</div> <div>Solution</div> <div>We use a Dynamic Programming (DP) approach where we create a DP table (or use the grid itself) that stores the minimum path sum to each cell. For each cell (i, j): – If i = 0 (first row), the only possible path is from the left. – If j = 0 (first column), the only possible path is from above. – For other cells, the minimum path sum is the cell value plus the minimum of the two possible previous cells (above and left). This ensures that by the time we reach the bottom-right cell, we have accumulated the minimum sum required to reach that cell.</div> <div>#73 Set Matrix Zeroes [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>The challenge is to mark rows and columns that need to be zeroed without using extra space.</li> <li>A common trick is to use the first row and first column as markers.</li> <li>Special care is needed for the first row and first column, as they are also used as markers.</li> </ul> </div> <div> <div>Matrix</div> <div>p.1428</div> </div> </div>                                                                                                                                               |
| <div> <ul style="list-style-type: none"> <li>The algorithm involves two passes: one for marking and one for updating the matrix based on the markers.</li> </ul> <div>Space &amp; Time</div> <div>Time Complexity: O(m * n)</div> <div>Space Complexity: O(1) (in-place, using only constant extra space)</div> <div>Solution</div> <div>We can solve this problem by using the first row and first column of the matrix as markers. First, determine if the first row or first column needs to be zeroed, then traverse the rest of the matrix and mark the corresponding first row and column positions if a zero is encountered. Finally, update the matrix backward using the markers, and finally update the first row and column if needed.</div> <div>The algorithm works in three main steps: – Check if the first row or first column should be zeroed (store this in two variables). – Iterate through the rest of the matrix; for any element that is zero, mark its row and column by setting the corresponding element in the first row and first column to zero. – Iterate through the matrix again (excluding the first row and column) and set elements to zero if their row or column marker is zero. – Finally, zero out the first row and/or first column if needed.</div> <div>This approach avoids extra space usage apart from the two boolean flags while ensuring that the matrix is</div> </div> <div> <div>Matrix</div> <div>p.1429</div> </div> | <div> <div>updated correctly.</div> <div>#74 Search a 2D Matrix [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>The matrix is essentially a flattened sorted array due to its properties.</li> <li>Use binary search to efficiently determine if the target exists.</li> <li>Map the 1D binary search index to the corresponding 2D matrix coordinates using division and modulus operations.</li> <li>Achieve O(log(m*n)) time complexity by navigating the matrix as if it were a single sorted list.</li> </ul> <div>Space &amp; Time</div> <div>Time Complexity: O(log(m * n))</div> <div>Space Complexity: O(1)</div> <div>Solution</div> <div>We can treat the matrix as a flattened sorted list without physically copying the elements. Given the total number of elements = m * n, perform binary search on indices ranging from 0 to (m*n - 1). For any index mid, calculate its corresponding row as mid // n and column as mid % n. Compare the element at matrix[row][col] with the target. If it equals target, return true; if it is less than target, search the right half; otherwise, search the left half.</div> </div> <div> <div>Matrix</div> <div>p.1430</div> </div> </div>                                                                                                                                        | <div> <div>#221 Maximal Square [Medium]</div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use dynamic programming to build a solution where each cell in the DP table represents the side length of the largest square ending at that cell.</li> <li>If the current cell is '1', its DP value is the minimum of the top, left, and top-left neighbors plus one.</li> <li>Update a global maximum side length during the process to determine the area.</li> <li>The square's area is the square of its maximum side length.</li> </ul> <div>Space &amp; Time</div> <div>Time Complexity: O(m * n), where m is the number of rows and n is the number of columns.</div> <div>Space Complexity: O(m * n) for the DP table (this can be optimized to O(n) with space reduction techniques).</div> <div>Solution</div> <div>We use a two-dimensional dynamic programming approach. We define a DP matrix with one extra row and column to simplify boundary conditions. For each cell (i, j) in the input matrix, if the value is '1', then we calculate the DP value as the minimum of its top, left, and top-left neighboring DP values plus one. This value represents the side length of the square ending at (i, j). We also keep track of the maximum side length found and finally compute the area of the maximal square by squaring the</div> </div> <div> <div>Matrix</div> <div>p.1431</div> </div> </div> |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <div>maximum side length.</div> <div>#311 Sparse Matrix Multiplication <span>Medium</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Exploit sparsity by iterating only over non-zero elements.</li> <li>For each non-zero element in mat1, multiply it with corresponding non-zero elements in mat2.</li> <li>Accumulate the product into the result matrix.</li> <li>This approach reduces unnecessary multiplication operations when many elements are zero.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: <math>O(m * k * n)</math> in the worst-case, but significantly lower when many elements are zero.</p> <p>Space Complexity: <math>O(m * n)</math> for the resulting matrix.</p> </div> <div> <div>Solution</div> <p>We loop through each row of mat1 and for every non-zero element, we traverse the corresponding row in mat2. By checking if an element in both matrices is non-zero before multiplying, we avoid redundant calculations. This optimization leverages the sparsity of the matrices. We use standard nested loops along with conditional checks to implement this efficient multiplication.</p> </div> <div>#498 Diagonal Traverse <span>Medium</span></div> <div> <div>Matrix</div> <div>p.1432</div> </div> </div>                                                                         | <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Traverse the matrix diagonally with alternating directions: upward (top-right) and downward (bottom-left).</li> <li>When the traversal reaches a boundary (top, bottom, left, or right), adjust the position and change the direction.</li> <li>Use simulation to process each element exactly once while managing the indices based on the current direction.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: <math>O(m * n)</math> where m is the number of rows and n is the number of columns, as every element is processed once.</p> <p>Space Complexity: <math>O(m * n)</math> for the output array (excluding the input matrix), while additional space usage remains constant.</p> </div> <div> <div>Solution</div> <p>The solution simulates the diagonal traversal using a direction flag (1 for upward and -1 for downward).</p> <p>– Begin at the matrix's top-left corner.</p> <p>– Append the current element to the result.</p> <p>– Depending on the current direction: If moving upward, decrease the row and increase the column. If moving downward, increase the row and decrease the column.</p> <p>– Adjust for boundary hits: When moving upward: If at the last column, increment the row and switch direction. If at the first</p> </div> <div> <div>Matrix</div> <div>p.1433</div> </div>                       | <div> <div>row, increment the column and switch direction. When moving downward: If at the last row, increment the column and switch direction. If at the first column, increment the row and switch direction. This process continues until all elements are added to the result array.</div> <div>#531 Lonely Pixel I <span>Medium</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Count the number of 'B' pixels in each row.</li> <li>Count the number of 'B' pixels in each column.</li> <li>A pixel is considered lonely if the count in its corresponding row and column is exactly 1.</li> <li>Two passes through the matrix can efficiently solve the problem: one for counting and one for checking conditions.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: <math>O(m * n)</math>, where m and n are the dimensions of the picture.</p> <p>Space Complexity: <math>O(m + n)</math>, used to store the row and column counts.</p> </div> <div> <div>Solution</div> <p>The solution works by first scanning the entire matrix to compute the number of black pixels in each row and each column using two arrays/lists. Once these counts are available, a second pass through the matrix</p> </div> <div> <div>Matrix</div> <div>p.1434</div> </div> </div>                                                                                          |
| <div> <div>determines for each black pixel if it is lonely by checking if its corresponding row and column count equals 1. The approach uses simple array data structures for tallying counts and ensures that each element is processed only a constant number of times.</div> <div>#723 Candy Crush <span>Medium</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use a simulation approach: repeatedly detect crushable candies and simulate their removal.</li> <li>Traverse horizontally and vertically to mark candies that are in groups of three or more.</li> <li>Instead of removing candies immediately, mark them for removal and update the board once for the entire pass.</li> <li>After crushing, simulate gravity by shifting non-zero candies downward in each column.</li> <li>Continue the process until no further candy crushes can be detected.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: <math>O((m * n)^2)</math> in the worst case since each iteration includes scanning the m x n grid and then applying gravity.</p> <p>Space Complexity: <math>O(m * n)</math> for maintaining additional markers or flags (or using the board itself with in-place modifications).</p> </div> <div> <div>Matrix</div> <div>p.1435</div> </div> </div>                                                         | <div> <div>Solution</div> <p>The approach involves iteratively scanning the board to identify candy groups that should be crushed using two passes:</p> <p>– Horizontal Check – For each row, check segments of 3 consecutive candies.</p> <p>– Vertical Check – For each column, check segments of 3 consecutive candies. If any candy qualifies, mark its position for crushing. After marking, update the board by turning all marked positions into 0. Then, for each column, drop down non-zero candies such that all empty cells (zeros) are at the top. Repeat this process until no group of 3 or more adjacent candies is found.</p> <p>Key data structures include the board array and a temporary marker (can be implicit by using signs or a separate boolean matrix) for tracking candies to crush. The algorithm ensures a stable board before returning it.</p> </div> <div> <div>#835 Image Overlap <span>Medium</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>You only need to consider the positions of 1's in each matrix.</li> <li>The overlap count for a given translation can be computed by comparing the relative differences between the positions of 1's in the two images.</li> <li>Using a hash map (or dictionary) to count how many times each translation vector appears lets you compute</li> </ul> </div> <div> <div>Matrix</div> <div>p.1436</div> </div> </div> | <div> <div>the maximum overlap efficiently.</div> <ul style="list-style-type: none"> <li>The translation operation is equivalent to shifting the positions, so the best possible translation is the one that aligns the most pairs of 1's.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: <math>O(n^4)</math> in the worst-case when using a nested iteration over positions for both matrices, though the average situation is typically better since not all cells are 1.</p> <p>Space Complexity: <math>O(n^2)</math> for storing the positions of 1's and the counts for translation vectors.</p> </div> <div> <div>Solution</div> <p>The solution involves the following steps:</p> <p>– Traverse both images to record the coordinates of all 1's.</p> <p>– For every pair of 1's (one from the first image, one from the second), compute the translation vector (offset) that aligns them.</p> <p>– Use a hash map to count how many times each offset appears.</p> <p>– The maximum count recorded in the hash map is the largest overlap achievable by any translation. This approach avoids simulating every possible slide over the matrix and focuses only on relative alignments of 1's.</p> </div> <div> <div>#1198 Find Smallest Common Element in All Rows <span>Medium</span></div> <div> <div>Key Insights</div> </div> <div> <div>Matrix</div> <div>p.1437</div> </div> </div> |
| <div> <ul style="list-style-type: none"> <li>Since every row is strictly increasing, binary search can be used efficiently to check for the presence of an element in a row.</li> <li>Iterating over the first row's elements and verifying their presence in all other rows is an effective approach.</li> <li>Early termination is possible if an element is not found in any one row.</li> <li>Time complexity is manageable given the constraints, even using binary search for each candidate across rows.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: <math>O(m * n * \log(n))</math> where m is the number of rows and n is the number of columns (binary search in each row for each candidate from the first row).</p> <p>Space Complexity: <math>O(1)</math> (only a few auxiliary variables are used).</p> </div> <div> <div>Solution</div> <p>The solution iterates over each element (candidate) in the first row of the matrix. For each candidate, it checks if the candidate is present in every other row using binary search, taking advantage of the sorted order. If a candidate is found in every row, return it immediately as it will be the smallest such common element because the first row is processed in order. If no candidate is common across all rows, return -1.</p> </div> <div> <div>Matrix</div> <div>p.1438</div> </div> | <div> <div>#1074 Number of Submatrices That Sum to Target <span>Hard</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Transform the 2D submatrix sum problem into multiple 1D subarray sum problems.</li> <li>Precompute cumulative sums by fixing two row boundaries and then summing columns between these rows.</li> <li>Use a hash table (or dictionary) to count the number of subarrays with a given sum, reducing the problem to the classic "subarray sum equals k" question.</li> <li>Iterate over all possible row pairs and for each, treat the column sums as an array where the target sum is sought.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: <math>O(n^2 * m)</math>, where n is the number of rows and m is the number of columns.</p> <p>Space Complexity: <math>O(m)</math>, used by the hash table for storing cumulative sums along the columns.</p> </div> <div> <div>Solution</div> <p>The idea is to iterate over all pairs of rows (top and bottom) and, for each pair, compute the cumulative column sums between these rows. This converts the problem into finding the number of subarrays in a one-dimensional array (the column sums) that add up to the target. For each such 1D problem, use a hash table</p> </div> <div> <div>Matrix</div> <div>p.1439</div> </div> </div>                                                             | <div> <div>to track the frequency of cumulative sums seen so far and count how many times the difference (current sum – target) has been seen. This count gives the number of subarrays ending at the current index which sum to target. Combining counts from all possible row pairs yields the final result.</div> <div>Linked List</div> <div>#16 Linked List</div> <div>By Pattern · Python Only · 23 questions</div> <div> <div>Matrix</div> <div>p.1440</div> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

#21 Merge Two Sorted Lists

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List · Recursion

Lists: Grind 169

Time Complexity:  $O(n + m)$ , where  $n$  and  $m$  are the lengths of the two lists. Space Complexity:  $O(1)$  for the iterative solution  $O(n + m)$  stack space for the recursive solution).

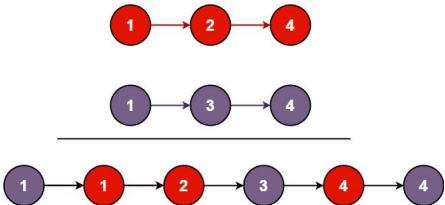
Problem

You are given the heads of two sorted linked lists `list1` and `list2`.

Merge the two lists into one sorted list. The list should be made by splicing together the nodes of the first two lists.

Return the head of the merged linked list.

Example 1:



Input: list1 = [1,2,4], list2 = [1,3,4]  
Output: [1,1,2,3,4,4]

Example 2:

Input: list1 = [], list2 = []  
Output: []

Example 3:

Input: list1 = [], list2 = [0]  
Output: [0]

Constraints:

- The number of nodes in both lists is in the range  $[0, 50]$ .
- $-100 \leq \text{Node.val} \leq 100$
- Both `list1` and `list2` are sorted in non-decreasing order.

Community Solutions (Python)

Linked List

p.1442

• WalkCC

```
class Solution:
 def mergeTwoLists(
 self,
 list1: ListNode | None,
 list2: ListNode | None,
) -> ListNode | None:
 if not list1 or not list2:
 return list1 if list1 else list2
 if list1.val > list2.val:
 list1, list2 = list2, list1

 list1.next = self.mergeTwoLists(list1.next, list2)
 return list1
```

• LeetDoocs

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def mergeTwoLists(
 self, list1: Optional[ListNode], list2:
 Optional[ListNode]
) -> Optional[ListNode]:
 if list1 is None or list2 is None:
```

Linked List

p.1443

```
 return list1 or list2
 if list1.val <= list2.val:
 list1.next = self.mergeTwoLists(list1.next,
list2)
 return list1
 else:
 list2.next = self.mergeTwoLists(list1,
list2.next)
 return list2
```

• SimplyLeet

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

def mergeTwoLists(list1: ListNode, list2: ListNode) ->
ListNode:
 # Create a dummy node to serve as the start of the
merged list
 dummy = ListNode(-1)
 current = dummy
```

```
Iterate while both lists have nodes
while list1 and list2:
 # Compare the values of the current nodes of both
lists
 if list1.val <= list2.val:
 current.next = list1 # Link the smaller node
to the merged list
 list1 = list1.next # Move the pointer in
list1
 else:
 current.next = list2
 list2 = list2.next
```

```
 current = current.next # Move the pointer in the
merged list
```

```
Once one list is exhausted, attach the remaining
nodes from the other list
if list1:
 current.next = list1
else:
 current.next = list2
```

```
The merged list head is at dummy.next
return dummy.next
```

• LC.ca

Linked List

p.1445

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
without ops after while loop
class Solution:
 def mergeTwoLists(self, list1: Optional[ListNode],
list2: Optional[ListNode]) -> Optional[ListNode]:
 dummy = ListNode()
 current = dummy
```

```
 while list1 or list2:
 v1 = list1.val if list1 else float('inf')
 v2 = list2.val if list2 else float('inf')
```

```
 if v1 < v2:
 current.next = list1
 list1 = list1.next
 else:
 current.next = list2
 list2 = list2.next
```

```
 current = current.next

 return dummy.next
```

```
#####

Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
```

Linked List

p.1446

```
self.next = next
class Solution:
 def mergeTwoLists(
 self, list1: Optional[ListNode], list2:
 Optional[ListNode]
) -> Optional[ListNode]:
 dummy = ListNode()
 curr = dummy
 while list1 and list2:
 if list1.val <= list2.val:
 curr.next = list1
```

```
 list1 = list1.next
 else:
 curr.next = list2
 list2 = list2.next
 curr = curr.next
 curr.next = list1 or list2
 return dummy.next
```

```
 curr.next = list1 or list2
```

```
better than:

if list1:
 current.next = list1
if list2:
 current.next = list2
```

```
#####

recursion
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

class Solution:

 def mergeTwoLists(self, l1: ListNode, l2: ListNode)
-> ListNode:
 if not l1: # If l1 is empty, return l2
 return l2
 if not l2: # If l2 is empty, return l1
 return l1

 if l1.val < l2.val:
 l1.next = self.mergeTwoLists(l1.next, l2)
 return l1
 else:

 l2.next = self.mergeTwoLists(l1, l2.next)
 return l2
```

Linked List

p.1448

Linked List

#141 Linked List Cycle

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · Linked List · Two Pointers

Lists: Grind 169

Time Complexity:  $O(n)$  where  $n$  is the number of nodes in the linked list. Space Complexity:  $O(1)$  as only two pointers are used.

Problem

Given head, the head of a linked list, determine if the linked list has a cycle in it.

There is a cycle in a linked list if there is some node in the list that can be reached again by continuously following the next pointer. Internally, `pos` is used to denote the index of the node that `tail's` next pointer is connected to. Note that `pos` is not passed as a parameter.

Return `true` if there is a cycle in the linked list. Otherwise, return `false`.

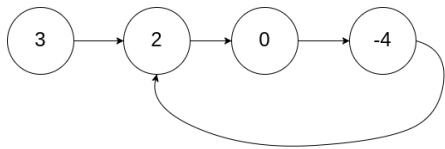
Example 1:

Linked List

p.1449

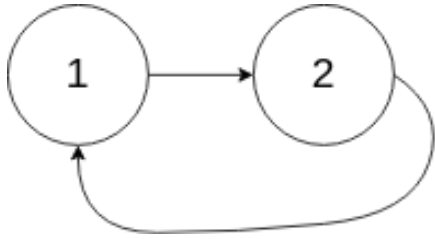
Linked List

p.1447



Input: head = [3,2,0,-4], pos = 1  
Output: true  
Explanation: There is a cycle in the linked list, where the tail connects to the 1st node (0-indexed).

Example 2:



Input: head = [1,2], pos = 0  
Output: true  
Explanation: There is a cycle in the linked list, where the tail connects to the 0th node.

Linked List p.1450

Example 3:



Input: head = [1], pos = -1  
Output: false  
Explanation: There is no cycle in the linked list.

Constraints:

- The number of the nodes in the list is in the range [0, 104].
- $-105 \leq \text{Node.val} \leq 105$
- pos is -1 or a valid index in the linked-list.

Follow up: Can you solve it using  $O(1)$  (i.e. constant) memory?

Community Solutions (Python)

- WalkCC

Linked List p.1451

```
class Solution:
 def hasCycle(self, head: ListNode) -> bool:
 slow = head
 fast = head

 while fast and fast.next:
 slow = slow.next
 fast = fast.next.next
 if slow == fast:
 return True

 return False
```

• LeetCode

```
Definition for singly-linked list.
class ListNode:
def __init__(self, x):
self.val = x
self.next = None

class Solution:
 def hasCycle(self, head: Optional[ListNode]) -> bool:
 s = set()

 while head:
 if head in s:
 return True
 s.add(head)
 head = head.next
 return False
```

Linked List p.1452

• SimplyLeet

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next

class Solution:
 def hasCycle(self, head: Optional[ListNode]) -> bool:
 # Initialize two pointers, slow and fast,
 # starting from the head.
 slow = head

 fast = head

 # Traverse the list until fast pointer reaches
 # the end.
 while fast and fast.next:
 slow = slow.next # Move slow pointer one
 step.
 fast = fast.next.next # Move fast pointer two
 steps.

 # If both pointers meet, there is a cycle.
 if slow == fast:
 return True

 # If loop terminates, there is no cycle.
 return False
```

• LC.ca

Linked List p.1453

```
Definition for singly-linked list.
class ListNode:
def __init__(self, x):
self.val = x
self.next = None

class Solution:
 def hasCycle(self, head: ListNode) -> bool:
 slow = fast = head
 while fast and fast.next:
 slow, fast = slow.next, fast.next.next
 if slow == fast:
 return True
 return False
```

Linked List p.1454

Linked List

## #206 Reverse Linked List

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List · Recursion

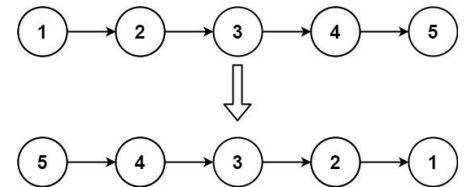
Lists: Grind 169

Time Complexity:  $O(n)$  for both iterative and recursive approaches, where  $n$  is the number of nodes. Space Complexity:  $O(1)$  for the iterative approach and  $O(n)$  for the recursive approach because of the

Problem

Given the head of a singly linked list, reverse the list, and return the reversed list.

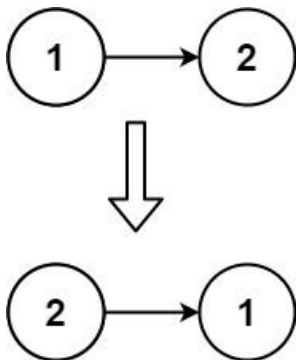
Example 1:



Input: head = [1,2,3,4,5]  
Output: [5,4,3,2,1]

Linked List p.1455

Example 2:



Input: head = [1,2]  
Output: [2,1]

Example 3:

Input: head = []  
Output: []

Constraints:

- The number of nodes in the list is the range [0, 5000].
- $-5000 \leq \text{Node.val} \leq 5000$

Linked List p.1456

Follow up: A linked list can be reversed either iteratively or recursively. Could you implement both?

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def reverseList(self, head):
 # Brute Force $O(n)$ space - collect values,
 # rebuild in reverse
 vals = []
 curr = head
 while curr:
 vals.append(curr.val)
 curr = curr.next
 dummy = ListNode(0)

 curr = dummy
 for v in reversed(vals):
 curr.next = ListNode(v)
 curr = curr.next
 return dummy.next
```

Community Solutions (Python)

- WalkCC

Linked List p.1457

```
class Solution:
 def reverseList(self, head: ListNode | None) -> ListNode | None:
 if not head or not head.next:
 return head

 newHead = self.reverseList(head.next)
 head.next.next = head
 head.next = None
 return newHead
```

• LeetCode

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next

class Solution:
 def reverseList(self, head: ListNode) -> ListNode:
 dummy = ListNode()
 curr = head
 while curr:
 next = curr.next
 curr.next = dummy.next
 dummy.next = curr
 curr = next
 return dummy.next
```

• SimplyLeet

Linked List p.1458

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val # Store the node value
 self.next = next # Pointer to the next node

Iterative approach to reverse the linked list.
def reverseList(head):
 prev = None # Initially, there is no previous node
 current = head # Start with the head of the list

 while current:
 next_node = current.next # Temporarily store the
 next node
 current.next = prev # Reverse the current node's
 pointer
 prev = current # Move previous pointer one step
 forward
 current = next_node # Move to the next node in
 the list
 return prev # The new head of the reversed list

Recursive approach to reverse the linked list.
def reverseListRecursive(head):
 # Base case: empty list or single node
```

Linked List

p.1459

```
if not head or not head.next:
 return head
new_head = reverseListRecursive(head.next) # Reverse
the rest of the list
head.next.next = head # Reverse the pointer for the
current node
head.next = None # Set the next pointer of the
current node to None
return new_head # Return the new head of the reversed
list
```

#### • LC.ca

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def reverseList(self, head: Optional[ListNode]) ->
Optional[ListNode]:
 pre, p = None, head
 while p:
 q = p.next
```

Linked List

p.1460

```
p.next = pre
pre = p
p = q
return pre

class Solution(object): # recursive
def reverseList(self, root):
 if not root or not root.next:
 return root
```

```
ret = self.reverseList(root.next)
root.next.next = root # root.next is end of the newly
reversed list
ret
root.next = None
return ret
```

```
class Solution:
 def reverseList(self, head: ListNode) -> ListNode:
 dummy = ListNode()
```

```
curr = head
while curr:
 next = curr.next
 curr.next = dummy.next
 dummy.next = curr
 curr = next
return dummy.next
```

Linked List

p.1461

#### Linked List

### #234 Palindrome Linked List

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List · Two Pointers · Recursion

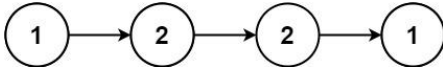
Lists: Grind 169

Time Complexity:  $O(n)$  – each node is visited a constant number of times. Space Complexity:  $O(1)$  – only constant extra space is used (in-place reversal).

#### Problem

Given the head of a singly linked list, return true if it is a palindrome or false otherwise.

Example 1:

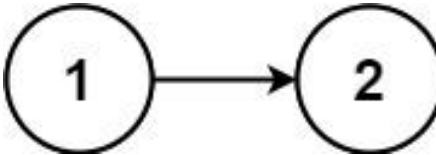


Input: head = [1,2,2,1]  
Output: true

Example 2:

Linked List

p.1462



Input: head = [1,2]  
Output: false

#### Constraints:

- The number of nodes in the list is in the range [1, 105].
- $0 \leq \text{Node.val} \leq 9$

Follow up: Could you do it in  $O(n)$  time and  $O(1)$  space?

#### Community Solutions (Python)

##### • WalkCC

```
class Solution:
 def isPalindrome(self, head: ListNode) -> bool:
 def reverseList(head: ListNode) -> ListNode:
 prev = None
 curr = head
 while curr:
 next = curr.next
 curr.next = prev
 prev = curr
 curr = next
```

Linked List

p.1463

```
return prev

slow = head
fast = head

while fast and fast.next:
 slow = slow.next
 fast = fast.next.next
```

if fast:

```
slow = slow.next
slow = reverseList(slow)

while slow:
 if slow.val != head.val:
 return False
 slow = slow.next
 head = head.next
```

return True

##### • LeetCode

Linked List

p.1464

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def isPalindrome(self, head: Optional[ListNode]) ->
bool:
 slow, fast = head, head.next
 while fast and fast.next:
 slow, fast = slow.next, fast.next.next

 pre, cur = None, slow.next
 while cur:
 t = cur.next
 cur.next = pre
 pre, cur = cur, t
 while pre:
 if pre.val != head.val:
 return False
 pre, head = pre.next, head.next
 return True
```

##### • SimplyLeet

Linked List

p.1465

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def isPalindrome(self, head: ListNode) -> bool:
 # Find the midpoint using slow and fast pointers
 slow, fast = head, head

 while fast and fast.next:
 slow = slow.next # Move slow pointer by 1
 fast = fast.next.next # Move fast pointer by 2

 # Reverse the second half of the list
 prev = None
 while slow:
 next_node = slow.next # Store the next node
 slow.next = prev # Reverse the current node's
 pointer
 prev = slow # Move prev to current node
 slow = next_node
```

Linked List

p.1466

```
slow = next_node # Move to next node

Compare the first half and the reversed second
half node by node
left, right = head, prev
while right: # Only need to compare till the end
of second half
 if left.val != right.val:
 return False # Mismatch found, not a
 palindrome
 left = left.next # Move left pointer
 right = right.next # Move right pointer

return True # All nodes matched, it's a
palindrome
```

##### • LC.ca

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def isPalindrome(self, head: Optional[ListNode]) ->
bool:
 slow, fast = head, head.next
 while fast and fast.next:
 slow, fast = slow.next, fast.next.next
```

Linked List

p.1467



```
pre, cur = None, slow.next
while cur:
 t = cur.next
 cur.next = pre
 pre, cur = cur, t
while pre:
 if pre.val != head.val:
 return False
 pre, head = pre.next, head.next
return True
```

Linked List

#706 Design HashMapEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Array · Hash Table · Linked List · Design · Hash Function  
Lists: Denny Zhang

Time Complexity: Average O(1) for put, get, and remove; worst-case O(n) when many keys hash to the same bucket. Space Complexity: O(n), where n is the number of key-value pairs stored.

Problem  
Design a HashMap without using any built-in hash table libraries.

Implement the MyHashMap class:

- MyHashMap() initializes the object with an empty map.
- void put(int key, int value) inserts a (key, value) pair into the HashMap. If the key already exists in the map, update the corresponding value.
- int get(int key) returns the value to which the specified key is mapped, or -1 if this map contains no mapping for the key.
- void remove(key) removes the key and its corresponding value if the map contains the mapping for the key.

Example 1:

Input

["MyHashMap", "put", "put", "get", "get", "put", "get", "remove", "get"]  
[[], [1, 1], [2, 2], [1], [3], [2, 1], [2], [2], [2]]  
Output  
[null, null, null, 1, -1, null, 1, null, -1]

Explanation

MyHashMap myHashMap = new MyHashMap();

Constraints:

- 0 <= key, value <= 106
- At most 104 calls will be made to put, get, and remove.

Community Solutions (Python)

```
class MyHashMap:
 def __init__(self):
 self.data = [-1] * 1000001

 def put(self, key: int, value: int) -> None:
 self.data[key] = value

 def get(self, key: int) -> int:
 return self.data[key]
```

```
def remove(self, key: int) -> None:
 self.data[key] = -1

Your MyHashMap object will be instantiated and called
as such:
obj = MyHashMap()
obj.put(key,value)
param_2 = obj.get(key)
obj.remove(key)

• SimplyLeet

ListNode class for handling collisions in each bucket
class ListNode:
 def __init__(self, key, value, next=None):
 self.key = key # Store key
 self.value = value # Store value
 self.next = next # Pointer to the next node

class MyHashMap:
 def __init__(self):
 self.size = 1000 # Define the bucket size
```

```
self.buckets = [None] * self.size # Initialize
buckets as an array of None

def hash(self, key):
 # Simple modulo based hash function to map keys
 to bucket indices
 return key % self.size

def put(self, key: int, value: int) -> None:
 index = self.hash(key)
 if self.buckets[index] is None:
 # If the bucket is empty, create a new node
 and place it here

 self.buckets[index] = ListNode(key, value)
 else:
 current = self.buckets[index]
 while True:
 if current.key == key:
 # If key exists, update its value
 current.value = value
 return
 if current.next is None:
 break
```

```
current = current.next
Append a new node at the end of the list if
key not found
current.next = ListNode(key, value)

def get(self, key: int) -> int:
 index = self.hash(key)
 current = self.buckets[index]
 while current:
 if current.key == key:
 # Key found; return its value

 return current.value
 current = current.next
 # Key not found; return -1
 return -1

def remove(self, key: int) -> None:
 index = self.hash(key)
 current = self.buckets[index]
 if current is None:
 return
```

```
if current.key == key:
 # Remove head node if it contains the key
 self.buckets[index] = current.next
else:
 prev = current
 current = current.next
 while current:
 if current.key == key:
 # Bypass the node to remove it from
 the list

 prev.next = current.next

 return
 prev, current = current, current.next

• LC.ca

class MyHashMap:
 def __init__(self):
 self.data = [-1] * 1000001

 def put(self, key: int, value: int) -> None:
 self.data[key] = value

 def get(self, key: int) -> int:
 return self.data[key]
```

```
def remove(self, key: int) -> None:
 self.data[key] = -1

Your MyHashMap object will be instantiated and called
as such:
obj = MyHashMap()
obj.put(key,value)
param_2 = obj.get(key)
obj.remove(key)

#####

class MyHashMap: # hash to bucket
 def __init__(self):
 """
 Initialize your data structure here.
 """
 self.size = 1000
 self.buckets = [[] for _ in range(self.size)]

 def _hash(self, key: int) -> int:
 """
 Generate a hash for a given key.
 """
 return key % self.size

 def put(self, key: int, value: int) -> None:
 """
 Value will always be non-negative.
 """
```

```
hash_key = self._hash(key)
for i, (k, v) in
enumerate(self.buckets[hash_key]):
 if k == key:
 self.buckets[hash_key][i] = (key, value)
 return
self.buckets[hash_key].append((key, value))

def get(self, key: int) -> int:
 """
 Returns the value to which the specified key is
mapped, or -1 if this map contains no mapping for the
key.

 """
 hash_key = self._hash(key)
 for k, v in self.buckets[hash_key]:
 if k == key:
 return v
 return -1

def remove(self, key: int) -> None:
 """
 Removes the mapping of the specified value key if
this map contains a mapping for the key.
```



```
"""
 hash_key = self._hash(key)
 for i, (k, v) in
enumerate(self.buckets[hash_key]):
 if k == key:
 del self.buckets[hash_key][i]
 return

#####

class MyHashMap: # open addressing with linear probing

 def __init__(self):
 self.capacity = 10000
 self.data = [None] * self.capacity
 self.DELETED = (None, None) # Marker for deleted
 entries

 def _hash(self, key):
 return key % self.capacity

 def put(self, key: int, value: int) -> None:
 idx = self._hash(key)
```

Linked List

p.1477

```
while self.data[idx] not in (None, self.DELETED,
key):
 idx = (idx + 1) % self.capacity
 self.data[idx] = (key, value)

def get(self, key: int) -> int:
 idx, found_key = self._find(key)
 if found_key:
 return self.data[idx][1]
 return -1

def remove(self, key: int) -> None:
 idx, found_key = self._find(key)
 if found_key:
 self.data[idx] = self.DELETED

def _find(self, key):
 original_idx = idx = self._hash(key)
 while self.data[idx] is not None:
 if self.data[idx] != self.DELETED and
self.data[idx][0] == key:
 return idx, True

 idx = (idx + 1) % self.capacity
 if idx == original_idx: # Came full circle
 break
 return original_idx, False
```

Linked List

p.1478

Linked List

## #876 Middle of the Linked List

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List · Two Pointers

Lists: Grind 169

Time Complexity:  $O(n)$ , where  $n$  is the number of nodes in the list. Space Complexity:  $O(1)$ , constant extra space.

### Problem

Given the head of a singly linked list, return the middle node of the linked list.

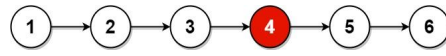
If there are two middle nodes, return the second middle node.

Example 1:



Input: head = [1,2,3,4,5]  
Output: [3,4,5]  
Explanation: The middle node of the list is node 3.

Example 2:



Linked List

p.1479

Input: head = [1,2,3,4,5,6]  
Output: [4,5,6]  
Explanation: Since the list has two middle nodes with values 3 and 4, we return the second one.

### Constraints:

- The number of nodes in the list is in the range [1, 100].
- $1 \leq \text{Node.val} \leq 100$

### Community Solutions (Python)

#### • WalkCC

```
class Solution:
 def middleNode(self, head: ListNode) -> ListNode:
 slow = head
 fast = head

 while fast and fast.next:
 slow = slow.next
 fast = fast.next.next

 return slow
```

#### • LeetDoocs

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def middleNode(self, head: ListNode) -> ListNode:
 slow = fast = head
 while fast and fast.next:
 slow, fast = slow.next, fast.next.next

 return slow
```

#### • SimplyLeet

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def middleNode(self, head: ListNode) -> ListNode:
 # Initialize slow and fast pointers at the head
 # of the list
 slow = head
```

Linked List

p.1481

```
fast = head
Loop until fast reaches the end of the list
while fast and fast.next:
 # Move slow pointer one step
 slow = slow.next
 # Move fast pointer two steps
 fast = fast.next.next
When loop ends, slow pointer points to the
middle node
return slow
```

#### • LC.ca

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def middleNode(self, head: ListNode) -> ListNode:
 slow = fast = head
 while fast and fast.next:
 slow, fast = slow.next, fast.next.next

 return slow
```

Linked List

p.1482

Linked List

## #1474 Delete N Nodes After M Nodes of Linked List

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List

Lists: Premium 98

Time Complexity:  $O(N)$ , where  $N$  is the number of nodes in the list. Space Complexity:  $O(1)$  since the modification is done in-place.

### Problem

You are given the head of a linked list and two integers  $m$  and  $n$ .

Traverse the linked list and remove some nodes in the following way:

- Start with the head as the current node.
- Keep the first  $m$  nodes starting with the current node.
- Remove the next  $n$  nodes
- Keep repeating steps 2 and 3 until you reach the end of the list.

Return the head of the modified list after removing the mentioned nodes.

Example 1:



Input: head = [1,2,3,4,5,6,7,8,9,10,11,12,13],  $m = 2$ ,  $n = 3$   
Output: [1,2,6,7,11,12]  
Explanation: Keep the first ( $m = 2$ ) nodes starting from the head of the linked list (1 -> 2) show in black nodes.  
Delete the next ( $n = 3$ ) nodes (3 -> 4 -> 5) show in red nodes.  
Continue with the same procedure until reaching the tail of the Linked List.  
Head of the linked list after removing nodes is returned.

Example 2:



Input: head = [1,2,3,4,5,6,7,8,9,10,11],  $m = 1$ ,  $n = 3$   
Output: [1,5,9]  
Explanation: Head of linked list after removing nodes is returned.

### Constraints:

- The number of nodes in the list is in the range [1, 104].

Linked List

p.1484

Linked List

Follow up: Could you solve this problem by modifying the list in-place?

### Community Solutions (Python)

#### • WalkCC

```
class Solution:
 def deleteNodes(
 self,
 head: ListNode | None,
 m: int,
 n: int,
) -> ListNode | None:
 curr = head
 prev = None # prev.next == curr
```

```
while curr:
 # Set the m-th node as 'prev'.
 for _ in range(m):
 if not curr:
 break
 prev = curr
 curr = curr.next
 # Set the (m + n + 1)-th node as 'curr'.
 for _ in range(n):
 if not curr:
```

Linked List

p.1485

Linked List

p.1483

```
 break
 curr = curr.next
 # Delete the nodes [m + 1..n - 1].
 prev.next = curr

 return head

• LeetCode
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def deleteNodes(self, head: ListNode, m: int, n: int)
-> ListNode:
 pre = head
 while pre:
 for _ in range(m - 1):

 if pre:
 pre = pre.next
 if pre is None:
 return head
 cur = pre
 for _ in range(n):
 if cur:
 cur = cur.next
 pre.next = None if cur is None else cur.next
 pre = pre.next

 return head
```

```
• SimplyLeet
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

class Solution:
 def deleteNodes(self, head: ListNode, m: int, n: int)
-> ListNode:
 # Initialize the current pointer at the head of
 the list.
 current = head

 while current:
 # Keep m nodes: traverse m-1 next pointers
 for i in range(1, m):
 if current is None: # if list ends, break
 break
 current = current.next
 # If we have reached the end of the list,
 break the loop.
 if current is None:
 break
 # Start deletion from the next node after the
 m-th node.
```

```
 temp = current.next
 # Delete next n nodes.
 for j in range(n):
 if temp is None:
 break
 temp = temp.next
 # Connect the m-th node to the node after the
 n deleted nodes.
 current.next = temp
 # Move current pointer to continue from the
 next kept node.
 current = temp

 return head

• LC.ca
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def deleteNodes(self, head: ListNode, m: int, n: int)
-> ListNode:
 pre = head
 while pre:
 for _ in range(m - 1):
```

```
 if pre:
 pre = pre.next
 if pre is None:
 return head
 cur = pre
 for _ in range(n):
 if cur:
 cur = cur.next
 pre.next = None if cur is None else cur.next
 pre = pre.next

 return head
```

Linked List

#2 Add Two NumbersMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List · Math · Recursion

Lists: Grind 169

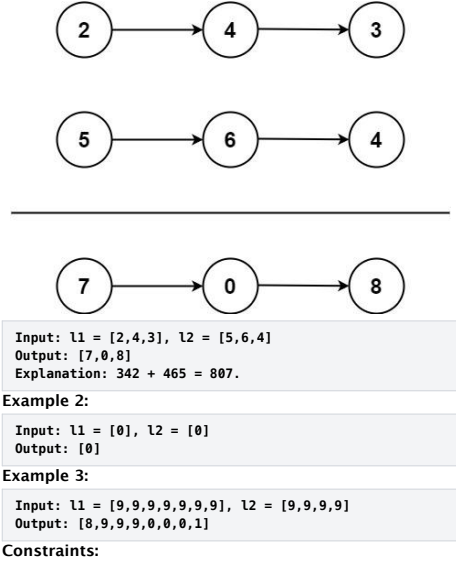
Time Complexity:  $O(\max(n, m))$ , where  $n$  and  $m$  are the lengths of the two linked lists. Space Complexity:  $O(\max(n, m) + 1)$  for the result list in the worst case.

Problem

You are given two non-empty linked lists representing two non-negative integers. The digits are stored in reverse order, and each of their nodes contains a single digit. Add the two numbers and return the sum as a linked list.

You may assume the two numbers do not contain any leading zero, except the number 0 itself.

Example 1:



• The number of nodes in each linked list is in the range  $[1, 100]$ .

•  $0 \leq \text{Node.val} \leq 9$

• It is guaranteed that the list represents a number that does not have leading zeros.

Brute Force (Python)

# Brute Force Approach

class Solution:

def addTwoNumbers(self, l1: ListNode, l2: ListNode):

# Brute Force - convert both lists to integers,

add, convert back

def to\_int(node):

num, mul = 0, 1

while node:

num += node.val \* mul

mul \*= 10

node = node.next

return num

total = to\_int(l1) + to\_int(l2)

dummy = curr = ListNode(0)

if total == 0: return ListNode(0)

while total:

curr.next = ListNode(total % 10)

curr = curr.next

total //= 10

return dummy.next

Community Solutions (Python)

• WalkCC

```
class Solution:
 def addTwoNumbers(self, l1: ListNode, l2: ListNode) ->
 ListNode:
 dummy = ListNode(0)
 curr = dummy
 carry = 0

 while carry or l1 or l2:
 if l1:
 carry += l1.val
 l1 = l1.next

 if l2:
 carry += l2.val
 l2 = l2.next
 curr.next = ListNode(carry % 10)
 carry //= 10
 curr = curr.next

 return dummy.next

• LeetCode
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def addTwoNumbers(
 self, l1: Optional[ListNode], l2:
 Optional[ListNode]
) -> Optional[ListNode]:
 dummy = ListNode()
```

```
 carry, curr = 0, dummy
 while l1 or l2 or carry:
 s = (l1.val if l1 else 0) + (l2.val if l2
 else 0) + carry
 carry, val = divmod(s, 10)
 curr.next = ListNode(val)
 curr = curr.next
 l1 = l1.next if l1 else None
 l2 = l2.next if l2 else None
 return dummy.next

• SimplyLeet
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

def addTwoNumbers(l1, l2):
 # Initialize a dummy head node to simplify the result
 list construction.
 dummy = ListNode()
 current = dummy
```

```
carry = 0

Traverse through both linked lists until all nodes
and any carry are processed.
while l1 or l2 or carry:
 # Get the current values; if a node is missing,
 use 0.
 val1 = l1.val if l1 else 0
 val2 = l2.val if l2 else 0

 # Calculate the sum of current digits and carry.
 total = val1 + val2 + carry

 carry = total // 10 # Update carry for next
iteration.
 digit = total % 10 # Compute current digit.

 # Append the digit as a new node in the result
list.
 current.next = ListNode(digit)
 current = current.next

 # Move to the next nodes if available.
 if l1:
 l1 = l1.next
 if l2:
 l2 = l2.next

 # Return the head of the newly formed linked list.
 return dummy.next
```

• LC.ca

Linked List p.1495

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def addTwoNumbers(
 self, l1: Optional[ListNode], l2:
Optional[ListNode]
) -> Optional[ListNode]:
 dummy = ListNode()

 carry, curr = 0, dummy
 while l1 or l2 or carry:
 s = (l1.val if l1 else 0) + (l2.val if l2
else 0) + carry
 carry, val = divmod(s, 10)
 curr.next = ListNode(val)
 curr = curr.next
 l1 = l1.next if l1 else None
 l2 = l2.next if l2 else None
 return dummy.next
```

Linked List p.1496

Linked List

#19 Remove Nth Node From End of List MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List · Two Pointers

Lists: Grind 169

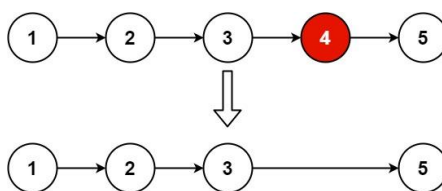
Time Complexity: O(L), where L is the length of the list

Space Complexity: O(1)

Problem

Given the head of a linked list, remove the nth node from the end of the list and return its head.

Example 1:



Input: head = [1,2,3,4,5], n = 2

Output: [1,2,3,5]

Example 2:

Linked List p.1497

```
Input: head = [1], n = 1
Output: []

Example 3:
Input: head = [1,2], n = 1
Output: [1]
```

- Constraints:
- The number of nodes in the list is sz.
  - 1 <= sz <= 30
  - 0 <= Node.val <= 100
  - 1 <= n <= sz
- Follow up: Could you do this in one pass?

Community Solutions (Python)

```
• WalkCC

class Solution:
 def removeNthFromEnd(self, head: ListNode, n: int) ->
ListNode:
 slow = head
 fast = head

 for _ in range(n):
 fast = fast.next
 if not fast:
 return head.next
```

Linked List p.1498

```
while fast.next:
 slow = slow.next
 fast = fast.next
 slow.next = slow.next.next

return head

• LeetDoocs

Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def removeNthFromEnd(self, head: Optional[ListNode],
n: int) -> Optional[ListNode]:
 dummy = ListNode(next=head)
 fast = slow = dummy
 for _ in range(n):

 fast = fast.next
 while fast.next:
 slow, fast = slow.next, fast.next
 slow.next = slow.next.next
 return dummy.next

• SimplyLeet
```

Linked List p.1499

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val # Holds the value of the node
 self.next = next # Points to the next node in the
list

class Solution:
 def removeNthFromEnd(self, head: ListNode, n: int) ->
ListNode:
 # Create a dummy node to simplify edge cases
 # (like removing the head)
 dummy = ListNode(0)

 dummy.next = head

 # Initialize two pointers starting at the dummy
node
 first = dummy
 second = dummy

 # Advance first pointer n+1 steps ahead to
maintain a gap of n nodes
 for _ in range(n + 1):
 first = first.next
```

Linked List p.1500

```
Move both pointers until first reaches the end
of the list
while first:
 first = first.next
 second = second.next

Skip the node to remove it from the list
second.next = second.next.next

Return the adjusted list starting from
dummy.next
return dummy.next
```

```
• LC.ca

Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def removeNthFromEnd(self, head: Optional[ListNode],
n: int) -> Optional[ListNode]:
 dummy = ListNode(next=head)
 fast = slow = dummy
 for _ in range(n):

 fast = fast.next
 while fast.next:
 slow, fast = slow.next, fast.next
 slow.next = slow.next.next
 return dummy.next
```

Linked List p.1501

Linked List

#24 Swap Nodes in Pairs MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List · Recursion

Lists: Grind 169

Time Complexity: O(n) – Each node is processed once.

Space Complexity: O(1) – Only constant extra space is used.

Problem

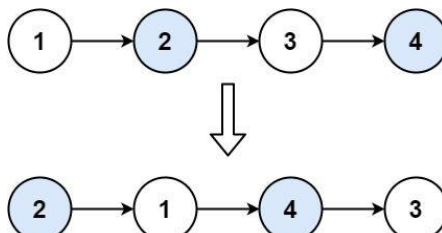
Given a linked list, swap every two adjacent nodes and return its head. You must solve the problem without modifying the values in the list's nodes (i.e., only nodes themselves may be changed.)

Example 1:

Input: head = [1,2,3,4]

Output: [2,1,4,3]

Explanation:



Linked List p.1502

Linked List

Example 2:

Input: head = []

Output: []

Example 3:

Input: head = [1]

Output: [1]

Example 4:

Input: head = [1,2,3]

Output: [2,1,3]

Constraints:

- The number of nodes in the list is in the range [0, 100].
- 0 <= Node.val <= 100

Community Solutions (Python)

Linked List p.1503

• WalkCC

```
class Solution:
 def swapPairs(self, head: ListNode) -> ListNode:
 def getLength(head: ListNode) -> int:
 length = 0
 while head:
 length += 1
 head = head.next
 return length

 length = getLength(head)

 dummy = ListNode(0, head)
 prev = dummy
 curr = head

 for _ in range(length // 2):
 next = curr.next
 curr.next = next.next
 prev.next = next
 prev = curr
 curr = curr.next

 return dummy.next
```

• LeetCode

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def swapPairs(self, head: Optional[ListNode]) -> Optional[ListNode]:
 if head is None or head.next is None:
 return head
 t = self.swapPairs(head.next.next)

 p = head.next
 p.next = head
 head.next = t
 return p
```

• SimplyLeet

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

class Solution:
 def swapPairs(self, head: ListNode) -> ListNode:
 # Create a dummy node and point it to the head of the list.
 dummy = ListNode(0)
```

```
dummy.next = head
Initialize the pointer 'prev' to dummy.
prev = dummy

Traverse the list while at least two nodes remain.
while head and head.next:
 # Identify the two nodes to swap.
 first_node = head
 second_node = head.next

 # Swapping the nodes by reassigning pointers.
 prev.next = second_node # Link previous node to second node.
 first_node.next = second_node.next # Link first node to node after second.
 second_node.next = first_node # Link second node to first to complete swap.

 # Update the pointers for next pair processing.
 prev = first_node # Move 'prev' pointer to the end of the swapped pair.
 head = first_node.next # Move 'head' pointer to the next pair.

Return the new head of the list.

return dummy.next
```

• LC.ca

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def swapPairs(self, head: Optional[ListNode]) -> Optional[ListNode]:
 dummy = ListNode(next=head)
 pre, cur = dummy, head
 while cur and cur.next:

 t = cur.next
 cur.next = t.next
 t.next = cur
 pre.next = t
 pre, cur = cur, cur.next
 return dummy.next
```

Linked List

#61 Rotate List

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List · Two Pointers

Lists: Grind 169

Time Complexity: O(n), where n is the number of nodes in the list, since we traverse the list a few times. Space Complexity: O(1), as the rotation is done in place without extra data structures.

Problem

Given the head of a linked list, rotate the list to the right by k places.

Example 1:

rotate 1

rotate 2

Input: head = [1,2,3,4,5], k = 2  
Output: [4,5,1,2,3]

Example 2:

rotate 1

rotate 2

rotate 3

rotate 4

Input: head = [0,1,2], k = 4  
Output: [2,0,1]

Constraints:

- The number of nodes in the list is in the range [0, 500].
- -100 <= Node.val <= 100

• 0 <= k <= 2 \* 109

Community Solutions (Python)

• WalkCC

```
class Solution:
 def rotateRight(self, head: ListNode, k: int) -> ListNode:
 if not head or not head.next or k == 0:
 return head

 tail = head
 length = 1
 while tail.next:
 tail = tail.next
 length += 1

 tail.next = head # Circle the list.

 t = length - k % length
 for _ in range(t):
 tail = tail.next
 newHead = tail.next
 tail.next = None

 return newHead
```

• LeetCode

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def rotateRight(self, head: Optional[ListNode], k: int) -> Optional[ListNode]:
 if head is None or head.next is None:
 return head
 cur, n = head, 0

 while cur:
 n += 1
 cur = cur.next
 k %= n
 if k == 0:
 return head
 fast = slow = head
 for _ in range(k):
 fast = fast.next
 while fast.next:
 fast, slow = fast.next, slow.next

 ans = slow.next
 slow.next = None
 fast.next = head
 return ans
```

• SimplyLeet

```
Definition for singly-linked list.
class ListNode:
def __init__(self, x):
self.val = x
self.next = None
class Solution:
 def rotateRight(self, head: ListNode, k: int) -> ListNode:
 if not head or not head.next or k == 0: # Check if rotation is necessary
 return head

 # Compute the length of the list and locate the tail node
 length = 1
 tail = head
 while tail.next:
 tail = tail.next
 length += 1

 # Calculate the effective number of rotations
 k = k % length
```

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------|
| <pre>        if k == 0:             return head          # Find the new tail: (length - k - 1) steps from the head         stepsToNewTail = length - k - 1         newTail = head         for _ in range(stepsToNewTail):             newTail = newTail.next          # Set the new head and break the list  newHead = newTail.next newTail.next = None  # Connect the old tail to the original head tail.next = head  return newHead</pre> <p>• LC.ca</p> <pre># Definition for singly-linked list. # class ListNode: #     def __init__(self, val=0, next=None): #         self.val = val #         self.next = next class Solution:     def rotateRight(self, head: Optional[ListNode], k: int) -&gt; Optional[ListNode]:         if head is None or head.next is None:             return head         cur, n = head, 0</pre>                                                                                                                                                                                                                                                                                                     | Linked List | p.1513 |
| <pre>while cur:     n += 1     cur = cur.next k %= n if k == 0:     return head fast = slow = head for _ in range(k):     fast = fast.next while fast.next:  fast, slow = fast.next, slow.next  ans = slow.next slow.next = None fast.next = head return ans</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Linked List | p.1514 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Linked List | p.1515 |
| <pre>Input ["LRUCache", "put", "put", "get", "put", "get", "put", "get", "get", "get"] [[2], [1, 1], [2, 2], [1], [3, 3], [2], [4, 4], [1], [3], [4]] Output [null, null, null, 1, null, -1, null, -1, 3, 4]</pre> <p>Explanation</p> <pre>LRUCache lruCache = new LRUCache(2);</pre> <p>Constraints:</p> <ul style="list-style-type: none"><li>• 1 &lt;= capacity &lt;= 3000</li><li>• 0 &lt;= key &lt;= 104</li><li>• 0 &lt;= value &lt;= 105</li><li>• At most 2 * 105 calls will be made to get and put.</li></ul> <p>Community Solutions (Python)</p> <p>• WalkCC</p> <pre>class Node:     def __init__(self, key: int, value: int):         self.key = key         self.value = value         self.prev = None         self.next = None  class LRUCache:     def __init__(self, capacity: int):</pre>                                                                                                                                                                                                                                                                                                                           | Linked List | p.1516 |
| <pre>        self.capacity = capacity # maximum capacity of the cache         self.cache = {} # dictionary mapping key -&gt; node         # Create dummy head and tail nodes.         self.head = Node(0, 0)         self.tail = Node(0, 0)         self.head.next = self.tail         self.tail.prev = self.head          # Helper function to remove a node from the linked list.         def _remove(self, node):              prev_node = node.prev             next_node = node.next             prev_node.next = next_node             next_node.prev = prev_node          # Helper function to add a node right after the head.         def _add(self, node):             node.prev = self.head             node.next = self.head.next             self.head.next.prev = node</pre>                                                                                                                                                                                                                                                                                                                                            | Linked List | p.1519 |
| <pre>        self.head.next = node          # Returns the value of the key if it exists; otherwise, returns -1.         def get(self, key):             if key in self.cache:                 node = self.cache[key]                 # Move the accessed node to the head (marking it as recently used)                 self._remove(node)                 self._add(node)                 return node.value              return -1          # Updates the value of the key if it exists; otherwise, adds the key-value pair.         # Removes the least recently used key if the cache exceeds its capacity.         def put(self, key, value):             if key in self.cache:                 # Remove the old node.                 self._remove(self.cache[key])                 node = Node(key, value)                 self._add(node)              self.cache[key] = node             if len(self.cache) &gt; self.capacity:                 # Remove the least recently used element.                 lru = self.tail.prev                 self._remove(lru)                 del self.cache[lru.key]</pre> <p>• LC.ca</p> | Linked List | p.1520 |
| <pre>class Node:     def __init__(self, key=0, val=0):         self.key = key         self.val = val         self.prev = None         self.next = None  class LRUCache:     def __init__(self, capacity: int):          self.cache = {} # key ==&gt; Node(val)         self.head = Node() # dummy node         self.tail = Node() # dummy node         self.capacity = capacity         self.size = 0         self.head.next = self.tail # note: key setup         self.tail.prev = self.head          def get(self, key: int) -&gt; int:             if key not in self.cache:                  return -1             node = self.cache[key]             self.move_to_head(node)             return node.val          def put(self, key: int, value: int) -&gt; None:             if key in self.cache:                 node = self.cache[key]                 node.val = value                 self.move_to_head(node)</pre>                                                                                                                                                                                                        | Linked List | p.1521 |

```
else:
 node = Node(key, value)
 self.cache[key] = node
 self.add_to_head(node)
 self.size += 1
 if self.size > self.capacity:
 tail = self.remove_tail()
 self.cache.pop(tail.key)
 self.size -= 1

def move_to_head(self, node):
 self.remove_node(node)
 self.add_to_head(node)

def remove_node(self, node):
 node.prev.next = node.next
 node.next.prev = node.prev

def add_to_head(self, node):
 node.next = self.head.next
 self.head.next = node
 node.prev = self.head
 self.head.next.prev = node

def remove_tail(self):
 node = self.tail.prev
 self.remove_node(node)
 return node
```

Linked List p.1522

```
Your LRUCache object will be instantiated and called as
such:
obj = LRUCache(capacity)
param_1 = obj.get(key)
obj.put(key,value)

#####

...

example:

>>> od = collections.OrderedDict()
>>>
>>> od[1]=1
>>> od[2]=2
>>> od[3]=3
>>>
>>> od
OrderedDict([(1, 1), (2, 2), (3, 3)])
>>> od.move_to_end(1)
>>> od
OrderedDict([(2, 2), (3, 3), (1, 1)])
>>>
>>> od.get(1)
1
>>> od.popitem()
(1, 1)
>>> od
OrderedDict([(2, 2), (3, 3)])
>>>
>>> od[1]=1
```

Linked List p.1523

```
>>> od
OrderedDict([(2, 2), (3, 3), (1, 1)])
>>>
>>> od.popitem(last=False)
(2, 2)
>>> od
OrderedDict([(3, 3), (1, 1)])
...

import collections

class LRUCache:
 def __init__(self, capacity: 'int'):
 self.cache = collections.OrderedDict()
 self.remain = capacity

 def get(self, key: 'int') -> 'int':
 if key not in self.cache:
 return -1

 self.cache.move_to_end(key) # meaning end is the
 most recently used
 return self.cache.get(key)

 def put(self, key: 'int', value: 'int') -> 'None':
 if key not in self.cache:
 if self.remain > 0:
 self.remain -= 1
 else:
 self.cache.popitem(last=False) # pop
 start position
 self.cache[key] = value
 self.remain += 1
```

Linked List p.1524

```
self.cache.pop(key)
self.cache[key] = value # add to end of dict,
meaning most recently used

#####

below solution with no ordered-dict
class DLinkedList:

 def __init__(self, key=0, value=0):

 self.key = key
 self.val = value
 self.next = None
 self.prev = None

class DLinkedList:

 def __init__(self):
 self.head = DLinkedList() # dummy head, its next
 is real head
```

Linked List p.1525

```
self.tail = DLinkedList() # dummy tail, its prev
is real tail
self.head.next, self.tail.prev = self.tail,
self.head

def add_node(self, node): # add to head
 node.prev, node.next = self.head, self.head.next
 node.next.prev, node.prev.next = node, node

def remove_node(self, node):
 node.prev.next, node.next.prev = node.next,
node.prev
return node.key

def remove_tail(self):
 return self.remove_node(self.tail.prev) # dummy
tail's prev is real tail

def move_to_head(self, node):
 self.remove_node(node)
 self.add_node(node)

def __repr__(self):
ans = []
```

Linked List p.1526

```
h = self.head
while h:
ans.append(str(h.val))
h = h.next
return '<DLinkedList: {}>'.format('-->'.join(ans))

class LRUCache:

 def __init__(self, capacity: int):
 self.capacity = capacity

 self.nodes_map = {}
 self.cache_list = DLinkedList()

 def get(self, key: int) -> int:
 node = self.nodes_map.get(key, None)
 if node:
 self.cache_list.move_to_head(node)
 return node.val
 else:
 return -1

 def put(self, key: int, value: int) -> None:
 node = self.nodes_map.get(key, None)
 if not node:
 if self.capacity > 0:
 self.capacity -= 1
 else:
 rm_key = self.cache_list.remove_tail()
 self.nodes_map.pop(rm_key) # note api
 new_node = DLinkedList(key, value)
```

Linked List p.1527

```
self.nodes_map[key] = new_node
self.cache_list.add_node(new_node)

else:
 self.cache_list.move_to_head(node)
 node.val = value
```

Linked List p.1528

Linked List

#148 Sort List MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List · Two Pointers · Divide and Conquer · Sorting · Merge Sort

Lists: Grind 169

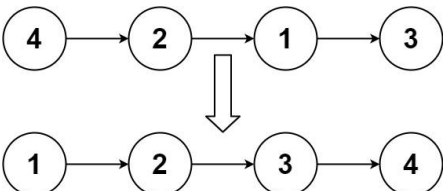
Time Complexity:  $O(n \log n)$  Space Complexity:  $O(1)$

extra space for an iterative bottom-up approach ( $O(\log n)$  for recursive calls if using recursion)

Problem

Given the head of a linked list, return the list after sorting it in ascending order.

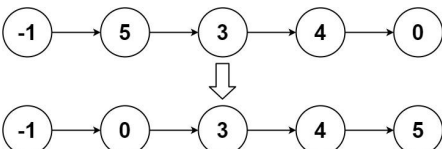
Example 1:



Linked List p.1529

Input: head = [4,2,1,3]  
Output: [1,2,3,4]

Example 2:



Input: head = [-1,5,3,4,0]  
Output: [-1,0,3,4,5]

Example 3:

Input: head = []  
Output: []

Constraints:

- The number of nodes in the list is in the range  $[0, 5 * 10^4]$ .
- $-105 \leq \text{Node.val} \leq 105$

Follow up: Can you sort the linked list in  $O(n \log n)$  time and  $O(1)$  memory (i.e. constant space)?

Community Solutions (Python)

- WalkCC

Linked List p.1530



```
class Solution:
 def sortList(self, head: ListNode) -> ListNode:
 def split(head: ListNode, k: int) -> ListNode:
 while k > 1 and head:
 head = head.next
 k -= 1
 rest = head.next if head else None
 if head:
 head.next = None
 return rest

 def merge(l1: ListNode, l2: ListNode) -> tuple:
 dummy = ListNode(0)
 tail = dummy

 while l1 and l2:
 if l1.val < l2.val:
 l1, l2 = l2, l1
 tail.next = l1
 l1 = l1.next
 else:
 tail.next = l2
 l2 = l2.next
 tail = tail.next

 return dummy.next, tail

 length = 0
 curr = head
 while curr:
 length += 1
 curr = curr.next

 dummy = ListNode(0, head)
 curr = dummy
 while curr:
 l = curr
 r = split(l, k)
 curr = split(r, k)
 mergedHead, mergedTail = merge(l, r)
 tail.next = mergedHead
 tail = mergedTail
 k -= 2

 return dummy.next
```

Linked List p.1531

```
length += 1
curr = curr.next

dummy = ListNode(0, head)

k = 1
while k < length:
 curr = dummy.next
 tail = dummy
 while curr:
 l = curr
 r = split(l, k)
 curr = split(r, k)
 mergedHead, mergedTail = merge(l, r)
 tail.next = mergedHead
 tail = mergedTail
 k -= 2

return dummy.next
```

• LeetCode

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def sortList(self, head: Optional[ListNode]) -> Optional[ListNode]:
 if head is None or head.next is None:
 return head
 slow, fast = head, head.next
```

Linked List p.1532

```
while fast and fast.next:
 slow = slow.next
 fast = fast.next.next
l1, l2 = head, slow.next
slow.next = None
l1, l2 = self.sortList(l1), self.sortList(l2)
dummy = ListNode()
tail = dummy
while l1 and l2:
 if l1.val <= l2.val:
 tail.next = l1
 l1 = l1.next
 else:
 tail.next = l2
 l2 = l2.next
 tail = tail.next
tail.next = l1 or l2
return dummy.next
```

• SimplyLeet

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

class Solution:
 def sortList(self, head: ListNode) -> ListNode:
 # Base case: if list is empty or contains a
 # single element.
 if not head or not head.next:
```

Linked List p.1533

```
return head

Function to split the list into two halves
def getMid(head):
 slow, fast = head, head.next
 # Move fast twice as fast as slow to locate
 middle

 while fast and fast.next:
 slow = slow.next
 fast = fast.next.next
 return slow

Split the linked list into two halves
mid = getMid(head)
right_head = mid.next
mid.next = None # cut the list into two parts

Recursive calls to sort each half
left = self.sortList(head)
right = self.sortList(right_head)

Merge two sorted lists
return merge(left, right)

def merge(l1, l2):
 dummy = ListNode(0)
 tail = dummy
 # Merge the lists by comparing nodes
 while l1 and l2:
 if l1.val < l2.val:
 tail.next = l1
 l1 = l1.next
 else:
 tail.next = l2
 l2 = l2.next
 tail = tail.next
 if l1:
 tail.next = l1
 if l2:
 tail.next = l2
 return tail.next
```

Linked List p.1534

```
l1 = l1.next
else:
 tail.next = l2
 l2 = l2.next
 tail = tail.next
Append the remaining nodes
tail.next = l1 if l1 else l2
return dummy.next
```

• LC.ca

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def sortList(self, head: ListNode) -> ListNode:
 if head is None or head.next is None:
 return head
 slow, fast = head, head.next

 while fast and fast.next:
 slow, fast = slow.next, fast.next.next
 t = slow.next
 l1, l2 = self.sortList(head), self.sortList(t)
 dummy = ListNode()
 cur = dummy
 while l1 and l2:
 if l1.val <= l2.val:
 cur.next = l1
 l1 = l1.next
 else:
 cur.next = l2
 l2 = l2.next
 cur = cur.next
 if l1:
 cur.next = l1
 if l2:
 cur.next = l2
 return cur.next
```

Linked List p.1535

```
l1 = l1.next
else:
 cur.next = l2
 l2 = l2.next
 cur = cur.next
cur.next = l1 or l2 # add the rest
return dummy.next
```

Linked List p.1536

Linked List

#328 Odd Even Linked List

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List

Lists: Grind 169

Time Complexity: O(n) since the list is traversed once.

Space Complexity: O(1) as only a few pointers are used.

Problem

Given the head of a singly linked list, group all the nodes with odd indices together followed by the nodes with even indices, and return the reordered list. The first node is considered odd, and the second node is even, and so on.

Note that the relative order inside both the even and odd groups should remain as it was in the input.

You must solve the problem in O(1) extra space complexity and O(n) time complexity.

Example 1:

Linked List p.1537

Input: head = [1,2,3,4,5]  
Output: [1,3,5,2,4]

Example 2:

Input: head = [2,1,3,5,6,4,7]  
Output: [2,3,6,7,1,5,4]

Constraints:

- The number of nodes in the linked list is in the range [0, 104].
- 106 <= Node.val <= 106

Community Solutions (Python)

Linked List p.1538

• WalkCC

```
class Solution:
 def oddEvenList(self, head: ListNode) -> ListNode:
 oddHead = ListNode(0)
 evenHead = ListNode(0)
 odd = oddHead
 even = evenHead
 isOdd = True

 while head:
 if isOdd:
 odd.next = head
 odd = head
 else:
 even.next = head
 even = head
 head = head.next
 isOdd = not isOdd

 even.next = None
 odd.next = evenHead.next

 return oddHead.next
```

• LeetCode

Linked List p.1539

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre># Definition for singly-linked list. # class ListNode: # def __init__(self, val=0, next=None): # self.val = val # self.next = next class Solution:     def oddEvenList(self, head: Optional[ListNode]) -&gt; Optional[ListNode]:         if head is None:             return None         a = head          b = c = head.next         while b and b.next:             a.next = b.next             a = a.next             b.next = a.next             b = b.next             a.next = c             return head</pre> <p>• SimplyLeet</p> <pre># Definition for singly-linked list. class ListNode:     def __init__(self, val=0, next=None):         self.val = val         self.next = next  class Solution:     def oddEvenList(self, head: ListNode) -&gt; ListNode:         # Check for empty list or list with one node.         if not head or not head.next:</pre> |        | <pre>        return head          odd = head # Point to the first (odd indexed)         node.         even = head.next # Point to the second (even         indexed) node.         even_head = even # Save the head of the even         list.          # Reorder the list by rearranging next pointers.         while even and even.next:             odd.next = even.next # Connect current odd to             next odd.             odd = odd.next # Move odd pointer.              even.next = odd.next # Connect current even             to next even.             even = even.next # Move even pointer.              # Append the even list after the odd list.             odd.next = even_head             return head</pre> <p>• LC.ca</p> |        | <pre># Definition for singly-linked list. # class ListNode: # def __init__(self, val=0, next=None): # self.val = val # self.next = next class Solution:     def oddEvenList(self, head: Optional[ListNode]) -&gt; Optional[ListNode]:         if head is None:             return None         a = head          b = c = head.next         while b and b.next:             a.next = b.next             a = a.next             b.next = a.next             b = b.next             a.next = c             return head</pre> |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | p.1540 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | p.1541 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1542 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <div>Linked List</div> <div>#369 Plus One Linked ListMED</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Linked List · Math · Recursion</div> <div>Lists: Premium 98</div> <div>Time Complexity: O(n), where n is the number of nodes in the list<br/>Space Complexity: O(1) extra space when using an iterative reversal method. (O(n) if recursion is used)</div> <div>Problem</div> <div>Given a non-negative integer represented as a linked list of digits, plus one to the integer.</div> <div>The digits are stored such that the most significant digit is at the head of the list.</div> <div>Example 1:</div> <div>Input: head = [1,2,3]<br/>Output: [1,2,4]</div> <div>Example 2:</div> <div>Input: head = [0]<br/>Output: [1]</div> <div>Constraints:</div> <ul style="list-style-type: none"><li>The number of nodes in the linked list is in the range [1, 100].</li><li>0 &lt;= Node.val &lt;= 9</li></ul> |        | <p>• The number represented by the linked list does not contain leading zeros except for the zero itself.</p> <p>Brute Force (Python)</p> <pre># Brute Force Approach class Solution:     def reverseList(self, head):         # Brute Force - collect into list, manipulate,         rebuild          vals = []         curr = head         while curr:             vals.append(curr.val)             curr = curr.next         dummy = ListNode(0)          curr = dummy         for v in vals:             curr.next = ListNode(v)             curr = curr.next         return dummy.next</pre> <p>Community Solutions (Python)</p> <p>• WalkCC</p> |        | <pre>class Solution:     def plusOne(self, head: ListNode) -&gt; ListNode:         if not head:             return ListNode(1)         if self._addOne(head) == 1:             return ListNode(1, head)         return head      def _addOne(self, node: ListNode) -&gt; int:         carry = self._addOne(node.next) if node.next else 1          summ = node.val + carry         node.val = summ % 10         return summ // 10  class Solution:     def plusOne(self, head: ListNode) -&gt; ListNode:         dummy = ListNode(0)         curr = dummy         dummy.next = head          while head:             if head.val != 9:                 curr = head                 head = head.next</pre> |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | p.1543 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | p.1544 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1545 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre># `curr` now points to the rightmost non-9 node.  curr.val += 1 while curr.next:     curr.next.val = 0     curr = curr.next  return dummy.next if dummy.val == 0 else dummy</pre> <p>• LeetCode</p> <pre># Definition for singly-linked list. # class ListNode: # def __init__(self, val=0, next=None): # self.val = val # self.next = next class Solution:     def plusOne(self, head: Optional[ListNode]) -&gt; Optional[ListNode]:         dummy = ListNode(0, head)         target = dummy         while head:             if head.val != 9:                 target = head                 head = head.next             target.val += 1             target = target.next         while target:             target.val = 0             target = target.next         return dummy if dummy.val else dummy.next</pre> <p>• SimplyLeet</p> |        | <pre># Definition for singly-linked list. class ListNode:     def __init__(self, val=0, next=None):         self.val = val         self.next = next  # Helper function to reverse the linked list def reverseList(head):     prev = None     curr = head      while curr:         nxt = curr.next # store next node         curr.next = prev # reverse pointer         prev = curr # move prev forward         curr = nxt # move to next node     return prev  def plusOne(head: ListNode) -&gt; ListNode:     # Reverse the linked list to start addition from the     least significant digit     head = reverseList(head)</pre> |        | <pre>        current = head         carry = 1 # initialize carry as the plus one         while current and carry:             current.val += carry # add carry to the current             node              carry = current.val // 10 # calculate new carry             current.val %= 10 # update the node's value             # If we're at the end and there's still a carry,             create a new node.             if not current.next and carry:                 current.next = ListNode(0)              current = current.next          # Reverse the list back to its original order before         returning.         return reverseList(head)</pre> <p>• LC.ca</p> <pre># Definition for singly-linked list. # class ListNode: # def __init__(self, val=0, next=None): # self.val = val # self.next = next class Solution:     def plusOne(self, head: ListNode) -&gt; ListNode:         dummy = ListNode(0, head)         target = dummy         while head:</pre> |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | p.1546 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | p.1547 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | p.1548 |

```
if head.val != 9:
 target = head
 head = head.next
target.val += 1
target = target.next
while target:
 target.val = 0
 target = target.next
return dummy if dummy.val else dummy.next
```

Linked List

#430 Flatten a Multilevel Doubly Linked List

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List · DFS · Doubly Linked List

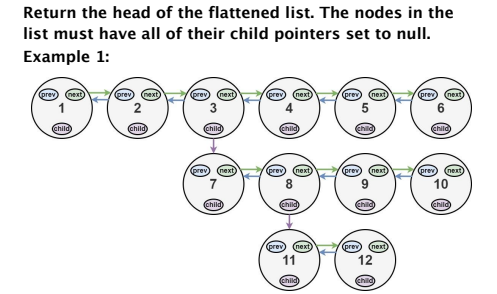
Lists: Premium 98

Time Complexity:  $O(n)$ , where  $n$  is the total number of nodes, since each node is processed exactly once. Space Complexity:  $O(n)$  in the worst-case due to recursion stack in case of a highly nested list.

**Problem**

You are given a doubly linked list, which contains nodes that have a next pointer, a previous pointer, and an additional child pointer. This child pointer may or may not point to a separate doubly linked list, also containing these special nodes. These child lists may have one or more children of their own, and so on, to produce a multilevel data structure as shown in the example below.

Given the head of the first level of the list, flatten the list so that all the nodes appear in a single-level, doubly linked list. Let curr be a node with a child list. The nodes in the child list should appear after curr and before curr.next in the flattened list.



Input: head = [1,2,3,4,5,6,null,null,null,7,8,9,10,null,null,11,12]  
Output: [1,2,3,7,8,11,12,9,10,4,5,6]  
Explanation: The multilevel linked list in the input is shown.  
After flattening the multilevel linked list it becomes:

Example 2:

Input: head = [1,2,null,3]  
Output: [1,3,2]  
Explanation: The multilevel linked list in the input is shown.  
After flattening the multilevel linked list it becomes:

Example 3:

Input: head = []  
Output: []  
Explanation: There could be empty list in the input.

**Constraints:**

- The number of Nodes will not exceed 1000.
- $1 \leq \text{Node.val} \leq 105$

How the multilevel linked list is represented in test cases:

We use the multilevel linked list from Example 1 above:

```
1---2---3---4---5---6--NULL
|
7---8---9---10--NULL
|
11--12--NULL
```

The serialization of each level is as follows:

```
[1,2,3,4,5,6,null]
[7,8,9,10,null]
[11,12,null]
```

To serialize all levels together, we will add nulls in each level to signify no node connects to the upper node of the previous level. The serialization becomes:

```
[1, 2, 3, 4, 5, 6, null]
|
[null, null, 7, 8, 9, 10, null]
|
[null, 11, 12, null]
```

**Merging the serialization of each level and removing trailing nulls we obtain:**

```
[1,2,3,4,5,6,null,null,null,7,8,9,10,null,null,11,12]
```

**Brute Force (Python)**

```
Brute Force Approach
class Solution:
 def flatten(self, head):
 # Brute Force - collect into list, manipulate, rebuild
 vals = []
 curr = head
 while curr:
 vals.append(curr.val)
 curr = curr.next
 dummy = ListNode(0)
 curr = dummy
 for v in vals:
 curr.next = ListNode(v)
 curr = curr.next
 return dummy.next
```

**Community Solutions (Python)**

- WalkCC

```
class Solution:
 def flatten(self, head: 'Node') -> 'Node':
 def flatten(head: 'Node', rest: 'Node') -> 'Node':
 if not head:
 return rest
 head.next = flatten(head.child, flatten(head.next, rest))
 if head.next:
 head.next.prev = head
 head.child = None
 return head
 return flatten(head, None)
```

```
class Solution:
 def flatten(self, head: 'Node') -> 'Node':
 curr = head
 while curr:
 if curr.child:
 cachedNext = curr.next
 curr.next = curr.child
 curr.child.prev = curr
 curr.child = None
```

```
tail = curr.next
while tail.next:
 tail = tail.next
tail.next = cachedNext
if cachedNext:
 cachedNext.prev = tail
curr = curr.next

return head
```

• LeetCode

```
"""
Definition for a Node.
class Node:
 def __init__(self, val, prev, next, child):
 self.val = val
 self.prev = prev
 self.next = next
 self.child = child
"""

class Solution:
 def flatten(self, head: 'Node') -> 'Node':
 def preorder(pre, cur):
 if cur is None:
 return pre
 cur.prev = pre
 pre.next = cur
 t = cur.next
```

```
tail = preorder(cur, cur.child)
cur.child = None
return preorder(tail, t)
```

if head is None:
 return None
dummy = Node(0, None, head, None)
preorder(dummy, head)
dummy.next.prev = None
return dummy.next

• SimplyLeet

```
Definition for a Node.
class Node:
 def __init__(self, val, prev=None, next=None, child=None):
 self.val = val
 self.prev = prev
 self.next = next
 self.child = child

def flatten(head):
 # If the list is empty, simply return None.
```

```
if not head:
 return head

Helper function to perform DFS and return the tail of the flattened list.
def dfs(node):
 current = node
 last = None
 while current:
 next_node = current.next # Store current.next before modifications
 # If current node has a child, we flatten the child list recursively.
 if current.child:
 # Recursively flatten the child list.
 child_tail = dfs(current.child)
 # Connect current node with the start of the child list.
 current.next = current.child
 current.child.prev = current
 # If there was a next node, attach it after the child list.
 if next_node:
 child_tail.next = next_node
 next_node.prev = child_tail
```

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------|
| <div><pre># Set child pointer to None as required. current.child = None # Update the tail pointer. last = child_tail  else:     last = current # Move to the next node (which might have been updated after flattening a child). current = next_node return last</pre></div> <div><pre>dfs(head) # Begin DFS traversal from the head. return head  # Example to illustrate usage: # head = Node(1) # head.next = Node(2) # head.child = Node(3) # flatten(head)</pre></div> <div><div>LC.ca</div><pre>""" # Definition for a Node. class Node:     def __init__(self, val, prev, next, child):         self.val = val         self.prev = prev         self.next = next         self.child = child """</pre></div>                                                                                                                                                                                            | Linked List | p.1558 |
| <div><pre>class Solution:     def flatten(self, head: 'Node') -&gt; 'Node':         def preorder(pre, cur):             if cur is None:                 return pre             cur.prev = pre             pre.next = cur              t = cur.next              tail = preorder(cur, cur.child)             cur.child = None             return preorder(tail, t)          if head is None:             return None         dummy = Node(0, None, head, None)         preorder(dummy, head)         dummy.next.prev = None         return dummy.next</pre></div>                                                                                                                                                                                                                                                                                                                                              | Linked List | p.1559 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Linked List | p.1560 |
| <div><ul style="list-style-type: none"><li>MyCircularQueue(k) Initializes the object with the size of the queue to be k.</li><li>int Front() Gets the front item from the queue. If the queue is empty, return -1.</li><li>int Rear() Gets the last item from the queue. If the queue is empty, return -1.</li><li>boolean enqueue(int value) Inserts an element into the circular queue. Return true if the operation is successful.</li><li>boolean dequeue() Deletes an element from the circular queue. Return true if the operation is successful.</li><li>boolean isEmpty() Checks whether the circular queue is empty or not.</li><li>boolean isFull() Checks whether the circular queue is full or not.</li></ul></div> <div>You must solve the problem without using the built-in queue data structure in your programming language.</div> <div>Example 1:</div>                                     | Linked List | p.1561 |
| <div><pre>Input ["MyCircularQueue", "enqueue", "enqueue", "enqueue", "enqueue", "Rear", "isFull", "dequeue", "enqueue", "Rear"] [[3], [1], [2], [3], [4], [], [], [], [4], []] Output [null, true, true, true, false, 3, true, true, true, 4]</pre></div> <div><pre>Explanation MyCircularQueue myCircularQueue = new MyCircularQueue(3);</pre></div> <div>Constraints:</div> <ul style="list-style-type: none"><li>1 &lt;= k &lt;= 1000</li><li>0 &lt;= value &lt;= 1000</li><li>At most 3000 calls will be made to enqueue, dequeue, Front, Rear, isEmpty, and isFull.</li></ul> <div>Community Solutions (Python)</div> <div><div>LeetCode</div><pre>class MyCircularQueue:      def __init__(self, k: int):         self.q = [0] * k         self.size = 0         self.capacity = k         self.front = 0          def enqueue(self, value: int) -&gt; bool:             if self.isFull():</pre></div>  | Linked List | p.1562 |
| <div><pre>def isEmpty(self) -&gt; bool:     return self.size == 0  def isFull(self) -&gt; bool:     return self.size == self.capacity</pre></div> <div><pre># Your MyCircularQueue object will be instantiated and called as such: # obj = MyCircularQueue(k) # param_1 = obj.enqueue(value)  # param_2 = obj.dequeue() # param_3 = obj.front() # param_4 = obj.rear() # param_5 = obj.isEmpty() # param_6 = obj.isFull()</pre></div> <div><div>SimplyLeet</div><pre># Python implementation of MyCircularQueue  class MyCircularQueue:     def __init__(self, k):         # Initialize the queue with capacity k         self.capacity = k         self.queue = [0] * k # fixed-size array         self.front = 0 # pointer to the front element         self.rear = 0 # pointer to the next insertion         position         self.count = 0 # current number of elements in         the queue</pre></div> | Linked List | p.1564 |
| <div><pre>def enqueue(self, value):     # Check if the queue is full     if self.isFull():         return False     # Insert element at rear pointer and update     pointer using modulo for circularity     self.queue[self.rear] = value     self.rear = (self.rear + 1) % self.capacity     self.count += 1     return True</pre></div> <div><pre>def dequeue(self):     # Check if the queue is empty     if self.isEmpty():         return False     # Move front pointer to the next element using     modulo for circularity     self.front = (self.front + 1) % self.capacity     self.count -= 1     return True</pre></div>                                                                                                                                                                                                                                                                         | Linked List | p.1565 |
| <div><pre>def Front(self):     # Return -1 if the queue is empty, else return     the front element     if self.isEmpty():         return -1     return self.queue[self.front]  def Rear(self):     # Return -1 if the queue is empty     if self.isEmpty():         return -1      # Since rear points to the next insertion     position, get the last element index     return self.queue[(self.rear - 1 + self.capacity)     % self.capacity]  def isEmpty(self):     # Check if count is 0     return self.count == 0  def isFull(self):     # Check if the queue has reached its capacity     return self.count == self.capacity</pre></div> <div><div>LC.ca</div></div>                                                                                                                                                                                                                                | Linked List | p.1566 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |             |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|--------|
| <pre>class MyCircularQueue:     def __init__(self, k: int):         self.q = [0] * k         self.front = 0         self.size = 0         self.capacity = k      def enqueue(self, value: int) -&gt; bool:         if self.isFull():             return False          idx = (self.front + self.size) % self.capacity         self.q[idx] = value         self.size += 1         return True      def dequeue(self) -&gt; bool:         if self.isEmpty():             return False         self.front = (self.front + 1) % self.capacity         self.size -= 1</pre> | <pre>        return True      def Front(self) -&gt; int:         return -1 if self.isEmpty() else         self.q[self.front]      def Rear(self) -&gt; int:         if self.isEmpty():             return -1         idx = (self.front + self.size - 1) %         self.capacity         return self.q[idx]      def isEmpty(self) -&gt; bool:         return self.size == 0      def isFull(self) -&gt; bool:         return self.size == self.capacity  # Your MyCircularQueue object will be instantiated and called as such: # obj = MyCircularQueue(k)</pre> | <pre># param_1 = obj.enqueue(value) # param_2 = obj.dequeue() # param_3 = obj.Front() # param_4 = obj.Rear() # param_5 = obj.isEmpty() # param_6 = obj.isFull()  #####  class MyCircularQueue(object):      def __init__(self, k):         """         Initialize your data structure here. Set the size of the queue to be k. :type k: int         """         self.queue = []         self.size = k         self.front = 0         self.rear = 0</pre> |        |             |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | p.1567                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                              | p.1568 | Linked List | p.1569 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |             |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|--------|
| <pre>def enqueue(self, value):     """     Insert an element into the circular queue. Return true if the operation is successful. :type value: int :rtype: bool     """     if self.rear - self.front &lt; self.size:         self.queue.append(value)         self.rear += 1          return True     else:         return False  def dequeue(self):     """     Delete an element from the circular queue. Return true if the operation is successful. :rtype: bool     """     if self.rear - self.front &gt; 0:</pre> | <pre>        self.front += 1         return True     else:         return False  def Front(self):     """     Get the front item from the queue. :rtype: int     """      if self.isEmpty():         return -1     else:         return self.queue[self.front]  def Rear(self):     """     Get the last item from the queue. :rtype: int     """      if self.isEmpty():         return -1     else:         return self.queue[self.rear - 1]  def isEmpty(self):     """     Checks whether the circular queue is empty or not. :rtype: bool     """</pre> | <pre>        return self.front == self.rear  def isFull(self):     """     Checks whether the circular queue is full or not. :rtype: bool     """     return self.rear - self.front == self.size  # Your MyCircularQueue object will be instantiated and called as such: # obj = MyCircularQueue(k) # param_1 = obj.enqueue(value) # param_2 = obj.dequeue() # param_3 = obj.Front() # param_4 = obj.Rear() # param_5 = obj.isEmpty() # param_6 = obj.isFull()</pre> |        |             |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1570                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                          | p.1571 | Linked List | p.1572 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |             |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------|--------|
| <div>Linked List</div> <div>#707 Design Linked List</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Linked List · Design</div> <div>Lists: Denny Zhang</div> <div>Time Complexity:</div> <div>Problem</div> <div>Design your implementation of the linked list. You can choose to use a singly or doubly linked list. A node in a singly linked list should have two attributes: val and next. val is the value of the current node, and next is a pointer/reference to the next node. If you want to use the doubly linked list, you will need one more attribute prev to indicate the previous node in the linked list. Assume all nodes in the linked list are 0-indexed. Implement the MyLinkedList class:</div> <div><ul style="list-style-type: none"><li>MyLinkedList() Initializes the MyLinkedList object.</li><li>int get(int index) Get the value of the indexth node in the linked list. If the index is invalid, return -1.</li><li>void addAtHead(int val) Add a node of value val before the first element of the linked list. After the insertion, the new node will be the first node of the linked list.</li></ul></div> | <div><ul style="list-style-type: none"><li>void addAtTail(int val) Append a node of value val as the last element of the linked list.</li><li>void addAtIndex(int index, int val) Add a node of value val before the indexth node in the linked list. If index equals the length of the linked list, the node will be appended to the end of the linked list. If index is greater than the length, the node will not be inserted.</li><li>void deleteAtIndex(int index) Delete the indexth node in the linked list, if the index is valid.</li></ul></div> <div>Example 1:</div> <div><div>Input["MyLinkedList", "addAtHead", "addAtTail", "addAtIndex", "get", "deleteAtIndex", "get"][[], [1], [3], [1, 2], [1], [1], [1]]</div><div>Output[null, null, null, null, 2, null, 3]</div><div>ExplanationMyLinkedList myLinkedList = new MyLinkedList();</div></div> <div>Constraints:</div> <div><ul style="list-style-type: none"><li>0 &lt;= index, val &lt;= 1000</li><li>Please do not use the built-in Linked List library.</li><li>At most 2000 calls will be made to get, addAtHead, addAtTail, addAtIndex and deleteAtIndex.</li></ul></div> <div>Community Solutions (Python)</div> <div><ul style="list-style-type: none"><li>WalkCC</li></ul></div> | <div>from dataclasses import dataclass</div> <div>@dataclassclass ListNode:    val: int    next: ListNode   None = None</div> <div>class MyLinkedList:</div> <div>    def __init__(self):        self.length = 0        self.dummy = ListNode(0)      def get(self, index: int) -&gt; int:        if index &lt; 0 or index &gt;= self.length:            return -1        curr = self.dummy.next        for _ in range(index):            curr = curr.next          return curr.val      def addAtHead(self, val: int) -&gt; None:        curr = self.dummy.next        self.dummy.next = ListNode(val)        self.dummy.next.next = curr        self.length += 1      def addAtTail(self, val: int) -&gt; None:        curr = self.dummy</div> |        |             |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | p.1573                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | p.1574 | Linked List | p.1575 |

```

while curr.next:
 curr = curr.next
curr.next = ListNode(val)
self.length += 1

def addAtIndex(self, index: int, val: int) -> None:
 if index > self.length:
 return
 curr = self.dummy
 for _ in range(index):

 curr = curr.next
 temp = curr.next
 curr.next = ListNode(val)
 curr.next.next = temp
 self.length += 1

def deleteAtIndex(self, index: int) -> None:
 if index < 0 or index >= self.length:
 return
 curr = self.dummy

 for _ in range(index):
 curr = curr.next
 temp = curr.next
 curr.next = temp.next
 self.length -= 1

```

• LeetCode

Linked List

p.1576

```

class MyLinkedList:
 def __init__(self):
 self.dummy = ListNode()
 self.cnt = 0

 def get(self, index: int) -> int:
 if index < 0 or index >= self.cnt:
 return -1
 cur = self.dummy.next
 for _ in range(index):

 cur = cur.next
 return cur.val

 def addAtHead(self, val: int) -> None:
 self.addAtIndex(0, val)

 def addAtTail(self, val: int) -> None:
 self.addAtIndex(self.cnt, val)

 def addAtIndex(self, index: int, val: int) -> None:

 if index > self.cnt:
 return
 pre = self.dummy
 for _ in range(index):
 pre = pre.next
 pre.next = ListNode(val, pre.next)
 self.cnt += 1

 def deleteAtIndex(self, index: int) -> None:
 if index >= self.cnt:

```

Linked List

p.1577

```

 return
 pre = self.dummy
 for _ in range(index):
 pre = pre.next
 t = pre.next
 pre.next = t.next
 t.next = None
 self.cnt -= 1

Your MyLinkedList object will be instantiated and
called as such:
obj = MyLinkedList()
param_1 = obj.get(index)
obj.addAtHead(val)
obj.addAtTail(val)
obj.addAtIndex(index, val)
obj.deleteAtIndex(index)

• SimplyLeet

Define a node for the singly linked list
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val # The value of the node
 self.next = next # Pointer to the next node

class MyLinkedList:
 def __init__(self):
 # Initialize size and a dummy head node to
simplify operations
 self.size = 0

```

Linked List

p.1578

```

 self.head = ListNode(0) # dummy head

 def get(self, index: int) -> int:
 # If index is out of bounds, return -1
 if index < 0 or index >= self.size:
 return -1
 current = self.head.next # start from the actual
head
 for _ in range(index):
 current = current.next
 return current.val

 def addAtHead(self, val: int) -> None:
 self.addAtIndex(0, val) # add at index 0

 def addAtTail(self, val: int) -> None:
 self.addAtIndex(self.size, val) # add at the end

 def addAtIndex(self, index: int, val: int) -> None:
 # Normalize negative index to 0 (insert at head)
 if index < 0:

 index = 0
 # Only insert if index is valid (<= size)
 if index > self.size:
 return
 self.size += 1
 # Start from the dummy head node
 pre = self.head
 for _ in range(index):
 pre = pre.next
 # Insert new node

```

Linked List

p.1579

```

 new_node = ListNode(val)
 new_node.next = pred.next
 pred.next = new_node

 def deleteAtIndex(self, index: int) -> None:
 # If index is out of bounds, do nothing
 if index < 0 or index >= self.size:
 return
 self.size -= 1
 pred = self.head

 for _ in range(index):
 pred = pred.next
 # Remove the node by skipping it
 pred.next = pred.next.next

• LC.ca

doubly linked node
class ListNode:
 def __init__(self, value=0, prev=None, next=None):
 self.value = value
 self.prev = prev
 self.next = next

class MyLinkedList:
 def __init__(self):
 self.head = ListNode(0) # Sentinel node as
pseudo-head

```

Linked List

p.1580

```

 self.tail = ListNode(0) # Sentinel node as
pseudo-tail
 self.head.next = self.tail
 self.tail.prev = self.head
 self.size = 0

 def get(self, index: int) -> int:
 if index < 0 or index >= self.size:
 return -1
 if index + 1 < self.size - index: # decide start
from head or tail
 curr = self.head

 for _ in range(index + 1):
 curr = curr.next
 else:
 curr = self.tail
 for _ in range(self.size - index):
 curr = curr.prev
 return curr.value

 def addAtHead(self, val: int) -> None:
 pred, succ = self.head, self.head.next

```

Linked List

p.1581

```

 self.size += 1
 to_add = ListNode(val, pred, succ)
 pred.next = to_add
 succ.prev = to_add

 def addAtTail(self, val: int) -> None:
 succ, pred = self.tail, self.tail.prev
 self.size += 1
 to_add = ListNode(val, pred, succ)
 pred.next = to_add

 succ.prev = to_add

 def addAtIndex(self, index: int, val: int) -> None:
 if index > self.size:
 return
 if index < 0:
 index = 0
 if index < self.size - index: # decide start from
head or tail
 pred = self.head
 for _ in range(index):

 pred = pred.next
 succ = pred.next
 else:
 succ = self.tail
 for _ in range(self.size - index):
 succ = succ.prev
 pred = succ.prev
 self.size += 1
 to_add = ListNode(val, pred, succ)
 pred.next = to_add

```

Linked List

p.1582

```

 succ.prev = to_add

 def deleteAtIndex(self, index: int) -> None:
 if index < 0 or index >= self.size:
 return
 if index < self.size - index:
 pred = self.head
 for _ in range(index):
 pred = pred.next
 succ = pred.next.next

 else:
 succ = self.tail
 for _ in range(self.size - index - 1):
 succ = succ.prev
 pred = succ.prev.prev
 self.size -= 1
 pred.next = succ
 succ.prev = pred

Your MyLinkedList object will be instantiated and
called as such:

obj = MyLinkedList()
param_1 = obj.get(index)
obj.addAtHead(val)
obj.addAtTail(val)
obj.addAtIndex(index, val)
obj.deleteAtIndex(index)

#####

singly linked node

```

Linked List

p.1583

```

class MyLinkedList:
 def __init__(self):
 self.dummy = ListNode()
 self.cnt = 0

 def get(self, index: int) -> int:
 if index < 0 or index >= self.cnt:
 return -1
 cur = self.dummy.next
 for _ in range(index):

 cur = cur.next
 return cur.val

 def addAtHead(self, val: int) -> None:
 self.addAtIndex(0, val)

 def addAtTail(self, val: int) -> None:
 self.addAtIndex(self.cnt, val)

 def addAtIndex(self, index: int, val: int) -> None:

 if index > self.cnt:
 return
 pre = self.dummy
 for _ in range(index):
 pre = pre.next
 pre.next = ListNode(val, pre.next) # insert
in-between
 self.cnt += 1

 def deleteAtIndex(self, index: int) -> None:
 if index >= self.cnt:

```

Linked List

p.1584



```
 return
 pre = self.dummy
 for _ in range(index):
 pre = pre.next
 t = pre.next
 pre.next = t.next
 t.next = None # don't forget to cutoff
 self.cnt -= 1

Your MyLinkedList object will be instantiated and
called as such:
obj = MyLinkedList()
param_1 = obj.get(index)
obj.addAtHead(val)
obj.addAtTail(val)
obj.addAtIndex(index, val)
obj.deleteAtIndex(index)
```

Linked List

#708 Insert into a Sorted Circular Linked List MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List

Lists: Premium 98

Time Complexity: O(n) in the worst-case when a full loop is required. Space Complexity: O(1) since we only use a few extra pointers.

Problem

Given a Circular Linked List node, which is sorted in non-descending order, write a function to insert a value insertVal into the list such that it remains a sorted circular list. The given node can be a reference to any single node in the list and may not necessarily be the smallest value in the circular list.

If there are multiple suitable places for insertion, you may choose any place to insert the new value. After the insertion, the circular list should remain sorted.

If the list is empty (i.e., the given node is null), you should create a new single circular list and return the reference to that single node. Otherwise, you should return the originally given node.

Example 1:

Linked List

Diagram showing a circular linked list with nodes 1, 3, and 4. Node 3 is labeled HEAD. Arrows indicate the sequence: 1 points to 3, 3 points to 4, and 4 points back to 1.

Input: head = [3,4,1], insertVal = 2  
Output: [3,4,1,2]  
Explanation: In the figure above, there is a sorted circular list of three elements. You are given a reference to the node with value 3, and we need to insert 2 into the list. The new node should be inserted between node 1 and node 3. After the insertion, the list should look like this, and we should still return node 3.

Example 2:

Input: head = [], insertVal = 1  
Output: [1]  
Explanation: The list is empty (given head is null). We create a new single circular list and return the reference to that single node.

Example 3:

Input: head = [1], insertVal = 0  
Output: [1,0]

Constraints:

- The number of nodes in the list is in the range [0, 5 \* 10<sup>4</sup>].
- 106 <= Node.val, insertVal <= 106

Community Solutions (Python)

• LeetDoocs

"""  
# Definition for a Node.  
class Node:  
 def \_\_init\_\_(self, val=None, next=None):  
 self.val = val  
 self.next = next  
"""  
  
class Solution:

def insert(self, head: 'Optional[Node]', insertVal: int) -> 'Node':  
 node = Node(insertVal)  
 if head is None:  
 node.next = node  
 return node  
 prev, curr = head, head.next  
 while curr != head:  
 if prev.val <= insertVal <= curr.val or (  
 prev.val > curr.val and (insertVal >= prev.val or insertVal <= curr.val)  
 ):  
 break  
 prev, curr = curr, curr.next  
 prev.next = node  
 node.next = curr  
 return head

• SimplyLeet

# Definition for a Node.  
class Node:  
 def \_\_init\_\_(self, val, next=None):  
 self.val = val  
 self.next = next  
  
class Solution:  
 def insert(self, head: 'Node', insertVal: int) -> 'Node':  
 # If the list is empty, create a new node pointing to itself and return it.  
 if not head:

newNode = Node(insertVal)  
newNode.next = newNode  
return newNode

curr = head  
inserted = False

while True:  
 # Case 1: insert between two nodes where insertVal is between curr.val and curr.next.val.  
 if curr.val <= insertVal <= curr.next.val:  
 inserted = True  
 # Case 2: curr.val > curr.next.val, i.e., we are at the turning point.  
 # In this case, insert if insertVal is greater than curr.val (largest)  
 # or insertVal is less than curr.next.val (smallest).  
 elif curr.val > curr.next.val:  
 if insertVal >= curr.val or insertVal <= curr.next.val:  
 inserted = True  
 if inserted:  
 newNode = Node(insertVal, curr.next)

curr.next = newNode  
return head

curr = curr.next  
# If we have traversed the entire circle and came back to head, break.  
if curr == head:  
 break

# Case 3: All values are the same or no valid index was found, insert arbitrarily.  
newNode = Node(insertVal, curr.next)

curr.next = newNode  
return head

• LC.ca

"""  
# Definition for a Node.  
class Node:  
 def \_\_init\_\_(self, val=None, next=None):  
 self.val = val  
 self.next = next  
"""  
  
class Solution:

def insert(self, head: 'Optional[Node]', insertVal: int) -> 'Node':  
 node = Node(insertVal)  
 if head is None:  
 node.next = node  
 return node  
 prev, curr = head, head.next  
 while curr != head:  
 if prev.val <= insertVal <= curr.val or (  
 prev.val > curr.val and (insertVal >= prev.val or insertVal <= curr.val)  
 ):  
 break  
 prev, curr = curr, curr.next  
 prev.next = node  
 node.next = curr  
 return head

Linked List

#1171 Remove Zero Sum Consecutive Nodes from Linked List MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · Linked List

Lists: Denny Zhang

Time Complexity: O(n) – Two passes over the list. Space Complexity: O(n) – Hash table storing at most n prefix sums.

Problem

Given the head of a linked list, we repeatedly delete consecutive sequences of nodes that sum to 0 until there are no such sequences.

After doing so, return the head of the final linked list. You may return any such answer.

(Note that in the examples below, all sequences are serializations of ListNode objects.)

Example 1:

Input: head = [1,2,-3,3,1]  
Output: [3,1]  
Note: The answer [1,2,1] would also be accepted.

Example 2:

Input: head = [1,2,3,-3,4]  
Output: [1,2,4]

Example 3:

Input: head = [1,2,3,-3,-2]  
Output: [1]

- Constraints:
- The given linked list will contain between 1 and 1000 nodes.
  - Each node in the linked list has  $-1000 \leq \text{node.val} \leq 1000$ .

Community Solutions (Python)

• WalkCC

```
class Solution:
 def removeZeroSumSublists(self, head: ListNode) ->
 ListNode:
 dummy = ListNode(0, head)
 prefix = 0
 prefixToNode = {0: dummy}

 while head:
 prefix += head.val
 prefixToNode[prefix] = head
 head = head.next
```

Linked List p.1594

```
prefix = 0
head = dummy

while head:
 prefix += head.val
 head.next = prefixToNode[prefix].next
 head = head.next

return dummy.next
```

• LeetCode

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def removeZeroSumSublists(self, head:
 Optional[ListNode]) -> Optional[ListNode]:
 dummy = ListNode(next=head)
 last = {}
 s, cur = 0, dummy
```

Linked List p.1595

```
while cur:
 s += cur.val
 last[s] = cur
 cur = cur.next
s, cur = 0, dummy
while cur:
 s += cur.val
 cur.next = last[s].next
 cur = cur.next
return dummy.next
```

• SimplyLeet

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

class Solution:
 def removeZeroSumSublists(self, head: ListNode) ->
 ListNode:
 # Create a dummy node to simplify cases where
 removal starts at the head.
 dummy = ListNode(0)
```

Linked List p.1596

```
dummy.next = head

Dictionary to store prefix sum to node mapping.
sum_to_node = {}
prefix_sum = 0
current = dummy

First pass: record the last occurrence of each
prefix sum.
while current:
 prefix_sum += current.val

 sum_to_node[prefix_sum] = current
 current = current.next

Second pass: remove nodes that are part of
zero-sum sequences.
prefix_sum = 0
current = dummy
while current:
 prefix_sum += current.val
 # Link current node's next pointer to skip
 the zero-sum subsequence.
 current.next = sum_to_node[prefix_sum].next

 current = current.next

return dummy.next
```

• LC.ca

Linked List p.1597

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def removeZeroSumSublists(self, head:
 Optional[ListNode]) -> Optional[ListNode]:
 dummy = ListNode(next=head)
 last = {}
 s, cur = 0, dummy

 while cur:
 s += cur.val
 last[s] = cur
 cur = cur.next
 s, cur = 0, dummy
 while cur:
 s += cur.val
 cur.next = last[s].next
 cur = cur.next
 return dummy.next
```

Linked List p.1598

Linked List

## #25 Reverse Nodes in k-Group

LeetCode · SimplyLeet · WalkCC · LeetCode

Linked List · Recursion

Lists: Grind 169

Time Complexity:  $O(n)$  where  $n$  is the number of nodes (each node is processed a constant number of times)

Space Complexity:  $O(1)$  (only constant extra space is used)

**Problem**

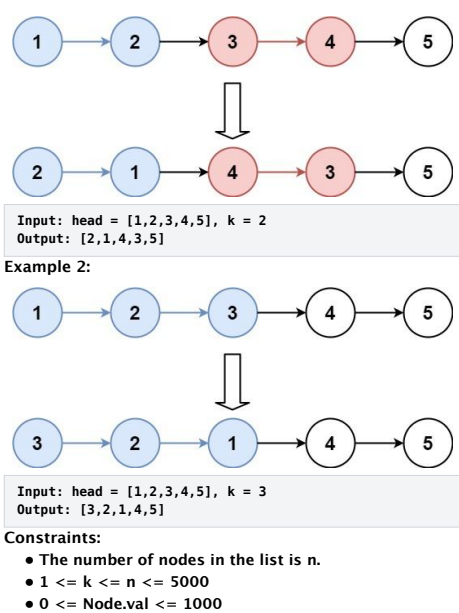
Given the head of a linked list, reverse the nodes of the list  $k$  at a time, and return the modified list.

$k$  is a positive integer and is less than or equal to the length of the linked list. If the number of nodes is not a multiple of  $k$  then left-out nodes, in the end, should remain as it is.

You may not alter the values in the list's nodes, only nodes themselves may be changed.

Example 1:

Linked List p.1599



Linked List p.1600

Follow-up: Can you solve the problem in  $O(1)$  extra memory space?

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def getKthNode(self, curr, k):
 # Brute Force - collect into list, manipulate,
 rebuild
 vals = []
 curr = curr
 while curr:
 vals.append(curr.val)
 curr = curr.next
 dummy = ListNode(0)

 curr = dummy
 for v in vals:
 curr.next = ListNode(v)
 curr = curr.next
 return dummy.next
```

Community Solutions (Python)

• WalkCC

Linked List p.1601

```
class Solution:
 def reverseKGroup(self, head: ListNode | None, k: int)
 -> ListNode | None:
 if not head:
 return None

 tail = head

 for _ in range(k):
 # There are less than k nodes in the list, do
 nothing.
 if not tail:

 return head
 tail = tail.next

 newHead = self._reverse(head, tail)
 head.next = self.reverseKGroup(tail, k)
 return newHead

 def _reverse(
 self,
 head: ListNode | None,
```

Linked List p.1602

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>        tail: ListNode   None,     ) -&gt; ListNode   None:         """Reverses [head, tail)."""         prev = None         curr = head         while curr != tail:             next = curr.next             curr.next = prev             prev = curr             curr = next      return prev  class Solution:     def reverseKGroup(self, head: ListNode, k: int) -&gt;     ListNode:         if not head or k == 1:             return head          def getLength(head: ListNode) -&gt; int:             length = 0             while head:                 length += 1                 head = head.next</pre> |        | <pre>        return length          length = getLength(head)         dummy = ListNode(0, head)         prev = dummy         curr = head          for _ in range(length // k):             for _ in range(k - 1):                 next = curr.next                  curr.next = next.next                 next.next = prev.next                 prev.next = next                 prev = curr                 curr = curr.next              return dummy.next      • LeetCode      # Definition for singly-linked list.     # class ListNode:     #     def __init__(self, val=0, next=None):     #         self.val = val     #         self.next = next     class Solution:         def reverseKGroup(self, head: Optional[ListNode], k:         int) -&gt; Optional[ListNode]:             def reverse(head: Optional[ListNode]) -&gt;             Optional[ListNode]:                 dummy = ListNode()                 cur = head</pre> |        | <pre>        while cur:             nxt = cur.next             cur.next = dummy.next             dummy.next = cur             cur = nxt          return dummy.next          dummy = pre = ListNode(next=head)         while pre:             cur = pre              for _ in range(k):                 cur = cur.next                 if cur is None:                     return dummy.next              node = pre.next             nxt = cur.next             cur.next = None             pre.next = reverse(node)             node.next = nxt             pre = node          return dummy.next      • SimplyLeet</pre> |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | p.1603 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | p.1604 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | p.1605 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre># Definition for singly-linked list. class ListNode:     def __init__(self, val=0, next=None):         self.val = val         self.next = next  class Solution:     def reverseKGroup(self, head: ListNode, k: int) -&gt;     ListNode:         # Create a dummy node to simplify edge cases         dummy = ListNode(0)          dummy.next = head         group_prev = dummy          while True:             # Find the kth node from group_prev             kth = self.getKthNode(group_prev, k)             if not kth:                 break             group_next = kth.next</pre> |        | <pre>        # Reverse the group starting at         group_prev.next up to kth         prev = kth.next         curr = group_prev.next         while curr != group_next:             temp = curr.next # Store next node             curr.next = prev # Reverse pointer             prev = curr # Move prev forward             curr = temp # Move curr forward          # Reconnect the reversed group with the         previous part          temp = group_prev.next # temp is the start of         the original group, becomes the new tail         group_prev.next = kth # Connect previous part         to the kth node (new head of reversed group)         group_prev = temp # Move group_prev to the         end of the reversed group          return dummy.next          # Helper function to get the kth node from the         current node         def getKthNode(self, curr, k):             while curr and k &gt; 0:                 curr = curr.next              k -= 1             return curr</pre> <p>• LC.ca</p> |        | <pre># Definition for singly-linked list. # class ListNode: #     def __init__(self, val=0, next=None): #         self.val = val #         self.next = next class Solution:     def reverseKGroup(self, head: ListNode, k: int) -&gt;     ListNode:         ...          for reverse 1-&gt;2-&gt;3-&gt;4-&gt;5, process is like         1-&gt;None, 2-&gt;3-&gt;4-&gt;5          2-&gt;1-&gt;None, 3-&gt;4-&gt;5         3-&gt;2-&gt;1-&gt;None, 4-&gt;5         4-&gt;3-&gt;2-&gt;1-&gt;None, 5         5-&gt;4-&gt;3-&gt;2-&gt;1-&gt;None, None          ...         def reverseList(head):             pre, p = None, head             while p:                 pnext = p.next                 p.next = pre                 pre = p                 p = pnext              return pre          dummy = ListNode(next=head)         pre = cur = dummy         while cur.next:             for _ in range(k):                 cur = cur.next                 if cur is None:</pre> |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | p.1606 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | p.1607 | Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | p.1608 |

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| <pre>        return dummy.next     t = cur.next     cur.next = None # cut from next k-group, so     to reverseList() for current k-group     start = pre.next     pre.next = reverseList(start)     start.next = t # so now 'start' is the last     node of k-group     pre = cur = start # same as the reset dummy     before while loop     return dummy.next  #####  # Definition for singly-linked list. # class ListNode(object): #     def __init__(self, x): #         self.val = x #         self.next = None  class Solution(object):     def reverseKGroup(self, head, k):         """</pre> | <pre>:type head: ListNode :type k: int :rtype: ListNode """  def reverseList(head, k):     pre = None     cur = head     while cur and k &gt; 0:         tmp = cur.next          cur.next = pre         pre = cur         cur = tmp         k -= 1     head.next = cur     return cur, pre  length = 0 p = head while p:      length += 1     p = p.next if length &lt; k:     return head step = length / k ret = None pre = None p = head while p and step:     next, newHead = reverseList(p, k)</pre> | <pre>if ret is None:     ret = newHead if pre:     pre.next = newHead pre = p p = next step -= 1 return ret</pre> |        |             |        |
| Linked List                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | p.1609                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Linked List                                                                                                       | p.1610 | Linked List | p.1611 |

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| <div> <div>Linked List</div> <div>#432 All O'one Data Structure</div> <div> <div> LeetDoocs · SimplyLeet · WalkCC · LeetCode Hash Table · Design · Doubly-Linked List · Linked List Lists: Denny Zhang Time Complexity: O(1) average for inc, dec, getMaxKey, and getMinKey. Space Complexity: O(n) where n is the number of unique keys in the data structure. Problem Design a data structure to store the strings' count with the ability to return the strings with minimum and maximum counts. Implement the AllOne class: <ul style="list-style-type: none"> <li>AllOne() Initializes the object of the data structure.</li> <li>inc(String key) Increments the count of the string key by 1. If key does not exist in the data structure, insert it with count 1.</li> <li>dec(String key) Decrements the count of the string key by 1. If the count of key is 0 after the decrement, remove it from the data structure. It is guaranteed that key exists in the data structure before the decrement.</li> <li>getMaxKey() Returns one of the keys with the maximal count. If no element exists, return an empty string "".</li> </ul> </div> </div> </div> <div> <div>Linked List</div> <div>p.1612</div> </div> | <div> <div></div> <div> <ul style="list-style-type: none"> <li>getMinKey() Returns one of the keys with the minimum count. If no element exists, return an empty string "".</li> </ul> </div> <div> Note that each function must run in O(1) average time complexity. Example 1: <div> <div> Input ["AllOne", "inc", "inc", "getMaxKey", "getMinKey", "inc", "getMaxKey", "getMinKey"] [[], ["hello"], ["hello"], [], [], ["leet"], [], []] Output [null, null, null, "hello", "hello", null, "hello", "leet"] </div> <div> Explanation AllOne allOne = new AllOne(); </div> </div> </div> <div> <div>Constraints:</div> <ul style="list-style-type: none"> <li>1 &lt;= key.length &lt;= 10</li> <li>key consists of lowercase English letters.</li> <li>It is guaranteed that for each call to dec, key is existing in the data structure.</li> <li>At most 5 * 104 calls will be made to inc, dec, getMaxKey, and getMinKey.</li> </ul> </div> <div> <div>Community Solutions (Python)</div> <ul style="list-style-type: none"> <li>WalkCC</li> </ul> </div> </div> <div> <div>Linked List</div> <div>p.1613</div> </div> | <div> <div></div> <div> <pre> from dataclasses import dataclass  @dataclass class Node:     def __init__(self, count: int, key: str   None = None):         self.count = count         self.keys: set[str] = {key} if key else set()         self.prev: Node   None = None         self.next: Node   None = None      def __eq__(self, other) -&gt; bool:         if not isinstance(other, Node):             return NotImplemented         return self.count == other.count and self.keys == other.keys  class AllOne:     def __init__(self):         self.keyToNode: dict[str, Node] = {}          self.head = Node(0)         self.tail = Node(0)         self.head.next = self.tail         self.tail.prev = self.head      def inc(self, key: str) -&gt; None:         if key in self.keyToNode:             self._incrementExistingKey(key)         else:             self._addNewKey(key) </pre> </div> </div> <div> <div>Linked List</div> <div>p.1614</div> </div> |
| <div> <div></div> <div> <pre> def dec(self, key: str) -&gt; None:     # It is guaranteed that key exists in the data     structure before the     # decrement.     self._decrementExistingKey(key)  def getMaxKey(self) -&gt; str:     return '' if self.tail.prev == self.head \         else next(iter(self.tail.prev.keys))  def getMinKey(self) -&gt; str:     return '' if self.head.next == self.tail \         else next(iter(self.head.next.keys))  def _addNewKey(self, key: str) -&gt; None:     """Adds a new node with frequency 1."""     if self.head.next.count == 1:         self.head.next.keys.add(key)     else:         self._insertAfter(self.head, Node(1, key))  self.keyToNode[key] = self.head.next  def _incrementExistingKey(self, key: str) -&gt; None:     """Increments the frequency of the key by 1."""     node = self.keyToNode[key]     node.keys.remove(key)     if node.next == self.tail or node.next.count &gt; node.count + 1:         self._insertAfter(node, Node(node.count + 1))     node.next.keys.add(key)     self.keyToNode[key] = node.next </pre> </div> </div> <div> <div>Linked List</div> <div>p.1615</div> </div>                                                  | <div> <div></div> <div> <pre> if not node.keys:     self._remove(node)  def _decrementExistingKey(self, key: str) -&gt; None:     """Decrements the count of the key by 1."""     node = self.keyToNode[key]     node.keys.remove(key)     if node.count &gt; 1:         if node.prev == self.head or node.prev.count != node.count - 1:             self._insertAfter(node.prev, Node(node.count - 1))      node.prev.keys.add(key)     self.keyToNode[key] = node.prev     else:         del self.keyToNode[key]         if not node.keys:             self._remove(node)  def _insertAfter(self, node: Node, newNode: Node) -&gt; None:     newNode.prev = node     newNode.next = node.next      node.next.prev = newNode     node.next = newNode  def _remove(self, node: Node) -&gt; None:     node.prev.next = node.next     node.next.prev = node.prev </pre> </div> </div> <div> <ul style="list-style-type: none"> <li>SimplyLeet</li> </ul> </div> <div> <div>Linked List</div> <div>p.1616</div> </div>                                                                                                         | <div> <div></div> <div> <pre> # Node class to represent a frequency bucket in the doubly linked list class Node:     def __init__(self, count):         self.count = count # frequency count for keys in this node         self.keys = set() # set of keys with this frequency         self.prev = None # previous node in the linked list         self.next = None # next node in the linked list  class AllOne:     def __init__(self):          # Use sentinel nodes for easier handling of edge cases.         self.head = Node(float('-inf'))         self.tail = Node(float('inf'))         self.head.next = self.tail         self.tail.prev = self.head         # Map each key to its corresponding node.         self.keyToNode = {}          # Helper: insert newNode after prevNode in the linked list.         def _insertNodeAfter(self, prevNode, newNode): </pre> </div> </div> <div> <div>Linked List</div> <div>p.1617</div> </div>                         |
| <div> <div></div> <div> <pre> newNode.prev = prevNode newNode.next = prevNode.next prevNode.next.prev = newNode prevNode.next = newNode  # Helper: remove a node if it has no keys. def _removeNode(self, node):     node.prev.next = node.next     node.next.prev = node.prev  def inc(self, key: str) -&gt; None:     # If key is new, treat current node as head.     if key not in self.keyToNode:         curr = self.head     else:         curr = self.keyToNode[key]         curr.keys.remove(key)     # Check if the next node has count = current count + 1.     targetCount = (0 if curr == self.head else curr.count) + 1     if curr.next == self.tail or curr.next.count != targetCount: </pre> </div> </div> <div> <div>Linked List</div> <div>p.1618</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                        | <div> <div></div> <div> <pre> # Create new node with targetCount. newNode = Node(targetCount) self._insertNodeAfter(curr, newNode)     else:         newNode = curr.next         newNode.keys.add(key)         self.keyToNode[key] = newNode         # Remove the current node if it is not a sentinel and has no keys.         if curr != self.head and len(curr.keys) == 0:             self._removeNode(curr)  def dec(self, key: str) -&gt; None:     curr = self.keyToNode[key]     curr.keys.remove(key)     # If decrementing causes the count to be 0, remove key completely.     if curr.count == 1:         del self.keyToNode[key]     else:         targetCount = curr.count - 1         # Check if the previous node has the target count. </pre> </div> </div> <div> <div>Linked List</div> <div>p.1619</div> </div>                                                                                                                                                                                                                                                                                          | <div> <div></div> <div> <pre> if curr.prev == self.head or curr.prev.count != targetCount:     newNode = Node(targetCount)     # Insert newNode before current.     self._insertNodeAfter(curr.prev, newNode)     else:         newNode = curr.prev         newNode.keys.add(key)         self.keyToNode[key] = newNode         # Clean up current node if empty.         if len(curr.keys) == 0:              self._removeNode(curr)  def getMaxKey(self) -&gt; str:     # The node before tail has the maximal count.     return next(iter(self.tail.prev.keys)) if self.tail.prev != self.head else ""  def getMinKey(self) -&gt; str:     # The node after head has the minimal count.     return next(iter(self.head.next.keys)) if self.head.next != self.tail else "" </pre> </div> </div> <div> <div>Linked List</div> <div>p.1620</div> </div>                                                                                                                      |

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| <pre># Example usage: # obj = AllOne() # obj.inc("hello") # obj.inc("hello") # print(obj.getMaxKey()) # "hello" # print(obj.getMinKey()) # "hello" # obj.inc("leet") # print(obj.getMaxKey()) # "hello" # print(obj.getMinKey()) # "leet"</pre> <p>• LC.ca</p> <pre>class Node:     def __init__(self, key='', cnt=0):         self.prev = None         self.next = None         self.cnt = cnt         self.keys = {key}      def insert(self, node):         node.prev = self         node.next = self.next          node.prev.next = node         node.next.prev = node         return node      def remove(self):         self.prev.next = self.next         self.next.prev = self.prev  class AllOne:</pre> | <pre>def __init__(self):     self.root = Node()     self.root.next = self.root     self.root.prev = self.root     self.nodes = {}      def inc(self, key: str) -&gt; None:         root, nodes = self.root, self.nodes         if key not in nodes:             if root.next == root or root.next.cnt &gt; 1:                  nodes[key] = root.insert(Node(key, 1))             else:                 root.next.keys.add(key)                 nodes[key] = root.next          else:             curr = nodes[key]             next = curr.next             if next == root or next.cnt &gt; curr.cnt + 1:                 nodes[key] = curr.insert(Node(key, curr.cnt + 1))             else:                  next.keys.add(key)                 nodes[key] = next                 curr.keys.discard(key)                 if not curr.keys:                     curr.remove()      def dec(self, key: str) -&gt; None:         root, nodes = self.root, self.nodes         curr = nodes[key]         if curr.cnt == 1:</pre> | <pre>        nodes.pop(key)     else:         prev = curr.prev         if prev == root or prev.cnt &lt; curr.cnt - 1:             nodes[key] = prev.insert(Node(key, curr.cnt - 1))         else:             prev.keys.add(key)             nodes[key] = prev             curr.keys.discard(key)             if not curr.keys:                  curr.remove()      def getMaxKey(self) -&gt; str:         return next(iter(self.root.prev.keys))      def getMinKey(self) -&gt; str:         return next(iter(self.root.next.keys))  # Your AllOne object will be instantiated and called as such:  # obj = AllOne() # obj.inc(key) # obj.dec(key) # param_3 = obj.getMaxKey() # param_4 = obj.getMinKey()</pre> |
| Linked Listp.1621                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Linked Listp.1622                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Linked Listp.1623                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

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| <div>Linked List</div> <div>#460 LFU Cache</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Hash Table · Linked List · Design · Doubly-Linked List</div> <div>Lists: Denny Zhang</div> <div>Time Complexity: O(1) average for both get and put.</div> <div>Space Complexity: O(capacity), where capacity is the maximum number of keys stored.</div> <div>Problem</div> <div>Design and implement a data structure for a Least Frequently Used (LFU) cache.</div> <div>Implement the LFUCache class:</div> <div><div>• LFUCache(int capacity) Initializes the object with the capacity of the data structure.</div><div>• int get(int key) Gets the value of the key if the key exists in the cache. Otherwise, returns -1.</div><div>• void put(int key, int value) Update the value of the key if present, or inserts the key if not already present. When the cache reaches its capacity, it should invalidate and remove the least frequently used key before inserting a new item. For this problem, when there is a tie (i.e., two or more keys with the same frequency), the least recently used key would be invalidated.</div></div> <div>To determine the least frequently used key, a use counter is maintained for each key in the cache. The key</div> | <div>with the smallest use counter is the least frequently used key.</div> <div>When a key is first inserted into the cache, its use counter is set to 1 (due to the put operation). The use counter for a key in the cache is incremented either a get or put operation is called on it.</div> <div>The functions get and put must each run in O(1) average time complexity.</div> <div>Example 1:</div> <div><div>Input</div><div>["LFUCache", "put", "put", "get", "put", "get", "get", "put", "get", "get", "get"]</div><div>[[2], [1, 1], [2, 2], [1], [3, 3], [2], [3], [4, 4], [1], [3], [4]]</div><div>Output</div><div>[null, null, null, 1, null, -1, 3, null, -1, 3, 4]</div><div>Explanation</div><div>// cnt(x) = the use counter for key x</div></div> <div>Constraints:</div> <div><div>• 1 &lt;= capacity &lt;= 104</div><div>• 0 &lt;= key &lt;= 105</div><div>• 0 &lt;= value &lt;= 109</div><div>• At most 2 * 105 calls will be made to get and put.</div></div> <div>Community Solutions (Python)</div> <div><div>• WalkCC</div></div> | <pre>class Node:     def __init__(self, key: int, value: int, freq: int, it):         self.key = key         self.value = value         self.freq = freq         self.it = it  class LFUCache:     def __init__(self, capacity: int):          self.capacity = capacity         self.minFreq = 0         self.keyToNode = {}         self.freqToList = {}      def get(self, key: int) -&gt; int:         if key not in self.keyToNode:             return -1          node = self.keyToNode[key]          self._touch(node)         return node.value      def put(self, key: int, value: int) -&gt; None:         if self.capacity == 0:             return          if key in self.keyToNode:             node = self.keyToNode[key]             node.value = value</pre> |
| Linked Listp.1624                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Linked Listp.1625                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Linked Listp.1626                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>        self._touch(node)         return      if len(self.keyToNode) == self.capacity:         # Evict an LRU key from `minFreq` list         keyToEvict = self.freqToList[self.minFreq][-1]         self.freqToList[self.minFreq].pop()         del self.keyToNode[keyToEvict]      self.minFreq = 1      if 1 not in self.freqToList:         self.freqToList[1] = []         self.freqToList[1].insert(0, key)         self.keyToNode[key] = Node(key, value, 1, 0) # Using index 0 as iterator      def _touch(self, node: Node) -&gt; None:         # Update the node's frequency         prevFreq = node.freq         node.freq += 1         newFreq = node.freq          # Remove the key from `prevFreq`'s list         self.freqToList[prevFreq].remove(node.key)         if not self.freqToList[prevFreq]:             del self.freqToList[prevFreq]         # Update `minFreq` if needed         if prevFreq == self.minFreq:             self.minFreq += 1          # Insert the key to the front of `newFreq`'s list</pre> | <pre>        if newFreq not in self.freqToList:             self.freqToList[newFreq] = []             self.freqToList[newFreq].insert(0, node.key)             node.it = 0 # Using index 0 as iterator  • SimplyLeet  from collections import OrderedDict  class LFUCache:     def __init__(self, capacity: int):         # Initialize cache capacity and helper variables.         self.capacity = capacity         self.min_freq = 0 # Track the minimum frequency         in cache.         # Map key -&gt; (value, frequency)         self.key_to_val_freq = {}         # Map frequency -&gt; OrderedDict of keys for LRU         ordering.          self.freq_to_keys = {}          def update_freq(self, key):             # Increase frequency of the key since it was             accessed.             value, freq = self.key_to_val_freq[key]             # Remove key from its current frequency list.             self.freq_to_keys[freq].pop(key)             # If current frequency list is empty, update             min_freq if necessary.             if not self.freq_to_keys[freq]:                 del self.freq_to_keys[freq]</pre> | <pre>        if self.min_freq == freq:             self.min_freq += 1             new_freq = freq + 1             # Insert key into the list corresponding to the             new frequency.             if new_freq not in self.freq_to_keys:                 self.freq_to_keys[new_freq] = OrderedDict()             self.freq_to_keys[new_freq][key] = None             # Update the frequency entry for the key.             self.key_to_val_freq[key] = (value, new_freq)      def get(self, key: int) -&gt; int:         # Return -1 if the key is not present.         if key not in self.key_to_val_freq:             return -1         # Update frequency due to access.         self.update_freq(key)         return self.key_to_val_freq[key][0]      def put(self, key: int, value: int) -&gt; None:         # If capacity is zero, nothing can be added.</pre> |
| Linked Listp.1627                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Linked Listp.1628                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Linked Listp.1629                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

```
if self.capacity <= 0:
 return
if key in self.key_to_val_freq:
 # Update value and frequency if key exists.
 self.key_to_val_freq[key] = (value,
self.key_to_val_freq[key][1])
 self.update_freq(key)
else:
 # If cache is full, evict the least
frequently used (and least recently used) key.
 if len(self.key_to_val_freq) >=
self.capacity:
 # Remove the first key from the lowest
frequency list.
```

```
key_to_remove, _ =
self.freq_to_keys[self.min_freq].popitem(last=False)
del self.key_to_val_freq[key_to_remove]
if not self.freq_to_keys[self.min_freq]:
 del self.freq_to_keys[self.min_freq]
Insert the new key with frequency 1.
self.key_to_val_freq[key] = (value, 1)
if 1 not in self.freq_to_keys:
 self.freq_to_keys[1] = OrderedDict()
self.freq_to_keys[1][key] = None
self.min_freq = 1 # Reset min_freq.
```

• LC.ca

Linked List

p.1630

```
from collections import defaultdict

class LFUCache:
 def __init__(self, capacity: int):
 self.capacity = capacity
 self.key_to_value = {}
 self.key_to_freq = defaultdict(int)
 self.freq_to_keys = defaultdict(OrderedDict)
 self.min_freq = 0
```

```
def get(self, key: int) -> int:
 if key not in self.key_to_value:
 return -1
```

```
Update the frequency
self._update_frequency(key)
return self.key_to_value[key]
```

```
def put(self, key: int, value: int) -> None:
 if self.capacity == 0:
```

```
return
```

```
if key in self.key_to_value:
 # Update the value and frequency
 self.key_to_value[key] = value
 self._update_frequency(key)
else:
 if len(self.key_to_value) >= self.capacity:
 # Evict the least frequently used key
 self._evict()
```

Linked List

p.1631

```
Add the new key-value pair
self.key_to_value[key] = value
self.key_to_freq[key] = 1
self.freq_to_keys[1][key] = None
self.min_freq = 1

def _update_frequency(self, key: int) -> None:
 freq = self.key_to_freq[key]
 del self.freq_to_keys[freq][key]
```

```
if not self.freq_to_keys[freq]:
 # If there are no keys with the previous
frequency, update min_freq
 if self.min_freq == freq:
 self.min_freq += 1
 del self.freq_to_keys[freq]
```

```
freq += 1
self.key_to_freq[key] = freq
self.freq_to_keys[freq][key] = None
```

Linked List

p.1632

```
def _evict(self) -> None:
 keys = self.freq_to_keys[self.min_freq]
 evict_key, _ = keys.popitem(last=False)
 del self.key_to_value[evict_key]
 del self.key_to_freq[evict_key]
```

```
Your LFUCache object will be instantiated and called as
such:
obj = LFUCache(capacity)
param_1 = obj.get(key)
```

```
obj.put(key,value)
```

```
#####
```

```
class Node:
 def __init__(self, key: int, value: int) -> None:
 self.key = key
 self.value = value
 self.freq = 1
```

```
self.prev = None
self.next = None
```

```
class DoublyLinkedList:
 def __init__(self) -> None:
 self.head = Node(-1, -1)
 self.tail = Node(-1, -1)
 self.head.next = self.tail
 self.tail.prev = self.head
```

Linked List

p.1633

```
def add_first(self, node: Node) -> None:
 node.prev = self.head
 node.next = self.head.next
 self.head.next.prev = node
 self.head.next = node
```

```
def remove(self, node: Node) -> Node:
 node.next.prev = node.prev
 node.prev.next = node.next
```

```
node.next, node.prev = None, None
return node
```

```
def remove_last(self) -> Node:
 return self.remove(self.tail.prev)
```

```
def is_empty(self) -> bool:
 return self.head.next == self.tail
```

```
class LFUCache:
 def __init__(self, capacity: int):
 self.capacity = capacity
 self.min_freq = 0
 self.map = defaultdict(Node)
 self.freq_map = defaultdict(DoublyLinkedList)
```

```
def get(self, key: int) -> int:
 if self.capacity == 0 or key not in self.map:
 return -1
```

```
node = self.map[key]
self.incr_freq(node)
return node.value
```

```
def put(self, key: int, value: int) -> None:
 if self.capacity == 0:
 return
 if key in self.map:
 node = self.map[key]
 node.value = value
```

```
self.incr_freq(node)
return
if len(self.map) == self.capacity:
 ls = self.freq_map[self.min_freq]
 node = ls.remove_last()
 self.map.pop(node.key)
node = Node(key, value)
self.add_node(node)
self.map[key] = node
self.min_freq = 1
```

```
def incr_freq(self, node: Node) -> None:
 freq = node.freq
 ls = self.freq_map[freq]
 ls.remove(node)
 if ls.is_empty():
 self.freq_map.pop(freq)
 if freq == self.min_freq:
 self.min_freq += 1
 node.freq += 1
```

Linked List

p.1635

```
self.add_node(node)
```

```
def add_node(self, node: Node) -> None:
 freq = node.freq
 ls = self.freq_map[freq]
 ls.add_first(node)
 self.freq_map[freq] = ls
```

```
Your LFUCache object will be instantiated and called as
such:
```

```
obj = LFUCache(capacity)
param_1 = obj.get(key)
obj.put(key,value)
```

Linked List

p.1636

```
from collections import defaultdict

class LFUCache:
 def __init__(self, capacity: int):
 self.capacity = capacity
 self.key_to_value = {}
 self.key_to_freq = defaultdict(int)
 self.freq_to_keys = defaultdict(OrderedDict)
 self.min_freq = 0

 def get(self, key: int) -> int:
 if key not in self.key_to_value:
 return -1

 # Update the value and frequency
 self.key_to_value[key] = value
 self._update_frequency(key)
 return self.key_to_value[key]

 def put(self, key: int, value: int) -> None:
 if self.capacity == 0:
 return

 if key in self.key_to_value:
 # Update the value and frequency
 self.key_to_value[key] = value
 self._update_frequency(key)
 else:
 if len(self.key_to_value) >= self.capacity:
 # Evict the least frequently used key
 self._evict()

 # Insert the new key with frequency 1
 self.key_to_value[key] = value
 self.key_to_freq[key] = 1
 self._update_frequency(key)

 def _update_frequency(self, key: int) -> None:
 freq = self.key_to_freq[key]
 del self.freq_to_keys[freq][key]

 if not self.freq_to_keys[freq]:
 # If there are no keys with the previous
 # frequency, update min_freq
 if self.min_freq == freq:
 self.min_freq += 1
 del self.freq_to_keys[freq]

 self.key_to_freq[key] = freq + 1
 self.freq_to_keys[freq + 1][key] = None
```

Linked List

p.1637

>> Quick Review — Linked List

23 questions · insights · complexity · solution

#21 Merge Two Sorted Lists [Easy]

Key Insights

- Leverage the fact that both lists are sorted.
- Use two pointers (one for each list) to compare nodes sequentially.
- Create a dummy node to simplify edge cases and build the resulting list.
- Iterate until one list is exhausted, then append the remaining nodes from the other list.
- There is also a recursive approach that can simplify the logic but may use more stack space.

Space & Time

Time Complexity:  $O(n + m)$ , where  $n$  and  $m$  are the lengths of the two lists.

Space Complexity:  $O(1)$  for the iterative solution ( $O(n + m)$  stack space for the recursive solution).

Solution

We use an iterative approach with a dummy node to merge two sorted linked lists. Two pointers traverse list1 and list2; at each step, the node with the smaller value is appended to our merged list. This process

Linked List

p.1638



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| <p>repeats until one of the lists is exhausted, after which we append the remaining nodes from the other list. This approach works in one pass over both lists, ensuring linear time complexity and constant extra space usage.</p> <p><b>#141 Linked List Cycle [Easy]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use two pointers (slow and fast) to traverse the list.</li> <li>The fast pointer moves two steps at a time and the slow pointer moves one step.</li> <li>If the fast pointer ever meets the slow pointer, a cycle exists.</li> <li>If the fast pointer reaches the end of the list (null), the list has no cycle.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) where n is the number of nodes in the linked list.</p> <p>Space Complexity: O(1) as only two pointers are used.</p> <p><b>Solution</b></p> <p>We use the Floyd's Cycle Detection algorithm (Tortoise and Hare approach). Two pointers, slow and fast, are initialized at the head of the list. The slow pointer moves one step at a time while the fast pointer moves two steps. If the linked list has a cycle, the fast pointer will eventually meet the slow pointer. If the fast pointer reaches the end (i.e., a null pointer), the linked list does</p> <div> <div>Linked List</div> <div>p.1639</div> </div> | <p>not have a cycle.</p> <p><b>#206 Reverse Linked List [Easy]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The problem can be solved using either an iterative or recursive approach.</li> <li>Iterative method uses constant space by updating pointers in-place.</li> <li>Recursive method leverages the call stack to reverse the list, which could use O(n) space in the worst case.</li> <li>Both approaches traverse each node exactly once, yielding linear time complexity.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) for both iterative and recursive approaches, where n is the number of nodes.</p> <p>Space Complexity: O(1) for the iterative approach and O(n) for the recursive approach because of the recursion call stack.</p> <p><b>Solution</b></p> <p>We can solve the problem by iterating through the list while reversing the next pointers of each node. We maintain two pointers (previous and current) and update them as we traverse the list, so that by the end, the previous pointer is at the new head of the reversed list. Alternatively, a recursive approach reverses the rest of the list first and then adjusts the pointers so that the</p> <div> <div>Linked List</div> <div>p.1640</div> </div>                                                                           | <p>original head becomes the tail.</p> <p><b>#234 Palindrome Linked List [Easy]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use two pointers (slow and fast) to reach the middle of the list.</li> <li>Reverse the second half of the list.</li> <li>Compare the nodes from the start of the list to the reversed half.</li> <li>This approach meets the O(n) time and O(1) space follow-up requirement.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) – each node is visited a constant number of times.</p> <p>Space Complexity: O(1) – only constant extra space is used (in-place reversal).</p> <p><b>Solution</b></p> <p>We first use the two-pointer technique to locate the midpoint of the linked list. The slow pointer moves one step at a time while the fast pointer moves two steps. Once the middle is found, we reverse the second half of the list in-place. Finally, we traverse both halves concurrently to compare the node values. If all corresponding values match, the linked list is a palindrome. A potential gotcha is handling the odd number of nodes; in that case, we ignore the middle</p> <div> <div>Linked List</div> <div>p.1641</div> </div>                                                     |
| <p>element when comparing.</p> <p><b>#706 Design HashMap [Easy]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a fixed-size array of buckets where each bucket uses a linked list to handle collisions.</li> <li>Choose a simple hash function (e.g., modulo operation) to map a key to its corresponding bucket.</li> <li>For each bucket, traverse the linked list to update an existing key or append a new node if the key does not exist.</li> <li>Removal requires updating pointers in the linked list to bypass the removed node.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: Average O(1) for put, get, and remove; worst-case O(n) when many keys hash to the same bucket.</p> <p>Space Complexity: O(n), where n is the number of key-value pairs stored.</p> <p><b>Solution</b></p> <p>We implement the HashMap using an array of fixed size (e.g., 1000). Each array element is the head of a linked list that manages collisions using chaining. The key is hashed using a modulo operation. For the put operation, if a key already exists in the chain, its value is updated; otherwise, a new node is appended. The get operation</p> <div> <div>Linked List</div> <div>p.1642</div> </div>                                                                                              | <p>traverses the chain at the computed bucket index to find and return the corresponding value, or returns -1 if not found. The remove operation also traverses the list, carefully reconnecting the links to remove the target node while handling edge cases such as removal at the head.</p> <p><b>#876 Middle of the Linked List [Easy]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use two pointers (slow and fast) to traverse the list.</li> <li>The fast pointer moves two steps at a time while the slow pointer moves one step.</li> <li>When the fast pointer reaches the end, the slow pointer will be at the middle node.</li> <li>This approach naturally handles both odd and even lengths of the list.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n), where n is the number of nodes in the list.</p> <p>Space Complexity: O(1), constant extra space.</p> <p><b>Solution</b></p> <p>The solution uses the two-pointer technique:</p> <ul style="list-style-type: none"> <li>Initialize two pointers, slow and fast, at the head.</li> <li>Traverse the list, moving slow by one step and fast by two steps during each iteration.</li> <li>When fast reaches the end or has no next node, slow will be at the middle.</li> </ul> <div> <div>Linked List</div> <div>p.1643</div> </div>                | <p>Return the slow pointer (the middle node). This strategy ensures linear time complexity and constant space, efficiently finding the middle node even for lists with an even number of elements.</p> <p><b>#1474 Delete N Nodes After M Nodes of Linked List [Easy]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Traverse the linked list while keeping count.</li> <li>Skip m nodes (i.e., keep them) and then proceed to delete the next n nodes.</li> <li>Update pointers to bypass the removed nodes.</li> <li>This is an in-place modification problem, so no additional list structure is needed.</li> <li>Be cautious about edge cases when the list has fewer than m or n nodes remaining.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(N), where N is the number of nodes in the list.</p> <p>Space Complexity: O(1) since the modification is done in-place.</p> <p><b>Solution</b></p> <p>The solution involves iterating through the linked list with a pointer. For each cycle:</p> <div> <div>Linked List</div> <div>p.1644</div> </div>                                                                                                                                                                                |
| <p>– Traverse m nodes and keep them. – Then, from the (m+1)th node, skip and remove the next n nodes by adjusting the 'next' pointer of the m-th node to point to the node after these n nodes. – Continue until you reach the end of the list. This in-place approach avoids using extra space and maintains O(N) time complexity.</p> <p><b>#2 Add Two Numbers [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The digits are stored in reverse order, meaning the first node is the least significant digit.</li> <li>A carry may need to propagate through subsequent nodes.</li> <li>The two linked lists might be of different lengths.</li> <li>Continue processing until both lists are fully traversed and any remaining carry is handled.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(max(n, m)), where n and m are the lengths of the two linked lists.</p> <p>Space Complexity: O(max(n, m) + 1) for the result list in the worst case.</p> <p><b>Solution</b></p> <p>The solution iterates through both linked lists node by node. For each pair of corresponding nodes, their values and any carry from the previous operation are summed. The result is decomposed into a current digit (modulo</p> <div> <div>Linked List</div> <div>p.1645</div> </div>                            | <p>10) and an updated carry (total divided by 10). A new node is created for the current digit and appended to the resulting linked list. This process continues until all nodes are processed and any leftover carry is added as a final node.</p> <p><b>#19 Remove Nth Node From End of List [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a dummy node at the start to simplify edge cases, especially when the head is removed.</li> <li>Utilize two pointers (fast and slow) and maintain a gap of n+1 nodes between them.</li> <li>By moving both pointers simultaneously, the slow pointer will point to the node immediately before the target node when the fast pointer reaches the end.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(L), where L is the length of the list</p> <p>Space Complexity: O(1)</p> <p><b>Solution</b></p> <p>We use a two-pointer approach to achieve a one-pass solution. A dummy node is created and linked to the head to handle edge cases gracefully (such as needing to remove the head). The fast pointer is advanced n+1 steps ahead of the slow pointer to maintain the required gap. Then, both pointers are moved simultaneously until the fast pointer reaches the end of the list. At that point,</p> <div> <div>Linked List</div> <div>p.1646</div> </div> | <p>the slow pointer is immediately before the node to be deleted. We then adjust the pointers to remove the target node from the list.</p> <p><b>#24 Swap Nodes in Pairs [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a dummy node to simplify edge cases, such as when the list is empty or has only one node.</li> <li>Process nodes in pairs and update the pointers accordingly.</li> <li>Ensure that the swapping does not lose the connection between the swapped pair and the rest of the list.</li> <li>The solution can be implemented iteratively or recursively.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) – Each node is processed once.</p> <p>Space Complexity: O(1) – Only constant extra space is used.</p> <p><b>Solution</b></p> <p>We use an iterative approach with a dummy node. The dummy node points to the head of the list. We maintain a pointer that traverses the list in pairs. For each pair, we adjust the next pointers to swap the two nodes. By the end, the dummy's next pointer will be the new head of the modified list. The main idea is to keep track of the previous node (prev) before the pair, and then adjust</p> <div> <div>Linked List</div> <div>p.1647</div> </div> |

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| <p>pointers accordingly using temporary variables to hold references to the nodes being swapped.</p> <p><b>#61 Rotate List [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Calculate the length of the list and locate the tail.</li> <li>Use k modulo length to handle rotations greater than the list size.</li> <li>Find the new tail of the list at position (length – k – 1) and update pointers accordingly.</li> <li>Link the original tail to the original head to form a circular list, then break the circle at the new tail.</li> </ul> <p><b>Space &amp; Time</b><br/> Time Complexity: O(n), where n is the number of nodes in the list, since we traverse the list a few times.<br/> Space Complexity: O(1), as the rotation is done in place without extra data structures.</p> <p><b>Solution</b><br/> We first traverse the list to determine its length and identify the tail node. The effective rotation is calculated by taking k modulo the length of the list. If the result is zero, the list remains unchanged. Otherwise, we locate the new tail node by moving (length – k – 1) steps from the head. The node after the new tail becomes the new head. Finally, we break the link after the new tail and connect the old tail to the original head to finalize the</p> <div>Linked Listp.1648</div>           | <p>rotated list.</p> <p><b>#146 LRU Cache [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a Hash Table (dictionary) to allow O(1) access to nodes by key.</li> <li>Use a Doubly Linked List to record the usage order. The head represents the most recently used element while the tail represents the least recently used.</li> <li>When a key is accessed or updated, move the corresponding node to the front (head) of the list.</li> <li>When the cache exceeds capacity, remove the node at the tail end which is the least recently used entry.</li> </ul> <p><b>Space &amp; Time</b><br/> Time Complexity: O(1) per get and put operation.<br/> Space Complexity: O(capacity), storing key–node pairs and linked list nodes.</p> <p><b>Solution</b><br/> The solution combines a hash table (or dictionary) and a doubly linked list:<br/> – The hash table stores keys and corresponding node addresses.<br/> – The doubly linked list maintains the order of usage with the most recently used items near the head.<br/> – For get(key), check the dictionary. If found, move the node to the head and return its value; if not, return –1.<br/> – For put(key, value), if the key exists, update its</p> <div>Linked Listp.1649</div>                               | <p>value and move it to the head. If not, create a new node and add it right after the head. If the capacity is reached, remove the tail node (least recently used) and update the dictionary accordingly.<br/> – Dummy head and tail nodes are used to simplify node addition and removal procedures without worrying about edge cases.</p> <p><b>#148 Sort List [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Merge sort is well–suited for linked lists; it naturally relies on splitting the list and merging sorted sublists.</li> <li>The slow and fast pointer technique is used to find the middle point of the list.</li> <li>A recursive merge sort implementation may use O(log n) space for recursion stack. To achieve O(1) extra space, an iterative bottom–up merge sort approach would be preferred.</li> <li>Merging two sorted lists is efficient with linked lists because of their sequential access.</li> </ul> <p><b>Space &amp; Time</b><br/> Time Complexity: O(n log n)<br/> Space Complexity: O(1) extra space for an iterative bottom–up approach (O(log n) for recursive calls if using recursion)</p> <p><b>Solution</b></p> <div>Linked Listp.1650</div>                                                                         |
| <p>The solution employs the merge sort algorithm tailored for linked lists. The overall steps are:<br/> – Use the slow and fast pointers to find the middle of the list and split it into two halves.<br/> – Recursively sort the two halves.<br/> – Merge the two sorted halves by comparing node values.<br/> – Ensure that during merging, pointers are adjusted properly to build the sorted list.<br/> For an O(1) space solution, an iterative bottom–up merge sort can be implemented. However, the recursive merge sort is easier to understand and implement, though it uses O(log n) space on the call stack.</p> <p><b>#328 Odd Even Linked List [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use two pointers: one for odd–indexed nodes and one for even–indexed nodes.</li> <li>The first node is considered odd, and the second node is even.</li> <li>Traverse the list once, reassigning the next pointers to segregate odd and even nodes.</li> <li>Finally, attach the even list at the end of the odd list.</li> <li>The solution must use O(1) extra space and O(n) time complexity.</li> </ul> <p><b>Space &amp; Time</b><br/> Time Complexity: O(n) since the list is traversed once.<br/> Space Complexity: O(1) as only a few pointers are used.</p> <div>Linked Listp.1651</div>                            | <p><b>Solution</b><br/> Maintain two pointers, odd and even, starting from the head and head.next, respectively. As you iterate, update odd.next to point to the next odd element (skipping an even node) and even.next to point to the next even element. Once the end is reached, attach the even list to the end of the odd list. This in–place rearrangement preserves the initial ordering within odd and even groups while meeting the space and time constraints.</p> <p><b>#369 Plus One Linked List [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The digits are stored such that the most significant digit comes first, but addition is easier starting from the least significant digit.</li> <li>Reversing the list allows simpler management of the carry during addition.</li> <li>After processing the addition, reversing the list back restores the original order.</li> <li>Consider special cases where all digits are 9, requiring a new head node to accommodate an additional digit.</li> </ul> <p><b>Space &amp; Time</b><br/> Time Complexity: O(n), where n is the number of nodes in the list<br/> Space Complexity: O(1) extra space when using an iterative reversal method. (O(n) if recursion is used)</p> <div>Linked Listp.1652</div>            | <p><b>Solution</b><br/> We solve the problem by first reversing the linked list, which converts it to a format where the least significant digit is processed first. We then iterate over the reversed list, adding one and managing any carry. If we encounter a carry at the end of the list, a new node is appended to handle the overflow. Finally, we reverse the list again to restore the original order. This method efficiently handles the addition with constant extra space during processing.</p> <p><b>#430 Flatten a Multilevel Doubly Linked List [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use a depth–first traversal to process child lists before proceeding to the next pointer.</li> <li>When a node with a child is found, recursively flatten the child list.</li> <li>Splice the flattened child list in between the current node and the next node.</li> <li>Maintain previous and next pointers correctly to preserve the doubly linked list properties.</li> <li>Ensure that all child pointers are set to null after flattening.</li> </ul> <p><b>Space &amp; Time</b></p> <div>Linked Listp.1653</div>                                                                                                                       |
| <p>Time Complexity: O(n), where n is the total number of nodes, since each node is processed exactly once.<br/> Space Complexity: O(n) in the worst–case due to recursion stack in case of a highly nested list.</p> <p><b>Solution</b><br/> We can solve this problem using a recursive depth–first search (DFS) approach. The main idea is to traverse the list and, whenever a node with a child pointer is encountered, recursively flatten its child list. Once the child list is flattened, it can be spliced into the main list between the current node and the node originally following it. We then continue processing the next pointer. This ensures that the entire list is flattened in depth–first order. The recursive helper function returns the tail of the flattened segment, which is used to connect the subsequent parts of the list correctly.</p> <p><b>#622 Design Circular Queue [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Leverage a fixed–size array to store the queue elements.</li> <li>Use two pointers: one for the front and one for the rear.</li> <li>Maintain a count variable (or use pointer arithmetic) to differentiate between an empty and a full queue.</li> <li>Use modulo arithmetic to wrap pointers when they reach the end of the array.</li> </ul> <div>Linked Listp.1654</div> | <ul style="list-style-type: none"> <li>Ensure all operations (enqueue, dequeue, front, rear, isEmpty, isFull) have O(1) time complexity.</li> </ul> <p><b>Space &amp; Time</b><br/> Time Complexity: O(1) per operation for enqueue, dequeue, front, rear, isEmpty, and isFull.<br/> Space Complexity: O(k), where k is the capacity of the circular queue.</p> <p><b>Solution</b><br/> We implement the circular queue using a fixed–size array alongside two pointers, front and rear, and a counter to keep track of the number of elements in the queue. The front pointer indicates the index of the first element, and the rear pointer indicates the index where the next element will be enqueued. When enqueueing an element, we insert the element at the rear index and then increment the rear pointer modulo the size of the array. Similarly, when dequeuing, we remove the element at the front index and increment the front pointer modulo the size. The counter helps in quickly checking if the queue is empty or full (count equals 0 or count equals capacity, respectively). This design ensures all operations remain O(1) and correctly wraps around, forming a circular structure.</p> <p><b>#707 Design Linked List [Medium]</b></p> <p><b>Key Insights</b></p> <div>Linked Listp.1655</div> | <ul style="list-style-type: none"> <li>Use a dummy head node to simplify insertion and deletion at the head.</li> <li>Maintain a size counter to quickly validate index–based operations.</li> <li>For every operation, traverse the list up to the required point since random access is not available.</li> <li>Ensure robustness by handling edge cases like negative indices and indices larger than the current length.</li> </ul> <p><b>Space &amp; Time</b><br/> Time Complexity:<br/> – get, addAtIndex, and deleteAtIndex operations require O(n) time due to the need to traverse the list.<br/> – addAtHead and addAtTail (if tail pointer is not maintained) are O(n) in worst case; however, these can be optimized further if a tail pointer is added.<br/> Space Complexity:<br/> – O(n) space for storing n nodes.</p> <p><b>Solution</b><br/> The solution employs a singly linked list with a dummy head node to ease edge–case handling. A size counter is maintained for quick index validations. Each node holds a value and a pointer to the next node. Operations operate by traversing the list to the correct insertion or deletion point. This design makes it simple to add the functionality of getting values, inserting at specified</p> <div>Linked Listp.1656</div> |

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| <div>indices, and removing nodes.</div> <div>#708 Insert into a Sorted Circular Linked List [Medium]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Handle the empty list edge-case by creating a new node that points to itself.</li> <li>Traverse the list until the proper insertion point is found.</li> <li>Consider the "turning point" where the maximum value is followed by the minimum value.</li> <li>If no proper place is found in one full rotation, insert the new node arbitrarily (e.g., after the head).</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n) in the worst-case when a full loop is required.</div> <div>Space Complexity: O(1) since we only use a few extra pointers.</div> </div> <div>Solution <div>We traverse the circular linked list starting from the arbitrary given node. For each pair of consecutive nodes (curr and next), we check if the new value can be inserted between them by confirming that: <ul style="list-style-type: none"> <li>The new value lies between curr and next (<math>\text{curr.val} \leq \text{insertVal} \leq \text{next.val}</math>).</li> <li>Or if we are at the pivot point where the maximum value meets the minimum value</li> </ul> </div> </div> <div>Linked Listp.1657</div>                                                                                                                                         | <div>(i.e., <math>\text{curr.val} &gt; \text{next.val}</math>) and then either the new value is larger than or equal to curr.val (should be appended after the maximum) or less than or equal to next.val (should be inserted before the minimum).</div> <div>If no proper spot is found after one complete loop, we insert anywhere (all values in the list are the same). A pointer reference is updated accordingly to maintain the circular structure.</div> <div>#1171 Remove Zero Sum Consecutive Nodes from Linked List [Medium]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Use a dummy node to handle edge cases such as the removal of nodes at the head.</li> <li>Compute a running (prefix) sum while traversing the list.</li> <li>Use a hash table (or hashmap) to map each prefix sum to the most recent node where that sum occurred.</li> <li>If the same prefix sum is encountered again, the nodes in between sum to 0 and can be removed by adjusting pointers.</li> <li>Perform two passes: the first to record prefix sums and their corresponding nodes, and the second to remove zero-sum sequences.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: O(n) – Two passes over the list.</div> </div> <div>Linked Listp.1658</div> | <div>Space Complexity: O(n) – Hash table storing at most n prefix sums.</div> <div>Solution <div>The solution uses a two-pass algorithm with a hash table. In the first pass, compute the prefix sum for each node and store the mapping from prefix sum to the node (overwriting with the latest occurrence). This ensures that for any repeated prefix sum, the nodes in between sum to 0. In the second pass, traverse the list again and use the hash table to skip the zero-sum subsequences by redirecting the next pointer of a node to the next pointer of the stored node for that prefix sum. Key data structures are the hash table and a dummy node to simplify pointer adjustments.</div> </div> <div>#25 Reverse Nodes in k-Group [Hard]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Use a dummy node to simplify edge cases.</li> <li>Identify groups of k nodes using a helper function.</li> <li>Reverse the nodes in each group using pointer manipulation.</li> <li>If a remaining group has fewer than k nodes, leave it unchanged.</li> <li>The reversal is done in-place with O(1) extra space.</li> </ul> </div> <div>Space &amp; Time</div> <div>Linked Listp.1659</div>                                                                                                                                                                                  |
| <div>Time Complexity: O(n) where n is the number of nodes (each node is processed a constant number of times)</div> <div>Space Complexity: O(1) (only constant extra space is used)</div> <div>Solution <div>The solution involves iterating through the linked list by identifying groups of k consecutive nodes. For each group: <ul style="list-style-type: none"> <li>Find the kth node from the current starting point. If it does not exist, the remainder of the list is not reversed.</li> <li>Reverse the nodes in the identified group in-place by adjusting the next pointers.</li> <li>Reconnect the reversed group with the previous part of the list and update pointers to move to the next group.</li> </ul> The approach leverages a dummy node for easier handling of head modifications and uses a helper function to fetch the kth node. The in-place reversal ensures that the extra memory usage is constant.</div> </div> <div>#432 All O'one Data Structure [Hard]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Use a Doubly-Linked List (DLL) where each node holds a unique count value and a set of keys with that count.</li> <li>Maintain a Hash Map (keyToNode) to quickly locate the node corresponding to each key.</li> <li>Insert new nodes in the DLL when a key's count is incremented or decremented and the adjacent node does</li> </ul> </div> <div>Linked Listp.1660</div> | <div>not have the target count.</div> <ul style="list-style-type: none"> <li>Remove nodes from the DLL when no keys remain in that count.</li> <li>The head of the DLL holds the minimum count and the tail holds the maximum count.</li> </ul> <div>Space &amp; Time <div>Time Complexity: O(1) average for inc, dec, getMaxKey, and getMinKey.</div> <div>Space Complexity: O(n) where n is the number of unique keys in the data structure.</div> </div> <div>Solution <div>The solution uses a combination of a doubly-linked list and hash maps. Each node in the DLL represents a frequency and maintains a set of keys that have that frequency. A hash map maps keys to their corresponding node. When inc(key) or dec(key) is called, we update the key's frequency and move the key to the appropriate node in the DLL. If the adjacent node with the required frequency doesn't exist, we create a new node and insert it in the correct position. Removal of a key from a node and deletion of empty nodes ensures that getMinKey() and getMaxKey() can be returned directly from the head and tail of the DLL.</div> </div> <div>#460 LRU Cache [Hard]</div> <div>Key Insights</div> <div>Linked Listp.1661</div>                                                           | <ul style="list-style-type: none"> <li>Maintain a mapping from key to its value and frequency.</li> <li>Use an additional mapping from frequency to keys (maintaining the order of insertion or recent use) for quick eviction.</li> <li>Keep a global minFreq variable to track the lowest frequency present in the cache.</li> <li>When a key is accessed or updated, increase its frequency and relocate it in the frequency mapping.</li> <li>Evictions occur by removing the oldest key from the lowest frequency bucket.</li> </ul> <div>Space &amp; Time <div>Time Complexity: O(1) average for both get and put.</div> <div>Space Complexity: O(capacity), where capacity is the maximum number of keys stored.</div> </div> <div>Solution <div>The solution utilizes two main data structures: <ul style="list-style-type: none"> <li>A hash map (keyToValFreq) that maps keys to a pair (value, frequency).</li> <li>A second hash map (freqToKeys) mapping frequencies to an ordered list/set of keys.</li> </ul> This data structure preserves the order in which keys were added to allow removal of the least recently used key when multiple keys share the same frequency. A global minFreq variable is maintained to quickly identify the bucket with the least frequency. When a key is accessed or updated, the key is removed from its</div> </div> <div>Linked Listp.1662</div> |
| <div>current frequency list and placed in the next higher frequency list. If the frequency list for the minimal frequency becomes empty, minFreq is updated.</div> <div>This design guarantees that both get and put operations execute in constant time on average.</div> <div>DFS</div> <div>#17 DFS</div> <div>By Pattern · Python Only · 23 questions</div> <div>Linked Listp.1663</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <div>DFS</div> <div>#463 Island Perimeter EAS</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Array · DFS · BFS · Matrix</div> <div>Lists: Denny Zhang</div> <div>Time Complexity: O(n * m), where n is the number of rows and m is the number of columns. Space Complexity: O(1) for the iterative solution (excluding the input storage).</div> <div>Problem <div>You are given row x col grid representing a map where grid[i][j] = 1 represents land and grid[i][j] = 0 represents water.</div> <div>Grid cells are connected horizontally/vertically (not diagonally). The grid is completely surrounded by water, and there is exactly one island (i.e., one or more connected land cells).</div> <div>The island doesn't have "lakes", meaning the water inside isn't connected to the water around the island. One cell is a square with side length 1. The grid is rectangular, width and height don't exceed 100.</div> <div>Determine the perimeter of the island.</div> <div>Example 1:</div> </div> <div>DFS</div> <div>Sheet 185 / 317 · LeetMastery By Pattern · Python Only · 3-up Portraitp.1664</div>                                                                                                                                                 | <div> </div> <div>Input: grid = [[0,1,0,0],[1,1,1,0],[0,1,0,0],[1,1,0,0]]</div> <div>Output: 16</div> <div>Explanation: The perimeter is the 16 yellow stripes in the image above.</div> <div>Example 2:</div> <div>Input: grid = [[1]]</div> <div>Output: 4</div> <div>Example 3:</div> <div>DFS</div> <div>p.1665</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

DFS p.1674

```

class Solution:
 def floodFill(
 self, image: List[List[int]], sr: int, sc: int,
 color: int
) -> List[List[int]]:
 def dfs(i: int, j: int):
 image[i][j] = color
 for a, b in pairwise(dirs):
 x, y = i + a, j + b
 if 0 <= x < len(image) and 0 <= y <
len(image[0]) and image[x][y] == oc:
 dfs(x, y)

```

```

oc = image[sr][sc]
if oc != color:
 dirs = (-1, 0, 1, 0, -1)
 dfs(sr, sc)
return image

```

#### • SimplyLeet

```

def floodFill(image, sr, sc, color):
 # Get the original color of the starting pixel
 orig_color = image[sr][sc]
 # If the new color is the same as the original,
 return the image as is
 if orig_color == color:
 return image
 rows, cols = len(image), len(image[0])

 # Recursive DFS helper function to fill the image
 def dfs(r, c):

```

DFS

p.1675

```

Check boundaries and if the current pixel is
the same as the original color
if r < 0 or r >= rows or c < 0 or c >= cols or
image[r][c] != orig_color:
 return
Update the current pixel's color
image[r][c] = color
Recursively call DFS in all four directions
dfs(r - 1, c) # up
dfs(r + 1, c) # down
dfs(r, c - 1) # left
dfs(r, c + 1) # right

```

```

dfs(sr, sc)
return image

```

#### • LC.ca

```

class Solution:
 def floodFill(
 self, image: List[List[int]], sr: int, sc: int,
 color: int
) -> List[List[int]]:
 def dfs(i, j):
 if (
 not 0 <= i < m
 or not 0 <= j < n
 or image[i][j] != oc
 or image[i][j] == color

```

DFS

p.1676

```

):
 return
 image[i][j] = color
 for a, b in pairwise(dirs):
 dfs(i + a, j + b)

dirs = (-1, 0, 1, 0, -1)
m, n = len(image), len(image[0])
oc = image[sr][sc]
dfs(sr, sc)

return image

```

DFS

p.1677

DFS

## #130 Surrounded Regions

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · DFS · BFS · Union Find · Matrix

Lists: Denny Zhang

Time Complexity:  $O(m * n)$  Space Complexity:  $O(m * n)$  in the worst case due to recursion stack or queue storage for DFS/BFS

### Problem

You are given an  $m \times n$  matrix board containing letters 'X' and 'O', capture regions that are surrounded:

- Connect: A cell is connected to adjacent cells horizontally or vertically.
- Region: To form a region connect every 'O' cell.
- Surround: A region is surrounded if none of the 'O' cells in that region are on the edge of the board. Such regions are completely enclosed by 'X' cells.

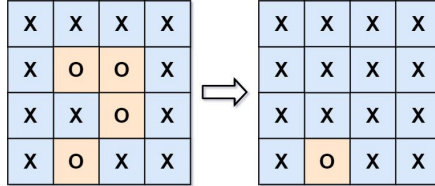
To capture a surrounded region, replace all 'O's with 'X's in-place within the original board. You do not need to return anything.

Example 1:

Input: board = `[["X","X","X","X"],["X","O","O","X"],["X","X","O","X"],["X","O","X","X"]]`

Output: `[["X","X","X","X"],["X","X","X","X"],["X","X","X","X"],["X","O","X","X"]]`

Explanation:



In the above diagram, the bottom region is not captured because it is on the edge of the board and cannot be surrounded.

Example 2:

Input: board = `[["X"]]`

Output: `[["X"]]`

Constraints:

- $m == \text{board.length}$
- $n == \text{board}[i].\text{length}$
- $1 \leq m, n \leq 200$
- $\text{board}[i][j]$  is 'X' or 'O'.

Brute Force (Python)

DFS

p.1679

```

Brute Force Approach
class Solution:
 def solve(self, board):
 # Brute Force $O(m \cdot n)$ - BFS from border 'O's, mark
 safe, flip the rest
 from collections import deque
 if not board: return
 m, n = len(board), len(board[0])
 safe = set()
 queue = deque()
 for r in range(m):

```

```

 for c in range(n):
 if (r==0 or r==m-1 or c==0 or c==n-1) and
board[r][c]=='O':
 queue.append((r,c))
 while queue:
 r,c = queue.popleft()
 if (r,c) in safe or board[r][c]!='O':
 continue
 safe.add((r,c))
 for dr,dc in [(1,0),(-1,0),(0,1),(0,-1)]:
 nr,nc = r+dr,c+dc
 if 0<=nr<m and 0<=nc<n:
 queue.append((nr,nc))

```

```

 for r in range(m):
 for c in range(n):
 if board[r][c]!='O' and (r,c) not in
safe:
 board[r][c]='X'

```

Community Solutions (Python)

DFS

p.1680

#### • WalkCC

```

class Solution:
 def solve(self, board: List[List[str]]) -> None:
 if not board:
 return

 DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0))
 m = len(board)
 n = len(board[0])
 q = collections.deque()

```

```

 for i in range(m):
 for j in range(n):
 if i * j == 0 or i == m - 1 or j == n - 1:
 if board[i][j] == 'O':
 q.append((i, j))
 board[i][j] = '*'

```

```

 # Mark the grids that stretch from the four sides
 with '*'.
 while q:
 i, j = q.popleft()

```

```

 for dx, dy in DIRS:
 x = i + dx
 y = j + dy
 if x < 0 or x == m or y < 0 or y == n:
 continue
 if board[x][y] != 'O':
 continue
 q.append((x, y))
 board[x][y] = '*'

```

DFS

p.1681

```

for row in board:
 for i, c in enumerate(row):
 row[i] = 'O' if c == '*' else 'X'

```

#### • LeetDoocs

```

class Solution:
 def solve(self, board: List[List[str]]) -> None:
 def dfs(i: int, j: int):
 if not (0 <= i < m and 0 <= j < n and
board[i][j] == "O"):
 return
 board[i][j] = "."
 for a, b in pairwise((-1, 0, 1, 0, -1)):
 dfs(i + a, j + b)

 m, n = len(board), len(board[0])

```

```

 for i in range(m):
 dfs(i, 0)
 dfs(i, n - 1)
 for j in range(n):
 dfs(0, j)
 dfs(m - 1, j)
 for i in range(m):
 for j in range(n):
 if board[i][j] == ".":
 board[i][j] = "O"

```

```

 elif board[i][j] == "O":
 board[i][j] = "X"

```

#### • SimplyLeet

DFS

p.1682

```

def solve(board):
 if not board or not board[0]:
 return

 rows, cols = len(board), len(board[0])

```

```

 def dfs(r, c):
 # Return if out of bounds or current cell is not
 'O'
 if r < 0 or c < 0 or r >= rows or c >= cols or
board[r][c] != 'O':
 return

```

```

 # Mark the cell as safe
 board[r][c] = '*'
 # Explore four possible directions
 dfs(r + 1, c)
 dfs(r - 1, c)
 dfs(r, c + 1)
 dfs(r, c - 1)

```

```

 # Traverse the board borders
 for i in range(rows):

```

```

 if board[i][0] == 'O':
 dfs(i, 0)
 if board[i][cols - 1] == 'O':
 dfs(i, cols - 1)
 for j in range(cols):
 if board[0][j] == 'O':
 dfs(0, j)
 if board[rows - 1][j] == 'O':
 dfs(rows - 1, j)

```

DFS

p.1683



```
Flip surrounded '0's and restore safe cells
for i in range(rows):
 for j in range(cols):
 if board[i][j] == '0':
 board[i][j] = 'X'
 elif board[i][j] == '*':
 board[i][j] = '0'
```

• LC.ca

```
class Solution:
 def solve(self, board: List[List[str]]) -> None:
 """
 Do not return anything, modify board in-place
 instead.
 """

 def dfs(i, j):
 board[i][j] = '.'
 for a, b in [[0, -1], [0, 1], [1, 0], [-1,
0]]:
 x, y = i + a, j + b
```

```
if 0 <= x < m and 0 <= y < n and
board[x][y] == '0':
 dfs(x, y)

m, n = len(board), len(board[0])
for i in range(m):
 for j in range(n):
 if board[i][j] == '0' and (
 i == 0 or i == m - 1 or j == 0 or j ==
n - 1
):
 dfs(i, j)

for i in range(m):
 for j in range(n):
 if board[i][j] == '0':
 board[i][j] = 'X'
 elif board[i][j] == '.':
 board[i][j] = '0'
```

DFS

#133 Clone Graph

LeetCode · SimplyLeet · WalkCC · LeetCode

Hash Table · DFS · BFS · Graph

Lists: Grind 169

Time Complexity:  $O(N + E)$ , where  $N$  is the number of nodes and  $E$  is the number of edges, because each node and each edge are traversed once.

Problem

Given a reference of a node in a connected undirected graph.

Return a deep copy (clone) of the graph.

Each node in the graph contains a value (int) and a list (List(Node)) of its neighbors.

class Node {
 public int val;
 public List<Node> neighbors;
}

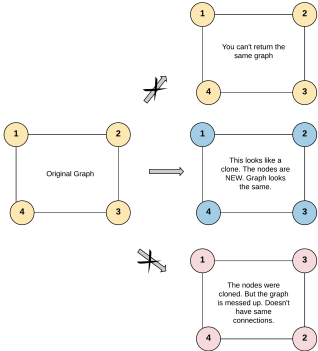
Test case format:

For simplicity, each node's value is the same as the node's index (1-indexed). For example, the first node with val == 1, the second node with val == 2, and so on. The graph is represented in the test case using an adjacency list.

An adjacency list is a collection of unordered lists used to represent a finite graph. Each list describes the set of neighbors of a node in the graph.

The given node will always be the first node with val = 1. You must return the copy of the given node as a reference to the cloned graph.

Example 1:



```
Input: adjList = [[2,4],[1,3],[2,4],[1,3]]
Output: [[2,4],[1,3],[2,4],[1,3]]
Explanation: There are 4 nodes in the graph.
1st node (val = 1)'s neighbors are 2nd node (val = 2)
and 4th node (val = 4).
2nd node (val = 2)'s neighbors are 1st node (val = 1)
and 3rd node (val = 3).
3rd node (val = 3)'s neighbors are 2nd node (val = 2)
and 4th node (val = 4).
4th node (val = 4)'s neighbors are 1st node (val = 1)
and 3rd node (val = 3).
```

Example 2:



```
Input: adjList = [[]]
Output: [[]]
Explanation: Note that the input contains one empty
list. The graph consists of only one node with val = 1
and it does not have any neighbors.
```

Example 3:

```
Input: adjList = []
Output: []
Explanation: This an empty graph, it does not have any
nodes.
```

- Constraints:
- The number of nodes in the graph is in the range  $[0, 100]$ .
  - $1 \leq \text{Node.val} \leq 100$
  - Node.val is unique for each node.
  - There are no repeated edges and no self-loops in the graph.
  - The Graph is connected and all nodes can be visited starting from the given node.

Community Solutions (Python)

• WalkCC

```
class Solution:
 def cloneGraph(self, node: 'Node') -> 'Node':
 if not node:
 return None

 q = collections.deque([node])
 map = {node: Node(node.val)}

 while q:
 u = q.popleft()

 for v in u.neighbors:
 if v not in map:
 map[v] = Node(v.val)
 q.append(v)
 map[u].neighbors.append(map[v])

 return map[node]
```

• LeetCode

```
"""
Definition for a Node.
class Node:
 def __init__(self, val = 0, neighbors = None):
 self.val = val
 self.neighbors = neighbors if neighbors is not
None else []
"""

from typing import Optional

class Solution:
 def cloneGraph(self, node: Optional["Node"]) ->
Optional["Node"]:
 def dfs(node):
 if node is None:
 return None
 if node in g:
 return g[node]
 cloned = Node(node.val)
 g[node] = cloned

 for nxt in node.neighbors:
 cloned.neighbors.append(dfs(nxt))
 return cloned

 g = defaultdict()
 return dfs(node)
```

• SimplyLeet

```
Definition for a Node.
class Node:
 def __init__(self, val, neighbors=None):
 self.val = val
 # Initialize neighbors with empty list if None is
provided
 self.neighbors = neighbors if neighbors is not
None else []

def cloneGraph(node: 'Node') -> 'Node':
 # Hash map to track original nodes and their
corresponding clones
 old_to_new = {}

 def dfs(current):
 # Base case: if the current node is None, return
None
 if current is None:
 return None
 # If the node has already been cloned, return the
clone to avoid duplication and cycle issues
 if current in old_to_new:
 return old_to_new[current]
 # Create a clone for the current node
 copy = Node(current.val)
```



```
old_to_new[current] = copy
Recursively clone all neighbors
for neighbor in current.neighbors:
 copy.neighbors.append(dfs(neighbor))
return copy

return dfs(node)
```

```
• LC.ca

"""
Definition for a Node.
class Node:
 def __init__(self, val = 0, neighbors = None):
 self.val = val
 self.neighbors = neighbors if neighbors is not
None else []
"""

class Solution:

 def cloneGraph(self, node: 'Node') -> 'Node':
 visited = defaultdict()

 def clone(node):
 if node is None:
 return None
 if node in visited:
 return visited[node]
 c = Node(node.val)
 visited[node] = c
```

```
for e in node.neighbors:
 c.neighbors.append(clone(e))
return c

return clone(node)
```

DFS

#200 Number of Islands

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · DFS · BFS · Union Find · Matrix

Lists: Grind 169

Time Complexity:  $O(m * n)$ , as each cell is visited at most once. Space Complexity:  $O(m * n)$  in the worst-case scenario due to recursion stack (DFS) or auxiliary data structures.

**Problem**

Given an  $m \times n$  2D binary grid grid which represents a map of '1's (land) and '0's (water), return the number of islands.

An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are all surrounded by water.

Example 1:

```
Input: grid = [
["1","1","1","1","0"],
["1","1","0","1","0"],
["1","1","0","0","0"],
["0","0","0","0","0"]
]
Output: 1

Example 2:

Input: grid = [
["1","1","0","0","0"],
["1","1","0","0","0"],
["0","0","1","0","0"],
["0","0","0","1","1"]
]
Output: 3

Constraints:
• m == grid.length
• n == grid[i].length
• 1 <= m, n <= 300
• grid[i][j] is '0' or '1'.

Brute Force (Python)
```

```
Brute Force Approach
class Solution:
 def numIslands(self, grid):
 # Brute Force $O(m^2 \cdot n^2)$ - DFS from every
unvisited land cell
 if not grid: return 0
 m, n = len(grid), len(grid[0])
 visited = set()
 def dfs(r, c):
 if r<0 or r>=m or c<0 or c>=n: return
 if (r,c) in visited or grid[r][c]=='0':
 return

 visited.add((r,c))
 for dr,dc in [(1,0),(-1,0),(0,1),(0,-1)]:
 dfs(r+dr, c+dc)
 count += 1
 for r in range(m):
 for c in range(n):
 if grid[r][c]=='1' and (r,c) not in
visited:
 dfs(r, c)
 count += 1
 return count
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def numIslands(self, grid: list[list[str]]) -> int:
 DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0))
 m = len(grid)
 n = len(grid[0])

 def bfs(r, c):
 q = collections.deque([(r, c)])
 grid[r][c] = '2' # Mark '2' as visited.
 while q:

 i, j = q.popleft()
 for dx, dy in DIRS:
 x = i + dx
 y = j + dy
 if x < 0 or x == m or y < 0 or y == n:
 continue
 if grid[x][y] != '1':
 continue
 q.append((x, y))
 grid[x][y] = '2' # Mark '2' as visited.

 ans = 0

 for i in range(m):
 for j in range(n):
 if grid[i][j] == '1':
 bfs(i, j)
 ans += 1

 return ans
```

```
class Solution:
 def numIslands(self, grid: list[list[str]]) -> int:
 m = len(grid)
 n = len(grid[0])

 def dfs(i: int, j: int) -> None:
 if i < 0 or i == m or j < 0 or j == n:
 return
 if grid[i][j] != '1':
 return

 grid[i][j] = '2' # Mark '2' as visited.
 dfs(i + 1, j)
 dfs(i - 1, j)
 dfs(i, j + 1)
 dfs(i, j - 1)

 ans = 0

 for i in range(m):

 for j in range(n):
 if grid[i][j] == '1':
 dfs(i, j)
 ans += 1

 return ans

• LeetDoocs
```

```
class Solution:
 def numIslands(self, grid: List[List[str]]) -> int:
 def dfs(i, j):
 grid[i][j] = '0'
 for a, b in pairwise(dirs):
 x, y = i + a, j + b
 if 0 <= x < m and 0 <= y < n and
grid[x][y] == '1':
 dfs(x, y)

 ans = 0

 dirs = (-1, 0, 1, 0, -1)
 m, n = len(grid), len(grid[0])
 for i in range(m):
 for j in range(n):
 if grid[i][j] == '1':
 dfs(i, j)
 ans += 1
 return ans

• SimplyLeet

Function to count the number of islands in the grid
def numIslands(grid):
 # If the grid is empty, return 0
 if not grid:
 return 0
 rows, cols = len(grid), len(grid[0])
 islands = 0

 # Depth-First Search helper to mark the entire island
as visited
 def dfs(r, c):
```

```
Check if the current cell is out of bounds or
water
 if r < 0 or r >= rows or c < 0 or c >= cols or
grid[r][c] == "0":
 return
 # Mark the cell as visited by setting it to water
 grid[r][c] = "0"
 # Explore all four adjacent cells (up, down,
left, right)
 dfs(r + 1, c)
 dfs(r - 1, c)
 dfs(r, c + 1)
 dfs(r, c - 1)

Iterate over every cell in the grid
for r in range(rows):
 for c in range(cols):
 # If a land cell is found, conduct DFS to
visit the entire island
 if grid[r][c] == "1":
 islands += 1
 dfs(r, c)

 return islands

• LC.ca
```

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre># dfs class Solution:     def numIslands(self, grid: List[List[str]]) -&gt; int:         def dfs(i, j):             if not (0 &lt;= i &lt; m and 0 &lt;= j &lt; n and                 grid[i][j] == '1'):                 return             grid[i][j] = '0'             for a, b in pairwise(dirs):                 x, y = i + a, j + b                 dfs(x, y)          ans = 0         dirs = (-1, 0, 1, 0, -1)         m, n = len(grid), len(grid[0])         for i in range(m):             for j in range(n):                 if grid[i][j] == '1':                     dfs(i, j)                     ans += 1          return ans  #####  # bfs from collections import deque  class Solution:     def numIslands(self, grid: List[List[str]]) -&gt; int:         if not grid:             return 0</pre> | p.1702 |
| <pre>m, n = len(grid), len(grid[0]) directions = [(0, 1), (0, -1), (1, 0), (-1, 0)] islands = 0  for i in range(m):     for j in range(n):         if grid[i][j] == "1":             islands += 1             q = deque([(i, j)]) # no need to             reset q, q already drained from previous bfs              while q:                 x, y = q.popleft()                 for dx, dy in directions:                     nx, ny = x + dx, y + dy                     if 0 &lt;= nx &lt; m and 0 &lt;= ny &lt; n and grid[nx][ny] == "1":                         q.append((nx, ny))                         grid[nx][ny] = "0" # mark              as visited             return islands  #####</pre>                                                                                                              | p.1703 |
| <pre># union find # similar to UF in https://leetcode.ca/2016-09-30-305-Number-of-Islands-II/ class Solution:     def numIslands(self, grid: List[List[str]]) -&gt; int:         def check(i, j):             return 0 &lt;= i &lt; m and 0 &lt;= j &lt; n and                 grid[i][j] == "1" # Check for "1" instead of 1          def find(x):             if p[x] != x:                  p[x] = find(p[x])             return p[x]          m, n = len(grid), len(grid[0])         p = list(range(m * n))         cur = 0 # Initialize cur as 0 instead of n         for i in range(m):             for j in range(n):                 if grid[i][j] == "1":                     cur += 1 # Increment cur when                     encountering "1"</pre>                                                          | p.1704 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>for x, y in [(-1, 0), (1, 0), (0, -1), (0, 1)]:     if check(i + x, j + y) and find(i * n + j) != find((i + x) * n + j + y):         p[find(i * n + j)] = find((i + x) * n + j + y)         cur -= 1      return cur  #####  ... &gt;&gt;&gt; a = set() &gt;&gt;&gt; &gt;&gt;&gt; a  = {(1, 1)} &gt;&gt;&gt; a {(1, 1)} &gt;&gt;&gt; &gt;&gt;&gt; a  = {(2, 2)} &gt;&gt;&gt; a {(1, 1), (2, 2)}</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | p.1705 |
| <pre>&gt;&gt;&gt; &gt;&gt;&gt; a.add((3, 3)) &gt;&gt;&gt; a {(1, 1), (3, 3), (2, 2)} &gt;&gt;&gt; &gt;&gt;&gt; (1,1) in a True &gt;&gt;&gt; (10,10) in a False ...  class Solution(object):     def numIslands(self, grid):         """"         :type grid: List[List[str]]         :rtype: int         """"         visited = set()         ans = 0          def dfs(grid, i, j, visited):              if i &lt; 0 or i &gt;= len(grid) or j &lt; 0 or j &gt;= len(grid[0]) or grid[i][j] == "0" or (i, j) in visited:                 return False             visited  = {(i, j)}             for di, dj in [(-1, 0), (1, 0), (0, 1), (0, -1)]:                 newi, newj = i + di, j + dj                 dfs(grid, newi, newj, visited)             return True          for i in range(0, len(grid)):             for j in range(0, len(grid[0])):</pre> | p.1706 |
| <pre>if dfs(grid, i, j, visited):     ans += 1 return ans</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | p.1707 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <div>DFS</div> <div>#207 Course ScheduleMED</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>DFS · BFS · Graph · Topological Sort</div> <div>Lists: Grind 169</div> <div>Time Complexity: O(V + E) Space Complexity: O(V + E)</div> <div>Problem</div> <div>There are a total of numCourses courses you have to take, labeled from 0 to numCourses - 1. You are given an array prerequisites where prerequisites[i] = [ai, bi] indicates that you must take course bi first if you want to take course ai.</div> <div>• For example, the pair [0, 1], indicates that to take course 0 you have to first take course 1.</div> <div>Return true if you can finish all courses. Otherwise, return false.</div> <div>Example 1:</div> <div>Input: numCourses = 2, prerequisites = [[1,0]]</div> <div>Output: true</div> <div>Explanation: There are a total of 2 courses to take. To take course 1 you should have finished course 0. So it is possible.</div> <div>Example 2:</div>                                              | p.1708 |
| <div>Input: numCourses = 2, prerequisites = [[1,0],[0,1]]</div> <div>Output: false</div> <div>Explanation: There are a total of 2 courses to take. To take course 1 you should have finished course 0, and to take course 0 you should also have finished course 1. So it is impossible.</div> <div>Constraints:</div> <div>• 1 &lt;= numCourses &lt;= 2000</div> <div>• 0 &lt;= prerequisites.length &lt;= 5000</div> <div>• prerequisites[i].length == 2</div> <div>• 0 &lt;= ai, bi &lt; numCourses</div> <div>• All the pairs prerequisites[i] are unique.</div> <div>Brute Force (Python)</div> <div># Brute Force Approach</div> <div>class Solution:</div> <div>def canFinish(self, numCourses, prerequisites):</div> <div># Brute Force O(V+E) - DFS cycle detection on adjacency list</div> <div>from collections import defaultdict</div> <div>graph = defaultdict(list)</div> <div>for a, b in prerequisites:</div> <div>graph[b].append(a)</div> <div># 0=unvisited, 1=in-stack, 2=done</div> <div>state = [0] * numCourses</div> | p.1709 |
| <div>def has_cycle(node):</div> <div>if state[node] == 1: return True</div> <div>if state[node] == 2: return False</div> <div>state[node] = 1</div> <div>for nei in graph[node]:</div> <div>if has_cycle(nei): return True</div> <div>state[node] = 2</div> <div>return False</div> <div>return not any(has_cycle(c) for c in range(numCourses))</div> <div>Community Solutions (Python)</div> <div>• WalkCC</div> <div>from enum import Enum</div> <div>class State(Enum):</div> <div>INIT = 0</div> <div>VISITING = 1</div> <div>VISITED = 2</div> <div>class Solution:</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | p.1710 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div> <pre> def canFinish(self, numCourses: int, prerequisites: List[List[int]]) -&gt; bool:     graph = [[] for _ in range(numCourses)]     states = [State.INIT] * numCourses      for v, u in prerequisites:         graph[u].append(v)      def hasCycle(u: int) -&gt; bool:         if states[u] == State.VISITING:             return True          if states[u] == State.VISITED:             return False          if states[u] == State.VISITED:             return False         states[u] = State.VISITING         if any(hasCycle(v) for v in graph[u]):             return True         states[u] = State.VISITED         return False      return not any(hasCycle(i) for i in range(numCourses)) </pre> </div> <div> <div>LeetDoocs</div> </div> <div> <div>DFS</div> <div>p.1711</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <div> <pre> class Solution:     def canFinish(self, numCourses: int, prerequisites: List[List[int]]) -&gt; bool:         g = [[] for _ in range(numCourses)]         indeg = [0] * numCourses         for a, b in prerequisites:             g[b].append(a)             indeg[a] += 1         q = [i for i, x in enumerate(indeg) if x == 0]         for i in q:             numCourses -= 1          for j in g[i]:             indeg[j] -= 1             if indeg[j] == 0:                 q.append(j)         return numCourses == 0 </pre> </div> <div> <div>SimplyLeet</div> </div> <div> <pre> from collections import deque  def canFinish(numCourses, prerequisites):     # Create a list to hold the number of prerequisites (indegree) for each course.     indegree = [0] * numCourses     # Create an adjacency list for each course.     graph = {i: [] for i in range(numCourses)}      # Build the graph and update indegree for each course.     for course, prereq in prerequisites: </pre> </div> <div> <div>DFS</div> <div>p.1712</div> </div> | <div> <pre> graph[prereq].append(course) indegree[course] += 1  # Initialize a queue with courses that have no prerequisites. queue = deque([i for i in range(numCourses) if indegree[i] == 0])  # Counter for courses that can be completed. visited = 0  while queue:      course = queue.popleft() # Remove course with no remaining prerequisites.     visited += 1 # Increment count of completed courses.     # For each neighbor course, reduce its indegree.     for neighbor in graph[course]:         indegree[neighbor] -= 1 # Remove dependency.         # If no more prerequisites, add to queue.         if indegree[neighbor] == 0:             queue.append(neighbor)  # If we processed all courses, it's possible to finish.  return visited == numCourses  # Example usage: print(canFinish(2, [[1,0]])) # Expected output: True print(canFinish(2, [[1,0],[0,1]])) # Expected output: False </pre> </div> <div> <div>DFS</div> <div>p.1713</div> </div>                                                                                                                                                                                                                                                                                                               |
| <div> <div>LC.ca</div> <pre> ''' &gt;&gt;&gt; from collections import defaultdict  &gt;&gt;&gt; a = defaultdict(int) &gt;&gt;&gt; a defaultdict(&lt;type 'int'&gt;, {}) &gt;&gt;&gt; a['hehehe'] 0 &gt;&gt;&gt; a defaultdict(&lt;type 'int'&gt;, {'hehehe': 0})  &gt;&gt;&gt; a = defaultdict(list) &gt;&gt;&gt; a defaultdict(&lt;type 'list'&gt;, {}) &gt;&gt;&gt; a['hehehe'] [] &gt;&gt;&gt; a defaultdict(&lt;type 'list'&gt;, {'hehehe': []}) </pre> </div> <div> <div>DFS</div> <div>p.1714</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <div> <pre> ... class Solution:     def canFinish(self, numCourses: int, prerequisites: List[List[int]]) -&gt; bool:         g = defaultdict(list)         indeg = [0] * numCourses         for a, b in prerequisites:             g[b].append(a)             indeg[a] += 1         cnt = 0         q = deque([i for i, v in enumerate(indeg) if v == 0])          while q:             i = q.popleft()             cnt += 1             for j in g[i]:                 indeg[j] -= 1                 if indeg[j] == 0:                     q.append(j)         return cnt == numCourses </pre> </div> <div> <div>DFS</div> <div>p.1715</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                              | <div> <div>DFS</div> </div> <div> <div>#210 Course Schedule II</div> <div>MED</div> </div> <div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>DFS · BFS · Graph · Topological Sort</div> <div>Lists: Grind 169</div> <div>Time Complexity: <math>O(V + E)</math> where <math>V</math> is the number of courses and <math>E</math> is the number of prerequisite pairs. Space Complexity: <math>O(V + E)</math> for storing the graph and in-degree array.</div> <div>Problem</div> <div>There are a total of <code>numCourses</code> courses you have to take, labeled from 0 to <code>numCourses - 1</code>. You are given an array prerequisites where <code>prerequisites[i] = [ai, bi]</code> indicates that you must take course <code>bi</code> first if you want to take course <code>ai</code>.</div> <div> <ul style="list-style-type: none"> <li>For example, the pair <code>[0, 1]</code>, indicates that to take course 0 you have to first take course 1.</li> </ul> </div> <div>Return the ordering of courses you should take to finish all courses. If there are many valid answers, return any of them. If it is impossible to finish all courses, return an empty array.</div> <div>Example 1:</div> </div> <div> <div>DFS</div> <div>p.1716</div> </div> |
| <div> <div>Input: numCourses = 2, prerequisites = [[1,0]]<br/>Output: [0,1]<br/>Explanation: There are a total of 2 courses to take. To take course 1 you should have finished course 0. So the correct course order is [0,1].</div> <div>Example 2:</div> <div> <div>Input: numCourses = 4, prerequisites = [[1,0],[2,0],[3,1],[3,2]]<br/>Output: [0,2,1,3]<br/>Explanation: There are a total of 4 courses to take. To take course 3 you should have finished both courses 1 and 2. Both courses 1 and 2 should be taken after you finished course 0.<br/>So one correct course order is [0,1,2,3]. Another correct ordering is [0,2,1,3].</div> </div> <div>Example 3:</div> <div> <div>Input: numCourses = 1, prerequisites = []<br/>Output: [0]</div> </div> <div> <div>Constraints:</div> <ul style="list-style-type: none"> <li><math>1 \leq \text{numCourses} \leq 2000</math></li> <li><math>0 \leq \text{prerequisites.length} \leq \text{numCourses} * (\text{numCourses} - 1)</math></li> <li><math>\text{prerequisites}[i].\text{length} == 2</math></li> <li><math>0 \leq a_i, b_i &lt; \text{numCourses}</math></li> <li><math>a_i \neq b_i</math></li> <li>All the pairs <code>[ai, bi]</code> are distinct.</li> </ul> </div> <div>Brute Force (Python)</div> <div> <div>DFS</div> <div>p.1717</div> </div> </div> | <div> <div># Brute Force Approach</div> <pre> class Solution:     def findOrder(self, numCourses, prerequisites):         # Brute Force <math>O(V+E)</math> - DFS topological sort         from collections import defaultdict         graph = defaultdict(list)         for a, b in prerequisites:             graph[b].append(a)         state = [0] * numCourses         result = []          def dfs(node):             if state[node] == 1: return False # cycle             if state[node] == 2: return True             state[node] = 1             for nei in graph[node]:                 if not dfs(nei): return False             state[node] = 2             result.append(node)             return True         for c in range(numCourses):              if not dfs(c): return []         return result[::-1] </pre> </div> <div> <div>Community Solutions (Python)</div> </div> <div> <div>WalkCC</div> </div> <div> <div>DFS</div> <div>Sheet 191 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait</div> <div>p.1718</div> </div>      | <div> <pre> from enum import Enum  class State(Enum):     INIT = 0     VISITING = 1     VISITED = 2  class Solution:      def findOrder(         self,         numCourses: int,         prerequisites: List[List[int]],     ) -&gt; List[int]:         ans = []         graph = [[] for _ in range(numCourses)]         states = [State.INIT] * numCourses          for v, u in prerequisites:              graph[u].append(v)          def hasCycle(u: int) -&gt; bool:             if states[u] == State.VISITING:                 return True             if states[u] == State.VISITED:                 return False             states[u] = State.VISITING             if any(hasCycle(v) for v in graph[u]):                 return True </pre> </div> <div> <div>DFS</div> <div>p.1719</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

```
states[u] = State.VISITED
ans.append(u)
return False

if any(hasCycle(i) for i in range(numCourses)):
 return []

return ans[::-1]

class Solution:
 def findOrder(
 self,
 numCourses: int,
 prerequisites: List[List[int]],
) -> List[int]:
 ans = []
 graph = [[] for _ in range(numCourses)]
 inDegrees = [0] * numCourses
 q = collections.deque()

 # Build the graph.
 for v, u in prerequisites:
 graph[u].append(v)
 inDegrees[v] += 1

 # Perform topological sorting.
 q = collections.deque([i for i, d in
 enumerate(inDegrees) if d == 0])

 while q:
```

```
u = q.popleft()
ans.append(u)
for v in graph[u]:
 inDegrees[v] -= 1
 if inDegrees[v] == 0:
 q.append(v)

return ans if len(ans) == numCourses else []

• LeetCode

class Solution:
 def findOrder(self, numCourses: int, prerequisites:
 List[List[int]]) -> List[int]:
 g = defaultdict(list)
 indeg = [0] * numCourses
 for a, b in prerequisites:
 g[b].append(a)
 indeg[a] += 1
 ans = []
 q = deque(i for i, x in enumerate(indeg) if x ==
 0)

 while q:

 i = q.popleft()
 ans.append(i)
 for j in g[i]:
 indeg[j] -= 1
 if indeg[j] == 0:
 q.append(j)
 q.append(j)

 return ans if len(ans) == numCourses else []

• SimplyLeet
```

```
Python solution using BFS (Kahn's Algorithm)
from collections import deque

def findOrder(numCourses, prerequisites):
 # Build the graph and the in-degree array
 in_degree = [0] * numCourses
 graph = {i: [] for i in range(numCourses)}
 for course, pre in prerequisites:
 graph[pre].append(course)
 in_degree[course] += 1

 # Initialize the queue with all courses having zero
 in-degree
 queue = deque([i for i in range(numCourses) if
 in_degree[i] == 0])
 order = []

 # Process courses
 while queue:
 course = queue.popleft()
 order.append(course)
 for neighbor in graph[course]:
```

```
in_degree[neighbor] -= 1
if in_degree[neighbor] == 0:
 queue.append(neighbor)

If the ordering includes all courses, return it;
otherwise, return an empty list
return order if len(order) == numCourses else []

Example usage:
print(findOrder(4, [[1,0],[2,0],[3,1],[3,2]]))

• LC.ca

class Solution:
 def findOrder(self, numCourses: int, prerequisites:
 List[List[int]]) -> List[int]:
 g = defaultdict(list)
 indeg = [0] * numCourses
 for a, b in prerequisites:
 g[b].append(a)
 indeg[a] += 1
 q = deque([i for i, v in enumerate(indeg) if v ==
 0])
 ans = [] # added from previous question
 while q:
```

```
i = q.popleft()
ans.append(i)
assumption is only one path
as in question 'You may assume that there
are no duplicate edges in the input prerequisites.'
for j in g[i]:
 indeg[j] -= 1
 if indeg[j] == 0:
 q.append(j)
return ans if len(ans) == numCourses else []
```

DFS

#261 Graph Valid Tree

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

DFS · BFS · Union Find · Graph

Lists: Grind 169 | Premium 98

Time Complexity:  $O(n + e)$ , where  $e$  is the number of edges  
Space Complexity:  $O(n + e)$  for storing the graph and traversal/union-find structures

Problem

You have a graph of  $n$  nodes labeled from 0 to  $n - 1$ . You are given an integer  $n$  and a list of edges where  $edges[i] = [a_i, b_i]$  indicates that there is an undirected edge between nodes  $a_i$  and  $b_i$  in the graph. Return true if the edges of the given graph make up a valid tree, and false otherwise.

Example 1:

Input:  $n = 5$ ,  $edges = [[0,1],[0,2],[0,3],[1,4]]$   
Output: true

Example 2:

Input:  $n = 5$ ,  $edges = [[0,1],[1,2],[2,3],[1,3],[1,4]]$   
Output: false

Constraints:

- $1 \leq n \leq 2000$
- $0 \leq edges.length \leq 5000$
- $edges[i].length == 2$
- $0 \leq a_i, b_i < n$
- $a_i \neq b_i$
- There are no self-loops or repeated edges.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def validTree(self, n, edges):
 # Brute Force - DFS from every unvisited node
 visited = set()
 count = 0
 def dfs(node):
 visited.add(node)
 for nb in graph.get(node, []):
 if nb not in visited:
 dfs(nb)

 for node in range(n):
 if node not in visited:
 dfs(node); count += 1
 return count

Community Solutions (Python)

• WalkCC

class Solution:
 def validTree(self, n: int, edges: List[List[int]]) ->
 bool:
 if n == 0 or len(edges) != n - 1:
 return False

 graph = [[] for _ in range(n)]
 q = collections.deque([0])
 seen = {0}

 for u, v in edges:
```

```
graph[u].append(v)
graph[v].append(u)

while q:
 u = q.popleft()
 for v in graph[u]:
 if v not in seen:
 q.append(v)
 seen.add(v)

return len(seen) == n

class UnionFind:
 def __init__(self, n: int):
 self.count = n
 self.id = list(range(n))
 self.rank = [0] * n

 def unionByRank(self, u: int, v: int) -> None:
 i = self._find(u)
 j = self._find(v)
 if i == j:
 return

 if self.rank[i] < self.rank[j]:
 self.id[i] = j
 elif self.rank[i] > self.rank[j]:
 self.id[j] = i
 else:
 self.id[i] = j
 self.rank[j] += 1
 self.count -= 1
```

DFS p.1729

```
def _find(self, u: int) -> int:
 if self.id[u] != u:
 self.id[u] = self._find(self.id[u])
 return self.id[u]

class Solution:
 def validTree(self, n: int, edges: List[List[int]]) -> bool:
 if n == 0 or len(edges) != n - 1:
 return False

 uf = UnionFind(n)

 for u, v in edges:
 uf.unionByRank(u, v)

 return uf.count == 1

• LeetCode
class Solution:
 def validTree(self, n: int, edges: List[List[int]]) -> bool:
 def find(x: int) -> int:
 if p[x] != x:
 p[x] = find(p[x])
 return p[x]

 p = list(range(n))
 for a, b in edges:
 pa, pb = find(a), find(b)
```

DFS p.1730

```
if pa == pb:
 return False
p[pa] = pb
n -= 1
return n == 1

• SimplyLeet
Python solution with DFS

def validTree(n, edges):
 # If the number of edges is not exactly n - 1, it cannot be a tree
 if len(edges) != n - 1:
 return False

 # Build the graph as an adjacency list
 graph = {}
 for i in range(n):
 graph[i] = []
 for a, b in edges:
 graph[a].append(b)
 graph[b].append(a)

 visited = set()

 # DFS helper function that accepts current node and its parent node
 def dfs(node, parent):
 # Add the current node to visited
 visited.add(node)
 # Traverse all the neighboring nodes
```

DFS p.1731

```
for neighbor in graph[node]:
 # Skip the neighbor if it is the parent to avoid trivial cycle detection
 if neighbor == parent:
 continue
 # If neighbor is already visited, a cycle is detected
 if neighbor in visited:
 return False
 # Recurse deeper; if cycle is found in recursion, propagate False upward
 if not dfs(neighbor, node):
 return False

return True

Start DFS from node 0, with parent set to -1 (non-existent)
if not dfs(0, -1):
 return False

Check if all nodes were visited (i.e., the graph is connected)
return len(visited) == n

Example usage:
print(validTree(5, [[0, 1], [0, 2], [0, 3], [1, 4]])) # Expected output: True
print(validTree(5, [[0, 1], [1, 2], [2, 3], [1, 3], [1, 4]])) # Expected output: False

• LC.ca
```

DFS p.1732

```
class Solution:
 def validTree(self, n: int, edges: List[List[int]]) -> bool:
 def find(x):
 if p[x] != x:
 p[x] = find(p[x])
 return p[x]

 p = list(range(n))
 for a, b in edges:
 if find(a) == find(b):
 return False
 p[find(a)] = find(b)
 n -= 1
 return n == 1
```

DFS p.1733

DFS

#310 Minimum Height Trees

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

DFS · BFS · Graph · Topological Sort

Lists: Grind 169

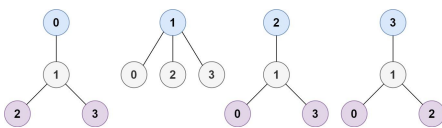
Time Complexity:  $O(n)$  where  $n$  is the number of nodes, since we process each node and edge once. Space Complexity:  $O(n)$  for storing the graph and additional data structures like the leaves list.

Problem

A tree is an undirected graph in which any two vertices are connected by exactly one path. In other words, any connected graph without simple cycles is a tree. Given a tree of  $n$  nodes labelled from  $0$  to  $n - 1$ , and an array of  $n - 1$  edges where  $edges[i] = [a_i, b_i]$  indicates that there is an undirected edge between the two nodes  $a_i$  and  $b_i$  in the tree, you can choose any node of the tree as the root. When you select a node  $x$  as the root, the result tree has height  $h$ . Among all possible rooted trees, those with minimum height (i.e.  $\min(h)$ ) are called minimum height trees (MHTs). Return a list of all MHTs' root labels. You can return the answer in any order.

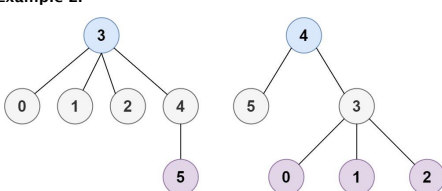
DFS p.1734

The height of a rooted tree is the number of edges on the longest downward path between the root and a leaf. Example 1:



Input:  $n = 4$ ,  $edges = [[1,0],[1,2],[1,3]]$   
Output: [1]  
Explanation: As shown, the height of the tree is 1 when the root is the node with label 1 which is the only MHT.

Example 2:



Input:  $n = 6$ ,  $edges = [[3,0],[3,1],[3,2],[3,4],[5,4]]$   
Output: [3,4]

Constraints:

DFS p.1735

- $1 \leq n \leq 2 \cdot 10^4$
- $edges.length == n - 1$
- $0 \leq a_i, b_i < n$
- $a_i \neq b_i$
- All the pairs  $(a_i, b_i)$  are distinct.
- The given input is guaranteed to be a tree and there will be no repeated edges.

Community Solutions (Python)

• WalkCC

```
class Solution:
 def findMinHeightTrees(self, n: int, edges: List[List[int]]) -> List[int]:
 if n == 1 or not edges:
 return [0]

 ans = []
 graph = collections.defaultdict(set)

 for u, v in edges:
 graph[u].add(v)
```

DFS p.1736

```
graph[v].add(u)

for label, children in graph.items():
 if len(children) == 1:
 ans.append(label)

while n > 2:
 n -= len(ans)
 nextLeaves = []
 for leaf in ans:
 u = next(iter(graph[leaf]))
 graph[u].remove(leaf)
 if len(graph[u]) == 1:
 nextLeaves.append(u)
 ans = nextLeaves

 return ans

• LeetCode
class Solution:
 def findMinHeightTrees(self, n: int, edges: List[List[int]]) -> List[int]:
 if n == 1:
 return [0]
 g = [[] for _ in range(n)]
 degree = [0] * n
 for a, b in edges:
 g[a].append(b)
 g[b].append(a)
 degree[a] += 1
```

DFS p.1737

```
degree[b] += 1
q = deque(i for i in range(n) if degree[i] == 1)
ans = []
while q:
 ans.clear()
 for _ in range(len(q)):
 a = q.popleft()
 ans.append(a)
 for b in g[a]:
 degree[b] -= 1

if degree[b] == 1:
 q.append(b)

return ans
```

• SimplyLeet

```
Python solution with line-by-line comments
class Solution:
 def findMinHeightTrees(self, n: int, edges: List[List[int]]) -> List[int]:
 # Special case: if there's only one node, return it as the only MHT root.
 if n == 1:
 return [0]

 # Build the graph using an adjacency list with sets for neighbors.
 graph = [set() for _ in range(n)]
 for u, v in edges:
```

```
graph[u].add(v)
graph[v].add(u)

Initialize the first layer of leaves (nodes with one neighbor).
leaves = [i for i in range(n) if len(graph[i]) == 1]

Remove leaves layer by layer until at most 2 nodes remain.
remaining_nodes = n
while remaining_nodes > 2:
 remaining_nodes -= len(leaves)

new_leaves = []
Remove current leaves.
for leaf in leaves:
 # Since it's a leaf, there is only one neighbor.
 neighbor = graph[leaf].pop()
 # Remove the connection from the neighbor to the leaf.
 graph[neighbor].remove(leaf)
 # If the neighbor becomes a leaf, add it to the new leaves list.
 if len(graph[neighbor]) == 1:
 new_leaves.append(neighbor)

leaves = new_leaves

The remaining nodes are the roots of the MHTs.
return leaves
```

```
from typing import List, Set
from collections import defaultdict

class Solution:
 def findMinHeightTrees(self, n: int, edges: List[List[int]]) -> List[int]:
 if n == 1:
 return [0]

 graph = defaultdict(set)

 for a, b in edges:
 graph[a].add(b)
 graph[b].add(a)

 # just check neighbours[] size, not indgree[] for each node
 leaves = [i for i in range(n) if len(graph[i]) == 1]

 while n > 2:
 n -= len(leaves)
 new_leaves = []
```

```
for leaf in leaves:
 # should be only one in hashset pop(), because it's a leaf node
 neighbor = graph[leaf].pop()
 graph[neighbor].remove(leaf)
 if len(graph[neighbor]) == 1:
 new_leaves.append(neighbor)
leaves = new_leaves

return leaves

#####

class Solution:
 def findMinHeightTrees(self, n: int, edges: List[List[int]]) -> List[int]:
 if n == 1:
 return [0]
 g = defaultdict(list)
 degree = [0] * n
 for a, b in edges:
 g[a].append(b)
```

```
g[b].append(a)
degree[a] += 1 # not needed, just len(g[a]) is good enough, will update it in next round updating
degree[b] += 1
q = deque()
for i in range(n):
 if degree[i] == 1:
 q.append(i)

ans = []
while q:
 n = len(q)

 ans.clear()
 for _ in range(n):
 a = q.popleft()
 ans.append(a)
 for b in g[a]:
 degree[b] -= 1
 if degree[b] == 1: # final round only
 q.append(b)

return ans
```

DFS

#323 Number of Connected Components in an Undirected Graph

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

DFS · BFS · Union Find · Graph

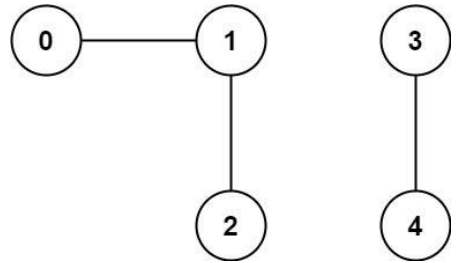
Lists: Grind 169 | Premium 98

Time Complexity:  $O(n + e)$ , where  $n$  is the number of nodes and  $e$  is the number of edges. Space Complexity:  $O(n + e)$  due to the storage of the graph and tracking of visited nodes.

Problem

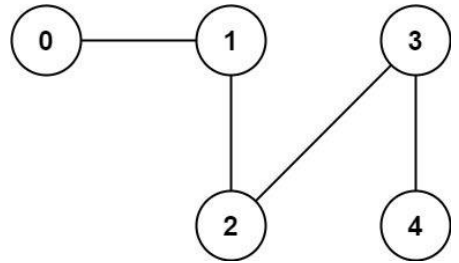
You have a graph of  $n$  nodes. You are given an integer  $n$  and an array `edges` where `edges[i] = [ai, bi]` indicates that there is an edge between `ai` and `bi` in the graph. Return the number of connected components in the graph.

Example 1:



Input:  $n = 5$ ,  $edges = [[0,1],[1,2],[3,4]]$   
Output: 2

Example 2:



Input:  $n = 5$ ,  $edges = [[0,1],[1,2],[2,3],[3,4]]$   
Output: 1

Constraints:

- $1 \leq n \leq 2000$
- $1 \leq edges.length \leq 5000$
- $edges[i].length == 2$
- $0 \leq ai \leq bi < n$
- $ai \neq bi$
- There are no repeated edges.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def count_components(self, n, edges):
 # Brute Force - DFS from every unvisited node
 visited = set()
 count = 0
 def dfs(node):
 visited.add(node)
 for nb in graph.get(node, []):
 if nb not in visited:
 dfs(nb)

 for node in range(n):
 if node not in visited:
 dfs(node); count += 1

 return count
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def countComponents(self, n: int, edges: List[List[int]]) -> int:
 ans = 0
 graph = [[] for _ in range(n)]
 seen = set()

 for u, v in edges:
 graph[u].append(v)
 graph[v].append(u)
```



```
def bfs(node: int, seen: set[int]) -> None:
 q = collections.deque([node])
 seen.add(node)

 while q:
 u = q.pop()
 for v in graph[u]:
 if v not in seen:
 q.append(v)
 seen.add(v)

for i in range(n):
 if i not in seen:
 bfs(i, seen)
 ans += 1

return ans

class UnionFind:
 def __init__(self, n: int):
 self.count = n
 self.id = list(range(n))
 self.rank = [0] * n

 def unionByRank(self, u: int, v: int) -> None:
 i = self._find(u)
 j = self._find(v)
 if i == j:
```

DFS p.1747

```
 return
 if self.rank[i] < self.rank[j]:
 self.id[i] = j
 elif self.rank[i] > self.rank[j]:
 self.id[j] = i
 else:
 self.id[i] = j
 self.rank[j] += 1
 self.count -= 1

 def _find(self, u: int) -> int:
 if self.id[u] != u:
 self.id[u] = self._find(self.id[u])
 return self.id[u]

class Solution:
 def countComponents(self, n: int, edges: List[List[int]]) -> int:
 uf = UnionFind(n)

 for u, v in edges:
 uf.unionByRank(u, v)

 return uf.count
```

• LeetCode

DFS p.1748

```
class Solution:
 def countComponents(self, n: int, edges: List[List[int]]) -> int:
 def dfs(i: int) -> int:
 if i in vis:
 return 0
 vis.add(i)
 for j in g[i]:
 dfs(j)
 return 1

 g = [[] for _ in range(n)]
 for a, b in edges:
 g[a].append(b)
 g[b].append(a)
 vis = set()
 return sum(dfs(i) for i in range(n))

class UnionFind:
 def __init__(self, n):
 self.p = list(range(n))
 self.size = [1] * n

 def find(self, x):
 if self.p[x] != x:
 self.p[x] = self.find(self.p[x])
 return self.p[x]
```

DFS p.1749

```
def union(self, a, b):
 pa, pb = self.find(a), self.find(b)
 if pa == pb:
 return False
 if self.size[pa] > self.size[pb]:
 self.p[pb] = pa
 self.size[pa] += self.size[pb]
 else:
 self.p[pa] = pb
 self.size[pb] += self.size[pa]

 return True

class Solution:
 def countComponents(self, n: int, edges: List[List[int]]) -> int:
 uf = UnionFind(n)
 for a, b in edges:
 n -= uf.union(a, b)
 return n
```

• SimplyLeet

DFS p.1750

```
Function to count the number of connected components in an undirected graph.
def count_components(n, edges):
 # Build the adjacency list for the graph.
 graph = {i: [] for i in range(n)}
 for a, b in edges:
 graph[a].append(b)
 graph[b].append(a)

 # Set to keep track of visited nodes.
 visited = set()

 # Helper function to perform DFS traversal starting from a node.
 def dfs(node):
 stack = [node]
 while stack:
 current = stack.pop()
 if current not in visited:
 visited.add(current)
 for neighbor in graph[current]:
 if neighbor not in visited:
```

DFS p.1751

```
 stack.append(neighbor)

 # Count connected components by initiating DFS on unvisited nodes.
 count = 0
 for i in range(n):
 if i not in visited:
 dfs(i)
 count += 1
 return count

Example usage:
print(count_components(5, [[0,1],[1,2],[3,4]])) # Expected output: 2

• LC.ca
...
example-2: [[0, 1], [1, 2], [2, 3], [3, 4]]

for loop

[0, 1]
 0 parent set to 1
 1 parent set to 1

[1, 2]
```

DFS p.1752

```
1 parent set to 2
2 parent set to 2

[2, 3]
2 parent set to 3
3 parent set to 3

final:
0 parent set to 1
1 parent set to 2

2 parent set to 3
3 parent set to 3

==> so filter for 'i == find(i)'
...
class Solution:
 def countComponents(self, n: int, edges: List[List[int]]) -> int:
 def find(x):
 if p[x] != x:
 p[x] = find(p[x])
```

DFS p.1753

```
 return p[x]

 p = list(range(n))
 for a, b in edges:
 p[find(a)] = find(b)
 return sum(i == find(i) for i in range(n)) # or len(set(parents))

#####

class Solution: # more literal

 def countComponents(self, n: int, edges: List[List[int]]) -> int:
 # Initialize parent array for union-find
 parent = [i for i in range(n)]

 # Perform union-find on each edge
 for u, v in edges:
 self.union(u, v, parent)

 # Count the number of distinct parents (connected components)
 count = len(set(self.find(x, parent) for x in range(n)))
```

DFS p.1754

```
 return count

 def union(self, x: int, y: int, parent: List[int]) -> None:
 rootX = self.find(x, parent)
 rootY = self.find(y, parent)
 if rootX != rootY:
 parent[rootX] = rootY

 def find(self, x: int, parent: List[int]) -> int:
 if parent[x] != x:
 parent[x] = self.find(parent[x], parent)
 return parent[x]
```

DFS p.1755

DFS

#417 Pacific Atlantic Water FlowMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · DFS · BFS · Matrix

Lists: Grind 169

Time Complexity:  $O(m * n)$  where m is the number of rows and n is the number of columns. Each cell is processed up to twice (once from each ocean). Space Complexity:  $O(m * n)$  for the visited matrices a

Problem

There is an  $m \times n$  rectangular island that borders both the Pacific Ocean and Atlantic Ocean. The Pacific Ocean touches the island's left and top edges, and the Atlantic Ocean touches the island's right and bottom edges. The island is partitioned into a grid of square cells. You are given an  $m \times n$  integer matrix heights where heights[r][c] represents the height above sea level of the cell at coordinate (r, c). The island receives a lot of rain, and the rain water can flow to neighboring cells directly north, south, east, and west if the neighboring cell's height is less than or equal to the current cell's height. Water can flow from any cell adjacent to an ocean into the ocean.

DFS

p.1756

Return a 2D list of grid coordinates result where result[i] = [ri, ci] denotes that rain water can flow from cell (ri, ci) to both the Pacific and Atlantic oceans. Example 1:

DFS

p.1757

Input: heights = [[1,2,2,3,5],[3,2,3,4,4],[2,4,5,3,1],[6,7,1,4,5],[5,1,1,2,4]]  
Output: [[0,4],[1,3],[1,4],[2,2],[3,0],[3,1],[4,0]]  
Explanation: The following cells can flow to the Pacific and Atlantic oceans, as shown below:  
[0,4]: [0,4] -> Pacific Ocean  
          [0,4] -> Atlantic Ocean  
[1,3]: [1,3] -> [0,3] -> Pacific Ocean  
          [1,3] -> [1,4] -> Atlantic Ocean  
[1,4]: [1,4] -> [1,3] -> [0,3] -> Pacific Ocean

Example 2:  
Input: heights = [[1]]  
Output: [[0,0]]  
Explanation: The water can flow from the only cell to the Pacific and Atlantic oceans.

Constraints:

- $m == \text{heights.length}$
- $n == \text{heights[r].length}$
- $1 \leq m, n \leq 200$
- $0 \leq \text{heights[r][c]} \leq 105$

Community Solutions (Python)

- WalkCC

DFS

p.1758

```
class Solution:
 def pacificAtlantic(self, heights: List[List[int]]) -> List[List[int]]:
 m = len(heights)
 n = len(heights[0])
 seenP = [[False] * n for _ in range(m)]
 seenA = [[False] * n for _ in range(m)]

 def dfs(i: int, j: int, h: int, seen: List[List[bool]]) -> None:
 if i < 0 or i == m or j < 0 or j == n:
 return

 if seen[i][j] or heights[i][j] < h:
 return

 seen[i][j] = True
 dfs(i + 1, j, heights[i][j], seen)
 dfs(i - 1, j, heights[i][j], seen)
 dfs(i, j + 1, heights[i][j], seen)
 dfs(i, j - 1, heights[i][j], seen)

 for i in range(m):
```

DFS

p.1759

```
dfs(i, 0, 0, seenP)
dfs(i, n - 1, 0, seenA)

for j in range(n):
 dfs(0, j, 0, seenP)
 dfs(m - 1, j, 0, seenA)

return [[i, j]
 for i in range(m)
 for j in range(n)
 if seenP[i][j] and seenA[i][j]]
```

• SimplyLeet

# Python solution for Pacific Atlantic Water Flow

class Solution:
 def pacificAtlantic(self, heights: List[List[int]]) -> List[List[int]]:
 if not heights or not heights[0]:
 return []

 rows, cols = len(heights), len(heights[0])
 pacific\_reachable = [[False for \_ in range(cols)] for \_ in range(rows)]
 atlantic\_reachable = [[False for \_ in range(cols)] for \_ in range(rows)]

DFS

p.1760

```
Directions: up, down, left, right
directions = [(-1, 0), (1, 0), (0, -1), (0, 1)]

DFS function to mark reachable cells
def dfs(r, c, reachable):
 reachable[r][c] = True
 for dr, dc in directions:
 new_r, new_c = r + dr, c + dc
 # Check bounds and conditions to flow

water

 if (0 <= new_r < rows and 0 <= new_c <
 cols and
 not reachable[new_r][new_c] and
 heights[new_r][new_c] >=
 heights[r][c]):
 dfs(new_r, new_c, reachable)

DFS from Pacific ocean borders (top row and left column)
for i in range(rows):
 dfs(i, 0, pacific_reachable)
for j in range(cols):
 dfs(0, j, pacific_reachable)
```

DFS

p.1761

```
DFS from Atlantic ocean borders (bottom row and right column)
for i in range(rows):
 dfs(i, cols - 1, atlantic_reachable)
for j in range(cols):
 dfs(rows - 1, j, atlantic_reachable)

Collect cells that can reach both oceans
result = []
for r in range(rows):

 for c in range(cols):
 if pacific_reachable[r][c] and atlantic_reachable[r][c]:
 result.append([r, c])

 return result
```

• LC.ca

class Solution:
 def pacificAtlantic(self, heights: List[List[int]]) -> List[List[int]]:
 def bfs(q, vis):
 while q:
 for \_ in range(len(q)):
 i, j = q.popleft()
 for a, b in [[0, -1], [0, 1], [1, 0], [-1, 0]]:
 x, y = i + a, j + b
 if (
 0 <= x < m

DFS

p.1762

```
and 0 <= y < n
and (x, y) not in vis
and heights[x][y] >=
heights[i][j]
):
 vis.add((x, y))
 q.append((x, y))

m, n = len(heights), len(heights[0])
vis1, vis2 = set(), set()
q1 = deque()

q2 = deque()
for i in range(m):
 for j in range(n):
 if i == 0 or j == 0:
 vis1.add((i, j))
 q1.append((i, j))
 if i == m - 1 or j == n - 1:
 vis2.add((i, j))
 q2.append((i, j))

bfs(q1, vis1)

bfs(q2, vis2)
return [
 (i, j)
 for i in range(m)
 for j in range(n)
 if (i, j) in vis1 and (i, j) in vis2
]
```

DFS

p.1763

DFS

#490 The MazeMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · DFS · BFS · Matrix

Lists: Premium 98

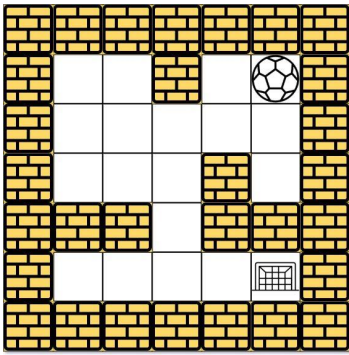
Time Complexity:  $O(m * n)$  where m and n are the dimensions of the maze. Space Complexity:  $O(m * n)$  due to the storage used by the visited matrix and the BFS/DFS queue or recursion stack.

Problem

There is a ball in a maze with empty spaces (represented as 0) and walls (represented as 1). The ball can go through the empty spaces by rolling up, down, left or right, but it won't stop rolling until hitting a wall. When the ball stops, it could choose the next direction. Given the  $m \times n$  maze, the ball's start position and the destination, where start = [startrow, startcol] and destination = [destinationrow, destinationcol], return true if the ball can stop at the destination, otherwise return false. You may assume that the borders of the maze are all walls (see examples). Example 1:

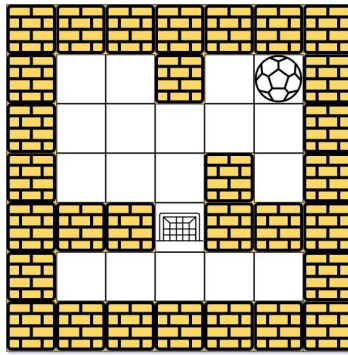
DFS

p.1764



Input: maze = [[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]], start = [0,4], destination = [4,4]  
Output: true  
Explanation: One possible way is : left -> down -> left -> down -> right -> down -> right.

Example 2:



Input: maze = [[0,0,1,0,0],[0,0,0,0,0],[0,0,0,1,0],[1,1,0,1,1],[0,0,0,0,0]], start = [0,4], destination = [3,2]  
Output: false  
Explanation: There is no way for the ball to stop at the destination. Notice that you can pass through the destination but you cannot stop there.

Example 3:

Input: maze = [[0,0,0,0,0],[1,1,0,0,1],[0,0,0,0,0],[0,1,0,0,1],[0,1,0,0,0]], start = [4,3], destination = [0,1]  
Output: false

Constraints:

- m == maze.length
- n == maze[i].length
- 1 <= m, n <= 100
- maze[i][j] is 0 or 1.
- start.length == 2
- destination.length == 2
- 0 <= startrow, destinationrow < m
- 0 <= startcol, destinationcol < n
- Both the ball and the destination exist in an empty space, and they will not be in the same position initially.
- The maze contains at least 2 empty spaces.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def hasPath(self, maze, start, destination):
 # Brute Force - DFS from every unvisited node
 visited = set()
 count = 0
 def dfs(node):
 visited.add(node)
 for nb in graph.get(node, []):
 if nb not in visited: dfs(nb)
```

DFS

p.1765

DFS

p.1766

DFS

p.1767

```
for node in range(n):
 if node not in visited:
 dfs(node); count += 1
return count
```

## Community Solutions (Python)

### • WalkCC

```
class Solution:
 def hasPath(
 self,
 maze: List[List[int]],
 start: List[int],
 destination: List[int],
) -> bool:
 DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0))
 m = len(maze)
 n = len(maze[0])

 q = collections.deque([(start[0], start[1])])
 seen = {(start[0], start[1])}

 def isValid(x: int, y: int) -> bool:
 return 0 <= x < n and 0 <= y < m and maze[x][y] == 0

 while q:
 i, j = q.popleft()
 for dx, dy in DIRS:
 x = i
```

DFS

p.1768

```
y = j
while isValid(x + dx, y + dy):
 x += dx
 y += dy
 if [x, y] == destination:
 return True
 if (x, y) in seen:
 continue
 q.append((x, y))
 seen.add((x, y))
```

return False

### • LeetCode

```
class Solution:
 def hasPath(
 self, maze: List[List[int]], start: List[int],
 destination: List[int]
) -> bool:
 def dfs(i, j):
 if vis[i][j]:
 return
 vis[i][j] = True
 if [i, j] == destination:
 return
```

DFS

p.1769

```
for a, b in [(0, -1), (0, 1), (1, 0), (-1, 0)]:
 x, y = i, j
 while 0 <= x + a < m and 0 <= y + b < n
 and maze[x + a][y + b] == 0:
 x, y = x + a, y + b
 dfs(x, y)

m, n = len(maze), len(maze[0])
vis = [[False] * n for _ in range(m)]
dfs(start[0], start[1])
return vis[destination[0]][destination[1]]
```

### • SimplyLeet

```
Function to determine if there is a valid path in the
maze from start to destination
def hasPath(maze, start, destination):
 from collections import deque
 rows, cols = len(maze), len(maze[0])
 # Create a visited 2D array to keep track of
 processed cells
 visited = [[False] * cols for _ in range(rows)]
 queue = deque([tuple(start)])
 visited[start[0]][start[1]] = True

 # Directions: up, down, left, right
```

DFS

p.1770

```
directions = [(-1, 0), (1, 0), (0, -1), (0, 1)]

Process nodes in BFS fashion
while queue:
 current = queue.popleft()
 # Check if the current position is the
 destination
 if [current[0], current[1]] == destination:
 return True
 # Explore all four possible directions
 for dx, dy in directions:
```

```
 x, y = current
 # Roll the ball until it hits a wall or
 boundary
 while 0 <= x + dx < rows and 0 <= y + dy <
 cols and maze[x + dx][y + dy] == 0:
 x += dx
 y += dy
 # If this new stop has not been visited, add
 it to the queue
 if not visited[x][y]:
 visited[x][y] = True
 queue.append((x, y))
 return False
```

DFS

p.1771

```
Example usage:
maze = [[0,0,1,0,0],
 [0,0,0,0,0],
 [0,0,0,1,0],
 [1,1,0,1,1],
 [0,0,0,0,0]]
start = [0,4]
destination = [4,4]
print(hasPath(maze, start, destination)) # Expected
output: True
```

### • LC.ca

```
class Solution:
 def hasPath(
 self, maze: List[List[int]], start: List[int],
 destination: List[int]
) -> bool:
 m, n = len(maze), len(maze[0])
 q = deque([start])
 rs, cs = start
 vis = {(rs, cs)}
 while q:
 i, j = q.popleft()
```

DFS

```
for a, b in [(0, -1), (0, 1), (-1, 0), (1, 0)]:
 x, y = i, j
 while 0 <= x + a < m and 0 <= y + b < n
 and maze[x + a][y + b] == 0:
 x, y = x + a, y + b
 if [x, y] == destination:
 return True
 if (x, y) not in vis:
 vis.add((x, y))
 q.append((x, y))
 return False
```

DFS

p.1773

DFS

#694 Number of Distinct IslandsMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Hash Table · DFS · BFS · Union Find · Hash Function

Lists: Premium 98

Time Complexity:  $O(m * n)$  as each cell is visited once.  
Space Complexity:  $O(m * n)$  in the worst-case due to recursion stack and storing island shapes.

Problem

You are given an  $m \times n$  binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical.) You may assume all four edges of the grid are surrounded by water.  
An island is considered to be the same as another if and only if one island can be translated (and not rotated or reflected) to equal the other.  
Return the number of distinct islands.

Example 1:

DFS

p.1774

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 |

Input: grid =  
[[1,1,0,0,0],[1,1,0,0,0],[0,0,0,1,1],[0,0,0,1,1]]  
Output: 1

Example 2:

DFS

p.1775

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | 1 |

Input: grid =  
[[1,1,0,1,1],[1,0,0,0,0],[0,0,0,1,1],[1,1,0,1,1]]  
Output: 3

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid[i].length}$
- $1 \leq m, n \leq 50$
- $\text{grid[i][j]}$  is either 0 or 1.

Brute Force (Python)

DFS

p.1776

# Brute Force Approach

class Solution:

def numDistinctIslands(self, grid):

# Brute Force - DFS from every unvisited node

visited = set()

count = 0

def dfs(node):

visited.add(node)

for nb in graph.get(node, []):

if nb not in visited: dfs(nb)

for node in range(n):

if node not in visited:

dfs(node); count += 1

return count

Community Solutions (Python)

- WalkCC

class Solution:

def numDistinctIslands(self, grid: List[List[int]]) -> int:

seen = set()

def dfs(i: int, j: int, i0: int, j0: int):

if i < 0 or i == len(grid) or j < 0 or j == len(grid[0]):

return

if grid[i][j] == 0 or (i, j) in seen:

return

DFS

p.1777

seen.add((i, j))

island.append((i - i0, j - j0))

dfs(i + 1, j, i0, j0)

dfs(i - 1, j, i0, j0)

dfs(i, j + 1, i0, j0)

dfs(i, j - 1, i0, j0)

islands = set() # all the different islands

for i in range(len(grid)):

for j in range(len(grid[0])):

island = []

dfs(i, j, i, j)

if island:

islands.add(frozenset(island))

return len(islands)

LeetCode

class Solution:

def numDistinctIslands(self, grid: List[List[int]]) -> int:

def dfs(i: int, j: int, k: int):

grid[i][j] = 0

path.append(str(k))

dirs = (-1, 0, 1, 0, -1)

for h in range(1, 5):

x, y = i + dirs[h - 1], j + dirs[h]

if 0 <= x < m and 0 <= y < n and

grid[x][y]:

dfs(x, y, h)

DFS

p.1778

path.append(str(-k))

paths = set()

path = []

m, n = len(grid), len(grid[0])

for i, row in enumerate(grid):

for j, x in enumerate(row):

if x:

dfs(i, j, 0)

paths.add("".join(path))

path.clear()

return len(paths)

SimplyLeet

def numDistinctIslands(grid):

# Get the number of rows and columns

rows, cols = len(grid), len(grid[0])

# Set to store unique island shapes

distinct\_shapes = set()

# DFS function to explore the island and record its relative shape

def dfs(r, c, origin\_r, origin\_c, shape):

# Check boundaries and if the cell is unvisited

land

if r < 0 or c < 0 or r >= rows or c >= cols or grid[r][c] == 0:

DFS

p.1779

return

# Mark the cell as visited by setting it to 0

grid[r][c] = 0

# Record the relative position from the origin of this island

shape.append((r - origin\_r, c - origin\_c))

# Explore all 4-directionally adjacent cells

dfs(r + 1, c, origin\_r, origin\_c, shape)

dfs(r - 1, c, origin\_r, origin\_c, shape)

dfs(r, c + 1, origin\_r, origin\_c, shape)

dfs(r, c - 1, origin\_r, origin\_c, shape)

# Iterate over every cell in the grid

for i in range(rows):

for j in range(cols):

if grid[i][j] == 1:

shape = []

dfs(i, j, i, j, shape)

# Convert the shape list to a tuple of sorted coordinates for canonical form

distinct\_shapes.add(tuple(sorted(shape)))

DFS

p.1780

# The number of distinct islands is the size of the set

set

return len(distinct\_shapes)

# Example usage:

grid = [[1,1,0,0,0],

[1,1,0,0,0],

[0,0,0,1,1],

[0,0,0,1,1]]

print(numDistinctIslands(grid)) # Output: 1

LC.ca

class Solution:

def numDistinctIslands(self, grid: List[List[int]]) -> int:

def dfs(i: int, j: int, k: int):

grid[i][j] = 0

path.append(str(k))

dirs = (-1, 0, 1, 0, -1)

for h in range(1, 5):

x, y = i + dirs[h - 1], j + dirs[h]

if 0 <= x < m and 0 <= y < n and

grid[x][y]:

dfs(x, y, h)

DFS

p.1781

path.append(str(-k))

paths = set()

path = []

m, n = len(grid), len(grid[0])

for i, row in enumerate(grid):

for j, x in enumerate(row):

if x:

dfs(i, j, 0)

paths.add("".join(path))

path.clear()

return len(paths)

DFS

p.1782

DFS

#695 Max Area of IslandMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · DFS · BFS · Union Find · Matrix

Lists: Denny Zhang

Time Complexity:  $O(m * n)$  where  $m$  and  $n$  are the dimensions of the grid since every cell is visited at most once. Space Complexity:  $O(m * n)$  in the worst-case scenario for the recursion stack (or auxil

Problem

You are given an  $m \times n$  binary matrix grid. An island is a group of 1's (representing land) connected 4-directionally (horizontal or vertical). You may assume all four edges of the grid are surrounded by water. The area of an island is the number of cells with a value 1 in the island. Return the maximum area of an island in grid. If there is no island, return 0.

Example 1:

DFS

p.1783

Input: grid = [[0,0,1,0,0,0,0,1,0,0,0,0,0],[0,0,0,0,0,0,1,1,0,0,0,0,0],[0,1,1,0,0,0,0,0,0,0,0,0,0],[0,1,0,0,1,0,0,0,1,0,0,0,0],[0,0,0,0,0,0,0,0,0,0,0,0,0],[0,0,0,0,0,0,0,0,0,0,0,0,0],[0,0,0,0,0,0,0,0,0,0,0,0,0],[0,0,0,0,0,0,0,0,0,0,0,0,0],[0,0,0,0,0,0,0,0,0,0,0,0,0]]

Output: 6

Explanation: The answer is not 11, because the island must be connected 4-directionally.

Example 2:

Input: grid = [[0,0,0,0,0,0,0,0,0]]

Output: 0

Constraints:

- $m == \text{grid.length}$
- $n == \text{grid[i].length}$

DFS

p.1784

- $1 \leq m, n \leq 50$
- $\text{grid[i][j]}$  is either 0 or 1.

Brute Force (Python)

# Brute Force Approach

class Solution:

def maxAreaOfIsland(self, grid):

# Brute Force  $O(m^2 \cdot n^2)$  – DFS from every unvisited land cell, track size

m, n = len(grid), len(grid[0])

visited = set()

def dfs(r, c):

if r < 0 or r >= m or c < 0 or c >= n: return 0

if (r, c) in visited or grid[r][c] == 0: return 0

visited.add((r, c))

return 1 + dfs(r+1, c) + dfs(r-1, c) + dfs(r, c+1) + dfs(r, c-1)

return max((dfs(r, c) for r in range(m) for c in range(n)), default=0)

Community Solutions (Python)

- WalkCC

DFS

p.1785

class Solution:

def maxAreaOfIsland(self, grid: List[List[int]]) -> int:

def dfs(i: int, j: int) -> int:

if i < 0 or i == len(grid) or j < 0 or j == len(grid[0]):

return 0

if grid[i][j] != 1:

return 0

grid[i][j] = 2

return 1 + dfs(i + 1, j) + dfs(i - 1, j) + dfs(i, j + 1) + dfs(i, j - 1)

return max(dfs(i, j) for i in range(len(grid)) for j in range(len(grid[0])))

• LeetDoocs

class Solution:

def maxAreaOfIsland(self, grid: List[List[int]]) -> int:

def dfs(i: int, j: int) -> int:

if grid[i][j] == 0:

return 0

ans = 1

grid[i][j] = 0

dirs = (-1, 0, 1, 0, -1)

for a, b in pairwise(dirs):

x, y = i + a, j + b

DFS

p.1786

if 0 <= x < m and 0 <= y < n:

ans += dfs(x, y)

return ans

m, n = len(grid), len(grid[0])

return max(dfs(i, j) for i in range(m) for j in range(n))

• SimplyLeet

# Function to compute the maximum area of an island in a grid using DFS

def maxAreaOfIsland(grid):

# Determine the dimensions of the grid

rows = len(grid)

cols = len(grid[0]) if rows else 0

# Helper function for performing DFS on the grid

def dfs(r, c):

# Check if the current cell is out-of-bounds or is water

if r < 0 or c < 0 or r >= rows or c >= cols or grid[r][c] == 0:

DFS

p.1787

return 0

# Mark the current cell as visited by setting it to 0 (water)

grid[r][c] = 0

area = 1 # Count the current cell as part of the island

# Recursively explore all four directions

area += dfs(r + 1, c)

area += dfs(r - 1, c)

area += dfs(r, c + 1)

area += dfs(r, c - 1)

return area

max\_area = 0

# Iterate through each cell in the grid for r in range(rows):

for c in range(cols):

if grid[r][c] == 1:

# Update max\_area with the area obtained from the DFS

max\_area = max(max\_area, dfs(r, c))

return max\_area

• LC.ca

DFS

p.1788

class Solution:

def maxAreaOfIsland(self, grid: List[List[int]]) -> int:

def dfs(i: int, j: int) -> int:

if grid[i][j] == 0:

return 0

ans = 1

grid[i][j] = 0

dirs = (-1, 0, 1, 0, -1)

for a, b in pairwise(dirs):

x, y = i + a, j + b

if 0 <= x < m and 0 <= y < n:

ans += dfs(x, y)

return ans

m, n = len(grid), len(grid[0])

return max(dfs(i, j) for i in range(m) for j in range(n))

DFS

p.1789

DFS

#721 Accounts MergeMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · String · DFS · BFS · Union Find

Lists: Grind 169

Time Complexity:  $O(N * M * \log(M))$  where  $N$  is the number of accounts and  $M$  is the number of emails per account (owing to sorting and union operations). Space Complexity:  $O(N * M)$ , used for storing ema

Problem

Given a list of accounts where each element accounts[i] is a list of strings, where the first element accounts[i][0] is a name, and the rest of the elements are emails representing emails of the account.

Now, we would like to merge these accounts. Two accounts definitely belong to the same person if there is some common email to both accounts. Note that even if two accounts have the same name, they may belong to different people as people could have the same name. A person can have any number of accounts initially, but all of their accounts definitely have the same name.

After merging the accounts, return the accounts in the following format: the first element of each account is the name, and the rest of the elements are emails in sorted

DFS

p.1790

order. The accounts themselves can be returned in any order.

Example 1:

Input: accounts = [[["John", "johnsmith@mail.com", "john\_newyork@mail.com"], ["John", "johnsmith@mail.com", "john00@mail.com"], ["Mary", "mary@mail.com"], ["John", "johnnybravo@mail.com"]]]

Output: [[["John", "john00@mail.com", "john\_newyork@mail.com"], ["John", "johnsmith@mail.com", "john00@mail.com"], ["Mary", "mary@mail.com"], ["John", "johnnybravo@mail.com"]]]

Explanation:

The first and second John's are the same person as they have the common email "johnsmith@mail.com".

The third John and Mary are different people as none of their email addresses are used by other accounts.

We could return these lists in any order, for example the answer [["Mary", "mary@mail.com"], ["John", "johnnybravo@mail.com"], ["John", "john00@mail.com", "john\_newyork@mail.com"], ["John", "johnsmith@mail.com"]]] would still be accepted.

Example 2:

DFS

p.1791

```
Input: accounts = [{"Gabe", "Gabe0@m.co", "Gabe3@m.co", "Gabe1@m.co"}, {"Kevin", "Kevin3@m.co", "Kevin5@m.co", "Kevin0@m.co"}, {"Ethan", "Ethan5@m.co", "Ethan4@m.co", "Ethan0@m.co"}, {"Hanzo", "Hanzo3@m.co", "Hanzo1@m.co", "Hanzo0@m.co"}, {"Fern", "Fern5@m.co", "Fern1@m.co", "Fern0@m.co"}]
Output: [{"Ethan", "Ethan0@m.co", "Ethan4@m.co", "Ethan5@m.co"}, {"Gabe", "Gabe0@m.co", "Gabe1@m.co", "Gabe3@m.co"}, {"Hanzo", "Hanzo0@m.co", "Hanzo1@m.co", "Hanzo3@m.co"}, {"Kevin", "Kevin0@m.co", "Kevin3@m.co", "Kevin5@m.co"}, {"Fern", "Fern0@m.co", "Fern1@m.co", "Fern5@m.co"}]
```

Constraints:

- 1 <= accounts.length <= 1000
- 2 <= accounts[i].length <= 10
- 1 <= accounts[i][j].length <= 30
- accounts[i][0] consists of English letters.
- accounts[i][j] (for j > 0) is a valid email.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def accountsMerge(self, accounts):
 # Brute Force - DFS from every unvisited node
 visited = set()
 count = 0
 def dfs(node):
 visited.add(node)
 for nb in graph.get(node, []):
 if nb not in visited: dfs(nb)
```

DFS

p.1792

```
for node in range(n):
 if node not in visited:
 dfs(node); count += 1
return count
```

Community Solutions (Python)

• SimplyLeet

```
Python solution for Accounts Merge

Define the UnionFind class for disjoint set operations
class UnionFind:
 def __init__(self):
 self.parent = {}

 # Find with path compression
 def find(self, x):
 if self.parent.setdefault(x, x) != x:
 self.parent[x] = self.find(self.parent[x])
 return self.parent[x]

 # Union two nodes
 def union(self, x, y):
 self.parent[self.find(x)] = self.find(y)

class Solution:
 def accountsMerge(self, accounts):
 uf = UnionFind()
```

DFS

p.1793

```
email_to_name = {}

Iterate over each account and union emails
for account in accounts:
 name = account[0]
 first_email = account[1]
 for email in account[1:]:
 uf.union(first_email, email)
 email_to_name[email] = name

Group emails by their root parent
groups = {}
for email in email_to_name:
 parent = uf.find(email)
 groups.setdefault(parent, []).append(email)

Return the merged account list with sorted emails
result = []
for parent, emails in groups.items():
 result.append([email_to_name[parent]] + sorted(emails))
```

DFS

p.1794

return result

```
Example usage:
accounts =
[{"John", "johnsmith@mail.com", "john_newyork@mail.com"},
[{"John", "johnsmith@mail.com", "john00@mail.com"},
[{"Mary", "mary@mail.com"}],
[{"John", "johnnybravo@mail.com"}]]
solution = Solution()
print(solution.accountsMerge(accounts))
```

• LC.ca

```
class Solution:
 def accountsMerge(self, accounts: List[List[str]]) -> List[List[str]]:
 def find(x):
 if p[x] != x:
 p[x] = find(p[x])
 return p[x]

 n = len(accounts)
 p = list(range(n))
 email_id = {}
```

DFS

p.1795

```
for i, account in enumerate(accounts):
 name = account[0]
 for email in account[1:]:
 if email in email_id:
 p[find(i)] = find(email_id[email])
 else:
 email_id[email] = i
mp = defaultdict(set)
for i, account in enumerate(accounts):
 for email in account[1:]:
 mp[find(i)].add(email)

ans = []
for i, emails in mp.items():
 t = [accounts[i][0]]
 t.extend(sorted(emails))
 ans.append(t)
return ans
```

DFS

p.1796

DFS

## #785 Is Graph Bipartite?

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

DFS · BFS · Union Find · Graph

Lists: Denny Zhang

Time Complexity:  $O(V + E)$  where  $V$  is the number of nodes and  $E$  is the number of edges, since every node and edge is traversed at most once. Space Complexity:  $O(V)$ , for the color array and the queue ( $o$ )

Problem

There is an undirected graph with  $n$  nodes, where each node is numbered between 0 and  $n - 1$ . You are given a 2D array graph, where graph[u] is an array of nodes that node u is adjacent to. More formally, for each  $v$  in graph[u], there is an undirected edge between node u and node v. The graph has the following properties:

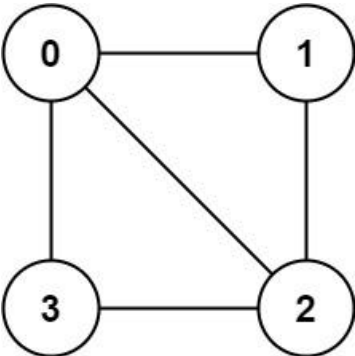
- There are no self-edges (graph[u] does not contain u).
- There are no parallel edges (graph[u] does not contain duplicate values).
- If  $v$  is in graph[u], then u is in graph[v] (the graph is undirected).
- The graph may not be connected, meaning there may be two nodes u and v such that there is no path between them.

DFS

p.1797

A graph is bipartite if the nodes can be partitioned into two independent sets A and B such that every edge in the graph connects a node in set A and a node in set B. Return true if and only if it is bipartite.

Example 1:

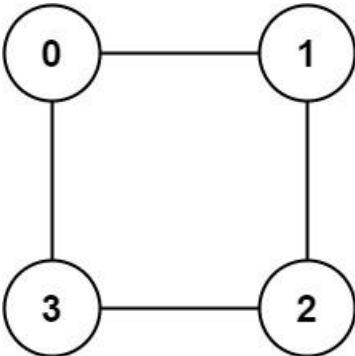


```
Input: graph = [[1,2,3],[0,2],[0,1,3],[0,2]]
Output: false
Explanation: There is no way to partition the nodes into two independent sets such that every edge connects a node in one and a node in the other.
```

DFS

p.1798

Example 2:



```
Input: graph = [[1,3],[0,2],[1,3],[0,2]]
Output: true
Explanation: We can partition the nodes into two sets: {0, 2} and {1, 3}.
```

Constraints:

- graph.length == n
- 1 <= n <= 100
- 0 <= graph[u].length < n
- 0 <= graph[u][i] <= n - 1

DFS

p.1799

DFS

p.1800



```
from enum import Enum

class Color(Enum):
 WHITE = 0
 RED = 1
 GREEN = 2

class Solution:

 def isBipartite(self, graph: list[list[int]]) -> bool:
 colors = [Color.WHITE] * len(graph)

 for i in range(len(graph)):
 # This node has been colored, so do nothing.
 if colors[i] != Color.WHITE:
 continue
 # Always paint red for a white node.
 colors[i] = Color.RED
 # BFS.

 q = collections.deque([i])
 while q:
 for _ in range(len(q)):
 u = q.popleft()
 for v in graph[u]:
 if colors[v] == colors[u]:
 return False
 if colors[v] == Color.WHITE:
 colors[v] = Color.RED if colors[u] == Color.GREEN else Color.GREEN
 q.append(v)
```

DFS

p.1801

```
return True

• LeetCode

class Solution:
 def isBipartite(self, graph: List[List[int]]) -> bool:
 def dfs(a: int, c: int) -> bool:
 color[a] = c
 for b in graph[a]:
 if color[b] == c or (color[b] == 0 and not dfs(b, -c)):
 return False
 return True

 n = len(graph)

 color = [0] * n
 for i in range(n):
 if color[i] == 0 and not dfs(i, 1):
 return False
 return True

• SimplyLeet
```

DFS

p.1802

```
Using Breadth-First Search (BFS)
from collections import deque

def isBipartite(graph):
 n = len(graph)
 # color array: -1 means uncolored, 0 and 1 are the two colors.
 colors = [-1] * n

 # Iterate through each node to handle disconnected components
 for i in range(n):
 if colors[i] == -1:
 # Start BFS from node i with color 0
 queue = deque([i])
 colors[i] = 0
 while queue:
 node = queue.popleft()
 # Check each neighbor of the current node for neighbor in graph[node]:
 if colors[neighbor] == -1:
 # Assign the opposite color to the neighbor
```

DFS

p.1803

```
 colors[neighbor] = 1 -
 queue.append(neighbor)
 elif colors[neighbor] ==

colors[node]:
 # Found a neighbor with the same

color: not bipartite
 return False

 return True

Example usage:
if __name__ == '__main__':
 graph_example = [[1,3],[0,2],[1,3],[0,2]]

 print(isBipartite(graph_example)) # Expected output:
 True

• LC.ca

class Solution:
 def isBipartite(self, graph: List[List[int]]) -> bool:
 def dfs(u, c):
 color[u] = c
 for v in graph[u]:
 if not color[v]:
 if not dfs(v, 3 - c):
 return False
 elif color[v] == c:
 return False
```

DFS

p.1804

```
return True

n = len(graph)
color = [0] * n
for i in range(n):
 if not color[i] and not dfs(i, 1):
 return False
return True
```

DFS

p.1805

DFS

#1236 Web Crawler

LeetCode • SimplyLeet • WalkCC • LeetCode

String • DFS • BFS • Interactive

Lists: Premium 98

Time Complexity:  $O(N + E)$  where  $N$  is the number of nodes (URLs) and  $E$  is the number of edges (links between URLs) encountered during the crawl. Space Complexity:  $O(N)$  for storing the visited URLs and

Problem

Given a url startUrl and an interface HtmlParser, implement a web crawler to crawl all links that are under the same hostname as startUrl. Return all urls obtained by your web crawler in any order.

Your crawler should:

- Start from the page: startUrl
- Call HtmlParser.getUrls(url) to get all urls from a webpage of given url.
- Do not crawl the same link twice.
- Explore only the links that are under the same hostname as startUrl.

DFS

p.1806

https://example.org:8888/foo/bar#bang

hostname

As shown in the example url above, the hostname is example.org. For simplicity sake, you may assume all urls use http protocol without any port specified. For example, the urls http://leetcode.com/problems and http://leetcode.com/contest are under the same hostname, while urls http://example.org/test and http://example.com/abc are not under the same hostname.

The HtmlParser interface is defined as such:

```
interface HtmlParser {
 // Return a list of all urls from a webpage of given url.
 public List<String> getUrls(String url);
}
```

Below are two examples explaining the functionality of the problem, for custom testing purposes you'll have three variables urls, edges and startUrl. Notice that you will only have access to startUrl in your code, while urls and edges are not directly accessible to you in code.

Note: Consider the same URL with the trailing slash "/" as a different URL. For example,

DFS

p.1807

"http://news.yahoo.com", and "http://news.yahoo.com/" are different urls.

Example 1:

```
Input:
urls = [
 "http://news.yahoo.com",
 "http://news.yahoo.com/news",
 "http://news.yahoo.com/news/topics/",
 "http://news.google.com",
 "http://news.yahoo.com/us"
]
```

Example 2:

DFS

Sheet 201 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait

p.1808

```
Input:
urls = [
 "http://news.yahoo.com",
 "http://news.yahoo.com/news",
 "http://news.yahoo.com/news/topics/",
 "http://news.google.com"
]
edges = [[0,2],[2,1],[3,2],[3,1],[3,0]]
```

Constraints:

- $1 \leq \text{urls.length} \leq 1000$
- $1 \leq \text{urls}[i].\text{length} \leq 300$
- startUrl is one of the urls.
- Hostname label must be from 1 to 63 characters long, including the dots, may contain only the ASCII letters

DFS

p.1809

from 'a' to 'z', digits from '0' to '9' and the hyphen-minus character ('-').

- The hostname may not start or end with the hyphen-minus character ('-').
- See: [https://en.wikipedia.org/wiki/Hostname#Restrictions\\_on\\_valid\\_hostnames](https://en.wikipedia.org/wiki/Hostname#Restrictions_on_valid_hostnames)
- You may assume there're no duplicates in url library.

Community Solutions (Python)

• WalkCC

```
""""
This is HtmlParser's API interface.
You should not implement it, or speculate about its
implementation
""""
class HtmlParser(object):
def getUrls(self, url: str) -> List[str]:

class Solution:
 def crawl(self, startUrl: str, htmlParser:
'HtmlParser') -> List[str]:
 q = collections.deque([startUrl])
```

```
seen = {startUrl}
hostname = startUrl.split('/')[2]

while q:
 currUrl = q.popleft()
 for url in htmlParser.getUrls(currUrl):
 if url in seen:
 continue
 if hostname in url:
 q.append(url)

seen.add(url)

return seen
```

• LeetDoocs

```
""""
This is HtmlParser's API interface.
You should not implement it, or speculate about its
implementation
""""
class HtmlParser(object):
def getUrls(self, url):
""""
:type url: str
:rtype List[str]
""""
```

```
class Solution:
 def crawl(self, startUrl: str, htmlParser:
'HtmlParser') -> List[str]:
 def host(url):
 url = url[7:]
 return url.split('/')[0]

 def dfs(url):
 if url in ans:
 return
 ans.add(url)
 for next in htmlParser.getUrls(url):
 if host(url) == host(next):
 dfs(next)

ans = set()
dfs(startUrl)
return list(ans)
```

• SimplyLeet

# Python Code Solution

```
class Solution:
 def crawl(self, startUrl: str, htmlParser:
'HtmlParser') -> List[str]:
 from collections import deque

 # Helper function to extract hostname from a url
 string.
 def get_hostname(url: str) -> str:
 # Remove protocol prefix "http://"
 hostname = url.split("://")[1]

 # The hostname is the part before the first
 "/" (if exists)
 return hostname.split("/")[0]

 # Extract the hostname from the starting URL.
 start_hostname = get_hostname(startUrl)

 # Initialize a set to keep track of visited urls
 and a queue for BFS.
 visited = set([startUrl])
 queue = deque([startUrl])
```

```
Perform BFS traversal of the web links.
while queue:
 current_url = queue.popleft()
 # Retrieve all URLs found at the current URL.
 for url in htmlParser.getUrls(current_url):
 # Check if URL is under the same hostname
 and has not been visited.
 if url not in visited and
 get_hostname(url) == start_hostname:
 visited.add(url)
 queue.append(url)

Return all visited URLs as a list.
return list(visited)
```

• LC.ca

```
""""
This is HtmlParser's API interface.
You should not implement it, or speculate about its
implementation
""""
class HtmlParser(object):
def getUrls(self, url):
""""
:type url: str
:rtype List[str]
""""
```

```
from collections import deque

class Solution:
 def crawl(self, startUrl: str, htmlParser:
'HtmlParser') -> List[str]:
 visited = set()
 domain = self.get_domain(startUrl)
 queue = deque([startUrl])
 visited.add(startUrl)

 while queue:
 current_url = queue.popleft()
 for url in htmlParser.getUrls(current_url):
 if url not in visited and
 self.get_domain(url) == domain:
 visited.add(url)
 queue.append(url)

 return list(visited)

def get_domain(self, url: str) -> str:

 return url.split('/')[2]
```

DFS

#1242 Web Crawler Multithreaded

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

DFS · BFS · Concurrency

Lists: Denny Zhang

Time Complexity: O(N) where N is the number of URLs processed, with additional constant factors due to threading overhead. Space Complexity: O(N) for storing the visited URLs and the work queue.

Problem

Given a URL startUrl and an interface HtmlParser, implement a Multi-threaded web crawler to crawl all links that are under the same hostname as startUrl. Return all URLs obtained by your web crawler in any order.

Your crawler should:

- Start from the page: startUrl
- Call HtmlParser.getUrls(url) to get all URLs from a webpage of a given URL.
- Do not crawl the same link twice.
- Explore only the links that are under the same hostname as startUrl.



As shown in the example URL above, the hostname is example.org. For simplicity's sake, you may assume all URLs use HTTP protocol without any port specified. For example, the URLs <http://leetcode.com/problems> and <http://leetcode.com/contest> are under the same hostname, while URLs <http://example.org/test> and <http://example.com/abc> are not under the same hostname.

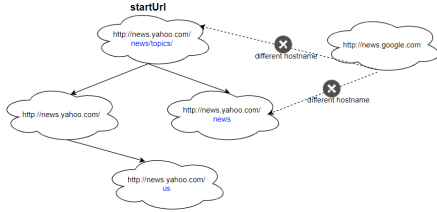
The HtmlParser interface is defined as such:

```
interface HtmlParser {
 // Return a list of all urls from a webpage of given url.
 // This is a blocking call, that means it will do HTTP
 // request and return when this request is finished.
 public List<String> getUrls(String url);
}
```

Note that getUrls(String url) simulates performing an HTTP request. You can treat it as a blocking function call that waits for an HTTP request to finish. It is guaranteed that getUrls(String url) will return the URLs within 15ms. Single-threaded solutions will exceed the time limit so, can your multi-threaded web crawler do better?

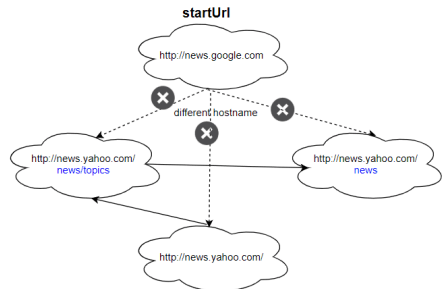
Below are two examples explaining the functionality of the problem. For custom testing purposes, you'll have three variables urls, edges and startUrl. Notice that you will only have access to startUrl in your code, while urls and edges are not directly accessible to you in code.

Example 1:



```
Input:
urls = [
 "http://news.yahoo.com",
 "http://news.yahoo.com/news",
 "http://news.yahoo.com/news/topics/",
 "http://news.google.com",
 "http://news.yahoo.com/us"
]
```

Example 2:



```

Input:
urls = [
 "http://news.yahoo.com",
 "http://news.yahoo.com/news",
 "http://news.yahoo.com/news/topics/",
 "http://news.google.com"
]
edges = [[0,2],[2,1],[3,2],[3,1],[3,0]]

```

Constraints:

- 1 <= urls.length <= 1000
- 1 <= urls[i].length <= 300
- startUrl is one of the urls.
- Hostname label must be from 1 to 63 characters long, including the dots, may contain only the ASCII letters

DFS p.1819

from 'a' to 'z', digits from '0' to '9' and the hyphen-minus character ('-').

- The hostname may not start or end with the hyphen-minus character ('-').
- See: [https://en.wikipedia.org/wiki/Hostname#Restrictions\\_on\\_valid\\_hostnames](https://en.wikipedia.org/wiki/Hostname#Restrictions_on_valid_hostnames)
- You may assume there're no duplicates in the URL library.

Follow up:

1. Assume we have 10,000 nodes and 1 billion URLs to crawl. We will deploy the same software onto each node. The software can know about all the nodes. We have to minimize communication between machines and make sure each node does equal amount of work. How would your web crawler design change?
2. What if one node fails or does not work?
3. How do you know when the crawler is done?

Brute Force (Python)

```

Brute Force Approach
class Solution:
 def crawl(self, startUrl, htmlParser):
 # Brute Force - DFS from every unvisited node
 visited = set()
 count = 0
 def dfs(node):
 visited.add(node)
 for nb in graph.get(node, []):
 if nb not in visited: dfs(nb)

```

DFS p.1820

```

for node in range(n):
 if node not in visited:
 dfs(node); count += 1
return count

```

Community Solutions (Python)

• SimplyLeet

```

import threading
from queue import Queue
from urllib.parse import urlparse

def crawl(startUrl, htmlParser):
 # Extract hostname of the startUrl
 start_host = urlparse(startUrl).netloc
 # Thread-safe set for visited URLs and a lock for safety
 visited = set()
 visited_lock = threading.Lock()

 # Queue to process URLs; initialized with startUrl
 url_queue = Queue()
 url_queue.put(startUrl)

 def worker():
 while True:
 try:
 current_url = url_queue.get(timeout=1) # timeout to avoid blocking indefinitely
 except:

```

DFS p.1821

```

 break
 # Retrieve URLs from the current page using
 htmlParser
 for next_url in
 htmlParser.getUrls(current_url):
 # Check if next_url shares the same
 hostname as startUrl
 if urlparse(next_url).netloc ==
 start_host:
 with visited_lock:
 if next_url not in visited:
 visited.add(next_url)
 url_queue.put(next_url)
 url_queue.task_done()

```

```

Mark startUrl as visited
with visited_lock:
 visited.add(startUrl)

Launch worker threads (8 threads used as example)
threads = []
for _ in range(8):
 t = threading.Thread(target=worker)
 t.start()

```

DFS p.1822

```

 threads.append(t)

 # Wait until all URLs have been processed
 url_queue.join()

 # Ensure all threads terminate
 for t in threads:
 t.join()

 return list(visited)

```

• LC.ca

```

from threading import Thread, Lock
from urllib.parse import urlparse # To parse URLs and
extract hostname for comparison
from typing import List

class Solution:
 def crawl(self, startUrl: str, htmlParser:
 'HtmlParser') -> List[str]:
 hostname = urlparse(startUrl).hostname
 seen = set([startUrl]) # Keep track of seen URLs
 to avoid re-crawling the same URL
 lock = Lock() # Lock for thread-safe operations
 on the seen set

```

DFS p.1823

```

 def dfs(url):
 for next_url in htmlParser.getUrls(url):
 if urlparse(next_url).hostname ==
 hostname and next_url not in seen:
 with lock: # Ensure thread-safe write
 if next_url not in seen:
 seen.add(next_url)
 dfs(next_url)

```

```

 def worker():
 while True:

```

```

 with lock:
 if not unseen:
 return
 url = unseen.pop()
 dfs(url)

```

```

 unseen = [startUrl] # Stack of URLs to be crawled
 threads = []
 num_threads = 10 # Or any number you deem
 appropriate based on the problem constraints

```

```

 # Start threads
 for _ in range(num_threads):
 thread = Thread(target=worker)
 thread.start()
 threads.append(thread)

```

```

 # Wait for all threads to finish
 for thread in threads:
 thread.join()

```

DFS p.1824

```

 return list(seen)

#####

reference https://levelup.gitconnected.com/multi-thread
ed-python-web-crawler-for-https-pages-e103f0839b71

class Crawler(threading.Thread):
 def __init__(self, base_url,
 links_to_crawl, have_visited, error_links, url_lock):
 threading.Thread.__init__(self)
 print(f"Web Crawler worker
 {threading.current_thread()} has Started")
 self.base_url = base_url
 self.links_to_crawl = links_to_crawl

 self.have_visited = have_visited
 self.error_links = error_links
 self.url_lock = url_lock

 def run(self):
 # we create a ssl context so that our script
 can crawl
 # the https sties with ssl_handshake.

 #Create a SSLContext object with default
 settings.
 my_ssl = ssl.create_default_context()

```

DFS p.1825

```

 # by default when creating a default ssl
 context and making an handshake
 # we verify the hostname with the certificate
 but our objective is to crawl
 # the webpage so we will not be checking the
 validity of the certificate.
 my_ssl.check_hostname = False

 # in this case we are not verifying the
 certificate and any
 # certificate is accepted in this mode.
 my_ssl.verify_mode = ssl.CERT_NONE

 # we are defining an infinite while loop so
 that all the links in our
 # queue are processed.

 while True:

 # In this part of the code we create a
 global lock on our queue of
 # links so that no two threads can access
 the queue at same time
 self.url_lock.acquire()
 print(f"Queue Size:
 {self.links_to_crawl.qsize()}")
 link = self.links_to_crawl.get()

```

DFS p.1826

```

 self.url_lock.release()

 # if the link is None the queue is
 exhausted or the threads are yet
 # process the links.

 if link is None:
 break

 # if The link is already visited we break
 the execution.
 if link in self.have_visited:

 print(f"The link {link} is already
 visited")
 break

 try:
 # This method constructs a full
 "absolute" URL by combining the
 # base url with other url. this uses
 components of the base URL,
 # in particular the addressing
 scheme, the network
 # location and the path, to provide
 missing components
 # in the relative URL.
 # in short we repair our relative url
 if it is broken.

```

DFS p.1827

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="43 44 496 623" data-label="Code-Block"> <pre> link = urljoin(self.base_url,link)  # we use the header parameter to "spoof" the "User-Agent" header # value which is used by the browser to identify itself. This is # because some servers will only allow the connection if it comes # from a verified browser. In this case we are using FireFox header. req = Request(link, headers= {'User-Agent': 'Mozilla/5.0'})  # we are opening the url using a ssl handshake. response = urlopen(req, context=my_ssl)  print(f"The URL {response.geturl()} status {response.getcode()}")  # this returns the html representation of the webpage soup = BeautifulSoup(response.read(),"html.parser")  # in this case we are finding all the links in the page. for a_tag in soup.find_all('a'): # we are checking of the link is already visited and (network location part) is our </pre> </div> <div data-bbox="27 663 513 682" data-label="Text"> <div>DFS</div> <div>p.1828</div> </div> | <div data-bbox="583 44 1036 506" data-label="Code-Block"> <pre> # base url itself. if (a_tag.get('href') not in self.have_visited) and (urlparse(link).netloc == "www.python.org"): self.links_to_crawl.put(a_tag .get("href"))  else: print(f"The link {a_tag.get('href')} is already visited or is not part \ of the website")  print(f"Adding {link} to the crawled list")  self.have_visited.add(link)  except URLError as e: print(f"URL {link} threw this error {e.reason} while trying to parse")  self.error_links.append(link)  finally: self.links_to_crawl.task_done() </pre> </div> <div data-bbox="566 663 1053 682" data-label="Text"> <div>DFS</div> <div>p.1829</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <div data-bbox="1123 44 1576 455" data-label="Code-Block"> <pre> if __name__ == '__main__': print("The Crawler is started") base_url = input("Please Enter Website to Crawl &gt; ") number_of_threads = input("Please Enter number of Threads &gt; ")  links_to_crawl = queue.Queue() url_lock = threading.Lock() links_to_crawl.put(base_url)  have_visited = set()  crawler_threads = [] error_links = [] #base_url, links_to_crawl,have_visited, error_links,url_lock for i in range(int(number_of_threads)): crawler = Crawler(base_url = base_url, links_to_crawl= links_to_crawl, have_visited= have_visited, error_links= error_links, url_lock=url_lock) </pre> </div> <div data-bbox="1107 663 1594 682" data-label="Text"> <div>DFS</div> <div>p.1830</div> </div>                                                                                                                                                                                                                                                                                                                       |
| <div data-bbox="43 743 496 1192" data-label="Code-Block"> <pre> crawler.start() crawler_threads.append(crawler)  for crawler in crawler_threads: crawler.join()  print(f"Total Number of pages visited are {len(have_visited)}") print(f"Total Number of Errornous links: {len(error_links)}")  #####  import multiprocessing from bs4 import BeautifulSoup from queue import Queue, Empty from concurrent.futures import ThreadPoolExecutor from urllib.parse import urljoin, urlparse import requests  # ThreadPoolExecutor class MultiThreadedCrawler: </pre> </div> <div data-bbox="27 1362 513 1381" data-label="Text"> <div>DFS</div> <div>p.1831</div> </div>                                                                                                                                                                                                                                                                                                                                  | <div data-bbox="583 743 1036 1138" data-label="Code-Block"> <pre> def __init__(self, seed_url): self.seed_url = seed_url self.root_url = '{}'.format(urlparse(self.seed_url).scheme, urlparse(self.seed_url).netloc) self.pool = ThreadPoolExecutor(max_workers=5) self.scraped_pages = set([]) self.crawl_queue = Queue() self.crawl_queue.put(self.seed_url)  def parse_links(self, html):  soup = BeautifulSoup(html, 'html.parser') Anchor_Tags = soup.find_all('a', href=True) for link in Anchor_Tags: url = link['href'] if url.startswith('/') or url.startswith(self.root_url): url = urljoin(self.root_url, url) if url not in self.scraped_pages: self.crawl_queue.put(url)  def scrape_info(self, html): </pre> </div> <div data-bbox="566 1362 1053 1381" data-label="Text"> <div>DFS</div> <div>p.1832</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div data-bbox="1123 743 1576 1325" data-label="Code-Block"> <pre> soup = BeautifulSoup(html, "html5lib") web_page_paragraph_contents = soup('p') text = '' for para in web_page_paragraph_contents: if not ('https:' in str(para.text)): text = text + str(para.text).strip() print(f'\n &lt;---Text Present in The WebPage is ----&gt;\n', text, '\n') return  def post_scrape_callback(self, res):  result = res.result() if result and result.status_code == 200: self.parse_links(result.text) self.scrape_info(result.text)  def scrape_page(self, url): try: res = requests.get(url, timeout=(3, 30)) return res except requests.RequestException:  return  def run_web_crawler(self): while True: try: print("\n Name of the current executing process: ", multiprocessing.current_process().name, '\n') target_url = self.crawl_queue.get(timeout=60) if target_url not in self.scraped_pages: print("Scraping URL: {}".format(target_url)) </pre> </div> <div data-bbox="1107 1362 1594 1381" data-label="Text"> <div>DFS</div> <div>p.1833</div> </div>                                 |
| <div data-bbox="43 1442 496 1820" data-label="Code-Block"> <pre> self.current_scraping_url = "{}".format(target_url) self.scraped_pages.add(target_url) job = self.pool.submit(self.scrape_page, target_url) job.add_done_callback(self.post_scrape_callback)  except Empty: return except Exception as e: print(e) continue  def info(self): print('\n Seed URL is: ', self.seed_url, '\n') print('Scraped pages are: ', self.scraped_pages, '\n')  if __name__ == '__main__': cc = MultiThreadedCrawler("https://www.leetcode.ca") cc.run_web_crawler() cc.info() </pre> </div> <div data-bbox="27 2062 513 2081" data-label="Text"> <div>DFS</div> <div>p.1834</div> </div>                                                                                                                                                                                                                                                                                                                        | <div data-bbox="583 1442 1036 2024" data-label="Complex-Block"> <div>DFS</div> <div>#269 Alien Dictionary</div> <div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Array · String · DFS · BFS · Graph · Topological Sort</div> <div>Lists: Grind 169   Premium 98</div> </div> <div>Time Complexity: <math>O(N * L + V + E)</math> where N is the number of words, L is the average length of each word, V is the number of unique characters, and E is the number of edges derived from adjacent word co</div> <div>Problem</div> <div>There is a new alien language that uses the English alphabet. However, the order of the letters is unknown to you.</div> <div>You are given a list of strings words from the alien language's dictionary. Now it is claimed that the strings in words are sorted lexicographically by the rules of this new language.</div> <div>If this claim is incorrect, and the given arrangement of string in words cannot correspond to any order of letters, return "".</div> <div>Otherwise, return a string of the unique letters in the new alien language sorted in lexicographically increasing order by the new language's rules. If there are multiple solutions, return any of them.</div> </div> <div data-bbox="566 2062 1053 2081" data-label="Text"> <div>DFS</div> <div>Sheet 204 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait</div> </div> | <div data-bbox="1123 1442 1576 2003" data-label="Complex-Block"> <div>Example 1:</div> <div> <div>Input: words = ["wrt","wrf","er","ett","rftt"]</div> <div>Output: "wertf"</div> </div> <div>Example 2:</div> <div> <div>Input: words = ["z","x","z"]</div> <div>Output: "zx"</div> </div> <div>Example 3:</div> <div> <div>Input: words = ["z","x","z"]</div> <div>Output: ""</div> <div>Explanation: The order is invalid, so return "".</div> </div> <div>Constraints:</div> <div> <ul style="list-style-type: none"> <li>1 &lt;= words.length &lt;= 100</li> <li>1 &lt;= words[i].length &lt;= 100</li> <li>words[i] consists of only lowercase English letters.</li> </ul> </div> <div>Brute Force (Python)</div> <div> <pre> # Brute Force Approach class Solution: def alienOrder(self, words): # Brute Force - DFS from every unvisited node visited = set() count = 0 def dfs(node): visited.add(node) for nb in graph.get(node, []): if nb not in visited: dfs(nb) </pre> </div> </div> <div data-bbox="1107 2062 1594 2081" data-label="Text"> <div>DFS</div> <div>p.1836</div> </div> |

```
for node in range(n):
 if node not in visited:
 dfs(node); count += 1
return count
```

Community Solutions (Python)

```
• WalkCC

class Solution:
 def alienOrder(self, words: list[str]) -> str:
 graph = {}
 inDegrees = [0] * 26

 self._buildGraph(graph, words, inDegrees)
 return self._topology(graph, inDegrees)

 def _buildGraph(
 self,

 graph: dict[str, set[str]],
 words: list[str],
 inDegrees: list[int],
) -> None:
 # Create a node for each character in each word.
 for word in words:
 for c in word:
 if c not in graph:
 graph[c] = set()
```

DFS p.1837

```
for first, second in zip(words, words[1:]):
 length = min(len(first), len(second))
 for j in range(length):
 u = first[j]
 v = second[j]
 if u != v:
 if v not in graph[u]:
 graph[u].add(v)
 inDegrees[ord(v) - ord('a')] += 1
 break # The order of characters after this are
meaningless.

First = 'ab', second = 'a' . invalid
if j == length - 1 and len(first) > len(second):
 graph.clear()
 return

def _topology(self, graph: dict[str, set[str]],
inDegrees: list[int]) -> str:
 s = ''
 q = collections.deque()

 for c in graph:
```

DFS p.1838

```
if inDegrees[ord(c) - ord('a')] == 0:
 q.append(c)

while q:
 u = q.pop()
 s += u
 for v in graph[u]:
 inDegrees[ord(v) - ord('a')] -= 1
 if inDegrees[ord(v) - ord('a')] == 0:
 q.append(v)

Words = ['z', 'x', 'y', 'x']
return s if len(s) == len(graph) else ''

• SimplyLeet

Python solution for Alien Dictionary
from collections import defaultdict, deque

def alienOrder(words):
 # Create data structures: graph and in_degree for
each unique character.
 graph = defaultdict(set)
 in_degree = {char: 0 for word in words for char in
word}

 # Build the graph.
 for i in range(len(words) - 1):
```

DFS p.1839

```
word1, word2 = words[i], words[i+1]
Check for invalid case where prefix of next
word is the same but word1 is longer.
if len(word1) > len(word2) and
word1.startswith(word2):
 return ""
Compare the words character by character.
for c1, c2 in zip(word1, word2):
 if c1 != c2:
 # If we haven't already recorded this
edge, add it.
 if c2 not in graph[c1]:
 graph[c1].add(c2)

 in_degree[c2] += 1
 break

Prepare for topological sort by adding all nodes
with in_degree == 0.
queue = deque([char for char in in_degree if
in_degree[char] == 0])
order = []

Kahn's algorithm for topological sort.
while queue:
 current = queue.popleft()
```

DFS p.1840

```
order.append(current)
for neighbor in graph[current]:
 in_degree[neighbor] -= 1
 if in_degree[neighbor] == 0:
 queue.append(neighbor)

Check if topological sort is possible.
if len(order) == len(in_degree):
 return "".join(order)
else:

 return ""

Example usage:
print(alienOrder(["wrt","wrf","er","ett","rftt"]))

• LC.ca
'''
>>> a = ['a','b','c']
>>> zip(a, a[1:])
[('a', 'b'), ('b', 'c')]

same as pairwise()
>>> from itertools import pairwise
>>> pairwise(a)
<itertools.pairwise object at 0x108458730>
```

DFS p.1841

```
>>> list(pairwise(a))
[('a', 'b'), ('b', 'c')]
'''

...
>>> inDegree = {'a':11, 'b':22, 'c':33}

>>> [print(c) for c in inDegree.keys()]
a
b

c
[None, None, None]

same as for c in inDegree.keys()
>>> [print(c) for c in inDegree]
a
b
c
[None, None, None]

>>> [print(c) for c in inDegree.values()]
11
22
33
[None, None, None]

>>> [print(c) for c in inDegree.items()]
('a', 11)
('b', 22)
('c', 33)
```

DFS p.1842

```
[None, None, None]
'''
from collections import defaultdict, deque
from typing import Dict, List, Set

class Solution:
 def alienOrder(self, words: List[str]) -> str:
 graph = defaultdict(set)
 inDegree = defaultdict(int)

 self._buildGraph(graph, words, inDegree)
 return self._topology(graph, inDegree)

 def _buildGraph(self, graph: Dict[str, Set[str]],
words: List[str], inDegree: Dict[str, int]) -> None:
 # Create a node for each character in each word
 for word in words:
 for c in word:
 inDegree[c] = 0 # necessary for final
char counting

 for first, second in zip(words, words[1:]): # or
pairwise(words)
```

DFS p.1843

```
length = min(len(first), len(second))
for j in range(length):
 u = first[j]
 v = second[j]
 if u != v:
 if v not in graph[u]:
 graph[u].add(v)
 inDegree[v] += 1
 break # Later characters' order is
meaningless

if j == length - 1 and len(first) >
len(second):

invalid order

If 'ab' comes before 'a', it's an
graph.clear()
return

def _topology(self, graph: Dict[str, Set[str]],
inDegree: Dict[str, int]) -> str:
 result = ''
 q = deque([c for c in inDegree if inDegree[c] ==
0])

 while q:
 u = q.popleft()
```

DFS p.1844

```
result += u
for v in graph[u]:
 inDegree[v] -= 1
 if inDegree[v] == 0:
 q.append(v)

If there are remaining characters in inDegree,
it means there's a cycle
if any(inDegree.values()):
 return ''

Words = ['z', 'x', 'y', 'x']
return result if len(result) == len(indegree)
else ''

#####

import collections

class Node(object):
 def __init__(self, val):

 self.val = val
 self.neighbors = []

 def connect(self, node):
 self.neighbors.append(node)

 def getNbrs(self):
 return self.neighbors
```

DFS p.1845

```
class Solution(object):
 def alienOrder(self, words):
 """
 :type words: List[str]
 :rtype: str
 """

 def dfs(root, graph, visited):
 visited[root] = 1
 for nbr in graph[root].getNbrs():

 if visited[nbr.val] == 0:
 if not dfs(nbr.val, graph, visited):
 return False
 elif visited[nbr.val] == 1:
 return False

 visited[root] = 2
 self.ans += root
 return True
```

```
self.ans = ""
graph = {}
visited = collections.defaultdict(int)
self.topNum = 0
for i in range(0, len(words) - 1):
 a = words[i]
 b = words[i + 1]
 i = 0
 while i < len(a) and i < len(b):
 if a[i] != b[i]:
```

DFS p.1846

```
nodeA = nodeB = None
if a[i] not in graph:
 nodeA = Node(a[i])
 graph[a[i]] = nodeA
else:
 nodeA = graph[a[i]]
if b[i] not in graph:
 nodeB = Node(b[i])
 graph[b[i]] = nodeB
else:
 nodeB = graph[b[i]]
nodeA.connect(nodeB)
break
i += 1
if i < len(a) and i >= len(b):
 return ""
```

```
for c in graph:
 if visited[c] == 0:
 if not dfs(c, graph, visited):
```

```
return ""

unusedSet = set()
for word in words:
 for c in word:
 unusedSet.add(c)

for c in unusedSet:
 if c not in graph:
 self.ans += c
```

DFS p.1847

```
return self.ans[::-1]
```

DFS p.1848

DFS

## #329 Longest Increasing Path in a Matrix

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · DFS · BFS · Graph · Topological Sort · Memoization · Matrix

Lists: Grind 169

Time Complexity:  $O(m * n)$  – each cell is computed once due to memoization. Space Complexity:  $O(m * n)$  – space is used for the memoization cache and recursion stack.

### Problem

Given an  $m \times n$  integers matrix, return the length of the longest increasing path in matrix.

From each cell, you can either move in four directions: left, right, up, or down. You may not move diagonally or move outside the boundary (i.e., wrap-around is not allowed).

Example 1:

|   |   |   |
|---|---|---|
| 9 | 9 | 4 |
| 6 | 6 | 8 |
| 2 | 1 | 1 |

Input: matrix = [[9,9,4],[6,6,8],[2,1,1]]  
Output: 4  
Explanation: The longest increasing path is [1, 2, 6, 9].

Example 2:

DFS p.1849

- $1 \leq m, n \leq 200$
- $0 \leq \text{matrix}[i][j] \leq 231 - 1$

### Community Solutions (Python)

#### • WalkCC

```
class Solution:
 def longestIncreasingPath(self, matrix: List[List[int]]) -> int:
 m = len(matrix)
 n = len(matrix[0])

 @functools.lru_cache(None)
 def dfs(i: int, j: int, prev: int) -> int:
 if i < 0 or i == m or j < 0 or j == n:
 return 0
 if matrix[i][j] <= prev:
```

```
return 0

curr = matrix[i][j]
return 1 + max(dfs(i + 1, j, curr),
 dfs(i - 1, j, curr),
 dfs(i, j + 1, curr),
 dfs(i, j - 1, curr))

return max(dfs(i, j, -math.inf) for i in range(m) for j in range(n))
```

#### • LeetCode

DFS p.1852

```
class Solution:
 def longestIncreasingPath(self, matrix: List[List[int]]) -> int:
 @cache
 def dfs(i: int, j: int) -> int:
 ans = 0
 for a, b in pairwise((-1, 0, 1, 0, -1)):
 x, y = i + a, j + b
 if 0 <= x < m and 0 <= y < n and matrix[x][y] > matrix[i][j]:
 ans = max(ans, dfs(x, y))
 return ans + 1
```

```
m, n = len(matrix), len(matrix[0])
return max(dfs(i, j) for i in range(m) for j in range(n))
```

#### • SimplyLeet

# Python implementation for Longest Increasing Path in a Matrix

```
class Solution:
 def longestIncreasingPath(self, matrix: List[List[int]]) -> int:
 # Return 0 if matrix is empty
 if not matrix or not matrix[0]:
 return 0
```

```
Dimensions of the matrix
m, n = len(matrix), len(matrix[0])
```

DFS p.1853

|   |   |   |
|---|---|---|
| 3 | 4 | 5 |
| 3 | 2 | 6 |
| 2 | 2 | 1 |

Input: matrix = [[3,4,5],[3,2,6],[2,2,1]]  
Output: 4  
Explanation: The longest increasing path is [3, 4, 5, 6]. Moving diagonally is not allowed.

Example 3:

Input: matrix = [[1]]  
Output: 1

### Constraints:

- $m == \text{matrix.length}$
- $n == \text{matrix}[i].length$

DFS p.1851

```
Create a memoization table (same dimensions as the matrix) initialized to 0
memo = [[0] * n for _ in range(m)]
```

```
Define the DFS function to explore the matrix
def dfs(i, j):
 # If result is already computed, return it
 if memo[i][j]:
 return memo[i][j]
```

```
Start with a path length of 1 (the cell itself)
```

```
max_len = 1
Define directions: up, down, left, right
directions = [(-1, 0), (1, 0), (0, -1), (0, 1)]
```

```
Explore each direction
for dx, dy in directions:
 ni, nj = i + dx, j + dy
 # If neighbor is within bounds and its value is greater
 if 0 <= ni < m and 0 <= nj < n and matrix[ni][nj] > matrix[i][j]:
 length = 1 + dfs(ni, nj)
```

DFS p.1854



```
max_len = max(max_len, length)

Store computed longest length in
memoization table
memo[i][j] = max_len
return max_len

Compute the longest increasing path starting
from each cell
longest_path = 0
for i in range(m):
 for j in range(n):

 longest_path = max(longest_path, dfs(i,
j))

return longest_path
```

- LC.ca

```
class Solution:
 def longestIncreasingPath(self, matrix:
List[List[int]]) -> int:
 @cache
 def dfs(i: int, j: int) -> int:
 ans = 0
 for a, b in pairwise((-1, 0, 1, 0, -1)):
 x, y = i + a, j + b
 if 0 <= x < m and 0 <= y < n and
matrix[x][y] > matrix[i][j]:
 ans = max(ans, dfs(x, y))
 return ans + 1
```

DFS

p.1855

```
m, n = len(matrix), len(matrix[0])
return max(dfs(i, j) for i in range(m) for j in
range(n))
```

DFS

p.1856

DFS

#685 Redundant Connection II

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

DFS · BFS · Union Find · Graph

Lists: Denny Zhang

Time Complexity: O(n) Space Complexity: O(n)

Problem

In this problem, a rooted tree is a directed graph such that, there is exactly one node (the root) for which all other nodes are descendants of this node, plus every node has exactly one parent, except for the root node which has no parents.

The given input is a directed graph that started as a rooted tree with n nodes (with distinct values from 1 to n), with one additional directed edge added. The added edge has two different vertices chosen from 1 to n, and was not an edge that already existed.

The resulting graph is given as a 2D-array of edges. Each element of edges is a pair [ui, vi] that represents a directed edge connecting nodes ui and vi, where ui is a parent of child vi.

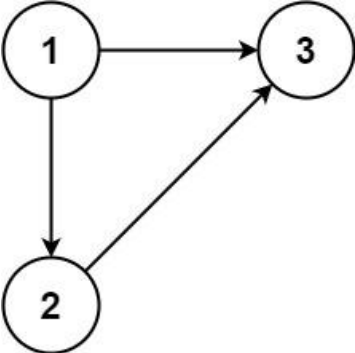
Return an edge that can be removed so that the resulting graph is a rooted tree of n nodes. If there are multiple answers, return the answer that occurs last in

DFS

p.1857

the given 2D-array.

Example 1:

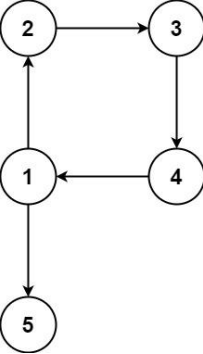


Input: edges = [[1,2],[1,3],[2,3]]  
Output: [2,3]

Example 2:

DFS

p.1858



Input: edges = [[1,2],[2,3],[3,4],[4,1],[1,5]]  
Output: [4,1]

Constraints:

- n == edges.length
- 3 <= n <= 1000
- edges[i].length == 2
- 1 <= ui, vi <= n
- ui != vi

Brute Force (Python)

DFS

p.1859

# Brute Force Approach

class Solution:

def findRedundantDirectedConnection(self, edges):

# Brute Force - DFS from every unvisited node

visited = set()

count = 0

def dfs(node):

visited.add(node)

for nb in graph.get(node, []):

if nb not in visited: dfs(nb)

for node in range(n):

if node not in visited:

dfs(node); count += 1

return count

Community Solutions (Python)

• WalkCC

class UnionFind:

def \_\_init\_\_(self, n: int):

self.id = list(range(n))

self.rank = [0] \* n

def unionByRank(self, u: int, v: int) -> bool:

i = self.\_find(u)

j = self.\_find(v)

if i == j:

return False

DFS

p.1860

if self.rank[i] < self.rank[j]:

self.id[i] = j

elif self.rank[i] > self.rank[j]:

self.id[j] = i

else:

self.id[i] = j

self.rank[j] += 1

return True

def \_find(self, u: int) -> int:

if self.id[u] != u:

self.id[u] = self.\_find(self.id[u])

return self.id[u]

class Solution:

def findRedundantDirectedConnection(self, edges: List[List[int]]) -> List[int]:

ids = [0] \* (len(edges) + 1)

nodeWithTwoParents = 0

for \_, v in edges:

ids[v] += 1

if ids[v] == 2:

nodeWithTwoParents = v

def findRedundantDirectedConnection(skippedEdgeIndex: int) -> List[int]:

uf = UnionFind(len(edges) + 1)

DFS

p.1861

for i, edge in enumerate(edges):

if i == skippedEdgeIndex:

continue

if not uf.unionByRank(edge[0], edge[1]):

return edge

return []

# If there is no edge with two ids, don't skip any edge.

if nodeWithTwoParents == 0:

return findRedundantDirectedConnection(-1)

for i in reversed(range(len(edges))):

\_, v = edges[i]

if v == nodeWithTwoParents:

# Try to delete the edges[i].

if not findRedundantDirectedConnection(i):

return edges[i]

• LeetDoocs

class Solution:

def findRedundantDirectedConnection(self, edges: List[List[int]]) -> List[int]:

def find(x: int) -> int:

if p[x] != x:

p[x] = find(p[x])

return p[x]

n = len(edges)

ind = [0] \* n

for \_, v in edges:

DFS

p.1862

ind[v - 1] += 1

dup = [i for i, (\_, v) in enumerate(edges) if ind[v - 1] == 2]

p = list(range(n))

if dup:

for i, (u, v) in enumerate(edges):

if i == dup[1]:

continue

pu, pv = find(u - 1), find(v - 1)

if pu == pv:

return edges[dup[0]]

p[pu] = pv

return edges[dup[1]]

for i, (u, v) in enumerate(edges):

pu, pv = find(u - 1), find(v - 1)

if pu == pv:

return edges[i]

p[pu] = pv

• SimplyLeet

# Python solution with line-by-line comments

class Solution:

def findRedundantDirectedConnection(self, edges):

n = len(edges)

candidate1 = None

candidate2 = None

# Dictionary to record the parent for each child node.

childParent = {}

# First pass: detect if a node has two parents.

for u, v in edges:

DFS

p.1863

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| <pre>        if v in childParent:             candidate1 = [childParent[v], v] # earlier edge providing parent             candidate2 = [u, v] # later edge providing parent         else:             childParent[v] = u  # Initialize Union-Find structure: each node is its own parent. root = [i for i in range(n+1)]  # Find function with path compression.  def find(x):     while root[x] != x:         root[x] = root[root[x]]         x = root[x]     return x  cycle_edge = None # Second pass: union-find to detect cycle, skip candidate2 if conflict exists. for u, v in edges:     # If candidate2 exists and current edge is candidate2, skip it.</pre>                                                                                                                               |        |        |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | p.1864 |        |
| <pre>        if candidate2 and [u, v] == candidate2:             continue         parent_u = find(u)         parent_v = find(v)         # If they have the same root, a cycle is detected.          if parent_u == parent_v:             cycle_edge = [u, v]         else:             root[parent_v] = parent_u  # Decide which edge to remove based on conflict and cycle detection. if candidate1:     return candidate1 if cycle_edge else candidate2 return cycle_edge  • LC.ca  class UnionFind:     def __init__(self, n):         self.p = list(range(n))         self.n = n      def union(self, a, b):         if self.find(a) == self.find(b):             return False         self.p[self.find(a)] = self.find(b)         self.n -= 1</pre>                                              |        |        |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | p.1865 |        |
| <pre>        return True  def find(self, x):     if self.p[x] != x:         self.p[x] = self.find(self.p[x])     return self.p[x]  class Solution:     def findRedundantDirectedConnection(self, edges: List[List[int]]) -&gt; List[int]:          n = len(edges)         p = list(range(n + 1))         uf = UnionFind(n + 1)         conflict = cycle = None         for i, (u, v) in enumerate(edges):             if p[v] != v:                 conflict = i             else:                 p[v] = u                 if not uf.union(u, v):                      cycle = i                     if conflict is None:                         return edges[cycle]         v = edges[conflict][1]         if cycle is not None:             return [p[v], v]         return edges[conflict]</pre> |        |        |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        | p.1866 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>DFS</div> <div>#1036 Escape a Large Maze</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Array · Hash Table · DFS · BFS</div> <div>Lists: Denny Zhang</div> <div>Time Complexity: <math>O(B^2)</math> in the worst case, where B is the number of blocked cells (with <math>B \leq 200</math>). Space Complexity: <math>O(B^2)</math> due to the storage of visited cells in the worst-case scenario.</div> <div>Problem</div> <div>There is a 1 million by 1 million grid on an XY-plane, and the coordinates of each grid square are (x, y). We start at the source = [sx, sy] square and want to reach the target = [tx, ty] square. There is also an array of blocked squares, where each blocked[i] = [xi, yi] represents a blocked square with coordinates (xi, yi). Each move, we can walk one square north, east, south, or west if the square is not in the array of blocked squares. We are also not allowed to walk outside of the grid. Return true if and only if it is possible to reach the target square from the source square through a sequence of valid moves.</div> <div>Example 1:</div> | <div>HAR</div> | <div>Input: blocked = [[0,1],[1,0]], source = [0,0], target = [0,2]<br/>Output: false<br/>Explanation: The target square is inaccessible starting from the source square because we cannot move. We cannot move north or east because those squares are blocked. We cannot move south or west because we cannot go outside of the grid.</div> <div>Example 2:</div> <div>Input: blocked = [], source = [0,0], target = [999999,999999]<br/>Output: true<br/>Explanation: Because there are no blocked cells, it is possible to reach the target square.</div> <div>Constraints:</div> <ul style="list-style-type: none"><li>0 &lt;= blocked.length &lt;= 200</li><li>blocked[i].length == 2</li><li>0 &lt;= xi, yi &lt; 106</li><li>source.length == target.length == 2</li><li>0 &lt;= sx, sy, tx, ty &lt; 106</li><li>source != target</li><li>It is guaranteed that source and target are not blocked.</li></ul> <div>Brute Force (Python)</div> |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | p.1867         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div>DFS</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                | <div>Input: blocked = [[0,1],[1,0]], source = [0,0], target = [0,2]<br/>Output: false<br/>Explanation: The target square is inaccessible starting from the source square because we cannot move. We cannot move north or east because those squares are blocked. We cannot move south or west because we cannot go outside of the grid.</div> <div>Example 2:</div> <div>Input: blocked = [], source = [0,0], target = [999999,999999]<br/>Output: true<br/>Explanation: Because there are no blocked cells, it is possible to reach the target square.</div> <div>Constraints:</div> <ul style="list-style-type: none"><li>0 &lt;= blocked.length &lt;= 200</li><li>blocked[i].length == 2</li><li>0 &lt;= xi, yi &lt; 106</li><li>source.length == target.length == 2</li><li>0 &lt;= sx, sy, tx, ty &lt; 106</li><li>source != target</li><li>It is guaranteed that source and target are not blocked.</li></ul> <div>Brute Force (Python)</div> |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | p.1868         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <div>DFS</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                | <div># Brute Force Approach<br/>class Solution:<br/>    def isEscapePossible(self, blocked, source, target):<br/>        # Brute Force - DFS from every unvisited node<br/>        visited = set()<br/>        count = 0<br/>        def dfs(node):<br/>            visited.add(node)<br/>            for nb in graph.get(node, []):<br/>                if nb not in visited: dfs(nb)<br/><br/>        for node in range(n):<br/>            if node not in visited:<br/>                dfs(node); count += 1<br/>        return count</div>                                                                                                                                                                                                                                                                                                                                                                                                      |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                | p.1869                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |                |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------|
| <pre>        if (i, j) in blocked or (i, j) in seen:             return False         seen.add((i, j))         return (len(seen) &gt; (1 + 199) * 199 // 2 or [i, j] == target or             dfs(i + 1, j, target, seen) or             dfs(i - 1, j, target, seen) or             dfs(i, j + 1, target, seen) or             dfs(i, j - 1, target, seen))          blocked = set(tuple(b) for b in blocked)          return (dfs(source[0], source[1], target, set()) and dfs(target[0], target[1], source, set()))  • LeetDoocs  class Solution:     def isEscapePossible(         self, blocked: List[List[int]], source: List[int], target: List[int]     ) -&gt; bool:         def dfs(source: List[int], target: List[int], vis: set) -&gt; bool:             vis.add(tuple(source))             if len(vis) &gt; m:                 return True             for a, b in pairwise(dirs):                 x, y = source[0] + a, source[1] + b</pre>        |        |                |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | p.1870 |                |
| <pre>        if 0 &lt;= x &lt; n and 0 &lt;= y &lt; n and (x, y) not in s and (x, y) not in vis:             if [x, y] == target or dfs([x, y], target, vis):                 return True             return False          return False          s = {(x, y) for x, y in blocked}         dirs = (-1, 0, 1, 0, -1)         n = 10**6         m = len(blocked) ** 2 // 2         return dfs(source, target, set()) and dfs(target, source, set())  • SimplyLeet  from collections import deque  def isEscapePossible(blocked, source, target):     blocked_set = {(x, y) for x, y in blocked}     n = len(blocked)     if n &lt; 2:         return True     max_steps = n * (n - 1) // 2      def bfs(start, finish):</pre>                                                                                                                                                                                                                                      |        |                |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | p.1871 |                |
| <div>Sheet 208 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        | <div>DFS</div> |
| <pre>        queue = deque([start])         seen = {tuple(start)}         directions = [(0, 1), (1, 0), (0, -1), (-1, 0)]         steps = 0          while queue:             x, y = queue.popleft()             if [x, y] == finish:                 return True             for dx, dy in directions:                  nx, ny = x + dx, y + dy                 if 0 &lt;= nx &lt; 10**6 and 0 &lt;= ny &lt; 10**6 and (nx, ny) not in blocked_set and (nx, ny) not in seen:                     queue.append((nx, ny))                     seen.add((nx, ny))                     steps += 1                     if steps &gt; max_steps:                         return True              return False          return bfs(source, target) and bfs(target, source)  # Example usage: print(isEscapePossible([[0,1],[1,0]], [0,0], [0,2])) # Expected output: false print(isEscapePossible([], [0,0], [999999,999999])) # Expected output: true  • LC.ca</pre> |        |                |
| DFS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        | p.1872         |

```

class Solution:
 def isEscapePossible(
 self, blocked: List[List[int]], source:
 List[int], target: List[int]
) -> bool:
 def dfs(source, target, seen):
 x, y = source
 if (
 not (0 <= x < 10**6 and 0 <= y < 10**6)
 or (x, y) in blocked
 or (x, y) in seen
):
 return False
 seen.add((x, y))
 if len(seen) > 20000 or source == target:
 return True
 for a, b in [[0, -1], [0, 1], [1, 0], [-1,
0]]:
 next = [x + a, y + b]
 if dfs(next, target, seen):
 return True
 return False

 blocked = set((x, y) for x, y in blocked)
 return dfs(source, target, set()) and dfs(target,
source, set())

```

DFS p.1873

DFS

## #1192 Critical Connections in a Network

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

DFS · Graph · Biconnected Component

Lists: Denny Zhang

Time Complexity:  $O(n + m)$  where  $n$  is the number of nodes and  $m$  is the number of connections. Space Complexity:  $O(n + m)$  for the graph representation and DFS recursion stack.

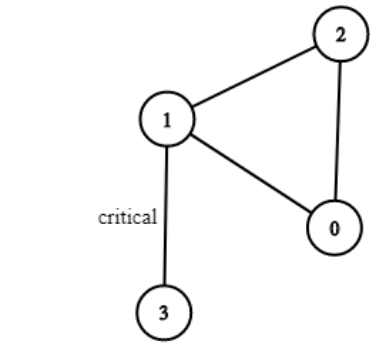
### Problem

There are  $n$  servers numbered from 0 to  $n - 1$  connected by undirected server-to-server connections forming a network where  $\text{connections}[i] = [a_i, b_i]$  represents a connection between servers  $a_i$  and  $b_i$ . Any server can reach other servers directly or indirectly through the network.

A critical connection is a connection that, if removed, will make some servers unable to reach some other server.

Return all critical connections in the network in any order.

Example 1:



Input:  $n = 4$ ,  $\text{connections} = [[0,1],[1,2],[2,0],[1,3]]$   
Output:  $[[1,3]]$   
Explanation:  $[[3,1]]$  is also accepted.

Example 2:

Input:  $n = 2$ ,  $\text{connections} = [[0,1]]$   
Output:  $[[0,1]]$

Constraints:

- $2 \leq n \leq 105$
- $n - 1 \leq \text{connections.length} \leq 105$
- $0 \leq a_i, b_i \leq n - 1$

DFS p.1875

- $a_i \neq b_i$
- There are no repeated connections.

### Brute Force (Python)

```

Brute Force Approach
class Solution:
 def criticalConnections(self, n, connections):
 # Brute Force - DFS from every unvisited node
 visited = set()
 count = 0
 def dfs(node):
 visited.add(node)
 for nb in graph.get(node, []):
 if nb not in visited: dfs(nb)

 for node in range(n):
 if node not in visited:
 dfs(node); count += 1
 return count

```

### Community Solutions (Python)

- LeetDoocs

```

class Solution:
 def criticalConnections(
 self, n: int, connections: List[List[int]]
) -> List[List[int]]:
 def tarjan(a: int, fa: int):
 nonlocal now
 now += 1
 dfn[a] = low[a] = now
 for b in g[a]:
 if b == fa:
 continue
 if not dfn[b]:
 tarjan(b, a)
 low[a] = min(low[a], low[b])
 if low[b] > dfn[a]:
 ans.append([a, b])
 else:
 low[a] = min(low[a], dfn[b])

 g = [[] for _ in range(n)]

 for a, b in connections:
 g[a].append(b)
 g[b].append(a)

 dfn = [0] * n
 low = [0] * n
 now = 0
 ans = []
 tarjan(0, -1)
 return ans

```

- SimplyLeet

DFS p.1877

```

def criticalConnections(n, connections):
 # Build the adjacency list for the graph
 graph = [[] for _ in range(n)]
 for u, v in connections:
 graph[u].append(v)
 graph[v].append(u)

 # Initialize discovery time and low arrays
 disc = [-1] * n # Discovery times, -1 indicates
 unvisited
 low = [0] * n # Low values for each node

 time = 0 # Global timer
 bridges = [] # List to store critical connections
 (bridges)

 def dfs(u, parent):
 nonlocal time
 disc[u] = low[u] = time
 time += 1

 # Traverse all adjacent nodes
 for v in graph[u]:

```

DFS p.1878

```

 if disc[v] == -1: # If v is not visited
 dfs(v, u)
 low[u] = min(low[u], low[v])
 # If the smallest reachable vertex from v
 is after u's discovery, edge (u,v) is a bridge
 if low[v] > disc[u]:
 bridges.append([u, v])
 elif v != parent: # v is visited and is not
 the parent (back edge)
 low[u] = min(low[u], disc[v])

 # Start DFS from node 0

 dfs(0, -1)
 return bridges

Example usage:
n = 4
connections = [[0,1],[1,2],[2,0],[1,3]]
print(criticalConnections(n, connections)) # Output: [[1,
3]]

```

- LC.ca

```

class Solution:
 def criticalConnections(
 self, n: int, connections: List[List[int]]
) -> List[List[int]]:
 def tarjan(a: int, fa: int):
 nonlocal now
 now += 1
 dfn[a] = low[a] = now
 for b in g[a]:
 if b == fa:

```

DFS p.1879

```

 continue
 if not dfn[b]:
 tarjan(b, a)
 low[a] = min(low[a], low[b])
 if low[b] > dfn[a]:
 ans.append([a, b])
 else:
 low[a] = min(low[a], dfn[b])

 g = [[] for _ in range(n)]

 for a, b in connections:
 g[a].append(b)
 g[b].append(a)

 dfn = [0] * n
 low = [0] * n
 now = 0
 ans = []
 tarjan(0, -1)
 return ans

```

DFS p.1880

DFS p.1881

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
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| <div>&gt;&gt; Quick Review — DFS</div> <div>23 questions · insights · complexity · solution</div> <div>#463 Island Perimeter <span>[Easy]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Each land cell contributes 4 edges if isolated.</li> <li>For every adjacent pair of land cells (neighbors horizontally or vertically), the shared edge should be subtracted from the total perimeter.</li> <li>Instead of DFS/BFS, we can iterate the grid and check the four neighbors for each land cell.</li> <li>Alternatively, one may use DFS/BFS to visit all connected cells if needed, but an iterative solution is straightforward with <math>O(\text{rows} \times \text{cols})</math> time complexity.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(n \times m)</math>, where <math>n</math> is the number of rows and <math>m</math> is the number of columns.</div> <div>Space Complexity: <math>O(1)</math> for the iterative solution (excluding the input storage).</div> </div> <div>Solution <div>We iterate through every cell in the grid. For each land cell, we initially add 4 to the perimeter. Then we check its four neighbors (up, down, left, right). For every neighbor that is also land, we subtract 1 from the</div> </div> <div>DFS<div>p.1882</div></div>                | <div>current cell's contribution since that side is not exposed. This simple approach ensures that shared borders between adjacent land cells are not over-counted.</div> <div>#733 Flood Fill <span>[Easy]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>The flood fill operation can be executed using either DFS (Depth First Search) or BFS (Breadth First Search).</li> <li>A check is needed at the beginning to determine if the new color is the same as the original; if so, no changes are needed.</li> <li>The algorithm recursively or iteratively visits each pixel that is directly adjacent and matches the original color, then updates its color.</li> <li>The problem has a worst-case scenario where all pixels are the same color, leading to a full traversal of the grid.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(m \times n)</math>, where <math>m</math> and <math>n</math> are the dimensions of the image grid, because in the worst-case all cells are visited.</div> <div>Space Complexity: <math>O(m \times n)</math> in the worst-case due to the recursion stack (or BFS queue) if the entire image needs to be filled.</div> </div> <div>Solution</div> <div>DFS<div>p.1883</div></div>                                                                                                                                                                                                                                                                                   | <div>We use a DFS approach to perform the flood fill. The algorithm starts at the given starting pixel and uses recursion to fill connected pixels with the same original color. The following steps are key: <ul style="list-style-type: none"> <li>Check if the starting pixel's color is already the target color; if so, return the image immediately.</li> <li>Define a recursive helper function that: Checks boundary conditions. Updates the current pixel's color. Recursively calls itself on the four adjacent directions (up, down, left, right).</li> <li>Return the modified image after the recursion completes.</li> </ul> </div> <div>Data structures used: <ul style="list-style-type: none"> <li>Recursion call stack for the DFS implementation.</li> <li>Variables to store the image dimensions and original color.</li> </ul> </div> <div>#130 Surrounded Regions <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Regions connected to the border are not captured.</li> <li>Use DFS/BFS from border 'O's to mark safe cells.</li> <li>After marking, flip all remaining 'O's to 'X' and then revert the marked cells back to 'O'.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(m \times n)</math></div> <div>Space Complexity: <math>O(m \times n)</math> in the worst case due to recursion stack or queue storage for DFS/BFS</div> </div> <div>DFS<div>p.1884</div></div> |
| <div>Solution <div>The solution involves two main steps. First, traverse the board's borders and run a DFS or BFS from any encountered 'O', marking it (e.g., using ' ') to indicate that it should not be flipped. Next, go through each cell in the board: flip any remaining 'O' (i.e., not marked safe) to 'X' and change the temporary marker ' ' back to 'O'. This approach ensures only the regions completely surrounded by 'X' are captured.</div> </div> <div>#133 Clone Graph <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>The graph is undirected and connected.</li> <li>Each node must be cloned exactly once to prevent cycles and repetitions.</li> <li>Use a mapping/dictionary to keep track of cloned nodes.</li> <li>A depth-first search (DFS) or breadth-first search (BFS) can be used to traverse and clone the graph.</li> <li>Handle edge cases where the graph might be empty.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(N + E)</math>, where <math>N</math> is the number of nodes and <math>E</math> is the number of edges, because each node and each edge is traversed once.</div> <div>Space Complexity: <math>O(N)</math>, to store the mapping from original nodes to their clones, and for the</div> </div> <div>DFS<div>p.1885</div></div> | <div>recursion/queue used in traversal.</div> <div>Solution <div>We employ a depth-first search (DFS) approach to clone the graph. The main idea is to create a mapping (hash table) that links each original node to its corresponding cloned node. During the DFS traversal, if a node is encountered for the first time, a new node is created and stored in the mapping; then, the algorithm recursively clones all of its neighbors. For an already visited node, the cloned node is directly returned from the mapping. This ensures that every node is cloned only once and correctly handles cycles in the graph.</div> </div> <div>#200 Number of Islands <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>Use graph traversal (DFS/BFS) to explore connected components of land.</li> <li>Iterate over the matrix and start a traversal (e.g., DFS) when encountering an unvisited "1".</li> <li>During traversal, mark visited cells to avoid duplicate counting.</li> <li>Each DFS/BFS call completely explores one island.</li> <li>Alternative approach can use Union Find to group connected lands.</li> </ul> </div> <div>Space &amp; Time</div> <div>DFS<div>p.1886</div></div>                                                                                                                                                                                                                                                                                                                            | <div>Time Complexity: <math>O(m \times n)</math>, as each cell is visited at most once.</div> <div>Space Complexity: <math>O(m \times n)</math> in the worst-case scenario due to recursion stack (DFS) or auxiliary data structures.</div> <div>Solution <div>We solve the problem using a DFS-based approach. The algorithm iterates over each cell in the grid; when a "1" (land) is encountered, the DFS is initiated to mark the entire island (all connected "1"s) as visited (by setting them to "0"). This prevents recounting the same island. The island count is incremented each time a new island is discovered and fully explored.</div> </div> <div>#207 Course Schedule <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>This problem can be modeled as a directed graph where courses are nodes and prerequisites are edges.</li> <li>If the graph has a cycle, then it is not possible to complete all courses.</li> <li>We can detect cycles using either Topological Sort (BFS/Kahn's algorithm) or DFS-based cycle detection.</li> <li>The time complexity is <math>O(V + E)</math>, where <math>V</math> is the number of courses and <math>E</math> is the number of prerequisite pairs.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(V + E)</math></div> </div> <div>DFS<div>p.1887</div></div>                                                                               |
| <div>Space Complexity: <math>O(V + E)</math></div> <div>Solution <div>We solve the problem using the Topological Sort algorithm (BFS/Kahn's algorithm). First, we build a graph (using an adjacency list) and track the indegree (number of prerequisites) for each course. We then find all courses with an indegree of zero (courses with no prerequisites) and add them to a processing queue. While the queue is not empty, we remove a course from the queue, reduce the indegree for its neighboring courses, and if any neighbor's indegree becomes zero, we add it to the queue as well. If we successfully process all courses, then it means that the graph was acyclic and a valid schedule exists, returning true. If not, a cycle exists, and we return false.</div> </div> <div>#210 Course Schedule II <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>This is a topological sort problem on a directed graph.</li> <li>The courses are nodes, and prerequisites form directed edges.</li> <li>We can solve the problem by either: <ul style="list-style-type: none"> <li>Using DFS with cycle detection and a stack for topological ordering.</li> <li>Using BFS (Kahn's Algorithm) to process nodes with zero in-degrees.</li> </ul> </li> </ul> </div> <div>DFS<div>p.1888</div></div>       | <ul style="list-style-type: none"> <li>If at any point no node with a zero in-degree is available, a cycle exists and no valid order can be produced.</li> </ul> <div>Space &amp; Time <div>Time Complexity: <math>O(V + E)</math> where <math>V</math> is the number of courses and <math>E</math> is the number of prerequisite pairs.</div> <div>Space Complexity: <math>O(V + E)</math> for storing the graph and in-degree array.</div> </div> <div>Solution <div>We treat the problem as a topological sort on a directed acyclic graph (DAG). For a BFS approach using Kahn's algorithm: <ul style="list-style-type: none"> <li>Build an adjacency list for the graph and an in-degree array to record the number of prerequisites (incoming edges) for each course.</li> <li>Initialize a queue with all courses that have an in-degree of zero (i.e., courses that can be taken without prerequisites).</li> <li>Dequeue nodes from the queue, add them to the ordering, and reduce the in-degree of their neighbors.</li> <li>If a neighbor's in-degree becomes zero, add it to the queue.</li> <li>If the final ordering includes all courses, return the ordering; otherwise, return an empty array indicating a cycle.</li> </ul> </div> <div>The chosen data structures are: <ul style="list-style-type: none"> <li>A list (or array) for the in-degree counts.</li> <li>A dictionary (or array of lists) for the adjacency list.</li> <li>A queue to process nodes with zero in-degree.</li> </ul> </div> <div>DFS<div>p.1889</div></div> </div> | <div>#261 Graph Valid Tree <span>[Medium]</span></div> <div>Key Insights <ul style="list-style-type: none"> <li>A tree must be acyclic and connected.</li> <li>For <math>n</math> nodes, a valid tree must have exactly <math>n - 1</math> edges.</li> <li>A DFS/BFS traversal can check connectivity and simultaneously detect cycles.</li> <li>The Union-Find (disjoint set) approach provides an alternative for cycle detection and connectivity verification.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(n + e)</math>, where <math>e</math> is the number of edges</div> <div>Space Complexity: <math>O(n + e)</math> for storing the graph and traversal/union-find structures</div> </div> <div>Solution <div>We can solve the problem using either a DFS/BFS approach or a Union-Find algorithm. For the DFS/BFS method, we first check if the number of edges is exactly <math>n - 1</math>. If not, it cannot be a tree. Then, we build an adjacency list for the graph and perform a DFS starting from node 0 while keeping track of visited nodes and parent information to avoid false-positive cycle detections. If during DFS we encounter a node that has been visited (and is not the parent), then a cycle exists. Finally, if all nodes are reached from node 0 and no cycle</div> </div> <div>DFS<div>p.1890</div></div>                                                                                           |

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| <p>is found, the graph is a valid tree.<br/>Using Union-Find, we iterate over each edge and try to union the two nodes. If they already share the same parent (i.e., are in the same set), a cycle exists. After processing all edges, we confirm that exactly one connected component exists.</p> <p><b>#310 Minimum Height Trees <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• A tree can have at most two centers that will serve as the roots for its minimum height trees.</li> <li>• Instead of trying all possible roots, we can remove leaves layer by layer until we are left with the central node(s).</li> <li>• The process is similar to a topological sort on the tree.</li> <li>• When there are 2 or fewer nodes left, these nodes are the centers (roots) of MHTs.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(n) where n is the number of nodes, since we process each node and edge once.<br/>Space Complexity: O(n) for storing the graph and additional data structures like the leaves list.</p> <p><b>Solution</b><br/>The solution builds an undirected graph from the input, then identifies and removes node-leaves iteratively. A leaf is defined as a node connected to only one other</p> <div>DFS<span>p.1891</span></div>            | <p>node. By repeatedly removing the leaves and updating the graph (decreasing neighbor degrees), the algorithm peels away the outer layers of the tree. When only one or two nodes remain, these nodes are the centers of the tree and form the roots of the minimum height trees.</p> <p><b>#323 Number of Connected Components in an Undirected Graph <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use an adjacency list to represent the graph.</li> <li>• A DFS (or BFS) traversal from an unvisited node will traverse an entire connected component.</li> <li>• Each new DFS/BFS call from an unvisited node signals a separate connected component.</li> <li>• Alternatively, the Union-Find data structure can be used to union connected nodes and count distinct sets.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(n + e), where n is the number of nodes and e is the number of edges.<br/>Space Complexity: O(n + e) due to the storage of the graph and tracking of visited nodes.</p> <p><b>Solution</b><br/>We solve the problem by iterating over each node and performing a DFS for any node that hasn't been visited. The DFS uses an iterative approach with a stack to traverse all nodes in a connected component. Each DFS</p> <div>DFS<span>p.1892</span></div>                                                            | <p>initiation corresponds to one connected component. This method efficiently explores the graph using an adjacency list and marks nodes as visited to prevent redundant traversals.</p> <p><b>#417 Pacific Atlantic Water Flow <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Reverse the typical DFS/BFS perspective: start from the oceans and work inward.</li> <li>• Maintain two matrices (or sets) to keep track of cells reachable from the Pacific and Atlantic edges.</li> <li>• A cell can be added to the result if it is reached in both traversals.</li> <li>• The height constraint ensures water flows only in non-increasing order.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(m * n) where m is the number of rows and n is the number of columns. Each cell is processed up to twice (once from each ocean).<br/>Space Complexity: O(m * n) for the visited matrices and the recursion/queue used in DFS/BFS.</p> <p><b>Solution</b><br/>The solution uses depth-first search (DFS) from the ocean boundaries. We initialize two boolean matrices, one for each ocean. For the Pacific, we start DFS from the leftmost column and top row; for the Atlantic, we start</p> <div>DFS<span>p.1893</span></div>                                        |
| <p>from the rightmost column and bottom row. In each DFS, if the next cell has a height equal to or greater than the current cell and hasn't been visited, we continue the search. Finally, we scan through the grid to collect cells that are reachable from both oceans. This reverse search simplifies the path validations and leverages the allowed water flow conditions.</p> <p><b>#490 The Maze <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The ball continues moving in one direction until it hits a wall (or the boundary, which is considered a wall).</li> <li>• You must simulate the ball's rolling rather than taking a single step.</li> <li>• Even if the ball passes through the destination, it is only valid if it can stop exactly there.</li> <li>• A breadth-first search (BFS) or depth-first search (DFS) strategy is ideal because we need to explore all stopping positions.</li> <li>• Use a visited structure to avoid revisiting positions and thus prevent infinite loops.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(m * n) where m and n are the dimensions of the maze.<br/>Space Complexity: O(m * n) due to the storage used by the visited matrix and the BFS/DFS queue or recursion</p> <div>DFS<span>p.1894</span></div> | <p>stack.</p> <p><b>Solution</b><br/>We simulate the ball's movement using a breadth-first search (BFS) approach. Each time we pick a stop position, we try to roll in all four directions until the ball hits a wall. At that point, if the new stopping position has not been visited, we mark it as visited and add it to our queue. If we ever reach the destination exactly, we return true. The key is simulating the "roll" by iteratively moving in a direction until no further move is possible. We store the coordinates in a queue (or stack for DFS) along with a 2D visited array to monitor which positions have been processed. The directions are maintained in a list/array, and for each direction, we keep moving until the next cell would be out of bounds or a wall.</p> <p><b>#694 Number of Distinct Islands <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use DFS/BFS to traverse and mark each island.</li> <li>• For each island, record its shape relative to a starting coordinate to allow translation invariance.</li> <li>• Represent the island's shape as a unique signature (e.g., tuple or string of relative positions).</li> <li>• Use a set (or hash table) to store unique shapes.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(m * n) as each cell is visited once.</p> <div>DFS<span>p.1895</span></div> | <p>Space Complexity: O(m * n) in the worst-case due to recursion stack and storing island shapes.</p> <p><b>Solution</b><br/>Use depth-first search (DFS) to explore the grid. When an unvisited land cell (1) is encountered, initiate a DFS to traverse the entire island. Record each land cell's coordinates relative to the starting position of the island. This relative representation is invariant under translation, making it possible to compare island shapes. Store the canonical shape signature (for example, as a tuple or string) in a set to ensure uniqueness. Finally, return the size of the set as the number of distinct islands.</p> <p><b>#695 Max Area of Island <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Traverse each cell in the grid.</li> <li>• Use Depth-First Search (DFS) or Breadth-First Search (BFS) to explore connected 1's.</li> <li>• Mark visited cells to prevent double counting (e.g., by setting them to 0).</li> <li>• Maintain and update a maximum area count during traversal.</li> <li>• The grid's boundaries are naturally water, so no extra edge processing is needed.</li> </ul> <p><b>Space &amp; Time</b></p> <div>DFS<span>p.1896</span></div>                                                                        |
| <p>Time Complexity: O(m * n) where m and n are the dimensions of the grid since every cell is visited at most once.<br/>Space Complexity: O(m * n) in the worst-case scenario for the recursion stack (or auxiliary data structures) when the grid is entirely land.</p> <p><b>Solution</b><br/>The solution leverages a DFS approach to iterate over each cell in the given grid. When a cell with a value of 1 is encountered, a DFS is initiated from that cell to calculate the area of the island connected to it. Cells are marked as visited (by changing them from 1 to 0) during the DFS to avoid revisiting. The DFS explores all four directions (up, down, left, right) and accumulates the count of connected cells. The maximum area found among all islands is returned at the end. This approach efficiently computes the largest island area with a simple traversal mechanism.</p> <p><b>#721 Accounts Merge <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Two accounts are merged if they share at least one common email.</li> <li>• Treat each email as a node in a graph; accounts represent connected components.</li> <li>• Both DFS/BFS and Union-Find can be used to find connected components.</li> </ul> <div>DFS<span>p.1897</span></div>               | <p> <ul style="list-style-type: none"> <li>• Keep a mapping from email to name because all emails in a connected component belong to the same person.</li> <li>• After grouping, sort the emails and prepend the name to form the final account list.</li> </ul> </p> <p><b>Space &amp; Time</b><br/>Time Complexity: O(N * M * log(M)) where N is the number of accounts and M is the number of emails per account (owing to sorting and union operations).<br/>Space Complexity: O(N * M), used for storing email mappings and union-find structures.</p> <p><b>Solution</b><br/>The solution involves using a Union-Find (Disjoint Set Union) data structure to merge emails that are connected by being on the same account. For each account, we union the first email with every other email in that account. We also maintain a mapping from email to owner name. After processing all accounts, we group the emails by their root parent derived from the union-find structure. For each group, we sort the emails and prepend the owner's name. Finally, we return the merged list of accounts.</p> <p><b>#785 Is Graph Bipartite? <span>[Medium]</span></b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• A graph is bipartite if you can assign two colors to each node such that no two adjacent nodes share the</li> </ul> <div>DFS<span>p.1898</span></div>                     | <p>same color.</p> <ul style="list-style-type: none"> <li>• The challenge may require exploring disconnected components.</li> <li>• BFS or DFS can be used to attempt the two-coloring, and a conflict during coloring indicates the graph is not bipartite.</li> <li>• Union Find is an alternative approach, but BFS/DFS is more intuitive for graph coloring problems.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(V + E) where V is the number of nodes and E is the number of edges, since every node and edge is traversed at most once.<br/>Space Complexity: O(V), for the color array and the queue (or recursion stack) used in BFS (or DFS).</p> <p><b>Solution</b><br/>We use a BFS/DFS approach to assign colors (0 and 1) to nodes. A color array, initially set to -1 for all nodes, is maintained. For each unvisited node (color -1), we start a BFS (or DFS) and assign it an initial color (e.g., 0). For every visited neighbor, if it has not been colored, assign it the opposite color; if it is already colored with the same color as the current node, a conflict has occurred, and the graph is not bipartite. This method gracefully handles disconnected components by iterating over all nodes.</p> <p><b>#1236 Web Crawler <span>[Medium]</span></b></p> <div>DFS<span>p.1899</span></div> |



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| <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Treat the set of webpages as a graph where each URL is a node and an edge exists from one URL to another if it is returned by <code>HtmlParser.getUrls(url)</code>.</li> <li>• Use either breadth-first search (BFS) or depth-first search (DFS) to traverse the webpages.</li> <li>• Extract the hostname from a URL to ensure that only URLs on the same domain are crawled.</li> <li>• Maintain a visited set (or equivalent) to avoid processing the same URL multiple times.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(N + E)</math> where N is the number of nodes (URLs) and E is the number of edges (links encountered during the crawl).</p> <p>Space Complexity: <math>O(N)</math> for storing the visited URLs and the queue/stack used for traversal.</p> <p><b>Solution</b></p> <p>The solution uses a BFS (or DFS) approach to traverse the web graph:</p> <ul style="list-style-type: none"> <li>- Extract the hostname from <code>startUrl</code> (for instance, by splitting on <code>"/"</code> and then the first <code>"/"</code> that follows).</li> <li>- Initialize a visited set to keep track of URLs that have been crawled.</li> <li>- Use a queue to implement the BFS.</li> <li>Enqueue the <code>startUrl</code>.</li> <li>- While the queue is not empty, dequeue a URL and retrieve its outgoing links using <code>HtmlParser.getUrls(url)</code>.</li> <li>- For each URL obtained: Extract</li> </ul> | <p>its hostname and compare it with the hostname of <code>startUrl</code>. If it matches and it has not been visited before, add it to the visited set and enqueue it. <li>- Finally, return the set of visited URLs.</li> <p>This approach guarantees that each URL is processed only once and that only URLs from the same hostname are crawled.</p> <p><b>#1242 Web Crawler Multithreaded [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use multiple threads to perform concurrent crawling, which speeds up the process over a single-threaded approach.</li> <li>• Maintain a thread-safe set (visited) to ensure no URL is processed twice.</li> <li>• Utilize a work queue to manage URLs pending processing.</li> <li>• Filter URLs based on the hostname extracted from <code>startUrl</code>.</li> <li>• Apply synchronization (locks or concurrent data structures) to avoid race conditions when multiple threads access shared data.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(N)</math> where N is the number of URLs processed, with additional constant factors due to threading overhead.</p> </p>                                                                                                                                                                            | <p>Space Complexity: <math>O(N)</math> for storing the visited URLs and the work queue.</p> <p><b>Solution</b></p> <p>The solution initializes a shared work queue with the <code>startUrl</code> and a thread-safe visited set. Each worker thread dequeues a URL, retrieves its linked URLs via the blocking <code>HtmlParser.getUrls()</code> call, and filters the resulting URLs by comparing their hostnames with the start URL's hostname. New URLs that have not yet been visited are added to both the visited set and the work queue. The use of a thread pool or multiple worker threads along with synchronization ensures that the crawler processes URLs concurrently while avoiding race conditions. The approach follows a BFS-like pattern spread across multiple threads.</p> <p><b>#269 Alien Dictionary [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Build a graph where each unique character is a node.</li> <li>• Compare adjacent words to determine the first different character, which gives the ordering (directed edge).</li> <li>• Ensure all characters, even those without edges, are represented.</li> <li>• Use a topological sort (using DFS or BFS) to determine a valid character ordering.</li> </ul>                                                                                                                                                                                                                                                 |
| <p>DFS</p> <p>p.1900</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>DFS</p> <p>p.1901</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>DFS</p> <p>p.1902</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>• Handle edge cases, such as words where a longer word appears before its prefix which is invalid.</p> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(N * L + V + E)</math> where N is the number of words, L is the average length of each word, V is the number of unique characters, and E is the number of edges derived from adjacent word comparisons.</p> <p>Space Complexity: <math>O(V + E)</math> for storing the graph and additional data structures for topological sorting.</p> <p><b>Solution</b></p> <p>We use a graph-based approach to solve the problem. First, we initialize a graph (adjacency list) and in-degree dictionary for all unique characters. We then iterate through each pair of consecutive words, comparing them character by character to find the first pair that differs. This difference implies an order: the first character should come before the second character. We add a directed edge accordingly and update the in-degree for the target character.</p> <p>To get the final ordering, we perform a topological sort using Kahn's algorithm:</p> <ul style="list-style-type: none"> <li>- Enqueue characters with in-degree zero.</li> <li>- Remove characters from the queue and append them to our result.</li> <li>- Decrement the in-degree of neighboring nodes.</li> <li>- If a neighbor's in-degree becomes zero, enqueue it.</li> <li>- If the result's length equals the number of unique nodes,</li> </ul>                                                     | <p>we have a valid ordering; otherwise, a cycle exists, and we return an empty string.</p> <p><b>#329 Longest Increasing Path in a Matrix [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The problem can be modeled as a graph where each cell is a node connected to its neighbors if the neighbor's value is greater.</li> <li>• A Depth First Search (DFS) with memoization is effective to avoid recalculations.</li> <li>• Dynamic Programming (DP) is used to cache the longest path found from each cell.</li> <li>• Since each cell can be a start point, compute DFS for every cell and update the global maximum.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(m * n)</math> - each cell is computed once due to memoization.</p> <p>Space Complexity: <math>O(m * n)</math> - space is used for the memoization cache and recursion stack.</p> <p><b>Solution</b></p> <p>We solve the problem using a DFS approach combined with memoization to keep track of already computed longest paths from each cell. For each cell in the matrix, a DFS is initiated to explore all four valid directions (up, down, left, right). If a neighboring cell has a larger value, the DFS recurses from that neighbor. The result for each</p>                                                         | <p>cell is cached in a 2D memoization table, ensuring that each cell is processed only once. After processing all cells, the maximum value in the memo table represents the longest increasing path in the entire matrix.</p> <p><b>#685 Redundant Connection II [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The extra edge introduces one of two issues:</li> <li>• A node has two parents.</li> <li>• A cycle exists in the graph.</li> <li>• First, scan through edges to detect if any node gets two parents. If found, there are two candidate edges.</li> <li>• Use the Union-Find data structure (Disjoint Set Union) to detect cycles.</li> <li>• Depending on whether a conflict (two parents) exists and/or a cycle is detected, decide which edge to remove:</li> <li>• If a node has two parents, test by ignoring the second candidate edge during union-find. If no cycle forms, the second candidate is removable; otherwise, remove the first candidate.</li> <li>• If no conflict exists, simply remove the edge that causes a cycle.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n)</math></p> <p>Space Complexity: <math>O(n)</math></p> <p><b>Solution</b></p>                                                                                                                                                                                                                                                                               |
| <p>DFS</p> <p>p.1903</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>DFS</p> <p>p.1904</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>DFS</p> <p>p.1905</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p>The solution consists of two main steps:</p> <ul style="list-style-type: none"> <li>- Traverse the list of edges to detect whether any node has two parents. If such a node is found, mark the two candidate edges (candidate1: the earlier edge, candidate2: the later edge).</li> <li>- Use the Union-Find algorithm to iterate over the edges: If there is a conflict (i.e., a node with two parents), skip candidate2 initially.</li> </ul> <p>While processing edges with Union-Find, check if adding an edge would create a cycle. Finally, based on the presence of a cycle and the two-parent conflict, decide which candidate edge to return.</p> <p><b>Key Data Structures:</b></p> <ul style="list-style-type: none"> <li>- An array (or hash map) to track the parent of each node.</li> <li>- The Union-Find (Disjoint Set Union) structure to union sets and detect cycles.</li> </ul> <p><b>#1036 Escape a Large Maze [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The maximum number of blocked cells is 200, which limits the size of a potential completely enclosed region.</li> <li>• Even if the target is unreachable by direct search, if the BFS (or DFS) explores more than a calculated threshold (based on the blocked cells count), then the region is not fully enclosed.</li> <li>• Perform two BFS explorations: one starting from the source and one from the target. If neither search is trapped within a small region, then an escape/path</li> </ul>        | <p>exists.</p> <ul style="list-style-type: none"> <li>• Early exit in BFS if the target is reached or if the number of visited nodes exceeds the maximum possible enclosed area.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(B^2)</math> in the worst case, where B is the number of blocked cells (with <math>B \leq 200</math>).</p> <p>Space Complexity: <math>O(B^2)</math> due to the storage of visited cells in the worst-case scenario.</p> <p><b>Solution</b></p> <p>The solution uses the Breadth-First Search (BFS) algorithm. A key observation here is that the blocked cells can only enclose a limited area. We compute a threshold (<code>maxSteps</code>) based on the number of blocked cells (<math>n * (n - 1) / 2</math>) to determine if the search is confined. The BFS from the source (and similarly from the target) stops in two cases: if the target is reached or if the visited area exceeds the threshold, indicating that the region is not fully enclosed by blocked cells. Running BFS from both the source and target ensures that neither is trapped inside a small banned region.</p> <p><b>#1192 Critical Connections in a Network [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The problem is about identifying bridges in an undirected graph.</li> </ul> | <ul style="list-style-type: none"> <li>• Use Depth-First Search (DFS) to explore the graph.</li> <li>• During DFS, track discovery times and low-link values for each node.</li> <li>• For an edge (u, v), if <code>low[v] &gt; disc[u]</code> then (u, v) is a critical connection.</li> <li>• Tarjan's algorithm is well-suited for solving this problem in <math>O(n + m)</math> time.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: <math>O(n + m)</math> where n is the number of nodes and m is the number of connections.</p> <p>Space Complexity: <math>O(n + m)</math> for the graph representation and DFS recursion stack.</p> <p><b>Solution</b></p> <p>We solve this problem using a DFS-based approach (Tarjan's algorithm). First, we transform the list of connections into an adjacency list for efficient graph traversal. We then start a DFS from an arbitrary node (node 0, since the graph is connected) while maintaining two arrays: one for discovery times (<code>disc</code>) and one for low-link values (<code>low</code>). The discovery time is the time when the node is first visited, and the low-link value is the smallest discovery time that can be reached from that node (including via back edges). For each edge (u, v), after a DFS call on v, if <code>low[v] &gt; disc[u]</code> then the edge (u, v) is a bridge because v (and its descendants) cannot reach u or any of its ancestors without this edge. This edge is then recorded as a critical connection.</p> |



Binary Search

#18 Binary Search

By Pattern · Python Only · 24 questions

DFS

p.1909

Binary Search

#35 Search Insert Position

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Binary Search

Lists: Denny Zhang

Time Complexity:  $O(\log n)$  due to binary search. Space Complexity:  $O(1)$  as no extra space is used.

Problem

Given a sorted array of distinct integers and a target value, return the index if the target is found. If not, return the index where it would be if it were inserted in order.

You must write an algorithm with  $O(\log n)$  runtime complexity.

Example 1:

Input: nums = [1,3,5,6], target = 5  
Output: 2

Example 2:

Input: nums = [1,3,5,6], target = 2  
Output: 1

Example 3:

Binary Search

p.1910

Input: nums = [1,3,5,6], target = 7  
Output: 4

Constraints:

- 1 <= nums.length <= 104
- 104 <= nums[i] <= 104
- nums contains distinct values sorted in ascending order.
- 104 <= target <= 104

Brute Force (Python)

# Brute Force Approach  
class Solution:  
 def searchInsert(self, nums, target):  
 # Brute Force O(n) - linear scan instead of binary search  
 for i, val in enumerate(nums):  
 if val == target:  
 return i  
 return -1

Community Solutions (Python)

- WalkCC

Binary Search

p.1911

class Solution:  
 def searchInsert(self, nums: List[int], target: int) -> int:  
 l = 0  
 r = len(nums)  
  
 while l < r:  
 m = (l + r) // 2  
 if nums[m] == target:  
 return m  
 if nums[m] < target:  
  
 l = m + 1  
 else:  
 r = m  
  
 return l

- LeetCode

class Solution:  
 def searchInsert(self, nums: List[int], target: int) -> int:  
 l, r = 0, len(nums)  
 while l < r:  
 mid = (l + r) // 2  
 if nums[mid] >= target:  
 r = mid  
 else:  
 l = mid + 1  
 return l

- SimplyLeet

Binary Search

p.1912

# Python solution using binary search  
def searchInsert(nums, target):  
 # Initialize low and high pointers for binary search  
 low, high = 0, len(nums) - 1  
 # Perform binary search until low pointer exceeds high pointer  
 while low <= high:  
 mid = (low + high) // 2 # Calculate the middle index  
 if nums[mid] == target:  
 return mid # Return mid if target is found  
 elif target < nums[mid]:  
  
 high = mid - 1 # Narrow search on the left side  
 else:  
 low = mid + 1 # Narrow search on the right side  
 # Return low pointer as the insertion index when target is not found  
 return low  
  
# Example usage  
print(searchInsert([1, 3, 5, 6], 5)) # Output: 2

- LC.ca

Binary Search

p.1913

class Solution:  
 def searchInsert(self, nums: List[int], target: int) -> int:  
 left, right = 0, len(nums)  
 while left < right:  
 mid = (left + right) >> 1  
 if nums[mid] >= target:  
 right = mid  
 else:  
 left = mid + 1  
 return left

Binary Search

p.1914

Binary Search

#69 Sqrt(x)

LeetCode · SimplyLeet · WalkCC · LeetCode

Math · Binary Search

Lists: Denny Zhang

Time Complexity:  $O(\log x)$  Space Complexity:  $O(1)$

Problem

Given a non-negative integer x, return the square root of x rounded down to the nearest integer. The returned integer should be non-negative as well.

You must not use any built-in exponent function or operator.

- For example, do not use pow(x, 0.5) in c++ or x \*\* 0.5 in python.

Example 1:

Input: x = 4  
Output: 2  
Explanation: The square root of 4 is 2, so we return 2.

Example 2:

Input: x = 8  
Output: 2  
Explanation: The square root of 8 is 2.82842..., and since we round it down to the nearest integer, 2 is returned.

Binary Search

p.1915

Constraints:

- 0 <= x <= 231 - 1

Brute Force (Python)

# Brute Force Approach  
class Solution:  
 def mySqrt(self, x):  
 # Brute Force O(n) - linear scan instead of binary search  
 for i, val in enumerate(x):  
 if val == target:  
 return i  
 return -1

Community Solutions (Python)

- WalkCC

class Solution:  
 def mySqrt(self, x: int) -> int:  
 return bisect.bisect\_right(range(x + 1), x, key=lambda m: m \* m) - 1

- LeetCode

Binary Search

p.1916

class Solution:  
 def mySqrt(self, x: int) -> int:  
 l, r = 0, x  
 while l < r:  
 mid = (l + r + 1) >> 1  
 if mid > x // mid:  
 r = mid - 1  
 else:  
 l = mid  
 return l

- SimplyLeet

# Binary search solution to find the integer square root  
def mySqrt(x):  
 # Edge case for small numbers  
 if x < 2:  
 return x  
  
 low, high = 0, x  
 ans = 0  
 # Binary search loop  
 while low <= high:

Binary Search

p.1917

```
mid = (low + high) // 2
If mid squared equals x, we found the exact
square root
if mid * mid == x:
 return mid
If mid squared is less than x, store mid and
search right half
elif mid * mid < x:
 ans = mid # candidate answer
 low = mid + 1
else:
 # If mid squared is more than x, search left
 half

high = mid - 1
return ans

Example usage:
print(mySqrt(4)) # Output: 2
print(mySqrt(8)) # Output: 2
```

```
• LC.ca

class Solution:
 def mySqrt(self, x: int) -> int:
 l, r = 0, x
 while l < r:
 mid = (l + r + 1) >> 1
 if mid > x // mid:
 r = mid - 1
 else:
 l = mid
 return l
```

Binary Search

#278 First Bad VersionEAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Binary Search · Interactive

Lists: Grind 169

Time Complexity: O(log n) Space Complexity: O(1)

Problem

You are a product manager and currently leading a team to develop a new product. Unfortunately, the latest version of your product fails the quality check. Since each version is developed based on the previous version, all the versions after a bad version are also bad. Suppose you have n versions [1, 2, ..., n] and you want to find out the first bad one, which causes all the following ones to be bad.

You are given an API bool isBadVersion(version) which returns whether version is bad. Implement a function to find the first bad version. You should minimize the number of calls to the API.

Example 1:

```
Input: n = 5, bad = 4
Output: 4
Explanation:
call isBadVersion(3) -> false
call isBadVersion(5) -> true
call isBadVersion(4) -> true
Then 4 is the first bad version.
```

Example 2:

```
Input: n = 1, bad = 1
Output: 1
```

Constraints:

- 1 <= bad <= n <= 231 - 1

Community Solutions (Python)

```
• WalkCC

class Solution:
 def firstBadVersion(self, n: int) -> int:
 l = 1
 r = n

 while l < r:
 m = (l + r) >> 1
 if isBadVersion(m):
 r = m
 else:
 l = m + 1

 return l
```

```
• LeetCode

The isBadVersion API is already defined for you.
def isBadVersion(version: int) -> bool:

class Solution:
 def firstBadVersion(self, n: int) -> int:
 l, r = 1, n
 while l < r:
 mid = (l + r) >> 1
 if isBadVersion(mid):
 r = mid
 else:
 l = mid + 1
 return l
```

```
• SimplyLeet

Python implementation of First Bad Version
class Solution:
 def firstBadVersion(self, n: int) -> int:
 # Initialize the left pointer to the first
 version and right pointer to the last version
 left, right = 1, n
 while left < right:
 # Calculate the mid point to avoid potential
 overflow
 mid = left + (right - left) // 2
 # Call the API isBadVersion to check if mid
 version is bad
 if isBadVersion(mid):
```

```
If mid is bad, search the left half
including mid
right = mid
else:
 # If mid is not bad, search the right
 half excluding mid
 left = mid + 1
When left equals right, we found the first bad
version
return left
```

```
• LC.ca

The isBadVersion API is already defined for you.
@param version, an integer
@return an integer
def isBadVersion(version):

class Solution:
 def firstBadVersion(self, n):
 """
 :type n: int

 :rtype: int
 """
 left, right = 1, n
 while left < right:
 mid = (left + right) >> 1
 if isBadVersion(mid):
 right = mid
 else:
 left = mid + 1
 return left
```

Binary Search

#374 Guess Number Higher or LowerEAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Math · Binary Search · Interactive

Lists: Denny Zhang

Time Complexity: O(log n) since we binary search the range. Space Complexity: O(1) as we use a constant amount of extra space.

Problem

We are playing the Guess Game. The game is as follows: I pick a number from 1 to n. You have to guess which number I picked (the number I picked stays the same throughout the game).

Every time you guess wrong, I will tell you whether the number I picked is higher or lower than your guess.

You call a pre-defined API int guess(int num), which returns three possible results:

- 1: Your guess is higher than the number I picked (i.e. num > pick).
- 1: Your guess is lower than the number I picked (i.e. num < pick).
- 0: your guess is equal to the number I picked (i.e. num == pick).

Return the number that I picked.

Example 1:

```
Input: n = 10, pick = 6
Output: 6
```

Example 2:

```
Input: n = 1, pick = 1
Output: 1
```

Example 3:

```
Input: n = 2, pick = 1
Output: 1
```

Constraints:

- 1 <= n <= 231 - 1
- 1 <= pick <= n

Community Solutions (Python)

```
• WalkCC

The guess API is already defined for you.
@param num, your guess
@return -1 if num is higher than the picked number
1 if num is lower than the picked number
otherwise return 0
def guess(num: int) -> int:

class Solution:
 def guessNumber(self, n: int) -> int:
 l = 1
```

```
r = n

Find the first guess number that >= the target
number
while l < r:
 m = (l + r) // 2
 if guess(m) <= 0: # -1, 0
 r = m
 else:
 l = m + 1

return l
```

```
• LeetCode

The guess API is already defined for you.
@param num, your guess
@return -1 if num is higher than the picked number
1 if num is lower than the picked number
otherwise return 0
def guess(num: int) -> int:

class Solution:
 def guessNumber(self, n: int) -> int:

 return bisect.bisect(range(1, n + 1), 0,
 key=lambda x: -guess(x))
```

• SimplyLeet

```
Python solution using binary search
class Solution:
 def guessNumber(self, n: int) -> int:
 # Initialize low and high pointers
 low, high = 1, n
 # Continue binary search until low exceeds high
 while low <= high:
 # Calculate midpoint safely to prevent
 overflow
 mid = low + (high - low) // 2
 # Call the guess API with mid

 result = guess(mid)
 # If the guess is correct, return mid
 if result == 0:
 return mid
 # If guess was too high, adjust the high
 pointer
 elif result == -1:
 high = mid - 1
 # If guess was too low, adjust the low
 pointer
 else:
 low = mid + 1

 # Return low (ideally should never reach here if
 pick exists in [1, n])
 return low
```

• LC.ca

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| <div><pre># The guess API is already defined for you. # @param num, your guess # @return -1 if num is higher than the picked number # 1 if num is lower than the picked number # otherwise return 0 # def guess(num: int) -&gt; int:  class Solution:     def guessNumber(self, n: int) -&gt; int:          return bisect.bisect(range(1, n + 1), 0, key=lambda x: -guess(x))</pre></div> <div>Binary Searchp.1927</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div><div>Binary SearchEAS</div><div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div><div>Array · Binary Search</div><div>Lists: Grind 169</div><div>Time Complexity: O(log n) Space Complexity: O(1)</div><div>Problem</div><div>Given an array of integers nums which is sorted in ascending order, and an integer target, write a function to search target in nums. If target exists, then return its index. Otherwise, return -1.</div><div>You must write an algorithm with O(log n) runtime complexity.</div><div>Example 1:</div><div>Input: nums = [-1,0,3,5,9,12], target = 9<br/>Output: 4<br/>Explanation: 9 exists in nums and its index is 4</div><div>Example 2:</div><div>Input: nums = [-1,0,3,5,9,12], target = 2<br/>Output: -1<br/>Explanation: 2 does not exist in nums so return -1</div><div>Constraints:</div><div><ul style="list-style-type: none"><li>1 &lt;= nums.length &lt;= 104</li></ul></div></div> <div>Binary Searchp.1928</div> | <div><ul style="list-style-type: none"><li>-104 &lt; nums[i], target &lt; 104</li><li>All the integers in nums are unique.</li><li>nums is sorted in ascending order.</li></ul></div> <div>Brute Force (Python)</div> <div><pre># Brute Force Approach class Solution:     def search(self, nums, target):         # Brute Force O(n) - linear scan ignoring sorted property         for i, val in enumerate(nums):             if val == target:                 return i         return -1</pre></div> <div>Community Solutions (Python)</div> <div><ul style="list-style-type: none"><li>WalkCC</li></ul><div><pre>class Solution:     def search(self, nums: List[int], target: int) -&gt; int:         i = bisect.bisect_left(nums, target)         return -1 if i == len(nums) or nums[i] != target else i</pre></div><ul style="list-style-type: none"><li>LeetDoocs</li></ul></div> <div>Binary Searchp.1929</div>                                                                                                                                                 |
| <div><pre>class Solution:     def search(self, nums: List[int], target: int) -&gt; int:         l, r = 0, len(nums) - 1         while l &lt; r:             mid = (l + r) &gt;&gt; 1             if nums[mid] &gt;= target:                 r = mid             else:                 l = mid + 1         return l if nums[l] == target else -1</pre></div> <div>• SimplyLeet</div> <div><pre># Python code for binary search  def search(nums, target):     # Set initial boundaries for the search     left, right = 0, len(nums) - 1      # Loop until the search space is empty     while left &lt;= right:         # Calculate middle index to avoid overflow         mid = left + (right - left) // 2</pre></div> <div>Binary Searchp.1930</div>                                                                                                                                                                                                                       | <div><pre># Check if the mid element matches the target if nums[mid] == target:     return mid # If target is smaller than mid element, narrow search to left half elif target &lt; nums[mid]:     right = mid - 1 # Else, narrow search to right half else:     left = mid + 1  # Target not found, return -1 return -1  # Example usage: print(search([-1,0,3,5,9,12], 9)) # Expected output: 4</pre></div> <div>• LC.ca</div> <div><pre>class Solution:     def search(self, nums: List[int], target: int) -&gt; int:         left, right = 0, len(nums) - 1         while left &lt; right:             mid = (left + right) &gt;&gt; 1             if nums[mid] &gt;= target:                 right = mid             else:                 left = mid + 1         return left if nums[left] == target else -1</pre></div> <div>Binary Searchp.1931</div>                                                                                                       | <div><div>Binary SearchMED</div><div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div><div>Array · Binary Search</div><div>Lists: Grind 169</div><div>Time Complexity: O(log n) Space Complexity: O(1)</div><div>Problem</div><div>There is an integer array nums sorted in ascending order (with distinct values).</div><div>Prior to being passed to your function, nums is possibly left rotated at an unknown index k (1 &lt;= k &lt; nums.length) such that the resulting array is [nums[k], nums[k+1], ..., nums[n-1], nums[0], nums[1], ..., nums[k-1]] (0-indexed). For example, [0,1,2,4,5,6,7] might be left rotated by 3 indices and become [4,5,6,7,0,1,2].</div><div>Given the array nums after the possible rotation and an integer target, return the index of target if it is in nums, or -1 if it is not in nums.</div><div>You must write an algorithm with O(log n) runtime complexity.</div><div>Example 1:</div></div> <div>Binary Searchp.1932</div>                                                                                                  |
| <div><div>Input: nums = [4,5,6,7,0,1,2], target = 0<br/>Output: 4</div><div>Example 2:</div><div>Input: nums = [4,5,6,7,0,1,2], target = 3<br/>Output: -1</div><div>Example 3:</div><div>Input: nums = [1], target = 0<br/>Output: -1</div><div>Constraints:</div><div><ul style="list-style-type: none"><li>1 &lt;= nums.length &lt;= 5000</li><li>-104 &lt;= nums[i] &lt;= 104</li><li>All values of nums are unique.</li><li>nums is an ascending array that is possibly rotated.</li><li>-104 &lt;= target &lt;= 104</li></ul></div><div>Brute Force (Python)</div><div><pre># Brute Force Approach class Solution:     def search(self, nums, target):         # Brute Force O(n) - linear scan         for i, val in enumerate(nums):             if val == target:                 return i         return -1</pre></div><div>Community Solutions (Python)</div><div><ul style="list-style-type: none"><li>WalkCC</li></ul></div><div>Binary Searchp.1933</div></div> | <div><pre>class Solution:     def search(self, nums: List[int], target: int) -&gt; int:         l = 0         r = len(nums) - 1          while l &lt;= r:             m = (l + r) // 2             if nums[m] == target:                 return m             if nums[l] &lt;= nums[m]: # nums[l..m] are sorted.                  if nums[l] &lt;= target &lt; nums[m]:                     r = m - 1                 else:                     l = m + 1             else: # nums[m..n - 1] are sorted.                 if nums[m] &lt; target &lt;= nums[r]:                     l = m + 1                 else:                     r = m - 1          return -1</pre></div> <div>• LeetDoocs</div> <div>Binary Searchp.1934</div>                                                                                                                                                                                                                               | <div><pre>class Solution:     def search(self, nums: List[int], target: int) -&gt; int:         n = len(nums)         left, right = 0, n - 1         while left &lt; right:             mid = (left + right) &gt;&gt; 1             if nums[0] &lt;= nums[mid]:                 if nums[0] &lt;= target &lt;= nums[mid]:                     right = mid                 else:                     left = mid + 1             else:                 if nums[mid] &lt; target &lt;= nums[n - 1]:                     left = mid + 1                 else:                     right = mid         return left if nums[left] == target else -1</pre></div> <div>• SimplyLeet</div> <div><pre># Define a function to search for the target in a rotated sorted array. def search(nums, target):     # Initialize pointers for binary search.     left, right = 0, len(nums) - 1      # Loop until the search space is exhausted.     while left &lt;= right:         # Compute the middle index.         mid = (left + right) // 2</pre></div> <div>Binary Searchp.1935</div> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
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| <pre># If the mid element is the target, return its index. if nums[mid] == target:     return mid  # Check if the left part of the array is sorted. if nums[left] &lt;= nums[mid]:     # If the target is within the sorted left     half, adjust the right pointer.     if nums[left] &lt;= target &lt; nums[mid]:         right = mid - 1     else:          # Otherwise, search the right half.         left = mid + 1 else:     # The right half must be sorted.     if nums[mid] &lt; target &lt;= nums[right]:         left = mid + 1     else:         right = mid - 1  # If the target is not found, return -1.  return -1  # Example usage: # result = search([4,5,6,7,0,1,2], 0) # print(result) # Expected output: 4</pre> <div>• LC.ca</div> | <pre># below 2 solutions, diff is while condition: left ('&lt;' or '&lt;=') right  # I like this better class Solution:     def search(self, nums: List[int], target: int) -&gt;     int:          n = len(nums)         left, right = 0, n - 1         while left &lt;= right:             mid = (left + right) &gt;&gt; 1             if nums[mid] == target:                  return mid             elif nums[0] &lt;= nums[mid]: # left half sorted                 if nums[0] &lt;= target &lt;= nums[mid]:                     right = mid - 1                 else:                     left = mid + 1             else: # right half sorted                 if nums[mid] &lt; target &lt;= nums[n - 1]:                     left = mid + 1                 else:</pre> | <pre>        right = mid - 1      return -1  #####  class Solution:     def search(self, nums: List[int], target: int) -&gt;     int:          n = len(nums)         left, right = 0, n - 1         while left &lt; right:              mid = (left + right) &gt;&gt; 1             if nums[0] &lt;= nums[mid]: # left half sorted                 if nums[0] &lt;= target &lt;= nums[mid]:                     right = mid                 else:                     left = mid + 1             else: # right half sorted                 if nums[mid] &lt; target &lt;= nums[n - 1]:                     left = mid + 1                 else:                      right = mid             return left if nums[left] == target else -1</pre> |
| Binary Searchp.1936                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Binary Searchp.1937                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Binary Searchp.1938                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

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| <div>Binary Search</div> <h2>#34 Find First and Last Position of Element in Sorted Array</h2> <div>MED</div> <p>LeetDoocs · SimplyLeet · WalkCC · LeetCode</p> <p>Array · Binary Search</p> <p>Lists: Denny Zhang</p> <p>Time Complexity: O(log n) Space Complexity: O(1)</p> <p><b>Problem</b></p> <p>Given an array of integers nums sorted in non-decreasing order, find the starting and ending position of a given target value.</p> <p>If target is not found in the array, return [-1, -1].</p> <p>You must write an algorithm with O(log n) runtime complexity.</p> <p><b>Example 1:</b></p> <div>Input: nums = [5,7,7,8,8,10], target = 8<br/>Output: [3,4]</div> <p><b>Example 2:</b></p> <div>Input: nums = [5,7,7,8,8,10], target = 6<br/>Output: [-1,-1]</div> <p><b>Example 3:</b></p> <div>Input: nums = [], target = 0<br/>Output: [-1,-1]</div> | <p><b>Constraints:</b></p> <ul style="list-style-type: none"><li>0 &lt;= nums.length &lt;= 105</li><li>-109 &lt;= nums[i] &lt;= 109</li><li>nums is a non-decreasing array.</li><li>-109 &lt;= target &lt;= 109</li></ul> <p><b>Brute Force (Python)</b></p> <pre># Brute Force Approach class Solution:     def searchRange(self, nums, target):         # Brute Force O(n) - linear scan instead of         binary search         for i, val in enumerate(nums):             if val == target:                 return i         return -1</pre> <p><b>Community Solutions (Python)</b></p> <div>• WalkCC</div> <pre>class Solution:     def searchRange(self, nums: list[int], target: int) -&gt;     list[int]:         l = bisect_left(nums, target)         if l == len(nums) or nums[l] != target:             return -1, -1         r = bisect_right(nums, target) - 1         return l, r</pre> <div>• LeetDoocs</div> | <pre>class Solution:     def searchRange(self, nums: List[int], target: int)     -&gt; List[int]:         l = bisect_left(nums, target)         r = bisect_left(nums, target + 1)         return [-1, -1] if l == r else [l, r - 1]</pre> <div>• SimplyLeet</div> <pre>def searchRange(nums, target):     # Helper function to find the boundary index     (leftmost or rightmost) for target.     def findBoundary(isLeft):         left, right = 0, len(nums) - 1         boundary = -1         while left &lt;= right:             mid = (left + right) // 2             # If the target is found, update boundary and             continue to search on one side.             if nums[mid] == target:                 boundary = mid          if isLeft:             right = mid - 1 # Continue search on         left half         else:             left = mid + 1 # Continue search on         right half         elif nums[mid] &lt; target:             left = mid + 1         else:             right = mid - 1         return boundary</pre> |
| Binary Searchp.1939                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Binary Searchp.1940                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Binary Searchp.1941                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| <pre># Find the leftmost index leftIndex = findBoundary(True) if leftIndex == -1:     return [-1, -1] # Find the rightmost index rightIndex = findBoundary(False) return [leftIndex, rightIndex]  # Example usage: # nums = [5,7,7,8,8,10]  # target = 8 # print(searchRange(nums, target)) # Output: [3,4]</pre> <div>• LC.ca</div> <pre>... &gt;&gt;&gt; bisect.bisect_left([1,1,1,2,2,2,7,7,7], 2) 3 &gt;&gt;&gt; bisect.bisect_left([1,1,1,2,2,2,7,7,7], 2+1) 6  &gt;&gt;&gt; bisect.bisect_right([1,1,1,2,2,2,7,7,7], 2) 6 &gt;&gt;&gt; bisect.bisect_right([1,1,1,2,2,2,7,7,7], 2+1)</pre> | <pre>6  &gt;&gt;&gt; bisect.bisect_left([1,1,1,2,2,2,7,7,7], 0) 0 &gt;&gt;&gt; bisect.bisect_left([1,1,1,2,2,2,7,7,7], 1) 0  # below, even single value for 2, after +1 the index will be different &gt;&gt;&gt; bisect.bisect_left([1,2,3,4,5], 2)  1 &gt;&gt;&gt; bisect.bisect_left([1,2,3,4,5], 2+1) 2 ...  import bisect  class Solution:     def searchRange(self, nums: List[int], target: int)     -&gt; List[int]:         l = bisect_left(nums, target)</pre> | <pre>        r = bisect_left(nums, target + 1)         return [-1, -1] if l == r else [l, r - 1]  #####  class Solution:     def searchRange(self, nums: List[int], target: int)     -&gt; List[int]:         def findRange(to_left):             l, r = 0, len(nums) - 1             m = 0              while l &lt;= r:                 m = l + (r - l) // 2                 if nums[m] &lt; target:                     l = m + 1                 elif nums[m] &gt; target:                     r = m - 1                 elif to_left and m &gt; 0 and nums[m - 1] ==                 nums[m]: # so now nums[m] == target, but maybe not the                 leftmost or rightmost                     r = m - 1                 elif not to_left and m + 1 &lt; len(nums)                 and nums[m + 1] == nums[m]: # so now nums[m] == target,                 but maybe not the leftmost or rightmost                     l = m + 1</pre> |
| Binary Searchp.1942                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Binary Searchp.1943                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Binary Searchp.1944                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

```

 else:
 return m
 if m < len(nums) and nums[m] == target:
 # 1. cover not-found,
 # or 2. cover input=[3] and target=3, first
 and last both 0, return [0,0]
 return m
 else:
 return -1
 return [findRange(True), findRange(False)]

```

Binary Search

#153 Find Minimum in Rotated Sorted ArrayMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Binary Search

Lists: Grind 169

Time Complexity:  $O(\log n)$  Space Complexity:  $O(1)$

**Problem**

Suppose an array of length  $n$  sorted in ascending order is rotated between  $1$  and  $n$  times. For example, the array  $\text{nums} = [0,1,2,4,5,6,7]$  might become:

- $[4,5,6,7,0,1,2]$  if it was rotated 4 times.
- $[0,1,2,4,5,6,7]$  if it was rotated 7 times.

Notice that rotating an array  $[a[0], a[1], a[2], \dots, a[n-1]]$  1 time results in the array  $[a[n-1], a[0], a[1], a[2], \dots, a[n-2]]$ .

Given the sorted rotated array  $\text{nums}$  of unique elements, return the minimum element of this array.

You must write an algorithm that runs in  $O(\log n)$  time.

**Example 1:**

```

Input: nums = [3,4,5,1,2]
Output: 1
Explanation: The original array was [1,2,3,4,5] rotated
3 times.

```

Example 2:

Input: nums = [4,5,6,7,0,1,2]
Output: 0
Explanation: The original array was [0,1,2,4,5,6,7] and
it was rotated 4 times.

Example 3:

Input: nums = [11,13,15,17]
Output: 11
Explanation: The original array was [11,13,15,17] and
it was rotated 4 times.

- Constraints:**
- $n == \text{nums.length}$
  - $1 \leq n \leq 5000$
  - $-5000 \leq \text{nums}[i] \leq 5000$
  - All the integers of  $\text{nums}$  are unique.
  - $\text{nums}$  is sorted and rotated between  $1$  and  $n$  times.

Brute Force (Python)

```

Brute Force Approach
class Solution:
 def findMin(self, nums):
 # Brute Force O(n) - just return the minimum
 (ignores sorted structure)
 return min(nums)

```

Community Solutions (Python)

- WalkCC

```

class Solution:
 def findMin(self, nums: List[int]) -> int:
 l = 0
 r = len(nums) - 1

 while l < r:
 m = (l + r) // 2
 if nums[m] < nums[r]:
 r = m
 else:
 l = m + 1

 return nums[l]

```

• LeetDoocs

class Solution:
def findMin(self, nums: List[int]) -> int:
if nums[0] <= nums[-1]:
return nums[0]
left, right = 0, len(nums) - 1
while left < right:
mid = (left + right) >> 1
if nums[0] <= nums[mid]:
left = mid + 1
else:
right = mid
return nums[left]

- SimplyLeet

```

def findMin(nums):
 # Initialize left and right pointers
 left, right = 0, len(nums) - 1
 # Continue while the search space is valid
 while left < right:
 mid = (left + right) // 2 # Compute middle index
 # If middle element is greater than the rightmost
 element,
 # the minimum must be in the right half
 if nums[mid] > nums[right]:
 left = mid + 1
 else:
 # Otherwise, the minimum is in the left half
 (including mid)
 right = mid
 # When loop ends, left == right, pointing to the
 minimum element
 return nums[left]

```

• LC.ca

class Solution:
def findMin(self, nums: List[int]) -> int:
if nums[0] <= nums[-1]:
return nums[0]
left, right = 0, len(nums) - 1
while left < right:
mid = (left + right) >> 1
if nums[0] <= nums[mid]:
left = mid + 1
else:
right = mid
return nums[left]

```

 right = mid
 return nums[left]

```

Binary Search

#300 Longest Increasing SubsequenceMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Binary Search · Dynamic Programming

Lists: Grind 169

Time Complexity:  $O(n \log n)$  where  $n$  is the length of  $\text{nums}$  Space Complexity:  $O(n)$  for storing the  $\text{dp}$  array

**Problem**

Given an integer array  $\text{nums}$ , return the length of the longest strictly increasing subsequence.

**Example 1:**

```

Input: nums = [10,9,2,5,3,7,101,18]
Output: 4
Explanation: The longest increasing subsequence is
[2,3,7,101], therefore the length is 4.

```

**Example 2:**

```

Input: nums = [0,1,0,3,2,3]
Output: 4

```

**Example 3:**

```

Input: nums = [7,7,7,7,7,7]
Output: 1

```

**Constraints:**

- $1 \leq \text{nums.length} \leq 2500$
  - $-104 \leq \text{nums}[i] \leq 104$
- Follow up: Can you come up with an algorithm that runs in  $O(n \log(n))$  time complexity?
- Brute Force (Python)**

```

Brute Force Approach
class Solution:
 def lengthOfLIS(self, nums):
 # Brute Force $O(2^n)$ - generate every
 subsequence, find longest increasing
 def lis(i, prev_val):
 if i == len(nums): return 0
 skip = lis(i + 1, prev_val)
 take = 0
 if nums[i] > prev_val:
 take = 1 + lis(i + 1, nums[i])
 return max(skip, take)
 return lis(0, float('-inf'))

```

Community Solutions (Python)

- WalkCC

```

class Solution:
 def lengthOfLIS(self, nums: List[int]) -> int:
 if not nums:
 return 0

 # dp[i] := the length of LIS ending in nums[i]
 dp = [1] * len(nums)

 for i in range(1, len(nums)):
 for j in range(i):

 if nums[j] < nums[i]:
 dp[i] = max(dp[i], dp[j] + 1)

 return max(dp)

class Solution:
 def lengthOfLIS(self, nums: List[int]) -> int:
 # tails[i] := the minimum tails of all the increasing
 subsequences having
 # length i + 1
 tails = []

 for num in nums:
 if not tails or num > tails[-1]:
 tails.append(num)
 else:
 tails[bisect.bisect_left(tails, num)] = num

 return len(tails)

```

- LeetDoocs

```
class Solution:
 def lengthOfLIS(self, nums: List[int]) -> int:
 n = len(nums)
 f = [1] * n
 for i in range(1, n):
 for j in range(i):
 if nums[j] < nums[i]:
 f[i] = max(f[i], f[j] + 1)
 return max(f)
```

```
class BinaryIndexedTree:
 def __init__(self, n: int):
 self.n = n
 self.c = [0] * (n + 1)

 def update(self, x: int, v: int):
 while x <= self.n:
 self.c[x] = max(self.c[x], v)
 x += x & -x
```

```
 def query(self, x: int) -> int:
 mx = 0
 while x:
 mx = max(mx, self.c[x])
 x -= x & -x
 return mx
```

```
class Solution:
 def lengthOfLIS(self, nums: List[int]) -> int:
```

Binary Search

p.1954

```
s = sorted(set(nums))
m = len(s)
tree = BinaryIndexedTree(m)
for x in nums:
 x = bisect_left(s, x) + 1
 t = tree.query(x - 1) + 1
 tree.update(x, t)
return tree.query(m)
```

• SimplyLeet

```
import bisect

def lengthOfLIS(nums):
 dp = [] # dp array stores smallest tail for each
 subsequence length
 for num in nums:
 # Find the leftmost insertion point using binary
 search
 idx = bisect.bisect_left(dp, num)
 if idx == len(dp):
 dp.append(num) # Extend dp if no larger tail
 exists
 else:
```

```
 dp[idx] = num # Replace to maintain the
 smallest tail
 return len(dp)
```

```
Example usage:
print(lengthOfLIS([10,9,2,5,3,7,101,18])) # Output: 4
```

• LC.ca

Binary Search

p.1955

```
find the largest end element in tails that is smaller
than nums[i]
and then replace it with nums[i] and discard the list
in the same length
which is implemented by `tail[idx] = num`
```

```
class Solution(object):
 def lengthOfLIS(self, nums):
 """"
 :type nums: List[int]
 :rtype: int
 """
```

```
 tail = []
 for num in nums:
 # if using bisect_right(tail, num),
 # then input=[7,7,7,7,7,7,7] will output 7
 but expected result is 1
 idx = bisect.bisect_left(tail, num)
 if idx == len(tail): # same as in java: if (i
 == len) len++;
 tail.append(num)
 else:
 tail[idx] = num
 return len(tail)
```

Binary Search

p.1956

```
implementation of bisect.bisect_left()
similar to Leetcode-302, find left/right/top/bottom
callable
def bisect_left(a, x, lo=0, hi=None):
 if hi is None:
 hi = len(a)
```

```
 while lo < hi:
 mid = (lo + hi) // 2
```

```
 if a[mid] < x:
 lo = mid + 1
 else:
 hi = mid
```

```
 return lo
```

```
#####
```

```
class BinaryIndexedTree:
```

```
 def __init__(self, n: int):
 self.n = n
 self.c = [0] * (n + 1)

 def update(self, x: int, v: int):
 while x <= self.n:
 self.c[x] = max(self.c[x], v)
 x += x & -x
```

```
 def query(self, x: int) -> int:
```

Binary Search

p.1957

```
mx = 0
while x:
 mx = max(mx, self.c[x])
 x -= x & -x
return mx
```

```
class Solution:
 def lengthOfLIS(self, nums: List[int]) -> int:
 s = sorted(set(nums))
```

```
m = len(s)
tree = BinaryIndexedTree(m)
for x in nums:
 x = bisect_left(s, x) + 1
 t = tree.query(x - 1) + 1
 tree.update(x, t)
return tree.query(m)
```

Binary Search

p.1958

Binary Search

## #362 Design Hit Counter

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Binary Search · Design · Queue

Lists: Grind 169

Time Complexity: O(1) per hit and getHits operation on average, since each hit is enqueued and later dequeued only once. Space Complexity: O(W) where W is the window size (300 seconds) in the worst ca

### Problem

Design a hit counter which counts the number of hits received in the past 5 minutes (i.e., the past 300 seconds).

Your system should accept a timestamp parameter (in seconds granularity), and you may assume that calls are being made to the system in chronological order (i.e., timestamp is monotonically increasing). Several hits may arrive roughly at the same time.

Implement the HitCounter class:

- HitCounter() Initializes the object of the hit counter system.
- void hit(int timestamp) Records a hit that happened at timestamp (in seconds). Several hits may happen at the same timestamp.

Binary Search

p.1959

• int getHits(int timestamp) Returns the number of hits in the past 5 minutes from timestamp (i.e., the past 300 seconds).

Example 1:

```
Input
["HitCounter", "hit", "hit", "hit", "getHits", "hit",
"getHits", "getHits"]
[[], [1], [2], [3], [4], [300], [300], [301]]
Output
[null, null, null, null, 3, null, 4, 3]
```

```
Explanation
HitCounter hitCounter = new HitCounter();
```

Constraints:

- 1 <= timestamp <= 2 \* 109
- All the calls are being made to the system in chronological order (i.e., timestamp is monotonically increasing).
- At most 300 calls will be made to hit and getHits.

Follow up: What if the number of hits per second could be huge? Does your design scale?

### Community Solutions (Python)

- WalkCC

Binary Search

p.1960

```
class HitCounter:
 def __init__(self):
 self.timestamps = [0] * 300
 self.hits = [0] * 300

 def hit(self, timestamp: int) -> None:
 i = timestamp % 300
 if self.timestamps[i] == timestamp:
 self.hits[i] += 1
 else:

 self.timestamps[i] = timestamp
 self.hits[i] = 1 # Reset the hit count to 1.

 def getHits(self, timestamp: int) -> int:
 return sum(h for t, h in zip(self.timestamps,
 self.hits)
 if timestamp - t < 300)
```

• LeetDoocs

```
class HitCounter:

 def __init__(self):
 self.ts = []

 def hit(self, timestamp: int) -> None:
 self.ts.append(timestamp)

 def getHits(self, timestamp: int) -> int:
 return len(self.ts) - bisect_left(self.ts,
 timestamp - 300 + 1)
```

Binary Search

p.1961

```
Your HitCounter object will be instantiated and called
as such:
obj = HitCounter()
obj.hit(timestamp)
param_2 = obj.getHits(timestamp)
```

• SimplyLeet

# Python implementation of HitCounter

```
from collections import deque
```

```
class HitCounter:
 def __init__(self):
 # Initialize the deque to store (timestamp,
 count) pairs.
 self.hits_queue = deque()
```

```
 def hit(self, timestamp: int) -> None:
```

Binary Search

p.1962



```
If the last recorded hit is at the same
timestamp, increment its count.
if self.hits_queue and self.hits_queue[-1][0] ==
timestamp:
 self.hits_queue[-1][1] += 1
else:
 # Append new (timestamp, count) pair.
 self.hits_queue.append([timestamp, 1])

def getHits(self, timestamp: int) -> int:
 # Remove outdated hits from the front of the
 queue.
 while self.hits_queue and self.hits_queue[0][0]
 <= timestamp - 300:
```

```
 self.hits_queue.popleft()
 # Sum counts of the remaining hit entries.
 return sum(count for _, count in self.hits_queue)
```

• LC.ca

```
from collections import deque
```

```
class HitCounter: # queue
```

```
 def __init__(self):
 """
 Initialize your data structure here.
 """
 self.hits = deque()
```

Binary Search

p.1963

```
def hit(self, timestamp: int) -> None:
 """
 Record a hit.
 @param timestamp - The current timestamp (in
 seconds granularity).
 """
 self.hits.append(timestamp)

def getHits(self, timestamp: int) -> int:
 """
 Return the number of hits in the past 5 minutes.

 @param timestamp - The current timestamp (in
 seconds granularity).
 """
 while self.hits and self.hits[0] <= timestamp -
 300:
 self.hits.popleft()
 return len(self.hits)

class HitCounter: # follow-up
 def __init__(self):
 self.times = [0] * 300
```

Binary Search

p.1964

```
self.hits = [0] * 300

def hit(self, timestamp: int) -> None:
 idx = timestamp % 300
 if self.times[idx] != timestamp:
 self.times[idx] = timestamp
 self.hits[idx] = 1
 else:
 self.hits[idx] += 1

def getHits(self, timestamp: int) -> int:
 res = 0
 for i in range(300):
 if timestamp - self.times[i] < 300:
 res += self.hits[i]
 return res

Your HitCounter object will be instantiated and called
as such:
obj = HitCounter()
obj.hit(timestamp)

param_2 = obj.getHits(timestamp)

#####

class HitCounter: # extra o(N) space
 def __init__(self):
 """
 Initialize your data structure here.
 """
 self.counter = Counter()
```

Binary Search

p.1965

```
def hit(self, timestamp: int) -> None:
 """
 Record a hit.
 @param timestamp - The current timestamp (in
 seconds granularity).
 """
 self.counter[timestamp] += 1

def getHits(self, timestamp: int) -> int:
 """
 Return the number of hits in the past 5 minutes.
 @param timestamp - The current timestamp (in
 seconds granularity).
 """
 return sum([v for t, v in self.counter.items() if
 t + 300 > timestamp])

Your HitCounter object will be instantiated and called
as such:
obj = HitCounter()
obj.hit(timestamp)
param_2 = obj.getHits(timestamp)
```

Binary Search

p.1966

Binary Search

## #528 Random Pick with Weight

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Math · Binary Search · Prefix Sum · Randomized

Lists: Grind 169

Time Complexity: O(n) for constructing the prefix sum array, and O(log n) per pickIndex() call. Space Complexity: O(n) for storing the prefix sum array.

### Problem

You are given a 0-indexed array of positive integers w where w[i] describes the weight of the ith index.

You need to implement the function pickIndex(), which randomly picks an index in the range [0, w.length - 1] (inclusive) and returns it. The probability of picking an index i is w[i] / sum(w).

- For example, if w = [1, 3], the probability of picking index 0 is 1 / (1 + 3) = 0.25 (i.e., 25%), and the probability of picking index 1 is 3 / (1 + 3) = 0.75 (i.e., 75%).

Example 1:

Binary Search

p.1967

```
Brute Force Approach
class Solution:
 def __init__(self, w):
 # Brute Force O(n) - linear scan instead of
 binary search
 for i, val in enumerate(w):
 if val == target:
 return i
 return -1
```

### Community Solutions (Python)

• WalkCC

```
class Solution:
 def __init__(self, w: List[int]):
 self.prefix = list(itertools.accumulate(w))

 def pickIndex(self) -> int:
 target = random.randint(0, self.prefix[-1] - 1)
 return bisect.bisect_right(range(len(self.prefix)),
 target,
 key=lambda m:
 self.prefix[m])
```

• LeetDoocs

```
class Solution:
 def __init__(self, w: List[int]):
 self.s = [0]
 for c in w:
 self.s.append(self.s[-1] + c)

 def pickIndex(self) -> int:
 x = random.randint(1, self.s[-1])
 left, right = 1, len(self.s) - 1
 while left < right:

 mid = (left + right) >> 1
 if self.s[mid] >= x:
 right = mid
 else:
 left = mid + 1
 return left - 1

Your Solution object will be instantiated and called as
such:
obj = Solution(w)

param_1 = obj.pickIndex()
```

• SimplyLeet

Binary Search

```
import random
import bisect

class Solution:
 def __init__(self, w):
 # Build the prefix sum array.
 self.prefix_sums = []
 current_sum = 0
 for weight in w:
 current_sum += weight

 self.prefix_sums.append(current_sum)
 # Store the total sum of weights.
 self.total_sum = current_sum

 def pickIndex(self):
 # Get a random target in the range [1, total_sum]
 target = random.randint(1, self.total_sum)
 # Use bisect_left to find the proper index.
 index = bisect.bisect_left(self.prefix_sums,
 target)

 return index

Example usage:
solution = Solution([1, 3])
print(solution.pickIndex())
```

• LC.ca

Binary Search

p.1971

Binary Search

p.1969

```
class Solution:
 def __init__(self, w: List[int]):
 self.s = [0]
 for c in w:
 self.s.append(self.s[-1] + c)

 def pickIndex(self) -> int:
 x = random.randint(1, self.s[-1])
 left, right = 1, len(self.s) - 1
 while left < right:

 mid = (left + right) >> 1
 if self.s[mid] >= x:
 right = mid
 else:
 left = mid + 1
 return left - 1
```

# Your Solution object will be instantiated and called as such:  
# obj = Solution(w)

# param\_1 = obj.pickIndex()

Binary Search

p.1972

Binary Search

## #852 Peak Index in a Mountain Array

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Binary Search

Lists: Denny Zhang

Time Complexity:  $O(\log(n))$  – The binary search approach reduces the search interval by half at each step.  
Space Complexity:  $O(1)$  – Only constant extra space is used.

Problem

You are given an integer mountain array `arr` of length `n` where the values increase to a peak element and then decrease.

Return the index of the peak element.

Your task is to solve it in  $O(\log(n))$  time complexity.

Example 1:

Input: `arr = [0,1,0]`

Output: 1

Example 2:

Input: `arr = [0,2,1,0]`

Output: 1

Example 3:

Input: `arr = [0,10,5,2]`

Binary Search

p.1973

Output: 1

Constraints:

- $3 \leq \text{arr.length} \leq 105$
- $0 \leq \text{arr}[i] \leq 106$
- `arr` is guaranteed to be a mountain array.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def peakIndexInMountainArray(self, arr):
 # Brute Force $O(n)$ – linear scan instead of
 binary search
 for i, val in enumerate(arr):
 if val == target:
 return i
 return -1
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def peakIndexInMountainArray(self, arr: List[int]) -> int:
 l = 0
 r = len(arr) - 1

 while l < r:
 m = (l + r) // 2
 if arr[m] >= arr[m + 1]:
 r = m
 else:
 l = m + 1
```

Binary Search

p.1974

l = m + 1

return l

• LeetCode

```
class Solution:
 def peakIndexInMountainArray(self, arr: List[int]) -> int:
 left, right = 1, len(arr) - 2
 while left < right:
 mid = (left + right) >> 1
 if arr[mid] > arr[mid + 1]:
 right = mid
 else:
 left = mid + 1
 return left
```

• SimplyLeet

```
def peakIndexInMountainArray(arr):
 # Initialize pointers for the binary search
 left, right = 0, len(arr) - 1
 # Continue while there is more than one element in
 the search space
 while left < right:
 mid = (left + right) // 2 # Find the mid index
 # If the sequence is still increasing, move the
 left pointer to mid+1
 if arr[mid] < arr[mid + 1]:
 left = mid + 1
 else:
 right = mid
```

Binary Search

p.1975

```
Otherwise, the peak is at mid or to the
left of mid
 right = mid
 # When left and right converge, we have the peak
 index
 return left

Example usage:
print(peakIndexInMountainArray([0, 1, 0])) # Output: 1
```

• LC.ca

```
class Solution:
 def peakIndexInMountainArray(self, arr: List[int]) -> int:
 left, right = 1, len(arr) - 2
 while left < right:
 mid = (left + right) >> 1
 if arr[mid] > arr[mid + 1]:
 right = mid
 else:
 left = mid + 1
 return left
```

Binary Search

p.1976

Binary Search

## #981 Time Based Key-Value Store

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Hash Table · String · Binary Search · Design

Lists: Grind 169

Time Complexity:

Problem

Design a time-based key-value data structure that can store multiple values for the same key at different time stamps and retrieve the key's value at a certain timestamp.

Implement the `TimeMap` class:

- `TimeMap()` Initializes the object of the data structure.
- `void set(String key, String value, int timestamp)` Stores the key `key` with the value `value` at the given time `timestamp`.
- `String get(String key, int timestamp)` Returns a value such that `set` was called previously, with `timestamp_prev <= timestamp`. If there are multiple such values, it returns the value associated with the largest `timestamp_prev`. If there are no values, it returns `""`.

Example 1:

Binary Search

p.1977

```
Input
["TimeMap", "set", "get", "get", "set", "get", "get"]
[[], ["foo", "bar", 1], ["foo", 1], ["foo", 3], ["foo", "bar2", 4], ["foo", 4], ["foo", 5]]
Output
[null, null, "bar", "bar", null, "bar2", "bar2"]
```

Explanation  
`TimeMap timeMap = new TimeMap();`

Constraints:

- $1 \leq \text{key.length}, \text{value.length} \leq 100$
- `key` and `value` consist of lowercase English letters and digits.
- $1 \leq \text{timestamp} \leq 107$
- All the timestamps `timestamp` of `set` are strictly increasing.
- At most  $2 * 105$  calls will be made to `set` and `get`.

Community Solutions (Python)

• WalkCC

```
class TimeMap:
 def __init__(self):
 self.values = collections.defaultdict(list)
 self.timestamps = collections.defaultdict(list)

 def set(self, key: str, value: str, timestamp: int) -> None:
 self.values[key].append(value)
 self.timestamps[key].append(timestamp)

 def get(self, key: str, timestamp: int) -> str:
```

```
 if key not in self.timestamps:
 return ''
 i = bisect.bisect(self.timestamps[key], timestamp)
 return self.values[key][i - 1] if i > 0 else ''
```

• LeetCode

```
class TimeMap:
 def __init__(self):
 self.ktv = defaultdict(list)

 def set(self, key: str, value: str, timestamp: int) -> None:
 self.ktv[key].append((timestamp, value))

 def get(self, key: str, timestamp: int) -> str:
 if key not in self.ktv:
 return ''
```

Binary Search

p.1979

Sheet 220 / 317 · LeetCode Mastery By Pattern · Python Only · 3-up Portrait

Binary Search

p.1980

```
tv = self.ktv[key]
i = bisect_right(tv, (timestamp, chr(127)))
return tv[i - 1][1] if i else ''
```

# Your TimeMap object will be instantiated and called as such:  
# obj = TimeMap()  
# obj.set(key,value,timestamp)  
# param\_2 = obj.get(key,timestamp)

• SimplyLeet

```
TimeMap class in Python with line-by-line comments.
class TimeMap:
 def __init__(self):
 # Initialize the dictionary to hold key and list
 of (timestamp, value) pairs.
 self.store = {}
```

```
 def set(self, key: str, value: str, timestamp: int) -> None:
 # Append the new (timestamp, value) pair for the
 given key.
 if key not in self.store:
 self.store[key] = []
```

```
self.store[key].append((timestamp, value))

def get(self, key: str, timestamp: int) -> str:
 # Return the value for the largest timestamp <=
 the given timestamp.
 if key not in self.store:
 return ""

 values = self.store[key]
 left, right = 0, len(values) - 1

 # Binary search for the rightmost index with
 timestamp <= given timestamp.
 while left <= right:
 mid = (left + right) // 2
 if values[mid][0] <= timestamp:
 left = mid + 1
 else:
 right = mid - 1

 # If no valid timestamp exists, right will be -1.
 if right == -1:
```

Binary Search

p.1981

```
return ""
return values[right][1]

Example usage:
timeMap = TimeMap()
timeMap.set("foo", "bar", 1)
print(timeMap.get("foo", 1)) # Output: "bar"
print(timeMap.get("foo", 3)) # Output: "bar"
timeMap.set("foo", "bar2", 4)
print(timeMap.get("foo", 4)) # Output: "bar2"

print(timeMap.get("foo", 5)) # Output: "bar2"
```

• LC.ca

```
class TimeMap:
 def __init__(self):
 self.ktv = defaultdict(list)

 def set(self, key: str, value: str, timestamp: int)
 -> None:
 self.ktv[key].append((timestamp, value))

 def get(self, key: str, timestamp: int) -> str:
 if key not in self.ktv:
 return ''
```

Binary Search

p.1982

```
tv = self.ktv[key]
i = bisect_right(tv, (timestamp, chr(127)))
return tv[i - 1][1] if i else ''

Your TimeMap object will be instantiated and called as
such:
obj = TimeMap()
obj.set(key,value,timestamp)
param_2 = obj.get(key,timestamp)
```

Binary Search

p.1983

Binary Search

#1011 Capacity to Ship Packages Within D Days

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Binary Search

Lists: Denny Zhang

Time Complexity:  $O(n * \log(\text{sum}(\text{weights})))$  where  $n$  is the number of packages. Space Complexity:  $O(1)$  additional space.

Problem

A conveyor belt has packages that must be shipped from one port to another within days days.

The  $i$ th package on the conveyor belt has a weight of  $\text{weights}[i]$ . Each day, we load the ship with packages on the conveyor belt (in the order given by  $\text{weights}$ ). We may not load more weight than the maximum weight capacity of the ship.

Return the least weight capacity of the ship that will result in all the packages on the conveyor belt being shipped within days days.

Example 1:

```
Input: weights = [1,2,3,4,5,6,7,8,9,10], days = 5
Output: 15
Explanation: A ship capacity of 15 is the minimum to
ship all the packages in 5 days like this:
1st day: 1, 2, 3, 4, 5
2nd day: 6, 7
3rd day: 8
4th day: 9
5th day: 10
```

Example 2:

```
Input: weights = [3,2,2,4,1,4], days = 3
Output: 6
Explanation: A ship capacity of 6 is the minimum to
ship all the packages in 3 days like this:
1st day: 3, 2
2nd day: 2, 4
3rd day: 1, 4
```

Example 3:

```
Input: weights = [1,2,3,1,1], days = 4
Output: 3
Explanation:
1st day: 1
2nd day: 2
3rd day: 3
4th day: 1, 1
```

Constraints:

- $1 \leq \text{days} \leq \text{weights.length} \leq 5 * 10^4$
- $1 \leq \text{weights}[i] \leq 500$

Binary Search

p.1985

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def shipWithinDays(self, weights, days):
 # Brute Force $O(n)$ - linear scan instead of
 binary search
 for i, val in enumerate(weights):
 if val == target:
 return i
 return -1
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def shipWithinDays(self, weights: List[int], days: int)
 -> int:
 def shipDays(shipCapacity: int) -> int:
 shipDays = 1
 capacity = 0
 for weight in weights:
 if capacity + weight > shipCapacity:
 shipDays += 1
 capacity = weight
 else:
 return shipDays
```

Binary Search

p.1986

```
capacity += weight
return shipDays

l = max(weights)
r = sum(weights)
return bisect.bisect_left(range(l, r), True,
 key=lambda m: shipDays(m)
 <= days) + 1
```

• LeetCode

```
class Solution:
 def shipWithinDays(self, weights: List[int], days:
 int) -> int:
 def check(mx):
 ws, cnt = 0, 1
 for w in weights:
 ws += w
 if ws > mx:
 cnt += 1
 ws = w
 return cnt <= days
```

```
left, right = max(weights), sum(weights) + 1
return left + bisect.bisect_left(range(left, right),
True, key=check)
```

• SimplyLeet

```
def shipWithinDays(weights, days):
 # Helper function to determine if a given capacity
 can ship packages within 'days' days.
 def canShip(capacity):
 current_load = 0 # Current total weight on the
 ship for the current day.
 required_days = 1 # Start with the first day.
 for weight in weights:
 # If adding the current weight exceeds
 capacity, increment the day counter.
 if current_load + weight > capacity:
 required_days += 1
 current_load = 0
```

```
 current_load += weight
 return required_days <= days

Establish lower and upper bounds for binary search.
low, high = max(weights), sum(weights)
while low < high:
 mid = (low + high) // 2
 # If mid capacity can ship all packages within
 the allowed days, try a lower capacity.
 if canShip(mid):
 high = mid
```

Binary Search

p.1988

```
else:
 low = mid + 1
 return low

Example usage:
weights = [1,2,3,4,5,6,7,8,9,10]
days = 5
print(shipWithinDays(weights, days))
```

• LC.ca

```
class Solution:
 def shipWithinDays(self, weights: List[int], days:
 int) -> int:
 def check(mx):
 ws, cnt = 0, 1
 for w in weights:
 ws += w
 if ws > mx:
 cnt += 1
 ws = w
 return cnt <= days
```

```
left, right = max(weights), sum(weights) + 1
return left + bisect.bisect_left(range(left, right),
True, key=check)
```

Binary Search

p.1989

Binary Search

p.1987

Binary Search

#1060 Missing Element in Sorted Array

LeetCode · SimplyLeetCode · WalkCC · LeetCode

Array · Binary Search

Lists: Premium 98

Time Complexity: O(log(n)) Space Complexity: O(1)

Problem

Given an integer array nums which is sorted in ascending order and all of its elements are unique and given also an integer k, return the kth missing number starting from the leftmost number of the array.

Example 1:

Input: nums = [4,7,9,10], k = 1  
Output: 5  
Explanation: The first missing number is 5.

Example 2:

Input: nums = [4,7,9,10], k = 3  
Output: 8  
Explanation: The missing numbers are [5,6,8,...], hence the third missing number is 8.

Example 3:

LC.ca

Binary Search

p.1990

Input: nums = [1,2,4], k = 3  
Output: 6  
Explanation: The missing numbers are [3,5,6,7,...], hence the third missing number is 6.

Constraints:

- 1 <= nums.length <= 5 \* 10<sup>4</sup>
- 1 <= nums[i] <= 10<sup>7</sup>
- nums is sorted in ascending order, and all the elements are unique.
- 1 <= k <= 10<sup>8</sup>

Follow up: Can you find a logarithmic time complexity (i.e., O(log(n))) solution?

Brute Force (Python)

# Brute Force Approach  
class Solution:  
 def missingElement(self, nums, k):  
 # Brute Force O(n) - linear scan instead of binary search  
 for i, val in enumerate(nums):  
 if val == target:  
 return i  
 return -1

Community Solutions (Python)

- LeetCode

Binary Search

p.1991

class Solution:  
 def missingElement(self, nums: List[int], k: int) -> int:  
  
 def missing(i: int) -> int:  
 return nums[i] - nums[0] - i  
  
 n = len(nums)  
 if k > missing(n - 1):  
 return nums[n - 1] + k - missing(n - 1)  
 l, r = 0, n - 1  
 while l < r:  
  
 mid = (l + r) >> 1  
 if missing(mid) >= k:  
 r = mid  
 else:  
 l = mid + 1  
 return nums[l - 1] + k - missing(l - 1)

- SimplyLeetCode

# Python solution with comprehensive comments  
  
def missingElement(nums, k):  
 # Calculate the missing count until index i:  
 missing(i) = nums[i] - nums[0] - i.  
 def missing\_count(index):  
 return nums[index] - nums[0] - index  
  
 n = len(nums)  
  
 # If kth missing number is beyond the last element.

Binary Search

p.1992

if k > missing\_count(n - 1):  
 return nums[-1] + k - missing\_count(n - 1)  
  
# Perform binary search to find the smallest index where missing\_count(index) >= k.  
left, right = 0, n - 1  
while left < right:  
 mid = left + (right - left) // 2  
 if missing\_count(mid) < k:  
 left = mid + 1  
 else:  
  
 right = mid  
  
# After binary search, left is the smallest index such that missing\_count(left) >= k.  
# The kth missing number is after nums[left - 1].  
return nums[left - 1] + k - missing\_count(left - 1)  
  
# Example usage:  
print(missingElement([4,7,9,10], 3)) # Output: 8

LC.ca

Binary Search

p.1993

class Solution:  
 def missingElement(self, nums: List[int], k: int) -> int:  
  
 def missing(i: int) -> int:  
 return nums[i] - nums[0] - i  
  
 n = len(nums)  
 if k > missing(n - 1):  
 return nums[n - 1] + k - missing(n - 1)  
 l, r = 0, n - 1  
 while l < r:  
  
 mid = (l + r) >> 1  
 if missing(mid) >= k:  
 r = mid  
 else:  
 l = mid + 1  
 return nums[l - 1] + k - missing(l - 1)

Binary Search

p.1994

Binary Search

#1062 Longest Repeating Substring

LeetCode · SimplyLeetCode · WalkCC · LeetCode

String · Binary Search · Dynamic Programming · Rolling Hash · Suffix Array · Hash Function

Lists: Denny Zhang

Time Complexity: O(n log n) on average, where n is the length of the string. (Each binary search step takes O(n) time for computing rolling hashes.) Space Complexity: O(n) for storing hash values in t

Problem

Given a string s, return the length of the longest repeating substrings. If no repeating substring exists, return 0.

Example 1:

Input: s = "abcd"  
Output: 0  
Explanation: There is no repeating substring.

Example 2:

Input: s = "abbaba"  
Output: 2  
Explanation: The longest repeating substrings are "ab" and "ba", each of which occurs twice.

Example 3:

Binary Search

p.1995

Input: s = "aabcaabdaab"  
Output: 3  
Explanation: The longest repeating substring is "aab", which occurs 3 times.

Constraints:

- 1 <= s.length <= 2000
- s consists of lowercase English letters.

Brute Force (Python)

# Brute Force Approach  
class Solution:  
 def longestRepeatingSubstring(self, s):  
 # Brute Force O(n) - linear scan instead of binary search  
 for i, val in enumerate(s):  
 if val == target:  
 return i  
 return -1

Community Solutions (Python)

- WalkCC

Binary Search

p.1996

class Solution:  
 def longestRepeatingSubstring(self, s: str) -> int:  
 n = len(s)  
 ans = 0  
 # dp[i][j] := the number of repeating characters of s[0..i] and s[0..j]  
 dp = [[0] \* (n + 1) for \_ in range(n + 1)]  
  
 for i in range(1, n + 1):  
 for j in range(i + 1, n + 1):  
 if s[i - 1] == s[j - 1]:  
  
 dp[i][j] = 1 + dp[i - 1][j - 1]  
 ans = max(ans, dp[i][j])  
  
 return ans

- LeetCode

class Solution:  
 def longestRepeatingSubstring(self, s: str) -> int:  
 n = len(s)  
 f = [[0] \* n for \_ in range(n)]  
 ans = 0  
 for i in range(1, n):  
 for j in range(i):  
 if s[i] == s[j]:  
 f[i][j] = 1 + (f[i - 1][j - 1] if j else 0)  
 else 0  
 ans = max(ans, f[i][j])  
  
 return ans

- SimplyLeetCode

Binary Search

p.1997

def longestRepeatingSubstring(s):  
 n = len(s)  
 # Convert characters to integers for rolling hash computation.  
 nums = [ord(c) - ord('a') for c in s]  
  
 # Base value for rolling hash.  
 a = 26  
 # Modulus value to avoid overflow.  
 mod = 2\*\*32  
  
 # Helper function to check if any substring of given length L repeats.  
 def search(L):  
 h = 0  
 for i in range(L):  
 h = (h \* a + nums[i]) % mod  
 seen = {h}  
 # Precompute a^L % mod for removing the first digit in the rolling hash.  
 aL = pow(a, L, mod)  
 for start in range(1, n - L + 1):  
 h = (h \* a - nums[start - 1] \* aL + nums[start + L - 1]) % mod

Binary Search

p.1998

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><pre>        if h in seen:             return True         seen.add(h)         return False      left, right = 1, n     ans = 0     while left &lt;= right:         mid = left + (right - left) // 2         if search(mid):              ans = mid             left = mid + 1         else:             right = mid - 1     return ans  # Example usage: s = "aabcaabdaab" print(longestRepeatingSubstring(s)) # Expected output: 3</pre></div> <div><div>• LC.ca</div><div><pre>class Solution:     def longestRepeatingSubstring(self, s: str) -&gt; int:         n = len(s)         dp = [[0] * n for _ in range(n)]         ans = 0         for i in range(n):             for j in range(i + 1, n):                 if s[i] == s[j]:                     dp[i][j] = dp[i - 1][j - 1] + 1 if i                 else 1              ans = max(ans, dp[i][j])</pre></div></div> <div>Binary Searchp.1999</div>                                                                                                                                                                                                                                                                                                                                                                      | <div><pre>        return ans</pre></div> <div>Binary Searchp.2000</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <div>Binary SearchMED</div> <div>#1533 Find the Index of the Large Integer</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Array · Binary Search · Interactive</div> <div>Lists: Premium 98</div> <div>Time Complexity: <math>O(\log n)</math> - because binary search is performed over the range. Space Complexity: <math>O(1)</math> - only a few pointers are maintained, no additional space is required.</div> <div>Problem</div> <div>We have an integer array <code>arr</code>, where all the integers in <code>arr</code> are equal except for one integer which is larger than the rest of the integers. You will not be given direct access to the array, instead, you will have an API <code>ArrayReader</code> which have the following functions:</div> <div><ul style="list-style-type: none"><li><code>int compareSub(int l, int r, int x, int y):</code> where <math>0 \leq l, r, x, y &lt; \text{ArrayReader.length()}</math>, <math>l \leq r</math> and <math>x \leq y</math>. The function compares the sum of sub-array <code>arr[l..r]</code> with the sum of the sub-array <code>arr[x..y]</code> and returns: 1 if <code>arr[l]+arr[l+1]+...+arr[r] &gt; arr[x]+arr[x+1]+...+arr[y]</code>.</li><li>0 if <code>arr[l]+arr[l+1]+...+arr[r] == arr[x]+arr[x+1]+...+arr[y]</code>.</li><li>-1 if <code>arr[l]+arr[l+1]+...+arr[r] &lt; arr[x]+arr[x+1]+...+arr[y]</code>.</li></ul></div> <div>Binary Searchp.2001</div> |
| <div>You are allowed to call <code>compareSub()</code> 20 times at most. You can assume both functions work in <math>O(1)</math> time. Return the index of the array <code>arr</code> which has the largest integer.</div> <div>Example 1:</div> <div><pre>Input: arr = [7,7,7,7,10,7,7,7] Output: 4 Explanation: The following calls to the API reader.compareSub(0, 0, 1, 1) // returns 0 this is a query comparing the sub-array (0, 0) with the sub array (1, 1), (i.e. compares arr[0] with arr[1]). Thus we know that arr[0] and arr[1] doesn't contain the largest element. reader.compareSub(2, 2, 3, 3) // returns 0, we can exclude arr[2] and arr[3]. reader.compareSub(4, 4, 5, 5) // returns 1, thus for sure arr[4] is the largest element in the array. Notice that we made only 3 calls, so the answer is valid.</pre></div> <div>Example 2:</div> <div><pre>Input: nums = [6,6,12] Output: 2</pre></div> <div>Constraints:</div> <div><ul style="list-style-type: none"><li><math>2 \leq \text{arr.length} \leq 5 * 10^5</math></li><li><math>1 \leq \text{arr}[i] \leq 100</math></li><li>All elements of <code>arr</code> are equal except for one element which is larger than all other elements.</li></ul></div> <div>Follow up:</div> <div>Binary Searchp.2002</div> | <div><ul style="list-style-type: none"><li>What if there are two numbers in <code>arr</code> that are bigger than all other numbers?</li><li>What if there is one number that is bigger than other numbers and one number that is smaller than other numbers?</li></ul></div> <div>Brute Force (Python)</div> <div><pre># Brute Force Approach class Solution:     def getIndex(self, reader):         # Brute Force O(n) - linear scan instead of         binary search         for i, val in enumerate(reader):             if val == target:                 return i         return -1</pre></div> <div>Community Solutions (Python)</div> <div><ul style="list-style-type: none"><li>WalkCC</li></ul></div> <div>Binary Searchp.2003</div>                                            | <div><pre># """ # This is ArrayReader's API interface. # You should not implement it, or speculate about its implementation # """ # class ArrayReader(object): #     # Compares the sum of arr[l..r] with the sum of arr[x..y] #     # return 1 if sum(arr[l..r]) &gt; sum(arr[x..y]) #     # return 0 if sum(arr[l..r]) == sum(arr[x..y]) #     # return -1 if sum(arr[l..r]) &lt; sum(arr[x..y]) #     def compareSub(self, l: int, r: int, x: int, y: int) -&gt; int:</pre></div> <div><pre># # # Returns the length of the array # def length(self) -&gt; int: #</pre></div> <div><pre>class Solution:     def getIndex(self, reader: 'ArrayReader') -&gt; int:         l = 0         r = reader.length() - 1</pre></div> <div>Binary Searchp.2004</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <div><pre>while l &lt; r:     m = (l + r) // 2     if (r - l) % 2 == 0:         res = reader.compareSub(l, m - 1, m + 1, r)         if res == 0:             return m         if res == 1:             r = m - 1         else: # res == -1              l = m + 1     else:         res = reader.compareSub(l, m, m + 1, r)         # res is either 1 or -1.         if res == 1:             r = m         else: # res == -1             l = m + 1      return l</pre></div> <div><ul style="list-style-type: none"><li>LeetDoocs</li></ul></div> <div>Binary Searchp.2005</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <div><pre># """ # This is ArrayReader's API interface. # You should not implement it, or speculate about its implementation # """ # class ArrayReader(object): #     # Compares the sum of arr[l..r] with the sum of arr[x..y] #     # return 1 if sum(arr[l..r]) &gt; sum(arr[x..y]) #     # return 0 if sum(arr[l..r]) == sum(arr[x..y]) #     # return -1 if sum(arr[l..r]) &lt; sum(arr[x..y]) #     def compareSub(self, l: int, r: int, x: int, y: int) -&gt; int:</pre></div> <div><pre># # # Returns the length of the array # def length(self) -&gt; int: #</pre></div> <div><pre>class Solution:     def getIndex(self, reader: 'ArrayReader') -&gt; int:         left, right = 0, reader.length() - 1         while left &lt; right:</pre></div> <div>Binary Searchp.2006</div> | <div><pre>t1, t2, t3 = (     left,     left + (right - left) // 3,     left + ((right - left) // 3) * 2 + 1, ) cmp = reader.compareSub(t1, t2, t2 + 1, t3) if cmp == 0:     left = t3 + 1 elif cmp == 1:     right = t2  else:     left, right = t2 + 1, t3 return left</pre></div> <div><ul style="list-style-type: none"><li>SimplyLeet</li></ul></div> <div><pre># Python solution with line-by-line comments class Solution:     def getIndex(self, reader):         # Initialize search bounds         left, right = 0, reader.length() - 1          # Binary search for the candidate index         while left &lt; right:             length = right - left + 1             if length % 2 == 0:</pre></div> <div>Binary Searchp.2007</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre># Even length: split into two equal halves.  mid = left + length // 2 # Compare left half [left, mid-1] with right half [mid, right] comp = reader.compareSub(left, mid - 1, mid, right)  if comp == 1:     # Large integer is in the left half     right = mid - 1 else:     # Large integer is in the right half     left = mid  else:     # Odd length: divide into two halves of equal size, ignoring middle element.     half = length // 2     # Compare left half [left, left+half-1] with right half [right-half+1, right]     comp = reader.compareSub(left, left + half - 1, right - half + 1, right)     if comp == 1:         # Large integer in left half         right = left + half - 1     elif comp == -1:         # Large integer in right half</pre>                                                                                                                              |        | <pre>left = right - half + 1 else:     # Halves are equal, so the middle element is the large one.     return left + half return left</pre> <p>• LC.ca</p> <pre># "" # This is ArrayReader's API interface. # You should not implement it, or speculate about its implementation # "" # class ArrayReader(object): # # Compares the sum of arr[l..r] with the sum of arr[x..y] # # return 1 if sum(arr[l..r]) &gt; sum(arr[x..y]) # # return 0 if sum(arr[l..r]) == sum(arr[x..y]) # # return -1 if sum(arr[l..r]) &lt; sum(arr[x..y]) # def compareSub(self, l: int, r: int, x: int, y: int) -&gt; int:  # # # Returns the length of the array # def length(self) -&gt; int: #  class Solution:     def getIndex(self, reader: 'ArrayReader') -&gt; int:         left, right = 0, reader.length() - 1         while left &lt; right:</pre>                                                                                                                                                                                                                         |        | <pre>t1, t2, t3 = (     left,     left + (right - left) // 3,     left + ((right - left) // 3) * 2 + 1, ) cmp = reader.compareSub(t1, t2, t2 + 1, t3) if cmp == 0:     left = t3 + 1 elif cmp == 1:     right = t2  else:     left, right = t2 + 1, t3 return left</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | p.2008 | Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | p.2009 | Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | p.2010 |
| <div>Binary Search</div> <div>#4 Median of Two Sorted Arrays</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>Array · Binary Search · Divide and Conquer</div> <div>Lists: Grind 169</div> <div>Time Complexity: O(log(min(m, n))) Space Complexity: O(1)</div> <div>Problem</div> <div>Given two sorted arrays nums1 and nums2 of size m and n respectively, return the median of the two sorted arrays.</div> <div>The overall run time complexity should be O(log (m+n)).</div> <div>Example 1:</div> <div>Input: nums1 = [1,3], nums2 = [2]</div> <div>Output: 2.00000</div> <div>Explanation: merged array = [1,2,3] and median is 2.</div> <div>Example 2:</div> <div>Input: nums1 = [1,2], nums2 = [3,4]</div> <div>Output: 2.50000</div> <div>Explanation: merged array = [1,2,3,4] and median is (2 + 3) / 2 = 2.5.</div> <div>Constraints:</div> <div>• nums1.length == m</div> | HAR    | <p>• nums2.length == n</p> <p>• 0 &lt;= m &lt;= 1000</p> <p>• 0 &lt;= n &lt;= 1000</p> <p>• 1 &lt;= m + n &lt;= 2000</p> <p>• -106 &lt;= nums1[i], nums2[i] &lt;= 106</p> <p>Brute Force (Python)</p> <pre># Brute Force Approach class Solution:     def findMedianSortedArrays(self, nums1, nums2):         # Brute Force O((m+n) log(m+n)) - merge both arrays, sort, find median         merged = sorted(nums1 + nums2)         n = len(merged)         if n % 2 == 1:             return float(merged[n // 2])         return (merged[n//2 - 1] + merged[n//2]) / 2.0</pre> <p>Community Solutions (Python)</p> <p>• WalkCC</p> <pre>class Solution:     def findMedianSortedArrays(self, nums1: List[int], nums2: List[int]) -&gt; float:         n1 = len(nums1)         n2 = len(nums2)         if n1 &gt; n2:             return self.findMedianSortedArrays(nums2, nums1)          l = 0         r = n1</pre>                                                                                                                                             |        | <pre>while l &lt;= r:     partition1 = (l + r) // 2     partition2 = (n1 + n2 + 1) // 2 - partition1     maxLeft1 = -2**31 if partition1 == 0 else nums1[partition1 - 1]     maxLeft2 = -2**31 if partition2 == 0 else nums2[partition2 - 1]     minRight1 = 2**31 - 1 if partition1 == n1 else nums1[partition1]     minRight2 = 2**31 - 1 if partition2 == n2 else nums2[partition2]     if maxLeft1 &lt;= minRight2 and maxLeft2 &lt;= minRight1:         return (max(maxLeft1, maxLeft2) + min(minRight1, minRight2)) * 0.5 if (n1 + n2) % 2 == 0 else max(maxLeft1, maxLeft2)     elif maxLeft1 &gt; minRight2:         r = partition1 - 1     else:         l = partition1 + 1</pre> <p>• LeetDoocs</p> <pre>class Solution:     def findMedianSortedArrays(self, nums1: List[int], nums2: List[int]) -&gt; float:         def f(i: int, j: int, k: int) -&gt; int:             if i &gt;= m:                 return nums2[j + k - 1]             if j &gt;= n:                 return nums1[i + k - 1]             if k == 1:                 return min(nums1[i], nums2[j])             p = k // 2</pre> |        |
| Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | p.2011 | Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | p.2012 | Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | p.2013 |
| <pre>x = nums1[i + p - 1] if i + p - 1 &lt; m else inf y = nums2[j + p - 1] if j + p - 1 &lt; n else inf return f(i + p, j, k - p) if x &lt; y else f(i, j + p, k - p)</pre> <p>m, n = len(nums1), len(nums2)</p> <p>a = f(0, 0, (m + n + 1) // 2)</p> <p>b = f(0, 0, (m + n + 2) // 2)</p> <p>return (a + b) / 2</p> <p>• SimplyLeet</p> <pre># Function to find the median of two sorted arrays def findMedianSortedArrays(nums1, nums2):     # Ensure nums1 is the smaller array for faster binary search     if len(nums1) &gt; len(nums2):         nums1, nums2 = nums2, nums1      m, n = len(nums1), len(nums2)     low, high = 0, m      while low &lt;= high:</pre>                                                                                                                                                                                                                              |        | <pre># Partition indexes for nums1 and nums2 partition_nums1 = (low + high) // 2 partition_nums2 = (m + n + 1) // 2 - partition_nums1  # If partition is 0 then there are no elements on left side. Use -infinity for comparison. maxLeftNums1 = float('-inf') if partition_nums1 == 0 else nums1[partition_nums1 - 1] # If partition is length then there are no elements on right side. Use infinity for comparison. minRightNums1 = float('inf') if partition_nums1 == m else nums1[partition_nums1]  # Similarly for nums2  maxLeftNums2 = float('-inf') if partition_nums2 == 0 else nums2[partition_nums2 - 1] minRightNums2 = float('inf') if partition_nums2 == n else nums2[partition_nums2]  # Check if we have found the correct partition if maxLeftNums1 &lt;= minRightNums2 and maxLeftNums2 &lt;= minRightNums1:     # If total number of elements is even     if (m + n) % 2 == 0:         return (max(maxLeftNums1, maxLeftNums2) + min(minRightNums1, minRightNums2)) / 2.0     else:         return float(max(maxLeftNums1, maxLeftNums2))</pre> |        | <pre># If not, adjust binary search range elif maxLeftNums1 &gt; minRightNums2:     # Move partition in nums1 to left     high = partition_nums1 - 1 else:     # Move partition in nums1 to right     low = partition_nums1 + 1  # Example usage: print(findMedianSortedArrays([1,3], [2])) # Output: 2.0  print(findMedianSortedArrays([1,2], [3,4])) # Output: 2.5</pre> <p>• LC.ca</p> <pre>class Solution:     def findMedianSortedArrays(self, nums1: List[int], nums2: List[int]) -&gt; float:          def findKth(i: int, j: int, k: int) -&gt; float:             if i &gt;= m:                 return nums2[j + k - 1]             if j &gt;= n:                 return nums1[i + k - 1]             if k == 1:                 return min(nums1[i], nums2[j])</pre>                                                                                                                                                                                                                                                                                                                                   |        |
| Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | p.2014 | Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | p.2015 | Binary Search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | p.2016 |



```
Division a // b : floordiv(a, b)
midVal1 = nums1[i + k // 2 - 1] if i + k // 2
- 1 < m else math.inf
midVal2 = nums2[j + k // 2 - 1] if j + k // 2
- 1 < n else math.inf

if midVal1 < midVal2:
 return findKth(i + k // 2, j, k - k // 2)
'+' or '-' k//2
else:
 return findKth(i, j + k // 2, k - k // 2)

m = len(nums1)
n = len(nums2)
Division a // b : floordiv(a, b)
left = (m + n + 1) // 2
right = (m + n + 2) // 2
return (findKth(0, 0, left) + findKth(0, 0,
right)) / 2.0

#####

iteration version
```

```
class Solution(object):
 def findMedianSortedArrays(self, nums1, nums2):
 a, b = sorted((nums1, nums2), key=len)
 m, n = len(a), len(b)
 after = (m + n - 1) / 2
 lo, hi = 0, m
 while lo < hi:
 i = (lo + hi) / 2
 if after - i - 1 < 0 or a[i] >= b[after - i - 1]:
 hi = i

 else:
 lo = i + 1
 i = lo
 nextfew = sorted(a[i:i + 2] + b[after - i:after - i +
2])
 return (nextfew[0] + nextfew[1 - (m + n) % 2]) / 2.0
```

Binary Search

#315 Count of Smaller Numbers After Self

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Binary Search · Divide and Conquer · Binary Indexed Tree · Segment Tree · Merge Sort · Ordered Set

Lists: Denny Zhang

Time Complexity: O(n log n) on average due to the merge sort based approach. Space Complexity: O(n) for auxiliary arrays used during sorting and counting.

Problem

Given an integer array nums, return an integer array counts where counts[i] is the number of smaller elements to the right of nums[i].

Example 1:

Input: nums = [5,2,6,1]  
Output: [2,1,1,0]  
Explanation:  
To the right of 5 there are 2 smaller elements (2 and 1).  
To the right of 2 there is only 1 smaller element (1).  
To the right of 6 there is 1 smaller element (1).  
To the right of 1 there is 0 smaller element.

Example 2:

Input: nums = [-1]  
Output: [0]

Example 3:

Input: nums = [-1,-1]  
Output: [0,0]

Constraints:

- 1 <= nums.length <= 105
- 104 <= nums[i] <= 104

Brute Force (Python)

# Brute Force Approach  
class Solution:  
 def countSmaller(self, nums):  
 # Brute Force O(n) - linear scan instead of binary search  
 for i, val in enumerate(nums):  
 if val == target:  
 return i  
 return -1

```
class FenwickTree:
 def __init__(self, n: int):
 self.sums = [0] * (n + 1)

 def add(self, i: int, delta: int) -> None:
 while i < len(self.sums):
 self.sums[i] += delta
 i += FenwickTree.lowbit(i)

 def get(self, i: int) -> int:

 summ = 0
 while i > 0:
 summ += self.sums[i]
 i -= FenwickTree.lowbit(i)
 return summ

 @staticmethod
 def lowbit(i: int) -> int:
 return i & -i

class Solution:
 def countSmaller(self, nums: list[int]) -> list[int]:
 ans = []
 ranks = self._getRanks(nums)
 tree = FenwickTree(len(ranks))

 for num in reversed(nums):
 ans.append(tree.get(ranks[num] - 1))
 tree.add(ranks[num], 1)
```

```
return ans[::-1]

def _getRanks(self, nums: list[int]) -> dict[int, int]:
 ranks = collections.Counter()
 rank = 0
 for num in sorted(set(nums)):
 rank += 1
 ranks[num] = rank
 return ranks

• LeetDoocs

class BinaryIndexedTree:
 def __init__(self, n):
 self.n = n
 self.c = [0] * (n + 1)

 @staticmethod
 def lowbit(x):
 return x & -x

 def update(self, x, delta):

 while x <= self.n:
 self.c[x] += delta
 x += BinaryIndexedTree.lowbit(x)

 def query(self, x):
 s = 0
 while x > 0:
 s += self.c[x]
 x -= BinaryIndexedTree.lowbit(x)
 return s
```

class Solution:

def countSmaller(self, nums: List[int]) -> List[int]:

all = sorted(set(nums))

m = {v: i for i, v in enumerate(all, 1)}

tree = BinaryIndexedTree(len(m))

ans = []

for v in nums[::-1]:

x = m[v]

tree.update(x, 1)

ans.append(tree.query(x - 1))

return ans[::-1]

• SimplyLeet

# Python solution using merge sort approach

def countSmaller(nums):

n = len(nums)

# Initialize counts array for results

counts = [0] \* n

# Pair each element with its original index

enum = [(num, i) for i, num in enumerate(nums)]

# Recursive function for merge sort with counting

```
def mergeSort(arr):
 if len(arr) <= 1:
 return arr
 # Split the array into two halves
 mid = len(arr) // 2
 left = mergeSort(arr[:mid])
 right = mergeSort(arr[mid:])
 merged = []
 l = r = 0
 # r_count counts number of elements taken from right array

 r_count = 0
 while l < len(left) and r < len(right):
 # If left element is less or equal, it means current right elements are smaller
 if left[l][0] <= right[r][0]:
 # update count for original index
 counts[left[l][1]] += r_count
 merged.append(left[l])
 l += 1
 else:
 # right element is smaller, so increment r_count and add it to merged
```

```
merged.append(right[r])
r += 1
r_count += 1
Append remaining elements from left reduce count by accumulated right elements
while l < len(left):
 counts[left[l][1]] += r_count
 merged.append(left[l])
 l += 1
Append any remaining elements from right side
while r < len(right):

 merged.append(right[r])
 r += 1
 return merged

Start the merge sort process
mergeSort(enum)
return counts

Example test
print(countSmaller([5, 2, 6, 1])) # Expected output: [2, 1, 1, 0]
```

• LC.ca

|                                                                                                                               |                                                                                                                                   |                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <div data-bbox="43 44 496 489"></div> <div data-bbox="27 663 513 682"> <div>Binary Search</div> <div>p.2026</div> </div>      | <div data-bbox="583 44 1037 611"></div> <div data-bbox="566 663 1052 682"> <div>Binary Search</div> <div>p.2027</div> </div>      | <div data-bbox="1123 44 1578 594"></div> <div data-bbox="1107 663 1593 682"> <div>Binary Search</div> <div>p.2028</div> </div>      |
| <div data-bbox="43 745 496 1022"></div> <div data-bbox="27 1362 513 1381"> <div>Binary Search</div> <div>p.2029</div> </div>  | <div data-bbox="583 745 1037 1304"></div> <div data-bbox="566 1362 1052 1381"> <div>Binary Search</div> <div>p.2030</div> </div>  | <div data-bbox="1123 745 1578 1186"></div> <div data-bbox="1107 1194 1593 1381"> <div>Binary Search</div> <div>p.2031</div> </div>  |
| <div data-bbox="43 1444 496 1976"></div> <div data-bbox="27 2064 513 2083"> <div>Binary Search</div> <div>p.2032</div> </div> | <div data-bbox="583 1444 1037 1976"></div> <div data-bbox="566 2064 1052 2083"> <div>Binary Search</div> <div>p.2033</div> </div> | <div data-bbox="1123 1444 1578 1869"></div> <div data-bbox="1107 2064 1593 2083"> <div>Binary Search</div> <div>p.2034</div> </div> |

```
Python solution for Split Array Largest Sum

def splitArray(nums, k):
 # Helper function to determine if we can split array
 into <= k subarrays with max subarray sum <= max_sum
 def canSplit(max_sum):
 subarray_count = 1 # start with one subarray
 current_sum = 0
 for num in nums:
 if current_sum + num > max_sum:
 subarray_count += 1 # start a new
 subarray
 current_sum = num # reset the sum with
 the current number
 else:
 current_sum += num # add num to current
 subarray
 return subarray_count <= k # valid if we used k
 or fewer subarrays

 low, high = max(nums), sum(nums) # boundaries for
 binary search
 while low < high:
 mid = (low + high) // 2 # candidate maximum
 subarray sum
 if canSplit(mid):
 high = mid # candidate is valid, try to find
 a smaller one
```

Binary Search

p.2035

```
 else:
 low = mid + 1 # candidate too small, increase
 it
 return low

Example usage:
nums = [7,2,5,10,8]
k = 2
print(splitArray(nums, k)) # Expected output: 18

• LC.ca

class Solution:
 def splitArray(self, nums: List[int], k: int) -> int:
 def check(mx):
 s, cnt = inf, 0
 for x in nums:
 s += x
 if s > mx:
 s = x
 cnt += 1
 return cnt <= k

 left, right = max(nums), sum(nums)
 return left + bisect_left(range(left, right + 1),
 True, key=check)
```

Binary Search

p.2036

Binary Search

## #668 Kth Smallest Number in Multiplication Table

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Math · Binary Search

Lists: Denny Zhang

Time Complexity:  $O(m * \log(m*n))$  Space Complexity:  $O(1)$

### Problem

Nearly everyone has used the Multiplication Table. The multiplication table of size  $m \times n$  is an integer matrix mat where  $mat[i][j] == i * j$  (1-indexed).

Given three integers  $m$ ,  $n$ , and  $k$ , return the  $k$ th smallest element in the  $m \times n$  multiplication table.

Example 1:

Binary Search

p.2037

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 2 | 4 | 6 |
| 3 | 6 | 9 |

|   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 2 | 3 | 3 | 4 | 6 | 6 | 9 |
|---|---|---|---|---|---|---|---|---|

Input:  $m = 3, n = 3, k = 5$   
Output: 3  
Explanation: The 5th smallest number is 3.

Example 2:

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 2 | 4 | 6 |

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 2 | 2 | 3 | 4 | 6 |
|---|---|---|---|---|---|

Binary Search

p.2038

Input:  $m = 2, n = 3, k = 6$   
Output: 6  
Explanation: The 6th smallest number is 6.

### Constraints:

- $1 \leq m, n \leq 3 * 10^4$
- $1 \leq k \leq m * n$

### Community Solutions (Python)

#### • LeetDoocs

```
class Solution:
 def findKthNumber(self, m: int, n: int, k: int) -> int:
 left, right = 1, m * n
 while left < right:
 mid = (left + right) >> 1
 cnt = 0
 for i in range(1, m + 1):
 cnt += min(mid // i, n)
 if cnt >= k:
 right = mid
 else:
 left = mid + 1
 return left
```

#### • SimplyLeet

Binary Search

p.2039

```
Python solution using binary search

class Solution:
 def findKthNumber(self, m: int, n: int, k: int) -> int:
 def count_less_equal(x: int) -> int:
 # Count numbers in the m x n multiplication
 table <= x
 count = 0
 for i in range(1, m + 1):
 # For each row, count of numbers <= x is
 minimum of n or x // i
 count += min(x // i, n)

 return count

 # Binary search range from 1 to m*n
 low, high = 1, m * n
 while low < high:
 mid = (low + high) // 2
 if count_less_equal(mid) >= k:
 high = mid # mid might be the answer
 else:
 low = mid + 1 # answer must be greater
 than mid

 return low
```

#### • LC.ca

Binary Search

p.2040

```
class Solution:
 def findKthNumber(self, m: int, n: int, k: int) -> int:
 left, right = 1, m * n
 while left < right:
 mid = (left + right) >> 1
 cnt = 0
 for i in range(1, m + 1):
 cnt += min(mid // i, n)
 if cnt >= k:
 right = mid
 else:
 left = mid + 1
 return left
```

Binary Search

p.2041

Binary Search

## #710 Random Pick with Blacklist

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Math · Binary Search · Sorting · Randomized

Lists: Denny Zhang

Time Complexity: constructor –  $O(b)$  for initializing data structures,  $pick()$  –  $O(1)$  per call. Space Complexity:  $O(b)$  for storing the blacklist set and mapping.

### Problem

You are given an integer  $n$  and an array of unique integers `blacklist`. Design an algorithm to pick a random integer in the range  $[0, n - 1]$  that is not in `blacklist`. Any integer that is in the mentioned range and not in `blacklist` should be equally likely to be returned.

Optimize your algorithm such that it minimizes the number of calls to the built-in random function of your language.

Implement the Solution class:

- `Solution(int n, int[] blacklist)` Initializes the object with the integer  $n$  and the blacklisted integers `blacklist`.
- `int pick()` Returns a random integer in the range  $[0, n - 1]$  and not in `blacklist`.

Example 1:

Binary Search

p.2042

Input  
[["Solution", "pick", "pick", "pick", "pick", "pick", "pick", "pick"]  
[[7, [2, 3, 5]], [], [], [], [], [], [], []]  
Output  
[null, 0, 4, 1, 6, 1, 0, 4]

Explanation  
Solution solution = new Solution(7, [2, 3, 5]);

### Constraints:

- $1 \leq n \leq 109$
- $0 \leq \text{blacklist.length} \leq \min(105, n - 1)$
- $0 \leq \text{blacklist}[i] < n$
- All the values of `blacklist` are unique.
- At most  $2 * 10^4$  calls will be made to `pick`.

### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def __init__(self, n: int, blacklist: list):
 # Brute Force $O(n)$ – linear scan instead of
 binary search
 for i, val in enumerate(blacklist):
 if val == target:
 return i
 return -1
```

### Community Solutions (Python)

#### • WalkCC

Binary Search

p.2043

```
class Solution:
 def __init__(self, n: int, blacklist: list[int]):
 self.validRange = n - len(blacklist)
 self.dict = {}

 maxAvailable = n - 1

 for b in blacklist:
 self.dict[b] = -1

 for b in blacklist:
 if b < self.validRange:
 # Find the slot that haven't been used.
 while maxAvailable in self.dict:
 maxAvailable -= 1
 self.dict[b] = maxAvailable
 maxAvailable -= 1

 def pick(self) -> int:
 value = random.randint(0, self.validRange - 1)

 if value in self.dict:
 return self.dict[value]

 return value
```

• LeetCode

Binary Search

p.2044

```
class Solution:
 def __init__(self, n: int, blacklist: List[int]):
 self.k = n - len(blacklist)
 self.d = {}
 i = self.k
 black = set(blacklist)
 for b in blacklist:
 if b < self.k:
 while i in black:
 i += 1

 self.d[b] = i
 i += 1

 def pick(self) -> int:
 x = randrange(self.k)
 return self.d.get(x, x)

 # Your Solution object will be instantiated and called as
 # such:
 # obj = Solution(n, blacklist)

 # param_1 = obj.pick()
```

• SimplyLeet

Binary Search

p.2045

```
import random

class Solution:
 def __init__(self, n: int, blacklist: list):
 # Total numbers valid for picking
 self.W = n - len(blacklist)
 # Mapping for blacklisted numbers in the valid
 range

 self.mapping = {}
 # Set to hold blacklisted numbers that are >= W
 blacklist_set = set(blacklist)

 # Pointer to find available numbers in [W, n-1]
 available = self.W
 for b in blacklist:
 if b < self.W:
 # Find next available non-blacklisted
 number in the tail region.
 while available in blacklist_set:
 available += 1
 self.mapping[b] = available
 available += 1

 def pick(self) -> int:
 # Generate a random number in the range [0,
 self.W-1]
 idx = random.randint(0, self.W - 1)
 # If the number is mapped, return the mapping;
 else it is valid.
 return self.mapping.get(idx, idx)
```

• LC.ca

Binary Search

p.2046

```
class Solution:
 def __init__(self, n: int, blacklist: List[int]):
 self.k = n - len(blacklist)
 self.d = {}
 i = self.k
 black = set(blacklist)
 for b in blacklist:
 if b < self.k:
 while i in black:
 i += 1

 self.d[b] = i
 i += 1

 def pick(self) -> int:
 x = randrange(self.k)
 return self.d.get(x, x)

 # Your Solution object will be instantiated and called as
 # such:
 # obj = Solution(n, blacklist)

 # param_1 = obj.pick()
```

Binary Search

p.2047

Binary Search

## #774 Minimize Max Distance to Gas Station

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Binary Search

Lists: Denny Zhang

Time Complexity:  $O(n * \log(\text{max\_gap} / \text{precision}))$ , where  $n$  is the number of stations. Space Complexity:  $O(1)$  (ignoring input storage)

### Problem

You are given an integer array `stations` that represents the positions of the gas stations on the  $x$ -axis. You are also given an integer  $k$ .

You should add  $k$  new gas stations. You can add the stations anywhere on the  $x$ -axis, and not necessarily on an integer position.

Let `penalty()` be the maximum distance between adjacent gas stations after adding the  $k$  new stations.

Return the smallest possible value of `penalty()`. Answers within  $10^{-6}$  of the actual answer will be accepted.

Example 1:

```
Input: stations = [1,2,3,4,5,6,7,8,9,10], k = 9
Output: 0.50000
```

Example 2:

Binary Search

p.2048

```
Input: stations = [23,24,36,39,46,56,57,65,84,98], k = 1
Output: 14.00000
```

Constraints:

- $10 \leq \text{stations.length} \leq 2000$
- $0 \leq \text{stations}[i] \leq 108$
- stations is sorted in a strictly increasing order.
- $1 \leq k \leq 106$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def minmaxGasDist(self, stations: list, k: int):
 # Brute Force $O(n)$ - linear scan instead of
 binary search
 for i, val in enumerate(stations):
 if val == target:
 return i
 return -1
```

Community Solutions (Python)

• WalkCC

Binary Search

p.2049

```
class Solution:
 def minmaxGasDist(self, stations: list[int], k: int) -> float:
 ERR = 1e-6
 l = 0
 r = stations[-1] - stations[0]

 def possible(k: int, m: float) -> bool:
 """
 Returns True if can use <= k gas stations to ensure
 that each adjacent
 distance between gas stations <= m.

 """
 for a, b in zip(stations, stations[1:]):
 diff = b - a
 if diff > m:
 k -= math.ceil(diff / m) - 1
 if k < 0:
 return False
 return True

 while r - l > ERR:
 m = (l + r) / 2
 if possible(k, m):
 r = m
 else:
 l = m

 return l
```

• LeetCode

Binary Search

p.2050

```
class Solution:
 def minmaxGasDist(self, stations: List[int], k: int) -> float:
 def check(x):
 return sum(int((b - a) / x) for a, b in
 pairwise(stations)) <= k

 left, right = 0, 1e8
 while right - left > 1e-6:
 mid = (left + right) / 2
 if check(mid):
 right = mid
 else:
 left = mid
 return left
```

• SimplyLeet

```
import math

def minmaxGasDist(stations, k):
 # Initialize the search range between 0 and the
 maximum gap between existing stations
 low = 0.0
 high = max(stations[i+1] - stations[i] for i in
 range(len(stations)-1))

 # Use binary search with a tolerance of 1e-6
 while high - low > 1e-6:
 mid = (low + high) / 2.0
```

Binary Search

p.2051

```
used = 0
Calculate additional stations needed for each
gap

for i in range(1, len(stations)):
 gap = stations[i] - stations[i-1]
 used += math.ceil(gap / mid) - 1
If the used stations are less than or equal to
k, try a smaller distance
if used <= k:
 high = mid
else:
 low = mid

return high

Example usage:
if __name__ == '__main__':
 print("{:.5f}".format(minmaxGasDist([1,2,3,4,5,6,7,8,
 9,10], 9)))
 print("{:.5f}".format(minmaxGasDist([23,24,36,39,46,5
 6,57,65,84,98], 1)))
```

• LC.ca

Binary Search

p.2052

```
class Solution:
 def minmaxGasDist(self, stations: List[int], k: int)
-> float:
 def check(x):
 return sum(int((b - a) / x) for a, b in
pairwise(stations)) <= k

 left, right = 0, 1e8
 while right - left > 1e-6:
 mid = (left + right) / 2
 if check(mid):
 right = mid

 else:
 left = mid
 return left
```

Binary Search

#1235 Maximum Profit in Job Scheduling HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Binary Search · Dynamic Programming · Sorting

Lists: Grind 169

Time Complexity: O(n log n) due to sorting and binary search for each job. Space Complexity: O(n) for the dp arrays.

Problem

We have n jobs, where every job is scheduled to be done from startTime[i] to endTime[i], obtaining a profit of profit[i].

You're given the startTime, endTime and profit arrays, return the maximum profit you can take such that there are no two jobs in the subset with overlapping time range.

If you choose a job that ends at time X you will be able to start another job that starts at time X.

Example 1:

profit = 40

profit = 10

profit = 50

profit = 70

1 2 3 4 5 6

Input: startTime = [1,2,3,3], endTime = [3,4,5,6], profit = [50,10,40,70]

Output: 120

Explanation: The subset chosen is the first and fourth job.

Time range [1-3]+[3-6] , we get profit of 120 = 50 + 70.

Example 2:

Example 3:

profit = 4

profit = 6

profit = 5

time 1 2 3 4

Input: startTime = [1,1,1], endTime = [2,3,4], profit = [5,6,4]

Output: 6

Constraints:

- 1 <= startTime.length == endTime.length == profit.length <= 5 \* 10<sup>4</sup>
- 1 <= startTime[i] < endTime[i] <= 10<sup>9</sup>
- 1 <= profit[i] <= 10<sup>4</sup>

Brute Force (Python)

# Brute Force Approach

class Solution:

def jobScheduling(self, startTime, endTime, profit):

# Brute Force O(n) - linear scan instead of binary search

for i, val in enumerate(startTime):

if val == target:

return i

return -1

Community Solutions (Python)

• WalkCC

class Solution:

def jobScheduling(self, startTime: List[int], endTime: List[int], profit: List[int]) -> int:

@cache

def dfs(i):

if i >= n:

return 0

\_, e, p = jobs[i]

j = bisect\_left(jobs, e, lo=i + 1, key=lambda x: x[0])

return max(dfs(i + 1), p + dfs(j))

jobs = sorted(zip(startTime, endTime, profit))

n = len(jobs)

return dfs(0)

• LeetCode

class Solution:

def jobScheduling(self, startTime: List[int], endTime: List[int], profit: List[int]) -> int:

@cache

def dfs(i):

if i >= n:

return 0

\_, e, p = jobs[i]

j = bisect\_left(jobs, e, lo=i + 1, key=lambda x: x[0])

return max(dfs(i + 1), p + dfs(j))

jobs = sorted(zip(startTime, endTime, profit))

n = len(jobs)

return dfs(0)

• SimplyLeet

import bisect

def jobScheduling(startTime, endTime, profit):

# Combine the jobs and sort them by end time.

jobs = sorted(zip(startTime, endTime, profit), key=lambda x: x[1])

# dp\_end\_times stores the end times and dp\_profits stores the corresponding maximum profit.

dp\_end\_times = [0]

dp\_profits = [0]

# Process each job in order.

for s, e, p in jobs:

# Find the index of the last job that finishes before or at the start of the current job using binary search.

index = bisect.bisect\_right(dp\_end\_times, s)

current\_profit = dp\_profits[index - 1] + p

Profit if we take current job.

# If taking current job increases the overall profit, update the dp arrays.

if current\_profit > dp\_profits[-1]:

dp\_end\_times.append(e)

dp\_profits.append(current\_profit)

return dp\_profits[-1]

# Example usage:

print(jobScheduling([1,2,3,3], [3,4,5,6], [50,10,40,70]))

• LC.ca

class Solution:

def jobScheduling(self, startTime: List[int], endTime: List[int], profit: List[int]) -> int:

@cache

def dfs(i):

if i >= n:

return 0

\_, e, p = jobs[i]

j = bisect\_left(jobs, e, lo=i + 1, key=lambda x: x[0])

return max(dfs(i + 1), p + dfs(j))

jobs = sorted(zip(startTime, endTime, profit))

n = len(jobs)

return dfs(0)

class Solution:

def jobScheduling(self, startTime: List[int], endTime: List[int], profit: List[int]) -> int:

@cache

def dfs(i):

if i >= n:

return 0

\_, e, p = jobs[i]

j = bisect\_left(jobs, e, lo=i + 1, key=lambda x: x[0])

return max(dfs(i + 1), p + dfs(j))

jobs = sorted(zip(startTime, endTime, profit))

n = len(jobs)

return dfs(0)

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| <div>&gt;&gt; Quick Review — Binary Search</div> <div>24 questions · insights · complexity · solution</div> <div>#35 Search Insert Position [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>The array is sorted, enabling the use of binary search.</li> <li>Binary search achieves a time complexity of <math>O(\log n)</math> which meets the problem constraints.</li> <li>When the target is not found during the search, the final low pointer represents the correct insertion index.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log n)</math> due to binary search.</div> <div>Space Complexity: <math>O(1)</math> as no extra space is used.</div> </div> <div>Solution <div>We use binary search to find the target in the sorted array. Initialize two pointers, low and high. At each iteration, compute the mid index. If the value at mid matches the target, return mid. If the target is less than the mid value, adjust the high pointer; if greater, adjust the low pointer. When the iterations finish without finding the target, return the low pointer, which represents the correct insertion index.</div> </div> <div>#69 Sqrt(x) [Easy]</div> <div>Binary Searchp.2062</div>                                                                                                                                                     | <div>Key Insights <ul style="list-style-type: none"> <li>The problem can be solved efficiently using binary search.</li> <li>Instead of iterating from 0 to x, binary search narrows down the candidate square roots quickly.</li> <li>The algorithm should handle edge cases like <math>x = 0</math> and <math>x = 1</math>.</li> <li>By checking <math>mid * mid</math> against x, we can determine if we need to search in the lower or upper half.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log x)</math></div> <div>Space Complexity: <math>O(1)</math></div> </div> <div>Solution <div>We use a binary search approach to find the integer square root. Initialize two pointers, low and high, where low is set to 0 and high is set to x. For each iteration, calculate the mid value. If <math>mid^2</math> equals x, we have found the square root. If <math>mid^2</math> is less than x, record mid as a potential answer and move the low pointer to <math>mid + 1</math>. Otherwise, move the high pointer to <math>mid - 1</math>. Continue the search until low exceeds high, then return the last recorded candidate.</div> </div> <div>#278 First Bad Version [Easy]</div> <div>Key Insights</div> <div>Binary Searchp.2063</div>                                                                          | <div> <ul style="list-style-type: none"> <li>The API returns true for bad versions and all versions after a bad version.</li> <li>The problem is essentially about finding a transition point in a sorted sequence which calls for a binary search approach.</li> <li>The aim is to minimize the number of calls to isBadVersion by leveraging the ordered nature of the versions.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log n)</math></div> <div>Space Complexity: <math>O(1)</math></div> </div> <div>Solution <div>We use binary search to efficiently locate the first bad version. Initialize two pointers, left and right, at 1 and n respectively. At each step, compute the mid point and call isBadVersion(mid). If mid is bad, it indicates that the first bad version may be mid or to its left, so adjust right to mid. Otherwise, adjust left to <math>mid + 1</math>. Continue the process until the pointers converge. The converged pointer will be the first bad version. Key structures include simple integer variables for left, right, and mid, with binary search being the fundamental algorithm to reduce the search space logarithmically.</div> </div> <div>#374 Guess Number Higher or Lower [Easy]</div> <div>Key Insights</div> <div>Binary Searchp.2064</div>                                                                                                                                                                                                         |
| <div> <ul style="list-style-type: none"> <li>The problem can be effectively solved using binary search since the search space is sorted by nature (1 to n).</li> <li>Each API call helps to halve the search space, leading to an efficient solution.</li> <li>Be cautious with potential overflow when calculating the mid-point by using <math>low + (high - low) // 2</math>.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log n)</math> since we binary search the range.</div> <div>Space Complexity: <math>O(1)</math> as we use a constant amount of extra space.</div> </div> <div>Solution <div>We use a binary search algorithm to efficiently locate the target number. We initialize two pointers: low (1) and high (n). In each iteration, we compute the mid-point, then call the guess API. Based on the response: <ul style="list-style-type: none"> <li>If guess(mid) returns 0, we have found the number.</li> <li>If it returns -1, it indicates that our guess is higher than the pick, so we update the high pointer to <math>mid - 1</math>.</li> <li>If it returns 1, it indicates that our guess is lower than the pick, so we update the low pointer to <math>mid + 1</math>.</li> </ul> This approach guarantees a logarithmic time solution.</div> </div> <div>#704 Binary Search [Easy]</div> <div>Key Insights</div> <div>Binary Searchp.2065</div> | <div> <ul style="list-style-type: none"> <li>Utilize binary search to achieve <math>O(\log n)</math> runtime on a sorted array.</li> <li>Use two pointers to maintain the current search region.</li> <li>Narrow down the search range by comparing the middle element with the target.</li> <li>Return the index when the target is found; otherwise, return -1 if not found.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log n)</math></div> <div>Space Complexity: <math>O(1)</math></div> </div> <div>Solution <div>We use an iterative binary search algorithm. We initialize two pointers, left and right, to represent the current segment of the array that could contain the target. At each iteration, we compute the middle index and compare the element there with the target. If they match, we return the index. Otherwise, based on whether the target is smaller or larger than the middle element, we adjust our pointers to narrow the search to either the left or right half of the array. This process continues until the target is found or until the pointers cross, indicating that the target is not present. No extra data structures are used, making the space complexity <math>O(1)</math>.</div> </div> <div>#33 Search in Rotated Sorted Array [Medium]</div> <div>Binary Searchp.2066</div> | <div>Key Insights <ul style="list-style-type: none"> <li>The array is originally sorted and then rotated at an unknown pivot.</li> <li>Use a modified binary search to accommodate the rotation.</li> <li>At every step, determine which half of the array is properly sorted.</li> <li>Narrow down the search based on the sorted half and the target's value.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log n)</math></div> <div>Space Complexity: <math>O(1)</math></div> </div> <div>Solution <div>We perform a binary search while accounting for the possibility that the array is rotated. At each iteration, we check if the left-to-mid portion is sorted. If it is, and the target falls within that range, we search the left half; otherwise, we search the right half. If the left half is not sorted, then the right half must be sorted, and similarly, we check if the target is in that sorted half. This approach ensures logarithmic time complexity while handling the rotation correctly.</div> </div> <div>#34 Find First and Last Position of Element in Sorted Array [Medium]</div> <div>Key Insights</div> <div>Binary Searchp.2067</div>                                                                                                                                                                                                                                                                                                                                      |
| <div> <ul style="list-style-type: none"> <li>The array is sorted, allowing the use of binary search.</li> <li>Two binary searches can efficiently find the leftmost (first) and rightmost (last) indices of the target.</li> <li>Handling edge cases where the target is not present by checking bounds.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log n)</math></div> <div>Space Complexity: <math>O(1)</math></div> </div> <div>Solution <div>The approach is to use binary search twice: <ul style="list-style-type: none"> <li>The first binary search finds the leftmost occurrence of the target by continuing to search in the left half even after finding the target.</li> <li>The second binary search finds the rightmost occurrence by searching in the right half.</li> </ul> The algorithm maintains pointers for the current search bounds and narrows the search space using comparisons. If the target is not found during the first binary search, the solution immediately returns [-1, -1]. This method leverages the sorted property of the array, ensuring <math>O(\log n)</math> time complexity with constant extra space.</div> </div> <div>#153 Find Minimum in Rotated Sorted Array [Medium]</div> <div>Key Insights</div> <div>Binary Searchp.2068</div>                                                                                          | <div> <ul style="list-style-type: none"> <li>The rotated sorted array consists of two sorted subarrays.</li> <li>A modified binary search can be used to locate the pivot point where the rotation occurs.</li> <li>By comparing the middle element to the rightmost element, we can decide which half of the array contains the minimum.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(\log n)</math></div> <div>Space Complexity: <math>O(1)</math></div> </div> <div>Solution <div>We use a binary search approach. Initialize two pointers, left and right, at the start and end of the array. While left is less than right, calculate the middle index. If the middle element is greater than the right element, then the minimum is in the right half, so move the left pointer to <math>mid + 1</math>; otherwise, the minimum must be in the left half including mid, so adjust the right pointer to mid. When the loop terminates, left equals right and points to the minimum element. This approach leverages the properties of the rotated sorted array to achieve logarithmic time complexity.</div> </div> <div>#300 Longest Increasing Subsequence [Medium]</div> <div>Key Insights</div> <div>Binary Searchp.2069</div>                                                                                         | <div> <ul style="list-style-type: none"> <li>A dynamic programming solution with <math>O(n^2)</math> time complexity exists, but we focus on an optimized approach.</li> <li>Using binary search and a dp array, we can achieve <math>O(n \log n)</math> time.</li> <li>The dp array maintains the smallest possible tail for any increasing subsequences of various lengths.</li> <li>For each number, perform binary search to decide if it can extend or replace a value in dp.</li> </ul> </div> <div>Space &amp; Time <div>Time Complexity: <math>O(n \log n)</math> where n is the length of nums</div> <div>Space Complexity: <math>O(n)</math> for storing the dp array</div> </div> <div>Solution <div>We use a dynamic programming approach combined with binary search. The idea is to maintain a dp array where <math>dp[i]</math> holds the smallest tail value for any increasing subsequence with length <math>i+1</math>. For each number in the array, we perform a binary search on dp to find its correct insertion position: <ul style="list-style-type: none"> <li>If the number is larger than all elements, we append it, effectively extending the current longest subsequence.</li> <li>Otherwise, we replace the first element in dp that is greater than or equal to the current number to keep the tail as small as possible. This process ensures that dp remains sorted and its length gives the length of the longest increasing subsequence.</li> </ul> </div> </div> <div>Binary Searchp.2070</div> |



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| <div>#362 Design Hit Counter <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• Use a sliding window of 300 seconds to count hits.</li> <li>• Since timestamps are monotonically increasing, outdated hits can be removed from the front.</li> <li>• A queue (or circular array) is an efficient data structure to maintain and remove expired hit records.</li> <li>• Grouping consecutive hits at the same timestamp can optimize space.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(1) per hit and getHits operation on average, since each hit is enqueued and later dequeued only once.</p> <p>Space Complexity: O(W) where W is the window size (300 seconds) in the worst case.</p> </div> <div> <div>Solution</div> <p>The solution uses a queue (or deque) data structure where each element stores a (timestamp, count) pair. When a hit is recorded, if the last element of the queue shares the same timestamp, then its count is incremented; otherwise, a new element is added. When getHits(timestamp) is called, the algorithm removes elements from the front of the queue whose timestamp is out of the 300-second window relative to the current timestamp, and then sums the counts of the remaining</p> </div> <div> <div>Binary Search</div> <div>p.2071</div> </div>                                                               | <div>elements to return the result. This ensures that only relevant hits within the past 5 minutes are counted. The design scales well when hit frequency is high because hits at the same timestamp are grouped together instead of inserting individual records.</div> <div>#528 Random Pick with Weight <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• Build a prefix sum array from the weights.</li> <li>• Calculate the total sum of the weights.</li> <li>• Generate a random number in the range [1, total sum] and use binary search to find the corresponding index in the prefix sum array.</li> <li>• Using binary search ensures efficient selection in O(log n) time for each call of pickIndex().</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(n) for constructing the prefix sum array, and O(log n) per pickIndex() call.</p> <p>Space Complexity: O(n) for storing the prefix sum array.</p> </div> <div> <div>Solution</div> <p>The solution leverages a prefix sum array where each entry at index i represents the cumulative sum of weights up to i. After computing the total weight, a random integer is generated between 1 and the total sum (inclusive). A binary search is then used to find the first index where the prefix sum is greater than or equal</p> </div> <div> <div>Binary Search</div> <div>p.2072</div> </div> | <div>to the random number. This approach ensures that the index is selected with a probability proportional to its weight.</div> <div>#852 Peak Index in a Mountain Array <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The array is strictly increasing up to a peak and then strictly decreasing.</li> <li>• A binary search technique is applicable because the array exhibits a single transition point (peak) between increasing and decreasing order.</li> <li>• By comparing mid and mid+1, we can decide which half of the array to continue the search in.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(log(n)) – The binary search approach reduces the search interval by half at each step.</p> <p>Space Complexity: O(1) – Only constant extra space is used.</p> </div> <div> <div>Solution</div> <p>We use a binary search strategy:</p> <ul style="list-style-type: none"> <li>– Initialize two pointers, left and right, at the beginning and end of the array.</li> <li>– In each iteration, calculate mid. If arr[mid] is less than arr[mid + 1], then the peak is in the right half; otherwise, it is in the left half.</li> <li>– When left equals right, we have found the peak index. This</li> </ul> </div> <div> <div>Binary Search</div> <div>p.2073</div> </div>                                                                              |
| <div>approach leverages the mountain array properties and ensures an O(log(n)) solution.</div> <div>#981 Time Based Key-Value Store <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• Use a dictionary (hash table) to map each key to a list of (timestamp, value) pairs.</li> <li>• Timestamps are strictly increasing when setting the values, which allows efficient appending.</li> <li>• For getting the value, perform a binary search on the list of timestamps to find the largest timestamp that is &lt;= the queried timestamp.</li> <li>• If no valid timestamp exists, return an empty string.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity:</p> <ul style="list-style-type: none"> <li>– set operation: O(1) per call (amortized).</li> <li>– get operation: O(log N) per call, where N is the number of timestamps stored for the key.</li> </ul> <p>Space Complexity: O(total number of set calls), to store all (timestamp, value) pairs.</p> </div> <div> <div>Solution</div> <p>We use a hash table to store keys mapping to a list of their timestamps and corresponding values. Since the timestamps are inserted in strictly increasing order, the list remains sorted. For the get operation, a binary search is applied to quickly locate the appropriate value</p> </div> <div> <div>Binary Search</div> <div>p.2074</div> </div>   | <div>for a given timestamp. This approach minimizes the lookup time and efficiently handles the problem constraints.</div> <div>#1011 Capacity to Ship Packages Within D Days <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The minimal possible capacity is the maximum value in weights; no ship can carry a package heavier than its capacity.</li> <li>• The maximum possible capacity is the sum of all weights; this allows shipping all packages in one day.</li> <li>• The problem is monotonic: if a capacity C works, any capacity greater than C will also work.</li> <li>• Use binary search on the capacity range and a greedy approach to simulate shipping for each candidate capacity.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(n * log(sum(weights))) where n is the number of packages.</p> <p>Space Complexity: O(1) additional space.</p> </div> <div> <div>Solution</div> <p>The solution employs binary search to find the minimum ship capacity. Start by setting the search bounds: the lower bound is the maximum weight (since each package must be shipped individually if needed) and the</p> </div> <div> <div>Binary Search</div> <div>p.2075</div> </div>                                                                                                                                                             | <div>upper bound is the sum of all weights (shipping everything in one go). For each midpoint capacity in the binary search, use a greedy method to simulate the shipping process—accumulating weights until adding another package would exceed the candidate ship capacity, at which point you increment the day count. If the number of days required exceeds D, the capacity is insufficient and the lower bound is increased; otherwise, adjust the upper bound. Continue until the bounds converge.</div> <div>#1060 Missing Element in Sorted Array <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The array is sorted and every element is unique.</li> <li>• For any index i, the number of missing numbers up to that point can be computed as nums[i] – nums[0] – i.</li> <li>• When missing(i) is less than k, the kth missing element is further to the right.</li> <li>• Use binary search to find the smallest index where the count of missing numbers is greater than or equal to k.</li> <li>• If all missing numbers before the last array element are counted and the kth missing still hasn't been reached, calculate the result by continuing past the last element.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(log(n))</p> </div> <div> <div>Binary Search</div> <div>p.2076</div> </div>                                                        |
| <div>Space Complexity: O(1)</div> <div>Solution</div> <p>We compute the number of missing elements until an index i with the formula: missing(i) = nums[i] – nums[0] – i. We then use binary search over the indices to find the smallest index where missing(i) &gt;= k. If the binary search index equals the length of the array, it means the kth missing number lies beyond the last element. Otherwise, the kth missing number is derived as: nums[index – 1] + (k – missing(index – 1)). This approach uses binary search for logarithmic time and does not require extra auxiliary data structures.</p> <div>#1062 Longest Repeating Substring <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• Use binary search to efficiently check for the existence of a repeating substring of a given length.</li> <li>• Apply a rolling hash technique to compute hash values for substrings in O(1) time after initial processing.</li> <li>• Use a hash set to detect duplicate hash values (indicating a repeating substring).</li> <li>• Adjust binary search bounds based on whether a repeating substring of the current length exists.</li> <li>• Alternative approaches include dynamic programming and suffix arrays, but the binary search with rolling hash is efficient and conceptually clean.</li> </ul> </div> <div> <div>Binary Search</div> <div>p.2077</div> </div> | <div>Space &amp; Time</div> <p>Time Complexity: O(n log n) on average, where n is the length of the string. (Each binary search step takes O(n) time for computing rolling hashes.)</p> <p>Space Complexity: O(n) for storing hash values in the hash set.</p> <div>Solution</div> <p>The solution uses binary search to decide on the candidate length L of the repeating substring. For each candidate length L, we use a rolling hash to compute hash values for all substrings of length L and store them in a hash set. If any hash is repeated, it confirms the presence of a repeating substring of length L, and we try a longer length. Otherwise, we decrease the candidate length. The rolling hash is computed using a base (26, for lowercase letters) and a modulus (2^32 to limit integer overflow) which allows efficient hash recomputation for sliding windows. This approach efficiently narrows down the maximum repeating substring length.</p> <div>#1533 Find the Index of the Large Integer <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• Use binary search to efficiently narrow down the position of the unique large value.</li> <li>• At each step, divide the candidate range into two halves (or almost two halves when the range size is odd)</li> </ul> </div> <div> <div>Binary Search</div> <div>p.2078</div> </div>                                | <div>and use compareSub to determine which half contains the large integer.</div> <ul style="list-style-type: none"> <li>• Comparing two subarrays of equal length will reveal the region that contains the extra value via the sum difference.</li> <li>• Handle odd-length ranges carefully by isolating the middle element if needed.</li> </ul> <div>Space &amp; Time</div> <p>Time Complexity: O(log n) – because binary search is performed over the range.</p> <p>Space Complexity: O(1) – only a few pointers are maintained, no additional space is required.</p> <div>Solution</div> <p>The solution applies a binary search approach over the array indices. At every iteration:</p> <ul style="list-style-type: none"> <li>– Calculate the current segment length.</li> <li>– If the length is even, split the segment evenly and compare the left half versus the right half using compareSub.</li> <li>– If the length is odd, compare two halves of equal size while ignoring the middle element. If the sums are equal, then the middle index holds the large value.</li> <li>– Update the search bounds based on the result of compareSub.</li> <li>– Continue until the search space is reduced to one index, which is the answer.</li> </ul> <p>This method ensures that you locate the large integer with O(log n) calls to compareSub, well within the limit of 20 calls.</p> <div> <div>Binary Search</div> <div>p.2079</div> </div> |

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| <div>#4 Median of Two Sorted Arrays <span>[Hard]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The median divides the sorted array into two equal halves.</li> <li>• We can perform binary search on the smaller array to partition both arrays correctly.</li> <li>• We need to ensure that every element in the left partition is less than or equal to every element in the right partition.</li> <li>• Edge cases must be handled when partitions are at the boundaries of either array.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(log(min(m, n)))</div> <div>Space Complexity: O(1)</div> </div> <div> <div>Solution</div> <p>The solution uses a binary search technique on the smaller array. We define a partition index i for nums1 and compute the corresponding index j for nums2 such that i + j equals half of the total number of elements. The algorithm checks if the largest element on the left side of nums1 (if any) is less than or equal to the smallest element on the right side of nums2, and vice versa for nums2. If the condition holds, we have found the correct partition and compute the median. If not, adjust the binary search range until the correct partition</p> </div> <div> <div>Binary Search</div> <div>p.2080</div> </div>                                                                                                              | <div>is found. This partitioning strategy allows us to find the median in logarithmic time.</div> <div>#315 Count of Smaller Numbers After Self <span>[Hard]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The brute force solution takes O(n<sup>2</sup>) time, which is too slow for large input arrays.</li> <li>• An efficient approach leverages the concept of Merge Sort to both sort the array and count the number of smaller elements by tracking the number of elements that "jump over" during the merge step.</li> <li>• Other efficient methods include Binary Indexed Trees (Fenwick Tree) or Balanced Binary Search Trees.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n log n) on average due to the merge sort based approach.</div> <div>Space Complexity: O(n) for auxiliary arrays used during sorting and counting.</div> </div> <div> <div>Solution</div> <p>We use a modified Merge Sort to sort the array while counting the number of smaller elements to the right. We keep track of the original indices by pairing each element with its index. During the merge process, when an element from the left subarray is placed into the temporary array, we add the count of elements from the right subarray that have already been placed (since</p> </div> <div> <div>Binary Search</div> <div>p.2081</div> </div> | <div>these represent smaller elements that were to its right). This efficient divide and conquer strategy guarantees an overall O(n log n) time complexity. Care must be taken to update the counts based on indices and during the merge step.</div> <div>#410 Split Array Largest Sum <span>[Hard]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• Use binary search over the answer: the candidate value is the maximum subarray sum.</li> <li>• The lower bound of the search is max(nums) and the upper bound is sum(nums).</li> <li>• A greedy approach is used to check if a candidate maximum sum allows splitting nums into at most k subarrays.</li> <li>• Adjust the binary search bounds based on the feasibility check.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n * log(sum(nums))) – n elements are processed for each binary search iteration.</div> <div>Space Complexity: O(1) – Constant extra space is used.</div> </div> <div> <div>Solution</div> <p>The solution employs binary search to determine the smallest maximum subarray sum possible. We set the initial lower bound to the maximum element and the upper bound to the total sum of the array. For each</p> </div> <div> <div>Binary Search</div> <div>p.2082</div> </div>                                                                                                                                         |
| <div>candidate value (mid), we check if the array can be split into k or fewer subarrays without any subarray sum exceeding mid using a greedy approach. If possible, we try to lower the candidate; if not, we increase the candidate. This continues until the optimal solution is found.</div> <div>#668 Kth Smallest Number in Multiplication Table <span>[Hard]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The table elements are not fully sorted row-wise or column-wise globally, but there is a pattern that can be exploited.</li> <li>• For any candidate number x, we can count how many numbers in the table are less than or equal to x by summing over rows: for row i, there are min(x // i, n) numbers.</li> <li>• Binary search can be used over the range [1, m*n] to find the smallest x such that the count is at least k.</li> <li>• The counting function is the key to connecting the candidate value with the order statistic.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(m * log(m*n))</div> <div>Space Complexity: O(1)</div> </div> <div> <div>Solution</div> </div> <div> <div>Binary Search</div> <div>p.2083</div> </div>                                                                                                                                                                                                                          | <div>We use binary search over the possible values in the multiplication table, which range from 1 to m*n. For each midpoint x in the binary search, we compute the number of elements in the table that are less than or equal to x by iterating over each row and summing up min(x // i, n). This count tells us whether the kth smallest element is to the left or right of x. We then adjust the binary search boundaries accordingly. The approach leverages the multiplicative structure of the table and the monotonicity of the counting function.</div> <div>#710 Random Pick with Blacklist <span>[Hard]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The valid output range has a total count of available numbers W = n – len(blacklist).</li> <li>• Instead of repeatedly generating random numbers and checking against a set, we can transform the range into a contiguous block [0, W – 1] and map any blacklist number within that block to a valid number in the upper range.</li> <li>• Mapping is precomputed: for any blacklisted number x in [0, W-1], assign it a valid number from [W, n-1] that is not blacklisted.</li> <li>• This approach ensures O(1) pick time after an O(b) precomputation where b = length of blacklist.</li> </ul> </div> <div> <div>Space &amp; Time</div> </div> <div> <div>Binary Search</div> <div>p.2084</div> </div>   | <div>Time Complexity: constructor – O(b) for initializing data structures, pick() – O(1) per call.</div> <div>Space Complexity: O(b) for storing the blacklist set and mapping.</div> <div> <div>Solution</div> <p>We compute the count of valid numbers, W = n – len(blacklist). For numbers in the blacklist that are less than W, we need to map them to a valid number from the range [W, n-1]. To do this, we first store all blacklist numbers in the higher range [W, n-1] in a set for quick lookup. Then for each blacklisted value in [0, W-1], we iterate from W upwards to find a number that is not blacklisted and assign it in a mapping. During the pick() function, we generate a random integer r in [0, W-1]. If r is in our mapping, we return its corresponding mapped value; otherwise, we return r as it is already a valid number.</p> </div> <div>#774 Minimize Max Distance to Gas Station <span>[Hard]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• The optimal "penalty" (i.e., the maximum distance between adjacent stations after placement) can be found using binary search over a continuous distance range.</li> <li>• For each candidate distance (mid), simulate the number of gas stations required by iterating over each adjacent station gap and calculating:</li> <li>• stations_required = ceil(gap / mid) – 1</li> </ul> </div> <div> <div>Binary Search</div> <div>p.2085</div> </div> |
| <div> <ul style="list-style-type: none"> <li>• If the total number of required additional stations is less than or equal to k, then mid is a feasible candidate; otherwise, it is too small.</li> <li>• Continue the binary search until the gap between low and high is within the desired tolerance (1e-6).</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n * log(max_gap/precision)), where n is the number of stations.</div> <div>Space Complexity: O(1) (ignoring input storage)</div> </div> <div> <div>Solution</div> <p>We use binary search over the possible maximum distances. The algorithm sets low = 0 and high = the maximum gap between adjacent stations. For a candidate distance mid, we count how many new stations are required for all gaps by computing ceil(gap / mid) – 1. If the total required is within k, we try a smaller candidate; otherwise, we increase the candidate distance. This approach efficiently zooms into the minimum possible maximum distance (penalty).</p> </div> <div>#1235 Maximum Profit in Job Scheduling <span>[Hard]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>• Sort jobs based on their end times to simplify finding non-conflicting jobs.</li> <li>• Use dynamic programming where the state represents the maximum profit until a given point.</li> </ul> </div> <div> <div>Binary Search</div> <div>p.2086</div> </div> | <div> <ul style="list-style-type: none"> <li>• For each job, perform a binary search to find the last job that ends before the current job's start time.</li> <li>• The recurrence relation is: dp[i] = max(dp[i-1], profit[i] + dp[index]) where dp[index] is from the latest non-conflicting job.</li> </ul> </div> <div> <div>Space &amp; Time</div> <div>Time Complexity: O(n log n) due to sorting and binary search for each job.</div> <div>Space Complexity: O(n) for the dp arrays.</div> </div> <div> <div>Solution</div> <p>The solution begins by sorting the jobs by their end times. For each job, a binary search is used to quickly locate the last job that does not conflict with the current job (i.e., whose end time is less than or equal to the current job's start time). Then, using dynamic programming, the maximum profit is updated by choosing whether to include the current job or skip it. Two parallel arrays (or lists) are maintained: one for the end times of the jobs considered so far, and another to store the corresponding maximum profit at those times. This approach efficiently computes the answer while keeping time complexity within acceptable bounds.</p> </div> <div> <div>Dynamic Programming</div> <div>p.2087</div> </div>                                                                                                                                 | <div>#19 Dynamic Programming</div> <div>By Pattern · Python Only · 24 questions</div> <div> <div>Dynamic Programming</div> <div>p.2088</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| <div><div>Dynamic Programming</div><div>#70 Climbing StairsEAS</div><div>LeetCode · SimplyLeet · WalkCC · LeetCode</div><div>Math · Dynamic Programming · Memoization</div><div>Lists: Grind 169</div><div>Time Complexity: O(n) Space Complexity: O(n) (can be optimized to O(1) using constant space)</div><div>Problem</div><div>You are climbing a staircase. It takes n steps to reach the top.</div><div>Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?</div><div>Example 1:</div><div>Input: n = 2<br/>Output: 2<br/>Explanation: There are two ways to climb to the top.<br/>1. 1 step + 1 step<br/>2. 2 steps</div><div>Example 2:</div><div></div><div>Dynamic Programmingp.2089</div></div>                                                                                                                                                                                                         | <div><div>Input: n = 3<br/>Output: 3<br/>Explanation: There are three ways to climb to the top.<br/>1. 1 step + 1 step + 1 step<br/>2. 1 step + 2 steps<br/>3. 2 steps + 1 step</div><div>Constraints:</div><div>• 1 &lt;= n &lt;= 45</div><div>Brute Force (Python)</div><div># Brute Force Approach<br/>class Solution:<br/>    def climbStairs(self, n):<br/>        # Brute Force O(2^n) - pure recursion, recomputes identical subproblems<br/>        def climb(n):<br/>            if n &lt;= 2: return n<br/>            return climb(n - 1) + climb(n - 2)<br/>        return climb(n)</div><div>Community Solutions (Python)</div><div>• WalkCC</div><div>Dynamic Programmingp.2090</div></div> | <div><div>class Solution:<br/>    def climbStairs(self, n: int) -&gt; int:<br/>        # dp[i] := the number of ways to climb to the i-th stair<br/>        dp = [1, 1] + [0] * (n - 1)<br/><br/>        for i in range(2, n + 1):<br/>            dp[i] = dp[i - 1] + dp[i - 2]<br/><br/>        return dp[n]</div><div>class Solution:<br/>    def climbStairs(self, n: int) -&gt; int:<br/>        prev1 = 1 # dp[i - 1]<br/>        prev2 = 1 # dp[i - 2]<br/><br/>        for _ in range(2, n + 1):<br/>            dp = prev1 + prev2<br/>            prev2 = prev1<br/>            prev1 = dp<br/><br/>        return prev1</div><div>• LeetCode</div><div>class Solution:<br/>    def climbStairs(self, n: int) -&gt; int:<br/>        a, b = 0, 1<br/>        for _ in range(n):<br/>            a, b = b, a + b<br/>        return b</div><div>Dynamic Programmingp.2091</div></div>                                                                                                                                                                                    |
| <div><div>import numpy as np</div><div>class Solution:<br/>    def climbStairs(self, n: int) -&gt; int:<br/>        res = np.asmatrix([[1, 1]], np.dtype("0"))<br/>        factor = np.asmatrix([[1, 1], (1, 0)],<br/>                                np.dtype("0"))<br/>        n -= 1<br/>        while n:<br/>            if n &amp; 1:<br/><br/>                res *= factor<br/>                factor *= factor<br/>                n &gt;&gt;= 1<br/>            return res[0, 0]</div><div>• SimplyLeet</div><div># Define a function to compute the number of ways to climb stairs.<br/>def climbStairs(n):<br/>    # Base case: if there is only one step, return 1.<br/>    if n == 1:<br/>        return 1<br/>    # Initialize the dp array with n+1 elements.<br/>    dp = [0] * (n + 1)<br/>    # There is 1 way to climb 1 step.<br/>    dp[1] = 1<br/>    # There are 2 ways to climb 2 steps.</div><div>Dynamic Programmingp.2092</div></div> | <div><div>dp[2] = 2<br/># Compute the number of ways for each step from 3 to n.<br/>for i in range(3, n + 1):<br/>    dp[i] = dp[i - 1] + dp[i - 2] # Sum of ways from the previous two steps.<br/>    return dp[n]</div><div># Example usage:<br/>print(climbStairs(3)) # Output: 3</div><div>• LC.ca</div><div>class Solution:<br/>    def climbStairs(self, n: int) -&gt; int:<br/>        a, b = 0, 1<br/>        for _ in range(n):<br/>            a, b = b, a + b<br/>        return b</div><div>Dynamic Programmingp.2093</div></div>                                                                                                                                                             | <div><div>Dynamic Programming</div><div>#121 Best Time to Buy and Sell StockEAS</div><div>LeetCode · SimplyLeet · WalkCC · LeetCode</div><div>Array · Dynamic Programming</div><div>Lists: Grind 169</div><div>Time Complexity: O(n) - requires one pass through the prices array. Space Complexity: O(1) - only a few extra variables are used.</div><div>Problem</div><div>You are given an array prices where prices[i] is the price of a given stock on the ith day.</div><div>You want to maximize your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock.</div><div>Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0.</div><div>Example 1:</div><div>Input: prices = [7,1,5,3,6,4]<br/>Output: 5<br/>Explanation: Buy on day 2 (price = 1) and sell on day 5 (price = 6), profit = 6-1 = 5.<br/>Note that buying on day 2 and selling on day 1 is not allowed because you must buy before you sell.</div><div>Dynamic Programmingp.2094</div></div> |
| <div><div>Example 2:</div><div>Input: prices = [7,6,4,3,1]<br/>Output: 0<br/>Explanation: In this case, no transactions are done and the max profit = 0.</div><div>Constraints:</div><div>• 1 &lt;= prices.length &lt;= 105<br/>• 0 &lt;= prices[i] &lt;= 104</div><div>Brute Force (Python)</div><div># Brute Force Approach<br/>class Solution:<br/>    def maxProfit(self, prices):<br/>        # Brute Force O(n^2) - try every buy/sell pair<br/>        n = len(prices)<br/>        res = 0<br/>        for i in range(n):<br/>            for j in range(i + 1, n):<br/>                res = max(res, prices[j] - prices[i])<br/>        return res</div><div>Community Solutions (Python)</div><div>• WalkCC</div><div>Dynamic Programmingp.2095</div></div>                                                                                                                                                                                            | <div><div>class Solution:<br/>    def maxProfit(self, prices: List[int]) -&gt; int:<br/>        sellOne = 0<br/>        holdOne = -math.inf<br/><br/>        for price in prices:<br/>            sellOne = max(sellOne, holdOne + price)<br/>            holdOne = max(holdOne, -price)<br/><br/>        return sellOne</div><div>• LeetCode</div><div>class Solution:<br/>    def maxProfit(self, prices: List[int]) -&gt; int:<br/>        ans, mi = 0, inf<br/>        for v in prices:<br/>            ans = max(ans, v - mi)<br/>            mi = min(mi, v)<br/>        return ans</div><div>• SimplyLeet</div><div>Dynamic Programmingp.2096</div></div>                                          | <div><div>def maxProfit(prices):<br/>    # Initialize min_price as infinity and max_profit as 0<br/>    min_price = float('inf')<br/>    max_profit = 0<br/>    # Iterate through each price in the list<br/>    for price in prices:<br/>        # Update the minimum price if current price is lower<br/>        if price &lt; min_price:<br/>            min_price = price<br/>        else:<br/><br/>            # Calculate current profit and update max_profit if it is higher<br/>            max_profit = max(max_profit, price - min_price)<br/>        return max_profit</div><div># Example usage:<br/># result = maxProfit([7,1,5,3,6,4])<br/># print(result) # Output should be 5</div><div>• LC.ca</div><div>Dynamic Programmingp.2097</div></div>                                                                                                                                                                                                                                                                                                                 |

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| <pre>... &gt;&gt;&gt; import math &gt;&gt;&gt; a = math.inf &gt;&gt;&gt; a inf  &gt;&gt;&gt; x = float('-inf') &gt;&gt;&gt; y = float('-inf') &gt;&gt;&gt; print(x &lt; y) # Output: True True  &gt;&gt;&gt; &gt;&gt;&gt; z = -math.inf &gt;&gt;&gt; x==z True  ... class Solution:     def maxProfit(self, prices: List[int]) -&gt; int:         ans, mi = 0, inf         for v in prices:              mi = min(mi, v)             ans = max(ans, v - mi)         return ans  #####  class Solution(object):     def maxProfit(self, prices):         """         :type prices: List[int]</pre>                                                                                                                                                                                                                                                                                                                                                                                                    | <pre>:rtype: int """ if not prices:     return 0 ans = 0 pre = prices[0] for i in range(1, len(prices)):     pre = min(pre, prices[i])     ans = max(prices[i] - pre, ans) return ans</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <div>Dynamic Programming</div> <div>#5 Longest Palindromic Substring</div> <div>MED</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div> <div>String · Dynamic Programming</div> <div>Lists: Grind 169</div> <div>Time Complexity: <math>O(n^2)</math> in the worst case as we expand around each center. Space Complexity: <math>O(1)</math>, excluding the space required for the output substring.</div> <div>Problem</div> <div>Given a string s, return the longest palindromic substring in s.</div> <div>Example 1:</div> <div>Input: s = "babad"<br/>Output: "bab"<br/>Explanation: "aba" is also a valid answer.</div> <div>Example 2:</div> <div>Input: s = "cbabd"<br/>Output: "bb"</div> <div>Constraints:</div> <div><ul style="list-style-type: none"><li>1 &lt;= s.length &lt;= 1000</li><li>s consist of only digits and English letters.</li></ul></div> <div>Brute Force (Python)</div> |
| <div>Dynamic Programming</div> <div>p.2098</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div>Dynamic Programming</div> <div>p.2099</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div>Dynamic Programming</div> <div>p.2100</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <pre># Brute Force Approach class Solution:     def expand_from_center(self, s, left, right):         # Brute Force <math>O(n^3)</math> - check every substring for         # palindrome property         def is_palindrome(sub):             return sub == sub[::-1]         res = ''         for i in range(len(s)):             for j in range(i + 1, len(s) + 1):                 sub = s[i:j]                  if is_palindrome(sub) and len(sub) &gt; len(res):                     res = sub         return res</pre> <div>Community Solutions (Python)</div> <div>• WalkCC</div> <pre>class Solution:     def longestPalindrome(self, s: str) -&gt; str:         if not s:             return ''          # (start, end) indices of the longest palindrome in s         indices = [0, 0]          def extend(s: str, i: int, j: int) -&gt; tuple[int, int]:             """</pre>                                                                                                            | <pre>Returns the (start, end) indices of the longest palindrome extended from the substring s[i..j]. """ while i &gt;= 0 and j &lt; len(s):     if s[i] != s[j]:         break     i -= 1     j += 1     return i + 1, j - 1  for i in range(len(s)):     l1, r1 = extend(s, i, i)     if r1 - l1 &gt; indices[1] - indices[0]:         indices = l1, r1     if i + 1 &lt; len(s) and s[i] == s[i + 1]:         l2, r2 = extend(s, i, i + 1)         if r2 - l2 &gt; indices[1] - indices[0]:             indices = l2, r2  return s[indices[0]:indices[1] + 1]  class Solution:     def longestPalindrome(self, s: str) -&gt; str:         t = '#'.join('@' + s + '\$')         p = self._manacher(t)         maxPalindromeLength, bestCenter = max(extend, i) for i, extend in enumerate(p)     l = (bestCenter - maxPalindromeLength) // 2     r = (bestCenter + maxPalindromeLength) // 2     return s[l:r]</pre> | <pre>def _manacher(self, t: str) -&gt; list[int]:     """     Returns an array 'p' s.t. 'p[i]' is the length of the     longest palindrome     centered at 't[i]', where 't' is a string with     delimiters and sentinels.     """     p = [0] * len(t)     center = 0     for i in range(1, len(t) - 1):         rightBoundary = center + p[center]         mirrorIndex = center - (i - center)          if rightBoundary &gt; i:             p[i] = min(rightBoundary - i, p[mirrorIndex])             # Try to expand the palindrome centered at i.             while t[i + 1 + p[i]] == t[i - 1 - p[i]]:                 p[i] += 1             # If a palindrome centered at i expands past             'rightBoundary', adjust             # the center based on the expanded palindrome.             if i + p[i] &gt; rightBoundary:                 center = i     return p  • LeetDoocs</pre>           |
| <div>Dynamic Programming</div> <div>p.2101</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div>Dynamic Programming</div> <div>p.2102</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div>Dynamic Programming</div> <div>p.2103</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <pre>class Solution:     def longestPalindrome(self, s: str) -&gt; str:         n = len(s)         f = [[True] * n for _ in range(n)]         k, mx = 0, 1         for i in range(n - 2, -1, -1):             for j in range(i + 1, n):                 f[i][j] = False                 if s[i] == s[j]:                     f[i][j] = f[i + 1][j - 1]              if f[i][j] and mx &lt; j - i + 1:                 k, mx = i, j - i + 1         return s[k : k + mx]  class Solution:     def longestPalindrome(self, s: str) -&gt; str:         def f(l, r):             while l &gt;= 0 and r &lt; n and s[l] == s[r]:                 l, r = l - 1, r + 1             return r - l - 1          n = len(s)         start, mx = 0, 1         for i in range(n):              a = f(i, i)             b = f(i, i + 1)             t = max(a, b)             if mx &lt; t:                 mx = t                 start = i - ((t - 1) &gt;&gt; 1)             return s[start : start + mx]</pre> | <div>• SimplyLeet</div> <pre># Function to expand from the center and return the palindrome substring def expand_from_center(s, left, right):     # Expand while the characters on both sides are the     same     while left &gt;= 0 and right &lt; len(s) and s[left] == s[right]:         left -= 1         right += 1     # Return the palindromic substring found after     expansion     return s[left + 1:right]  def longest_palindromic_substring(s):      if not s:         return ""     longest = ""     # Iterate over each index in string as potential     center     for i in range(len(s)):         # Check for odd-length palindrome         odd_palindrome = expand_from_center(s, i, i)         if len(odd_palindrome) &gt; len(longest):             longest = odd_palindrome         # Check for even-length palindrome</pre>                                                                   | <pre>even_palindrome = expand_from_center(s, i, i + 1) if len(even_palindrome) &gt; len(longest):     longest = even_palindrome return longest  # Example usage print(longest_palindromic_substring("babad"))  • LC.ca  class Solution:     def longestPalindrome(self, s: str) -&gt; str:         mlen = 0         start = end = 0         n = len(s)         dp = [ [False] * n for i in range(n) ]          for j in range(n):             for i in range(j + 1):                 dp[i][j] = (i == j) or (s[i] == s[j] and j - i == 1) or (s[i] == s[j] and dp[i + 1][j - 1])                  if dp[i][j] is True and j - i + 1 &gt; mlen:                     mlen = j - i + 1                     start = i                     end = j              return s[start: end + 1]  #####  class Solution:</pre>                                                                                                |
| <div>Dynamic Programming</div> <div>p.2104</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <div>Dynamic Programming</div> <div>p.2105</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <div>Dynamic Programming</div> <div>p.2106</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

```
def longestPalindrome(self, s: str) -> str:
 n = len(s)
 f = [[True] * n for _ in range(n)]
 k, mx = 0, 1
 for i in range(n - 2, -1, -1):
 for j in range(i + 1, n):
 f[i][j] = False
 if s[i] == s[j]:
 f[i][j] = f[i + 1][j - 1]
 if f[i][j] and mx < j - i + 1:
 k, mx = i, j - i + 1
 return s[k : k + mx]
```

Dynamic Programming

p.2107

Dynamic Programming

#53 Maximum SubarrayMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming · Divide and Conquer

Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(1)

Problem

Given an integer array nums, find the subarray with the largest sum, and return its sum.

Example 1:

```
Input: nums = [-2,1,-3,4,-1,2,1,-5,4]
Output: 6
Explanation: The subarray [4,-1,2,1] has the largest sum 6.
```

Example 2:

```
Input: nums = [1]
Output: 1
Explanation: The subarray [1] has the largest sum 1.
```

Example 3:

```
Input: nums = [5,4,-1,7,8]
Output: 23
Explanation: The subarray [5,4,-1,7,8] has the largest sum 23.
```

Dynamic Programming

p.2108

Constraints:

- 1 <= nums.length <= 105
- -104 <= nums[i] <= 104

Follow up: If you have figured out the O(n) solution, try coding another solution using the divide and conquer approach, which is more subtle.

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def maxSubArray(self, nums):
 # Brute Force O(n^2) - compute sum of every contiguous subarray
 res = nums[0]
 for i in range(len(nums)):
 curr = 0
 for j in range(i, len(nums)):
 curr += nums[j]
 res = max(res, curr)

 return res
```

Community Solutions (Python)

- WalkCC

Dynamic Programming

p.2109

```
class Solution:
 def maxSubArray(self, nums: List[int]) -> int:
 # dp[i] := the maximum sum subarray ending in i
 dp = [0] * len(nums)

 dp[0] = nums[0]
 for i in range(1, len(nums)):
 dp[i] = max(nums[i], dp[i - 1] + nums[i])

 return max(dp)

class Solution:
 def maxSubArray(self, nums: List[int]) -> int:
 ans = -math.inf
 summ = 0

 for num in nums:
 summ = max(num, summ + num)
 ans = max(ans, summ)

 return ans

from dataclasses import dataclass

@dataclass(frozen=True)
class T:
 summ: int
 # the sum of the subarray starting from the first number
 maxSubarraySumLeft: int
 # the sum of the subarray ending in the last number
 maxSubarraySumRight: int
```

Dynamic Programming

p.2110

```
maxSubarraySum: int

class Solution:
 def maxSubArray(self, nums: List[int]) -> int:
 def divideAndConquer(l: int, r: int) -> T:
 if l == r:
 return T(nums[l], nums[l], nums[l], nums[l])
 m = (l + r) // 2
 left = divideAndConquer(l, m)

 right = divideAndConquer(m + 1, r)
 maxSubarraySumLeft = max(left.maxSubarraySumLeft,
 left.summ +
 right.maxSubarraySumLeft)
 maxSubarraySumRight = max(
 left.maxSubarraySumRight + right.summ,
 right.maxSubarraySumRight)
 maxSubarraySum = max(left.maxSubarraySumRight +
 right.maxSubarraySumLeft,
 left.maxSubarraySum,
 right.maxSubarraySum)
 summ = left.summ + right.summ
 return T(summ, maxSubarraySumLeft,
 maxSubarraySumRight, maxSubarraySum)

 return divideAndConquer(0, len(nums) - 1).maxSubarraySum
```

- LeetCode

Dynamic Programming

p.2111

```
class Solution:
 def maxSubArray(self, nums: List[int]) -> int:
 ans = f = nums[0]
 for x in nums[1:]:
 f = max(f, 0) + x
 ans = max(ans, f)
 return ans

class Solution:
 def maxSubArray(self, nums: List[int]) -> int:
 def crossMaxSub(nums, left, mid, right):
 lsum = rsum = 0
 lmx = rmx = -inf
 for i in range(mid, left - 1, -1):
 lsum += nums[i]
 lmx = max(lmx, lsum)
 for i in range(mid + 1, right + 1):
 rsum += nums[i]
 rmx = max(rmx, rsum)
 return lmx + rmx

 def maxSub(nums, left, right):
 if left == right:
 return nums[left]
 mid = (left + right) >> 1
 lsum = maxSub(nums, left, mid)
 rsum = maxSub(nums, mid + 1, right)
 csum = crossMaxSub(nums, left, mid, right)
```

Dynamic Programming

p.2112

```
return max(lsum, rsum, csum)

left, right = 0, len(nums) - 1
return maxSub(nums, left, right)

• SimplyLeet

Python solution using Kadane's Algorithm
def maxSubArray(nums):
 # Initialize currentSum and maxSum to the first element
 current_sum = max_sum = nums[0]
 # Traverse through the array starting from the second element
 for num in nums[1:]:
 # If current_sum + num is less than num, restart at num
 current_sum = max(num, current_sum + num)
 # Update max_sum if current_sum is greater
 max_sum = max(max_sum, current_sum)

 return max_sum

Example usage
if __name__ == "__main__":
 sample = [-2,1,-3,4,-1,2,1,-5,4]
 print(maxSubArray(sample)) # Output: 6

• LC.ca
```

Dynamic Programming

p.2113

```
class Solution:
 def maxSubArray(self, nums: List[int]) -> int:
 res = cur_sum = nums[0]
 for num in nums[1:]:
 cur_sum = num + max(cur_sum, 0)
 res = max(res, cur_sum)
 return res
```

Dynamic Programming

p.2114

Dynamic Programming

#55 Jump GameMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming · Greedy

Lists: Grind 169

Time Complexity: O(n) where n is the length of the array. Space Complexity: O(1)

Problem

You are given an integer array nums. You are initially positioned at the array's first index, and each element in the array represents your maximum jump length at that position.

Return true if you can reach the last index, or false otherwise.

Example 1:

```
Input: nums = [2,3,1,1,4]
Output: true
Explanation: Jump 1 step from index 0 to 1, then 3 steps to the last index.
```

Example 2:

Dynamic Programming

p.2115

Input: nums = [3,2,1,0,4]  
Output: false  
Explanation: You will always arrive at index 3 no matter what. Its maximum jump length is 0, which makes it impossible to reach the last index.

Constraints:

- 1 <= nums.length <= 104
- 0 <= nums[i] <= 105

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def canJump(self, nums):
 # Brute Force O(2^n) - recursion: try every
 # reachable position
 def can_reach(i):
 if i >= len(nums) - 1: return True
 for step in range(1, nums[i] + 1):
 if can_reach(i + step): return True
 return False
 return can_reach(0)
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def canJump(self, nums: list[int]) -> bool:
 i = 0
 reach = 0

 while i < len(nums) and i <= reach:
 reach = max(reach, i + nums[i])
 i += 1

 return i == len(nums)
```

• LeetCode

```
class Solution:
 def canJump(self, nums: List[int]) -> bool:
 mx = 0
 for i, x in enumerate(nums):
 if mx < i:
 return False
 mx = max(mx, i + x)
 return True
```

• SimplyLeet

```
Function to determine if you can reach the last index
of the array
def canJump(nums):
 # Initialize maxReach which is the max index
 # reachable at start (index 0)
 maxReach = 0
 # Iterate through each element and its index
 for i, jump in enumerate(nums):
 # If current index is greater than maximum
 # reachable index, return False
 if i > maxReach:
 return False
 # Update maxReach with the maximum of current
 # maxReach and i + jump length at index i
 maxReach = max(maxReach, i + jump)
```

```
If current maxReach is already at or beyond the
last index, return True
if maxReach >= len(nums) - 1:
 return True

Return True if the end is reachable after
processing all indices
return True
```

```
Example usage:
print(canJump([2,3,1,1,4])) # Expected output: True
print(canJump([3,2,1,0,4])) # Expected output: False
```

- LC.ca

Dynamic Programming

p.2116

Dynamic Programming

p.2117

Dynamic Programming

p.2118

```
class Solution:
 def canJump(self, nums: List[int]) -> bool:
 mx = 0
 for i, x in enumerate(nums):
 if mx < i:
 return False
 mx = max(mx, i + x)
 return True
```

Dynamic Programming

#62 Unique Paths

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Math · Dynamic Programming · Combinatorics

Lists: Grind 169

Time Complexity: O(m \* n) Space Complexity: O(m \* n) - can be optimized to O(n) using a 1D DP array.

Problem

There is a robot on an m x n grid. The robot is initially located at the top-left corner (i.e., grid[0][0]). The robot tries to move to the bottom-right corner (i.e., grid[m - 1][n - 1]). The robot can only move either down or right at any point in time.

Given the two integers m and n, return the number of possible unique paths that the robot can take to reach the bottom-right corner.

The test cases are generated so that the answer will be less than or equal to 2 \* 10<sup>9</sup>.

Example 1:



Input: m = 3, n = 7  
Output: 28

Example 2:

Input: m = 3, n = 2  
Output: 3  
Explanation: From the top-left corner, there are a total of 3 ways to reach the bottom-right corner:  
1. Right -> Down -> Down  
2. Down -> Down -> Right  
3. Down -> Right -> Down

Constraints:

- 1 <= m, n <= 100

Brute Force (Python)

Dynamic Programming

p.2119

Dynamic Programming

p.2120

Dynamic Programming

p.2121

```
Brute Force Approach
class Solution:
 def uniquePaths(self, m, n):
 # Brute Force O(2^(m+n)) - recursion without
 # memo; each step right or down
 def paths(r, c):
 if r == m - 1 and c == n - 1: return 1
 if r >= m or c >= n: return 0
 return paths(r + 1, c) + paths(r, c + 1)
 return paths(0, 0)
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def uniquePaths(self, m: int, n: int) -> int:
 # dp[i][j] := the number of unique paths from (0, 0)
 # to (i, j)
 dp = [[1] * n for _ in range(m)]

 for i in range(1, m):
 for j in range(1, n):
 dp[i][j] = dp[i - 1][j] + dp[i][j - 1]

 return dp[-1][-1]
```

```
class Solution:
 def uniquePaths(self, m: int, n: int) -> int:
 dp = [1] * n

 for _ in range(1, m):
 for j in range(1, n):
 dp[j] += dp[j - 1]

 return dp[-1]
```

• LeetCode

```
class Solution:
 def uniquePaths(self, m: int, n: int) -> int:
 f = [[0] * n for _ in range(m)]
 f[0][0] = 1
 for i in range(m):
 for j in range(n):
 if i:
 f[i][j] += f[i - 1][j]
 if j:
 f[i][j] += f[i][j - 1]
```

return f[-1][-1]

```
class Solution:
 def uniquePaths(self, m: int, n: int) -> int:
 f = [1] * n
 for _ in range(1, m):
 for j in range(1, n):
 f[j] += f[j - 1]
 return f[-1]
```

• SimplyLeet

# Python solution using Dynamic Programming

```
def uniquePaths(m, n):
 # Initialize a 2D list with 1's for the first row and
 # first column
 dp = [[1] * n for _ in range(m)]
```

```
Fill in the DP table
for i in range(1, m):
 for j in range(1, n):
 # The number of ways to reach this cell is
 # the sum of the ways from the cell above and the cell on
 # the left
```

```
dp[i][j] = dp[i-1][j] + dp[i][j-1]
Return the value in the bottom-right cell
return dp[m-1][n-1]
```

```
Example usage:
print(uniquePaths(3, 7))
```

- LC.ca

```
class Solution:
 def uniquePaths(self, m: int, n: int) -> int:
 # avoid setting dp[i][j] to 1 for i==0 or j==0 as
 # initialization
 dp = [[1] * n for _ in range(m)]
 for i in range(1, m):
 for j in range(1, n):
 dp[i][j] = dp[i - 1][j] + dp[i][j - 1]
 return dp[-1][-1]
```

Dynamic Programming

p.2122

Dynamic Programming

p.2123

Dynamic Programming

p.2124



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| <div><div>Dynamic Programming</div><div>#72 Edit DistanceMED</div><div>LeetCode · SimplyLeet · WalkCC · LeetCode</div><div>String · Dynamic Programming</div><div>Lists: Denny Zhang</div><div>Time Complexity: <math>O(m \cdot n)</math>, where <math>m</math> and <math>n</math> are the lengths of word1 and word2 respectively. Space Complexity: <math>O(m \cdot n)</math> due to the DP table used for storing computed distances.</div><div>Problem</div><div>Given two strings word1 and word2, return the minimum number of operations required to convert word1 to word2.</div><div>You have the following three operations permitted on a word:</div><div><ul style="list-style-type: none"><li>Insert a character</li><li>Delete a character</li><li>Replace a character</li></ul></div><div>Example 1:</div></div> <div><div>Dynamic Programming</div><div>p.2125</div></div>                                                                                                                                                                                                                                               | <div><div>Input: word1 = "horse", word2 = "ros"<br/>Output: 3<br/>Explanation:<br/>horse -&gt; rorse (replace 'h' with 'r')<br/>rorse -&gt; rose (remove 'r')<br/>rose -&gt; ros (remove 'e')</div><div>Example 2:</div><div>Input: word1 = "intention", word2 = "execution"<br/>Output: 5<br/>Explanation:<br/>intention -&gt; inention (remove 't')<br/>inention -&gt; enention (replace 'i' with 'e')<br/>enention -&gt; exention (replace 'n' with 'x')<br/>exention -&gt; execution (replace 'n' with 'c')<br/>execution -&gt; execution (insert 'u')</div><div>Constraints:</div><div><ul style="list-style-type: none"><li><math>0 \leq \text{word1.length}, \text{word2.length} \leq 500</math></li><li>word1 and word2 consist of lowercase English letters.</li></ul></div><div>Brute Force (Python)</div></div> <div><div>Dynamic Programming</div><div>p.2126</div></div>                                                                                                                                                                                                                                                                                                                                                                                                      | <div><div># Brute Force Approach<br/>class Solution:<br/>    def minDistance(self, word1, word2):<br/>        # Brute Force <math>O(3^{(m+n)})</math> - recursion without memo;<br/>        try insert/delete/replace<br/>        def dp(i, j):<br/>            if i == len(word1): return len(word2) - j<br/>            if j == len(word2): return len(word1) - i<br/>            if word1[i] == word2[j]:<br/>                return dp(i + 1, j + 1)<br/>            return 1 + min(dp(i + 1, j), # delete<br/>                           dp(i, j + 1), # insert<br/>                           dp(i + 1, j + 1)) # replace<br/>        return dp(0, 0)</div><div>Community Solutions (Python)</div><div><ul style="list-style-type: none"><li>WalkCC<br/>class Solution:<br/>    def minDistance(self, word1: str, word2: str) -&gt; int:<br/>        m = len(word1)<br/>        n = len(word2)<br/>        # dp[i][j] := the minimum number of operations to convert word1[0..i) to word2[0..j)<br/>        dp = [[0] * (n + 1) for _ in range(m + 1)]<br/><br/>        for i in range(1, m + 1):<br/>            dp[i][0] = i</li></ul></div></div> <div><div>Dynamic Programming</div><div>p.2127</div></div> |
| <div><div>for j in range(1, n + 1):<br/>    dp[0][j] = j<br/><br/>    for i in range(1, m + 1):<br/>        for j in range(1, n + 1):<br/>            if word1[i - 1] == word2[j - 1]:<br/>                dp[i][j] = dp[i - 1][j - 1]<br/>            else:<br/>                dp[i][j] = min(dp[i - 1][j - 1], dp[i - 1][j], dp[i][j - 1]) + 1<br/><br/>    return dp[m][n]</div><div><ul style="list-style-type: none"><li>LeetCode<br/>class Solution:<br/>    def minDistance(self, word1: str, word2: str) -&gt; int:<br/>        m, n = len(word1), len(word2)<br/>        f = [[0] * (n + 1) for _ in range(m + 1)]<br/>        for j in range(1, n + 1):<br/>            f[0][j] = j<br/>        for i, a in enumerate(word1, 1):<br/>            f[i][0] = i<br/>            for j, b in enumerate(word2, 1):<br/>                if a == b:<br/>                    f[i][j] = f[i - 1][j - 1]<br/>                else:<br/>                    f[i][j] = min(f[i - 1][j], f[i][j - 1], f[i - 1][j - 1]) + 1<br/>            return f[m][n]</li></ul></div></div> <div><div>Dynamic Programming</div><div>p.2128</div></div> | <div><ul style="list-style-type: none"><li>SimplyLeet<br/># Python solution to compute the Edit Distance<br/>def minDistance(word1, word2):<br/>    m = len(word1)<br/>    n = len(word2)<br/><br/>    # Initialize the DP table with size (m+1) x (n+1)<br/>    dp = [[0] * (n + 1) for _ in range(m + 1)]<br/><br/>    # Base cases: transform word1[0:i] to empty word2 requires i deletions<br/>    for i in range(m + 1):<br/>        dp[i][0] = i<br/>    # Base cases: transform empty word1 to word2[0:j] requires j insertions<br/>    for j in range(n + 1):<br/>        dp[0][j] = j<br/><br/>    # Fill up the dp table<br/>    for i in range(1, m + 1):<br/>        for j in range(1, n + 1):<br/>            # If the characters are the same, no new operation is needed.<br/>            if word1[i - 1] == word2[j - 1]:<br/>                dp[i][j] = dp[i - 1][j - 1]</li></ul></div> <div><div>Dynamic Programming</div><div>p.2129</div></div>                                                                                                                                                                                                                                                                                                                      | <div><div>dp[i][j] = dp[i - 1][j - 1]<br/>else:<br/>    # consider insertion, deletion, and replacement<br/>    dp[i][j] = 1 + min(dp[i][j - 1], # Insert<br/>                       dp[i - 1][j], # Delete<br/>                       dp[i - 1][j - 1]) #<br/><br/>Replace<br/>    return dp[m][n]<br/><br/># Example usage:<br/>print(minDistance("horse", "ros"))</div><div><ul style="list-style-type: none"><li>LC.ca<br/>class Solution:<br/>    def minDistance(self, word1: str, word2: str) -&gt; int:<br/>        m, n = len(word1), len(word2)<br/>        dp = [[0] * (n + 1) for _ in range(m + 1)]<br/>        for i in range(m + 1):<br/>            dp[i][0] = i<br/>        for j in range(n + 1):<br/>            dp[0][j] = j<br/>        for i in range(1, m + 1):<br/>            for j in range(1, n + 1):</li></ul></div></div> <div><div>Dynamic Programming</div><div>p.2130</div></div>                                                                                                                                                                                                                                                                                                     |
| <div><div>"b" distance,<br/># then replace either way for "c" or "f"<br/>dp[i][j] = 1 + min(dp[i][j - 1], dp[i - 1][j], dp[i - 1][j - 1])<br/>return dp[-1][-1]<br/><br/>#####<br/><br/>class Solution:<br/>    def minDistance(self, word1: str, word2: str) -&gt; int:<br/>        m, n = len(word1), len(word2)<br/>        f = [[0] * (n + 1) for _ in range(m + 1)]<br/>        for j in range(1, n + 1):<br/>            f[0][j] = j<br/>        for i, a in enumerate(word1, 1):<br/>            f[i][0] = i # merged for loops from above solution, but not good for readability<br/>            for j, b in enumerate(word2, 1):<br/>                if a == b:<br/>                    f[i][j] = f[i - 1][j - 1]<br/>                else:<br/>                    f[i][j] = min(f[i - 1][j], f[i][j - 1], f[i - 1][j - 1]) + 1<br/>            return f[m][n]</div><div><div>Dynamic Programming</div><div>p.2131</div></div></div>                                                                                                                                                                                           | <div><div>Dynamic Programming</div><div>#91 Decode WaysMED</div><div>LeetCode · SimplyLeet · WalkCC · LeetCode</div><div>String · Dynamic Programming</div><div>Lists: Grind 169</div><div>Time Complexity: <math>O(n)</math>, where <math>n</math> is the length of the string. Space Complexity: <math>O(n)</math>, due to the dp array (which can be optimized to <math>O(1)</math>).</div><div>Problem</div><div>You have intercepted a secret message encoded as a string of numbers. The message is decoded via the following mapping:<br/>"1" -&gt; 'A' "2" -&gt; 'B' ... "25" -&gt; 'Y' "26" -&gt; 'Z'<br/>However, while decoding the message, you realize that there are many different ways you can decode the message because some codes are contained in other codes ("2" and "5" vs "25").<br/>For example, "11106" can be decoded into:<br/><ul style="list-style-type: none"><li>"AAJF" with the grouping (1, 1, 10, 6)</li><li>"KJF" with the grouping (11, 10, 6)</li><li>The grouping (1, 11, 06) is invalid because "06" is not a valid code (only "6" is valid).</li></ul></div><div>Note: there may be strings that are impossible to decode. Given a string s containing only digits, return</div></div> <div><div>Dynamic Programming</div><div>p.2132</div></div> | <div><div>the number of ways to decode it. If the entire string cannot be decoded in any valid way, return 0.<br/>The test cases are generated so that the answer fits in a 32-bit integer.</div><div>Example 1:<br/>Input: s = "12"<br/>Output: 2<br/>Explanation:<br/>"12" could be decoded as "AB" (1 2) or "L" (12).</div><div>Example 2:<br/>Input: s = "226"<br/>Output: 3<br/>Explanation:<br/>"226" could be decoded as "BZ" (2 26), "VF" (22 6), or "BBF" (2 2 6).</div><div>Example 3:<br/>Input: s = "06"<br/>Output: 0<br/>Explanation:<br/>"06" cannot be mapped to "F" because of the leading zero ("6" is different from "06"). In this case, the string is not a valid encoding, so return 0.</div><div>Constraints:</div><div><ul style="list-style-type: none"><li><math>1 \leq s.length \leq 100</math></li><li>s contains only digits and may contain leading zero(s).</li></ul></div></div> <div><div>Dynamic Programming</div><div>p.2133</div></div>                                                                                                                                                                                                                                           |

Community Solutions (Python)

• WalkCC

```
class Solution:
 def numDecodings(self, s: str) -> int:
 n = len(s)
 # dp[i] := the number of ways to decode s[i..n)
 dp = [0] * n + [1]

 def isValid(a: str, b=None) -> bool:
 if b:
 return a == '1' or a == '2' and b < '7'
 return a != '0'

 if isValid(s[-1]):
 dp[n - 1] = 1

 for i in reversed(range(n - 1)):
 if isValid(s[i]):
 dp[i] += dp[i + 1]
 if isValid(s[i], s[i + 1]):
 dp[i] += dp[i + 2]

 return dp[0]
```

• LeetCode

```
class Solution:
 def numDecodings(self, s: str) -> int:
 n = len(s)
 f = [1] + [0] * n
 for i, c in enumerate(s, 1):
 if c != "0":
 f[i] = f[i - 1]
 if i > 1 and s[i - 2] != "0" and int(s[i - 2 : i]) <= 26:
 f[i] += f[i - 2]
 return f[n]

class Solution:
 def numDecodings(self, s: str) -> int:
 f, g = 0, 1
 for i, c in enumerate(s, 1):
 h = g if c != "0" else 0
 if i > 1 and s[i - 2] != "0" and int(s[i - 2 : i]) <= 26:
 h += g
 f, g = g, h
 return g
```

• SimplyLeet

```
Python solution using dynamic programming
def num_decodings(s: str) -> int:
 # If the string starts with '0', it's not decodable
 if not s or s[0] == '0':
 return 0

 # dp array where dp[i] represents the number of ways
 # to decode s[:i]
 n = len(s)
 dp = [0] * (n + 1)

 # Base case: an empty string has one decoding
 dp[0] = 1

 for i in range(1, n + 1):
 # Check for single digit decoding (s[i-1] is
 # valid if not '0')
 if s[i - 1] != '0':
 dp[i] += dp[i - 1]
 # Check for two-digit decoding (if i>=2, s[i-2:i]
 # forms a valid number between 10 and 26)
 if i >= 2 and s[i - 2] != '0':
 two_digit = int(s[i - 2:i])
 dp[i] += dp[i - 2]

 if 10 <= two_digit <= 26:
 dp[i] += dp[i - 2]

 return dp[n]

Example usage:
print(num_decodings("12")) # Expected output: 2
```

• LC.ca

```
...
>>> s = "abcdef"
>>> [(i,c) for i, c in enumerate(s, 1)]
[(1, 'a'), (2, 'b'), (3, 'c'), (4, 'd'), (5, 'e'), (6, 'f')]
...
class Solution:
 def numDecodings(self, s: str) -> int:
 # f: i-2
 # g: i-1
 f, g = 0, 1

 for i, c in enumerate(s, 1):
 h = g if c != "0" else 0
 if i > 1 and s[i - 2] != "0" and int(s[i - 2 : i]) <= 26:
 h += g
 f, g = g, h
 return g

#####

class Solution:
```

```
def numDecodings(self, s: str) -> int:
 if len(s) == 0:
 return 0

 length = len(s)
 dp = [0] * (length + 1) # dp[i] => at index i, its
 decode ways

 dp[length] = 1 # initiator, just make forloop
 flow working
 dp[length - 1] = 0 if s[length - 1] == '0' else 1
 # assumption is starting at i, so starting as 0 is not
 decodable

 # start at 'len-2'
 # or else below 'dp[i+2]' will out of index
 boundary
 for i in range(length - 2, -1, -1):
 if s[i] == '0':
 continue
 tem = int(s[i:i + 2])

 if tem > 26:
 dp[i] = dp[i + 1]
 else:
```

```
dp[i] = dp[i + 1] + dp[i + 2]
return dp[0]

optimized from above solution, no dp array
class Solution:
 def numDecodings(self, s: str) -> int:
 n = len(s)
 # a: dp[i-2], b: dp[i-1], c: count for current
 index i
 a, b, c = 0, 1, 0
 for i in range(1, n + 1):

 c = 0
 if s[i - 1] != '0':
 c += b
 if i > 1 and s[i - 2] != '0' and (int(s[i - 2]
) * 10 + int(s[i - 1])) <= 26):
 c += a
 a, b = b, c
 return c

#####
```

```
class Solution:
 def numDecodings(self, s: str) -> int:
 n = len(s)
 dp = [0] * (n + 1)
 dp[0] = 1
 for i in range(1, n + 1):
 if s[i - 1] != '0':
 dp[i] += dp[i - 1]
 if i > 1 and s[i - 2] != '0' and (int(s[i - 2]
) * 10 + int(s[i - 1])) <= 26):
 dp[i] += dp[i - 2]

 return dp[n]
```

Dynamic Programming

#152 Maximum Product SubarrayMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming

Lists: Grind 169

Time Complexity: O(n) - The solution iterates through the array once. Space Complexity: O(1) - Constant space is used for tracking products.

Problem

Given an integer array nums, find a subarray that has the largest product, and return the product.

The test cases are generated so that the answer will fit in a 32-bit integer.

Note that the product of an array with a single element is the value of that element.

Example 1:

Input: nums = [2,3,-2,4]  
Output: 6  
Explanation: [2,3] has the largest product 6.

Example 2:

Input: nums = [-2,0,-1]  
Output: 0  
Explanation: The result cannot be 2, because [-2,-1] is not a subarray.

Constraints:

• 1 <= nums.length <= 2 \* 10<sup>4</sup>  
• -10 <= nums[i] <= 10  
• The product of any subarray of nums is guaranteed to fit in a 32-bit integer.

Brute Force (Python)

# Brute Force Approach  
class Solution:  
 def maxProduct(self, nums):  
 # Brute Force O(n^2) - compute product of every  
 contiguous subarray  
 res = nums[0]  
 for i in range(len(nums)):  
 prod = 1  
 for j in range(i, len(nums)):  
 prod \*= nums[j]  
 res = max(res, prod)

return res

Community Solutions (Python)

• WalkCC

```
class Solution:
 def maxProduct(self, nums: List[int]) -> int:
 ans = nums[0]
 dpMin = nums[0] # the minimum so far
 dpMax = nums[0] # the maximum so far

 for i in range(1, len(nums)):
 num = nums[i]
 prevMin = dpMin # dpMin[i - 1]
 prevMax = dpMax # dpMax[i - 1]

 if num < 0:
 dpMin = min(prevMax * num, num)
 dpMax = max(prevMin * num, num)
 else:
 dpMin = min(prevMin * num, num)
 dpMax = max(prevMax * num, num)

 ans = max(ans, dpMax)

 return ans
```

#### • LeetCode

```
class Solution:
 def maxProduct(self, nums: List[int]) -> int:
 ans = f = g = nums[0]
 for x in nums[1:]:
 ff, gg = f, g
 f = max(x, ff * x, gg * x)
 g = min(x, ff * x, gg * x)
 ans = max(ans, f)

 return ans
```

Dynamic Programming

p.2143

#### • SimplyLeet

```
def maxProduct(nums):
 # If the input array is empty, return 0 (though
 # problem guarantees at least one element)
 if not nums:
 return 0
 # Initialize maximum, minimum product ending here and
 # the result.
 max_ending_here = nums[0]
 min_ending_here = nums[0]
 global_max = nums[0]

 # Iterate over the array starting from the second
 # element.

 for num in nums[1:]:
 # If the current number is negative, swap max and
 # min.
 if num < 0:
 max_ending_here, min_ending_here =
 min_ending_here, max_ending_here
 # Update the max product at the current index.
 max_ending_here = max(num, max_ending_here * num)
 # Update the min product at the current index.
 min_ending_here = min(num, min_ending_here * num)
 # Update the global maximum product.
 global_max = max(global_max, max_ending_here)

 return global_max

Example usage:
print(maxProduct([2,3,-2,4])) # Output: 6
```

Dynamic Programming

p.2144

#### • LC.ca

```
class Solution:
 def maxProduct(self, nums: List[int]) -> int:
 ans = f = g = nums[0]
 for x in nums[1:]:
 ff, gg = f, g
 f = max(x, ff * x, gg * x)
 g = min(x, ff * x, gg * x)
 ans = max(ans, f)

 return ans
```

Dynamic Programming

p.2145

### Dynamic Programming

MED

## #198 House Robber

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming

Lists: Grind 169

Time Complexity:  $O(n)$ , where  $n$  is the number of houses.

Space Complexity:  $O(1)$ , using only two variables to track the previous decisions.

#### Problem

You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed, the only constraint stopping you from robbing each of them is that adjacent houses have security systems connected and it will automatically contact the police if two adjacent houses were broken into on the same night.

Given an integer array `nums` representing the amount of money of each house, return the maximum amount of money you can rob tonight without alerting the police.

Example 1:

```
Input: nums = [1,2,3,1]
Output: 4
Explanation: Rob house 1 (money = 1) and then rob house
3 (money = 3).
Total amount you can rob = 1 + 3 = 4.
```

#### Example 2:

```
Input: nums = [2,7,9,3,1]
Output: 12
Explanation: Rob house 1 (money = 2), rob house 3
(money = 9) and rob house 5 (money = 1).
Total amount you can rob = 2 + 9 + 1 = 12.
```

#### Constraints:

- $1 \leq \text{nums.length} \leq 100$
- $0 \leq \text{nums}[i] \leq 400$

#### Community Solutions (Python)

#### • WalkCC

```
class Solution:
 def rob(self, nums: List[int]) -> int:
 if not nums:
 return 0
 if len(nums) == 1:
 return nums[0]

 # dp[i] := max money of robbing nums[0..i]
 dp = [0] * len(nums)
 dp[0] = nums[0]
```

Dynamic Programming

p.2147

```
dp[1] = max(nums[0], nums[1])

for i in range(2, len(nums)):
 dp[i] = max(dp[i - 1], dp[i - 2] + nums[i])

return dp[-1]
```

```
class Solution:
 def rob(self, nums: List[int]) -> int:
 prev1 = 0 # dp[i - 1]
 prev2 = 0 # dp[i - 2]
```

```
for num in nums:
 dp = max(prev1, prev2 + num)
 prev2 = prev1
 prev1 = dp
```

return prev1

#### • LeetCode

```
class Solution:
 def rob(self, nums: List[int]) -> int:
 @cache
 def dfs(i: int) -> int:
 if i >= len(nums):
 return 0
 return max(nums[i] + dfs(i + 2), dfs(i + 1))

 return dfs(0)
```

Dynamic Programming

p.2148

```
class Solution:
 def rob(self, nums: List[int]) -> int:
 n = len(nums)
 f = [0] * (n + 1)
 f[1] = nums[0]
 for i in range(2, n + 1):
 f[i] = max(f[i - 1], f[i - 2] + nums[i - 1])
 return f[n]
```

```
class Solution:
 def rob(self, nums: List[int]) -> int:
 f = g = 0
 for x in nums:
 f, g = max(f, g), f + x
 return max(f, g)
```

#### • SimplyLeet

```
Definition of the Solution class
class Solution:
 def rob(self, nums: List[int]) -> int:
 # Edge case: if there are no houses, return 0
 if not nums:
 return 0

 # Initialize two variables to keep track of the
 # max amounts
 prev1, prev2 = 0, 0
```

Dynamic Programming

p.2149

```
Iterate over each house's money
for money in nums:
 # Calculate the maximum money that can be
 # robbed if we consider the current house
 current = max(prev1, prev2 + money) # Either
 # skip current house or rob it
 # Update variables: prev2 becomes previous
 # best, and prev1 becomes current best
 prev2 = prev1
 prev1 = current

Return the maximum money that can be robbed
return prev1
```

#### • LC.ca

```
greedy
class Solution:
 def rob(self, nums: List[int]) -> int:
 not_rob, rob = 0, nums[0]
 for num in nums[1:]:
 # must max check
 # eg. first robbed 99999, then following ones
 # are just 3,3,3,3,3
 not_rob, rob = rob, max(num + not_rob, rob)
 return rob
```

Dynamic Programming

p.2150

```
#####

class Solution(object):
 def rob(self, nums):
 """
 :type nums: List[int]
 :rtype: int
 """
 if len(nums) == 0:
 return 0

 if len(nums) <= 2:
 return max(nums)
 dp = [0 for i in range(0, 2)]
 dp[0] = nums[0]
 dp[1] = max(nums[1], nums[0])
 for i in range(2, len(nums)):
 dp[i % 2] = max(dp[(i - 1) % 2], dp[(i - 2) % 2] +
 nums[i])
 return dp[(len(nums) - 1) % 2]
```

Dynamic Programming

p.2151

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| <div> <div>Dynamic Programming</div> <div>#256 Paint House</div> <div>MED</div> </div> <div> <div> LeetCode · SimplyLeet · WalkCC · LeetCode </div> <div> Array · Dynamic Programming </div> <div> Lists: Premium 98 </div> <div> Time Complexity: O(n), where n is the number of houses.<br/> Space Complexity: O(1) extra space if the input matrix is modified in place. </div> <div> <b>Problem</b> <p>There is a row of n houses, where each house can be painted one of three colors: red, blue, or green. The cost of painting each house with a certain color is different. You have to paint all the houses such that no two adjacent houses have the same color.</p> <p>The cost of painting each house with a certain color is represented by an n x 3 cost matrix costs.</p> <ul style="list-style-type: none"> <li>For example, costs[0][0] is the cost of painting house 0 with the color red; costs[1][2] is the cost of painting house 1 with color green, and so on...</li> </ul> <p>Return the minimum cost to paint all houses.</p> <p>Example 1:</p> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2152</div> </div> | <div> <div> <div>Input: costs = [[17,2,17],[16,16,5],[14,3,19]]</div> <div>Output: 10</div> <div>Explanation: Paint house 0 into blue, paint house 1 into green, paint house 2 into blue.</div> <div>Minimum cost: 2 + 5 + 3 = 10.</div> </div> <div> <div>Example 2:</div> <div> <div>Input: costs = [[7,6,2]]</div> <div>Output: 2</div> </div> </div> <div> <div>Constraints:</div> <ul style="list-style-type: none"> <li>costs.length == n</li> <li>costs[i].length == 3</li> <li>1 &lt;= n &lt;= 100</li> <li>1 &lt;= costs[i][j] &lt;= 20</li> </ul> </div> <div> <div>Brute Force (Python)</div> <div> <pre># Brute Force Approach class Solution:     def minCost(self, costs):         # Brute Force O(2^n) - plain recursion, no         memoisation         def recurse(i):             if i &gt;= len(costs): return 0             take = costs[i] + recurse(i + 1)             skip = recurse(i + 1)             return max(take, skip)         return recurse(0)</pre> </div> </div> <div> <div>Community Solutions (Python)</div> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2153</div> </div> | <div> <div> <div>• WalkCC</div> <div> <pre>class Solution:     def minCost(self, costs: List[List[int]]) -&gt; int:         list[list[int]]:             for i in range(1, len(costs)):                 costs[i][0] += min(costs[i - 1][1], costs[i - 1][2])                 costs[i][1] += min(costs[i - 1][0], costs[i - 1][2])                 costs[i][2] += min(costs[i - 1][0], costs[i - 1][1])          return min(costs[-1])</pre> </div> </div> <div> <div>• LeetCode</div> <div> <pre>class Solution:     def minCost(self, costs: List[List[int]]) -&gt; int:         a = b = c = 0         for ca, cb, cc in costs:             a, b, c = min(b, c) + ca, min(a, c) + cb, min(a, b) + cc         return min(a, b, c)</pre> </div> </div> <div> <div>• SimplyLeet</div> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2154</div> </div> |
| <div> <div> <div># Python solution for the Paint House problem</div> <div> <pre>def minCost(costs):     # Get the number of houses     n = len(costs)     # Base case: If there are no houses, return 0 cost     if n == 0:         return 0      # Iterate from the second house to the last house      for i in range(1, n):         # Update the cost for painting the current house         red             costs[i][0] += min(costs[i-1][1], costs[i-1][2])         # Update the cost for painting the current house         blue             costs[i][1] += min(costs[i-1][0], costs[i-1][2])         # Update the cost for painting the current house         green             costs[i][2] += min(costs[i-1][0], costs[i-1][1])      # The answer is the minimum cost of painting the last house with any of the colors     return min(costs[-1])</pre> </div> </div> <div> <div># Example usage:</div> <div> <pre>example_costs = [[17,2,17],[16,16,5],[14,3,19]] print(minCost(example_costs)) # Output: 10</pre> </div> </div> <div> <div>• LC.ca</div> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2155</div> </div>       | <div> <div> <pre>class Solution:     def minCost(self, costs: List[List[int]]) -&gt; int:         a = b = c = 0         for ca, cb, cc in costs:             a, b, c = min(b, c) + ca, min(a, c) + cb, min(a, b) + cc         return min(a, b, c)  ##### ...  Two-dimensional dynamic array dp, where dp[i][j] represents the minimum cost of brushing to the i+1th house with color j,  and the state transition equation is: dp[i][j] = cost[i][j] + min(dp[i-1][(j + 1)% 3], dp[i-1][(j + 2)% 3])  ...  class Solution(object):     def minCost(self, costs):         """"</pre> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2156</div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <div> <div> <pre>:type costs: List[List[int]] :rtype: int """ if not costs:     return 0 dp = [0] * (len(costs[0])) dp[:] = costs[0] for i in range(1, len(costs)):     d0 = d1 = d2 = 0     for j in range(0, 3):          if j == 0:             d0 = min(dp[1], dp[2]) + costs[i][0]         elif j == 1:             d1 = min(dp[0], dp[2]) + costs[i][1]         else:             d2 = min(dp[0], dp[1]) + costs[i][2]     dp[0], dp[1], dp[2] = d0, d1, d2     return min(dp)</pre> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2157</div> </div>                                                                                                                                                                                                                                                                                          |
| <div> <div> <div>Dynamic Programming</div> <div>#309 Best Time to Buy and Sell Stock with Cooldown</div> <div>MED</div> </div> <div> <div> LeetCode · SimplyLeet · WalkCC · LeetCode </div> <div> Array · Dynamic Programming </div> <div> Lists: Denny Zhang </div> <div> Time Complexity: O(n), where n is the number of days.<br/> Space Complexity: O(1), since only a few variables are maintained for the states. </div> <div> <b>Problem</b> <p>You are given an array prices where prices[i] is the price of a given stock on the ith day.</p> <p>Find the maximum profit you can achieve. You may complete as many transactions as you like (i.e., buy one and sell one share of the stock multiple times) with the following restrictions:</p> <ul style="list-style-type: none"> <li>After you sell your stock, you cannot buy stock on the next day (i.e., cooldown one day).</li> </ul> <p>Note: You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).</p> <p>Example 1:</p> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2158</div> </div> </div>              | <div> <div> <div>Input: prices = [1,2,3,0,2]</div> <div>Output: 3</div> <div>Explanation: transactions = [buy, sell, cooldown, buy, sell]</div> </div> <div> <div>Example 2:</div> <div> <div>Input: prices = [1]</div> <div>Output: 0</div> </div> </div> <div> <div>Constraints:</div> <ul style="list-style-type: none"> <li>1 &lt;= prices.length &lt;= 5000</li> <li>0 &lt;= prices[i] &lt;= 1000</li> </ul> </div> <div> <div>Brute Force (Python)</div> <div> <pre># Brute Force Approach class Solution:     def maxProfit(self, prices):         # Brute Force O(3^n) - recursion: at each day         choose buy/sell/rest             def dfs(i, holding, cooldown):                 if i &gt;= len(prices): return 0                 if cooldown: return dfs(i+1, holding, False)          # must rest             if holding:                 sell = prices[i] + dfs(i+1, False, True)          # sell today             rest = dfs(i+1, True, False) # hold</pre> </div> </div> </div> <div> <div>Dynamic Programming</div> <div>p.2159</div> </div>                                                            | <div> <div> <pre>        return max(sell, rest)     else:         buy = -prices[i] + dfs(i+1, True, False)  # buy today         rest = dfs(i+1, False, False) # wait         return max(buy, rest)     return dfs(0, False, False)</pre> </div> </div> <div> <div>Community Solutions (Python)</div> </div> <div> <div>• WalkCC</div> <div> <pre>class Solution:     def maxProfit(self, prices: list[int]) -&gt; int:         sell = 0         hold = -math.inf         prev = 0          for price in prices:             cache = sell             sell = max(sell, hold + price)             hold = max(hold, prev - price)          prev = cache          return sell</pre> </div> </div> <div> <div>• LeetCode</div> </div> <div> <div>Dynamic Programming</div> <div>p.2160</div> </div>                                                                  |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|
| <pre>class Solution:     def maxProfit(self, prices: List[int]) -&gt; int:         @cache         def dfs(i: int, j: int) -&gt; int:             if i &gt;= len(prices):                 return 0             ans = dfs(i + 1, j)             if j:                 ans = max(ans, prices[i] + dfs(i + 2, 0))             else:                 ans = max(ans, -prices[i] + dfs(i + 1, 1))          return ans      return dfs(0, 0)  class Solution:     def maxProfit(self, prices: List[int]) -&gt; int:         n = len(prices)         f = [[0] * 2 for _ in range(n)]         f[0][1] = -prices[0]         for i in range(1, n):             f[i][0] = max(f[i - 1][0], f[i - 1][1] + prices[i])             f[i][1] = max(f[i - 1][1], f[i - 2][0] - prices[i])         return f[n - 1][0]</pre>                                                                                                                                                                                                                                     | Dynamic Programming | p.2161 |
| <pre>class Solution:     def maxProfit(self, prices: List[int]) -&gt; int:         f, f0, f1 = 0, 0, -prices[0]         for x in prices[1:]:             f, f0, f1 = f0, max(f0, f1 + x), max(f1, f - x)          return f0</pre> <p>• SimplyLeet</p> <p># Python Solution with line-by-line comments</p> <pre>def maxProfit(prices):     # Handle edge case: if there are no prices, profit is 0     if not prices:         return 0      # Initialize states:     # s0: Maximum profit when ready to buy     # s1: Maximum profit when holding a stock (bought)      # s2: Maximum profit during cooldown after a sale     s0, s1, s2 = 0, -prices[0], 0      # Iterate over prices starting from day 1     for price in prices[1:]:         # Store previous state values         prev_s0, prev_s1, prev_s2 = s0, s1, s2         # s0 can come either from staying in s0 or coming from a cooldown (s2)         s0 = max(prev_s0, prev_s2)         # s1 is updated by either keeping the stock or buying today with profit from s0</pre> | Dynamic Programming | p.2162 |
| <pre>s1 = max(prev_s1, prev_s0 - price) # s2 is achieved by selling the stock that we were holding (prev_s1 + price) s2 = prev_s1 + price # The final maximum profit is the maximum of not holding a stock (s0 or s2) return max(s0, s2)  # Example usage: print(maxProfit([1,2,3,0,2]))</pre> <p>• LC.ca</p> <pre>class Solution:     def maxProfit(self, prices: List[int]) -&gt; int:         f1, f2, f3 = -prices[0], 0, 0         for price in prices[1:]:             pf1, pf2, pf3 = f1, f2, f3             f1 = max(pf1, pf3 - price)             f2 = max(pf2, pf1 + price)             f3 = max(pf3, pf2) # cooldown          return f2  #####  class Solution:     def maxProfit(self, prices: List[int]) -&gt; int:         f, f0, f1 = 0, 0, -prices[0]         for x in prices[1:]:             f, f0, f1 = f0, max(f0, f1 + x), max(f1, f - x)          return f0  #####</pre>                                                                                                                                               | Dynamic Programming | p.2163 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                     |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|
| <pre>class Solution(object):     def maxProfit(self, prices):         """         :type prices: List[int]         :rtype: int         """         if len(prices) &lt; 2:             return 0         buy = [0] * len(prices)          sell = [0] * len(prices)         buy[0] = -prices[0]         buy[1] = max(-prices[1], buy[0])         sell[0] = 0         sell[1] = max(prices[1] - prices[0], 0)         for i in range(2, len(prices)):             buy[i] = max(sell[i - 2] - prices[i], buy[i - 1])             sell[i] = max(prices[i] + buy[i - 1], sell[i - 1])         return max(sell)</pre>                                                                                                                                                                                                      | Dynamic Programming | p.2164 |
| <p>Dynamic Programming</p> <h3>#377 Combination Sum IV</h3> <p>LeetDoocs · SimplyLeet · WalkCC · LeetCode</p> <p>Array · Dynamic Programming</p> <p>Lists: Grind 169</p> <p>Time Complexity: <math>O(\text{target} * n)</math>, where <math>n</math> is the length of nums. Space Complexity: <math>O(\text{target})</math>, for the dp array of size target + 1.</p> <p><b>Problem</b></p> <p>Given an array of distinct integers nums and a target integer target, return the number of possible combinations that add up to target.</p> <p>The test cases are generated so that the answer can fit in a 32-bit integer.</p> <p>Example 1:</p> <pre>Input: nums = [1,2,3], target = 4 Output: 7 Explanation: The possible combination ways are: (1, 1, 1, 1) (1, 1, 2) (1, 2, 1) (1, 3)</pre> <p>Example 2:</p> | Dynamic Programming | p.2165 |
| <pre>class Solution:     def combinationSum4(self, nums: List[int], target: int) -&gt; int:         dp = [1] + [-1] * target          def dfs(target: int) -&gt; int:             if target &lt; 0:                 return 0             if dp[target] != -1:                 return dp[target]              dp[target] = sum(dfs(target - num) for num in nums)             return dp[target]          return dfs(target)</pre> <p>• LeetDoocs</p> <pre>class Solution:     def combinationSum4(self, nums: List[int], target: int) -&gt; int:         f = [1] + [0] * target         for i in range(1, target + 1):             for x in nums:                 if i &gt;= x:                     f[i] += f[i - x]         return f[target]</pre> <p>• SimplyLeet</p>                                            | Dynamic Programming | p.2166 |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                     |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|
| <pre>class Solution:     def combinationSum4(self, nums: List[int], target: int) -&gt; int:         dp = [1] + [-1] * target          def dfs(target: int) -&gt; int:             if target &lt; 0:                 return 0             if dp[target] != -1:                 return dp[target]              dp[target] = sum(dfs(target - num) for num in nums)             return dp[target]          return dfs(target)</pre> <p>• LeetDoocs</p> <pre>class Solution:     def combinationSum4(self, nums: List[int], target: int) -&gt; int:         f = [1] + [0] * target         for i in range(1, target + 1):             for x in nums:                 if i &gt;= x:                     f[i] += f[i - x]         return f[target]</pre> <p>• SimplyLeet</p>                                                                                                                                                                                          | Dynamic Programming | p.2167 |
| <p># Python solution using dynamic programming</p> <pre>class Solution:     def combinationSum4(self, nums, target):         # Initialize dp array where dp[i] stores the number of ways to reach sum i         dp = [0] * (target + 1)         # Base case: there is one way to form a sum of 0 (use no elements)         dp[0] = 1          # Fill the dp table for each sum from 1 to target          for i in range(1, target + 1):             for num in nums:                 if i - num &gt;= 0:                     dp[i] += dp[i - num]         return dp[target]</pre> <p>• LC.ca</p> <pre>class Solution:     def combinationSum4(self, nums: List[int], target: int) -&gt; int:         f = [1] + [0] * target         for i in range(1, target + 1):             for x in nums:                 if i &gt;= x:                     f[i] += f[i - x]         return f[target]</pre>                                                                 | Dynamic Programming | p.2168 |
| <p>Dynamic Programming</p> <h3>#416 Partition Equal Subset Sum</h3> <p>LeetDoocs · SimplyLeet · WalkCC · LeetCode</p> <p>Array · Dynamic Programming</p> <p>Lists: Grind 169</p> <p>Time Complexity: <math>O(n * \text{target})</math>, where target is half of the total sum. Space Complexity: <math>O(\text{target})</math>, using a one-dimensional DP array.</p> <p><b>Problem</b></p> <p>Given an integer array nums, return true if you can partition the array into two subsets such that the sum of the elements in both subsets is equal or false otherwise.</p> <p>Example 1:</p> <pre>Input: nums = [1,5,11,5] Output: true Explanation: The array can be partitioned as [1, 5, 5] and [11].</pre> <p>Example 2:</p> <pre>Input: nums = [1,2,3,5] Output: false Explanation: The array cannot be partitioned into equal sum subsets.</pre> <p><b>Constraints:</b></p> <ul style="list-style-type: none"><li>1 &lt;= nums.length &lt;= 200</li></ul> | Dynamic Programming | p.2169 |

• 1 <= nums[i] <= 100

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def canPartition(self, nums):
 # Brute Force O(2^n) - try including/excluding
 each element for target sum
 total = sum(nums)
 if total % 2: return False
 target = total // 2
 def can_reach(i, rem):
 if rem == 0: return True
 if i == len(nums) or rem < 0: return False
```

```
 return can_reach(i+1, rem - nums[i]) or
 can_reach(i+1, rem)
 return can_reach(0, target)
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def canPartition(self, nums: list[int]) -> bool:
 summ = sum(nums)
 if summ % 2 == 1:
 return False
 return self.knapsack_(nums, summ // 2)

 def knapsack_(self, nums: list[int], subsetSum: int) ->
 bool:
 n = len(nums)
 # dp[i][j] := True if j can be formed by nums[0..i]

 dp = [[False] * (subsetSum + 1) for _ in range(n + 1)]
 dp[0][0] = True

 for i in range(1, n + 1):
 num = nums[i - 1]
 for j in range(subsetSum + 1):
 if j < num:
 dp[i][j] = dp[i - 1][j]
 else:
 dp[i][j] = dp[i - 1][j] or dp[i - 1][j - num]

 return dp[n][subsetSum]
```

• LeetCode

```
class Solution:
 def canPartition(self, nums: List[int]) -> bool:
 m, mod = divmod(sum(nums), 2)
 if mod:
 return False
 n = len(nums)
 f = [[False] * (m + 1) for _ in range(n + 1)]
 f[0][0] = True
 for i, x in enumerate(nums, 1):
 for j in range(m + 1):
 f[i][j] = f[i - 1][j] or (j >= x and f[i
 - 1][j - x])
 return f[n][m]
```

• SimplyLeet

# Python solution for Partition Equal Subset Sum

```
def canPartition(nums):
 # Calculate the total sum of the array
 total_sum = sum(nums)

 # If the total sum is odd, equal partition is not
 possible
 if total_sum % 2 != 0:
 return False
```

Dynamic Programming

p.2170

Dynamic Programming

p.2171

Dynamic Programming

p.2172

```
Set the target sum to half of the total sum
target = total_sum // 2

Initialize dp array where dp[i] is True if a subset
sum i is achievable
dp = [False] * (target + 1)
dp[0] = True # Base case: subset sum of 0 is always
possible
```

```
Iterate through each number in the array
for num in nums:
 # Iterate backwards to avoid overwriting dp
 values needed for current iteration
```

```
 for i in range(target, num - 1, -1):
 # Update dp[i] if dp[i-num] is True
 if dp[i - num]:
 dp[i] = True
```

```
dp[target] is True if a subset adds up to target
sum
return dp[target]
```

```
Example usage:
print(canPartition([1, 5, 11, 5]))
```

• LC.ca

```
class Solution:
 def canPartition(self, nums: List[int]) -> bool:
 s = sum(nums)
 if s % 2 != 0:
 return False
 n = s >> 1
 dp = [False] * (n + 1)
 dp[0] = True
 for v in nums:
 for target in range(n, v - 1, -1): #
 including v itself
```

```
 dp[target] = dp[target] or dp[target - v]
 return dp[-1]
```

```
class Solution: # recursive, but over time limit in OJ
 def canPartition(self, nums: List[int]) -> bool:
 total_sum = sum(nums)
 if total_sum % 2 != 0:
 return False
 target_sum = total_sum // 2
```

```
 memo = {} # ==> or just use @cache for dfs()
```

```
 def dfs(idx, curr_sum):
 if curr_sum == target_sum:
 return True
 if curr_sum > target_sum or idx >= len(nums):
 return False
 if (idx, curr_sum) in memo:
 return memo[(idx, curr_sum)]
```

```
Explore two options: include the current
number or exclude it
include = dfs(idx + 1, curr_sum + nums[idx])
exclude = dfs(idx + 1, curr_sum)

memo[(idx, curr_sum)] = include or exclude
return memo[(idx, curr_sum)]

return dfs(0, 0)
```

#####

```
class Solution:
 def canPartition(self, nums: List[int]) -> bool:
 m, mod = divmod(sum(nums), 2)
 if mod:
 return False
 f = [True] + [False] * m
 for x in nums:
 for j in range(m, x - 1, -1):
 f[j] = f[j] or f[j - x]
```

```
 return f[m]
```

Dynamic Programming

p.2173

Dynamic Programming

p.2174

Dynamic Programming

p.2175

Dynamic Programming

#435 Non-overlapping Intervals

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Dynamic Programming · Greedy · Sorting

Lists: Grind 169

Time Complexity: O(n log n) due to sorting the intervals.  
Space Complexity: O(n) if considering the storage for the  
sorted list (or O(1) if sorting in-place).

Problem

Given an array of intervals intervals where intervals[i] =  
[starti, endi], return the minimum number of intervals  
you need to remove to make the rest of the intervals  
non-overlapping.

Note that intervals which only touch at a point are  
non-overlapping. For example, [1, 2] and [2, 3] are  
non-overlapping.

Example 1:

```
Input: intervals = [[1,2],[2,3],[3,4],[1,3]]
Output: 1
Explanation: [1,3] can be removed and the rest of the
intervals are non-overlapping.
```

Example 2:

```
Input: intervals = [[1,2],[1,2],[1,2]]
Output: 2
Explanation: You need to remove two [1,2] to make the
rest of the intervals non-overlapping.
```

Example 3:

```
Input: intervals = [[1,2],[2,3]]
Output: 0
Explanation: You don't need to remove any of the
intervals since they're already non-overlapping.
```

Constraints:

- 1 <= intervals.length <= 105
- intervals[i].length == 2
- -5 \* 104 <= starti < endi <= 5 \* 104

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def eraseOverlapIntervals(self, intervals):
 # Brute Force O(2^n) - try removing every subset,
 keep smallest valid
 # (In practice: sort by end, greedy is O(n log
 n); brute shown for intuition)
 intervals.sort(key=lambda x: x[1])
 removed = 0
 end = float('-inf')
 for iv in intervals:
 if iv[0] >= end:
```

```
 end = iv[1] # keep this interval
 else:
 removed += 1 # drop overlapping interval
 return removed
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def eraseOverlapIntervals(self, intervals:
 list[list[int]]) -> int:
 ans = 0
 currentEnd = -math.inf

 for interval in sorted(intervals, key=lambda x:
 x[1]):
 if interval[0] >= currentEnd:
 currentEnd = interval[1]
 else:
 ans += 1

 return ans
```

• LeetCode

Dynamic Programming

p.2176

Dynamic Programming

p.2177

Dynamic Programming

p.2178



```
class Solution:
 def eraseOverlapIntervals(self, intervals:
List[List[int]]) -> int:
 intervals.sort(key=lambda x: x[1])
 ans = len(intervals)
 pre = -inf
 for l, r in intervals:
 if pre <= l:
 ans -= 1
 pre = r
 return ans
```

• SimplyLeet

# Python Implementation for Non-overlapping Intervals

```
def eraseOverlapIntervals(intervals):
 # Sort intervals based on the end time
 intervals.sort(key=lambda x: x[1])

 # Initialize the count of intervals in the
non-overlapping set
 count = 0
 # Initialize the end of the last added interval to
the minimum possible value
 last_included_end = float('-inf')
```

Dynamic Programming

p.2179

```
Iterate over every interval in the sorted list
for interval in intervals:
 # If the current interval does not overlap with
the last added interval
 if interval[0] >= last_included_end:
 count += 1 # We include this interval
 last_included_end = interval[1] # Update end
for subsequent comparisons
 # The minimum number of removals is total intervals
minus non-overlapping intervals count
 return len(intervals) - count
```

```
Example usage:
print(eraseOverlapIntervals([[1,2],[2,3],[3,4],[1,3]])) #
Expected output: 1
```

• LC.ca

```
class Solution:
 def eraseOverlapIntervals(self, intervals:
List[List[int]]) -> int:
 intervals.sort(key=lambda x: x[1])
 ans, t = 0, intervals[0][1]
 for s, e in intervals[1:]:
 if s >= t:
 t = e
 else:
 ans += 1
 return ans
```

Dynamic Programming

p.2180

Dynamic Programming

## #516 Longest Palindromic Subsequence

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

String · Dynamic Programming

Lists: Denny Zhang

Time Complexity:  $O(n^2)$  where  $n$  is the length of the string. Space Complexity:  $O(n^2)$  due to the usage of a 2D DP table.

### Problem

Given a string  $s$ , find the longest palindromic subsequence's length in  $s$ .

A subsequence is a sequence that can be derived from another sequence by deleting some or no elements without changing the order of the remaining elements.

Example 1:

```
Input: s = "bbbab"
Output: 4
Explanation: One possible longest palindromic
subsequence is "bbbb".
```

Example 2:

```
for i in range(n - d):
 j = i + d
 if s[i] == s[j]:
 dp[i][j] = 2 + dp[i + 1][j - 1]
 else:
 dp[i][j] = max(dp[i + 1][j], dp[i][j - 1])

return dp[0][n - 1]
```

• LeetDoocs

```
class Solution:
 def longestPalindromeSubseq(self, s: str) -> int:
 n = len(s)
 f = [[0] * n for _ in range(n)]
 for i in range(n):
 f[i][i] = 1
 for i in range(n - 1, -1, -1):
 for j in range(i + 1, n):
 if s[i] == s[j]:
 f[i][j] = f[i + 1][j - 1] + 2
 else:
 f[i][j] = max(f[i + 1][j], f[i][j - 1])

 return f[0][-1]
```

• SimplyLeet

Dynamic Programming

p.2184

```
Input: s = "cbdd"
Output: 2
Explanation: One possible longest palindromic
subsequence is "bb".
```

### Constraints:

- $1 \leq s.length \leq 1000$
- $s$  consists only of lowercase English letters.

### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def longest_palindromic_subseq(self, s):
 # Brute Force $O(2^n)$ - plain recursion, no
memoisation
 def recurse(i):
 if i >= len(s): return 0
 take = s[i] + recurse(i + 1)
 skip = recurse(i + 1)
 return max(take, skip)
 return recurse(0)
```

### Community Solutions (Python)

• WalkCC

```
class Solution:
 def longestPalindromeSubseq(self, s: str) -> int:
 @functools.lru_cache(None)
 def dp(i: int, j: int) -> int:
 """Returns the length of LPS(s[i..j])."""
 if i > j:
 return 0
 if i == j:
 return 1
 if s[i] == s[j]:
```

```
 return 2 + dp(i + 1, j - 1)
 return max(dp(i + 1, j), dp(i, j - 1))

 return dp(0, len(s) - 1)
```

```
class Solution:
 def longestPalindromeSubseq(self, s: str) -> int:
 n = len(s)
 # dp[i][j] := the length of LPS(s[i..j])
 dp = [[0] * n for _ in range(n)]

 for i in range(n):
 dp[i][i] = 1

 for d in range(1, n):
```

Dynamic Programming

p.2183

```
Function to compute the longest palindromic subsequence
length
def longest_palindromic_subseq(s):
 n = len(s)
 # Create a 2D DP table initialized to 0
 dp = [[0] * n for _ in range(n)]

 # Base case: every single character is a palindrome
of length 1
 for i in range(n):
 dp[i][i] = 1
```

```
Build the table. 'sub_len' is the current length of
substring we are considering.
for sub_len in range(2, n+1):
 for i in range(n - sub_len + 1):
 j = i + sub_len - 1 # Compute the end index of
the substring
 if s[i] == s[j]:
 # Characters match, add 2 to the result
from the inner substring if available
 dp[i][j] = 2 + (dp[i+1][j-1] if sub_len >
2 else 0)
 else:
 # Characters don't match, choose the
maximum from either excluding left or right char
 dp[i][j] = max(dp[i+1][j], dp[i][j-1])
```

Dynamic Programming

p.2185

```
The answer for the full string is stored at
dp[0][n-1]
return dp[0][n-1]

Example usage:
s = "bbbab"
print(longest_palindromic_subseq(s)) # Expected output: 4
```

• LC.ca

```
class Solution:
 def longestPalindromeSubseq(self, s: str) -> int:
 n = len(s)
 dp = [[0] * n for _ in range(n)]
 for i in range(n - 1, -1, -1):
 dp[i][i] = 1 # pre-set before inner loop
 for j in range(i + 1, n):
 if s[i] == s[j]:
 dp[i][j] = dp[i + 1][j - 1] + 2
 else:
 dp[i][j] = max(dp[i + 1][j], dp[i][j
- 1])
 return dp[0][-1]

#####

class Solution:
 def longestPalindromeSubseq(self, s: str) -> int:
 n = len(s)
 dp = [[0] * n for _ in range(n)]
 for i in range(n):
```

Dynamic Programming

p.2186

```
dp[i][i] = 1 # cannot move it inside 'for j
in range(1, n)'
because below 'for j in range' starting at
1
so cannot put dp[j][j]=1 inside below for
loop
for j in range(1, n):
 for i in range(j - 1, -1, -1):
 if s[i] == s[j]:
 dp[i][j] = dp[i + 1][j - 1] + 2
 # "aa" => i=0,j=1 => dp[1][0] is 0,
still works
 else:
 dp[i][j] = max(dp[i + 1][j], dp[i][j
- 1])

return dp[0][-1]
```

Dynamic Programming

p.2187

## #576 Out of Boundary Paths

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Dynamic Programming

Lists: Denny Zhang

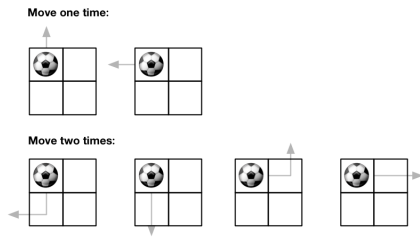
Time Complexity:  $O(\text{maxMove} * m * n)$ , since we compute each state at most once. Space Complexity:  $O(\text{maxMove} * m * n)$  for the memoization table.

## Problem

There is an  $m \times n$  grid with a ball. The ball is initially at the position  $[\text{startRow}, \text{startColumn}]$ . You are allowed to move the ball to one of the four adjacent cells in the grid (possibly out of the grid crossing the grid boundary). You can apply at most  $\text{maxMove}$  moves to the ball.

Given the five integers  $m, n, \text{maxMove}, \text{startRow}, \text{startColumn}$ , return the number of paths to move the ball out of the grid boundary. Since the answer can be very large, return it modulo  $10^9 + 7$ .

Example 1:



Input:  $m = 2, n = 2, \text{maxMove} = 2, \text{startRow} = 0, \text{startColumn} = 0$   
Output: 6

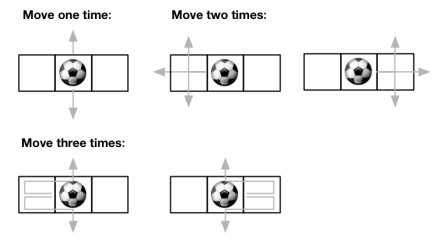
Example 2:

Dynamic Programming

p.2188

Dynamic Programming

p.2189



Input:  $m = 1, n = 3, \text{maxMove} = 3, \text{startRow} = 0, \text{startColumn} = 1$   
Output: 12

## Constraints:

- $1 \leq m, n \leq 50$
- $0 \leq \text{maxMove} \leq 50$
- $0 \leq \text{startRow} < m$
- $0 \leq \text{startColumn} < n$

Brute Force (Python)

Dynamic Programming

p.2190

```
Brute Force Approach
class Solution:
 def findPaths(self, m, n, maxMove, startRow, startColumn):
 # Brute Force $O(2^n)$ - plain recursion, no memoisation
 def recurse(i):
 if i >= len(maxMove): return 0
 take = maxMove[i] + recurse(i + 1)
 skip = recurse(i + 1)
 return max(take, skip)
 return recurse(0)
```

## Community Solutions (Python)

## • WalkCC

```
class Solution:
 def findPaths(self, m, n, maxMove, startRow, startColumn):
 MOD = 1000000007

 @functools.lru_cache(None)
 def dp(k: int, i: int, j: int) -> int:
 """
 Returns the number of paths to move the ball at (i, j) out-of-bounds with k moves.
 """
```

Dynamic Programming

p.2191

```
if i < 0 or i == m or j < 0 or j == n:
 return 1
if k == 0:
 return 0
return (dp(k - 1, i + 1, j) + dp(k - 1, i - 1, j) +
 dp(k - 1, i, j + 1) + dp(k - 1, i, j - 1)) % MOD

MOD
return dp(maxMove, startRow, startColumn)

class Solution:
 def findPaths(self, m: int, n: int, maxMove: int, startRow: int, startColumn: int) -> int:
 DIRS = ((0, 1), (1, 0), (0, -1), (-1, 0))

 MOD = 1_000_000_007
 ans = 0
 # dp[i][j] := the number of paths to move the ball (i, j) out-of-bounds
 dp = [[0] * n for _ in range(m)]
 dp[startRow][startColumn] = 1

 for _ in range(maxMove):
 newDp = [[0] * n for _ in range(m)]
 for i in range(m):
 for j in range(n):
```

Dynamic Programming

p.2192

```
if dp[i][j] > 0:
 for dx, dy in DIRS:
 x = i + dx
 y = j + dy
 if x < 0 or x == m or y < 0 or y == n:
 ans = (ans + dp[i][j]) % MOD
 else:
 newDp[x][y] = (newDp[x][y] + dp[i][j]) % MOD

MOD
dp = newDp

return ans
```

## • LeetCode

```
class Solution:
 def findPaths(self, m: int, n: int, maxMove: int, startRow: int, startColumn: int) -> int:
 @cache
 def dfs(i: int, j: int, k: int) -> int:
 if not 0 <= i < m or not 0 <= j < n:
 return int(k >= 0)
 if k <= 0:
 return 0
```

Dynamic Programming

p.2193

```
ans = 0
for a, b in pairwise(dirs):
 x, y = i + a, j + b
 ans = (ans + dfs(x, y, k - 1)) % mod
return ans

mod = 10**9 + 7
dirs = (-1, 0, 1, 0, -1)
return dfs(startRow, startColumn, maxMove)
```

## • SimplyLeet

```
MOD = 10**9 + 7

def findPaths(m, n, maxMove, startRow, startColumn):
 # Initialize memo dictionary to cache the states
 memo = {}

 def dp(i, j, movesLeft):
 # If out of grid bounds, it's a valid path
 if i < 0 or i >= m or j < 0 or j >= n:
 return 1

 # No moves left and still inside the grid means no valid path out
 if movesLeft == 0:
 return 0
 # Check if we already computed this state
 if (i, j, movesLeft) in memo:
 return memo[(i, j, movesLeft)]

 count = 0
 # Four possible directions: up, down, left, right
 count = (dp(i - 1, j, movesLeft - 1) +
```

Dynamic Programming

p.2194

```
dp(i + 1, j, movesLeft - 1) +
dp(i, j - 1, movesLeft - 1) +
dp(i, j + 1, movesLeft - 1)) % MOD

memo[(i, j, movesLeft)] = count
return count

return dp(startRow, startColumn, maxMove)

Example usage

if __name__ == "__main__":
 print(findPaths(2, 2, 2, 0, 0)) # Expected output: 6
```

## • LC.ca

```
class Solution:
 def findPaths(self, m: int, n: int, maxMove: int, startRow: int, startColumn: int) -> int:
 @cache
 def dfs(i, j, k):
 if i < 0 or j < 0 or i >= m or j >= n:
 return 1
 if k <= 0:
 return 0
```

Dynamic Programming

p.2195

```
res = 0
for a, b in [(-1, 0), (1, 0), (0, 1), (0, -1)]:
 x, y = i + a, j + b
 res += dfs(x, y, k - 1)
 res %= mod
return res

mod = 10**9 + 7
return dfs(startRow, startColumn, maxMove)
```

Dynamic Programming

p.2196

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <div><div>Dynamic Programming</div><div>#1143 Longest Common SubsequenceMED</div><div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div><div>String · Dynamic Programming</div><div>Lists: Denny Zhang</div><div>Time Complexity: <math>O(m * n)</math>, where <math>m</math> and <math>n</math> are the lengths of <code>text1</code> and <code>text2</code>. Space Complexity: <math>O(m * n)</math> for maintaining the 2D DP table. Space can be optimized to <math>O(\min(m, n))</math> if needed.</div><div>Problem</div><div>Given two strings <code>text1</code> and <code>text2</code>, return the length of their longest common subsequence. If there is no common subsequence, return 0.</div><div>A subsequence of a string is a new string generated from the original string with some characters (can be none) deleted without changing the relative order of the remaining characters.</div><div><ul style="list-style-type: none"><li>For example, "ace" is a subsequence of "abcde".</li></ul></div><div>A common subsequence of two strings is a subsequence that is common to both strings.</div><div>Example 1:</div><div></div><div>Dynamic Programmingp.2197</div></div> | <div><div>Input: <code>text1 = "abcde", text2 = "ace"</code><br/>Output: 3<br/>Explanation: The longest common subsequence is "ace" and its length is 3.</div><div>Example 2:</div><div>Input: <code>text1 = "abc", text2 = "abc"</code><br/>Output: 3<br/>Explanation: The longest common subsequence is "abc" and its length is 3.</div><div>Example 3:</div><div>Input: <code>text1 = "abc", text2 = "def"</code><br/>Output: 0<br/>Explanation: There is no such common subsequence, so the result is 0.</div><div>Constraints:</div><div><ul style="list-style-type: none"><li>1 &lt;= <code>text1.length</code>, <code>text2.length</code> &lt;= 1000</li><li><code>text1</code> and <code>text2</code> consist of only lowercase English characters.</li></ul></div><div>Community Solutions (Python)</div><div><ul style="list-style-type: none"><li>WalkCC</li></ul></div><div></div><div>Dynamic Programmingp.2198</div></div>                                                                                                                                                                                   | <div><div>class Solution:<br/>    def longestCommonSubsequence(self, text1: str, text2: str) -&gt; int:<br/>        m = len(text1)<br/>        n = len(text2)<br/>        # dp[i][j] := the length of LCS(text1[0..i], text2[0..j])<br/>        dp = [[0] * (n + 1) for _ in range(m + 1)]<br/><br/>        for i in range(m):<br/>            for j in range(n):<br/>                dp[i + 1][j + 1] = (1 + dp[i][j] if text1[i] == text2[j] else max(dp[i][j + 1], dp[i + 1][j]))<br/><br/>        return dp[m][n]</div><div>• LeetDoocs</div><div>class Solution:<br/>    def longestCommonSubsequence(self, text1: str, text2: str) -&gt; int:<br/>        m, n = len(text1), len(text2)<br/>        f = [[0] * (n + 1) for _ in range(m + 1)]<br/>        for i in range(1, m + 1):<br/>            for j in range(1, n + 1):<br/>                if text1[i - 1] == text2[j - 1]:<br/>                    f[i][j] = f[i - 1][j - 1] + 1<br/>                else:<br/>                    f[i][j] = max(f[i - 1][j], f[i][j - 1])</div><div></div><div>Dynamic Programmingp.2199</div></div>                                                                                                                         |
| <div><div>return f[m][n]</div><div>• SimplyLeet</div><div># Function to find the longest common subsequence length<br/>def longestCommonSubsequence(text1: str, text2: str) -&gt; int:<br/>    m, n = len(text1), len(text2)<br/>    # Initialize a (m+1) x (n+1) DP table filled with zeros<br/>    dp = [[0] * (n + 1) for _ in range(m + 1)]<br/><br/>    # Build the DP table bottom-up<br/>    for i in range(1, m + 1):<br/>        for j in range(1, n + 1):<br/>            # If characters match, extend the subsequence length by 1<br/><br/>            if text1[i - 1] == text2[j - 1]:<br/>                dp[i][j] = dp[i - 1][j - 1] + 1<br/>            else:<br/>                # Otherwise, choose the maximum from the left or top cell<br/>                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])<br/><br/>    return dp[m][n]<br/><br/># Example usage<br/>if __name__ == "__main__":</div><div></div><div>Dynamic Programmingp.2200</div></div>                                                                                                                                                                                                                     | <div><div>text1 = "abcde"<br/>text2 = "ace"<br/>print(longestCommonSubsequence(text1, text2)) #<br/>Output: 3</div><div>• LC.ca</div><div>class Solution:<br/>    def longestCommonSubsequence(self, text1: str, text2: str) -&gt; int:<br/>        m, n = len(text1), len(text2)<br/>        f = [[0] * (n + 1) for _ in range(m + 1)]<br/>        for i in range(1, m + 1):<br/>            for j in range(1, n + 1):<br/>                if text1[i - 1] == text2[j - 1]:<br/>                    f[i][j] = f[i - 1][j - 1] + 1<br/>                else:<br/>                    f[i][j] = max(f[i - 1][j], f[i][j - 1])<br/><br/>    return f[m][n]</div><div></div><div>Dynamic Programmingp.2201</div></div>                                                                                                                                                                                                                                                                                                                                                                                                        | <div><div>Dynamic Programming</div><div>#10 Regular Expression MatchingHAR</div><div>LeetDoocs · SimplyLeet · WalkCC · LeetCode</div><div>String · Dynamic Programming · Recursion</div><div>Lists: Denny Zhang</div><div>Time Complexity: <math>O(m * n)</math> where <math>m</math> = length(s) and <math>n</math> = length(p), due to exploring each subproblem combination. Space Complexity: <math>O(m * n)</math> for storing the memoization table in the dynamic programming or re</div><div>Problem</div><div>Given an input string <code>s</code> and a pattern <code>p</code>, implement regular expression matching with support for <code>'.'</code> and <code>'*'</code> where:</div><div><ul style="list-style-type: none"><li><code>'.'</code> Matches any single character.</li><li><code>'*'</code> Matches zero or more of the preceding element.</li></ul></div><div>Return a boolean indicating whether the matching covers the entire input string (not partial).</div><div>Example 1:</div><div>Input: <code>s = "aa", p = "a*"</code><br/>Output: true<br/>Explanation: "a*" does not match the entire string "aa".</div><div>Example 2:</div><div></div><div>Dynamic Programmingp.2202</div></div> |
| <div><div>Input: <code>s = "aa", p = "a*"</code><br/>Output: true<br/>Explanation: <code>'*'</code> means zero or more of the preceding element, 'a'. Therefore, by repeating 'a' once, it becomes "aa".</div><div>Example 3:</div><div>Input: <code>s = "ab", p = ".*"</code><br/>Output: true<br/>Explanation: <code>".*"</code> means "zero or more (*) of any character (.)".</div><div>Constraints:</div><div><ul style="list-style-type: none"><li>1 &lt;= <code>s.length</code> &lt;= 20</li><li>1 &lt;= <code>p.length</code> &lt;= 20</li><li><code>s</code> contains only lowercase English letters.</li><li><code>p</code> contains only lowercase English letters, <code>'.'</code>, and <code>'*'</code>.</li><li>It is guaranteed for each appearance of the character <code>'*'</code>, there will be a previous valid character to match.</li></ul></div><div>Community Solutions (Python)</div><div><ul style="list-style-type: none"><li>WalkCC</li></ul></div><div></div><div>Dynamic Programmingp.2203</div></div>                                                                                                                                                         | <div><div>class Solution:<br/>    def isMatch(self, s: str, p: str) -&gt; bool:<br/>        m = len(s)<br/>        n = len(p)<br/>        # dp[i][j] := True if s[0..i] matches p[0..j]<br/>        dp = [[False] * (n + 1) for _ in range(m + 1)]<br/>        dp[0][0] = True<br/><br/>        def isMatch(i: int, j: int) -&gt; bool:<br/>            return j &gt;= 0 and p[j] == '.' or s[i] == p[j]<br/><br/>        for j, c in enumerate(p):<br/>            if c == '*' and dp[0][j - 1]:<br/>                dp[0][j + 1] = True<br/><br/>        for i in range(m):<br/>            for j in range(n):<br/>                if p[j] == '*':<br/>                    # The minimum index of '*' is 1.<br/>                    noRepeat = dp[i + 1][j - 1]<br/><br/>                    doRepeat = isMatch(i, j - 1) and dp[i][j + 1]<br/>                    dp[i + 1][j + 1] = noRepeat or doRepeat<br/>                elif isMatch(i, j):<br/>                    dp[i + 1][j + 1] = dp[i][j]<br/><br/>        return dp[m][n]</div><div>• LeetDoocs</div><div></div><div>Dynamic Programmingp.2204</div></div> | <div><div>class Solution:<br/>    def isMatch(self, s: str, p: str) -&gt; bool:<br/>        @cache<br/>        def dfs(i, j):<br/>            if j &gt;= n:<br/>                return i == m<br/>            if j + 1 &lt; n and p[j + 1] == '*':<br/>                return dfs(i, j + 2) or (<br/>                    i &lt; m and (s[i] == p[j] or p[j] == '.') and dfs(i + 1, j)<br/>                )<br/>            return i &lt; m and (s[i] == p[j] or p[j] == '.') and dfs(i + 1, j + 1)<br/><br/>        m, n = len(s), len(p)<br/>        return dfs(0, 0)<br/><br/>class Solution:<br/>    def isMatch(self, s: str, p: str) -&gt; bool:<br/>        m, n = len(s), len(p)<br/>        f = [[False] * (n + 1) for _ in range(m + 1)]<br/>        f[0][0] = True<br/>        for i in range(m + 1):<br/>            for j in range(1, n + 1):<br/>                if p[j - 1] == '*':<br/>                    f[i][j] = f[i][j - 2]<br/>                    if i &gt; 0 and (p[j - 2] == "." or s[i - 1] == p[j - 2]):</div><div></div><div>Dynamic Programmingp.2205</div></div>                                                                                                                              |

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| <pre>        f[i][j] != f[i - 1][j]         elif i &gt; 0 and (p[j - 1] == "." or s[i - 1] == p[j - 1]):             f[i][j] = f[i - 1][j - 1]             return f[m][n]</pre> <p>• <b>SimplyLeet</b></p> <p># Python solution using recursion with memoization</p> <pre>def isMatch(s: str, p: str) -&gt; bool:     memo = {}      def dp(i: int, j: int) -&gt; bool:         # If result is already computed, return it.         if (i, j) in memo:             return memo[(i, j)]          # If we have reached the end of the pattern, s         # must also be at the end.          if j == len(p):             ans = i == len(s)         else:             # Check if current characters match             first_match = i &lt; len(s) and (p[j] == s[i] or             p[j] == '.')              # If there is a '*' coming after the current             # pattern character             if j + 1 &lt; len(p) and p[j+1] == '*':                 # Two cases:                 # 1. '*' stands for zero occurrence -&gt;                 # skip current pattern char and '*'                 ans = dp(i, j+2) or (first_match and                 dp(i+1, j))             else:                 # No '*' means we directly check next                 # characters in both s and p.                 ans = first_match and dp(i+1, j+1)          memo[(i, j)] = ans         return ans      return dp(0, 0)</pre> <p># Example test run</p> <pre>#print(isMatch("aa", "a*")) # Expected output: True</pre> <p>• <b>LC.ca</b></p> <pre>class Solution:     def isMatch(self, s: str, p: str) -&gt; bool:         m, n = len(s), len(p)         f = [[False] * (n + 1) for _ in range(m + 1)]         f[0][0] = True         for i in range(m + 1):             for j in range(1, n + 1):                 if p[j - 1] == "*":                     f[i][j] = f[i][j - 2]                     if i &gt; 0 and (p[j - 2] == "." or s[i - 1] == p[j - 2]):                         f[i][j] = f[i - 1][j - 1]</pre> | Dynamic Programming | p.2206 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Dynamic Programming | p.2207 |
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| <p>Dynamic Programming</p> <h2>#115 Distinct Subsequences</h2> <p>LeetCode · SimplyLeet · WalkCC · LeetCode</p> <p>String · Dynamic Programming</p> <p>Lists: Denny Zhang</p> <p>Time Complexity: <math>O(n * m)</math>, where <math>n</math> is the length of <math>s</math> and <math>m</math> is the length of <math>t</math>. Space Complexity: <math>O(n * m)</math>, which can be optimized to <math>O(m)</math> with careful row reuse.</p> <p><b>Problem</b></p> <p>Given two strings <math>s</math> and <math>t</math>, return the number of distinct subsequences of <math>s</math> which equals <math>t</math>.</p> <p>The test cases are generated so that the answer fits on a 32-bit signed integer.</p> <p><b>Example 1:</b></p> <pre>Input: s = "rabbbit", t = "rabbit" Output: 3 Explanation: As shown below, there are 3 ways you can generate "rabbit" from s. rabbbit rabbbit rabbbit</pre> <p><b>Example 2:</b></p> |  |  |
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| <p>Dynamic Programming</p> <h2>#123 Best Time to Buy and Sell Stock III</h2> <p>LeetCode · SimplyLeet · WalkCC · LeetCode</p> <p>Array · Dynamic Programming</p> <p>Lists: Denny Zhang</p> <p>Time Complexity: <math>O(n)</math> where <math>n</math> is the number of days</p> <p>Space Complexity: <math>O(n)</math> due to extra arrays used for forward and backward computations (this can be optimized to <math>O(1)</math> with a state variable approach)</p> <p><b>Problem</b></p> <p>You are given an array prices where prices[i] is the price of a given stock on the <math>i</math>th day.</p> <p>Find the maximum profit you can achieve. You may complete at most two transactions.</p> <p><b>Note:</b> You may not engage in multiple transactions simultaneously (i.e., you must sell the stock before you buy again).</p> <p><b>Example 1:</b></p> |  |  |
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| <div>Input: prices = [3,3,5,0,0,3,1,4]<br/>Output: 6<br/>Explanation: Buy on day 4 (price = 0) and sell on day 6 (price = 3), profit = 3-0 = 3.<br/>Then buy on day 7 (price = 1) and sell on day 8 (price = 4), profit = 4-1 = 3.</div> <div>Example 2:<br/><br/>Input: prices = [1,2,3,4,5]<br/>Output: 4<br/>Explanation: Buy on day 1 (price = 1) and sell on day 5 (price = 5), profit = 5-1 = 4.<br/>Note that you cannot buy on day 1, buy on day 2 and sell them later, as you are engaging multiple transactions at the same time. You must sell before buying again.</div> <div>Example 3:<br/><br/>Input: prices = [7,6,4,3,1]<br/>Output: 0<br/>Explanation: In this case, no transaction is done, i.e. max profit = 0.</div> <div>Constraints:<ul style="list-style-type: none"><li>1 &lt;= prices.length &lt;= 105</li><li>0 &lt;= prices[i] &lt;= 105</li></ul></div> <div>Brute Force (Python)</div> | <div># Brute Force Approach<br/>class Solution:<br/>    def max_profit(self, prices):<br/>        # Brute Force <math>O(2^n)</math> - plain recursion, no memoisation<br/>        def recurse(i):<br/>            if i &gt;= len(prices): return 0<br/>            take = prices[i] + recurse(i + 1)<br/>            skip = recurse(i + 1)<br/>            return max(take, skip)<br/>        return recurse(0)</div> <div>Community Solutions (Python)</div> <div><ul style="list-style-type: none"><li>WalkCC<br/><pre>class Solution:<br/>    def maxProfit(self, prices: list[int]) -&gt; int:<br/>        sellTwo = 0<br/>        holdTwo = -math.inf<br/>        sellOne = 0<br/>        holdOne = -math.inf<br/><br/>        for price in prices:<br/>            sellTwo = max(sellTwo, holdTwo + price)<br/>            holdTwo = max(holdTwo, sellOne - price)<br/><br/>            sellOne = max(sellOne, holdOne + price)<br/>            holdOne = max(holdOne, -price)<br/><br/>        return sellTwo</pre></li><li>LeetDoocs</li></ul></div> | <div>class Solution:<br/>    def maxProfit(self, prices: List[int]) -&gt; int:<br/>        # <math>0 \leq i &lt; j &lt; \text{len}(\text{prices})</math><br/>        f1, f2, f3, f4 = -prices[0], 0, -prices[0], 0<br/>        for price in prices[1:]:<br/>            f1 = max(f1, -price)<br/>            f2 = max(f2, f1 + price)<br/>            f3 = max(f3, f2 - price)<br/>            f4 = max(f4, f3 + price)<br/>        return f4</div> <div>• SimplyLeet<br/># Python solution for Best Time to Buy and Sell Stock III<br/><br/>def max_profit(prices):<br/>    # If there are less than 2 days, no transaction can be made.<br/>    if not prices or len(prices) &lt; 2:<br/>        return 0<br/><br/>    n = len(prices)<br/>    # Initialize left and right profit arrays with zeros.<br/>    left_profits = [0] * n</div> |
| Dynamic Programmingp.2215                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Dynamic Programmingp.2216                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Dynamic Programmingp.2217                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

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| <pre>right_profits = [0] * n<br/><br/># Forward pass: compute max profit until day i with one transaction.<br/>min_price = prices[0]<br/>for i in range(1, n):<br/>    # Update the min price encountered so far.<br/>    min_price = min(min_price, prices[i])<br/>    # Maximum profit if sold on day i.<br/>    left_profits[i] = max(left_profits[i - 1], prices[i] - min_price)<br/><br/># Backward pass: compute max profit from day i to end with one transaction.<br/>max_price = prices[-1]<br/>for i in range(n - 2, -1, -1):<br/>    # Update max price seen from the right.<br/>    max_price = max(max_price, prices[i])<br/>    # Maximum profit possible starting from day i.<br/>    right_profits[i] = max(right_profits[i + 1], max_price - prices[i])<br/><br/># Combine the two profits to find the maximum total profit.<br/>max_total_profit = 0</pre> | <pre>for i in range(n):<br/>    max_total_profit = max(max_total_profit, left_profits[i] + right_profits[i])<br/><br/>return max_total_profit<br/><br/># Example usage:<br/>prices = [3,3,5,0,0,3,1,4]<br/>print(max_profit(prices)) # Expected output: 6</pre> <div><ul style="list-style-type: none"><li>LC.ca<br/><pre>...<br/>- `f1` <math>0 \leq i &lt; j &lt; \text{len}(\text{prices})</math><br/>- `f2` <math>0 \leq i &lt; j &lt; \text{len}(\text{prices})</math><br/>- `f3` <math>0 \leq i &lt; j &lt; \text{len}(\text{prices})</math><br/>- `f4` <math>0 \leq i &lt; j &lt; \text{len}(\text{prices})</math><br/>...</pre></li></ul></div> <div>class Solution:<br/>    def maxProfit(self, prices: List[int]) -&gt; int:<br/>        # <math>0 \leq i &lt; j &lt; \text{len}(\text{prices})</math><br/>        f1, f2, f3, f4 = -prices[0], 0, -prices[0], 0<br/><br/>        for price in prices[1:]:<br/>            f1 = max(f1, -price)<br/>            f2 = max(f2, f1 + price)<br/>            f3 = max(f3, f2 - price)<br/>            f4 = max(f4, f3 + price)<br/>        return f4<br/><br/>class Solution:<br/>    def maxProfit(self, prices: List[int]) -&gt; int:</div> | <pre>if not prices:<br/>    return 0<br/><br/># 0 to i, max profit<br/>left_max = [0] * len(prices)<br/><br/>tmp_min = prices[0]<br/>for i in range(len(prices)):<br/>    tmp_min = min(tmp_min, prices[i])<br/>    left_max[i] = max(left_max[i-1], prices[i]-tmp_min) if i &gt; 0 else 0<br/><br/># i to end, max profit<br/>right_max = [0] * len(prices)<br/><br/>tmp_max = prices[-1]<br/>for i in range(len(prices)-1, -1, -1):<br/>    tmp_max = max(tmp_max, prices[i])<br/>    right_max[i] = max(right_max[i+1], tmp_max-prices[i]) if i &lt; len(prices)-1 else 0<br/><br/>return max(left_max[i] + right_max[i] for i in range(len(prices)))</pre> |
| Dynamic Programmingp.2218                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Dynamic Programmingp.2219                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Dynamic Programmingp.2220                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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| <div>#####</div> <div>class Solution(object):<br/>    def maxProfit(self, prices):<br/>        """"<br/>        :type prices: List[int]<br/>        :rtype: int<br/>        """"<br/>        buy1 = buy2 = float("-inf")<br/><br/>        sell1 = sell2 = 0<br/><br/>        for i in range(len(prices)):<br/>            sell1 = max(prices[i] + buy1, sell1)<br/>            buy1 = max(buy1, -prices[i])<br/>            sell2 = max(sell2, prices[i] + buy2)<br/>            buy2 = max(sell1 - prices[i], buy2)<br/>        return max(sell1, sell2)</div> | <div>Dynamic Programming</div> <div>#132 Palindrome Partitioning IIHAR</div> <div>LeetDoocs · SimplyLeet · WalkCC · LeetCode<br/>String · Dynamic Programming<br/>Lists: Denny Zhang<br/>Time Complexity: <math>O(n^2)</math> Space Complexity: <math>O(n^2)</math> (due to the 2D table for palindrome checks, plus <math>O(n)</math> for the dp array)<br/>Problem<br/>Given a string s, partition s such that every substring of the partition is a palindrome.<br/>Return the minimum cuts needed for a palindrome partitioning of s.<br/>Example 1:<br/><div>Input: s = "aab"<br/>Output: 1<br/>Explanation: The palindrome partitioning ["aa","b"] could be produced using 1 cut.</div><br/>Example 2:<br/><div>Input: s = "a"<br/>Output: 0</div><br/>Example 3:</div> | <div>Input: s = "ab"<br/>Output: 1</div> <div>Constraints:<ul style="list-style-type: none"><li>1 &lt;= s.length &lt;= 2000</li><li>s consists of lowercase English letters only.</li></ul></div> <div>Community Solutions (Python)</div> <div><ul style="list-style-type: none"><li>WalkCC<br/><pre>class Solution:<br/>    def minCut(self, s: str) -&gt; int:<br/>        n = len(s)<br/>        # isPalindrome[i][j] := True if s[i..j] is a palindrome<br/>        isPalindrome = [[True] * n for _ in range(n)]<br/>        # dp[i] := the minimum cuts needed for a palindrome partitioning of s[0..i]<br/>        dp = [n] * n<br/><br/>        for l in range(2, n + 1):<br/>            i = 0</pre></li></ul></div> |
| Dynamic Programmingp.2221                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Dynamic Programmingp.2222                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Dynamic Programmingp.2223                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| <pre>        for j in range(l - 1, n):             isPalindrome[i][j] = s[i] == s[j] and isPalindrome[i + 1][j - 1]             i += 1          for i in range(n):             if isPalindrome[0][i]:                 dp[i] = 0                 continue          # Try all the possible partitions.          for j in range(i):             if isPalindrome[j + 1][i]:                 dp[i] = min(dp[i], dp[j] + 1)          return dp[-1]</pre> <p>• LeetCode</p> <pre>class Solution:     def minCut(self, s: str) -&gt; int:         n = len(s)         g = [[True] * n for _ in range(n)]         for i in range(n - 1, -1, -1):             for j in range(i + 1, n):                 g[i][j] = s[i] == s[j] and g[i + 1][j - 1]          f = list(range(n))         for i in range(1, n):             for j in range(i + 1):</pre>                                                                                                                                                                                         | Dynamic Programming | p.2224 |
| <pre>        if g[j][i]:             f[i] = min(f[i], 1 + f[j - 1] if j else 0)             return f[-1]</pre> <p>• SimplyLeet</p> <pre>def minCut(s: str) -&gt; int:     n = len(s)     # Precompute palindrome table where isPal[i][j] is True if s[i:j+1] is a palindrome.     isPal = [[False] * n for _ in range(n)]     # dp[i] will store the minimum cuts required for substring s[0...i]     dp = [0] * n     for i in range(n):         min_cut = i # maximum cuts (cut every character)         for j in range(i + 1):             # Check if s[j...i] is a palindrome:              if s[j] == s[i] and (i - j &lt; 2 or isPal[j+1][i-1]):                 isPal[j][i] = True # Mark substring as palindrome.                 # If the palindrome starts at index 0, no cut is needed.                 min_cut = 0 if j == 0 else min(min_cut, dp[j-1] + 1)             dp[i] = min_cut # Record minimum cuts needed for s[0...i]             return dp[-1]  # Example usage: # print(minCut("aab")) # Output: 1</pre> | Dynamic Programming | p.2225 |
| <p>• LC.ca</p> <pre>class Solution:     def minCut(self, s: str) -&gt; int:         pal = [[False for j in range(0, len(s))] for i in range(0, len(s))]         dp = [len(s) for _ in range(0, len(s) + 1)]         for i in range(0, len(s)):             for j in range(0, i + 1):                 # also ok if, (j + 1 &gt;= i - 1)                 if (s[i] == s[j]) and ((j + 1 &gt; i - 1) or (pal[i - 1][j + 1])):                     pal[i][j] = True                     dp[i + 1] = min(dp[i + 1], dp[j] + 1)         if j != 0 else 0          # 'if j != 0' to ensure from start to i is a palindrome         return dp[-1]  #####  class Solution: # clear cache     def minCut(self, s: str) -&gt; int:         @cache         def dfs(i):             if i &gt;= n - 1:</pre>                                                                                                                                                                                                                                      | Dynamic Programming | p.2226 |

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| <pre>        return 0         ans = inf         for j in range(i, n):             if g[i][j]:                 ans = min(ans, dfs(j + 1) + (j &lt; n - 1))          return ans          n = len(s)         g = [[True] * n for _ in range(n)]         for i in range(n - 1, -1, -1):          for j in range(i + 1, n):             g[i][j] = s[i] == s[j] and g[i + 1][j - 1]          ans = dfs(0)         dfs.cache_clear() # nice, clear cache !!!         return ans  #####  class Solution:     def minCut(self, s: str) -&gt; int:</pre>                                                                                                                                                                                                                                                                                                                                                                             | Dynamic Programming | p.2227 |
| <pre>        n = len(s)         is_pal = [[False] * n for _ in range(n)]         for i in range(n - 1, -1, -1):             for j in range(i, n):                 is_pal[i][j] = s[i] == s[j] and (j - i &lt; 3 or is_pal[i + 1][j - 1])             min_cut = [i for i in range(n)]             for i in range(n):                 for j in range(0, i + 1):                     if is_pal[j][i]:                         min_cut[i] = min(min_cut[i], min_cut[j - 1] + 1) if j &gt; 0 else 0          return min_cut[-1]</pre>                                                                                                                                                                                                                                                                                                                                                                                           | Dynamic Programming | p.2228 |
| <p>Dynamic Programming</p> <p>#312 Burst Balloons</p> <p>LeetCode · SimplyLeet · WalkCC · LeetCode</p> <p>Array · Dynamic Programming</p> <p>Lists: Denny Zhang</p> <p>Time Complexity: <math>O(n^3)</math> - Three nested loops: one for the interval length, one for the starting index, and one for the final balloon choice. Space Complexity: <math>O(n^2)</math> - A 2D dynamic programming table i</p> <p>Problem</p> <p>You are given n balloons, indexed from 0 to n - 1. Each balloon is painted with a number on it represented by an array nums. You are asked to burst all the balloons. If you burst the ith balloon, you will get <math>nums[i - 1] * nums[i] * nums[i + 1]</math> coins. If i - 1 or i + 1 goes out of bounds of the array, then treat it as if there is a balloon with a 1 painted on it. Return the maximum coins you can collect by bursting the balloons wisely.</p> <p>Example 1:</p> | Dynamic Programming | p.2229 |

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| <p>Input: nums = [3,1,5,8]</p> <p>Output: 167</p> <p>Explanation:</p> <p>nums = [3,1,5,8] --&gt; [3,5,8] --&gt; [3,8] --&gt; [8] --&gt; []</p> <p>coins = 3*1*5 + 3*5*8 + 1*3*8 + 1*8*1 = 167</p> <p>Example 2:</p> <p>Input: nums = [1,5]</p> <p>Output: 10</p> <p>Constraints:</p> <ul style="list-style-type: none"><li>• n == nums.length</li><li>• 1 &lt;= n &lt;= 300</li><li>• 0 &lt;= nums[i] &lt;= 100</li></ul> <p>Brute Force (Python)</p> <pre># Brute Force Approach class Solution:     def maxCoins(self, nums):         # Brute Force O(n! * n) - try every order to burst all balloons         from itertools import permutations         nums = [1] + nums + [1]         best = 0         # Only feasible for tiny n; actual brute is exponential         def burst(balloons, coins):             nonlocal best</pre>     | Dynamic Programming | p.2230 |
| <pre>        if not balloons:             best = max(best, coins)             return         for i in range(len(balloons)):             left = balloons[i-1] if i &gt; 0 else 1             right = balloons[i+1] if i+1 &lt; len(balloons) else 1             gained = left * balloons[i] * right             burst(balloons[:i] + balloons[i+1:], coins + gained)             interior = list(nums[1:-1])             burst(interior, 0)          return best</pre> <p>Community Solutions (Python)</p> <p>• WalkCC</p> <pre>class Solution:     def maxCoins(self, nums: list[int]) -&gt; int:         n = len(nums)         nums = [1] + nums + [1]          @functools.lru_cache(None)         def dp(i: int, j: int) -&gt; int:             ""Returns maxCoins(nums[i..j]).""             if i &gt; j:                 return 0</pre> | Dynamic Programming | p.2231 |
| <pre>        return max(dp(i, k - 1) + dp(k + 1, j) + nums[i - 1] * nums[k] * nums[j + 1] for k in range(i, j + 1))          return dp(1, n)</pre> <p>• LeetCode</p> <pre>class Solution:     def maxCoins(self, nums: List[int]) -&gt; int:         n = len(nums)         arr = [1] + nums + [1]         f = [[0] * (n + 2) for _ in range(n + 2)]         for i in range(n - 1, -1, -1):             for j in range(i + 2, n + 2):                 for k in range(i + 1, j):                     f[i][j] = max(f[i][j], f[i][k] + f[k][j] + arr[i] * arr[k] * arr[j])             return f[0][-1]</pre> <p>• SimplyLeet</p>                                                                                                                                                                                                               | Dynamic Programming | p.2232 |



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| <pre># Define the function to compute maximum coins def maxCoins(nums):     # Add virtual balloons with value 1 at both     boundaries     n = len(nums)     extended = [1] + nums + [1]     # Initialize dp table with 0's; dimensions: (n+2) x     (n+2)     dp = [[0] * (n + 2) for _ in range(n + 2)]      # Calculate dp values for intervals with length 2 to     n+1     for length in range(2, n + 2): # interval length from     2 to n+1          for left in range(0, n + 2 - length):             right = left + length             # Iterate through all possible last burst             balloons in the interval (left, right)             for k in range(left + 1, right):                 # Coins when bursting balloon k last in                 the interval:                 # value from bursting: extended[left] *                 extended[k] * extended[right]                 # plus coins from subintervals [left, k]                 and [k, right]                 coins = extended[left] * extended[k] *                 extended[right] + dp[left][k] + dp[k][right]                 dp[left][right] = max(dp[left][right],                 coins)</pre> |        | <pre># The final answer is stored in dp[0][n+1] which represents bursting all real balloons. return dp[0][n+1]  # Example test cases print(maxCoins([3,1,5,8])) # Expected output: 167 print(maxCoins([1,5])) # Expected output: 10</pre>                                                                                                                                                                                                                                                                                 | <div>Dynamic Programming</div> <div>HAR</div> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        | <pre>• LC.ca  class Solution:     def maxCoins(self, nums: List[int]) -&gt; int:         nums = [1] + nums + [1]         n = len(nums)         dp = [[0] * n for _ in range(n)]         for l in range(2, n):             for i in range(n - l):                 j = i + l                 for k in range(i + 1, j):                     dp[i][j] = max(                         dp[i][j], dp[i][k] + dp[k][j] +                         nums[i] * nums[k] * nums[j]                     )         return dp[0][-1]</pre> |                                               |
| Dynamic Programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | p.2233 | Dynamic Programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | p.2234                                        |

|                                                                                                                                                                                                                                                                                                                                                      |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <pre>Input: stones = [3,2,4,1], k = 2 Output: 20 Explanation: We start with [3, 2, 4, 1]. We merge [3, 2] for a cost of 5, and we are left with [5, 4, 1]. We merge [4, 1] for a cost of 5, and we are left with [5, 5]. We merge [5, 5] for a cost of 10, and we are left with [10]. The total cost was 20, and this is the minimum possible.</pre> |        | <pre>• 1 &lt;= n &lt;= 30 • 1 &lt;= stones[i] &lt;= 100 • 2 &lt;= k &lt;= 30  Community Solutions (Python)  • LeetCode  class Solution:     def mergeStones(self, stones: List[int], K: int) -&gt;     int:         n = len(stones)         if (n - 1) % (K - 1):             return -1         s = list(accumulate(stones, initial=0))         f = [[[inf] * (K + 1) for _ in range(n + 1)] for _         in range(n + 1)]         for i in range(1, n + 1):             f[i][i][1] = 0             for l in range(2, n + 1):                  for i in range(1, n - l + 2):                     j = i + l - 1                     for k in range(1, K + 1):                         for h in range(i, j):                             f[i][j][k] = min(f[i][j][k],                             f[i][h][1] + f[h + 1][j][k - 1])                             f[i][j][1] = f[i][j][K] + s[j] - s[i - 1]                             return f[1][n][1]</pre> |        | <pre># Python solution for merging stones with detailed comments. class Solution:     def mergeStones(self, stones: List[int], k: int) -&gt;     int:         n = len(stones)         # Check for merge feasibility.         if (n - 1) % (k - 1) != 0:             return -1          # Precompute prefix sums for fast range sum         queries.         prefix = [0] * (n + 1)          for i in range(n):             prefix[i + 1] = prefix[i] + stones[i]          # Initialize dp table, dp[i][j] represents         minimal cost to merge stones[i...j].         dp = [[0] * n for _ in range(n)]          # Process all possible subarrays with lengths &gt;=         k.         for length in range(k, n + 1):             for i in range(n - length + 1):                 j = i + length - 1</pre> |  |
| Dynamic Programming                                                                                                                                                                                                                                                                                                                                  | p.2236 | Dynamic Programming                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | p.2237 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|
| <pre>dp[i][j] = float('inf') # Try all valid partitions with step size (k-1). for mid in range(i, j, k - 1):     dp[i][j] = min(dp[i][j], dp[i][mid] +     dp[mid+1][j]) # If this segment can be merged into one pile, add cost. if (length - 1) % (k - 1) == 0:     dp[i][j] += prefix[j+1] - prefix[i]     return dp[0][n-1]  # Example usage:  # solution = Solution() # print(solution.mergeStones([3, 2, 4, 1], 2)) # Expected output: 20</pre> |        | <pre>for i in range(1, n - l + 2):     j = i + l - 1     for k in range(1, K + 1):         for h in range(i, j):             f[i][j][k] = min(f[i][j][k],             f[i][h][1] + f[h + 1][j][k - 1])             f[i][j][1] = f[i][j][K] + s[j] - s[i - 1]             return f[1][n][1]</pre> |                                                  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |                                                                                                                                                                                                                                                                                                  | <div>Dynamic Programming</div> <div>p.2241</div> |
| Dynamic Programming                                                                                                                                                                                                                                                                                                                                                                                                                                   | p.2239 | Dynamic Programming                                                                                                                                                                                                                                                                              | p.2240                                           |

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|  |  | <div>Dynamic Programming</div> <div>p.2241</div> |
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24 questions · insights · complexity · solution

### #70 Climbing Stairs [Easy]

#### Key Insights

- The problem can be solved using dynamic programming because the solution for a given step depends on the solutions of the previous two steps.
- The recurrence relation is:  $\text{ways}[i] = \text{ways}[i-1] + \text{ways}[i-2]$ .
- Base cases: when there is 1 step, there is 1 way; when there are 2 steps, there are 2 ways.
- This is analogous to the Fibonacci sequence, shifting indices accordingly.

#### Space & Time

Time Complexity:  $O(n)$   
Space Complexity:  $O(n)$  (can be optimized to  $O(1)$  using constant space)

#### Solution

We use a dynamic programming approach to build a solution gradually. The idea is that to reach step  $i$ , you could have come from step  $i-1$  (taking one step) or from step  $i-2$  (taking two steps). Therefore, the total number of ways to reach step  $i$  is the sum of ways to reach step

palindrome found during these expansions. This method is efficient given the input constraints, and no extra data structures are needed besides variables to store indices or substrings.

### #53 Maximum Subarray [Medium]

#### Key Insights

- Use Kadane's Algorithm to solve the problem in  $O(n)$  time.
- Keep track of a running sum (currentSum) and update it with the maximum between the current element and the sum including the current element.
- The maximum sum encountered during the iteration (maxSum) is the answer.
- A divide and conquer approach also exists, splitting the array and merging solutions, but Kadane's method is optimal for this problem.

#### Space & Time

Time Complexity:  $O(n)$   
Space Complexity:  $O(1)$

#### Solution

The solution uses Kadane's Algorithm. Iterate over the array while maintaining two variables: currentSum, which tracks the maximum subarray sum ending at the current position, and maxSum, which stores the best sum seen so far. For each element, update currentSum to

– Initialize  $\text{dp}[0][j] = 1$  for all  $j$  (only one way to move right in the first row). – Initialize  $\text{dp}[i][0] = 1$  for all  $i$  (only one way to move down in the first column). – For each other cell, compute  $\text{dp}[i][j] = \text{dp}[i-1][j] + \text{dp}[i][j-1]$ . The final answer will be at  $\text{dp}[m-1][n-1]$ . The combinatorial alternative computes  $C(m+n-2, m-1)$  or  $C(m+n-2, n-1)$  using basic factorial arithmetic, which is efficient for the given constraints.

### #72 Edit Distance [Medium]

#### Key Insights

- This is a classic dynamic programming problem.
- Use a 2D DP table where  $\text{dp}[i][j]$  represents the minimum edit distance between  $\text{word1}[0\dots i-1]$  and  $\text{word2}[0\dots j-1]$ .
- Base cases: converting an empty string to a string of length  $j$  requires  $j$  insertions, and vice versa.
- Transition: If characters match then  $\text{dp}[i][j] = \text{dp}[i-1][j-1]$ , otherwise take minimum among insertion, deletion, and replacement operations and add 1.

#### Space & Time

Time Complexity:  $O(mn)$ , where  $m$  and  $n$  are the lengths of word1 and word2 respectively.  
Space Complexity:  $O(m)$

$i-1$  and  $i-2$ . We initialize  $\text{dp}[1] = 1$  and  $\text{dp}[2] = 2$ , and iterate from 3 to  $n$  to fill  $\text{dp}[]$ . Each code solution below uses this approach with comments to explain the logic.

### #121 Best Time to Buy and Sell Stock [Easy]

#### Key Insights

- Traverse the list of prices once while keeping track of the minimum price encountered so far.
- At each day, calculate the profit if you sold the stock on that day.
- Update the maximum profit when the current profit exceeds the previous maximum.
- The optimal solution makes a single pass through the array with constant space usage.

#### Space & Time

Time Complexity:  $O(n)$  – requires one pass through the prices array.  
Space Complexity:  $O(1)$  – only a few extra variables are used.

#### Solution

We solve the problem using a simple linear scan. We maintain two key variables: one for the minimum price seen so far and another for the maximum profit calculated so far. As we iterate, we continuously update the minimum price if a lower price is found, and then compute the potential profit at the current price. If this

be either the current element or the sum of currentSum and the current element, whichever is higher. Update maxSum accordingly. This approach efficiently computes the maximum contiguous subarray sum in a single pass.

### #55 Jump Game [Medium]

#### Key Insights

- The greedy approach is optimal for this problem.
- Track the furthest index reachable as you iterate through the array.
- If the current index exceeds the maximum reachable index, it is impossible to proceed further.
- The solution only requires constant extra space.

#### Space & Time

Time Complexity:  $O(n)$  where  $n$  is the length of the array.  
Space Complexity:  $O(1)$

#### Solution

The solution uses a greedy algorithm. As you iterate through each array element, you maintain a variable (maxReach) that represents the furthest index you can reach so far. For each index  $i$ , if  $i$  is greater than maxReach then it is not possible to reach index  $i$  and hence return false immediately. Otherwise, update maxReach to the maximum of its current value and  $i + \text{nums}[i]$ . If you can iterate through the array without  $i$  exceeding maxReach, then you can reach the end and

$n$ ) due to the DP table used for storing computed distances.

#### Solution

We use a dynamic programming approach by constructing a 2D table  $\text{dp}$  of size  $(m+1) \times (n+1)$ , where  $m$  and  $n$  are the lengths of word1 and word2. Each  $\text{dp}[i][j]$  holds the minimum number of operations needed to convert the first  $i$  characters of word1 to the first  $j$  characters of word2. The algorithm includes the following steps:  
– Initialize the base cases:  $\text{dp}[0][j] = j$  (converting an empty word1 to  $j$  characters of word2 requires  $j$  insertions)  $\text{dp}[i][0] = i$  (converting  $i$  characters of word1 to an empty word2 requires  $i$  deletions) – Iterate over each character in word1 and word2: If the current characters are equal, no additional operation is needed and  $\text{dp}[i][j] = \text{dp}[i-1][j-1]$ . Otherwise, choose the best operation among insertion ( $\text{dp}[i][j-1]$ ), deletion ( $\text{dp}[i-1][j]$ ), or substitution ( $\text{dp}[i-1][j-1]$ ) and add 1. – The answer is found in  $\text{dp}[m][n]$ . This method reliably computes the edit distance using a systematic DP approach, ensuring that all subproblems are solved and reused efficiently.

### #91 Decode Ways [Medium]

#### Key Insights

profit exceeds the current maximum profit, we update the maximum profit. The final maximum profit is returned after iterating through all prices.

### #5 Longest Palindromic Substring [Medium]

#### Key Insights

- Every palindrome is centered around a character (for odd-length) or a pair of characters (for even-length).
- Expand from each center until the substring is no longer a palindrome.
- Update the longest palindrome when a longer one is found during the expansion.
- The solution employs a two-pointer technique without extra storage for dynamic programming tables.

#### Space & Time

Time Complexity:  $O(n^2)$  in the worst case as we expand around each center.  
Space Complexity:  $O(1)$ , excluding the space required for the output substring.

#### Solution

The approach is based on expanding around potential centers of palindromes. For each character in the string, consider it as the center for an odd-length palindrome and the gap between it and the next character for an even-length palindrome. Using two pointers, expand outward until the characters differ. Track the longest

return true.

### #62 Unique Paths [Medium]

#### Key Insights

- The robot is constrained to only move right or down.
- Any valid path consists of exactly  $(m-1)$  down moves and  $(n-1)$  right moves.
- The problem can be solved using Dynamic Programming where each cell is the sum of the ways to reach the cell from the top and left.
- Alternatively, a combinatorial solution is possible by calculating the number of unique sequences of moves, equivalent to choosing  $(m-1)$  moves from  $(m+n-2)$  moves.

#### Space & Time

Time Complexity:  $O(mn)$   
Space Complexity:  $O(m)$  – can be optimized to  $O(n)$  using a 1D DP array.

#### Solution

We can solve the problem using a dynamic programming approach. The idea is to create a 2D DP table where each cell  $(i, j)$  represents the number of ways to reach that cell. Since the robot can only come from above or from the left, we update the DP table as follows:

- A digit '0' cannot be mapped on its own; it must be part of "10" or "20".
- Use dynamic programming where  $\text{dp}[i]$  represents the number of ways to decode the substring  $\text{s}[0\dots i-1]$ .
- For each index  $i$ , if the digit at position  $i-1$  is non-zero, add  $\text{dp}[i-1]$  to  $\text{dp}[i]$ .
- If the two-digit number formed by  $\text{s}[i-2]$  and  $\text{s}[i-1]$  is between 10 and 26, add  $\text{dp}[i-2]$  to  $\text{dp}[i]$ .
- Initialize  $\text{dp}[0]$  as 1 (an empty string has one way to be decoded).

#### Space & Time

Time Complexity:  $O(n)$ , where  $n$  is the length of the string.  
Space Complexity:  $O(n)$ , due to the dp array (which can be optimized to  $O(1)$ ).

#### Solution

We solve the problem using a dynamic programming approach. We initialize a dp array where  $\text{dp}[i]$  indicates the number of ways to decode the substring  $\text{s}[0\dots i-1]$ . We set  $\text{dp}[0] = 1$  since an empty string is trivially decodable. We loop through the string, and at each step:  
– If  $\text{s}[i-1]$  is not '0', we add  $\text{dp}[i-1]$  to  $\text{dp}[i]$  because  $\text{s}[i-1]$  can be mapped to a valid letter. – If the substring  $\text{s}[i-2:i]$  (for  $i \geq 2$ ) forms a valid two-digit number between 10 and 26, we add  $\text{dp}[i-2]$  to  $\text{dp}[i]$ . This cumulative approach yields the total ways to decode the

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| <p>string, and if the string starts with '0' or contains invalid sequences, dp[n] will eventually be 0.</p> <p><b>#152 Maximum Product Subarray [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Maintain both the maximum and minimum product at the current index because a negative number can swap the max and min.</li> <li>• A subarray must be contiguous.</li> <li>• Use dynamic programming to update both the maximum product and the minimum product ending at each index.</li> <li>• Resetting the product in case of a zero is crucial since the product becomes 0.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) – The solution iterates through the array once.</p> <p>Space Complexity: O(1) – Constant space is used for tracking products.</p> <p><b>Solution</b></p> <p>The solution uses an iterative dynamic programming approach. At each index, compute the maximum product (maxEndingHere) and minimum product (minEndingHere) ending at that index. If the current element is negative, swap these two values, because multiplying a negative with a minimum product</p> <div> <div>Dynamic Programming</div> <div>p.2251</div> </div>                                                                                                                                                                                              | <p>(possibly negative) might yield a larger product. After processing each element, update the global maximum. This method handles edge cases like zeros and negatives efficiently with constant additional space.</p> <p><b>#198 House Robber [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The problem can be solved using Dynamic Programming.</li> <li>• For each house, decide whether to rob it or not.</li> <li>• Recurrence relation: <math>dp[i] = \max(dp[i-1], dp[i-2] + \text{currentHouseMoney})</math></li> <li>• Use iterative computation to keep track of the optimal solutions without needing extra space for the entire dp array.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n), where n is the number of houses.</p> <p>Space Complexity: O(1), using only two variables to track the previous decisions.</p> <p><b>Solution</b></p> <p>We use a dynamic programming approach by iterating through each house in the array. At each step, we consider two scenarios: either we rob the current house (and add its money to the total from two houses ago) or we skip it (keeping the optimal solution up to the previous house). By using two variables to store these</p> <div> <div>Dynamic Programming</div> <div>p.2252</div> </div>                                                                                                                                  | <p>intermediate results, we achieve an optimized space complexity of O(1).</p> <p><b>#256 Paint House [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The choice for painting each house depends on the previous house's color choices.</li> <li>• Use dynamic programming to accumulate the cost while ensuring adjacent houses do not have the same color.</li> <li>• The state of each house can be represented by the minimum cost when the house is painted a specific color, computed using the minimal costs of the previous house from the other two colors.</li> <li>• The problem only requires updating the cost matrix in-place for optimal space usage.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n), where n is the number of houses.</p> <p>Space Complexity: O(1) extra space if the input matrix is modified in place.</p> <p><b>Solution</b></p> <p>We use a dynamic programming approach with iterative in-place updating of the cost matrix. For each house starting from the second one, we update the cost for each color by adding the minimal cost of painting the previous house with either of the other two colors. This</p> <div> <div>Dynamic Programming</div> <div>p.2253</div> </div>                                                                              |
| <p>guarantees we never use the same color for adjacent houses while accumulating the minimal total cost. The final answer is the minimum value for the last house.</p> <p><b>#309 Best Time to Buy and Sell Stock with Cooldown [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use Dynamic Programming to solve the problem.</li> <li>• Maintain three states to represent: <ul style="list-style-type: none"> <li>• s0: the state where you are ready to buy (not holding any stock).</li> <li>• s1: the state where you are holding a stock.</li> <li>• s2: the cooldown state encountered right after selling a stock.</li> </ul> </li> <li>• Transition between states: <ul style="list-style-type: none"> <li>• Transition into s0 can come from staying in s0 or coming out from the cooldown state s2.</li> <li>• To enter s1, you either continue holding your stock, or buy a stock using the profit available from s0.</li> <li>• The only way to reach s2 is by selling the stock from s1.</li> </ul> </li> <li>• The final answer is determined by the maximum profit when you are not holding a stock (i.e., max(s0, s2)).</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n), where n is the number of days.</p> <div> <div>Dynamic Programming</div> <div>p.2254</div> </div>                                                                    | <p>Space Complexity: O(1), since only a few variables are maintained for the states.</p> <p><b>Solution</b></p> <p>The solution involves updating three state variables for each day:</p> <ul style="list-style-type: none"> <li>– s0 (ready to buy) is updated as the maximum of the previous s0 and the previous cooldown s2. – s1 (holding a stock) is updated as the maximum of the previous s1 or buying today with the profit from s0 minus the current price. – s2 (cooldown) is updated by selling the stock held in s1 (i.e., previous s1 plus the current price).</li> </ul> <p>At the end of the iteration, the maximum profit will be in state s0 (ready to buy) or s2 (cooldown), since remaining in s1 means you are holding a stock, which does not constitute a realized profit.</p> <p><b>#377 Combination Sum IV [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use dynamic programming to count the number of valid combinations.</li> <li>• The order in which numbers are picked matters, making this akin to permutation counting.</li> <li>• Initialize a DP table where dp[i] represents the number of ways to form the sum i.</li> <li>• dp[0] is initialized to 1 since there is one way to form a sum of 0 (by choosing nothing).</li> </ul> <div> <div>Dynamic Programming</div> <div>p.2255</div> </div>                                                                                | <ul style="list-style-type: none"> <li>• For each sum i from 1 to target, iterate over nums and update dp[i] based on dp[i – num] when i &gt;= num.</li> <li>• For negative numbers, the problem becomes unbounded due to potential infinite combinations; hence a constraint on the combination length is required to make it well-defined.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(target * n), where n is the length of nums.</p> <p>Space Complexity: O(target), for the dp array of size target + 1.</p> <p><b>Solution</b></p> <p>The problem is solved using a bottom-up dynamic programming approach. We create a dp array where each dp[i] represents the number of possible combinations to reach the sum i. We iterate from 1 up to target, and for each sum, we check every number in nums. If the number can contribute to the current sum (i – num &gt;= 0), we add the number of ways to form the remainder (i – num) to dp[i]. This ensures that every possible permutation is counted. When negative numbers are allowed in the input, the potential for an infinite number of combinations arises unless we impose additional constraints, such as limiting the length of the combination.</p> <p><b>#416 Partition Equal Subset Sum [Medium]</b></p> <div> <div>Dynamic Programming</div> <div>p.2256</div> </div> |
| <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The total sum of the array must be even, otherwise, partitioning into two equal subsets is impossible.</li> <li>• The problem can be transformed into a "subset sum" problem where we need to check if there exists a subset of numbers that sums to half of the total sum.</li> <li>• A dynamic programming approach can efficiently decide if such a subset exists.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n * target), where target is half of the total sum.</p> <p>Space Complexity: O(target), using a one-dimensional DP array.</p> <p><b>Solution</b></p> <p>First, compute the total sum of the array. If the total is odd, return false since an odd number cannot be equally partitioned. Next, set the target sum to half of the total. Use a dynamic programming array (dp) where dp[i] represents whether a subset with sum i is achievable. Initialize dp[0] as true because a subset sum of 0 is always possible. Then, for each number in the array, iterate through the dp array backwards from target to the number's value and update dp[i] to true if dp[i – num] is true. Ultimately, if dp[target] is true, the array can be partitioned into two subsets with equal sum.</p> <p><b>#435 Non-overlapping Intervals [Medium]</b></p> <div> <div>Dynamic Programming</div> <div>p.2257</div> </div> | <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Sorting is the key: By sorting intervals by their end times, we can greedily choose the optimal set of non-overlapping intervals.</li> <li>• Greedy approach: Always select the interval that ends the earliest, as it leaves maximum room for subsequent intervals.</li> <li>• Counting removals: The answer is the total number of intervals minus the count of intervals selected by this greedy approach.</li> <li>• Edge observation: Intervals that touch at the boundary are not overlapping, so the check uses "&gt;=" instead of "&gt;".</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n log n) due to sorting the intervals.</p> <p>Space Complexity: O(n) if considering the storage for the sorted list (or O(1) if sorting in-place).</p> <p><b>Solution</b></p> <p>The solution involves first sorting the intervals by their end times. Initialize a variable to track the end of the last chosen interval. Iterate over the sorted intervals: if the current interval's start is greater than or equal to the last chosen interval's end, update the last end and include this interval in the non-overlapping set. Otherwise, count the interval as one that must be removed. The final answer is the removal count, which is computed as the total number of intervals minus the</p> <div> <div>Dynamic Programming</div> <div>p.2258</div> </div> | <p>count of non-overlapping intervals selected.</p> <p><b>#516 Longest Palindromic Subsequence [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The problem can be solved efficiently using dynamic programming.</li> <li>• A 2D DP table is used where dp[i][j] represents the length of the longest palindromic subsequence in the substring s[i..j].</li> <li>• If the characters at the two ends of the substring match (s[i] == s[j]), they contribute to the palindrome length.</li> <li>• Otherwise, the solution is the maximum sequence length by either excluding the left or the right character.</li> <li>• The approach builds the solution from smaller substrings (length 1) to the entire string.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n^2) where n is the length of the string.</p> <p>Space Complexity: O(n^2) due to the usage of a 2D DP table.</p> <p><b>Solution</b></p> <p>We solve this problem using a bottom-up dynamic programming approach. We create a 2D table dp[][] such that dp[i][j] stores the longest palindromic subsequence length for the substring s[i..j]. For any single character,</p> <div> <div>Dynamic Programming</div> <div>p.2259</div> </div>                                                                                                |

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>dp[i][i] is 1. For substrings of length greater than 1, if the characters at positions i and j (s[i] and s[j]) match, then dp[i][j] = dp[i+1][j-1] + 2. If they do not match, we take the maximum of dp[i+1][j] and dp[i][j-1]. Finally, the dp[0][n-1] holds the result for the entire string.</p> <p><b>#576 Out of Boundary Paths [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use Dynamic Programming (DP) as the state is defined by (current row, current column, moves remaining).</li> <li>Transitions are made from a cell to its four adjacent cells.</li> <li>If the ball moves out of bounds, it contributes to one valid path.</li> <li>States can be memoized to avoid recomputation and optimize the recursive solution.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(maxMove * m * n), since we compute each state at most once.</p> <p>Space Complexity: O(maxMove * m * n) for the memoization table.</p> <p><b>Solution</b></p> <p>We use a recursive dynamic programming (or memoization) approach. Define a function dp(i, j, movesLeft) that returns the number of ways to move out of bounds starting from cell (i, j) with movesLeft moves</p> <p>Dynamic Programmingp.2260</p>                                                                                                                                 | <p>remaining. For each move from the current cell, check:</p> <ul style="list-style-type: none"> <li>If the cell is out of bounds, return 1 (base case).</li> <li>If no moves are left (movesLeft == 0) and we are still inside, return 0.</li> <li>Otherwise, use recursion to explore all four directions and sum the paths. Store results in a memoization table to avoid recomputation, then return the computed value modulo 10^9 + 7.</li> </ul> <p><b>#1143 Longest Common Subsequence [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The problem is well-suited for a dynamic programming approach.</li> <li>Use a 2D table where dp[i][j] stores the length of the longest common subsequence for text1[0..i-1] and text2[0..j-1].</li> <li>If the characters match at the current position, extend the subsequence by 1 from dp[i-1][j-1]; otherwise, take the maximum from dp[i-1][j] or dp[i][j-1].</li> <li>Filling the table in a bottom-up manner ensures that all subproblems are solved optimally.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(m * n), where m and n are the lengths of text1 and text2.</p> <p>Space Complexity: O(m * n) for maintaining the 2D DP table. Space can be optimized to O(min(m, n)) if needed.</p> <p><b>Solution</b></p> <p>Dynamic Programmingp.2261</p>                                                                                                                       | <p>This solution uses a dynamic programming approach with a 2D array (dp) to record the lengths of common subsequences at each substring pair. We initialize a table of dimensions (m+1) x (n+1) with zeros, where m and n are the lengths of text1 and text2. We iterate over both strings, comparing characters; if they match, we set dp[i][j] = dp[i-1][j-1] + 1, otherwise we choose the maximum from dp[i-1][j] and dp[i][j-1]. The final answer is found at dp[m][n].</p> <p><b>#10 Regular Expression Matching [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use recursion or dynamic programming to simulate matching as the '*' operator can affect the current and following characters.</li> <li>The '.' operator is a wildcard that matches any single character.</li> <li>When encountering a '*' in the pattern, consider both skipping the preceding element or consuming one character from the string if it matches.</li> <li>Using memoization helps to save intermediate results and efficiently process overlapping subproblems.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(mn) where m = length(s) and n = length(p), due to exploring each subproblem combination.</p> <p>Dynamic Programmingp.2262</p>                                                                                                                                     |
| <p>Space Complexity: O(mn) for storing the memoization table in the dynamic programming or recursive approach.</p> <p><b>Solution</b></p> <p>The solution uses a top-down recursive strategy with memoization to efficiently match the string against the pattern. For each character position in s and p, we check if they match directly or if the pattern character is '.' to allow any match. When encountering '*' in the pattern, two possibilities are explored: either '.' stands for zero occurrences of the preceding element, or if there is a valid match, it accounts for one occurrence and we proceed further in the string while maintaining the same pattern index. The memoization aspect prevents repeated computations of the same (i, j) state in the recursion, resulting in an improved overall runtime.</p> <p><b>#115 Distinct Subsequences [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Use dynamic programming to build the solution by comparing characters of s and t.</li> <li>Define a 2D dp table where dp[i][j] represents the number of ways to form the first j characters of t using the first i characters of s.</li> <li>The recurrence relation:</li> <li>If s[i-1] equals t[j-1], then include both the cases of matching this character as well as skipping it: dp[i][j] =</li> </ul> <p>Dynamic Programmingp.2263</p> | <p>dp[i-1][j-1] + dp[i-1][j].</p> <ul style="list-style-type: none"> <li>If they do not match, then simply skip the current character of s: dp[i][j] = dp[i-1][j].</li> <li>Initialize dp[i][0] = 1 for all i since an empty t can always be formed.</li> <li>The final answer is dp[s.length][t.length].</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n * m), where n is the length of s and m is the length of t.</p> <p>Space Complexity: O(n * m), which can be optimized to O(m) with careful row reuse.</p> <p><b>Solution</b></p> <p>We use a dynamic programming approach with a 2D table. The table has dimensions (len(s) + 1) x (len(t) + 1). Each cell dp[i][j] stores the number of distinct subsequences from the first i characters of s that match the first j characters of t.</p> <p>Steps:</p> <ul style="list-style-type: none"> <li>Initialize dp[i][0] = 1 for every i, because an empty t is a subsequence of any prefix of s.</li> <li>Iterate over each character of s (outer loop) and for each, iterate over t (inner loop).</li> <li>For every character in s, if it equals the current character in t, update dp[i][j] as the sum of the ways by using or skipping this character. Otherwise, carry over the previous count.</li> <li>Return dp[len(s)][len(t)] as the result.</li> </ul> <p>Dynamic Programmingp.2264</p>                                                                                                               | <p><b>#123 Best Time to Buy and Sell Stock III [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>The problem requires the optimal selection of at most two buy-sell transactions.</li> <li>One effective strategy is to split the problem into two parts:</li> <li>For each day i, calculate the maximum profit obtainable from day 0 to i with one transaction.</li> <li>For each day i, calculate the maximum profit obtainable from day i to the end with one transaction.</li> <li>Finally, for each day, the sum of these two profits (left and right) gives a candidate answer.</li> <li>Alternatively, a state-based dynamic programming approach can track 4 states corresponding to the two transactions.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) where n is the number of days</p> <p>Space Complexity: O(n) due to extra arrays used for forward and backward computations (this can be optimized to O(1) with a state variable approach)</p> <p><b>Solution</b></p> <p>We use a two-pass dynamic programming approach. In the first pass (forward), we compute an array "leftProfits" where leftProfits[i] is the maximum profit from day 0 to day i with one transaction. We do this by keeping track of the minimum price observed so far and</p> <p>Dynamic Programmingp.2265</p>                                                                 |
| <p>calculating the profit if sold on day i. In the second pass (backward), we compute an array "rightProfits" where rightProfits[i] is the maximum profit obtainable from day i to the last day with one transaction by tracking the maximum price seen in reverse order. Finally, for every index i, the sum leftProfits[i] + rightProfits[i] represents the maximum profit achievable by completing the first transaction on or before day i and the second transaction starting from day i. The answer is the maximum sum over all i.</p> <p><b>#132 Palindrome Partitioning II [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Precompute a 2D table (isPal) where isPal[i][j] indicates if the substring s[i..j] is a palindrome.</li> <li>Use dynamic programming: dp[i] represents the minimum cuts needed for the substring s[0..i].</li> <li>For every index i, iterate over all possible starting indices j. If s[j..i] is a palindrome, update dp[i] using dp[j-1] + 1 (or 0 if j is 0).</li> <li>Time complexity is dominated by the O(n^2) palindrome checking and state transitions.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n^2)</p> <p>Space Complexity: O(n^2) (due to the 2D table for palindrome checks, plus O(n) for the dp array)</p> <p>Dynamic Programmingp.2266</p>                                               | <p><b>Solution</b></p> <p>The solution relies on dynamic programming with a two-step approach. First, precompute a 2D table (isPal) where each cell indicates if a substring is a palindrome. This is done in O(n^2) time by leveraging the fact that a substring s[j..i] is a palindrome if the characters at the ends match and the inner substring s[j+1..i-1] is also a palindrome (or if the length is less than 3). Then, use a dp array where dp[i] holds the minimum number of cuts needed for the substring s[0..i]. For each i, check every index j (0 ≤ j ≤ i) and if s[j..i] is a palindrome, update dp[i] to be the minimum of its current value and dp[j-1] + 1 (with a special case when j is 0, meaning no cut is needed). This two-phase approach efficiently breaks down the problem and computes the answer while avoiding redundant palindrome checks.</p> <p><b>#312 Burst Balloons [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>Transform the problem by adding virtual balloons with value 1 at both ends to simplify edge handling.</li> <li>Use interval dynamic programming where dp[i][j] represents the maximum coins that can be obtained by bursting all balloons between indices i and j (exclusive).</li> <li>Realize that the order of bursting balloons matters; choose the last balloon to burst in each interval to break dependencies into independent subproblems.</li> </ul> <p>Dynamic Programmingp.2267</p> | <ul style="list-style-type: none"> <li>The solution involves iterating over all possible intervals and determining the optimal burst order.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n^3) - Three nested loops: one for the interval length, one for the starting index, and one for the final balloon choice.</p> <p>Space Complexity: O(n^2) - A 2D dynamic programming table is used to store the maximum coins for each interval.</p> <p><b>Solution</b></p> <p>The problem is solved using interval dynamic programming. First, extend the input array by adding 1 at both ends to handle edge cases seamlessly. Define a dp table where dp[i][j] represents the maximum coins that can be obtained by bursting all the balloons between i and j (not including i and j, as these are the virtual boundaries). For every interval [i, j], iterate through the possible choices for the last balloon k to burst in that interval. The value for bursting balloon k will be determined by the coins from bursting it last (i.e., nums[i] * nums[k] * nums[j]) plus the maximum coins from the two resulting subintervals: (i, k) and (k, j). By filling the dp table in increasing interval lengths, you build up to the solution for the overall interval that covers the entire (extended) array.</p> <p><b>#1000 Minimum Cost to Merge Stones [Hard]</b></p> <p>Dynamic Programmingp.2268</p> |



**Key Insights**

- You can only merge stones into one final pile if (n – 1) is divisible by (k – 1). If not, return –1.
- A dynamic programming approach is used where dp[i][j] represents the minimum cost to merge the subarray stones[i..j] into as few piles as possible.
- A prefix sum array is precomputed to quickly calculate the sum of stones in any subarray.
- When a segment can be merged into one pile (i.e. when (length – 1) % (k – 1) == 0), add the sum of stones in that segment to the dp value.

**Space & Time**

Time Complexity: O(n^3) due to the triple nested loop over segments and splits.

Space Complexity: O(n^2) from the dp table and O(n) for the prefix sum array.

**Solution**

The problem is solved using a range DP approach. We first verify that merging all piles into one is feasible by checking if (n – 1) % (k – 1) == 0. Next, a prefix sum array is computed to optimize range sum calculations. A 2D dp table dp[i][j] is then used to store the minimum cost to merge stones[i..j]. For each interval, we consider valid split points, updating dp[i][j] by combining sub-problems. If the current segment can form a single merged pile (determined by the condition on the interval length), the cost of merging that segment (the sum of

Dynamic Programmingp.2269

stones) is added to dp[i][j].

Stack

## #20 Stack

By Pattern · Python Only · 25 questions

Dynamic Programmingp.2270

Stack

## #20 Valid Parentheses

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

String · Stack

Lists: Grind 169

Time Complexity: O(n), where n is the length of the string, since we process each character once. Space Complexity: O(n) in the worst case for the stack when all characters are opening brackets.

**Problem**

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

An input string is valid if:

1. Open brackets must be closed by the same type of brackets.
2. Open brackets must be closed in the correct order.
3. Every close bracket has a corresponding open bracket of the same type.

**Example 1:**

Input: s = "()"

Output: true

**Example 2:**

Input: s = "()[]{}"

Output: true

Stackp.2271

**Example 3:**  
Input: s = "()"   
Output: false  
**Example 4:**  
Input: s = "([])"

Output: true  
**Example 5:**  
Input: s = "([)]"   
Output: false

- Constraints:**
- 1 <= s.length <= 104
  - s consists of parentheses only '()[]{}'.

Community Solutions (Python)

• WalkCC

```
class Solution:
 def isValid(self, s: str) -> bool:
 stack = []

 for c in s:
 if c == '(':
 stack.append('(')
 elif c == '{':
 stack.append('{')
 elif c == '[':
 stack.append('[')
 elif c == ')':
 if not stack or stack[-1] != '(':
 return False
 stack.pop()
 elif c == '}':
 if not stack or stack[-1] != '{':
 return False
 stack.pop()
 elif c == ']':
 if not stack or stack[-1] != '[':
 return False
 stack.pop()

 return not stack
```

Stackp.2272

```
stack.append('(')
elif not stack or stack.pop() != c:
 return False

return not stack
```

• LeetDoocs

```
class Solution:
 def isValid(self, s: str) -> bool:
 stk = []
 d = {'(': ')', '[': ']', '{': '}' }
 for c in s:
 if c in '({[':
 stk.append(c)
 elif not stk or stk.pop() + c not in d:
 return False
 return not stk
```

• SimplyLeet

```
def isValid(s: str) -> bool:
 # Mapping from closing brackets to their
 # corresponding opening brackets.
 paren_map = {'(': ')', '[': ']', '{': '}' }
 # Initialize an empty list to use as a stack.
 stack = []
 # Process each character in the string.
 for char in s:
 # If the character is a closing bracket.
 if char in paren_map:
 # Pop the top element from the stack if
 # available, else use a placeholder.
```

Stackp.2273

```
top_element = stack.pop() if stack else '#'
Check if the mapping does not match the top
element.

if paren_map[char] != top_element:
 return False

else:
 # For opening brackets, push onto the stack.
 stack.append(char)
 # Return True if all brackets are matched (i.e., the
 # stack is empty).
 return not stack
```

• LC.ca

```
class Solution:
 def isValid(self, s: str) -> bool:
 stk = []
 d = {'(': ')', '[': ']', '{': '}' }
 for c in s:
 if c in '({[':
 stk.append(c)
 elif not stk or stk.pop() + c not in d:
 return False
 return not stk
```

Stackp.2274

Stack

## #232 Implement Queue using Stacks

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Stack · Design · Queue

Lists: Grind 169

Time Complexity: Amortized O(1) per operation Space Complexity: O(n)

**Problem**

Implement a first in first out (FIFO) queue using only two stacks. The implemented queue should support all the functions of a normal queue (push, peek, pop, and empty).

Implement the MyQueue class:

- void push(int x) Pushes element x to the back of the queue.
- int pop() Removes the element from the front of the queue and returns it.
- int peek() Returns the element at the front of the queue.
- boolean empty() Returns true if the queue is empty, false otherwise.

**Notes:**

- You must use only standard operations of a stack, which means only push to top, peek/pop from top, size,

Stackp.2275

and is empty operations are valid.

- Depending on your language, the stack may not be supported natively. You may simulate a stack using a list or deque (double-ended queue) as long as you use only a stack's standard operations.

**Example 1:**

**Input**  
["MyQueue", "push", "push", "peek", "pop", "empty"]  
[[], [1], [2], [], [], []]  
**Output**  
[null, null, null, 1, 1, false]

**Explanation**  
MyQueue myQueue = new MyQueue();

**Constraints:**

- 1 <= x <= 9
- At most 100 calls will be made to push, pop, peek, and empty.
- All the calls to pop and peek are valid.

**Follow-up:** Can you implement the queue such that each operation is amortized O(1) time complexity? In other words, performing n operations will take overall O(n) time even if one of those operations may take longer.

Community Solutions (Python)

• WalkCC

Stackp.2276

Sheet 253 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait

Stackp.2277

```
 return self.stk2.pop()

def peek(self) -> int:
 self.move()
 return self.stk2[-1]

def empty(self) -> bool:
 return not self.stk1 and not self.stk2

def move(self):

 if not self.stk2:
 while self.stk1:
 self.stk2.append(self.stk1.pop())

Your MyQueue object will be instantiated and called as
such:
obj = MyQueue()
obj.push(x)
param_2 = obj.pop()
param_3 = obj.peek()

param_4 = obj.empty()
```

• SimplyLeet

```
class MyQueue:
 def __init__(self):
 # Stack for enqueue operations.
 self.input_stack = []
 # Stack for dequeue operations.
 self.output_stack = []

 def push(self, x: int) -> None:
 # Always push new elements onto the input stack.
 self.input_stack.append(x)

 def pop(self) -> int:
 # Ensure the output stack has the current queue
 order.
 if not self.output_stack:
 while self.input_stack:
 # Move all elements from input_stack to
 output_stack.
 self.output_stack.append(self.input_stack
 .pop())
 # The top element of output_stack is the front of
 the queue.
 return self.output_stack.pop()
```

```
def peek(self) -> int:
 # Ensure the output stack has the current queue
 order.
 if not self.output_stack:
 while self.input_stack:
 # Move all elements from input_stack to
 output_stack.
 self.output_stack.append(self.input_stack
 .pop())
 # Peek the top of output_stack without popping.
 return self.output_stack[-1]

def empty(self) -> bool:

 # The queue is empty if both stacks are empty.
 return not self.input_stack and not
 self.output_stack

• LC.ca

class MyQueue:
 def __init__(self):
 self.stk1 = []
 self.stk2 = [] # reversed order

 def push(self, x: int) -> None:
 self.stk1.append(x)

 def pop(self) -> int:
 self.move()
```

```
 return self.stk2.pop()

def peek(self) -> int:
 self.move()
 return self.stk2[-1]

def empty(self) -> bool:
 return not self.stk1 and not self.stk2

def move(self):

 if not self.stk2: # only when skt2 is empty
 while self.stk1:
 self.stk2.append(self.stk1.pop())

Your MyQueue object will be instantiated and called as
such:
obj = MyQueue()
obj.push(x)
param_2 = obj.pop()
param_3 = obj.peek()

param_4 = obj.empty()

#####

class MyQueue:

 def __init__(self):
 self.sk = []
 self.rsk = [] # reversed
```

```
def push(self, x: int) -> None:
 self.sk.append(x);

def pop(self) -> int:
 self.peek()
 return self.rsk.pop()

def peek(self) -> int:
 if self.rsk:

 return self.rsk[-1]
 else:
 while self.sk:
 self.rsk.append(self.sk.pop())
 return self.rsk[-1]

def empty(self) -> bool:

 return not (self.sk or self.rsk)

Your MyQueue object will be instantiated and called as
such:
obj = MyQueue()
obj.push(x)
param_2 = obj.pop()
param_3 = obj.peek()
param_4 = obj.empty()
```

Stack

EAS

#844 Backspace String Compare

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Two Pointers · String · Stack · Simulation

Lists: Grind 169

Time Complexity: O(n) where n is the length of the longer string. Space Complexity: O(1) using the two-pointer strategy (O(n) if using stacks).

Problem

Given two strings s and t, return true if they are equal when both are typed into empty text editors. '#' means a backspace character.

Note that after backspacing an empty text, the text will continue empty.

Example 1:

Input: s = "ab#c", t = "ad#c"

Output: true

Explanation: Both s and t become "ac".

Example 2:

Input: s = "ab##", t = "c#d#"

Output: true

Explanation: Both s and t become "".

Example 3:

Input: s = "a#c", t = "b"

Output: false

Explanation: s becomes "c" while t becomes "b".

Constraints:

• 1 <= s.length, t.length <= 200

• s and t only contain lowercase letters and '#' characters.

Follow up: Can you solve it in O(n) time and O(1) space?

Community Solutions (Python)

• WalkCC

class Solution:
 def backspaceCompare(self, s: str, t: str) -> bool:
 def backspace(s: str) -> str:
 stack = []
 for c in s:
 if c != '#':
 stack.append(c)
 elif stack:
 stack.pop()
 return stack

 return backspace(s) == backspace(t)

class Solution:
 def backspaceCompare(self, s: str, t: str) -> bool:
 i = len(s) - 1 # s' index
 j = len(t) - 1 # t's index

 while True:
 # Delete characters of s if needed.
 backspace = 0
 while i >= 0 and (s[i] == '#' or backspace > 0):
 backspace += 1 if s[i] == '#' else -1
 i -= 1

 i -= 1
 # Delete characters of t if needed.
 backspace = 0
 while j >= 0 and (t[j] == '#' or backspace > 0):
 backspace += 1 if t[j] == '#' else -1
 j -= 1

 j -= 1
 if i >= 0 and j >= 0 and s[i] == t[j]:
 i -= 1
 j -= 1
 else:
 break

 return i == -1 and j == -1

• LeetDoocs

class Solution:
 def backspaceCompare(self, s: str, t: str) -> bool:
 i, j, skip1, skip2 = len(s) - 1, len(t) - 1, 0, 0
 while i >= 0 or j >= 0:
 while i >= 0:
 if s[i] == '#':
 skip1 += 1
 i -= 1
 elif skip1:
 skip1 -= 1
 i -= 1

 while j >= 0:
 if t[j] == '#':
 skip2 += 1
 j -= 1
 elif skip2:
 skip2 -= 1
 j -= 1

 if i >= 0 and j >= 0:
 if s[i] != t[j]:
 return False
 elif i >= 0 or j >= 0:
 return False
 i, j = i - 1, j - 1
 return True

• SimplyLeet



```
Function to compare two strings with backspace
processing
def backspaceCompare(s: str, t: str) -> bool:
 # Pointers for both strings starting at the end
 i, j = len(s) - 1, len(t) - 1

 # Process both strings until all characters are
 compared
 while i >= 0 or j >= 0:
 # Process string s: skip backspaced characters
 skip = 0
 while i >= 0:

 if s[i] == '#':
 skip += 1
 i -= 1
 elif skip > 0:
 skip -= 1
 i -= 1
 else:
 break

 # Process string t: skip backspaced characters
```

Stack p.2287

```
 skip = 0
 while j >= 0:
 if t[j] == '#':
 skip += 1
 j -= 1
 elif skip > 0:
 skip -= 1
 j -= 1
 else:
 break

 # Compare the current valid characters from s and
 t
 if i >= 0 and j >= 0:
 if s[i] != t[j]:
 return False
 elif i >= 0 or j >= 0:
 return False

 i -= 1
 j -= 1

 return True

Example test cases
print(backspaceCompare("ab#c", "ad#c")) # Expected
output: True
print(backspaceCompare("ab##", "c#d#")) # Expected
output: True
print(backspaceCompare("a#c", "b")) # Expected output:
False
```

Stack p.2288

```
• LC.ca
class Solution:
 def backspaceCompare(self, s: str, t: str) -> bool:
 i, j, skip1, skip2 = len(s) - 1, len(t) - 1, 0, 0
 while i >= 0 or j >= 0:
 while i >= 0:
 if s[i] == '#':
 skip1 += 1
 i -= 1
 elif skip1:
 skip1 -= 1
 i -= 1
 else:
 break
 while j >= 0:
 if t[j] == '#':
 skip2 += 1
 j -= 1
 elif skip2:
 skip2 -= 1
 j -= 1
 else:
 break
 if i >= 0 and j >= 0:
 if s[i] != t[j]:
 return False
 elif i >= 0 or j >= 0:
 return False
 i, j = i - 1, j - 1
 return True
```

Stack p.2289

Stack

## #143 Reorder List

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Linked List · Two Pointers · Stack · Recursion

Lists: Grind 169

Time Complexity:  $O(n)$  - Each of the three main steps (finding the middle, reversing, merging) takes  $O(n)$  time. Space Complexity:  $O(1)$  - The algorithm is done in-place with only a few pointers.

### Problem

You are given the head of a singly linked-list. The list can be represented as:

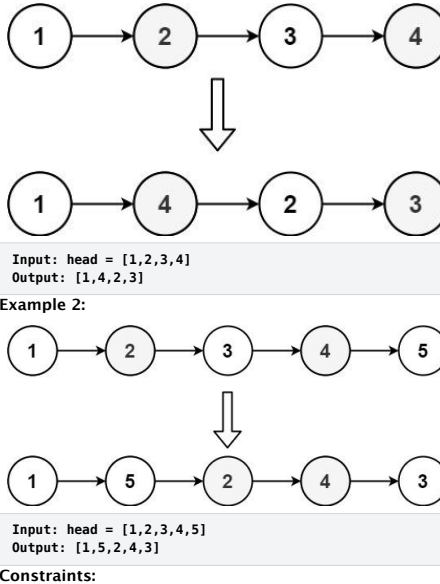
$L_0 \rightarrow L_1 \rightarrow \dots \rightarrow L_n - 1 \rightarrow L_n$

Reorder the list to be on the following form:

$L_0 \rightarrow L_n \rightarrow L_1 \rightarrow L_n - 1 \rightarrow L_2 \rightarrow L_n - 2 \rightarrow \dots$

You may not modify the values in the list's nodes. Only nodes themselves may be changed.

Example 1:



Stack p.2291

- The number of nodes in the list is in the range  $[1, 5 * 10^4]$ .
- $1 \leq \text{Node.val} \leq 1000$

### Community Solutions (Python)

#### • WalkCC

```
class Solution:
 def reorderList(self, head: ListNode) -> None:
 def findMid(head: ListNode):
 prev = None
 slow = head
 fast = head

 while fast and fast.next:
 prev = slow
 slow = slow.next
 fast = fast.next.next
 prev.next = None

 return slow

 def reverse(head: ListNode) -> ListNode:
 prev = None
 curr = head

 while curr:
 curr.next = prev
 prev = curr
 curr = curr.next
```

Stack p.2292

```
 next = curr.next
 curr.next = prev
 prev = curr
 curr = next

 return prev

def merge(l1: ListNode, l2: ListNode) -> None:
 while l2:
 next = l1.next
 l1.next = l2
 l1 = l2
 l2 = next

 if not head or not head.next:
 return

 mid = findMid(head)
 reversed = reverse(mid)
 merge(head, reversed)
```

• LeetDoocs

Stack p.2293

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def reorderList(self, head: Optional[ListNode]) -> None:
 fast = slow = head
 while fast.next and fast.next.next:
 slow = slow.next
 fast = fast.next.next

 cur = slow.next
 slow.next = None

 pre = None
 while cur:
 t = cur.next
 cur.next = pre
 pre = cur
 cur = t

 cur = head

 while pre:
 t = pre.next
 pre.next = cur.next
 cur.next = pre
 cur, pre = pre.next, t
```

• SimplyLeet

Stack p.2294

```
Definition for singly-linked list.
class ListNode:
 def __init__(self, val=0, next=None):
 self.val = val
 self.next = next

class Solution:
 def reorderList(self, head: ListNode) -> None:
 if not head or not head.next:
 return

 # Step 1: Find the middle of the linked list
 # using slow and fast pointers.
 slow, fast = head, head
 while fast and fast.next:
 slow = slow.next
 fast = fast.next.next

 # Step 2: Reverse the second half of the list.
 prev, curr = None, slow
 slow.next = None # Split the list into two
 halves.
```

Stack p.2295

```
while curr:
 next_temp = curr.next # Save next node.
 curr.next = prev # Reverse current pointer.
 prev = curr # Move prev to current.
 curr = next_temp # Move to the next node.

Step 3: Merge the two halves, alternating
nodes.
first, second = head, prev
while second:
 temp1 = first.next # Save next node from
 first half.

 temp2 = second.next # Save next node from
 reversed second half.
 first.next = second # Link first node to
 second list
 node.
 second.next = temp1 # Link second node to the
 next node in first half.
 first = temp1 # Move pointer in first half.
 second = temp2 # Move pointer in second half.
```

• LC.ca

```
Definition for singly-linked list.
class ListNode:
def __init__(self, val=0, next=None):
self.val = val
self.next = next
class Solution:
 def reorderList(self, head: Optional[ListNode]) -> None:

 fast = slow = head
 while fast.next and fast.next.next:
 slow = slow.next

 fast = fast.next.next

 cur = slow.next
 slow.next = None

 pre = None
 while cur:
 t = cur.next
 cur.next = pre
 pre, cur = cur, t

 cur = head

 while pre:
 t = pre.next
 pre.next = cur.next
 cur.next = pre
 cur, pre = pre.next, t
```

Stack

#150 Evaluate Reverse Polish NotationMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Math · Stack

Lists: Grind 169

Time Complexity: O(n), where n is the number of tokens (each token is processed once). Space Complexity: O(n), in the worst-case scenario where all tokens are numbers and are stored in the stack.

Problem

You are given an array of strings tokens that represents an arithmetic expression in a Reverse Polish Notation. Evaluate the expression. Return an integer that represents the value of the expression.

Note that:

- The valid operators are '+', '-', '\*', and '/'.
- Each operand may be an integer or another expression.
- The division between two integers always truncates toward zero.
- There will not be any division by zero.
- The input represents a valid arithmetic expression in a reverse polish notation.

• The answer and all the intermediate calculations can be represented in a 32-bit integer.

Example 1:

Input: tokens = ["2","1","+","3","\*"]  
Output: 9  
Explanation: ((2 + 1) \* 3) = 9

Example 2:

Input: tokens = ["4","13","5","/","+"]  
Output: 6  
Explanation: (4 + (13 / 5)) = 6

Example 3:

Input: tokens = ["10","6","9","3","+","-11","\*","/","\*","17","+","5","+"]  
Output: 22  
Explanation: ((10 \* (6 / ((9 + 3) \* -11))) + 17) + 5 = ((10 \* (6 / (12 \* -11))) + 17) + 5 = ((10 \* (6 / -132)) + 17) + 5 = ((10 \* 0) + 17) + 5 = (0 + 17) + 5 = 17 + 5

Constraints:

- 1 <= tokens.length <= 104
- tokens[i] is either an operator: "+", "-", "\*", or "/", or an integer in the range [-200, 200].

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def evalRPN(self, tokens):
 # Brute Force (stack is already optimal) -
 simulate token by token
 stack = []
 ops = {'+': lambda a,b: a+b, '-': lambda a,b: a-b,
 '*': lambda a,b: a*b, '/': lambda a,b: int(a/b)}

 for tok in tokens:
 if tok in ops:
 b, a = stack.pop(), stack.pop()

 stack.append(ops[tok](a, b))
 else:
 stack.append(int(tok))
 return stack[0]
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def evalRPN(self, tokens: list[str]) -> int:
 stack = []
 op = {
 '+': lambda a, b: a + b,
 '-': lambda a, b: a - b,
 '*': lambda a, b: a * b,
 '/': lambda a, b: int(a / b),
 }
```

```
for token in tokens:
 if token in op:
 b = stack.pop()
 a = stack.pop()
 stack.append(op[token](a, b))
 else:
 stack.append(int(token))

return stack.pop()
```

• LeetDoocs

```
import operator

class Solution:
 def evalRPN(self, tokens: List[str]) -> int:
 opt = {
 "+": operator.add,
 "-": operator.sub,
 "*": operator.mul,
 "/": operator.truediv,
 }
 s = []
 for token in tokens:
 if token in opt:
 s.append(int(opt[token](s.pop(-2), s.pop(-1))))
 else:
 s.append(int(token))
 return s[0]
```

```
class Solution:
 def evalRPN(self, tokens: List[str]) -> int:
 nums = []
 for t in tokens:
 if len(t) > 1 or t.isdigit():
 nums.append(int(t))
 else:
 if t == "+":
 nums[-2] += nums[-1]
 elif t == "-":
 nums[-2] -= nums[-1]
 elif t == "*":
 nums[-2] *= nums[-1]
 else:
 nums[-2] = int(nums[-2] / nums[-1])
 nums.pop()
 return nums[0]
```

• SimplyLeet

```
def evalRPN(tokens):
 # Initialize a stack to hold numbers
 stack = []
 # Loop through each token in the input list
 for token in tokens:
 if token in {"+", "-", "*", "/"}:
 # Pop the two topmost values; note the order
 is important
 b = stack.pop()
 a = stack.pop()
 # Apply the operation based on the token
```

```
if token == "+":
 result = a + b
elif token == "-":
 result = a - b
elif token == "*":
 result = a * b
else:
 # Use int division to truncate towards
 zero
 result = int(a / b)
 # Push the result back onto the stack

stack.append(result)
else:
 # Token is a number; push its integer value
 onto the stack
 stack.append(int(token))
 # The final result is the only element left in the
 stack
 return stack.pop()
```

• Example usage:

# tokens = ["2", "1", "+", "3", "\*"]

# print(evalRPN(tokens)) # Output: 9

• LC.ca

```
class Solution:
 def evalRPN(self, tokens: List[str]) -> int:
 nums = []
 for t in tokens:
 if len(t) > 1 or t.isdigit():
 nums.append(int(t))
 else:
 if t == "+":
 nums[-2] += nums[-1]
 elif t == "-":
 nums[-2] -= nums[-1]
 elif t == "*":
 nums[-2] *= nums[-1]
 else:
 nums[-2] = int(nums[-2] / nums[-1])
 nums.pop()
 return nums[0]
```

Stack

#155 Min StackMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Stack · Design

Lists: Grind 169

Time Complexity: O(1) per operation (push, pop, top, getMin) Space Complexity: O(n), where n is the number of elements in the stack, since an additional stack is used to keep track of minima.

Problem

Design a stack that supports push, pop, top, and retrieving the minimum element in constant time.

Implement the MinStack class:

- MinStack() initializes the stack object.
- void push(int val) pushes the element val onto the stack.
- void pop() removes the element on the top of the stack.
- int top() gets the top element of the stack.
- int getMin() retrieves the minimum element in the stack.

You must implement a solution with O(1) time complexity for each function.

Example 1:

Stackp.2305

Input

["MinStack", "push", "push", "push", "getMin", "pop", "top", "getMin"]  
[[], [-2], [0], [-3], [], [], [], []]

Output

[null, null, null, null, -3, null, 0, -2]

Explanation

Constraints:

- 231 <= val <= 231 - 1
- Methods pop, top and getMin operations will always be called on non-empty stacks.
- At most 3 \* 10^4 calls will be made to push, pop, top, and getMin.

Community Solutions (Python)

- WalkCC

class MinStack:  
 def \_\_init\_\_(self):  
 self.stack = []  
  
 def push(self, x: int) -> None:  
 mn = x if not self.stack else min(self.stack[-1][1], x)  
 self.stack.append([x, mn])  
  
 def pop(self) -> None:  
 self.stack.pop()

Stackp.2306

def top(self) -> int:  
 return self.stack[-1][0]  
  
def getMin(self) -> int:  
 return self.stack[-1][1]

- LeetCode

class MinStack:  
 def \_\_init\_\_(self):  
 self.stk1 = []  
 self.stk2 = [inf]  
  
 def push(self, val: int) -> None:  
 self.stk1.append(val)  
 self.stk2.append(min(val, self.stk2[-1]))  
  
 def pop(self) -> None:  
  
 self.stk1.pop()  
 self.stk2.pop()  
  
 def top(self) -> int:  
 return self.stk1[-1]  
  
 def getMin(self) -> int:  
 return self.stk2[-1]

Stackp.2307

# Your MinStack object will be instantiated and called as such:  
# obj = MinStack()  
# obj.push(val)  
# obj.pop()  
# param\_3 = obj.top()  
# param\_4 = obj.getMin()

- SimplyLeet

# Python implementation of MinStack with detailed comments  
  
class MinStack:  
 def \_\_init\_\_(self):  
 # Main stack to store all values  
 self.stack = []  
 # Auxiliary stack to store minimum values  
 self.min\_stack = []  
  
 def push(self, val: int) -> None:

Stackp.2308

# Push value onto main stack  
self.stack.append(val)  
# If min\_stack is empty or the new value is less than or equal to current minimum, push it onto min\_stack  
if not self.min\_stack or val <= self.min\_stack[-1]:  
 self.min\_stack.append(val)  
  
def pop(self) -> None:  
 # Pop the top value from main stack  
 popped\_value = self.stack.pop()  
 # If the popped value is the current minimum, pop it from min\_stack as well  
  
 if popped\_value == self.min\_stack[-1]:  
 self.min\_stack.pop()  
  
def top(self) -> int:  
 # Return the top value from main stack without removing it  
 return self.stack[-1]  
  
def getMin(self) -> int:  
 # Return the minimum value from min\_stack  
 return self.min\_stack[-1]

Stackp.2309

# Example usage:  
# minStack = MinStack()  
# minStack.push(-2)  
# minStack.push(0)  
# minStack.push(-3)  
# print(minStack.getMin()) # -> -3  
# minStack.pop()  
# print(minStack.top()) # -> 0  
# print(minStack.getMin()) # -> -2

- LC.ca

class MinStack:  
 def \_\_init\_\_(self):  
 self.sk = []  
 self.minsk = [inf] # trick, avoid later empty-check for min-stack  
  
 def push(self, x: int) -> None:  
 self.sk.append(x)  
 self.minsk.append(min(x, self.minsk[-1]))  
  
 def pop(self) -> None:

Stackp.2310

if not self.sk:  
 return  
 self.sk.pop()  
 self.minsk.pop()  
  
def top(self) -> int:  
 return self.sk[-1]  
  
def getMin(self) -> int:  
 return self.minsk[-1]

# Your MinStack object will be instantiated and called as such:  
# obj = MinStack()  
# obj.push(x)  
# obj.pop()  
# param\_3 = obj.top()  
# param\_4 = obj.getMin()

#####

class MinStack:  
  
def \_\_init\_\_(self):  
 self.sk = []  
 self.minsk = []  
  
def push(self, val: int) -> None:  
 self.sk.append(val)

Stackp.2311

# empty check  
self.minsk.append(val if not self.minsk else min(val, self.minsk[-1]))  
  
def pop(self) -> None:  
if not self.sk:  
 return  
  
self.sk.pop()  
self.minsk.pop()  
  
def top(self) -> int:  
 return self.sk[-1]  
  
def getMin(self) -> int:  
 return self.minsk[-1]  
  
# optimize above, using only 1 stack, with stack element as tuple: (val, its associated min)  
class MinStack:  
  
 def \_\_init\_\_(self):  
  
 self.\_stack = []  
  
 def push(self, x: int) -> None:  
 cur\_min = self.getMin()  
 if x < cur\_min:  
 cur\_min = x  
 self.\_stack.append((x, cur\_min))  
  
 def pop(self) -> None:  
 self.\_stack.pop()

Stackp.2312

def top(self) -> int:  
 if not self.\_stack:  
 return None  
 else:  
 return self.\_stack[-1][0]  
  
def getMin(self) -> int:  
 if not self.\_stack:  
 return float('inf')

else:  
 return self.\_stack[-1][1]

Stackp.2313

Stack

#173 Binary Search Tree IteratorMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Stack · Tree · Design · Binary Search Tree · Iterator

Lists: Denny Zhang

Time Complexity: O(1) on average per operation (amortized) Space Complexity: O(h), where h is the height of the tree

Problem

Implement the BSTIterator class that represents an iterator over the in-order traversal of a binary search tree (BST):

- BSTIterator(TreeNode root) Initializes an object of the BSTIterator class. The root of the BST is given as part of the constructor. The pointer should be initialized to a non-existent number smaller than any element in the BST.
- boolean hasNext() Returns true if there exists a number in the traversal to the right of the pointer, otherwise returns false.
- int next() Moves the pointer to the right, then returns the number at the pointer.

Notice that by initializing the pointer to a non-existent smallest number, the first call to next() will return the smallest element in the BST.

Stackp.2314

You may assume that next() calls will always be valid. That is, there will be at least a next number in the in-order traversal when next() is called. Example 1:

```
graph TD; 7((7)) --- 3((3)); 7 --- 15((15)); 15 --- 9((9)); 15 --- 20((20))
```

Stackp.2315

Input

["BSTIterator", "next", "next", "hasNext", "next", "hasNext", "next", "hasNext", "next", "hasNext"]  
[[[7, 3, 15, null, null, 9, 20]], [], [], [], [], [], [], []], []]  
Output  
[null, 3, 7, true, 9, true, 15, true, 20, false]

Explanation

BSTIterator bSTIterator = new BSTIterator([7, 3, 15, null, null, 9, 20]);

Constraints:

- The number of nodes in the tree is in the range [1, 105].
- 0 <= Node.val <= 106
- At most 105 calls will be made to hasNext, and next.

Follow up:

- Could you implement next() and hasNext() to run in average O(1) time and use O(h) memory, where h is the height of the tree?

Community Solutions (Python)

- WalkCC

Stackp.2316

```
class BSTIterator:
 def __init__(self, root: TreeNode | None):
 self.i = 0
 self.vals = []
 self._inorder(root)

 def next(self) -> int:
 self.i += 1
 return self.vals[self.i - 1]

 def hasNext(self) -> bool:
 return self.i < len(self.vals)

 def _inorder(self, root: TreeNode | None) -> None:
 if not root:
 return
 self._inorder(root.left)
 self.vals.append(root.val)
 self._inorder(root.right)
```

```
class BSTIterator:
 def __init__(self, root: TreeNode | None):
 self.stack = []
 self._pushLeftsUntilNull(root)

 def next(self) -> int:
 root = self.stack.pop()
 self._pushLeftsUntilNull(root.right)
 return root.val
```

Stackp.2317

```
def hasNext(self) -> bool:
 return self.stack

def _pushLeftsUntilNull(self, root: TreeNode | None) -> None:
 while root:
 self.stack.append(root)
 root = root.left
```

- LeetCode

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class BSTIterator:
 def __init__(self, root: TreeNode):
 def inorder(root):
 if root:
 inorder(root.left)
 self.vals.append(root.val)
 inorder(root.right)

 self.cur = 0
 self.vals = []
 inorder(root)

 def next(self) -> int:
 res = self.vals[self.cur]
 self.cur += 1
 return res
```

Stackp.2318

```
self.cur += 1
return res

def hasNext(self) -> bool:
 return self.cur < len(self.vals)

Your BSTIterator object will be instantiated and called as such:
obj = BSTIterator(root)
param_1 = obj.next()

param_2 = obj.hasNext()
```

- SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

class BSTIterator:
 def __init__(self, root):
 # Initialize an empty stack to simulate in-order traversal
```

Stackp.2319

```
self.stack = []
Push all left nodes starting from the root
self._push_left_branch(root)

def _push_left_branch(self, node):
 # Push node and all its left children to the stack
 while node:
 self.stack.append(node)
 node = node.left

 def next(self):
 # Pop the top node which represents the next smallest element
 node = self.stack.pop()
 # If the node has a right child, push its left branch onto the stack
 if node.right:
 self._push_left_branch(node.right)
 # Return the node's value
 return node.val

 def hasNext(self):
 # The iterator has a next element if the stack is not empty
 return len(self.stack) > 0
```

- LC.ca

Stackp.2320

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class BSTIterator:
 def __init__(self, root: TreeNode):
 self.stack = []
 self._leftmost_inorder(root)

 def _leftmost_inorder(self, node):
 while node:
 self.stack.append(node)
 node = node.left

 def next(self) -> int:
 cur = self.stack.pop()
 node = cur.right
 self._leftmost_inorder(node)

 return cur.val

 def hasNext(self) -> bool:
 return len(self.stack) > 0

Your BSTIterator object will be instantiated and called as such:
obj = BSTIterator(root)
param_1 = obj.next()
param_2 = obj.hasNext()
```

Stackp.2321

```
#####

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class BSTIterator:
 def __init__(self, root: TreeNode):

 self.stack = []
 while root:
 self.stack.append(root)
 root = root.left

 def next(self) -> int:
 cur = self.stack.pop()
 node = cur.right
 while node: # duplicated while block
 self.stack.append(node)

 node = node.left
 return cur.val

 def hasNext(self) -> bool:
 return len(self.stack) > 0

Your BSTIterator object will be instantiated and called as such:
obj = BSTIterator(root)
param_1 = obj.next()
```

Stackp.2322

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```
param_2 = obj.hasNext()
```

Stack

## #227 Basic Calculator II

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Math · String · Stack

Lists: Grind 169

Time Complexity: O(n) where n is the length of the expression, since the algorithm processes each character once. Space Complexity: O(n) in the worst-case scenario when all numbers are stored in the s

### Problem

Given a string s which represents an expression, evaluate this expression and return its value.

The integer division should truncate toward zero.

You may assume that the given expression is always valid. All intermediate results will be in the range of [-231, 231 - 1].

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as eval().

### Example 1:

```
Input: s = "3+2+2"
Output: 7
```

### Example 2:

Stack

p.2323

Stack

p.2324

Stack

p.2325

```
if not c.isdigit() and c != ' ' or i == len(s) - 1:
 if op == '+' or op == '-':
 ans += prevNum
 prevNum = currNum if op == '+' else -currNum
 elif op == '*':
 prevNum = prevNum * currNum
 elif op == '/':
 if prevNum < 0:
 prevNum = math.ceil(prevNum / currNum)
 else:
 prevNum = prevNum // currNum
 op = c
 currNum = 0

return ans + prevNum
```

### • LeetDoocs

```
class Solution:
 def calculate(self, s: str) -> int:
 v, n = 0, len(s)
 sign = '+'
 stk = []
 for i, c in enumerate(s):
 if c.isdigit():
 v = v * 10 + int(c)
 if i == n - 1 or c in '+-*/':
 match sign:
 case '+':
 stk.append(v)
 case '-':
 stk.append(-v)
 case '*':
 stk.append(stk.pop() * v)
 case '/':
 sign = c
 v = 0

 return sum(stk)
```

Stack

p.2326

Stack

p.2327

Stack

p.2328

```
return sum(stack)
```

```
Example usage:
print(calculate("3+2+2")) # Output: 7
print(calculate(" 3/2 ")) # Output: 1
print(calculate(" 3+5 / 2 ")) # Output: 5
```

### • LC.ca

```
class Solution:
 def calculate(self, s: str) -> int:
 v, n = 0, len(s)
 sign = '+'
 stk = []
 for i, c in enumerate(s):
 if c.isdigit():
 v = v * 10 + int(c)
 if i == n - 1 or c in '+-*/':
 if sign == '+':
 stk.append(v)
 # for "10-2+5": when '-' encountered, var
 # 'sign' is still '+'
 # so '10' will be pushed to stk before
 # setting sign to '-'
 elif sign == '-':
 stk.append(-v)
 elif sign == '*':
 stk.append(stk.pop() * v)
 elif sign == '/':
 stk.append(int(stk.pop() / v))
 else:
 sign = c
 v = 0
```

Stack

p.2329

Stack

p.2330

Stack

p.2331

```
print("operator not supported")
```

```
sign = c # reset inside 'if'
v = 0
return sum(stk)
```

```
Input: s = " 3/2 "
Output: 1
```

### Example 3:

```
Input: s = " 3+5 / 2 "
Output: 5
```

### Constraints:

- 1 <= s.length <= 3 \* 10<sup>5</sup>
- s consists of integers and operators ('+', '-', '\*', '/') separated by some number of spaces.
- s represents a valid expression.
- All the integers in the expression are non-negative integers in the range [0, 2<sup>31</sup> - 1].
- The answer is guaranteed to fit in a 32-bit integer.

### Community Solutions (Python)

### • WalkCC

```
class Solution:
 def calculate(self, s: str) -> int:
 ans = 0
 prevNum = 0
 currNum = 0
 op = '+'

 for i, c in enumerate(s):
 if c.isdigit():
 currNum = currNum * 10 + int(c)
```

```
if char.isdigit():
 # Build the current number by shifting
 # previous digits.
 current_number = current_number * 10 +
 int(char)
 # If the character is an operator (or we're at
 # the end), process the previous operator.
 if char in "+-*/" or i == len(s) - 1:
 if previous_operator == '+':
 stack.append(current_number)
 elif previous_operator == '-':
 stack.append(-current_number)
 elif previous_operator == '*':
```

```
last_number = stack.pop()
stack.append(last_number *
current_number)
elif previous_operator == '/':
 last_number = stack.pop()
 # Python division truncates toward
 # negative infinity by default, so we use int() to truncate
 # toward zero.
 stack.append(int(last_number /
current_number))
Reset for the next number and update the
operator.
previous_operator = char
current_number = 0

Sum up all numbers in the stack which gives the
final result.
```

Stack

## #255 Verify Preorder Sequence in Binary Search Tree

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Tree · BST · Stack · Monotonic Stack

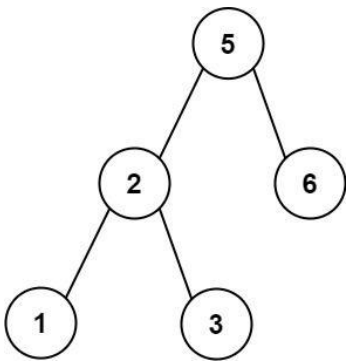
Lists: Premium 98

Time Complexity: O(n) – Each element is processed once. Space Complexity: O(n) – In the worst-case an explicit stack is used; can be optimized to O(1) by reusing the input array.

### Problem

Given an array of unique integers preorder, return true if it is the correct preorder traversal sequence of a binary search tree.

### Example 1:



Input: preorder = [5,2,1,3,6]  
Output: true

Example 2:

Input: preorder = [5,2,6,1,3]  
Output: false

Constraints:

- $1 \leq \text{preorder.length} \leq 104$
- $1 \leq \text{preorder}[i] \leq 104$
- All the elements of preorder are unique.

Stack p.2332

Follow up: Could you do it using only constant space complexity?

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def verifyPreorder(self, preorder):
 # Brute Force $O(n^2)$ - simulate without stack via inner loop
 n = len(preorder)
 result = [-1] * n
 for i in range(n):
 for j in range(i + 1, n):
 if preorder[j] > preorder[i]:
 result[i] = preorder[j]
```

```
 break
 return result
```

Community Solutions (Python)

• WalkCC

```
class Solution:
 def verifyPreorder(self, preorder: List[int]) -> bool:
 i = 0

 def dfs(min: int, max: int) -> None:
 nonlocal i
 if i == len(preorder):
 return
 if preorder[i] < min or preorder[i] > max:
 return
 return
```

Stack p.2333

```
 val = preorder[i]
 i += 1
 dfs(min, val)
 dfs(val, max)

 dfs(-math.inf, math.inf)
 return i == len(preorder)
```

```
class Solution:
 def verifyPreorder(self, preorder: List[int]) -> bool:
 low = -math.inf
 stack = []

 for p in preorder:
 if p < low:
 return False
 while stack and stack[-1] < p:
 low = stack.pop()
```

```
 stack.append(p)
```

```
 return True
```

Stack p.2334

```
class Solution:
 def verifyPreorder(self, preorder: List[int]) -> bool:
 low = -math.inf
 i = -1

 for p in preorder:
 if p < low:
 return False
 while i >= 0 and preorder[i] < p:
 low = preorder[i]
```

```
 i -= 1
 i += 1
 preorder[i] = p
```

```
 return True
```

• LeetDoocs

```
class Solution:
 def verifyPreorder(self, preorder: List[int]) -> bool:
 stk = []
 last = -inf
 for x in preorder:
 if x < last:
 return False
 while stk and stk[-1] < x:
 last = stk.pop()
 stk.append(x)
```

```
 return True
```

• SimplyLeet

Stack p.2335

```
Python solution using a monotonic stack with detailed comments
def verifyPreorder(preorder):
 lower_bound = float('-inf') # Set initial lower bound to negative infinity
 stack = [] # Use a stack to track the BST construction
 for value in preorder:
 # If the current value is less than lower_bound, the BST property is violated
 if value < lower_bound:
 return False
 # If current value is greater than the top of stack, pop elements and update lower_bound
 while stack and value > stack[-1]:
```

```
 lower_bound = stack.pop() # Update lower_bound as we move to the right subtree
 # Push the current value onto the stack
 stack.append(value)
 return True

Example usage:
print(verifyPreorder([5,2,1,3,6])) # Expected output: True
```

• LC.ca

Stack p.2336

```
class Solution:
 def verifyPreorder(self, preorder: List[int]) -> bool:
 stk = []
 last = -inf
 for x in preorder:
 if x < last:
 return False
 while stk and stk[-1] < x:
 last = stk.pop()
 stk.append(x)
```

```
 return True
```

```
class Solution: # o(N) but modifying input list
 def verifyPreorder(self, preorder: List[int]) -> bool:
 low = -math.inf
 i = -1

 for p in preorder:
 if p < low:
```

Stack p.2337

```
 return False
 while i >= 0 and preorder[i] < p:
 low = preorder[i]
 i -= 1
 i += 1
 preorder[i] = p
```

```
 return True
```

#####

class Solution(object):

```
 # O(n) sapce complexity, **stack**
 def verifyPreorder(self, preorder):
 """
 :type preorder: List[int]
 :rtype: bool
 """
 if len(preorder) <= 1:
```

```
 return True
 stack, lastElem = [preorder[0]], None
 for i in range(1, len(preorder)):
 if lastElem > preorder[i]:
 return False
 while len(stack) > 0 and preorder[i] > stack[-1]:
 lastElem = stack.pop()
 stack.append(preorder[i])
```

```
 return True
```

Stack p.2338

Stack

## #394 Decode String

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

String · Stack · Recursion

Lists: Grind 169

Time Complexity:  $O(n)$ , where  $n$  is the length of the string. Space Complexity:  $O(n)$ , for the recursion stack or the auxiliary stack used.

Problem

Given an encoded string, return its decoded string.

The encoding rule is:  $k[\text{encoded\_string}]$ , where the encoded\\_string inside the square brackets is being repeated exactly  $k$  times. Note that  $k$  is guaranteed to be a positive integer.

You may assume that the input string is always valid; there are no extra white spaces, square brackets are well-formed, etc. Furthermore, you may assume that the original data does not contain any digits and that digits are only for those repeat numbers,  $k$ . For example, there will not be input like 3a or 2[4].

The test cases are generated so that the length of the output will never exceed 105.

Example 1:

Stack p.2339

Sheet 260 / 317 · LeetMastery By Pattern · Python Only · 9-up Portrait

Input: s = "3[a]2[bc]"  
Output: "aaabcbc"

Example 2:

Input: s = "3[a2[c]]"  
Output: "accaccacc"

Example 3:

Input: s = "2[abc]3[cd]ef"  
Output: "abccabccdcdef"

Constraints:

- $1 \leq s.length \leq 30$
- $s$  consists of lowercase English letters, digits, and square brackets '['].
- $s$  is guaranteed to be a valid input.
- All the integers in  $s$  are in the range [1, 300].

Community Solutions (Python)

• WalkCC

```
class Solution:
 def decodeString(self, s: str) -> str:
 stack = [] # (prevStr, repeatCount)
 currStr = ''
 currNum = 0

 for c in s:
 if c.isdigit():
 currNum = currNum * 10 + int(c)
 else:
```

Stack p.2340



```
 if c == '[':
 stack.append((currStr, currNum))
 currStr = ''
 currNum = 0
 elif c == ']':
 prevStr, num = stack.pop()
 currStr = prevStr + num * currStr
 else:
 currStr += c

 return currStr

class Solution:
 def decodeString(self, s: str) -> str:
 ans = ''

 while self.i < len(s) and s[self.i] != '':
 if s[self.i].isdigit():
 k = 0
 while self.i < len(s) and s[self.i].isdigit():
 k = k * 10 + int(s[self.i])
 self.i += 1

 self.i += 1 # '['
 decodedString = self.decodeString(s)
 self.i += 1 # ']'
 ans += k * decodedString
 else:
 ans += s[self.i]
 self.i += 1

 return ans
```

Stackp.2341

```
 i = 0

 • LeetCode

 class Solution:
 def decodeString(self, s: str) -> str:
 s1, s2 = [], []
 num, res = 0, ''
 for c in s:
 if c.isdigit():
 num = num * 10 + int(c)
 elif c == '[':
 s1.append(num)
 s2.append(res)

 num, res = 0, ''
 elif c == ']':
 res = s2.pop() + res * s1.pop()
 else:
 res += c
 return res

 • SimplyLeet
```

Stackp.2342

```
def decodeString(s: str) -> str:
 # index pointer as a list so it can be mutable within recursion
 def helper(i: int) -> (str, int):
 decoded = ""
 while i < len(s):
 if s[i].isdigit():
 # build the number (it might be more than one digit)
 k = 0
 while i < len(s) and s[i].isdigit():
 k = k * 10 + int(s[i])

 i += 1
 # skip the '[' character
 i += 1 # skip '['
 # recursively solve the substring inside the brackets
 sub, i = helper(i)
 # append the substring repeated k times
 decoded += sub * k
 elif s[i] == ']':
 # returning decoded string and new index after ']'
 return decoded, i + 1
```

Stackp.2343

```
 else:
 # accumulate alphabetic characters
 decoded += s[i]
 i += 1
 return decoded, i

 result, _ = helper(0)
 return result

Example usage

#print(decodeString("3[a2[c]]")) # Output: "accaccacc"

 • LC.ca

 class Solution:
 def decodeString(self, s: str) -> str:
 num_stk, str_stk = [], []
 num, res = 0, ''
 for c in s:
 if c.isdigit():
 num = num * 10 + int(c)
 elif c == '[':
 num_stk.append(num)
 str_stk.append(res) # initial, res being pushed is empty str ''

 num, res = 0, ''
 elif c == ']':
 res = str_stk.pop() + res * num_stk.pop()
 else:
 res += c
 return res
```

Stackp.2344

Stack

#439 Ternary Expression ParserMED

LeetCode · SimplyLeet · WalkCC · LeetCode

String · Stack · Recursion

Lists: Premium 98

Time Complexity: O(n), where n is the length of the expression, as we process each character exactly once. Space Complexity: O(n), due to the use of a stack for storing characters during evaluation.

Problem

Given a string expression representing arbitrarily nested ternary expressions, evaluate the expression, and return the result of it.

You can always assume that the given expression is valid and only contains digits, '?', ',', 'T', and 'F' where 'T' is true and 'F' is false. All the numbers in the expression are one-digit numbers (i.e., in the range [0, 9]).

The conditional expressions group right-to-left (as usual in most languages), and the result of the expression will always evaluate to either a digit, 'T' or 'F'.

Example 1:

Stackp.2345

```
Input: expression = "T?2:3"
Output: "2"
Explanation: If true, then result is 2; otherwise result is 3.
```

Example 2:

```
Input: expression = "F?1:T?4:5"
Output: "4"
Explanation: The conditional expressions group right-to-left. Using parenthesis, it is read/evaluated as:
"(F ? 1 : (T ? 4 : 5))" -> "(F ? 1 : 4)" -> "4"
or "(F ? 1 : (T ? 4 : 5))" -> "(T ? 4 : 5)" -> "4"
```

Example 3:

```
Input: expression = "T?T?F:5:3"
Output: "F"
Explanation: The conditional expressions group right-to-left. Using parenthesis, it is read/evaluated as:
"(T ? (T ? F : 5) : 3)" -> "(T ? F : 3)" -> "F"
"(T ? (T ? F : 5) : 3)" -> "(T ? F : 5)" -> "F"
```

Constraints:

- 5 <= expression.length <= 104
- expression consists of digits, 'T', 'F', '?', and ','.
- It is guaranteed that expression is a valid ternary expression and that each number is a one-digit number.

Community Solutions (Python)

Stackp.2346

```
 • WalkCC

 class Solution:
 def parseTernary(self, expression: str) -> str:
 c = expression[self.i]

 if self.i + 1 == len(expression) or expression[self.i + 1] == ':':
 self.i += 2
 return str(c)

 self.i += 2
 first = self.parseTernary(expression)

 second = self.parseTernary(expression)

 return first if c == 'T' else second

 i = 0

 • LeetCode

 class Solution:
 def parseTernary(self, expression: str) -> str:
 stk = []
 cond = False
 for c in expression[::-1]:
 if c == ':':
 continue
 if c == '?':
 cond = True
 else:
```

Stackp.2347

```
 if cond:
 if c == 'T':
 x = stk.pop()
 stk.pop()
 stk.append(x)
 else:
 stk.pop()
 cond = False
 else:
 stk.append(c)

 return stk[0]

 • SimplyLeet

 def parseTernary(expression: str) -> str:
 # Initialize stack to hold characters/results during evaluation
 stack = []
 # Iterate backwards over the expression
 i = len(expression) - 1
 while i >= 0:
 if expression[i] == '?':
 # Pop the true and false parts from the stack
 true_expr = stack.pop()
 false_expr = stack.pop()
```

Stackp.2348

```
 i -= 1 # Move to the condition character
 # Choose based on the condition
 if expression[i] == 'T':
 stack.append(true_expr)
 else:
 stack.append(false_expr)
 elif expression[i] == ':':
 # Skip the colon as it is only a separator
 pass
 else:
 # Push digits, 'T', and 'F' onto stack as is
 stack.append(expression[i])
 i -= 1 # Move to the next character
 return stack[-1]

 # Example usage:
 print(parseTernary("T?2:3")) # Expected output "2"

 • LC.ca

 class Solution:
 def parseTernary(self, expression: str) -> str:
 stk = []
 cond = False
 for c in expression[::-1]:
 if c == ':':
 continue
 if c == '?':
 cond = True
 else:
```

Stackp.2349

```

 if cond:
 if c == 'T':
 x = stk.pop()
 stk.pop()
 stk.append(x)
 else:
 stk.pop()
 cond = False
 else:
 stk.append(c)

return stk[0]
```

Stack

#484 Find PermutationMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · String · Stack · Greedy

Lists: Premium 98

Time Complexity: O(n) – Each element is pushed and popped exactly once. Space Complexity: O(n) – For the stack and the output array.

**Problem**

A permutation perm of n integers of all the integers in the range [1, n] can be represented as a string s of length n – 1 where:

- s[i] == 'I' if perm[i] < perm[i + 1], and
- s[i] == 'D' if perm[i] > perm[i + 1].

Given a string s, reconstruct the lexicographically smallest permutation perm and return it.

**Example 1:**

Input: s = "I"  
Output: [1,2]  
Explanation: [1,2] is the only legal permutation that can be represented by s, where the number 1 and 2 construct an increasing relationship.

**Example 2:**

```

Input: s = "DI"
Output: [2,1,3]
Explanation: Both [2,1,3] and [3,1,2] can be represented as "DI", but since we want to find the smallest Lexicographical permutation, you should return [2,1,3]
```

- Constraints:**
- 1 <= s.length <= 105
  - s[i] is either 'I' or 'D'.

```

Brute Force Approach
class Solution:
 def findPermutation(self, s):
 # Brute Force O(n^2) – simulate without stack via inner loop
 n = len(s)
 result = [-1] * n
 for i in range(n):
 for j in range(i + 1, n):
 if s[j] > s[i]:
 result[i] = s[j]
 break
 return result
```

- Community Solutions (Python)**
- WalkCC

```

class Solution:
 def findPermutation(self, s: str) -> List[int]:
 ans = []
 stack = []

 for i, c in enumerate(s):
 stack.append(i + 1)
 if c == 'I':
 while stack: # Consume all decreasings
 ans.append(stack.pop())

 stack.append(len(s) + 1)

 while stack:
 ans.append(stack.pop())

 return ans

class Solution:
 def findPermutation(self, s: str) -> List[int]:
 ans = [i for i in range(1, len(s) + 2)]

 # For each D* group (s[i..j]), reverse ans[i..j + 1].
 i = -1
 j = -1

 def getNextIndex(c: str, start: int) -> int:
 for i in range(start, len(s)):

```

```

 if s[i] == c:
 return i
 return len(s)

 while True:
 i = getNextIndex('D', j + 1)
 if i == len(s):
 break
 j = getNextIndex('I', i + 1)
 ans[i:j + 1] = ans[i:j + 1][::-1]

 return ans

• LeetCode

class Solution:
 def findPermutation(self, s: str) -> List[int]:
 n = len(s)
 ans = list(range(1, n + 2))
 i = 0
 while i < n:
 j = i
 while j < n and s[j] == 'D':
 j += 1
 ans[i : j + 1] = ans[i : j + 1][::-1]

 i = max(i + 1, j)
 return ans

• SimplyLeet
```

```

Function to find the lexicographically smallest permutation
def findPermutation(s):
 stack = [] # Use a stack to hold numbers during a sequence of 'D's
 result = [] # Result list for the final permutation
 # Iterate over 0 to len(s) (inclusive) to process all characters plus one final number.
 for i in range(len(s) + 1):
 # Push i+1 since our numbers range from 1 to n
 stack.append(i + 1)
 # If we are at the end or the current character is 'I', flush the stack.
 if i == len(s) or s[i] == 'I':
 while stack:
 result.append(stack.pop())

 # Example usage:
 print(findPermutation("DI"))

• LC.ca

class Solution:
 def findPermutation(self, s: str) -> List[int]:
 n = len(s)
 ans = list(range(1, n + 2))
 i = 0
 while i < n:
 j = i
 while j < n and s[j] == 'D':
 j += 1
 ans[i : j + 1] = ans[i : j + 1][::-1]
```

```

 i = max(i + 1, j)
 return ans
```

Stack

#735 Asteroid CollisionMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Stack · Simulation

Lists: Grind 169

Time Complexity: O(n) where n is the number of asteroids, since each asteroid is pushed and popped at most once. Space Complexity: O(n) in the worst case, when no collisions occur and all asteroids re

**Problem**

We are given an array asteroids of integers representing asteroids in a row. The indices of the asteroid in the array represent their relative position in space. For each asteroid, the absolute value represents its size, and the sign represents its direction (positive meaning right, negative meaning left). Each asteroid moves at the same speed.

Find out the state of the asteroids after all collisions. If two asteroids meet, the smaller one will explode. If both are the same size, both will explode. Two asteroids moving in the same direction will never meet.

**Example 1:**

```

Input: asteroids = [5,10,-5]
Output: [5,10]
Explanation: The 10 and -5 collide resulting in 10. The 5 and 10 never collide.
```

```

Input: asteroids = [8,-8]
Output: []
Explanation: The 8 and -8 collide exploding each other.
```

```

Input: asteroids = [10,2,-5]
Output: [10]
Explanation: The 2 and -5 collide resulting in -5. The 10 and -5 collide resulting in 10.
```

```

Input: asteroids = [3,5,-6,2,-1,4]
Output: [-6,2,4]
Explanation: The asteroid -6 makes the asteroid 3 and 5 explode, and then continues going left. On the other side, the asteroid 2 makes the asteroid -1 explode and then continues going right, without reaching asteroid 4.
```

- Constraints:**
- 2 <= asteroids.length <= 104
  - -1000 <= asteroids[i] <= 1000
  - asteroids[i] != 0

**Brute Force (Python)**

```
Brute Force Approach
class Solution:
 def asteroidCollision(self, asteroids):
 # Brute Force O(n^2) - simulate without stack via
 inner loop
 n = len(asteroids)
 result = [-1] * n
 for i in range(n):
 for j in range(i + 1, n):
 if asteroids[j] > asteroids[i]:
 result[i] = asteroids[j]

 break
 return result
```

Community Solutions (Python)

```
• WalkCC
class Solution:
 def asteroidCollision(self, asteroids: list[int]) ->
list[int]:
 stack = []

 for a in asteroids:
 if a > 0:
 stack.append(a)
 else: # a < 0
 # Destroy the previous positive one(s).
 while stack and stack[-1] > 0 and stack[-1] < -a:
```

```
 stack.pop()
 if not stack or stack[-1] < 0:
 stack.append(a)
 elif stack[-1] == -a:
 stack.pop() # Both asteroids explode.
 else: # stack[-1] > the current asteroid.
 pass # Destroy the current asteroid, so do
 nothing.

 return stack

• LeetCode
class Solution:
 def asteroidCollision(self, asteroids: List[int]) ->
List[int]:
 stk = []
 for x in asteroids:
 if x > 0:
 stk.append(x)
 else:
 while stk and stk[-1] > 0 and stk[-1] <
-x:
 stk.pop()
 if stk and stk[-1] == -x:

 stk.pop()
 elif not stk or stk[-1] < 0:
 stk.append(x)

 return stk
```

- SimplyLeet

```
Function to simulate asteroid collisions
def asteroidCollision(asteroids):
 # Initialize an empty list as a stack to hold
 surviving asteroids
 stack = []
 # Iterate through each asteroid in the input list
 for ast in asteroids:
 # Flag to check if the current asteroid has
 exploded
 exploded = False
 # Check for potential collision while there is an
 asteroid in stack and a collision condition holds
 (right-moving on stack and left-moving current)
 while stack and ast < 0 and stack[-1] > 0:

 # If the incoming left-moving asteroid is
 larger than the right-moving one on stack, pop the top
 and continue checking
 if abs(ast) > abs(stack[-1]):
 stack.pop()
 continue # Continue the loop to check
 next potential collision
 # If they are equal, both explode: pop the
 stack and mark explosion, break the loop
 elif abs(ast) == abs(stack[-1]):
 stack.pop()
 # Mark the incoming asteroid as exploded and
 break out of the loop
 exploded = True
 break
```

```
 # If the asteroid did not explode during
 collisions, push it onto the stack
 if not exploded:
 stack.append(ast)
 # Return the surviving asteroids
 return stack

Example usage:
print(asteroidCollision([5,10,-5])) # Output: [5,10]

• LC.ca
class Solution:
 def asteroidCollision(self, asteroids: List[int]) ->
List[int]:
 stk = []
 for x in asteroids:
 if x > 0:
 stk.append(x)
 else:
 while stk and stk[-1] > 0 and stk[-1] <
-x:
 stk.pop()
 if stk and stk[-1] == -x:

 stk.pop()
 elif not stk or stk[-1] < 0:
 stk.append(x)

 return stk
```

Stack

#739 Daily TemperaturesMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Stack · Monotonic Stack

Lists: Grind 169

Time Complexity: O(n), where n is the number of temperatures, because each index is pushed and popped from the stack at most once. Space Complexity: O(n) for storing the stack and the answer array.

Problem

Given an array of integers temperatures represents the daily temperatures, return an array answer such that answer[i] is the number of days you have to wait after the ith day to get a warmer temperature. If there is no future day for which this is possible, keep answer[i] == 0 instead.

Example 1:

Input: temperatures = [73,74,75,71,69,72,76,73]  
Output: [1,1,4,2,1,1,0,0]

Example 2:

Input: temperatures = [30,40,50,60]  
Output: [1,1,1,0]

Example 3:

```
Input: temperatures = [30,60,90]
Output: [1,1,0]

Constraints:
 • 1 <= temperatures.length <= 105
 • 30 <= temperatures[i] <= 100

Brute Force (Python)

Brute Force Approach
class Solution:
 def dailyTemperatures(self, temperatures):
 # Brute Force O(n^2) - for each day scan forward
 for warmer day
 n = len(temperatures)
 res = [0] * n
 for i in range(n):
 for j in range(i + 1, n):
 if temperatures[j] > temperatures[i]:
 res[i] = j - i

 break
 return res
```

Community Solutions (Python)

- WalkCC

```
class Solution:
 def dailyTemperatures(self, temperatures: list[int]) ->
list[int]:
 ans = [0] * len(temperatures)
 stack = [] # a decreasing stack

 for i, temperature in enumerate(temperatures):
 while stack and temperature >
temperatures[stack[-1]]:
 index = stack.pop()
 ans[index] = i - index
 stack.append(i)

 return ans
```

```
• LeetCode
class Solution:
 def dailyTemperatures(self, temperatures: List[int])
-> List[int]:
 stk = []
 n = len(temperatures)
 ans = [0] * n
 for i in range(n - 1, -1, -1):
 while stk and temperatures[stk[-1]] <=
temperatures[i]:
 stk.pop()
 if stk:
 ans[i] = stk[-1] - i

 stk.append(i)
 return ans
```

```
• SimplyLeet
def dailyTemperatures(temperatures):
 n = len(temperatures)
 # Initialize answer list with zeros
 answer = [0] * n
 # Stack to keep indices of temperatures with
 unresolved next warmer days
 stack = []
 # Iterate through the temperature list
 for i, temp in enumerate(temperatures):
 # Check if the current temperature resolves any
 previous indices
 while stack and temperatures[stack[-1]] < temp:

 prev_index = stack.pop() # Get the index of
 the previous colder day
 answer[prev_index] = i - prev_index # Update
 the number of days waited
 # Push the current index onto the stack
 stack.append(i)
 return answer

Example usage:
if __name__ == "__main__":
 temps = [73,74,75,71,69,72,76,73]
 print(dailyTemperatures(temps))
```

- LC.ca

```
class Solution:
 def dailyTemperatures(self, temperatures: List[int])
-> List[int]:
 ans = [0] * len(temperatures)
 stk = []
 for i, t in enumerate(temperatures):
 while stk and temperatures[stk[-1]] < t:
 j = stk.pop()
 ans[j] = i - j
 stk.append(i)
 return ans
```

Stack

#962 Maximum Width RampMED

LeetDooocs · SimplyLeet · WalkCC · LeetCode

Array · Stack · Monotonic Stack

Lists: Denny Zhang

Time Complexity: O(n) Space Complexity: O(n)

Problem

A ramp in an integer array nums is a pair (i, j) for which i < j and nums[i] <= nums[j]. The width of such a ramp is j - i.

Given an integer array nums, return the maximum width of a ramp in nums. If there is no ramp in nums, return 0.

Example 1:

Input: nums = [6,0,8,2,1,5]

Output: 4

Explanation: The maximum width ramp is achieved at (i, j) = (1, 5): nums[1] = 0 and nums[5] = 5.

Example 2:

Input: nums = [9,8,1,0,1,9,4,0,4,1]

Output: 7

Explanation: The maximum width ramp is achieved at (i, j) = (2, 9): nums[2] = 1 and nums[9] = 1.

Constraints:

Stackp.2368

• 2 <= nums.length <= 5 \* 104

• 0 <= nums[i] <= 5 \* 104

Brute Force (Python)

# Brute Force Approach

class Solution:

def maxWidthRamp(self, nums):

# Brute Force O(n^2) - check every pair (i,j)

where i<j and nums[i]<=nums[j]

res = 0

for i in range(len(nums)):

for j in range(i, len(nums)):

if nums[j] >= nums[i]:

res = max(res, j - i)

return res

Community Solutions (Python)

• WalkCC

class Solution:

def maxWidthRamp(self, nums: list[int]) -> int:

ans = 0

stack = []

for i, num in enumerate(nums):

if stack == [] or num <= nums[stack[-1]]:

stack.append(i)

for i, num in reversed(list(enumerate(nums))):

Stackp.2369

while stack and num >= nums[stack[-1]]:

ans = max(ans, i - stack.pop())

return ans

• LeetCode

class Solution:

def maxWidthRamp(self, nums: List[int]) -> int:

stk = []

for i, v in enumerate(nums):

if not stk or nums[stk[-1]] > v:

stk.append(i)

ans = 0

for i in range(len(nums) - 1, -1, -1):

while stk and nums[stk[-1]] <= nums[i]:

ans = max(ans, i - stk.pop())

if not stk:

break

return ans

• SimplyLeet

Stackp.2370

# Python solution for Maximum Width Ramp

def maxWidthRamp(nums):

n = len(nums)

stack = []

# Build a stack of indices where array values are in decreasing order

for i in range(n):

if not stack or nums[i] < nums[stack[-1]]:

stack.append(i)

max\_width = 0

# Iterate backwards to find the maximum width ramp

for j in range(n - 1, -1, -1):

while stack and nums[j] >= nums[stack[-1]]:

max\_width = max(max\_width, j - stack.pop())

return max\_width

# Example usage:

nums = [6, 0, 8, 2, 1, 5]

print(maxWidthRamp(nums)) # Output: 4

• LC.ca

class Solution:

def maxWidthRamp(self, nums: List[int]) -> int:

stk = []

for i, v in enumerate(nums):

if not stk or nums[stk[-1]] > v:

stk.append(i)

ans = 0

for i in range(len(nums) - 1, -1, -1):

while stk and nums[stk[-1]] <= nums[i]:

ans = max(ans, i - stk.pop())

Stackp.2371

if not stk:

break

return ans

Stackp.2372

Stack

#1214 Two Sum BSTsMED

LeetDooocs · SimplyLeet · WalkCC · LeetCode

Two Pointers · Binary Search · Stack · Tree · DFS · BST · BFS

Lists: Premium 98

Time Complexity: O(n + m) where n and m are the number of nodes in the first and second trees respectively. Space Complexity: O(n) for storing the node values from the first tree (in worst-case scenar

Problem

Given the roots of two binary search trees, root1 and root2, return true if and only if there is a node in the first tree and a node in the second tree whose values sum up to a given integer target.

Example 1:

Stackp.2373

|   |   |   |   |
|---|---|---|---|
| 9 | 9 | 8 | 1 |
| 5 | 6 | 2 | 6 |
| 8 | 2 | 6 | 4 |
| 6 | 2 | 2 | 2 |

|   |   |
|---|---|
| 9 | 9 |
| 8 | 6 |

Input: root1 = [2,1,4], root2 = [1,0,3], target = 5

Output: true

Explanation: 2 and 3 sum up to 5.

Example 2:

Stackp.2374

```
graph TD
 0((0)) --> -10((-10))
 0 --> 10((10))
 5((5)) --> 1((1))
 5 --> 7((7))
 1 --> 0((0))
 1 --> 2((2))
```

Input: root1 = [0,-10,10], root2 = [5,1,7,0,2], target = 18

Output: false

Constraints:

• The number of nodes in each tree is in the range [1, 5000].

• -109 <= Node.val, target <= 109

Community Solutions (Python)

• WalkCC

Stackp.2375

class BSTIterator:

def \_\_init\_\_(self, root: TreeNode | None, leftToRight: bool):

self.stack = []

self.leftToRight = leftToRight

self.\_pushUntilNone(root)

def hasNext(self) -> bool:

return len(self.stack) > 0

def next(self) -> int:

node = self.stack.pop()

if self.leftToRight:

self.\_pushUntilNone(node.right)

else:

self.\_pushUntilNone(node.left)

return node.val

def \_pushUntilNone(self, root: TreeNode | None):

while root:

self.stack.append(root)

Stackp.2376

```
 root = root.left if self.leftToRight else
root.right

class Solution:
 def twoSumBSTs(
 self,
 root1: TreeNode | None,
 root2: TreeNode | None,
 target: int,
) -> bool:

 bst1 = BSTIterator(root1, True)
 bst2 = BSTIterator(root2, False)

 l = bst1.next()
 r = bst2.next()
 while True:
 summ = l + r
 if summ == target:
 return True
 if summ < target:

 if not bst1.hasNext():
 return False
 l = bst1.next()
 else:
 if not bst2.hasNext():
 return False
 r = bst2.next()
```

LeetCode

Stackp.2377

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def twoSumBSTs(
 self, root1: Optional[TreeNode], root2:
Optional[TreeNode], target: int
) -> bool:

 def dfs(root: Optional[TreeNode], i: int):
 if root is None:
 return
 dfs(root.left, i)
 nums[i].append(root.val)
 dfs(root.right, i)

 nums = [], []
 dfs(root1, 0)
 dfs(root2, 1)

 i, j = 0, len(nums[1]) - 1
 while i < len(nums[0]) and ~j:
 x = nums[0][i] + nums[1][j]
 if x == target:
 return True
 if x < target:
 i += 1
 else:
 j -= 1
 return False
```

Stackp.2378

• SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

def twoSumBSTs(root1: TreeNode, root2: TreeNode, target:
int) -> bool:
 # Create a set to store values from the first BST.
 values_set = set()

 # Helper function to perform DFS and add node values
 to the set.
 def dfs_collect(node):
 if not node:
 return
 values_set.add(node.val) # Add current node value
 dfs_collect(node.left) # Traverse left subtree
 dfs_collect(node.right) # Traverse right subtree

 # Collect all values from first tree.
```

Stackp.2379

```
def dfs_collect(root1)

Helper function to traverse second tree and check
for complement.
def dfs_check(node):
 if not node:
 return False
 # If the complement exists in the first tree
 values, return True.
 if target - node.val in values_set:
 return True
 # Recursively check left and right children.

 return dfs_check(node.left) or
dfs_check(node.right)

Check the second tree.
return dfs_check(root2)

Example usage:
Construct trees and call twoSumBSTs as needed.
```

LC.ca

Stackp.2380

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def twoSumBSTs(
 self, root1: Optional[TreeNode], root2:
Optional[TreeNode], target: int
) -> bool:

 def dfs(root: Optional[TreeNode], i: int):
 if root is None:
 return
 dfs(root.left, i)
 nums[i].append(root.val)
 dfs(root.right, i)

 nums = [], []
 dfs(root1, 0)
 dfs(root2, 1)

 i, j = 0, len(nums[1]) - 1
 while i < len(nums[0]) and ~j:
 x = nums[0][i] + nums[1][j]
 if x == target:
 return True
 if x < target:
 i += 1
 else:
 j -= 1
 return False
```

Stackp.2381

Stack

#1762 Buildings With an Ocean ViewMED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Stack · Monotonic Stack

Lists: Premium 98

Time Complexity: O(n) where n is the number of buildings. Space Complexity: O(n) for storing the result in the worst-case scenario.

Problem

There are n buildings in a line. You are given an integer array heights of size n that represents the heights of the buildings in the line.

The ocean is to the right of the buildings. A building has an ocean view if the building can see the ocean without obstructions. Formally, a building has an ocean view if all the buildings to its right have a smaller height.

Return a list of indices (0-indexed) of buildings that have an ocean view, sorted in increasing order.

Example 1:

```
Input: heights = [4,2,3,1]
Output: [0,2,3]
Explanation: Building 1 (0-indexed) does not have an
ocean view because building 2 is taller.
```

Stackp.2382

Example 2:

```
Input: heights = [4,3,2,1]
Output: [0,1,2,3]
Explanation: All the buildings have an ocean view.
```

Example 3:

```
Input: heights = [1,3,2,4]
Output: [3]
Explanation: Only building 3 has an ocean view.
```

Constraints:

- 1 <= heights.length <= 105
- 1 <= heights[i] <= 109

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findBuildingsWithOceanView(self, heights):
 # Brute Force O(n^2) - simulate without stack via
inner loop
 n = len(heights)
 result = [-1] * n
 for i in range(n):
 for j in range(i + 1, n):
 if heights[j] > heights[i]:
 result[i] = heights[j]

 break
 return result
```

Community Solutions (Python)

Stackp.2383

• WalkCC

```
class Solution:
 def findBuildings(self, heights: list[int]) ->
list[int]:
 stack = []

 for i, height in enumerate(heights):
 while stack and heights[stack[-1]] <= height:
 stack.pop()
 stack.append(i)

 return stack
```

• LeetCode

```
class Solution:
 def findBuildings(self, heights: List[int]) ->
List[int]:
 ans = []
 mx = 0
 for i in range(len(heights) - 1, -1, -1):
 if heights[i] > mx:
 ans.append(i)
 mx = heights[i]
 return ans[::-1]
```

• SimplyLeet

Stackp.2384

```
def findBuildingsWithOceanView(heights):
 n = len(heights)
 result = []
 max_height = 0
 # Iterate from the rightmost building to the leftmost
 building.
 for i in range(n - 1, -1, -1):
 if heights[i] > max_height:
 result.append(i)
 max_height = heights[i]
 # Reverse the result to return indices in increasing
 order.

 result.reverse()
 return result

Example usage:
heights = [4,2,3,1]
print(findBuildingsWithOceanView(heights)) # Output:
[0,2,3]
```

• LC.ca

```
class Solution:
 def findBuildings(self, heights: List[int]) ->
List[int]:
 ans = []
 mx = 0
 for i in range(len(heights) - 1, -1, -1):
 if heights[i] > mx:
 ans.append(i)
 mx = heights[i]
 return ans[::-1]
```

Stackp.2385

Stack

#32 Longest Valid Parentheses

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode  
String · Dynamic Programming · Stack  
Lists: Grind 169  
Time Complexity: O(n) where n is the length of the string. Space Complexity: O(n) for the stack data structure.  
**Problem**  
Given a string containing just the characters '(' and ')', return the length of the longest valid (well-formed) parentheses substring.  
**Example 1:**  

Input: s = "()()"

Output: 2

Explanation: The longest valid parentheses substring is "()".

  
**Example 2:**  

Input: s = ")()())"

Output: 4

Explanation: The longest valid parentheses substring is "()()".

  
**Example 3:**

Stackp.2386

Input: s = ""  
Output: 0

**Constraints:**

- 0 <= s.length <= 3 \* 10<sup>4</sup>
- s[i] is '(', or ')'

**Community Solutions (Python)**  

- WalkCC  

```
class Solution:
 def longestValidParentheses(self, s: str) -> int:
 s2 = ')' + s
 # dp[i] := the length of the longest valid
 # parentheses in the substring
 # s2[1..i]
 dp = [0] * len(s2)

 for i in range(1, len(s2)):
 if s2[i] == ')' and s2[i - dp[i] - 1] == '(':
 dp[i] = dp[i - 1] + dp[i - dp[i] - 1] + 2

 return max(dp)
```
- LeetCode

Stackp.2387

```
class Solution:
 def longestValidParentheses(self, s: str) -> int:
 n = len(s)
 f = [0] * (n + 1)
 for i, c in enumerate(s, 1):
 if c == "(":
 if i > 1 and s[i - 2] == "(":
 f[i] = f[i - 2] + 2
 else:
 j = i - f[i - 1] - 1

 if j and s[j - 1] == "(":
 f[i] = f[i - 1] + 2 + f[j - 1]

 return max(f)
```

```
class Solution:
 def longestValidParentheses(self, s: str) -> int:
 stack = [-1]
 ans = 0
 for i in range(len(s)):
 if s[i] == '(':
 stack.append(i)
 else:
 stack.pop()
 if not stack:

 stack.append(i)
 else:
 ans = max(ans, i - stack[-1])

 return ans
```

SimplyLeet

Stackp.2388

```
Python solution using a stack
def longestValidParentheses(s: str) -> int:
 # Initialize the stack with the base index -1
 stack = [-1]
 max_length = 0

 # Iterate over the string with index
 for i, char in enumerate(s):
 if char == '(':
 # Push the index of '(' onto the stack

 stack.append(i)
 else:
 # Pop the last index for ')'
 stack.pop()
 if not stack:
 # If stack is empty, push current index
 # as the new base
 stack.append(i)
 else:
 # Calculate the length of the current
 # valid substring
 current_length = i - stack[-1]

 max_length = max(max_length,
 current_length)

 return max_length

Example usage:
print(longestValidParentheses("()()())") # Output: 4
```

LC.ca

Stackp.2389

```
class Solution:
 def longestValidParentheses(self, s: str) -> int:
 left = right = 0
 res = 0
 for c in s: # from left to right, '()' => will
 never hit left==right
 if c == '(':
 left += 1
 else:
 right += 1
 if left == right:

 res = max(res, 2 * left)
 if left < right:
 left = right = 0

 left = right = 0 # dont forget to reset
 for c in reversed(s): # from right to left, '()'
 => will never hit left==right
 if c == '(':
 left += 1
 else:
 right += 1
```

Stackp.2390

```
if left == right:
 res = max(res, 2 * left)
if left > right: # reverse '<' to '>'
 left = right = 0

return res

#####

class Solution:
 def longestValidParentheses(self, s: str) -> int:
 n = len(s)
 if n < 2:
 return 0
 dp = [0] * n
 for i in range(1, n):
 if s[i] == ')':
 if s[i - 1] == '(':
 dp[i] = 2 + (dp[i - 2] if i > 1 else
 0)
 else:

 j = i - dp[i - 1] - 1
 if j >= 0 and s[j] == '(':
 dp[i] = 2 + dp[i - 1] + dp[j - 1]

 return max(dp)

 #####

 >>> s="abcdefg"
 >>> [print(i,"",c) for i, c in enumerate(s, 1)]
```

Stackp.2391

```
1, a
2, b
3, c
4, d
5, e
6, f
7, g
[None, None, None, None, None, None, None]
>>>
...
```

```
class Solution:
 def longestValidParentheses(self, s: str) -> int:
 n = len(s)
 f = [0] * (n + 1)
 for i, c in enumerate(s, 1): # starting i from 1,
 not 0
 if c == "(":
 if i > 1 and s[i - 2] == "(":
 f[i] = f[i - 2] + 2
 else:
 j = i - f[i - 1] - 1

 if j and s[j - 1] == "(":
 f[i] = f[i - 1] + 2 + f[j - 1]

 return max(f)
```

Stackp.2392

Stack

#42 Trapping Rain Water

HAR

LeetCode · SimplyLeet · WalkCC · LeetCode  
Array · Two Pointers · Dynamic Programming · Stack · Monotonic Stack  
Lists: Grind 169  
Time Complexity: O(n) Space Complexity: O(1) when using the two-pointer approach  
**Problem**  
Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.  
**Example 1:**

Stackp.2393

Input: height = [0,1,0,2,1,0,1,3,2,1,2,1]  
Output: 6  
Explanation: The above elevation map (black section) is represented by array [0,1,0,2,1,0,1,3,2,1,2,1]. In this case, 6 units of rain water (blue section) are being trapped.

**Example 2:**  

Input: height = [4,2,0,3,2,5]  
Output: 9

**Constraints:**

- n == height.length
- 1 <= n <= 2 \* 10<sup>4</sup>
- 0 <= height[i] <= 10<sup>5</sup>

**Brute Force (Python)**  

```
Brute Force Approach
class Solution:
 def trap(self, height):
 # Brute Force O(n^2) - for each bar scan
 # left/right for max heights
 n = len(height)
 res = 0
 for i in range(n):
 left_max = max(height[:i + 1])
 right_max = max(height[i:])
 res += min(left_max, right_max) - height[i]

 return res
```

Stackp.2394

Stack

Sheet 266 / 317 · LeetCode Mastery By Pattern · Python Only · 3-up Portrait

Stack



Community Solutions (Python)

```
• WalkCC

class Solution:
 def trap(self, height: List[int]) -> int:
 n = len(height)
 l = [0] * n # l[i] := max(height[0..i])
 r = [0] * n # r[i] := max(height[i..n])

 for i, h in enumerate(height):
 l[i] = h if i == 0 else max(h, l[i - 1])

 for i, h in reversed(list(enumerate(height))):
 r[i] = h if i == n - 1 else max(h, r[i + 1])

 return sum(min(l[i], r[i]) - h
 for i, h in enumerate(height))

class Solution:
 def trap(self, height: List[int]) -> int:
 if not height:
 return 0

 ans = 0
 l = 0
 r = len(height) - 1
 maxL = height[l]
 maxR = height[r]
```

```
while l < r:
 if maxL < maxR:
 ans += maxL - height[l]
 l += 1
 maxL = max(maxL, height[l])
 else:
 ans += maxR - height[r]
 r -= 1
 maxR = max(maxR, height[r])

return ans

• LeetCode

class Solution:
 def trap(self, height: List[int]) -> int:
 n = len(height)
 left = [height[0]] * n
 right = [height[-1]] * n
 for i in range(1, n):
 left[i] = max(left[i - 1], height[i])
 right[n - i - 1] = max(right[n - i], height[n - i - 1])
 return sum(min(l, r) - h for l, r, h in zip(left, right, height))

• SimplyLeet
```

```
Python solution using the two pointers approach
def trap(height):
 # Initialize left and right pointers and water accumulator
 left, right = 0, len(height) - 1
 left_max, right_max = 0, 0
 water = 0

 # Traverse the elevation map from both ends
 while left < right:
 # Update left_max and right_max as we encounter new bars
 if height[left] < height[right]:
 # left side is the determining side
 if height[left] >= left_max:
 left_max = height[left] # update left max
 since current height is higher
 else:
 water += left_max - height[left] # water trapped at left position
 left += 1 # move left pointer to the right
 else:
 # right side is the determining side
 if height[right] >= right_max:
```

```
right_max = height[right] # update right max since current height is higher
else:
 water += right_max - height[right] # water trapped at right position
 right -= 1 # move right pointer to the left
return water

Example usage:
print(trap([0,1,0,2,1,0,1,3,2,1,2,1])) # Expected output: 6

• LC.ca

class Solution:
 def trap(self, height: List[int]) -> int:
 n = len(height)
 if n < 3:
 return 0

 lmx = [height[0]] * n
 rmx = [height[n - 1]] * n
 for i in range(1, n):
 lmx[i] = max(lmx[i - 1], height[i]) # i self compare
```

```
rmx[n - 1 - i] = max(rmx[n - i], height[n - 1 - i])

no negative, worst is, min(left,right) is itself
return sum(min(lmx[i], rmx[i]) - height[i] for i in range(n))

#####

class Solution: # zip
 def trap(self, height: List[int]) -> int:
 n = len(height)

 left = [height[0]] * n
 right = [height[-1]] * n
 for i in range(1, n):
 left[i] = max(left[i - 1], height[i])
 right[n - i - 1] = max(right[n - i], height[n - i - 1])
 return sum(min(l, r) - h for l, r, h in zip(left, right, height))

#####

class Solution: # find the highest bar first
```

```
def trap(self, height: List[int]) -> int:
 if not height:
 return 0

 total_water = 0
 max_index = 0
 max_height = float('-inf')

 # Find the index of the maximum height
 for i in range(len(height)):

 if height[i] > max_height:
 max_height = height[i]
 max_index = i

 # Calculate water trapped on the left side of the maximum height
 left_max = height[0]
 for i in range(1, max_index):
 if left_max < height[i]:
 left_max = height[i]
 total_water += left_max - height[i]

 # Calculate water trapped on the right side of the maximum height
 right_max = height[-1]
 for i in range(len(height) - 2, max_index, -1):
 if right_max < height[i]:
 right_max = height[i]
 total_water += right_max - height[i]

 return total_water
```

Stack

#84 Largest Rectangle in Histogram HAR

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Stack · Monotonic Stack

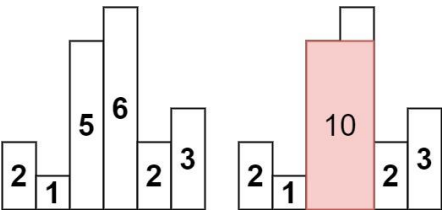
Lists: Grind 169

Time Complexity: O(n) Space Complexity: O(n)

Problem

Given an array of integers heights representing the histogram's bar height where the width of each bar is 1, return the area of the largest rectangle in the histogram.

Example 1:



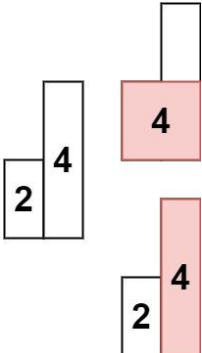
Stack p.2401

Input: heights = [2,1,5,6,2,3]

Output: 10

Explanation: The above is a histogram where width of each bar is 1. The largest rectangle is shown in the red area, which has an area = 10 units.

Example 2:



Input: heights = [2,4]

Output: 4

Constraints:

Stack p.2402

• 1 <= heights.length <= 105

• 0 <= heights[i] <= 104

Brute Force (Python)

# Brute Force Approach

class Solution:

def largestRectangleArea(self, heights):

# Brute Force O(n^2) - fix left bar, expand right, track min height

n = len(heights)

res = 0

for i in range(n):

min\_h = heights[i]

for j in range(i, n):

min\_h = min(min\_h, heights[j])

res = max(res, min\_h \* (j - i + 1))

return res

Community Solutions (Python)

• WalkCC

Stack p.2403

```
class Solution:
 def largestRectangleArea(self, heights: List[int]) ->
int:
 ans = 0
 stack = []

 for i in range(len(heights) + 1):
 while stack and (i == len(heights) or
heights[stack[-1]] > heights[i]):
 h = heights[stack.pop()]
 w = i - stack[-1] - 1 if stack else i
 ans = max(ans, h * w)

 stack.append(i)

 return ans
```

• LeetCode

```
class Solution:
 def largestRectangleArea(self, heights: List[int]) ->
int:
 n = len(heights)
 stk = []
 left = [-1] * n
 right = [n] * n
 for i, h in enumerate(heights):
 while stk and heights[stk[-1]] >= h:
 right[stk[-1]] = i
 stk.pop()
```

Stackp.2404

```
if stk:
 left[i] = stk[-1]
 stk.append(i)
 return max(h * (right[i] - left[i] - 1) for i, h
in enumerate(heights))
```

• SimplyLeet

```
Python solution

def largestRectangleArea(heights):
 # Append a zero to flush out the remaining bars in
the stack
 heights.append(0)
 stack = [] # Stack to store indices of bars
 max_area = 0

 # Iterate over each index and bar height
 for i, h in enumerate(heights):

 # Process bars that are taller than the current
bar
 while stack and heights[stack[-1]] > h:
 height = heights[stack.pop()] # Height of the
bar at the index popped from the stack
 # Calculate width: if stack is empty, width
is the current index; otherwise, it's the difference
width = i if not stack else i - stack[-1] - 1
 max_area = max(max_area, height * width)
 stack.append(i)

 return max_area
```

Stackp.2405

```
Example usage:
if __name__ == "__main__":
 histogram = [2, 1, 5, 6, 2, 3]
 print(largestRectangleArea(histogram))
```

• LC.ca

```
...
 h = heights[right_i]
 left_i = right_i

 h * (right_i - left_i - 1)

class Solution:
 def largestRectangleArea(self, heights: List[int]) ->
int:
 n = len(heights)
 stk = []
 left = [-1] * n

 right = [n] * n
 for i, h in enumerate(heights):
 while stk and heights[stk[-1]] >= h:
 right[stk[-1]] = i
 stk.pop()
 if stk:
 left[i] = stk[-1] # same as below:
stk[-1] in 'i - stack[-1] - 1'
 stk.append(i)

 # valid for one element input [3]
```

Stackp.2406

```
return max(h * (right[i] - left[i] - 1) for i, h
in enumerate(heights))

#####

class Solution:
 def largestRectangleArea(self, heights: List[int]) ->
int:
 if not heights:
 return 0

 heights.append(-1) # make for loop running
stack = []
ans = 0
 for i in range(0, len(heights)):
 while stack and heights[i] <
heights[stack[-1]]:
 currentLowestBarIndex = stack.pop()
 h = heights[currentLowestBarIndex]
 # stack[-1] is after pop(). why not
i-currentLowestBarIndex?
 # because not working for tie case in the
stack,
 # eg. for input=[4,2,0,3,2,5], expected=6
but output=5
```

Stackp.2407

```
heights[i] < heights[stack[-1]]
heights[i] < heights[stack[-1]]
heights[i] < heights[stack[-1]]
index = stack[-1] - 1
w = i - stack[-1] - 1 if stack else i
ans = max(ans, h * w)
stack.append(i)

heights.pop() # restore, pop -1
return ans
```

Stackp.2408

Stack#224 Basic CalculatorHAR

LeetCode · SimplyLeet · WalkCC · LeetCode  
Math · String · Stack · Recursion  
Lists: Grind 169  
Time Complexity: O(n) where n is the length of the input string, since we process each character once. Space Complexity: O(n) in the worst-case scenario due to the use of a stack for nested parentheses

Problem  
Given a string s representing a valid expression, implement a basic calculator to evaluate it, and return the result of the evaluation.  
Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as eval().

Example 1:  
Input: s = "1 + 1"  
Output: 2  
Example 2:  
Input: s = " 2-1 + 2 "  
Output: 3  
Example 3:

Stackp.2409

Input: s = "(1+(4+5+2)-3)+(6+8)"  
Output: 23  
Constraints:  
• 1 <= s.length <= 3 \* 10<sup>5</sup>  
• s consists of digits, '+', '-', '(', ')', and ' '.  
• s represents a valid expression.  
• '+' is not used as a unary operation (i.e., "+1" and "(+2 + 3)" is invalid).  
• '-' could be used as a unary operation (i.e., "-1" and "-(2 + 3)" is valid).  
• There will be no two consecutive operators in the input.  
• Every number and running calculation will fit in a signed 32-bit integer.

Community Solutions (Python)

• WalkCC

```
class Solution:
 def calculate(self, s: str) -> int:
 ans = 0
 num = 0
 sign = 1
 stack = [sign] # stack[-1]: the current environment's
sign
 for c in s:
 if c.isdigit():
 num = num * 10 + int(c)
```

Stackp.2410

```
elif c == '(':
 stack.append(sign)
elif c == ')':
 stack.pop()
elif c == '+' or c == '-':
 ans += sign * num
 sign = (1 if c == '+' else -1) * stack[-1]
 num = 0

return ans + sign * num
```

• LeetCode

```
class Solution:
 def calculate(self, s: str) -> int:
 stk = []
 ans, sign = 0, 1
 i, n = 0, len(s)
 while i < n:
 if s[i].isdigit():
 x = 0
 j = i
 while j < n and s[j].isdigit():
 x = x * 10 + int(s[j])
 j += 1
 ans += sign * x
 i = j - 1
 elif s[i] == "+":
 sign = 1
 elif s[i] == "-":
 sign = -1
 elif s[i] == "(":
 stk.append(ans)
```

Stackp.2411

```
stk.append(sign)
ans, sign = 0, 1
elif s[i] == "(":
 ans = stk.pop() * ans + stk.pop()
 i += 1
return ans
```

• SimplyLeet

```
Python solution for the Basic Calculator problem

def calculate(s: str) -> int:
 # Stack to store pairs of (current result, current
sign) before a '('
 stack = []
 result = 0 # Running total
 number = 0 # Current number being processed
 sign = 1 # Current sign (1 or -1)

 # Iterate through each character in the string

 for char in s:
 if char.isdigit():
 # Build number digit by digit
 number = number * 10 + int(char)
 elif char in "+-":
 # Add the complete number with its sign to
result
 result += sign * number
 # Reset number for the next potential number
 number = 0
 # Update the sign based on the operator
```

Stackp.2412

```
 sign = 1 if char == '+' else -1
 elif char == '(':
 # Push the current result and sign onto the
stack
 stack.append(result)
 stack.append(sign)
 # Reset result and sign for the new
sub-expression
 result = 0
 sign = 1
 elif char == ')':
 # Complete the number before the parentheses
close

 result += sign * number
 number = 0
 # The top of the stack is the sign before the
parenthesis
 prev_sign = stack.pop()
 # The next value on the stack is the result
before the parenthesis
 prev_result = stack.pop()
 # Combine the result of the previous
expression with the sub-expression evaluated
 result = prev_result + prev_sign * result
 # Skip spaces implicitly when no if condition is
met
```

Stackp.2413

```
 # Add any number left after the loop ends
 result += sign * number
 return result

Example usage:
print(calculate("1 + 1")) # Expected output: 2
print(calculate(" 2-1 + 2 ")) # Expected output: 3
print(calculate("(1+(4+5+2)-3)+(6+8)") # Expected
output: 23
```

• LC.ca

```
only + and -, no *, no /
class Solution:
 def calculate(self, s: str) -> int:
 stk = []
 ans, sign = 0, 1
 i, n = 0, len(s)
 while i < n:
 if s[i].isdigit():
 x = 0
 j = i

 # with this while, no need to do final
 calculation like below solution
 while j < n and s[j].isdigit():
 x = x * 10 + int(s[j])
 j += 1
 ans += sign * x
 i = j - 1
 elif s[i] == "+":
 sign = 1
 elif s[i] == "-":
 sign = -1
```

Stackp.2414

```
 elif s[i] == "(":
 stk.append(ans)
 stk.append(sign)
 ans, sign = 0, 1
 elif s[i] == ")":
 ans = stk.pop() * ans + stk.pop()
 i += 1
 return ans

#####

class Solution:
 def calculate(self, s: str) -> int:
 ans = 0
 num = 0
 sign = 1
 stack = [sign] # stack[-1]: current env's sign

 for c in s:
 if c.isdigit():
```

Stackp.2415

```
 num = num * 10 + int(c)
 # num = num * 10 + (ord(c) - ord('0')) => also
works
 elif c == '(':
 stack.append(sign)
 elif c == ')':
 stack.pop() # pop pairing sign for this () pair
 elif c == '+' or c == '-':
 ans += sign * num
 sign = (1 if c == '+' else -1) * stack[-1] # after
all + or -, the sign for current num
 num = 0

 return ans + sign * num # final calculation
```

Stackp.2416

Stack

#239 Sliding Window MaximumHAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Queue · Sliding Window · Monotonic Queue

Lists: Grind 169

Time Complexity: O(n) where n is the number of elements in nums, since each element is processed at most twice. Space Complexity: O(k), as the deque stores indices for at most k elements at any time.

Problem

You are given an array of integers nums, there is a sliding window of size k which is moving from the very left of the array to the very right. You can only see the k numbers in the window. Each time the sliding window moves right by one position.

Return the max sliding window.

Example 1:

```
Input: nums = [1,3,-1,-3,5,3,6,7], k = 3
Output: [3,3,5,5,6,7]
Explanation:
Window position Max

[1 3 -1] -3 5 3 6 7 3
1 [3 -1 -3] 5 3 6 7 3
1 3 [-1 -3 5] 3 6 7 5
```

Stackp.2417

Example 2:

```
Input: nums = [1], k = 1
Output: [1]
```

Constraints:

- 1 <= nums.length <= 105
- -104 <= nums[i] <= 104
- 1 <= k <= nums.length

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def maxSlidingWindow(self, nums, k):
 # Brute Force O(n^2) - simulate without stack via
inner loop
 n = len(nums)
 result = [-1] * n
 for i in range(n):
 for j in range(i + 1, n):
 if nums[j] > nums[i]:
 result[i] = nums[j]

 break
 return result
```

Community Solutions (Python)

- WalkCC

Stackp.2418

```
class Solution:
 def maxSlidingWindow(self, nums: List[int], k: int) ->
List[int]:
 ans = []
 maxQ = collections.deque()

 for i, num in enumerate(nums):
 while maxQ and maxQ[-1] < num:
 maxQ.pop()
 maxQ.append(num)
 if i >= k and nums[i - k] == maxQ[0]: #
out-of-bounds

 maxQ.popleft()
 if i >= k - 1:
 ans.append(maxQ[0])

 return ans
```

• LeetDoocs

```
class Solution:
 def maxSlidingWindow(self, nums: List[int], k: int)
-> List[int]:
 q = [(-v, i) for i, v in enumerate(nums[: k -
1])]
 heapify(q)
 ans = []
 for i in range(k - 1, len(nums)):
 heappush(q, (-nums[i], i))
 while q[0][1] <= i - k:
 heappop(q)
 ans.append(-q[0][0])
```

Stackp.2419

```
 return ans
```

• SimplyLeet

```
from collections import deque

def maxSlidingWindow(nums, k):
 # Create a deque to store indices of elements in the
current window
 deq = deque()
 result = []

 for i, num in enumerate(nums):
 # Remove indices from the front if they are out
of the current window
 if deq and deq[0] == i - k:

 deq.popleft()

 # Remove elements from the back while the current
number is larger
 # than the number at the indices stored in the
deque
 while deq and nums[deq[-1]] < num:
 deq.pop()

 # Append the current index to the deque
 deq.append(i)
```

Stackp.2420

```
 # If we've hit the window size, append the
current max to result
 if i >= k - 1:
 result.append(nums[deq[0]])

 return result

Example usage:
print(maxSlidingWindow([1,3,-1,-3,5,3,6,7], 3))
```

• LC.ca

```
from collections import deque

class Solution:
 def maxSlidingWindow(self, nums: List[int], k: int)
-> List[int]:
 q = deque()
 ans = []
 for i, v in enumerate(nums):
 if q and i - k + 1 > q[0]:
 q.popleft() # remove index if out of
window left
 while q and nums[q[-1]] <= v: # `<=`, not
`<`, to ensure the bigger index stored in deque
```

Stackp.2421

```
q.pop()
q.append(i)
if i >= k - 1:
 ans.append(nums[q[0]])
return ans

#####

class Solution(object):
 def maxSlidingWindow(self, nums, k):

 """
 :type nums: List[int]
 :type k: int
 :rtype: List[int]
 """
 if k == 0:
 return []
 ans = [0 for _ in range(len(nums) - k + 1)]
 stack = collections.deque([])
 for i in range(0, k):

 while stack and nums[stack[-1]] < nums[i]:
 stack.pop()
 stack.append(i)
 ans[0] = nums[stack[0]]
 idx = 0
 for i in range(k, len(nums)):
 idx += 1
 if stack and stack[0] == i - k:
 stack.popleft()
 while stack and nums[stack[-1]] < nums[i]:
 stack.pop()
 stack.append(i)
 ans[idx] = nums[stack[0]]
```

Stack

p.2422

```
stack.pop()
stack.append(i)
ans[idx] = nums[stack[0]]

return ans
```

Stack

p.2423

Stack

#716 Max Stack

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Stack · Design · Doubly-Linked List · Ordered Set

Lists: Denny Zhang

Time Complexity:

Problem

Design a max stack data structure that supports the stack operations and supports finding the stack's maximum element.

Implement the MaxStack class:

- MaxStack() Initializes the stack object.
- void push(int x) Pushes element x onto the stack.
- int pop() Removes the element on top of the stack and returns it.
- int top() Gets the element on the top of the stack without removing it.
- int peekMax() Retrieves the maximum element in the stack without removing it.
- int popMax() Retrieves the maximum element in the stack and removes it. If there is more than one maximum element, only remove the top-most one.

You must come up with a solution that supports O(1) for each top call and O(logn) for each other call.

Stack

p.2424

Example 1:

Input

["MaxStack", "push", "push", "push", "top", "popMax", "top", "peekMax", "pop", "top"]

[[], [5], [1], [5], [1], [1], [1], [1], [1], [1]]

Output

[null, null, null, null, 5, 5, 1, 5, 1, 5]

Explanation

MaxStack stk = new MaxStack();

Constraints:

- -107 <= x <= 107
- At most 105 calls will be made to push, pop, top, peekMax, and popMax.
- There will be at least one element in the stack when pop, top, peekMax, or popMax is called.

Community Solutions (Python)

- SimplyLeet

Stack

p.2425

```
Python solution implementing the MaxStack data structure
We use collections.OrderedDict for demonstration.
However, in Python we do not have a built in balanced BST;
one can use "sortedcontainers" or manually maintain a heap and dictionary.
For clarity, we simulate an ordered map with a TreeMap-like structure using sorted list of keys.
In production, one may use "SortedDict" from sortedcontainers.

from collections import defaultdict

class Node:

 def __init__(self, val):
 self.val = val
 self.prev = None
 self.next = None

class DoubleLinkedList:
 def __init__(self):
 # Create dummy head and tail nodes.
 self.head = Node(0)
 self.tail = Node(0)

 return self.sortedKeys[-1]

 def popMax(self) -> int:
 max_val = self.peekMax()
 # Get the most recent node with max_val.
 node = self.valToNodes[max_val].pop()
 if not self.valToNodes[max_val]:
 self.sortedKeys.pop() # removes the last element
 # Remove the node from the doubly-linked list.
 self.dll.pop(node)

 return max_val

Example usage:
stk = MaxStack()
stk.push(5)
stk.push(1)
stk.push(5)
print(stk.top()) # returns 5
print(stk.popMax()) # returns 5
print(stk.top()) # returns 1

print(stk.peekMax()) # returns 5
print(stk.pop()) # returns 1
print(stk.top()) # returns 5
```

Stack

p.2426

```
self.head.next = self.tail
self.tail.prev = self.head

def add_last(self, node):
 # Add node right before the tail.
 node.prev = self.tail.prev
 node.next = self.tail
 self.tail.prev.next = node
 self.tail.prev = node

def pop(self, node=None):
 # If no node is provided, pop from the end.
 if not node:
 node = self.tail.prev
 node.prev.next = node.next
 node.next.prev = node.prev
 return node

def top(self):
 # Return the last node's value.

 return self.tail.prev.val

class MaxStack:
 def __init__(self):
 self.dll = DoubleLinkedList()
 # Dictionary mapping value to list of nodes
 self.valToNodes = defaultdict(list)
 # Sorted list to simulate ordered set of keys.
 self.sortedKeys = []
```

Stack

p.2427

```
def push(self, x: int) -> None:
 node = Node(x)
 self.dll.add_last(node)
 self.valToNodes[x].append(node)
 # Insert x into sortedKeys if new, otherwise maintain count.
 if len(self.valToNodes[x]) == 1:
 # binary insertion into sortedKeys
 lo, hi = 0, len(self.sortedKeys)
 while lo < hi:
 mid = (lo + hi) // 2
 if self.sortedKeys[mid] < x:
 lo = mid + 1
 else:
 hi = mid
 self.sortedKeys.insert(lo, x)

 def pop(self) -> int:
 node = self.dll.pop()
 x = node.val
 self.valToNodes[x].pop()

 if not self.valToNodes[x]:
 # Remove x from sortedKeys
 self.sortedKeys.remove(x)
 return x

 def top(self) -> int:
 return self.dll.top()

 def peekMax(self) -> int:
 # The maximum is the last element in sortedKeys.
```

Stack

p.2428

```
return self.sortedKeys[-1]

def popMax(self) -> int:
 max_val = self.peekMax()
 # Get the most recent node with max_val.
 node = self.valToNodes[max_val].pop()
 if not self.valToNodes[max_val]:
 self.sortedKeys.pop() # removes the last element
 # Remove the node from the doubly-linked list.
 self.dll.pop(node)

 return max_val

Example usage:
stk = MaxStack()
stk.push(5)
stk.push(1)
stk.push(5)
print(stk.top()) # returns 5
print(stk.popMax()) # returns 5
print(stk.top()) # returns 1

print(stk.peekMax()) # returns 5
print(stk.pop()) # returns 1
print(stk.top()) # returns 5
```

Stack

p.2429

```
class MaxStack(object):

 def __init__(self):
 """
 initialize your data structure here.
 """
 self.stack = []
 self.max_stack = []
 # or the same trick in min-stack
 # self.max_stack = [-math.inf]

 def push(self, x):
 """
 :type x: int
 :rtype: void
 """
 self.stack.append(x)
 self.max_stack.append(max(x, self.max_stack[-1]))
 if self.max_stack else x)

 def pop(self):
 """
 :rtype: int
 """
 if len(self.stack) != 0:
 self.max_stack.pop(-1)
 return self.stack.pop(-1)

 def top(self):
 """
 :rtype: int
 """
```

Stack

p.2430

Sheet 270 / 317 · LeetMastery By Pattern · Python Only · 3-up Portrait

```

 """
 return self.stack[-1] if self.stack else None

def peekMax(self):
 """
 :rtype: int
 if len(self.max_stack) != 0:
 return self.max_stack[-1]

def popMax(self):
 """
 :rtype: int
 """
 val = self.peakMax()
 buff = []
 while self.top() != val:
 # re-use pop(), ops on both max_stack and
stack
 buff.append(self.pop())
 self.pop()

 while len(buff) != 0:
 # re-use push(), no need to save buffer for
max-stack
 self.push(buff.pop(-1))
 return val

#####

from sortedcontainers import SortedList

class Node:
```

Stackp.2431

```

def __init__(self, val=0):
 self.val = val
 self.prev: Union[Node, None] = None
 self.next: Union[Node, None] = None

class DoubleLinkedList:
 def __init__(self):
 self.head = Node()
 self.tail = Node()

 self.head.next = self.tail
 self.tail.prev = self.head

 def append(self, val) -> Node:
 node = Node(val)
 node.next = self.tail
 node.prev = self.tail.prev
 self.tail.prev = node
 node.prev.next = node
 return node

 @staticmethod
 def remove(node) -> Node:
 node.prev.next = node.next
 node.next.prev = node.prev
 return node

 def pop(self) -> Node:
 return self.remove(self.tail.prev)
```

Stackp.2432

```

def peek(self):
 return self.tail.prev.val

class MaxStack:
 def __init__(self):
 self.stk = DoubleLinkedList()
 self.sl = SortedList(key=lambda x: x.val)

 def push(self, x: int) -> None:

 node = self.stk.append(x)
 self.sl.add(node)

 def pop(self) -> int:
 node = self.stk.pop()
 self.sl.remove(node)
 return node.val

 def top(self) -> int:
 return self.stk.peak()

 def peekMax(self) -> int:
 return self.sl[-1].val

 def popMax(self) -> int:
 node = self.sl.pop()
 DoubleLinkedList.remove(node)
 return node.val
```

Stackp.2433

```

Your MaxStack object will be instantiated and called as
such:
obj = MaxStack()
obj.push(x)
param_2 = obj.pop()
param_3 = obj.top()
param_4 = obj.peakMax()
param_5 = obj.popMax()
```

Stackp.2434

Stack

#772 Basic Calculator IIIHAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Math · String · Stack · Recursion

Lists: Premium 98

Time Complexity: O(n) where n is the length of the string, as every character is processed. Space Complexity: O(n) due to the recursion stack or auxiliary stack used for intermediate results.

Problem

Implement a basic calculator to evaluate a simple expression string.

The expression string contains only non-negative integers, '+', '-', '\*', '/' operators, and open '(' and closing parentheses ')'. The integer division should truncate toward zero.

You may assume that the given expression is always valid. All intermediate results will be in the range of [-231, 231 - 1].

Note: You are not allowed to use any built-in function which evaluates strings as mathematical expressions, such as eval().

Example 1:

Stackp.2435

```

Input: s = "1+1"
Output: 2

Example 2:
Input: s = "6-4/2"
Output: 4

Example 3:
Input: s = "2*(5+5*2)/3+(6/2+8)"
Output: 21

Constraints:
• 1 <= s <= 104
• s consists of digits, '+', '-', '*', '/', '(', and ')'.
• s is a valid expression.

Community Solutions (Python)
```

• WalkCC

```

class Solution:
 def calculate(self, s: str) -> int:
 nums = []
 ops = []

 def calc():
 b = nums.pop()
 a = nums.pop()
 op = ops.pop()
 if op == '+':
```

Stackp.2436

```

 nums.append(a + b)
 elif op == '-':
 nums.append(a - b)
 elif op == '*':
 nums.append(a * b)
 else: # op == '/'
 nums.append(int(a / b))

def precedes(prev: str, curr: str) -> bool:
 """

 Returns True if the previous character is a
 operator and the priority of
 the previous operator >= the priority of the
 current character (operator).
 """
 if prev == '(':
 return False
 return prev in '*/' or curr in '+-'

i = 0
hasPrevNum = False

while i < len(s):
 c = s[i]
 if c.isdigit():
 num = int(c)
 while i + 1 < len(s) and s[i + 1].isdigit():
 num = num * 10 + int(s[i + 1])
 i += 1
 nums.append(num)
 hasPrevNum = True
 elif c == '(':
```

Stackp.2437

```

 ops.append('(')
 hasPrevNum = False
 elif c == ')':
 while ops[-1] != '(':
 calc()
 ops.pop() # Pop '('
 elif c in '+-*/':
 if not hasPrevNum: # Handle input like "-1-(-1)"
 num.append(0)
 while ops and precedes(ops[-1], c):

 calc()
 ops.append(c)
 i += 1

 while ops:
 calc()

 return nums.pop()
```

• LeetDoocs

```

class Solution:
 def calculate(self, s: str) -> int:
 def dfs(q):
 num, sign, stk = 0, "+", []
 while q:
 c = q.popleft()
 if c.isdigit():
 num = num * 10 + int(c)
 if c == "(":
 num = dfs(q)
 return num
```

Stackp.2438

```

 if c in "+-*/" or not q:
 match sign:
 case "+":
 stk.append(num)
 case "-":
 stk.append(-num)
 case "*":
 stk.append(stk.pop() * num)
 case "/":
 stk.append(int(stk.pop() /
num))

 num, sign = 0, c
 if c == ")":
 break
 return sum(stk)

 return dfs(deque(s))
```

• SimplyLeet

```

def calculate(s: str) -> int:
 # Recursive helper function that processes the
 expression starting from index 'i'
 def helper(i):
 stack = []
 num = 0
 sign = '+'
 while i < len(s):
 char = s[i]
 # Build the current number if the character
 is a digit
 if char.isdigit():
```

Stackp.2439

```
 num = num * 10 + int(char)
 # If '(' is encountered, evaluate the
 subexpression recursively
 if char == '(':
 num, i = helper(i + 1)
 # When encountering an operator or closing
 parenthesis, process the sign and number
 if (not char.isdigit()) and char != ' ' or i
 == len(s) - 1:
 if sign == '+':
 stack.append(num)
 elif sign == '-':
 stack.append(-num)

 elif sign == '*':
 prev = stack.pop()
 stack.append(prev * num)
 elif sign == '/':
 prev = stack.pop()
 stack.append(int(prev / num)) #
Truncate toward zero
 sign = char
 num = 0
 # If a closing parenthesis is reached, end
 this level of recursion
 if char == ')':
```

Stackp.2440

```
 return sum(stack), i
 i += 1
 return sum(stack), i

 result, _ = helper(0)
 return result

Example test
print(calculate("2*(5+5*2)/3+(6/2+8)"))

• LC.ca

good one, based on solution of Basic Calculator II
class Solution: # dfs recursion
 def calculate(self, s: str) -> int:
 stack = []
 num = 0
 op = "+"
 i = 0
 while i < len(s):
 c = s[i]
 if c.isdigit():

 num = num * 10 + int(c)
 elif c == "(":
 j = self.findClosing(s, i)
 num = self.calculate(s[i+1:j]) # j is ')'
 i = j # out if, i will +1 below
 if c in "+-*/" or i == len(s) - 1:
 if op == "+":
 stack.append(num)
 elif op == "-":
 stack.append(-num)
```

Stackp.2441

```
 elif op == "*":
 stack[-1] *= num
 else:
 stack[-1] = int(stack[-1] / num)
 op = c
 num = 0
 i += 1
 return sum(stack)

def findClosing(self, s: str, i: int) -> int:

 level = 0
 while i < len(s):
 if s[i] == "(":
 level += 1
 elif s[i] == ")":
 level -= 1
 if level == 0:
 return i
 i += 1
 return i

#####

class Solution: # iterative
 def calculate(self, s: str) -> int:
 def evaluate_expression(stack):
 res = stack.pop() if stack else 0
 # Evaluate the expression till we get '(' or
 empty stack
```

Stackp.2442

```
 while stack and stack[-1] != '(':
 sign = stack.pop()
 if sign == '+':
 res += stack.pop()
 elif sign == '-':
 res -= stack.pop()
 elif sign == '*':
 res *= stack.pop()
 elif sign == '/':
 res //= stack.pop()

 return res

 stack = [] # mix of num and operator
 n = len(s)
 i = 0
 while i < n:
 if s[i].isdigit():
 # Get the complete number and push it to
 the stack
 j = i
 while j < n and s[j].isdigit():
```

Stackp.2443

```
 j += 1
 stack.append(int(s[i:j]))
 i = j - 1
 elif s[i] in '+-*/':
 stack.append(s[i])
 elif s[i] == '(':
 # Evaluate the expression within the
 parentheses
 res = evaluate_expression(stack)
 stack.pop() # Remove the opening '('
 stack.append(res)

 i += 1

 return evaluate_expression(stack)

#####

class Solution:
 def calculate(self, s: str) -> int:
 if not s:
```

Stackp.2444

```
 return 0

 # remove leading and trailing spaces and white
 spaces.
 s = s.strip().replace(" ", "")

 if not s:
 return 0

 op_stack = []
 num_stack = []

 i = 0
 while i < len(s):
 if s[i].isdigit():
 num = 0
 while i < len(s) and s[i].isdigit():
 num = num * 10 + int(s[i])
 i += 1
 num_stack.append(num)
 else:

 op = s[i]
 if not op_stack:
 op_stack.append(op)
 i += 1
 elif op == '+' or op == '-':
 top = op_stack[-1]
 if top == '(':
 op_stack.append(op)
 i += 1
 else:
```

Stackp.2445

```
 self.calculate_helper(num_stack,
 op_stack)

 elif op == '*' or op == '/':
 top = op_stack[-1]
 if top == '(' or top == '+' or top ==
 '-':

 op_stack.append(op)
 i += 1
 else:
 self.calculate_helper(num_stack,
 op_stack)

 elif op == '(':
 op_stack.append(op)

 i += 1
 elif op == ')':
 while op_stack[-1] != '(':
 self.calculate_helper(num_stack,
 op_stack)

 op_stack.pop()
 i += 1

 while op_stack:
 self.calculate_helper(num_stack, op_stack)
```

Stackp.2446

```
 return num_stack[-1]

 def calculate_helper(self, num_stack, op_stack):
 num2 = num_stack.pop()
 num1 = num_stack.pop()

 op = op_stack.pop()

 if op == '+':
 ans = num1 + num2

 elif op == '-':
 ans = num1 - num2
 elif op == '*':
 ans = num1 * num2
 else:
 ans = int(num1 / num2)

 num_stack.append(ans)
```

Stackp.2447

Stack

#895 Maximum Frequency Stack

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · Stack · Design

Lists: Grind 169

Time Complexity: O(1) per push and pop. Space Complexity: O(n), where n is the number of elements pushed onto the stack.

Problem

Design a stack-like data structure to push elements to the stack and pop the most frequent element from the stack.

Implement the FreqStack class:

- FreqStack() constructs an empty frequency stack.
- void push(int val) pushes an integer val onto the top of the stack.
- int pop() removes and returns the most frequent element in the stack. If there is a tie for the most frequent element, the element closest to the stack's top is removed and returned.

Example 1:

Stackp.2448



|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><div>Input</div><div>["FreqStack", "push", "push", "push", "push", "push", "push", "pop", "pop", "pop", "pop"]<div>[[], [5], [7], [5], [7], [4], [5], [], [], [], []]</div></div><div><div>Output</div><div>[null, null, null, null, null, null, 5, 7, 5, 4]</div></div><div><div>Explanation</div><div>FreqStack freqStack = new FreqStack();</div></div><div>Constraints:<ul style="list-style-type: none"><li>0 &lt;= val &lt;= 109</li><li>At most 2 * 104 calls will be made to push and pop.</li><li>It is guaranteed that there will be at least one element in the stack before calling pop.</li></ul></div><div>Community Solutions (Python)</div><div><div>• WalkCC</div><div><div>class FreqStack:<div>def __init__(self):<div>self.maxFreq = 0</div><div>self.count = collections.Counter()</div><div>self.countToStack = collections.defaultdict(list)</div></div><div>def push(self, val: int) -&gt; None:<div>self.count[val] += 1</div><div>self.countToStack[self.count[val]].append(val)</div><div>self.maxFreq = max(self.maxFreq, self.count[val])</div></div></div></div></div><div>Stack</div><div>p.2449</div></div> | <div><div>def pop(self) -&gt; int:<div>val = self.countToStack[self.maxFreq].pop()</div><div>self.count[val] -= 1</div><div>if not self.countToStack[self.maxFreq]:<div>self.maxFreq -= 1</div></div><div>return val</div></div><div>• LeetCode</div><div><div>class FreqStack:<div>def __init__(self):<div>self.cnt = defaultdict(int)</div><div>self.q = []</div><div>self.ts = 0</div></div><div>def push(self, val: int) -&gt; None:<div>self.ts += 1</div><div>self.cnt[val] += 1</div><div>heappush(self.q, (-self.cnt[val], -self.ts, val))</div></div><div>def pop(self) -&gt; int:<div>val = heappop(self.q)[2]</div><div>self.cnt[val] -= 1</div><div>return val</div></div><div># Your FreqStack object will be instantiated and called as such:<div># obj = FreqStack()</div><div># obj.push(val)</div></div><div># param_2 = obj.pop()</div></div></div><div>Stack</div><div>p.2450</div></div> | <div><div>• SimplyLeet</div><div># Python implementation of FreqStack</div><div>class FreqStack:<div>def __init__(self):<div># Dictionary mapping value to its frequency</div><div>self.freq = {}</div><div># Dictionary mapping frequency to a list (stack)</div></div><div>of values<div>self.group = {}</div><div># Variable tracking the current maximum frequency</div></div><div>in the stack<div>self.maxfreq = 0</div></div><div>def push(self, val: int) -&gt; None:<div># Update frequency count for the value</div><div>f = self.freq.get(val, 0) + 1</div><div>self.freq[val] = f</div><div># Update the maximum frequency if necessary</div><div>if f &gt; self.maxfreq:<div>self.maxfreq = f</div></div><div># Append the value to the appropriate frequency</div><div>group</div><div>if f not in self.group:</div></div></div></div> <div>Stack</div> <div>p.2451</div> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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| <div><div>self.group[f] = []<div>self.group[f].append(val)</div></div><div>def pop(self) -&gt; int:<div># Remove the value from the stack of the current maximum frequency</div><div>val = self.group[self.maxfreq].pop()</div><div># Decrement the frequency count for the value</div><div>self.freq[val] -= 1</div><div># If no more elements have the current max frequency, decrease maxfreq</div><div>if not self.group[self.maxfreq]:</div></div><div>self.maxfreq -= 1</div><div>return val</div></div> <div>• LC.ca</div> <div><div>class FreqStack:<div>def __init__(self):<div>self.cnt = defaultdict(int)</div><div>self.d = defaultdict(list)</div><div>self.mx = 0</div></div><div>def push(self, val: int) -&gt; None:<div>self.cnt[val] += 1</div><div>self.d[self.cnt[val]].append(val)</div><div>self.mx = max(self.mx, self.cnt[val])</div></div></div></div> <div>Stack</div> <div>p.2452</div> | <div><div>def pop(self) -&gt; int:<div>val = self.d[self.mx].pop()</div><div>self.cnt[val] -= 1</div><div>if not self.d[self.mx]:<div>self.mx -= 1</div></div><div>return val</div></div><div># Your FreqStack object will be instantiated and called as such:</div><div># obj = FreqStack()</div><div># obj.push(val)</div><div># param_2 = obj.pop()</div></div> <div>Stack</div> <div>p.2453</div> |  |
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| <div>&gt;&gt; Quick Review — Stack</div> <div>25 questions · insights · complexity · solution</div> <div>#20 Valid Parentheses [Easy]</div> <div>Key Insights<ul style="list-style-type: none"><li>The problem can be solved using a stack data structure.</li><li>When encountering an opening bracket, push it onto the stack.</li><li>When encountering a closing bracket, check if the top of the stack is its matching opening bracket.</li><li>If the stack is empty before finding a match or if there is a mismatch, the string is invalid.</li><li>At the end, the string is valid only if the stack is empty, meaning all brackets were properly matched.</li></ul></div> <div>Space &amp; Time<div>Time Complexity: O(n), where n is the length of the string, since we process each character once.</div><div>Space Complexity: O(n) in the worst case for the stack when all characters are opening brackets.</div></div> <div>Solution<div>We use a stack to check for valid parentheses. The algorithm iterates through the string, pushing opening brackets onto the stack and popping them when a</div></div> <div>Stack</div> <div>p.2455</div> | <div>corresponding closing bracket is encountered. A mapping of closing to opening brackets helps in validating the matching pairs. By ensuring that the stack is empty at the end, we confirm that every opening bracket was properly closed.</div> <div>#232 Implement Queue using Stacks [Easy]</div> <div>Key Insights<ul style="list-style-type: none"><li>Use two stacks:</li><li>One (input stack) for enqueue operations.</li><li>The other (output stack) for dequeue operations.</li><li>When performing pop or peek, if the output stack is empty, transfer all elements from the input stack to the output stack. This reversal makes the oldest element accessible.</li><li>Each element is moved at most once between the stacks, ensuring an amortized O(1) cost per operation.</li><li>The space complexity is O(n) where n is the number of elements in the queue.</li></ul></div> <div>Space &amp; Time<div>Time Complexity: Amortized O(1) per operation</div><div>Space Complexity: O(n)</div></div> <div>Solution<div>We maintain two stacks: an input stack for pushing new elements and an output stack for popping or peeking the front element of the queue. When we need to access</div></div> <div>Stack</div> <div>Sheet 273 / 317 · LeetCode Mastery By Pattern · Python Only · 3-up Portrait</div> <div>p.2456</div> | <div>the front of the queue (either during a pop or peek), we check if the output stack is empty. If it is, we transfer all elements from the input stack to the output stack. This reversal of order allows us to simulate the FIFO nature of the queue. When performing a push, the element is simply added to the input stack.</div> <div>#844 Backspace String Compare [Easy]</div> <div>Key Insights<ul style="list-style-type: none"><li>'#' represents a backspace operation that deletes the previous character.</li><li>A direct simulation using a stack can reconstruct the final string.</li><li>A two-pointer approach allows comparison from the end of the strings while skipping backspaced characters.</li><li>The two-pointer technique can achieve O(n) time and O(1) space by processing the strings in reverse.</li></ul></div> <div>Space &amp; Time<div>Time Complexity: O(n) where n is the length of the longer string.</div><div>Space Complexity: O(1) using the two-pointer strategy (O(n) if using stacks).</div></div> <div>Solution<div>The solution uses two pointers starting from the end of each string. For each string, maintain a skip counter to account for backspace characters. As you iterate</div></div> <div>Stack</div> <div>p.2457</div> |
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| <p>backwards:</p> <ul style="list-style-type: none"> <li>- If a '#' is encountered, increment the skip counter.</li> <li>- If a normal character is encountered while the skip counter is positive, skip that character by decrementing the counter.</li> <li>- Once a valid character is found, compare the characters from both strings. If all corresponding characters match and both pointers finish processing at the same time, the strings are considered equal.</li> </ul> <p>#143 Reorder List <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Find the middle of the linked list using the fast and slow pointer technique.</li> <li>• Reverse the second half of the list.</li> <li>• Merge the two halves, alternating nodes from the first half and the reversed second half.</li> <li>• The challenge is done in-place, ensuring a linear time solution with constant extra space.</li> </ul> <p>Space &amp; Time</p> <p>Time Complexity: O(n) - Each of the three main steps (finding the middle, reversing, merging) takes O(n) time.</p> <p>Space Complexity: O(1) - The algorithm is done in-place with only a few pointers.</p> <p>Solution</p> <p>The solution involves three key steps:</p> <div>Stackp.2458</div> | <ul style="list-style-type: none"> <li>- Use two pointers (slow and fast) to identify the middle of the list.</li> <li>- Reverse the second half of the list. This can be done iteratively by adjusting the next pointers.</li> <li>- Merge the two lists: One starting from the head and the other from the head of the reversed second half. Interleave the nodes by alternating between the two lists until complete. This approach uses in-place pointer manipulation, thus ensuring no extra space is used aside from a few temporary variables.</li> </ul> <p>#150 Evaluate Reverse Polish Notation <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Use a stack to process the tokens.</li> <li>• Push numbers onto the stack.</li> <li>• When an operator is encountered, pop the top two numbers, perform the operation, and push the result back.</li> <li>• Handle division carefully to ensure it truncates toward zero.</li> <li>• The final result remains as the only value in the stack.</li> </ul> <p>Space &amp; Time</p> <p>Time Complexity: O(n), where n is the number of tokens (each token is processed once).</p> <p>Space Complexity: O(n), in the worst-case scenario where all tokens are numbers and are stored in the stack.</p> <div>Stackp.2459</div>                                 | <p>Solution</p> <p>We use a stack data structure to evaluate the RPN expression. Iterate through each token:</p> <ul style="list-style-type: none"> <li>- If the token is a number, convert and push it onto the stack.</li> <li>- If the token is an operator, pop the top two numbers (be careful about order), perform the corresponding arithmetic operation, and push the result back onto the stack.</li> <li>- At the end of processing, the stack contains the result of the expression. Special attention is needed for division to ensure it truncates toward zero (using language-specific functions when necessary).</li> </ul> <p>#155 Min Stack <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Use an auxiliary data structure to track the current minimum element.</li> <li>• A common approach is to use two stacks: one regular stack for all elements and a second stack to keep track of the minimum values.</li> <li>• When pushing a new element, push it onto the main stack and also update the min stack if the element is less than or equal to the current minimum.</li> <li>• When popping, if the popped element is equal to the top of the min stack, pop from the min stack as well.</li> <li>• This design guarantees that all operations (push, pop, top, getMin) run in constant time, O(1).</li> </ul> <div>Stackp.2460</div>                                 |
| <p>Space &amp; Time</p> <p>Time Complexity: O(1) per operation (push, pop, top, getMin)</p> <p>Space Complexity: O(n), where n is the number of elements in the stack, since an additional stack is used to keep track of minima.</p> <p>Solution</p> <p>The solution involves using two stacks:</p> <ul style="list-style-type: none"> <li>- Main Stack: Stores all the elements.</li> <li>- Min Stack: Stores the minimum elements encountered so far.</li> </ul> <p>When an element is pushed, we compare it with the top of the min stack. If it is smaller than or equal to the current minimum, we also push it onto the min stack.</p> <p>For the pop operation, if the element being popped is the same as the top of the min stack, we pop from the min stack as well. This approach ensures that the getMin operation always returns the top element of the min stack, which is the current minimum in constant time.</p> <p>#173 Binary Search Tree Iterator <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Use an iterative in-order traversal strategy with a stack.</li> <li>• In-order traversal for BST yields sorted order.</li> </ul> <div>Stackp.2461</div>                                                          | <ul style="list-style-type: none"> <li>• Pre-load the stack with all left descendants from the root.</li> <li>• After returning a node, process its right subtree by pushing its left branch.</li> <li>• This method maintains an average time complexity of O(1) per operation and uses O(h) memory, where h is the tree height.</li> </ul> <p>Space &amp; Time</p> <p>Time Complexity: O(1) on average per operation (amortized)</p> <p>Space Complexity: O(h), where h is the height of the tree</p> <p>Solution</p> <p>We use a stack to simulate an in-order traversal. The process involves initially pushing all left-child nodes from the root into the stack. For each call to next(), we pop the top element (the current smallest node) and then push all the left descendants of the node's right child. This ensures that the next smallest element is always at the top of the stack. The hasNext() function then simply checks if the stack is non-empty.</p> <p>#227 Basic Calculator II <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• The expression can be processed in one pass while using a stack to handle operator precedence.</li> </ul> <div>Stackp.2462</div>                                                                                                                           | <ul style="list-style-type: none"> <li>• When encountering '+' or '-' operators, the current number is directly added or subtracted (as a positive or negative value).</li> <li>• For '*' and '/' operators, compute immediately with the last number from the stack and update it.</li> <li>• Handle spaces and multi-digit numbers by building the number while iterating over the string.</li> <li>• Division truncation toward zero must be carefully applied, which aligns with typical integer division in most languages.</li> </ul> <p>Space &amp; Time</p> <p>Time Complexity: O(n) where n is the length of the expression, since the algorithm processes each character once.</p> <p>Space Complexity: O(n) in the worst-case scenario when all numbers are stored in the stack.</p> <p>Solution</p> <p>We use a stack-based approach. Iterate over each character of the string while maintaining a variable for the current number and a previous operator (initialized to '+'). When an operator or the end of the string is reached, perform an action based on the previous operator:</p> <ul style="list-style-type: none"> <li>- For '+' add the current number to the stack.</li> <li>- For '-' add the negative of the current number.</li> <li>- For '*' and '/' pop the last number from the stack, compute the result by performing multiplication or division, and then push</li> </ul> <div>Stackp.2463</div> |
| <p>the result back to the stack. After processing the entire string, sum the elements in the stack to get the final result. This approach ensures that multiplication and division are handled before addition and subtraction.</p> <p>#255 Verify Preorder Sequence in Binary Search Tree <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• A BST has the property that every node's left subtree contains values less than the node, and every right subtree contains values greater.</li> <li>• Preorder traversal processes nodes as: root, left subtree, then right subtree.</li> <li>• Using a monotonic stack, we can simulate constructing the BST while tracking a lower bound for allowable node values.</li> <li>• When moving to a right child, update the lower bound to ensure subsequent nodes are valid.</li> </ul> <p>Space &amp; Time</p> <p>Time Complexity: O(n) - Each element is processed once.</p> <p>Space Complexity: O(n) - In the worst-case an explicit stack is used; can be optimized to O(1) by reusing the input array.</p> <p>Solution</p> <p>We use a monotonic stack to simulate the BST construction from the preorder sequence. The process</p> <div>Stackp.2464</div>                               | <p>is:</p> <ul style="list-style-type: none"> <li>- Initialize a variable lower_bound to -∞.</li> <li>- Iterate through each value in the preorder array: If a value is less than lower_bound, it violates BST properties and the sequence is invalid. While the stack is not empty and the current value is greater than the element at the top of the stack, pop from the stack and update lower_bound to the popped value. This simulates moving to the right subtree. Push the current value onto the stack.</li> <li>- If the sequence is processed without violations, return true. This approach efficiently ensures that each value adheres to BST requirements during preorder traversal.</li> </ul> <p>#394 Decode String <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Use a stack or recursion to manage nested encoded patterns.</li> <li>• Process the string character by character, accumulating digits for the repeat count.</li> <li>• When encountering a '[', start a new context (recursively or via a new stack frame) for the encoded substring.</li> <li>• When encountering a ']', finish the current context and repeat the accumulated substring as needed.</li> <li>• Careful handling of multi-digit numbers is essential.</li> </ul> <p>Space &amp; Time</p> <div>Stackp.2465</div> | <p>Time Complexity: O(n), where n is the length of the string.</p> <p>Space Complexity: O(n), for the recursion stack or the auxiliary stack used.</p> <p>Solution</p> <p>We can solve this problem using a recursive approach. When we see a digit, we determine the full repeat count. Upon encountering a '[', we recursively read the encoded substring until the corresponding ']'. Each recursive call returns the decoded substring until it hits a ']', which is then repeated based on the previously computed count. Alternatively, an iterative solution using a stack is also feasible. As we parse the string, push the current string and number onto the stack when encountering '[', then pop and combine when ']' is reached. This approach correctly handles nested patterns.</p> <p>#439 Ternary Expression Parser <span>Medium</span></p> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• The ternary operator groups right-to-left. This property can be exploited by processing the string from the end to the beginning.</li> <li>• Using a stack allows us to "collapse" nested ternary expressions as soon as their three components (condition, true branch, false branch) are available.</li> <li>• An iterative approach is efficient and avoids building an explicit syntax tree for the expression.</li> </ul> <div>Stackp.2466</div>                                                       |

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| <div> <ul style="list-style-type: none"> <li>• The condition is not stored on the stack; instead, it is encountered immediately preceding the '?' operator when traversing backwards.</li> </ul> </div> <div> <p>Space &amp; Time</p> <p>Time Complexity: O(n), where n is the length of the expression, as we process each character exactly once.</p> <p>Space Complexity: O(n), due to the use of a stack for storing characters during evaluation.</p> </div> <div> <p>Solution</p> <p>The solution uses an iterative approach with a stack. We traverse the expression string from right to left. For every character:</p> <ul style="list-style-type: none"> <li>– If it is a digit, 'T', or 'F' (and not a '!' or part of a pending operator), we push it onto the stack. – When we encounter a '?' character, we know that the next available items on the stack represent the true and false results of the ternary expression. We then skip the '!' separator that comes between them. – We then use the condition (the character immediately preceding the '?') to choose which result to push back on the stack. This method “collapses” each ternary expression into a single result, and by the time we finish traversing the string, the stack contains the final evaluated result.</li> </ul> </div> <div> <p>#484 Find Permutation <span>Medium</span></p> </div> <div> <p>Key Insights</p> </div> <div> <div>Stack</div> <div>p.2467</div> </div> | <div> <ul style="list-style-type: none"> <li>• The answer is a permutation of numbers from 1 to n, where n = len(s) + 1.</li> <li>• A common strategy is to use a stack to reverse the segments when a decrease ('D') is encountered.</li> <li>• When an 'I' is encountered (or at the end), pop all numbers from the stack to output them in reverse order, ensuring the smallest lexicographical order.</li> <li>• This greedy approach guarantees that as soon as a rising pattern is detected, we “flush” the pending decreases correctly.</li> </ul> </div> <div> <p>Space &amp; Time</p> <p>Time Complexity: O(n) – Each element is pushed and popped exactly once.</p> <p>Space Complexity: O(n) – For the stack and the output array.</p> </div> <div> <p>Solution</p> <p>We use a greedy approach with a stack. Iterate from 0 to n (where n = len(s) + 1):</p> <ul style="list-style-type: none"> <li>– Push the current number (i+1) onto the stack. – If the current character is 'I' or we have reached the end of the string, pop all elements from the stack and append them to the result. By doing so, segments corresponding to series of 'D's are reversed, while 'I's trigger the flush, ensuring that the final permutation is the lexicographically smallest.</li> </ul> </div> <div> <div>Stack</div> <div>p.2468</div> </div> | <div> <p>#735 Asteroid Collision <span>Medium</span></p> </div> <div> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Use a stack to keep track of the surviving asteroids.</li> <li>• Only a collision happens when a right-moving asteroid (positive) is on the stack and a left-moving asteroid (negative) is encountered.</li> <li>• Compare the absolute values to decide if one or both asteroids explode.</li> <li>• Continue processing collisions until no further collisions are possible with the current asteroid.</li> </ul> </div> <div> <p>Space &amp; Time</p> <p>Time Complexity: O(n) where n is the number of asteroids, since each asteroid is pushed and popped at most once.</p> <p>Space Complexity: O(n) in the worst case, when no collisions occur and all asteroids remain on the stack.</p> </div> <div> <p>Solution</p> <p>The solution uses a stack to simulate the asteroid collision process. For each asteroid in the input:</p> <ul style="list-style-type: none"> <li>– If it is moving to the right or the stack is empty, simply add it to the stack. – If it is moving to the left, check the asteroid on top of the stack: If the top asteroid is moving to the right, they might collide. Compare sizes: If the left-moving asteroid is larger (in absolute terms), pop the stack and continue checking. If both are equal, pop the stack and do not add the current asteroid.</li> </ul> </div> <div> <div>Stack</div> <div>p.2469</div> </div> |
| <div> <p>If the right-moving asteroid is larger, the current asteroid explodes. – If no collision occurs for the current asteroid, push it onto the stack. Finally, the stack represents the surviving asteroids order.</p> </div> <div> <p>#739 Daily Temperatures <span>Medium</span></p> </div> <div> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Use a monotonic stack to keep track of indices of days with unresolved warmer temperatures.</li> <li>• As you iterate through the temperatures, resolve previous days that have found a warmer temperature.</li> <li>• The index differences yield the number of days waited until a warmer temperature.</li> </ul> </div> <div> <p>Space &amp; Time</p> <p>Time Complexity: O(n), where n is the number of temperatures, because each index is pushed and popped from the stack at most once.</p> <p>Space Complexity: O(n) for storing the stack and the answer array.</p> </div> <div> <p>Solution</p> <p>The solution uses a monotonic stack to track the indices whose next warmer temperature has not been found yet. As you iterate through the temperature list, compare the current temperature with the temperature at the index stored at the top of the stack. If the current temperature is higher, it means you've found a warmer day for the</p> </div> <div> <div>Stack</div> <div>p.2470</div> </div>                                                                        | <div> <p>day represented by the index from the stack. Pop that index, calculate the difference between the current index and the popped index, and store the result. Push the current index onto the stack for future comparisons. This approach efficiently finds the answer without nested iterations.</p> </div> <div> <p>#962 Maximum Width Ramp <span>Medium</span></p> </div> <div> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• We need to identify pairs (i, j) where i &lt; j and nums[i] &lt;= nums[j].</li> <li>• A monotonic decreasing stack can be constructed to maintain candidate starting indices.</li> <li>• By iterating from left to right, we push indices onto the stack only when they have a smaller value than the last.</li> <li>• Then, a backward pass from the end of the array allows us to efficiently match each element with the smallest possible candidate index from the stack.</li> </ul> </div> <div> <p>Space &amp; Time</p> <p>Time Complexity: O(n)</p> <p>Space Complexity: O(n)</p> </div> <div> <p>Solution</p> <p>The solution is built in two phases:</p> <ul style="list-style-type: none"> <li>– Build a decreasing stack of indices by iterating through the array (only push an index if its value is less</li> </ul> </div> <div> <div>Stack</div> <div>p.2471</div> </div>       | <div> <p>than the value at the current top of the stack). – Iterate through the array backwards. For each index j, check and pop elements from the stack while nums[j] is greater than or equal to the nums value at the popped index. For each valid ramp found, compute the width (j – popped index) and update the maximum width accordingly. This approach guarantees we examine each element a constant number of times ensuring an efficient solution.</p> </div> <div> <p>#1214 Two Sum BSTs <span>Medium</span></p> </div> <div> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Traverse the first tree to collect all node values.</li> <li>• Traverse the second tree to check for each node if the complement (target – current node value) exists in the first tree's values.</li> <li>• Using a hash set for the first tree values allows efficient O(1) lookups.</li> <li>• The BST property is not essential for the hash set solution, but can be exploited if using iterator techniques for a two-pointer approach.</li> </ul> </div> <div> <p>Space &amp; Time</p> <p>Time Complexity: O(n + m) where n and m are the number of nodes in the first and second trees respectively.</p> <p>Space Complexity: O(n) for storing the node values from the first tree (in worst-case scenario).</p> </div> <div> <div>Stack</div> <div>p.2472</div> </div>                                                                                                                  |
| <div> <p>Solution</p> <p>We perform a DFS (or any tree traversal) on the first BST to collect all its values in a hash set. Then, we traverse the second BST and for each node, we calculate the complement (target – node value) and check whether this complement exists in the hash set. If we find such a pair, we return true. If no such pair exists after traversing the second tree completely, we return false. This approach is straightforward and leverages extra space (the set) for constant time lookups, yielding an overall linear time complexity relative to the number of nodes.</p> </div> <div> <p>#1762 Buildings With an Ocean View <span>Medium</span></p> </div> <div> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Processing the buildings from right to left simplifies the problem because the rightmost building always has an ocean view.</li> <li>• Keep track of the highest building encountered while iterating from right to left.</li> <li>• A building has an ocean view if its height is greater than the maximum height seen so far.</li> <li>• Append indices meeting the criteria, then reverse the order of indices for the solution since they were collected in reverse order.</li> </ul> </div> <div> <p>Space &amp; Time</p> </div> <div> <div>Stack</div> <div>p.2473</div> </div>                                                                                                                 | <div> <p>Time Complexity: O(n) where n is the number of buildings.</p> <p>Space Complexity: O(n) for storing the result in the worst-case scenario.</p> </div> <div> <p>Solution</p> <p>Iterate over the "heights" array from right to left while maintaining a variable to store the maximum height encountered so far. For each building, if its height is greater than this maximum height, add its index to the results list and update the maximum height. Since indices are collected in reverse order, reverse the list before returning it. This approach ensures that each building is visited only once.</p> </div> <div> <p>#32 Longest Valid Parentheses <span>Hard</span></p> </div> <div> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• Use a stack to keep track of indices where potential valid substrings begin.</li> <li>• Start by pushing a dummy index (–1) to handle edge cases.</li> <li>• For every '(', push its index onto the stack.</li> <li>• For every ')', pop from the stack. If the stack becomes empty, push the current index as a new base. Otherwise, compute the length of the valid substring using the current index minus the top index of the stack.</li> </ul> </div> <div> <div>Stack</div> <div>p.2474</div> </div>                                                                      | <div> <ul style="list-style-type: none"> <li>• An alternative is the dynamic programming approach where dp[i] represents the length of the longest valid substring ending at index i.</li> </ul> </div> <div> <p>Space &amp; Time</p> <p>Time Complexity: O(n) where n is the length of the string.</p> <p>Space Complexity: O(n) for the stack data structure.</p> </div> <div> <p>Solution</p> <p>The stack-based solution works by maintaining indices for unmatched parentheses. Initially, a dummy index (–1) is pushed onto the stack to serve as the base for calculating subsequent valid substring lengths. As we iterate through the string:</p> <ul style="list-style-type: none"> <li>– If the character is '(', we push its index onto the stack.</li> <li>– If the character is ')', we pop from the stack. If the stack is empty afterwards, that means there is no valid starting position, so we push the current index to reset the base. Otherwise, we calculate the length of the current valid substring by subtracting the current index with the new top index of the stack and update the maximum length accordingly.</li> </ul> </div> <div> <p>#42 Trapping Rain Water <span>Hard</span></p> </div> <div> <p>Key Insights</p> <ul style="list-style-type: none"> <li>• The water trapped at any index is determined by the minimum of the maximum height on its left and right</li> </ul> </div> <div> <div>Stack</div> <div>p.2475</div> </div>                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <p>minus the height at that index.</p> <ul style="list-style-type: none"> <li>• A two-pointers approach can efficiently solve the problem in one pass by maintaining left and right boundaries.</li> <li>• Alternative approaches include dynamic programming with pre-computed left max and right max arrays, or using a monotonic stack to determine boundaries.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(n)<br/>Space Complexity: O(1) when using the two-pointer approach</p> <p><b>Solution</b><br/>We use a two-pointers strategy by initializing pointers at the beginning and the end of the array. Keep track of the maximum height seen so far from the left (left_max) and the right (right_max). At each step, move the pointer with the lesser height because the water trapped is determined by the shorter boundary. Compute the water at that position as the difference between the boundary (left_max or right_max) and the current height, accumulating it into the total. This approach ensures linear time scanning with constant extra space.</p> <p><b>#84 Largest Rectangle in Histogram [Hard]</b></p> <p><b>Key Insights</b></p> | <ul style="list-style-type: none"> <li>• Each bar can potentially extend left and right as long as the neighboring bars are not shorter.</li> <li>• A monotonic (increasing) stack can be used to efficiently track indices of bars.</li> <li>• When encountering a bar lower than the one on the top of the stack, pop the stack and calculate the area of the rectangle formed with the popped bar height.</li> <li>• A dummy bar (of height 0) is appended at the end to ensure all bars are processed.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(n)<br/>Space Complexity: O(n)</p> <p><b>Solution</b><br/>The solution employs a monotonic stack to traverse the histogram once. For each bar, while the current bar's height is less than the height at the top index of the stack, it pops from the stack and calculates the area using the popped height as the smallest bar. The width for the rectangle is determined by the difference between the current index and the index from the stack (or the current index if the stack is empty). This effectively finds the maximum rectangular area possible. A sentinel value (0) is added at the end of the histogram to flush any remaining bars from the stack.</p> <p><b>#224 Basic Calculator [Hard]</b></p>                                                 | <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a stack to save previous results and signs when encountering parentheses.</li> <li>• Process the string left-to-right by accumulating multi-digit numbers and applying the current sign.</li> <li>• When encountering a '(', push the current result and sign onto the stack and reset them.</li> <li>• When encountering a ')', complete the current sub-expression then pop the previous state to incorporate the computed value.</li> <li>• Handle whitespaces by simply skipping them.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(n) where n is the length of the input string, since we process each character once.<br/>Space Complexity: O(n) in the worst-case scenario due to the use of a stack for nested parentheses.</p> <p><b>Solution</b><br/>The solution uses an iterative approach with a stack:<br/>– Initialize variables: result to keep the running total, sign to handle addition/subtraction, and index to iterate over the string.<br/>– As you iterate, build multi-digit numbers by checking for consecutive numerical characters.<br/>– Adjust the total by adding the product of the current number and its sign.<br/>– When encountering a '(', push the current result and sign onto the stack, then reset results and sign for the new sub-expression.<br/>–</p> |
| <p>Stackp.2476</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Stackp.2477</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Stackp.2478</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>When encountering a ')', complete the sub-expression by popping the previous result and sign, then combine. – Skip over spaces. – The final result is the computed value after the end of the string.</p> <p>This approach effectively simulates recursion and correctly handles nested expressions and negation.</p> <p><b>#239 Sliding Window Maximum [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a deque (double-ended queue) to maintain the indices of potential maximum elements in a decreasing order.</li> <li>• Ensure that the deque always has the current window's indices by removing elements that fall outside the current window.</li> <li>• Only add elements that are greater than the ones at the back of the deque, thus preserving a monotonic decreasing order.</li> <li>• The maximum for the current window is at the front of the deque.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(n) where n is the number of elements in nums, since each element is processed at most twice.<br/>Space Complexity: O(k), as the deque stores indices for at most k elements at any time.</p>      | <p><b>Solution</b><br/>The solution uses a monotonic deque to maintain indices of potential maximum elements in the current window. As the window slides, we:<br/>– Remove indices from the front if they are out of the window.<br/>– Remove indices from the back if the current element is larger, because they cannot be the maximum if the current element is in the window.<br/>– Append the current index. – After the first k elements, the element at the front of the deque is the maximum for that window. This approach ensures that each element is pushed and popped from the deque only once, achieving an overall O(n) time complexity.</p> <p><b>#716 Max Stack [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a doubly-linked list to support fast insertion and removal from the stack.</li> <li>• Maintain a balanced tree (or ordered dictionary/multiset) to quickly locate the maximum element in O(log n) time.</li> <li>• Map each value to its corresponding node(s) in the doubly-linked list, so that when popMax is called, you can immediately remove the corresponding node from the list.</li> <li>• The double-linking allows removal of arbitrary nodes in O(1), once you have the reference.</li> </ul>                                                          | <p><b>Space &amp; Time</b><br/>Time Complexity:<br/>– push: O(log n) due to insertion in the ordered structure.<br/>– pop: O(1) for removal from the doubly-linked list, plus O(log n) for updating the tree.<br/>– top: O(1)<br/>– peekMax: O(1) or O(log n) depending on the tree implementation.<br/>– popMax: O(log n) for looking up and removal from the tree, O(1) for doubly-linked list deletion.<br/>Space Complexity:<br/>– O(n) for storing all nodes in the doubly-linked list and the tree structure mapping values to nodes.</p> <p><b>Solution</b><br/>We use two data structures:<br/>– A doubly-linked list to store the elements in stack order. This supports fast push and pop operations and allows O(1) removal when given a reference. – An ordered structure (like a balanced BST, TreeMap in Java, or SortedList in Python) that maps each value to a list of nodes that hold that value in the linked list. This allows us to quickly determine the maximum element. When there are duplicates, we remove the one closest to the top by storing nodes in order. The trick is to keep these two structures synchronized. When an element is pushed, add its node to both the doubly-linked list and</p>                                                                                                                                                          |
| <p>Stackp.2479</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Stackp.2480</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Stackp.2481</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <p>the ordered structure. When an element is removed (either via pop or popMax), remove its node from both structures.</p> <p><b>#772 Basic Calculator III [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use recursion to handle nested expressions defined by parentheses.</li> <li>• A stack can be used to correctly evaluate the expression and respect operator precedence.</li> <li>• Multiplication and division have higher precedence than addition and subtraction.</li> <li>• When encountering an open parenthesis, recursively compute its value until a closing parenthesis is found.</li> <li>• Use immediate calculation for multiplication and division by updating the top of the stack before moving to the next operator.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(n) where n is the length of the string, as every character is processed.<br/>Space Complexity: O(n) due to the recursion stack or auxiliary stack used for intermediate results.</p> <p><b>Solution</b><br/>We use a recursive approach combined with a stack to solve the problem. As we iterate through the string:</p>      | <p>– Convert digits into numbers. – When encountering an operator or parenthesis, process the previously accumulated number based on the preceding operator. – For multiplication or division, pop the last number from the stack, apply the operation, and push the result back. – When a '(' is encountered, recursively evaluate the substring until the corresponding ')'. – After processing all tokens, sum the values in the stack to get the final result. This method correctly handles both operator precedence and nested expressions.</p> <p><b>#895 Maximum Frequency Stack [Hard]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a hash map to count the frequency of each element.</li> <li>• Maintain another hash map that maps frequencies to stacks (lists) of elements.</li> <li>• Track the current maximum frequency to enable O(1) retrieval of the most frequent element.</li> <li>• When pushing an element, update its frequency and add it to the corresponding frequency stack.</li> <li>• When popping an element, remove it from the stack corresponding to the maximum frequency and update the frequency count; if that stack becomes empty, decrease the current maximum frequency.</li> </ul> <p><b>Space &amp; Time</b><br/>Time Complexity: O(1) per push and pop.</p> | <p>Space Complexity: O(n), where n is the number of elements pushed onto the stack.</p> <p><b>Solution</b><br/>The solution involves using two main data structures:<br/>– A hash map (freq) to store the frequency count for each value.<br/>– A hash map (group) that maps a frequency to a stack (list) of values that have that frequency.</p> <p>During a push:<br/>– Increment the frequency count of the value. – Push the value onto the stack corresponding to its updated frequency. – Update the maximum frequency if necessary.</p> <p>During a pop:<br/>– Pop the top element from the stack corresponding to the current maximum frequency. – Decrement its frequency count in the freq map. – If the frequency stack becomes empty after popping, decrement the current maximum frequency. This design enables the retrieval of the most frequent element in constant time while dealing with frequency ties by returning the element closest to the top.</p> <p>Trees &amp; BST</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| <p>Stackp.2482</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <p>Stackp.2483</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>Stackp.2484</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |



## #21 Trees & BST

By Pattern · Python Only · 37 questions

Trees & BST

p.2485

Trees & BST

### #100 Same Tree

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BFS

Lists: Grind 169

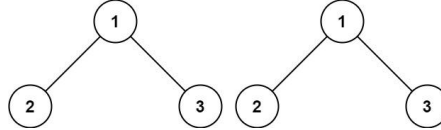
Time Complexity:  $O(n)$ , where  $n$  is the number of nodes in the trees, since every node is visited once. Space Complexity:  $O(n)$  in the worst-case (unbalanced tree), due to recursion call stack usage. In

#### Problem

Given the roots of two binary trees  $p$  and  $q$ , write a function to check if they are the same or not.

Two binary trees are considered the same if they are structurally identical, and the nodes have the same value.

Example 1:

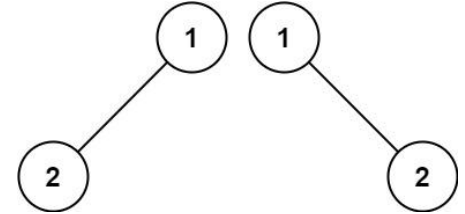


Trees & BST

p.2486

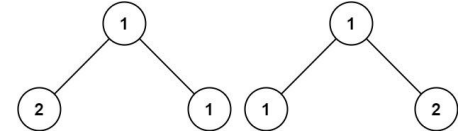
Input:  $p = [1,2,3]$ ,  $q = [1,2,3]$   
Output: true

Example 2:



Input:  $p = [1,2]$ ,  $q = [1,null,2]$   
Output: false

Example 3:



Input:  $p = [1,2,1]$ ,  $q = [1,1,2]$   
Output: false

Constraints:

Trees & BST

p.2487

- The number of nodes in both trees is in the range  $[0, 100]$ .
- $-104 \leq \text{Node.val} \leq 104$

#### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def isSameTree(self, p, q):
 # Brute Force O(n) - recursive structural comparison
 if not p and not q: return True
 if not p or not q or p.val != q.val: return False
 return self.isSameTree(p.left, q.left) and self.isSameTree(p.right, q.right)
```

#### Community Solutions (Python)

##### • WalkCC

```
class Solution:
 def isSameTree(self, p: TreeNode | None, q: TreeNode | None) -> bool:
 if not p or not q:
 return p == q
 return (p.val == q.val and self.isSameTree(p.left, q.left) and self.isSameTree(p.right, q.right))
```

##### • LeetCode

Trees & BST

p.2488

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def isSameTree(self, p: Optional[TreeNode], q: Optional[TreeNode]) -> bool:
 if p == q:
 return True
```

```
 if p is None or q is None or p.val != q.val:
 return False
 return self.isSameTree(p.left, q.left) and self.isSameTree(p.right, q.right)
```

##### • SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
def isSameTree(p, q):
 # If both nodes are None, trees are identical at this branch.
 if not p and not q:
```

Trees & BST

p.2489

```
 return True
 # If one of the nodes is None, trees are not identical.
 if not p or not q:
 return False
 # Check if current nodes have the same value.
 if p.val != q.val:
 return False
 # Recursively compare the left subtrees and the right subtrees.
 return isSameTree(p.left, q.left) and isSameTree(p.right, q.right)
```

##### • LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
self.val = x
```

Trees & BST

p.2490

```
self.left = None
self.right = None
```

```
class Solution(object):
 def isSameTree(self, p, q):
 """
 :type p: TreeNode
 :type q: TreeNode
 :rtype: bool
 """
```

```
 if not p or not q:
 return p == q # covers: p=None and q!=None, q=None and p!=None, both None
 return p.val == q.val and self.isSameTree(p.left, q.left) and self.isSameTree(p.right, q.right)
```

```
iteration
class Solution:
 def isSameTree(self, p: TreeNode, q: TreeNode) -> bool:
 if p is None:
 return q is None
```

Trees & BST

p.2491

```
 if q is None:
 return p is None
```

```
 stack1 = [p]
 stack2 = [q]
```

```
 while stack1 and stack2:
 current1 = stack1.pop()
 current2 = stack2.pop()
```

```
 if current1 is None and current2 is None:
 continue
 elif current1 is None or current2 is None:
 return False
 elif current1.val != current2.val:
 return False
```

```
 stack1.append(current1.left)
 stack2.append(current2.left)
```

```
 stack1.append(current1.right)
 stack2.append(current2.right)
```

```
 return not stack1 and not stack2
```

Trees & BST

p.2492

Trees & BST

### #101 Symmetric Tree

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BFS

Lists: Grind 169

Time Complexity:  $O(n)$ , where  $n$  is the number of nodes in the tree, since we traverse each node once. Space Complexity:  $O(h)$  for recursive call stack (worst-case  $O(n)$  for a skewed tree) or  $O(n)$  for the

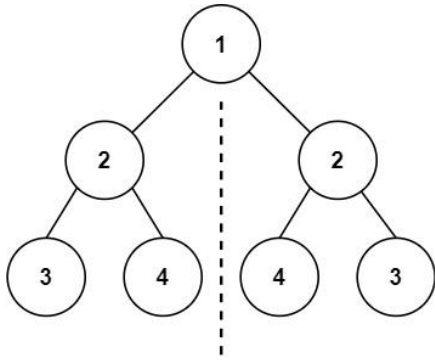
#### Problem

Given the root of a binary tree, check whether it is a mirror of itself (i.e., symmetric around its center).

Example 1:

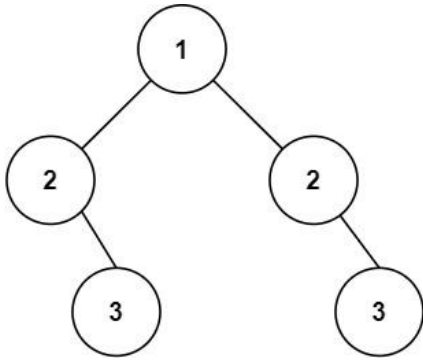
Trees & BST

p.2493



Input: root = [1,2,2,3,4,4,3]  
Output: true

Example 2:



Input: root = [1,2,2,null,3,null,3]  
Output: false

Constraints:

- The number of nodes in the tree is in the range [1, 1000].
- -100 <= Node.val <= 100

Follow up: Could you solve it both recursively and iteratively?

Community Solutions (Python)

• WalkCC

```
class Solution:
 def isSymmetric(self, root: TreeNode | None) -> bool:
 def isSymmetric(p: TreeNode | None, q: TreeNode | None) -> bool:
 if not p or not q:
 return p == q
 return (p.val == q.val and
 isSymmetric(p.left, q.right) and
 isSymmetric(p.right, q.left))
 return isSymmetric(root, root)
```

• LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def isSymmetric(self, root: Optional[TreeNode]) -> bool:
 def dfs(root1: Optional[TreeNode], root2: Optional[TreeNode]) -> bool:
 if root1 == root2:
 return True
 if not root1 or not root2:
 return False
 if root1.val != root2.val:
 return False
 return dfs(root1.left, root2.right) and dfs(root1.right, root2.left)
```

Trees & BST

p.2494

Trees & BST

p.2495

Trees & BST

p.2496

```
return True
if root1 is None or root2 is None or
root1.val != root2.val:
 return False
return dfs(root1.left, root2.right) and
dfs(root1.right, root2.left)

return dfs(root.left, root.right)
```

• SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # current node's value
 self.left = left # pointer to left child
 self.right = right # pointer to right child

class Solution:
 def isSymmetric(self, root: TreeNode) -> bool:
 # Helper function to check mirror symmetry
 # between two trees.
```

```
def isMirror(t1: TreeNode, t2: TreeNode) -> bool:
 # Both nodes are null, so they are mirrors.
 if t1 is None and t2 is None:
 return True
 # If one is null or the values do not match,
 # symmetry fails.
 if t1 is None or t2 is None or t1.val !=
t2.val:
 return False
 # Recursively check mirror symmetry for
 # children:
 # t1's left with t2's right and t1's right
 # with t2's left.
 return isMirror(t1.left, t2.right) and
isMirror(t1.right, t2.left)

Start recursion comparing the tree with itself.
return isMirror(root, root)
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def isSymmetric(self, root: Optional[TreeNode]) -> bool:
 def dfs(root1, root2):
 if root1 is None and root2 is None:
```

```
return True
if root1 is None or root2 is None or
root1.val != root2.val:
 return False
return dfs(root1.left, root2.right) and
dfs(root1.right, root2.left)

return dfs(root, root)
```

Trees & BST

p.2497

Trees & BST

p.2498

Trees & BST

p.2499

Trees & BST

## #104 Maximum Depth of Binary Tree

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BFS

Lists: Grind 169

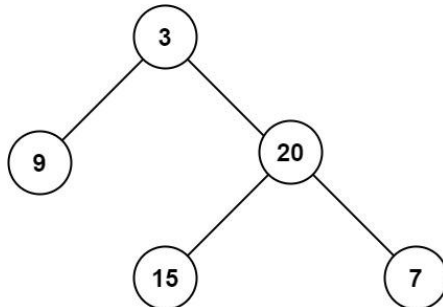
Time Complexity:  $O(n)$ , where  $n$  is the number of nodes, since every node is visited once. Space Complexity:  $O(h)$  for recursion stack in DFS, where  $h$  is the height of the tree (worst-case  $O(n)$  for a skewed tree).

Problem

Given the root of a binary tree, return its maximum depth.

A binary tree's maximum depth is the number of nodes along the longest path from the root node down to the farthest leaf node.

Example 1:



Input: root = [3,9,20,null,null,15,7]  
Output: 3

Example 2:

Input: root = [1,null,2]  
Output: 2

Constraints:

- The number of nodes in the tree is in the range [0, 104].
- -100 <= Node.val <= 100

Community Solutions (Python)

• WalkCC

```
class Solution:
 def maxDepth(self, root: TreeNode | None) -> int:
 if not root:
 return 0
 return 1 + max(self.maxDepth(root.left),
 self.maxDepth(root.right))
```

• LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def maxDepth(self, root: TreeNode) -> int:
 if root is None:
 return 0
 l, r = self.maxDepth(root.left),
self.maxDepth(root.right)
 return 1 + max(l, r)
```

• SimplyLeet

Trees & BST

p.2500

Trees & BST

p.2501

Trees & BST

p.2502



```
Python implementation using DFS

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
```

class Solution:

```
def maxDepth(self, root: Optional[TreeNode]) -> int:
 # Base case: if the node is null, return 0.
 if not root:
 return 0

 # Recursively compute the maximum depth of the
 left subtree.
 left_depth = self.maxDepth(root.left)
 # Recursively compute the maximum depth of the
 right subtree.
 right_depth = self.maxDepth(root.right)
```

```
Return the greater depth plus one for the
current node.
return 1 + max(left_depth, right_depth)
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def maxDepth(self, root: TreeNode) -> int:
 if root is None:
 return 0
```

```
l, r = self.maxDepth(root.left),
self.maxDepth(root.right)
return 1 + max(l, r)
```

Trees & BST

## #108 Convert Sorted Array to Binary Search Tree

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Divide and Conquer · Tree

Lists: Grind 169

Time Complexity:  $O(n)$  – Every element is processed exactly once. Space Complexity:  $O(n)$  –  $O(n)$  space is used to store the BST, and additional  $O(\log n)$  space may be used for the recursion stack.

### Problem

Given an integer array `nums` where the elements are sorted in ascending order, convert it to a height-balanced binary search tree.

Example 1:

Trees & BST

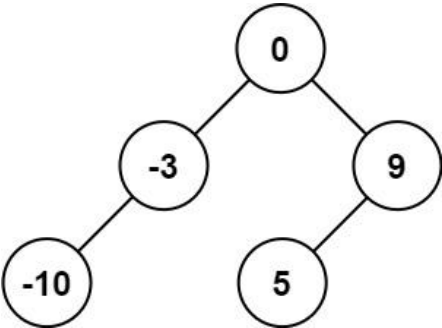
p.2503

Trees & BST

p.2504

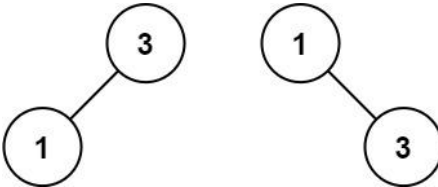
Trees & BST

p.2505



```
Input: nums = [-10,-3,0,5,9]
Output: [0,-3,9,-10,null,5]
Explanation: [0,-10,5,null,-3,null,9] is also accepted:
```

Example 2:



```
Input: nums = [1,3]
Output: [3,1]
Explanation: [1,null,3] and [3,1] are both
height-balanced BSTs.
```

### Constraints:

- $1 \leq \text{nums.length} \leq 104$
- $-104 \leq \text{nums}[i] \leq 104$
- `nums` is sorted in a strictly increasing order.

### Brute Force (Python)

Trees & BST

p.2506

Trees & BST

p.2507

Trees & BST

p.2508

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def sortedArrayToBST(self, nums: List[int]) ->
Optional[TreeNode]:
 def dfs(l: int, r: int) -> Optional[TreeNode]:
 if l > r:
```

```
 return None
 mid = (l + r) >> 1
 return TreeNode(nums[mid], dfs(l, mid - 1),
dfs(mid + 1, r))

 return dfs(0, len(nums) - 1)
```

• SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 # Initializes a tree node with a value and optional
 left/right children.
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
 def sortedArrayToBST(self, nums):
```

```
Helper function to build BST from subarray
indices.
def buildBST(left, right):
 # Base condition: if no elements to process,
 return None.
 if left > right:
 return None
 # Choose the middle element as the root for
 balance.
```

```
 mid = (left + right) // 2
 node = TreeNode(nums[mid])
 # Recursively build the left subtree.
 node.left = buildBST(left, mid - 1)
```

```
Recursively build the right subtree.
node.right = buildBST(mid + 1, right)
return node
Build and return the BST from the entire array.
return buildBST(0, len(nums) - 1)
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def sortedArrayToBST(self, nums: List[int]) ->
Optional[TreeNode]:
 def dfs(l, r):
 if l > r:
```

```
Brute Force Approach
class Solution:
 def sortedArrayToBST(self, nums):
 # Brute Force - collect inorder, then process
sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)
```

```
 collect(nums)
 return vals[0] if vals else 0
```

### Community Solutions (Python)

• WalkCC

```
class Solution:
 def sortedArrayToBST(self, nums: List[int]) -> TreeNode
| None:
 def build(l: int, r: int) -> TreeNode | None:
 if l > r:
 return None
 m = (l + r) // 2
 return TreeNode(nums[m],
 build(l, m - 1),
 build(m + 1, r))

 return build(0, len(nums) - 1)
```

• LeetDoocs

```
 return None
 mid = (l + r) >> 1
 left = dfs(l, mid - 1)
 right = dfs(mid + 1, r)
 return TreeNode(nums[mid], left, right)
```

```
 return dfs(0, len(nums) - 1)
```

#####

```
Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
self.val = x
self.left = None
self.right = None
```

```
class Solution(object):
 def sortedArrayToBST(self, nums):

:type nums: List[int]
:rtype: TreeNode
"""
 if nums:
 midPos = len(nums) / 2
 mid = nums[midPos]
 root = TreeNode(mid)
 root.left = self.sortedArrayToBST(nums[:midPos])
 root.right = self.sortedArrayToBST(nums[midPos +
1:])
 return root
```

Trees & BST

p.2509

Trees & BST

p.2510

Trees & BST

p.2511

Trees & BST

#110 Balanced Binary Tree

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Tree · DFS  
Lists: Grind 169  
Time Complexity:  $O(n)$  where  $n$  is the number of nodes in the tree, since we visit each node once. Space Complexity:  $O(h)$  where  $h$  is the height of the tree, which corresponds to the recursion stack  $O(n)$   
Problem  
Given a binary tree, determine if it is height-balanced.  
Example 1:

Trees & BSTp.2512

```
graph TD
 3((3)) --- 9((9))
 3 --- 20((20))
 20 --- 15((15))
 20 --- 7((7))
```

Input: root = [3,9,20,null,null,15,7]  
Output: true

Example 2:

Trees & BSTp.2513

```
graph TD
 1((1)) --- 2L((2))
 1 --- 2R((2))
 2L --- 3L((3))
 2L --- 3R((3))
 3L --- 4L((4))
 3L --- 4R((4))
```

Input: root = [1,2,2,3,3,null,null,4,4]  
Output: false

Example 3:

Input: root = []  
Output: true

Constraints:

- The number of nodes in the tree is in the range [0, 5000].
- $-104 \leq \text{Node.val} \leq 104$

Community Solutions (Python)

- WalkCC

Trees & BSTp.2514

```
class Solution:
 def isBalanced(self, root: TreeNode | None) -> bool:
 if not root:
 return True

 def maxDepth(root: TreeNode | None) -> int:
 if not root:
 return 0
 return 1 + max(maxDepth(root.left),
 maxDepth(root.right))

 return (abs(maxDepth(root.left) -
 maxDepth(root.right)) <= 1 and
 self.isBalanced(root.left) and
 self.isBalanced(root.right))

class Solution:
 def isBalanced(self, root: TreeNode | None) -> bool:
 ans = True

 def maxDepth(root: TreeNode | None) -> int:
 """Returns the height of root and sets ans to false
 if root unbalanced."""
 nonlocal ans
 if not root or not ans:
 return 0
 left = maxDepth(root.left)
 right = maxDepth(root.right)
 if abs(left - right) > 1:
 ans = False
 return 1 + max(left, right)
```

Trees & BSTp.2515

```
right = maxDepth(root.right)
if abs(left - right) > 1:
 ans = False
return max(left, right) + 1

maxDepth(root)
return ans

class Solution:
 def isBalanced(self, root: TreeNode | None) -> bool:
 def maxDepth(root: TreeNode | None) -> int:
 """Returns the height of root if root is balanced;
 otherwise, returns -1."""
 if not root:
 return 0

 left = maxDepth(root.left)
 if left == -1:
 return -1

 right = maxDepth(root.right)
 if right == -1:
 return -1
 if abs(left - right) > 1:
 return -1

 return 1 + max(maxDepth(root.left),
 maxDepth(root.right))

 return maxDepth(root) != -1
```

- LeetDoocs

Trees & BSTp.2516

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def isBalanced(self, root: Optional[TreeNode]) -> bool:
 def height(root):
 if root is None:
 return 0
 l, r = height(root.left), height(root.right)
 if l == -1 or r == -1 or abs(l - r) > 1:
 return -1
 return 1 + max(l, r)

 return height(root) >= 0
```

- SimplyLeet

Trees & BSTp.2517

```
def checkHeight(node: TreeNode) -> int:
 # An empty tree has height 0 and is balanced.
 if not node:
 return 0

 # Recursively get the height of the left
 subtree.
 leftHeight = checkHeight(node.left)
 # If left subtree is unbalanced, return -1.
 if leftHeight == -1:
 return -1

 # Recursively get the height of the right
 subtree.
 rightHeight = checkHeight(node.right)
 # If right subtree is unbalanced, return -1.
 if rightHeight == -1:
 return -1

 # If the current node's subtrees differ in
 height by more than 1, it's unbalanced.
 if abs(leftHeight - rightHeight) > 1:
 return -1

 # Return the height of the current node.
 return max(leftHeight, rightHeight) + 1

The tree is balanced if checkHeight does not
return -1.
return checkHeight(root) != -1
```

- LC.ca

Trees & BSTp.2518

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def isBalanced(self, root: Optional[TreeNode]) -> bool:
 def height(root):
 if root is None:
 return 0
 l, r = height(root.left), height(root.right)
 if l == -1 or r == -1 or abs(l - r) > 1:
 return -1
 return 1 + max(l, r)

 return height(root) >= 0
```

Trees & BSTp.2519

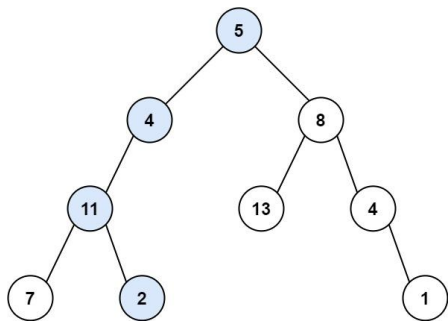
Trees & BST

#112 Path Sum

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Tree · DFS · BFS  
Lists: Denny Zhang  
Time Complexity:  $O(N)$  where  $N$  is the number of nodes, since every node may need to be visited. Space Complexity:  $O(H)$  where  $H$  is the height of the tree, accounting for the recursion stack (worst-case)  
Problem  
Given the root of a binary tree and an integer targetSum, return true if the tree has a root-to-leaf path such that adding up all the values along the path equals targetSum.  
A leaf is a node with no children.  
Example 1:

Trees & BSTp.2520

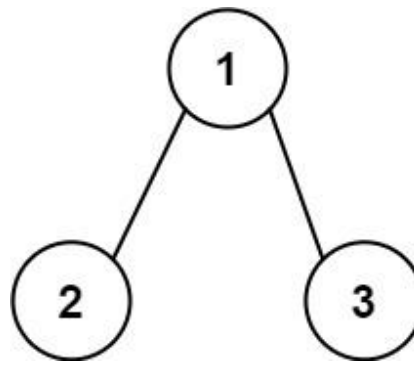


Input: root = [5,4,8,11,null,13,4,7,2,null,null,null,1], targetSum = 22  
Output: true  
Explanation: The root-to-leaf path with the target sum is shown.

Example 2:

Trees & BST

p.2521



Input: root = [1,2,3], targetSum = 5  
Output: false  
Explanation: There are two root-to-leaf paths in the tree:  
(1 --> 2): The sum is 3.  
(1 --> 3): The sum is 4.  
There is no root-to-leaf path with sum = 5.

Example 3:

Trees & BST

p.2522

Input: root = [], targetSum = 0  
Output: false  
Explanation: Since the tree is empty, there are no root-to-leaf paths.

Constraints:

- The number of nodes in the tree is in the range [0, 5000].
- $-1000 \leq \text{Node.val} \leq 1000$
- $-1000 \leq \text{targetSum} \leq 1000$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def hasPathSum(self, root, targetSum):
 # Brute Force - collect inorder, then process sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else 0
```

Community Solutions (Python)

- WalkCC

Trees & BST

p.2523

```
class Solution:
 def hasPathSum(self, root: TreeNode, summ: int) -> bool:
 if not root:
 return False
 if root.val == summ and not root.left and not root.right:
 return True
 return (self.hasPathSum(root.left, summ - root.val) or
 self.hasPathSum(root.right, summ - root.val))
```

• LeetCode

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def hasPathSum(self, root: Optional[TreeNode], targetSum: int) -> bool:
 def dfs(root, s):
 if root is None:
 return False
 s += root.val
 if root.left is None and root.right is None and s == targetSum:
 return True
 return dfs(root.left, s) or dfs(root.right, s)
 return dfs(root, 0)
```

Trees & BST

p.2524

```
return False
s += root.val
if root.left is None and root.right is None and s == targetSum:
 return True
return dfs(root.left, s) or dfs(root.right, s)

return dfs(root, 0)
```

• SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # Node's value
 self.left = left # Pointer to left child
 self.right = right # Pointer to right child

def hasPathSum(self, root: Optional[TreeNode], targetSum: int) -> bool:
 # If the current node is None, no valid path exists
 if not root:
 return False
```

Trees & BST

p.2525

```
return False
Check if the current node is a leaf node
if not root.left and not root.right:
 # Return true if the remaining targetSum equals the leaf value
 return targetSum == root.val
Recursively check the left and right subtree with the updated targetSum
return hasPathSum(root.left, targetSum - root.val) or hasPathSum(root.right, targetSum - root.val)
```

```
Example usage:
root = TreeNode(5, TreeNode(4, TreeNode(11, TreeNode(7), TreeNode(2))),
TreeNode(8, TreeNode(13), TreeNode(4, None, TreeNode(1))))

print(hasPathSum(root, 22)) # Output: True
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def hasPathSum(self, root: Optional[TreeNode], targetSum: int) -> bool:
 def dfs(root, s):
 if root is None:
 return False
```

Trees & BST

p.2526

```
return False
s += root.val
if root.left is None and root.right is None and s == targetSum:
 return True
return dfs(root.left, s) or dfs(root.right, s)

return dfs(root, 0)
```

Trees & BST

p.2527

Trees & BST

## #226 Invert Binary Tree

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BFS

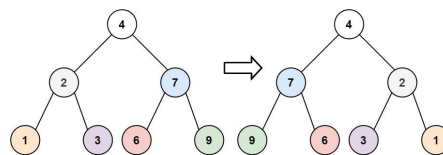
Lists: Grind 169

Time Complexity:  $O(n)$  where  $n$  is the number of nodes, since each node is visited once. Space Complexity:  $O(h)$  where  $h$  is the height of the tree, which is the recursion stack space in the recursive app

Problem

Given the root of a binary tree, invert the tree, and return its root.

Example 1:

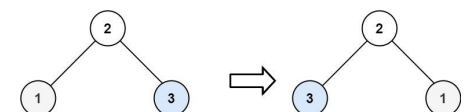


Input: root = [4,2,7,1,3,6,9]  
Output: [4,7,2,9,6,3,1]

Example 2:

Trees & BST

p.2528



Input: root = [2,1,3]  
Output: [2,3,1]

Example 3:

Input: root = []  
Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 100].
- $-100 \leq \text{Node.val} \leq 100$

Community Solutions (Python)

- WalkCC

Trees & BST

p.2529

```
class Solution:
 def invertTree(self, root: TreeNode | None) -> TreeNode | None:
 if not root:
 return None

 left = root.left
 right = root.right
 root.left = self.invertTree(right)
 root.right = self.invertTree(left)
 return root
```

• LeetCode

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def invertTree(self, root: Optional[TreeNode]) -> Optional[TreeNode]:
 if root is None:
 return None
```

```
l, r = self.invertTree(root.left),
self.invertTree(root.right)
root.left, root.right = r, l
return root
```

• SimplyLeet

Trees & BST

p.2530

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # Initialize node value
 self.left = left # Pointer to left child
 self.right = right # Pointer to right child

class Solution:
 def invertTree(self, root: TreeNode) -> TreeNode:
 # Base case: if the node is None, return None.
```

```
 if not root:
 return None
 # Recursively invert the left and right subtrees.
 left_inverted = self.invertTree(root.left)
 right_inverted = self.invertTree(root.right)
 # Swap the left and right children.
 root.left, root.right = right_inverted,
 left_inverted
 return root
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def invertTree(self, root: Optional[TreeNode]) -> Optional[TreeNode]:
 def dfs(root):
 if root is None:
```

Trees & BST

p.2531

```
 return
 root.left, root.right = root.right, root.left
 dfs(root.left)
 dfs(root.right)

 def dfs(self, root):
 return root
```

Trees & BST

p.2532

Trees & BST

## #270 Closest Binary Search Tree Value

EAS

LeetCode · SimplyLeet · WalkCC · LeetCode

Binary Search · Tree · DFS · BST

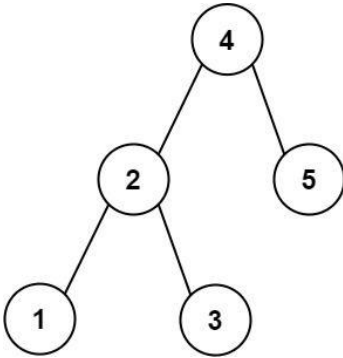
Lists: Premium 98

Time Complexity:  $O(h)$  on average, where  $h$  is the height of the tree ( $O(\log n)$  for a balanced tree, and  $O(n)$  in the worst-case scenario for a skewed tree). Space Complexity:  $O(1)$  for an iterative solut

### Problem

Given the root of a binary search tree and a target value, return the value in the BST that is closest to the target. If there are multiple answers, print the smallest.

Example 1:



```
Input: root = [4,2,5,1,3], target = 3.714286
Output: 4
```

Example 2:

```
Input: root = [1], target = 4.428571
Output: 1
```

Constraints:

- The number of nodes in the tree is in the range  $[1, 104]$ .
- $0 \leq \text{Node.val} \leq 109$
- $-109 \leq \text{target} \leq 109$

Trees & BST

p.2534

### Community Solutions (Python)

• WalkCC

```
class Solution:
 def closestValue(self, root: TreeNode | None, target: float) -> int:
 if not root:
 return None
 # If target < root.val, search the left subtree.
 if target < root.val and root.left:
 left = self.closestValue(root.left, target)
 if abs(left - target) <= abs(root.val - target):
 return left

 # If target > root.val, search the right subtree.
 if target > root.val and root.right:
 right = self.closestValue(root.right, target)
 if abs(right - target) < abs(root.val - target):
 return right

 return root.val
```

• LeetCode

Trees & BST

p.2533

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def closestValue(self, root: Optional[TreeNode], target: float) -> int:
 def dfs(node: Optional[TreeNode]):
 if node is None:
 return None

 return
 n = abs(target - node.val)
 nonlocal ans, diff
 if n < diff or (n == diff and node.val < ans):
 ans = node.val
 diff = n
 if target < node.val:
 dfs(node.left)
 else:
 dfs(node.right)

 ans = 0
 dfs(root)
 return ans
```

• SimplyLeet

Trees & BST

p.2536

```
Python solution for Closest Binary Search Tree Value

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

class Solution:
 def closestValue(self, root: TreeNode, target: float) -> int:
 # Initialize the best value with the value of the root
 best_value = root.val

 # Traverse the BST iteratively
 current = root
 while current:
 # Calculate the difference between current node's value and target
 current_diff = abs(current.val - target)
 best_diff = abs(best_value - target)
```

```
 class Solution:
 def closestValue(self, root: TreeNode, target: float) -> int:
 # Initialize the best value with the value of the root
 best_value = root.val

 # Traverse the BST iteratively
 current = root
 while current:
 # Calculate the difference between current node's value and target
 current_diff = abs(current.val - target)
 best_diff = abs(best_value - target)
```

Trees & BST

```
 # Update best_value if a closer difference is found, or if equal and current value is smaller
 if current_diff < best_diff or (current_diff == best_diff and current.val < best_value):
 best_value = current.val

 # Decide the traversal direction based on the target comparison
 if target < current.val:
 current = current.left
 else:
 current = current.right

 def closestValue(self, root: Optional[TreeNode], target: float) -> int:
 return best_value
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
self.val = x
self.left = None
self.right = None
```

```
class Solution(object):
 def closestValue(self, root, target):
 """
```

Trees & BST

p.2538

```
:type root: TreeNode
:type target: float
:rtype: int
"""
ans = p.val
while p:
 if abs(target - p.val) < abs(ans - target):
 ans = p.val
 if target < p.val:
 p = p.left
 else:
 p = p.right
return ans

#####

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):

self.val = val
self.left = left
self.right = right
class Solution:
 def closestValue(self, root: Optional[TreeNode],
target: float) -> int:
 ans, mi = root.val, inf
 while root:
 t = abs(root.val - target)
 if t < mi:
 mi = t
```

Trees & BSTp.2539

```
ans = root.val
if root.val > target:
 root = root.left
else:
 root = root.right
return ans
```

Trees & BSTp.2540

Trees & BST

#501 Find Mode in Binary Search TreeEAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BST

Lists: Denny Zhang

Time Complexity: O(n), where n is the number of nodes, because each node is processed twice (once in each traversal). Space Complexity: O(h) where h is the height of the tree due to the recursion stac

Problem

Given the root of a binary search tree (BST) with duplicates, return all the mode(s) (i.e., the most frequently occurred element) in it.

If the tree has more than one mode, return them in any order.

Assume a BST is defined as follows:

- The left subtree of a node contains only nodes with keys less than or equal to the node's key.
- The right subtree of a node contains only nodes with keys greater than or equal to the node's key.
- Both the left and right subtrees must also be binary search trees.

Example 1:

Trees & BSTp.2541

```
graph TD
 1((1)) --- 2L((2))
 1 --- 2R((2))
```

Input: root = [1,null,2,2]  
Output: [2]

Example 2:

Input: root = [0]  
Output: [0]

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- -105 <= Node.val <= 105

Trees & BSTp.2542

Follow up: Could you do that without using any extra space? (Assume that the implicit stack space incurred due to recursion does not count).

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findMode(self, root):
 # Brute Force - collect inorder, then process sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else 0
```

Community Solutions (Python)

- WalkCC

Trees & BSTp.2543

```
class Solution:
 def findMode(self, root: TreeNode | None) -> List[int]:
 self.ans = []
 self.pred = None
 self.count = 0
 self.maxCount = 0

 def updateCount(root: TreeNode | None) -> None:
 if self.pred and self.pred.val == root.val:
 self.count += 1

 else:
 self.count = 1

 if self.count > self.maxCount:
 self.maxCount = self.count
 self.ans = [root.val]
 elif self.count == self.maxCount:
 self.ans.append(root.val)

 self.pred = root

 def inorder(root: TreeNode | None) -> None:
 if not root:
 return

 inorder(root.left)
 updateCount(root)
 inorder(root.right)

 inorder(root)
```

Trees & BSTp.2544

```
return self.ans
```

- LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def findMode(self, root: TreeNode) -> List[int]:
 def dfs(root):
 if root is None:

 return
 nonlocal mx, prev, ans, cnt
 dfs(root.left)
 cnt = cnt + 1 if prev == root.val else 1
 if cnt > mx:
 ans = [root.val]
 mx = cnt
 elif cnt == mx:
 ans.append(root.val)
 prev = root.val

 dfs(root.right)

 prev = None
 mx = cnt = 0
 ans = []
 dfs(root)
 return ans
```

Trees & BSTp.2545

- SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

class Solution:
 def findMode(self, root):
 # Initialize state variables: prev value, current frequency count, maximum frequency, and modes list.

 self.prev = None
 self.currentCount = 0
 self.maxCount = 0
 self.modes = []

 # First in-order traversal to find the maximum frequency (maxCount).
 def inorder_first(node):
 if not node:
 return
 inorder_first(node.left)
```

Trees & BSTp.2546

```
Update the count for the current node.
if self.prev is None or node.val !=
self.prev:
 self.currentCount = 1
else:
 self.currentCount += 1
self.prev = node.val
Update maxCount if currentCount exceeds it.
self.maxCount = max(self.maxCount,
self.currentCount)
inorder_first(node.right)

inorder_first(root)

Second in-order traversal to collect all node values with frequency equal to maxCount.
self.prev = None # Reset previous value.
self.currentCount = 0 # Reset count.

def inorder_second(node):
 if not node:
 return
 inorder_second(node.left)
```

Trees & BSTp.2547

```
 if self.prev is None or node.val !=
self.prev:
 self.currentCount = 1
 else:
 self.currentCount += 1
 self.prev = node.val
 if self.currentCount == self.maxCount:
 self.modes.append(node.val)
 inorder_second(node.right)

 inorder_second(root)

 return self.modes
```

```
• LC.ca
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def findMode(self, root: TreeNode) -> List[int]:
 def dfs(root):
 if root is None:
```

```
 return
 nonlocal mx, prev, ans, cnt
 dfs(root.left)
 cnt = cnt + 1 if prev == root.val else 1
 if cnt > mx:
 ans = [root.val]
 mx = cnt
 elif cnt == mx:
 ans.append(root.val)
 prev = root.val

 dfs(root.right)

 prev = None
 mx = cnt = 0
 ans = []
 dfs(root)
 return ans
```

Trees & BST

#543 Diameter of Binary Tree

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS

Lists: Grind 169

Time Complexity:  $O(n)$ , where  $n$  is the number of nodes in the tree. Space Complexity:  $O(n)$ , in the worst-case scenario due to the recursion stack (skewed tree).

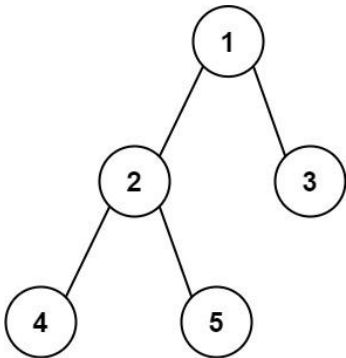
**Problem**

Given the root of a binary tree, return the length of the diameter of the tree.

The diameter of a binary tree is the length of the longest path between any two nodes in a tree. This path may or may not pass through the root.

The length of a path between two nodes is represented by the number of edges between them.

Example 1:



Input: root = [1,2,3,4,5]  
Output: 3  
Explanation: 3 is the length of the path [4,2,1,3] or [5,2,1,3].

Example 2:

Input: root = [1,2]  
Output: 1

Constraints:

- The number of nodes in the tree is in the range [1, 104].

```
• -100 <= Node.val <= 100
Community Solutions (Python)

• WalkCC
class Solution:
 def diameterOfBinaryTree(self, root: TreeNode | None)
-> int:
 ans = 0

 def maxDepth(root: TreeNode | None) -> int:
 nonlocal ans
 if not root:
 return 0

 l = maxDepth(root.left)

 r = maxDepth(root.right)
 ans = max(ans, l + r)
 return 1 + max(l, r)

 maxDepth(root)
 return ans

• LeetDoocs
```

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def diameterOfBinaryTree(self, root: TreeNode) ->
int:
 def dfs(root):
 if root is None:

 return 0
 nonlocal ans
 left, right = dfs(root.left), dfs(root.right)
 ans = max(ans, left + right)
 return 1 + max(left, right)

 ans = 0
 dfs(root)
 return ans

• SimplyLeet
```

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

class Solution:
 def diameterOfBinaryTree(self, root: TreeNode) ->
int:
 # Global variable to track the maximum diameter
 found

 self.max_diameter = 0

 def dfs(node):
 # Base case: if node is None, depth is 0.
 if not node:
 return 0
 # Compute the depth of left and right
 subtrees.
 left_depth = dfs(node.left)
 right_depth = dfs(node.right)
 # Update the global diameter: the path
 through this node
```

```
 self.max_diameter = max(self.max_diameter,
left_depth + right_depth)
 # Return the depth of the tree rooted at this
 node.

 return max(left_depth, right_depth) + 1

 dfs(root)
 return self.max_diameter

• LC.ca
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def diameterOfBinaryTree(self, root: TreeNode) ->
int:
 def dfs(root):
 if root is None:

 return 0
 nonlocal ans
 left, right = dfs(root.left), dfs(root.right)
 ans = max(ans, left + right)
 return 1 + max(left, right)

 ans = 0
 dfs(root)
 return ans
```

Trees & BST

#572 Subtree of Another Tree

EAS

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS · String Matching

Lists: Grind 169

Time Complexity:  $O(m * n)$  in the worst case, where  $m$  is the number of nodes in the main tree and  $n$  is the number of nodes in the subRoot tree. Space Complexity:  $O(\max(\text{depth of main tree}, \text{depth of subR}))$

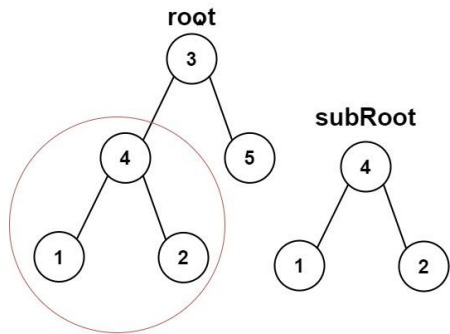
**Problem**

Given the roots of two binary trees root and subRoot, return true if there is a subtree of root with the same structure and node values of subRoot and false otherwise.

A subtree of a binary tree tree is a tree that consists of a node in tree and all of this node's descendants. The tree tree could also be considered as a subtree of itself.

Example 1:



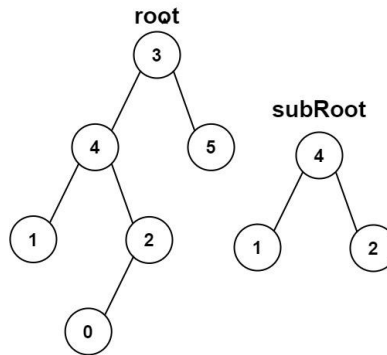


Input: root = [3,4,5,1,2], subRoot = [4,1,2]  
Output: true

Example 2:

Trees & BST

p.2557



Input: root = [3,4,5,1,2,null,null,null,null,0],  
subRoot = [4,1,2]  
Output: false

Constraints:

- The number of nodes in the root tree is in the range [1, 2000].
- The number of nodes in the subRoot tree is in the range [1, 1000].
- $-104 \leq \text{root.val} \leq 104$
- $-104 \leq \text{subRoot.val} \leq 104$

Trees & BST

p.2558

## Community Solutions (Python)

### • LeetCode

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def isSubtree(self, root: Optional[TreeNode],
subRoot: Optional[TreeNode]) -> bool:
 def same(p: Optional[TreeNode], q:
Optional[TreeNode]) -> bool:
 if p is None or q is None:
```

```
 return p is q
 return p.val == q.val and same(p.left,
q.left) and same(p.right, q.right)
```

```
 if root is None:
 return False
 return (
 same(root, subRoot)
 or self.isSubtree(root.left, subRoot)
 or self.isSubtree(root.right, subRoot)
)
```

### • SimplyLeet

Trees & BST

p.2559

# Python solution for checking if subRoot is a subtree of root.

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
```

class Solution:

```
 def isSubtree(self, root: TreeNode, subRoot:
TreeNode) -> bool:
 # Helper function to check if two trees are
 identical.
 def isSameTree(s, t):
 # Both nodes are None, trees match.
 if not s and not t:
 return True
 # One node is None or values differ.
 if not s or not t or s.val != t.val:
 return False
 # Recurse on left and right subtrees.
```

Trees & BST

p.2560

```
 return isSameTree(s.left, t.left) and
isSameTree(s.right, t.right)

 # Base case: if root is None, then no subtree
 exists.
 if not root:
 return False
 # Check if current subtree is identical to
 subRoot.
 if isSameTree(root, subRoot):
 return True
 # Recursively check left and right subtrees.
 return self.isSubtree(root.left, subRoot) or
self.isSubtree(root.right, subRoot)
```

### • LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def isSubtree(self, root: TreeNode, subRoot:
TreeNode) -> bool:
 def dfs(root1, root2):
 if root1 is None and root2 is None:
```

Trees & BST

p.2561

```
 return True
 if root1 is None or root2 is None:
 return False
 return (
 root1.val == root2.val
 and dfs(root1.left, root2.left)
 and dfs(root1.right, root2.right)
)
```

```
 if root is None:
```

```
 return False
 return (
 dfs(root, subRoot)
 or self.isSubtree(root.left, subRoot)
 or self.isSubtree(root.right, subRoot)
)
```

Trees & BST

p.2562

Trees & BST

## #98 Validate Binary Search Tree

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BST

Lists: Grind 169

Time Complexity:  $O(N)$ , where  $N$  is the number of nodes, as each node is visited once. Space Complexity:  $O(H)$ , where  $H$  is the height of the tree, due to the recursion stack.

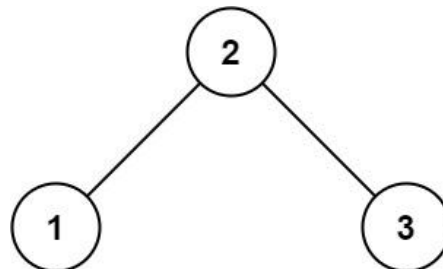
### Problem

Given the root of a binary tree, determine if it is a valid binary search tree (BST).

A valid BST is defined as follows:

- The left subtree of a node contains only nodes with keys strictly less than the node's key.
- The right subtree of a node contains only nodes with keys strictly greater than the node's key.
- Both the left and right subtrees must also be binary search trees.

Example 1:

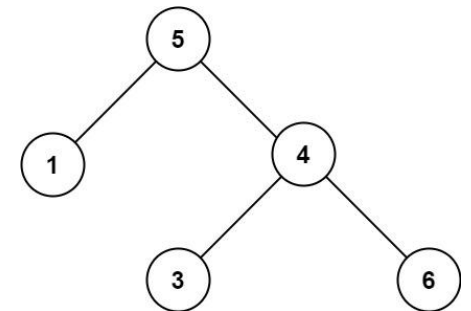


Input: root = [2,1,3]  
Output: true

Example 2:

Trees & BST

p.2564



Input: root = [5,1,4,null,null,3,6]  
Output: false  
Explanation: The root node's value is 5 but its right child's value is 4.

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- $-231 \leq \text{Node.val} \leq 231 - 1$

## Community Solutions (Python)

### • WalkCC

Trees & BST

p.2563

Trees & BST

p.2565

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>class Solution:     def isValidBST(self, root: TreeNode   None) -&gt; bool:         def isValidBST(root: TreeNode   None,                         minNode: TreeNode   None, maxNode:                         TreeNode   None) -&gt; bool:             if not root:                 return True             if minNode and root.val &lt;= minNode.val:                 return False             if maxNode and root.val &gt;= maxNode.val:                 return False              return (isValidBST(root.left, minNode, root) and                     isValidBST(root.right, root, maxNode))          return isValidBST(root, None, None)  class Solution:     def isValidBST(self, root: TreeNode   None) -&gt; bool:         stack = []         pred = None          while root or stack:             while root:                 stack.append(root)                 root = root.left             root = stack.pop()              return (isValidBST(root.left, minNode, root) and                     isValidBST(root.right, root, maxNode))</pre> | p.2566 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>        if pred and pred.val &gt;= root.val:             return False         pred = root         root = root.right          return True  class Solution:     # Definition for a binary tree node.     # class TreeNode:     #     def __init__(self, val=0, left=None, right=None):     #         self.val = val     #         self.left = left     #         self.right = right     class Solution:         def isValidBST(self, root: Optional[TreeNode]) -&gt;         bool:             def dfs(root: Optional[TreeNode]) -&gt; bool:                 if root is None:                     return True                 if not dfs(root.left):                     return False                 nonlocal prev                 if prev &gt;= root.val:                     return False                 prev = root.val                 return dfs(root.right)              prev = -inf              return dfs(root)</pre> | p.2567 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>• SimplyLeet  # Definition for a binary tree node. class TreeNode:     def __init__(self, val=0, left=None, right=None):         self.val = val # Node's value         self.left = left # Left child         self.right = right # Right child  class Solution:     def isValidBST(self, root: TreeNode) -&gt; bool:         # Helper function to recursively validate nodes          def validate(node, low, high):             if not node:                 return True # An empty node/subtree is                 valid             # If current node's value is out of the             allowed range, return False             if not (low &lt; node.val &lt; high):                 return False             # Recursively check the left subtree and             right subtree             return validate(node.left, low, node.val) and             validate(node.right, node.val, high)              return validate(root, float("-inf"),                             float("inf"))  • LC.ca</pre> | p.2568 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre># Definition for a binary tree node. # class TreeNode: #     def __init__(self, val=0, left=None, right=None): #         self.val = val #         self.left = left #         self.right = right # Definition for a binary tree node. # class TreeNode: #     def __init__(self, val=0, left=None, right=None): #         self.val = val  # self.left = left # self.right = right  class Solution: # recursive     def isValidBST(self, root: Optional[TreeNode]) -&gt;     bool:         def dfs(root, mi, ma):             if not root:                 return True             return mi &lt; root.val &lt; ma and dfs(root.left,             mi, root.val) and dfs(root.right, root.val, ma)</pre> | p.2569 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |

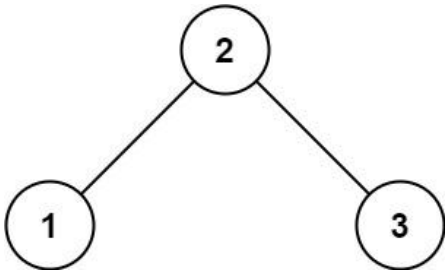
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>        return dfs(root, -math.inf, math.inf)  class Solution: # iterative     def isValidBST(self, root: TreeNode) -&gt; bool:         # store (node, lower, upper) tuples in a single         queue          queue = [(root, None, None)]          # another option:         # queue = [(root, -math.inf, math.inf)]          while queue:             node, lower, upper = queue.pop(0)              if node is None:                 continue              val = node.val             if lower is not None and val &lt;= lower:                 return False              if upper is not None and val &gt;= upper:                 return False              queue.append((node.right, val, upper))             queue.append((node.left, lower, val))              return True  #####</pre> | p.2570 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>class Solution:     def isValidBST(self, root: Optional[TreeNode]) -&gt; bool:         def isValidBST(root: Optional[TreeNode],                         minNode: Optional[TreeNode], maxNode:                         Optional[TreeNode]) -&gt; bool:             if not root:                 return True             if minNode and root.val &lt;= minNode.val:                 return False             if maxNode and root.val &gt;= maxNode.val:                 return False              return isValidBST(root.left, minNode, root) and             isValidBST(root.right, root, maxNode)              return isValidBST(root, None, None)  #####  # Definition for a binary tree node. # class TreeNode: #     def __init__(self, val=0, left=None, right=None):</pre> | p.2571 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre># self.val = val # self.left = left # self.right = right class Solution:     def isValidBST(self, root: TreeNode) -&gt; bool:         def dfs(root):             nonlocal prev             if root is None:                 return True             if not dfs(root.left):                  return False             if prev &gt;= root.val:                 return False             prev = root.val             if not dfs(root.right):                 return False             return True              prev = -inf             return dfs(root)  #####  # Definition for a binary tree node. # class TreeNode(object): #     def __init__(self, x): #         self.val = x #         self.left = None #         self.right = None</pre> | p.2572 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <pre>class Solution(object):     def isValidBST(self, root):         """         :type root: TreeNode         :rtype: bool         """         prev = -float("inf")         stack = [(1, root)]         while stack:             p = stack.pop()              if not p[1]:                 continue             if p[0] == 0:                 if p[1].val &lt;= prev:                     return False                 prev = p[1].val             else:                 stack.append((1, p[1].right))                 stack.append((0, p[1]))                 stack.append((1, p[1].left))              return True</pre> | p.2573 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| <div>Trees &amp; BST</div> <div>#102 Binary Tree Level Order TraversalMED</div> <div>LeetCode · SimplyLeet · WalkCC · LeetCode</div> <div>Tree · BFS</div> <div>Lists · Grind 169</div> <div>Time Complexity: O(n), where n is the number of nodes, because each node is processed once. Space Complexity: O(n), which accounts for the space used by the queue in the worst-case scenario.</div> <div>Problem</div> <div>Given the root of a binary tree, return the level order traversal of its nodes' values. (i.e., from left to right, level by level).</div> <div>Example 1:</div> | p.2574 |
| Trees & BST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |



Input: root = [3,9,20,null,null,15,7]  
Output: [[3],[9,20],[15,7]]

Example 2:

Input: root = [1]  
Output: [[1]]

Example 3:

Input: root = []  
Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 2000].
- $-1000 \leq \text{Node.val} \leq 1000$

Trees & BST

p.2575

## Community Solutions (Python)

### • WalkCC

```
class Solution:
 def levelOrder(self, root: TreeNode | None) -> List[List[int]]:
 if not root:
 return []

 ans = []
 q = collections.deque([root])

 while q:
 currLevel = []

 for _ in range(len(q)):
 node = q.popleft()
 currLevel.append(node.val)
 if node.left:
 q.append(node.left)
 if node.right:
 q.append(node.right)
 ans.append(currLevel)

 return ans
```

### • LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def levelOrder(self, root: Optional[TreeNode]) -> List[List[int]]:
 ans = []
 if root is None:
 return ans
 q = deque([root])
 while q:
 t = []
 for _ in range(len(q)):
 node = q.popleft()
 t.append(node.val)
 if node.left:
 q.append(node.left)
 if node.right:
 q.append(node.right)
 ans.append(t)
 return ans
```

### • SimplyLeet

```
level_size = len(queue) # Number of nodes at the current level.
level_nodes = [] # List to hold node values at the current level.

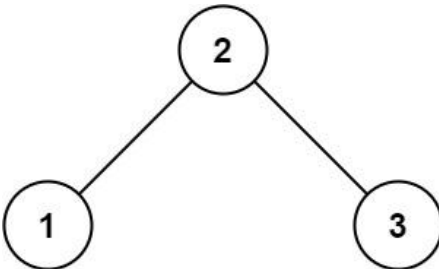
for _ in range(level_size):
 node = queue.popleft() # Dequeue the next node.

 level_nodes.append(node.val)
 # Enqueue left child if it exists.
 if node.left:
 queue.append(node.left)
 # Enqueue right child if it exists.
 if node.right:
 queue.append(node.right)
Append the current level's values to the result.
result.append(level_nodes)
return result
```

### • LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def levelOrder(self, root: Optional[TreeNode]) -> List[List[int]]:
 ans = []
 if root is None:
```

```
return ans
q = deque([root])
while q:
 t = []
 for _ in range(len(q)):
 node = q.popleft()
 t.append(node.val)
 if node.left:
 q.append(node.left)
 if node.right:
 q.append(node.right)
 ans.append(t)
return ans
```



Input: root = [3,9,20,null,null,15,7]  
Output: [[3],[20,9],[15,7]]

Example 2:

Input: root = [1]  
Output: [[1]]

Example 3:

Input: root = []  
Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 2000].
- $-100 \leq \text{Node.val} \leq 100$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def zigzagLevelOrder(self, root):
 # Brute Force - collect inorder, then process sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else 0
```

## Community Solutions (Python)

### • WalkCC

```
class Solution:
 def zigzagLevelOrder(self, root: TreeNode | None) -> List[List[int]]:
 if not root:
 return []

 ans = []
 dq = collections.deque([root])
 isLeftToRight = True

 while dq:
```

Trees & BST

## #103 Binary Tree Zigzag Level Order Traversal

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · BFS

Lists: Grind 169

Time Complexity:  $O(n)$ , where  $n$  is the number of nodes, because each node is visited exactly once. Space Complexity:  $O(n)$ , as in the worst-case scenario (e.g., a complete binary tree) the queue used for

Problem

Given the root of a binary tree, return the zigzag level order traversal of its nodes' values. (i.e., from left to right, then right to left for the next level and alternate between).

Example 1:

Trees & BST

p.2581

Trees & BST

p.2582

Trees & BST

p.2583

```
currLevel = []
for _ in range(len(dq)):
 if isLeftToRight:
 node = dq.popleft()
 currLevel.append(node.val)
 if node.left:
 dq.append(node.left)
 if node.right:
 dq.append(node.right)
 else:
 node = dq.pop()
 currLevel.append(node.val)
 if node.right:
 dq.appendleft(node.right)
 if node.left:
 dq.appendleft(node.left)
 ans.append(currLevel)
 isLeftToRight = not isLeftToRight

return ans
```

• LeetCode

Trees & BST

p.2584

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def zigzagLevelOrder(self, root: Optional[TreeNode])
-> List[List[int]]:
 ans = []
 if root is None:
 return ans
 q = deque([root])
 ans = []
 left = 1
 while q:
 t = []
 for _ in range(len(q)):
 node = q.popleft()
 t.append(node.val)
 if node.left:
 q.append(node.left)
 if node.right:
 q.append(node.right)
 ans.append(t if left else t[::-1])
 left ^= 1
 return ans
```

• SimplyLeet

Trees & BST

p.2585

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

def zigzagLevelOrder(root):
 # If the tree is empty, return an empty list
 if not root:
 return []

 # Initialize the result list and the queue with the
 root node.
 result = []
 queue = [root]
 left_to_right = True # Flag to indicate the current
 direction of traversal

 # Process the tree level by level.
 while queue:
 level_size = len(queue)
```

Trees & BST

p.2586

```
current_level = []

Process all nodes for the current level.
for i in range(level_size):
 node = queue.pop(0) # Dequeue the node from
the front of the queue
 current_level.append(node.val) # Append the
node's value

Enqueue the left child if it exists.
if node.left:
 queue.append(node.left)

Enqueue the right child if it exists.
if node.right:
 queue.append(node.right)

If the current level's order is right-to-left,
reverse the list.
if not left_to_right:
 current_level.reverse()

Append the current level's values to the
result.
result.append(current_level)

Toggle the direction for the next level.
left_to_right = not left_to_right

return result
```

• LC.ca

Trees & BST

p.2587

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
'''
can also use list.insert()

>>> my_list = [2, 3, 4]
>>> my_list.insert(0, 1) # Insert 1 at the head of the
list
>>> print(my_list) # Output: [1, 2, 3, 4]

>>> a = deque([])
>>> a
deque([])
>>> a.append(1)
>>> a.append(2)

>>> a.append(3)
>>> a
deque([1, 2, 3])
>>>
>>> a.append(0, 555)
Traceback (most recent call last):
 File "<stdin>", line 1, in <module>
TypeError: deque.append() takes exactly one argument (2
given)
>>> a.insert(0, 555)
>>> a
```

Trees & BST

p.2588

```
deque([555, 1, 2, 3])
'''

from collections import deque

class Solution:
 def zigzagLevelOrder(self, root: Optional[TreeNode])
-> List[List[int]]:
 ans = []
 if root is None:
 return ans

 q = deque([root])
 ans = []
 left = True
 while q:
 t = []
 for _ in range(len(q)):
 node = q.popleft()
 t.append(node.val)
 if node.left:
 q.append(node.left)

 if node.right:
 q.append(node.right)
 ans.append(t if left else t[::-1])
 left = (not left)
 return ans

#####

Definition for a binary tree node.
class TreeNode(object):
```

Trees & BST

p.2589

```
def __init__(self, x):
self.val = x
self.left = None
self.right = None
from collections import deque

class Solution(object):
 def zigzagLevelOrder(self, root):
 """

:type root: TreeNode
:rtype: List[List[int]]
"""
 stack = deque([root])
 ans = []
 odd = True
 while stack:
 level = []
 for k in range(0, len(stack)):
 top = stack.popleft()

 if top is None:
 continue
 level.append(top.val)
 stack.append(top.left)
 stack.append(top.right)
 if level:
 if odd:
 ans.append(level)
 else:
 ans.append(level[::-1])
```

Trees & BST

p.2590

```
odd = not odd
return ans

#####

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left

self.right = right
class Solution:
 def zigzagLevelOrder(self, root: Optional[TreeNode])
-> List[List[int]]:
 ans = []
 if root is None:
 return ans
 q = deque([root])
 ans = []
 left = 1
 while q:
 t = []
 for _ in range(len(q)):
 node = q.popleft()
 t.append(node.val)
 if node.left:
 q.append(node.left)
 if node.right:
 q.append(node.right)
 ans.append(t if left else t[::-1])
 left ^= 1
```

Trees & BST

p.2591

```
return ans
```

Trees & BST

p.2592

## #105 Construct Binary Tree from Preorder and Inorder Traversal

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Tree · DFS

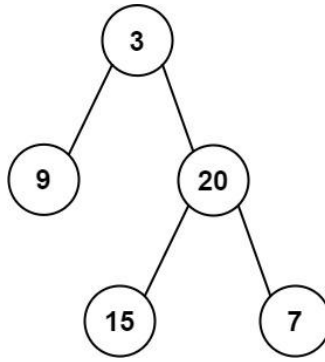
Lists: Grind 169

Time Complexity:  $O(n)$  since we process every node exactly once. Space Complexity:  $O(n)$  due to the recursion stack and the hash map used for inorder indices.

### Problem

Given two integer arrays preorder and inorder where preorder is the preorder traversal of a binary tree and inorder is the inorder traversal of the same tree, construct and return the binary tree.

Example 1:



Input: preorder = [3,9,20,15,7], inorder = [9,3,15,20,7]  
Output: [3,9,20,null,null,15,7]

Example 2:

Input: preorder = [-1], inorder = [-1]  
Output: [-1]

Constraints:

- $1 \leq \text{preorder.length} \leq 3000$
- $\text{inorder.length} == \text{preorder.length}$
- $-3000 \leq \text{preorder}[i], \text{inorder}[i] \leq 3000$

Trees &amp; BST

p.2593

Trees &amp; BST

p.2594

Trees &amp; BST

p.2595

```

root = TreeNode(rootVal)
root.left = build(preStart + 1, preStart +
leftSize,
inStart, rootInIndex - 1)
root.right = build(preStart + leftSize + 1,
preEnd, rootInIndex + 1, inEnd)
return root

return build(0, len(preorder) - 1, 0, len(inorder) -
1)

```

### • LeetCode

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
def buildTree(self, preorder: List[int], inorder:
List[int]) -> Optional[TreeNode]:
def dfs(i: int, j: int, n: int) ->
Optional[TreeNode]:
if n <= 0:

```

Trees &amp; BST

p.2596

```

return None
v = preorder[i]
k = d[v]
l = dfs(i + 1, j, k - j)
r = dfs(i + 1 + k - j, k + 1, n - k + j - 1)
return TreeNode(v, l, r)

d = {v: i for i, v in enumerate(inorder)}
return dfs(0, 0, len(preorder))

```

```

class Solution:
def getBinaryTrees(self, preOrder: List[int],
inOrder: List[int]) -> List[TreeNode]:
def dfs(i: int, j: int, n: int) ->
List[TreeNode]:
if n <= 0:
return [None]
v = preOrder[i]
ans = []
for k in d[v]:
if j <= k < j + n:
for l in dfs(i + 1, j, k - j):

```

```

for r in dfs(i + 1 + k - j, k + 1,
n - 1 - (k - j)):
ans.append(TreeNode(v, l, r))

return ans

d = defaultdict(list)
for i, x in enumerate(inOrder):
d[x].append(i)
return dfs(0, 0, len(preOrder))

```

### • SimplyLeet

Trees &amp; BST

p.2597

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

```

```

class Solution:
def buildTree(self, preorder: List[int], inorder:
List[int]) -> TreeNode:
Create a hashmap to store value->index for
inorder traversal

```

```

inorder_index = {value: idx for idx, value in
enumerate(inorder)}

Recursive helper function to build the tree
def helper(pre_start, in_left, in_right):
Base case: if there are no elements to
construct subtree
if in_left > in_right:
return None

Pick up pre_start element as a root
root_val = preorder[pre_start]

```

Trees &amp; BST

p.2598

```

root = TreeNode(root_val)

Get the inorder index of the root
in_index = inorder_index[root_val]

Number of nodes in left subtree
left_subtree_size = in_index - in_left

Recursively build the left subtree
root.left = helper(pre_start + 1, in_left,
in_index - 1)

Recursively build the right subtree
root.right = helper(pre_start +
left_subtree_size + 1, in_index + 1, in_right)

return root

Initiate recursion from the start of preorder
and full inorder range
return helper(0, 0, len(inorder) - 1)

```

### • LC.ca

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

best
class Solution:
def buildTree(self, preorder: List[int], inorder:
List[int]) -> Optional[TreeNode]:

if not preorder: # or, not inorder
return None
v = preorder[0]
i = inorder.index(v)
root = TreeNode(v)
root.left = self.buildTree(preorder[1: i+1],
inorder[:i])
root.right = self.buildTree(preorder[i+1:],
inorder[i+1:])
return root

```

```

inorder.index(preorder_val)
class Solution:
def buildTree(self, preorder: List[int], inorder:
List[int]) -> Optional[TreeNode]:
def dfs(pleft, pright, ileft, iright):
if pleft > pright or ileft > iright:
return None
k = inorder.index(preorder[pleft])
root = TreeNode(preorder[pleft])
root.left = dfs(pleft + 1, pleft + (k -
ileft), ileft, k - 1)
root.right = dfs(pleft + 1 + (k - ileft),
pright, k + 1, iright)

return root

n = len(preorder)
return dfs(0, n - 1, 0, n - 1)

build dict for val => index
class Solution:
def buildTree(self, preorder: List[int], inorder:
List[int]) -> Optional[TreeNode]:
def dfs(pleft, pright, ileft, iright):

```

Trees &amp; BST

p.2599

Trees &amp; BST

p.2600

Trees &amp; BST

p.2601

```

 if pleft > pright or ileft > iright:
 return None
 v = preorder[pleft]
 k = d[v]
 root = TreeNode(v)
 root.left = dfs(pleft + 1, pleft + (k - ileft), ileft, k - 1)
 root.right = dfs(pleft + 1 + (k - ileft), pright, k + 1, iright)
 return root

 d = {v: i for i, v in enumerate(inorder)}

 n = len(preorder)
 return dfs(0, n - 1, 0, n - 1)

#####

...
search for index, also use .index(val)
>>> a = [1,2,3,4,5]
>>> a.index(3)
2

```

Trees & BST

p.2602

```

...
class Solution(object):
 def buildTree(self, preorder, inorder):
 """
 :type preorder: List[int]
 :type inorder: List[int]
 :rtype: TreeNode
 """

 self.preindex = 0

 ind = {v: i for i, v in enumerate(inorder)}
 head = self.dc(0, len(preorder) - 1, preorder,
inorder, ind)
 return head

 def dc(self, start, end, preorder, inorder, ind):
 if start >= end:
 mid = ind[preorder[self.preindex]]
 self.preindex += 1
 root = TreeNode(inorder[mid])
 root.left = self.dc(start, mid - 1, preorder,
inorder, ind)

```

Trees & BST

p.2603

```

 root.right = self.dc(mid + 1, end, preorder,
inorder, ind)
 return root

#####

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left

self.right = right
class Solution:
 def buildTree(self, preorder: List[int], inorder:
List[int]) -> Optional[TreeNode]:
 def dfs(i: int, j: int, n: int):
 if n <= 0:
 return None
 v = preorder[i]
 k = d[v]
 l = dfs(i + 1, j, k - j)
 r = dfs(i + 1 + k - j, k + 1, n - k + j - 1)

 return TreeNode(v, l, r)

 d = {v: i for i, v in enumerate(inorder)}
 return dfs(0, 0, len(preorder))

```

Trees & BST

p.2604

Trees & BST

### #199 Binary Tree Right Side View

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BFS

Lists: Grind 169

Time Complexity: O(n) – we visit every node in the tree.  
Space Complexity: O(n) – in the worst case, the queue (or call stack for DFS) may hold all nodes of the tree.

#### Problem

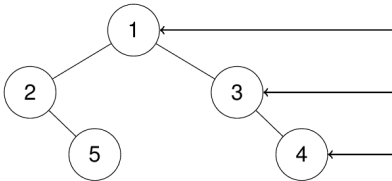
Given the root of a binary tree, imagine yourself standing on the right side of it, return the values of the nodes you can see ordered from top to bottom.

#### Example 1:

Input: root = [1,2,3,null,5,null,4]

Output: [1,3,4]

Explanation:

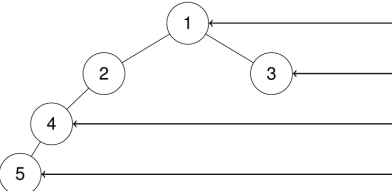


#### Example 2:

Input: root = [1,2,3,4,null,null,null,5]

Output: [1,3,4,5]

Explanation:



Trees & BST

p.2605

#### Example 3:

Input: root = [1,null,3]

Output: [1,3]

#### Example 4:

Input: root = []

Output: []

#### Constraints:

- The number of nodes in the tree is in the range [0, 100].
- -100 <= Node.val <= 100

#### Brute Force (Python)

```

Brute Force Approach
class Solution:
 def rightSideView(self, root):
 # Brute Force O(n) – BFS, take last node of each
level
 from collections import deque
 if not root: return []
 result, queue = [], deque([root])
 while queue:
 level_size = len(queue)
 for i in range(level_size):

 node = queue.popleft()
 if i == level_size - 1:
 result.append(node.val)
 if node.left: queue.append(node.left)
 if node.right: queue.append(node.right)
 return result

```

Trees & BST

p.2607

### Community Solutions (Python)

#### • WalkCC

```

class Solution:
 def rightSideView(self, root: TreeNode | None) ->
list[int]:
 if not root:
 return []

 ans = []
 q = collections.deque([root])

 while q:
 size = len(q)

 for i in range(size):
 root = q.popleft()
 if i == size - 1:
 ans.append(root.val)
 if root.left:
 q.append(root.left)
 if root.right:
 q.append(root.right)

 return ans

```

#### • LeetDoocs

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def rightSideView(self, root: Optional[TreeNode]) ->
List[int]:
 ans = []
 if root is None:

 return ans
 q = deque([root])
 while q:
 ans.append(q[0].val)
 for _ in range(len(q)):
 node = q.popleft()
 if node.right:
 q.append(node.right)
 if node.left:
 q.append(node.left)

 return ans

```

#### • SimplyLeet

```

Python solution using BFS
from collections import deque

class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # Node value
 self.left = left # Left child
 self.right = right # Right child

def rightSideView(root):

 # If the tree is empty, return an empty list
 if not root:
 return []

 right_view = [] # List to store right side view
 queue = deque([root]) # Initialize queue with the
root node

 while queue:
 level_size = len(queue) # Number of nodes in
current level
 for i in range(level_size):

```

Trees & BST

p.2608

Trees & BST

p.2609

Trees & BST

p.2610



```

node = queue.popleft() # Get the front node
from the queue
if it's the last node in the current level,
add it.
if i == level_size - 1:
 right_view.append(node.val)
add left child if exists
if node.left:
 queue.append(node.left)
add right child if exists
if node.right:
 queue.append(node.right)

```

return right\_view

#### • LC.ca

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def rightSideView(self, root: Optional[TreeNode]) ->
List[int]:
 ans = []
 if root is None:

```

```

return ans
q = deque([root])
while q:
 ans.append(q[-1].val) # add last node of
previous level traversal results
 for _ in range(len(q)):
 node = q.popleft()
 if node.left:
 q.append(node.left)
 if node.right:
 q.append(node.right)

```

return ans

#####

```

Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
self.val = x
self.left = None
self.right = None

```

from collections import deque

```

class Solution(object):
 def rightSideView(self, root):
 """"
 :type root: TreeNode
 :rtype: List[int]
 """

```

```

def dfs(root, h):
 if root:
 if h == len(ans):
 ans.append(root.val)
 # pre-order, all the way to the right
 dfs(root.right, h + 1)
 dfs(root.left, h + 1)

```

```

ans = []
dfs(root, 0)

```

return ans

Trees & BST

p.2611

Trees & BST

p.2612

Trees & BST

p.2613

Trees & BST

## #230 Kth Smallest Element in a BST MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BST

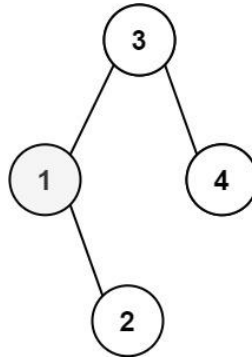
Lists: Grind 169

Time Complexity:  $O(n)$  in the worst-case scenario (when  $k \approx n$ ), but on average the traversal may stop early. Space Complexity:  $O(h)$  where  $h$  is the height of the tree ( $O(n)$  in the worst-case for a skewed tree)

### Problem

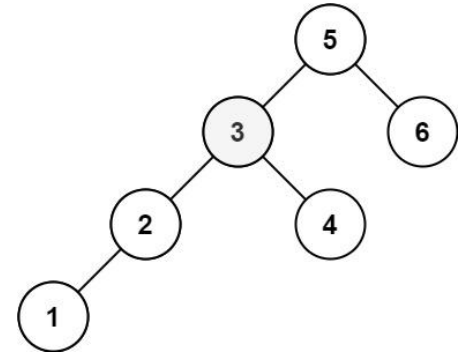
Given the root of a binary search tree, and an integer  $k$ , return the  $k$ th smallest value (1-indexed) of all the values of the nodes in the tree.

Example 1:



Input: root = [3,1,4,null,2], k = 1  
Output: 1

Example 2:



Input: root = [5,3,6,2,4,null,null,1], k = 3  
Output: 3

### Constraints:

- The number of nodes in the tree is  $n$ .
- $1 \leq k \leq n \leq 104$
- $0 \leq \text{Node.val} \leq 104$

Follow up: If the BST is modified often (i.e., we can do insert and delete operations) and you need to find the  $k$ th smallest frequently, how would you optimize?

Trees & BST

p.2614

Trees & BST

p.2615

Trees & BST

p.2616

## Community Solutions (Python)

### • WalkCC

```

class Solution:
 def kthSmallest(self, root: TreeNode | None, k: int) ->
int:
 def countNodes(root: TreeNode | None) -> int:
 if not root:
 return 0
 return 1 + countNodes(root.left) +
countNodes(root.right)

 leftCount = countNodes(root.left)
 if leftCount == k - 1:
 return root.val
 if leftCount >= k:
 return self.kthSmallest(root.left, k)
 return self.kthSmallest(root.right, k - 1 -
leftCount) # leftCount < k

```

### • LeetCode

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def kthSmallest(self, root: Optional[TreeNode], k:
int) -> int:
 stk = []
 while root or stk:

```

```

 if root:
 stk.append(root)
 root = root.left
 else:
 root = stk.pop()
 k -= 1
 if k == 0:
 return root.val
 root = root.right

```

### • SimplyLeet

```

Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
 def kthSmallest(self, root: TreeNode, k: int) -> int:
 # Initialize counter and result

```

```

self.k = k
self.ans = None

In-order traversal: left -> node -> right
def inorder(node):
 if not node:
 return
 # Traverse left subtree
 inorder(node.left)
 # Process current node

 self.k -= 1
 if self.k == 0:
 self.ans = node.val
 return
 # Traverse right subtree if kth element not
 found
 inorder(node.right)

 inorder(root)
 return self.ans

```

### • LC.ca

Trees & BST

p.2617

Trees & BST

p.2618

Trees & BST

p.2619

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def kthSmallest(self, root: Optional[TreeNode], k:
int) -> int:
 stk = []
 while root or stk:
```

```
 if root:
 stk.append(root)
 root = root.left
 else:
 root = stk.pop()
 k -= 1
 if k == 0:
 return root.val
 root = root.right
```

```
#####

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution_followUp:
 def kthSmallest(self, root: TreeNode, k: int) -> int:
```

Trees & BST

p.2620

```
 node = self.build(root)
 return self.dfs(node, k)

class MyTreeNode:
 def __init__(self, x):
 self.val = x
 self.count = 1 # default 1
 self.left = None
 self.right = None

 def build(self, root: TreeNode) -> MyTreeNode:
 if not root:
 return None
 node = self.MyTreeNode(root.val) # default count
 already is 1
 node.left = self.build(root.left)
 node.right = self.build(root.right)
 if node.left:
 node.count += node.left.count
 if node.right:
 node.count += node.right.count

 return node

 def dfs(self, node: MyTreeNode, k: int) -> int:
 if node.left:
 cnt = node.left.count
 if k < cnt + 1:
 return self.dfs(node.left, k)
 elif k > cnt + 1:
 return self.dfs(node.right, k - 1 - cnt)
 else: # k == cnt+1
```

Trees & BST

p.2621

```
 return node.val
 else:
 if k == 1: # cannot move to beginning of
dfs()
 return node.val
 return self.dfs(node.right, k - 1)
```

Trees & BST

p.2622

Trees & BST

## #235 Lowest Common Ancestor of a Binary Search Tree

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS · BST

Lists: Grind 169

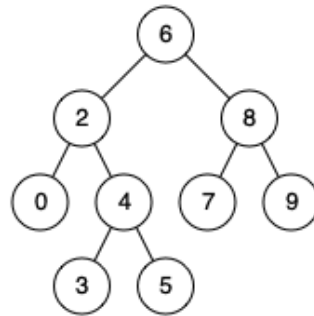
Time Complexity:  $O(h)$ , where  $h$  is the height of the tree ( $O(\log n)$  for balanced trees and  $O(n)$  in the worst-case skewed tree). Space Complexity:  $O(1)$  with an iterative approach ( $O(h)$  if using recursion)

### Problem

Given a binary search tree (BST), find the lowest common ancestor (LCA) node of two given nodes in the BST.

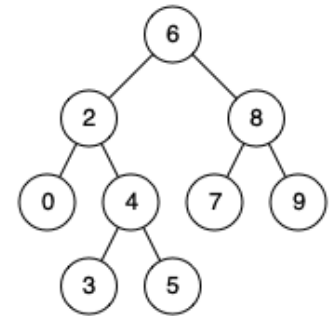
According to the definition of LCA on Wikipedia: "The lowest common ancestor is defined between two nodes  $p$  and  $q$  as the lowest node in  $T$  that has both  $p$  and  $q$  as descendants (where we allow a node to be a descendant of itself)."

Example 1:



```
Input: root = [6,2,8,0,4,7,9,null,null,3,5], p = 2, q = 8
Output: 6
Explanation: The LCA of nodes 2 and 8 is 6.
```

Example 2:



```
Input: root = [6,2,8,0,4,7,9,null,null,3,5], p = 2, q = 4
Output: 2
Explanation: The LCA of nodes 2 and 4 is 2, since a node can be a descendant of itself according to the LCA definition.
```

Example 3:

```
Input: root = [2,1], p = 2, q = 1
Output: 2
```

Constraints:

Trees & BST

p.2623

Trees & BST

p.2624

Trees & BST

p.2625

- The number of nodes in the tree is in the range [2, 105].
- $-109 \leq \text{Node.val} \leq 109$
- All  $\text{Node.val}$  are unique.
- $p \neq q$
- $p$  and  $q$  will exist in the BST.

### Brute Force (Python)

```
Brute Force Approach
class Solution:
 def lowestCommonAncestor(self, root, p, q):
 # Brute Force $O(n)$ - find root-to-node paths,
 find last common node
 def path_to(node, target):
 path = []
 while node:
 path.append(node)
 if target.val < node.val: node =
node.left
 elif target.val > node.val: node =
node.right
 else: break
 return path
 p_path = path_to(root, p)
 q_path = path_to(root, q)
 p_set = {n.val for n in p_path}
 for node in reversed(q_path):
 if node.val in p_set:
 return node
```

Trees & BST

p.2626

### Community Solutions (Python)

#### • WalkCC

```
class Solution:
 def lowestCommonAncestor(
 self,
 root: 'TreeNode',
 p: 'TreeNode',
 q: 'TreeNode',
) -> 'TreeNode':
 if root.val > max(p.val, q.val):
 return self.lowestCommonAncestor(root.left, p, q)
 if root.val < min(p.val, q.val):
 return self.lowestCommonAncestor(root.right, p, q)
 return root
```

#### • LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, x):
self.val = x
self.left = None
self.right = None

class Solution:
 def lowestCommonAncestor(
```

Trees & BST

p.2627

Trees & BST

p.2628

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, x):
self.val = x
self.left = None
self.right = None

class Solution:
 def lowestCommonAncestor(
 self, root: 'TreeNode', p: 'TreeNode', q:
 'TreeNode'

) -> 'TreeNode':
 while 1:
 if root.val < min(p.val, q.val): # no =, so
root is p or q is in else block
 root = root.right
 elif root.val > max(p.val, q.val):
 root = root.left
 else:
 return root

#####
```

Trees & BST

p.2629

```
class Solution:
 def lowestCommonAncestor(self, root, p, q):
 if root is None:
 return root

 if root.val > max(p.val, q.val):
 return self.lowestCommonAncestor(root.left,
p, q)
 elif root.val < min(p.val, q.val):
 return self.lowestCommonAncestor(root.right,
p, q)
 else:
 return root

#####

class Solution(object):
 def lowestCommonAncestor(self, root, p, q):
 """"
 :type root: TreeNode
 :type p: TreeNode
```

Trees & BST

p.2630

```
:type q: TreeNode
:rtype: TreeNode
"""
a, b = sorted([p.val, q.val])
while not a <= root.val <= b:
 if a > root.val:
 root = root.right
 else:
 root = root.left
return root
```

Trees & BST

p.2631

Trees & BST

## #236 Lowest Common Ancestor of a Binary Tree

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS

Lists: Grind 169

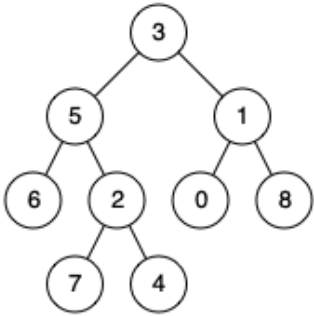
Time Complexity:  $O(n)$ , where  $n$  is the number of nodes in the tree since each node is visited once. Space Complexity:  $O(h)$ , where  $h$  is the height of the tree due to the recursion stack ( $O(n)$  in the wor

### Problem

Given a binary tree, find the lowest common ancestor (LCA) of two given nodes in the tree.

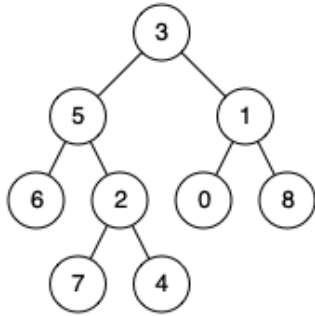
According to the definition of LCA on Wikipedia: "The lowest common ancestor is defined between two nodes  $p$  and  $q$  as the lowest node in  $T$  that has both  $p$  and  $q$  as descendants (where we allow a node to be a descendant of itself)."

Example 1:



Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 1  
Output: 3  
Explanation: The LCA of nodes 5 and 1 is 3.

Example 2:



Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 4  
Output: 5  
Explanation: The LCA of nodes 5 and 4 is 5, since a node can be a descendant of itself according to the LCA definition.

Example 3:

Input: root = [1,2], p = 1, q = 2  
Output: 1

Constraints:

Trees & BST

p.2632

Trees & BST

p.2633

Trees & BST

p.2634

- The number of nodes in the tree is in the range [2, 105].
- $-109 \leq \text{Node.val} \leq 109$
- All  $\text{Node.val}$  are unique.
- $p \neq q$
- $p$  and  $q$  will exist in the tree.

### Community Solutions (Python)

• WalkCC

```
class Solution:
 def lowestCommonAncestor(
 self,
 root: 'TreeNode',
 p: 'TreeNode',
 q: 'TreeNode',
) -> 'TreeNode':
 if not root or root == p or root == q:
 return root
 left = self.lowestCommonAncestor(root.left, p, q)

 right = self.lowestCommonAncestor(root.right, p, q)
 if left and right:
 return root
 return left or right
```

Trees & BST

p.2635

```
class Solution:
 def lowestCommonAncestor(
 self,
 root: TreeNode | None,
 p: TreeNode | None,
 q: TreeNode | None,
) -> TreeNode | None:
 q_ = collections.deque([root])
 parent = {root: None}
 ancestors = set() # p's ancestors
```

```
Iterate until we find both p and q.
while p not in parent or q not in parent:
 root = q_.popleft()
 if root.left:
 parent[root.left] = root
 q_.append(root.left)
 if root.right:
 parent[root.right] = root
 q_.append(root.right)
```

```
Insert all the p's ancestors.
while p:
 ancestors.add(p)
 p = parent[p] # 'p' becomes None in the end.

Go up from q until we meet any of p's ancestors.
while q not in ancestors:
 q = parent[q]
```

return q

Trees & BST

• LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, x):
self.val = x
self.left = None
self.right = None
```

```
class Solution:
 def lowestCommonAncestor(
 self, root: "TreeNode", p: "TreeNode", q:
 "TreeNode"
) -> "TreeNode":
 if root in (None, p, q):
 return root
 left = self.lowestCommonAncestor(root.left, p, q)
 right = self.lowestCommonAncestor(root.right, p,
q)
 return root if left and right else (left or
right)
```

• SimplyLeet

Trees & BST

p.2637

Trees &amp; BST p.2638

Trees &amp; BST p.2639

Trees &amp; BST p.2640

Trees & BST p.2641Trees & BST p.2642

Trees &amp; BST p.2644

Trees &amp; BST p.2645

Trees &amp; BST p.2646

```

rightUnival = isUnival(node.right)

Check if left child exists and its value
does not match current node's value
if node.left and node.left.val != node.val:
 leftUnival = False
Check if right child exists and its value
does not match current node's value
if node.right and node.right.val != node.val:
 rightUnival = False

If either left or right subtree is not
univalued, then the current subtree is not univalued

if leftUnival and rightUnival:
 # Increment the count if the subtree
 rooted at the current node is univalued
 self.count += 1
 return True
else:
 return False

Start recursion from root
isUnival(root)
return self.count

```

```

Example usage:
Construct the binary tree: [5,1,5,5,5,null,5]
root = TreeNode(5)
root.left = TreeNode(1, TreeNode(5), TreeNode(5))
root.right = TreeNode(5, None, TreeNode(5))
print(Solution().countUnivalSubtrees(root)) # Output: 4

```

Trees & BST

p.2647

```

• LC.ca

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

class Solution:
 def countUnivalSubtrees(self, root: TreeNode) -> int:
 count = 0

 def is_unival(node):
 nonlocal count
 if not node:
 return True

 left_unival = is_unival(node.left)
 right_unival = is_unival(node.right)

 if left_unival and right_unival:

```

Trees & BST

p.2648

```

 if node.left and node.val !=
node.left.val:
 return False
 if node.right and node.val !=
node.right.val:
 return False
 count += 1
 return True

 return False

is_unival(root)

return count

class Solution: # return 2 values
 def countUnivalSubtrees(self, root: TreeNode) -> int:
 def is_unival(node):
 if not node:
 return True, 0

 left_unival, left_count =
is_unival(node.left)

```

Trees & BST

p.2649

```

right_unival, right_count =
is_unival(node.right)

if left_unival and right_unival:
 if node.left and node.val !=
node.left.val:
 return False, 0
 if node.right and node.val !=
node.right.val:
 return False, 0
 return True, left_count + right_count + 1

return False, 0

_, count = is_unival(root)
return count

```

Trees & BST

p.2650

Trees & BST

#285 Inorder Successor in BST MED

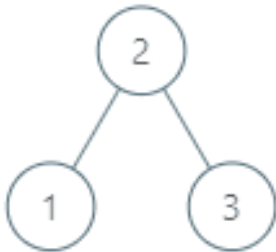
LeetDoocs · SimplyLeet · WalkCC · LeetCode  
Tree · DFS · BST  
Lists: Grind 169

Time Complexity:  $O(h)$ , where  $h$  is the height of the tree.  
In the worst-case (skewed tree), it can be  $O(n)$ . Space Complexity:  $O(1)$  as only a constant amount of extra space is used.

**Problem**  
Given the root of a binary search tree and a node  $p$  in it, return the in-order successor of that node in the BST. If the given node has no in-order successor in the tree, return null.  
The successor of a node  $p$  is the node with the smallest key greater than  $p.val$ .  
Example 1:

Trees & BST

p.2651



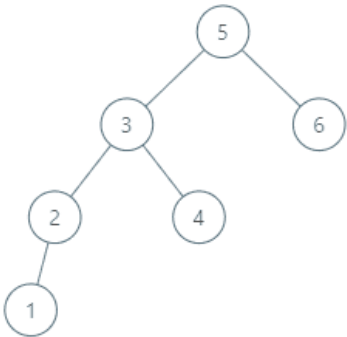
```

Input: root = [2,1,3], p = 1
Output: 2
Explanation: 1's in-order successor node is 2. Note that both p and the return value is of TreeNode type.
Example 2:

```

Trees & BST

p.2652



```

Input: root = [5,3,6,2,4,null,null,1], p = 6
Output: null
Explanation: There is no in-order successor of the current node, so the answer is null.

```

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- $-105 \leq \text{Node.val} \leq 105$
- All Nodes will have unique values.

Trees & BST

p.2653

Community Solutions (Python)

• WalkCC

```

class Solution:
 def inorderSuccessor(
 self,
 root: TreeNode | None,
 p: TreeNode | None,
) -> TreeNode | None:
 if not root:
 return None
 if root.val <= p.val:
 return self.inorderSuccessor(root.right, p)

 return self.inorderSuccessor(root.left, p) or root

```

• LeetDoocs

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, x):
self.val = x
self.left = None
self.right = None

class Solution:
 def inorderSuccessor(self, root: TreeNode, p:
TreeNode) -> Optional[TreeNode]:

```

Trees & BST

p.2654

```

ans = None
while root:
 if root.val > p.val:
 ans = root
 root = root.left
 else:
 root = root.right
return ans

• SimplyLeet

Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
 def inorderSuccessor(self, root: TreeNode, p:
TreeNode) -> TreeNode:
 # If p has a right subtree, the successor is the
left-most node in that subtree.

```

Trees & BST

p.2655

```
if p.right:
 successor = p.right
 while successor.left:
 successor = successor.left
 return successor

Otherwise, start from the root and keep track
of a potential successor.
successor = None
current = root
while current:

 if p.val < current.val:
 # current is a candidate, so record it
 and move left to find a smaller candidate
 successor = current
 current = current.left
 else:
 # If p.val is greater or equal, go right.
 current = current.right
 return successor
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, x):
self.val = x
self.left = None
self.right = None

class Solution:
 def inorderSuccessor(self, root: TreeNode, p:
TreeNode) -> Optional[TreeNode]:

 ans = None
 while root:
 if root.val > p.val:
 ans = root
 root = root.left
 else:
 root = root.right
 return ans
```

Trees & BST

#298 Binary Tree Longest Consecutive Sequence

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · DFS

Lists: Premium 98

Time Complexity: O(N), where N is the number of nodes in the tree, since each node is visited once. Space Complexity: O(H), where H is the height of the tree. In the worst-case of a skewed tree, the r

Problem

Given the root of a binary tree, return the length of the longest consecutive sequence path.

A consecutive sequence path is a path where the values increase by one along the path.

Note that the path can start at any node in the tree, and you cannot go from a node to its parent in the path.

Example 1:

Trees & BST

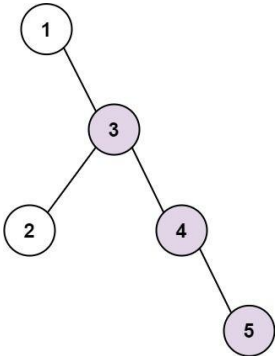
p.2656

Trees & BST

p.2657

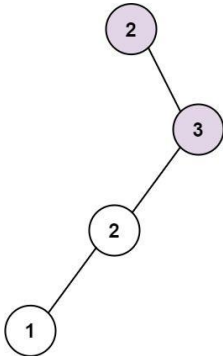
Trees & BST

p.2658



Input: root = [1,null,3,2,4,null,null,5]  
Output: 3  
Explanation: Longest consecutive sequence path is 3-4-5, so return 3.

Example 2:



Input: root = [2,null,3,2,null,1]  
Output: 2  
Explanation: Longest consecutive sequence path is 2-3, not 3-2-1, so return 2.

- Constraints:
- The number of nodes in the tree is in the range [1, 3 \* 104].
  - -3 \* 104 <= Node.val <= 3 \* 104

Community Solutions (Python)

```
• WalkCC

class Solution:
 def longestConsecutive(self, root: TreeNode | None) -> int:
 if not root:
 return 0

 def dfs(root: TreeNode | None, target: int, length: int, maxLength: int) -> int:
 if not root:
 return maxLength
 if root.val == target:
 length += 1

 maxLength = max(maxLength, length)
 else:
 length = 1
 return max(dfs(root.left, root.val + 1, length, maxLength),
 dfs(root.right, root.val + 1, length, maxLength))

 return dfs(root, root.val, 0, 0)

• LeetDoocs
```

Trees & BST

p.2659

Trees & BST

p.2660

Trees & BST

p.2661

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

class Solution:
 def longestConsecutive(self, root:
Optional[TreeNode]) -> int:
 def dfs(root: Optional[TreeNode]) -> int:
 if root is None:
 return 0
 l = dfs(root.left) + 1
 r = dfs(root.right) + 1
 if root.left and root.left.val - root.val != 1:
 l = 1
 if root.right and root.right.val - root.val != 1:
 r = 1
 t = max(l, r)
 nonlocal ans
 ans = max(ans, t)

 return t

 ans = 0
 dfs(root)
 return ans
```

• SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
 def longestConsecutive(self, root: TreeNode) -> int:
 # Initialize a variable to keep track of the maximum sequence length.

 self.max_length = 0

 # Helper function to perform DFS traversal.
 def dfs(node, parent_val, current_length):
 if not node:
 return
 # Check if the current node's value is consecutive.
 if node.val == parent_val + 1:
 current_length += 1
 else:
 current_length = 1
```

```
 current_length = 1
 # Update the maximum sequence length found so far.
 self.max_length = max(self.max_length, current_length)
 # Recursively traverse the left and right children.
 dfs(node.left, node.val, current_length)
 dfs(node.right, node.val, current_length)

 # Start DFS with an initial condition that forces the first node to count.
 dfs(root, root.val - 1, 0)
 return self.max_length

• LC.ca

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

dfs
class Solution:
 def longestConsecutive(self, root: TreeNode) -> int:
```

Trees & BST

p.2662

Trees & BST

p.2663

Trees & BST

p.2664



```

maxLen = 0

curLen: consecutive length until parent node
def dfs(root: TreeNode, lastVal: int, curLen:
int) -> None:
 nonlocal maxLen
 if not root:
 return

 curLen = (curLen + 1) if (not lastVal) and
root.val == lastVal + 1 else 1
 maxLen = max(maxLen, curLen)

 dfs(root.left, root.val, curLen)
 dfs(root.right, root.val, curLen)

#####

Definition for a binary tree node.

```

Trees & BST

p.2665

```

class TreeNode(object):
def __init__(self, x):
self.val = x
self.left = None
self.right = None

from collections import deque

iterative, bfs
class Solution:

 def longestConsecutive(self, root: TreeNode) -> int:
 if not root:
 return 0

 max_length = 0
 # Queue elements are tuples: (current node,
length of consecutive sequence)
 queue = deque([(root, 1)])

 while queue:
 current, length = queue.popleft()

```

Trees & BST

p.2666

```

max_length = max(max_length, length)

 for neighbor in [current.left,
current.right]:
 if neighbor:
 # Check if neighbor is consecutive
 if neighbor.val == current.val + 1:
 queue.append((neighbor, length +
1))

 else:
 queue.append((neighbor, 1))

return max_length

#####

class Solution:
 def longestConsecutive(self, root:
Optional[TreeNode]) -> int:
 def dfs(root: Optional[TreeNode]) -> int:
 if root is None:
 return 0
 l = dfs(root.left) + 1

```

Trees & BST

p.2667

```

r = dfs(root.right) + 1
if root.left and root.left.val - root.val !=
1:
 l = 1
 if root.right and root.right.val - root.val
!= 1:
 r = 1
 t = max(l, r)
 nonlocal ans
 ans = max(ans, t)
 return t

ans = 0
dfs(root)
return ans

```

Trees & BST

p.2668

Trees & BST

### #314 Binary Tree Vertical Order Traversal

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Hash Table · Tree · BFS · Sorting

Lists: Premium 98

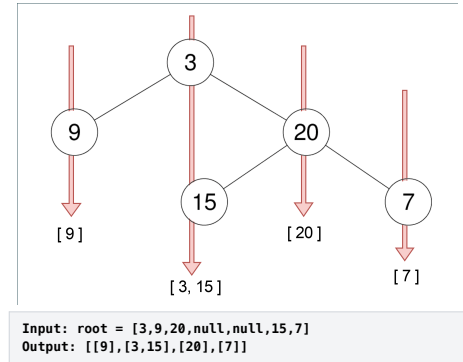
Time Complexity:  $O(n \log n)$  in the worst-case due to sorting the column indices ( $n$  nodes in the worst-case if each node is on a unique column). Space Complexity:  $O(n)$  for storing the node information

Problem

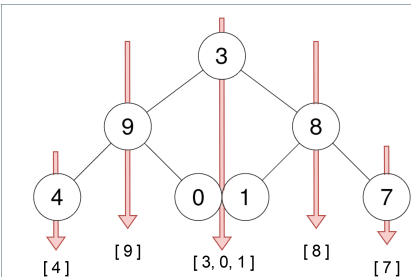
Given the root of a binary tree, return the vertical order traversal of its nodes' values. (i.e., from top to bottom, column by column).

If two nodes are in the same row and column, the order should be from left to right.

Example 1:



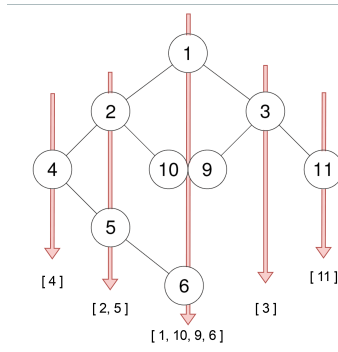
Example 2:



Example 3:

Trees & BST

p.2671



Constraints:

- The number of nodes in the tree is in the range  $[0, 100]$ .
- $-100 \leq \text{Node.val} \leq 100$

Brute Force (Python)

Trees & BST

p.2672

```

Brute Force Approach
class Solution:
 def verticalOrder(self, root: TreeNode):
 # Brute Force - collect inorder, then process
sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else 0

```

Community Solutions (Python)

• WalkCC

```

class Solution:
 def verticalOrder(self, root: TreeNode | None) ->
list[list[int]]:
 if not root:
 return []

 range_ = [0] * 2

 def getRange(root: TreeNode | None, x: int) -> None:
 if not root:
 return

```

Trees & BST

p.2673

```

range_0] = min(range_0], x)
range_1] = max(range_1], x)

getRange(root.left, x - 1)
getRange(root.right, x + 1)

getRange(root, 0) # Get the leftmost and the
rightmost x index.

ans = [[] for _ in range(range_1] - range_0] + 1)]

q = collections.deque([(root, -range_0)]) #
(TreeNode, x)

while q:
 node, x = q.popleft()
 ans[x].append(node.val)
 if node.left:
 q.append((node.left, x - 1))
 if node.right:
 q.append((node.right, x + 1))

return ans

```

• LeetCode

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def verticalOrder(self, root: Optional[TreeNode]) ->
List[List[int]]:
 def dfs(root, depth, offset):
 if root is None:
 return
 d[offset].append((depth, root.val))
 dfs(root.left, depth + 1, offset - 1)
 dfs(root.right, depth + 1, offset + 1)

 d = defaultdict(list)
 dfs(root, 0, 0)
 ans = []
 for _, v in sorted(d.items()):
 v.sort(key=lambda x: x[0])

 ans.append([x[1] for x in v])
 return ans

```

• SimplyLeet

```

from collections import deque, defaultdict

Definition for a binary tree node.
class TreeNode:
 def __init__(self, x):
 self.val = x
 self.left = None
 self.right = None

class Solution:

 def verticalOrder(self, root: TreeNode):
 # Return empty list if the tree is empty
 if not root:
 return []

 # Dictionary to hold nodes for each column index
 col_table = defaultdict(list)
 # Use deque for efficient FIFO queue for BFS;
 store tuple (node, column)
 queue = deque([(root, 0)])

```

Trees & BST

p.2674

Trees & BST

p.2675

Trees & BST

p.2676

```

while queue:
 node, col = queue.popleft()
 # Append current node value to the list of
 its column
 col_table[col].append(node.val)
 # If there is a left child, assign column
 col-1
 if node.left:
 queue.append((node.left, col - 1))
 # If there is a right child, assign column
 col+1
 if node.right:
 queue.append((node.right, col + 1))

```

```

Extract the columns in sorted order to build
the final result
sorted_cols = sorted(col_table.keys())
result = [col_table[col] for col in sorted_cols]
return result

```

• LC.ca

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
...
>>> d = {1:'a', -1:'b', 10:'c', -100:'d'}
>>>

```

```

>>> d.items()
{1: 'a', 10: 'c', -100: 'd', -1: 'b'}

>>> ds = sorted(d.items())
>>> ds
[(-100, 'd'), (-1, 'b'), (1, 'a'), (10, 'c')]

>>> sorted(d.items(), key=lambda l: l[1])
[(1, 'a'), (-1, 'b'), (10, 'c'), (-100, 'd')]
>>> dict(sorted(d.items(), key=lambda l: l[1]))

{1: 'a', 10: 'c', -100: 'd', -1: 'b'}

>>> [v for i,v in sorted(d.items(), key=lambda l: l[0])]
['d', 'b', 'a', 'c']
'''

class Solution:
 def verticalOrder(self, root: Optional[TreeNode]) ->
List[List[int]]:
 if root is None:
 return []

 q = deque([(root, 0)])
 d = defaultdict(list)
 while q:
 for _ in range(len(q)):
 root, offset = q.popleft()
 d[offset].append(root.val)
 if root.left:
 q.append((root.left, offset - 1))
 if root.right:
 q.append((root.right, offset + 1))

```

```

return [v for _, v in sorted(d.items())]

#####

from sortedcontainers import SortedDict
from typing import List
from collections import deque

similar to above, but use existing data structure
class Solution:

 def verticalOrder(self, root: TreeNode) ->
List[List[int]]:
 if not root:
 return []

 # Use SortedDict to automatically sort by column
index
 sorted_dict = SortedDict()

 # Queue for breadth-first search; stores pairs of
(node, column_index)
 queue = deque([(0, root)])

```

Trees & BST

p.2677

Trees & BST

p.2678

Trees & BST

p.2679

```

while queue:
 node, column = queue.popleft()

 if column not in sorted_dict:
 sorted_dict[column] = [node.val]
 else:
 sorted_dict[column].append(node.val)

 # Add left and right children to the queue
 if node.left:
 queue.append((node.left, column - 1))
 if node.right:
 queue.append((node.right, column + 1))

Return the values from sorted_dict as a list of
lists
return list(sorted_dict.values())

#####

another SortedDict, but different handling on None
node, before enqueue vs after enqueue

```

```

class Solution:
 def verticalOrder(self, root: TreeNode):
 if not root:
 return []

 # Initialize a SortedDict to maintain columns in
sorted order.
 column_table = SortedDict()
 queue = deque([(root, 0)]) # (node, column_index)

 while queue:

 node, column = queue.popleft()

 if node is not None:
 if column not in column_table:
 column_table[column] = []
 column_table[column].append(node.val)

 # Add left and right children to the
queue.
 queue.append((node.left, column - 1))
 queue.append((node.right, column + 1))

```

```

Extract the values from the sorted dictionary
and return them.
return list(column_table.values())

#####

from collections import defaultdict

class Solution(object):
 def verticalOrder(self, root):
 """
 :type root: TreeNode
 :rtype: List[List[int]]
 """

 def dfs(p, i, j, res): # i -> depth, j ->
vertical-shift
 if p:
 res[j].append((p.val, i))

 self.leftMost = min(j, self.leftMost)
 dfs(p.left, i + 1, j - 1, res)
 dfs(p.right, i + 1, j + 1, res)

 self.leftMost = float("inf")
 ans = []
 res = defaultdict(list)
 dfs(root, 0, 0, res)
 i = self.leftMost
 while True:

```

Trees & BST

p.2680

Trees & BST

p.2681

Trees & BST

p.2682

```
if not res[i]:
 break
ans.append([item[0] for item in sorted(res[i],
key=lambda a: a[1])])
i += 1
return ans
```

Trees & BST

p.2683

Trees & BST

### #333 Largest BST Subtree

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Dynamic Programming · Tree · DFS · BST

Lists: Premium 98

Time Complexity:  $O(n)$  (each node is visited once) Space Complexity:  $O(h)$ , where  $h$  is the height of the tree (due to recursion stack)

#### Problem

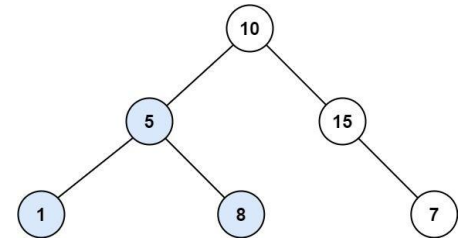
Given the root of a binary tree, find the largest subtree, which is also a Binary Search Tree (BST), where the largest means subtree has the largest number of nodes.

A Binary Search Tree (BST) is a tree in which all the nodes follow the below-mentioned properties:

- The left subtree values are less than the value of their parent (root) node's value.
- The right subtree values are greater than the value of their parent (root) node's value.

Note: A subtree must include all of its descendants.

Example 1:



Input: root = [10,5,15,1,8,null,7]

Output: 3

Explanation: The Largest BST Subtree in this case is the highlighted one. The return value is the subtree's size, which is 3.

Example 2:

Input: root =

[4,2,7,2,3,5,null,2,null,null,null,null,1]

Output: 2

Constraints:

- The number of nodes in the tree is in the range [0, 104].
- $-104 \leq \text{Node.val} \leq 104$

Follow up: Can you figure out ways to solve it with  $O(n)$  time complexity?

Trees & BST

p.2685

#### Community Solutions (Python)

##### • WalkCC

```
from dataclasses import dataclass
```

```
@dataclass(frozen=True)
```

```
class T:
```

```
 mn: int # the minimum value in the subtree
```

```
 mx: int # the maximum value in the subtree
```

```
 size: int # the size of the subtree
```

```
class Solution:
```

```
 def largestBSTSubtree(self, root: TreeNode | None) ->
```

```
 int:
```

```
 def dfs(root: TreeNode | None) -> T:
```

```
 if not root:
```

```
 return T(math.inf, -math.inf, 0)
```

```
 l = dfs(root.left)
```

```
 r = dfs(root.right)
```

```
 if l.mx < root.val < r.mn:
```

```
 return T(min(l.mn, root.val), max(r.mx,
root.val), 1 + l.size + r.size)

 # Mark one as invalid, but still record the size of
 # children.
 # Return (-inf, inf) because no node will be > inf
 # or < -inf.
 return T(-math.inf, math.inf, max(l.size, r.size))

 return dfs(root).size
```

##### • LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def largestBSTSubtree(self, root: Optional[TreeNode])
 -> int:
 def dfs(root):
 if root is None:
```

Trees & BST

p.2686

```
self.max_bst_size = 0

def helper(node):
 # Returns a tuple of: (isBST, size, min_val,
max_val)
 if not node:
 # An empty node is considered a BST of
size 0 with extreme min and max values.
 return (True, 0, float('inf'),
float('-inf'))

 left_is_bst, left_size, left_min, left_max =
helper(node.left)
 right_is_bst, right_size, right_min,
right_max = helper(node.right)
```

```
 # Check if the current node forms a BST with
its left and right subtrees.
 if left_is_bst and right_is_bst and node.val
> left_max and node.val < right_min:
 current_size = left_size + right_size + 1
 self.max_bst_size =
max(self.max_bst_size, current_size)
 current_min = min(left_min, node.val)
 current_max = max(right_max, node.val)
 return (True, current_size, current_min,
current_max)
 else:
 # Not a BST: Return false and the size of
the largest BST found so far in its subtrees.
```

Trees & BST

p.2689

```
 return (False, max(left_size,
right_size), 0, 0)

 helper(root)
 return self.max_bst_size
```

##### • LC.ca

```
'''
>>> float("inf")
inf
>>> float("inf") + 1
inf
>>> import math
>>> float("inf") == math.inf
True
'''

Definition for a binary tree node.

class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def largestBSTSubtree(self, root: Optional[TreeNode])
 -> int:
 def dfs(root):
 if root is None:
 return inf, -inf, 0
```

Trees & BST

p.2690

Sheet 299 / 317 · LeetMastery By Pattern · Python Only · 9-up Portrait

```
lmi, lmx, ln = dfs(root.left)
rmi, rmx, rn = dfs(root.right)
nonlocal ans
this if can check if left/right is a bst
eg. if left is not bst, the returned
lmi=-inf and lmx=inf, so 'lmx < root.val < rmi' is false
if lmx < root.val < rmi:
 ans = max(ans, ln + rn + 1)
 # lmi is guranteed < root.val, except
when left(right) is None
 return min(lmi, root.val), max(rmx,
root.val), ln + rn + 1
 return -inf, inf, 0 # if this one not bst,
then all parents will not be bst
```

```
ans = 0
dfs(root)
return ans
```

```
#####
```

```
Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
```

Trees & BST

p.2691

```
self.val = x
self.left = None
self.right = None

class Solution(object):
 def largestBSTSubtree(self, root):
 """
 :type root: TreeNode
 :rtype: int
 """

 def helper(root):
 if not root:
 return (0, 0, float("-inf"), float("-inf")) #
 numBST, maxNumBST, min, max
 lnumBST, lmaxNumBST, lmin, lmax = helper(root.left)
 rnumBST, rmaxNumBST, rmin, rmax =
 helper(root.right)
 numBST = -1
 if lmax < root.val < rmin and lnumBST != -1 and
 rnumBST != -1:
 numBST = 1 + lnumBST + rnumBST
 maxNumBST = max(1, lmaxNumBST, rmaxNumBST, numBST)

 return numBST, maxNumBST, min(lmin, rmin,
 root.val), max(lmax, rmax, root.val)

 return helper(root)[1]
```

Trees & BST

p.2692

Trees & BST

## #366 Find Leaves of Binary Tree

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS · Sorting

Lists: Premium 98

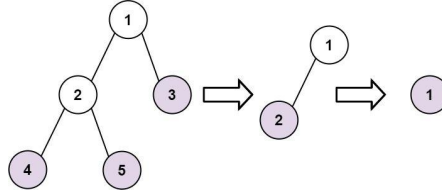
Time Complexity:  $O(n)$  — Each node is visited exactly once. Space Complexity:  $O(n)$  — Space used for the recursion stack and the result list.

### Problem

Given the root of a binary tree, collect a tree's nodes as if you were doing this:

- Collect all the leaf nodes.
- Remove all the leaf nodes.
- Repeat until the tree is empty.

Example 1:



Trees & BST

p.2693

Input: root = [1,2,3,4,5]  
Output: [[4,5,3],[2],[1]]

Explanation:  
[[3,5,4],[2],[1]] and [[3,4,5],[2],[1]] are also considered correct answers since per each level it does not matter the order on which elements are returned.

Example 2:

Input: root = [1]  
Output: [[1]]

### Constraints:

- The number of nodes in the tree is in the range [1, 100].
- $-100 \leq \text{Node.val} \leq 100$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def findLeaves(self, root: TreeNode):
 # Brute Force - collect inorder, then process
 sorted List
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else 0
```

Trees & BST

p.2694

## Community Solutions (Python)

### • WalkCC

```
class Solution:
 def findLeaves(self, root: TreeNode | None) -> List[List[int]]:
 ans = []

 def depth(root: TreeNode | None) -> int:
 """Returns the depth of the root (0-indexed)."""
 if not root:
 return -1

 l = depth(root.left)

 r = depth(root.right)
 h = 1 + max(l, r)

 if len(ans) == h: # Meet a leaf
 ans.append([])

 ans[h].append(root.val)
 return h

 depth(root)

 return ans
```

### • LeetCode

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def findLeaves(self, root: Optional[TreeNode]) -> List[List[int]]:
 def dfs(root: Optional[TreeNode]) -> int:
 if root is None:
 return 0
 l, r = dfs(root.left), dfs(root.right)
 h = max(l, r)
 if len(ans) == h:
 ans.append([])
 ans[h].append(root.val)
 return h + 1

 ans = []
 dfs(root)

 return ans
```

### • SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # The value of the node.
 self.left = left # Left child node.
 self.right = right # Right child node.

class Solution:
 def findLeaves(self, root: TreeNode):
 result = [] # This will store lists of nodes at each removal step.

 def getHeight(node):
 # Base case: when the node is None, return 0.
 if not node:
 return 0
 # Recursively find the height of left and right subtrees.
 left_height = getHeight(node.left)
 right_height = getHeight(node.right)
 # Current node's height is max of left/right heights plus one.
 current_height = max(left_height, right_height) + 1
```

Trees & BST

p.2695

Trees & BST

p.2696

Trees & BST

p.2697

```
Ensure the result list has a sublist for the current height.
if len(result) < current_height:
 result.append([])
Append the current node value to the corresponding height bucket.
result[current_height - 1].append(node.val)
return current_height

getHeight(root) # Start recursion from the root.
return result
```

### • LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def findLeaves(self, root: TreeNode) -> List[List[int]]:
 def dfs(root, prev, t):
 if root is None:
```

```
return
if root.left is None and root.right is None:
 t.append(root.val)
 if prev.left == root:
 prev.left = None
 else:
 prev.right = None
dfs(root.left, root, t)
dfs(root.right, root, t)
```

```
res = []
prev = TreeNode(left=root)
while prev.left:
 t = []
 dfs(prev.left, prev, t)
 res.append(t)
return res
```

Trees & BST

p.2699

Trees & BST

## #437 Path Sum III

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS · Prefix Sum

Lists: Grind 169

Time Complexity:  $O(n)$  on average (where  $n$  is the number of nodes), though worst-case can be  $O(n^2)$  in a skewed tree. Space Complexity:  $O(n)$  due to recursion stack and the hash map that stores prefix s

### Problem

Given the root of a binary tree and an integer targetSum, return the number of paths where the sum of the values along the path equals targetSum.

The path does not need to start or end at the root or a leaf, but it must go downwards (i.e., traveling only from parent nodes to child nodes).

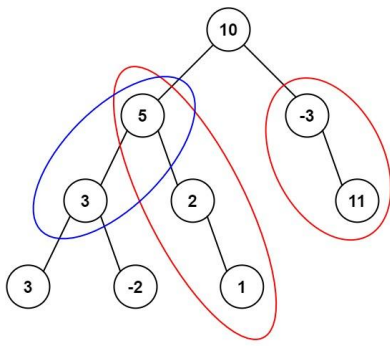
Example 1:

Trees & BST

p.2698

Trees & BST

p.2700



Input: root = [10,5,-3,3,2,null,11,3,-2,null,1], targetSum = 8  
Output: 3  
Explanation: The paths that sum to 8 are shown.

Example 2:

Input: root = [5,4,8,11,null,13,4,7,2,null,null,5,1], targetSum = 22  
Output: 3

Constraints:

- The number of nodes in the tree is in the range [0, 1000].
- $-109 \leq \text{Node.val} \leq 109$
- $-1000 \leq \text{targetSum} \leq 1000$

#### Community Solutions (Python)

##### • WalkCC

```
class Solution:
 def pathSum(self, root: TreeNode | None, summ: int) -> int:
 if not root:
 return 0

 def dfs(root: TreeNode, summ: int) -> int:
 if not root:
 return 0
 return (int(summ == root.val) +
 dfs(root.left, summ - root.val) +
 dfs(root.right, summ - root.val))

 return (dfs(root, summ) +
 self.pathSum(root.left, summ) +
 self.pathSum(root.right, summ))
```

##### • LeetCode

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def pathSum(self, root: Optional[TreeNode],
 targetSum: int) -> int:
 def dfs(node, s):
 if node is None:
 return 0
 s += node.val
 ans = cnt[s - targetSum]
 cnt[s] += 1
 ans += dfs(node.left, s)
 ans += dfs(node.right, s)
 cnt[s] -= 1
 return ans

 cnt = Counter({0: 1})

 return dfs(root, 0)
```

##### • SimplyLeet

Trees & BST

p.2701

Trees & BST

p.2702

Trees & BST

p.2703

```
Python solution for Path Sum III
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # value of the node
 self.left = left # left child
 self.right = right # right child

class Solution:
 def pathSum(self, root: TreeNode, targetSum: int) -> int:
 # Helper function to perform DFS traversal

 def dfs(node, current_sum, prefix_sums):
 if not node:
 return 0

 # Update current sum by including the current
 # node's value
 current_sum += node.val
 # Check the number of times (current_sum -
 # targetSum) has occurred
 num_paths = prefix_sums.get(current_sum -
 targetSum, 0)

 # Update the prefix sum hash map with the
 # current sum
 prefix_sums[current_sum] = prefix_sums.get(
 current_sum, 0) + 1

 # Recursively search in the left and right
 # subtrees
 num_paths += dfs(node.left, current_sum,
 prefix_sums)
 num_paths += dfs(node.right, current_sum,
 prefix_sums)

 # Backtrack: remove the current sum count
 # before returning to the parent node
 prefix_sums[current_sum] -= 1
 return num_paths

 # Initial prefix sum hashmap with 0 sum seen once
 # (empty path)
 prefix_sums = {0: 1}
 return dfs(root, 0, prefix_sums)
```

Trees & BST

p.2704

```
prefix_sums[current_sum] =
prefix_sums.get(current_sum, 0) + 1

Recursively search in the left and right
subtrees
num_paths += dfs(node.left, current_sum,
 prefix_sums)
num_paths += dfs(node.right, current_sum,
 prefix_sums)

Backtrack: remove the current sum count
before returning to the parent node
prefix_sums[current_sum] -= 1
return num_paths

Initial prefix sum hashmap with 0 sum seen once
(empty path)
prefix_sums = {0: 1}
return dfs(root, 0, prefix_sums)
```

##### • LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def pathSum(self, root: Optional[TreeNode],
 targetSum: int) -> int:
 def dfs(node, s):
 if node is None:
 return 0
```

Trees & BST

p.2705

Trees & BST

p.2706

Trees & BST

## #545 Boundary of Binary Tree

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS

Lists: Premium 98

Time Complexity:  $O(n)$  - Every node is visited at most once. Space Complexity:  $O(n)$  -  $O(n)$  space is used for the output list and recursion stack (in worst case tree is skewed).

Problem

The boundary of a binary tree is the concatenation of the root, the left boundary, the leaves ordered from left-to-right, and the reverse order of the right boundary.

The left boundary is the set of nodes defined by the following:

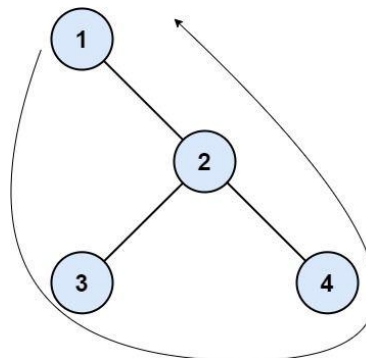
- The root node's left child is in the left boundary. If the root does not have a left child, then the left boundary is empty.
- If a node is in the left boundary and has a left child, then the left child is in the left boundary.
- If a node is in the left boundary, has no left child, but has a right child, then the right child is in the left boundary.
- The leftmost leaf is not in the left boundary.

The right boundary is similar to the left boundary, except it is the right side of the root's right subtree. Again, the leaf is not part of the right boundary, and the right boundary is empty if the root does not have a right child.

The leaves are nodes that do not have any children. For this problem, the root is not a leaf.

Given the root of a binary tree, return the values of its boundary.

Example 1:

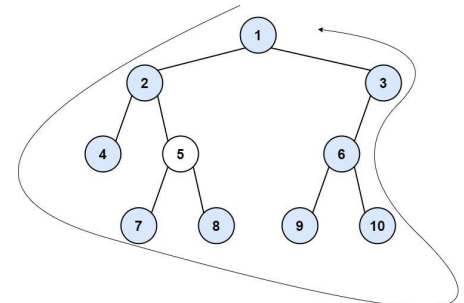


Trees & BST

p.2708

Input: root = [1,null,2,3,4]  
Output: [1,3,4,2]  
Explanation:  
- The left boundary is empty because the root does not have a left child.  
- The right boundary follows the path starting from the root's right child 2 -> 4.  
4 is a leaf, so the right boundary is [2].  
- The leaves from left to right are [3,4].  
Concatenating everything results in [1] + [] + [3,4] + [2] = [1,3,4,2].

Example 2:



Trees & BST

p.2707

Trees & BST

p.2709

Input: root = [1,2,3,4,5,6,null,null,7,8,9,10]  
Output: [1,2,4,7,8,9,10,6,3]  
Explanation:  
- The left boundary follows the path starting from the root's left child 2 -> 4.  
4 is a leaf, so the left boundary is [2].  
- The right boundary follows the path starting from the root's right child 3 -> 6 -> 10.  
10 is a leaf, so the right boundary is [3,6], and in reverse order is [6,3].  
- The leaves from left to right are [4,7,8,9,10].

Constraints:

- The number of nodes in the tree is in the range [1, 104].
- -1000 <= Node.val <= 1000

Community Solutions (Python)

• WalkCC

```
class Solution:
 def boundaryOfBinaryTree(self, root: TreeNode | None)
 -> list[int]:
 if not root:
 return []

 ans = [root.val]

 def dfs(root: TreeNode | None, lb: bool, rb: bool):
 """
 1. root.left is left boundary if root is left
 boundary.
```

Trees & BST

p.2710

```
 root.right if left boundary if root.left is
 None.
 2. Same appls for right boundary.
 3. If root is left boundary, add it before 2
 children - preorder.
 If root is right boundary, add it after 2
 children - postorder.
 4. A leaf that is neighter left/right boundary
 belongs to the bottom.
 """
 if not root:
 return
 if lb:
 ans.append(root.val)

 if not lb and not rb and not root.left and not
 root.right:
 ans.append(root.val)

 dfs(root.left, lb, rb and not root.right)
 dfs(root.right, lb and not root.left, rb)
 if rb:
 ans.append(root.val)

 dfs(root.left, True, False)
 dfs(root.right, False, True)

 return ans
```

• LeetDoocs

Trees & BST

p.2711

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def boundaryOfBinaryTree(self, root:
 Optional[TreeNode]) -> List[int]:
 def dfs(nums: List[int], root:
 Optional[TreeNode], i: int):
 if root is None:

 return
 if i == 0:
 if root.left != root.right:
 nums.append(root.val)
 if root.left:
 dfs(nums, root.left, i)
 else:
 dfs(nums, root.right, i)
 elif i == 1:
 if root.left == root.right:
```

Trees & BST

p.2712

```
 nums.append(root.val)
 else:
 dfs(nums, root.left, i)
 dfs(nums, root.right, i)

 else:
 if root.left != root.right:
 nums.append(root.val)
 if root.right:
 dfs(nums, root.right, i)
 else:
```

```
 dfs(nums, root.left, i)
```

```
 ans = [root.val]
 if root.left == root.right:
 return ans
 left, leaves, right = [], [], []
 dfs(left, root.left, 0)
 dfs(leaves, root, 1)
 dfs(right, root.right, 2)
 ans += left + leaves + right[::-1]
```

```
 return ans
```

• SimplyLeet

Trees & BST

p.2713

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
 def boundaryOfBinaryTree(self, root: TreeNode) ->
 list:

 if not root:

 return []

 # Check if a node is a leaf.
 def isLeaf(node):
 return node and not node.left and not
 node.right

 # Add left boundary nodes (excluding leaves).
 def addLeftBoundary(node, boundary):
 while node:
 if not isLeaf(node):
```

Trees & BST

p.2714

```
 boundary.append(node.val)
 # Prefer left child, if not present then
 right child.

 if node.left:
 node = node.left
 else:
 node = node.right

 # Add leaf nodes using DFS.
 def addLeaves(node, leaves):
 if not node:

 return
 if isLeaf(node):
 leaves.append(node.val)
 else:
 addLeaves(node.left, leaves)
 addLeaves(node.right, leaves)

 # Add right boundary nodes in reverse order
 (excluding leaves).
 def addRightBoundary(node, boundary):
 temp = []
```

Trees & BST

p.2715

```
while node:
 if not isLeaf(node):
 temp.append(node.val)
 if node.right:
 node = node.right
 else:
 node = node.left
Append in reverse order.
while temp:
 boundary.append(temp.pop())
```

```
boundary = []
Add root value if it is not a leaf.
if not isLeaf(root):
 boundary.append(root.val)

addLeftBoundary(root.left, boundary)
addLeaves(root, boundary)
addRightBoundary(root.right, boundary)
```

```
return boundary
```

```
Example usage:
Construct a binary tree and call
boundaryOfBinaryTree(root)
```

• LC.ca

Trees & BST

p.2716

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def boundaryOfBinaryTree(self, root: TreeNode) ->
 List[int]:
 self.res = []
 if not root:

 return self.res
 # root
 if not self.is_leaf(root):
 self.res.append(root.val)

 # left boundary
 t = root.left
 while t:
 if not self.is_leaf(t):
 self.res.append(t.val)

 t = t.left if t.left else t.right

 # leaves
 self.add_leaves(root)

 # right boundary(reverse order)
 s = []
 t = root.right
 while t:
 if not self.is_leaf(t):
```

Trees & BST

p.2717

```
 s.append(t.val)
 t = t.right if t.right else t.left
 while s:
 self.res.append(s.pop())

 # output
 return self.res

 def add_leaves(self, root):
 if self.is_leaf(root):

 self.res.append(root.val)
 return
 if root.left:
 self.add_leaves(root.left)
 if root.right:
 self.add_leaves(root.right)

 def is_leaf(self, node) -> bool:
 return node and node.left is None and node.right
 is None
```

Trees & BST

p.2718



## #549 Binary Tree Longest Consecutive Sequence II

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS

Lists: Premium 98

Time Complexity:  $O(n)$ , where  $n$  is the number of nodes since every node is visited once. Space Complexity:  $O(h)$ , where  $h$  is the height of the tree, due to the recursion stack.

## Problem

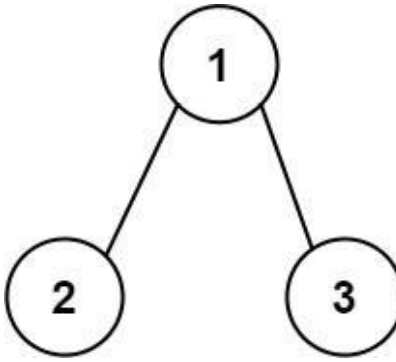
Given the root of a binary tree, return the length of the longest consecutive path in the tree.

A consecutive path is a path where the values of the consecutive nodes in the path differ by one. This path can be either increasing or decreasing.

- For example,  $[1,2,3,4]$  and  $[4,3,2,1]$  are both considered valid, but the path  $[1,2,4,3]$  is not valid.

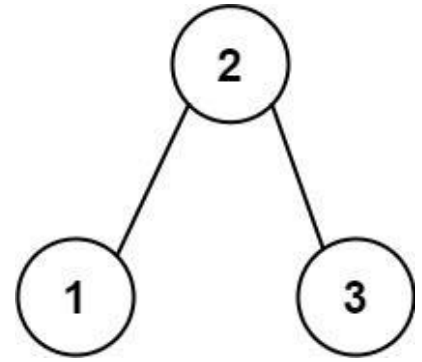
On the other hand, the path can be in the child-Parent-child order, where not necessarily be parent-child order.

Example 1:



Input: root = [1,2,3]  
Output: 2  
Explanation: The longest consecutive path is [1, 2] or [2, 1].

Example 2:



Input: root = [2,1,3]  
Output: 3  
Explanation: The longest consecutive path is [1, 2, 3] or [3, 2, 1].

Constraints:

- The number of nodes in the tree is in the range  $[1, 3 * 10^4]$ .
- $-3 * 10^4 \leq \text{Node.val} \leq 3 * 10^4$

Community Solutions (Python)

Trees &amp; BST

p.2719

Trees &amp; BST

p.2720

Trees &amp; BST

p.2721

## • LeetCode

```

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def longestConsecutive(self, root: TreeNode) -> int:
 def dfs(root):
 if root is None:
 return [0, 0]
 nonlocal ans
 incr = decr = 1
 i1, d1 = dfs(root.left)
 i2, d2 = dfs(root.right)
 if root.left:
 if root.left.val + 1 == root.val:
 incr = i1 + 1
 if root.left.val - 1 == root.val:
 decr = d1 + 1
 if root.right:
 if root.right.val + 1 == root.val:
 incr = max(incr, i2 + 1)
 if root.right.val - 1 == root.val:
 decr = max(decr, d2 + 1)
 ans = max(ans, incr + decr - 1)
 return [incr, decr]

 ans = 0
 dfs(root)

```

Trees &amp; BST

p.2722

return ans

## • SimplyLeet

```

Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
 def longestConsecutive(self, root: TreeNode) -> int:
 # Initialize global maximum path length variable

 self.max_length = 0

 def dfs(node):
 if not node:
 return (0, 0) # (increasing, decreasing)

 inc = dec = 1 # Each node counts as length 1

 # Process left child
 if node.left:

```

Trees &amp; BST

p.2723

```

 left_inc, left_dec = dfs(node.left)
 # If left child's value is consecutive
 # increasing with respect to node
 if node.left.val == node.val + 1:
 inc = max(inc, left_inc + 1)
 # If left child's value is consecutive
 # decreasing with respect to node
 elif node.left.val == node.val - 1:
 dec = max(dec, left_dec + 1)

 # Process right child
 if node.right:

 right_inc, right_dec = dfs(node.right)
 # If right child's value is consecutive
 # increasing with respect to node
 if node.right.val == node.val + 1:
 inc = max(inc, right_inc + 1)
 # If right child's value is consecutive
 # decreasing with respect to node
 elif node.right.val == node.val - 1:
 dec = max(dec, right_dec + 1)

 # Update the global maximum path by combining
 # increasing and decreasing paths
 self.max_length = max(self.max_length, inc +
 dec - 1)

 return (inc, dec)

 def dfs(self, root):
 return self.max_length

```

• LC.ca

Trees &amp; BST

p.2724

```

Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
self.val = x
self.left = None
self.right = None
...
>>> float('-inf')+1
-inf

>>> float('-inf')-1
-inf
...

class Solution(object):
 def longestConsecutive(self, root):
 ...

 :type root: TreeNode
 :rtype: int
 ...

```

Trees &amp; BST

p.2725

```

 def dfs(root):
 if not root:
 # val, increasing length, decreasing
 # length, max length
 return float('-inf'), 0, 0, 0
 # inc/dec starting from root
 inc = dec = 1
 left, leftInc, leftDec, leftMax =
 dfs(root.left)
 right, rightInc, rightDec, rightMax =
 dfs(root.right)

 if root.val + 1 == left:
 inc = max(leftInc + 1, inc)

 if root.val - 1 == left:
 dec = max(leftDec + 1, dec)
 if root.val + 1 == right:
 inc = max(rightInc + 1, inc)
 if root.val - 1 == right:
 dec = max(rightDec + 1, dec)
 # note: max may occur in children subtree,
 # not current root's tree
 return root.val, inc, dec, max(inc + dec - 1,
 leftMax, rightMax, inc, dec)
 # -1 because root is counted twice, even
 # left/right both null

```

Trees &amp; BST

p.2726

```

 return dfs(root)[3]

#####

Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right

class Solution:
 def longestConsecutive(self, root: TreeNode) -> int:
 def dfs(root):
 if root is None:
 return [0, 0]
 nonlocal ans
 incr = decr = 1
 i1, d1 = dfs(root.left)
 i2, d2 = dfs(root.right)
 if root.left:

```

Trees &amp; BST

p.2727

```

 if root.left.val + 1 == root.val:
 incr = i1 + 1
 if root.left.val - 1 == root.val:
 decr = d1 + 1
 if root.right:
 if root.right.val + 1 == root.val:
 incr = max(incr, i2 + 1) # incr1 and
incr2 must be in different tree branches, eg. left child
cannot be both increasing and decreasing. Same as decr1
and decr2
 if root.right.val - 1 == root.val:
 decr = max(decr, d2 + 1)
 ans = max(ans, incr + decr - 1) # -1 because
root is counted twice, even left/right both null

```

```

 return [incr, decr]

```

```

 ans = 0
 dfs(root)
 return ans

```

#####

```

class Solution: # issue with this solution, duplicated
dfs() search
def longestConsecutive(self, root: TreeNode) -> int:

```

Trees & BST

p.2728

```

def helper(node: TreeNode, diff: int) -> int:
 if not node:
 return 0

 left = right = 0
 if node.left and node.val - node.left.val ==
diff:
 left = 1 + helper(node.left, diff)
 if node.right and node.val - node.right.val
== diff:
 right = 1 + helper(node.right, diff)

```

```

 return max(left, right)
 if not root:
 return 0

```

```

 res = helper(root, 1) + helper(root, -1) + 1 # +1
is increasing, -1 is decreasing
 return max(res,
max(longestConsecutive(root.left),
longestConsecutive(root.right)))

```

Trees & BST

p.2729

Trees & BST

## #582 Kill Process

MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Array · Hash Table · Tree · DFS · BFS

Lists: Premium 98

Time Complexity:  $O(n)$ , where  $n$  is the number of processes (both building the mapping and the tree traversal take linear time). Space Complexity:  $O(n)$ , for storing the mapping and the traversal queue/s

### Problem

You have  $n$  processes forming a rooted tree structure. You are given two integer arrays `pid` and `ppid`, where `pid[i]` is the ID of the  $i$ th process and `ppid[i]` is the ID of the  $i$ th process's parent process.

Each process has only one parent process but may have multiple children processes. Only one process has `ppid[i] = 0`, which means this process has no parent process (the root of the tree).

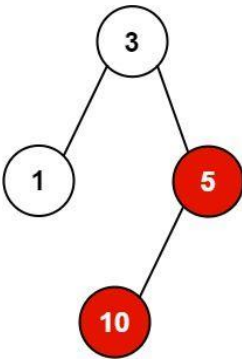
When a process is killed, all of its children processes will also be killed.

Given an integer `kill` representing the ID of a process you want to kill, return a list of the IDs of the processes that will be killed. You may return the answer in any order.

Trees & BST

p.2730

### Example 1:



Input: `pid = [1,3,10,5]`, `ppid = [3,0,5,3]`, `kill = 5`  
Output: `[5,10]`  
Explanation: The processes colored in red are the processes that should be killed.

### Example 2:

Input: `pid = [1]`, `ppid = [0]`, `kill = 1`  
Output: `[1]`

### Constraints:

- $n == \text{pid.length}$

Trees & BST

p.2731

- $n == \text{ppid.length}$
- $1 \leq n \leq 5 \times 10^4$
- $1 \leq \text{pid}[i] \leq 5 \times 10^4$
- $0 \leq \text{ppid}[i] \leq 5 \times 10^4$
- Only one process has no parent.
- All the values of `pid` are unique.
- kill is guaranteed to be in `pid`.

### Brute Force (Python)

```

Brute Force Approach
class Solution:
 def killProcess(self, pid, ppid, kill):
 # Brute Force - collect inorder, then process
sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

```

```

 collect(pid)
 return vals[0] if vals else 0

```

### Community Solutions (Python)

- WalkCC

Trees & BST

p.2732

```

class Solution:
 def killProcess(self, pid: List[int], ppid:
List[int], kill: int) -> List[int]:
 def dfs(i: int):
 ans.append(i)
 for j in g[i]:
 dfs(j)

 g = defaultdict(list)
 for i, p in zip(pid, ppid):
 g[p].append(i)

```

```

 ans = []
 dfs(kill)
 return ans

```

### • SimplyLeet

# Python solution using BFS

```

from collections import deque

```

```

def killProcess(pid, ppid, kill):
 # Build a mapping from parent process id to list of
child process ids
 tree = {}
 for child, parent in zip(pid, ppid):
 if parent not in tree:
 tree[parent] = []

```

```

tree[parent].append(child)

Use a deque for BFS traversal
result = []
queue = deque([kill])

while queue:
 current = queue.popleft() # Get process from the
left of the queue
 result.append(current) # Add the current process
to the result
 if current in tree:

```

```

 # Enqueue all children of the current process
for child in tree[current]:
 queue.append(child)

return result

```

```

Example usage:
pid = [1, 3, 10, 5]
ppid = [3, 0, 5, 3]
kill = 5

```

```

Expected output: [5, 10]

```

- LC.ca

Trees & BST

p.2735

```

class Solution:
 def killProcess(self, pid: List[int], ppid:
List[int], kill: int) -> List[int]:
 def dfs(i: int):
 ans.append(i)
 for j in g[i]:
 dfs(j)

 g = defaultdict(list)
 for i, p in zip(pid, ppid):
 g[p].append(i)

```

```

 ans = []
 dfs(kill)
 return ans

```

Trees & BST

p.2736

Trees & BST

#662 Maximum Width of Binary TreeMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Tree · BFS

Lists: Grind 169

Time Complexity: O(N), where N is the number of nodes in the tree. Space Complexity: O(N), due to the queue used for BFS which in the worst case holds nodes of the last level.

Problem

Given the root of a binary tree, return the maximum width of the given tree.

The maximum width of a tree is the maximum width among all levels.

The width of one level is defined as the length between the end-nodes (the leftmost and rightmost non-null nodes), where the null nodes between the end-nodes that would be present in a complete binary tree extending down to that level are also counted into the length calculation.

It is guaranteed that the answer will in the range of a 32-bit signed integer.

Example 1:

Trees & BSTp.2737

```
graph TD
 1((1)) --> 3L((3))
 1 --> 2((2))
 3L --> 5L((5))
 3L --> 3R((3))
 2 --> 9((9))
```

Input: root = [1,3,2,5,3,null,9]  
Output: 4  
Explanation: The maximum width exists in the third level with length 4 (5,3,null,9).

Example 2:

Trees & BSTp.2738

```
graph TD
 1((1)) --> 3((3))
 1 --> 2((2))
 3 --> 5((5))
 5 --> 6((6))
 2 --> 9((9))
 9 --> 7((7))
```

Input: root = [1,3,2,5,null,null,9,6,null,7]  
Output: 7  
Explanation: The maximum width exists in the fourth level with length 7 (6,null,null,null,null,null,7).

Example 3:

Trees & BSTp.2739

```
graph TD
 1((1)) --> 3((3))
 1 --> 2((2))
 3 --> 5((5))
```

Input: root = [1,3,2,5]  
Output: 2  
Explanation: The maximum width exists in the second level with length 2 (3,2).

Constraints:

- The number of nodes in the tree is in the range [1, 3000].
- 100 <= Node.val <= 100

Community Solutions (Python)

Trees & BSTp.2740

• LeetDoocs

# Definition for a binary tree node.  
# class TreeNode:  
# def \_\_init\_\_(self, val=0, left=None, right=None):  
# self.val = val  
# self.left = left  
# self.right = right  
class Solution:  
 def widthOfBinaryTree(self, root: Optional[TreeNode]) -> int:  
 ans = 0  
 q = deque([(root, 1)])  
  
 while q:  
 ans = max(ans, q[-1][1] - q[0][1] + 1)  
 for \_ in range(len(q)):  
 root, i = q.popleft()  
 if root.left:  
 q.append((root.left, i << 1))  
 if root.right:  
 q.append((root.right, i << 1 | 1))  
 return ans

• SimplyLeet

Trees & BSTp.2741

# Definition for a binary tree node.  
# class TreeNode:  
# def \_\_init\_\_(self, val=0, left=None, right=None):  
# self.val = val  
# self.left = left  
# self.right = right  
from collections import deque  
  
class Solution:  
  
 def widthOfBinaryTree(self, root: TreeNode) -> int:  
 if not root:  
 return 0  
  
 max\_width = 0  
 # Initialize queue with a tuple of (node, index)  
 queue = deque([(root, 1)])  
  
 while queue:  
 level\_length = len(queue)

Trees & BSTp.2742

```
level # Get the index of first node at current
_, first_index = queue[0]
Process nodes at current level
for i in range(level_length):
 node, index = queue.popleft()
 # Check and add left child with index 2 *
index
 if node.left:
 queue.append((node.left, 2 * index))
 # Check and add right child with index 2
* index + 1
 if node.right:
 queue.append((node.right, 2 * index +
1))
For the last node, calculate width of
the level
 if i == level_length - 1:
 current_width = index - first_index +
1
 max_width = max(max_width,
current_width)
 return max_width
```

• LC.ca

Trees & BSTp.2743

# Definition for a binary tree node.  
# class TreeNode:  
# def \_\_init\_\_(self, val=0, left=None, right=None):  
# self.val = val  
# self.left = left  
# self.right = right  
class Solution:  
 def widthOfBinaryTree(self, root: Optional[TreeNode]) -> int:  
 ans = 0  
 q = deque([(root, 1)])  
  
 while q:  
 ans = max(ans, q[-1][1] - q[0][1] + 1)  
 for \_ in range(len(q)):  
 root, i = q.popleft()  
 if root.left:  
 q.append((root.left, i << 1))  
 if root.right:  
 q.append((root.right, i << 1 | 1))  
 return ans

Trees & BSTp.2744

Trees & BST

#863 All Nodes Distance K in Binary TreeMED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · Tree · DFS · BFS

Lists: Grind 169

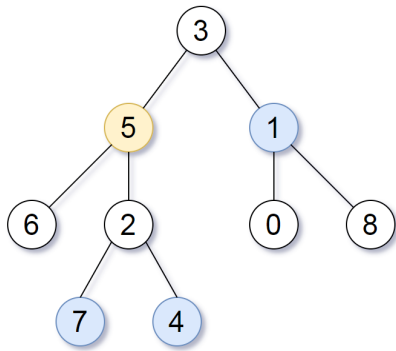
Time Complexity: O(N), where N is the number of nodes in the tree. Each node is visited once when constructing the graph and once in the BFS. Space Complexity: O(N), for storing the graph and the addi

Problem

Given the root of a binary tree, the value of a target node target, and an integer k, return an array of the values of all nodes that have a distance k from the target node. You can return the answer in any order.

Example 1:

Trees & BSTp.2745



Input: root = [3,5,1,6,2,0,8,null,null,7,4], target = 5, k = 2  
 Output: [7,4,1]  
 Explanation: The nodes that are a distance 2 from the target node (with value 5) have values 7, 4, and 1.

Example 2:

Input: root = [1], target = 1, k = 3  
 Output: []

Constraints:

- The number of nodes in the tree is in the range [1, 500].
- $0 \leq \text{Node.val} \leq 500$
- All the values `Node.val` are unique.
- target is the value of one of the nodes in the tree.
- $0 \leq k \leq 1000$

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def distanceK(self, root: TreeNode, target: int, k: int):
 # Brute Force - collect inorder, then process sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else []
```

Community Solutions (Python)

- LeetCode

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, x):
self.val = x
self.left = None
self.right = None
```

```
class Solution:
 def distanceK(self, root: TreeNode, target: TreeNode, k: int) -> List[int]:
```

```
def dfs(root, fa):
 if root is None:
 return
 g[root] = fa
 dfs(root.left, root)
 dfs(root.right, root)
```

```
def dfs2(root, fa, k):
 if root is None:
 return
```

```
if k == 0:
 ans.append(root.val)
 return
for nxt in (root.left, root.right, g[root]):
 if nxt != fa:
 dfs2(nxt, root, k - 1)
```

```
g = {}
dfs(root, None)
ans = []
```

Trees & BST

p.2746

Trees & BST

p.2747

Trees & BST

p.2748

```
dfs2(target, None, k)
return ans
```

• SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right
```

```
def distanceK(self, root: TreeNode, target: int, k: int):
 from collections import defaultdict, deque
```

```
Build an adjacency list to represent the graph
graph = defaultdict(list)
```

```
Helper function to build the graph using DFS
def buildGraph(node, parent):
 if node is None:
 return
 if parent is not None:
 # Add bidirectional connection between node and parent
 graph[node.val].append(parent.val)
```

```
graph[parent.val].append(node.val)
buildGraph(node.left, node)
buildGraph(node.right, node)
```

```
buildGraph(root, None)
```

```
BFS from target node
visited = set([target])
queue = deque([target])
current_distance = 0

while queue:
 if current_distance == k:
 # Return all nodes at distance k
 return list(queue)
 size = len(queue)
 for _ in range(size):
 current = queue.popleft()
 for neighbor in graph[current]:
 if neighbor not in visited:
```

```
visited.add(neighbor)
queue.append(neighbor)
current_distance += 1
```

```
return [] # If no nodes at distance k
```

```
Example usage:
Construct tree or use a tree builder before calling distanceK function.
```

- LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, x):
self.val = x
self.left = None
self.right = None
```

```
class Solution:
 def distanceK(self, root: TreeNode, target: TreeNode, k: int) -> List[int]:
```

```
def parents(root, prev):
 nonlocal p
 if root is None:
 return
 p[root] = prev
 parents(root.left, root)
 parents(root.right, root)
```

```
def dfs(root, k):
 nonlocal ans, vis
```

```
if root is None or root.val in vis:
 return
vis.add(root.val)
if k == 0:
 ans.append(root.val)
 return
dfs(root.left, k - 1)
dfs(root.right, k - 1)
dfs(p[root], k - 1)
```

Trees & BST

p.2749

Trees & BST

p.2750

Trees & BST

p.2751

```
p = {}
parents(root, None)
ans = []
vis = set()
dfs(target, k)
return ans
```

Trees & BST

## #1120 Maximum Average Subtree

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS

Lists: Premium 98

Time Complexity:  $O(n)$ , where  $n$  is the number of nodes (each node is visited once). Space Complexity:  $O(h)$ , where  $h$  is the height of the tree (space for recursion stack).

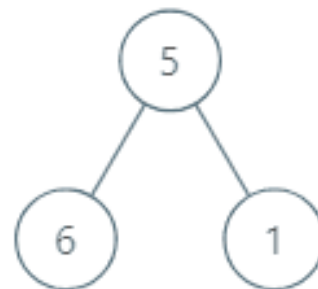
Problem

Given the root of a binary tree, return the maximum average value of a subtree of that tree. Answers within  $10^{-5}$  of the actual answer will be accepted.

A subtree of a tree is any node of that tree plus all its descendants.

The average value of a tree is the sum of its values, divided by the number of nodes.

Example 1:



Input: root = [5,6,1]  
 Output: 6.00000  
 Explanation:  
 For the node with value = 5 we have an average of  $(5 + 6 + 1) / 3 = 4$ .  
 For the node with value = 6 we have an average of  $6 / 1 = 6$ .  
 For the node with value = 1 we have an average of  $1 / 1 = 1$ .  
 So the answer is 6 which is the maximum.

Example 2:

Input: root = [0,null,1]  
 Output: 1.00000

Trees & BST

p.2752

Trees & BST

p.2753

Trees & BST

p.2754

- Constraints:
- The number of nodes in the tree is in the range [1, 104].
  - 0 <= Node.val <= 105

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def maximumAverageSubtree(self, root):
 # Brute Force - collect inorder, then process
sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else 0
```

Community Solutions (Python)

- WalkCC

```
from dataclasses import dataclass

@dataclass(frozen=True)
class T:
 summ: int
 count: int
 maxAverage: int

class Solution:
 def maximumAverageSubtree(self, root: TreeNode | None)
-> float:
 def maximumAverage(root: TreeNode | None) -> T:
 if not root:
 return T(0, 0, 0)

 left = maximumAverage(root.left)
 right = maximumAverage(root.right)

 summ = root.val + left.summ + right.summ

 count = 1 + left.count + right.count
 maxAverage = max(summ / count, left.maxAverage,
right.maxAverage)
 return T(summ, count, maxAverage)

 return maximumAverage(root).maxAverage
```

- LeetDoocs

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def maximumAverageSubtree(self, root:
Optional[TreeNode]) -> float:
 def dfs(root):
 if root is None:

 return 0, 0
 ls, ln = dfs(root.left)
 rs, rn = dfs(root.right)
 s = root.val + ls + rs
 n = 1 + ln + rn
 nonlocal ans
 ans = max(ans, s / n)
 return s, n

 ans = 0

 dfs(root)
 return ans
```

- SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # value of the node
 self.left = left # left child
 self.right = right # right child

class Solution:
 def maximumAverageSubtree(self, root):
 # Initialize the maximum average to negative
infinity

 self.maxAvg = float('-inf')

 # Recursive DFS function that returns a tuple
(subtree sum, node count)
 def dfs(node):
 if not node:
 return (0, 0) # return sum 0 and count 0
if node is null
 leftSum, leftCount = dfs(node.left) #
traverse left subtree
 rightSum, rightCount = dfs(node.right) #
traverse right subtree
 currentSum = node.val + leftSum + rightSum #
total sum at current node
 currentCount = 1 + leftCount + rightCount #
total node count
```

```
 currentAvg = currentSum / currentCount #
average value for current subtree
 self.maxAvg = max(self.maxAvg, currentAvg) #
update maximum average if necessary
 return (currentSum, currentCount)

 def dfs(self, root):
 return self.maxAvg

class Solution:
 def maximumAverageSubtree(self, root:
Optional[TreeNode]) -> float:
 def dfs(root):
 if root is None:

 return 0, 0
 ls, ln = dfs(root.left)
 rs, rn = dfs(root.right)
 s = root.val + ls + rs
 n = 1 + ln + rn
 nonlocal ans
 ans = max(ans, s / n)
 return s, n

 ans = 0
```

- LC.ca

```
def dfs(self, root):
 return ans
```

Trees & BST

#1490 Clone N-ary Tree

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · Tree · DFS · BFS

Lists: Premium 98

Time Complexity: O(N), where N is the number of nodes in the tree, since each node is visited exactly once. Space Complexity: O(N), accounting for the recursive call stack (or an explicit queue in BFS)

Problem

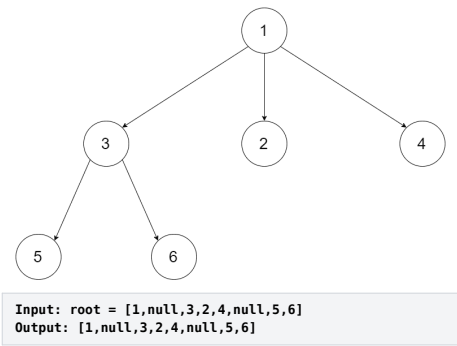
Given a root of an N-ary tree, return a deep copy (clone) of the tree.

Each node in the n-ary tree contains a val (int) and a list (List[Node]) of its children.

class Node {
 public int val;
 public List<Node> children;
}

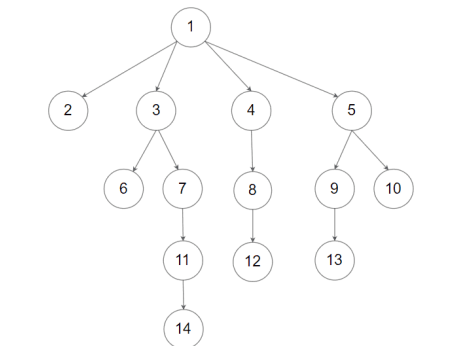
Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value (See examples).

Example 1:



Input: root = [1,null,3,2,4,null,5,6]  
Output: [1,null,3,2,4,null,5,6]

Example 2:



Input: root = [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]  
Output: [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]

- Constraints:
- The depth of the n-ary tree is less than or equal to 1000.
  - The total number of nodes is between [0, 104].
- Follow up: Can your solution work for the graph problem?

Brute Force (Python)

```
Brute Force Approach
class Solution:
 def cloneTree(self, root):
 # Brute Force - collect inorder, then process
 sorted list
 vals = []
 def collect(node):
 if not node: return
 collect(node.left)
 vals.append(node.val)
 collect(node.right)

 collect(root)
 return vals[0] if vals else 0
```

Community Solutions (Python)

```
• LeetCode
"""
Definition for a Node.
class Node:
 def __init__(self, val=None, children=None):
 self.val = val
 self.children = children if children is not None
 else []
"""

class Solution:
```

```
def cloneTree(self, root: 'Node') -> 'Node':
 if root is None:
 return None
 children = [self.cloneTree(child) for child in
root.children]
 return Node(root.val, children)
```

SimplyLeet

```
Definition for a Node.
class Node:
 def __init__(self, val=None, children=None):
 self.val = val
 self.children = children if children is not None
 else []

def cloneTree(root):
 # Base case: if root is None, return None.
 if root is None:
 return None
```

```
Create a clone for the current node.
cloned_node = Node(root.val)

Recursively clone each child and add it to the
cloned_node's children list.
for child in root.children:
 cloned_child = cloneTree(child)
 cloned_node.children.append(cloned_child)

Return the cloned node.
```

```
return cloned_node

Example usage:
original_tree = Node(1, [Node(3, [Node(5), Node(6)]),
Node(2), Node(4)])
cloned_tree = cloneTree(original_tree)
```

LC.ca

```
"""
Definition for a Node.
class Node:
 def __init__(self, val=None, children=None):
 self.val = val
 self.children = children if children is not None
 else []
"""
```

```
class Solution:

 def cloneTree(self, root: 'Node') -> 'Node':
 if root is None:
 return None
 children = [self.cloneTree(child) for child in
root.children]
 return Node(root.val, children)
```

Trees & BST

#1522 Diameter of N-Ary Tree

MED

LeetCode · SimplyLeet · WalkCC · LeetCode

Tree · DFS

Lists: Premium 98

Time Complexity: O(n), where n is the number of nodes, since each node is visited once. Space Complexity: O(h), where h is the height of the tree, due to the recursion stack (worst-case O(n) for a ske

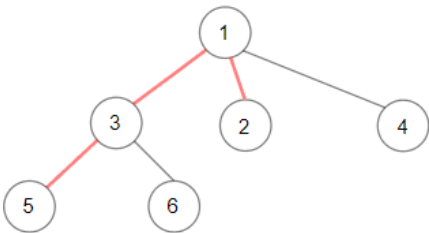
Problem

Given a root of an N-ary tree, you need to compute the length of the diameter of the tree.

The diameter of an N-ary tree is the length of the longest path between any two nodes in the tree. This path may or may not pass through the root.

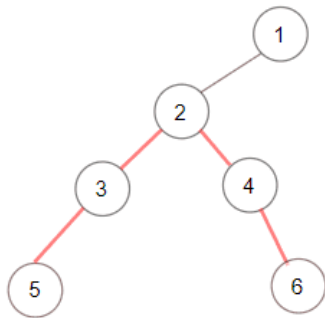
(Nary-Tree input serialization is represented in their level order traversal, each group of children is separated by the null value.)

Example 1:



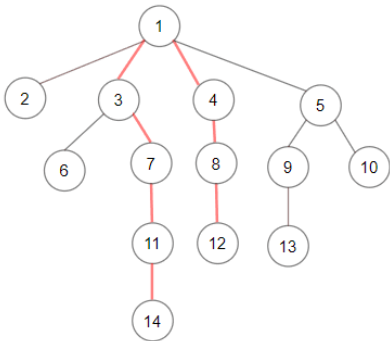
Input: root = [1,null,3,2,4,null,5,6]  
Output: 3  
Explanation: Diameter is shown in red color.

Example 2:



Input: root = [1,null,2,null,3,4,null,5,null,6]  
Output: 4

Example 3:



Input: root = [1,null,2,3,4,5,null,null,6,7,null,8,null,9,10,null,null,11,null,12,null,13,null,null,14]  
Output: 7

Constraints:

- The depth of the n-ary tree is less than or equal to 1000.
- The total number of nodes is between [1, 104].

Community Solutions (Python)

• WalkCC

```
class Solution:
 def diameter(self, root: 'Node') -> int:
 ans = 0

 def maxDepth(root: 'Node') -> int:
 """Returns the maximum depth of the subtree rooted
at 'root'."""
 nonlocal ans
 maxSubDepth1 = 0
 maxSubDepth2 = 0
 for child in root.children:
 maxSubDepth = maxDepth(child)
 if maxSubDepth > maxSubDepth1:
 maxSubDepth1 = maxSubDepth
 elif maxSubDepth > maxSubDepth2:
 maxSubDepth2 = maxSubDepth
 ans = max(ans, maxSubDepth1 + maxSubDepth2)
 return 1 + maxSubDepth1

 maxDepth(root)

 return ans
```

LeetCode

```
"""
Definition for a Node.
class Node:
 def __init__(self, val=None, children=None):
 self.val = val
 self.children = children if children is not None
 else []
"""

class Solution:

 def diameter(self, root: 'Node') -> int:
 """
 :type root: 'Node'
 :rtype: int
 """

 def dfs(root):
 if root is None:
 return 0
 nonlocal ans

 m1 = m2 = 0
 for child in root.children:
 t = dfs(child)
 if t > m1:
 m2, m1 = m1, t
 elif t > m2:
 m2 = t
 ans = max(ans, m1 + m2)
 return 1 + m1
```



```
ans = 0
dfs(root)
return ans

• SimplyLeet

Definition for a Node.
class Node:
def __init__(self, val=None, children=None):
self.val = val
self.children = children if children is not None else []

class Solution:
 def diameter(self, root: 'Node') -> int:
 # Global variable to keep track of the maximum
 # diameter found so far.
 self.max_diameter = 0

 def dfs(node):
 if not node:
 return 0 # Base case: if the node is
 None, return depth 0

 # Variables to hold the top two maximum
 # depths from the children.
 max_depth1, max_depth2 = 0, 0

 # Iterate over each child to compute their
 # maximum depths.
 for child in node.children:
```

```
child_depth = dfs(child) # Recurse on the
child

Update the two largest depths
if child_depth > max_depth1:
 max_depth2 = max_depth1
 max_depth1 = child_depth
elif child_depth > max_depth2:
 max_depth2 = child_depth

Update the maximum diameter. The candidate
is the sum of the two deepest paths.
self.max_diameter = max(self.max_diameter,
max_depth1 + max_depth2)

Return the maximum depth from this node.
return max_depth1 + 1

dfs(root)
return self.max_diameter

• LC.ca

"""
Definition for a Node.
class Node:
 def __init__(self, val=None, children=None):
 self.val = val
 self.children = children if children is not None
 else []
"""

class Solution:
```

```
def diameter(self, root: 'Node') -> int:
 """
 :type root: 'Node'
 :rtype: int
 """

 def dfs(root):
 if root is None:
 return 0
 nonlocal ans

 m1 = m2 = 0
 for child in root.children:
 t = dfs(child)
 if t > m1:
 m2, m1 = m1, t
 elif t > m2:
 m2 = t
 ans = max(ans, m1 + m2)

 return 1 + m1

ans = 0
dfs(root)
return ans
```

Trees & BST

#1650 Lowest Common Ancestor of a Binary Tree III MED

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Hash Table · Tree · Two Pointers

Lists: Premium 98

Time Complexity: O(h), where h is the height of the tree.

Space Complexity: O(1)

Problem

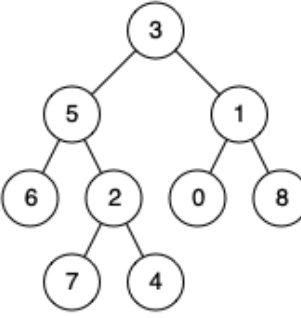
Given two nodes of a binary tree p and q, return their lowest common ancestor (LCA).

Each node will have a reference to its parent node. The definition for Node is below:

class Node {
 public int val;
 public Node left;
 public Node right;
 public Node parent;
}

According to the definition of LCA on Wikipedia: "The lowest common ancestor of two nodes p and q in a tree T is the lowest node that has both p and q as descendants (where we allow a node to be a descendant of itself)."

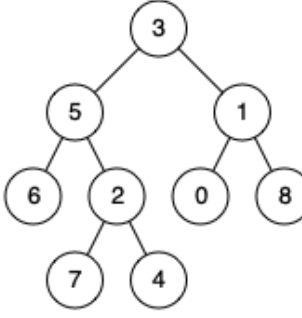
Example 1:



Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 1

Output: 3

Explanation: The LCA of nodes 5 and 1 is 3.



Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 4

Output: 5

Explanation: The LCA of nodes 5 and 4 is 5 since a node can be a descendant of itself according to the LCA definition.

Example 3:

Input: root = [1,2], p = 1, q = 2

Output: 1

Constraints:

• The number of nodes in the tree is in the range [2, 105].

• -109 <= Node.val <= 109

• All Node.val are unique.

• p != q

• p and q exist in the tree.

Brute Force (Python)

# Brute Force Approach

class Solution:

def lowestCommonAncestor(self, p, q):

# Brute Force - collect inorder, then process sorted list

vals = []

def collect(node):

if not node: return

collect(node.left)

vals.append(node.val)

collect(node.right)

collect(p)

return vals[0] if vals else 0

Community Solutions (Python)

• WalkCC

```
class Solution:
 # Same as 160. Intersection of Two Linked Lists
 def lowestCommonAncestor(self, p: 'Node', q: 'Node') -> 'Node':
 a = p
 b = q

 while a != b:
 a = a.parent if a else q
 b = b.parent if b else p

 return a

• LeetDoocs

"""
Definition for a Node.
class Node:
 def __init__(self, val):
 self.val = val
 self.left = None
 self.right = None
 self.parent = None
"""
```

```
class Solution:
 def lowestCommonAncestor(self, p: "Node", q: "Node")
 -> "Node":
 vis = set()
 node = p
 while node:
 vis.add(node)
 node = node.parent
 node = q
 while node not in vis:
 node = node.parent
 return node

• SimplyLeet

Definition for a Node.
class Node:
def __init__(self, val=0, left=None, right=None,
parent=None):
self.val = val
self.left = left
self.right = right
self.parent = parent

def lowestCommonAncestor(p, q):
 # Compute the depth for node p
```

```
def getDepth(node):
 depth = 0
 while node:
 depth += 1
 node = node.parent
 return depth

depthP = getDepth(p)
depthQ = getDepth(q)

Bring both nodes to the same depth level
while depthP > depthQ:
 p = p.parent
 depthP -= 1
while depthQ > depthP:
 q = q.parent
 depthQ -= 1

Traverse up until we find the common ancestor
while p != q:

 p = p.parent
 q = q.parent

return p
```

• LC.ca

Trees & BST

p.2782

```
"""
Definition for a Node.
class Node:
 def __init__(self, val):
 self.val = val
 self.left = None
 self.right = None
 self.parent = None
"""

class Solution:
 def lowestCommonAncestor(self, p: 'Node', q: 'Node')
 -> 'Node':
 a, b = p, q
 while a != b:
 a = a.parent if a.parent else q
 b = b.parent if b.parent else p
 return a

#####

ref: https://iamageek.medium.com/leetcode-1650-lowest-c
ommon-ancestor-of-a-binary-tree-iii-6d008b93376c

def lowestCommonAncestor(self, a: 'Node', b: 'Node') ->
'Node':
 ancestors = set()

 while a is not None:
 ancestors.add(a)
 a = a.parent
```

Trees & BST

p.2783

```
while b is not None:
 if b in ancestors:
 return b
 b = b.parent

or another one
def lowestCommonAncestor(self, a: 'Node', b: 'Node') ->
'Node':
 pointerA, pointerB = a, b

 while pointerA != pointerB:

 pointerA = pointerA.parent if pointerA else b
 pointerB = pointerB.parent if pointerB else a

 return pointerA
```

Trees & BST

p.2784

Trees & BST

## #124 Binary Tree Maximum Path Sum

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

Dynamic Programming · Tree · DFS

Lists: Grind 169

Time Complexity:  $O(n)$ , where  $n$  is the number of nodes in the tree, since each node is visited exactly once. Space Complexity:  $O(n)$  in the worst-case scenario due to the recursion stack ( $O(H)$  where  $H$  is the height of the tree).

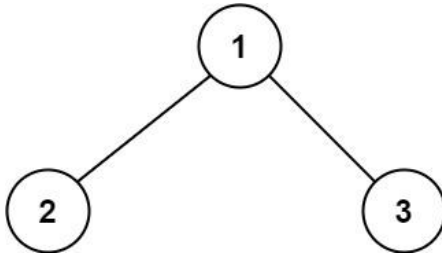
### Problem

A path in a binary tree is a sequence of nodes where each pair of adjacent nodes in the sequence has an edge connecting them. A node can only appear in the sequence at most once. Note that the path does not need to pass through the root.

The path sum of a path is the sum of the node's values in the path.

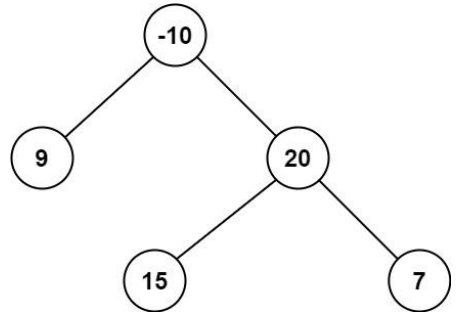
Given the root of a binary tree, return the maximum path sum of any non-empty path.

Example 1:



Input: root = [1,2,3]  
Output: 6  
Explanation: The optimal path is 2 -> 1 -> 3 with a path sum of 2 + 1 + 3 = 6.

Example 2:



Input: root = [-10,9,20,null,null,15,7]  
Output: 42  
Explanation: The optimal path is 15 -> 20 -> 7 with a path sum of 15 + 20 + 7 = 42.

### Constraints:

- The number of nodes in the tree is in the range  $[1, 3 \times 10^4]$ .
- $-1000 \leq \text{Node.val} \leq 1000$

### Community Solutions (Python)

• WalkCC

Trees & BST

p.2785

Trees & BST

p.2786

Trees & BST

p.2787

```
class Solution:
 def maxPathSum(self, root: TreeNode | None) -> int:
 ans = -math.inf

 def maxPathSumDownFrom(root: TreeNode | None) -> int:
 """
 Returns the maximum path sum starting from the
 current root, where
 root.val is always included.
 """
 nonlocal ans

 if not root:
 return 0

 l = max(0, maxPathSumDownFrom(root.left))
 r = max(0, maxPathSumDownFrom(root.right))
 ans = max(ans, root.val + l + r)
 return root.val + max(l, r)

 maxPathSumDownFrom(root)
 return ans
```

• LeetDoocs

Trees & BST

p.2788

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def maxPathSum(self, root: Optional[TreeNode]) ->
 int:
 def dfs(root: Optional[TreeNode]) -> int:
 if root is None:
 return 0
 left = max(0, dfs(root.left))
 right = max(0, dfs(root.right))
 nonlocal ans
 ans = max(ans, root.val + left + right)
 return root.val + max(left, right)

 ans = -inf
 dfs(root)
 return ans
```

• SimplyLeet

Trees & BST

p.2789

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val
 self.left = left
 self.right = right

class Solution:
 def maxPathSum(self, root: TreeNode) -> int:
 # Initialize global maximum path sum with a very
 small number

 self.global_max = float('-inf')

 def dfs(node: TreeNode) -> int:
 if not node:
 return 0

 # Recursively calculate max gain from left
 and right children, discard negative gains
 left_gain = max(dfs(node.left), 0)
 right_gain = max(dfs(node.right), 0)
```

Trees & BST

p.2790

```
Current maximum including both left and
right gains
current_max = node.val + left_gain +
right_gain

Update global maximum path sum if
current_sum is higher
self.global_max = max(self.global_max,
current_max)

Return the node's value plus the max of one
child gain (only one side can be chosen for parent path)
return node.val + max(left_gain, right_gain)

dfs(root)

return self.global_max
```

• LC.ca

```
Definition for a binary tree node.
class TreeNode:
def __init__(self, val=0, left=None, right=None):
self.val = val
self.left = left
self.right = right
class Solution:
 def maxPathSum(self, root: Optional[TreeNode]) ->
int:
 def dfs(root: Optional[TreeNode]) -> int:
 if root is None:
```

Trees & BST

p.2791

```
return 0
left = max(0, dfs(root.left))
right = max(0, dfs(root.right))
nonLocal ans
ans = max(ans, root.val + left + right)
return root.val + max(left, right)

ans = -inf
dfs(root)
return ans
```

Trees & BST

p.2792

Trees & BST

## #297 Serialize and Deserialize Binary Tree

HAR

LeetDoocs · SimplyLeet · WalkCC · LeetCode

String · Tree · DFS · BFS · Design

Lists: Grind 169

Time Complexity: O(n) for both serialization and deserialization, where n is the number of nodes in the tree. Space Complexity: O(n) due to the storage used for the serialized string and additional da

### Problem

Serialization is the process of converting a data structure or object into a sequence of bits so that it can be stored in a file or memory buffer, or transmitted across a network connection link to be reconstructed later in the same or another computer environment.

Design an algorithm to serialize and deserialize a binary tree. There is no restriction on how your serialization/deserialization algorithm should work. You just need to ensure that a binary tree can be serialized to a string and this string can be deserialized to the original tree structure.

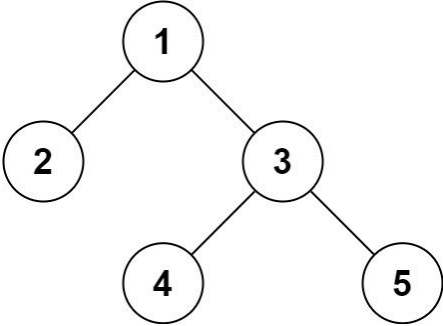
Clarification: The input/output format is the same as how LeetCode serializes a binary tree. You do not necessarily need to follow this format, so please be

Trees & BST

p.2793

creative and come up with different approaches yourself.

Example 1:



Input: root = [1,2,3,null,null,4,5]  
Output: [1,2,3,null,null,4,5]

Example 2:

Input: root = []  
Output: []

Constraints:

- The number of nodes in the tree is in the range [0, 104].

Trees & BST

p.2794

- -1000 <= Node.val <= 1000

### Community Solutions (Python)

- WalkCC

```
class Codec:
 def serialize(self, root: 'TreeNode') -> str:
 """Encodes a tree to a single string."""
 if not root:
 return ''

 s = ''
 q = collections.deque([root])

 while q:

 node = q.popleft()
 if node:
 s += str(node.val) + ' '
 q.append(node.left)
 q.append(node.right)
 else:
 s += 'n '

 return s
```

Trees & BST

p.2795

```
def deserialize(self, data: str) -> 'TreeNode':
 """Decodes your encoded data to tree."""
 if not data:
 return None

 vals = data.split()
 root = TreeNode(vals[0])
 q = collections.deque([root])

 for i in range(1, len(vals), 2):

 node = q.popleft()
 if vals[i] != 'n':
 node.left = TreeNode(vals[i])
 q.append(node.left)
 if vals[i + 1] != 'n':
 node.right = TreeNode(vals[i + 1])
 q.append(node.right)

 return root
```

- LeetDoocs

```
Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
self.val = x
self.left = None
self.right = None
```

```
class Codec:
 def serialize(self, root):
```

Trees & BST

p.2796

```
"""Encodes a tree to a single string.

:type root: TreeNode
:rtype: str
"""
if root is None:
 return ""
q = deque([root])
ans = []
while q:
```

```
node = q.popleft()
if node:
 ans.append(str(node.val))
 q.append(node.left)
 q.append(node.right)
else:
 ans.append("#")
return ",".join(ans)
```

```
def deserialize(self, data):
```

```
"""Decodes your encoded data to tree.

:type data: str
:rtype: TreeNode
"""
if not data:
 return None
vals = data.split(",")
root = TreeNode(int(vals[0]))
q = deque([root])
```

Trees & BST

p.2797

```
i = 1
while q:
 node = q.popleft()
 if vals[i] != "#":
 node.left = TreeNode(int(vals[i]))
 q.append(node.left)
 i += 1
 if vals[i] != "#":
 node.right = TreeNode(int(vals[i]))
 q.append(node.right)
```

```
i += 1
return root
```

```
Your Codec object will be instantiated and called as
such:
codec = Codec()
codec.deserialize(codec.serialize(root))
```

- SimplyLeet

```
Definition for a binary tree node.
class TreeNode:
 def __init__(self, val=0, left=None, right=None):
 self.val = val # value of the node
 self.left = left # left child
 self.right = right # right child

class Codec:
 # Encodes a tree to a single string using BFS.
 def serialize(self, root):
```

Trees & BST

p.2798

```
if not root:
 return "" # Return empty string for an empty
tree

result = []
queue = [root] # Initialize the queue with the
root node
while queue:
 node = queue.pop(0)
 if node:
 result.append(str(node.val)) # Append the
value of the node
 queue.append(node.left) # Enqueue left
child
 queue.append(node.right) # Enqueue right
child
```

```
else:
 result.append("null") # Placeholder for
null node
return ",".join(result) # Join list with commas
to form the string

Decodes the encoded data to reconstruct the tree
using BFS.
def deserialize(self, data):
 if not data:
 return None # Return None for empty string
 nodes = data.split(",")
 root = TreeNode(int(nodes[0])) # Create root node
from first element
```

Trees & BST

p.2799

```
queue = [root]
i = 1 # Index for traversing nodes list
while queue:
 current = queue.pop(0)
 if i < len(nodes) and nodes[i] != "null":
 current.left = TreeNode(int(nodes[i]))
 queue.append(current.left)
 i += 1
 if i < len(nodes) and nodes[i] != "null":
 current.right = TreeNode(int(nodes[i]))
 queue.append(current.right)
 i += 1
return root
```

```
• LC.ca

Definition for a binary tree node.
class TreeNode(object):
def __init__(self, x):
self.val = x
self.left = None
self.right = None

class Codec:
 def serialize(self, root):
```

```
"""Encodes a tree to a single string.

:type root: TreeNode
:rtype: str
"""

if root is None:
 return ''
res = []

def preorder(root):
 if root is None:
 res.append("#,")
 return
 res.append(str(root.val) + ",")
 preorder(root.left)
 preorder(root.right)

preorder(root)
return ''.join(res)

def deserialize(self, data):
 """Decodes your encoded data to tree.

 :type data: str
 :rtype: TreeNode
 """
 if not data:
 return None
 vals = data.split(',')
```

```
def inner():
 first = vals.pop(0)
 if first == '#':
 return None
 return TreeNode(int(first), inner(), inner())
seems using a constructor __init__(val,
left, right)

return inner()

Your Codec object will be instantiated and called as
such:
ser = Codec()
deser = Codec()
ans = deser.deserialize(ser.serialize(root))

#####

from collections import deque

bfs, each level based
class Codec:
 def serialize(self, root):
 """Encodes a tree to a single string.

 :type root: TreeNode
 :rtype: str
 """
 ret = []
 queue = deque([root])
```

```
while queue:
 top = queue.popleft()
 if not top:
 ret.append("None")
 continue
 else:
 ret.append(str(top.val))
 queue.append(top.left)
 queue.append(top.right)
return ",".join(ret)

def deserialize(self, data):
 """Decodes your encoded data to tree.

 :type data: str
 :rtype: TreeNode
 """
 data = data.split(",")
 if data[0] == "None":
 return None

 root = TreeNode(int(data[0]))
 queue = deque([root])
 i = 0
 while queue and i < len(data):
 top = queue.popleft()
 i += 1
 left = right = None
 if i < len(data) and data[i] != "None":
 left = TreeNode(int(data[i]))
 queue.append(left)
```

```
i += 1
if i < len(data) and data[i] != "None":
 right = TreeNode(int(data[i]))
 queue.append(right)

top.left = left
top.right = right

return root

Your Codec object will be instantiated and called as
such:
codec = Codec()
codec.deserialize(codec.serialize(root))
```

>> Quick Review — Trees & BST

37 questions · insights · complexity · solution

#100 Same Tree [Easy]

Key Insights

- Use recursion to compare nodes in both trees.
- If both nodes being compared are null, they are the same.
- If one node is null while the other is not, the trees differ.
- Check if current nodes' values match, then recursively verify the left and right subtrees.

Space & Time

Time Complexity: O(n), where n is the number of nodes in the trees, since every node is visited once.  
Space Complexity: O(n) in the worst-case (unbalanced tree), due to recursion call stack usage. In a balanced tree, the space would be O(log n).

Solution

The problem can be solved using a recursive approach (Depth-First Search) where we compare the current nodes from both trees. If the current nodes are both null, they match; if one is null, then the trees are not identical. For non-null nodes, check if the values are

equal. Then, recursively apply the same logic on the left and right children of the nodes. This approach ensures that both the structure and the node values are identical across the two trees.

#101 Symmetric Tree [Easy]

Key Insights

- Use recursion (or iteration) to compare pairs of nodes.
- For two trees (or two nodes) to be mirror images:
- Their root values must be the same.
- The left subtree of one must match the right subtree of the other.
- The right subtree of one must match the left subtree of the other.
- Alternatively, iterative approaches using a queue can be applied to compare the nodes in pairs.

Space & Time

Time Complexity: O(n), where n is the number of nodes in the tree, since we traverse each node once.  
Space Complexity: O(h) for recursive call stack (worst-case O(n) for a skewed tree) or O(n) for the iterative approach using a queue.

Solution

We solve the problem using recursion by defining a helper function that takes two nodes and checks if they are mirrors of each other. The helper function:

– Returns true if both nodes are null. – Returns false if one node is null or if their values differ. – Recursively compares the left child of the first node with the right child of the second node, and vice-versa. This method ensures each comparison verifies symmetry at every level. A similar logic can be applied iteratively by using a queue to compare pairs of nodes.

#104 Maximum Depth of Binary Tree [Easy]

Key Insights

- Use a recursive strategy (DFS) to traverse the binary tree.
- At each node, compute the maximum depth of its left and right subtree.
- The maximum depth at the current node is 1 (current node) plus the maximum of the left and right subtree depths.
- Alternatively, an iterative approach using BFS (level-order traversal) can be used.

Space & Time

Time Complexity: O(n), where n is the number of nodes, since every node is visited once.  
Space Complexity: O(h) for recursion stack in DFS, where h is the height of the tree (worst-case O(n) for a skewed tree); O(n) for BFS in the worst-case scenario if the tree is balanced.

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| <div>Solution</div> <div>The problem is approached using Depth-First Search (DFS) with recursion. The idea is to visit every node and calculate the maximum depth of the left and right subtrees, then add 1 for the current node. This method naturally handles the base case when a null node (empty tree) is reached by returning 0. Alternatively, a Breadth-First Search (BFS) approach can determine the tree level by level, with the total number of levels equaling the maximum depth. Key data structures include recursion call stack for DFS or a queue for BFS.</div> <div>#108 Convert Sorted Array to Binary Search Tree [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>The array is sorted in ascending order, which allows the use of a divide and conquer strategy.</li> <li>Choosing the middle element of the array as the root ensures that the left and right subtrees are roughly equal in size, yielding a balanced BST.</li> <li>A recursive approach is natural for building the tree: the base case is when the subarray is empty.</li> <li>The algorithm splits the array into two halves to recursively build the left and right subtrees.</li> </ul> </div> <div>Space &amp; Time</div> <div>Trees &amp; BSTp.2809</div> | <div>Time Complexity: O(n) – Every element is processed exactly once.</div> <div>Space Complexity: O(n) – O(n) space is used to store the BST, and additional O(log n) space may be used for the recursion stack.</div> <div>Solution</div> <div>We use a divide and conquer approach with recursion. The main idea is to select the middle element of the current subarray as the root node so that the left subarray elements form the left subtree and the right subarray elements form the right subtree, ensuring the tree is balanced. The recursion continues until the subarray is empty, which serves as the base case. This strategy naturally maintains the BST property since all elements in the left subarray are smaller than the root, and all in the right subarray are larger.</div> <div>#110 Balanced Binary Tree [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Use a bottom-up DFS (Depth-First Search) traversal to compute the height of each subtree.</li> <li>If any subtree is found unbalanced, propagate an error indicator (e.g., -1) upward, allowing early termination.</li> <li>An empty tree is considered balanced.</li> <li>The key check is to ensure that for each node, the absolute difference between the heights of its left and right subtrees is no more than 1.</li> </ul> </div> <div>Trees &amp; BSTp.2810</div> | <div>Space &amp; Time</div> <div>Time Complexity: O(n) where n is the number of nodes in the tree, since we visit each node once.</div> <div>Space Complexity: O(h) where h is the height of the tree, which corresponds to the recursion stack (O(n) in the worst-case of a skewed tree).</div> <div>Solution</div> <div>The solution uses a recursive DFS approach. A helper function recursively computes the height of a subtree. If a subtree is found to be unbalanced (the height difference between its left and right subtrees exceeds 1), the helper returns a special value (e.g., -1) to indicate that the subtree is unbalanced. This value is then propagated up the recursion to terminate further unnecessary computations. The main function checks the result of the helper function and returns true if the tree is balanced (i.e., the helper did not return -1) and false otherwise.</div> <div>#112 Path Sum [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Use a depth-first search (DFS) strategy to traverse the tree from the root to the leaves.</li> <li>At each node, subtract the node's value from the targetSum.</li> <li>Check if the node is a leaf; if yes and the remaining targetSum equals the node's value then a valid path is</li> </ul> </div> <div>Trees &amp; BSTp.2811</div>                                                                                                                                                                                          |
| <div>found.</div> <div> <ul style="list-style-type: none"> <li>If the tree is empty, return false as no path exists.</li> </ul> </div> <div>Space &amp; Time</div> <div>Time Complexity: O(N) where N is the number of nodes, since every node may need to be visited.</div> <div>Space Complexity: O(H) where H is the height of the tree, accounting for the recursion stack (worst-case O(N) for a skewed tree).</div> <div>Solution</div> <div>The approach uses a recursive depth-first search. The algorithm subtracts the current node's value from targetSum as it moves down the tree. When it reaches a leaf, it checks if the modified targetSum is equal to the leaf's value. If yes, it returns true. The recursion continues until either a valid path is found or all paths are exhausted. The primary data structure used here is the call stack for recursion.</div> <div>#226 Invert Binary Tree [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>The problem can be solved using recursion (depth-first search) by swapping the children of every node.</li> <li>Alternatively, an iterative approach (using a queue for breadth-first search) can be used.</li> </ul> </div> <div>Trees &amp; BSTp.2812</div>                        | <div> <ul style="list-style-type: none"> <li>The recursion terminates when a null node is encountered.</li> <li>The operation is performed in-place, modifying the original tree structure.</li> </ul> </div> <div>Space &amp; Time</div> <div>Time Complexity: O(n) where n is the number of nodes, since each node is visited once.</div> <div>Space Complexity: O(h) where h is the height of the tree, which is the recursion stack space in the recursive approach. In worst-case (skewed tree), this could be O(n).</div> <div>Solution</div> <div>We solve the problem using recursion (depth-first search). The approach is to: <ul style="list-style-type: none"> <li>Check the base case: if the node is null, simply return null.</li> <li>Recursively invert the left subtree and the right subtree.</li> <li>Swap the left and right children of the current node.</li> <li>Return the current node after the swap.</li> </ul> The recursive approach elegantly handles all nodes and ensures the inversion process covers the complete tree.</div> <div>#270 Closest Binary Search Tree Value [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>BST property allows us to traverse the tree efficiently by comparing the target with node values.</li> </ul> </div> <div>Trees &amp; BSTp.2813</div>                                                 | <div> <ul style="list-style-type: none"> <li>We can iteratively traverse the tree, moving left if the target is smaller than the current node's value and moving right if greater.</li> <li>At each node, calculate the absolute difference between node's value and the target, and update the answer if the difference is smaller—or equal with a smaller candidate value.</li> <li>This approach minimizes the search area by leveraging the inherent ordering in BSTs.</li> </ul> </div> <div>Space &amp; Time</div> <div>Time Complexity: O(h) on average, where h is the height of the tree (O(log n) for a balanced tree, and O(n) in the worst-case scenario for a skewed tree).</div> <div>Space Complexity: O(1) for an iterative solution (or O(h) if using recursion due to the call stack).</div> <div>Solution</div> <div>We solve the problem by performing an iterative traversal of the BST. Starting at the root, we maintain a variable to store the value closest to the target. For every node encountered, we: <ul style="list-style-type: none"> <li>Compute the absolute difference between the node's value and the target.</li> <li>Update the stored closest value if the computed difference is less than the current smallest difference; if the difference is the same, choose the smaller node value.</li> <li>Depending on whether the target is less than or greater than the current node's value, move to the left or right child accordingly. This</li> </ul> </div> <div>Trees &amp; BSTp.2814</div> |
| <div>approach efficiently defines a search path that is consistent with the properties of a BST.</div> <div>#501 Find Mode in Binary Search Tree [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>The BST in-order traversal yields a sorted order, which helps in counting consecutive duplicates.</li> <li>A two-pass in-order traversal can be used: one pass to determine the maximum frequency (maxCount) and a second pass to collect all values that match this frequency.</li> <li>The solution can be implemented without extra space (apart from recursion stack) by leveraging constant space counters and state variables.</li> </ul> </div> <div>Space &amp; Time</div> <div>Time Complexity: O(n), where n is the number of nodes, because each node is processed twice (once in each traversal).</div> <div>Space Complexity: O(h) where h is the height of the tree due to the recursion stack. O(1) extra space is used aside from this.</div> <div>Solution</div> <div>We perform an in-order traversal of the BST twice. The first traversal determines the maximum frequency of any value in the BST by comparing the current node's value with the previously visited node. Since the</div> <div>Trees &amp; BSTp.2815</div>        | <div>in-order traversal processes nodes in sorted order, duplicate values are consecutive, making the counting simple. After calculating maxCount, we reset our state and perform a second in-order traversal to collect all node values with frequency equal to maxCount. The key trick is to leverage the BST property (sorted in-order traversal) to count frequencies in one pass without using additional data structures like hash maps.</div> <div>#543 Diameter of Binary Tree [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Use depth-first search (DFS) to explore the tree.</li> <li>The diameter at a node is the sum of the depths of its left and right subtrees.</li> <li>Track the maximum diameter found during the DFS traversal.</li> <li>Recursively calculating the depth of subtrees provides an efficient solution.</li> </ul> </div> <div>Space &amp; Time</div> <div>Time Complexity: O(n), where n is the number of nodes in the tree.</div> <div>Space Complexity: O(n), in the worst-case scenario due to the recursion stack (skewed tree).</div> <div>Solution</div> <div>The solution employs a DFS traversal of the binary tree. At every node in the tree, compute the depth of the left</div> <div>Trees &amp; BSTp.2816</div>                                                                                               | <div>and right subtrees. The sum of these depths gives the potential diameter (longest path through that node). Update a global maximum diameter variable if this sum exceeds the current maximum. Finally, return the maximum diameter after completing the DFS.</div> <div>#572 Subtree of Another Tree [Easy]</div> <div>Key Insights <ul style="list-style-type: none"> <li>Use Depth-First Search (DFS) to visit each node in the main tree.</li> <li>At each node, check if the subtree starting from that node is identical to subRoot.</li> <li>A helper function is used to compare the structure and node values of two trees.</li> <li>Consider recursion as the primary approach to traverse and compare trees.</li> <li>Early exit when a mismatch is found can optimize average-case performance.</li> </ul> </div> <div>Space &amp; Time</div> <div>Time Complexity: O(m * n) in the worst case, where m is the number of nodes in the main tree and n is the number of nodes in the subRoot tree.</div> <div>Space Complexity: O(max(depth of main tree, depth of subRoot)) due to recursive call stack.</div> <div>Solution</div> <div>Trees &amp; BSTp.2817</div>                                                                                                                                                                                                                                                                                                                                                       |



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| <p>We solve the problem by performing a DFS traversal on the main tree. For each visited node, a helper function checks whether the subtree starting at that node is identical to subRoot by comparing node values and recursively checking both left and right subtrees. This method leverages recursion effectively and provides early termination if mismatches occur, allowing for efficient traversal and comparison.</p> <p><b>#98 Validate Binary Search Tree [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Every node must adhere to a range: all nodes in the left subtree should be less than the current node, and all nodes in the right subtree should be greater.</li> <li>• A recursive depth-first search approach works well by passing down the valid range (lower and upper bounds) for each node.</li> <li>• Using the limits (negative and positive infinity) simplifies handling edge cases at the tree root.</li> <li>• An in-order traversal would alternatively yield a sorted order for node values, but updating ranges directly is more intuitive for recursive implementation.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(N), where N is the number of nodes, as each node is visited once.</p> <p>Space Complexity: O(H), where H is the height of the tree, due to the recursion stack.</p> <p>Trees &amp; BSTp.2818</p> | <p><b>Solution</b></p> <p>We use a recursive approach to validate the BST property. We maintain two variables, low and high, representing the valid range for the current node's value. For the left child, the valid range becomes (low, node.val) and for the right child, it becomes (node.val, high). By checking if each node's value falls within its respective range, we can ensure that the tree satisfies the BST properties.</p> <p><b>#102 Binary Tree Level Order Traversal [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use Breadth-First Search (BFS) to traverse the tree.</li> <li>• A queue data structure is ideal for maintaining the order of nodes at each level.</li> <li>• For every level, process all nodes currently in the queue and add their children for the next level.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n), where n is the number of nodes, because each node is processed once.</p> <p>Space Complexity: O(n), which accounts for the space used by the queue in the worst-case scenario.</p> <p><b>Solution</b></p> <p>This solution leverages a BFS approach using a queue. The algorithm starts by enqueueing the root node. Then, it enters a loop that continues until the queue is empty.</p> <p>Trees &amp; BSTp.2819</p>                                                                                       | <p>For each iteration—representing one level of the tree—it computes the number of nodes at that level (levelSize), processes these nodes by dequeuing them, and collects their values. Their non-null children are then enqueued. The list of values for each level is added to the final result list. Key data structures include the queue (for managing nodes) and a list (to store level values).</p> <p><b>#103 Binary Tree Zigzag Level Order Traversal [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a breadth-first search (BFS) approach to traverse the tree level by level.</li> <li>• Alternate the direction of traversal on each level to achieve the zigzag pattern.</li> <li>• A flag or counter can be used to determine the order of nodes for the current level.</li> <li>• Using a double-ended data structure or simply reversing the list at every alternate level is an effective approach.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n), where n is the number of nodes, because each node is visited exactly once.</p> <p>Space Complexity: O(n), as in the worst-case scenario (e.g., a complete binary tree) the queue used for BFS may hold O(n) nodes at once.</p> <p>Trees &amp; BSTp.2820</p>                                  |
| <p><b>Solution</b></p> <p>We can solve this problem using a breadth-first search (BFS). Start by initializing a queue to hold nodes for each level. For each level:</p> <ul style="list-style-type: none"> <li>- Determine the number of nodes at the current level. - Iterate over these nodes, appending their children for the next level. - Depending on a flag (or level count), either append node values in the natural order (left to right) or reverse the order (right to left) before adding to the result. Key data structures:</li> <li>- A queue for BFS traversal. - A list (or similar container) to store current level values. The main trick is toggling the order of insertion or reversing the list at alternate levels to achieve the "zigzag" pattern.</li> </ul> <p><b>#105 Construct Binary Tree from Preorder and Inorder Traversal [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Preorder traversal always starts with the root node.</li> <li>• Inorder traversal splits the tree into left subtree and right subtree based on the root.</li> <li>• A hash table (map/dictionary) can be used to quickly locate the index of the root in the inorder array.</li> <li>• Use recursion to construct left and right subtrees.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Trees &amp; BSTp.2821</p>                                                | <p>Time Complexity: O(n) since we process every node exactly once.</p> <p>Space Complexity: O(n) due to the recursion stack and the hash map used for inorder indices.</p> <p><b>Solution</b></p> <p>We solve the problem using a recursive divide-and-conquer approach. First, create a hash map to store each value's index in the inorder array for quick lookups. The preorder array always provides the current root. Use the inorder array to divide the tree into left and right subtrees. Recursively construct the subtrees by adjusting the preorder index and the boundaries of the inorder array. The algorithm leverages recursion and a hash map to achieve O(n) time complexity.</p> <p><b>#199 Binary Tree Right Side View [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• The problem is essentially about capturing the last node at each level (or the rightmost node) in a binary tree.</li> <li>• A Breadth-First Search (BFS) approach naturally works by processing nodes level by level.</li> <li>• Alternatively, Depth-First Search (DFS) can be adapted by prioritizing the right subtree first, ensuring that the first node encountered at each depth is the visible one.</li> <li>• The number of nodes is capped at 100, so both approaches are efficient.</li> </ul> <p>Trees &amp; BSTp.2822</p>                                                          | <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) - we visit every node in the tree.</p> <p>Space Complexity: O(n) - in the worst case, the queue (or call stack for DFS) may hold all nodes of the tree.</p> <p><b>Solution</b></p> <p>We can solve this problem using BFS (level-order traversal). The idea is to traverse the tree level by level using a queue. For each level, the last element added to the level represents the rightmost view from that level. We then append that element's value to our result list. Alternatively, a DFS approach can be used where we visit the right subtree first. By keeping track of the current depth and checking if it's the first time we've encountered this depth, we can add the first node visited at that depth (which will be the rightmost one) to the result.</p> <p>In both approaches, appropriate data structures (a queue for BFS or recursion with a list for DFS) are used to store nodes and track the traversal state.</p> <p><b>#230 Kth Smallest Element in a BST [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• An in-order traversal of a BST visits nodes in increasing order.</li> <li>• Use a counter to keep track of the number of nodes visited.</li> </ul> <p>Trees &amp; BSTp.2823</p>              |
| <ul style="list-style-type: none"> <li>• As soon as the counter reaches k, the current node's value is the kth smallest.</li> <li>• For frequent modifications (insertions/deletions) and queries, consider augmenting the BST with subtree counts.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n) in the worst-case scenario (when k ≈ n), but on average the traversal may stop early.</p> <p>Space Complexity: O(h) where h is the height of the tree (O(n) in the worst-case for a skewed tree).</p> <p><b>Solution</b></p> <p>We perform an in-order traversal of the BST which visits nodes in ascending order. During the traversal, we maintain a counter (or decrement k) to check when we have encountered the kth smallest element. When the counter reaches 0 (or k becomes 0), we capture the current node's value as our answer and return it. This approach is both simple and efficient for the given constraints.</p> <p><b>#235 Lowest Common Ancestor of a Binary Search Tree [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Utilize the BST property: for any node, values in the left subtree are less and values in the right subtree are greater.</li> </ul> <p>Trees &amp; BSTp.2824</p>                                                                                                                                          | <ul style="list-style-type: none"> <li>• If both p and q have values less than the current node, the LCA must lie in the left subtree.</li> <li>• If both p and q have values greater than the current node, the LCA must lie in the right subtree.</li> <li>• When one node value is less than and the other is greater than (or equal to) the current node, the current node is the lowest common ancestor.</li> <li>• An iterative approach is efficient and uses constant extra space.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(h), where h is the height of the tree (O(log n) for balanced trees and O(n) in the worst-case skewed tree).</p> <p>Space Complexity: O(1) with an iterative approach (O(h) if using recursion due to call stack).</p> <p><b>Solution</b></p> <p>The solution leverages the BST properties. Algorithm:</p> <ul style="list-style-type: none"> <li>- Start at the root node. - While the current node exists: If both p and q have values less than the current node's value, move to the left child. If both p and q have values greater than the current node's value, move to the right child. Otherwise, the current node is the split point where p and q diverge, so it is the lowest common ancestor. - Return the current node.</li> </ul> <p>This approach efficiently finds the LCA using a single traversal from the root.</p> <p>Trees &amp; BSTp.2825</p> | <p><b>#236 Lowest Common Ancestor of a Binary Tree [Medium]</b></p> <p><b>Key Insights</b></p> <ul style="list-style-type: none"> <li>• Use a depth-first search (DFS) traversal to explore the tree.</li> <li>• If the current node is either p or q, it could be the LCA.</li> <li>• Recursively find p and q in the left and right subtrees.</li> <li>• If both left and right recursive calls return a non-null value, then the current node is the LCA.</li> <li>• If only one side returns a non-null value, propagate that value up the recursion.</li> </ul> <p><b>Space &amp; Time</b></p> <p>Time Complexity: O(n), where n is the number of nodes in the tree since each node is visited once.</p> <p>Space Complexity: O(h), where h is the height of the tree due to the recursion stack (O(n) in the worst case of a skewed tree).</p> <p><b>Solution</b></p> <p>We solve the problem using recursion with depth-first search. The algorithm checks if the current node matches p or q. It then recursively searches the left and right subtrees. If both subtrees contain one of p or q, the current node is the LCA. Otherwise, the algorithm forwards the non-null result up the recursion. This approach naturally handles cases where one node is an ancestor of the other.</p> <p>Trees &amp; BSTp.2826</p> |



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| <div>#250 Count Univalue Subtrees <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Perform a postorder traversal (left, right, root) to determine whether each subtree is univalue.</li> <li>Treat null nodes as univalue to ease the recursion.</li> <li>A node's subtree is univalue if both left and right subtrees are univalue and, if they exist, their values equal the current node's value.</li> <li>Use a global or external counter that increments every time a univalue subtree is confirmed.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(n), where n is the number of nodes, since each node is visited once.</p> <p>Space Complexity: O(h), where h is the height of the tree, due to the recursion stack.</p> </div> <div> <div>Solution</div> <p>We use a recursive DFS approach (postorder traversal) to solve the problem. For each node, we recursively check its left and right subtrees to decide if they are univalue. By default, if a child is null, it is considered univalue. Then, we verify the current node's value against its non-null children. If both conditions hold (subtrees are univalue and matching values), we count the current node's subtree as univalue. This DFS checks in a bottom-up manner, ensuring that the properties</p> </div> <div>Trees &amp; BSTp.2827</div> | <div>required for a subtree to be univalue are maintained. The key trick is to return a boolean indicating if the subtree at each node is univalue and, while backtracking, increment the counter appropriately.</div> <div>#285 Inorder Successor in BST <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>In a BST, an in-order traversal yields nodes in ascending order.</li> <li>If the node p has a right subtree, the in-order successor is the left-most node in that right subtree.</li> <li>If p has no right subtree, the successor is one of its ancestors; traverse from the root towards p while keeping track of the last node that is greater than p.</li> <li>An iterative approach is efficient and avoids recursion stack overhead.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(h), where h is the height of the tree. In the worst-case (skewed tree), it can be O(n).</p> <p>Space Complexity: O(1) as only a constant amount of extra space is used.</p> </div> <div> <div>Solution</div> <p>We solve the problem using two cases:</p> <ul style="list-style-type: none"> <li>If node p has a right subtree, we simply find the left-most node in that right subtree.</li> <li>If node p does not have a right subtree, initiate a search from the BST</li> </ul> </div> <div>Trees &amp; BSTp.2828</div> | <div>root. As we traverse: If the current node's value is greater than p.val, the current node is a candidate for the successor; move to the left subtree to look for a smaller candidate. Otherwise, move to the right subtree. By the end of the traversal, the candidate (if exists) is the in-order successor. This approach benefits from the BST property, limiting traversal to O(h) time.</div> <div>#298 Binary Tree Longest Consecutive Sequence <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use a Depth-First Search (DFS) to traverse the binary tree.</li> <li>Pass the current node's value and the length of the current consecutive sequence in the recursive call.</li> <li>When a node's value is exactly one greater than its parent's value, increment the sequence length; otherwise, reset it to 1.</li> <li>Keep track of the maximum sequence length found during the traversal.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(N), where N is the number of nodes in the tree, since each node is visited once.</p> <p>Space Complexity: O(H), where H is the height of the tree. In the worst-case of a skewed tree, the recursion stack can be O(N).</p> </div> <div>Trees &amp; BSTp.2829</div>                                                                                                                  |
| <div>Solution</div> <p>The solution involves using DFS to explore every node in the binary tree while keeping track of the length of the consecutive sequence. For each node, if the node's value is one greater than its parent's value, we increment the sequence length; otherwise, we reset the sequence length to 1. During the traversal, a global variable is updated if the current sequence length exceeds the previously recorded maximum. This approach ensures that every possible consecutive path is considered.</p> <div>#314 Binary Tree Vertical Order Traversal <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use Breadth-First Search (BFS) to traverse the tree level by level.</li> <li>Track the column index for each node where root is column 0, left child is column-1, and right child is column+1.</li> <li>Group nodes by their column indices using a hash table (or dictionary).</li> <li>BFS traversal naturally handles left-to-right ordering within the same level.</li> <li>Finally, sort the keys (column indices) to generate results from leftmost column to rightmost column.</li> </ul> </div> <div> <div>Space &amp; Time</div> </div> <div>Trees &amp; BSTp.2830</div>                                                                                                                                 | <div>Time Complexity: O(n log n) in the worst-case due to sorting the column indices (n nodes in the worst-case if each node is on a unique column).</div> <div>Space Complexity: O(n) for storing the node information and the resultant structure.</div> <div> <div>Solution</div> <p>The solution uses BFS with a queue that stores nodes along with their corresponding column index. For each visited node, add its value to a dictionary keyed by its column index. When processing children, update the column index appropriately (left child: col-1, right child: col+1). After the traversal, sort the keys of the dictionary and compile the node values in order from smallest to largest column index. This approach guarantees that nodes at the same level will be processed in left-to-right order, matching the required output.</p> </div> <div>#333 Largest BST Subtree <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use a bottom-up DFS (post-order traversal) to process each node.</li> <li>For each node, determine whether its left and right subtrees are BSTs.</li> <li>Maintain additional data per subtree: size (number of nodes), minimum value, and maximum value.</li> <li>A subtree rooted at the current node is a BST if:</li> </ul> </div> <div>Trees &amp; BSTp.2831</div>                                          | <ul style="list-style-type: none"> <li>Its left subtree is a BST.</li> <li>Its right subtree is a BST.</li> <li>The maximum value in the left subtree is less than the current node's value.</li> <li>The current node's value is less than the minimum value in the right subtree.</li> <li>Track the size of the largest BST identified during the traversal.</li> </ul> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(n) (each node is visited once)</p> <p>Space Complexity: O(h), where h is the height of the tree (due to recursion stack)</p> </div> <div> <div>Solution</div> <p>Perform a post-order traversal of the binary tree. For each node, return a tuple (or object in some languages) that provides:</p> <ul style="list-style-type: none"> <li>Whether the subtree rooted at this node is a BST.</li> <li>The size of the subtree if it is a BST (or the size of the largest BST found within it).</li> <li>The minimum value in the subtree.</li> <li>The maximum value in the subtree.</li> </ul> <p>At each node, if both left and right children form valid BSTs and the current node's value fits the BST condition (greater than left's max and less than right's min), then the subtree rooted at the node is also a BST. Calculate its size and update the current global maximum if necessary. If the condition fails, return the maximum</p> </div> <div>Trees &amp; BSTp.2832</div> |
| <div>BST size found in its subtrees.</div> <div>#366 Find Leaves of Binary Tree <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Use a postorder traversal (bottom-up approach) to process the tree.</li> <li>Determine the "height" of each node (distance from the deepest leaf) where leaves have a height of 1.</li> <li>Group nodes by their computed height to simulate the layer-wise removal of leaves.</li> <li>The same node may become a leaf after its children are removed.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(n) — Each node is visited exactly once.</p> <p>Space Complexity: O(n) — Space used for the recursion stack and the result list.</p> </div> <div> <div>Solution</div> <p>The solution uses a recursive helper function that performs a postorder traversal of the tree. For each node, it computes its height as max(height of left, height of right) + 1. Nodes with the same height are collected together in a list. This idea mirrors the process of removing leaves level by level, where leaves (height 1) are removed first, then their parents become new leaves, and so on. Data Structures used:</p> </div> <div>Trees &amp; BSTp.2833</div>                                                                                                                             | <div>– An array or list to maintain the groups of node values by their height. Algorithmic Approach: – Use Depth-First Search (DFS) via recursion. – Postorder traversal ensures that leaf nodes (base cases) are processed before their parents. – Append the node's value to the correct list based on the computed height.</div> <div>#437 Path Sum III <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>A path can start and end at any nodes, as long as it goes downwards.</li> <li>Using DFS helps traverse the tree and explore all possible paths.</li> <li>A prefix sum technique with a hash map allows quick calculation of the sum of any subpath.</li> <li>Backtracking is used to remove a path's prefix sum when moving back up the recursion tree.</li> <li>Although a brute-force approach exists, it is inefficient compared to the prefix sum method.</li> </ul> </div> <div> <div>Space &amp; Time</div> <p>Time Complexity: O(n) on average (where n is the number of nodes), though worst-case can be O(n^2) in a skewed tree.</p> <p>Space Complexity: O(n) due to recursion stack and the hash map that stores prefix sums.</p> </div> <div> <div>Solution</div> </div> <div>Trees &amp; BSTp.2834</div>                                                                                                                            | <div>The solution employs a DFS traversal of the tree along with a hash map that records the frequency of prefix sums encountered so far. At each node, the current prefix sum is updated by adding the node's value. We then check the hash map for how many times (current prefix sum – targetSum) has occurred, since this value indicates the existence of a subpath summing to targetSum. We update the hash map with the current prefix sum, continue the DFS to traverse child nodes, and then backtrack by decrementing the current prefix sum's count before retreating to the parent node. This ensures that sums from other branches are not incorrectly combined.</div> <div>#545 Boundary of Binary Tree <span>[Medium]</span></div> <div> <div>Key Insights</div> <ul style="list-style-type: none"> <li>Break down the problem into three parts:</li> <li>Left boundary (excluding leaves).</li> <li>All leaves (do a DFS to find leaves).</li> <li>Right boundary (excluding leaves, then reverse the collected list).</li> <li>Special handling is needed when the tree doesn't have a left or right subtree.</li> <li>The root is always included and is never considered as a leaf, even if it has no children.</li> <li>Avoid duplication of leaf nodes in the left/right boundaries.</li> </ul> </div> <div>Trees &amp; BSTp.2835</div>                                                                 |

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| <div> <div>Space &amp; Time</div> <div> Time Complexity: <math>O(n)</math> – Every node is visited at most once.<br/> Space Complexity: <math>O(n)</math> – <math>O(n)</math> space is used for the output list and recursion stack (in worst case tree is skewed). </div> </div> <div> <div>Solution</div> <div> The solution uses a Depth-First Search (DFS) strategy to traverse the tree and collect the boundary nodes in three steps:<br/> – For the left boundary, start at root.left and traverse down, adding nodes (skipping leaves).<br/> – For the leaves, perform a DFS from the root and add any node that has no children.<br/> – For the right boundary, start at root.right and traverse down, adding nodes (skipping leaves).<br/> Since the right boundary must be added in reverse order, we reverse the collected list after traversal.<br/> – Finally, concatenate these lists in the order: root, left boundary, leaves, reversed right boundary.<br/> The approach makes use of helper functions to clearly separate the logic for gathering left boundary, leaves, and right boundary. This modular approach simplifies reasoning and avoids edge case mistakes such as including duplicate leaf nodes. </div> </div> <div> <div>#549 Binary Tree Longest Consecutive Sequence II</div> <div>[Medium]</div> </div> <div> <div>Trees &amp; BST</div> <div>p.2836</div> </div>             | <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Use Depth-First Search (DFS) to visit every node.</li> <li>For each node, compute: <ul style="list-style-type: none"> <li>The longest increasing consecutive path starting from that node.</li> <li>The longest decreasing consecutive path starting from that node.</li> </ul> </li> <li>Combine the increasing and decreasing paths (subtracting one to avoid double counting the current node) to potentially form a longer consecutive sequence through the node.</li> <li>Update a global maximum to record the longest path seen.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div> Time Complexity: <math>O(n)</math>, where n is the number of nodes since every node is visited once.<br/> Space Complexity: <math>O(h)</math>, where h is the height of the tree, due to the recursion stack. </div> </div> <div> <div>Solution</div> <div> The solution employs a DFS traversal. At every node, we calculate two values:<br/> – inc – the length of the longest increasing consecutive sequence starting from that node.<br/> – dec – the length of the longest decreasing consecutive sequence starting from that node.<br/> For each child of the current node: </div> </div> <div> <div>Trees &amp; BST</div> <div>p.2837</div> </div>                                                                                                | <div> <div>– If the child's value is node.val + 1, update the increasing sequence. – If the child's value is node.val – 1, update the decreasing sequence.</div> <div>Then, the longest path through the current node is inc + dec – 1 (subtracting one to avoid counting the current node twice). A global variable maintains the maximum length encountered. This process works regardless of the path direction because it combines both increasing and decreasing paths.</div> </div> <div> <div>#582 Kill Process</div> <div>[Medium]</div> </div> <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Build a mapping (dictionary/hash table) from a parent process ID to its list of child process IDs.</li> <li>Use Depth-First Search (DFS) or Breadth-First Search (BFS) starting from the kill process to traverse the tree and collect all descendant processes.</li> <li>The problem requires handling a tree structure that may have multiple children per node.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div> Time Complexity: <math>O(n)</math>, where n is the number of processes (both building the mapping and the tree traversal take linear time).<br/> Space Complexity: <math>O(n)</math>, for storing the mapping and the traversal queue/stack. </div> </div> <div> <div>Solution</div> <div></div> </div> <div> <div>Trees &amp; BST</div> <div>p.2838</div> </div> |
| <div> <div>We first construct a mapping from each process's parent ID to its list of child IDs. This allows us to quickly access all children of any process. Once the mapping is built, we perform a tree traversal (either BFS or DFS) starting from the process to kill. During the traversal, we record every process encountered. This collected list represents all processes that will be terminated. The data structures used are a dictionary (for the mapping) and either a stack (for DFS) or a queue (for BFS), making the solution efficient and straightforward.</div> </div> <div> <div>#662 Maximum Width of Binary Tree</div> <div>[Medium]</div> </div> <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Perform a level order traversal (BFS) of the tree.</li> <li>Assign an index to each node as if the tree were complete (e.g., root is index 1, left child is 2</li> <li>i, right child is 2</li> <li>i+1).</li> <li>For each level, compute the width as (last_index – first_index + 1).</li> <li>Using indices allows capturing gaps between nodes due to nulls in between.</li> <li>Edge cases include handling very skewed trees.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div> Time Complexity: <math>O(N)</math>, where N is the number of nodes in the tree. </div> </div> <div> <div>Trees &amp; BST</div> <div>p.2839</div> </div> | <div> <div>Space Complexity: <math>O(N)</math>, due to the queue used for BFS which in the worst case holds nodes of the last level.</div> </div> <div> <div>Solution</div> <div> We use a Breadth-First Search (BFS) approach with a queue to traverse the tree level by level. At each level, we pair each node with an index representing its position assuming a complete binary tree. This allows us to compute the width of each level using the formula: width = last_index – first_index + 1. We then track the maximum width seen throughout the traversal. The use of indices handles cases where there are null gaps between nodes. </div> </div> <div> <div>#863 All Nodes Distance K in Binary Tree</div> <div>[Medium]</div> </div> <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Convert the binary tree into an undirected graph so that we can easily traverse both down to the children and up to the parent.</li> <li>Use a DFS or simple recursive traversal to create an adjacency list representing the graph.</li> <li>Perform a BFS starting from the target node to explore nodes level-by-level until reaching distance k.</li> <li>Use a visited set (or equivalent) to avoid revisiting nodes and to prevent cycles.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div></div> </div> <div> <div>Trees &amp; BST</div> <div>p.2840</div> </div>                         | <div> <div>Time Complexity: <math>O(N)</math>, where N is the number of nodes in the tree. Each node is visited once when constructing the graph and once in the BFS.</div> <div>Space Complexity: <math>O(N)</math>, for storing the graph and the additional data structures used in BFS and DFS.</div> </div> <div> <div>Solution</div> <div> The solution involves two main steps:<br/> – Build an undirected graph representation of the binary tree using a DFS. For each node, add the node's value as a key in an adjacency list and connect it with its children (if any). This creates bidirectional links between parent and child.<br/> – Use a BFS starting from the target node and explore nodes level-by-level. Keep a counter for the current distance, and once the counter reaches k, return all node values found at that level. To avoid processing nodes more than once, maintain a set of visited nodes. </div> </div> <div> <div>#1120 Maximum Average Subtree</div> <div>[Medium]</div> </div> <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>Use a depth-first search (DFS) to traverse every subtree.</li> <li>At each node, compute the sum and count of nodes in its subtree.</li> <li>Calculate the average for the current subtree and update a global maximum if needed.</li> </ul> </div> </div> <div> <div>Trees &amp; BST</div> <div>p.2841</div> </div>                  |
| <div> <div> <ul style="list-style-type: none"> <li>Processing every node exactly once yields an efficient solution.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div> Time Complexity: <math>O(n)</math>, where n is the number of nodes (each node is visited once).<br/> Space Complexity: <math>O(h)</math>, where h is the height of the tree (space for recursion stack). </div> </div> <div> <div>Solution</div> <div> We perform a post-order DFS traversal, calculating at each node both the cumulative sum and the count of nodes in its subtree. This enables us to compute the average value in constant time for any subtree. We maintain a global variable to track the maximum average encountered during the traversal. The primary technique is recursive DFS combined with tuple (or pair) aggregation. The simplicity of the recursion and in-place aggregation ensures an efficient and clear solution. </div> </div> <div> <div>#1490 Clone N-ary Tree</div> <div>[Medium]</div> </div> <div> <div>Key Insights</div> <div> <ul style="list-style-type: none"> <li>The tree is N-ary, meaning every node can have multiple children.</li> <li>A deep copy requires creating new nodes for each node in the tree.</li> </ul> </div> </div> <div> <div>Trees &amp; BST</div> <div>p.2842</div> </div>                                                                           | <div> <div> <ul style="list-style-type: none"> <li>Both DFS (recursive) or BFS (iterative) traversals can be used to clone the tree.</li> <li>The overall time complexity is <math>O(N)</math> since each node is visited once.</li> <li>We can follow similar cloning techniques used in cloning graphs, using either recursion or an auxiliary data structure like a queue.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div> Time Complexity: <math>O(N)</math>, where N is the number of nodes in the tree, since each node is visited exactly once.<br/> Space Complexity: <math>O(N)</math>, accounting for the recursive call stack (or an explicit queue in BFS) and the space needed to store the clone. </div> </div> <div> <div>Solution</div> <div> We perform a DFS traversal starting from the root node. For each node we encounter, we:<br/> – Create a new node with the same value.<br/> – Recursively clone all its children and attach the cloned children to the new node. This approach ensures that every original node is copied into a new node and that the parent-child relationships are preserved. Edge cases such as an empty tree (null root) are also handled by returning null immediately. </div> </div> <div> <div>#1522 Diameter of N-Ary Tree</div> <div>[Medium]</div> </div> <div> <div>Key Insights</div> <div></div> </div> <div> <div>Trees &amp; BST</div> <div>p.2843</div> </div> | <div> <div> <ul style="list-style-type: none"> <li>The diameter of the tree can be found by considering the two largest depths from any node among its children.</li> <li>Use a depth-first search (DFS) to compute the maximum depth from each node.</li> <li>For each node, the potential diameter is the sum of the two longest depths among its children.</li> <li>Update a global variable tracking the maximum diameter as you recurse through the tree.</li> </ul> </div> </div> <div> <div>Space &amp; Time</div> <div> Time Complexity: <math>O(n)</math>, where n is the number of nodes, since each node is visited once.<br/> Space Complexity: <math>O(h)</math>, where h is the height of the tree, due to the recursion stack (worst-case <math>O(n)</math> for a skewed tree). </div> </div> <div> <div>Solution</div> <div> Use a DFS approach to traverse the tree. At each node, calculate the longest and second longest depth among its children. The sum of these two depths gives the candidate diameter through that node. Maintain a global variable to track the maximum diameter found so far. Return the maximum depth from the node (i.e., one plus the largest depth among its children) to be used in parent's calculations. </div> </div> <div> <div>Trees &amp; BST</div> <div>p.2844</div> </div>                                                                                                      |

#1650 Lowest Common Ancestor of a Binary Tree III [Medium]

Key Insights

- Each node has a parent pointer, so you can traverse from any node to the root.
- By determining the depth of each node, you can “align” the two nodes to the same depth.
- When both nodes are aligned, traverse upward simultaneously until they meet.
- This approach guarantees finding the LCA with O(h) time complexity, where h is the height of the tree.

Space & Time

Time Complexity: O(h), where h is the height of the tree.  
Space Complexity: O(1)

Solution

We first compute the depth of both nodes by moving up their parent pointers. If one node is deeper, move it upward until both nodes are at the same level. Then, move both nodes upward one step at a time until they meet; the meeting point is the lowest common ancestor. This approach uses constant extra space and leverages the parent pointer for an efficient solution.

#124 Binary Tree Maximum Path Sum [Hard]

Key Insights

- Use Depth-First Search (DFS) to explore the tree in a post-order fashion.
- For each node, calculate the maximum gain including that node and optionally one of its subtrees.
- A node's best contribution to its parent can be either just its own value or its value plus the maximum gain from either the left or right child.
- Update a global maximum path sum that considers the possibility of combining both left and right gains with the current node.
- Handle negative values by comparing gains against 0 to avoid decreasing the path sum.

Space & Time

Time Complexity: O(n), where n is the number of nodes in the tree, since each node is visited exactly once.  
Space Complexity: O(n) in the worst-case scenario due to the recursion stack (O(H) where H is the height of the tree, which can be O(n) for skewed trees).

Solution

We use a recursive DFS approach (post-order traversal) to compute the maximum gain from each subtree. For every node:  
– Recursively compute the maximum gain from the left and right child, ensuring that negative gains are treated as 0. – The maximum gain that can be provided to the node's parent is the node's value plus the larger gain

from its left or right child. – Simultaneously, the potential maximum path sum including the current node as the highest (or root) node is the node's value plus both left and right gains. – Update a global variable that holds the maximum path sum found so far. – Return the maximum gain (node value plus one best child) to be used by the node's parent.  
This strategy ensures that every possible path sum is considered while avoiding double-counting nodes.

#297 Serialize and Deserialize Binary Tree [Hard]

Key Insights

- The tree can be traversed using either depth-first search (DFS) or breadth-first search (BFS).
- Using BFS (level-order traversal) simplifies reconstruction since nodes are processed level-by-level.
- Placeholder markers (e.g., "null") are used to indicate missing children.
- Both serialization and deserialization processes should handle empty trees and edge cases gracefully.
- Ensuring a one-to-one mapping between the tree structure and the serialized string is critical.

Space & Time

Time Complexity: O(n) for both serialization and deserialization, where n is the number of nodes in the

Trees & BST p.2845

Trees & BST p.2846

Trees & BST p.2847

tree.  
Space Complexity: O(n) due to the storage used for the serialized string and additional data structures (queue) during processing.

Solution

This solution uses level-order traversal (BFS) for both serialization and deserialization. During serialization, each node is processed sequentially with missing children recorded as "null". For deserialization, the string is split by commas, and a queue is used to rebuild the tree level by level. The root is created from the first token, then subsequent tokens are assigned as left and right children accordingly, ensuring an accurate reconstruction of the original tree.

Trees & BST p.2848